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Lecture Notes on Quantum Field Theory

Alexander Cassem[a]

[a] *Institute of Cosmology, Tufts University, Medford, MA 02155, USA*

E-mail: alexander.cassem@tufts.edu

Abstract: Presented here are lecture notes I took when taking quantum field theory (QFT) 1 and 2 (253a and 253b) at Harvard University taught by Xi Yin, as well as QFT 3 (253c) by Xi Yin, but this was a self-study not taken for credit. However, I also decided to attach my *other* QFT 3 notes (253c') on axion physics by Matthew Reece (which I did attend), as well as my string theory notes taught by Daniel Jafferis (287A).

Update: as of January 2026, I have finished QFT 1 and QFT 2 notes with the psets included while the axions, CFT, and string notes are put together, they just need to be written; also some appendices need to be included such as Morse theory, index theorems in QM, updated integral table, some useful theorems, and some notes on classical field theory.

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Preface 1: What Are These Notes?

I wanted to give a quick preface as to what these notes on, "quantum field theory" are and are not. These notes are an *extremely* theoretical and in-depth exposition into a "field theoretic description of quantum mechanics." Therefore, they are *not* a particle physics description or, let's say, an experimental-practical view of QFT. Yet, these notes are still very much applicable and use-able for almost every situation (with enough "elbow-grease" applied). It should also be noted that these QFT notes are *not* the standard approach. These notes *do not* follow a traditional "historical approach" to learning QFT, and instead we dive into the material via the best known formulation of QFTs: the path integral. The canonical quantization method of QFT will only be mentioned in passing.

For QFT 1 and 2, the primary textbook that was used in the course was Weinberg's QFT volume 1 and 2 [1, 2]. However, a nice complementary book that is easier to read is Schwartz's QFT textbook [3]. There is no textbook reference for QFT 3 (either version), but for string theory the primary textbook was Polchinski's string theory volumes 1 and 2 [4, 5]. I have tried my best to fill-in derivations where necessary such that they add clarity to the derivations and explanations (so, those side-margin notes/derivations taken during lecture), as well as I have included all of the problem sets (psets) from the course (with some updates now in hindsight compared to my original solutions). The psets here should be taken with care, and should be thought of as an *extension* of the lecture notes and not just some "calculational practice," (well, sometimes but typically not).

Now onto the *real* preface for these notes introducing the setup for QFT 1 and 2.

Question: What is Quantum Field Theory?

An question you may have when entering a QFT course is, quite literally, "what is a qft?" And you may find the answer lack-luster since a qft is, "quantum mechanics (QM) plus locality." That's it. And QFT is essentially broken up into two categories: local QFT's and effective field theory:

1. Local, "ultraviolet (UV) complete," QFT:
   (a) a QM theory with Poincaré symmetric and that is necessarily infinity many degrees of freedom.
   (b) The theory contains local "field" operators.
   (c) The theory obeys microcausality.
2. Effective field theory:
   (a) EFTs are QFTs that are defined perturbatively based on a path integral over a space of fields.
   (b) The EFT captures the long-distance, low-energy observables.

Examples of local QFTs are ϕ^4 theory in $D = 2, 3$, the statistical Ising model in $D = 2, 3$ near criticality, and Yang-Mills/quantum chromodynamics (QCD). Examples of EFTs side of QFTs are ϕ^4 theory in $D = 4$, quantum electrodynamics (QED), the entire standard model (SM), chiral perturbation theory, and general relativity (GR). In QFT 1, we will be looking at ϕ^4 theory in $D = 2, 3, 4$, and QED, and look at the rest in QFT 2 except for GR. Among the examples listed under EFTs, ϕ^4 in $D = 4$, QED, and the SM are all considered "renormalizable" theories, while chiral perturbation theory and GR are not. With all of this in mind, here is the plan for QFT 1:

1. Lagrangian formulation of QM
   (a) path integral and regularization thereof
   (b) perturbation theory and Feynamn diagrams
   (c) renormalization and counter terms
2. Relativistic particles and fields
   (a) ϕ^4 theory
   (b) Green functions
   (c) asymptotic states
   (d) S-matrix and LSZ reduction
3. Particles and fields with spin
   (a) classification of relativistic particles

(b) fermions and gauge bosons

(c) QED

(d) applications: electron g-factor and Lamb shift

1 Quantum Field Theory 1 (253a)

Prelude into QFT 1

A question we can ask is, "why do we need a "field theory" to describe QM of particles with relativistic symmetry?" Well, let us consider an Hilbert space $\mathcal{H}$ that contains a vacuum state $|\Omega\rangle$ such that an Hamiltonian obeys $H|\Omega\rangle = 0$ on the vacuum. In this Hilbert space, there are single particle states denotes $|\mathbf{p}\rangle$ where the momentum here is in D-dimensions $p^\mu \equiv (p^0, \mathbf{p})$. The Hamiltonian acting on one particle states is just a Hamiltonian for a single relativistic particle

$$H|\mathbf{p}\rangle = \sqrt{\mathbf{p}^2 + m^2}\,|\mathbf{p}\rangle\,. \tag{1.1}$$

For a multi-particle non-interacting states, the Hamiltonian is then just a sum of relativistic particle energies with states denotes by $|\mathbf{p}_1, \ldots, \mathbf{p}_i\rangle$ and Hamiltonian

$$H = \sum_{i=1}^{n} \sqrt{\mathbf{p}^2 + m^2}. \tag{1.2}$$

In an equivalent manner, we can introduce creation and annhilation operators $a^\dagger_{\mathbf{p}}$ and $a_{\mathbf{p}}$ such that one particle states are generated from the vacuum $|\mathbf{p}\rangle = a^\dagger_{\mathbf{p}}|\Omega\rangle$, and two particle states are also $|\mathbf{p}_1, \mathbf{p}_2\rangle = a^\dagger_{\mathbf{p}_1} a^\dagger_{\mathbf{p}_2}|\Omega\rangle$, and so on and so on. With the introduction of the operators, they obey the usual commutator algebra $[a, a] = [a^\dagger, a^\dagger] = 0$ and[1] $[a_{\mathbf{p}}, a^\dagger_{\mathbf{p}'}] = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}')$. With these operators, you can show that the 1-particle states obey $\langle \mathbf{p}|\mathbf{p}'\rangle = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}')$. Then, with the combination of the Hamiltonain and the creation and annhilation algebra, we can re-write the Hamiltonian as

$$H = \int d^{D-1}\mathbf{p}\,\sqrt{\mathbf{p}^2 + m^2}\,a^\dagger_{\mathbf{p}} a_{\mathbf{p}}. \tag{1.3}$$

This Hamiltonian is a QM system of free relativistic particles. But now how do we include interactions between (relativistic) particles? Well, we can "expand" the Hamiltonian into the free part and the interaction part as $H = H_0 + H_{\text{int}}$ where the interaction Hamiltonain can contain operators of the form $H_{\text{int}} \supset \{a^\dagger a^\dagger a, aaa^\dagger, \cdots\}$. However, it is not easy to find an interacting Hamiltonian that respects relativistic symmetry! The primary issue? Causality.

[1] Note that the typical factors of $(2\pi)^3\omega_{\mathbf{p}}$ and such are just normalization conventions which we don't address at this stage.

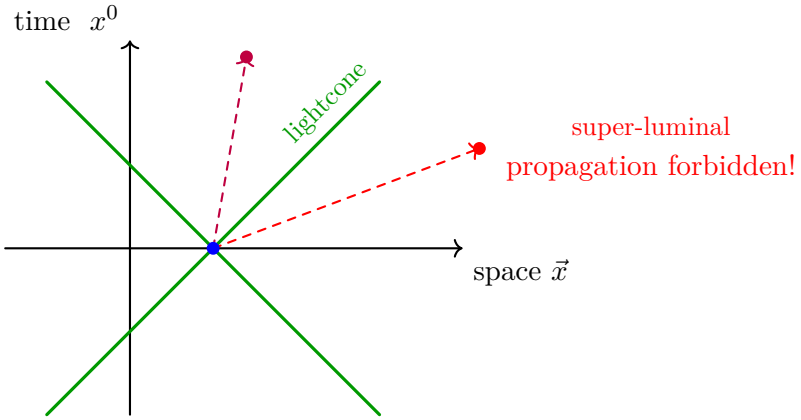


Figure 1. Example of subluminal and superluminal propagation in a lightcone.

As an example of what we mean by causality, take a look at 1 that has both subluminal and superluminal propagation. In a generic QM system, signal propagation can be instantaneous, and appear as the red signal above. To remedy this, we could imagine that in a relativistic system, we could expect local disturbances to be represented by a "field operator" denotes as $\hat{\phi}(x)$ where x here is really $x^\mu = (x^0, \mathbf{x})$ and now in the relativistic case, the position here is *just* a set of parameters, not operators themselves. The same field operator at two different spacetime points then can be related by Poincaré symmetry, the underlying symmetry that makes systems relativistic. Poincaré symmetry is a combination of the Lorentz symmetry we know and love already plus translations, which are encoded in the following transformation

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu \tag{1.4}$$

where the Lorentz matrices obeys the usual $\eta^{\alpha\beta}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta = \eta^{\mu\nu}$, and the Minkowski metric $\eta^{\mu\nu}$ has the mostly plus signature, namely $\eta^{\mu\nu} = \mathrm{diag}(-1, +1, +1, \ldots, +1)$ in D-dimensions. In QM, we know that symmetries are related to the operators of the system by putting them into a unitary representation, and so there is a corresponding unitary operator that holds the Poincaré symmetry $U(\Lambda, a)$ such that

$$\hat{\phi}(\Lambda x + a) = U(\Lambda, a)\hat{\phi}(x)(U(\Lambda, a))^{-1}. \tag{1.5}$$

It is sufficient enough to study the infinitesimal version to which we can expand the Lorentz and translation symmetry as $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$ and $a^\mu = \epsilon^\mu$ where you can think of $\omega^\mu{}_\nu$ and a^μ as small perturbations that we shall only keep to *first order*. In this sense, we can expand/write the unitary operator as

$$U(\Lambda, a) = 1 - i\epsilon^\mu \hat{P}_\mu + \frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu} \tag{1.6}$$

where $\hat{P}_\mu$ and $\hat{J}^{\mu\nu}$ correspond to the (relativistic) energy-momentum and boost plus angular momentum operators. You may be wondering where these operators came from, and it comes from the fact that the set of physical operators that correspond to translations and

Lorentz transformations is extremely limit, and is essentially only those two. Thus, the only index-contracted unitary operator we can construct is of the form given in (1.6). The correspond (what one may call) group algebra, is the following

$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda a + a') \tag{1.7}$$

$$[P, P] = 0 \tag{1.8}$$

$$[P^\mu, J^{\rho\sigma}] = -i\left(\eta^{\mu\rho}P^\sigma - (\rho \leftrightarrow \sigma)\right) \tag{1.9}$$

$$[J^{\mu\nu}, J^{\rho\sigma}] = -i\left(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - (\rho \leftrightarrow \sigma)\right). \tag{1.10}$$

In our QM-relativistic system, to enforce causality, we instead enforce *microcausality*,

$$\left[\hat{\phi}(x), \hat{\phi}(y)\right] = 0 \tag{1.11}$$

which is true whenever x, y are spacelike separated as can be originally seen in 1, or in other words, spacelike separated is when

$$(x-y)^2 \equiv -(x^0 - y^0)^2 + (\mathbf{x} - \mathbf{y})^2 > 0. \tag{1.12}$$

But does our model of free relativistic particles obey microcausality? Well, to check this, we need to first ask, does our field operator even exist though? And the answer is[2] yes!

$$\hat{\phi}(x) = \int \frac{d^{D-1}\mathbf{p}}{\sqrt{(2\pi)^{D-1}2\omega_{\mathbf{p}}}}\left(a_{\mathbf{p}}e^{ip\cdot x} + a^\dagger_{\mathbf{p}}e^{-ip\cdot x}\right). \tag{1.13}$$

The "dot-product" here in the mostly plus notation is written as $p\cdot x = -\omega_{\mathbf{p}}x^0 + \mathbf{p}\cdot\mathbf{x}$. Now, we can check microcauslity directly utilizing the algebra of the creation and annhilation operators

$$\left[\hat{\phi}(x), \hat{\phi}(y)\right] = \int \frac{d^{D-1}\mathbf{p}}{\sqrt{(2\pi)^{D-1}2\omega_{\mathbf{p}}}}\left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)}\right] \tag{1.14}$$

$$= \int \frac{d^D p^\mu}{(2\pi)^{D-1}}\theta(p^0)\delta(p^2+m^2)\left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)}\right] \tag{1.15}$$

where we actually "re-did"the time-integral namely $(2\omega_{\mathbf{p}})^{-1} = \int dp^0\,\theta(p^0)\delta(p^2+m^2)$. The measure here including the step-function and delta function is invariant under $p^\mu \to \Lambda^\mu{}_\nu p^\nu$ while the entire integrand plus measure is invariant under $x \to \Lambda x$ and $y \to \Lambda y$. The result is then only a function of $(x-y)^2$. Now, *if* $(x-y)^2 > 0$ then without loss of generality (WLOG), we can take $x^0 - y^0 = 0$ and $\mathbf{x} - \mathbf{y} \neq 0$. We also notice that the integrand is odd under the exchange of $\mathbf{p} \to -\mathbf{p}$, and so the integral vanishes. Thus, the commutator of the field operator expansion vanishes and obeys microcausality for when $(x-y)^2 > 0$. It is also useful to verify the following for practice

$$U(\Lambda, a)\hat{\phi}(x)(U(\Lambda, a))^{-1} = \hat{\phi}(\Lambda x + a). \tag{1.16}$$

And so after all of this analysis, we are left with what we must postulate are the properties for a QM relativistic system that obeys causality:

[2] Again, this is just a preface for what is to come, and everything will be shown later

1. Must contain an Hilbert space $\mathcal{H}$
2. Is invariant under Poincaré symmetry $U(\Lambda, a)$
3. Contains a Poincaré invariant vacuum $|\Omega\rangle$
4. there exists local "field" operators $\hat{\phi}(x)$ that obey Poincaré covariance and micro-causality

These postulates also hold for interacting relativistic particles as well! But then how do we construct such theories? It will prove useful to work with a formalism in which Poincaré symmetry is manifest, and thus we finally start our journey by learning the path integral method of quantization.

1.1 Lagrangian Formalism

1.1.1 Lagrangian Formulation of QM

To start studying the path integral quantization method, we will start with a quantum mechanical approach. But, it is best to remember from classical mechanics how we discuss states and equations of motion, namely the two different ways, Lagrangian vs Hamiltonian:

1. Lagrangian:
   (a) The state for a Lagrangian description is described by q and $\dot{q}$.
   (b) The equations of motion are found by extremizing the action $S = \int dt\, L(q, \dot{q})$.
2. Hamiltonain:
   (a) The state for a Hamltonian is described by q and p.
   (b) The equations of motion are found by solving/writing down $\dot{f} = \{H, f\}_{\text{Poisson}}$ where $f \equiv f(q, p, t)$. For example, $\dot{q} = \partial H/\partial p$ and $\dot{p} = -\partial H/\partial q$.

The relation between the two methods for studying a classical system are related by $H(p, q) = p\dot{q} - L(q, \dot{q})|_{\partial L/\partial \dot{q} = p}$. In quantum mechanics, we have a similar dichotmoy

1. Lagrangian:
   (a) The state of a QM system in the Lagrangian formalism are given by $S = \int dt\, L(q, \dot{q})$.
   (b) The time evolution of the states are given by the expectation value of an operator $U(T)$
   $$\langle q_f | U(T) | q_i \rangle = \int [Dq] \Big|_{q(0)=q_i}^{q(T)=q_f} e^{\frac{i}{\hbar} S} \tag{1.17}$$
2. Hamiltonian:
   (a) The state of a Hamiltonian is part of Hilbert space $|\psi\rangle \in \mathcal{H}$ with the Hamiltonain being promoted to an operator $H(p, q) \to \hat{H}$.

(b) The time evolution for states written in terms of the Hamiltonian are given by the unitary operator $U(t) = e^{-\frac{i}{\hbar}\hat{H}t}$ and, thanks to Heisenberg, states evolve as

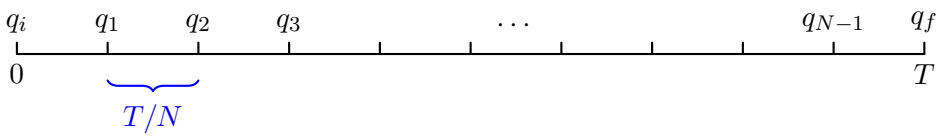
$$\mathcal{O} \to (U(t))^{-1}\mathcal{O}U(t) \equiv \mathcal{O}(t). \tag{1.18}$$

(c) The time evolution then with respect to the Hamiltonian for an operator is given by the following

$$\dot{\mathcal{O}}(t) = \frac{i}{\hbar}\left[\hat{H}, \mathcal{O}(t)\right]. \tag{1.19}$$

In the Lagrangian formulation, practically it's entire existent lies in understanding what exactly is the measure $[Dq]|_{q(0)=q_i}^{q(T)=q_f}$. Much of the subtleties with the Lagrangian formulation of QM have to do with the path integral measure $[Dq]$. To understand where it came from, let us revisit it's derivation.

We first consider the time evolution over some time T of an operator $U(T)$ with the time evolution being separated into N steps in the following manner:



The time evolution of the operator is then written as

$$\begin{aligned}
\langle q_f|\, U(T)\, |q_i\rangle &= \int d^D q_1 \cdots d^D q_{N-1}\, \langle q_f|\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_{N-1}\rangle \cdots \langle q_1|\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_i\rangle \\
&= \int \prod_{n=1}^{N-1} d^D q_n \prod_{n=0}^{N-1} \langle q_{n+1}|\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_n\rangle \\
&= \int \prod_{n=1}^{N-1} d^D q_n \prod_{n=0}^{N-1} \frac{d^D p_n}{(2\pi\hbar)^D} \exp\left(\frac{i}{\hbar}\sum_{n=0}^{N-1}\left[p_n \cdot (q_{n+1} - q_n) - H(q_n, p_n)\frac{T}{N} + \mathcal{O}(\hbar, T^2/N^2)\right]\right)
\end{aligned} \tag{1.20}$$

where in the second line of (1.20) the bra-ket expression was evaluated by inserting a factor of unity in momentum, and evaluating the exponential on the left ket, namely

$$\langle q_{n+1}|\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_n\rangle = \int \frac{d^D p_n}{(2\pi\hbar)^D} \langle q_{n+1}|p_n\rangle\, |p_n\rangle\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_n\rangle \tag{1.21}$$

where $\langle q_{n+1}|p_n\rangle = \exp\left(\frac{i}{\hbar}p_n \cdot q_{n+1}\right)$ and

$$\langle p_n|\, e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}}\, |q_n\rangle \approx e^{-\frac{i}{\hbar}H(q_n,p_n)\frac{T}{N}} e^{-\frac{i}{\hbar}p_n q_n}. \tag{1.22}$$

In (1.20), we can "formally" take the $N \to \infty$ "limit" to find

$$\langle q_f|\, U(T)\, |q_i\rangle = \int Dq\, Dp\, e^{\frac{i}{\hbar}\int_0^T dt(p\cdot\dot{q} - H(q,p))}. \tag{1.23}$$

There is a lot to unpack in (1.23) that you may have not seen on your first pass with learning the path integral. The differentials here $DqDp$ are only defined via a regularization technique such as discretizing time, otherwise they are not well defined[3]. The Hamiltonain in the exponential in (1.23) contains $\mathcal{O}(\hbar)$ ambiguities such as things called "counter-terms." These ambiguities are tied to the scheme of regularization used in the path integral.

Side note: as an example of an alternative regularization scheme, we can consider *time-lattice* regularization in which the integral over time in the exponential changes as

$$\int_0^T dt\, p\dot{q} \to \sum_{n=0}^{N-1} p_n \cdot (q_{n+1} - q_n) \tag{1.24}$$

to which we compare to $\sum_{n=0}^{N-1} p_{n+1} \cdot (q_{n+1} - q_n)$. Then, the exact representation of the time evolution is then given by (in the integrand)

$$\cdots \langle q_{n+1}|p_n\rangle \langle p_n| e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}} |q_n\rangle \cdots \tag{1.25}$$

to which the Hamiltonian in the exponential is evaluated as $\langle p_n| \hat{H} |q_n\rangle = H(p_n, q_n)\langle p_n|q_n\rangle$. But then, consider the $n+1$ shift in the ket namely

$$\cdots \langle p_{n+1}|q_n\rangle \langle q_n| e^{-\frac{i}{\hbar}\hat{H}\frac{T}{N}} |p_n\rangle \cdots \tag{1.26}$$

where in this case, the Hamiltonian is evaluated as $\langle q_n| \hat{H} |q_n\rangle = H'(p_n, q_n)\langle q_n|p_n\rangle$. Now, we have two Hamiltonian that can differ at order $\mathcal{O}(\hbar)$.

Coming back to (1.23), if $H(q,p)$ is quadratic in p such as if the Hamiltonian has the form $H(q,p) = p^2/2m + V(q)$, then we can integrate out $p(t)$ as a Gaussian integral leaving behind[4]

$$\langle q_f| U(T) |q_i\rangle = \int [Dq]\, e^{\frac{i}{\hbar}\int_0^T dt\,(p\cdot q - H)}\Big|_{\dot{q}=\partial H/\partial p} \tag{1.27}$$

where the measure now denotes the absorption of the determinant that arises from integrating out $p(t)$, and we can see that we recover the Lagrangian in the argument of the exponential.

Side note: On more general grounds, the Lagrangian can be written in any generalized coordinates which typically takes the following form

$$L = \frac{1}{2}G_{\alpha\beta}(q)\dot{q}^\alpha\dot{q}^\beta - V(q) \leftrightarrow H = \frac{1}{2}(G(q)^{-1})^{\alpha\beta}p_\alpha p_\beta + V(q) + \mathcal{O}(\hbar), \tag{1.28}$$

of which an example for would be rotating 3D rigid body. Inevitably when we put this Hamiltonian into the path integral and then integrate out the momentum, we find the following

$$\int Dp\, Dq\, e^{\frac{i}{\hbar}\int dt\,(p_\alpha\dot{q}^\alpha - H(p,q))} = \int [Dq]\, e^{\frac{i}{\hbar}\int dt\, L(q,\dot{q})} \tag{1.29}$$

[3]Again, we are just looking at QM for now, but things will naturally carry over much easier when we get to fields since the same subtleties will carry over, they just change form.

[4]There is an appendix to these notes that has common integrals we will be using such as (A.1).

where the measure is now

$$[Dq] \to \prod_{n=1}^{N} \frac{dq_n^\alpha}{\sqrt{2\pi i\hbar T/N}} \sqrt{\det G_{\alpha\beta}(q_n)}. \tag{1.30}$$

Coming back to (1.27), let us look at how the path integral is utilized in computing observables, such as correlation functions in QM. First, let us consider in the Hamiltonian formalism on the vacuum, a correlation function in some ground state $|0\rangle$ where we want to analyze the energy states of some system (where $\hat{H}\,|n\rangle = E_n\,|n\rangle$),

$$\begin{aligned}
G(t) &\equiv \langle 0|\,\hat{q}(t)\hat{q}(0)\,|0\rangle \\
&= \sum_n \langle 0|\,\hat{q}(t)\,|n\rangle\,\langle n|\,\hat{q}(0)\,|0\rangle && \left(\mathbb{1} = \sum_n |n\rangle\,\langle n|\right) \\
&= \sum_n \langle 0|\,e^{i\hat{H}t/\hbar}\hat{q}(0)e^{-i\hat{H}t/\hbar}\,|n\rangle\,\langle n|\,\hat{q}(0)\,|0\rangle && \left(\hat{q}(t) = e^{i\hat{H}t/\hbar}\hat{q}(0)e^{-i\hat{H}t/\hbar}\right) \\
&= \sum_n \langle 0|\,\hat{q}(0)\,|n\rangle\,\langle n|\,\hat{q}(0)\,|0\rangle\,e^{-i(E_n-E_0)t/\hbar} \\
&= \sum_n \rho_n e^{-i(E_n-E_0)t/\hbar}
\end{aligned} \tag{1.31}$$

where ρ_n is some number defined as $\rho_n \equiv \langle 0|\,\hat{q}(0)\,|n\rangle\,\langle n|\,\hat{q}(0)\,|0\rangle$. Notice that as $E_n - E_0 \geq 0$, we can analytically continue $G(t)$ as a function of complex t provided that $\mathrm{Im}t < 0$ such that the sum $\sum_n$ converges. Namely, we analytically continue such that to ensure convergence of the summation. This is known as the Paley-Weiner theorem. As an example of analytic continuation that will be prevelant and used throughout the entirety of these notes is called the *Wick rotation* to which it is so important and widely used, we give it it's own equation

$$t = i\tau \tag{1.32}$$

where $\tau \in \mathbb{R}_+$ (only the positive real numbers). Then, the correlation function $G(t)$ becomes

$$\langle 0|\,\hat{q}(t)\hat{q}(0)\,|0\rangle = \sum_n \langle 0|\,\hat{q}(0)\,|n\rangle\,\langle n|\,\hat{q}(0)\,|0\rangle\,e^{-(E_n-E_0)\tau/\hbar}. \tag{1.33}$$

To move forward to think of more generic states that we can probe other than the vacuum, denoted as $|\psi\rangle$, we will assume that there is a gap $E_1 - E_0 \geq 0$ and so we can replace

$$|0\rangle \to \lim_{T\to\infty} e^{-(H-E_0)T/\hbar}\,|\psi\rangle \tag{1.34}$$

that is defined up to an overall normalization for any generic state $|\psi\rangle$ (that has a non-zero overlap with $|0\rangle$). Then, the correlation function becomes[5]

$$\langle 0|\,\hat{q}(t)\hat{q}(0)\,|0\rangle = \lim_{T\to\infty} \frac{\langle\psi|\,e^{-\hat{H}T/\hbar}\hat{q}(t)\hat{q}(0)e^{-\hat{H}T/\hbar}\,|\psi\rangle}{\langle\psi|\,e^{-2\hat{H}T/\hbar}|\psi\rangle}. \tag{1.35}$$

Now, it is beneficial to pay attention to the time-path that is being taken here in the correlation function. Well, before we try to explain anything, let me present the graph first:

[5] The ending ket $|\psi\rangle$ should not be colored, I have yet to fix it [**AC:**] 12/21/2025.

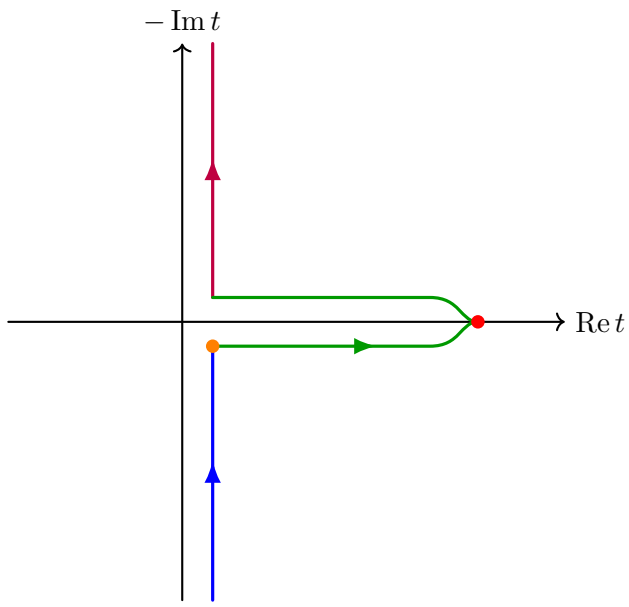
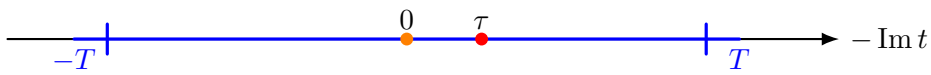


Figure 2. Time-path taken by the two-point correlation function.

The colors in (1.35) match with those drawn on the graph in figure 2. It should be clear now then how to think of where operators in QM when computing correlators live on the time-graph, and how we "hit them" in order to observe them. Notice that the $\hat{q}(t)$ is slightly off the real axis, and could still have an imaginary component which we could have placed anywhere, but for visualization, made it simple. However, we can alternatively think of the of the Wick rotated version

$$\begin{aligned} G(t=-i\tau) &= \langle -0| \, e^{H\tau/\hbar}\hat{q}(0)e^{-H\tau/\hbar}\hat{q}(0) \, |0\rangle \\ &= \lim_{T\to\infty} \frac{\langle \psi| \, e^{-H(T-\tau)/\hbar}\hat{q}(0)e^{-H\tau/\hbar}\hat{q}(0)e^{-HT/\hbar}}{\langle \psi| \, e^{-2HT/\hbar} \, |\psi\rangle}. \end{aligned} \tag{1.36}$$

In this case, all of the operators lie solely along the same line such as:



Notice that now, all of the operators lie along the same line such that when we integrate, there are no ambiguities. But now let us do the same calculation but from the point of view of the path integral. This is written as

$$\langle 0| \, \hat{q}(t)\hat{q}(0) \, |0\rangle = \lim_{(\operatorname{Im} t_f, \operatorname{Im} t_i)\to(-\infty,+\infty)} \frac{\int [Dq] \, e^{\frac{i}{\hbar}\int_{t_i}^{t_f} dt \, L[q,\dot{q}]} q(t)q(0)}{\int [Dq] \, e^{\frac{i}{\hbar}\int_{t_i}^{t_f} dt \, L[q,\dot{q}]}} \tag{1.37}$$

where, we want to emphasize again, that in the path integral, $q(t)$'s are now classical variables and not "hatted" operators. We now perform the same analytic contiuation or Wick rotation via $q(t) \to q(\tau)$ with $t = i\tau$ (1.32), and now we must Wick rotate the

Lagrangian which contains time derivatives namely $\partial_t q = i\partial_\tau q$, and we now *define* the Wick rotated Lagrangian or *Euclidean Lagrangian* as

$$L[q, \dot{q}] \equiv -L^E[q, \partial_\tau q]. \tag{1.38}$$

In this manner, we then get the *Euclidean action*

$$e^{\frac{i}{\hbar}\int dt\, L[q,\dot{q}]} \to e^{-\frac{1}{\hbar}\int d\tau\, L^E[q,\partial_\tau q]} \tag{1.39}$$

where the exponential is now $\exp\Big(-S^E\Big)$. As an example, the QM Lagrangian turns into

$$L[q, \dot{q}] = \frac{1}{2}\dot{q}^2 - V(q) \to L^E[q, \partial_\tau q] = \frac{1}{2}(\partial_\tau q)^2 + V(q). \tag{1.40}$$

Something to notice is that S^E is decaying and damps any wildly fluctuating terms. But now how do we even evaluate the path integral in (1.37) since all we have done so far is analytically continue our variables...[6] The key to evaluating the path integral, as hinted at prior, is to find a *conveniently* regularized expression of the measure $[Dq]$. For example, what if we expand the variables $q(\tau)$ on a suitable basis of functions, such as a Fourier expansion $q(\tau) = \sum_n q_n f_n(\tau)$? Well then, the measure would become something like $[Dq] \rightsquigarrow \prod_{n=1}^{\infty} dq_n$? Well, let's try this with a specific example of a harmonic oscillator with Hamiltonian (setting $m = 1$)

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 \leftrightarrow L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}\omega^2 q^2. \tag{1.41}$$

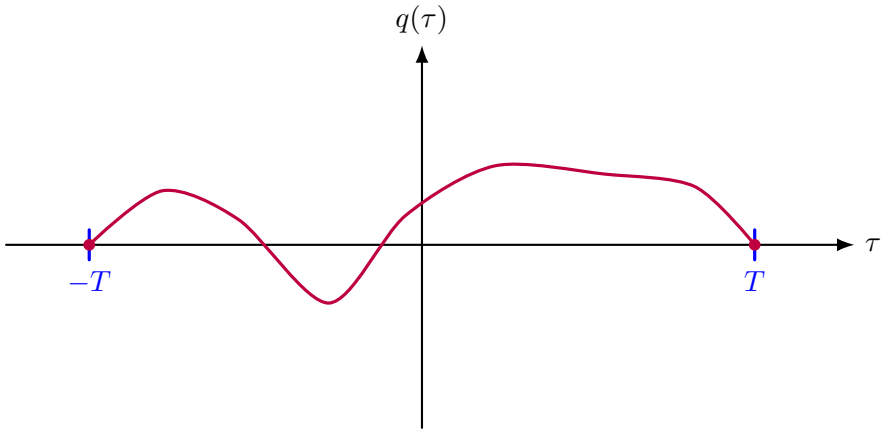
Let us Wick rotate this to obtain the Euclidean Lagrangian as (using (1.38))

$$L^E[q, \partial_\tau q] = \frac{1}{2}(\partial_\tau q)^2 + \frac{1}{2}\omega^2 q^2 \tag{1.42}$$

that will allow us to evaluate $\int [Dq] \exp\Big(-S^E/\hbar\Big)$. To do so, we must first choose a suitable boundary condition in imaginary time. From our sketch above that is below (1.36), it makes sense to choose $q(-T) = q(T) = 0$ (notice that the Hamiltonian informed us of where to place the boundary for our variables $q(T)$). As a sketch of the integration path our path integral could take[7] consider the graph below. Notice that this is for a *single* $q(\tau)$ that could behave wildly different for any of the possible *infinite* number of variables we are integrating over. Hence, the reasoning for wanting a suitable basis of functions such as a Fourier expansion.

[6] Note that this Wick rotation carries extremely naturally to fields, and actually almost nothing changes in our calculations except dimensionality showing how nice the path integral is.

[7] That sounds funny hehe.



Given our boundary condition, for however we expand our variables $q(\tau)$, the basis on which we expand must also obey the boundary condition as well of course. So, if we choose $q(\tau) = \sum_{n=1}^{\infty} q_n f_n(\tau)$ where f_n is a Fourier basis of functions, then it too must vanish on the boundary. So, with some hindsight, we can pick

$$f_n(\tau) = \frac{1}{\sqrt{T}} \sin \frac{n\pi(\tau + T)}{2T} \tag{1.43}$$

from which we can see that $f_n(-T) = 0$ and $f_n(T) = T^{-1/2} \sin n\pi$ which vanishes for all $n \in \mathbb{Z}$. And so, we now have a basis of functions that obeys the boundary condition, and is orthonormal with respect to a well-defined inner product such as[8]

$$\begin{aligned}(f_n, f_m) &:= \int_{-T}^{T} d\tau\, f_n^*(\tau) f_m(\tau) \\ &= \delta_{nm}.\end{aligned} \tag{1.44}$$

Also in this "clever" choice for basis of functions, we conveniently diagonalize the action as well

$$\begin{aligned}S^E &= \int_{-T}^{T} d\tau \left(\frac{1}{2}(\partial_\tau q)^2 + \frac{1}{2}\omega^2 q^2\right) \\ &= \frac{1}{2} \sum_{n,m=1}^{\infty} \int_{-T}^{T} d\tau \left(\frac{mn\pi^2}{4T^2} \cos \frac{m\pi(\tau+T)}{2T} \cos \frac{n\pi(\tau+T)}{2T} + \frac{\omega^2}{T^2} \sin \frac{n\pi(\tau+T)}{2T} \sin \frac{m\pi(\tau+T)}{2T}\right) q_n q_m \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \left(\left(\frac{n\pi}{2T}\right)^2 + \omega^2\right) q_n^2.\end{aligned} \tag{1.45}$$

Then, the path integral becomes

$$\int [Dq] e^{-S^E/\hbar} \to \int \prod_{n=1}^{\infty} dq_n\, e^{-\frac{1}{\hbar} \sum_{n=1}^{\infty} \frac{1}{2} \left(\left(\frac{n\pi}{2T}\right)^2 + \omega^2\right) q_n^2} \tag{1.46}$$

[8] To show this, put $f_n(\tau)$ into it's exponential form using $\sin(z) = (e^{iz} - e^{-iz})/2i$, perform the integration, and then notice that only when the values of n, m are equal, that the inner product becomes one, otherwise zero.

which we recognize as a Gaussian function we can perform! And in general for correlation functions we write schematically

$$\langle \cdots \rangle = \frac{\int [Dq]\, e^{-S^E/\hbar} \cdots}{\int [Dq]\, e^{-S^E/\hbar}} \tag{1.47}$$

and so we have the following

$$\begin{aligned}
\langle q(\tau_1) q(\tau_2) \rangle &= \frac{\int [Dq]\, e^{-S^E/\hbar} q(\tau_1) q(\tau_2)}{\int [Dq]\, e^{-S^E/\hbar}} \\
&= \sum_{n,m=1} \frac{\int [Dq]\, e^{-S^E/\hbar} q_n q_m f_n(\tau_1) f_m(\tau_2)}{\int [Dq]\, e^{-S^E/\hbar}} \\
&= \sum_{n,m=1} \langle q_n q_m \rangle f_n(\tau_1) f_m(\tau_2)
\end{aligned} \tag{1.48}$$

where we are able to separate out the two point correlation function $\langle q_n q_m \rangle$ and the $f_n(\tau)$'s since the measure is independent of τ. We then need to evaluate $\langle q_n q_m \rangle$. Since this may be the first time doing one of these integrals, let us first work out the denominator. Schematically, it has the form of

$$\int \prod_{n=1}^{\infty} dq_n\, e^{-\frac{1}{\hbar} \sum_{n=1}^{\infty} \frac{1}{2} \left(\left(\frac{n\pi}{2T} \right)^2 + \omega^2 \right)} \sim \int \prod e^{-\frac{1}{2} q_n A_{nm} q_m} \tag{1.49}$$

where we have un-done the delta-function $q_n^2 = \delta_{nm} q_n q_m$ to give us (also note that here, $\delta_{nm} = \delta^{nm}$ i.e. it is it's own inverse)

$$A_{nm} \equiv \delta_{nm} \left(\frac{\hbar}{\left(\frac{n\pi}{2T} \right)^2 + \omega^2} \right)^{-1} \tag{1.50}$$

which has the following solution as $\sqrt{(2\pi)^N / \det A_{nm}}$ (take $N \to \infty$ but we are glossing over that detail for now). Now, notice that the numerator is twice the derivative of the denominator, and so when we do so, we get a factor of A_{nm}^{-1}! The left over from the numerator and denominator cancel out (hence why I glossed over $N \to \infty$ aspect...) and we get

$$\langle q_n q_m \rangle = \delta_{nm} \frac{\hbar}{\left(\frac{n\pi}{2T} \right)^2 + \omega^2}. \tag{1.51}$$

Now, we can combine (1.48) and (1.51) to find

$$\begin{aligned}
\langle q(\tau_1)q(\tau_2)\rangle &= \sum_{n,m=1} \langle q_n q_m\rangle f_n(\tau_1) f_m(\tau_2) \\
&= \sum_{n=1}^{\infty} \frac{\hbar}{\left(\frac{n\pi}{2T}\right)^2+\omega^2}\frac{1}{T}\sin\frac{n\pi(\tau_1+T)}{2T}\sin\frac{n\pi(\tau_2+T)}{2T} \\
&= \frac{2\hbar}{2\pi}\int_0^{\infty} dk\,\frac{\sin k(\tau_1+T)\sin k(\tau_2+T)}{k^2+\omega^2} \\
&= \frac{\hbar}{4\pi}\int_{-\infty}^{\infty}\frac{dk}{k^2+\omega^2}\left(e^{ik(\tau_1-\tau_2)}+e^{-ik(\tau_1-\tau_2)}-e^{ik(\tau_1+\tau_2+2T)}-e^{-ik(\tau_1+\tau_2+2T)}\right)
\end{aligned} \tag{1.52}$$

where in the final steps of (1.52), we took the $T\to\infty$ limit which allowed us to write $\sum_n \frac{\pi}{2T} \rightsquigarrow \int dk$ where we defined $k\equiv n\pi/2T$, and then expanded the sines in terms of their exponential definitions. We also used the integration trick of $\int_0^\infty = \frac{1}{2}\int_{-\infty}^{\infty}$. You may have never seen the type of exponential integral above, so let us first consider the following integral

$$I \equiv \int_{-\infty}^{\infty} dk\,\frac{e^{ik\tau}}{k^2+\omega^2}. \tag{1.53}$$

The integral in (1.53) is inherently complex, and so we will be using our residue calculus. Notice the integrand diverges when $k^2+\omega^2=0$ or specifically at $k=\pm i\omega$. These points of divergence are known as *poles* and if you graphed it in the complex plane, it would look something like this:

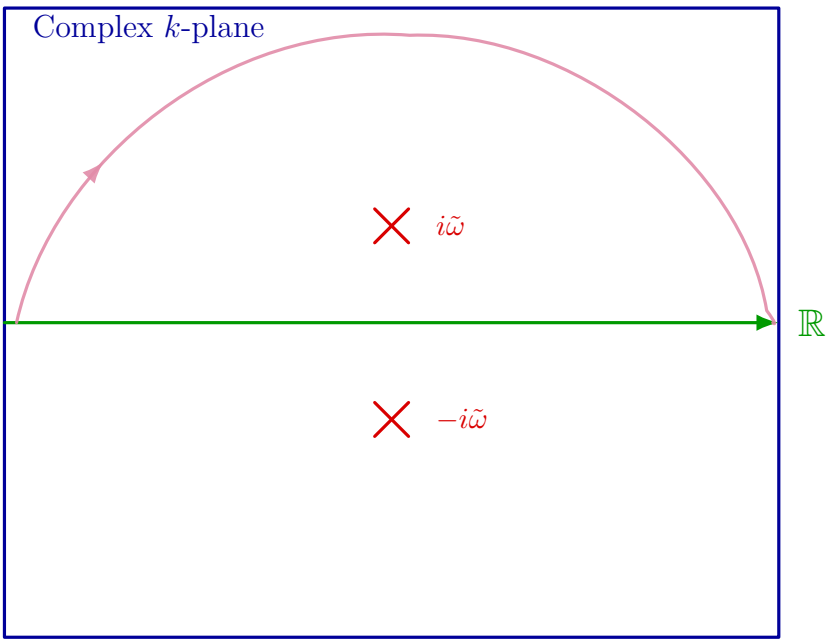


Figure 3. Complex graph for the integral in (1.53).

Notice that *if* we take $\tau > 0$, then when we split the argument of the exponential into $k\tau = \operatorname{Re} k\tau + i \operatorname{Im} k\tau$ then the argument of the exponential becomes

$$|e^{ik\tau}| = e^{-\operatorname{Im} k\tau} \tag{1.54}$$

which is exponentially suppressed at large $\operatorname{Im} k$. This allows us to use the residue theorem since the contour at complex infinity will then vanish. We are then allowed to close the contour via a large semicircle in the upper half plane (UHP). Thus, the integral becomes (evaluated for $\tau > 0$)

$$\begin{aligned}\int_{-\infty}^{\infty} dk \frac{e^{-ik\tau}}{k^2+\omega^2} &= 2\pi i \operatorname{Res}_{k\to i\omega} \frac{e^{ik\tau}}{(k+i\omega)(k-i\omega)} \\ &= 2\pi i \frac{e^{ik\tau}}{i\omega + i\omega} \\ &= \frac{\pi}{\omega} e^{-k\tau}\end{aligned} \tag{1.55}$$

and similarly for if $\tau < 0$ and we closed below which results in $\pi e^{\omega\tau}/\omega$. Coming back to (1.52) we have

$$\begin{aligned}\langle q(\tau_1) q(\tau_2)\rangle &= \frac{\hbar}{4\pi} \int_{-\infty}^{\infty} \frac{dk}{k^2+\omega^2} \left(e^{ik(\tau_1-\tau_2)} + e^{-ik(\tau_1-\tau_2)} - e^{ik(\tau_1+\tau_2+2T)} - e^{-ik(\tau_1+\tau_2+2T)}\right) \\ &= \frac{\hbar}{2\omega} \left(e^{-\omega|\tau_{12}|} - e^{-\omega(\tau_1+\tau_2+2T)}\right) \\ &= \frac{\hbar}{2\omega} e^{-\omega|\tau_{12}|}\end{aligned} \tag{1.56}$$

where the last term in (1.56) vanishes as $T \to \infty$ and $\tau_{12} \equiv \tau_1 - \tau_2$. Now, recall what we found prior namely (1.31) when using the regular Hamiltonian formalism. We can see here that our frequency ω is the energy difference of the system, and that when we Wick-rotate back we get the proper sign and factor of i in the exponential. We can also see what ρ_n is supposed to be proportional to once, in the Hamiltonian formalism, you would have picked a suitable basis as well. Thus, the two results match!

1.1.2 n-Point Correlation Functions & the Anharmonic Oscillator (in QM)

Now, as is common in every physics class, since we did the nice example of single variable mechanics with $q(\tau)$, let us consider a much more general class of theories where $q(\tau)$ now is an N-dimensional vector, and thus will produce a general Gaussian integral of the form

$$\begin{aligned}Z[J] &= \int d^N\mathbf{q}\, e^{-\frac{1}{2}q^T A q + \mathbf{J}\cdot\mathbf{q}} \\ &= (2\pi)^{N/2} (\det A)^{-1/2} e^{\frac{1}{2}J^T A^{-1} J}\end{aligned} \tag{1.57}$$

where A in (1.57) is a symmetric matrix. Then, the correlation function for some product *or even functions* of $\mathbf{q}$, $F(\mathbf{q})$ is given by

$$\begin{aligned}\langle F(\mathbf{q})\rangle &\coloneqq \frac{1}{Z[0]}\int d^N\mathbf{q}\, e^{-\frac{1}{2}q^TAq}\cdot F(\mathbf{q})\\ &= \frac{1}{Z[0]}F\left(\frac{\partial}{\partial\mathbf{J}}\right)\cdot Z[\mathbf{J}]\,\Big|_{J=0}\,.\end{aligned} \tag{1.58}$$

Let us consider some examples. If we wish to compute $\langle q_n\rangle$, then this would only generate a single derivative with respect to J, and thus when evaluated at $J=0$, would vanish. Thus, $\langle q_n\rangle = 0$. The two-point correlation function can be found by taking two derivatives or namely

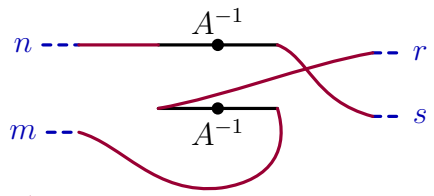
$$\begin{aligned}\langle q_nq_m\rangle &= \frac{\partial}{\partial J_n}\frac{\partial}{\partial J_m}e^{\frac{1}{2}J_p(A^{-1})_{pq}J_q}\,\Big|_{J=0}\\ &= \frac{\partial}{\partial J_n}\left(\frac{1}{2}\frac{\partial J_p}{\partial J_m}(A^{-1})_{pq}J_q+\frac{1}{2}J_p(A^{-1})_{pq}\frac{\partial J_q}{\partial J_m}\right)e^{\frac{1}{2}J_p(A^{-1})_{pq}J_q}\,\Big|_{J=0}\qquad\left(\frac{\partial J_a}{\partial J_b}=\delta_{ab}\right)\\ &= \frac{\partial}{\partial J_n}\left(\frac{1}{2}\delta_{pm}(A^{-1})_{pq}J_q+\frac{1}{2}J_p(A^{-1})_{pq}\delta_{pm}\right)e^{\frac{1}{2}J_p(A^{-1})_{pq}J_q}\,\Big|_{J=0}\\ &= \left(\frac{1}{2}\delta_{pm}(A^{-1})_{pq}\delta_{pn}+\frac{1}{2}\delta_{pn}(A^{-1})_{pq}\delta_{pn}\right)e^{\frac{1}{2}J_p(A^{-1})_{pq}J_q}\,\Big|_{J=0}\\ &= (A^{-1})_{nm}\end{aligned} \tag{1.59}$$

where in the last line of (1.59), we can see that now, when we evaluate at $J=0$, only the exponential vanishes to 1, and we are left with an answer! Something to also notice is that if we have more J-derivatives than present on whatever we are acting on, it will always vanish. Luckily, we are working with an exponential, and so we can expand to whatever order in J's that we prefer to get the first order term that would give a finite answer. For example, if we had 4-J-derivatives, then just the expansion of the exponential $\exp\left(\frac{1}{2}J^TA^{-1}A\right)\approx 1+\frac{1}{2}J^TA^{-1}J$ would vanish. Thus, we need to go to the next higher order term in the expansion. But then, the number of possible ways to perform the derivatives and match the indices explodes! Explicitly, this is

$$\begin{aligned}\langle q_nq_mq_rq_s\rangle &= \frac{\partial}{\partial J_n}\frac{\partial}{\partial J_m}\frac{\partial}{\partial J_r}\frac{\partial}{\partial J_s}e^{\frac{1}{2}J^TA^{-1}J}\,\Big|_{J=0}\\ &= \frac{\partial}{\partial J_n}\frac{\partial}{\partial J_m}\frac{\partial}{\partial J_r}\frac{\partial}{\partial J_s}\frac{1}{2}\left(\frac{1}{2}J^TA^{-1}J\right)^2\end{aligned}$$

which can instead be re-written as:

$\frac{1}{8}\times\Big($ $\sum$



$\Big)$

all possible
ways to connect

where the summation of all the possible ways to connect the 4 different points amounts to $4! = 24$ different terms! But then we have the $1/8$ giving us $24/8 = 3$ possible terms, namely

$$\langle q_n q_m q_r q_s \rangle = (A^{-1})_{nm}(A^{-1})_{rs} + (A^{-1})_{nr}(A^{-1})_{ms} + (A^{-1})_{ns}(A^{-1})_{mr} \tag{1.60}$$

or all the possible unique combinations of the indices. What we are describing is called the *Wick contraction* rule, and in general is given as

$$\langle q_n q_m q_r \cdots \rangle = \sum_{\text{all inequivalent Wick contractions}} \overbrace{q_n q_m q_r \cdots \cdots}^{} \tag{1.61}$$

where a Wick contraction is denoted and given as $\overbrace{q_n q_m}^{} := (A^{-1})_{nm}$. Now, coming back to our generalization of the Gaussian *functional* integral (the N-dimensional q), we have $\vec{q} \rightsquigarrow q(\tau) \in$ a linear space of real functions that is defined with a suitable measure. We also have the generalization of $\vec{q} \cdot \vec{J} \rightsquigarrow \int d\tau\, q(\tau) J(\tau)$. But, what we don't have yet is a generalization of the operator A that we encountered...

So now, consider a linear operator A that we encountered in-directly above (we just didn't call it as such) $A \cdot q(\tau) := (-\partial_\tau^2 + \omega^2) q(\tau)$ with a corresponding inverse found by deriving the Green's function $A^{-1} q(\tau) = \int d\tau'\, G(\tau, \tau') q(\tau')$ where G obeys the 1-dimensional Green's function

$$(-\partial_\tau^2 + \omega^2) G(\tau, \tau') = \delta(\tau - \tau'). \tag{1.62}$$

In this manner, we can write the Euclidean action as an inner product

$$S^E = \frac{1}{2}(q, Aq) = \frac{1}{2} \int d\tau\, q(\tau) A \cdot q(\tau) \tag{1.63}$$

which translates to a two point correlation function of the form

$$\langle q(\tau_1) q(\tau_2) \rangle = \frac{\int [Dq]\, e^{-\frac{1}{2}(q, Aq)} q(\tau_1) q(\tau_2)}{\int [Dq]\, e^{-\frac{1}{2}(q, Aq)}}. \tag{1.64}$$

In order to avoid any confusion with the normalization in discretizing the integral in τ, we can obtain the answer by using an integration by parts trick. Namely, we will write the numerator as a derivative of the exponential that we did prior but *just once*, and then integrate by parts to move the derivative off of the exponential and place it on the remaining q-variable. What this looks like is the following

$$\begin{aligned} \int [Dq]\, e^{-\frac{1}{2}\int d\tau\, q(\tau)(Aq)(\tau)} q(\tau_1) q(\tau_2) &= \int [Dq]\, q(\tau_2) \left(-A^{-1} \frac{\delta}{\delta q(\tau_1)} \right) e^{-\frac{1}{2}\int d\tau\, q(\tau)(Aq)(\tau)} \\ &= \int [Dq]\, e^{-\frac{1}{2}\int d\tau\, q(\tau)(Aq)(\tau)} A^{-1} \frac{\delta}{\delta q(\tau_1)} q(\tau_2). \end{aligned} \tag{1.65}$$

Now, the piece not in the exponential in (1.65) can be written as

$$A^{-1} \delta(\tau_1 - \tau_2) = \int d\tau'\, G(\tau, \tau') \delta(\tau' - \tau_2) = G(\tau_1, \tau_2). \tag{1.66}$$

So what we have is that, with this integration by parts trick, the numerator and denominator when computing the correlation function will now drop out, and all we are left with is the Green's function but also the Wick contraction between the operators,

$$\langle q(\tau_1)q(\tau_2)\rangle = \overbracket{q(\tau_1)q}(\tau_2) = G(\tau_1, \tau_2). \tag{1.67}$$

Now, you can think of doing all of this in *position space* QM. But, we can of course Fourier transform to which you will seem some familar looking results. We can Fourier transform the coordinate $q(\tau)$ as

$$q(\tau) = \int_{-\infty}^{\infty} \frac{dk}{2\pi} \widetilde{q}(k) e^{ik\tau} \tag{1.68}$$

where $\widetilde{q}^*(k) = \widetilde{q}(-k)$. The Fourier transform then of the operator A is $\widetilde{A \cdot q}(k) = (k^2 + \omega^2)\widetilde{q}(k)$ and similar for the inverse $\widetilde{A^{-1}q}(k) = (k^2+\omega^2)^{-1}\widetilde{q}(k)$. This leads to the following Green's function in "momentum space"

$$G(\tau, \tau') = \int_{-\infty}^{\infty} \frac{dk}{2\pi} \frac{e^{ik(\tau-\tau')}}{k^2+\omega^2} \tag{1.69}$$

to which you can check is exactly the same residue we found prior $e^{-\omega|\tau_{12}|}/2\omega$ in equation (1.56). Performing the same steps we did in position space to find the two-point correlation function, we can find the two-point correlation function in momentum space. Starting with the action, since A is diagonal in the Fourier transform basis, then

$$S^E = \frac{1}{2}(q, Aq) = \frac{1}{2}\int_{-\infty}^{\infty} d\tau\, q(\tau)Aq(\tau) = \frac{1}{2}\int_{-\infty}^{\infty} \frac{dk}{2\pi}\widetilde{q}(-k)(k^2+\omega^2)\widetilde{q}(k). \tag{1.70}$$

Then, by using the same integration by parts trick we just used earlier, the two point function becomes

$$\langle \widetilde{q}(k_1)\widetilde{q}(k_2)\rangle = \frac{2\pi\delta(k_1+k_2)}{k_1^2+\omega^2}. \tag{1.71}$$

We just learned so much formalism and found so many results, we should probably *finally* apply this to an example, so let us consider an "anharmonic oscillator," and answer[9], "what is the energy spectrum, or specifically, the ground state and corrections to the ground state energy to a QM that obeys the anharmonic oscillator?"

Like in any physics problem, let us first write down the Hamiltonian, and then the Lagrangian for the system

$$H = \frac{p^2}{2} + \frac{q^2}{2} + \frac{g}{4!}q^4 \leftrightarrow L = \frac{1}{2}\dot{q}^2 - \frac{q^2}{2} - \frac{g}{4!}q^4. \tag{1.72}$$

[9] I apologize in advanced since in doing this example, we will find even *more* general results that are extremely important.

To study it's spectrum, let us consider the two point correlation function[10] assuming $\tau > 0$

$$\begin{aligned}\langle 0|\,\hat{q}(\tau)\hat{q}(0)\,|0\rangle &= \sum_n |\,\langle n|\,\hat{q}(0)\,|0\rangle\,|^2 e^{-E_n\tau} \\ &= \frac{1}{Z}\int [Dq]\, e^{-\int d\tau L^E} q(\tau)q(0)\end{aligned} \tag{1.73}$$

where in blue, I wrote out the usual way correlation functions are evaluated using the Hamiltonian so that we have a reference for reading off the energy of the ground state, and recall that Z which we shall now officially call the *partition function* is given as

$$Z = \int [Dq]\, e^{-\int d\tau\, L^E}. \tag{1.74}$$

The Wick rotation Lagrangian is given as

$$L^E = \frac{1}{2}(\partial_\tau q)^2 + \frac{q^2}{2} + \frac{g}{4!}q^4. \tag{1.75}$$

The difference in (1.75) compared to our usual "free" or non-interacting Lagrangian we were looking at prior is that we only had fields quadractic in q. To handle the q^4 here, we will re-write *that* part of the exponential as a series, namely[11]

$$\begin{aligned}Z_g &= \int [Dq]\, e^{-\int d\tau \left(\frac{1}{2}(\partial_\tau q)^2 + \frac{q^2}{2} + \frac{g}{4!}q^4\right)} \\ &= \int [Dq]\, e^{-\int d\tau \left(\frac{1}{2}(\partial_\tau q)^2 + \frac{q^2}{2}\right)} \cdot \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{g}{4!}\int d\tau'\, q^4(\tau)\right)^4.\end{aligned} \tag{1.77}$$

We will assume that in (1.77) that we can exchange the order of the path integral integration measure $\int [Dq]$ with the summation expansion in g (which is actually problematic, but we will see why later). This allows us to write the correlation function for the interacting theory as the interacting part Z_g divided out by Z_0 (no interactions), and will cancel out the denominator and the non-interacting piece in (1.77) which can be written as

$$\frac{Z_g}{Z_0} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{g}{4!}\right)^n \int d\tau_1 \cdots d\tau_n \cdot \langle (q(\tau_1))^4 \cdots (q(\tau_n))^4 \rangle_0 \tag{1.78}$$

where the partition function attached at the end is computed using "free" Wick contractions (also note that (1.78) is for an n-point interacting function, we have not specificed to 2-

[10] From now on, the "correlation function" will often be just stated as the "n-point function" where it is understood we are talking about correlation functions with n-insertions.

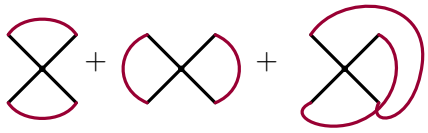
[11] An exponential that has a matrix-values in the argument are expanded as

$$e^X = \sum_{k=0}^{\infty} \frac{1}{k!} X^k \tag{1.76}$$

where X is a real or complex matrix of $n \times n$ size.

points). For example, at first order in g we have

$$-\frac{g}{4!}\int d\tau\,\langle (q(\tau_1))^4\rangle_0 =$$



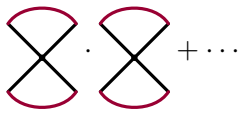
$$= -\frac{g}{8}\int d\tau\,(G_0(\tau,\tau))^2 \tag{1.79}$$

where in the first line of (1.79) I drew what the corresponding diagramatic representations would look like, to which let me explain: notice that the interaction we are looking at has 4-factors of $q(\tau)$ in it. This can be thought of as a diagram piece that has 4-legs sticking out of it. Via Wick's theorem (namely we consider all possible contractions/pairs of the fields), this is almost *exactly* as the case we looked at in (1.60), except in this case, we have q's that are ontop of each other, i.e. all at the same point, but still three terms, and thus three[12] diagrams. The single dots in the diagrams represent the interaction $\sim g$, while the connected line (purple here) are Green's functions or what we now call *propagators*. Let us go to higher order in g and see what happens.

At order g^2, we have the following expansion

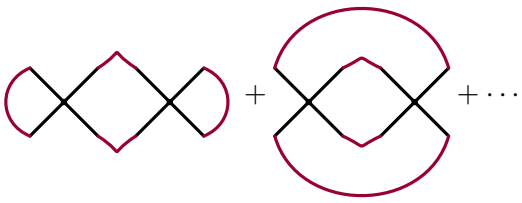
$$\frac{1}{2}\left(-\frac{g}{4!}\right)^2\int d\tau_1 d\tau_2\,\langle (q(\tau_1))^4 (q(\tau_2))^4\rangle_0. \tag{1.80}$$

We now have *two* vertices and must consider all possible ways to connect the lines aka, all the possible number of Wick contractions. But here, we find someting different, namely two classes of diagrams: connected and disconneted. The disconnected diagrams look like



$$+\cdots \tag{1.81}$$

while the connected diagrams are exactly as they sound



$$+\cdots \tag{1.82}$$

The combinatorics for higher order diagrams can be tedious, so here is a general formula for the number or set of Wick contractions that can be represented by a graph. Let us consider a graph with K-connected components, and that among them there are m_ℓ connected graphs that contain ℓ vertices each where $\ell = 1, 2, 3, \ldots$

$$\sum_{\ell\geq 1} m_\ell = K \quad \text{and} \quad \sum_{\ell\geq 1} m_\ell\cdot\ell = n. \tag{1.83}$$

[12] We will actually see later that these three diagrams are exactly those representing the Mandelstam variables/channels s, t, and u with the diagrams going in the same order.

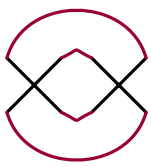
For a given such partition of n that is specified by m_ℓ's, the number of ways of grouping the n-vertices in this pattern is given by $n!(\prod_{\ell\geq 1}(\ell!)^{m_\ell}m_\ell!)^{-1}$ with the total contribution being

$$\frac{1}{n!}\frac{n!}{\prod_{\ell\geq 1}(\ell!)^{m_\ell}m_\ell!}\prod_{\ell\geq 1}(G^{(\ell)})^{m_\ell} = \prod_{\ell\geq 1}\frac{1}{m_\ell!}\left(\frac{1}{\ell!}G^{(\ell)}\right)^{m_\ell}. \tag{1.84}$$

The $G^{(\ell)}$ here is the contribution from the connected Wick contractions of ℓ-vertices. What this means for our n-point function is that diagramtically, we have the following contributions for the anharmonic oscillator

$$\begin{aligned}\frac{Z_g}{Z_0} &= \sum_{\{m_\ell\}}\prod_{\ell\geq 1}\frac{1}{m_\ell!}\left(\frac{1}{\ell!}G^{(\ell)}\right)^{m_\ell} = \prod_{\ell\geq 1}\exp\left(\frac{1}{\ell!}G^{(\ell)}\right)\\ &= \exp\left(\sum_{\ell\geq 1}\frac{1}{\ell!}G^{(\ell)}\right).\end{aligned} \tag{1.85}$$

The argument of the exponential contains all possible contributions from connected graphs such as [**AC:**] this diagram is missing a node τ_3



$$= \frac{1}{3!}\left(-\frac{g}{4!}\right)^3\int d\tau_1 d\tau_2 d\tau_3\,(G_0(\tau_1,\tau_2))^2(G_0(\tau_2,\tau_3))^2(G_0(\tau_3,\tau_1))^2 \tag{1.86}$$

but note that this still diverges as $\int d\tau$ (constant). Now, all of this machinery is just to setup to any order in g the computation of the partition function for the anharmonic oscillator, and I recently called Z the partition function, but is that *really* the best physical interpretation of it? Well, we have the following two different way to compute Z,

$$Z = \int [Dq]\, e^{-\int d\tau\, L^E} = \langle q_f|\, e^{-HT}\, |q_i\rangle \tag{1.87}$$

where the states here, $\langle q_f|\cdots|q_i\rangle$, are written such that there is a boundary condition at intial and final time $\tau = \{0, T\}$ which is left unspecified, but we will take $T\to\infty$. Now, notice what happens if we take the logarithm of the far RHS of Z,

$$\log Z = -E_0 T + \mathcal{O}(1) \tag{1.88}$$

where E_0 is the ground state energy. If we take the same logarithim but with the path integral version using (1.85) we instead get

$$\log\frac{Z_g}{Z_0} = \sum_{\ell=1}^{\infty}\frac{1}{\ell!}G^{(\ell)} \tag{1.89}$$

where $G^{(\ell)}$ is, again, the sum of all Wick contractions of ℓ vertices via *connected* graphs. As a note: one often absorbs the combinatorical factors of $1/\ell!$ into the "symmetry factor"

of the graph itself. Notice that each $G^{(\ell)}$ is of the form $\int d\tau$ (constant) $\propto T$, and thus we can relate the ground state energy as

$$E_0 = -\lim_{T\to\infty} \frac{1}{T} \log Z. \tag{1.90}$$

And so essentially we have found that the partition function does in fact encode information of the QM system, and is even related in the usual statistical mechanics manner to the energy! Let us put (1.90) into action by starting at order g or $\ell = 1$ to which we have already seen is

$$G^{(1)} = 3 \cdot$$

$$= -\frac{g}{8} \int d\tau \, (G_0(\tau,\tau))^2 \tag{1.91}$$

where the factor of 3 arises since all three diagrams are symmetrically the same, but this will not always be the case (in fact typically never). Now, recall we have already computed the Green's functions at unequal times, $G_0(\tau_1,\tau_2) = e^{-\omega|\tau_{12}|}/2\omega$, and so when $\tau_1 = \tau_2$ we get $G_0(\tau,\tau) = 1/2\omega$. And so from (1.90) at order g, we can see that we have

$$\Delta E_0 = \frac{g}{8}\left(\frac{1}{2\omega}\right)^2 + \mathcal{O}(g^2) \tag{1.92}$$

where ΔE_0 is a correction to the ground state energy. Let us compare our answer to typical Hamiltonian perturbation theory seen in quantum mechanics courses. Our Hamiltonian for the anharmonic oscillator is

$$H = \frac{\hat{p}^2}{2} + \frac{\hat{q}^2}{2} + \frac{g}{4!}\hat{q}^4 = H_0 + H_1. \tag{1.93}$$

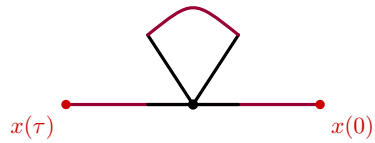
Let $|\psi_0\rangle$ be the ground state of H_0, then leading order in g, $\Delta E_0 \approx \langle\psi_0| H_1 |\psi_0\rangle = g\langle\hat{q}^4\rangle_0/4! = g/32$, which matches what we found above. What we just looked at will eventually be called *zero-point energy* calculations, and now we are going to move onto the full 2-point function which will essentially be the same setup, just now we have two insertions of the variable $q(\tau)$ (assuming τ?0),

$$\begin{aligned}\langle 0| \hat{q}(\tau)\hat{q}(0) |0\rangle &= \frac{1}{Z}\int [Dq]\, e^{-\int d\tau\, L^E} q(\tau)q(0) \\ &= \frac{1}{Z}\langle q(\tau)q(0) \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{g}{4!}\int d\tau'\, q^4(\tau_1')\right)^n \rangle_0\end{aligned} \tag{1.94}$$

where we expanded the exponential with the interacting term as we did prior, and then group things together into correlation functions by definition. (1.94) is computed in the exact same manner/trend as we have been seeing so far, by summing up all possible Wick

contractions (they just get nastier now) which consists of terms like

$$\langle 0|\,\hat{q}(\tau)\hat{q}(0)\,|0\rangle \supset \overbrace{q(\tau)q(0)\left(-\frac{g}{4!}\int d\tau' q^4(\tau')\right)}\,(\cdots)(\cdots)(\cdots)$$

$=$ 

$\times$ (disconnected graphs) . (1.95)

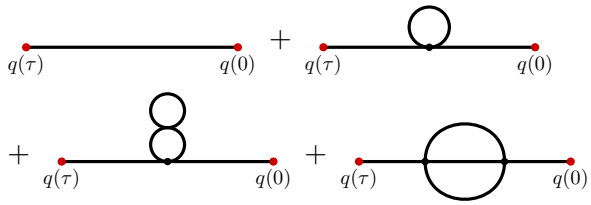
1.1.3 Loops and Self-Energy $\Sigma(k)$ in QM

Let me explain what is happening in (1.95). Recall that to find the total contribution to the correlation functions, we take all possible Wick contractions. However, we are only looking at the numerator in this case and are not taking into account the Z^{-1} factor seen in (1.94). In the non-interacting partition function $Z[0]$ is where the disconnected diagrams/graphs lay. And so, in the numerator, the graphs organize as the connected graphs times the disconnected ones. Besides that, the endpoints of the graph here (in position space) represent $q(\tau)$ and $q(0)$ positioning, so we can think of a particle traveling from $x(0)$ to $x(\tau)$ and in-between it does some weird stuff. But, notice that with any leftover variables $q(\tau)$ that are given from the interaction, they must be Wick contracted as well, giving us a *1-loop correction*. The *entire* sum in (1.94) factorizes in this way as well

$Z[0]\,\langle 0|\,\hat{q}(\tau)\hat{q}(0)\,|0\rangle =$

$\times$ (disconnected graphs) . (1.96)

And so, after dividing out the disconnected graphs, the summation in (1.94) and thus (1.96) can be summed up into the following form[13]

$\langle 0|\,\hat{q}(\tau)\hat{q}(0)\,|0\rangle =$ 

(1.97)

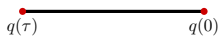
Side note: recall that we only considered when $\tau > 0$, but the same path integral expression is used to compute instead $\langle 0|\,\hat{q}(0)\hat{q}(\tau)\,|0\rangle$ for $\tau < 0$. The actual *full* two-point correlation function must actually take into account both of these to produce the full Euclidean Green function,

$$\langle q(\tau)q(0)\rangle = \begin{cases} \langle 0|\,\hat{q}(\tau)\hat{q}(0)\,|0\rangle & \tau \geq 0 \\ \langle 0|\,\hat{q}(0)\hat{q}(\tau)\,|0\rangle & \tau < 0 \end{cases}, \tag{1.98}$$

so we actually only have "half" of the necessary information, but the steps are the same. We will see the same effect in field theory.

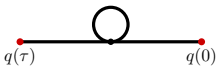
[13] And yes, all of these diagrams have fun names! For example, the first diagram is just the propagator, the second diagram is called a bubble, the third is a double stacked bubble, and the last one is a sunrise!

Let us actually compute the quantities (which you will find out we already have done a good amount of work for!) in (1.97). The first term is just the propagator which we should know and love by now,



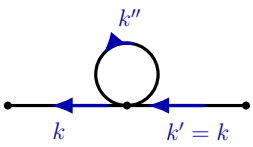
$$= G_0(\tau) = \frac{1}{2}e^{-|\tau|} = \int \frac{dk}{2\pi}\widetilde{G}_0(k)e^{ik\tau} \tag{1.99}$$

where the Fourier transformed propagator is again (with $\omega = 1$) $\widetilde{G}_0(k) = (k^2+1)^{-1}$. And actually, the Fourier transformation will be the most useful one to utilize from here on out that we shall show now for the next diagram, the bubble, in (1.97),



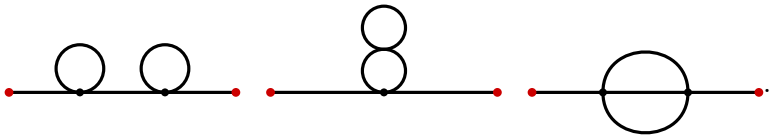
$$\begin{aligned}
&= -\frac{g}{2}\int d\tau'\, G_0(\tau-\tau')G_0(\tau')G_0(0) \\
&= -\frac{g}{2}\int d\tau' \int \frac{dk}{2\pi}\frac{dk'}{2\pi}\widetilde{G}_0(k)\widetilde{G}_0(k')e^{ik(\tau-\tau')+ik'\tau'}\int \frac{dk''}{2\pi}\widetilde{G}_0(k'') \\
&= -\frac{g}{2}\int d\tau' \int \frac{dk}{2\pi}\frac{dk'}{2\pi}\widetilde{G}_0(k)\widetilde{G}_0(k')e^{ik(\tau-\tau')+ik'\tau'}\frac{2\pi i}{(2\pi)(2i)} \\
&= -\frac{g}{2}\frac{1}{2}\int \frac{dk}{2\pi}\frac{e^{ik\tau}}{(k^2+1)^2},
\end{aligned} \tag{1.100}$$

to which leaving (1.100) un-evaluated will be beneficial later on. I should clarify that the way you write a general diagram representation of (1.100) is in the following manner,



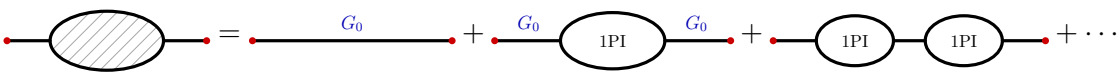
$$= -\frac{g}{2}\frac{1}{2}\int \frac{dk}{2\pi}\frac{e^{ik\tau}}{(k^2+1)^2} \tag{1.101}$$

where you can see that if we label each leg by a different momentum, k, k', k'', etc., that by momentum conservation, $k' = k$. At order g^2 (to which is your problem on the homework[14] we have the following diagrams to compute,



(1.102)

The full *perturbative* 2-point function $\langle q(\tau)q(0)\rangle$ has the following structure



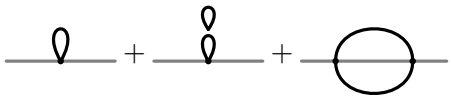
$$\begin{aligned}
&= \int \frac{dk}{2\pi}e^{ik\tau}\widetilde{G}_0(k)\sum_{n=0}^{\infty}\left(\widetilde{G}_0(k)\cdot\Sigma(k)\right)^n \\
&= \int \frac{dk}{2\pi}\frac{e^{ik\tau}}{k^2+1-\Sigma(k)}
\end{aligned} \tag{1.103}$$

[14] Well ok, *our* problem (also, is what type of "voice" is this, first, second or third?) *insert laughing emoji later.*

where we have defined $\Sigma(k)$ as[15]

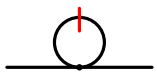
$$\Sigma(k) =$$

$$= \quad + \quad + \quad + \cdots \tag{1.104}$$

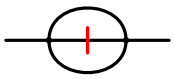


$$= -\frac{g}{4} + \frac{g^2}{8}\left(\frac{1}{4} + \frac{1}{k^2+9}\right) + \mathcal{O}(g^3). \tag{1.105}$$

The greyed out legs here in (1.105) denotes that was is being computed is *only* the loop-diagram in the middle. In other words, the external propagators are not included in the calculation, and this is considered an *amputated graph*. I also introduced some new terminology namely a "1PI" which is called a *one-particle irreducible* diagram. These are very important diagrams that, by definition, would remain connected after *cutting* one propagator, namely



and



(1.106)

where the lines or *cuts* for now shall just signify just that, lines that help us distinguish between non-1PI diagrams and those that are[16]. And so in conclusion, we can write the 2-point function as

$$\mathbb{G}(\tau) \equiv \langle q(\tau)q(0)\rangle = \int \frac{dk}{2\pi}\widetilde{\mathbb{G}}(k)e^{ik\tau} \tag{1.107}$$

where $\widetilde{\mathbb{G}}(k)$ contains corrections to it's momentum k (instead of the usual propagator $G(k)$)

$$\widetilde{\mathbb{G}}(k) = \frac{1}{k^1 + 1 + \frac{9}{4} - \frac{g^2}{8}\left(\frac{1}{4} + \frac{1}{k^2+9}\right) + \mathcal{O}(g^3)}. \tag{1.108}$$

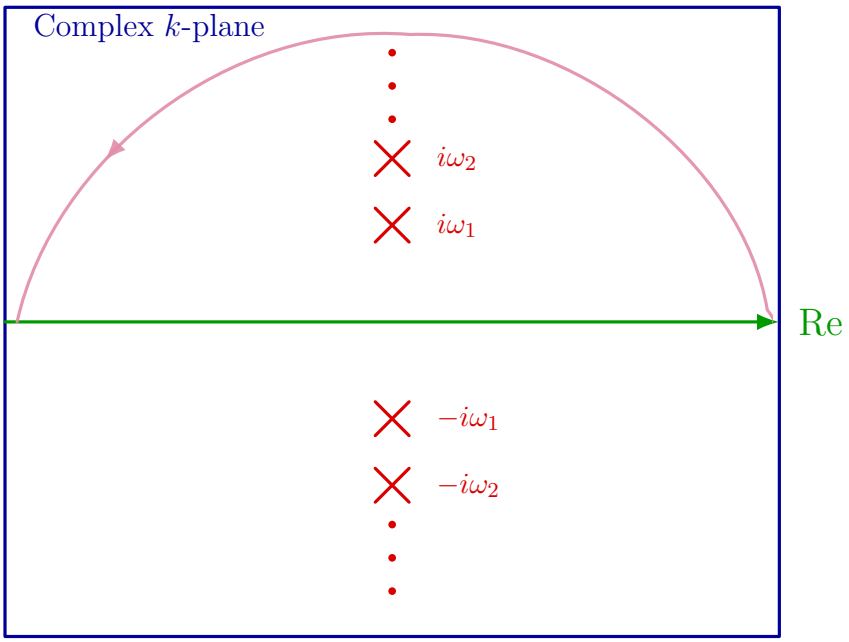
However, on the other hand we also know from the Hamiltonian formalism that (for $\tau > 0$)

$$\langle 0|\, \hat{q}(\tau)\hat{q}(0)\, |0\rangle = \sum_n |\, \langle n|\, \hat{q}(0)\, |0\rangle\, |^2 e^{-(E_n - E_0)\tau}. \tag{1.109}$$

And so the question becomes, how can we relate (1.108) back to the energy of excited states seen in (1.109)? Well, let us consider the analytic continuation of $\widetilde{\mathbb{G}}(k)$ (thinking about the (1.109)) to the complex k-plane.

[15] **[AC:]** Those loops in the second diagram should be connected and the third diagram should have a black propagator in the middle.

[16] We will see later on that there are specific cuts called *unitarity cuts* which help us compute Feynman diagrams.



We can see that there are poles in both the UHP and lower half plane (LHP), but let us consider closing the contour in the UHP such that we compute the residue at each of those poles ω_n,

$$\int \frac{dk}{2\pi} \widetilde{\mathbb{G}}(k) e^{ik\tau} = i \sum_n e^{-\omega_n \tau} \mathrm{Res}_{k \to i\omega_n} \widetilde{\mathbb{G}}(k) \tag{1.110}$$

where we see that

$$\omega_n = E_n - E_0. \tag{1.111}$$

This relationship should not be taken lightly. We can see that poles in the Green's function correspond to the energy differences between different excited states and the ground state for a quantum mechanical system *in general*. If we now compare this with our derive result in (1.108), then we would expect the *first* pole (i.e. the *first excited state*) to be neat $k \approx i$ where

$$\omega_1 = 1 + \frac{g}{8} - \frac{g^2}{32} + \mathcal{O}(g^3). \tag{1.112}$$

This result is truly amazing; the location of the poles (in our QM system) describe the energy differences between the ground states and the next excited state, and with the path integral, we can even derive them to be "functions" of the coupling strength g. In other words, "you tell me what g is, and I can predict what ΔE_n is."

Let us consider another instructive example except this time our interaction term has *time*-derivatives attached to it,

$$L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2 - \frac{g}{8}q^2\dot{q}^2 \tag{1.113}$$

where the corresponding momentum and Hamiltonian are

$$p = \frac{\partial L}{\partial \dot{q}} = \dot{q}\left(1 - \frac{g}{4}q^2\right) \tag{1.114}$$

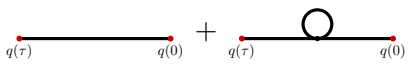
$$\begin{aligned} H &= \dot{q}p - L \\ &= \frac{1}{2}\left(1 - \frac{g}{4}q^2\right)^{-1} p^2 + \frac{1}{2}q^2. \end{aligned} \tag{1.115}$$

Side note: we could also consider a change of variables to the above Lagrangian and Hamiltonian to make things nicer. For example, consider letting $\dot{y} = \sqrt{1 - g\dot{q}^2/4}\dot{q}$, and then $y = q - gq^3/24 + \cdots$. The Lagrangian can then be re-written as $L \rightsquigarrow L(y, \dot{y}) = \dot{y}^2/2 - V(y)$ with $V(y) = y^2/2 + gy^4/24 + \cdots$, and so we are left with the same as an anharmonic oscillator up to $\mathcal{O}(g^2)$ corrections to the potential. *However*, we will insist on working with the q-variable in the path integral since what happens will be of great interest.

The Euclidean version of (1.113) is

$$\begin{aligned} L^E &= \frac{1}{2}(\partial_\tau q)^2 + \frac{1}{2}q^2 - \frac{g}{8}q^2(\partial_\tau q)^2 \\ &= \frac{1}{2}(\partial_\tau q)^2 + \frac{1}{2}q^2 - \frac{g}{24}\partial_\tau(q^3\partial_\tau q) + \frac{g}{24}q^3\partial_\tau^2 q \end{aligned} \tag{1.116}$$

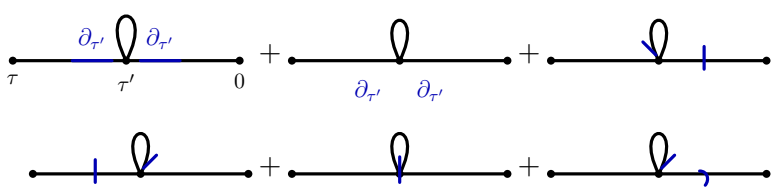
where in the last line, since we know it is going to be put into the action S^E, we can integrate by parts right now on the last term and observe that the purple term is a total derivative and thus will not contribute to S^E while the orange term is what we keep. Then, if we only keep up to order g, the 2-point function will have the following contributions

$\langle q(\tau)q(0)\rangle =$ 

$+ \cdots$

$$= G_0(\tau) + q(\tau)q(0)\frac{g}{8}\int d\tau'(q(\tau'))^2(\partial_{\tau'}q(\tau'))^2 + \cdots. \tag{1.117}$$

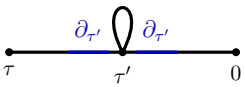
The last term in (1.117) has several diagram representations since we can always perform an integration by parts to move one of the time-derivatives to a different $q(\tau')$, and so we have the following *permutations* of the diagram,



(1.118)

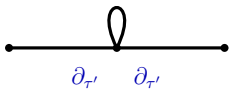
The blue cuts in (1.118) represent derivatives on the interaction terms. This first diagram

here translates to the following expression[17]

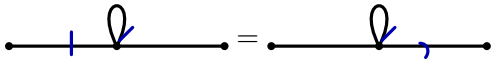


$$
\begin{aligned}
&= 2 \times q(\tau)q(0)\frac{g}{8}\int d\tau'\, q(\tau')q(\tau')\partial_{\tau'}q(\tau')\partial_{\tau'}q(\tau') \\
&= \frac{g}{4}\int d\tau'\, G_0(\tau-\tau')G_0(\tau')(-\partial^2 G_0(0)) \\
&= \frac{g}{4}\int \frac{dk}{2\pi}\frac{e^{ik\tau}}{(k^2+1)^2}\int \frac{dk'}{2\pi}\frac{(k')^2}{(k')^2+1}.
\end{aligned}
\tag{1.119}
$$

But now wait a second... is that last term divergent (it is, hence I colored it blue)?! But ok, the other diagrams are finite since we can check them directly



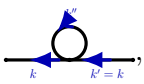
$$
= \frac{g}{4}\int \frac{dk}{2\pi}\frac{e^{ik\tau}k^2}{(k^2+1)^2}\int \frac{dk'}{2\pi}\frac{1}{(k')^2+1} \tag{1.120}
$$



$$
\begin{aligned}
&= \frac{g}{2}\int \frac{dk}{2\pi}\frac{e^{ik\tau}ik}{(k^2+1)^2}\int \frac{dk'}{2\pi}\frac{ik'}{(k')^2+1} \\
&= 0
\end{aligned}
\tag{1.121}
$$

where the first diagram is finite by power counting since the exponential does a lot of heavy lifting in damping the powers, and the last two vanish by symmetry since each integration is an odd power divided by an even power. The same goes for the rest of the diagrams. The divergence in (1.119) is due evaluating the integral for large k (i.e. $k \to \infty$ and so that is pretty large). The origin of this divergence however lies not within the interaction itself but within the path integral measure since the measure informed us to integrate out all values of k! Recall that the measure can be written as $[Dq] = \prod_k d\tilde{q}_k$. If we were to think of a *regularized* measure, the idea is that we simply do not want those specific values of k that diverge; so let's just cut them off[18]

$$
\prod_k d\tilde{q}_k \to \prod_{|k|<\Lambda} d\tilde{q}_k. \tag{1.122}
$$

In reference to 

, this *effectively* cuts off the k' "loop" integral up to values of $|k'| < \Lambda$; we will eventually take $\Lambda \to \infty$ since no physical observable should be dependent upon specific regularization schemes of course (this would just be bad physics). Returing to the 2-point function but now *with* the cutoff, recall that the total Green's function

[17] The blue markings are supposed to be on the loop itself, two of them, but it's slightly hard to do that as you can see in the last few diagrams.

[18] This ad hoc manner of, "well if we don't want it, let's just get rid of it," feels very wrong physically at a first glance. We will see however that this process is tied to the ambigous nature of quantization. In QFT 2, a more satisfactory answer will be given.

$\mathbb{G}(k) = (k^2+1-\Sigma(k))^{-1}$ contains the loop diagrams in $\Sigma(k)$. With the cutoff regularization, $\Sigma(k)$ turns into

$$\begin{aligned}\Sigma(k) &= \frac{g}{4}\int\limits_{|k'|<\Lambda} \frac{dk'}{2\pi}\frac{(k')^2}{(k')^2+1} + \frac{g}{4}k^2\int\frac{dk'}{2\pi}\frac{1}{(k')^2+1} + \mathcal{O}(g^2) \\ &= \frac{g}{4}\left(\frac{k^2}{2}+\frac{\Lambda}{\pi}-\frac{1}{2}+\mathcal{O}(\Lambda^{-1})\right)+\mathcal{O}(g^2). \end{aligned} \tag{1.123}$$

But now, "what is the physical interpretation of $\Lambda \to \infty$?" Well, let us where we started at with the Hamiltonian. There is nothing wrong with the H we wrote in (1.115) (except for a possible ordering ambiguity when promoting q and p to operators...). This H is well-defined at least for $g < 0$. So that's fine. So ok, let us restore $\hbar$-dependence and track it throughout our calculation. As should almost be normal at this point, with the path integral we start with the partition function $Z = \int [Dq]\, \exp\Big(-S^E/\hbar\Big)$. At the level of the Lagrangian we have

$$L^E = \frac{1}{2}(\partial_\tau q)^2 + \frac{q^2}{2} - \frac{g}{8}q^2(\partial_\tau q)^2 + \mathcal{O}(\hbar)? \tag{1.124}$$

The blue piece contains possible corrections at order $\hbar$ when writing down a *quantum* Lagrangian which may be necessary corrections that are needed to match the quantum Hamiltonian. These terms are dubbed *counter-terms*. Ok, in a typical calculation is the Green's function $\widetilde{\mathbb{G}}(k) = \hbar(k^2+1-\Sigma(k))^{-1}$, so then to make units work in the denominator, there must be a factor of $\hbar$ multiplied to $\Sigma(k)$ namely

$$\Sigma(k) = \frac{\hbar g}{4}\left(\frac{k^2}{2}+\frac{\Lambda}{\pi}-\frac{1}{2}+\mathcal{O}(\Lambda^{-1})\right)+\mathcal{O}(\hbar^2 g^2). \tag{1.125}$$

Well, one thing that this analysis has done so far is to show us that if want to get rid of Λ, we could do it by utilizing the quantum ambiguities that arise in L^E and "cook-up" a new term to add ΔL^E that is of order $\hbar$, namely

$$\Delta L^E = cg\hbar\frac{q^2}{2}. \tag{1.126}$$

This new term in (1.126) will then contribute to $\Sigma(k)$ of the form $-cg\hbar$ at order g, giving us the following result

$$\Sigma(k) = \hbar g\left(\frac{k^2}{8}+\frac{\Lambda}{4\pi}-\frac{1}{8}-c+\mathcal{O}(\Lambda^{-1})\right)+\mathcal{O}(g^2). \tag{1.127}$$

Let us now state an apparent fact (that is very important), *the path integral is only* ***defined*** *provided a regularization scheme via the cutoff* Λ. This does not sound great for computing observables, but we now notice that if we utilize those QM ambiguities in writing down a quantum Lagrangian, these new additional terms we can write in such a matter that absorb the cutoff. Therefore, let us choose

$$c = \frac{\Lambda}{4\pi} + \text{finite}, \tag{1.128}$$

which allows us to cancel out the cutoff and unphysical constant found in (1.127). Then, if we take $\Lambda \to \infty$ at the very end, we get a finite result for $\Sigma(k)$! The finite part of c represents an ambiguity in quantization itself, i.e. the possibility of adjusting the quantum Hamiltonian by a term of order $\hbar$. Therefore, after this procedure, we have the following correction to the Green's function

$$\widetilde{\mathbb{G}}(k) = \frac{\hbar}{k^1 + 1 - \Sigma(k)} \tag{1.129}$$

which we know has poles at $k = i\omega_1$ where ω_1 is

$$\begin{aligned} \omega_1 &= 1 + \hbar g \left(-\frac{\Lambda}{8\pi} + \frac{c}{2} + \frac{1}{8} \right) + \mathcal{O}(g^2) \\ &= E_1 - E_0. \end{aligned} \tag{1.130}$$

This energy gap is an unambiguous physical observable regardless of the interpretation of $q(\tau)$. The counter-terms here, together with the regularization scheme, should be *viewed* as part of the definition of the path integral.

This however is still unsatisfactory since we would like to be able to fix the finite part of the counter-term at least by some physical principle. Well, to fix the finte part of the counter-terms that arise requires additional physical inputs, for example, symmetric of the quantum Hamiltonian.

In general QM, practically anything can go into the Hamiltonian. However, in QFT, Poincaré symmetry plus causality will make the choices of finite terms and counter-terms highly restrictive! This is where we end our analysis of quantum mechanics via the path integral.

Something I want to emphasize before we transition over to field theory is that, with our path integral method, practically everything we have just learned will transfer over almost exactly! Literally: the Wick rotation, Wick contractions, interpretation of Z as the partition function, regularization scheme, interpretation of poles as energy-mass states, and counter-terms all naturally transfer over! The primary difference is that now we are working with *field* operators that have spacetime points x^μ as parameters now (not operators as in QM, namely the position and momentum operators), and the dimensionality increases.

1.2 Relativistic Particles and Field

Prelude: So, what does a "manifestly relativistic theory of 'fields'," appear as or is constructed? Well, if we think back to the classical mechanics, we had generalized coordinates $q^i(t)$. In a field theory, we now instead have quantities that are coordiante dependent, including time. So the idea is to take all of the results/language in 1.1, and transfer them over to fields via the replacement

$$q^i(t) \rightsquigarrow \phi(t, \vec{x}). \tag{1.131}$$

Here, $\vec{x} \in \mathbb{R}^{D-1}$ for $D = d + 1$ spacetime coordinates. The $\vec{x}$ here should be viewed as parameters labeling the continuous family of variables (namely the spatial spacetime coordinates). We will also be writing the *spacetime/relativistic coordinate* as

$$x^\mu \equiv (x^0, \vec{x}). \tag{1.132}$$

The relativistic action then, following (1.131) becomes

$$S = \int dt\, d^{d-1}\vec{x}\, \mathcal{L}\left[\phi(x), \partial_\mu \phi(x), \cdots\right] \tag{1.133}$$

where many times we will denote $dtd^{D-1}\vec{x} \equiv d^D x$, and $\mathcal{L}$ is called the *Lagrangian density*. And so quite literally, the only difference between what we did in 1.1 and here is a factor of $d^{D-1}\vec{x}$ coordinates (which will lead to some great physics)! We shall assume for now that $\mathcal{L}$ depends only upon $\phi(x)$ (where $\phi(x) \equiv \phi(x^\mu)$) and its corresponding first order derivative $\partial_\mu \phi(x)$ where the notation here is

$$\partial_\mu \phi(x) \equiv \frac{\partial}{\partial x^\mu}\phi(x) \tag{1.134}$$

and therefore $\partial^\mu \equiv \frac{\partial}{\partial x_\mu}$. Therefore, the variation of the action is

$$\begin{aligned} \delta S &= \int d^D x \left(\delta\phi(x)\frac{\partial \mathcal{L}}{\partial \phi(x)} + \delta(\partial_\mu \phi(x))\frac{\partial \mathcal{L}}{\partial \phi(x)}\right) \\ &= \int d^D x \left(\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)}\right)\delta\phi(x) \end{aligned} \tag{1.135}$$

where in the first line the "derivatives" commute $\delta\partial_\mu\phi(x) = \partial_\mu(\delta\phi(x))$ to which on the same term, we performed integration by parts assuming proper boundary conditions (Hamilton's) to which from here on out, a nice trick for doing integration by parts is

$$A\partial_\mu B = -B\partial_\mu A. \tag{1.136}$$

Then, from (1.135) we can read off the field-version of the Euler-Lagrange equations of motion

$$\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} = 0. \tag{1.137}$$

A note before moving on is that now with field theory, we are working on a spacetime background that is flat or in other words, *Minkowskian* where the it has the following values

$$\eta^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.138}$$

where the inverse, $\eta_{\mu\nu}$ also has the same values (the generalization to D-dimensions is just more 1's in the diagonal-spatial components). A very helpful formula is that the "inner product in Minkowski space" (just a fancy dot-product being careful with signs) between two spacetime vectors is

$$A \cdot B \equiv A^\mu B_\mu = \eta^{\mu\nu} A_\mu B_\nu = -A^0 B^0 + \vec{A}\cdot\vec{B}. \tag{1.139}$$

So, with (1.139), a quick calculation that should be become routine is that $p^2 \equiv p_\mu p^\mu = -p_0^2 + (\vec{p})^2$.

As our first example of a relativistic field theory, let us consider a massive *scalar* field theory (where a scalar is a spinless particle) where the Lagrangian density (from now on just referred to as *the Lagrangian*) is given as

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}(\partial_t\phi)^2 - \frac{1}{2}\sum_{i=1}^{D-1}(\partial_{\vec{x}}\phi)^2 - \frac{1}{2}m^2\phi^2 \\ &\equiv -\frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2 \end{aligned} \tag{1.140}$$

where in the first line of (1.140), the explicit summation notation of the inner product is shown, but we will often adopt *Einstein notation* in which the summation is simply dropped (but it is simplified or still there). The equations of motion using (1.137) are given as[19]

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)} &= -\frac{1}{2}2m^2\phi - \partial_\mu\left(-\frac{1}{2}\frac{\partial}{\partial\partial_\mu\phi}\partial_\nu\phi\partial^\nu\phi\right) \\ &= -m^2\phi + \frac{1}{2}\partial_\mu\left(\delta^\mu_\nu\partial^\nu\phi + \partial^\nu\phi\delta^\mu_\nu\right) \\ 0 &= -m^2\phi^2 + \partial^\mu\partial_\mu\phi \end{aligned} \tag{1.141}$$

where in deriving (1.141), I used the fact that $\partial_\mu\phi/(\partial_\nu\phi) = \delta^\nu_\mu$ and similar for other index structures. (1.141) is known as the *Klein-Gordan equation*. This δ^ν_μ here should be viewed as the usual delta-function from QM/classical field theory where δ^ν_μ has a value of 1 if the indices are the same, otherwise 0. Hence, we can use it to swap, for example, and ν index for a μ index whatever the delta-function is multiplied to. We can consider a general solution to (1.141) as

$$\phi(x) = \int d^Dx\, f(k)e^{ik\cdot x} \tag{1.142}$$

where $k\cdot x \equiv -k^0x^0 + \vec{k}\cdot\vec{x}$, which when we plug into (1.141) we find

$$\begin{aligned}(\partial^\mu\partial_\mu - m^2)\int d^Dx\, f(k)e^{ik_\nu x^\nu} &= \int d^Dx\, \left(\partial^\mu\partial_\mu e^{ik\cdot x} - m^2e^{ik\cdot x}\right)f(k) \\ &= \int d^Dx\, \left(\partial^\mu(ik_\alpha\delta^\alpha_\mu)e^{ik\cdot x} - m^2e^{ik\cdot x}\right)f(k) \\ &= \int d^Dx\, \left(i^2k_\alpha\delta^\alpha_\mu k^\beta\delta^\mu_\beta - m^2\right)e^{ik\cdot x}f(k) \\ 0 &= (k^2+m^2)f(k) \end{aligned} \tag{1.143}$$

where $f(k)$ here is a *distribution* supported at

$$k^2 + m^2 = 0 \tag{1.144}$$

[19] It should be noted that the spacetime indices here can *never* be repeated. So, if we have an equation that is $(\text{stuff})^\mu = A^\mu$, this is a vector-type equation and must have the same index structure on both the RHS and LHS. However, within "stuff" can be wild *index contractions* such as stuff $= R_{\alpha\beta}R^{\alpha\beta} + R_{\gamma\delta\alpha\beta}R^{\gamma\delta\alpha\beta} + \cdots$ where we notice that in every *individual term*, no index is left alone and everything is *summed over*, but we can re-use these *dummy indices* on other terms as long as they are not multipled together.

where (1.144) is known as the *mass-shell* i.e. if we expand k^2 in (1.144) we can solve for k^0 namely $k^0 = \pm\sqrt{(\vec{k})^2 + m^2}$ where we shall denote $\omega_k \equiv +\sqrt{(\vec{k})^2 + m^2}$ which shall turn into the *energy* of the scalar. If our "Fourier expansion" of the scalar field has support on the mass-shell, then we can re-write (1.142) as

$$\phi(x) = \int d^{D-1}\vec{k}\left(\widetilde{f}(\vec{k})e^{i\vec{k}\cdot\vec{x}-i\omega_k x^0} + \widetilde{f}^*(\vec{k})e^{-i\vec{k}\cdot\vec{x}+i\omega_k x^0}\right). \tag{1.145}$$

Given (1.145), the question now becomes how to determine $\widetilde{f}$'s, or in other words, an initial value problem: given $\phi(t=0,\vec{x})$ and $\partial_t\phi(t=0,\vec{x})$, how do we determine $\phi(t>0,\vec{x})$? The initial values will actually give us everything we need to determine the $\widetilde{f}$'s,

$$\phi(t=0,\vec{x}) = \int d^{D-1}\vec{k}\, e^{i\vec{k}\cdot\vec{c}}\left(\widetilde{f}(\vec{k}) + \widetilde{f}^*(-\vec{k})\right) \tag{1.146}$$

$$\partial_t\phi(t=0,\vec{x}) = \int d^{D-1}\vec{k}\, e^{i\vec{k}\cdot\vec{x}}(-i\omega_k)\left(\widetilde{f}(\vec{k}) - \widetilde{f}^*(-\vec{k})\right). \tag{1.147}$$

Together, these two fully determine $\widetilde{f}(\vec{k})$ and $\widetilde{f}^*(-\vec{k})$, and hence $\phi(x)$ at any time $t \equiv x^0$.

This is essentially the entirety of (classical) relativistic field theory: write down the Lagrangian, find the equations of motion, and compute the *mode-function* for said corresponding field. This also has been a prelude of sorts to give you some of the necessary tools to do the computations we shall see shortly.

Let us finally move onto addressing the question, " what is a relativistic *quantum* field theory," (since we just went over the bare bones of what a relativistic (classical) field theory is).

1.2.1 Hamiltonian and Path Integral Quantization of QFT

A relativistic quantum theory may be *constructed* via a "quantization" of the classical[20] field theory. There are (primarily) two approaches to such a procedure:

1. Canonical quantization:
   (a) This is performed by constructing an Hilbert space using wave functions in variable $\phi(t=0,\vec{x})$ and the Hamiltonian H.
2. Path integral:
   (a) Quite literally, the method is summed up in understanding the partition function

$$Z = \int [D\phi(x)]\, e^{\frac{i}{\hbar}\int d^D x\, \mathcal{L}[\phi,\partial_\mu\phi]} \tag{1.148}$$

   where the primary objective is to understand what in the world is the measure $[D\phi(x)]$, namely making sense of the functional measure that typically requires regularization (as we have seen in QM).

[20] Note that even if a "classical" field theory for a quantum field theory exists, that does *not* mean that there is a corresponding classical effect. As an example we will see in QFT 2, there is a classical field theory for the strong force, but there is no classical strong interaction, only a working quantum mechanical understanding.

Let us first take a look at the canonical quantization method for multiple reasons, first, to connect to the standard way QFT is taught, second is to give us a physical reference back to known physics when we compute observables with the path integral (exactly what we did in QM in 1.1), and finally to get some very important quantities and relationships such as Noether's theorem.

In the canonical formalism, the *total* Lagrangian is written as $L = \int d^{D-1}\vec{x}\,\mathcal{L}[\phi, \partial_\mu\phi]$ where as we have seen for a *free* (non-interacting) scalar, the Lagrangian density is $\mathcal{L} = -(\partial_\mu\phi)^2/2 - m^2\phi^2/2$, and the dynamical variable here is the field $\phi(t, \vec{x})$ where the coordinates are (again) just some label (but keeping track of the time coordinates is extremely important–the spatial coordinates don't really matter, they just make the calculus harder). In the regular Hamiltonian formalism in QM where the variables were canonical coordinates, here we replace all canonical coordinates with the field variables. Namely the *conjugate canonical momentum density* is given as

$$\pi(t, \vec{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(t, \vec{x})} = \dot{\phi}(t, \vec{x}) \tag{1.149}$$

where in blue is our working example of a free scalar field. The (classical) Hamiltonian is given as

$$\begin{aligned} H &= \int d^{D-1}\vec{x}\,\mathcal{H} \\ &= \int d^{D-1}\vec{x}\,\left(\dot{\phi}(t, \vec{x})\pi(t, \vec{x}) - \mathcal{L}\right) \end{aligned} \tag{1.150}$$

where $\mathcal{H}$ is the Hamiltonian density. For a free scalar field, the Hamiltonian density is

$$\begin{aligned} \mathcal{H} &= \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\partial_{\vec{x}}\phi)^2 + \frac{1}{2}m^2\phi^2 \\ &= \frac{1}{2}\pi^2 + \frac{1}{2}(\partial_{\vec{x}}\phi)^2 + \frac{1}{2}m^2\phi^2 \end{aligned} \tag{1.151}$$

where the first term in (1.151) is the kinetic term, while the last two are considered the potential. In the classical theory, we have the following Poisson brackets (omitting time)

$$\{\phi(\vec{x}), \phi(\vec{x}')\} = \{\pi(\vec{x}), \pi(\vec{x}')\} \tag{1.152}$$

$$\{\pi(\vec{x}), \phi(\vec{x}')\} = \delta^{D-1}(\vec{x} - \vec{x}'). \tag{1.153}$$

The general Poisson bracket for some variables $\mathcal{F}$ and $\mathcal{G}$ are

$$\{\mathcal{F}, \mathcal{G}\} := \int d^{D-1}\vec{x}\left(\frac{\delta\mathcal{F}}{\delta\pi(\vec{x})}\frac{\delta\mathcal{G}}{\delta\phi(\vec{x})} - \frac{\delta\mathcal{F}}{\delta\phi(\vec{x})}\frac{\delta\mathcal{G}}{\delta\pi(\vec{x})}\right) \tag{1.154}$$

and therefore have the following equations of motion as usual

$$\dot{\mathcal{F}} = \{\mathcal{H}, \mathcal{F}\}. \tag{1.155}$$

In order to perform the canonical quantization, we promote ϕ, π to operators $\hat{\phi}, \hat{\pi}$ (but again, unlike in QM, their coordinates are *not* promoted to operators) that obey the *equal*

time commutators (just take the Poisson brackets above and multiply by $i\hbar$)

$$\left[\phi(t,\vec{x}),\hat{\phi}(t,\vec{x}')\right] = \left[\hat{\pi}(t,\vec{x}),\hat{\pi}(t,\vec{x}')\right] = 0 \tag{1.156}$$

$$\left[\hat{\phi}(t,\vec{x}),\hat{\pi}(t,\vec{x}')\right] = i\hbar\delta^{D-1}(\vec{x}-\vec{x}'). \tag{1.157}$$

It will be convenient to perform a spatial Fourier transformation, let's say, at $t = 0$ (notice the signs on the creation and annihilation operators and how it flips the sign in the exponential),

$$\begin{aligned}\hat{\phi}(t=0,\vec{x}) &= \int \frac{d^{D-1}\vec{k}}{\sqrt{(2\pi)^{D-1}2\nu(\vec{k})}}(a_{\vec{k}}e^{i\vec{k}\cdot\vec{x}} + a^{\dagger}_{-\vec{k}}e^{-i\vec{k}\cdot\vec{x}}) \\ &= \int \frac{d^{D-1}\vec{k}}{\sqrt{(2\pi)^{D-1}2\nu(\vec{k})}}(a_{\vec{k}} + a^{\dagger}_{-\vec{k}})e^{i\vec{k}\cdot\vec{x}} \end{aligned} \tag{1.158}$$

$$\pi(t=0,\vec{x}) = \int \frac{d^{D-1}\vec{k}}{\sqrt{(2\pi)^{D-1}2\nu(\vec{k})}}(-i\nu(\vec{k})(a_{\vec{k}} - a^{\dagger}_{-\vec{k}})e^{i\vec{k}\cdot\vec{x}} \tag{1.159}$$

where for now $\nu(\vec{k})$ is just a normalization convention, and we should view the $a_{\vec{k}}$ and $a^{\dagger}_{-\vec{k}}$ as annihilation and creation operators as usual. It then follows from the commutation relations in (1.157) that the creation and annihilation operators obey

$$\left[a_{\vec{k}} + a^{\dagger}_{-\vec{k}}, a_{\vec{k}'} + a^{\dagger}_{-\vec{k}'}\right] = \left[a_{\vec{k}} - a^{\dagger}_{-\vec{k}}, a_{\vec{k}'} - a^{\dagger}_{-\vec{k}'}\right] = 0 \tag{1.160}$$

$$\left[a_{\vec{k}} + a^{\dagger}_{-\vec{k}}, a_{\vec{k}'} - a^{\dagger}_{-\vec{k}'}\right] = \left[a_{\vec{k}} - a^{\dagger}_{-\vec{k}}, a_{\vec{k}'} + a^{\dagger}_{-\vec{k}'}\right] = 2\delta^{D-1}(\vec{k}+\vec{k}') \tag{1.161}$$

Note that $a^{\dagger}_{\vec{k}}$ being the Hermitian conjugate of $a_{\vec{k}}$ ensure that $\hat{\phi}$ and $\hat{\pi}$ are Hermitian too. The commutation relation in (1.161) are equivalent to the normal ones you may have seen

$$\left[a_{\vec{k}}, a_{\vec{k}'}\right] = \left[a^{\dagger}_{\vec{k}}, a^{\dagger}_{\vec{k}'}\right] = 0 \tag{1.162}$$

$$\left[a_{\vec{k}}, a^{\dagger}_{\vec{k}'}\right] = \delta^{D-1}(\vec{k}-\vec{k}'). \tag{1.163}$$

These are precisely the commutation relations obeyed by particle annihilation and creation operators that we have seen prior, just in a field-context. However, this does not mean $a_{\vec{k}}$ and $a^{\dagger}_{\vec{k}}$ are actually annihilation and creation operators... We haven't used the Hamiltonian nor specified what $\nu(\vec{k})$ is and so their exact physical interpretation is not actually set yet.

Side note: considered if we had actually defined $b_{\vec{k}'}$ and $b^{\dagger}_{\vec{k}'}$ via

$$b_{\vec{k}} + b^{\dagger}_{-\vec{k}} = \lambda(\vec{k})(a_{\vec{k}} + a^{\dagger}_{-\vec{k}}) \tag{1.164}$$

$$b_{\vec{k}} - b^{\dagger}_{-\vec{k}} = \lambda^{-1}(\vec{k})(a_{\vec{k}} - a^{\dagger}_{-\vec{k}}) \tag{1.165}$$

where b and $b^{\dagger}$ also obey the commutation relation of creation and annihilation operators as in (1.161)! The operators we have written here, namely

$$b_{\vec{k}} = \frac{\lambda+\lambda^{-1}}{2}a_{\vec{k}} + \frac{\lambda-\lambda^{-1}}{2}a^{\dagger}_{-\vec{k}} \tag{1.166}$$

is called a *Bogoliubov transformation*. The question that I then pose to you is, "is the vacuum state $|\Omega\rangle$ annihilated by $a_{\vec{k}}$? If so, $b_{\vec{k}}|\Omega\rangle \neq 0$ for $\lambda \neq 1$."

Let us actually use the Hamiltonian for the free scalar and see if our interpretation of $a_{\vec{k}}$ and $a^\dagger_{\vec{k}}$ holds (where $\hat{H}$ should vanish on the vacuum). The Hamiltonian can be written as[21]

$$\begin{aligned}
\hat{H} &= \int d^{D-1}\vec{x}\left(\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\partial_{\vec{x}}\phi)^2 + \frac{1}{2}m^2\hat{\phi}^2\right) \\
&= \int d^{D-1}\vec{k}\left(\frac{1}{2}\frac{\nu(\vec{k})}{2}(a_{\vec{k}} - a^\dagger_{-\vec{k}}(a^\dagger_{\vec{k}} - a_{-\vec{k}}) + \frac{1}{2}\frac{1}{\nu(\vec{k})}(\vec{k})^2 + m^2)(a_{\vec{k}} + a^\dagger_{-\vec{k}} + a_{-\vec{k}}\right) \\
&= \int d^{D-1}\vec{k}\,\omega_{\vec{k}}\left(a^\dagger_{\vec{k}}a_{\vec{k}} + \frac{1}{2}\delta^{D-1}(0)\right)
\end{aligned} \tag{1.167}$$

where we have set $\nu(\vec{k}) = \omega_{\vec{k}}$ where $\omega_{\vec{k}} \equiv \sqrt{(\vec{k})^2 + m^2}$. However, we have a major problem, the orange term is infinite! This is an ordering ambiguity of the creation and annihilation operators that is of order $\hbar$... (This sort of thing has already popped up and explained in QM in 1.1). The *true* quantum Hamiltonian must be a well-defined self-adjoint operator however to obey the postulates of QM.

We will remove the normal ordering constant by a constant shift of $\hat{H}$. This will be justified as a consequence of the assumption of Poincaré invariance of the vacuum $|\Omega\rangle$. But for now, we shall leave it and look at the t-dependence of the scalar field.

Consider the (quantum) canonical equation of motion $\dot{\mathcal{F}} = i[\hat{H}, \mathcal{F}]/\hbar$, from which we can determine the t-dependence of $\hat{\phi}$ and $\hat{\pi}$, for example how the mode expansion of (1.158) behaves. As we have seen previously $\hat{\phi}(x)$ is a Poincaré-covariant local operator in the sense that

$$\hat{\phi}(\Lambda x + a) = U(\Lambda, a)\hat{\phi}(x)(U(\Lambda, a))^{-1}. \tag{1.168}$$

We also know that the vacuum obeys $a_{\vec{k}}|\Omega\rangle = 0$ for all $\vec{k}$ and that $H|\Omega\rangle = 0$. But beyond this, we can't say much, and it won't be useful for us when we get to the path integral.

Moving onto symmetries of fields, recall that in *classical* field theory, the action $S[\phi]$ is considered invariant under a transformation of the field if $\delta S = 0$, namely for a scalar that transforms as

$$\phi(x) \rightsquigarrow \phi'(x) = \phi(x) + \epsilon\delta\phi(x) \tag{1.169}$$

where ϵ is an infinitesimal parameter and $\delta\phi(x)$ is determined by $\phi(x)$ or $\partial_\mu\phi(x)$ or derivative thereof. As an example of an explicit symmetry, we can consider the Poincaré symmetry which for a scalar field, is said to be a symmetry of said scalar if $\phi(x) \rightsquigarrow \phi'(x)$ where $\phi'(x)$ is such that

$$\phi'(\Lambda \cdot x + a) = \phi(x). \tag{1.170}$$

Recall that the infinitesimal form for the Lorentz transformation is $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$ where $\omega_{\mu\nu} \equiv \eta_{\mu\rho}\omega^\rho{}_\nu$ is antisymmetric in $[\mu\nu]$ namely that $\omega_{\mu\nu} = -\omega_{\nu\mu}$. Here, $\omega_{\mu\nu}$ and

[21] I urge that you try out (at least once or twice) the creation-annihilation algebra that is going on here just to get a feel for it. The homework questions will also do this as well.

the infinitesimal form of a^μ, ϵ^μ should be treated as infinitesimal. Therefore, if we series expand (1.170) we find

$$\phi'(\Lambda \cdot x + a) = \phi'(x) + (\omega^\mu{}_\nu x^\nu + \epsilon^\mu)\partial_\mu \phi'(x) \implies \delta\phi(x) = -(\omega^\mu{}_\nu x^\nu + \epsilon^\mu)\partial_\mu \phi(x). \quad (1.171)$$

What this implies for the action under a Poincaré transformation is the following

$$\begin{aligned} S[\phi'] &= \int d^D x\, \mathcal{L}\left[\phi'(x), \partial_\mu \phi'(x), \cdots\right] \\ &= \int d^D x'\, \mathcal{L}\left[\phi'(x'), \frac{\partial \phi'(x')}{\partial x'^\mu}, \cdots\right] \\ &= \int d^D x\, \mathcal{L}\left[\phi(x), (\Lambda^\mu)_\mu{}^\nu \partial_\nu \phi(x) \cdots\right] \\ &= S[\phi] \end{aligned} \quad (1.172)$$

where in the second line we renamed the variable $x \to x'$, and then performed Poincaré transformation which shows that the scalar field and derivatives of transformed covariantly under Poincaré transformations, and thus the entire action is invariant under Poincaré transformations *provided* that $\mathcal{L}$ is covariant. With this brief review of symmetries in field theory, let us move onto Noether's theorem which is a state about *continuous* symmetries[22].

Noether's theorem is the statement that for every continuous symmetry, there is a corresponding conserved current $j^\mu(x)$ such that it is conserved $\partial_\mu j^\mu(x) = 0$ that is a consequence of the equations of motion, and there is a corresponding charge $Q = \int d^{D-1}\vec{x} j^0(x)$.

To see this, consider a *non*-symmetry transformation

$$\widetilde{\delta}\phi(x) = \phi(x) + \varepsilon(x)\delta\phi(x) \quad (1.173)$$

where $\varepsilon(x)$ is a coordinate dependent quantity, and thus no longer a symmetry, and $\delta\phi(x)$ here is the usual symmetry variation. The action then transforms as

$$\widetilde{\delta} S[\phi] \equiv S[\phi + \widetilde{\delta}\phi] - S[\phi] \quad (1.174)$$

which is not expected to vanish, but it should take the form of

$$\widetilde{\delta} S = \int d^D x\, \partial_\mu \varepsilon(x)\, j^\mu(x). \quad (1.175)$$

The $\partial_\mu \varepsilon(x)$ here is setup such that if $\varepsilon(x)$ is a constant, then the variation of S under the $\widetilde{\delta}$ transformation vanishes. Now, assume that $\varepsilon(x)$ has compact support which just means that $\varepsilon(x) = 0$ for x's that are not in some set of point $\mathcal{D}$ where $\mathcal{D} \supset \mathbb{R}^{1,D-1}$ where $\mathcal{D}$ is called a *compact domain*. We can re-write $\widetilde{\delta} S$ by integrating by parts to move the derivative onto j^μ

$$\widetilde{\delta} S = -\int_{\mathcal{D}} d^D x\, \varepsilon(x)\, \partial_\mu j^\mu(x). \quad (1.176)$$

[22] There is an analogous Noether's theorem for *finite* discrete symmetries but there it is just that there are no conserved continuous symmetries but there is a conserved charge.

However by construction, $j^\mu(x)$ is built out of the field $\phi(x)$ and derivatives thereof. Now, *if* we restrict ourselves to $\phi(x)$ such that it obeys the equations of motion (1.141) or $\delta' S = 0$ for *any* variation $\delta'\phi$, then $\widetilde{\delta} S = 0$ for *any* $\varepsilon(x)$ i.e.

$$\partial_\mu j^\mu(x) = 0 \tag{1.177}$$

giving us a conservation law. And then by construction, the charge is also conserved namely

$$Q = \int d^{D-1}\vec{x}\, j^0(x) \tag{1.178}$$

which under a time-derivative becomes

$$\partial_t Q = \int d^{D-1}\vec{x}\, \partial_0 j^0 = -\int d^{D-1}\vec{x} \sum_{i=1}^{D-1} \frac{\partial j^i}{\partial x^i} = 0 \tag{1.179}$$

by the conservation law (1.177). There is even more here that is sometimes not clear. If we express Q in terms of the canonical variables ϕ and π, then Q is also called a *generator* of the symmetry in the sense that the variation of the field is given by the Poisson bracket

$$\delta\phi(x) = \{Q, \phi(x)\}. \tag{1.180}$$

Since this isn't often seen in textbooks, I shall give a direct proof. Consider the same transformation $\widetilde{\delta}$ that we saw above on the action S,

$$\begin{aligned}
\widetilde{\delta} S &= \int d^D x \left[\frac{\partial \mathcal{L}}{\partial \phi} \varepsilon(x) \delta\phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\mu (\varepsilon(x)\delta\phi) \right] \\
&= \int d^D x \left[\partial_\mu \varepsilon(x) \cdot \delta\phi \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} + \varepsilon(x) \left(\delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + \partial_\mu \delta\phi \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) \right] \\
&= \int d^D x \left[\partial_\mu \varepsilon(x) \cdot \delta\phi \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} + \varepsilon(x) \left(\partial_\mu \delta \mathcal{U}^\mu \right) \right].
\end{aligned} \tag{1.181}$$

We also have that the Noether current and charge can be written as

$$j^\mu = \delta\phi \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} - \delta \mathcal{U}^\mu \tag{1.182}$$

$$Q = \int d^{D-1}\vec{x} \left(\delta\phi \cdot \pi - \delta \mathcal{U}^0 \right). \tag{1.183}$$

Now, if $\delta\phi$ is independent of $\dot{\phi}$, and thus of π, $\delta\mathcal{U}^0$ must also vanish. We also have that

$$\{Q, \phi(x)\} = \frac{\delta Q}{\delta \pi(x)} = \delta\phi(x) \tag{1.184}$$

and thus re-produces the variation of ϕ. We also have the case where if $\delta\phi$ depends on $\dot{\phi}$, for example as $\delta\phi \propto \dot{\phi}$ in the case of a time-translation symmetry such as for the general SO(N) scalar theory

$$\mathcal{L} = \frac{1}{2} G_{\alpha\beta}(\phi) \dot{\phi}^\alpha \dot{\phi}^\beta - V(\phi) \tag{1.185}$$

then we have $\delta\mathcal{U}^0 = \frac{1}{2}\delta\phi^\alpha\pi_\alpha$ plus an independent term of $\dot{\phi}$ or π. In this case, $\{Q, \phi\} = \delta\phi$ still holds by direct computation.

Let us also consider the Noether current associated with translation symmetry where the scalar transformation rule is $\delta\phi(x) = -a^\mu\partial_\mu\phi(x)$, which has a simple Noether current $j^\mu(x) = a^\nu T^\mu{}_\nu(x)$ where $T^\mu{}_\nu(x)$ is called the *stress-energy tensor* (also called the energy-momentum tensor). As a direct example, we can consider the scalar field theory again but with $m = 0$ namely

$$S[\phi] = \int d^D x \left(-\frac{1}{2}(\partial_\mu\phi)^2 - V(\phi)\right) \tag{1.186}$$

which, under the general transformation $\widetilde{\delta}\phi = -\varepsilon^\mu(x)\partial_\mu\phi(x)$ becomes

$$\begin{aligned}\widetilde{\delta}S &= \int d^D X \left(\partial^\mu\phi\partial_\mu(\varepsilon^\nu(x)\partial_\nu\phi) + \varepsilon^\mu(x)\partial_\mu\phi\frac{\partial V}{\partial\phi}\right)\\ &= \int d^D x \left(\partial_\mu\varepsilon^\nu\partial^\mu\phi\partial_\nu\phi + \varepsilon^\nu\partial_\nu\left(\frac{1}{2}\partial^\mu\phi\partial_\mu\phi\right) + V\right)\\ &= \int d^D x\, \partial_\mu\varepsilon^\nu(x)\, T^\mu{}_\nu(x).\end{aligned} \tag{1.187}$$

(1.187) allows us to read off directly what the energy-momentum tensor is

$$T^\mu{}_\nu = \partial^\mu\phi\partial_\nu\phi - \delta^\mu{}_\nu\left(\frac{1}{2}\partial^\rho\phi\partial_\rho\phi + V(\phi)\right) \tag{1.188}$$

where $\delta^\mu{}_\nu$ is often called the Euclidean metric and has completely diagonal components with values of only 1. But despite this being a 2-index object, it is still just a Noether current, and thus there is a corresponding Noether charge which happens to be the *Hamiltonian*

$$\begin{aligned}H &= \int d^{D-1}\vec{x}\, T^{00}\\ &= \int d^{D-1}\vec{x}\left(\frac{1}{2}(\partial_0\phi)^2 + \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2 + V(\phi)\right).\end{aligned} \tag{1.189}$$

But the interesting thing with a ranked-2 Noether current is that the 0th component also holds more conserved quantities, which in this case T^{0i} gives the conserved spatial momenta

$$P^i = \int d^{D-1}\vec{x}\, T^{0i} \tag{1.190}$$

where $i = 1, \ldots, D-1$. But, what about Lorentz symmetry since, combined with the translation symmetry we just looked at, makes up the Poincaré symmetry. For the scalar field, the Lorentz symmetry is given by $\delta\phi(x) = -\omega^\mu{}_\nu x^\nu\partial_\mu\phi(x)$, and following the same procedure (replace the ε^μ above)

$$j^\mu(x) = \omega^\nu{}_\rho\, x^\rho T^\mu{}_\nu(x). \tag{1.191}$$

The variation of such a current does not look trivial, so let's check

$$\partial^\mu j^\mu = \omega^\nu{}_\rho x^\rho\partial_\mu T^\mu{}_\nu + \omega^\nu{}_\mu T^\mu{}_\nu = \frac{1}{2}(\omega_{\nu\mu} + \omega_{\nu\mu})T^{\mu\nu} = 0 \tag{1.192}$$

where we separated the $\omega_{\mu\nu}$ tensor into two parts, and then recall that $\omega_{\mu\nu} = -\omega_{\nu\mu}$ which therefore vanishes *if* $T^{\mu\nu} = T^{\nu\mu}$.

Now, everything we just looked at is the *classical* Noether theorem, and so of course there is (or should be) a quantum mechanical version. Upon quantization, the Noether current $j^\mu(x)$ becomes a local *field* operator itself just like $\hat{\phi}(x)$. The Noether current now becomes a Hermitian operator that generates the symmetry via the commutator, and not the Poisson brackets

$$\delta\hat{\phi}(x) = -\frac{i}{\hbar}\left[Q, \hat{\phi}(x)\right]. \tag{1.193}$$

The full covariant stress-tensor is a unique operator as well. The energy-momentum tensor $T^{\mu\nu}(x)$ is *always* a well-defined local operator in a QFT. For example it can build the true momentum-operator

$$\hat{P} = \int d^{D-1}\vec{x}\, T^{0\mu}(x) \tag{1.194}$$

as well as the boost-angular momentum operator we saw at the very beginning

$$\hat{J}^{\mu\nu} = \int d^{D-1}\vec{x}\left(x^\mu T^{0\nu}(x) - (\mu \leftrightarrow \nu)\right). \tag{1.195}$$

Something helpful to recall is that the unitary operator that allows to transform field valued operators was constructed out of (1.194) and (1.195)

$$U(\Lambda, a) = 1 - i\epsilon^\mu \hat{P}_\mu + \frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu}. \tag{1.196}$$

Ok, so all of that was done via the canonical quantization method which really was just a method to get towards the Noether currents and symmetry discussions of the action that will be extremely useful for us when we get to the path integral... Which we finally intend to fully embark on now. From this point forward, our primary tool for exploring QFT will be the path integral, with occasionally references to the canonical quantization method we just saw.

The path integral quantization of a *scalar* field theory involves evaluating the following partition function

$$Z = \int D[\phi] : e^{i\int d^D x\, \mathcal{L}(\phi, \partial_\mu\phi)}. \tag{1.197}$$

Formally, extending the path integral we used in QM for a *finite* many degrees of freedom involves a state in the Hilbert space $\Psi \in \mathcal{H}$ that becomes a wave functional in the field formalism with a new argument of $\Psi[\phi(\vec{x}, t_0)]$ where t_0 is a fixed time (again, formally, not practically as we will see why). The path integral then becomes

$$\int [D\phi]\Big|_{\phi(\vec{x},t_i)=\phi_i(\vec{x})}^{\phi(\vec{x},t_f)=\phi_f(\vec{x})} e^{i\int_{t_i}^{t_f} dx^0 \int d^{D-1}\vec{x}\, \mathcal{L}(\phi, \partial_\mu\phi)} = \langle\phi_f|\, U(t_f, t_i)\, |\phi_i\rangle \tag{1.198}$$

where the unitary operator here is $U(t_f, t_i) = \exp\Big(-i\hat{H}(t_f - t_i)\Big)$. The state $|\phi_i\rangle$ is an eigenstate with respect to the operator $\hat{\phi}(x)$ defined via an insertion of $\phi(x)$ in the path integral namely

$$\hat{\phi}(\vec{x}, 0) \cdot \Psi[\phi(\vec{x}, 0)] = \phi(\vec{x}, 0)\Psi[\phi(\vec{x}, 0] \tag{1.199}$$

to which implies the following eigenstate (via operator-state relation)

$$|\phi_i\rangle \leftrightarrow \Psi_i[\phi(\vec{x},0)] \propto \delta[\phi(\vec{x},0) - \phi_i(\vec{x})] \tag{1.200}$$

where the δ-function here is called a *functional δ-distribution*. As always, to make all of this analysis precise requires regularization (just like in QM 1.1). For example either via discretizing space or using a cutoff on spatial wave number of $\phi(\vec{x},t)$. We will mostly avoid directly working with the wave functional[23] $\Psi[\phi(\vec{x},0)]$ which is much harder to regularize. It will be more convenient to extract physical observables from vacuum expectation values for example,

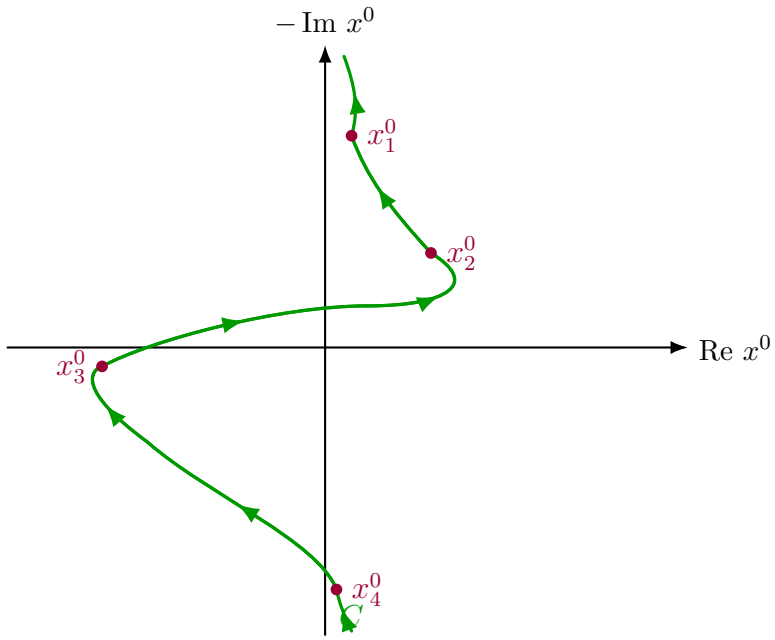
$$\langle\Omega|\,\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\,|\Omega\rangle\,. \tag{1.201}$$

To ensure convergence of the path integral Z, it is useful to analytically continue the time components of each field operator in (1.201), x_1^0, x_2^0, $\cdots$, x_n^0 to complex time but we *enforce* the specific ordering of *imaginary* time,

$$\operatorname{Im} x_1^0 < \operatorname{Im} x_2^0 < \cdots < \operatorname{Im} x_n^0 \tag{1.202}$$

which ensures that analyticity is maintained. This is us subtly inputting what the partition function must satisfy in order to ensure *causality*. This will be one of the first instances we see the connection between analyticity an causality.

The real parts of the time components like $\operatorname{Re} x_1^0$ and so on, are not subject to any order a priori as shown in this graph for example:



[23]The wave functional formalism is actually popular however when thinking about quantum field theory in curved spacetime where wave function *coefficients* arise that are directly related to correlation functions, such as those used to compute observables like the power spectrum for inflation.

where in 1.2.1, the contour C here is actually

$$S = \int_C dx^0 \int d^{D-1}\vec{x}\, \mathcal{L} \tag{1.203}$$

at the level of the path integral. It is mot convenient to start with the Euclidean path integral here which is defined as the analytically continued correlation function

$$\langle \Omega | \hat{\phi}(x_1) \cdots \hat{\phi}(x_n) | \Omega \rangle \equiv \langle \hat{\phi}_E(x_1^E) \cdots \hat{\phi}_E(x_n^E) \rangle \tag{1.204}$$

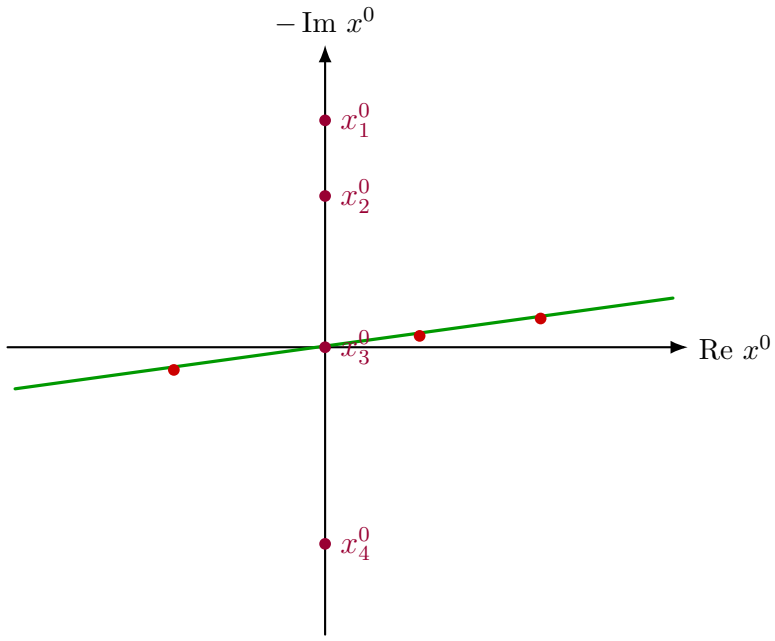
where for each time coordinate, it is defined to the Euclidean coordinate by $e_i^0 = -i(x_1^E)^D$, and $\vec{x}_i \equiv \vec{x}_i^E$ where the ordering of the Euclidean time coordinates is[24]

$$(x_1^E)^D > (x_2^E)^D > \cdots > (x_n^E)^D. \tag{1.205}$$

The computation of the Euclidean correlator is then just the path integral definition but with the Euclidean action

$$\langle \hat{\phi}(x_1^E) \cdots \hat{\phi}(x_n^E) \rangle = \frac{1}{Z} \int [D\phi] e^{-S^E[\phi]} \phi(x_1^E) \cdots \phi(x_n^E) \tag{1.206}$$

where the Euclidean action is something we have already seen a lot of in QM, $S^E = \int d^D x^E\, \mathcal{L}^E$ where $\mathcal{L}^E = -\mathcal{L}$, and so we are already well acquainted with it. What this *Wick rotation* looks like graphically is the following



We can see that when comparing 1.2.1 to 1.2.1, we are performing a *uniform* Wick rotation on all $i = 1, \ldots, n$ which simultaneously leads back to the correlator $\langle \Omega | \hat{\phi}(x_1) \cdots \hat{\phi}(x_n) | \Omega \rangle$. Specifically, we rotate by a general angle α each time coordinate as

$$-ix_i^0 = e^{-i\alpha} t_i. \tag{1.207}$$

[24] As a note on notation, later we will typically omit the Euclidean subscript and write $\hat{\phi}_E$ simply as $\hat{\phi}$ when in the context of Euclidean correlators.

However, if we want to maintain the inequality order of the time coordinates we established easier, we need to have it such that $\alpha = \pi/2$ which approach 0 from the positive side (denoted 0^+), namely we keep the order

$$x_1^0 > x_2^0 > \cdots > x_n^0. \tag{1.208}$$

The conclusion we reach then is, "the time-ordered real-time correlator is obtained from the Euclidean one by a *uniform* Wick rotation on all x_i^0's." As a concrete example, let us revisit the massive free scalar field that has the following Lagrangian and corresponding Euclidean Lagrangian,

$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2 \tag{1.209}$$

$$\mathcal{L}^E = \frac{1}{2}\sum_{\mu=1}^{D}(\partial_\mu\phi)^2 + \frac{1}{2}m^2\phi^2. \tag{1.210}$$

The Euclidean 2-point function is computed exactly as in the QM case with just higher dimensional derivatives,

$$\begin{aligned}\langle\phi(x)\phi(y)\rangle &= \frac{1}{Z}\int[D\phi]\, e^{-S^E}\phi(x)\phi(y)\\ &= G(x,y).\end{aligned} \tag{1.211}$$

From the equation of motion, $(-\Box_x + m^2)\phi(x) = 0$, where $\Box_x \equiv \partial^\mu\partial_\mu$, the Green's function must obey

$$(-\Box_x + m^2)G(x,y) = \delta^D(x-y) \tag{1.212}$$

We can Fourier transform (1.212) via $\Box_x \to -k^2$ since $\partial_\mu \to ik_\mu$, and then divide over the operator $(-\Box_x + m^2) = (k^2 + m^2)$ to find the momentum (Fourier) space version of the *Euclidean* Green's function,

$$G(x,y) = \int \frac{d^Dk}{(2\pi)^D}\frac{e^{ik\cdot(x-y)}}{k^2+m^2} \tag{1.213}$$

where it should be noted the inner product in (1.213) is an inner product using the *Euclidean* metric, not the Minkowski one (so all plus signs). We now perform an analytic continuation to real time (also called Lorentzian time) by setting $-ix^D = e^{-i\alpha}t$ and $-iy^D = e^{-i\alpha}t'$ such that $t > t'$ and $\alpha \to 0^+$. We also set $-ik^D = e^{i\alpha}k^0$. (1.213) then becomes

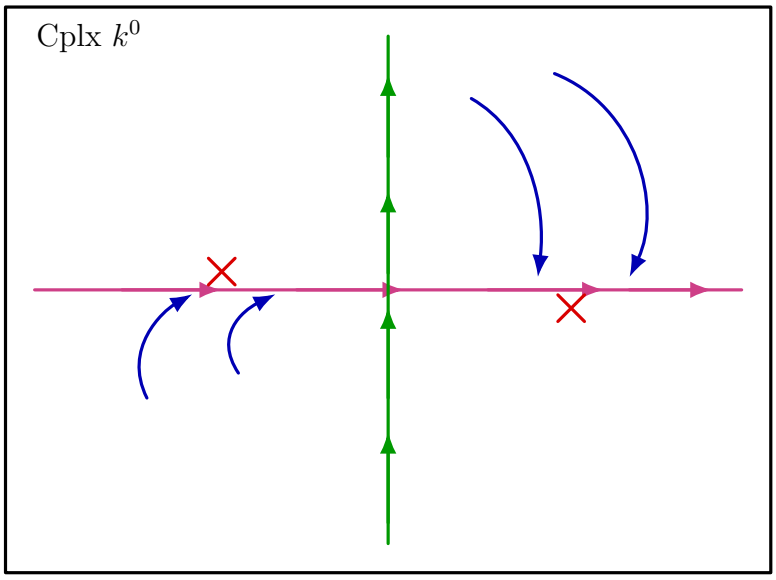
$$G(x,y) = -e^{i\alpha}\int \frac{dk^0 d^{D-1}\vec{k}}{(2\pi)^D}\frac{e^{i\vec{k}\cdot(\vec{x}-\vec{y})-ik^0(t-t')}}{-e^{2i\alpha}(k^0)^2 + (\vec{k})^2 + m^2}. \tag{1.214}$$

To clarify, the sign in (1.207) is from the orientation. Also, notice that the denominator never vanishes as α ranges from $(\pi/2, 0^+)$. But what about the $\alpha \to 0$ limit? Notice that if we set $\alpha = 0$ in the integrand,the denominator would vanish when $k^0 = \pm\sqrt{(\vec{k})^2 + m^2}$ (i.e. the poles of the function) and is thus ill-defined. Therefore, in order to maintain analyticity, we can add an $-i\epsilon$ to the denominator such that the imaginary part of (1.214)

never vanishes. Then at the end, we take the $\epsilon \to 0^+$ *after* evaluating the k-integrals. The result then from (1.214) is the *time-ordered* 2-point function

$$\begin{aligned}\langle\omega| T\hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle &= -i\int \frac{dk^0 d^{d-1}\vec{k}}{(2\pi)^D}\frac{e^{i\vec{k}\cdot(\vec{x}-\vec{y})-ik^0(x^0-y^0)}}{-(k^0)^2+(\vec{k})^2+m^2-i\epsilon} \\ &= -i\int \frac{d^Dk}{(2\pi)^D}\frac{e^{ik\cdot(x-y)}}{k^2+m^2-i\epsilon}\end{aligned} \tag{1.215}$$

where now any inner products in (1.215) are of *Lorentzian signature* namely $k^2 \equiv k_\mu k^\mu = -(k^0)^2 + (\vec{k})^2$ as we have seen before. I'd recommend practically memorizing (1.215) since we will not be seeing and using this result over and over and over... But, before we do so, it should be noted that the analytic structure of(1.215) appears as



The primary thing to notice here is that there are two poles in the Green's function, essentially giving us four different ways to perform the integration over k^0, which correspond to different Green's functions. However, so far we have seen *three* different types of correlation functions, so it may be best to lay them all out right now. There are three useful types of Green's functions in a general QFT. The first is called the *Wightman function* where in this case, the spacetime coordinates are $x_a = (t_a, \vec{x}_a) \in \mathbb{R}^{1,D-1}$ and is written as

$$W(x_1, \cdots, x_n) = \langle\Omega| \hat{\phi}(x_1)\cdots\hat{\phi}(x_n) |\Omega\rangle . \tag{1.216}$$

Another type is the *time-ordered* Greens function,

$$G(x_1, \cdots, x_n) = \langle\Omega| T\hat{\phi}(x_1)\cdots\hat{\phi}(x_n) |\Omega\rangle \tag{1.217}$$

where the operators in-between the states are time-ordered products with the time components obeying $x^0_{i_1} \geq x^0_{i_2} \geq \cdots \geq x^0_{i_n}$ for $(i_1 \cdots i_n)$ a permutation of $(1 \cdots n)$. The final one which we have the most work with so far is the *Euclidean* Greens function also known

as the *Schwinger* function where $x_a^E = (\tau_a, \vec{x}_a) \in \mathbb{R}^D$,

$$G_E(x_1^E, \cdots, x_n^E) = W(x_i, \cdots, x_{i_n}) \Big|_{t_a \to -i\tau_a} \tag{1.218}$$

where the Lorentzian time of the Wightman function is analytically continued to Euclidean time for $\tau_{i_1} \geq \cdots \tau_{i_n}$ for $(i_1, \cdots, i_n)$ a permutation of $(1 \cdots n)$ as we have seen before. Each of theses Greens functions have their own useability.

The Wightman functions (1.216) are most directly related to the *spectrum* of a given correlation function/system. For example, let us consider the 2-point Wightman function of scalars,

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle &= \int d\alpha \ \langle\Omega| \hat{\phi}(x) |\alpha\rangle \langle\alpha| \hat{\phi}(y) |\Omega\rangle \\
&= \int d\alpha \ \langle\Omega| e^{-i\hat{P}\cdot x}\hat{\phi}(0)e^{i\hat{P}\cdot x} |\alpha\rangle \langle\alpha| e^{-i\hat{P}\cdot y}\hat{\phi}(0)e^{i\hat{P}\cdot y} |\Omega\rangle \\
&= \int d\alpha \ e^{iP_\alpha\cdot(x-y)} |\langle\Omega| \hat{\phi}(0) |\alpha\rangle|^2 \\
&= \int d^Dk \ e^{ik\cdot(x-y)} \int d\alpha \ \delta^D(k - P_\alpha) |\langle\Omega| \hat{\phi}(0) |\alpha\rangle|^2.
\end{aligned} \tag{1.219}$$

In the first line, we inserted a complete basis of state $\int d\alpha \ |\alpha\rangle \langle\alpha| = \mathbb{1}$ where α here is taken to be a particle label for some different state related to the total momentum operator as $\hat{P}^\mu |\alpha\rangle = p_\alpha^\mu |\alpha\rangle$. We have also assumed that $\hat{\phi}(0)$ is Hermitian, allowing us to write the expectation value for it. In the final step, we "un-did" an integral over k^μ (the $\int d^Dk$) via a Dirac delta function. Now, I claim that the integral over $d\alpha$ is invariant under a Lorentz transformation $k^\mu \to \Lambda^\mu{}_\nu k^\nu$. As proof, recall that the Lorentz transformation acts as a unitary operator $U(\Lambda)$ as in (1.16), while on a given state $|\alpha\rangle$, the unitary Lorentz transformation behaves as

$$U(\Lambda) |\alpha\rangle = \left|\alpha^\Lambda\right\rangle \tag{1.220}$$

or just some transformed state. The momentum transforms similarly $P^\mu_{\alpha^\Lambda} = \Lambda^\mu{}_\nu P^\nu_\alpha$. Then, the $d\alpha$ integral under a Lorentz transformation becomes

$$\begin{aligned}
\int d\alpha \ \delta^D(\Lambda \cdot k - P_\alpha) |\langle\Omega| \hat{\phi}(0) |\alpha\rangle|^2 &= \int d\alpha \ \delta^D(\Lambda \cdot k - P_{\alpha^\Lambda}) |\langle\Omega| \hat{\phi}(0) \left|\alpha^\Lambda\right\rangle|^2 \\
&= \int d\alpha \ \delta^D(k - P_\alpha) |\langle\Omega| (U(\Lambda))^{-1}\hat{\phi}(0)U(\Lambda) |\alpha\rangle|^2 \\
&= \int d\alpha \ \delta^D(k - P_\alpha) |\langle\Omega| \hat{\phi}(0) |\alpha\rangle|^2
\end{aligned} \tag{1.221}$$

thus showing the integral over $d\alpha$ is invariant under Lorentz transformations. What did all of this buy us? Just a showing that after an insertion for a complete basis of states that the Wightman function is still Lorentz invariant. This also means that there are relativistic states of the corresponding spectrum lurking within it. Namely, we can write the integral over $d\alpha$ as

$$\int d\alpha \ \delta^D(k - P_\alpha) |\langle\Omega| \hat{\phi}(0) |\alpha\rangle|^2 := \frac{\theta(k^0)}{(2\pi)^{D-1}} \rho(-k^2) \tag{1.222}$$

where $\rho(-k^2)$ is some non-negative function. The appearance of the step-function $\theta(k^0)$ (which is equal to 1 for $k^0 \geq 0$ and zero if $k^0 < 0$) is due to our assumption that $P^0 \geq 0$ for *all* physical states (recall that P^0 is the Hamiltonian). The Wighman function then becomes

$$\begin{aligned}\langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle &= \int d^Dk\, e^{ik\cdot(x-y)}\frac{\theta(k^0)}{(2\pi)^{D-1}}\rho(-k^2)\\ &= \int_0^\infty d\mu^2\,\rho(\mu^2)\int\frac{d^Dk}{(2\pi)^{D-1}}\theta(k^0)\delta(k^2+\mu^2)e^{ik\cdot(x-y)}\end{aligned} \tag{1.223}$$

where we shall define the integral over k^μ as the *advanced propagator* Δ_+[25]

$$\begin{aligned}\Delta_+(x;\mu^2) &= \int\frac{d^Dk}{(2\pi)^{D-1}}\theta(k^0)\delta(k^2+\mu^2)e^{ik\cdot x}\\ &= \int\frac{d^{D-1}\vec{k}dk^0}{(2\pi)^{D-1}}\theta(k^0)\frac{1}{2\sqrt{(\vec{k})^2+\mu^2}}(\delta(k^0+\sqrt{(\vec{k})^2+\mu^2})+\delta(k^0-\sqrt{(\vec{k})^2+\mu^2}))e^{-ik^0x^0+i\vec{k}\cdot\vec{x}}\\ &= \int\frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}}\frac{1}{2\sqrt{(\vec{k})^2+\mu^2}}e^{i\vec{k}\cdot\vec{x}-i\sqrt{(\vec{k})^2+\mu^2}x^0}\end{aligned} \tag{1.225}$$

where in the last step, only the second delta function gives us a non-zero answer due to the step function. We have actually already seen something like this in the prelude in that $\Delta_+(x;\mu^2) = \Delta_+(-x;\mu^2)$ for $x^2 > 0$ or spacelike, but are not equal for timelike separation. Moving onto the time-ordered Greens function (1.217), is most closely related to the scattering amplitudes of particles (via the LSZ reduction which we will see later).

For now, let us just inspect its structure, to which we will find the Feynman propagator. So, starting off with the 2-point function as we have done so many times

$$\begin{aligned}G(x,y) &\coloneqq \langle\Omega|\,T\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle\\ &\equiv \theta(x^0-y^0)\,\langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle + \theta(y^0-x^0)\,\langle\Omega|\,\hat{\phi}(y)\hat{\phi}(x)\,|\Omega\rangle\\ &= \int_0^\infty d\mu^2\,\rho(\mu^2)\left(\theta(x^0-y^0)\Delta_+(x-y;\mu^2)+\theta(y^0-x^0)\Delta_+(y-x;\mu^2)\right)\end{aligned} \tag{1.226}$$

to which, similar to what we did with the Wightman function, everything after the $d\mu^2\rho(\mu^2)$ we will call the *Feynamn propagator*(but put it into a nicer form later)

$$\Delta_F(x;\mu^2) = \int\frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}}\frac{e^{i\vec{k}\cdot\vec{x}}}{2\sqrt{(\vec{k})^2+\mu^2}}\left(\theta(x^0)e^{-i\sqrt{(\vec{k})^2+\mu^2}x^0}+\theta(-x^0)e^{i\sqrt{(\vec{k})^2+\mu^2}x^0}\right). \tag{1.227}$$

[25] A property of the Dirac delta function used here is that

$$\delta(x^2-a^2) = \frac{1}{2|a|}(\delta(x+a)+\delta(x-a)) \tag{1.224}$$

but in our case it is a little tricky since $k^2+\mu^2 = -(k^0)^2+(\vec{k})^2+\mu^2$ and so if we integrate over k^0 then $a \rightsquigarrow \sqrt{(\vec{k})^2+\mu^2}$.

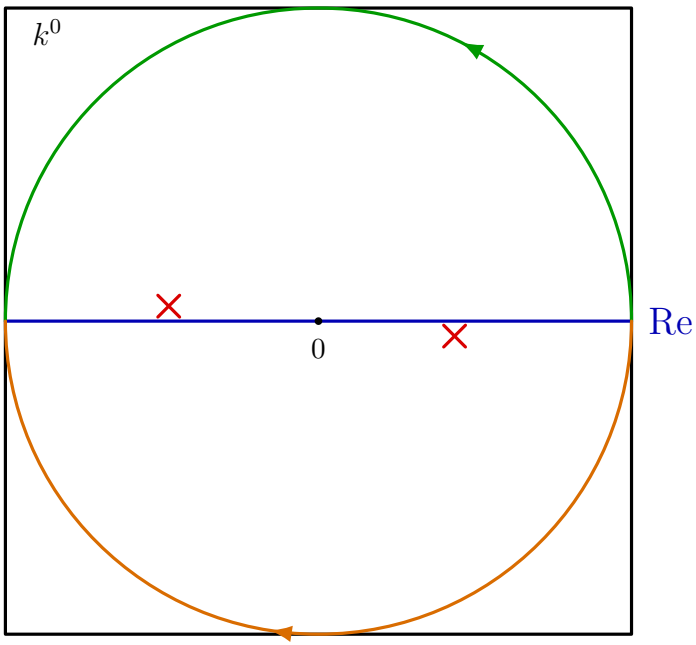
This propagator has a slightly different structure then the advanced one in (1.225) thanks to the step functions, and so we need to be a little more careful of the analytic structure. So, let us just consider the following integral that has a similar pole structure,

$$\int \frac{dk^0}{2\pi i} \frac{e^{ik^0x^0}}{-(k^0)^2 + (\vec{k})^2 + \mu^2 - i\epsilon}. \tag{1.228}$$

If we view this contour integral on the complex k^0 plane as we have done before, we can see that there are poles when

$$k^0 = \pm\sqrt{(\vec{k})^2 + \mu^2 - i\epsilon} \approx \pm\sqrt{(\vec{k})^2 + \mu^2}\left(1 - \frac{i\epsilon}{2((\vec{k})^2 + \mu^2)}\right) \tag{1.229}$$

from which we can plot to see the two possible contours enforced by the different step functions,



The contour in green would be us closing the contour at $k^0 \to +i\infty$ if $x^0 < 0$. The orange contour would be us closing the contour at $k^0 \to -\infty$ if $x^0 > 0$. We can separate out the integral into these two different contour integrations by inserting a step function when computing the residue

$$\begin{aligned}\int \frac{dk^0}{2\pi i} \frac{e^{ik^0x^0}}{k^2 + \mu^2 - i\epsilon} &= -\theta(x^0) \operatorname*{Res}_{k^0 \to \sqrt{(\vec{k})^2+\mu^2-i\epsilon}} \frac{e^{-ik^0x^0}}{k^2 + \mu^2 - i\epsilon} + \theta(x^0) \operatorname*{Res}_{k^0 \to -\sqrt{(\vec{k})^2+\mu^2-i\epsilon}} \frac{e^{-ik^0x^0}}{k^2 + \mu^2 - i\epsilon} \\ &= -\theta(x^0)\left(\frac{e^{-i\sqrt{(\vec{k})^2+\mu^2}x^0}}{-2\sqrt{(\vec{k})^2 + \mu^2}}\right) + \theta(-x^0)\left(\frac{e^{i\sqrt{(\vec{k})^2+\mu^2}x^0}}{2\sqrt{(\vec{k})^2 + \mu^2}}\right).\end{aligned} \tag{1.230}$$

If we compare with the definition of Δ_F above, we can see that we can write Δ_F equivalently as

$$\Delta_F(x; \mu^2) = -i \int \frac{d^Dk}{(2\pi)^D} \frac{e^{ik\cdot x}}{k^2 + \mu^2 - i\epsilon} \tag{1.231}$$

which is *thee* Feynman propagator that will be used often. We still have one more Greens function left, the Euclidean Greens function (1.218) to which we will see, it really is just a handy tool to compute correlators that have a complicated analytic structure (and for numerical computations) in all honesty. We start off with te Wightman function,

$$W(x,y) = \langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle = \int d\alpha\, e^{iP_\alpha\cdot(x-y)}|\,\langle\Omega|\,\hat{\phi}(0)\,|\alpha\rangle\,|^2 \tag{1.232}$$

where $\exp(iP_\alpha\cdot(x-y)) = \exp\Big(i\vec{P}_\alpha\cdot(\vec{x}-\vec{y}) - iP^0_\alpha(x^0-y^0)\Big)$ (again, assuming $P^0_\alpha \geq 0$). But now what about the analytic continuation to complex x^0 and y^0 to the Euclidean Green's function; is it valid? Well, $W(x,y)$ is an analytic function of complex x^0 and y^0 provided that $\exp\big(-iP^0_\alpha(x^0-y^0)\big)$ is exponentially suppressed at large P^0_α i.e. $\mathrm{Im}(x^0-y^0) < 0$. This is possible to main the inequality when going from real x^0 and y^0 to $x^0 = -i\tau_1$ and $y^0 = -i\tau_2$ with $\tau_1 > \tau_2$ or in other words, the Euclidean coordinates are $x^E\vec{(}\tau_1,\vec{x})$ and $y^E \equiv (\tau_2,\vec{y})$. The Euclidean greens function $G(x^E,y^E)$ is then defined as the analytic continuation of $W(x,y)$ for $(x^0,y^0) = (-i\tau_1,-i\tau_2)$ for $\tau_1 > \tau_2$. If we instead had the ordering $\tau_1 < \tau_2$ then $G_E(x^E,y^E)$ is just analytic continuation of instead $W(y,x)$. The result is then

$$\begin{aligned} G_E(x^E,y^E) &= \int_0^\infty d\mu^2\,\rho(\mu^2)\int\frac{d^Dk_E}{(2\pi)^D}\frac{e^{ik_E\cdot x^E}}{(k^E)^2+\mu^2} \\ &= \int_0^\infty d\mu^2\,\rho(\mu^2)\Delta_E(x^E-y^E;\mu^2) \end{aligned} \tag{1.233}$$

where, again to emphasize, the inner product here or dot product is over Euclidean space, which is just a metric with all 1's. The Euclidean Greens function actually has the simplest analytic structure and thus much easier to evaluate (which we will be using).

So, to give a brief recap:

1. Wightman function: directly related to the spectrum of particle states (will help us get into scattering)

2. Time-ordered Greens function: gave us the Feynman propagator and is the most physical due to it's Lorentzian signature.

3. Euclidean Green's function: an analytic continuation of the Wightman function (so, hold's the same importance) but is even easier to compute due to it's analytic structure.

We now look more into the structure of the Greens function that we mentioned prior with respect to the Wightman function

$$\langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle = \int_0^\infty d\mu^2\,\rho(\mu^2)\Delta_+(x-y;\mu^2) \tag{1.234}$$

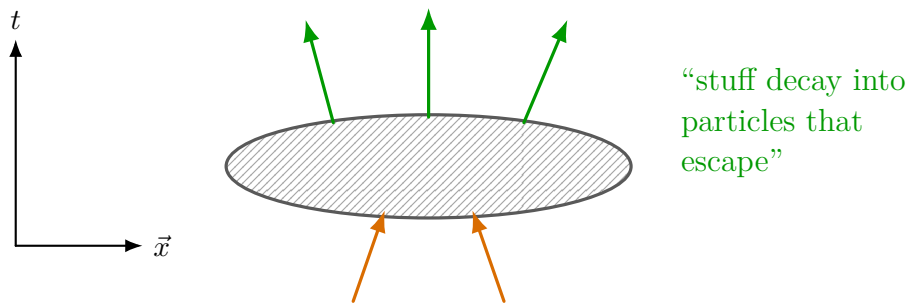
where the *density function* (for now we will call it as such) ρ is given as

$$\rho(\mu^2) = (2\pi)^{D-1} \int d\alpha\, \delta^D(k - P_\alpha) |\langle\alpha| \hat{\phi}(0) |\Omega\rangle|^2 \tag{1.235}$$

where $k^2 = -\mu^2$ and $k^0 > 0$. The question becomes, with the Wightman function written this way, "what is the [particle] content of $\hat{\phi}(x)|\Omega\rangle$?"

1.2.2 1-Particle State of a Scalar Field

At a given instance of time, a physical state may not admit any obvious particle interpretation, but it is expected to evolve into a superposition of states that consist of far-separated wave packets of individual particles as $t \to \infty$ or $t \to -\infty$. Let us take for example the incoming of 2 particles that come in, do something in a frame we can measure, and leaves as 3 particles.



An underlying assumption is that "particles" are weakly interacting at long distance. The particles could be strongly interacting at finite distances, but the invariant mass squared, i.e. $-p_\mu p^\mu = m^2$m of any state would be identical to that of its decay product.

Suppose there is only one species of particle in our theory of some mass m (such as a scalar but not necessarily) with possible invariant masses of $|\alpha\rangle$ being the vacuum $|\Omega\rangle$ where $-p^2 = 0$. The 1-particle state is then $-p^2 = m^2$, while states that decay into two or more particles has an invariant mass of

$$-p^2 \geq (2m)^2 \tag{1.236}$$

which is a *continuum* of possible values for the mass given the momentum squared. Without loss of generality, we can shift $\hat{\phi}(x)$ by a constant such that the field operator has a zero expectation value $\langle\Omega|\hat{\phi}(x)|\Omega\rangle = 0$. Then, we would expect that the spectrum of particles when the field operator acts on the vacuum would generate the following

$$\hat{\phi}(x)|\Omega\rangle = |\text{1-particle}\rangle + |\text{multi-particle stuff}\rangle \tag{1.237}$$

where in the multi-particle state has the mass-shell condition of $-p^2 \geq (2m)^2$. If we assume there are no internal degrees of freedom (such as spin), we can write a basis of 1-particle

states that is labeled by the spatial momentum $\left|\vec{k}\right\rangle$ with $P^i\left|\vec{k}\right\rangle = k^i\left|\vec{k}\right\rangle$ for $i = 1, \cdots, D-1$ and $P^0\left|\vec{k}\right\rangle = \sqrt{(\vec{k})^2 + m^2}\left|\vec{k}\right\rangle$. We choose a normalization of the 1-particle states such that

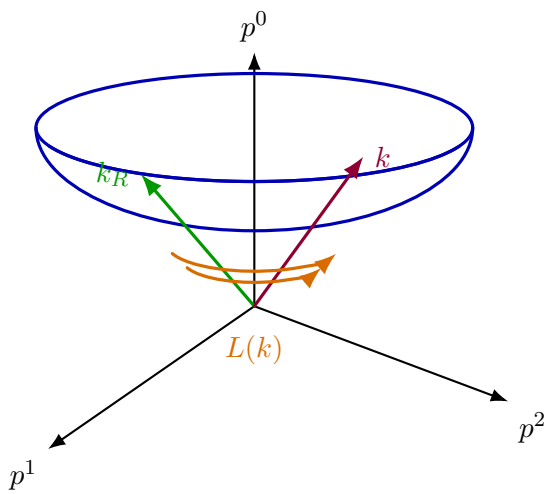
$$\langle\vec{k}|\vec{k}'\rangle = \delta^{D-1}(\vec{k} - \vec{k}'). \tag{1.238}$$

A question though we must answer (which brings into question, "is the particle or field fundamental[26]), "how does the unitary Lorentz symmetry operator $U(\Lambda)$ act on $\left|\vec{k}\right\rangle$?" (We will come back to $\hat{\phi}(x)\left|\Omega\right\rangle$ in a little bit).

We would expect the unitary operator to *transform* the 1-particle state itself up to a function that is dependent upon the Lorentz transformation and the momentum of the original state

$$U(\Lambda)\left|\vec{k}\right\rangle = C(\Lambda, \vec{k})\left|\Lambda\vec{k}\right\rangle \tag{1.239}$$

with $(\Lambda k)^\mu = \Lambda^\mu{}_\nu k^\nu$. To proceed, let us pick a reference momentum frame k_R^μ such that it obeys the mass-shell condition $k_R^2 \equiv k_R^\mu k_{R\mu} = -m^2$. Then, for every possible momentum k^μ of the particle, we will *choose* a "standard boost" denoted $L^\mu{}_\nu(\vec{k})$ such that the reference momentum can be related to another $k^\mu = L^\mu{}_\nu(\vec{k})k_R^\nu$. For a visualization of the *hyperbolic* mass-shell condition and the vectors on it and how they are transformed *around* the mass-shell, it looks like this:



From (1.2.2), it is easier to see that wherever we place the reference vector k_R, the boosts $L(k)$ will rotate it along the hyperboloid of the mass shell to generate a new vctor k.

We can specify the phase of $\left|\vec{k}\right\rangle$ by *defining* the relation to the reference momentum

$$\left|\vec{k}\right\rangle := N(\vec{k})U(L(\vec{k}))\left|\vec{k}_R\right\rangle. \tag{1.240}$$

[26] It's the particle...

For any Lorentz transformation Λ, we can transform (1.240) as

$$\begin{aligned} U(\Lambda)\left|\vec{k}\right\rangle &= N(\vec{k})U(\Lambda)U(L(\vec{k}))\left|\vec{k}_R\right\rangle \\ &= N(\vec{k})U(\Lambda L(\vec{k}))\left|\vec{k}_R\right\rangle \\ &= N(\vec{k})U(L(\vec{\Lambda k}))U((L(\vec{\Lambda k}))^{-1}\Lambda L(\vec{k}))\left|\vec{k}_R\right\rangle \\ &= N(\vec{k})U(L(\vec{\Lambda k}))U(W)\left|\vec{k}_R\right\rangle \end{aligned} \tag{1.241}$$

where we have defined W to be

$$W \equiv L(\vec{\Lambda k})^{-1}\Lambda L(\vec{k}). \tag{1.242}$$

The W object is very interesting. Reading it from right to left (operator style) we can see that W for some given state first boosts k_R by $L(\vec{k})$, then takes $L(\vec{k})k_R = k$ and "spins" it via Λ creating Λk, and then boosts-back by the inverse transformation L^{-1} recovering k_R. Therefore, $U(W)\left|\vec{k}_R\right\rangle \propto \left|\vec{k}_R\right\rangle$. We can see then that $U(W)$ acts on $\left|\vec{k}_R\right\rangle$ by multiplying by some phase $D(W)$ (i.e. to get rid of the proportionality sign to make it an equal, we multiply by $D(W)$) which gives

$$U(\Lambda)\left|\vec{k}\right\rangle = N(\vec{k})U(L(\vec{\Lambda k}))D(W)\left|\vec{k}_R\right\rangle = N(\vec{k})D(W)(N(\vec{\Lambda k}))^{-1}\left|\vec{\Lambda k}\right\rangle \tag{1.243}$$

where $C(\Lambda, \vec{k}) = N(\vec{k})D(W)(N(\vec{\Lambda k}))^{-1}$ that we were originally searching for. But then the question is, "what is $D(W)$?" For now, we will assume that $m > 0$, and without loss of generality, we can choose a reference momentum such that $k_R^\mu = (m, \vec{0})$. We already see that W is a Lorentz transformation that fixes k_R^μ i.e. a spatial rotation implying W is an element of $W \in \mathrm{SO}(D-1)$ and thus obeys the following group algebra $D(W_1W_2) = D(W_1)D(W_2)$ with normalization $|D(W)|^2 = 1$.

For $D \geq 4$, $\mathrm{SO}(D-1)$ is non-Abelian, and furthermore, every element of $\mathrm{SO}(D-1)$ can be written as a "commutator" $W = XYX^{-1}Y^{-1}$ for som $X, Y \in \mathrm{SO}(D-1)$. It then follows that

$$D(W) = D(X)D(Y)(D(X))^{-1}(D(Y))^{-1} = 1. \tag{1.244}$$

Now, such particles that transform in the following way are called *scalar particles*, and unironically at the moment, there is no obvious relation to "scalar fields" yet. The conclusion that we then come to is that

$$U(\Lambda)\left|\vec{k}\right\rangle = \frac{N(\vec{k})}{N(\vec{\Lambda k})}\left|\vec{\Lambda k}\right\rangle. \tag{1.245}$$

To determine the normalization of (1.245), we can compare to the normalization of the 1-particle state by computing

$$\begin{aligned} \left\langle\vec{k}'\right|(U(\Lambda))^\dagger U(\Lambda)\left|\vec{k}\right\rangle &= \frac{N(\vec{k}')}{N(\vec{\Lambda k}')}\frac{N(\vec{k})}{N(\vec{\Lambda k})}\langle\vec{\Lambda k}'|\vec{\Lambda k}\rangle \\ &= \left(\frac{N(\vec{k})}{N(\vec{\Lambda k})}\right)^2 \delta^{D-1}(\vec{\Lambda k}' - \vec{\Lambda k}) \end{aligned} \tag{1.246}$$

to which it should match (1.238). We can re-write the D-dimensional delta function as

$$\delta^D(k'-k) = 2k^0\delta((k')^2-k^2)\delta^{D-1}(\vec{k}'-\vec{k}) \tag{1.247}$$

to which the LHS and the $D=1$ delta function on the RHS are invariant under $k \to \Lambda k$ and $k' \to \Lambda k'$. We can then see that the ratio of the $D-1$ delta functions is

$$\frac{\delta^{D-1}(\vec{k}'-\vec{k})}{\delta^{D-1}(\Lambda\vec{k}'-\Lambda\vec{k})} = \frac{2(\Lambda k)^0}{2k^0} = \left(\frac{N(\vec{k})}{N(\Lambda\vec{k})}\right)^2 \tag{1.248}$$

which implies that $N(\vec{k}) \propto 1/\sqrt{k^0}$. Since $N(\vec{k}_R) = 1$ by definition, we conclude that the proper normalization coefficient is

$$N(\vec{k}) = \sqrt{\frac{m}{k^0}}. \tag{1.249}$$

Coming back to the general scalar-operator acting on the vacuum (1.237), let us now study the overlap it has with a single particle state $\left|\vec{k}\right\rangle$,

$$\left\langle\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle = \left\langle\vec{k}\right| e^{-i\hat{P}\cdot x}\hat{\phi}(0)e^{i\hat{P}\cdot x}\left|\Omega\right\rangle = e^{-ik\cdot x}\left\langle\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle, \tag{1.250}$$

this last expectation value can be written with an insertion of Unitary operators giving us a boosted 1-particle state and an overall normalization

$$\left\langle\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle = \left\langle\vec{k}\right|(U(\Lambda))^{-1}\hat{\phi}(0)U(\Lambda)\left|\Omega\right\rangle = \left(\frac{N(\vec{k})}{N(\Lambda\vec{k}}\right)^*\left\langle\Lambda\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle \tag{1.251}$$

where the normalization combination out front is $\sqrt{(\Lambda k)^0/k^0}$. The conclusion that we then reach is that

$$\left\langle\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle \propto \frac{1}{\sqrt{k^0}} = \frac{1}{\omega_{\vec{k}}} \tag{1.252}$$

where $\omega_{\vec{k}} = \sqrt{(\vec{k})^2+m^2}$. This implies that we can write the the expectation value of the scalar field operator with respect to the vacuum and a 1-particle state as

$$\left\langle\vec{k}\right|\hat{\phi}(0)\left|\Omega\right\rangle \equiv \left(\frac{\mathcal{Z}}{(2\pi)^{D-1}2\omega_{\vec{k}}}\right)^{\frac{1}{2}} \tag{1.253}$$

where $\mathcal{Z} \geq 0$ is a constant and is often called the *field renormalization constant*. For a *free* scalar field, we have already seen that $\mathcal{Z}=1$. People sometimes say that $\mathcal{Z}$ here is an "unphysical" quantity; this is completely **false**. $\mathcal{Z}$ is intrinsic to the field operator $\hat{\phi}(x)$, and is unambiguously defined provided that the field operator $\hat{\phi}(x)$ is defined.

1.2.3 Källen-Lehmann Spectral Representation & Mass-Renormalization

To see the implementation of $\mathcal{Z}$ and where it comes up, recall the *spectral function* (where $p^2 = -\mu^2$ here and $p^0 > 0$)

$$\begin{aligned}\rho(\mu^2) &= (2\pi)^{D-1}\int d\alpha\, \delta^D(p-P_\alpha)|\left\langle\alpha\right|\hat{\phi}(0)\left|\Omega\right\rangle|^2 \\ &= \int d^{D-1}\vec{k}\,\frac{\mathcal{Z}}{2\omega_{\vec{k}}}\delta^D(p-k)\Big|_{k^0=\omega_{\vec{k}}} + (\text{multi-particle } \left|\alpha\right\rangle \text{ contribution}).\end{aligned} \tag{1.254}$$

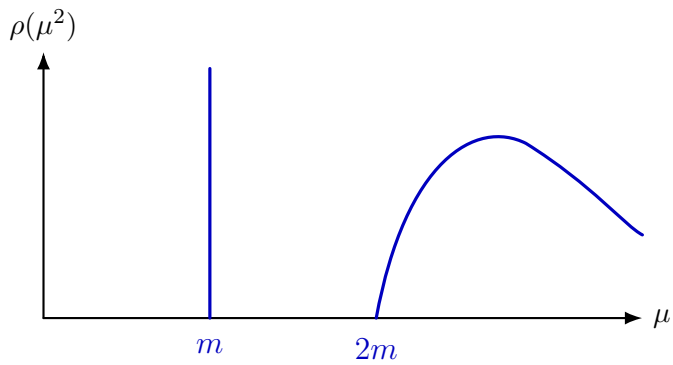
The first term here we can directly evaluate the delta function over $d^{D-1}\vec{k}$ leaving only the k^0 and p^0 contribution

$$\frac{\mathcal{Z}}{2\omega_{\vec{k}}}\delta(p^0 - \sqrt{(\vec{p})^2 + m^2}) = \mathcal{Z}\delta(p^2 + m^2) = Z\delta(\mu^2 - m^2). \tag{1.255}$$

Therefore, the spectral density is

$$\rho(\mu^2) = \mathcal{Z}\delta(\mu^2 - m^2) + \sigma(\mu^2) \tag{1.256}$$

where $\sigma(\mu^2)$ here denotes/contains all of the multi-particle contributions that are supported at invariant mass $\mu \geq 2m$. The typical graph for such a spectral density is:



It then follows that the 2-point time-ordered Greens function can be written as

$$\begin{aligned}\langle\Omega|\, T\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle &= \int_0^\infty d\mu^2\, \rho(\mu^2)\Delta_F(x-y;\mu^2) \\ &= \mathcal{Z}\Delta_F(x-y;m^2) + \int_{4m^2}^\infty d\mu^2\, \sigma(\mu^2)\Delta_F(x-y;\mu^2).\end{aligned} \tag{1.257}$$

The result we have just (and finally) derived is known as the *Källen-Lehmann spectral representation* and it is quite a pretty one; let me explain why. Firstly, the 2-point correlation function in QFT is not just a way to derive propagators of fields/particles. The correlation function here contains all the different ways a scalar particle (in this particular case, but holds for other types) can go in and leave as a scalar while doing all sort of crazy things in-between. These in-between states are what are being analyzed in the second term of (1.257) which can give rise to new states, even bound states, of the scalar particle in general. Hence, these states are only found at or beyond $\mu \geq 2m$. More of this will be analyzed in the problem sets found at the end of these lecture notes, but for now we move onto actually computing correlation functions in QFT via perturbation theory (i.e. corrections to the n-point correlation function). We will do this by looking at a particular example of what is called *scalar ϕ^4 theory* with corresponding Lagrangian and Euclidean

Lagrangian

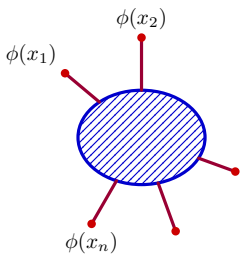
$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{4!}\phi^4 \tag{1.258}$$

$$\mathcal{L}^E = \frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{g}{4!}\phi^4. \tag{1.259}$$

We start with computing an n-point Euclidean function of scalars

$$\langle\phi(x_1^E)\cdots\phi(x_n^E)\rangle = \frac{\int[D\phi]\, e^{-S^E}\phi(x_1^E)\cdots\phi(x_n^E)}{\int[D\phi]\, e^{-S^E[\phi]}}$$

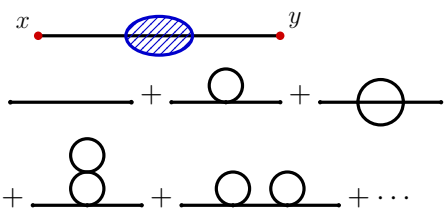
$$= \sum_{\mathcal{S}} \tag{1.260}$$



where the summation over $\mathcal{S}$ here denotes $\mathcal{S}$ = "graphs whose connected components are connected to extern
i.e., there are no reducible graphs like the ones we cut in the last QM section.

For $n = 2$, we get the 2-point function which (you will recognize the diagrams very quickly!) becomes

$$\langle\phi(x)\phi(y)\rangle =$$



(1.261)

Just as in the QM case, we can write down a Fourier transformed Euclidean Greens function that we can related to the spectral density (if need be) as

$$G_E(x,y) = \int \frac{d^Dk^E}{(2\pi)^D} e^{ik^E\cdot(x^E-y^E)}\widetilde{G}_E(k) \tag{1.262}$$

$$= \int_0^\infty d\mu^2\, \rho(\mu^2) \int \frac{d^Dk^E}{(2\pi)^D}\frac{e^{ik^E\cdot(x^E-y^E)}}{k^2+\mu^2} \tag{1.263}$$

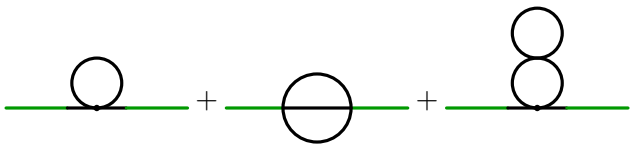
where the Euclidean Greens function may take on a familiar form,

$$\widetilde{G}_E(k) = \frac{1}{k^2+m^2-\Sigma(k)} = \int_0^\infty d\mu^2 \frac{\rho(\mu^2)}{k^2+\mu^2} \tag{1.264}$$

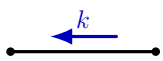
(the greyed out part is just a reminder that we *could* relate it back to the spectral density) where the $\Sigma(k)$ is a direct analog of the one we saw in QM in (1.105) which has the same

structure (but of course these are now fields)

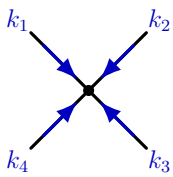
$\Sigma(k) =$

$=$  $+ \cdots$ (1.265)

Note that in (1.265) we could have reorganized the diagrams into more "blobs" like the 1-particle irreducible (1PI) diagram we had before where we have blobs on the propagators that *make* the loops themselves giving even *more* higher order corrections! But let us actually compute some of these diagrams (well, setup the diagrams since we don't have *all* of the tools yet to compute such integrals) to which we will need the *Feynman diagram*[27] rules. We have already seen the propagator which is (1.225) which in Euclidean space is $\Sigma(k)$ removed. The rule for the vertex in momentum space is actually extremely straightforward in Euclidean space, it is literally just reading off the 4-point interaction term in the Euclidean Lagrangian (1.259) and multiply by a momentum conserving delta function, to which we take all of the arrows going inwards. These rules can be summarized as



$$= \frac{1}{k^2 + m^2} \tag{1.266}$$

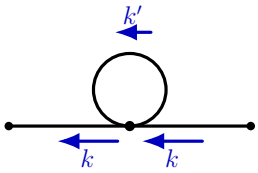


$$= -\frac{g}{4!}(2\pi)^D \delta^D(k_1 + k_2 + k_3 + k_4) \tag{1.267}$$

Also, for every loop diagram, again, as we have seen in QM, we integrate over all possible momentum by re-labeling the bubble-momentum. Let us write down two of the diagrams

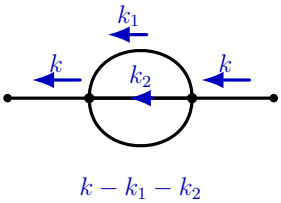
[27] Technically these are not the *original* Feynman diagram rules since those are formulated in Lorentzian signature and not in Euclidean. But, I am lazy and do not know what else to call them.

in (1.265), the bubble and sunrise diagrams,



$$= 4 \cdot 3 \left(-\frac{g}{4!}\right) \int \frac{d^D k'}{(2\pi)^D} \frac{1}{(k')^2 + m^2}$$

$$= -\frac{g}{2} \int \frac{d^D k'}{(2\pi)^D} \frac{1}{(k')^2 + m^2} \tag{1.268}$$



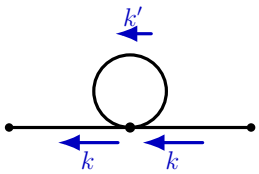
$$= 4 \cdot 4! \left(-\frac{g}{4!}\right)^2 \int \frac{d^D k_1}{(2\pi)^D} \frac{d^D k_2}{(2\pi)^D} \frac{1}{(k_1^2 + m^2)(k_2^2 + m^2)((k - k_1 - k_2)^2 + m^2)}$$

$$= \frac{g^2}{6} \int \frac{d^D k_1}{(2\pi)^D} \frac{d^D k_2}{(2\pi)^D} \frac{1}{(k_1^2 + m^2)(k_2^2 + m^2)((k - k_1 - k_2)^2 + m^2)} \tag{1.269}$$

where the extra factors in red denote an overall multiplicative factor that comes from all the possible contractions/configurations of the diagram in question such as all the possible ways to orientate the 4-different scalars, and then the different orientations of the diagram itself. Let us work with and interpret (1.268), namely, we first recognize that the loop integral is *divergent* for $D \geq 2$ by power counting: the integrand after expanding the $d^D k'$ goes as $k^{D-1} k^{-2} = k^{D-3}$, and so we see that if $D = 3$, then the integral is an infinite volume integrating over a constant, while if $D = 2$, then we get a logarithmic divergence. Only when $D = 1$ do we get a factor of $1/k^2$ which is convergent. *Just* as in quantum mechanics, we must regularize the path integral via

$$\int [D\phi] \to \int \prod_{|k|<\Lambda} d\widetilde{\phi}(k). \tag{1.270}$$

Then, (1.268) becomes



$$= -\frac{g}{2} \int_{|k|<\Lambda} \frac{d^D k'}{(2\pi)^D} \frac{1}{(k')^2 + m^2}$$

$$= -\frac{g}{2} \frac{\mathrm{vol}(S^{D-1})}{(2\pi)^D} \int_0^\Lambda dk \, \frac{k^{D-1}}{k^2 + m^2}$$

$$\equiv -g \cdot c_1. \tag{1.271}$$

If you haven't seen these types of integrals, let me explain. A D-dimensional measure can be re-written in terms of a $D-1$-sphere and one radial coordinate assuming we are on

some D-dimensional spherically symmetric momentum space. This allows us to write the measure as $d^D k' = k^{D-1} d\Omega_{D-1} dk$ where $d\Omega_{D-1}$ is the $D-1$ solid angle measure. The integration over this generates an semi-obvious volume of the $D-1$ sphere, just as if we integrate over the solid angle in a 3-dimensional space in spherical coordinates, we generate the volume of a sphere when integrating over. The volume of the $D-1$ sphere here is

$$\mathrm{vol}(S^{D-1}) = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \tag{1.272}$$

Finally, in the last step, we grouped everything together (except g) into a "constant" c_1. Notice that this divergent k-integral has the following behavior

$$\int_0^\Lambda dk \frac{k^{D-1}}{k^2+m^2} = \begin{cases} \frac{1}{2}\log\frac{\Lambda^2+m^2}{m^2} & D=2 \\ \frac{\Lambda^{D-2}}{D-2} + \cdots & D \geq 3 \end{cases} \tag{1.273}$$

Remember that we are supposed to send our cutoff to infinity as we learned in QM, and so both of the possible values we indicated diverge... what is going on and how can we solve this? In QM, we utilized the many different $\hbar$ deformations that can be made in the Hamiltonian to absorb the divergences thanks to the quantization ambiguities. But what do we do here?

Well, recall the Källen-Lehmann spectral representation of the Euclidean Greens function

$$\widetilde{G}_E(k) = \frac{1}{k^2+m^2-\Sigma(k)} = \frac{\mathcal{Z}}{k^2+m_*^2} + \int_{\mu^2 \geq 4m_*^2} d\mu^2 \frac{\sigma(\mu^2)}{k^2+\mu^2} \tag{1.274}$$

I have highlighted here the *new labeling* of the mass m_*^2 since here, we will interpret this as the *actual mass* of the particle which corresponds to the pole in k^2 (or just k^D at fixed $\vec{k} = (k^1, \cdots, k^{D-1})$) closest to the origin (the line in the graph of the spectral function). So then how would our c_1 term contribute to the actual mass of the particle? A constant term at order g in $\Sigma(k)$ shifts the pole position! This can be written as

$$m_*^2 = m^2 + gc_1 + \mathcal{O}(g^2) \tag{1.275}$$

where c_1 is either approximately $\log(\Lambda/m)$ or Λ^{D-2} depending on the number of spacetime dimensions. So, what is the interpretation?

1. m_* is the mass of the actual particle.

2. m is just *some* parameter in the Lagrangian; its relation to the physical mass is dependent on the regularization scheme.

3. It is standard terminology to refer to m as the "bare mass," and the corresponding Lagrangian that enters he regularized path integral as the "bare Lagrangian."

4. One might be slightly disturbed by the fact that while the difference between m^2 and m_*^2 is formally suppressed by the coupling g, the coefficient c_1 diverges as $\Lambda \to \infty$, and so gc_1 is not exactly what one may consider to be, "small."

One way to handle this in a consistent manner in perturbation theory is to write

$$m^2 = m_R^2 + \delta m^2 \tag{1.276}$$

where in blue, this is a finite "renormalized" mass m_R that is close to m_*. The mass in orange is viewed as a "counter-term" similar to what we have seen before that diverges as $\Lambda \to \infty$, but appears at order g. For general power of g,

$$\delta m^2 = \sum_{n=1}^{\infty} (\delta m^2)^{(n)} g^n \tag{1.277}$$

where $(\delta m^2)^{(n)}$ is chosen to cancel the divergent contribution to the pole position m_*^2 at order g^n in perturbation theory. For our example, and as an example, we have that $m_*^2 = m_R^2 + \delta m^2 + gc_1 + \mathcal{O}(g^2)$, and so we would choose that $(\delta m^2)^{(1)} = -c_1 + \text{finite}$. At the level of the Lagrangian, correspondingly, we split the bare Lagrangian as

$$\mathcal{L}^E = \mathcal{L}_R^E + \Delta\mathcal{L}^E \tag{1.278}$$

where $\mathcal{L}_R^E$ contains the renormalized mass (and renormalized terms) such as $m_R^2\phi^2/2$ while the $\Delta\mathcal{L}^E$ is the counter-term Lagrangian $\Delta\mathcal{L}^E = \delta m^2\phi^2/2$. We will treat $\Delta\mathcal{L}$ as producing interaction vertices themselves at order g^n such as

 $\leftrightarrow -(\delta m^2)^{(n)} g^n \phi^2.$ (1.279)

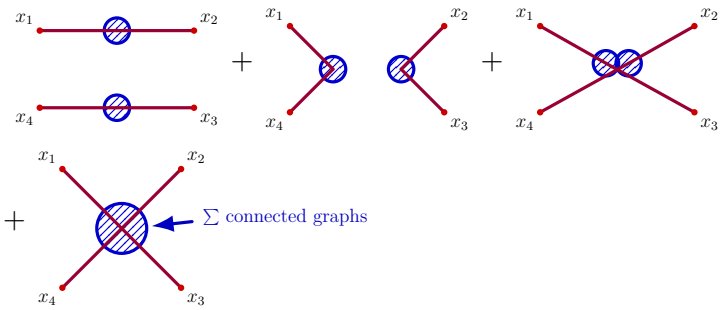
The separation of m^2 into m_R^2 and δm^2 is entirely ad hoc. It's only purpose is to reorganize perturbation theory such tat when we take $\Lambda \to \infty$, the divergences present cancel manifestly at each order in g. This is opposed to canceling divergences between different orders in g for example, between m^2 at zeroth order and $c_1 g$ at first order. To make meaningful predictions, we should pick a regularization scheme, find the relation between m and m_*, compute observables using the Lagrangian that involves $m^2 = m_R^2 + \delta m^2$, and express the result in terms of m_*, the physical mass.

All of this was under the analysis of a 2-point correlator, but now let us move onto the 4-point correlation function[28]. A 4-point correlator is different compared to any other lower-point correlator since there can be *disconnected* diagrams and *connected* diagrams. The disconnected graphs become products of lower point correlators as can be seen in the

[28]The 3-point correlator is tied to computing decay rates, some special identities you will see in the problem sets, and anomalies which we will explore in QFT 2.

following manner

$$\langle\phi(x_1^E)\phi(x_2^E)\phi(x_3^E)\phi(x_4^E)\rangle =$$



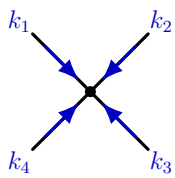
$$= \langle\phi(x_1^E)\phi(x_2^E)\rangle\langle\phi(x_3^E\phi(x_4^E)\rangle + (2 \leftrightarrow 4) + (2 \leftrightarrow 3)$$
$$+ \langle\phi(x_1^E)\phi(x_2^E)\phi(x_3^E)\phi(x_4^E)\rangle^{\text{conn.}} \tag{1.280}$$

where the last correlator contains just the connected pieces. The disconnected pieces are just products of 2-point functions which we know how to compute, but the connected piece is new. Similar to the 2-point function, we can Fourier transform it as

$$\begin{aligned} G_E^{\text{conn}}(x_1^E, \cdots, x_4^E) &= \langle\phi(x_1^E)\cdots\phi(x_4^E)\rangle^{\text{conn}} \\ &\equiv \int \frac{d^D k_1^E}{(2\pi)^D}\cdots\frac{d^D k_4^E}{(2\pi)^D} e^{ik_1\cdot x_1^E + \cdots + ik_4^E\cdot x_4^E}\,\widetilde{G}_E^{\text{conn}}(k_1^E, \cdots, k_4^E) \end{aligned} \tag{1.281}$$

where the Fourier transformed 4-point function to first order in g (again, in reference to our ϕ^4 theory)

$$\widetilde{G}_E^{\text{conn}}(k_1^E, \cdots, k_4^E) =$$



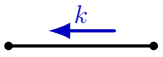
$$+ \cdots$$

$$= (2\pi)^D \delta^D(k_1^E + k_2^E + k_3^E + k_4^E)\left(-g\prod_{j=1}^{4}\frac{1}{(k_j^E)^2 + m^2} + \mathcal{O}(g^2)\right) \tag{1.282}$$

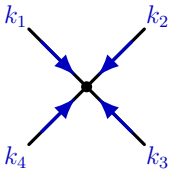
where the $\cdots$ denotes loop order corrections, and the m^2 here can again be written as $m^2 = m_R^2 + \delta m^2$, and expand in δm^2 and then regroup the terms according to powers of g. A convenient choice for the renormalized mass is taking $m_R = m_*$.

Now, everything we have said so far about the perturbative computations of the n-point Euclidean Greens functions $G_E(x_1^E, \cdots, x_n^E)$, can be *equally* applied to computing the time-ordered Greens function (1.217) with the following modifications of the Feynman

rules:

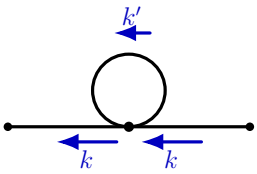


$$= \frac{-i}{k^2 + m^2 - i\epsilon} \tag{1.283}$$



$$= -i\frac{g}{4!}(2\pi)^D \delta^D(k_1 + k_2 + k_3 + k_4) \tag{1.284}$$

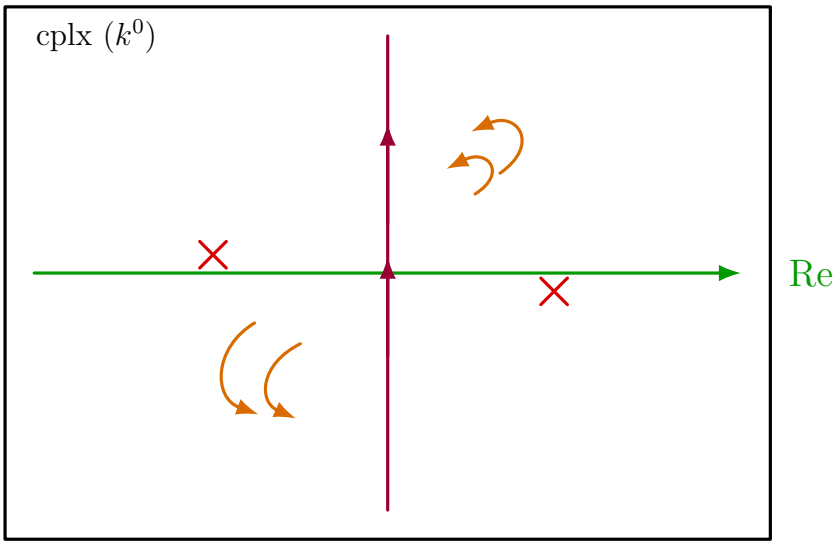
where we can see the only differences between these and the (1.266) and (1.267) are factors of i (and a sign). But, all the momentum vectors now are Lorentzian ones, namely $k^2 \equiv \eta_{\mu\nu}k^\mu k^\nu$. Let us do the same bubble diagram we computed for the Euclidean case but now for the Lorentzian case



$$= -\frac{ig}{2}\int \frac{d^D k'}{(2\pi)^D} \frac{-i}{(k')^2 + m^2 - i\epsilon}$$

$$= -\frac{ig}{2} i \int \frac{d^D k^E}{(2\pi)^D} \frac{-i}{(k^E)^2 + m^2} \tag{1.285}$$

where we *still* have to Wick rotate by $(k^0)' \equiv ik'^D$ (this is the red-piece), and when we do so, we can see that this is consistent with the Euclidean computation, but we notice that in the Lorentzian signature, this contributes a factor of $i\Sigma(k)$ to the mass-corrections. In the complex $(k')^0$ plane, this looks like the following by rotating from the real-green line to the purple (with the rotation direction denoted in orange)



The full time-ordered 2-point function takes the form (in Lorentzian signature)

$$\begin{aligned} G(x,y) &= \int \frac{d^D k}{(2\pi)^D} e^{ik\cdot(x-y)} \widetilde{G}(k) \\ &= \int \frac{d^D k}{(2\pi)^D} e^{ik\cdot(x-y)} \left(\frac{-i}{k^2 + m^2 - i\epsilon - \Sigma(k)} \right). \end{aligned} \tag{1.286}$$

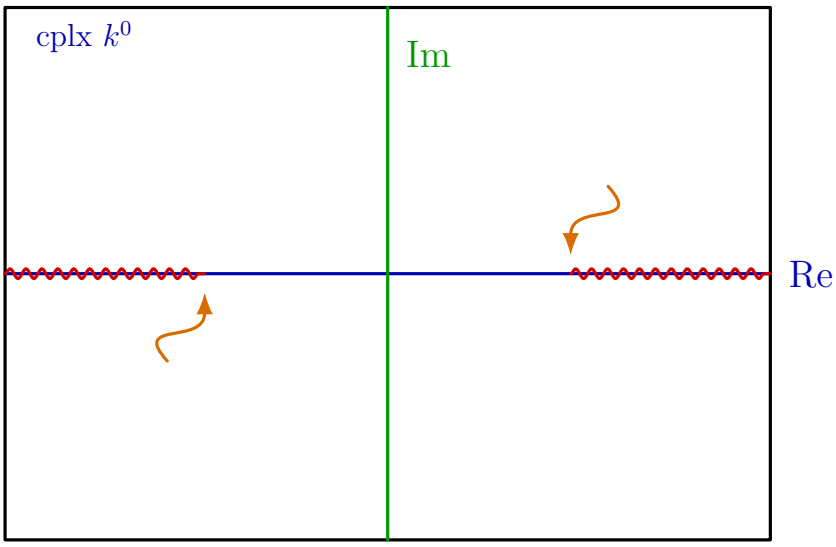
Note that $\Sigma(k)$ is expected to have a pair of *branch cuts* in k^0 starting at the multi-particle threshold $-k^2 = M^2$ where M is the invariant mass of lightest multi-particle state. The branch cuts are expected to arise since there are two different mappings for the same value, namely *since* we have a mass-shell condition $-k^2 = M^2$ this implies that

$$-\left(-(k^0)^2 + (\vec{k})^2\right) = M^2 \implies k^0 = \pm\sqrt{M^2 + (\vec{k})^2} \tag{1.287}$$

where the $\pm$ literally gives us two different sides of physical values on the real axis in the complex graph. These branch cuts may be chosen such that they extend along the real k^0-axis

$$k^0 > \sqrt{(\vec{k})^2 + M^2} \quad \text{and} \quad k^0 < -\sqrt{(\vec{k})^2 + M^2} \tag{1.288}$$

which on a complex graph, can be represented by the following red "zig-zags."



The actual value of $\Sigma(k)$ at a real point on k^0 is determined by our Wick rotation procedure relating G_E to G. For example, for real values of $k^0 > \sqrt{(\vec{k})^2 + M^2}$, $\Sigma(k)$ is given by analytic continuation from $k^0 \in i\mathbb{R}$, or more generally, on the *first-sheet* for complex k^0, to positive real k^0-axis from *above*.

Ok, let us take a quick step back and quickly review everything we have just learned so far about field theory before we get into our next topic, *scattering theory*!

1. The unitary Lorentz transformations are constructed out of the boost-angular momentum and total momentum operators, and the energy-momentum tensor is simply the current from translation symmetry informed to us via Noether's theorem.

2. There are three different types of Greens functions we can use to compute/investigate different physical quantities in QFT: time ordered, Euclidean, and Wightman Greens functions.

3. The uniform Wick rotation analyticity is enforced by causality/time-ordering of the temporal components.

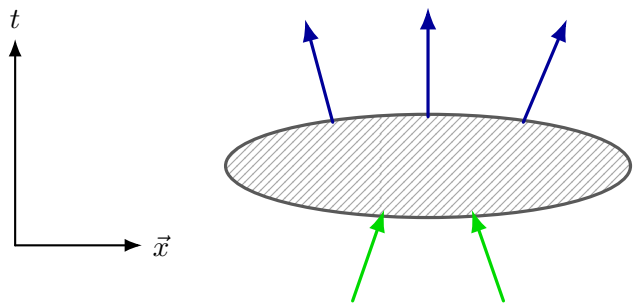
4. The Källen-Lehmann spectral representation can inform us about higher energy mass and bound states found in QFTs.

5. The scalar field operator when going from the vacuum to a 1-particle state (via the expectation value) defines a renormalization constant (we will see it's use later).

6. The infinities (and ad hoc manner) in which we renormalize the mass are what allow us to get finite observables.

1.2.4 The S-Matrix & Derivation of

Let us now move onto scattering theory, where we will thoroughly (and finally) analyze this image:



The green lines here are what we call an *in-basis*. These are states that look like far-separated particles in the far past (infinitely far away in the infinite past). The blue lines denote *out-basis*. These are states that look like far-separated wave packets of particles in the far future.

We can label in or out basis states *as if* they consist of free particles namely

$$\left|\vec{k}_1, \vec{k}_2, \cdots, \vec{k}_n\right\rangle^{\text{in}} \quad \text{and} \quad \left|\vec{k}_1, \vec{k}_2, \cdots, \vec{k}_n\right\rangle^{\text{out}}. \tag{1.289}$$

For now, we will assume there is only one particle species; otherwise we would also add labels of particle type such as helicity. Now, the entire Hilbert space $\mathcal{H}$ for scattering states is spanned by either the out-basis states $|\alpha\rangle^{\text{out}}$ or the in-basis states $|\alpha\rangle^{\text{in}}$ where we shall use $\alpha, \beta, \cdots$ to denote a generically prepared state.

In an interacting QFT,the in- and out-basis are differet, and are related by a linear change of basis

$$^{\text{out}}\langle\beta|\alpha\rangle^{\text{int}} \equiv S_{\beta\alpha} \tag{1.290}$$

where $S_{\beta\alpha}$ is *thee* S-matrix (take note that the ordering of the indices here matter; they are not spacetime indices, but particle labels). We can express the S-matrix in an equivalent

manner by inserting a complete basis of states, or by stating that the in-state is just the integral over all possible out-states evolved by the S-matrix

$$|\alpha\rangle^{\text{in}} = \int d\beta \, S_{\beta\alpha} \, |\beta\rangle^{\text{out}} \, . \tag{1.291}$$

Next, we will give an explicit construction of in- or out-basis states using the local field operator $\hat{\phi}(x)$, and thereby relating the S-matrix to correlators of $\hat{\phi}(x)$. This work was originally done by Haag in 1958 [6] and Ruelle in 1962 [7] (which generated Haag's theorem and Haag-Ruelle scattering theory[29]).

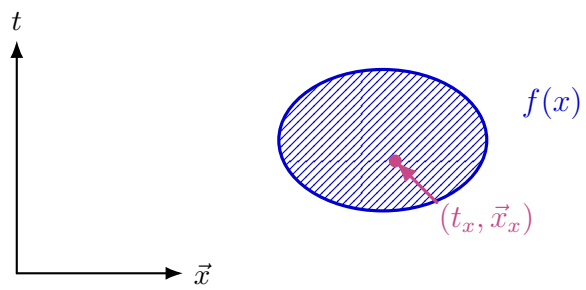
Recall that when the scalar operator acts on the interacting vacuum we have

$$\hat{\phi}(x) \, |\Omega\rangle = \int d^{D-1}\vec{k} \, e^{-i\vec{k}\cdot\vec{x}+i\omega_{\vec{k}}x^0} \left(\frac{\mathcal{Z}}{(2\pi)^{D-1}2\omega_{\vec{k}}} \right)^{1/2} \left|\vec{k}\right\rangle + \int d\alpha \, e^{-iP_\alpha\cdot x} \, \langle\alpha| \, \hat{\phi}(0) \, |\omega\rangle \, |\alpha\rangle \, . \tag{1.292}$$

Consider a *smearing function* $f(x)$ with $x \in \mathbb{R}^{1,D-1}$ with Fourier transformation

$$f(x) = \int \frac{d^D k}{(2\pi)^D} \widetilde{f}(x) e^{ik\cdot x} \tag{1.293}$$

such that the Fourier transform $\widetilde{f}(k)$ is supported near the mass-shell $k^0 > 0$ with $k^2 + m^2 = 0$. And, at the same time, $f(x)$ is concentrated in a finite domain in the spacetime; consider this pictorial representation for such a function:



An example of such a smearing function is a Gaussian profile,

$$f(x) = \frac{1}{(2\pi)^{\frac{D}{2}} L^D} \exp\left(-\frac{(x^0 - t_*)^2}{2L^2} - \frac{(\vec{x} - \vec{x}_*)^2}{2L^2} + i\vec{p}_* \cdot (\vec{x} - \vec{x}_*) - iE_*(x^0 - t_*) \right) \tag{1.294}$$

where $L > 0$ is some large constant distance (a wide Gaussian profile), which has the following Fourier transformation

$$\widetilde{f}(k) = \exp\left(-\frac{L^2(k^0 - E_*)^2}{2} - \frac{L^2(\vec{k} - \vec{p}_*)^2}{2} - i\vec{k} \cdot \vec{x}_* + ik^0 t^0 \right) \tag{1.295}$$

[29] Just to give an idea of the importance of this work, the CPT theorem, spin-statistics theorem in QFT, and no superluminality, are all a consequence of Haag-Ruelle theory (for QFT systems)

which is peaked near $(k^0, \vec{k}) \approx (E_*, \vec{p}_*)$. It should be noted that a more convenient choice would be to make $\widetilde{f}(k)$ compactly supported (within a momentum domain of size $\sim L^{-1}$), but still smooth, in which case $f(x)$ spread to a bloc of size $\sim L$ and diminishes outside the blob faster than $|x|^{-N}$ for any $N > 0$ as $|x| \to \infty$. We will further choose $(E_*, \vec{p}_*)$ to be such that $E_*^2 - \vec{p}_*^2 = m^2$. Now consider the *smeared operator* (naturally built out of the smeared function)

$$\hat{\phi}_f \equiv \int d^D x \, f(x) \hat{\phi}(x) \tag{1.296}$$

which when operating on the vacuum just gives a new overall multiple of $\widetilde{f}$,

$$\hat{\phi}_f \left|\Omega\right\rangle = \int d^{D-1}\vec{k} \, \widetilde{f}(\omega_{\vec{k}}, \vec{k}) \left(\frac{\mathcal{Z}}{(2\pi)^{D-1} 2\omega_{\vec{k}}} \right)^{\frac{1}{2}} \left|\vec{k}\right\rangle + \int\limits_{\text{m.p.}} d\alpha \widetilde{f}(p_\alpha) \left\langle\alpha\right| \hat{\phi}(0) \left|\Omega\right\rangle \left|\alpha\right\rangle \tag{1.297}$$

where m.p. denotes integration over multi-particle states while in the first term only the 1-particle state is retained. Note that $\widetilde{f}(p_\alpha)$ is suppressed by our choice of $\widetilde{f}(k)$. We now *define* the following transformation $f(x) \mapsto f^{(T)}(x)$ by setting

$$\widetilde{f}^{(T)}(k) \coloneqq e^{i(k^0 - \omega_{\vec{k}} T} \widetilde{f}(k) \tag{1.298}$$

You can check that the mapping (1.298) when applied to (1.2.4) is independent of T, $\widetilde{f}^{(T)}(\omega_{\vec{k}}, \vec{k}) = \widetilde{f}(\omega_{\vec{k}}, \vec{k})$. Therefore, the 1-particle piece of $\hat{\phi}_{f^{(T)}} \left|\Omega\right\rangle$ is independent of T. With the above prescription for $f(x)$, the multi-particle component of $\hat{\phi}_f \left|\Omega\right\rangle$ is negligible and therefore

$$\hat{\phi}_{f^{(T)}} \left|\Omega\right\rangle \approx \hat{\phi}_f \left|\Omega\right\rangle \tag{1.299}$$

where the approximation made here is uniformly valid for all, including arbitrarily large T. But what does $f^{(T)}(x)$ look like? Well, let's just write it out,

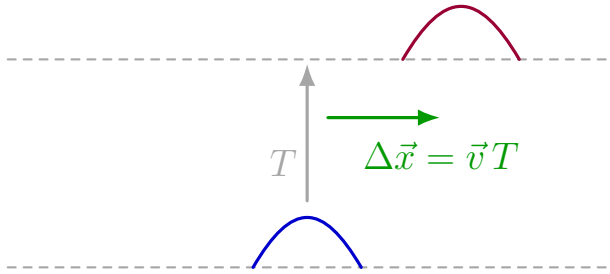
$$\begin{aligned} f^{(T)}(x) &= \int \frac{d^D k}{(2\pi)^D} e^{ik\cdot x + i(k^0 - \omega_{\vec{k}})T} \widetilde{f}(k) \\ &= \int d^D y \int \frac{d^D k}{(2\pi)^D} e^{ik\cdot x + i(k^0 - \omega_{\vec{k}})T} f(y) \\ &= \int d^D y Ł\delta(x^0 - y^0 - T) \int \frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}} e^{i\vec{k}\cdot(\vec{x}-\vec{y}) - i\omega_{\vec{k}} T} f(y) \\ &= \int d^{D-1}\vec{y} \, K(\vec{x} - \vec{y}; T) f(x^0 - T, \vec{y}) \end{aligned} \tag{1.300}$$

where in the second line we substituted in the Fourier transformation of $f(y)$, and then defined the function

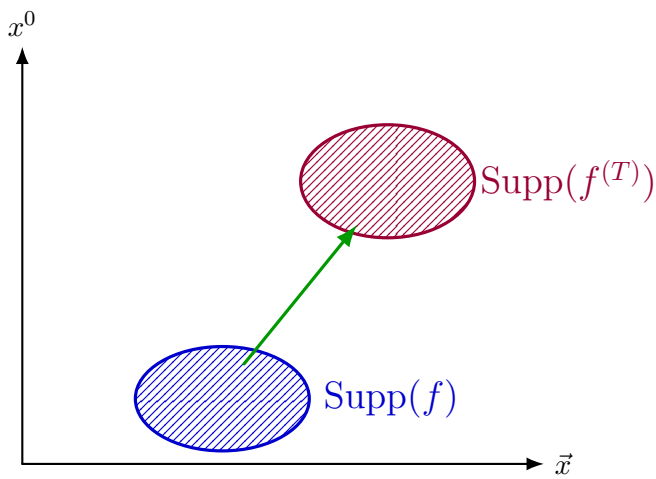
$$K(\vec{x} - \vec{y}; T) = \int \frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}} e^{i\vec{k}\cdot(\vec{x}-\vec{y}) - i\omega_{\vec{k}} T} \tag{1.301}$$

and then evaluated y^0 via the delta-function in the last line. Notice that $K(\vec{x}; T)$ obeys the Klein-Gordan equation $(\partial_{\vec{x}}^2 - \partial_T^2 - m^2) K(\vec{k}; T) = 0$ and the initial condition $K(\vec{x}; 0) = \delta^{D-1}(\vec{x})$. If $\widetilde{f}(k)$ is peaked at$k \approx (E_* = \omega_{\vec{p}_*}, \vec{p}_*)$, then $f^{(T)}(x^0, \vec{x})$ looks like the wave packet $f(x^0 - T, \vec{x})$ having moved with group velocity $\vec{v} = \partial\omega_{\vec{p}_*} / \partial\vec{p}_*$ for time T. There are two

pictures that go along with this. The first is how f moves with respect to $f^{(T)}$ over time T where the group velocity looks like a wave packet in the following manner:

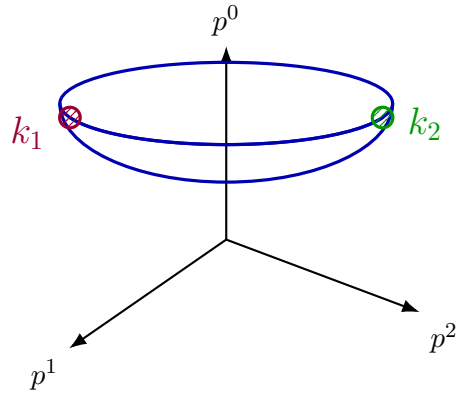


There is also the way the smeared functions move in position space along with their support in the following manner:

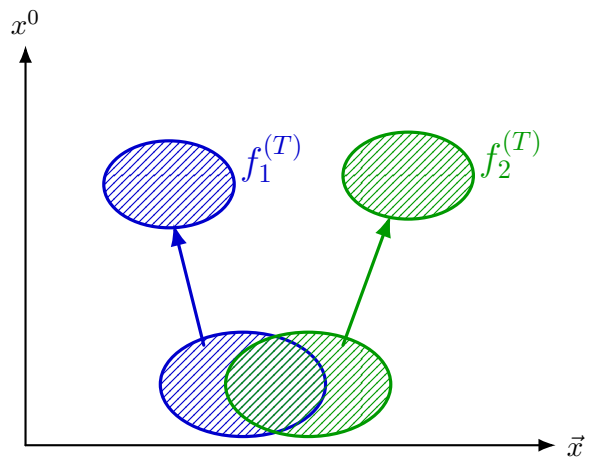


In evolving $f \rightsquigarrow f^{(T)}$, the support of $f(x)$, say in a region of spacetime that has a large size $\sim L$ centered around some $(t_*, \vec{x}_*)$, moves to a region centered around $(t_* + T, \vec{x}_* + \vec{v}T)$, and yet we still have $\hat{\phi}_{f^{(T)}}\,|\Omega\rangle \approx \hat{\phi}_f\,|\Omega\rangle$, which is the same 1-particle state! This may seem counter intuitive, but there is no contradiction since this shows us a very important fact:*there are a lot more operators than states*!

Now, consider a *pair* of function $f_1(x)$ and $f_2(x)$ of the same above prescribed form (smearing functions) who Fourier transforms $\widetilde{f}_1(k)$ and $\widetilde{f}_2(k)$ are supported near $k \approx k_1$ and $k \approx k_2$ respectfully, and obey the typical mass-shell condition $k_1^2 = k_2^2 = -m^2$. We shall also assume that $\vec{k} \neq \vec{k_2}$. On the mass-shell as given in figure 1.2.2, you can think of k_1 and k_2 as two blobs on some sides of the mass-shell hyperboloid, or quite literally as the following:



And as in the single $f(x)$ case, we can also transform both of these smearing functions as well, but now, we can place them close to each other and then send them far away:



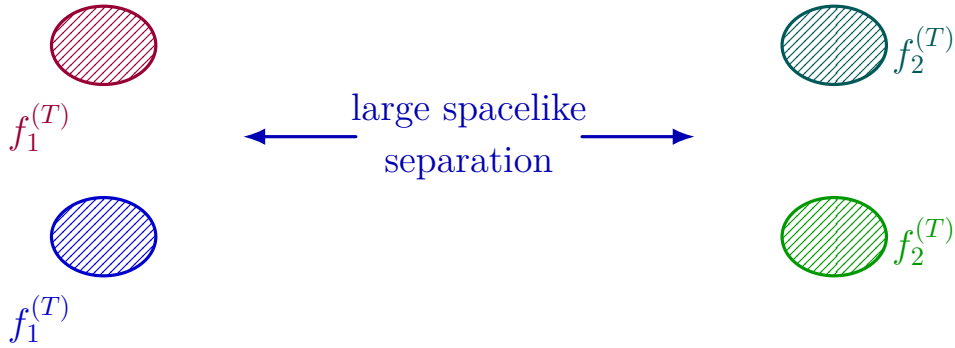
The regions of concentration of $f_1^{(T)}$ and $f_2^{(T)}$ move apart as $T \to \pm\infty$, and become far space-like separated. This was just for the smearing functions, but now what about multiple smearing operators? We can consider the following state

$$\left|\Psi^{(T)}\right\rangle = \hat{\phi}_{f_1^{(T)}} \hat{\phi}_{f_2^{(T)}} \left|\Omega\right\rangle . \tag{1.302}$$

Then, the overlap between two different states, one at T and another at T' would be

$$\langle \Psi^{(T)} | \Psi^{(T')} \rangle = \langle \Omega | \hat{\phi}_{f_2^{(T)*}} \hat{\phi}_{f_1^{(T)*}} \hat{\phi}_{f_1^{(T')}} \hat{\phi}_{f_2^{(T')}} \left|\Omega\right\rangle . \tag{1.303}$$

We now fix $T' - T \equiv \Delta T$, and we send $T \to \pm\infty$. In doing so, we expect (and can check explicitly if you'd like) that the 4-operators in (1.303) separate/factorize which gives the pairs a large spacelike separation namely:



We can write this explicitly as (again, in the limit $T \to \pm\infty$),

$$\langle\Omega|\, \hat{\phi}_{f_2^{(T)}*} \hat{\phi}_{f_1^{(T)}*} \hat{\phi}_{f_1^{(T')}} \hat{\phi}_{f_2^{(T')}} \,|\Omega\rangle \to \langle\Omega|\, \hat{\phi}_{f_1^{(T)}*} \hat{\phi}_{f_1^{(T')}} \,|\Omega\rangle \, \langle\Omega|\, \hat{\phi}_{f_2^{(T)}*} \hat{\phi}_{f_2^{(T')}} \,|\Omega\rangle \,. \tag{1.304}$$

This factorization is what is known as the *cluster property*. To see how this factorization came to be, recall that $\hat{\phi}_{f_1^{(T')}} |\Omega\rangle \approx \hat{\phi}_{f_1^{(T)}} |\Omega\rangle$ which is independent of T! It then follows from (1.304) that the difference in T and T' prepared states are

$$|| \left|\Psi^{(T')}\right\rangle - \left|\Psi^{(T)}\right\rangle || \to \infty \tag{1.305}$$

as $T \to \pm\infty$. Assuming this difference vanishes sufficiently fast[30], then the product on the vacuum

$$\lim_{T\to\pm\infty} \hat{\phi}_{f_1^{(T)}} \hat{\phi}_{f_2^{(T)}} |\Omega\rangle := |\phi_{f_1}, \phi_{f_2}\rangle^{\text{out/in}} \tag{1.306}$$

will be a well-defined state in the Hilbert space $\mathcal{H}$. If we now compare to how the scalar field operator acts on the vacuum to generate the 1-particle state (assuming that Supp $\widetilde{f}$ is concentrated near the 1-particle mass-shell with no overlap with multi-particle momentum region), we can write the in/out state as

$$|\phi_{f_1}, \phi_{f_2}\rangle^{\text{out/in}} = \int \prod_{j=1}^{2} d^{D-1}\vec{k}_j \; \widetilde{f}_j(\omega_{\vec{k}_j}, \vec{k}_j) \left(\frac{\mathcal{Z}}{(2\pi)^{D-1} 2\omega_{\vec{k}_j}} \right)^{\frac{1}{2}} \left|\vec{k}_1, \vec{k}_2\right\rangle^{\text{out/in}} . \tag{1.307}$$

Using the cluster property (1.304), one can verify that $\left|\vec{k}_1, \vec{k}_2\right\rangle^{\text{out/in}}$ obeys orthogonality

$$^{\text{out}}\langle \vec{k}_1, \vec{k}_2 | \vec{k}_1', \vec{k}_2' \rangle^{\text{out}} = \delta^{D-1}(\vec{k}_1 - \vec{k}_2')\delta^{D-1}(\vec{k}_2 - \vec{k}_2') + (k_1' \leftrightarrow k_2') \tag{1.308}$$

and a similar result for the in-basis orthogonality $^{\text{int}}\langle \vec{k}_1, \vec{k}_2 | \vec{k}_1', \vec{k}_2' \rangle^{\text{in}}$. The generalization to n-particle in/out-states $\left|\vec{k}_1, \cdots, \vec{k}_n\right\rangle^{\text{in/out}}$ should hopefully be clear and straightforward.

Armed with out well-defined multi-particle basis states, given some n number of smearing functions $f_1(x), \cdots, f_n(x)$ whose Fourier transform $\widetilde{f}_1(k), \cdots, \widetilde{f}_n(k)$ are supported near

[30] Actually in fact, with non-overlapping and smooth smearing functions $\widetilde{f}_i(k)$, the difference vanishes faster than$|T|^{-N}$ for any $N > 0$ as $T \to \pm\infty$ which was showed by Hepp in 1965 [8, 9].

the mass-shell $k^2+m^2=0$ with $k^0>0$ and do not overlap namely

$$\lim_{T\to\pm\infty}\hat{\phi}_{f_1^{(T)}}\cdots\hat{\phi}_{f_n^{(T)}}|\Omega\rangle := |\phi_{f_1},\cdots,\phi_{f_n}\rangle^{\text{out/in}}$$
$$=\int\prod_{j=1}^{n}d^{D-1}\vec{k}_j\,\widetilde{f}_j(\omega_{\vec{k}_j},\vec{k}_j)\left(\frac{\mathcal{Z}}{(2\pi)^{D-1}2\omega_{\vec{k}_j}}\right)^{\frac{1}{2}}\cdot\left|\vec{k}_1,\cdots,\vec{k}_n\right\rangle^{\text{out/in}} \tag{1.309}$$

the normalization for m out- and n out-state is just a multiple of δ_{nm}

$$^{\text{out}}\langle\vec{k}_1,\cdots,\vec{k}_n|\vec{k}'_n,\cdots,\vec{k}'_m\rangle^{\text{out}}=\delta_{nm}\sum_{\sigma\in S_n}\prod_{j=1}^{n}\delta^{D-1}(\vec{k}_j-\vec{k}'_{\sigma'_j}) \tag{1.310}$$

where S_n is the permutation group. The same normalization goes for in-states as well.

Now we are finally in a position to define the S-matrix elements, and not just for some ambiguous states as in (1.290). We shall define these as

$$S(\vec{k}_1,\cdots,\vec{k}_n|\vec{k}'_1,\cdots,\vec{k}'_m)={}^{\text{out}}\langle\vec{k}_1,\cdots,\vec{k}_n|\vec{k}'_1,\cdots,\vec{k}'_m\rangle^{\text{in}} \tag{1.311}$$

The generalization to more than one species of particles is semi-clear. If we just have different particle species, then we must include a different particle label on the corresponding particle momenta. By construction, the S-matrix elements are determined by a suitable limit of integrated Wightman functions. Let us inspect this relation in more detail for in-states with smeared functions g and out states with smeared functions f namely

$$^{\text{out}}\langle\phi_{f_1},\cdots|\phi_{g_1},\cdots\rangle^{\text{int}}=\int d^{D-1}\vec{k}_1\,\widetilde{f}_1^*(\omega_{\vec{k}_1},\vec{k}_1)\left(\frac{\mathcal{Z}}{(2\pi)^{D-1}2\omega_{\vec{k}_1}}\right)^{\frac{1}{2}}\cdots$$
$$\times\,d^{D-1}\vec{k}'_1\,g_1(\omega_{\vec{k}'_1},\vec{k}'_1)\left(\frac{\mathcal{Z}}{(2\pi)^{D-1}2\omega_{\vec{k}'_1}}\right)^{\frac{1}{2}}\cdots\times S(\vec{k}_1,\cdots|\vec{k}'_1,\cdots). \tag{1.312}$$

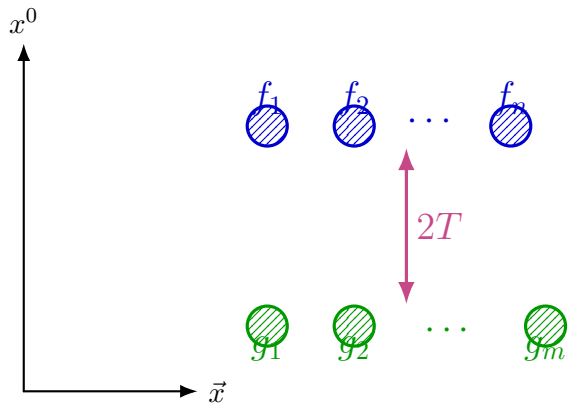
The LHS of (1.312) by definition are

$$\begin{aligned}
^{\text{out}}\langle\phi_{f_1},\cdots|\phi_{g_1},\cdots\rangle^{\text{in}}&=\lim_{T\to\infty}\langle\Omega|\,\hat{\phi}_{f_n^{(T)*}}\cdots\hat{\phi}_{f_1^{(T)*}}\hat{\phi}_{g_1^{(-T)}}\cdots\hat{\phi}_{g_m^{(-T)}}|\Omega\rangle\\
&=\lim_{T\to\infty}\langle\Omega|\,T\hat{\phi}_{f_1^{(T)*}}\cdots\hat{\phi}_{g_1^{(-T)}}\cdots|\Omega\rangle\\
&=\lim_{T\to\infty}\int d^Dx_1\,f_1^{(T)*}(x_1)\cdots d^Dy_1g_1^{(-T)}(y_1)\cdots\times\langle\Omega|\,T\hat{\phi}(x_1)\cdots\hat{\phi}(y_1)\cdots|\Omega\rangle\\
&=\lim_{T\to\infty}\int\frac{d^Dk_1}{(2\pi)^D}\widetilde{f}_1^*(k_1)e^{-(k_1^0-\omega_{\vec{k}_1})T}\cdots\times\frac{d^Dk'_1}{(2\pi)^D}\widetilde{g}_1(-k'_1)e^{i(-k_1{}'0-\omega_{\vec{k}'_1})(-T)}\cdots\\
&\quad\times\widetilde{G}(k_1,\cdots,k_n,k'_1,\cdots,k'_m)
\end{aligned} \tag{1.313}$$

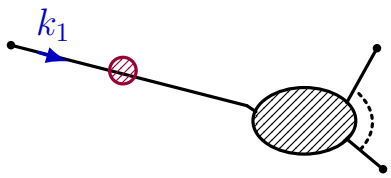
where in the second to last line, recall that the $n+m$-point correlation function can be Fourier transformed as

$$\begin{aligned}
\langle\Omega|\,T\hat{\phi}(x_1)\cdots\hat{\phi}(y_1)\cdots|\Omega\rangle&\equiv G(x_1,\cdots,x_n,y_1,\cdots,y_m)\\
&=\int\frac{d^Dk_1}{(2\pi)^D}e^{ik_1\cdot x_1}\cdots\frac{d^Dk'_1}{(2\pi)^D}e^{ik'_1\cdot y_1}\cdots\widetilde{G}(k_1,\cdots,k'_1,\cdots)
\end{aligned} \tag{1.314}$$

which allows us to complete the x_1 and y_1 integrals leaving us with what we found. A visualization of what exactly we are computing is to think of some n and m number of wave packets (smearing functions dressed up to be an operator) that come in and then leave with a difference of $2T$-time (which we take in the infinite limit:



The $T \to \infty$ limit of the k_i^0 and $k_j^{0'}$ integrals vanishes unless $\widetilde{G}$ is *singular* (has a pole structure) at $k_i^0 = \omega_{\vec{k}_i}$ and $k_j^{0'} = -\omega_{\vec{k}'_j}$. Now, we see that $\widetilde{G}$ is popping up in our calculations, but let us actually compare what computing $\widetilde{G}$ is like, and so we can consider looking at one of the external propagators going *into* some scattering process,



Here, our external leg has momentum k_1, and where that purple blob is located is

$$\sim \frac{-i}{k_1^2 + m_0^2 - i\epsilon - \Sigma(k_1)} \tag{1.315}$$

where m_0^2 here is the *bare mass*. But this has a pole/singular structure at $k_1^2 = -m_*^2$ for the *physical mass* m_*. In this case, if we took the infinite time limit, we can integrate over the k^0 integral

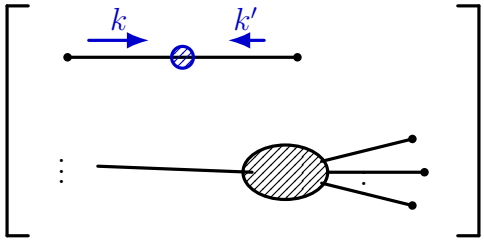
$$\lim_{T\to\infty} \int \frac{d^D k}{(2\pi)^D} \frac{\widetilde{g}(-k) e^{i(-k^0 - \omega_{\vec{k}})(-T)}}{k^2 + m^2 - i\epsilon} = \int \frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}} \widetilde{g}(\omega_{\vec{k}}, -\vec{k}) \lim_{T\to\infty} \int \frac{dk^0}{2\pi} \frac{e^{i(-k^0 - \omega_{\vec{k}})(-T)}}{k^2 + m^2 - i\epsilon} \tag{1.316}$$

of which the dk^0 integral has a residue at $k^0 = -\omega_{\vec{k}} + i\epsilon$ and thus evaluates to $i(2\omega_{\vec{k}})^{-1}$. In a similar manner, the f-term also integrates to

$$\lim_{T\to\infty} \int \frac{d^D k}{(2\pi)^D} \frac{\widetilde{f} - k) e^{i(-k^0 - \omega_{\vec{k}})(-T)}}{k^2 + m^2 - i\epsilon} = \int \frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}} \widetilde{f}^*(\omega_{\vec{k}}, \vec{k}) \frac{i}{2\omega_{\vec{k}}}. \tag{1.317}$$

The time-ordered Greens function $G(x_1, \cdots, x_n)$ takes the form $\sim \sum \prod G^{\text{conn}}$ where $G^{\text{conn}}(x_1, \cdots, x_n)$ decays at large spacelike separation (this is a re-statement of the cluster property). After

Fourier transforming, each $\widetilde{G}^{\text{conn}}(\{\text{subset of } k_i\text{'s}\})$ contains a momentum-delta-function. In taking the $T \to \infty$ limit above, the momentum-delta-function in $\widetilde{G}$ came along for a ride with one exceptional case:



where the square brackets denote that this is a *single* term we should consider. The top piece behaves as

$$(2\pi)^D \delta^D(k_1 + k_1') \left(\frac{-i\mathcal{Z}}{k_1^2 + m^2 - i\epsilon} + (\text{non-singular at } k_1^2 = -m^2) \right) \tag{1.318}$$

from which, the correspond S-matrix element becomes (where I am following the same steps when we started with (1.314) and then dwindled it down to (**??**), just in this case, we have the known Greens function to plug-in, not some general form)

$$\begin{aligned}
\lim_{T\to\infty} \int \frac{d^D k_1}{(2\pi)^D} \widetilde{f}_1^* e^{-i(k_1^0 - \omega_{\vec{k}_1})T} \times \frac{d^D k_1'}{(2\pi)^D} \widetilde{g}_1(-k_1') e^{i(k_1'^0 - \omega_{\vec{k}'})(-T)} (2\pi)^D \delta^D(k_1 + k_1') \frac{-i\mathcal{Z}}{k_1^2 + m^2 - i\epsilon} \\
= \lim_{T\to\infty} \int \frac{d^D k_1}{(2\pi)^D} \widetilde{f}_1^*(k) \widetilde{g}_1(k) e^{2i(k^0 - \omega_{\vec{k}})T} \frac{-i\mathcal{Z}}{k^2 + m^2 - i\epsilon} \\
= \int \frac{d^{D-1}\vec{k}}{(2\pi)^{D-1}} \widetilde{f}_1^*(\omega_{\vec{k}}, \vec{k}) \widetilde{g}(\omega_{\vec{k}}, \vec{k}) \frac{\mathcal{Z}}{2\omega_{\vec{k}}} \\
= \langle \phi_f | \phi_g \rangle
\end{aligned} \tag{1.319}$$

which is what we would expect.

We are now finally in a position to put all of our smeared functions/operators definitions and working knowledge with the S-matrix into an extremely important result, the *Lehmann-Symanzik-Zimmermann (LSZ) reduction formula.* This important result will show us how to go from correlation functions to S-matrix elements that will help us get observables, namely the amplitude (and decay rates, etc.). We can essentially read off the LSZ reduction formula now by comparing the S-matrix elements (1.311) with that of what we just learned about with correlation functions related to the in-out states in (1.312) combined with the integrals we performed over g and $\widetilde{f}$. The result is the following

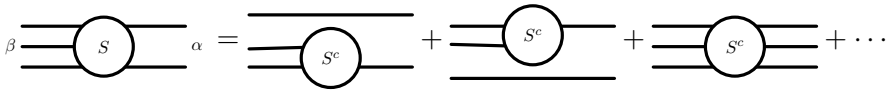
$$\begin{aligned}
S(\vec{k}_1, \cdots, \vec{k}_n | -\vec{k}_1', \cdots, -\vec{k}_m') = & \prod_{i=1}^m \left(\mathcal{Z}(2\pi)^{D-1} 2\omega_{\vec{k}_i} \right)^{-\frac{1}{2}} \prod_{j=1}^m \left(\mathcal{Z}(2\pi)^{D-1} 2\omega_{\vec{k}_j'}^{-\frac{1}{2}} \right) \\
& \times \lim_{(k_i^0, k_j^{0\prime}) \to (\omega_{k_i}, -\omega_{k_j'})} \prod_{i=1}^n i(k_i^2 + m^2) \prod_{j=1}^m i((k_j')^2 + m^2) \\
& \times \widetilde{G}(k_1, \cdots, k_n, k_1', \cdots, k_m')
\end{aligned} \tag{1.320}$$

The formula above, (1.320), is the LSZ reduction formula and it may look disgustingly long and complicated, but it really is not as we will see. Something to now though is that the LSZ has an exception namely that of disconnected diagrams/terms in $\widetilde{G}$ that contains $\delta^D(k_i + k'_j)$ which gives rise to disconnected factor $\delta^{D-1}(\vec{k}_i - \vec{k}'_j)$ in S as discussed above.

It follows from the cluster property (1.304) of Greens functions and LSZ that the S-matrix elements also obey a similar cluster property

$$S(\beta|\alpha) \equiv {}^{\text{out}}\langle\beta|\alpha\rangle^{\text{in}} = \sum_{(\alpha,\beta)=(\coprod \alpha_I, \coprod \beta_I)} \prod_I S^{\text{conn}}(\beta_I|\alpha_I) \tag{1.321}$$

where $\alpha = \coprod \alpha_I$ is the coproduct[31] of possible α-in-states. This makes much more sense pictorially/with diagrams since it says to take all possible combinations between the α and β states:



(1.322)

where that last diagram we have only the connected S-matrix elements now. Each connected S-matrix element comes with a momentum δ-function[32]

$$\begin{aligned} S^{\text{conn}}(\vec{k}_1, \cdots, \vec{k}_n|\vec{k}'_1, \cdots, \vec{k}'_m) =& i(2\pi)^D \delta^D(k_1 + \cdots + k_n - k'_1 - \cdots - k'_m) \\ & \times \mathcal{M}(\vec{k}_1, \cdots, \vec{k}_n|\vec{k}'_1, \cdots, \vec{k}'_m). \end{aligned} \tag{1.323}$$

This allows us to write a more precise version of the LSZ reduction formulain terms of only connected elements/components

$$\begin{aligned} S^{\text{conn}}(\vec{k}_1, \cdots, \vec{k}_n| - \vec{k}'_1, \cdots, -\vec{k}'_m) =& \prod_{i=1}^m \left(\mathcal{Z}(2\pi)^{D-1} 2\omega_{\vec{k}_i}\right)^{-\frac{1}{2}} \prod_{j=1}^m \left(\mathcal{Z}(2\pi)^{D-1} 2\omega_{\vec{k}'_j}^{-\frac{1}{2}}\right) \\ & \times \lim_{(k_i^0, k_j^{0\prime}) \to (\omega_{k_i}, -\omega_{k'_j})} \prod_{i=1}^n i(k_i^2 + m^2) \prod_{j=1}^m i((k'_j)^2 + m^2) \\ & \times \widetilde{G}^{\text{conn}}(k_1, \cdots, k_n, k'_1, \cdots, k'_m) \end{aligned} \tag{1.324}$$

which is valid for $n, m \geq 2$. $\mathcal{M}$ here contains the information about the correlation functions, as we can see when comparing (1.323) and (1.324). $\mathcal{M}$ here is the manifestly invariant form of the amplitude which we shall see soon. Due to our normalization convention of 1-particle states, it is convenient to further define $\mathcal{M}$ as

$$\mathcal{M}(\vec{k}_1, \cdots, \vec{k}_n|\vec{k}'_1, \cdots, \vec{k}'_m) \equiv \prod_{i=1}^n \frac{1}{(2\pi)^{\frac{D-1}{2}} \sqrt{2\omega_{\vec{k}_i}}} \prod_{j=1}^n \frac{1}{(2\pi)^{\frac{D-1}{2}} \sqrt{2\omega_{\vec{k}_j}}} \mathcal{A}(k_1, \ldots, k_n|k'_1, \ldots, k'_m) \tag{1.325}$$

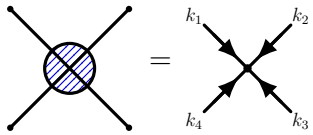
[31] The coproduct of some sets or categories is like a disjoint union that creates a new set with tagged elements. For example, the coproduct of sets $A = \{1, 2\}$ with $B = \{a, b\}$ denoted $A \coprod B = \{(1, 1), (2, 1), (a, 2), (b, 2)\}$.

[32] The factor of i here for the connected S-matrix elements is pure convention.

where $\mathcal{A}$ here is the typical amplitude we usually think about. $\mathcal{A}$ is invariant under Lorentz transformations $k \mapsto \Lambda k$. As an example of how all of this machinery works (or a start since the problem set will explore it much more with a 2-scalar species example), let us consider $2 \to 2$ scattering (and that you will now see over and over and over...). The incoming and outgoing sets of particles are $\alpha = \{\vec{k}_1', \vec{k}_2'\}$ and $\beta = \{\vec{k}_1, \vec{k}_2\}$. The S-matrix can then be written as (using (1.323))

$$\begin{aligned} S(\vec{k}_1, \vec{k}_2|\vec{k}_1', \vec{k}_2') \equiv & {}^{\text{out}}\langle \vec{k}_1, \vec{k}_2|\vec{k}_1', \vec{k}_2'\rangle^{\text{in}} \\ &= \delta^{D-1}(\vec{k}_1 - \vec{k}_1')\delta^{D-1}(\vec{k}_2 - \vec{k}_2') + \delta^{D-1}(\vec{k}_1 - \vec{k}_2')\delta^{D-1}(\vec{k}_2 - \vec{k}_1') \\ &+ i(2\pi)^D\delta^D(k_1 + k_2 - k_1' - k_2')\mathcal{M}(\vec{k}_1, \vec{k}_2; \vec{k}_1', \vec{k}_2') \end{aligned} \tag{1.326}$$

where the first two delta-functions represent particles hitting nothing (they just pass right on by... waving) and the last term is the good stuff, that is a scattering element with connected pieces. Let us consider ϕ^4 theory once again for 4-point scattering



$$= \qquad + \mathcal{O}(g^2) \tag{1.327}$$

$$\widetilde{G}^{\text{conn}}(k_1, k_2, k_3, k_4) = (2\pi)^D\delta^D(k_1 + \cdots + k_4)\left((-ig)\prod_{j=1}^{4}\frac{-i}{k_j^2 + m^2 - i\epsilon} + \mathcal{O}(g^2)\right) \tag{1.328}$$

where here the renormalization constant is $\mathcal{Z} = 1 + \mathcal{O}(g^2)$. Under the LSZ reduction formula (1.324) the S-matrix elements then becomes (combining (1.326) and (1.328))

$$\begin{aligned} S^{\text{conn}}(\vec{k}_1, \vec{k}_2| - \vec{k}_1', -\vec{k}_2') = & \prod_{i=1}^{2}\left(\mathcal{Z}(2\pi)^{D-1}2\omega_{\vec{k}_i}\right)^{-\frac{1}{2}}\prod_{j=1}^{2}\left(\mathcal{Z}(2\pi)^{D-1}2\omega_{\vec{k}_j}\right)^{-\frac{1}{2}} \\ & (2\pi)^D\delta^{D-1}(\vec{k}_1 + \vec{k}_2 + \vec{k}_1' + \vec{k}_2')\delta(\omega_{\vec{k}_1} + \omega_{\vec{k}_2} - \omega_{\vec{k}_1'} - \omega_{\vec{k}_2'}) \\ & \times\left(-ig + \mathcal{O}(g^2)\right) \end{aligned} \tag{1.329}$$

from which we can read of the amplitude using (1.325) finding

$$i\mathcal{A}(k_1, k_2|k_1', k_2]) = -ig + \mathcal{O}(g^2) \tag{1.330}$$

where the $\mathcal{O}(g^2)$ contains loop-corrections as well as corrections to the renormalization constant as factors of $\mathcal{Z}^{-2}$. Let us take a look at our first amplitude! We can see that really, all this amplitude is, is namely the correlation functions we have been computing but with the external legs stripped off! This is essentially what the LSZ reduction formula does. We can also notice that to compute the amplitude $\mathcal{A}$, all we need are the Feynman rules modulo multiplying by the external legs. Everything done here also is considered *on-shell* while if you did multiply by the external legs, we can see that this would just be the correlation function, but would *not* correspond to S-matrix elements and would be considered *off-shell* which is not physical since we shall see that observables such as cross-sections and decay rates all come from computing the S-matrix elements and thus

the amplitudes. These observables are what we take a look at now which naturally come from the properties of the S-matrix.

Something that more than likely has not been obvious is that the S-matrix is a unitary operator. Let us recall it's definition $S_{\beta\alpha} \equiv {}^{\text{out}}\langle\beta|\alpha\rangle^{\text{in}}$. Now, let us define the *operator* version of is via

$$\hat{S}\,|\alpha\rangle^{\text{out}} = |\alpha\rangle^{\text{in}} \tag{1.331}$$

and thus to recover $S_{\beta\alpha}$, we right-multiply by ${}^{\text{out}}\langle\beta|$,

$${}^{\text{out}}\langle\beta|\,\hat{S}\,|\alpha\rangle^{\text{out}} = S_{\beta}\alpha. \tag{1.332}$$

It can be seen that $\hat{S}$ preserves the inner product ${}^{\text{in}}\langle\alpha|\alpha'\rangle^{\text{in}} = {}^{\text{out}}\langle\alpha|\alpha'\rangle^{\text{out}}$ which implies that $\hat{S}^\dagger S = \mathbb{1} = \hat{S}\hat{S}^\dagger$, thus showing that the S-matrix is a unitary operator on the Hilbert space $\mathcal{H}$. Let us pick an orthonormal out-basis $|\alpha\rangle^{\text{out}}$ such that ${}^{\text{out}}\langle\beta|\alpha\rangle^{\text{out}} = \delta_{\beta\alpha}$ and that

$$\int d\alpha\ |\alpha\rangle\,{}^{\text{out}}{}^{\text{out}}\langle\alpha| = \mathbb{1}. \tag{1.333}$$

Then, we can take (1.333) and sandwich an inner product to get

$$\begin{aligned} {}^{\text{out}}\langle\beta|\mathbb{1}|\alpha\rangle^{\text{out}} &= \int d\gamma\ {}^{\text{out}}\langle\beta|\gamma\rangle^{\text{out}}{}^{\text{out}}\langle\gamma|\alpha\rangle^{\text{out}} \\ {}^{\text{out}}\langle\beta|\alpha\rangle^{\text{out}} &= \int d\gamma\ (S_{\gamma\beta})^* S_{\gamma\alpha}. \end{aligned} \tag{1.334}$$

This really puts it into perspective that we can interpret the S-matrix elements as transition amplitudes/probabilities. $|\alpha\rangle^{\text{in}}$ is the state that 'looks like a far-separated (incoming) particles with momenta $\alpha = \{\vec{k}_1, \cdots, \vec{k}_n\}$ in the far past." The probability of finding the state consisting of outgoing particles with momenta distributed in the range $\beta = \{\vec{k}'_1, \cdots, \vec{k}'_m\} \in \mathcal{D}$ is

$$\mathcal{P}_{\alpha\to\mathcal{D}} = \int_{\mathcal{D}} d\beta\ |S_{\beta\alpha}|^2. \tag{1.335}$$

Recall that we have defined the scattering amplitude $\mathcal{M}$ via

$$S_{\beta\alpha} \equiv \delta_{\beta\alpha} + i(2\pi)^D\delta^D(P_\beta - P_\alpha)\mathcal{M}_{\beta\alpha}. \tag{1.336}$$

We shall assume that the state α is not found within the outgoing states $\alpha \notin \mathcal{D}$, in which we can re-write the probability (1.335) in terms of the S matrix as

$$\begin{aligned} P_{\alpha\to\mathcal{D}} &= \int_{\mathcal{D}} d\beta\ |(2\pi)^D\delta^D(P_\beta - P_\alpha)\mathcal{M}_{\beta\alpha}|^2 \\ &= (2\pi)^D\delta^D(0)\int_{\mathcal{D}} d\beta\ (2\pi)^D\delta^D(P_\beta - P_\alpha)|\mathcal{M}_{\beta\alpha}|^2 \end{aligned} \tag{1.337}$$

where we shall take $(2\pi)^D\delta^D(0) = V \cdot T$ as the total volume of spacetime (yes, this is inherently infinite, just give it a second). This spacetime volume divergence is no contradiction since we assumed that $|\alpha\rangle^{\text{in}}$ has definite energy-momentum P_α and is hence *not* a

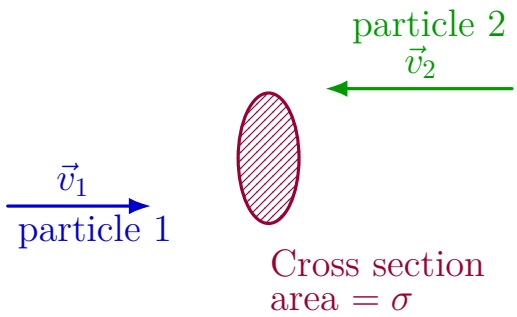
normalizable state. For example, the 2-particle in-state that is written as $|\alpha\rangle^{\text{in}} = \left|\vec{k}_1, \vec{k}_2\right\rangle^{\text{in}}$ is normalized as

$$^{\text{in}}\langle\vec{k}_1, \vec{k}_2|\vec{k}_1, \vec{k}_2\rangle^{\text{in}}\delta^{D-1}(0)\delta^{D-1}(0) = \left(\frac{V}{(2\pi)^{D-1}}\right)^2. \tag{1.338}$$

Since we have the probability, we can also write down the transition rate per unit volume

$$\Gamma_{\alpha\to\mathcal{D}} \equiv \frac{P_{\alpha\to\mathcal{D}}}{VT} = \int_{\mathcal{D}} d\beta\,(2\pi)^D\delta^D(P_\beta - P_\alpha)|\mathcal{M}_{\beta\alpha}|^2. \tag{1.339}$$

We can use this to find the cross-section. To derive/write down the cross-section, we can think of the direction of the particles in the lab frame pictorially looks like



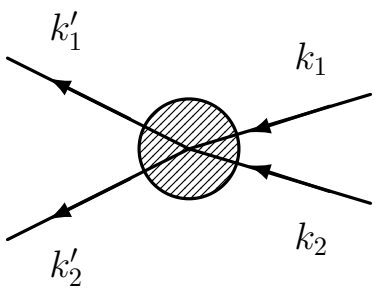
Here, the relative velocity in the lab frame is given as $v \equiv |\vec{v}_1 - \vec{v}_2|$. The scattering cross section, denoted $\sigma_{\mathcal{D}}$, is defined such that the probability ratio of scattering in $\beta \in \mathcal{D}$ is given as $\sigma_{\mathcal{D}} vT/V$ where the numerator is the effective volume that passes through σ while the denominator is the total volume of space (*not* volume of spacetime). Then, the transition rate is related to the cross-section via

$$\Gamma_{\alpha\to\mathcal{D}} = \frac{^{\text{in}}\langle\alpha|\alpha\rangle^{\text{in}} \cdot \frac{\sigma_{\mathcal{D}} vT}{V}}{VT} = \frac{\sigma_{\mathcal{D}} v}{(2\pi)^{2D-2}} \tag{1.340}$$

which implies the following form for the cross-section

$$\sigma_{\mathcal{D}} = \frac{(2\pi)^{2D-2}}{v}\int_{\mathcal{D}} d\beta\,(2\pi)^D\delta^D(P_\beta - P_\alpha)|M_{\beta\alpha}|^2. \tag{1.341}$$

As an example, if we consider $2 \to 2$ scattering in $D = 4$ with diagram,



which then has the corresponding total cross-section of 2-particle final state

$$\begin{aligned}\sigma_{2\to 2} =& \frac{(2\pi)^6}{v}\frac{1}{2}\int d^3k_1' d^3k_2'\,(2\pi)^4\delta^4(k_1'+k_2'-k_1-k_2)|\mathcal{M}(\vec{k}_1,',\vec{k}_2';\vec{k}_1,\vec{k}_2)|^2 \\ =& \frac{(2\pi)^6}{\frac{|\vec{k}_1|}{\omega_{k_1}}+\frac{|\vec{k}_2|}{\omega_{k_2}}}\frac{1}{2}\int d^3\vec{k}_1'\delta(\omega_{k_1'}+\omega_{k_2'}-E)\frac{|\mathcal{A}(k_1',k_2';k_1,k_2)}{(2\pi)^{12}2^4\omega_{k_1'}\omega_{k_2'}\omega_{k_1}\omega_{k_2}}\end{aligned} \tag{1.342}$$

where the factor of $1/2$ is inserted since we have identical particles. In our particular example, $\vec{k}_2' = \vec{k}_1 + \vec{k}_2 - \vec{k}_1'$ by conservation of energy. To reduce this further, in the center-of-mass (COM) frame, $\vec{k}_1+\vec{k}_2 = \vec{k}_1'+\vec{k}_2' = 0$, and thus the relative velocity becomes

$$v = \frac{|\vec{k}_1|}{\omega_{k_1}} + \frac{|\vec{k}_2|}{\omega_{k_2}} = \frac{|\vec{k}_1|E}{\omega_{k_1}\omega_{k_2}} \tag{1.343}$$

where E is the total energy. The integration over $d|\vec{k}_1'|$ is

$$\int |\vec{k}_1'|d|^2\vec{k}_1'|\,\delta(\omega_{k_1'}+\omega_{k_2'}-E) = \frac{|\vec{k}_1'|^2}{\frac{\partial\omega_{k_1'}}{\partial|\vec{k}_1'|}+\frac{\partial\omega_{k_2'}}{\partial|\vec{k}_1'|}} = \frac{|\vec{k}_1'|^2}{\frac{|\vec{k}_1'|}{\omega_{k_1'}}+\frac{|\vec{k}_2'|}{\omega_{k_2'}}} = \frac{|\vec{k}_1'|\omega_{k_1'}\omega_{k_2'}}{E}. \tag{1.344}$$

Then, the $2\to 2$ cross section is

$$\begin{aligned}\sigma_{2\to 2} =& \frac{(2\pi)^6}{\frac{|\vec{k}_1|E}{\omega_{k_1}\omega_{k_2}}}\frac{(2\pi)^4}{2}\int d\hat{k}_1'\,\frac{|\vec{k}_1'|\omega_{k_1'}\omega_{k_2'}}{E}\frac{|\mathcal{A}|}{(2\pi)^{12}2^4\omega_{k_1'}\omega_{k_2'}\omega_{k_1}\omega_{k_2}} \\ =& \frac{1}{2^7\pi^2E^2}\int d\hat{k}_1'\,|\mathcal{A}|^2\end{aligned} \tag{1.345}$$

If we are working with perturbative ϕ^4 theory where $\mathcal{A} = -g+\mathcal{O}(g^2)$, then the cross-section is

$$\sigma_{2\to 2} = \frac{g^2}{32\pi E^2} + \mathcal{O}(g^3) \tag{1.346}$$

for the COM frame.

1.2.5 Partial Wave Basis for Scattering Amplitudes

Now, something we have not done yet is to define or write down a specific basis for scattering amplitudes. A common choice taken and modified from QM is the *partial wave basis* that we specialize for 2-particle states (in $D = 4$). This will allow us to probe higher angular momentum states. The partial-wave basis can be written as $\left|E,\vec{P},\ell,m\right\rangle$ where $\vec{J}^2 = \ell(\ell+1)$ (a lot of this should feel familiar from QM), $J_3 = m$, $\ell = 0,1,2,4,\cdots$ (however, only *even* ℓ is allowed for identical particles) and $m = -\ell,-\ell+1,\cdots,\ell$. Previously, we were using the plane-wave basis $\left|\vec{k}_1,\vec{k}_2\right\rangle$, to which we can relate out partial wave basis to the plane-wave basis (in the COM frame $\vec{P} = 0$) as (this will allow us to probe higher angular momentum states, while also transferring over all of our prior results)

$$\langle\vec{k}_1,\vec{k}_2|E,\vec{P}=0,\ell,m\rangle = \delta^3(\vec{k}_1+\vec{k}_2)\delta(\omega_{k_1}+\omega_{k_2}-E)\mathcal{N}(E)Y_{\ell,m}(\hat{k}_1) \tag{1.347}$$

where $\mathcal{N}(E)$ is a normalization constant (to be found) and the $Y_{\ell,m}$ are our *oh most joyous* spherical harmonics. Let us set a normalization convention for the partial wave basis as

$$\langle E, \vec{P}, \ell, m | E', \vec{P}', \ell', m' \rangle = \delta(E - E')\delta^3(\vec{P} - \vec{P}')\delta_{\ell\ell'}\delta_{mm'}. \tag{1.348}$$

We can compare this normalization to that of the partial wave basis in the COM frame by computing

$$\begin{aligned}
\langle E, \vec{P}, \ell, m | E', \vec{P}' = 0, \ell', m' \rangle &= \frac{1}{2}\int d^3\vec{k}_1 d^3\vec{k}_2 \langle E, \vec{P}, \ell, m | \vec{k}_1, \vec{k}_2 \rangle \langle \vec{k}_1, \vec{k}_2 | \langle E', 0, \ell', m' \rangle \\
&= \frac{1}{2}\delta(E - E')\delta^3(\vec{P})(\mathcal{N}(E))^2 \int d^3\vec{k}_1\, \delta(\omega_{k_1} + \omega_{k_2} - E) Y^*_{\ell m}(\hat{k}_1) Y_{\ell' m'}(\hat{k}_1) \\
&= \frac{1}{2}\delta(E - E')\delta^3(\vec{P})(\mathcal{N}(E))^2 \int d\hat{k}_1 \frac{|\vec{k}_1|\omega_{k_1}\omega_{k_2}}{E} Y^*_{\ell m}(\hat{k}_1) Y_{\ell' m'}(\hat{k}_1) \\
&= \frac{1}{2}\delta(E - E')\delta^3(\vec{P})(\mathcal{N}(E))^2 \frac{|\vec{k}_1|\omega_{k_1}\omega_{k_2}}{E} \delta_{\ell\ell'}\delta_{mm'}
\end{aligned} \tag{1.349}$$

to which we compare with (1.348) to read off the normalization constant to be

$$\mathcal{N}(E) = \sqrt{\frac{2E}{|\vec{k}_1|\omega_{k_1}\omega_{k_2}}}. \tag{1.350}$$

This normalization formula without the factor of 2 is applicable to non-identical 2-particle state of possible different masses as well. The $2 \to 2$ S-matrix elements in the partial wave basis can be written in a similar form to what we had in the plane wave basis

$$\begin{aligned}
S(E', \vec{P}', \ell', m' | E, \vec{P}, \ell, m) &\equiv {}^{\text{out}}\langle E', \vec{P}', \ell', m' | E, \vec{P}, \ell, m \rangle^{\text{in}} \\
&= \delta(E' - E)\delta^3(\vec{P}' - \vec{P}) S_{\ell' m' | \ell m}(E, \vec{P}).
\end{aligned} \tag{1.351}$$

In the COM frame where $\vec{P} = 0$, the S-matrix becomes

$$S_{\ell' m' | \ell m}(E, \vec{P} = 0) = \delta_{\ell\ell'}\delta_{mm'} S_\ell(E) \tag{1.352}$$

where the delta-functions in ℓ and m arise from angular momentum conservation. The in-state for the partial wave basis can then be written in terms of the out-partial wave basis via the S-matrix

$$\left| E, \vec{P} = 0, \ell, m \right\rangle^{\text{in}} = S_\ell(E) \left| E, \vec{P} = 0, \ell, m \right\rangle^{\text{out}} + (\text{non-elastic scattering product}). \tag{1.353}$$

But, remember that we found out the S-matrix is a unitary operator, and therefore, unitarity implies the following *inequality*

$$|S_\ell(E)| \geq 1 \tag{1.354}$$

where $|S_\ell(E)| = 1$ for elastic scattering, for example, for $2m \geq E < 3m$ in the case of 1 species of particle. This inequality (1.354) should not be taken lightly since it implies that whatever S-matrix elements you are computing, they *all* must obey this inequality. If you

calculation does not, you either made a mistake, or have a non-unitary theory. Now, using our new knowledge of how the plane-wave basis is related to the partial wave basis namely

$$\left|\vec{k}_1, \vec{k}_2\right\rangle = \mathcal{N}(E) \sum_{\ell,m} Y^*_{\ell m}(\hat{k}_1) \left|E, \vec{P} = 0, \ell, m\right\rangle, \tag{1.355}$$

and thus the S-matrix becomes (the $2\mathbb{Z}^+$ just denotes only the even integers including zero)

$$\begin{aligned} S(\vec{k}_1', \vec{k}_2'|\vec{k}_1, \vec{k}_2) &= \delta^4(k_1' + k_2' - k_1 - k_2)(\mathcal{N}(E))^2 \sum_{\ell=2\mathbb{Z}^+}^{\infty} \sum_{m=-\ell}^{\ell} Y_{\ell,m}(\hat{k}_1') Y^*_{\ell,m}(\hat{k}_1) S_\ell(E) \\ &= \delta^4(k_1' + k_2' - k_1 - k_2)(\mathcal{N}(E))^2 \sum_{\ell=2\mathbb{Z}^+}^{\infty} \frac{1}{4\pi}(2\ell+1) P_\ell(\hat{k}_1' \cdot \hat{k}_1) \end{aligned} \tag{1.356}$$

where we used the the *addition theorem*[33] to complete the sum over the spherical harmonics, and $P_\ell(x)$ is the Legendre polynomial. *But*(!), remember that we already have a form of the S-matrix in terms of the (Lorentz invariant) amplitude $\mathcal{M}$ that I will re-write here for clarity (which is also equal to ${}^{\text{out}}\langle \vec{k}_1', \vec{k}_2'|\vec{k}_1, \vec{k}_2\rangle^{\text{in}}$

$$S(\vec{k}_1', \vec{k}_2'|\vec{k}_1, \vec{k}_2) = i(2\pi)^4 \delta^4(k_1' + k_2' - k_1 - k_2) \mathcal{M}(\vec{k}_1', \vec{k}_2,' |\vec{k}_1, \vec{k}_2). \tag{1.358}$$

We can match these two different ways to write the S-matrix to derive a relationship for the amplitude in terms of partial waves,

$$i(2\pi)^4 \mathcal{M}(\vec{k}_1', \vec{k}_2,' |\vec{k}_1, \vec{k}_2) = \frac{(\mathcal{N}(E))^2}{4\pi} \sum_{\ell=2\mathbb{Z}^+}^{\infty} (2\ell+1) P_\ell(\hat{k}_1' \cdot \hat{k}_1)(S_\ell(E) - 1). \tag{1.359}$$

We can further relate this to the total $2 \to 2$ cross section as

$$\begin{aligned} \sigma_{2\to 2} =& \frac{(2\pi)^6}{\frac{|\vec{k}_1|E}{\omega_{k_1}\omega_{k_2}}} \frac{1}{2} (2\pi)^4 \int d\hat{k}_1 \, \frac{|\vec{k}_1'|\omega_{k_1'}\omega_{k_2'}}{E} \\ &\times \left| (2\pi)^4 \frac{2E}{|\vec{k}_1|\omega_{k_1}\omega_{k_2}} \frac{1}{4\pi} \sum_\ell (2\ell+1) P_\ell(\hat{k}_1' \cdot \hat{k}_1)(S_\ell(E) - 1) \right|^2 \\ =& \frac{2}{4|\vec{k}_1|^2} \sum_{\ell,\ell'=2\mathbb{Z}^+}^{\infty} (2\ell+1)(2\ell'+1) \int d\hat{k}_1' \, P_\ell(\hat{k}_1' \cdot \hat{k}_1) P_{\ell'}(\hat{k}_1 \cdot \hat{k}_1) \\ &\times (S_\ell(E) - 1)(S_{\ell'}(E) - 1)^* \\ =& \frac{2\pi}{|\vec{k}_1|^2} \sum_{\ell=2\mathbb{Z}^+}^{\infty} (2\ell+1)|S_\ell(E) - 1|^2 \end{aligned} \tag{1.360}$$

[33] It just states (and I took this from Wikipedia by the way) that

$$P_\ell(\vec{x} \cdot \vec{y}) = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_\ell^m(\vec{y}) Y^m *_\ell (\vec{x}). \tag{1.357}$$

where in the second to last line we used the *completeness relation* for the Legendre polynomials namely (where the 4π comes from the solid angle for a 2-sphere)

$$\int dx\, P_\ell(x) P_{\ell'}(x) = \frac{4\pi}{2\ell+1}\delta_{\ell\ell'} \tag{1.361}$$

This wraps up the lectures on introducing the systematics of quantum field theory from the point of view of the path integral *only* looking at scalars however. Before I give a brief introduction on what to expect for the final third of these lecture notes, let us briefly review all that we went over:

1. We started off by introducing the basics to classical field theory and corresponding equations of motions. We then saw how to Fourier transform fields.

2. We then looked briefly into how canonical quantization with the Hamiltonian is performed in the field formalism that allowed us to write scalar field operators in terms of creation and annihilation operators. We also saw Noether's theorem.

3. We then moved onto the path integral quantization and saw how the proper ordering of the time coordinates for the field operators leads to a causal path integral as well as *three* different correlation functions.

4. The analytic structure of these correlation functions showed us that poles correspond to the mass of the particle while branch cuts correspond to higher energy/mass states beyond the mass of the particle.

5. We then saw how to write down 1-particle states for scalar particles which lead us to *Wigner's matrices* (which we will see the full power of Wigner's matrices when we get to spinning particles next) which lead us to the field renormalization constant $\mathcal{Z}$.

6. With the knowledge of 1-particle and multi-particle states combined with the correlation functions, we were able to find a spectral representation of the 2-point Greens function which lead us to the Källen-Lehamann spectral representation.

7. We then saw how the divergences in the 1-loop calculation can be absorbed (in an ad hoc manner) into the mass of the particle.

8. Finally, we formulated the basics of scattering theory via smearing functions and operators that lead us to define asymptotic states and thus the S-matrix. We then used this formalism to derive the cross-section in the plane-wave basis as well as in the partial wave basis.

All of these tools shall follow us and be essential when we start to look at particles with non-zero spin, especially our derivation of the 1-particle state since we will use Wigner's matrices in their full power to do something some textbooks don't cover, which is the derivation of Dirac's equation.

So, what can you look forward to? First, we will start a general group-theoretic discussion (granted, in a physics-language and not exact) on particles with non-zero spin.

With Wigner's particle language and matrices, we will specialize to $j = 1/2$ which will naturally lead us to *derive* from group-principles Dirac's equation. We will then construct spin-1/2 field operators, and then develop the path integral mathematics necessary to handle spin-1/2 particles called *Berezin integrals* and *Grassmann numbers*. We will then of course get the Feynman rules and compute some correlation functions and amplitudes.

We will then set $j = 1$ and discover spin-1 particles and their corresponding particle representation and naturally find that we can map our results to Maxwell's theory. We then see how when we have a classical theory with gauge redundancies. This will lead to us develop some new techniques in the path integral method that we won't get into here, but in QFT 2 will be called, *Faddeev-Popov method*.

We will then *finally* formulate *quantum electrodynamics* (QED) where we will compute Compton scattering, find the renormalized Lagrangian for QED, compute a full 1-loop integral by using *dimensional regularization* (dim-reg), and finally end with compute the electron g-factor correction to the famous $g - 2$ result.

1.3 Particles and Fields with Spin

1.3.1 Wigner's Little Group for Spin-1/2 & Dirac Equation

Let us now consider some particle with an internal degree of freedom that we shall call *spin*. We shall restrict to $D = 4$ and $m > 0$ for now; the primary reason for this is that there are some objects for spin-1/2 that are not well-defined in dimensions higher that 4, and massless particles always carry some baggage called *gauge invariance/symmetries* that we do not have the tools/technology to deal with gauge symmetries yet. As before with the spin-0 case, we shall pick a reference momentum $\vec{k}_R$ where $k_R = (\omega_{\vec{k}_R}, \vec{k}_R)$ such that it obeys the mass-shell condition $k_R^2 + m^2 = 0$. For example, we could pick $\vec{k}_R = \vec{0}$ and then $k_R = (m, \vec{0})$. We shall choose a "standard boost" $L^\mu{}_\nu(\vec{k})$ for any on-shell k^μ with $k^0 > 0$ such that

$$k^\mu = L^\mu{}_\nu(\vec{k}) k_R^\nu \tag{1.362}$$

which is just what we did before. We now shall define a 1-particle state again up to some normalization just as we did before

$$\left|\vec{k}, \sigma\right\rangle = N(\vec{k}) U(L(\vec{k})) \left|\vec{k}_R, \sigma\right\rangle \tag{1.363}$$

where if we set the normalization to something similar that we did before

$$\langle \vec{k}, \sigma | \vec{k}', \sigma' \rangle = \delta^3(\vec{k} - \vec{k}') \delta_{\sigma\sigma'} \tag{1.364}$$

then the normalization function we found actually stays the same $N(\vec{k}) = \sqrt{k_R^0 / k^0}$ as before. Now, for any Lorentz transformation Λ, we still expect that the unitary operator will boost, spin, then boost-back the 1-particle state as

$$U(\Lambda) \left|\vec{k}, \sigma\right\rangle = N(\vec{k}) U(L(\vec{\Lambda k})) U(L^{-1}(\vec{\Lambda k}) \Lambda L(\vec{k})) \left|\vec{k}_R, \sigma\right\rangle \tag{1.365}$$

where again, $W = L^{-1}(\vec{\Lambda k}) \Lambda L(\vec{k})$, but W fixed k_R which implies that for σ we instead have

$$U(W) \left|\vec{k}_R, \sigma\right\rangle \sum_{\sigma'} D_{\sigma\sigma'}(W) \left|\vec{k}_R, \sigma'\right\rangle \tag{1.366}$$

where $D(W)$ are often called Wigner matrices. Thus, a general Lorentz transformation acts as

$$U(\Lambda)\left|\vec{k},\sigma\right\rangle = \frac{N(\vec{k})}{N(\vec{\Lambda k})}\sigma_{\sigma'}D_{\sigma\sigma'}(W)\left|\vec{\Lambda k},\sigma'\right\rangle. \tag{1.367}$$

For $\vec{k}_R = 0$, the set of W's that leave $k_R = (m, \vec{0})$ invariant is the group of spatial rotations $G = \mathrm{SO}(3)$ which, in this context, is called *Wigner's little group*. For finding such a representation of SO(3), we would expect the unitary transformations and thus the Wigner matrices algebra to obey the following algebra

$$U(W_1W_2) = U(W_1)U(W_2) \tag{1.368}$$

$$D_{\sigma\sigma'}(W_1W_2) = \sum_{\sigma''} D_{\sigma\sigma''}(W_2)D_{\sigma''\sigma'}(W_1), \tag{1.369}$$

or i.e. as matrices $D(W) = (D_{\sigma\sigma'}(W))$ with size $1 \leq \sigma, \sigma' \leq N$, they obey

$$D(W_1W_2) = D(W_2)\cdot D(W_1) \tag{1.370}$$

which we should recognize is the opposite order of multiplication compared to that of the $U(W)$'s. Actually, we have made too strong of an assumption, namely how the multiplicative algebra behaves under the little group SO(3). It is possible that the rotation symmetry is represented *projectively* namely up to a phase

$$U(W_1)U(W_2) = e^{i\phi(W_1,W_2)}U(W_1\cdot W_2) \tag{1.371}$$

where $\phi(W_1, W_2)$ is some constant phase. We can specify $U(W)$ using the generators $\vec{J}$ by composing infinitesimal rotations along a path $\gamma \supset \mathrm{SO}(3)$, and then denote the resulting unitary operator as U_γ. Well ok, how would we construct such an operator. For this discussion, refer to the figure directly below.

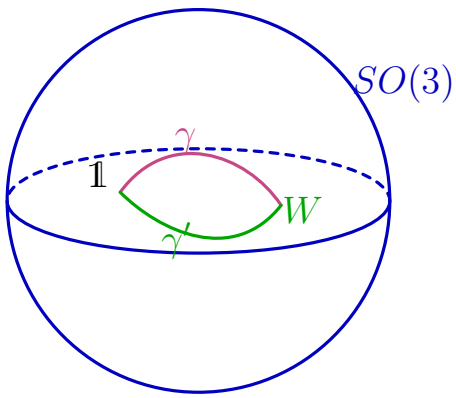
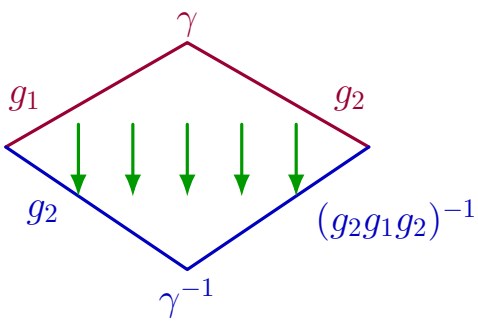


Figure 4. Paths taken from unit of SO(3) group to another element W.

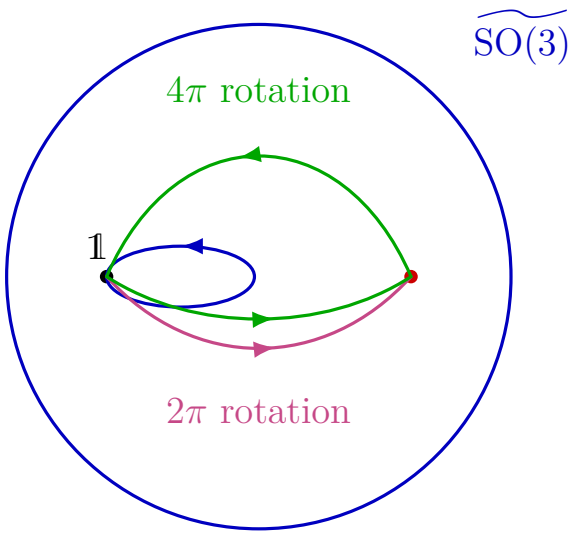
Suppose that we have two paths taken along SO(3) that both start at the unit element $\mathbb{1}$ and end at some element W[34] as seen in 4. We also know that the commutation relations above J_i's are such that $U_\gamma = U_{\gamma'}$ provided that γ can be deformed into γ' continuously. As an example, consider for some rotation group elements g_1 and g_2 that are close to one as seen in this image:



Here, we can see that we can compose the unitary operator taking of path γ can be decomposed via the group algebra into a product of rotation group elements and thus we find that

$$\text{“}U_\gamma = U(g_2)U(g_2)\text{”}=\text{“}U_{\gamma'} = U(g_2 g_1 g_2^{-1})U(g_2)\text{”} \tag{1.372}$$

The only way for $U_{\gamma'}$ to be different from U_γ would be if γ' cannot be continuously deformed into γ (fixing the end point W). In particular, for $W = \mathbb{1}$, $U_\gamma = 1$ if γ is considered a *contractible* path in SO(3). We then run into the following fact: *2π rotation is not contractible while 4π rotation is.* The semi-obvious nature of this can be seen in the following drawing:



[34]If you do not like me relating SO(3) to that of a sphere, do not worry you are **not** wrong. I am just doing this for illustrative purposes for now, and instead we will make this rigorous very soon. In essence, we will find that the unitary operators with elements W, $U(W)$ are not actually found within SO(3) but instead it's double cover! If you know a little bit about algebraic topology, then you'd say, "well, this is obvious," since you know that $\pi_1(\mathrm{SO}(3)) \simeq \mathbb{Z}/2\mathbb{Z}$ *but* $\pi_1(S^2) = 0$!

In the image above, an example of a contractible path is shown in blue. However, if we just apply a single rotation operator, then we only travel 2π which only takes us to some point denoted in red. Thus, we need to take a full 4π rotation around, to which we can see is indeed a contractible path. As a consequence of this fact, $U(W)$ or $D(W)$ is defined *not* on SO(3), but it's *2-fold covering* space $\widetilde{\mathrm{SO}(3)}$. The points of $\widetilde{\mathrm{SO}(3)}$ are equivalent classes of path $\gamma : \mathbb{1} \to W$ that are related by continuous deformations. Namely, these D-matrices are a mapping from $\widetilde{\mathrm{SO}(3)}$ to unitary matrices written as

$$D : \widetilde{\mathrm{SO}(3)} \to \text{unitary } N \times N \text{ matrices } D_{\sigma\sigma'(W)} \tag{1.373}$$

which is a *unitary representation* of $\widetilde{\mathrm{SO}(3)}$. We shall take the following theorem for granted (and not prove here, which the proof can be found in [1])

Theorem 1 *(Diagonal Spin-Matrices) We can find a basis $|\sigma\rangle$ such that*

$$D(W) = \begin{pmatrix} D^{(\ell_1)}(W) & & 0 \\ & D^{(\ell_2)}(W) & \\ 0 & & \ddots \end{pmatrix} \tag{1.374}$$

where $D^{(j)}(W)$ is a $(2j+1) \times (2j+1)$ matrix with $j = 0, 1/2, 1, 3/2, \cdots$ called a spin-j representation.

We have actually already seen the $j = 0$ case which was the trivial representation with the scalar above. We shall now move up in j and place our focus on $j = 1/2$ case. Consider the infinitesimal version of $W_{ij} = \delta_{ij} + \omega_{ij}$ where these ω_{ij} are the same as we saw before for the infinitesimal Lorentz transformations $\omega_{ij} = -\omega_{ji}$ with $i, j = 1, 2, 3$. For a finite rotation, the unitary representation of W_{ij} is given as

$$U(W) = e^{\frac{i}{2}\omega_{ij}S^{ij}} \rightsquigarrow D(W) = e^{-\frac{i}{2}\omega_{ij}S^{ij}} \tag{1.375}$$

where the matrix S^{ij} here is $S^{ij} = \epsilon^{ijk}\sigma_k/2$ where σ_k are the Pauli matrices that obey the same commutation relations as $\hat{J}^{ij}$'s.

Taking a quick step back, the way we were able to construct $D(W)$ in such a manner is that when $j = 1/2$ then the size of the matrices should be 2×2, out of which, those matrices that we know represent rotations about an axis are the Pauli matrices.

As a concrete example of what S^{ij} and $D(W)$ can represent, let us take $\omega_{12} = \theta$ and $\hat{J}^{12} = \hat{J}_3$. Then, $e^{i\theta\hat{J}_3}$ is the rotation angle around the x_3-axis by an angle θ. Therefore, the $D(W)$ operator becomes

$$e^{-i\theta S^{12}} = e^{-i\theta\sigma_3/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \tag{1.376}$$

which equals to the unit matrix $\mathbb{1}$ when $\theta = 2\pi$.

If we consider a *massive* spin-1/2 particle state, we label this as $\left|\vec{k}, \sigma\right\rangle$ where $\sigma = \{1, 2\}$ (for the two possible spin-state) with the unitary representation of the Lorentz matrix

acting on the 1-particle spin-1/2 spin state as

$$U(\Lambda)\left|\vec{k},\sigma\right\rangle = \sqrt{\frac{(\Lambda k)^0}{k^0}}\sum_{\sigma'} D_{\sigma\sigma'}(W)\left|\vec{\Lambda k},\sigma'\right\rangle \tag{1.377}$$

with the same W as above (the boost, rotate, boost back) and the $D_{\sigma\sigma'}(W)$ that was constructed above. We can of course ask the same question as we did with the scalars in that, "what kind of local field operator can create a spin-1/2 1-particle state?" Well, let us denote this generic state as $\hat{\Phi}_\alpha(x)$ where α denotes the spin-state of the particle and obeys a generic Lorentz transformation

$$U(\Lambda)\hat{\Phi}_\alpha(x)(U(\Lambda))^{-1} = (R(\Lambda))_\alpha{}^\beta\hat{\Phi}_\beta(\Lambda x) \tag{1.378}$$

where $R(\Lambda)$ is not necessarily a unitary matrix unlike $D(W)$, but is the matrix representation that transforms the field based on the little group. Note here that since $U(\Lambda_1\Lambda_2) = U(\Lambda_1)U(\Lambda_2)$, then the R's here must obey $R(\Lambda_1\Lambda_2) = R(\Lambda_2)\cdot R(\Lambda_1)$ just as was the case with $D(W)$, and similar to $D(W)$, we will allow $R(\Lambda)$ to be defined on the covering group $\widetilde{\mathrm{SO}(1,3)}$.

Let us now consider the overlap of $\hat{\Phi}_\alpha(x)\left|\Omega\right\rangle$ with the 1-particle state

$$\left\langle\vec{k},\sigma\right|\hat{\Phi}_\alpha(x)\left|\Omega\right\rangle = e^{-ik\cdot x}\left\langle\vec{k},\sigma\right|\hat{\Phi}_\alpha(0)\left|\Omega\right\rangle. \tag{1.379}$$

Again, following the scalar case, to determine the RHS of (1.379) we will write it up to some function $\mathcal{U}_\alpha^\sigma(\vec{k})$ that we will call the *spinorial polarization* multiplied by some constant C (which will be chosen later)

$$\left\langle\vec{k},\sigma\right|\hat{\Phi}_\alpha(0)\left|\Omega\right\rangle = C\cdot\mathcal{U}_\alpha^\sigma(\vec{k}). \tag{1.380}$$

We can write (1.380) into terms of transformed fields as (1.378) to get

$$\left\langle\vec{k},\sigma\right|(U(\Lambda))^\dagger U(\Lambda)\hat{\Phi}_\alpha(0)(U(\Lambda))^{-1}\left|\Omega\right\rangle = \sqrt{\frac{(\Lambda k)^0}{k^0}}\sum_{\sigma'} D^*_{\sigma\sigma'}(W)R_\alpha{}^\beta(\Lambda)\left\langle\vec{\Lambda k},\sigma'\right|\hat{\Phi}_\beta(0)\left|\Omega\right\rangle. \tag{1.381}$$

We can compare (1.380) with (1.381) to read off the unitary transformation of $\mathcal{U}_\alpha^\sigma$ to find

$$\mathcal{U}_\alpha^\sigma(\vec{k}) = \sqrt{\frac{(\Lambda k)^0}{k^0}}D^*_{\sigma\sigma'}(W)R_\alpha{}^\beta(\Lambda)\mathcal{U}_\beta^{\sigma'}(\vec{\Lambda k}). \tag{1.382}$$

For particular boosts and transformation, namely if we set $\Lambda k = k_R$ and $\Lambda = (L(\vec{k}))^{-1}$, and $W = \mathbb{1}$, then (1.382) becomes

$$\mathcal{U}_\alpha^\sigma(\vec{k}) = \sqrt{\frac{k_R^0}{k^0}}R_\alpha{}^\beta((L(\vec{k}))^{-1})\mathcal{U}_\beta^\sigma(\vec{k}_R). \tag{1.383}$$

If we instead chose that $\vec{k} = \vec{k}_R$ and $\Lambda = w$, then the unitary transformation of the spinors are

$$\mathcal{U}_\alpha^\sigma(\vec{k}_R) = D^*_{\sigma\sigma'}(W)R_\alpha{}^\beta(W)\mathcal{U}_\beta^{\sigma'}(\vec{k}_R). \tag{1.384}$$

If we assume that $\mathcal{U}^\sigma_\alpha(\vec{k}_R)$ is non-degenerate, then by *Schur's lemma*[35] implies that $R_\alpha{}^\beta(W)$ must also be a direct sum of spinor representations of $\widetilde{\mathrm{SO}(3)} \supset \widetilde{\mathrm{SO}(1,3)}$ where $W \in \widetilde{\mathrm{SO}(3)}$ and $\Lambda \in \widetilde{\mathrm{SO}(1,3)}$. To construct $R_\alpha{}^\beta(\Lambda)$ explicitly, we begin with an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu \tag{1.385}$$

which has the following finite unitary transformation

$$U(\Lambda) = e^{\frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu}} \rightsquigarrow R(\Lambda) = e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}. \tag{1.386}$$

Here, $S^{\mu\nu}$ is a matrix $((S^{\mu\nu\beta}_\alpha)$ that obeys the same commutation relation as $\hat{J}^{\mu\nu}$'s i.e.

$$[S^{\mu\nu}, S^{\rho\sigma}] = -i\left(\eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} - (\rho \leftrightarrow \sigma)\right). \tag{1.387}$$

Consider now that *if* we have a set of matrices γ^μ that obey an *anticommuting* relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}. \tag{1.388}$$

Then, we can re-write the spin-1/2 angular momentum-like operator as

$$S^{\mu\nu} \equiv -\frac{i}{4}\left[\gamma^\mu, \gamma^\nu\right] \tag{1.389}$$

will also satisfy the commutation relation in (1.387). A possible choice ofa γ^μ's are the following

$$\gamma^0 = -i\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad \gamma^j = -i\begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix} \tag{1.390}$$

where I_2 is the 2-dimensional identity matrix and $j = 1, 2, 3$. We could have also chosen $\widetilde{\gamma^\mu} = A\gamma^\mu A^{-1}$ for some matrix A would also do.

Now that we have specified $R(\Lambda) = e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}$, we have also determined the spinorial polarization $\mathcal{U}^\sigma_\alpha(\vec{k})$ in terms of $\mathcal{U}^\sigma_\beta(\vec{k}_R)$ given (1.382). A useful property to have between the spinor angular-momentum operator and the gamma-matrices is that since they obey the following commutation relation

$$[S^{\mu\nu}, \gamma^\rho] = -i\gamma^\mu\eta^{\nu\rho} + i\gamma^\nu\eta^{\mu\rho}, \tag{1.391}$$

then the transformation of γ^μ is

$$R(\Lambda)\gamma^\mu(R(\Lambda))^{-1} = \Lambda^\mu{}_\nu\gamma^\nu. \tag{1.392}$$

In particular, consider the transformation of the following Lorentz scalar $k_\mu\gamma^\mu$,

$$R(L(\vec{k}))k_\mu\gamma^\mu R((L(\vec{k}))^{-1}) = k_\mu(L(\vec{k}))^\mu{}_\nu\gamma^\nu = ((L(\vec{k}))^{-1}\cdot k)_\nu\gamma^\nu = (k_R)_\mu\gamma^\mu. \tag{1.393}$$

[35] Schur's lemma states that for groups M and N are two finite-dimensional irreducible representations of a group G and φ is a linear map from M to N that commutes with the action of the group, then either φ is invertible or $\varphi = 0$.

Very often from now, we will be using a shorthand notation of $\not{k} \equiv k_\mu \gamma^\mu$. Then, this Lorentz scalar multiplied by the spinor is related to the spinor in the reference frame momentum as

$$\not{k}\mathcal{U}^\sigma(\vec{k}) = \sqrt{\frac{k_R^0}{k^0}}\not{k}R((L(\vec{k}))^{-1})\mathcal{U}^\sigma(\vec{k}_R) = \sqrt{\frac{k_R^0}{k^0}}R((L(\vec{k}))^{-1})\not{k}_R\mathcal{U}^\sigma(\vec{k}_R). \tag{1.394}$$

The matrix $\not{k}_R$, which is equal to $-m\gamma^0$ for $k_R = (m, \vec{0})$, obeys the following

$$\not{k}_R^2 = \frac{1}{2}k_{R\mu}k_{R\nu}\{\gamma^\mu, \gamma^\nu\} = k_R^2 = -m^2 \tag{1.395}$$

The eigenvalues of $\not{k}_R$ are $\pm im$ which presents us with a possible basis for the spinors

$$\not{k}_R\mathcal{U}_R^\sigma = im\mathcal{U}_R^\sigma \tag{1.396}$$

$$\not{k}_R\mathcal{V}_R^\sigma = -im\mathcal{V}_R^\sigma. \tag{1.397}$$

It then follows that the corresponding spinor polarizations $\mathcal{U}_\alpha^\sigma(\vec{k})$ and $\mathcal{V}_\alpha^\sigma(\vec{k})$ obey the following equations

$$(\not{k} - im)_\alpha{}^\beta\mathcal{U}_\beta^\sigma(\vec{k}) = 0 \tag{1.398}$$

$$(\not{k} + im)_\alpha{}^\beta\mathcal{V}_\beta^\sigma(\vec{k}) = 0. \tag{1.399}$$

These two equations in (1.399) are a *fairly unknown* set of equations[36] called the *Dirac equation* (in the spinor basis). Some useful properties of the spinor $\mathcal{U}_\alpha^\sigma(\vec{k})$ are the following.

Consider setting $\vec{k}_R = 0$ from now on. Then, the transformation of the spinor is given as

$$\mathcal{U}_\alpha^\sigma(\vec{k}) = \sqrt{\frac{m}{k^0}}R_\alpha{}^\beta((L(\vec{k}))^{-1})\mathcal{U}_\beta^\sigma(\vec{0}) \tag{1.400}$$

but remember, $R(\Lambda)$ is *not* necessarily unitary, and in this case it is not!

For the general case to see that $R(\Lambda)$ is not always unitary, consider the definition

$$R_\alpha{}^\beta(\Lambda) = \left(e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}\right)_\alpha{}^\beta \tag{1.401}$$

where $S^{\mu\nu} = -\frac{i}{4}[\gamma^\mu, \gamma^\nu]$ here which is *not* Hermitian. Consider the choice of the gamma matrices made in (1.390) where their Hermitian adjoint is $(\gamma^\mu)^\dagger = \gamma_\mu = \gamma^0\gamma^\mu\gamma^0$. Let us define $\beta \equiv i\gamma^0$ such that $\beta^2 = \mathbb{1}$. Then,

$$(\gamma^\mu) = -\beta\gamma^\mu\beta \tag{1.402}$$

which implies that the Hermitian adjoint for $S^{\mu\nu}$, $R(\Lambda)$ and the spinor are[37]

$$(S^{\mu\nu})^\dagger = \beta S^{\mu\nu}\beta \tag{1.403}$$

$$(R(\Lambda))^\dagger = \beta R(\Lambda^{-1})\beta \tag{1.404}$$

$$\begin{aligned}(\mathcal{U}^\sigma(\vec{k}))^\dagger &= \sqrt{\frac{m}{k^0}}(\mathcal{U}^\sigma(\vec{0}))^\dagger(R((L(\vec{k}))^{-1}))^\dagger \\ &= \sqrt{\frac{m}{k^0}}(\mathcal{U}^\sigma(\vec{0}))^\dagger\beta R(L(\vec{k}))\beta.\end{aligned} \tag{1.405}$$

[36] This is pure sarcasm...

[37] You can show that these are in fact the case and are true by using their corresponding definitions of gamma-matrices, but the computation is not illuminating at all. In problem set 5, you will actually find something illuminating on how to construct spinors that have only one type of helicities.

In particular, a specific product of two spinors are

$$
\begin{aligned}
(\mathcal{U}^\sigma(\vec{k}))^\dagger \beta \mathcal{U}^{\sigma'}(\vec{k}) &= \frac{m}{k^0}(\mathcal{U}^\sigma(\vec{0}))^\dagger \beta R(L(\vec{k}))\beta \cdot \beta R((L(\vec{k}))^{-1})\mathcal{U}^{\sigma'}(\vec{0}) \\
&= \frac{m}{k^0}(\mathcal{U}^\sigma(\vec{0}))^\dagger \beta \mathcal{U}^{\sigma}(\vec{0}).
\end{aligned} \tag{1.406}
$$

We also have that if the reference momentum is $\not{k}_R = -m\gamma^0 = im\beta$, then the spinors transform under β as $\beta\mathcal{U}_R^\sigma = \mathcal{U}_R^\sigma$ and $\beta\mathcal{V}_R^\sigma = -\mathcal{V}_R^\sigma$.. We also normalize the spinors such that, with $\sigma, \sigma' = 1, 2$,

$$
(\mathcal{U}_R^\sigma)^\dagger \mathcal{U}_R^{\sigma'} = (\mathcal{V}_R^\sigma)^\dagger \mathcal{V}_R^{\sigma'} = \delta_{\sigma\sigma'}, \quad \text{and} \quad (\mathcal{U}_R^\sigma)^\dagger \mathcal{V}_R^{\sigma'} = 0. \tag{1.407}
$$

Now, in particular, we can state the following relations

$$
(\mathcal{U}^\sigma(\vec{k}))^\dagger \beta \mathcal{U}^{\sigma'}(\vec{k}) = -\frac{m}{k^0}\delta_{\sigma\sigma'} \tag{1.408}
$$

$$
(\mathcal{V}^\sigma(\vec{k}))^\dagger \beta \mathcal{V}^{\sigma'}(\vec{k}) = -\frac{m}{k^0}\delta_{\sigma\sigma'} \tag{1.409}
$$

$$
(\mathcal{U}^\sigma(\vec{k}))^\dagger \beta \mathcal{V}^{\sigma'}(\vec{k}) = 0. \tag{1.410}
$$

We also have the following *completeness relations* for the spinors[38]

$$
\sum_\sigma \mathcal{U}_R^\sigma (\mathcal{U}_R^\sigma)^\dagger = \frac{1+\beta}{2} \tag{1.411}
$$

$$
\sum_\sigma \mathcal{V}_R^\sigma (\mathcal{V}_R^\sigma)^\dagger = \frac{1-\beta}{2}. \tag{1.412}
$$

It then follows from the completeness relations that the product of spinors are

$$
\begin{aligned}
\sum_\sigma \mathcal{U}^\sigma(\vec{k})(\mathcal{U}^\sigma(\vec{k}'))^\dagger &= \frac{m}{k^0}\sum_\sigma R(L^{-1})\mathcal{U}_R^\sigma(\mathcal{U}_R^\sigma)^\dagger (R(L^{-1}))^{-1} \\
&= \frac{m}{k^0} R(L^{-1})\frac{1+\beta}{2}\beta R(L)\beta \\
&= \frac{1}{2k^0}(m - iR(L^{-1})\not{k}_R R(L)) \cdot \beta \\
&= \frac{m - i\not{k}}{2k^0}\beta.
\end{aligned} \tag{1.413}
$$

In a similar manner, the other spinors behave something similar

$$
\sum_\sigma \mathcal{V}^\sigma(\vec{k})(\mathcal{V}(\vec{k}))^\dagger = \frac{-m - i\not{k}}{2k^0}\beta. \tag{1.414}
$$

Now let us inspect the following relation again in (1.384) but with $\vec{k}_R = \vec{0}$

$$
\mathcal{U}_\alpha^\sigma(\vec{0}) = D^*_{\sigma\sigma'}(W) R_\alpha{}^\beta(W)\mathcal{U}_\beta^{\sigma'}(\vec{0}). \tag{1.415}
$$

[38] I know a lot of these relations are just being *stated* and not derived, but there is not much (if any) physical insight into their derivations. A lot of it is using relations over and over until we have a nice form. Many of the derivations are found in Peskin & Schroeder.

Let us consider $W = \exp(i\theta J_{12})$, and then $R(W)$ and $D(W)$ becomes

$$R(W) = e^{-i\theta\left(-\frac{i}{4}[\gamma^1,\gamma^2]\right)} = \begin{pmatrix} e^{-\frac{i\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{\frac{i\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{-\frac{i\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{\frac{i\theta}{2}} \end{pmatrix} \tag{1.416}$$

$$D(W) = e^{-i\theta\frac{\sigma_3}{2}} = \begin{pmatrix} e^{-\frac{i\theta}{2}} & 0 \\ 0 & e^{\frac{i\theta}{2}} \end{pmatrix}. \tag{1.417}$$

Recall hat we have chosen a basis such that $\left|\vec{k}_R, \sigma\right\rangle$ to be such that

$$\hat{J}_3 \left|\vec{0}, 1\right\rangle = -\frac{1}{2}\left|\vec{0}, 1\right\rangle \quad \text{and} \quad \hat{J}_3 \left|\vec{0}, 2\right\rangle = \frac{1}{2}\left|\vec{0}, 2\right\rangle \tag{1.418}$$

which allows us to write the spinors up to an overall phase as

$$\mathcal{U}_R^1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathcal{U}_R^2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \tag{1.419}$$

$$\mathcal{V}_R^1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad \mathcal{V}_R^2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}. \tag{1.420}$$

Next, let us *define* a *chirality matrix*

$$\gamma_5 \equiv -i\gamma^0\gamma^1\gamma^2\gamma^3 \tag{1.421}$$

where γ_5 obeys $\gamma_5^2 = 1$ and $\{\gamma_5, \gamma^\mu\} = 0$ where $\mu = 0, 1, 2, 3$. For our particular basis of gamma matrices (1.390), the chirality matrix takes the following form

$$\gamma_5 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} \tag{1.422}$$

which relates the two different spinors as $\mathcal{V}_R^\sigma = \gamma_5 \mathcal{U}_R^\sigma$. Now, after doing all of this, we are finally able to tackle the question that we posed in (1.380) namely, " what is the field $\hat{\Phi}_\alpha(x)$ that has the following property,"

$$\left\langle \vec{k}, \sigma \right| \hat{\Phi}_\alpha(x) \left|\Omega\right\rangle = e^{-ik\cdot x} \cdot C \cdot \mathcal{V}_\alpha^\sigma(\vec{k}) \text{ ??"} \tag{1.423}$$

1.3.2 Constructing Fermionic Fields

Well, now that we are armed with knowledge of the spinor variables, even how they interact with momentum 4-vectors, we can attempt to attempt a free spin-1/2 field

$$\hat{\Phi}_\alpha(x) = \int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(e^{-ik\cdot x}\mathcal{V}_\alpha^\sigma(\vec{k}) a_\sigma^\dagger(\vec{k}) + \lambda e^{ik\cdot x}\mathcal{U}_\alpha^\sigma(\vec{k})\widetilde{a}_\sigma(\vec{k}) \right)_{k^0=\omega_{\vec{k}}}. \tag{1.424}$$

Notice that we had the Dirac equation relations given in (1.399) of which if we Fourier transform the 4-vector $k^\mu \to i\partial^\mu$, then the Dirac *operator* acting on the spinors becomes $(\gamma^\mu k_\mu - im) \to i(\gamma^\mu \partial_\mu - m)$, to which we can combine the spinors to find the *full* Dirac equation

$$(\partial\!\!\!/ + m)\hat{\Phi}(x) = 0 \tag{1.425}$$

which is the Dirac equation acting on the spin-1/2 field and $\partial\!\!\!/ \equiv \gamma^\mu \partial_\mu$. The Hermitian conjugate of the spinor field (1.424) is given by

$$(\hat{\Phi}^\dagger(x)\beta)^\alpha = \int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(e^{-ik\cdot x}(\mathcal{V}^\sigma(\vec{k})\beta)^\alpha a^\dagger_\sigma(\vec{k}) + \lambda^* e^{ik\cdot x}(\mathcal{U}^{\sigma\dagger}\beta)^\alpha(\vec{k})\widetilde{a}_\sigma(\vec{k}) \right)_{k^0=\omega_{\vec{k}}}. \tag{1.426}$$

As with the case of the scalar, we can attempt to compute the commutation relation between these fields

$$\begin{aligned}\left[\hat{\Phi}_\alpha(x), (\hat{\Phi}^\dagger(y)\beta)^{\alpha'}\right] &= \int \frac{d^3\vec{k}}{(2\pi)^3} \Big(-e^{-ik\cdot(x-y)} \left(\mathcal{V}^\sigma(\vec{k})(\mathcal{V}^\sigma(\vec{k}))^\dagger\beta\right))_\alpha{}^{\alpha'} \\ &\quad + |\lambda|^2 e^{ik\cdot(x-y)} \left(\mathcal{U}^\sigma(\vec{k})(\mathcal{U}^\sigma(\vec{k}))^\dagger\beta\right)_\alpha{}^{\alpha'} \Big) \\ &= \int \frac{d^3\vec{k}}{(2\pi)^3} \left(-e^{-ik\cdot(x-y)} \frac{-m - i\not{k}}{2\omega_k} + |\lambda|^2 e^{ik\cdot(x-y)} \frac{m - i\not{k}}{2\omega_k} \right)_\alpha{}^{\alpha'}. \end{aligned} \tag{1.427}$$

For causal commutation relations, we want spacelike separated points x, y namely if we enforce $(x-y)^0 = 0$ and $(\vec{x} - \vec{y}) \neq 0$ we have

$$\left[\hat{\Phi}_\alpha(x), (\hat{\Phi}^\dagger(y)\beta)^{\alpha'}\right] = \int \frac{d^3\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \left(-\frac{-m + i\not{k}}{2\omega_k} + |\lambda|^2 \frac{m - i\not{k}}{2\omega_k} \right)_\alpha{}^{\alpha'} \tag{1.428}$$

which we can see does *not* equal zero! We have a term leftover that is proportional to $\propto 1 + |\lambda|^2$, and we can see that even if we set $\lambda = i$ then $\lambda\lambda^* = 1$ we still don't get something that can vanish, so we need a new approach. Let us instead set $\lambda = 1$ and try the following different spinor field expansion in terms of some new creation and annhilation operators

$$\hat{\psi}_\alpha(x) = \int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(e^{-ik\cdot x}\mathcal{V}^\sigma_\alpha(\vec{k}) b^\dagger_\sigma(\vec{k}) + e^{ik\cdot x}\mathcal{U}^\sigma_\alpha(\vec{k})\widetilde{b}_\sigma(\vec{k}) \right)_{k^0=\omega_{\vec{k}}} \tag{1.429}$$

and then *postulate* that these new creation and annihilation operators obey an *anti-commuting* relation

$$\{b_\sigma(\vec{k}), b^\dagger_{\sigma'}(\vec{k}')\} = \{\widetilde{b}_\sigma(\vec{k}), \widetilde{b}^\dagger_{\sigma'}(\vec{k}')\} = \delta_{\sigma\sigma'}\delta^{D-1}(\vec{k} - \vec{k}'). \tag{1.430}$$

Then, if we compute the anti-commutator of the spinor field (1.429) we find

$$\begin{aligned}\{\hat{\psi}_\alpha(x), (\hat{\psi}^\dagger(y)\beta)^{\alpha'}\} &= \int \frac{d^3\vec{k}}{2\pi)^3} \left(e^{-ik\cdot(x-y)} \frac{-m - i\not{k}}{2\omega_k} + e^{ik\cdot(x-y)} \frac{m - i\not{k}}{2\omega_k} \right)_\alpha{}^{\alpha'} \\ &= (-\partial\!\!\!/_x + m)_\alpha{}^{\alpha'} \left(\Delta_+(x-y; m^2) - \Delta_+(y-x; m^2) \right) \end{aligned} \tag{1.431}$$

for which the parenthesis vanishes for $(x-y)^2 > 0$! The conclusion that we reach is that the causal field operator that creates a spin-1/2 particle should be *fermionic* (based on the anti-commuting relation we had to used)! Now, even with our correct (what we can now *confidently* call) fermionic field operator that generates spin-1/2 particles, the question still remains, "how do we formulate a quantum theory of fermionic fields?" Notice that *nothing* that we just worked on was quantum mechanical in nature... And so, we return to our two methods of quantization. If we went with the canonical quantization, the question then becomes, "what is the classical theory to start with?" If we went with the path integral, then the question becomes, "what is the field space to integrate over?"

But, there is still a more subtle question we must tackle first which is, "exactly what are the spin-1/2 particles we are quantizing?" Unlike in the scalar case, the fermionic case has a much more richer quantity of particle content, namely we will differentiate between *Dirac vs Weyl vs Majorana* fermions (in $D = 4$).

A Dirac spinor field operator, denoted $\hat{\psi}_\alpha$ transforms under Lorentz symmetry according to the familiar rule

$$U(\Lambda)\hat{\psi}_\alpha(x)(U(\Lambda))^{-1} = (R(\Lambda))_\alpha{}^\beta \hat{\psi}_\beta(\Lambda \cdot x). \tag{1.432}$$

For infinitesimal Lorentz transformations $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$. Then the unitary operator encoding the spin-1/2 Lorentz transformation is $R(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}\right)$ where $S^{\mu\nu} = -\frac{i}{4}[\gamma^\mu, \gamma^\nu]$ (which are 4×4 matrices). A priori, there are no linear relations between the conjugate spinor $(\hat{\psi}^\dagger)^\alpha_x$ and the spinor $\hat{\psi}_\alpha(x)$ which each have four complex components. Now, since γ_5 commutes with $S^{\mu\nu}$, then the product $(\gamma_5\hat{\psi})_\alpha(x)$ transforms under Poincaré symmetry in the same way as $\hat{\psi}_\alpha(x)$ does. To uncover the behavior of this chiral matrix, we can *define* a new operator $\hat{\chi}^\pm_\alpha(x)$ as the chiral/anti-chiral projection

$$\hat{\chi}^\pm_\alpha(x) \equiv \left(\frac{1 \pm \gamma_5}{2}\hat{\psi}\right)_\alpha (x) \tag{1.433}$$

which essentially describes how the spin of the fermion is aligned with its corresponding momentum. In our standard basis with

$$\gamma_5 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \tag{1.434}$$

the sum $(1 + \gamma_5)/2$ becomes

$$\frac{1+\gamma_5}{2} = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}. \tag{1.435}$$

Here, either $\hat{\chi}^+$ or $\hat{\chi}^-$ retains only *two* components of $\hat{\psi}$.

A *Weyl spinor* field denoted $\hat{\chi}_\alpha(x)$ is one that obeys the following when multiplied by the chirality matrix

$$\gamma^5\hat{\chi} = \hat{\chi} \qquad \text{(chiral)} \tag{1.436}$$

$$\gamma_5\hat{\chi} = -\hat{\chi} \qquad \text{(ant-chiral).} \tag{1.437}$$

Now we have seen/learned that a Dirac spinor field $\hat{\Psi}$ contains a chiral Weyl spinor field $\hat{\chi}^+$ and an anti-chiral Weyl spinor field $\hat{\chi}^-$.

Before we get into our third and final type of fermion, there are a few technical things we need to discuss about how gamma matrices transform under some different representation. Recall that in our convention the gamma matrices take the form of (1.390), and then we have the conjugate of the gamma matrices can be written as

$$(\gamma^\mu)^* = B\gamma^\mu B^{-1} \tag{1.438}$$

where B may be chosen as $B = \gamma_2$, and so the conjugate of spinor-angular-momentum and unitary transformation of Lorentz symmetry can also be written as

$$(S^{\mu\nu})^* = -BS^{\mu\nu}B^{-1} \tag{1.439}$$

$$(R(\Lambda))^* = BR(\Lambda)B^{-1}. \tag{1.440}$$

It then follows that $(B^{-1}\hat{\psi}^*)_\alpha(x)$ (where we are taking specifically the Hermitian conjugate here, not the adjoint as well) has the same Poincaré transformation property as $\hat{\psi}_\alpha(x)$[39] Also, applying such an operation twice gives back the original $\hat{\psi}$ namely $B^{-1}(B^{-1}\hat{\psi}^*)^* = \hat{\psi}$ since $B^{-1}(B^{-1})^* = I$. Now we can talk about Majorana fermions.

A *Majorana spinor field* that we shall denote as $\hat{\zeta}_\alpha(x)$ is one such spinor that obeys

$$\hat{\zeta}^*_\alpha(x) = (B\zeta)_\alpha(x). \tag{1.442}$$

We can associate a Weyl spinor $\hat{\chi}^\dagger$ with a Majorana spinor via

$$\hat{\zeta} = \hat{\chi}^\dagger + B^*(\hat{\chi}^\dagger)^* \tag{1.443}$$

and going the other way

$$\hat{\chi}^\dagger = \frac{1+\gamma_5}{2}\hat{\zeta}. \tag{1.444}$$

As a quick check that this is true the following should vanish

$$\frac{1+\gamma_5}{2}B^*(\hat{\chi}^\dagger)^* = B^*\frac{1-\gamma_5}{2}(\hat{\chi}^\dagger)^* = B^*\left(\frac{1-\gamma_5}{2}\hat{\chi}^\dagger\right)^* = 0. \tag{1.445}$$

Also, note that given a chiral Weyl spinor $\hat{\chi}^\dagger$ and $B^*(\hat{\chi}^\dagger)^*$ are anti-chiral

$$\gamma_5 B^*(\hat{\chi}^\dagger)^* = -B^*\gamma_5(\hat{\chi}^\dagger)^* = -B^*(\hat{\chi}^\dagger)^*. \tag{1.446}$$

In summary, Weyl and Majorana spinor fields can be obtained by projections on a Dirac spinor field. These notions of spinor fields are useful in construvting operators and Lagrangians. Note though that there is *no a priori* logical connection between the existence of these different type of field operators and the particle content of the QFT. But, we would of course like to see how we can relate these particle types to ones in real life via a field formalism, namely a QFT, and thus we shall begin our dive into how to formulate fermionic fields into the path integral quanization.

[39] We can show this explicitly in the following manner

$$\begin{aligned} U(\Lambda)B^{-1}\hat{\Psi}^*(x)(U(\Lambda))^{-1} &= B^{-1}(R(\Lambda)\hat{\psi}(\Lambda x))^* = B^{-1}(BR(\Lambda)B^{-1})\hat{\psi}^*(\Lambda x) \\ &= R(\Lambda)B^{-1}\hat{\psi}^*(\Lambda x). \end{aligned} \tag{1.441}$$

1.3.3 Grassmann Numbers & Variables (yes, yes we do need them)

To understand the introduction of the new mathematics/number-system we are about to see, there is a great discussion/physical reasoning as to why in Weinberg's textbook chapter 9.5 [1], but the basic premise is that with finding out spin-1/2 field representations anticommute instead of commute with the normal commutation relations, this implies that the "generalized coordinates" for spinors will anticommute as well, and thus when operating on the vacuum you can generate a relationship $q_a q_b + q_b q_a = 0$, hence forcing us to use anticommuting *numbers/variables*. Thus, we need to begin an exposé into the classical mechanics of Grassmann variables and it's corresponding quantization.

Let us begin by introducting *Grassmann coordinates* denoted as η_α where $\alpha = 1, \cdots, N$ that obey the following two *very important* relations

$$\eta_\alpha \eta_\beta = -\eta_\beta \eta_\alpha \quad \text{and} \quad \eta_\alpha^2 = 0. \tag{1.447}$$

In contrast to ordinary coordinates that parameterize the configuration space of a system, the "configuration space" of Grassmann mechanics can only be characterized through relations among functions on the space $f(\eta_\alpha)$. Let us consider the following Lagrangian

$$L = \sum_{\alpha,\beta=1}^{N} M_{\alpha\beta} \eta_\alpha \dot{\eta}_\beta \tag{1.448}$$

with a corresponding action (and $\dot{\eta}_\alpha \equiv d\eta_\alpha/dt$)

$$S = \int dt \, M_{\alpha\beta} \eta_\alpha \dot{\eta}_\beta \tag{1.449}$$

where $M_{\alpha\beta}$ is some symmetric matrix in $(\alpha\beta)$. Notice the following feature of the action (1.449) under integration by parts

$$-\int dt \, M_{\alpha\beta} \dot{\eta}_\alpha \eta_\beta = \int dt \, M_{\alpha\beta} \eta_\beta \dot{\eta}_\alpha \tag{1.450}$$

since the Grassmann coordinates anticommute! We can derive the corresponding canonical momenta to η_α as

$$\pi_\alpha = L \frac{\overleftarrow{\partial}}{\partial \dot{\eta}_\alpha} = M_{\gamma\beta} \eta_\gamma \dot{\eta}_\beta \frac{\overleftarrow{\partial}}{\partial \dot{\eta}_\alpha} = M_{\gamma\beta} \eta_\gamma \delta_\alpha^\beta = M_{\alpha\gamma} \eta_\gamma \tag{1.451}$$

where opposite vector arrow here $\overleftarrow{\partial}_{\dot{\eta}_\alpha}$ denotes on operating the partial derivative from right to left, or in other words, we are used to operating derivatives that hit whatever is on the RHS side of them first, or reading LHS to RHS, but now we go backwards.

Now, we could proceed in the normal sense of writing down some sort of anti-commuting Poisson bracket like

$$\{\eta_\alpha, \eta_\beta\}_P = \{\pi_\alpha, \pi_\beta\}_P = 0 \tag{1.452}$$

$$\{\eta_\alpha, \pi_\beta\}_P = \{\pi_\alpha, \eta_\beta\}_P = \delta_{\alpha\beta} \tag{1.453}$$

but then our conjugate momentum of $\pi_\alpha = M_{\alpha\beta}\eta_\beta$ would not satisfy these Poisson brackets. So, is there some sort of contradiction? What follows is taken from Dirac's "lectures on QM" from 1964, which is still one of the best ways to understand how to handle quantizing spinors.

Dirac taught us that we should view the conjugate momenta of a grassmann coordinate as a *secondary-class primary constraint*

$$\chi_\alpha(\eta, \pi) \equiv \pi_\alpha - M_{\alpha\beta}\eta_\beta = 0 \tag{1.454}$$

where "second-class" here means non-degenerate Poisson bracket among themselves in contrast to "first-class" which has to do with "gauge redundancies." The wording "primary" here are in contrast to "seconday" constraints which follow from the equations of motion. The second-class constraint here in essence is computing

$$\begin{aligned} K_{\alpha\beta} &\equiv \{\chi_\alpha, \chi_\beta\}_P \\ &= \frac{\partial \chi_\alpha}{\partial \pi_\gamma}\frac{\partial \chi_\beta}{\partial \eta_\gamma} + \frac{\partial \chi_\alpha}{\partial \eta_\gamma}\frac{\partial \chi_\beta}{\partial \pi_\gamma} \\ &= -2M_{\alpha\beta} \end{aligned} \tag{1.455}$$

of which we assume *this* is non-degenerate. We shall now *define* a classical Dirac bracket

$$\{f, g\}_D \coloneqq \{f, g\}_P - \{f, \chi_\alpha\}_P (K^{-1})^{\alpha\beta}\{\chi_\beta, g\}_P \tag{1.456}$$

which shall replace the Poisson bracket in our usual quantization steps. The consistency of the constraint (1.454) is ensured by enforcing

$$\{f\,\chi_\alpha\}_D = 0 \tag{1.457}$$

for any $f(\eta, \pi)$. Now, let us see what happens when we compute the Dirac bracket for two Grassmann coordinates

$$\begin{aligned} \{\eta_\alpha, \eta_\beta\}_D &= \{\eta_\alpha, \eta_\beta\}_P - \{\eta_\alpha, \chi_\gamma\}_P (K^{-1})^{\gamma\delta}\{\chi_\delta, \eta_\beta\}_P \\ &= -(K^{-1})_{\alpha\beta} = \frac{1}{2}(M^{-1})_{\alpha\beta}. \end{aligned} \tag{1.458}$$

In a similar manner, $\{\eta_\alpha, \pi_\beta\}_D = \delta_{\alpha\beta}/2$. An toy-example that will become important is when we have two Grassmann coordinates i.e. $N = 2$ with an action

$$S = \int dt\, i(\eta_1\dot{\eta}_1 + \eta_2\dot{\eta}_2) = \int dt\, i\chi\dot{\eta} \tag{1.459}$$

where $\eta \equiv \eta_1 + i\eta_2$ and $\chi = \eta_1 - i\eta_2$. The corresponding Dirac brackets are

$$\{\eta, \chi\}_D = -i \quad \text{and} \quad \{\eta, \eta\}_D = \{\chi, \chi\}_D = 0 \tag{1.460}$$

which is consistent with $\{\eta_a, \eta_b\} = -i\delta_{ab}/2$ with $a, b = 1, 2$. To quantize this system (again, we are still dealing with just QM here and at first the canonical approach to get some sense

of what is going on), we can promote η_α to an ordinary linear operator $\hat{\eta}_\alpha$ that acts on an Hilbert space $\mathcal{H}$ such that

$$\{\hat{\eta}_\alpha, \hat{\eta}_\beta\} = i\hbar(M^{-1})_{\alpha\beta}/2. \tag{1.461}$$

As an example, let us consider an $N = 2$ system with Lagrangian and corresponding Hamiltonian

$$L = i\sum_{\alpha=1}^{2} \eta_\alpha \dot{\eta}_\alpha - i\omega\eta_1\eta_@ \tag{1.462}$$

$$H = \sum_{\alpha=1}^{2} \pi_\alpha \dot{\eta}_\alpha - L = i\omega_{\eta_1\eta_2}. \tag{1.463}$$

Under quantization with $\eta_\alpha \rightsquigarrow \hat{\eta}_\alpha$, the square of these new operators is actually not what we would expect, for example,

$$\{\hat{\eta}_1, \hat{\eta}_1\} = 2\hat{\eta}_1^2 = i\hbar(M^{-1})_{11}/2 \implies \hat{\eta}_1^2 = \frac{1}{4} \tag{1.464}$$

and similar for $\hat{\eta}_2^2 = 1/4$ which is wild! We also have that $\{\hat{\eta}_1, \hat{\eta}_2\} = 0$ and thus they anticommute (so if you switch their placements, they pick up a minus sign). We can pick a minimal choice for the dimension of our Hilbert space $\dim \mathcal{H} = 2$ such that

$$\hat{\eta}_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \hat{\eta}_2 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{1.465}$$

which givesa Hamiltonian of the following form

$$\hat{H} = i\omega\hat{\eta}_1\hat{\eta}_2 = \begin{pmatrix} -\frac{\omega}{4} & 0 \\ 0 & \frac{\omega}{4} \end{pmatrix} \tag{1.466}$$

giving us a 2-state system.

Now, moving onto the path integral quantization of our Grassmann (fermionic) system, let us consider the same Lagrangian that we worked with in the canonical quantization as we did prior

$$L = i\chi\dot{\eta} - H(\chi, \eta). \tag{1.467}$$

where we recall that the hatted-operators obey $\{\hat{\eta}, \hat{\chi}\} = 1$ and $\hat{\eta}^2 = \hat{\chi}^2 = 0$ where we can think of $\hat{\eta}$ as fermion annihilation and $\hat{\chi}$ and fermion creation, or in other words, these operators obey the following map:

$$\begin{tikzcd} |0\rangle & |0\rangle \arrow[l, "\hat{\eta}"'] \arrow[r, bend left, "\hat{\chi}"] & |1\rangle \arrow[l, bend left, "\hat{\eta}"] \arrow[r, "\hat{\chi}"] & |0\rangle \end{tikzcd} \tag{1.468}$$

We can define a "position eigenstate" as $|\eta\rangle = |1\rangle\,\eta + |0\rangle$ where η is a formal Grassmann parameter. The Grassmann operator then on this position eigenstate then becomes

$$\hat{\eta}\,\langle\eta| = |0\rangle\,\eta = |\eta\rangle\,\eta. \tag{1.469}$$

Let us also define a corresponding bra-state as

$$\langle\langle\chi| = -\langle 1| + \chi\langle 0| \tag{1.470}$$

which obeys the following property

$$\langle\langle\chi|\eta\rangle = \chi - \eta \tag{1.471}$$

and

$$\langle\langle\chi\hat{\eta} = \chi\langle 1| = -\chi\langle\langle\chi|. \tag{1.472}$$

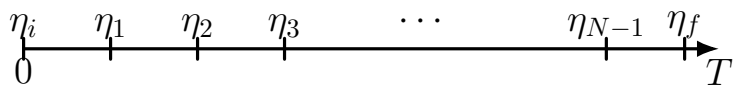
We shall now define a new *class* of integrals called *Berezin integrals* that are an integral over Grassmann variables (coordinates for now) $\int d\eta\, f(\eta)$ which have two defining features:

$$\int d\eta\, \eta = 1 \quad \text{and} \quad \int d\eta\, 1 = 0. \tag{1.473}$$

And by convention, since they are anti-commuting after all, $\eta d\eta = -d\eta\, \eta$. IT then follows that the proper insertion of a $\mathbb{1}$ for these integrals is the following

$$\int |\eta\rangle\, d\eta \langle\langle\eta| = \int (|1\rangle\, \eta + |0\rangle) d\eta (-\langle 1| + \eta\langle 0|) = |1\rangle\langle 1| + |0\rangle\langle 0| = \mathbb{1} \tag{1.474}$$

which you should think is the analog to $\int dx\ |x\rangle\langle x| = 1$. Now, as with the QM path integral derivation we did in 1.1, we can consider a discrete time evolution over time T of the Grassmann coordinates



The expectation value then of a unitary operator is given by

$$\langle\langle\eta_f|U(T)|\eta_i\rangle = \int \langle\langle\eta_f|e^{-i\hat{H}\frac{T}{N}}|\eta_{N-1}\rangle\, d\eta_{N-1}\langle\langle\eta_{N-1}|\cdots|\eta_1\rangle\, d\eta_1\langle\langle\eta_1|e^{-i\hat{H}\frac{T}{N}}|\eta_i\rangle\,. \tag{1.475}$$

Now, we can use the following two handy simplifications to reduce (1.475)

$$\langle\langle\eta_{n+1}|\eta_n\rangle = \eta_{n+1} - \eta_n = \int d\chi_n\, e^{-\chi_n(\eta_{n+1}-\eta_n)} \tag{1.476}$$

$$\langle\langle\eta_{n+1}|\hat{\chi}|\eta_n\rangle = (-\langle 1| + \eta_{n+1}\langle 0|)|1\rangle = -1 = -\int d\chi_n\, \chi_n e^{-\chi_n(\eta_{n+1}-\eta_n)}. \tag{1.477}$$

Then, with these two equations, the exponentials above that are sandwiched by η_{n+1} and η_n states in (1.475) can be approximated by

$$\langle\langle\eta_{n+1}|e^{-i\hat{H}\frac{T}{N}}|\eta_n\rangle \approx -\int d\chi_n\, e^{-iH(\chi_n,\eta_n)\frac{T}{N}-\chi_n(\eta_{n+1}-\eta_n)}. \tag{1.478}$$

Substituting this into (1.475) we find the following

$$\begin{aligned}\langle\langle\eta_f \equiv \eta_N|U(T)\,|\eta_i \equiv \eta_0\rangle &= -\int d\chi_0 \prod_{n=1}^{N-1} d\eta_n d\chi_n\, e^{\sum_{n=0}^{N-1}\left(-iH(\chi_n,\eta_n)\frac{T}{N}-\chi_n(\eta_{n+1}-\eta_n)\right)} \\ &\rightsquigarrow \int D\eta\, D\chi\, e^{\int_0^T dt(-\chi\dot{\eta}-iH(\chi,\eta))} \\ &= \int D\eta\, D\chi\, e^{iS} \end{aligned} \tag{1.479}$$

where the action here is

$$S = \int dt\, L = \int dt\,(i\chi\dot{\eta} - H(\chi,\eta)) \tag{1.480}$$

where equation (1.479) is formally known as the *Grassmann (Berezin) path integral*. And now of course after seeing the QM version of the anti-commuting path integral quantization, we are now ready to move onto the path integral of fermionic spinor *field* path integral quantization to which, since the steps are so similar in the sense that you combine the field-theory steps we took for quantizing the scalar and the anti-commuting subtleties we just encountered in the QM case, we wont re-derive the fermionic path integral and just state it. A full derivation is of course available in Weinberg's textbook [1]. With the fermionic path integral, we will take the same steps as we did with the scalar case: derive the (time-ordered Feynman) propagator, see how to compute the 1-particle state of a fermion, derive the Feynman rules from the LSZ reduction formula (1.324), and then finish up with fermions by computing a similar toy model to ϕ^4 theory that we saw prior.

1.3.4 Path Integral for Fermions

Recall that we have "two" fermionic spinor field variables, $\psi_\alpha(x)$ and $(\psi^\dagger)^\alpha(x)$. We will define the "bar"-version of the conjugate of the fermionic spinor as (which will become out *anti-fermion*)

$$\overline{\psi}^\alpha \equiv (\psi^\dagger)^{\alpha'}\beta_{\alpha'}{}^\alpha \tag{1.481}$$

where $\beta = i\gamma^0$ as before, and the Gamma matrices transform as $(\gamma^\mu)^\dagger = -\beta\gamma^\mu\beta$. Then, under quantization, they of course become operators $\hat{\psi}_\alpha(x)$ and $\hat{\overline{\psi}}^{\alpha'}(x)$. We also expect the fermionic spinors to obey the equal time commutator (1.431)

$$\begin{aligned}\{\hat{\psi}_\alpha(x), \hat{\overline{\psi}}^{\alpha'}(y)\}\Big|_{y^0=x^0} &= (-\partial\!\!\!/_x + m)_\alpha{}^{\alpha'}\Big(\Delta_+(x-y;m^2) - \Delta_+(y-x;m^2)\Big)\Big|_{y^0=x^0} \\ &= -(\gamma^0)_\alpha{}^{\alpha'}\frac{\partial}{\partial x^0}\Big(\Delta_+(x-y;m^2) - \Delta_+(y-x;m^2)\Big)\Big|_{y^0=x^0} \\ &= -(\gamma^0)_\alpha{}^{\alpha'}\Big(-i\delta^3(\vec{x}-\vec{y})\Big) \\ &= i(\gamma^0)_\alpha{}^{\alpha'} \end{aligned} \tag{1.482}$$

while the rest vanish,

$$\{\hat{\psi}(x), \hat{\psi}(y)\}\Big|_{y_0=x_0} = \{\hat{\psi}^\dagger, \hat{\psi}^\dagger(y)\}\Big|_{y^0=x^0} = 0. \tag{1.483}$$

We also have that the conjugate momentum for a fermionic spinor is $\pi_\psi(x) = -\overline{\psi}(x)\gamma^0$, and that the equations of motion are given by the Dirac equation (1.425), $(\partial\!\!\!/ + m)\psi(x) = 0$. The corresponding Lagrangian density that gives rise to the Dirac equationvia the Euler-Lagrange equation is (and written in it's full spinor-index for clarity)

$$\begin{aligned}\mathcal{L} &= -\overline{\psi}(x)(\partial\!\!\!/ + m)\psi(x) \\ &= -\overline{\psi}^\alpha(x)\left((\gamma^\mu)_\alpha{}^{\alpha'}\partial_\mu + m\delta_\alpha{}^{\alpha'}\right)\psi_{\alpha'}(x).\end{aligned} \tag{1.484}$$

The corresponding *Lorentzian* path integral is

$$Z = \int D\psi D\overline{\psi}\, e^{i\int d^4x\,\mathcal{L}}. \tag{1.485}$$

There is only a single combination of the fermionic spinor fields (up to a sign of course) that does not vanish. Namely, the following two correlation functions (or propagators) vanish

$$\langle\Omega|\, T\hat{\psi}_\alpha(x)\hat{\psi}_\beta(y)\,|\Omega\rangle = \langle\Omega|\, T\hat{\overline{\psi}}^{\alpha'}(x)\hat{\overline{\psi}}^{\beta'}(y)\,|\Omega\rangle = 0 \tag{1.486}$$

as a consequence of the anti-commutation relation (1.482). The only pair of fermionic fields that do not vanish gives the Dirac (Feynman) propagator/Greens function,

$$\begin{aligned}\langle\Omega|\, T\hat{\psi}_\alpha\hat{\overline{\psi}}^\beta(y)\,|\Omega\rangle &= (-\partial\!\!\!/_x + m)_\alpha{}^\beta\Delta_F(x-y;m^2) \\ &= \int\frac{d^4k}{(2\pi)^4}e^{ik\cdot(x-y)}\frac{-i(-ik\!\!\!/_\alpha{}^\beta + m\delta_\alpha^\beta)}{k^2+m^2-i\epsilon}.\end{aligned} \tag{1.487}$$

If we now compare with the "Gaussian" Berezin integral form with partition function

$$Z = \int\prod_a d\overline{\eta}_a d\eta\, e^{-\overline{\eta}^T\cdot A\cdot\eta} = \det\, A, \tag{1.488}$$

then the 2-point correlation function of Grassmann variables is

$$\begin{aligned}\langle\eta_a\overline{\eta}^b\rangle &= \frac{1}{Z}\int\prod d\overline{\eta}d\eta\, e^{-\overline{\eta}^T A\eta}\eta_a\overline{\eta}^b \\ &= \frac{1}{Z}\int\prod d\overline{\eta}d\eta\,\eta_a(A^{-1})_c{}^b\frac{\partial}{\partial\eta_c}e^{-\overline{\eta}^T\eta} \\ &= \frac{1}{Z}\int\prod d\overline{\eta}d\eta\,(A^{-1})_a{}^b e^{-\overline{\eta}^T A\eta} \\ &= (A^{-1})_a{}^b\end{aligned} \tag{1.489}$$

where in the second line, we used the derivative trick just as we did prior with scalars, and then in the third line we integrated by parts which left us with a spinor-index matrix valued object that was independent of the measure, and thus cancelled with the Z^{-1} infront. For higher point fermionic correlation functions with respect to Gaussian functional Berezin integrals, luckily for us, can still be computed via Wick contractions *provided* that we take into account the appropriate signs when moving Grassmann-variables. For instance,

$$\langle\cdots\overbrace{\eta_a\cdots\overline{\eta}^b}\cdots\rangle = (-1)^N\langle\eta_a\overline{\eta}^b\rangle\langle\cdots\rangle \tag{1.490}$$

where the $\cdots$ in (1.490) represents an N-number of η's or $\overline{\eta}$'s. More generally, in any QFT, a fermionic spinor field operator $\hat{\psi}_\alpha(x)$ obeys the following unitary-Poincaré transformation

$$U(\Lambda,a)\hat{\psi}_\alpha(x)(U(\Lambda,a))^{-1} = (R(\Lambda))_\alpha{}^\beta\hat{\psi}_\beta(\Lambda x+a) \tag{1.491}$$

which still have all of their normal infinitesimal forms we saw above, $\Lambda^\mu{}_\nu + \delta^\mu{}_\nu + \omega^\mu{}_\nu$, $R(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}\right)$, and $S^{\mu\nu} = -\frac{i}{4}[\gamma^\mu,\gamma^\nu]$. We shall now assume that there exists a U(1) *global* symmetry attributed to the fermions with unitary operator $U_\theta = e^{i\theta\hat{Q}}$ where $\hat{Q}$ is a Hermitian *charge* operator. Under this U(1) symmetry, the fermionic spinors transform as

$$U_\theta\hat{\psi}_\alpha(x)U_\theta^{-1} = e^{i\theta}\hat{\psi}_\alpha(x) \tag{1.492}$$

$$U_\theta\hat{\overline{\psi}}^\alpha(x)U_\theta^{-1} = e^{-i\theta}\hat{\overline{\psi}}^\alpha(x) \tag{1.493}$$

or written differently in terms of commutators, the charge-symmetry here generates the following

$$\left[\hat{Q},\hat{\psi}_\alpha(x)\right] = \hat{\psi}_\alpha(x) \tag{1.494}$$

$$\left[\hat{Q},\hat{\overline{\psi}}^\alpha(x)\right] = -\hat{\overline{\psi}}^\alpha(x). \tag{1.495}$$

We shall now assume that $\hat{Q}\left|\Omega\right\rangle = 0$, namely that the vacuum is chargeless. We shall now also state that the fermionic-spinor states take on 3-values $\left|\vec{k},\sigma,q=\pm1\right\rangle$ where q is the eigenvalue of $\hat{Q}$,

$$\hat{Q}\left|\vec{k},\sigma,q\right\rangle = q\left|\vec{k},\sigma,q\right\rangle. \tag{1.496}$$

Then, we can compute the 1-particle state, which implied by Lorentz symmetry, becomes

$$\left\langle\vec{k},\sigma,+\right|\hat{\psi}_\alpha(x)\left|\Omega\right\rangle = \frac{e^{-ik\cdot z}}{(2\pi)^{3/2}}\left(\mathcal{Z}_v^{1/2}\mathcal{V}_\alpha^\sigma(\vec{k}) + \mathcal{Z}_u^{1/2}\mathcal{U}_\alpha^\sigma(\vec{k})\right). \tag{1.497}$$

It should be noted that the combination $(e^{i\zeta\gamma_5}\hat{\psi})_\alpha(x)$ has the *same* Lorentz transformation property as $\hat{\psi}_\alpha(x)$ with $\mathcal{U}^\sigma \leftrightarrow \mathcal{V}^\sigma$ under a multiple of γ_5. This implies that we can always re-define $\hat{\psi}(x) \rightsquigarrow e^{i\zeta\gamma_5}\hat{\psi}(x)$ that allows us to set (1.497) to a simplified form

$$\left\langle\vec{k},\sigma,+\right|\hat{\psi}_\alpha(x)\left|\Omega\right\rangle = \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}}e^{-ik\cdot x}\mathcal{V}_\alpha^\sigma(\vec{k}) \tag{1.498}$$

with $Z_\psi = 1$ agreeing with our free fermion field convention. To find the 1-particle state but for $q=-1$, we can utilize microcausality in the sense that

$$\left\langle\Omega\right|\left\{\hat{\psi}_\alpha(x),\hat{\overline{\psi}}^\beta(y)\right\}\left|\Omega\right\rangle = 0 \tag{1.499}$$

which is true for spacelike separated points $(x-y)^2 > 0$, which then implies the following

$$\left\langle\vec{k},\sigma,-\right|\hat{\overline{\psi}}^\alpha(x)\left|\Omega\right\rangle = \frac{\mathcal{Z}_{\overline{\psi}}^{1/2}}{(2\pi)^{3/2}}e^{-ik\cdot x}\left(\overline{\mathcal{U}}^\sigma\right)^\alpha(\vec{k}). \tag{1.500}$$

The implication here arises from that since the anti-commutator vanishes on the vacuum, we can insert a 1-particle basis in-between the fermionic spinors and separate them. It should be noted that $|\mathcal{Z}_{\overline{\psi}}| = |\mathcal{Z}_\psi|$, which allows us to choose a phase convention of the 1-particle states to be such that $Z_\psi = Z_{\overline{\psi}} \geq 0$.

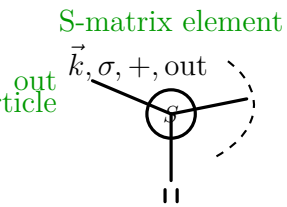
1.3.5 S-Matrix Elements for Fermions

Let us now consider the possible higher mass states to the 2-point correlation function to (1.487). To do so, there is a similar derivation to that of the Källén-Lehmann spectral representation that we found prior for scalar fields in (1.257), but for fermions it gives

$$\langle\Omega|\, T\hat{\psi}_\alpha(x)\hat{\overline{\psi}}^\beta(y)\,|\Omega\rangle = Z_\psi(-\partial\!\!\!/_x + m)_\alpha{}^\beta \Delta_F(x-y;m^2) + (\text{multi-particle contribution}) \tag{1.501}$$

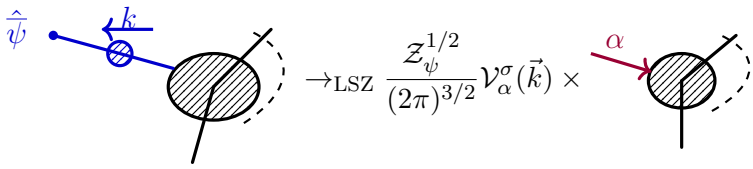
where m here is the *physical mass* of the spin-1/2 particle and the we should recall that the multi-particle contributions here do *not* affect the LSZ (mass) pole at $k^2 = -m^2$. Now, to relate the time-order Greens functions involving $\hat{\psi}_\alpha(x)$ or $\hat{\overline{\psi}}^\beta(x)$ to S-matrix elements involving in/out particle/anti-particle, we do this truncation by the LSZ reduction formula (1.324). We won't derive the fermionic LSZ reduction formula here, but it is in essence a truncation of the fermionic/Dirac propagators of the external legs of Feynman diagrams. Unlike the scalars though, we have **four** different possible external legs due to the particle/anti-particle nature of fermionic spinors. For general momentum space Greens function relation to S-matrix element, and we start with an out +-particle states,

$$\widetilde{G}(k,\cdots) = \int d^4x\, e^{-ik\cdot x}\langle\hat{\overline{\psi}}(x)\cdots\rangle \leftrightarrow$$



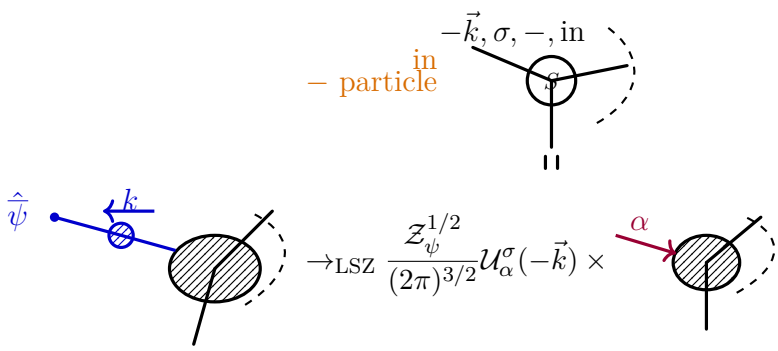
(1.502)

which has the following Feynman rule for the out anti-fermion



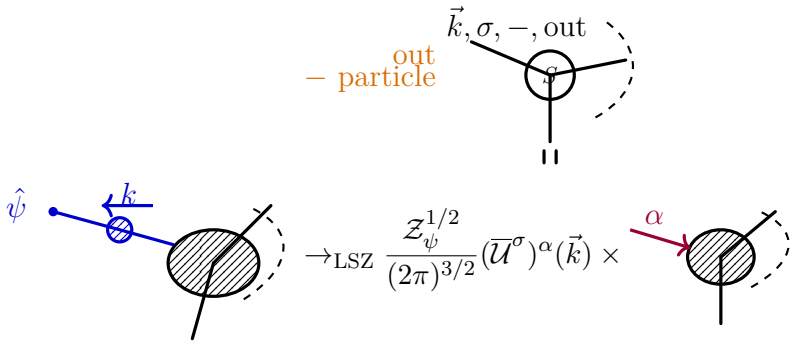
(1.503)

For a fermion going in, the Feynman diagram is



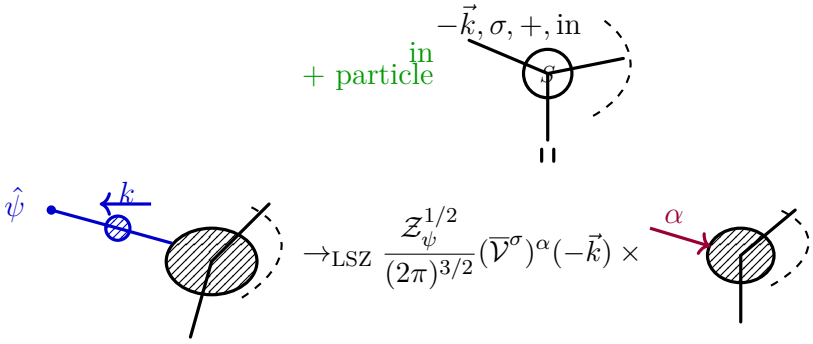
(1.504)

And similarly, the out fermion state is



$$\to_{\text{LSZ}} \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} (\overline{\mathcal{U}}^\sigma)^\alpha(\vec{k}) \times \tag{1.505}$$

while for a fermion $+$-charge fermion (the last one) is



$$\to_{\text{LSZ}} \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} (\overline{\mathcal{V}}^\sigma)^\alpha(-\vec{k}) \times \tag{1.506}$$

As an explicit example, we can consider a theory similar to ϕ^4 theory above namely

$$\mathcal{L} = -\overline{\psi}(\partial\!\!\!/ + m)\psi - \frac{g}{2}(\overline{\psi}\psi)^2. \tag{1.507}$$

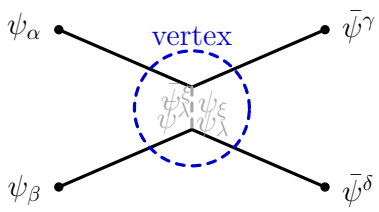
At order g, we have the following connected correlator

$$\langle \psi_\alpha(x_1)\psi_\beta(x_2)\overline{\psi}^\gamma(x_3)\overline{\psi}^\delta(x_4)\rangle^{\text{conn.}} \tag{1.508}$$

which is computed, when expanded in the path integral to order g, we have the following Wick contractions

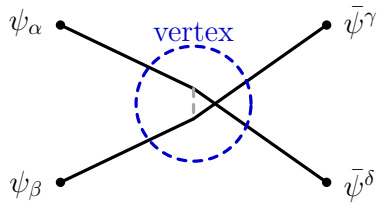
$$\psi_\alpha(x_1)\psi_\beta(x_2)\overline{\psi}^\gamma(x_3)\overline{\psi}^\delta(x_4)\left(-\frac{ig}{2}\right)\int d^4y\, \overline{\psi}^\xi(y)\psi_\xi(y)\overline{\psi}^\lambda(y)\psi_\lambda(y) \tag{1.509}$$

with three other possible contractions. They can be represented by the two diagrams:



$$= ig\int d^4y\, \langle \psi_\alpha\overline{\psi}^\xi\rangle\langle\psi_\xi\overline{\psi}^\gamma\rangle\langle\psi_\beta\overline{\psi}^\lambda\rangle\langle\psi_\lambda\overline{\psi}^\delta\rangle \tag{1.510}$$

where the change in sign is due to the order of ψ's. Also, we can imagine an *effective vertex*[40] that we drew in blue that contains the contractions with the interaction term in the Lagrangian. The other diagram is



$$= -ig \int d^4y \, \langle \psi_\alpha \overline{\psi}^\xi \rangle \langle \psi_\xi \overline{\psi}^\delta \rangle \langle \psi_\beta \overline{\psi}^\lambda \rangle \langle \psi_\lambda \overline{\psi}^\gamma \rangle. \tag{1.511}$$

After LSZ reduction, the S-matrix element, for example,

$$^{\text{out}}\langle \vec{k}_1, \sigma_1, -; \vec{k}_2, \sigma_2, -| - \vec{k}_3, \sigma_3, -; -\vec{k}_4, \sigma_4, - \rangle^{\text{in}}, \tag{1.512}$$

is computed at tree-level by stripping off the external propagators and attaching the spinor polarization factors, for example (take from the LSZ rules above)

$$\langle \psi_\alpha \overline{\psi}^\xi \rangle \to \frac{1}{(2\pi)^{3/2}} (\overline{\mathcal{U}}^{\sigma_1})^\xi (\vec{k}_1) \tag{1.513}$$

$$\langle \psi_\xi \overline{\psi}^\gamma \rangle \to \frac{1}{(2\pi)^{3/2}} \mathcal{U}_\xi^{\sigma_3} (-\vec{k}_3). \tag{1.514}$$

The completion of the computation of the amplitude is found in the problem set.

1.3.6 Wigner's Little Group for Massless Spinning Particles

Ok, after looking at first *massive* scalars and then *massive* fermions (spin-1/2), we now stop avoiding the possibility of *massless* particles for some general spin (from a fundamental group theoretic approach point of view). We will approach this as we have done before by working with a reference momentum, seeing how the Wigner matrices and elements behave for the given representation, and then see the consequences. However, in this case, we do not have a mass-shell condition, *well* we actually do, it is just that it is $k^2 = 0$. So, let us choose a reference momentum

$$k_R^\mu = (E, 0, 0, E) \tag{1.515}$$

which does obey the mass-shell condition $k_R^2 = E - E = 0$. The question becomes though, is there a Wigner element/matrix in $W \in \text{SO}(1,3)$ such that $W \cdot k_R = k_R$? Well, we need to work out then the components of W itself then. Recall that for a general Lorentz transformation $\Lambda^\mu{}_\nu$ can be written in terms of the angular-momentum generators $\Lambda = \exp\left(\frac{i}{2}\omega_{\mu\nu} M^{\mu\nu}\right)$ where $\omega_{\mu\nu}$ is the usual anti-symmetric infinitesimal Lorentz matrix and

$$(M^{\mu\nu})^\rho{}_\sigma = -i \left(\eta^{\mu\rho} \delta^\nu{}_\sigma - \eta^{\nu\rho} \delta^\mu{}_\sigma\right) \tag{1.516}$$

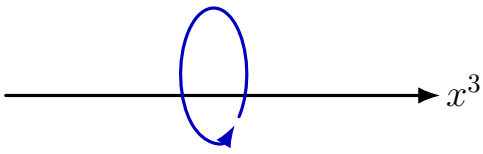
[40] If you have taken a particle physics course before taking QFT 1, you may actually recognize this Lagrangian as Fermi's 4-fermion theory to *effectively* describe beta-decay. Hence, we give a preview of this type of effective vertex before QFT 2.

which obeys the following commutation algebra with it's corresponding operator form $\hat{J}^{\mu\nu}$

$$[M^{\mu\nu}, M^{\rho\sigma}] = -i\left(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - (\rho \leftrightarrow \sigma)\right) \tag{1.517}$$

$$\left[\hat{J}^{\mu\nu}, \hat{J}^{\rho\sigma}\right] = -i\left(\eta^{\nu\rho}\hat{J}^{\mu\sigma} - \eta^{\mu\rho}\hat{J}^{\nu\sigma} - (\rho \leftrightarrow \sigma)\right). \tag{1.518}$$

Also recall that the unitary Lorentz transformation is $U(\Lambda) = \exp\left(\frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu}\right)$. To analyze the general spin-j case, we can consider aligning the spin around x^3 axis



and then take the angular momentum matrix to be

$$(M_{12})^{\mu}{}_{\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.519}$$

which turns the Wigner matrix into

$$W = e^{i\theta M_{12}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.520}$$

We could also construct another Wigner matrix of the form

$$W = e^{i\alpha A + i\beta B} = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix} \tag{1.521}$$

where $\zeta = (\alpha^2 + \beta)^2/2$ and A and B are constructed out of the generators of the Lorentz symmetry

$$A = M_{01} + M_{31} = -i \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \tag{1.522}$$

and

$$B = M_{02} + M_{32} = -i \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.523}$$

A general massless little group element takes the form of

$$W(\alpha,\beta,\theta) = e^{i\alpha A + i\beta B} e^{i\theta M_{12}}. \tag{1.524}$$

When we promote the generators to operators in the quantum theory, then A and B are also promoted to $\hat{A} = \hat{J}_{01} + \hat{J}_{31}$ and $\hat{B} = \hat{J}_{02} + \hat{J}_{32}$ where, since A and B commuted previously, their quantum commutator also commutes $[\hat{A}, \hat{B}] = 0$.

Within the subspace of 1-particle states that are carrying momentum $\vec{k}_R$, there exists a way to simultaneously diagonalize $\hat{A}$ and $\hat{B}$ with the basis of states $\left|\vec{k}_R, a, b\right\rangle$ where a and b are the eigenvalues with respect to $\hat{A}$ and $\hat{B}$. Furthermore, let us notice the following commutator relations

$$[M_{12}, A] = iB \implies [\hat{J}_{12}, \hat{A}] = i\hat{B} \tag{1.525}$$

$$[M_{12}, B] = -iA \implies [\hat{J}_{12}, \hat{B}] = -i\hat{A}, \tag{1.526}$$

then we can determine how $\hat{J}_{12}$ acts on the 1-particle state. It is not obvious, but the angular momentum operator will act on the 1-particle state as $ib\partial_a - ia\partial_b +$ non-derivative, which rotates the values of (a, b). This implies then the unitary operator $U(\Lambda)$ acts on the 1-particle state as

$$e^{i\theta\hat{J}_{12}} \left|\vec{k}_R, a, b\right\rangle \propto \left|\vec{k}_R, a\cos\theta - -b\sin\theta, a\sin\theta + b\cos\theta\right\rangle. \tag{1.527}$$

Notice that we would have a continuous family of massless particles at a given momentum *unless* $a = b = 0$. In the case of $a \neq 0$ and $b \neq 0$, we actually have an entire family of massless spin-s particles all on top of each other giving rise to a highly degenerate state with no discernible particle interpretation. However, from now on, we will *assume* that $a = b = 0$, i.e., we assume that $\hat{A}$ and $\hat{B}$ annihilate the 1-particle state of reference momentum $k_R = (E, 0, 0, E)$. With loss of generality, we can assume that the $\hat{J}_3$ operator acts on the 1-particle state as

$$\hat{J}_3 \left|\vec{k}_R, h\right\rangle = h \left|\vec{k}_R, h\right\rangle \tag{1.528}$$

where h here denotes and defines the *helicity* of our massless spinning particle. In an equivalent manner, we can consider the exponential version of $\hat{J}_3$

$$e^{i\theta\hat{J}_3} \left|\vec{k}_R, h\right\rangle = e^{ih\theta} \left|\vec{k}_R, h\right\rangle \tag{1.529}$$

or in other words, the unitary operator with Wigner matrix W is

$$U(W(\alpha,\beta,\theta)) \left|\vec{k}_R, h\right\rangle = e^{ih\theta} \left|\vec{k}_R, h\right\rangle \tag{1.530}$$

which completely determines the Lorentz transformation of the 1-particle state

$$\left|\vec{k}, h\right\rangle = \sqrt{\frac{k_R^0}{k^0}} U(L(\vec{k})) \left|\vec{k}_R, h\right\rangle. \tag{1.531}$$

As we have seen prior for the case of spin-1/2 above with the little group, we shall demand invariance under 4π rotations. What this implies is that $h \in \mathbb{Z}/2$. And so from here,

we can see that we could analyze a whole host of massless spinning particles, but we will restrict our attention to the $h = \pm 1$ case which we shall denote as, *the photon*. Now, for the last time in these lecture notes (I think...) we will ask the question, "what kind of field operator can create a 1-photon state?"

Well, if the scalar and fermion case taught us anything, we can guess that the 1-photon case (or 1-particle photon case) should have a field operator of the form $\hat{\Phi}_\mu(x)$. Then, it will transform under unitary Lorentz transformations as

$$U(\Lambda)\hat{\Phi}_\mu(x)(U(\Lambda))^{-1} = \Lambda^\nu{}_\mu \hat{\Phi}_\nu(\Lambda \cdot x). \tag{1.532}$$

As with the spin-1/2 case, we can construct a 1-particle photon state up to some constant C and then insert some unitary matrices to solve for the resulting function on the RHS which equates to the following

$$\left\langle \vec{k}, h \right| \hat{\Phi}_\mu(0) \left| \Omega \right\rangle = C\mathcal{U}^g_\mu(\vec{k}), \tag{1.533}$$

and then we insert two factors of the identity in terms of the unitary-Lorentz matrices

$$\left\langle \vec{k}, h \right| (U(\Lambda))^\dagger U(\Lambda)\hat{\Phi}_\mu(0)(U(\Lambda))^{-1} \left| \Omega \right\rangle = \sqrt{\frac{(\Lambda k)^0}{k^0}}(D(W))^* \Lambda^\nu{}_\mu \left\langle \vec{\Lambda k}, h \right| \hat{\Phi}_\nu(0) \left| \Omega \right\rangle \tag{1.534}$$

which allows us to compare and read off how the function $\mathcal{U}^h_\mu(\vec{k})$ transforms as

$$\mathcal{U}^h_\mu(\vec{k}) = \sqrt{\frac{(\Lambda k)^0}{k^0}}(D(W))^* \Lambda^\nu{}_\mu \mathcal{U}^h_\nu(\vec{\Lambda k}). \tag{1.535}$$

Using equation (1.534), we can read off what $D(W)$ is namely

$$D(W(\alpha, \beta, \theta)) = e^{ih\theta} \tag{1.536}$$

which combined with, if we pick a reference frame momentum such that $\vec{k} = \vec{k}_R$, then $\Lambda = W(\alpha, \beta, \theta)$, and then (1.535) becomes

$$\mathcal{U}^h_\mu(\vec{k}_R) = e^{-ih\theta}(W(\alpha, \beta, \theta))^\nu{}_\mu \mathcal{U}^h_\nu(\vec{k}_R). \tag{1.537}$$

If we pick the particular case where $\alpha = \beta = 0$, then the Wigner matrix becomes

$$W(\alpha, \beta, \theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.538}$$

to which yes, we can recognize as a rotation matrix about the z-axis. Now, for $h = \pm 1$ helicity, we can write down the following $\mathcal{U}^h_\mu$ representations (given W above and up to some normalization that's not important)

$$\mathcal{U}^{+1}(\vec{k}_R) \propto \begin{pmatrix} 0 \\ 1 \\ -i \\ 0 \end{pmatrix} \quad \text{and} \quad \mathcal{U}^{-1}(\vec{k}_R) \propto \begin{pmatrix} 0 \\ 1 \\ i \\ 0 \end{pmatrix} \tag{1.539}$$

to which these are commonly called the *polarization vectors of right/left-handed photon.*

I want to take a break and just recognize the awe-inspiring derivations we have performed using Wigner's representation of particles. With the scalar case, everything was fairly straightforward in what we were trying to do: find a field representation that transforms as a scalar on the 1-particle-ket state. For spin-1/2, doing the exact same thing lead us to particle/anti-particle representations as well as the Dirac equation as a *natural* consequence. And now, with spin-1, we were able to find out everything on the massless spin-1 particle (abelian and a photon) which just so happened to give us the polarization vectors and two states of the photon. But, here is the caveat now with the spin-1 case...

Let us consider the case of $\theta = 0$ in (1.535) and $\alpha\,\beta \neq 0$. Then, we get

$$\mathcal{U}^h_\mu(\vec{k}_R) = (W(\alpha,\beta,0))^\nu{}_\mu \mathcal{U}^h_\nu(\vec{k}_R) \tag{1.540}$$

where the Wigner matrix now is, with $\zeta = (\alpha^2+\beta^2)/2$,

$$W(\alpha,\beta,0) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}. \tag{1.541}$$

But now, if we did the matrix algebra with the polarization vectors above in (1.539) we get

$$\left((W(\alpha,\beta,0))^\nu{}_\mu \mathcal{U}^{\pm 1}_\nu(\vec{k}_R)\right) \propto \begin{pmatrix} \alpha \mp i\beta \\ 1 \\ \mp i \\ -\alpha \pm i\beta \end{pmatrix} \tag{1.542}$$

which **does not** equate to $\mathcal{U}^{\pm 1}_\mu(\vec{k}_R)$! So what is going on? The conclusion is that the operators $\hat{\Phi}_\mu(x)$ with the property that $\hat{\Phi}_\mu(x)\,|\Omega\rangle$ overlapping with the 1-photon state **does not exist**! But then, wait, can we never generate a single-photon state? Well, hang on.

Note however that the following combination

$$(W(\alpha,\beta,0))^\nu{}_\mu \mathcal{U}^h_\nu(k_R) - \mathcal{U}^h_\mu(k_R) \propto k_{R\mu} \tag{1.543}$$

could have the following interpretation. While the operator $\hat{\Phi}_\mu(x)$ does not exists, $\partial_\mu\hat{\Phi}_\nu(x) - \partial_\nu\hat{\Phi}_\mu(x)$ *does* exist and that when we try to generate a 1-particle state,

$$\left\langle \vec{k}, h \right| \left(\partial_\mu\hat{\Phi}_\nu(x) - \partial_\nu\hat{\Phi}_\mu(x)\right)|\Omega\rangle = Ce^{-ik\cdot x}\left(-ik_\mu\mathcal{U}^h_\nu + ik_\nu\mathcal{U}^h_\mu\right) \tag{1.544}$$

is well-defined. Now, in the literature (and by literature, I mean the entirety of the physics literature... I think), we typically denote $\hat{\Phi}_\mu(x)$ by $\hat{A}_\mu(x)$ which may be more familiar. With (1.544) there is an inherent *gauge redundancy* namely

$$\hat{A}_\mu(x) \sim \hat{A}_\mu(x) + \partial_\mu\zeta(x) \tag{1.545}$$

and the *field strength* that we just saw is written as (and is a well-defined local field tensor)

$$\hat{F}_{\mu\nu}(x) = \partial_\mu\hat{A}_\nu(x) - \partial_\nu\hat{A}_\mu(x). \tag{1.546}$$

1.3.7 Maxwell's Equations and the Photon

With having essentially discovered the photon in a field theoretic manner, we are lucky enough to already know the corresponding action and won't have to re-derive that. In classical Maxwell theory (by classical I literally mean functions without hats...), it describes a vector potential $A_\mu(x)$ subject to the gauge redundancy given in (1.545) where $\zeta(x)$ is equal to an arbitrary function of *spacetime* coordinates. The corresponding Lagrangian density for Maxwell's theory is

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \\ &= \frac{1}{2}\sum_{i=1}^{3}(\partial_i A_i - \partial_i A_0)^2 - \frac{1}{4}\sum_{ij}^{3}(\partial_i A_j - \partial_j A_i)^2. \end{aligned} \tag{1.547}$$

Note that there are no terms that contain time derivatives of the time-component of the vector potential $\dot{A}_0$. As an a priori, we can view $A_\mu(t,\vec{x})$ at a given time t as generalized coordinates whose coorresponding conjugate canonical momentum densities are

$$\pi^\mu(\vec{x}) = \frac{\partial\mathcal{L}}{\partial(\partial_0 A_\mu)} \tag{1.548}$$

with the components being

$$\pi_i = \partial_0 A_i - \partial_i A_0 = F_{0i} \tag{1.549}$$

$$\pi^0 = 0 \tag{1.550}$$

where this last equation is called a *primary constraint*, and from the equations of motion

$$\begin{aligned} 0 &= -\partial_i\frac{\partial\mathcal{L}}{\partial(\partial_i A_0)} + \frac{\partial\mathcal{L}}{\partial A_0} \\ &= -\partial_i(\partial_i A_0 - \partial_0 A_i) \implies \end{aligned} \tag{1.551}$$

$$0 = \partial_i\pi_i(\vec{x}) \tag{1.552}$$

where this last equation is now called a *secondary constraint*. The constrained quantities $\pi_0(\vec{x})$ and $\partial_i\pi_i(\vec{x})$ have vanishing Poisson bracket amongst themselves and therefore, Dirac would call them *first-class constraints*. The first class constraints are closely tied to gauge redundancy, hence we have to care about them. Consider the following variation of $A_\mu(x)$ under a general gauge transformation similar to one's we have done before (for a general ε)

$$\begin{aligned}\delta A_\mu(\vec{x}) &\equiv \int d^3\vec{y}\left(\varepsilon(\vec{y})\{\pi^0(\vec{y}), A_\mu(\vec{x}\}_P + \widetilde{\varepsilon}(\vec{y})\{\partial_i\pi_i(\vec{y}), A_\mu(\vec{x})\}_P\right) \\ &= \varepsilon(\vec{x})\delta^0_\mu - \partial_i\widetilde{\varepsilon}(\vec{x})\delta_{i\mu} \\ &= \partial_\mu\zeta(z)\Big|_{x^0=t} \end{aligned} \tag{1.553}$$

where we have written $\zeta(t_0,\vec{x}) = -\widetilde{\varepsilon}(\vec{x})$ and $\partial_t\zeta(t_0,\vec{x}) = \varepsilon(\vec{x})$, which matches our expectation for the gauge redundancy (1.545).

To define the phase space of a gauge theory, we need to *remove* the first class constraints by *gauge-fixing* i.e., we need to impose constraints directly on generalized coordinates and so, in return, we change the first-class constraints into second class ones.

As an example, we will consider the *axial gauge* which is done by setting

$$A_3(x) = 0. \tag{1.554}$$

By choice, this will eliminate A_3 and π_3 as canonical variables which changes the Lagrangian as

$$\begin{aligned}\mathcal{L} = &-\frac{1}{2}\sum_{i=1}^{2}(\partial_0 A_i - \partial_i A_0)^2 - \frac{1}{2}(\partial_3 A_0)^2 \\ &-\frac{1}{2}(\partial_1 A_2 - \partial_2 A_)^2 - \frac{1}{2}\sum_{i=1}^{2}(\partial_3 A_i)^2.\end{aligned} \tag{1.555}$$

Now, if we compute the equations of motion via the Euler-Lagrange equations by varying A_0 we obtain

$$\sum_{i=1}^{2}\partial_i(\partial_i A_0 - \partial_0 A_i) + \partial_3^2 A_0 = 0 \tag{1.556}$$

or in other words

$$\sum_{i=1}^{2}\partial_i \pi + i + \partial_3^2 A_0 = 0, \tag{1.557}$$

which is a second class constraint. Note that now the Poisson bracket with the other first class constraint from prior, π_0, becomes

$$\{\pi_0(\vec{x}), \sum_{i=1}^{2}\partial_i \pi_i(\vec{y}) + \partial_3^2 A_0(\vec{y}\}_P = \partial_3^2 \delta^2(\vec{x} - \vec{y}) \neq 0. \tag{1.558}$$

And so, the constraints $\pi_0(\vec{x}) = 0$ and (1.557) have become[41] *second class constraints*! We can proceed by replacing the Poisson brackets with the Dirac brackets (1.456), or equivalently, we can solve for A_0 from the constraint (1.557)

$$\partial_3^2 A_0 = -\sum_{i=1}^{2}\partial_i \pi_i(\vec{x}), \tag{1.559}$$

or at least for Fourier modes of non-zero wave number in the x_3 direction is this method valid (of bypassing the Dirac brackets). When we solve for A_0 from the constraint (1.559), we can eliminate A_0 and π^0 all together as canonical coordinates. The conclusion then is that the phase space of Maxwell theory is parameterized by $A_i(\vec{x})$ and $\pi_i(\vec{x})$ but *only* for $i = 1, 2$ only *in* the axial gauge.

[41] The idea of constraints and classes on theories may still seem weird, but here, the idea is that if the conjugate momentum itself vanishes, then that is a primary class while derivatives there of are secondary constraints. These then generate a first class constraint with the property of vanishing Poisson brackets. However, if a primary with a secondary does not have vanishing Poisson brackets, then they are part of a second class constraint.

1.3.8 Path Integral for Abelian Gauge Field (photon)

After gauge-fixing Maxwell's theory, we can then proceed with canonical quantization, but, as you may have already predicted, we will pursue the path integral method for convenience of calculations. Formally, the path integral for a massless photon is

$$Z = \int [DA_\mu]\, e^{i\int d^4x\, \mathcal{L}} \tag{1.560}$$

which, as we have seen, has the following gauge redundancy $A_\mu \to A_\mu + \partial_\mu \zeta$. But notice that if we shift the path integral measure in (1.560), the action S is, albeit, invariant under the gauge transformation, but wouldn't the measure over count the possible field configurations? The space of possible configurations can be seen in the cartoon-esc plot below which we shall explain afterwards with a toy model.

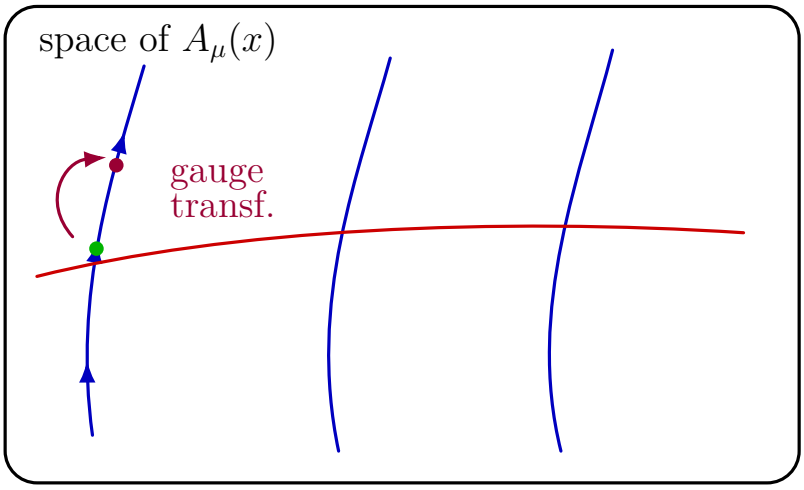


Figure 5. Here is a graph representing the entire space of possible configurations for $A_\mu(x)$. The blue lines are called *gauge orbits* and performing a gauge transformation of $A_\mu(x)$ moves it along the gauge orbit. However, the path integral measure integrates over *all* possible gauge orbits, hence we need to gauge fix otherwise we will be over counting. The gauge-fixing is choosing a *gauge-slice* which is the red curve seen here.

Consider the toy model with an "action" of the form

$$S(x, y) = (x - y)^2 \tag{1.561}$$

which has a corresponding "path integral" of

$$Z = \int dx\, dy\, e^{-S(x,y)}. \tag{1.562}$$

Notice that the action in (1.561) is invariant under the "gauge transformation"

$$(x, y) \to (x + \zeta, y + \zeta) \tag{1.563}$$

but the path integral measure in (1.562) is actually divergent under this *shift symmetry*. Let us consider a gauge fixing condition $\ell(x, y) = 0$ where $\ell(x, y)$ is some linear function

that fixes the gauge symmetry. Since $\ell(x,y)$ fixes the gauge, this corresponds to eliminating one of the variables in the action, say y for example, as we saw above with the Maxwell case. Integrating over dy then gives some constant N. The gauge fixed path integral is then

$$\begin{aligned} Z &= N \int dx\, e^{-S(x,y)|_{\ell(x,y)=0}} \\ &= N \int dx\, e^{-(x+ax+b)^2} \end{aligned} \tag{1.564}$$

where we chose $\ell(x,y) = y + ax + b = 0 \implies y = -(ax+b)$. But, we know that physical observabeles should be gauge invariant! So are we missing something or what isn't clear? Let us consider some "observable operator" $O(x,y)$ which is just some function. We assume that this observable is invariant under the gauge transformation i.e.

$$O(x+\zeta, y+\zeta) = O(x,y). \tag{1.565}$$

Then, the correlation function of such an observable via the path integral is

$$\begin{aligned} \langle O(x,y)\rangle &= \frac{\int dx\, e^{-S(x,y)|_{\ell=0}} O(x,y)|_{\ell=0}}{\int dx\, e^{-S(x,y)|_{\ell=0}}} \\ &= \frac{\int dx\, e^{-(x+ax+b)^2} \mathcal{O}(x,-ax+b)}{\int dx\, e^{-(x+ax+b)^2}} \\ &= \frac{\frac{1}{1+a}\int d\widetilde{x}\, e^{-\widetilde{x}^2} O(\widetilde{x},0)}{\frac{1}{1+a}\int d\widetilde{x}\, e^{-\widetilde{x}^2}} \end{aligned} \tag{1.566}$$

where in the second line, the operator is really $O(x,-ax-b) = O(x+ax+b,0)$, and in the third line, we defined $\widetilde{x} = x + ax + b$ which induces a Jacobian in the measures of each integral, which cancel nicely, as we find a result that is *independent* of a and b! To go slightly further, we can also use a weighted average of gauge conditions and not just a specific gauge. For example, we could re-write the path integral (1.562) as

$$Z = \int da\, db\, P(a,b) \int dx\, e^{-S(x,y)_{y=-ax-b}} \tag{1.567}$$

where $P(a,b)$ is some positive function. The expectation value of $O(x,y)$ is unaffected by this weighted average.

Let us now return to the Maxwell theory we were working with in (1.560). If we mimic the toy model we just used, then an example of a weighted average of gauge conditions is an insertion of the Dirac delta into our path integral,

$$Z = \int [DA_\mu]\, \delta(A_3(x)) e^{iS}. \tag{1.568}$$

Something we of course want out of this path integral is to compute the propagator for a photon, but notice that if we rewrite the action into the square-able form we need to

integrate we get

$$
\begin{aligned}
S &= -\frac{1}{4}\int d^4x\ \left(\partial_\mu A_\nu - \partial_\nu A_\mu\right)\left(\partial^\mu A^\nu - \partial^\nu A_\mu\right) \\
&= -\frac{1}{4}\int d^4x\ \left(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A_\mu - \partial_\nu A_\mu \partial^\mu A^\nu + \partial_\nu A_\mu \partial^\nu A^\mu\right) \\
&= \frac{1}{2}\int d^4x\ \left(A^\nu \partial_\mu \partial^\mu A_\nu - A_\mu \partial^\mu \partial^\nu A_\nu\right) \\
&= \frac{1}{2}\int d^4x\ A_\mu \left(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu\right) A_\nu \\
&= \frac{1}{2}\int d^4x\ A_\mu \mathbb{D}^{\mu\nu} A_\nu
\end{aligned}
\tag{1.569}
$$

where what we would want to be the inverse of the propagator here, $\mathbb{D}^{\mu\nu}$ is *not* invertible unless we fix the gauge. Working in the axial gauge where $A_3(x) = 0$, and so in (1.569) the only indices we are summing over are $\mu, \nu = 0, 1, 2$ but the box-operator still sums over all indices namely

$$
\Box = \sum_{\rho=0}^{3} \partial^\rho \partial_\rho. \tag{1.570}
$$

We can Fourier transform the vector potential as

$$
A_\mu(x) = \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot x} \widetilde{A}_\mu(x) \tag{1.571}
$$

which when we plug into the action (1.569) becomes

$$
S = \frac{1}{2}\int \frac{d^4k}{(2\pi)^4} \widetilde{A}_\mu(-k) \widetilde{\mathbb{D}}^{\mu\nu}(k) \widetilde{A}_\nu(k) \tag{1.572}
$$

where the matrix $\widetilde{\mathbb{D}}^{\mu\nu}(k) = -\eta^{\mu\nu}k^2 + k^\mu k^\nu$ which has the following inverse

$$
\left(\widetilde{\mathbb{D}}(k)\right)^{-1}_{\mu\nu} = -\frac{1}{k^2}\left(\eta_{\mu\nu} + \frac{k_\mu k_\nu}{k_3^2}\right). \tag{1.573}
$$

We can explicitly check this by computing the following

$$
\begin{aligned}
&\sum_{\rho-0}^{2} (-\eta_{\mu\rho}k^2 + k^\mu k^\rho)\left(-\frac{1}{k^2}\right)\left(\eta_{\rho\nu} + \frac{k_\rho k_\nu}{k_3^2}\right) \\
&\qquad = \delta^\mu_\nu - \frac{k^\mu k_\nu}{k^2} + \frac{k^\mu k_\nu}{k_3^2} - \frac{k^\mu k_\nu \sum_{\rho=0}^{2} k^\rho k_\rho}{k^2 k_3^2} \\
&\qquad = \delta^\mu_\nu.
\end{aligned}
\tag{1.574}
$$

We still need to take into account the $i\epsilon$ prescription for Lorentzian time-ordered Gree functions, from which we can then conclude that the photon propagator in the axial gauge is

$$
\langle\Omega| T\hat{A}_\mu(x)\hat{A}_\nu(y) |\Omega\rangle = \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} \frac{-i}{k^2 - i\epsilon}\left(\eta_{\mu\nu} + \frac{k_\mu k_\nu}{k_3^2}\right). \tag{1.575}
$$

Notice that this propagator is singular in $k_3 = 0$ which has to do with the axial gauge condition we chose, and so for modes that have $k_3 = 0$, this propagator is singular. This utter lack of manifest Lorentz invariance however is not a problem as the physically meaningful expectation values are those of *gauge-invariant* operators. For example, this is not the 2-point function of the vector potential but instead of the *field strength*

$$\langle\Omega| T\hat{F}_{\mu\nu}(x)\hat{F}_{\rho\sigma}(y)|\Omega\rangle = \int \frac{d^4k}{(2\pi)^4}\frac{-ie^{ik\cdot(x-y)}}{k^2-i\epsilon}\Bigg((ik_\mu)(-ik_\rho)\left(\eta_{\nu\sigma}+\frac{k_\nu k_\sigma}{k_3^2}\right) - (\mu\leftrightarrow\nu) - (\rho\leftrightarrow\sigma) + (\mu\leftrightarrow\nu,\rho\leftrightarrow\sigma)\Bigg) \tag{1.576}$$

where the terms multiplied with the Minkowski metric becomes (since we already know that terms with any number of momentum 4-vectors are Lorentz invariant)

$$\langle\Omega| T\hat{F}_{\mu\nu}(x)\hat{F}_{\rho\sigma}(y)|\Omega\rangle \supset \int \frac{d^4k}{(2\pi)^4}\frac{-ie^{ik\cdot(x-y)}}{k^2-i\epsilon}\left(k_\mu k_\rho\eta_{\nu\sigma} - k_\nu k_\rho\eta_{\mu\sigma} - k_\mu k_\sigma\eta_{\nu\rho} + k_\rho k_\sigma\eta_{\mu\nu}\right) \tag{1.577}$$

which is Lorentz invariant. It would be useful however to wk with a gauge condition that is manifestly Lorentz invariant. One such gauge is called the *Landau gauge*

$$\partial_\mu A^\mu(x) = 0. \tag{1.578}$$

To see how this arises, we can consider a generic gauge condition that sets $A^\mu(x)$ to some specified vector potential $A^\mu(x) = \mathcal{V}^\mu(x)$. Then, given the gauge transformation $A^\mu \to A^\mu + \partial^\mu\zeta$, we can choose ζ to be a value such that $= -\partial_\mu\mathcal{V}^\mu$ implying we can set $\partial_\mu A^\mu(x) = 0$, which fixes the gauge redundancy. We can go even further and generalize the Landau gauge (1.578) via

$$\partial_\mu A^\mu(x) = \varphi(x) \tag{1.579}$$

which can be realized in the *Euclidean* path integral by inserting an appropriate delta function

$$\delta(\partial_\mu A^\mu - \varphi) = \int [D\lambda]\, e^{i\int d^4x^E\, \lambda(x)(\partial_\mu A^\mu(x)-\varphi(x))}. \tag{1.580}$$

As with the toy model example, we want to average over the different possible values of $\varphi(x)$ with an appropriate Gaussian

$$\int [D\varphi]\, e^{-\frac{1}{2\zeta}\int d^4x^E\,(\varphi(x))^2} \tag{1.581}$$

where ξ is some constant that must obey $\xi > 0$ otherwise the Euclidean action has the wrong sign, and if it is zero, then there is a clear divergence *at this stage of the calculation* (give it a few more lines of physics). If we insert (1.581) into (1.580), then what this amounts to is the following

$$\begin{aligned}\int [D\varphi]\, e^{-\frac{1}{2\xi}\int d^4x^E\,\varphi^2(x)}\delta(\partial_\mu A^\mu - \varphi) &= \int [D\varphi D\lambda] e^{-\frac{1}{2\xi}\int d^4x^E\left(\varphi^2 + i\lambda(\partial_\mu A^\mu - \varphi)\right)} \\ &= \int [D\lambda]\, e^{-\frac{\xi}{2}\int\left(\lambda^2 + i\lambda\partial_\mu A^\mu\right)} \\ &= \mathcal{N}\, e^{-\frac{1}{2\xi}\int d^4x^E\,(\partial_\mu A^\mu)^2}\end{aligned} \tag{1.582}$$

where in the second line we integrated out ϕ, and then in the final line, we completed the square and integrated out λ giving us our final contribution to the Maxwell action for the Landau gauge. What this amounts to is a *deformation* of the Euclidean or Lorentzian path integral by including

$$\Delta S^E = \frac{1}{2\xi}\int d^4x^E\,(\partial_\mu A^\mu)^2 \tag{1.583}$$

$$\Delta S = -\frac{1}{2\xi}\int d^4x\,(\partial_\mu A^\mu)^2 \tag{1.584}$$

The final gauge-fixed form of the Maxwell action is then

$$\begin{aligned} S &= \int d^4x\,\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2\right) \\ &= \frac{1}{2}\int d^4x\,A_\mu\left(\eta^{\mu\nu}\Box - \left(1-\frac{1}{\xi}\right)\partial^\mu\partial^\nu\right)A_\nu \end{aligned} \tag{1.585}$$

where the term in the parenthesis is the new gauge-fixed propagator which we will denote as $\mathbb{D}_\xi^{\mu\nu}$ which is invertible to give us the propagator after a suitable Fourier transformation

$$\widetilde{\mathbb{D}}_\xi^{\mu\nu}(k) = -\eta^{\mu\nu}(k^2) + \left(1-\frac{1}{\xi}\right)k^\mu k^\nu \tag{1.586}$$

$$\left(\widetilde{\mathbb{D}}_\xi(k)\right)^{-1}_{\mu\nu} = -\frac{\eta^{\mu\nu}}{k^2} + (1-\xi)\frac{k^\mu k^\nu}{(k^2)^2}. \tag{1.587}$$

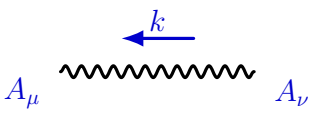
We can of course check to see if the inverse does in fact exist to which:

$$\begin{aligned} &\left(-\eta^{\mu\rho}k^2 + \left(1-\frac{1}{\xi}\right)k^\mu k^\rho\right)\left(\frac{\eta_{\rho\nu}}{k^2} + (1-\xi)\frac{k_\rho k_\nu}{(k^2)^2}\right) \\ &\qquad = \delta^\mu{}_\nu + \left(1-\frac{1}{\xi}\right)\frac{k^\mu k_\nu}{k^2} - (1-\xi)\frac{k^\mu k_\nu}{k^2} + \left(1-\frac{1}{\xi}\right)(1-\xi)\frac{k^\mu k_\nu}{k^2} \\ &\qquad = \delta^\mu{}_\nu \end{aligned} \tag{1.588}$$

the inverse is correct. Therefore, we can finally write down a manifestly Lorentz invariant time-ordered Greens function

$$\langle\Omega|\,T\hat{A}_\mu\hat{A}_\nu(y)\,|\Omega\rangle = \int\frac{d^4k}{(2\pi)^4}\frac{-ie^{ik\cdot(x-y)}}{k^2-i\epsilon}\left(\eta^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2-i\epsilon}\right). \tag{1.589}$$

There are two primary values for the gauge parameter ξ, if we set $\xi \to 0$, this is called the Landau gauge, while if we set $\xi = 1$, this is the "Feynman gauge," (the reason for the $\xi \to 0$ instead of just setting it to be is that if we set $\xi = 0$ at the level of the path integral or action, we get a divergence, while if set $\xi = 0$ now here at the propagator at the end of the derivation, all seems fine; this is just being cautious). We will be working with $\xi = 1$ since it gives the simlpest form of the photon propagator namely



$$= \frac{-i}{k^2-i\epsilon}\eta^{\mu\nu}. \tag{1.590}$$

The Maxwell theory of the photon is a free theory in the sense that there are no higher ordered vector potential terms beyond $\sim A^2$. $\hat{F}_{\mu\nu}|\Omega\rangle$ is a linear combination of 1-photon states in the sense that the 1-particle photon states are $\left|\vec{k}, h\right\rangle$ with $h = \pm 1$, and that

$$\left\langle \vec{k}, h \right| \hat{A}_\mu(x) |\Omega\rangle = \frac{e^{-ik\cdot x}}{\sqrt{(2\pi)^3 2|\vec{k}|}} \mathcal{Z}_A^{1/2} e_\mu^{h*}(\vec{k}) \tag{1.591}$$

where e_μ^h are the polarization vectors, and we can write the 1-particle states as a sum of the helicities/polarizations[42]. The polarization vector here is gauge-dependent

$$e_\mu^h \sim e_\mu^h + \alpha k_\mu. \tag{1.592}$$

Note that $\hat{A}_\mu(x)$ is *not* a well-defined local operator unlike $\hat{F}_{\mu\nu}(x)$. Indeed, the matrix element of the field strength operator

$$\left\langle \vec{k}, h \right| \hat{F}_{\mu\nu}(x) |\Omega\rangle = \frac{e^{-ik\cdot x}}{\sqrt{(2\pi)^3 2|\vec{k}|}} \mathcal{Z}_A^{\frac{1}{2}} \left(-ik_\mu e_\nu^{h*}(\vec{k}) + ik_\nu e_\mu^{h*}(\vec{k})\right) \tag{1.593}$$

which is well-defined and is invariant under the gauge transformation $e_\mu^h(\vec{k}) \to e_\mu^h(\vec{k}) + \alpha k_\mu$. We have already seen that at the reference momentum $k_E^\mu = (E, 0, 0, E)$, when we consider the rotation symmetry around x_3-axis will enforce (up to a normalization)

$$e^{\pm 1}(\vec{k}_R) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} \tag{1.594}$$

which is valid up to a shift by αk_R. In particular the rotation symmetry around x_3-axis (which again, can be traced back to Lorentz invariance via the Wigner matrices) demands the following condition $k_R^\mu e_\mu^h(\vec{k}_R) = 0$. For general light-like momentum with $k^\mu = (L(\vec{k}))^\mu{}_\nu k_R^\nu$, then the polarization vector transforms as $e_\mu^h(\vec{k}) = (L(\vec{k}))_\mu{}^\nu e_\nu^h(\vec{k}_R)$ which they must obey

$$k^\mu e_\mu^h(\vec{k}) = 0. \tag{1.595}$$

This condition is a manifestation of requiring Lorentz invariance for our massless gauge theory which induces transversality. The polarization vectors must also obey the following orthogonality condition

$$e_\mu^h(\vec{k}) e^{h'*\mu}(\vec{k}) = \delta_{hh'} \tag{1.596}$$

as well as the following completeness relation

$$\sum_{h=\pm 1} e_\mu^h(\vec{k}) e^{h*}{}_\nu(\vec{k}) = \eta_{\mu\nu} + k_\mu c_\nu + k_\nu c_\mu \tag{1.597}$$

for some c^μ that obeys $k \cdot c = -1$ and $c^2 = 0$. For example, we could pick $c^\mu = (2|\vec{k}|)^{-1}(k^0, \vec{k})$. We also need to make one final derivational-remark before we move onto

[42] Technically, polarization is a general spin in any direction (or shape) while helicity is a projection of the spin of the particle onto a particular axis in the direction of the momentum.

quantum electrodynamics which is determining the renormalization constant Z_A. Let us compare the (non-time ordered) 2-point function of the Maxwell field strength tensor to that of the Wightmann function obtained by analytic continuation. The first one is

$$\langle\Omega|\,\hat{F}_{\mu\nu}(x)\hat{F}_{\rho\sigma}(y)\,|\Omega\rangle = \sum_{h=\pm1}\int d^3\vec{k}\;\langle\Omega|\,\hat{F}_{\mu\nu}(x)\left|\vec{k},h\right\rangle\left\langle\vec{k},h\right|\hat{F}_{\rho\sigma}(y)\,|\Omega\rangle$$
$$= \sum_{h=\pm1}\int\frac{d^3\vec{k}}{(2\pi)^3 2|\vec{k}|}e^{ik\cdot(x-y)}\mathcal{Z}_A\left(k_\mu k_\rho e^h_\nu(\vec{k})e^{h*}_\sigma(\vec{k}) - (\mu\leftrightarrow\nu) - (\rho\leftrightarrow\sigma) + (\mu\leftrightarrow\nu,\rho\leftrightarrow\sigma)\right). \tag{1.598}$$

Now, if we compare this to the Wightman function obtained via analytic continuation from the time-ordered or Euclidean 2-point function of $\hat{F}_{\mu\nu}$

$$\langle\Omega|\,\hat{F}_{\mu\nu}(x)\hat{F}_{\rho\sigma}(y)\,|\Omega\rangle = \int\frac{d^3\vec{k}}{(2\pi)^3 2|\vec{k}|}e^{ik\cdot(x-y)}\Big(k_\mu k_\rho\eta_{\nu\sigma} - (\mu\leftrightarrow\nu) - (\rho\leftrightarrow\sigma) + (\mu\leftrightarrow\nu,\rho\leftrightarrow\sigma)\Big) \tag{1.599}$$

where we can see that these two 2-point functions can only be in agreement if $\mathcal{Z}_A = 1$ (up to renormalized corrections). A remark that should be made is that you may be wondering what actually goes wrong if we define $\hat{A}_\mu(x)$ via the insertion of $A_\mu(x)$ in the gauge-fixed form of the path integral, and consider the state $\hat{A}_\mu(x)\,|\Omega\rangle$. It would have to contain more than just the 1-particle photon state in order to reproduce the 2-point function $\langle\Omega|\,\hat{A}_\mu(x)\hat{A}_\nu(x)\,|\Omega\rangle$. In fact, $\hat{A}_\mu(x)\,|\Omega\rangle$ does contain *extra states* that are *unphysical*. A systematic treatment of the Hilbert space based on the gauged-fixed path integrals requires something called the *BRST formalism* which will be a topic of discussion in QFT 2.

1.3.9 Quantum Electrodynamics

Now, we finally move onto a *particular* QFT called *quantum electrodynamics* (QED) after all the work with fermionic spinors and now massless spin-1 particles, we will finally look at a QFT that is known to physically exist. QED is, from a field theory perspective, quite simply, it is the combination of Maxwell $A_\mu(x)$ and a charged fermion $\psi_\alpha(x)$ and $(\psi^\dagger)^\alpha(x)$. In this combination, the gauge redundancy of the massless spin-1 photon shows up in the unitary charge operator of the fermion

$$(A_\mu(x),\psi_\alpha(x)) \sim (A_\mu(x)+\partial_\mu\zeta(z), e^{iq\zeta(x)}\psi(x)) \tag{1.600}$$

where q here is the electric charge, which for the case of QED, $q=-e$ of which we will assign the name *electron* to the fermionic spinors. The corresponding Lagrangian density for QED is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \overline{\psi}(\gamma^\mu D_\mu + m)\psi \tag{1.601}$$

where the new derivative here is called a *covariant derivative* defined as[43]

$$D_\mu\psi \equiv \partial_\mu\psi - iqA_\mu\psi. \tag{1.602}$$

[43] I wanted to give a few clarifications to the name, "covariant derivative." For anyone who has taken a course on differential manifolds or general relativity, you may think that the covariant derivative is

Notice that under a gauge transformation,

$$D_\mu \psi \to \partial_\mu (e^{iq\zeta}\psi) - iq(A_\mu + \partial_\mu \zeta) e^{iq\zeta}\psi = e^{iq\zeta} D_\mu \psi, \tag{1.603}$$

and so notice in the action (1.601) that we get a factor of $e^{-iq\zeta}$ from the anti-particle, which cancels out with the $e^{iq\zeta}$ factor from the covariant derivative. Thus, $\mathcal{L}$ is gauge-invariant as needed.

We can carry out perturbation theory via the gauge-fixed path integral with the aforementioned deformation of the Maxwell equation (1.582)

$$Z = \int [DA_\mu D\psi]\, e^{i\int d^4x\, (\mathcal{L}+\Delta\mathcal{L})} \tag{1.604}$$

where $\Delta\mathcal{L} = -(\partial_\mu A^\mu)^2/2\xi$ as in pure Maxwell. As a cautionary note,$\hat{\psi}_\alpha(x)$ is *not* gauge-invariant and thus not a well-defined local operator. This may be fixed by instead by considering the operator

$$\hat{\psi}_\alpha(x) \to e^{iq\int_{C_x} \hat{A}_\mu(y) dy^\mu} \hat{\psi}_\alpha(x) \tag{1.605}$$

where this combination is gauge-invariant but *not local*. The contour here is called a *Wilson line* which starts at some spacetime point x and is continuous afterwards such as:

Consequently, correlation functions then of the fermionic-spinor operator in a QED theory such as $\langle\Omega|\, T\hat{\psi}_\alpha(x)\hat{\overline{\psi}}^\beta(y)\,|\Omega\rangle$ are not gauge-invariant and generally involve unphysical states (in the sense of BRST which will make more sense in QFT 2). However, a similar 2-point function built such as (1.605)

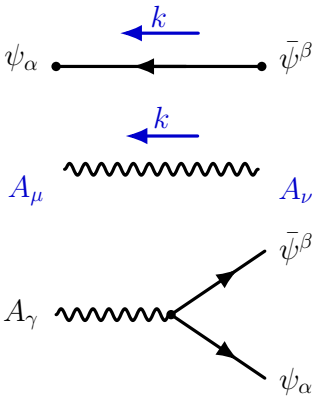
$$\langle\Omega|\, T\hat{\psi}_\alpha(x) e^{-iq\int_y^x \hat{A}_\mu(z) dz^\mu} \hat{\overline{\psi}}^\beta(y)\,|\Omega\rangle \tag{1.606}$$

is gauge-invariant and well-defined. Well, well-defined modulo possible regularization need to define the Wilson line operator itself (more about this in QFT 2; just a general remark for now). This issye is typically ignored in the formulation of scattering theory where one anticipates that the LSZ limit of the Greens function of gauge non-invariant operators nonetheless gives rise to physically meaningful S-matrix elements. One can in principle very the consistency of the resulting S-matrix elements such as unitarity and causality. This expectation is *almost* correct modulo the issue of *infrared (IR) divergences*.

$\nabla_\mu = \partial_\mu \pm \Gamma_{\mu\cdot}^{(\cdot)}$ acting on some tensor. Well, there is a larger class of covariant derivatives that involve *spin-connections* that are tangent bundles to a differentiable manifold. In our case here, the spin-connection is a U(1) abelian connection but the differential manifold is taken to be flat.

1.3.10 S-Matrix Elements (Feynman Rules) for QED

Nonetheless, the Feynman rules for QED coming from the path integral method using the QED Lagrangian in (1.601) are the following for the propagators and the 3-vertex between a photon, a fermion, and an anti-fermion:



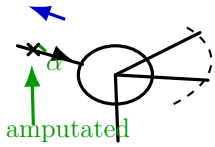
$$= \frac{-i(-i\not{k} + m)_\alpha{}^\beta}{k^2 + m^2 - i\epsilon} \tag{1.607}$$

$$= \frac{-i}{k^2 - i\epsilon}\left(\eta^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2 - i\epsilon}\right) \tag{1.608}$$

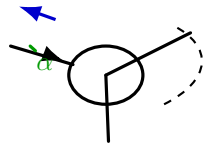
$$= -q(\gamma^\mu)_\beta{}^\alpha \tag{1.609}$$

where for a QED electron theory again we take $q = -e$. The in-out particle states for QED are the following:

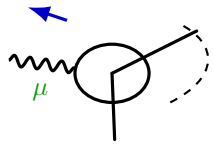
$$^{\text{out}}\left\langle e^-, \vec{k}, \sigma; \cdots \right| = \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} \overline{(\mathcal{U}^\sigma)}^\alpha(\vec{k}) \times \tag{1.610}$$



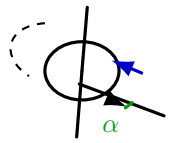
$$^{\text{out}}\left\langle e^+, \vec{k}, \sigma; \cdots \right| = \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} \mathcal{V}_\alpha^\sigma(\vec{k}) \times \tag{1.611}$$



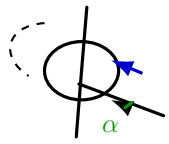
$$^{\text{out}}\left\langle \gamma, \vec{k}, h; \cdots \right| = \frac{\mathcal{Z}_A^{1/2}}{(2\pi)^{3/2}\sqrt{2|\vec{k}|}} e_\mu^{h*}(k) \times \tag{1.612}$$



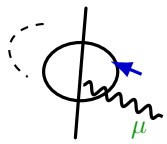
$$\left| \cdot, e^-, \vec{k}, \sigma \right\rangle^{\text{in}} = \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} \mathcal{U}_\alpha^\sigma(\vec{k}) \times \tag{1.613}$$



$$\left| \cdots, e^+, \vec{k}, \sigma \right\rangle = \frac{\mathcal{Z}_\psi^{1/2}}{(2\pi)^{3/2}} \overline{(\mathcal{V}^\sigma)}^\alpha(\vec{k}) \times \tag{1.614}$$

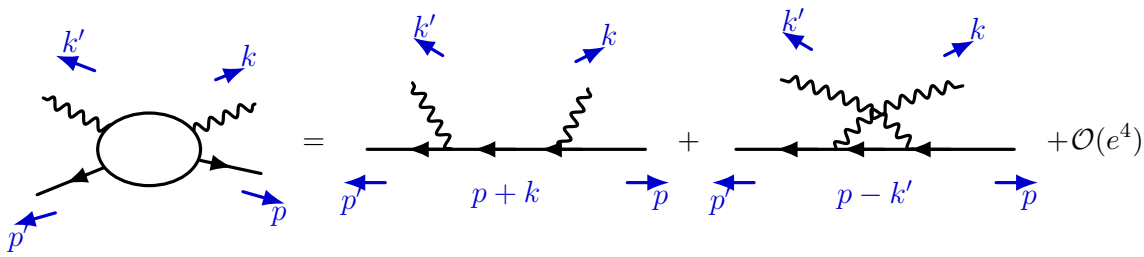


$$\left| \cdots, \gamma, \vec{k}, h \right\rangle^{\text{in}} = \frac{\mathcal{Z}_A^{1/2}}{(2\pi)^{3/2}\sqrt{2|\vec{k}|}} e_\mu^h(\vec{k}) \times \tag{1.615}$$



where these external *states* not *legs* of the diagrams (recall the LSZ reduction and how it truncates off the off-shell external legs (1.324)). As an example, let us consider *Compton*

scattering which is of the general form, and when expanded to tree level gives the following diagrams:



(1.616)

The scattering amplitude at tree level is

$$\begin{aligned}
i\mathcal{M}(\vec{p}',\sigma';\vec{k}',h'|\vec{p},\sigma;\vec{k},h) =& \frac{(\overline{\mathcal{U}}^{\sigma'})^{\beta}(\vec{p}')}{(2\pi)^{3/2}}\frac{e^{h'*}_{\nu}(\vec{k}')}{(2\pi)^{3/2}\sqrt{2|\vec{k}'|}}\frac{\mathcal{U}^{\sigma}_{\alpha}(\vec{p})}{(2\pi)^{3/2}}\frac{e^{h}_{\mu}(\vec{k})}{(2\pi)^{3/2}\sqrt{2|\vec{k}|}} \\
&\times\Bigg(e(\gamma^{\nu})_{\beta}{}^{\beta'}\frac{-i(-i(\not{p}+\not{k})+m)_{\beta'}{}^{\alpha'}}{(p+k)^2+m^2-i\epsilon}e(\gamma^{\mu})_{\alpha'}{}^{\alpha} \\
&+ e(\gamma^{\mu})_{\beta}{}^{\beta'}\frac{-i(-i(\not{p}-\not{k}')+m)_{\beta'}{}^{\alpha'}}{(p-k')^2+m^2-i\epsilon}e(\gamma^{\nu})_{\alpha'}{}^{\alpha}\Bigg) \\
=& \frac{-ie^2}{(2\pi)^6\sqrt{2|\vec{k}|}\sqrt{2|\vec{k}'}}\overline{\mathcal{U}}^{\sigma'}(\vec{p}')\Bigg(\not{e}^{h'*}(\vec{k}')\frac{-i(\not{p}+\not{k})+m}{(p+k)^2+m^2-i\epsilon}\not{e}^{h}(\vec{k}) \\
&+\not{e}^{h}(\vec{k})\frac{-i(\not{p}-\not{k}')+m}{(p-k')^2+m^2-i\epsilon}\not{e}^{h'*}(\vec{k}')\Bigg)\mathcal{U}^{\sigma}(\vec{p}).
\end{aligned} \tag{1.617}$$

Noet that $\mathcal{U}^{\sigma}(\vec{p})$ obeys the Dirac equation (1.399) $(\not{p}-im)\mathcal{U}^{\sigma}(\vec{p})=0$ for the one, while the other $\mathcal{U}^{\sigma'}(\vec{p}')$ obeys $(\not{p}'-im)\mathcal{U}^{\sigma'}(\vec{p}')=0$ and for the conjugate field (anti-particle) $\overline{\mathcal{U}}^{\sigma'}(\not{p}'-im)=0$. Also, we still have the transversality condition (1.595) $k\cdot e^{h}(\vec{k})=k'\cdot e^{h'}(\vec{k}')=0$. Now, under a "gauge transformation" $e^{h}_{\mu}\to e^{h}_{\mu}+\alpha k_{\mu}$, the term $\not{e}^{h}(\vec{k})\mathcal{U}^{\sigma}(\vec{p})$ for instance, is shifted by an overal factor of $\alpha\not{k}\mathcal{U}^{\sigma}(\vec{p})=\alpha(\not{k}+\not{p}-im)\mathcal{U}^{\sigma}(\vec{p})$. The "Dirac operator" $(\not{k}+\not{p}-im)$ commutes with the $\psi,\overline{\psi}$ propagator, and also by momentum conservation $\not{k}+\not{p}=\not{k}'+\not{p}'$. Then, the first term in (1.617) shifts by

$$-i\alpha\overline{\mathcal{U}}^{\sigma'}\not{e}^{h'*}(\vec{k}')\mathcal{U}^{\sigma}(\vec{p}) \tag{1.618}$$

while the second term shifts by a similar term

$$\alpha\overline{\mathcal{U}}^{\sigma'}(\vec{p}')(\not{k}-\not{p}'+im)\frac{-i(\not{p}-\not{k}')+m}{(p-k')^2+m^2-i\epsilon}\not{e}^{h'*}(\vec{k})\mathcal{U}^{\sigma}(\vec{p})=\alpha\overline{\mathcal{U}}^{\sigma'}(\vec{p}')(\,i\,)\not{e}^{h'*}(\vec{k})\mathcal{U}^{\sigma}(\vec{p}) \tag{1.619}$$

implying that the amplitude is gauge invariant! Now, the total cross section (recall from (1.360)) is given as

$$\sigma=\frac{(2\pi)^6}{v}\int d^3\vec{k}'d^3\vec{p}'\,(2\pi)^4\delta^4(k'+p'-k-p)\sum_{(h',\sigma')=(\pm1,1,2)}|\mathcal{M}(\vec{p}',\sigma';\vec{k},h'|\vec{p},\sigma;\vec{k},h)|^2. \tag{1.620}$$

We leave the evaluation of the polarization sum and momentum integrals to the Weinberg's textbook given in section 8.7 (since there is nothing physical to be learned from it; it is just tedious and Weinberg does it in amazing detail).

1.3.11 Renormalized QED (proper QED Lagrangian) & Ward Identity

However, I have a quick confession. When I called the Lagrangian in (1.601), the QED Lagrangian, it really is not since none of the operators are in terms of *renormalized fields* which are scaling of the fields in terms of the renormalization constants as we have seen time and time again. What we really wrote in (1.601) is for the *bare* fields, *bare* couplings (q), and *bare* mass. So, we should really re-label it as

$$\mathcal{L} = -\frac{1}{4}(\partial_\mu A^B_\nu - \partial_\nu A^B_\mu)(\partial^\mu A^{B\nu} - \partial^\nu A^{B\mu}) - \overline{\psi}^B(\gamma^\mu(\partial_\mu + ie^B A^B_\mu) + m^B)\psi^B. \tag{1.621}$$

The renormalized fields are defined as[44]

$$A_\mu \equiv \mathcal{Z}_A^{-1/2} A^B_\mu \quad \text{and} \quad \psi_\alpha \equiv \mathcal{Z}_\psi^{-1/2}\psi^B_\alpha. \tag{1.622}$$

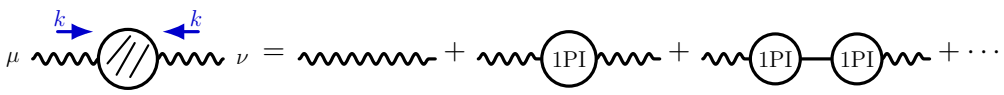
The renormalized fields are setup such that the 1-particle components of $\hat{F}_{\mu\nu}(x)\,|\Omega\rangle$ and $\hat{\psi}_\alpha(x)\,|\Omega\rangle$ are normalized as in free Maxwell and free fermion (Dirac) theories respectively. Then, we can re-write the bare Lagrangian in terms of A_μ and ψ_α as

$$\begin{aligned}
\mathcal{L} &= -\frac{1}{4}\mathcal{Z}_A F_{\mu\nu}F^{\mu\nu} - \mathcal{Z}_\psi\overline{\psi}(\gamma^\mu\partial_\mu + m^B)\psi - ie^B\mathcal{Z}_A^{1/2}\mathcal{Z}_\psi A_\mu\overline{\psi}\gamma^\mu\psi \\
&= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \overline{\psi}(\gamma^\mu\partial_\mu + m)\psi - ieA_\mu\overline{\psi}\gamma^\mu\psi \\
&\quad -\frac{1}{4}(\mathcal{Z}_A - 1)F^{\mu\nu}F_{\mu\nu} - (\mathcal{Z}_\psi - 1)\overline{\psi}(\gamma_\mu\partial^\mu + m)\psi - \mathcal{Z}_\psi\delta m\overline{\psi}\psi - ie(\mathcal{Z}_\psi - 1)A_\mu\overline{\psi}\gamma^\mu\psi
\end{aligned} \tag{1.623}$$

where this is really (in my opinion) the *true* QED Lagrangian since it is in terms of renormalized fields. Note here that, with some clever foresight, we $\pm$ some terms in $\mathcal{L}$ and gathered them into $\mathcal{Z} - 1$ factors, and also inserted the renormalized mass written as $m_B = m + \delta m$ where again we may choose m to be the physical mass. Also, written in purple, we have defined the *renormalized charge*

$$e = e^B\mathcal{Z}_A^{1/2}. \tag{1.624}$$

We shall begin by studying the 2-point function of the photon namely corrections to the propagators



(1.625)

[44] Again, here in QFT 1, these renormalized values are there to absorb divergences that correspond to their respective diagrams, namely the renormalized field and masses correspond to renormalizing the 2-point function and the renormalized coupling refers to renormalized the 3-point vertex (1.609). Also, recall that our reason previously was due to ambiguities in quantizing the Hamiltonian, but there will be a better reason/method in QFT 2.

where the photon propagator can be written as

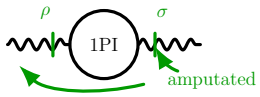
$$= -i\Delta_{\mu\nu}(k) = \frac{-i}{k^2 - i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - i\epsilon}\right). \tag{1.626}$$

Notice that the Fourier transformed (total) Greens function for the propagator in terms of 1PI's can be written as

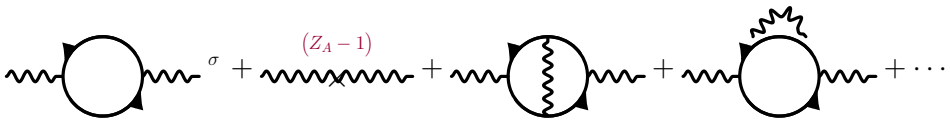
$$\begin{aligned}\widetilde{G}_{\mu\nu}(k) &= -i\left(\Delta_{\mu\nu}(k) + \Delta_{\mu\rho}(k)\Pi^{\rho\sigma}\Delta_{\sigma\nu}(k) + \cdots\right)\\ &= -i\left((\Delta^{-1}(k) - \Pi(k)^{-1}\right)_{\mu\nu}\\ &\equiv -i\left(\Delta_{\mu\nu}(k) + \Delta_{\mu\rho}(k)M^{\rho\sigma}(k)\Delta_{\sigma\nu}(k)\right)\end{aligned} \tag{1.627}$$

where we have defined the 1PI function $\Pi^{\rho\sigma}(k)$ as

$i\Pi^{\rho\sigma}(k) =$



$=$



(1.628)

and that $M^{\rho\sigma}$ here is the resulting matrix after completing the geometric sum in (1.627). This result will actually be of use when computing the gauge-invariant 2-point function of the Maxwell field strength tensor or derivatives thereof[45]. A useful result for computing this is[46]

$$2k_\alpha k^{[\mu}\Delta^{\alpha]\rho} = \frac{-i}{k^2 - i\epsilon}(k^\mu k^\rho - k^2\eta^{\mu\rho}) \tag{1.629}$$

which can be used to show that (I recommend you try this out for yourself)

$$\begin{aligned}\langle\Omega|\, T\partial_\alpha\hat{F}^{\mu\alpha}(x)\partial_\beta\hat{F}^{\nu\beta}(y)\,|\Omega\rangle = \int\frac{d^4k}{(2\pi)^4}e^{ik\cdot(x-y)}\Big(&-i(k^2\eta^{\mu\nu} - k^\mu k^\nu)\\ &- \frac{i(k^\mu k^\rho - k^2\eta^{\mu\rho})}{(k^2 - i\epsilon)^2}M_{\rho\sigma}(k)(k^\sigma k^\nu - k^2\eta^{\sigma\nu})\Big)\end{aligned} \tag{1.630}$$

where the first term can be disregarded since it is a *contact term* which are terms that would only contribute a delta function in the correlation function. Now, on the other hand, the equations of motion for A_μ with the corresponding source $\hat{j}^\mu$ is

$$\begin{aligned}\partial_\nu\hat{F}^{\mu\nu} &= \hat{j}^\mu\\ &= -i\mathcal{Z}_A^{-1}e\mathcal{Z}_\psi\hat{\bar{\psi}}\gamma^\mu\hat{\psi}\end{aligned} \tag{1.631}$$

[45] We are trying to get to a specific result, just wait

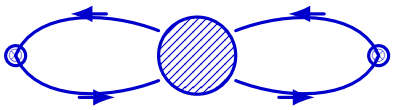
[46] Where we have used the following anti-symmetric notation $A^{[ab]} \equiv \frac{1}{2}(A^{ab} - A^{ba})$ which comes into play since the Maxwell field strength tensor is antisymmetric in it's indices, namely $F_{\mu\nu} = -F_{\nu\mu}$.

where the electromagnetic current here can be found via the usual Noether procedure (1.177). Then there is of course the corresponding Noether charge which is the overall electric charge (1.178)

$$\hat{Q} \equiv \int d^3\vec{x}\, \hat{j}^0(x). \tag{1.632}$$

The conservation of the charge is straightforward since the second order derivative of the Maxwell field strength tensor vanishes since it is totally antisymmetric $\partial_\mu \hat{j}^\mu = \partial_\mu \partial_\nu \hat{F}^{\mu\nu} = 0$. Then, what we really are computing in (1.630) is that of the 2-point function for the QED current which is equal to (up to a contact term)

$$\langle \Omega | T \hat{j}^\mu(x) \hat{j}^\nu(y) | \Omega \rangle =$$



$$= -i \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} M^{\mu\nu}(k). \tag{1.633}$$

But, we have that the current is conserved, and so we have

$$\partial_\mu \hat{j}^\mu = 0 \implies k^\rho M_{\rho\sigma}(k) = 0. \tag{1.634}$$

This apparently tiny looking equation is quite an amazing one called the *Ward identity* (for QED). This identity is a statement that describes the physically possible scattering processes (so on-shell) that can occur in the QED S-matrix. Physically then, this implies that the longitudinal polarization of the photon that arises in our ξ-gauged theory is unphysical and disappears from the S-matrix. This also constraints the possible form that the tensor structure $M_{\mu\nu}$ can possible take on for us computing the *vaccum polarization* (the photon 2-point function). Also in (1.634), ther eis no contact term is the derivative of the correlator $\langle \partial_\mu j^\mu(x) j^\nu(y) \rangle$ as $j^\nu(y)$ is an unchanged operator. It also follows that the 1PI obeys $k_\rho \Pi^{\rho\sigma} = 0$, and by Lorentz invariance, we must have the following structure for the 1PI

$$\Pi^{\rho\sigma}(k) = (\eta^{\rho\sigma} k^2 - k^\rho k^\sigma)\pi(k^2) \tag{1.635}$$

for some function $\pi(k^2)$. There is a nice format for writing the Fourier transformed Greens function that makes the contribution from loops that are contained in $\pi(k^2)$ semi-obvious. A neat result is that

$$\Delta_{\mu\rho} \Pi^{\rho\sigma} = \frac{1}{k^2 - i\epsilon} \Pi_\mu{}^\sigma \tag{1.636}$$

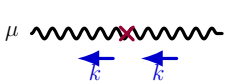
which allows us to compute

$$\begin{aligned} \widetilde{G}_{\mu\nu}(k) &= -i \left(\Delta_{\mu\nu}(k) + \frac{1}{(k^2 - i\epsilon)^2} \Pi_{\mu\nu}(k) + \frac{1}{(k^2 - i\epsilon)^3} \Pi_{\mu\rho}(k) \Pi^\rho{}_\nu(k) + \cdots \right) \\ &= \frac{-i}{k^2 - i\epsilon} \left(\frac{\eta_{\mu\nu}}{1 - \pi(k^2)} - \frac{k_\mu k_\nu}{k^2 - i\epsilon} \left(1 - \zeta + \frac{\pi(k^2)}{1 - \pi(k^2)} \right) \right). \end{aligned} \tag{1.637}$$

Now, we would expect (or at least hope) that since the photon is massless corresponding to $k^2 = 0$ then $\pi(k^2)$ is regular at $k^2 = 0$. Now, we should also have/choose the renoramlized photon field $\mathcal{Z}_A$ to be such that $\pi(k^2 = 0) = 0$ such that the 1-particle state $\hat{F}_{\mu\nu}(x) |\Omega\rangle$ (evaluated at the 1-particle state) is normalized as in the free Maxwell theory.

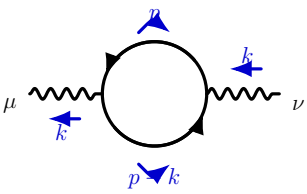
1.3.12 Loop Calculations in QED and $g-2$

Now, let us actually calculate $\pi(k^2)$ and $\mathcal{Z}_A$ at 1-loop order. The counter term vertex arising from the renormalized QED Lagrangian (1.623) is



$$= -i(\mathcal{Z}_A - 1)(k^2\eta_{\mu\nu} - k_\mu k_\nu) \tag{1.638}$$

which will contribute a factor of $1-\mathcal{Z}_A$ to $\pi(k^2)$. The 1-loop diagram is written as



$$\begin{aligned} &= -\int \frac{d^4p}{(2\pi)^4}\,\mathrm{Tr}\left[e\gamma_\mu \frac{-i(-i\not{p}+m)}{p^2+m^2-i\epsilon} e\gamma_\nu \frac{-i(-i(\not{p}-\not{k})+m)}{(p-k)^2+m^2-i\epsilon}\right] \\ &= -i\int \frac{d^4p^E}{(2\pi)^4}\,\mathrm{Tr}\left[e\gamma_\mu \frac{-i(-i\not{p}+m)}{p^2+m^2} e\gamma_\nu \frac{-i(-i(\not{p}-\not{k})+m)}{(p-k)^2+m^2}\right] \end{aligned} \tag{1.639}$$

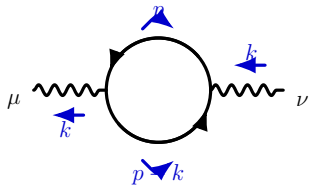
where the minus sign in the first line comes from when Fermions are in a loop (the Wick contract between the 4-fields generates a minus sign since they switch positions when contracted namely something like $\overline{\psi}\psi\overline{\psi}\psi$. In the second line, we wick rotated $p_0 = ip_0^E$ which allows us to drop the $i\epsilon$ prescription. The trace is over the spinor indices. Notice that this integral is divergent in momentum since the measure contributes as $d^4p^E \propto p^3\, d\Omega_2 dp$ and the integrand contributes as $1/p^2$ and so it is a linear divergence. To handle these divergences, we will now *and finally* introduce one of the most powerful techniques to hand such divergences in (practically) *any* loop diagram called *dimensional regularization* which was formulated by t'Hooft and Veltmann in 1972. The idea is this, we take the measure and extend the total number of dimensions to D dimensions by shifting $D \to 4-\epsilon$ where we are to take $\epsilon \to 0$ at the end (or absorb said divergences into renormalizable terms),

$$\int \frac{d^4p}{(2\pi)^4} \to \int \frac{d^Dp}{(2\pi)^D} \quad \text{and} \quad D = 4-\epsilon. \tag{1.640}$$

We maintain however that $\{\gamma_\mu, \gamma_\nu\} = \eta_{\mu\nu}$ but with $\eta^{\mu\nu}\eta_{\mu\nu} = D$. The trace/Dirac spinor index is still $\mathrm{Tr}\,\mathbb{1} = 4$[47]. By the Feynman trick given as (B.1), we can re-write the integral

[47] Actually, spinors in general dimension is an ongoing research discussion since there are uniquely defined objects in $D=4$ such as γ_5 that seemingly have no analog in higher dimensions.

as



$$= ie^2 \int \frac{d^D p^E}{(2\pi)^D} \int_0^1 dx \frac{Tr\left[\gamma_\mu(-i\not{p}+m)\gamma_\nu(-i(\not{p}-\not{k})+m)\right]}{(x(p-k)^2+(1-x)p^2+m^2)^2}$$

$$= ie^2 \int_0^1 \frac{d^D p}{(2\pi)^D} \frac{\mathrm{Tr}\left[\gamma_\mu(-i(\not{p}+x\not{k})+m)\gamma_\nu(-i(\not{p}-(1-x)\not{k})+m)\right]}{(p^2+x(1-x)k^2+m^2)^2} \tag{1.641}$$

where in going from the first to second line, we shifted the momentum via $p \to p + xk$ since it keeps the measure unchanged. There is a specific trace formula we need to continue which is that (we won't derive it here since it's derivation is unilluminating to any physical process, just handy)

$$\mathrm{Tr}\left[\gamma_\mu\gamma_\alpha\gamma_\nu\gamma_\beta\right] = 4(\eta_{\mu\alpha}\eta_{\nu\beta} - \eta_{\mu\nu}\eta_{\alpha\beta} + \eta_{\mu\beta}\eta_{\nu\alpha}). \tag{1.642}$$

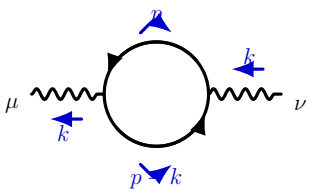
Then, the spinor-trace in the numerator becomes

$$\begin{aligned}\mathrm{Tr}[\cdots] =& 4\Big(\eta_{\mu\nu}((+p+xk)\cdot(p-(1-x)k)+m^2) \\ &- (p_\mu + xk_\mu)(p_\nu - (1-x)k_\nu) - (p_\nu + xk_\nu)(p_\mu - (1-x)k_\mu)\Big).\end{aligned} \tag{1.643}$$

By SO(D) rotation symmetry, terms that are proportional to $p_\mu k_\nu$, $p_\nu k_\mu$ or $p \cdot k$ integrate to zero (since the integrand is odd divided by even), while terms like $p_\mu p_\nu$ can be replaced with

$$p_\mu p_\nu \to \frac{1}{D}\eta_{\mu\nu}p^2. \tag{1.644}$$

We therefore end up with



$$= ie^2 \int_0^1 dx \int \frac{d^D p}{(2\pi)^D} \frac{4\left(\eta_{\mu\nu}(p^2 - x(1-x)k^2 + m^2) - \frac{2}{D}\eta_{\mu\nu}p^2 + 2x(1-x)k_\mu k_\nu\right)}{(p^2+x(1-x)k^2+m^2)^2}. \tag{1.645}$$

Now, we can break down the D-dimensional measure as

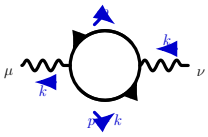
$$\int \frac{d^D p}{(2\pi)^D}\cdot = \frac{\mathrm{vol}(S^{D-1})}{(2\pi)^D}\int d|p|\,|p|^{D-1}\cdots \tag{1.646}$$

where the volume of a $D-1$ sphere is $\mathrm{vol}(S^{D-1}) = 2\pi^{D/2}\Gamma(D/2)$. The dp integrals that arise can be evaluated as (since there are terms with p^2 in the numerator)

$$\int_0^\infty dp \frac{p^{D-1}}{(p^2+M^2)^2} = \frac{1}{2}(M^2)^{\frac{D}{2}-2}\Gamma(D/2)\Gamma(2-D/2) \tag{1.647}$$

$$\int_0^\infty dp \frac{p^{D+1}}{(p^2+M^2)^2} = \frac{1}{2}(M^2)^{\frac{D}{2}-1}\Gamma(D/2+1)\Gamma(1-D/2) \tag{1.648}$$

where $\Gamma(x)$ here is the *Gamma function*. Note that the D-dimensional integrals here are defined by analytic continuation from $D<2$. Then, the 1-loop integral can be evaluated to find



$$\begin{aligned}
&= \frac{4ie^2\cdot 2}{(4\pi)^{D/2}\Gamma(D/2)}\frac{1}{2}\int_0^1 dx\,\Bigg(\left(1-\frac{2}{D}\right)\eta_{\mu\nu}(x(1-x)k^2+m^2)^{\frac{D}{2}-1}\Gamma\left(\frac{D}{2}+1\right)\Gamma\left(1-\frac{D}{2}\right)\\
&\quad + \Big(2x(1-x)k_\mu k_\nu + \eta_{\mu\nu}(-x(1-x)k^2+m^2)\Big)\\
&\quad \times (x(1-x)k^2+m^2)^{\frac{D}{2}-1}\Gamma\left(\frac{D}{2}\right)\Gamma\left(2-\frac{D}{2}\right)\Bigg)\\
&= -\frac{4ie^2}{(4\pi)^{D/2}\Gamma(D/2)}\Gamma\left(\frac{D}{2}\right)\Gamma\left(2-\frac{D}{2}\right)2(\eta_{\mu\nu}k^2-k_\mu k_\nu)\\
&\quad \times \int_0^1 dx\, x(1-x)\Big(x(1-x)k^2+m^2\Big)^{\frac{D}{2}-2}.
\end{aligned} \tag{1.649}$$

The total contribution to $\pi(k^2)$ is then

$$\pi(k^2) = 1 - \mathcal{Z}_A - \frac{8e^2}{(4\pi)^{\frac{D}{2}}}\Gamma\left(2-\frac{D}{2}\right)\int_0^1 dx\, x(1-x)\Big(x(1-x)k^2+m^2\Big)^{\frac{D}{2}-2}. \tag{1.650}$$

If we now set $\pi(0)=0$, this will fix the value of $\mathcal{Z}_A$ to be

$$\begin{aligned}
\mathcal{Z}_A &= 1 - \frac{8e^2}{(4\pi)^{\frac{D}{2}}}\Gamma\left(2-\frac{D}{2}\right)\frac{1}{6}m^{D-4}\\
&= 1 + e^2\left(-\frac{1}{6\pi^2}\frac{1}{\epsilon} + \text{finite}\right)
\end{aligned} \tag{1.651}$$

where in the last line, we took $D=4-\epsilon$ and series-expanded around $\epsilon\to 0$ to isolate the *ultraviolet divergence*. Notice that in the limit that $\epsilon\to 0$ $\mathcal{Z}_A$ may diverge, but $\pi(k^2)$ is finite,

$$\pi(k^2) = \frac{e^2}{2\pi^2}\int_0^1 dx\, x(1-x)\ln\frac{x(1-x)k^2+m^2}{m^2}. \tag{1.652}$$

So, we just looked at the loop effect to the photon propagator which gave us the Ward identity, but now let us look at a similar loop calculation but now with respect to a single

electron state that is generated by a current. We can denote the 1-electron state as $|\vec{p},\sigma\rangle$. The matrix element we are interested in is then

$$\langle\vec{p}',\sigma'|\,\hat{j}^\mu(x)\,|\vec{p},\sigma\rangle = e^{i(p-p')\cdot x}\,\langle\vec{p}',\sigma'|\,\hat{j}^\mu(0)\,|\vec{p},\sigma\rangle \tag{1.653}$$

where (1.653) is called the *electromagnetic form factor*. Lorentz invariance implies the following for (1.653)

$$\begin{aligned}\langle\vec{p}',\sigma'|\,\hat{j}^\mu(0)\,|\vec{p},\sigma\rangle &= \langle\vec{p}',\sigma'|\,(U(\Lambda))^{-1}\cdot U(\Lambda)\hat{j}^\mu(0)(U(\Lambda))^{-1}\cdot U(\Lambda)\,|\vec{p},\sigma\rangle \\ &= \sqrt{\frac{(\Lambda p')^0}{p'^0}}\sum_{\lambda'} D^*_{\sigma'\lambda}(W)\cdot\Lambda_\nu{}^\mu\sum_\lambda D_{\sigma\lambda}\left\langle\vec{\Lambda p}',\lambda'\right|\hat{j}^\nu(0)\left|\vec{\Lambda p},\lambda\right\rangle.\end{aligned} \tag{1.654}$$

The most general solution to (1.654) is

$$\langle\vec{p}',\sigma'|\,\hat{j}^\mu(0)\,|\vec{p},\sigma\rangle = \frac{iq}{(2\pi)^3}\overline{\mathcal{U}}^{\sigma'}(\vec{p}')M^\mu(p',p)\mathcal{U}^\sigma(\vec{p}) \tag{1.655}$$

where $M^\mu(p',p)$ is a Lorentz *covariant* combination of p'^ν, p^ν, and γ^ν's. Recall that we also have to obey Dirac's equation (1.399) and so the only independent Lorentz covariant expressions that M^μ can be built out of are a sum of p'^ν, p^ν, and γ^ν's. For example we could have $M^\mu = \gamma_5\gamma^\mu$ but this is not compatible with parity. We could also have $M^\mu = \gamma_5\gamma_\nu\epsilon^{\mu\nu\rho\sigma}p_\rho p'_\sigma$ which is compatible with parity (the totally antisymmetric-epsilon tensor times the γ_5 matrix that is parity odd gives a parity even result) but does not give a linearly independent result (thus not compatible with the Dirac equation).

We also have the constraint from the conservation law $\partial_\mu\hat{j}^\mu(x) = 0$ which implies that the matrix element (1.653) obeys

$$(p-p')_\mu\,\langle\vec{p}',\sigma'|\,\hat{j}^\mu(0)\,|\vec{p},\sigma\rangle = 0 \implies (p-p')_\mu\overline{\mathcal{U}}^{\sigma'}(\vec{p}')M^\mu(p',p)\mathcal{U}^\sigma(\vec{p}) = 0 \tag{1.656}$$

which implies the following general form of M^μ[48]

$$M^\mu(p',p) = \gamma^\mu F(k^2) - \frac{i}{2m}(p+p')^\mu G(k^2) \tag{1.657}$$

where $k^2 \equiv (p-p')^2$ for some real-valued functions $F(k^2)$ and $G(k^2)$. Note that in a free fermion theory, we would naturally have $F(k^2) = 1$ and $G(k^2) = 0$. We will generally take the $p' \to op$ limit which gives us the simplified form

$$\langle\vec{p}',\sigma'|\,\hat{j}^\mu(x)\,|\vec{p},\sigma\rangle = \frac{iq}{(2\pi)^3}\overline{\mathcal{U}}^{\sigma'}(\vec{p})\left(\gamma^\mu F(0) - \frac{i}{m}p^\mu G(0)\right)\mathcal{U}^\sigma(\vec{p}). \tag{1.658}$$

This form allows us to use the spinor relations such as (1.412) which can be recasted as

$$2p^\mu\overline{\mathcal{U}}^{\sigma'}(\vec{p})\mathcal{U}^\sigma(\vec{p}) = \overline{\mathcal{U}}^{\sigma'}(\vec{p})\{\gamma^\mu,\not{p}\}\mathcal{U}^\sigma(\vec{p}) = 2im\overline{\mathcal{U}}^{\sigma'}(\vec{p})\gamma^\mu\overline{\mathcal{U}}^\sigma(\vec{p}) \tag{1.659}$$

and we also have that $\overline{\mathcal{U}}^{\sigma'}(\vec{p})\mathcal{U}^\sigma(\vec{p}) = (m/p^0)\delta_{\sigma,\sigma'}$. We then have

$$\langle\vec{p}',\sigma'|\,\hat{j}^\mu(x)\,|\vec{p},\sigma\rangle = \frac{q}{(2\pi)^3}\frac{p^\mu}{p^0}\delta_{\sigma\sigma'}(F(0)+G(0)). \tag{1.660}$$

[48] In case the notation is not clear $(p+p')^\mu \equiv p^\mu + p'^\mu$.

Let us compare this to the total charge of QED namely the Noether charge $\hat{Q} = \int d^3\vec{x}\hat{j}^0(x)$ that acts on our 1-particle (electron) state as $\hat{Q}\,|\vec{p},\sigma\rangle = q\,|\vec{p},\sigma\rangle$ which implies the following matrix element when integrating over (1.660)

$$\langle\vec{p},\sigma'|\,\hat{Q}\,|\vec{p},\sigma\rangle = q\delta_{\sigma\sigma'}\delta^3(0) = \frac{q}{(2\pi)^3}\delta_{\sigma\sigma'}\int d^3\vec{x}\,1 \tag{1.661}$$

which implies that

$$F(0) + G(0) = 1. \tag{1.662}$$

The interpretation of $F(0)$ is that if we turn on a weak, slow-varying background electromagnetic field $A_\mu^{\text{classical}}(x)$ (a classical background field), the Hamiltonian receives the contribution

$$\Delta H = -\int d^3\vec{x}\,A_\mu^{\text{cl}}(x)\hat{j}^\mu(x). \tag{1.663}$$

The energy of an electron propagating in such a background electromagnetic field is given by $p' \to p$ limit of the corresponding matrix element

$$\begin{aligned}\langle\vec{p}',\sigma'|\,\Delta H\,|\vec{p},\sigma\rangle &= -\int d^3\vec{x}\,A_\mu^{\text{cl}}(x)\,\langle\vec{p}',\sigma'|\,\hat{j}^\mu(x)\,|\vec{p},\sigma\rangle \\ &= -\int d^3\vec{x}\,A_\mu^{\text{cl}}(x)e^{i(p-p')\cdot x}\frac{iq}{(2\pi)^3}\overline{\mathcal{U}}^{\sigma'}(\vec{p}')\left(\gamma^\mu F(k^2) - \frac{i(p+p')^\mu}{2m}G(k^2)\right)\mathcal{U}^\sigma(\vec{p}).\end{aligned} \tag{1.664}$$

We can rewrite the spinors acting on the 4-momentum term as

$$\begin{aligned}\overline{\mathcal{U}}^{\sigma'}(\vec{p}')[\gamma^\mu,\gamma^\nu](p'-p)_\nu\mathcal{U}^\sigma(\vec{p}) =& \overline{\mathcal{U}}^{\sigma'}(\vec{p}')[\gamma^\mu,\not{p}'-\not{p}]\mathcal{U}^\sigma(\vec{p}) \\ =& \overline{\mathcal{U}}^{\sigma'}(\vec{p}')\left(-2\not{p}'\gamma^\mu + \{\gamma^\mu,\not{p}'\} + \{\gamma^\mu,\not{p}\} - 2\gamma^\mu\not{p}\right)\mathcal{U}^\sigma(\vec{p}) \\ =& -4im\overline{\mathcal{U}}^{\sigma'}(\vec{p}')\gamma^\mu\mathcal{U}^\sigma(\vec{p}) + 2(p^\mu + p'^\mu)\overline{\mathcal{U}}^{\sigma'}(\vec{p})\mathcal{U}^\sigma(\vec{p}).\end{aligned} \tag{1.665}$$

We also have that the commutator of the gamma matrices is $-\frac{i}{4}[\gamma^\mu,\gamma^\nu] = S^{\mu\nu}$ and so we can re-write (1.664) as

$$\begin{aligned}\langle\vec{p}',\sigma'|\,\Delta H\,|\vec{p},\sigma\rangle = -\int d^3\vec{x}\,A_\mu^{\text{cl}}(x)e^{i(p-p')\cdot x}\frac{iq}{(2\pi)^3}\Bigg(&-\frac{i}{2m}(p+p')^\mu(F(k^2)+G(k^2))\overline{\mathcal{U}}^{\sigma'}(\vec{p}')\mathcal{U}^\sigma(\vec{p}) \\ &-\frac{1}{m}(p'-p)_\nu F(k^2)\overline{\mathcal{U}}^{\sigma'}(\vec{p}')S^{\mu\nu}\mathcal{U}^\sigma(\vec{p})\Bigg).\end{aligned} \tag{1.666}$$

In the case of a background magnetic field, we can set $A_0^{\text{cl}} = 0$ and we can pick the background spatial piece to be such that $\vec{A}^{\text{cl}}$ obey $\nabla\times\vec{A}^{\text{cl}} = \vec{B}$. The first term in (1.666) represents the effect of the Lorentz force since it is spin-independent in the non-relativistic limit, i.e. $|\vec{p}|,|\vec{p}'| \ll m$. The second term in (1.666) represents the spin-magnetic field coupling which we shall be interested in for the rest of the lectures. Let us denote this second term as

$$\begin{aligned}\langle\vec{p}',\sigma'|\,\Delta H^{\text{spin}}\,|\vec{p},\sigma\rangle =& \frac{iq}{(2\pi)^3 m}\int d^3\vec{x}\,A_i^{\text{cl}}(x)e^{i(p-p')\cdot x}(p'-p)_\nu F(k^2)\overline{\mathcal{U}}^{\sigma'}(\vec{p}')S^{i\nu}\mathcal{U}^\sigma(\vec{p}) \\ \approx& \frac{iq}{(2\pi)^3 m}\int d^3\vec{x}\,A_i^{\text{cl}}(x)e^{i(p-p')\cdot x}(p'-p)_j F(0)\overline{\mathcal{U}}^{\sigma'}(\vec{0}')S^{ij}\mathcal{U}^\sigma(\vec{0}) \\ =& -\frac{qF(0)}{(2\pi)^3 m}\int d^3\vec{x}\,e^{i(p-p')\cdot x}\vec{B}\cdot\left(\frac{\vec{\sigma}}{2}\right)_{\sigma'\sigma}\end{aligned} \tag{1.667}$$

where in the second line we first took the non-relativistic limit, and we can recognize that the blue term can be written as

$$A_i^{\rm cl}(x)e^{i(p-p')\cdot x}(p'-p)_j = iA_i^{\rm cl}(x)\partial_j e^{i(p-p')\cdot x} \tag{1.668}$$

which allows us to integrate by parts to move the ∂_j onto the background field, from which we can symmetrize to get $iF_{ij}^{\rm cl}(x)e^{i(p-p')\cdot x}$. The purple term is actually equal to

$$\overline{\mathcal{U}}^{\sigma'}(\vec{0}')S^{ij}\mathcal{U}^{\sigma}(\vec{0}) = \epsilon^{ijk}\left(\frac{\sigma^k}{2}\right)_{\sigma'\sigma} \tag{1.669}$$

leaving us with the final expression, which in other words is a matrix element that computes what happens when we are acting on an electron state at small spatial momentum namely

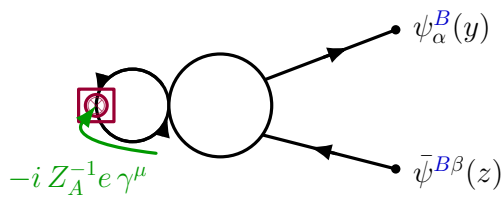
$$\Delta H^{\rm spin} = -\frac{qF(0)}{2m}\vec{B}\cdot\vec{\sigma} = -g\frac{q}{2m}\vec{B}\cdot\frac{\vec{\sigma}}{2} \tag{1.670}$$

where we can define the *magnetic moment* $\mu \equiv qF(0)/2m$, and this *g-factor* here is $g = 2F(0)$. We know that if there are no interactions, then $F(k^2) = 1$, and we have the usual $g = 2$ term from quantum mechanics, but let us see what the next order term looks like, i.e. the perturbative computation of the electromagnetic form factor. We start with the renormalized

$$j^\mu = -i\mathcal{Z}_A^{-1}e\mathcal{Z}_\psi\overline{\psi}\gamma^\mu\psi \tag{1.671}$$

where $\mathcal{Z}_\psi\overline{\psi}\gamma^\mu\psi = \overline{\psi}^B\gamma^\mu\psi^B$. To compute the 1-loop correction, we consider a Greens function of the following form

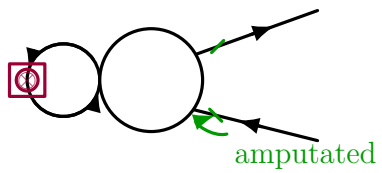
$$\langle\Omega|\,T\psi_\alpha^B(y)j^\mu(x)\overline{\psi}^{B\beta}(z)\,|\Omega\rangle =$$



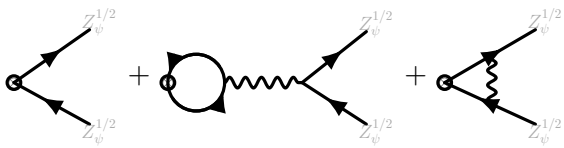
(1.672)

which is computed in perturbation theory via the diagram above. After taking the LSZ limit on the $\psi_\alpha^B(y)$ and $\overline{\psi}^{B\beta}(z)$, the form factor is computed via

$$\langle\vec{p}',\sigma'|\,j^\mu(0)\,|\vec{p},\sigma\rangle =$$



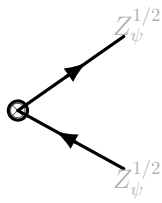
$$=$$



(1.673)

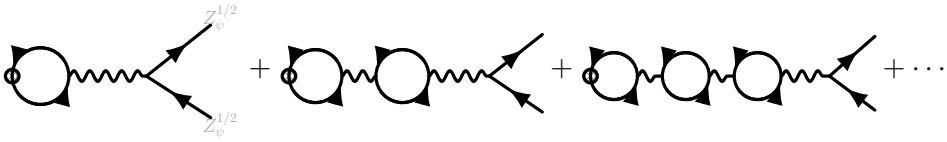
where the green-slashes denote the amputated legs, and the diagrams here are constructed via bare vertices and bare field propagators. Note that if we work with]*psi* instead of

ψ^B then the $Z_\psi^{1/2}$ factors on external lines would be absent, but we would need to include additional vertices corresponding to counter term, and the final result for the form factor would be identical. At tree level we have



$$= -i\mathcal{Z}_A^{-1} e \frac{\mathcal{Z}_\psi}{(2\pi)^3} \mathcal{Z}_\psi \overline{\mathcal{U}}^{\sigma'}(\vec{p}')\gamma^\mu \mathcal{U}^\sigma(\vec{p}) \tag{1.674}$$

which we can combine with other 1PI diagrams such as

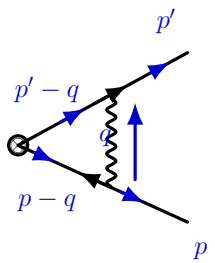


(1.675)

to give

$$-i\mathcal{Z}_A^{-1} e \frac{\mathcal{Z}_\psi}{(2\pi)^3} \frac{\mathcal{Z}_A}{1-\pi(k^2)} \overline{\mathcal{U}}^{\sigma'}(\vec{p}')\gamma^\mu \mathcal{U}(\vec{p}). \tag{1.676}$$

This combines with $\mathcal{Z}_\psi(1-\pi(k^2))^{-1}$ to $F(k)$ and does not contribute to $G(k)$. At order e^2, the only contribution to $G(k)$ comes from the diagram



$$= -i\mathcal{Z}_A^{-1} e \frac{\mathcal{Z}_\psi}{(2\pi)^3} \int \frac{d^4q}{(2\pi)^4} \frac{-i}{q^2 - i\epsilon} \tag{1.677}$$

$$\times \overline{\mathcal{U}}^{\sigma'}(\vec{p}') e\gamma^\rho \frac{-i(-i(\not{p}' - \not{q}) + m)}{(p'-q)^2 + m^2 - i\epsilon} \gamma^\mu \frac{-i(-i(\not{p} - \not{q}) + m)}{(p-q)^2 + m^2 - i\epsilon} e\gamma_\rho \mathcal{U}^\sigma(\vec{p}).$$

To proceed, as for any loop calculation really, the steps are the following (as we have seen). We first Wick rotate via $q^0 \to iq^4$. Then, we use dimensional regularization $D = 4 - \epsilon$. Then, we use the Feynman trick to combine the denominators into the q-integrand. After taking into account the Dirac equation namely

$$(i\not{p} + m)\mathcal{U}^\sigma(\vec{p}) = \overline{\mathcal{U}}^{\sigma'}(\vec{p}')(i\not{p}' + m) = 0, \tag{1.678}$$

we can simplify the q-integral into

$$\overline{\mathcal{U}}^{\sigma'}(\vec{p}')\gamma^\mu \mathcal{U}^\sigma(\vec{p}) A(k^2) + (p+p')^\mu \overline{\mathcal{U}}^{\sigma'}(\vec{p}')\mathcal{U}^\sigma(\vec{p}) B(k^2) \tag{1.679}$$

where $A(k^2)$ contributes to $F(k^2)$ where $A(k^2)$ contains both UV and IR divergences, but these cancel against those coming from $\mathcal{Z}_\psi$. The $B(k^2)$ function contributes to $G(k^2)$ and

turns out to be finite on it's own! The result is (see Weinberg section 11.3 for full details [1])

$$G(k^2) = -\frac{e^2 m^2}{4\pi^2} \int_0^1 dx \int_0^x dy \, \frac{x(1-x)}{m^2 x^2 + k^2 y(x-y)} \tag{1.680}$$

which at $k^2 = 0$ gives

$$G(0) = -\frac{e^2}{8\pi^2} = -\frac{\alpha}{2\pi} \tag{1.681}$$

where $\alpha \equiv e^2/(4\pi) \approx 1/137$ which contributes to the electron g-factor as (using (1.662))

$$g = 2F(0) = 2(1 - G(0)) = 2 + \frac{\alpha}{\pi} + \mathcal{O}(\alpha^2) \approx 2.002323 \cdots \tag{1.682}$$

to which if we compare with the current result from today $g \approx 2.0023193 \cdots$ which it should be noted we only computed to first loop order and only took into account QED! And yet, we are accurate to four decimal places or in other words, 99.99979% close to the accepted experimental value!

This completes the lecture notes for QFT 1.

1.4 Problem set 1:

1.4.1 Problem 1: Some nice field theory calculations

In the first lecture we introduced a scalar field operator in the system of free relativistic particles of mass m in D-dimensional Minkowskian spacetime, of the form

$$\hat{\phi}(x) = \int \frac{d^{D-1}\vec{k}}{\sqrt{(2\pi)^{D-1}2\omega_{\vec{k}}}} \left(a_{\vec{k}} e^{i\vec{k}\cdot\vec{x} - i\omega_{\vec{k}}x^0} + a^\dagger_{\vec{k}} e^{-i\vec{k}\cdot\vec{x} + i\omega_{\vec{k}}x^0} \right), \tag{1}$$

where $x^\mu \equiv (x^0, \vec{x})$ is the space-time coordinate, $\vec{x}$ is the $(D-1)$-dimensional spatial position vector, $\vec{k}$ is the spatial momentum vector, and $\omega_{\vec{k}} = \sqrt{\vec{k}^{\,2} + m^2}$ is the energy of a particle according to the relativistic dispersion relation. $a_{\vec{k}}$ and $a^\dagger_{\vec{k}}$ are annihilation and creation operators that obey the commutation relations

$$[a_{\vec{k}}, a_{\vec{k}'}] = 0 = [a^\dagger_{\vec{k}}, a^\dagger_{\vec{k}'}], \qquad [a_{\vec{k}}, a^\dagger_{\vec{k}'}] = \delta^{D-1}(\vec{k} - \vec{k}'). \tag{2}$$

(a) Verify the <u>equal-time</u> commutation relations

$$[\hat{\phi}(x^0, \vec{x}), \hat{\phi}(x^0, \vec{y})] = 0, \qquad \left[\hat{\phi}(x^0, \vec{x}), \frac{\partial}{\partial x^0}\hat{\phi}(x^0, \vec{y})\right] = i\,\delta^{D-1}(\vec{x} - \vec{y}). \tag{3}$$

(b) The stress-energy tensor operator $\hat{T}_{\mu\nu}(x)$ is given by

$$\hat{T}_{\mu\nu}(x) =: \partial_\mu\hat{\phi}(x)\,\partial_\nu\hat{\phi}(x) : -\eta_{\mu\nu}\left[\frac{1}{2} : \partial^\rho\hat{\phi}(x)\,\partial_\rho\hat{\phi}(x) : + \frac{1}{2}m^2 : (\hat{\phi}(x))^2 :\right], \tag{4}$$

where $: \cdots :$ stands for the normal ordering, defined simply by moving the $a^\dagger$'s to the left of the a's in a product of creation and annihilation operators. The derivation of this expression through the Noether procedure will be discussed in class later. For now you are asked to simply take the expression as given, and verify the conservation relation

$$\partial_\mu \hat{T}^\mu{}_\nu(x) = 0, \tag{5}$$

where $\partial_\mu \equiv \frac{\partial}{\partial x^\mu}$, and the Lorentz indices are raised or lowered by contracting with $\eta^{\mu\nu}$ or $\eta_{\mu\nu}$.[49] A consequence is that

$$\hat{P}_\mu = \int d^{D-1}\vec{x}\,\hat{T}^0{}_\mu(x) \tag{6}$$

is independent of time x^0, i.e. conserved. Verify that $\hat{P}^0$ is the Hamiltonian and $\hat{\vec{P}}$ is the total spatial momentum operator as mentioned in the first lecture.

Also verify that the boost-angular-momentum operator

$$\hat{J}^{\mu\nu} = \int d^{D-1}\vec{x}\left[x^\mu\hat{T}^{0\nu}(x) - x^\nu\hat{T}^{0\mu}(x)\right] \tag{7}$$

is conserved.

[49] In our convention, the Minkowskian metric $(\eta_{\mu\nu})$, as well as its inverse matrix $(\eta^{\mu\nu})$, is given by $\mathrm{diag}\{-1, 1, \cdots, 1\}$.

(c) In the lecture notes we asserted the Poincaré transformation property of $\hat{\phi}(x)$,

$$\hat{\phi}(\Lambda x + a) = U(\Lambda, a)\, \hat{\phi}(x)\, (U(\Lambda, a))^{-1}. \tag{8}$$

Verify this relation for the infinitesimal transformation and its corresponding unitary operator

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu}, \qquad U(\Lambda, a) = 1 - i a^{\mu} \hat{P}_{\mu} + \frac{i}{2} \omega_{\mu\nu} \hat{J}^{\mu\nu}, \tag{9}$$

where $\omega_{\mu\nu}$ and a^{μ} are infinitesimal constants (i.e. we keep only first order terms in the Taylor series in ω and a). *Hint: you may simplify your computations making use of the result from step* ***(a)****.*

1.4.2 Problem 2: Thermal partition function

Consider a free non-relativistic particle moving on a circle of radius R, whose Lagrangian is

$$L(q, \dot{q}) = \frac{1}{2} \dot{q}^{\,2}, \tag{10}$$

where the coordinate q is periodically valued, namely

$$q \sim q + 2\pi R. \tag{11}$$

In this problem you will study the energy spectrum by analyzing the thermal partition function[50]

$$Z(\beta) = \mathrm{Tr}\, e^{-\beta \hat{H}} = \int dq\, \langle q | e^{-\beta \hat{H}} | q \rangle, \tag{12}$$

where the trace is taken over the Hilbert space of the system, using the Euclidean path integral representation

$$Z(\beta) = \int [Dq] \exp \left[- \int_0^{\beta} d\tau\, \frac{1}{2} \left(\partial_{\tau} q \right)^2 \right], \tag{13}$$

where $q(\tau)$ is subject to the boundary condition

$$q(\beta) = q(0) \mod 2\pi R. \tag{14}$$

(a) A general real function $q(\tau)$ subject to (14) may be expanded in terms of Fourier modes as

$$q(\tau) = 2\pi R\, w\, \frac{\tau}{\beta} + \sum_{k=-\infty}^{\infty} q_k\, e^{2\pi i k \tau / \beta}, \tag{15}$$

where w is an integer, q_k's for $k \neq 0$ are complex coefficients that obey $q_{-k} = (q_k)^*$, and q_0 is a real periodically valued parameter, $q_0 \sim q_0 + 2\pi R$. One might attempt to replace the functional integration with integration over q_k's and sum over w:

$$\int [Dq] \overset{?}{\longrightarrow} \sum_{w=-\infty}^{\infty} \int_0^{2\pi R} dq_0 \int \prod_{k=1}^{\infty} dq_k\, dq_k^*. \tag{16}$$

[50] We have henceforth set $\hbar$ to 1.

In fact, this is almost correct, up to certain normalization factors that depend on β. To determine this normalization, one can start with the Hamiltonian form of the path integral,

$$Z(\beta) = \int Dp\, Dq\ \exp\left[\int_0^\beta d\tau\, \left(ip\,\partial_\tau q - H(p,q)\right)\right], \tag{17}$$

where $q(\tau)$ obeys (14) while $p(\tau)$ obeys $p(\beta) = p(0)$, and express the functional integral as one over the Fourier modes of q and p in the form

$$\int Dp\, Dq\ \to\ \mathcal{N} \sum_w \int \prod_k \frac{dq_k\, dp_k}{2\pi}, \tag{18}$$

where $\mathcal{N}$ is assumed to be independent of β. Your task is to perform the integral over q_k, p_k, and express the result in the form

$$Z(\beta) = \mathcal{N} \sum_{w=-\infty}^{\infty} f(w), \tag{19}$$

for some function $f(w)$ (which also depends on β).

(b) Using *Poisson resummation formula*

$$\sum_{w=-\infty}^{\infty} f(w) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} dw\ e^{2\pi i n w}\, f(w), \tag{20}$$

rewrite your result for (19) as a sum over n. Compare with what you expect from the eigenvalues of the quantum Hamiltonian of this system. What is the value of the normalization factor $\mathcal{N}$?

Remark: the RHS of (18), while reasonable looking, isn't quite the same as the integration over p, q at discrete time steps in the original derivation of the path integral from Hamiltonian formalism. The two different ways of doing the functional integral differ by a Jacobian factor. This difference can be absorbed by shifting the Lagrangian by a constant counter term that depends on the time step. You are not required to analyze this Jacobian factor, but you may have extra fun by doing so.

Problem 1 answer:

(a) Recall that the field operator in d-dimensions is

$$\hat{\phi}(x) = \int \frac{d^{d-1}k}{\sqrt{(2\pi)^{d-1}2\omega_{\vec{k}}}} \left(a_{\vec{k}} e^{i\vec{k}\cdot\vec{x} - i\omega_{\vec{k}}x^0} + a^\dagger_{\vec{k}} e^{-i\vec{k}\cdot\vec{x} + i\omega_{\vec{k}}x^0} \right) \tag{1.683}$$

where the 4-position-vector is $x^\mu = (x^0 \equiv t, \vec{x})$ and $\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. The creation and annhilation obeys the following commutator

$$\left[a_{\vec{k}}, a^\dagger_{\vec{p}} \right] = \delta^{(d-1)}(\vec{k} - \vec{p}) \tag{1.684}$$

and all other commutators vanish[51]. To compute the equal time commutator, $[\hat{\phi}(x), \hat{\phi}(y)]$ where we take the commutators to be at the same time, i.e. $x^0 = y^0$, we do not want to run head-first into what can be a long computation. We know that there are 4 possible products of a's. The only commutators that survive are $[a_{\vec{k}}, a^\dagger_{\vec{p}}]$ and $[a_{\vec{p}}, a^\dagger_{\vec{k}}]$. These omit delta-functions allowing us to integrate over $d^{d-1}\vec{p}$ giving us just an integral in $d^{d-1}\vec{k}$. The commutator is then

$$\left[\hat{\phi}(x), \hat{\phi}(y) \right] = \int \frac{d^{d-1}\vec{k}}{(2\pi)^{d-1}2\omega_{\vec{k}}} \left(e^{i\vec{k}\cdot(\vec{x}-\vec{y})} - e^{-i\vec{k}\cdot(\vec{x}-\vec{y})} \right). \tag{1.685}$$

Notice that this integrand is odd under $\vec{k} \to -\vec{k}$ flip while the integration domain is symmetric, and thus the integral will vanish.

In a similar fashion, only the cross term in the commutator will survive with respect to the time derivative of the field, and thus we find

$$\begin{aligned} \left[\hat{\phi}(t,\vec{x}), \partial_t \hat{\phi}(t,\vec{y}) \right] &= \int \frac{d^{D-1}k}{(2\pi)^{D-1}} \frac{i}{2} \left(e^{i\vec{k}\cdot(\vec{x}-\vec{y})} + e^{-i\vec{k}\cdot(\vec{x}-\vec{y})} \right) \\ &= i \int \frac{d^{D-1}k}{(2\pi)^{D-1}} e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \\ &= i\delta^{(D-1)}(\vec{x} - \vec{y}). \end{aligned} \tag{1.686}$$

(b) The energy momentum tensor for the scalar is

$$\hat{T}_{\mu\nu}(x) = :\partial_\mu \hat{\phi}(x) \partial_\nu \hat{\phi}(x): - \frac{1}{2}\eta_{\mu\nu} \left(:(\partial\hat{\phi})^2(x): + m^2 :\hat{\phi}(x)^2: \right) \tag{1.687}$$

where :...: is a signal to take all of the $a^\dagger$'s and move them to the left of a's. The goal is to check $\partial_\mu \hat{T}^{\mu\nu} = 0$. Note that when moving all of the indices up on the energy momentum tensor (EMT) that there is no sign change on any of the terms, but there *is* if you only lower one of the indices, then there is a sign change.

There are 3-ways to perform this problem. The first is to just assume that the classical conservation equation will obey the quantum mechanical one. For this case,

[51] The commutator is defined as $[A, B] = AB - BA$.

this is ok since it is true, but in general it does not have to be. The second way is to insert the field operators as given at a position x^μ of the form (1.683) and simplify using the commutator (1.684). This path helps greatly when finding $\hat{P}^\mu$ and $\hat{J}_{\mu\nu}$. The third and most rigorous method is a combination of the first two. The best calculational method is by computing $\partial_\mu \langle \hat{T}^{\mu\nu}(x) \rangle = 0$. This is what is known as an *operator equation* and is the most rigorous form of showing that a classic conservation equation is true in the quantum mechanical version, but this is **not** always the case [52].

The first method can be checked as follows

$$\begin{aligned}
\partial_\mu T^{\mu\nu} &= \Box\phi\partial^\nu\phi + \partial_\mu\phi\partial^\mu\partial^\nu\phi - \frac{1}{2}\left(2\partial^\rho\phi\partial^\mu\partial_\rho\phi + 2m^2\phi\partial^\nu\phi\right) \\
&= m^2\phi(\partial^\nu\phi) + \partial_\mu\phi\partial^\mu\partial^\nu\phi - \left(\partial^\rho\phi\partial^\mu\partial_\rho\phi + m^2\phi\partial^\nu\phi\right) \\
&= 0.
\end{aligned} \tag{1.688}$$

Note that the Klein-Gordon equation here, the equation of motion for a scalar field, is given as $(\Box - m^2)\phi = 0 \implies \Box\phi = m^2\phi$ which makes the first and fourth term vanish. The second and third term vanish since we can always re-name summed over indices i.e change the indices on the second term from $\mu \to \rho$ and then everything cancels.

The second way can seem rather a bore and tedious. However, we can actually guess the form of $\hat{T}^{\mu\nu}$ and then check the answer. $T^{\mu\nu}$ is a 2-index tensor and the only Lorentz invariant quantities we have are the 4-momentum p_μ. So, we can guess that

$$\hat{T}^{\mu\nu}(x) = \int \frac{d^d p}{(2\pi)^{d-1}\omega_{\vec{p}}} p^\mu p^\nu a^\dagger_{\vec{p}} a_{\vec{p}}. \tag{1.689}$$

Then, the derivative (divergence) of (1.689) can be computed. Instead of brute-forcing our way, notice that the fourier transform of $\partial_\mu \leftrightarrow -i\partial^\mu$, and then in the integrand of $d^{d-1}p$ we have $\propto p^\nu p^2$ with an integral over all of the possible momenta from $(-\infty, \infty)$ giving us zero.

The third way is reserved for QFT 2...

The 4-momentum operator can then be read of from (1.689) as

$$\hat{P}^\mu = \int \frac{d^{d-1}\vec{p}}{2(2\pi)^{d-1}\omega_{\vec{p}}} p^\mu a^\dagger_{\vec{p}} a_{\vec{p}}. \tag{1.690}$$

The Hamiltonian *operator* is then just the p^0 component being $\omega_{\vec{p}}$ and the 3-spatial momentum operator is then just $\vec{p}$ inserted.

If you did is the classical way you should have found that

$$H = \int d^{d-1}x \left(\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2\right) \tag{1.691}$$

$$\vec{p} = \int d^{d-1}x \dot{\phi}\nabla\phi. \tag{1.692}$$

[52] When this is not the case, i.e. that $\partial\langle\hat{T}^{\mu\nu}(x)\rangle \neq 0$ then the symmetry is *anomalous*.

Conservation of the angular momentum operator then follows from (1.689) in a straightforward manner since in terms of operators it is given as

$$\hat{J}^{\mu\nu} = i \int \frac{d^{d-1}\vec{p}}{2(2\pi)^{d-1}\omega_{\vec{p}}} \left(p^\mu \partial^\nu a^\dagger_{\vec{p}} a_{\vec{p}} - p^\nu \partial^\mu a^\dagger_{\vec{p}} a_{\vec{p}} \right). \tag{1.693}$$

You can of course Fourier tranform the partial derivatives in (1.693) to get a more clean version.

Also, take note that the angular momentum operator is conserved, namely that

$$\frac{d}{dt}\hat{J}^{\mu\nu} = \int d^{D-1}x \left(\delta^\mu_0 \hat{T}^{0\nu} - \delta^\nu_0 \hat{T}^{0\mu} + x^\mu \partial_0 \hat{T}^{0\nu} - x^\nu \partial_0 \hat{T}^{0\mu} \right) \tag{1.694}$$

from which, using $\partial_0 \hat{T}^{0\nu} = -\partial_i \hat{T}^{i\nu}$ since the energy-momentum tensor is conserved, after integrating by parts and dropping a boundary term, you will find that $d\hat{J}^{\mu\nu}/dt = 0$.

(c) Given the form for U above, to first order, the inverse matrix is

$$U^{-1} = 1 + ia^\mu \hat{P}_\mu - \frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu} + \cdots \tag{1.695}$$

Then, the transformation of the scalar field $\phi(x)$ can be written as

$$U\hat{\phi}(x)U^{-1} = \hat{\phi}(x) + \left[-ia^\mu \hat{P}_\mu + \frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu}, \hat{\phi}(x) \right]. \tag{1.696}$$

With these, we can very that this is indeed the correct transformation rule for infinitesimal transformations by checking that the transformation can be read off with the commutators for the generators.

With respect to translations, from part (a) using $[\phi, \pi] = i\delta^{D-1}(x-y)$, we also have that $\hat{P}_0 = -\hat{H}$ and $\hat{P}_i = \hat{P}^i$ as in part (b), and thus we get

$$[\hat{P}_\mu, \hat{\phi}(x)] = i\partial_\mu \hat{\phi}(x). \tag{1.697}$$

With respect to the Lorentz transformations, we have that

$$[\hat{J}^{\mu\nu}, \hat{\phi}(x)] = i(x^\mu \partial^\nu - x^\nu \partial^\mu)\hat{\phi}(x). \tag{1.698}$$

From which we can plug-into the unitary transformation of the scalar

$$U\hat{\phi}(x)U^{-1} = \hat{\phi}(x) + a^\mu \partial_\mu \hat{\phi}(x) + \frac{1}{2}\omega_{\mu\nu}(x^\mu \partial^\nu - x^\nu \partial^\mu)\hat{\phi}(x). \tag{1.699}$$

Since $\omega_{\mu\nu}$ is antisymmetric, we can rewrite the last terms as $\omega^\mu{}_\nu x^\nu \partial_\mu$, and thus we find that

$$U\hat{\phi}(x)U^{-1} = \hat{\phi}(x) + (a^\mu + \omega^\mu{}_\nu x^\nu)\partial_\mu \hat{\phi}(x) = \hat{\phi}(x + \delta x) \tag{1.700}$$

where we identify $\delta x^\mu = a^\mu + \omega^\mu{}_\nu x^\nu$.

Problem 2 answer:

(a) Let me first re-iterate that the mode functions for $q(\tau)$ and $p(\tau)$ can be written in the following manner

$$q(\tau) = \frac{2\pi R w \tau}{\beta} + \sum_{k=-\infty}^{\infty} q_k e^{2\pi i k \tau/\beta} \tag{1.701}$$

$$p(\tau) = \sum_{k=-\infty}^{\infty} p_k e^{2\pi i k \tau/\beta}. \tag{1.702}$$

Now let me explain. The $q(\tau)$ mode expansion arises from requiring that the position of the particle on a circle obeys $q(\beta) = q(0) \mod 2\pi R$, of which you can check (i.e. set $q(\tau = \beta)$ and watch the magic). The mode expansion for the momentum $p(\tau)$ in (1.702) may seem weird, but recall that the problem stated that $p(\tau) = p(\beta)$ and if we had included a finite part to $p(\tau)$'s mode expansion (something that is modular) then the boundary condition would not be correct (or vice versa with respect to the mode expansion). Also, if the momentum had a modular part on it, i.e. of the form $p(\tau) \supset 2\pi R w \tau/\beta$ then $q(\tau)$ would have to have a τ^2 dependence, which is weird (in non-relativistic mechanics). q_k obeys $q_{-k} = (q_k)^*$ and similar for p_k. The path integral can be written as (in terms of the position and conjugate momenta)

$$Z(\beta) = \int [Dq][Dp] \exp\left(\int_0^\beta d\tau \left(ip\partial_\tau q - H(p,q)\right)\right). \tag{1.703}$$

The Hamiltonian is just that of a free particle, where the Lagrangian is given as $\frac{1}{2}\dot{q}^2$ and so via $H = p \cdot q - L \implies H = p^2/2$. The action in (1.703) can be written as

$$S = \int_0^\beta d\tau \left(ip(\tau)\dot{q}(\tau) - \frac{1}{2}p^2(\tau)\right). \tag{1.704}$$

The action (1.704) when plugging in (1.702) and (1.701) becomes

$$S = \int_0^\beta d\tau \left(i\sum_{k_1} p_{k_1} e^{2\pi i k_2 \tau/\beta} \cdot \left(\frac{2\pi R w}{\beta} + \sum_{k_2} \frac{2\pi i k_2}{\beta} q_{k_2} e^{2\pi i k_2 \tau/\beta}\right) - \frac{1}{2}\sum_k p_k p_{-k} e^{\frac{2\pi i \tau}{\beta}(k+(-k))}\right). \tag{1.705}$$

The Hamiltonian on the far RHS of (1.705) is an integral of the form $\int_0^\beta d\tau e^{2\pi i(k+m)\tau/\beta}$ which is either $= \beta$ if $k + m = 0$ or 0 otherwise. We can then perform the $d\tau$ integration in (1.705) to find

$$S = 2\pi i R w p_0 - 2\pi \sum_{k\neq 0} k p_k q_{-k} - \frac{\beta}{2} p_k p_{-k} \tag{1.706}$$

where when we integrate over the $k \neq 0$ modes, the exponential gives us a delta function $\delta(p_k)$ effectively killing all modes functions not p_0. I should clarify, this

delta-function arises from integrating over

$$\int dq_k e^{2\pi i k p_k q_{-k}} \tag{1.707}$$

which is the path-integral measure integration. This integration just omits a delta function $\delta(p_k)$ and the reasoning above follows.

Now we integrate over the remaining p_0 modes. The integration is just a Gaussian integral as follows

$$\begin{aligned} Z(\beta) &= \mathcal{N} \sum_{w=-\infty}^{\infty} \int \frac{dp_0}{2\pi} \exp\left(2\pi i R w p_0 - \frac{\beta}{2} p_0^2\right) \\ &= \frac{\mathcal{N}}{\sqrt{2\pi\beta}} \sum_{w=-\infty}^{\infty} \exp\left(-\frac{2\pi^2 R^2 w^2}{\beta}\right). \end{aligned} \tag{1.708}$$

We can absorb the numerical factor into $\mathcal{N}$ if we wish to.

(b) Recall that the Poisson resummation formula is given as

$$\sum_{-\infty}^{\infty} f(w) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} dw\, e^{2\pi i n w} f(w). \tag{1.709}$$

Based on our integration in (1.708), we have to evaluate the following integral expression

$$\int_{-\infty}^{\infty} dw\, e^{2\pi i n w} \exp\left(-\frac{2\pi^2 R^2 w^2}{\beta}\right). \tag{1.710}$$

This is just a Gaussian integral of the form

$$\int_{-\infty}^{\infty} dx\, e^{-Ax^2+Bx} = \sqrt{\frac{\pi}{A}} \exp\left(\frac{B^2}{4A}\right). \tag{1.711}$$

For us, $A = 2\pi^2 R^2/\beta$ and $B = 2\pi i n$. If we plug this into (1.711), simplify the integral, and recast (1.708) into the Poisson resummation formula (4.5) we get

$$Z(\beta) = \mathcal{N}\sqrt{\frac{\beta}{2\pi R^2}} \sum_{n=-\infty}^{\infty} \exp\left(-\frac{\beta n^2}{2R^2}\right). \tag{1.712}$$

We can compare this to the typical QM-partition function where in the QM version we get $\mathcal{N}\sqrt{\beta/2\pi R^2} = 1$ (where the momentum for a free particle on a circle is $p = n/R$). If we solve for $\mathcal{N}$ then $\mathcal{N} = \sqrt{2\pi R^2/\beta}$. However, we *assumed* $\mathcal{N}$ is independent of β. This is where the absorbing of the factors in (1.708) comes back to haunt us. If we trace back to when we absorbed said factors, then $\mathcal{N} = 2\pi R$ which confirms our assumtion. This is just a lesson of when *and when not to* absorb factors and throw them away.

1.5 Problem set 2:

1.5.1 Problem 1: Quantum mechanical anharmonic oscillator

The quantum anharmonic oscillator is defined through the Hamiltonian

$$H = \frac{1}{2}\hat{p}^{\,2} + \frac{1}{2}\hat{q}^{\,2} + \frac{1}{4!}\,g\,\hat{q}^{\,4}. \tag{1}$$

In class we discussed the path integral representation of the system with the Lagrangian

$$L(q,\dot{q}) = \frac{1}{2}\dot{q}^{\,2} - \frac{1}{2}q^2 - \frac{1}{4!}\,g\,q^4, \tag{2}$$

or more conveniently, the Euclidean path integral with the Euclidean Lagrangian

$$L^E(q,\partial_\tau q) = \frac{1}{2}(\partial_\tau q)^2 + \frac{1}{2}q^2 + \frac{1}{4!}\,g\,q^4, \tag{3}$$

where the Euclidean time τ is related to the real time t by Wick rotation $t = -i\tau$. In this problem you will analyze the two-point Euclidean Green function $G(\tau)$, defined as

$$G(\tau) \equiv \begin{cases} \langle 0|\hat{q}(\tau)\hat{q}(0)|0\rangle, & \tau \geq 0, \\ \langle 0|\hat{q}(0)\hat{q}(\tau)|0\rangle, & \tau < 0, \end{cases} \tag{4}$$

where $|0\rangle$ stands for the ground state of the system, perturbatively through its path integral representation

$$G(\tau) = \frac{\displaystyle\int [Dq]\; e^{-\int d\tau\, L^E}\; q(\tau)\, q(0)}{\displaystyle\int [Dq]\; e^{-\int d\tau\, L^E}}. \tag{5}$$

(a) Compute $G(\tau)$, or its Fourier transform $\tilde{G}(k)$ related by

$$G(\tau) = \int \frac{dk}{2\pi}\,\tilde{G}(k)\,e^{ik\tau}, \tag{6}$$

at order g^2, using Feynman diagrams as described in class.

(b) In class, we argued that $\tilde{G}(k)$ takes the form

$$\tilde{G}(k) = \frac{1}{k^2 + 1 - \Sigma(k)}, \tag{7}$$

where $\Sigma(k)$ is computed by 1PI diagrams and that the poles of $\tilde{G}(k)$ on the positive imaginary k-axis correspond to possible excitation energies of the state $\hat{q}(0)|0\rangle$. What can you conclude about the energy of excited states using the expression for $\Sigma(k)$ up to order g^2?

Remark: You could have extra fun by checking your answer against Hamiltonian perturbation theory to second order, or even by solving Schrödinger's equation numerically, but you are not required to do so.

1.5.2 Problem 2: Mass spectrum of ϕ^3-theory

Consider a scalar field theory in D dimensions, defined through the path integral representation with the action

$$S[\phi] = \int d^D x \left[-\frac{1}{2}\, \eta^{\mu\nu}\, \partial_\mu \phi(x)\, \partial_\nu \phi(x) - \frac{1}{2}\, m^2 (\phi(x))^2 - \frac{1}{3!}\, g (\phi(x))^3 \right], \tag{8}$$

where $x^\mu = (x^0, x^1, \ldots, x^{D-1})$ and $(\eta^{\mu\nu}) = \mathrm{diag}(-1, 1, \ldots, 1)$, or more conveniently, the Euclidean path integral with the Euclidean action[53]

$$S^E[\phi] = \int d^D x \left[\frac{1}{2} (\partial_\mu \phi(x))^2 + \frac{1}{2}\, m^2 (\phi(x))^2 + \frac{1}{3!}\, g (\phi(x))^3 \right], \tag{9}$$

where $x = (x^1, \ldots, x^D)$, x^D being the Euclidean time related to the real time x^0 by the Wick rotation $x^0 = -i x^D$.

(a) We begin by analyzing the vacuum expectation value

$$\langle \phi \rangle \equiv \langle 0 | \hat{\phi}(x) | 0 \rangle = \frac{\displaystyle\int [D\phi]\; e^{-S^E[\phi]}\, \phi(x)}{\displaystyle\int [D\phi]\; e^{-S^E}}, \tag{10}$$

where $|0\rangle$ stands for the vacuum state, $\hat{\phi}(x)$ is a local field operator defined through the path integral quantization. Calculate $\langle \phi \rangle$ at order g. You should find a divergent result whenever $D \geq 2$. We will cure this problem by carefully defining the path integral using the following regularization:

$$[D\phi] \;\longrightarrow\; \prod_{|k|<\Lambda} d\tilde{\phi}(k), \tag{11}$$

where $\tilde{\phi}(k)$ are the Fourier modes of $\phi(x)$. This will effectively cut off the momentum integral in the diagram for $\langle \phi \rangle$ and result in an answer that depends on the cutoff Λ.

At this point we could proceed by redefining $\phi'(x) = \phi(x) - \langle \phi \rangle$, and work with $\phi'(x)$ which has vanishing vacuum expectation value. Alternatively, we may modify the action S^E by adding a counter term

$$\Delta S^E = \int d^D x\, c_1 \phi(x), \tag{12}$$

such that $\langle \phi \rangle = 0$ after taking into account the contribution from this counter term. These two procedures are equivalent up to a redefinition of m^2. It will be convenient to adopt the second point of view, and define the theory with the regularization (11) and the counter term (12) included in the action, and take the limit $\Lambda \to \infty$ in the end.

(b) Now you will analyze the two-point Euclidean Green function $G(x)$, defined as

$$G(x) \equiv \begin{cases} \langle 0 | \hat{\phi}(x) \hat{\phi}(0) | 0 \rangle, & x^D \geq 0, \\ \langle 0 | \hat{\phi}(0) \hat{\phi}(x) | 0 \rangle, & x^D < 0, \end{cases} \tag{13}$$

[53] This Euclidean action is not bounded from below due to the cubic coupling, but we will ignore this issue while working only perturbatively in g.

through its path integral representation

$$G(x) = \frac{\int [D\phi]\; e^{-S^E[\phi]}\, \phi(x)\phi(0)}{\int [D\phi]\; e^{-S^E}}. \tag{14}$$

Compute $G(x)$, or its Fourier transform $\tilde{G}(k)$ related by

$$G(x) = \int \frac{d^D k}{(2\pi)^D}\, \tilde{G}(k)\, e^{ik\cdot x}, \tag{15}$$

at order g^2. You should find a well-defined answer when $D = 2, 3$ (and a divergent result when $D = 4$).

(c) By similar diagrammatics as discussed in the anharmonic oscillator example, we expect $\tilde{G}(k)$ here to take the form

$$\tilde{G}(k) = \frac{1}{k^2 + m^2 - \Sigma(k)}, \tag{16}$$

and you have just determined $\Sigma(k)$ at order g^2 in step (b). Compare with the spectral decomposition

$$G(x) = \int d\alpha\, \langle 0|\hat{\phi}(0)|\alpha\rangle\, e^{i\vec{p}_\alpha\cdot\vec{x} - p^0_\alpha x^D}\, \langle\alpha|\hat{\phi}(0)|0\rangle \qquad (x^D > 0), \tag{17}$$

where $|\alpha\rangle$ is a basis of energy-momentum eigenstates that obey $\hat{P}^\mu|\alpha\rangle = P^\mu_\alpha|\alpha\rangle$, and the measure $d\alpha$ is normalized such that $\int d\alpha\, |\alpha\rangle\langle\alpha| = 1$. What can you conclude about the invariant masses of the states $|\alpha\rangle$ that contribute to this Green function?

Remark: unlike in problem 1 where $\tilde{G}(k)$ has only poles on the upper half imaginary k-plane, corresponding to discrete (normalizable) energy eigenstates, here $\tilde{G}(k)$ as a function of complex k^0 (say at fixed spatial momentum $\vec{k}$) may have poles as well as branch cuts, the latter corresponding to a continuum of (delta-function-normalizable) energy eigenstates.

Problem 1 answer:

The Euclidean Lagrangian for our quantum mechanical 4-point theory (i.e. a theory with an interaction of the form $\propto gq^4$) is given as

$$L^E(q, \partial_\tau q) = \frac{1}{2}\left(\partial_\tau q\right)^2 + \frac{1}{2}q^2 + \frac{1}{4!}gq^4 \tag{1.713}$$

where from now on, I shall denote $\partial_\tau q \equiv \dot{q}$ whenever it arises. We know what the Green's function is, or the corresponding 2-point function or the propagator, namley

$$G_0(\tau - \tau_1) = \int \frac{dk}{2\pi}\tilde{G}(k)e^{ik(\tau-\tau_1)} \tag{1.714}$$

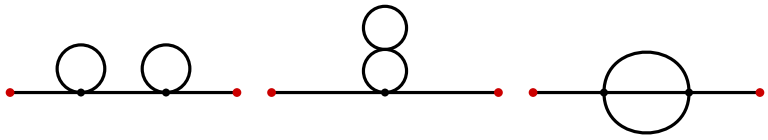
where the Fourier transformed Green's function $\tilde{G}(k)$ is given as

$$\tilde{G}(k) = \frac{1}{k^2+1}. \tag{1.715}$$

There are three diagrams that correspond to $\mathcal{O}(g^2)$ (order g^2) interactions for the 2-point function where the 2-point function we will use is

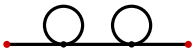
$$G(\tau) = \langle 0|\, \hat{q}(\tau)\hat{q}(0)\, |0\rangle \tag{1.716}$$

where (1.716) is valid for $\tau \geq 0$. The three diagrams are



(1.717)

comically called the "double-bubble" diagram, the "fish" diagram, and the "sunset" diagram. The double-bubble can be set-up as

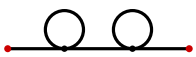


$$\begin{aligned} &= \left(-\frac{g}{2}\right)\int d\tau_1 d\tau_2 G_0(\tau-\tau_1)G_0(\tau_1-\tau_1)G_0(\tau_1-\tau_2)G_0(\tau_2-\tau_2)G_0(\tau_2-0) \\ &= \frac{g^2}{4}\int d\tau_1 d\tau_2 \int \frac{dk_1\cdots dk_5}{(2\pi)^5}\tilde{G}_0(k_1)e^{ik_1(\tau-\tau_1)}\tilde{G}_0(k_2)e^{ik_2(\tau_1-\tau_1)}\tilde{G}_0(k_3)e^{ik_3(\tau_1-\tau_2)}\tilde{G}_0(k_4)e^{ik_4\tau_2}\tilde{G}_0(k \end{aligned} \tag{1.718}$$

From (1.718), we can simplify the exponentials. The arguement of the exponential is of the form

$$\exp(ik_1\tau + i\tau_1(k_3-k_1) + i\tau_2(k_4-k_3)). \tag{1.719}$$

From (1.719) we can get two delta-functions of the form $\delta(k_4-k_3)\delta(k_3-k_1)$ where we can first integrate over k_1 to only get k_3 leftover, while the other delta function allows us to send $k_3 \to k_4$. We then have



$$\begin{aligned} &= \frac{g^2}{4}\int \frac{dk_2 dk_4 dk_5}{(2\pi)^3}e^{ik_4\tau}\tilde{G}_0(k_2)\left(\tilde{G}_0(k_4)\right)^3\tilde{G}_0(k_5) \\ &= \frac{g^2}{4(2\pi)^3}\pi^2\int dk\frac{e^{ik\tau}}{(k^2+1)^3} \end{aligned} \tag{1.720}$$

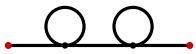
where we can integrate over $\tilde{G}_0(k_i)$ via the residue theorem namely

$$\int \frac{dk}{k^2+1^2} = \int \frac{dk}{(k-i)(k+i)} = 2\pi i \mathrm{Res}_{k\to i}\frac{1}{k+i} = 2\pi i\left(\frac{1}{2i}\right) = \pi. \tag{1.721}$$

To perform the remaining contour integral in (1.720), we can turn to *Mathematica*. There is a `Residue[]` function in *Mathematica* that we can tell to take the residue of around i, but make sure to multiply by a factor of $2\pi i$ since it can compute the residue, but it knows nothing about the *Residue Theorem*. You should find that

$$\int dk \frac{e^{ik\tau}}{(k^2+1)^3} = \frac{\pi}{8}e^{-\tau}(3+3\tau+\tau^2). \tag{1.722}$$

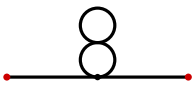
The final answer then is



$$= \frac{1}{8192}e^{-\tau}(3+3\tau+\tau^2). \tag{1.723}$$

However, with the other two diagrams we still have to compute, (1.723) makes absolutely no physical sense, and cannot contribute to the *quantum corrections* to the frequency (or mass) of the q-particle.

The next diagram is the "fish" which can be setup and computed in a similar manner

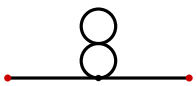


$$\begin{aligned} &= \frac{g^2}{4}\int d\tau_1 d\tau_2 G_0(\tau-\tau_1)G_0(\tau_1-\tau_2)G_0(\tau_2-\tau_2)G_0(\tau_2-\tau_1)G_0(\tau_1) \\ &= \frac{g^2}{4}\int d\tau_1 d\tau_2 \int \frac{dk_1\cdots dk_5}{(2\pi)^5}\tilde{G}_0(k_1)e^{ik_1(\tau-\tau_1)}\tilde{G}_0(k_2)e^{ik_2(\tau_1-\tau_2)}\tilde{G}_0(k_3)e^{ik_3(\tau_2-\tau_2)}\tilde{G}_0(k_4)e^{ik_4(\tau_2-\tau_1)} \end{aligned} \tag{1.724}$$

Hopefully we know the rules of the game for quantum mechanical loops, and so we know to now integration over $d\tau_1 d\tau_2$ from the exponentials which are of the form

$$e^{ik_1\tau}e^{i\tau_1(k_2-k_2-k_1)}e^{i\tau_2(k_2-k_2+k_4)}. \tag{1.725}$$

(1.725) simply tell us that the momentum of $k_1 = k_4 = 0$ since we get $\delta(k_1)\delta(k_4)$. The Green functions then containing those terms simply becomes 1. We are then left with

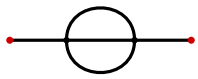


$$\begin{aligned} &= \frac{g^2}{4}\int \frac{dk_2 dk_3 dk_5}{(2\pi)^3}\tilde{G}_0(k_2)\tilde{G}_0(k_3)\tilde{G}_0(k_5) \\ &= \frac{g^2}{32} \end{aligned} \tag{1.726}$$

where in (1.726) we performed the same integration as prior.

Finally for the sunrise diagram, this is where things get tricky, but there are self-checks built-in place. I will be taking **all of the momentum going to the right**. This implies then at the vertices, for example, that at τ_1: $k_1 = k_2+k_3+k_4$ while at τ_2: $k_2+k_3+k_4 = k_5$,

but we should have $k_1 = k_5$. Lets see if our integrals tell us this. The sunrise (sr) diagram is



$$= \frac{g^2}{4}\int d\tau_1 d\tau_2 G_0(\tau-\tau_1)G_0(\tau_1-\tau_2)G_0(\tau_1-\tau_2)G_0(\tau_1-\tau_2)G_0(\tau_2)$$

$$= \frac{g^2}{4}\int d\tau_1 d\tau_2 \int \frac{dk_1\cdots dk_5}{(2\pi)^5}\tilde{G}_0(k_1)e^{ik_1(\tau-\tau_1)}\tilde{G}_0(k_2)e^{ik_2(\tau_1-\tau_2)}$$
$$\times \tilde{G}_0(k_3)e^{ik_3(\tau_1-\tau_2)}\tilde{G}_0(k_4)e^{ik_4(\tau_1-\tau_2)}\tilde{G}_0(k_5)e^{ik_5\tau_2}. \tag{1.727}$$

The argument of the exponentials give

$$e^{ik_1\tau}e^{i\tau_1(k_2-k_1+k_3+k_4)}e^{i\tau_2(k_5-k_2-k_3-k_4)}. \tag{1.728}$$

We can see we get two delta-functions $\delta(k_2-k_1+k_3+k_4)\delta(k_5-k_2-k_3-k_4)$ which enforces that $k_2 = k_1-k_3-k_4$ from the first one which when we plug into the second delta function we get $\delta(k_5-k_1+k_3+k_4-k_3-k_4) = \delta(k_5-k_1)$ which enforces that $k_1 = k_5$ just as we had hoped! Note that there are different schemes for setting up the momentum of the loop integration (in QM and QFT), but what matters is that the momentum *on-shell* is conserved always no matter what scheme you choose for the loop momenta (and by scheme, I mean direction). The algebra is tedious for the delta-functions, so I recommend putting them into *Mathematica*. The final answer is

$$\text{“sr”} = \frac{g^2}{8}\int \frac{dk}{2\pi}\frac{e^{ik\tau}}{k^2+9}. \tag{1.729}$$

The final integration just gives $\frac{\pi}{3}e^{-3\tau}$.

For part (b), we can re-sum all of the diagrams we have computed into the form of

$$\tilde{G}(k) = \frac{1}{k^1+1-\sum(k)} \tag{1.730}$$

where $\sum(k)$ contains all higher order loop corrections of the order $\mathcal{O}(g^n)$ for $n \geq 1$. The $\sum(k)$ however only contains terms that are **not disconnected** i.e. diagrams that remain connected after cutting any one-Green's function. The 1PI is formally just the loop diagrams without the external legs (i.e. without the factors of $G_0(\tau-\tau_1)$ and $G_0(\tau_2)$. This gives a momentum integral leftover allowing us to find a geometric sum of terms.

I should clarify, the only reason we consider terms that are 1PI to be associated to corrections to the frequency ω (which has been 1 the entire time in our calculations) is because non-1PI diagrams are typically acausal in the sense that they do not decay in the UHP of complex τ fast enough.

The final answer then for $\sum(k)$ is then

$$\sum(k) = -\frac{g}{4}+\frac{g^2}{8}\left(\frac{1}{4}+\frac{1}{k^2+9}\right)+\mathcal{O}(g^3) \tag{1.731}$$

where $-g/4$ comes from the 1-loop correction computed in lecture. This correction correspond to the pole near $k \approx i$ with

$$\omega_1 = 1+\frac{g}{8}-\frac{g^2}{32}+\mathcal{O}(g^3). \tag{1.732}$$

ω_1 corresponds to a **bound state** since $\omega_n = E_n - E_0$ Hamiltonian quantum mechanics. *This is an important fact!* **Bound states correspond to poles along the imaginary axis of the momentum!** Therefore, we can predict the bound state energies at any correction.

Problem 2 answer:

For this case, we are doing almost the *exact* same computation as in the first problem, but with a $\lambda\phi^3/3!$ interaction and not a gq^4 interaction. So, the diagrams will be different.

The 2-point Green's function in this case is the *scalar propagator* which can be solved to give

$$G(x,y) = \int \frac{d^D p}{(2\pi)^D} \frac{e^{ip\cdot(x-y)}}{p^2+m^2} \tag{1.733}$$

where we now are using D-dimensional spacetime vectors i.e. $x^\mu = (x^0, ..., x^D)$.

The only diagram that is available at $\mathcal{O}(g)$ is called the *tadpole* diagram. It is a particle that is created at a *singular point* and goes nowhere. This is *extremely* unpleasant since this obviously is not something that should occur, and requires a process called *regularization* which we perform with *counter-terms* just as you saw previously.

So, the tadpole diagram is a particle that is created at a singular point with momentum p^μ and creates a loop with momentum k^μ, but the particle has to vanish, so $p^\mu = 0$! This is bizarre, but these terms **do** occur in the expansion of the path integral, so we have to care about them. The setup for the tadpole is the following (with the particle being created at point y (the loop starts at one point, say z, and comes back to the same point)

$$\begin{aligned}
\langle\phi\rangle &= -\frac{g}{2}\int d^D y G(x,y)G(0) \\
&= -\frac{g}{2}\int d^D y \int \frac{d^D p}{(2\pi)^D}\frac{e^{ip\cdot(x-y)}}{p^2+m^2}\int \frac{d^D k}{(2\pi)^D}\frac{1}{k^2+m^2} \\
&= -\frac{g}{2}\int \frac{d^D k}{(2\pi)^D}\delta^{(D)}(p)\frac{e^{ip\cdot(x-y)}}{p^2+m^2}\frac{1}{k^2+m^2} \\
&= -\frac{g}{2}\int \frac{d^D k}{(2\pi)^D}\frac{1}{m^2}\frac{1}{k^2+m^2} \\
&= -\frac{g}{2m^2}\int d\Omega_D \int dk \frac{k^{D-1}}{k^2+m^2} \\
&= -\frac{g}{2m^2}\frac{2\pi^{D/2}}{\Gamma(D/2)}\int dk \frac{k^{D-1}}{k^2+m^2}.
\end{aligned} \tag{1.734}$$

Going from top to bottom to arrive at the form of (1.734), the first line is just the integral traveling over the position the particle takes from propagation to the loop. The next line is the Fourier transformed version of $G(x,y)$ which is $\tilde{\mathbb{G}}(k) = (k^2+m^2)^{-1}$. The next line is performing the $d^D y$ and $d^D p$ integral simultaneously where $(2\pi)^D\delta^{(D)}(p) = \int d^D y\, e^{ip\cdot y}$ (there is a leftover $e^{ip\cdot x}$ but this is just zero thanks to the delta-function). The next lines are simplifications where $\int d\Omega_D = \Omega_D$ which is the volume element of the D-sphere which is the quoted answer. We then arrive at a final answer. Notice that this is just an integral over k, which diverges when $k = 2$ since we acquire a logarithmic divergence. Thus, anything beyond $D \geq 2$ diverges. To remedy this, we regularize the differential of the path integral via

$$[D\phi] = \prod_{|k|<\Lambda} d\tilde{\phi}(k) \tag{1.735}$$

where Λ is some cutoff that dictates our ultraviolet or UV-physics (i.e. it is a point found via experiment for physically reasonable processes; the tadpole is not physical). The regularization of (1.735) simply changes (1.734) to

$$\langle\phi(x)\rangle = -\frac{g}{2m^2}\frac{2\pi^{D/2}}{\Gamma(D/2)}\int\limits_{|k|<\Lambda}^{\Lambda} dk\frac{k^{D-1}}{k^2+m^2}. \tag{1.736}$$

This regularizes the integral to give us something that $\propto \log(\Lambda^2/m^2)$, which still diverges. To finally cure this divergence that is not prominently coming from only our scheme, and not the physics, is by including a counter-term, since clearly the tadpole should vanish. To do so consider including in the Lagrangian $\Delta\mathcal{L}^E = c_\Lambda\phi$. This *solely* contributes to the tadpole or 1-point correlator and gives us

$$\langle\phi(x)\rangle = -\frac{c_\Lambda}{m^2} - \frac{g}{2m^2}G(0). \tag{1.737}$$

We want (1.737) to be 0, which if we enforce then we *choose* $c_\Lambda = -\frac{g}{2}G(0)$.

For part (b), there is a single diagram we need to contribute, *just* a bubble. Let us *just* focus on the bubble since in momentum space it directly contributes to $\sum(p)$ where p here is the physical momentum while I will take k to be the loop momentum (i.e. drop the external legs, they will never matter unless you put them off-shell to compute a potential). The loop integral can be written as

$$\sum(p) = (-g)^2\int\frac{d^Dk}{(2\pi)^D}\frac{1}{k^2+m^2}\frac{1}{((p-k)^2+m^2)}. \tag{1.738}$$

To handle this integral to put it into a known form, notice that

$$\frac{1}{AB} = \int\limits_0^1 dx\frac{1}{(A+(B-A)x)^2} \tag{1.739}$$

which when applied to (1.738) the denominator algebra becomes, with $A = k^2+m^2$ and $B = (p-k)^2+m^2$,

$$k^2+m^2+x((p-k)^2-k^2) = k^2+m^2+p^2x-2k\cdot px. \tag{1.740}$$

Notice that if we shift the momentum $k \to \ell + px$ where ℓ is now our *new* loop momentum variable since $d^Dk \to d^D(\ell+px) = d^D\ell$ since p is the physical momentum and x has its own differential, then (1.740) becomes

$$l^2+m^2-p^2(x-1)x = l^2+\Delta \tag{1.741}$$

where $\Delta \equiv m^2 - p^2(x-1)x$. (1.738) is then simply

$$\begin{aligned}\sum(p) &= g^2 \int_0^1 dx \int \frac{d^D\ell}{(2\pi)^D} \frac{1}{(p^2+\Delta)^2} \\ &= \frac{g^2}{(2\pi)^D} \int_0^1 dx \int d\Omega_D \int dk \frac{k^{D-1}}{(k^2+\Delta)^2} \\ &= \frac{g^2}{(2\pi)^D} \frac{2\pi^{D/2}}{\Gamma(D/2)} \frac{\Gamma(D/2)\Gamma((4-D)/2)}{2\Gamma(2)} \int_0^1 dx\, \Delta^{\frac{D-4}{2}} \end{aligned} \tag{1.742}$$

where

$$\int dk \frac{k^a}{(k^2+\Delta)^b} = \Delta^{\frac{a+1}{2}-b} \frac{\Gamma((a+1)/2)\Gamma(b-(a+1)/2)}{2\Gamma(b)}. \tag{1.743}$$

Notice that in (1.742) for $D = 1, 2, 3$ the integral does not diverge, but for $D = 4$ it does. Feel free to use *Mathematica* to graph the results for different dimensions and perform a numerical integration. Notice that the *integration* does not diverge for $D = 4$ but the Gamma-functions in front *do*.

Part (c) is lecture 10, but the main points are the following. No matter what you compute for a final answer, it will always be of the form of the following 1PI (in Euclidean signature)

$$\text{1PI of self-energy} \equiv \tilde{\mathbb{G}}(k) = \frac{1}{k^2 + m^2 - \sum(k)}. \tag{1.744}$$

Every propagator corresponds to a real particle where if $\sum(k) = 0$ then there is a simple pole at $k^2 = -m^2_{\text{physical}}$ where m_{physical} is the physical mass of the particle (what we measure). Notice that the 3-point interaction gives us the possibility at 1-loop to produce three real particles (due to the cubic interaction) which occurs when the incoming momentum of ϕ, k can decay into three particles with mass m. This means that the lowest possible start for a branch cut in $\sum(k^2)$ is

$$k^2 \geq (3m)^2 = 9m^2. \tag{1.745}$$

Then, the minimal possible energy to create such a triplet of particles in the center-of-mass (COM) frame is $\sqrt{p^2} = 3m$. (1.745) is what is called a *threshold*. Below this threshold, the scalar particle is stable with a well defined mass pole as stated prior. However above the threshold, the particle can decay via the ϕ^3 interaction giving an imaginary part in the self-energy.

The threshold is represented by a *branch cut* on the real axis of p which starts at $p^2 = 9m^2$ and goes to $p^2 \to \infty$.

1.6 Problem set 3:

1.6.1 Problem 1: Analytics of the Wightmann function

Consider in D-dimensional spacetime a free scalar field theory with the Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\,\partial^\mu\phi\,\partial_\mu\phi - \frac{1}{2}\,m^2\phi^2. \tag{1}$$

In class we have derived the Fourier representation of the time-ordered 2-point function

$$G(x,y) \equiv \langle\Omega|\,T\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle = -i\int\frac{d^Dk}{(2\pi)^D}\,\frac{e^{ik\cdot(x-y)}}{k^2+m^2-i\epsilon}, \tag{2}$$

where $k\cdot(x-y) = \vec{k}\cdot(\vec{x}-\vec{y}) - k^0(x^0-y^0)$ and $k^2 = -(k^0)^2 + \vec{k}^{\,2}$, as well as the Euclidean 2-point function

$$G_E(x_E,y_E) \equiv \langle\hat{\phi}_E(x_E)\hat{\phi}_E(y_E)\rangle = \int\frac{d^Dk_E}{(2\pi)^D}\,\frac{e^{ik_E\cdot(x_E-y_E)}}{k_E^2+m^2}, \tag{3}$$

where $k_E\cdot(x_E-y_E) = \sum_{\mu=1}^D k_E^\mu(x_E-y_E)^\mu$ and $k_E^2 = \sum_{\mu=1}^D(k_E^\mu)^2$, and $\hat{\phi}_E(x_E)$ is related to $\hat{\phi}(x)$ by the analytic continuation in x^0 that takes $x^0\to -ix_E^D$.

For general D, the k-integral can be expressed in terms of Bessel functions. When D is an odd integer, the result is an elementary function. In this problem, you are asked to consider the $D=3$ case, and write an elementary expression for $G(x,y)$, $G_E(x_E,y_E)$, as well as the Wightman function

$$W(x,y) \equiv \langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle\,. \tag{4}$$

Explain how $W(x,y)$ is related to $G_E(x_E,y_E)$ by analytic continuation, and how $W(x,y)$ differs from $W(y,x)$ for time-like separated x,y.

To be sure that you understand what's going on, make a plot for both the real and imaginary part of $W(x,0)$ as a function of (real time) x^0 in the case $m=0$ for some fixed spatial coordinate $\vec{x}\neq 0$.

1.6.2 Problem 2: Schwinger-Dyson or Ward identities

Consider in D-dimensional spacetime a free complex scalar field theory with the Lagrangian density

$$\mathcal{L} = -\partial^\mu\phi^*\,\partial_\mu\phi - m^2\phi^*\phi, \tag{5}$$

where $\phi = \phi_1 + i\phi_2$, $\phi^* = \phi_1 - i\phi_2$. The $U(1)$ symmetry that rotates the phase of ϕ is associated with a Noether current operator, of the form[54]

$$\hat{j}_\mu(x) = -i:\hat{\phi}^*(x)\overleftrightarrow{\partial_\mu}\hat{\phi}(x):, \tag{7}$$

[54] An equivalent expression for $\hat{j}_\mu$ in terms of $\hat{\phi}$ is

$$\hat{j}_\mu(x) = -i\lim_{y\to x}\left[\hat{\varphi}^*(y)\overleftrightarrow{\partial}_\mu\hat{\varphi}(x) - \langle\hat{\varphi}^*(y)\overleftrightarrow{\partial}_\mu\hat{\varphi}(x)\rangle\right]. \tag{6}$$

where $f\overleftrightarrow{\partial_\mu} g = f\,\partial_\mu g - (\partial_\mu f)\,g$.

(a) Verify that $\hat{j}^\mu$ is conserved, namely $\partial_\mu \hat{j}^\mu = 0$ as an operator.

(b) Calculate the time-ordered 3-point correlation function

$$\begin{aligned} H^\mu(x_1,x_2,x_3) &\equiv \langle\Omega|\, T\,\hat{\phi}^*(x_1)\hat{\phi}(x_2)\hat{j}^\mu(x_3)\,|\Omega\rangle \\ &= \int \frac{d^Dk_1}{(2\pi)^D}\frac{d^Dk_2}{(2\pi)^D}\,\tilde{H}^\mu(k_1,k_2)\,e^{ik_1\cdot x_1+ik_2\cdot x_2-i(k_1+k_2)\cdot x_3}. \end{aligned} \tag{8}$$

(c) Calculate

$$\frac{\partial}{\partial x_3^\mu}H^\mu(x_1,x_2,x_3) = \int \frac{d^Dk_1}{(2\pi)^D}\frac{d^Dk_2}{(2\pi)^D}\,\big(-i(k_1+k_2)_\mu\big)\,\tilde{H}^\mu(k_1,k_2)\,e^{ik_1\cdot x_1+ik_2\cdot x_2-i(k_1+k_2)\cdot x_3}. \tag{9}$$

Does the result vanish? Explain whether/how it is compatible with your answer to (a).

1.6.3 Problem 3: 2-particle state normalization

In class we introduced the 1-particle state $\left|\vec{k}\right\rangle$ for a massive particle with no internal degree of freedom, normalized according to $\langle\vec{k}|\vec{k}'\rangle = \delta^{D-1}(\vec{k}-\vec{k}')$, on which the Lorentz transformation acts by

$$U(\Lambda)\left|\vec{k}\right\rangle = \sqrt{\frac{(\Lambda k)^0}{k^0}}\,\left|\overrightarrow{\Lambda k}\right\rangle, \tag{10}$$

where $k^0 = \omega_{\vec{k}} = \sqrt{\vec{k}^{\,2}+m^2}$. Recall that for infinitesimal Lorentz transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$, the usual angular momentum operator $\vec{J} = (J_1,J_2,J_3)$ is related by $J_{ij} = \sum_{k=1}^3 \epsilon_{ijk}J_k$.

Now consider a two-particle state $\left|\vec{k}_1,\vec{k}_2\right\rangle$. For now you may assume that the two particles are not interacting, and that $U(\Lambda)$ acts by the obvious generalization of (10)[55]

$$U(\Lambda)\left|\vec{k}_1,\vec{k}_2\right\rangle = \sqrt{\frac{(\Lambda k_1)^0(\Lambda k_2)^0}{k_1^0k_2^0}}\,\left|\overrightarrow{\Lambda k_1},\overrightarrow{\Lambda k_2}\right\rangle. \tag{11}$$

Assuming the two particles are identical bosons, it is natural to normalize the two-particle state according to

$$\langle\vec{k}_1,\vec{k}_2|\vec{k}_1',\vec{k}_2'\rangle = \delta^3(\vec{k}_1-\vec{k}_1')\delta^3(\vec{k}_2-\vec{k}_2') + \delta^3(\vec{k}_1-\vec{k}_2')\delta^3(\vec{k}_2-\vec{k}_1'). \tag{12}$$

We can consider an alternative basis $\left|E,\vec{P},\ell,m\right\rangle$ for the two-particle state, that are eigen-states with respect to the total energy-momentum vector (of eigenvalue $(E,\vec{P})$), and with respect to the angular momentum squared $\vec{J}^2 = \ell(\ell+1)$ and the angular momentum component $J_3 = m$ defined in the center-of-mass frame. We will normalize these states according to the convention

$$\langle E,\vec{P},\ell,m|E',\vec{P}',\ell',m'\rangle = \delta(E-E')\delta^3(\vec{P}-\vec{P}')\delta_{\ell\ell'}\delta_{mm'}. \tag{13}$$

[55] We will see in class soon that this is true for the so-called in- and out- asymptotic scattering states in an interacting quantum field theory as well.

It is easy to see that the overlap between the two kinds of basis states are of the form

$$\langle \vec{k}_1, \vec{k}_2 | E, \vec{P}, \ell, m \rangle = \delta_(\omega_{\vec{k}_1} + \omega_{\vec{k}_2} - E)\delta^3(\vec{k}_1 + \vec{k}_2 - \vec{P}) f_{\ell m}(E, \vec{P}; \hat{k}_1) \tag{14}$$

where the function $f_{\ell m}$ depends only on $E, \vec{P}$, and the unit vector $\hat{k}_1 \equiv \vec{k}_1/|\vec{k}_1|$. Your task is to determine $f_{\ell m}$ in the special case $\vec{P} = 0$. In particular, show that

$$f_{\ell m}(E, 0, \hat{k}_1) = N(E)\, Y_{\ell m}(\hat{k}_1), \tag{15}$$

where $Y_{\ell m}$ are the spherical harmonics, and determine the function $N(E)$.

1.6.4 Problem 4: Wigner little group for massless particles

A key fact concerning the Lorentz transformation property of the 1-particle state, as discussed in class, is that the action of $U(\Lambda)$ on the 1-particle state is determined through that of $U(W)$, where W is a Lorentz transformation that leaves the momentum of the particle invariant. The set of all such W's for a given particle momentum is called the "little group". In this problem you will study the little group of a massless particle in $D = 4$ spacetime dimensions.

(a) We begin with a light-like reference momentum k_R^μ,

$$(k_R^0, k_R^1, k_R^2, k_R^3) = (E, 0, 0, E). \tag{16}$$

For any spatial momentum 3-vector $\vec{k}$, and the corresponding light-like 4-momentum vector $(k^0 = |\vec{k}|, \vec{k})$, we can construct a "standard" Lorentz transformation $L(\vec{k})$ that takes k_R to k, namely

$$L(\vec{k}) = \hat{M}(\vec{k}) \cdot R_1, \tag{17}$$

where

$$(\hat{M}(\vec{k}))^\mu{}_\nu = \delta^\mu{}_\nu + \frac{(k - k_R)^\mu (k - k_R)_\nu}{k \cdot k_R}, \tag{18}$$

and R_1 is the reflection in the x^1-direction. Verify that (17) is indeed a Lorentz transformation of the desired property.

(b) $W = (L(\overrightarrow{\Lambda k_R}))^{-1}\Lambda$, which leaves k_R invariant (i.e. $W k_R = k_R$) for any generic Lorentz transformation Λ, is an element of the little group. Find an example of W that is not merely a rotation about the x^3 axis nor a reflection.

Remark: The structure of the massless little group you find here, as we will see in a few weeks is closely related to why the photon is necessarily described by a gauge theory.

Problem 1 answer:

We have a few things to do: compute G_E, G, and W, and graph W with G_E via an analytic continuation. The computations are somewhat tedious.

Recall that the Euclidean Green's function is given as (in $D = 3$)

$$G_E(x_E, y_E) = \int \frac{d^3k_E}{(2\pi)^3} \frac{e^{ik_E\cdot(x_E - y_E)}}{k_E^2 + m^2}. \tag{1.746}$$

Rename $r \equiv |x_E - y_E|$. Then, we can switch to polar coordinates (with two spatial and one time dimension) by setting $k_E \cdot r = k_E r \cos(\theta)$ where k_E and r are just scalars now. The differential changes as (check the back of Schwartz's QFT)

$$\int d^3k_E = \int d\Omega_3 \int_0^\infty dk_E k_E^2. \tag{1.747}$$

The integration over $d\Omega_3$ is slightly involved due to the $\cos(\theta)$ in the exponential but yields (you should know how to do this)

$$\int d\Omega e^{ik_E r\cos(\theta)} = 4\pi \frac{\sin(k_E r)}{k_E r}. \tag{1.748}$$

(1.746) then becomes

$$\begin{aligned} G_E(r) &= \frac{1}{(2\pi)^2 r} \int_0^\infty dk_E \frac{k_E \sin(k_E r)}{k_E^2 + m^2} \\ &= \frac{1}{(2\pi)^2 r} \left(\frac{\pi}{2} e^{-mr} \right) \\ &= \frac{e^{-mr}}{4\pi r}. \end{aligned} \tag{1.749}$$

In Minkowski space, where we compute $G(x, y)$, the situation is more involved due to the complex integration. The Minkowski Green's function integral can be setup as

$$\begin{aligned} G(x - y) &= -i \int \frac{d^3k}{(2\pi)^3} \frac{e^{ik\cdot(x-y)}}{k^2 + m^2 + i\epsilon} \\ &= -i \int \frac{d^2\vec{k}}{(2\pi)^2}) e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \int \frac{dk^0}{2\pi} \frac{e^{ik^0(x^0-y^0)}}{k_0^2 - \omega_k^2 + i\epsilon} \end{aligned} \tag{1.750}$$

where $\omega^2 = \vec{k}^2 + m^2$. We can perform the k_0 (or k^0 depending on your notation) in the usual manner: there exists a pole at $k_0 = \pm\omega_k^2$, but this is purely real! Wiat, what is going on? Well ok, we have a pole on the real line, lets integrate over the positive one, and the $+i\epsilon$ tells us essentially to slap on a Heavy-side function

$$G(x - y) = -i\theta(t) \int \frac{d^2\vec{k}}{(2\pi)^2} \frac{e^{i\vec{k}\cdot(\vec{x}-\vec{y})} e^{-i\omega_k t}}{2\omega_k}. \tag{1.751}$$

Again, we switch to polar coordinates except this time, we only have a ϕ integral and no θ one (since we have 2 spatial dimensions) and so the integral changes as

$$\int_0^{2\pi} d\phi e^{ikr\cos(\phi)} = 2\pi J_0(kr) \tag{1.752}$$

where $J_0(x)$ is a Bessel function of the first kind. we are then left with a rather nice integral using some clever tricks namely

$$\int_0^{\infty} dk \frac{kJ_0(kr)}{\omega_k} e^{i\omega_k t} = \frac{2}{s} e^{ims} \tag{1.753}$$

where $s \equiv \sqrt{(x-y)^2 + i\epsilon}$. The final answer is then

$$G(x-y) = \frac{e^{ims}}{4\pi s}. \tag{1.754}$$

Now, the Wightman function is more involved since *you* have to find the explict form I am going to write down. Namely, the Wightman function is the following

$$\begin{aligned} W(x,y) &\equiv \langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle \\ &= \int \frac{d^D k}{(2\pi)^D} e^{ik\cdot(x-y)} 2\pi\theta(k^0)\delta(k^2+m^2) \end{aligned} \tag{1.755}$$

which is found by: setting up the expectation value, writing the fields in terms of their creation and annihlation operators, and then simplfying. In $D = 3$ (1.755) reduces to

$$W(x,y) = \int \frac{d^2\vec{k}}{(2\pi)^2} \int_0^{\infty} \frac{dk_0}{2\pi} e^{ik_0\cdot(x_0-y_0)+i\vec{k}\cdot(\vec{x}-\vec{y})} 2\pi\delta(k_0^2 - \omega_k^2). \tag{1.756}$$

To perform the dk_0 integral, we can utilize the delta function which has roots at $k_0 = \pm\omega_k$. But, $\theta(k_0)$ enforces positivity which implies that we only take the positive root of k_0. Then the integral over dk_0 is simply $\exp(-i\omega_k(x^0 - y^0))/2\omega_k$. The Wightman function is the

$$W(x,y) = \int \frac{d^2\vec{k}}{(2\pi)^2} \frac{e^{i\omega_k(x^0-y^0)+i\vec{k}\cdot(\vec{x}-\vec{y})}}{2\omega_k}. \tag{1.757}$$

We can simplify the calculation via $\mathbf{r} = \vec{x} - \vec{y}$ and letting $t = x_0 - y_0$. We just have polar coordinates leftover then namely $\vec{k}\cdot\mathbf{r} = kr\cos(\theta - \phi)$. the integral over θ yields

$$\int_0^{2\pi} \frac{d\theta}{(2\pi)^2} e^{ikr\cos(\theta-\phi)} = \frac{1}{2\pi} J_0(kr). \tag{1.758}$$

The integral can then be reduced to a representation of the Hankel functions

$$\begin{aligned} W(t,r) &= \frac{1}{2\pi} \int_0^{\infty} \frac{kdk}{2\omega_k} e^{i\omega_k t} J_0(kr) \\ &= \frac{1}{2\pi}\left(\frac{2}{i} K_0(ms)\right) \qquad \left(s = \sqrt{(t-i\epsilon)^2 - r^2}\right) \\ &= \frac{1}{2\pi} K_0\left(m\sqrt{-(x-y)^2 - i\epsilon}\right). \end{aligned} \tag{1.759}$$

Notice that if $(x-y)^2 > 0$, which is timelike separation, then K_0 has a real and positive argument giving a real result. If $(x-y)^2 < 0$, which is spacelike separation, then K_0 has an imaginary component and yet consistent with causality since the function decays rapidly in the UHP.

The relation between the Euclidean Green's function and the Wightman function can be seen via analytic continuations. Recall that in order to go from Minkowski to Euclidean, we perform a *Wick rotation* (which is just a fancy name for an analytic continuation) of the time coordinate $x_E^3 = -ix^0$. The Minkowski Greens function then becomes the Euclidean Greens function. But the connection to the Euclidean Greens function to that of the Wightman one is via *another* analytic continuation namely

$$W(x,y) = G_E(x_E \to -i(x^0 - i\epsilon), \vec{x}). \tag{1.760}$$

However, notice that the difference in the position of the Wightman functions yields

$$W(x,y) - W(y,x) = \langle\Omega| \left[\hat{\phi}(x), \hat{\phi}(y)\right] |\Omega\rangle \,. \tag{1.761}$$

If the commutator is zero, then causality is obeyed (since spacelike separated). Also in their difference can arise branch cuts which, if we are spacelike separated as we should be with $(x-y)^2 < 0$, then the square root in the Wightman function is imaginary and the $\pm i\epsilon$ prescription identifies which sides of the branch cut we are on which leads to a discontinuity.

Notice also that the delta functions that arise when we take the Wightman function at $W(x,0)$ enforces causality. There are also poles in the imaginary part of the light cone showing divergences associated with massless particles. The signs on the imaginary parts give the direction of time and the causal structure of the theory we are working with.

To plot the Wightman function, we simply cannot take $m = 0$ in (1.759) since we could be missing poles and branch cut structures from the original function. Instead, we can resolve the Wightman function in a similar manner setting $m = 0$ and $y = 0$. Then you should find

$$W(x, y = 0; m = 0) = \frac{1}{2\pi} \frac{-ix_0}{(x_0)^2 - r^2 + i\epsilon} \tag{1.762}$$

where the $i\epsilon$ prescription is implemented to avoid the poles, but if we want to see what happens when $\epsilon \to 0$ then the following limit is useful

$$\lim_{\epsilon \to 0} \frac{\epsilon}{x^2 + \epsilon^2} = \pi\delta(x). \tag{1.763}$$

Notice that when $\epsilon \to 0$ we have poles on the real axis (when plotting an imaginary graph of x_0) at $\pm r$. There are no branch cuts since we are in the massless limit.

Note that if we did not take the massless limit (and $y = 0$), there would be branch cuts extending from $\pm r$. However, with the $i\epsilon$ prescription, when we perform the contour integral, the poles are elevated above the branch cuts allowing us to miss them when integrating over. This branch cut (singularity) is known as the lightcone singularity.

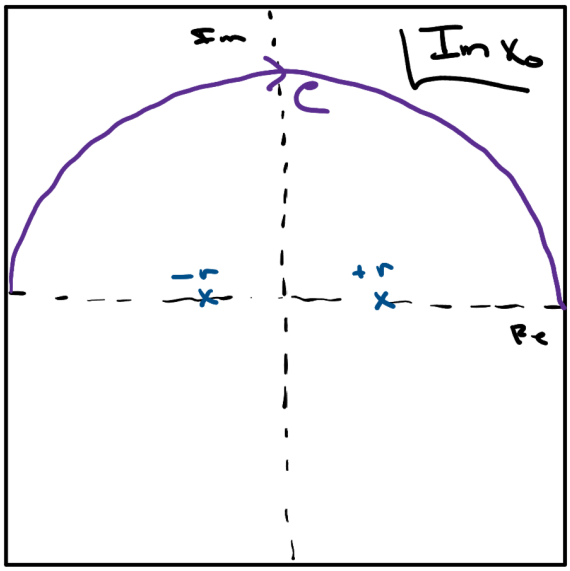


Figure 6. Complex graph of x_0 for the massless Wightman function.

Problem 2 answer:

This will be our first time working with complex scalars, so let me give a few pointers and quick rundowns prior. The complex scalar Lagrangian (without a gauge field) is given as

$$\mathcal{L} = -\eta^{\mu\nu}\partial_\mu\varphi^*\partial_\nu\varphi - m^2\varphi^*\varphi \tag{1.764}$$

where $\varphi = \phi_1 + i\phi_2$ where $\phi_{1,2}$ are *two different* scalar fields. This is key since we can treat φ^* and φ as two *different* fields. In that case, notice that there is no interaction term, and thus we are **not** doing Feynman diagram calculations! To drive this point home, notice that

$$m^2\varphi^*\varphi = m^2(\phi_1 - i\phi^2)(\phi_1 + i\phi_2) = m^2(\phi_1^2 + \phi_2^2) \tag{1.765}$$

indicating that we really do have two different scalar fields that share the same mass [56].

Another point to make is that the Wick contraction of like fields vanish for spacelike separated fields. Therefore, we only contract those from "different matter sectors" of the theory.

Recall that the current is

$$\begin{aligned} :\hat{j}^\mu: &= -i :\hat{\varphi}^*(x)\overleftrightarrow{\partial}_\mu\hat{\varphi}(x): \\ &= -i :(\hat{\varphi}^*\partial_\mu\hat{\varphi} - (\partial_\mu\hat{\varphi}^*)\hat{\varphi}): \end{aligned} \tag{1.766}$$

where $:\cdots:$ is the normal ordering of the fields (move all creation operators to the left). Now we can start. (Note: I will be lazy and drop the "hats" on all the field operators but it should be noted that **everything** should have a hat...)

[56] Even if they had a different mass, depending upon the setup, you can do something weird as $M \equiv m_1 + im_2$ but this is so bizarre it never happens.

For part (a), we can compute this directly. Notice first though that we have two equations of motion namely

$$\left(\Box - m^2\right)\varphi(x) = 0 \tag{1.767}$$

$$\left(\Box - m^2\right)\varphi^*(x) = 0. \tag{1.768}$$

Then, (where I am dropping the normal ordering for now)

$$\begin{aligned}\partial_\mu j^\mu &= -i\left(\partial_\mu\varphi^*\partial^\mu\varphi + \varphi^*\Box\varphi - (\Box\varphi^*)\varphi - \partial^\mu\varphi^*\partial_\mu\varphi\right)\\ &= 0.\end{aligned} \tag{1.769}$$

Done (if you do not like swapping up and down indices, just insert the equations of motion).

Now for part (b) this is slightly more tricky, but notice that

$$\begin{aligned}H^\mu(x_1,x_2,x_3) &\equiv \langle\Omega|\, T\varphi^*(x_1)\varphi(x_2)j^\mu(x_3)\,|\Omega\rangle\\ &= -i\,\langle\Omega|\, T\left(\varphi(x_1)\varphi(x_2)\varphi^*(x_3)\partial^\mu\varphi(x_3) - \varphi(x_1)\varphi(x_2)(\partial^\mu\varphi^*)(x_3)\varphi(x_3)\right).\end{aligned} \tag{1.770}$$

So we really have two 3-point correlation functions to compute with their Fourier transformation being

$$H^\mu(x_1,x_2,x_3) = \int \frac{d^Dk_1}{(2\pi)^D}\frac{d^Dk_2}{(2\pi)^D}\tilde{H}^\mu(k_1,k_2)e^{ik_1\cdot x_2 + ik_2\cdot x_2 - i(k_1+k_2)\cdot x_3} \tag{1.771}$$

where we only need to Fourier transform two of the momenta since we can use conservation of momentum $(k_1+k_2+k_3)^\mu = 0$. Notice in (1.770) that we have the following contractions (where for shorthand $\varphi(x_1) \equiv \varphi_1$)

$$H^\mu(x_1,x_2,x_3) = -i\,\langle\Omega|\, T\left(\varphi_1^*\partial^\mu\varphi_3\,\varphi_2\varphi_3^* - \varphi_1^*\partial^\mu\varphi_3^*\varphi_2\varphi_3\right)|\Omega\rangle\,. \tag{1.772}$$

The wick contraction of $\varphi_3\partial^\mu\varphi_3$ vanishes due to the normal ordering (which kills any tadpole diagrams). (1.772) is then just a product of four expectation values

$$H^\mu(x_1,x_2,x_3) = -i\langle T\varphi_1^*\partial^\mu\varphi_3\rangle\langle T\varphi_2\varphi_3^*\rangle + i\langle T\varphi_1^*\varphi_3\rangle\langle T\varphi_2\partial^\mu\varphi_3^*\rangle. \tag{1.773}$$

We know exactly what all of the products are in (1.773)! They are simply Green's functions or derivatives of (i.e. complex scalar propagators). Namely

$$\langle T\varphi(x)\varphi(y)\rangle \equiv G_F(x-y) = \int\frac{d^Dk}{(2\pi)^D}\frac{i}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)} \tag{1.774}$$

and for the derivative of the propagator

$$\partial^\mu G(x-y) = -\int\frac{d^Dp}{(2\pi)^D}\frac{p^\mu e^{-ip\cdot(x-y)}}{p^2-m^2+i\epsilon}. \tag{1.775}$$

Now it is just a matter of plugging in terms and simplifying which you should find to be

$$H^\mu(x_1,x_2,x_3) = i\int \frac{d^Dk_1}{(2\pi)^D}\frac{d^Dk_2}{(2\pi)^D}\frac{(k_1-k_2)^\mu}{(k_1^2-m^2+i\epsilon)(k_2^2-m^2+i\epsilon)}e^{ik_1\cdot(x_1-x_3)-ik_2(x_2-x_3)}. \tag{1.776}$$

implying that the Fourier transformed 3-point function is

$$\tilde{H}^\mu(k_1,k_2) = \frac{(k_1-k_2)^\mu}{(k_1^2-m^2+i\epsilon)(k_2^2-m^2+i\epsilon)}. \tag{1.777}$$

Now for part (c) we want to compute $\partial H^\mu/\partial x^3$. First recall that the fourier transform of a partial derivative is $\partial_\mu \Leftrightarrow -ik_\mu$. Therefore

$$\frac{\partial}{\partial x_3^\mu} \Leftrightarrow -ik_{\mu,3} = -i(k_1+k_2)_\mu. \tag{1.778}$$

Working with just the Fourier transform of the 3-point function, notice we get the product

$$\frac{(k_1+k_2)\cdot(k_1-k_2)}{(k_1^2-m^2+i\epsilon)(k_2^2-m^2+i\epsilon)} = \frac{k_1^2-k_2^2}{(k_1^2-m^2+i\epsilon)(k_2^2-m^2+i\epsilon)} = \frac{1}{k_1^2-m^2+i\epsilon}-\frac{1}{k_2^2-m^2+i\epsilon} \tag{1.779}$$

where in the last term I used partial fractions. The derivative of the 3-point function then splits into two different pieces

$$\frac{\partial}{\partial x_3^\mu}H^\mu(x_1,x_2,x_3) = \int\frac{d^Dk_1}{(2\pi)^D}\frac{e^{ik_1\cdot(x_1-x_2)}}{k_1^2-m^2+i\epsilon}\int\frac{d^Dk_2}{(2\pi)^D}e^{ik_2\cdot(x_2-x_3)} - \int\frac{d^Dk_2}{(2\pi)^D}\frac{e^{ik_2\cdot(x_2-x_3)}}{k_2^2-m^2+i\epsilon}\int\frac{d^Dk_1}{(2\pi)^D}e^{ik_1\cdot(x_1-x_3)} \tag{1.780}$$

but the $\int d^Dk\ \exp(ik_2\cdot(x_2-x_3)) \propto \delta^{(D)}(x_2-x_3)$ is present, but we cannot use the delta function present since it is unsure whether or not we would have support around any spatial points x_i^μ for $i=\{1,2,3\}$. Our final answer is then

$$\frac{\partial}{\partial x_3^\mu}H^\mu(x_1,x_2,x_3) = G(x_1-x_2)\delta^{(D)}(x_2-x_3) - G(x_2-x_3)\delta^{(D)}(x_1-x_3). \tag{1.781}$$

But, what does (1.781) mean in comparison with what we computed in part (a) namely $\partial_\mu\hat{j}^\mu(x)=0$? How can these two computations seemingly be computing the same thing, but get different answers? Well, the computation of current conservation is the statement that the divergence of $\hat{j}^\mu(x)$ is zero as an *operator* equation. We get the same result for classic fields as well. Therefore the current, which corresponds to the U(1) symmetry of the theory, is a conserved quantity (or conserved symmetry).

However, in computing $H^\mu(x_1,x_2,x_3)$, we are computing the *correlation function* of two scalar particles at different points in spacetime of which how they interact with the *production* of scalar-particles at another point in spacetime. Well, this is not very illuminating, but recall that $\partial_\mu j^\mu \equiv -(\delta S/\delta\phi(x))\delta\phi(x)$ where δS is the variation of the action, which is the same as finding the *equations of motion*. If we insert δS into the correlation function we computed, then this is the statement that the classical equations of motion **and** current conservation hold **only** up to the terms found in (1.781) which are called *contact terms*. These contact terms arise whenever insertions of the δS operator or $\partial_\mu j^\mu$ operator

is inserted into a correlation function; the contact terms describe how $\partial j(x)$ coincides with the point x of one of the other fields inside the correlator.

If you insert the equations of motion, you recover what is called the *Schwinger-Dyson equation* while the insertion of the current yields the Ward identity which in general are given as (Schwinger-Dyson first then Ward)

$$\langle\Omega| T\left(\frac{\delta S}{\delta\phi(x)}\right)\phi(x_1)\cdots\phi(x_n)|\Omega\rangle = i\sum_{j=1}^{n}\langle\Omega| T\phi(x_1)\cdots\phi(x_j)\delta^{(D)}(x-x_j)\cdots\phi(x_n)|\Omega\rangle \tag{1.782}$$

$$\partial_\mu\langle\Omega| Tj^\mu(x)\phi(x_1)\cdots\phi(x_n)|\Omega\rangle = -i\sum_{j=1}^{n}\langle\Omega|\phi(x_1)\cdots\phi(x_j)\left(\delta\phi(x_j)\delta^{(D)}(x-x_j)\right)\cdots\phi(x_n)|\Omega\rangle\,. \tag{1.783}$$

Note that, for example, in QED, the Ward identites (and correspondingly the Schwinger-Dyson equations) are always valid *classically* but not always quantum mechanically (i.e. in terms of operators). In the operator version, the correlation function $\langle\cdots\rangle$ will have divergences that need regulating (such as loops). The Ward identities only hold up to valid reguarlization schemes. In QED, these can be found, but in general not always. When a reguarlization scheme cannot handle the divergences, then the corresponding symmetry is considered *anomalous* (this is one of the subjects in QFT 2).

Here are some clarifying remarks on the Schwinger-Dyson equation (SD). The SD-equation should be thought of as the quantum mechanical version of the Euler-Lagrange equations where in Euler-Lagrange we have functionals in terms of classical fields ϕ while the SD-equation is an equation of motion in terms of Greens Functions. This can best be seen by taking the KG-operator and acting on the 2-point Greens function which omits a delta-function

$$(\Box - m^2)\langle\Omega|\hat{\phi}(x_1)\hat{\phi}(x_2)|\Omega\rangle = \delta^{(D)}(x_1-x_2). \tag{1.784}$$

These are exactly the *contact terms* we derived prior. A graph that is helpful, which forces you to remember that $\hat{\phi}(x)|\Omega\rangle$ can be described in terms of smearing functions that decay off to infinity can be seen below. In figure 7, there are 3 scalar field operators denoted as $\phi(x_i^\mu)$ (you can take the derivative of ϕ_3 and multiply by another ϕ_3 but the interpretation remains the same) with a shaded region around them showing their distributions we saw previously in lecture (the smearing functions). They each have some spacetime interval between them written in black, but also colored in are the terms we found in (1.782) in the pink, namely terms of the form $G(x_i - x_j)\delta^{(D)}(x_i - x_k)$. As you can see, the Greens functions are being smeared by the delta-functions (since Dirac-delta functions are really distributions which decay quickly outside of their argument's range). We can clearly see from the graph that *if* we tried to use the Dirac-deltas in the normal way (by switching out their arguments), then we would indeed find $\partial_\mu H^\mu(x_1, x_2, x_3) = 0$ but this can not be the case since we can only perform such a maneuver if there is local support between x_1 and x_3, which, as we can see from the spacetime graph, there is not.

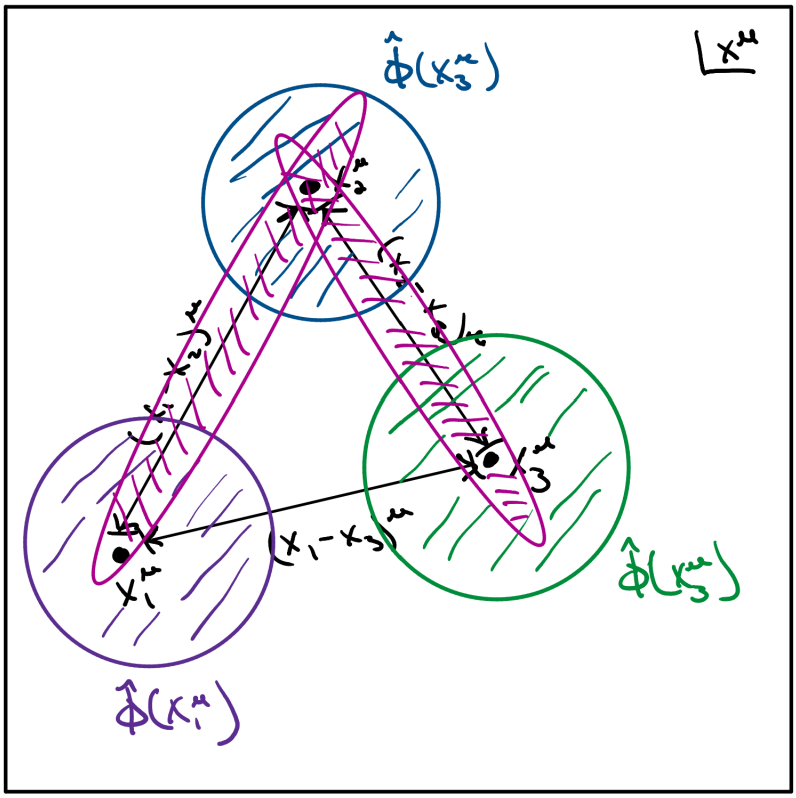


Figure 7. This is a graph of 3 operators $\phi(x_i^\mu)$ in three different colors that have their spacetime distances identified. Also identified, in pink, is $G(x_i - x_j)\delta^{(D)}(x_i - x_k)$.

Problem 3 answer:

Notice that in the basis $|E, \vec{p}, \ell, m\rangle$, when we act on it with operators P^μ (the total energy-momentum operator which we can write down instead as $\hat{P}^\mu = (\hat{E}, \hat{\vec{p}})$) acting on the state with eigenvalues $(E, \vec{p})$ while the angular momentum operator $\hat{\mathbf{J}}^2$ omits $\mathbf{J}^2 = \ell(\ell+1)$ and $J_3 = m$ just as in quantum mechanics. Note that this is all defined in the center-of-mass (COM) frame (the sum of the momenta is zero).

In the COM frame, $\vec{p} \equiv \vec{k}_1 + \vec{k}_2 = 0 \implies \vec{k}_1 = -\vec{k} \equiv -\vec{k}$ for convience. The total energy is then $E = \omega_{k_1} + \omega_{k_2} = 2\omega_k$. The COM frame forces the 2-particle state to become the ket (or bra) $\left|\vec{k}, -\vec{k}\right\rangle$. The normalization then of the states just gives a factor of 2 i.e. $\left\langle \vec{k}, -\vec{k} \middle| \vec{k}, -\vec{k} \right\rangle = 2$. Let us guess that the eigenstate in the COM frame can be written as

$$|E, 0, \ell, m\rangle = \mathcal{N}(E) \int \frac{d^3k}{(2\pi)^3} \delta(2\omega_k - E) \left|\vec{k}, -\vec{k}\right\rangle Y^*_{\ell m}(\hat{\vec{k}}). \tag{1.785}$$

To see if this is correct, consider the norm of the state (1.785) at the same angular momentum

$$\langle E, 0, \ell, m | E, 0, \ell, m\rangle = |\mathcal{N}(E)|^2 \int \frac{d^3k}{(2\pi)^3} \delta(2\omega_k - E) \left\langle \vec{k}, -\vec{k} \middle| \vec{k}, -\vec{k} \right\rangle \left|Y_{\ell m}(\hat{k})\right|^2. \tag{1.786}$$

Recall that $\omega_k = \sqrt{k^2 + m^2}$ and so $2\omega_k \delta(2\omega_k - E) = \delta(k - k_E)$ where we have defined $k = \sqrt{(E/2)^2 - m^2}$. The integral can then be computing using spherical coordinates with $d^3k = k^2 dk d\Omega$ where the $d\Omega$ integral can be completed with the spherical harmonics just giving 1. The norm then, along with $\omega = E/2$ is

$$\langle E, 0, \ell, m | E, 0, \ell, m \rangle = |\mathcal{N}(E)|^2 \frac{4k_E^2}{E} \tag{1.787}$$

where by convention I absorbed the factor of $1/(2\pi)^3$ when we did the k-integrals. The norm must be equal to 1, and therefore the normalization factor is then

$$\mathcal{N}(E) = \sqrt{\frac{E}{4k_E^2}}. \tag{1.788}$$

Note that this is not as general as the normalization factor found in lecture notes, i.e. in L16, but no matter what state you guess, it must obey the normalization conditions. This is just *one* way that you can define the angular momentum states.

Problem 4 answer:

To perform part (a), where we need to show that $L(\Lambda)$ given is a proper Lorentz transformation (i.e. part of the Lorentz group), there are two ways to do this. The first is algebraically a mess and requires some assumptions about how massless-spinning particles interact. This is by showing that $\Lambda^T \eta \Lambda = \eta$ or in words, the Lorentz transformation leaves the Minkowski metric invariant. We have not talked about interacting spinning particles yet however, so lets not do that way.

Instead, we just need to show that from the refernce momentum k_R^μ that when *boosted* by $L^\mu{}_\nu(k)$, we recover k^μ or in equation form

$$L^\mu{}_\nu k_R^\nu = k^\mu. \tag{1.789}$$

Recall that the reference momentum we have defined is $k_R^\mu = (E, 0, 0, E)$ in $(t \equiv x_0, x_1, x_2, x_3)$ notation. One condition that null or light-like particles must satisfy (massless particles) is that their 4-momentum squared is 0, where in our case $k_R^2 \equiv k_R^\mu \eta_{\mu\nu} k_R^\nu = -E^2 + E^2$ is the $(-,+,+,+)$ signature and $k^2 = -(k_0)^2 + \vec{k}^2 = 0 \implies k_0 = \left|\vec{k}\right|$. Then

$$\begin{aligned}
L^\mu{}_\nu k_R^\nu &= \left(\delta_\nu^\mu + \frac{(k-k_R)^\mu (k-k_R)_\nu}{(k \cdot k_R)}\right) k^\nu \\
&= \delta_\nu^\mu k_R^\nu + \frac{(k-k_R)^\mu \left(k_\nu k_R^\nu - k_{R,\nu} k_R^\nu\right)}{(k \cdot k_R)} \\
&= k_R^\mu + (k-k_R)^\mu \left(\frac{k \cdot k_R - k_R^2}{k \cdot k^R}\right) \\
&= k_R^\mu + (k-k_R)^\mu \\
&= k^\mu
\end{aligned} \tag{1.790}$$

where I used the fact that $k_R^2 = 0$. If you are wondering where the reflection went, a reflection about the x^1 axis (or just the x-axis) is a matrix of the form

$$R_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{1.791}$$

from which we can see that a reflection about the x-axis would not affect k_R^μ.

Part (b) has a typo in it namely the W-matrix should be

$$W = L^{-1}(\Lambda k)\Lambda L(k). \tag{1.792}$$

Notice what (1.792) actually does to some reference momentum k_R (we read the matrix right to left): first we boost it in the sense that $L(k) \cdot k_R = k$ by definition (as you showed in part (a)). Then we *spin* k via the Λ term, and then do an *inverse boost* with $L^{-1}(\Lambda k)$ (where the argument here is the "spinned-momentum." Therefore, at the end of that matrix mess, we should be left with

$$W \cdot k_R = k_R \tag{1.793}$$

or in index notation $W^\alpha{}_\beta k_R^\beta = k_R^\alpha$. But, before we do this problem, let me clarify a lot of what we have been doing in this problem.

Recall that we started with some reference momentum k_R^μ that we can boost with a Lorentz transformation $k^\mu = L^\mu{}_\nu(k, k_R)k^\nu$. Wigner, in his seminal paper in 1939 which everyone should read [10], *defined* that for some reference state $|k_R, \sigma\rangle$ (where sigma here can be any labels of the particle (spin, angular momentum, mass, etc) the boosted state as

$$|k, \sigma\rangle = U(L(k, k_R))\,|k_R, \sigma\rangle\,. \tag{1.794}$$

Notice that σ does not change. Now consider a unitary representation of the Lorentz group $U(\Lambda)$

$$\begin{aligned} U(\Lambda)\,|k, \sigma\rangle &= U(\Lambda)U(L(k, k_R))\,|k_R, \sigma\rangle \\ &= U(L(\Lambda k, k_R))U(L^{-1}(\Lambda k, k_R)\Lambda L(k, k_R))\,|k_R, \sigma\rangle \end{aligned} \tag{1.795}$$

where we now define $W = L^{-1}(\Lambda k, k_R)\Lambda L(k, k_R)$ which leaves k_R invariant which is *Wigner's Little Group.* Therefore $U(W)\,|k_R, \sigma\rangle$ cannot change k_R but it **can** change σ. Therefore

$$U(W)\,|k_R, \sigma\rangle = \mathcal{D}_{\sigma\sigma'}(W)\,|k_R, \sigma'\rangle \tag{1.796}$$

which answers what are the σ labels: they are representations of the Little Group. From here, the representations of the Little Group depend upon whether or not the particle is massive or massless. We just so happen to be working with the massless Little Group, and already know the form of W: it is just constructed out of the $L(k, k_R)$ we have been using, plus your favorite Lorentz transformation!

So just writing down what W is in terms of some generic Λ and the given form of L previously given, and then showing that $W \cdot k_R = k_R$ solves the problem. However, there are a numerous number of other ways to do this problem. For example, you could simply set W to be a rotation about some axis with components $W = (0, \mathbf{J})$ for any $\mathbf{J} = (J^1, J^2, J^3)$. But, the question specifically asks not to do this... (for some reason).

The other way is to write out $W = W(\theta, \alpha, \beta) = S(\alpha, \beta)R(\theta)$ where S is a paramterized Lorentz transformation you can even find in Goldstein's classical mechanics book, and $R(\theta)$ is a rotation about any axis. From here, I urge you to read chapter 2.5 of Weinberg's text [1] **in-depth**! The entire construction of Wigner's little group for massive and massless particles underlies everything we do! It is even a showing that (at least for massive particles, since massless particles requires new structure called *helicity* to be added on) that the entirety of quantum mechanics representations that use Clebsch-Gordon coefficients and such can simply be carried over as representations of massive particles in relativistic quantum mechanics.

1.7 Problem set 4:

Our convention for scattering amplitudes is such that the $2 \to 2$ S-matrix element (in the case of identical in- and out-particles) is written as

$$\begin{aligned} S(\vec{k}_1', \vec{k}_2'; \vec{k}_1, \vec{k}_2) &\equiv {}^{\text{out}}\Big\langle \vec{k}_1', \vec{k}_2' \,|\, \vec{k}_1, \vec{k}_2 \Big| \vec{k}_1', \vec{k}_2' \,|\, \vec{k}_1, \vec{k}_2 \Big\rangle^{\text{in}} \\ &= \delta^{D-1}(\vec{k}_1' - \vec{k}_1)\, \delta^{D-1}(\vec{k}_2' - \vec{k}_2) + \delta^{D-1}(\vec{k}_1' - \vec{k}_2)\, \delta^{D-1}(\vec{k}_2' - \vec{k}_1) \\ &\quad + i(2\pi)^D\, \delta^D(k_1' + k_2' - k_1 - k_2)\, \mathcal{M}(\vec{k}_1', \vec{k}_2'; \vec{k}_1, \vec{k}_2), \end{aligned} \tag{1}$$

where $\mathcal{M}$ is related to the manifestly Lorentz invariant form of the amplitude $\mathcal{A}$ by

$$\mathcal{M}(\vec{k}_1', \vec{k}_2'; \vec{k}_1, \vec{k}_2) \equiv \prod_{j=1}^{2} \frac{1}{(2\pi)^{\frac{D-1}{2}} \sqrt{2\omega_{\vec{k}_j'}}} \prod_{j=1}^{2} \frac{1}{(2\pi)^{\frac{D-1}{2}} \sqrt{2\omega_{\vec{k}_j}}} \mathcal{A}(k_1', k_2'; k_1, k_2), \tag{2}$$

where $\omega_{\vec{k}} \equiv \sqrt{\vec{k}^{\,2} + m^2}$.

Also recall that in $D = 4$ dimensions, the partial wave amplitude $S_\ell(E)$, at total energy E and angular momentum $\vec{J}^2 = \ell(\ell+1)$ in the center-of-mass frame ($\vec{k}_1 + \vec{k}_2 = \vec{k}_1' + \vec{k}_2' = 0$) is related by

$$i(2\pi)^4\, \mathcal{M}(\vec{k}_1', \vec{k}_2'; \vec{k}_1, \vec{k}_2) = \big(N(E)\big)^2\, \frac{1}{4\pi} \sum_\ell (2\ell + 1)\, P_\ell(\hat{k}_1 \cdot \hat{k}_1')(S_\ell(E) - 1), \tag{3}$$

where $P_\ell(x)$ is the ℓ-th Legendre polynomial. For identical particles, $N(E) = \sqrt{\frac{8}{|\vec{k}_1|\, E}}$ and the sum is restricted to even non-negative integer ℓ.

1.7.1 Problem 1: Multiple scalar species scattering

Consider a field theory in $D = 4$ dimensions described by the Lagrangian density

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi_1)^2 - \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2} m_1^2 \phi_1^2 - \frac{1}{2} m_2^2 \phi_2^2 - \frac{g}{2} \phi_1^2 \phi_2, \tag{4}$$

where $\phi_1(x)$ and $\phi_2(x)$ are a pair of scalar fields, with $m_2 \gg m_1 > 0$.

(a) Calculate the $2 \to 2$ scattering amplitude $\mathcal{A}(k_1', k_2'; k_1, k_2)$ of the lightest particle in perturbation theory up to order g^2 (tree level).

(b) Calculate the $\ell = 0$ partial wave (i.e. "s-wave") amplitude $S_0(E)$ up to order g^2. Below the energy threshold for inelastic processes, unitarity implies that $S_0(E)$ is a phase. Is your answer consistent with this expectation?

1.7.2 Problem 2: Spectral representation for multiple scalars

Consider the scalar field theory described by the same Lagrangian as in problem 1, but now in $D = 4 - \delta$ dimensions ($\delta > 0$), and with additional linear counter terms so that $\langle \phi_1 \rangle = \langle \phi_2 \rangle = 0$.

(a) Calculate the mass of the lightest particle, as well as the Z factor of ϕ_1, in perturbation theory up to order g^2.

(b) Calculate the self-energy $\Sigma_2(k)$ of ϕ_2 up to order g^2, related to the time-ordered 2-point function via

$$\langle\Omega|\, T\, \phi_2(x)\phi_2(0)\, |\Omega\rangle = -i \int \frac{d^D k}{(2\pi)^D} \frac{e^{ik\cdot x}}{k^2 + m_2^2 - i\epsilon - \Sigma_2(k)}. \tag{5}$$

Describe the spectral function $\rho_2(\mu^2)$ related via the Källén–Lehmann spectral representation

$$\frac{1}{k^2 + m_2^2 - i\epsilon - \Sigma_2(k)} = \int_{m_2^2}^{\infty} d\mu^2 \, \frac{\rho_2(\mu^2)}{k^2 + \mu^2 - i\epsilon}. \tag{6}$$

Note: there are two qualitatively distinct cases: (1) $m_2 < 2m_1$, and (2) $m_2 > 2m_1$.

(c) Interpret your results in the $D \to 4$ (i.e. $\delta \to 0$) limit.

1.7.3 Problem 3: Renormalized coupling for multiple scalars

Consider the ϕ^4 theory defined by the Lagrangian density

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 - \frac{g}{4!}\phi^4 \tag{7}$$

in $D = 4 - \delta$ spacetime dimensions.

(a) Calculate the $2 \to 2$ amplitude $\mathcal{A}(k_1', k_2'; k_1, k_2)$ up to order g^2 (1-loop order). Express your result as a function of the Mandelstam invariants

$$\begin{aligned} s &= -(k_1 + k_2)^2, \\ t &= -(k_1 - k_1')^2, \\ u &= -(k_1 - k_2')^2. \end{aligned} \tag{8}$$

Note: you may choose to express the result in terms of the physical mass; there is no need to compute the mass counter term explicitly.

(b) Now take the $D \to 4$ ($\delta \to 0$) limit. You should find a divergence that can be canceled by a counter term of the form

$$\Delta\mathcal{L} = \frac{C}{\delta} \frac{g^2}{4!}\phi^4, \tag{9}$$

where C is a numerical constant. The term (9) can be absorbed into the definition of the bare coupling in the Lagrangian. Define a renormalized coupling g_R via

$$-g_R \equiv \mathcal{A}\Big|_{s=t=u=\frac{4m^2}{3}}, \tag{10}$$

and then express the amplitude $\mathcal{A}$ at generic momenta as a function of the renormalized coupling. (*Hopefully, the result will be manifestly finite in the $\delta \to 0$ limit.*)

Note: the condition on the RHS of (10) requires analytic continuation of the momenta outside of the regime of physical kinematics; nonetheless, the mass-shell condition and the momentum conservation is preserved, and such an analytic continuation is unambiguous.

Problem 1 answer:

For reference, the Lagrangian for the 2-scalar model is

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi_1)^2 - \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}m_1^2\phi_1^2 - \frac{1}{2}m_2^2\phi_2^2 - \frac{g}{2}\phi_1^2\phi_2 \tag{1.797}$$

where we are taking the masses to be $m_2 > m_1 > 0$. At this point, we should be able to read off the Feynman rules. If we do *not* Wick rotation the action, then we get a factor of i in the path integral. We can also perform integration by parts on the kinetic terms to get [**AC:**] Insert figures for Feynman diagrams the propagator and interaction term which mediates 2 ϕ_1 scalars and 1 ϕ_2 scalar

$$\text{“propagators”} = \frac{-i}{p^2 - m_i^2 + i\epsilon} \tag{1.798}$$

$$\text{“vertex”} = ig. \tag{1.799}$$

At order g^2, there is only the tree level diagrams which exchange a ϕ_2 particle of mass m_2. For the s-channel diagram, the value is [**AC:**] insert s,t,u diagrams

$$\text{“s-channel diagram”} = \frac{ig^2}{(p_1+p_2)^2 - m_2^2 + i\epsilon} \tag{1.800}$$

where the exchanged momentum is running from left to right in the Feynman diagram. We can use Mandelstam variables which we define here as

$$s = (p_1+p_2)^2, \;\; t = (p_1-p_3)^2, \;\; u = (p_1-p_4)^2. \tag{1.801}$$

In this convention, conservation of momentum is $(p_1+p_2)^\mu = (p_3+p_4)^\mu$ (note that many people just drop the spacetime index μ when writing the momenta out). The t- and u-channel follow suit namely the entire tree level diagram is

$$\mathcal{A}(k_3,k_4|k_1,k_2) = ig^2\left(\frac{1}{s-m_2^2+i\epsilon} + \frac{1}{t-m_2^2+i\epsilon} + \frac{1}{u-m_2^2+i\epsilon}\right). \tag{1.802}$$

Moving onto part (b), notice that we want to position ourselves into the center-of-mass (COM) frame where, since we have identical particles (both scalars) $\omega_i = \omega_j = \omega \equiv E/2$. We are also in $D = 4$ dimensions. In COM, s is the total energy and we can write the t and u variables in terms of total energy, mass, and cosines

$$s = E^2, \;\; t = -\frac{E^2-4m_1^2}{2}(1-\cos(\theta)), \;\; u = -\frac{E^2-4m_1^2}{2}(1+\cos(\theta)). \tag{1.803}$$

Now consider the equations that have been given with respect to the Lorentz invariant amplitude $\mathcal{M}$. For 4-particle scattering, we can write the relationship between the amplitude $\mathcal{A}$ and the Lorentz invariant amplitude $\mathcal{M}$ as

$$\mathcal{M}(\cos(\theta)) = \frac{1}{(2\pi)^6 E^2}\mathcal{A}(\cos(\theta)) \implies i\mathcal{A} = i\mathcal{M}(\cos(\theta))(2\pi)^6 E^2. \tag{1.804}$$

For scalars, $\ell = 0$, and so $P_0(\hat{k}_1 \cdot \hat{k}_3) = 1$. The math actually enforces this on us to. For instance, consider integrating over $d\cos(\theta)$ on boths sides of the amplitude namely there is the term

$$\int_{-1}^{1} P_\ell(\cos(\theta)) d\cos(\theta) = 2\delta_{\ell,0} \tag{1.805}$$

where $\delta_{\ell,0} \neq 0$ only if $\ell = 0$. Then, on the partial wave expansion side (equation (3) in the homework outline)

$$(\mathcal{N}(E))^2 \frac{1}{4\pi} \cdot 2 \cdot (S_0(E) - 1). \tag{1.806}$$

In the defintion for the partial wave expansion of $\mathcal{M}$, there is a factor of $(2\pi)^4$, so lets only move over that term in (1.804) which gives us

$$\frac{i}{(2\pi)^2 E^2} \int_{-1}^{1} \mathcal{A} d\cos(\theta) = \frac{\mathcal{N}(E)^2}{2\pi}(S_0(E) - 1) \implies S_0(E) - 1 = \frac{i|k|}{16\pi E} \int_{-1}^{1} \mathcal{A}(\cos(\theta)) d\cos(\theta) \tag{1.807}$$

where I used $\mathcal{N}(E)^2 = 8/(|k|E)$. Switching to integrating over $\mathcal{A}(\cos(\theta))$, notice that the integrating is only over $\cos(\theta)$ and so the s-channel term stays (replace $s \to E^2$). Let $A = (E^2 - 4m_1^2)/2$. The $t - u$ channel terms are then of the form

$$I_{tu} = -\int_{-1}^{1} d\cos(\theta) \left(\frac{1}{m_2^2 + A(1 - \cos(\theta)} + \frac{1}{m_2^2 + A(1 + \cos(\theta)} \right). \tag{1.808}$$

Notice we can combine the fractions into a single term that is a well known integral namely

$$\frac{1}{C - A\cos(\theta)} + \frac{1}{C + A\cos(\theta)} = \frac{2C}{C^2 - A^2\cos^2(\theta)} \tag{1.809}$$

where $C \equiv m_2^2 + A$, which then becomes

$$I_{tu} = -2C \int_{-1}^{1} \frac{d\cos(\theta)}{C^2 - A^2\cos^2(\theta)} \tag{1.810}$$

which is an integral of the form

$$\int_{-1}^{1} \frac{d\cos(\theta)}{a^2 - b^2\cos^2(\theta)} = \frac{1}{ab} \ln\left(\frac{a+b}{a-b}\right) \tag{1.811}$$

assuming $C > A$. Then

$$I_{tu} = -\frac{2}{A} \ln\left(\frac{C+A}{C-a}\right) = \frac{4}{E^2 - 4m_1^2} \ln\left(1 + \frac{E^2 - 4m_1^2}{m_2^2}\right). \tag{1.812}$$

We can now solve for $S_0(E)$ and simplify to find that

$$S_0(E) = 1 - \frac{g^2|k|}{8\pi E} \left(\frac{1}{E^2 - m_2^2 + i\epsilon} - \frac{2}{E^2 - 4m_1^2} \ln\left(1 + \frac{E^2 - 4m_1^2}{m_2^2}\right)\right). \tag{1.813}$$

Recall that $|k| = \sqrt{E^2/4 - m_1^2} = i\sqrt{m_1^2 - E^2/4}$.

To compare with unitarity bounds, recall that if S is a unitary matrix, then $SS^\dagger = 1$ which implies that $|S_\ell(E)| \leq 1$ which implies itself that $S_\ell(E)$ itself should be a pure phase $S_\ell(E) = \exp(2i\delta_\ell(E))$. We can expand the S-matrix as

$$S = e^{2i\delta_\ell(E)} = 1 + 2i\delta_\ell(E) + \mathcal{O}(\delta_\ell). \tag{1.814}$$

Noticethat this allows us to read off the phase shift $\delta_0(E)$ as

$$\delta_0(E) = -\frac{g^2|k|}{16\pi E}\left(\frac{1}{E^2 - m_2^2 + i\epsilon} - \frac{2}{E^2 - 4m_1^2}\ln\left(1 + \frac{E^2 - 4m_1^2}{m_2^2}\right)\right) + \mathcal{O}(g^3). \tag{1.815}$$

Notice that all the arguments in the brackets is positive and real since $E < 2m_1^2 < 2m_2^2$ (since we are only probing tree level perturbations, we can only excite particles of certain energy at a thresholds $E = m_2^2$ and $E = 2m_1$ due to the pole-like structure of the denominators (otherwise the logarithm can diverge). Then, it would appear that $S_0(E)$ is a pure phase since δ_0 is a real phase. Notice that this interpretation stems from taking the threshold $|k|$ is imaginary, and we plot on complex momentum. Notice though that below the inelastic threshold, there are no physical processes that can lead to an imaginary part of the amplitude. [**AC:**] Insert graph of pole structure for s.

Problem 2 answer:

For part (a), to find the mass of the lightest particle, i.e. that of ϕ_1, recall that the Kallen-Lehmann representation allows us to rewrite the 2-point Greens function as

$$\tilde{G}_E(k) = \frac{1}{k^2 + m^2 - \Sigma(k)} = \frac{Z}{k^2 + m_*^2} + \int\limits_{\mu \geq 4m_*^2} d\mu^2 \frac{\sigma(\mu^2)}{k^2 + \mu^2} \tag{1.816}$$

where m_* is the actual mass which becomes a sum of higher order g^n terms at order n. It is typically written as $m_*^2 = m_R^2 + \delta m^2$ where m_R is the finite renormalized mass. The two mass terms are remedied via

$$m_*^2 = m_R^2 + \delta m^2 + \mathcal{O}(g^n) \tag{1.817}$$

where we typically choose δm^2 such that it cancels divergences that arises in the loop calculations.

Recall also that $\Sigma_2(k)$ is the 1-particle irreducible (1PI) corrections to the mass. This is essentially what we are computing when computing loop corrections for the self-energy of any particle. For us, since we have a coupling of $g\phi_1^2\phi_2$, then the loop will have a term containing both ϕ_1 and ϕ_2 namely we will compute and put into the following form

$$\begin{aligned} i\Sigma(k) &= g^2 \int \frac{d^D p}{(2\pi)^D} \frac{1}{p^2 - m_1^2 + i\epsilon} \frac{1}{(p-k)^2 - m_2^2 + i\epsilon} \\ &= g^2 \int\limits_0^1 dx \int \frac{d^D \ell}{(2\pi)^D} \frac{1}{(\ell^2 - \Delta)^2}. \end{aligned} \tag{1.818}$$

To get to the final form quoted, we use the Feynman parameter integration

$$\frac{1}{AB} = \int\limits_0^1 dx \frac{1}{[A + (B-A)x]^2} \tag{1.819}$$

where $A = p^2 - m_1^2 + i\epsilon$ and $B = (p-k)^2 - m_2^2 + i\epsilon$. For now, I will drop the $i\epsilon$ since it always ends up in the end by itself. You get to perform the algebra, but you should find that if you shift the loop momentum $p \to \ell + xk$, then you are left with a denominator of the form

$$\frac{1}{(\ell^2 - \Delta)^2} \tag{1.820}$$

where $\Delta \equiv x(m_2^2 + p^2(x-1)) + m_1^2(1-x)$. We then have the form of (1.818). The momentum integral is "well known" namely

$$\frac{d^D \ell}{(2\pi)^D} \frac{1}{(\ell^2 - \Delta + i\epsilon)^2} = \frac{i}{(4\pi)^{D/2}} \Delta^{\frac{D}{2}-2} \Gamma\left(\frac{4-D}{2}\right). \tag{1.821}$$

Now, we expand $D \to 4 - \delta$ which can be done in Mathematica which gives us the following answer (with a factor multiplied by μ^{4-D} where μ has mass dimensions of 1)

$$i\Sigma(k) = \frac{ig^2}{16\pi^2} \int\limits_0^1 dx \left(\frac{2}{\delta} + \ln(4\pi) - \gamma_E + \ln\left(\mu^2/\Delta\right)\right) \tag{1.822}$$

where γ_E is the Euler-Mascheroni constant. To make it more feasible as to what are the physics and what is not, consider defining

$$B(k^2, m_1^2, m_2^2) = \int_0^1 dx \ln\left(\frac{\mu^2}{\Delta}\right). \tag{1.823}$$

Then, we can relate the mass of ϕ_1 to that of the 1PI to be

$$m_1^2 = m_R^2 + \Sigma_1(k = m_1^2). \tag{1.824}$$

This arises from the fact that the propagator G must have a pole when $k^2 = m^2$ where m^2 here is the physical mass in question.

Actually evaluating the integral is not very illuminating, but the structure of the integral is since it depends on both m_1 and m_2, and thus the mass of the scalar field ϕ_1 gets contributions from ϕ_2. We will see a much deeper aspect of this in the next part of the problem. The field renormalization Z is found by taking the Källen-Lehman spectral representation given in (1.816)

$$Z = 1 - \frac{d\Sigma_1(k^2)}{dk^2}\bigg|_{p^2=m^2} = 1 - \frac{g^2}{(4\pi)^2}\int_0^1 dx \, \frac{x(1-x)}{xm_2^2 + (1-x)m_2^2 - x(1-x)m_1^2}. \tag{1.825}$$

Now, for part (b), you should recognized that this calculation is the exact same as part (a) but replace m_2 in the integral with m_1 so that the only mass terms present are m_1^2. Then, the math is exactly the same, it is just that now we are computing the self-energy for the ϕ_2 term (as a hint, notice that $\Delta = m_1^2 - x(1-x)k^2$ in this case). The self-energy is quite literally the same except that

$$\Sigma_2(k) = \frac{g^2}{16\pi^2}\left(\frac{2}{\delta} - \Gamma_E + \ln(4\pi) - \int_0^1 dx \ \ln\left(\frac{m_1^2 - x(1-x)k^2 - i\epsilon}{\mu^2}\right)\right). \tag{1.826}$$

This is related to the self-energy (propagator plus corrections) of ϕ_2 as

$$\langle\Omega| T\hat{\phi}_2(x)\hat{\phi}_2(0)|\Omega\rangle = i\int \frac{d^Dk}{(2\pi)^D}\frac{e^{ik\cdot x}}{k^2 - m_2^2 + i\epsilon + \Sigma_2(k)} = \int \frac{d^Dk}{(2\pi)^D} i\Pi(k^2)e^{ik\cdot x}. \tag{1.827}$$

Remember that it was hinted at earlier in the semester that the physical mass of the particle can be found at the pole of k, which should give us some indication on how to proceed. Consider setting the momentum k^2 inside of $\Sigma_2(k)$ on shell i.e. $k^2 = m_1^2$. Then *only* the loop contribution can be written as (changing $\mu \to \Lambda$ for clarity later)

$$\Sigma_2(k) \supset -\frac{g^2}{16\pi^2}\int_0^1 dx \ln\left(\frac{m_1^2 - xm_1^2(1-x) - i\epsilon}{\Lambda^2}\right) \tag{1.828}$$

where the finite terms can always be chosen to cancel and vanish by selecting the proper δm^2 term (just as in L12). Now notice that $x(1-x) \leq 1/4$ thus for $2m_1 < m2$ the

expression is *real* and thus $\mathrm{Im}\,\mathcal{M}_{\mathrm{loop}} = 0$. In this regime, the decay rate i.e. the possibility that ϕ_2 can decay into $\phi_1\phi_2$ pair is zero.

Ok, let me give some context instead of just saying "decay rate" which seems like it came out of nowhere. The decay rate comes from the *optical theorem*; an extremely important result in every area of scattering-physics.

Recall that the S-matrix is a unitary matrix since it must satisfy $SS^\dagger = 1$ where 1 is really the identity matrix in 4-dimensions. Now recall that the S-matrix elements can be related to the scattering matrix $\mathcal{T}$ for some incoming $|i\rangle$ and outgoing state $\langle f|$ defined as

$$\langle f|\,\mathcal{T}\,|i\rangle = (2\pi)^4\delta^{(4)}(p_i - p_f)\mathcal{M}(i \to f) \tag{1.829}$$

which is really just the statement that we can break down the S matrix into the no-scattering plus the scattered stuff as $S = 1 + i\mathcal{T}$. This simple relation however combined with unitarity gives $1 = SS^\dagger = (1 - i\mathcal{T}^\dagger)(1 + i\mathcal{T})$ omitting

$$i(\mathcal{T}^\dagger - \mathcal{T}) = \mathcal{T}^\dagger\mathcal{T}. \tag{1.830}$$

If we sandwich the initial and final states together while inserting a complete basis of states given as $1 = \sum_X \int d\Pi_X\,|X\rangle\,\langle X|$ where $d\Pi_X$ is a Lorentz invariant measure over states X (which can be anything really) then

$$\begin{aligned}\langle f|\,\mathcal{T}^\dagger\mathcal{T}\,|i\rangle &= \sum_X \int d\Pi_X\,\langle f|\,\mathcal{T}^\dagger\,|X\rangle\,\langle X|\,\mathcal{T}\,|i\rangle \\ &= \sum_X \int d\Pi_X (2\pi)^4\delta^{(4)}(p_f - p_X)(2\pi)^4\delta^{(4)}(p_i - p_X)\mathcal{M}(i \to X)\mathcal{M}^*(f \to X).\end{aligned} \tag{1.831}$$

In other words, we get the *generalized optical theorem*

$$\mathcal{M}(i \to f) - \mathcal{M}^*(f \to i) = i\sum_X \int d\Pi_X (2\pi)^4\delta^{(4)}(p_i - p_X))\mathcal{M}(i \to X)\mathcal{M}^*(f \to X). \tag{1.832}$$

A special case is when the initial and final state are the same, say $|i\rangle = |f\rangle = |A\rangle$. Then

$$2i\mathrm{Im}\mathcal{M}(A \to A) = i\sum_X \int d\Pi_X (2\pi)^4\delta^{(4)}(p_A - p_X)|\mathcal{M}(A \to X)|^2. \tag{1.833}$$

If $|A\rangle$ is a 1-particle state, then the decay rate is

$$\Gamma(A \to X) = \frac{1}{2m_A}\int d\Pi_X (2\pi)^4\delta^{(4)}(p_A - p_X)|\mathcal{M}(A \to X)|^2. \tag{1.834}$$

Thus,

$$\mathrm{Im}\mathcal{M}(A \to A) = m_A\sum_X \Gamma(A \to X) = m_A\Gamma_{\mathrm{total}} \tag{1.835}$$

where Γ_{total} is the total decay rate of a particle, equal to its inverse lifetime.

Now, we can move on further for the other case of the masses.

However, for the $2m_1 > m_2$ recall that from some of your mathematics class

$$\ln(-A - i\epsilon) = \ln A - i\pi. \tag{1.836}$$

The loop can then be integrated as

$$\Sigma_2(k) \supset \frac{g^2}{16\pi^2} \int_0^1 dx\, \theta(m_1^2 x(1-x) - m_2^2) = \frac{g^2}{16\pi^2} \sqrt{1 - \frac{m_1^2}{m_2^2}} \theta(m_2 - 2m_1). \tag{1.837}$$

Now recall that we can define the spectral representation or the Källén-Lehmann representation as

$$\Pi(k^2) \equiv \int_0^\infty dq^2 \frac{\rho(q^2)}{k^2 - q^2 + i\epsilon}. \tag{1.838}$$

We can then write the spectral density as

$$\rho(k^2) = -\frac{1}{\pi} \operatorname{Im} \Pi(k^2). \tag{1.839}$$

The following identity may be of some use

$$\operatorname{Im} \frac{1}{p^2 - m^2 + i\epsilon} = -\pi\delta(p^2 - m^2). \tag{1.840}$$

Then, the spectral function can be written as

$$\begin{aligned}
\rho(k^2) &= -\frac{1}{\pi} \operatorname{Im} \Pi(k^2) \\
&= -\frac{1}{\pi} \left(k^2 - m_2^2 + i\epsilon + \frac{ig^2}{16\pi^2} \sqrt{1 - \frac{4m_1^2}{k^2}} \theta(k^2 - 4m_1^2) + \cdots \right) \\
&= \delta(k^2 - m_2^2) + \theta(k^2 - 4m_1^2) \frac{g^2}{16\pi^2} \frac{1}{(k^2 - m_2^2)^2} \sqrt{1 - \frac{4m_1^2}{k^2}} + \cdots
\end{aligned} \tag{1.841}$$

Ok, that was a lot of math, but what the hell did we find? The spectral density function is observable (in principle) since it is base on the observable 2-point Greens function. Notice that there is a pole at the 1-particle states in m_2 in the complex k-plane (everything we discuss is in the complex k-plane). The theta-step function is an argument that ensures the spectral function is non-zero only when $k^2 > 4m_1^2$. There is a branch cut in the square root when

$$1 - \frac{4m_1^2}{k^2} = 0 \implies k^2 = 4m^2 \tag{1.842}$$

is when the branch cut singularity starts above the multi-particle threshold.

If we graphed $\rho(k^2)$, then below $k^2 = 4m_1^2$ then ρ vanishes, while above the threshold, it peaks at $k^2 = m_2^2$ and then decreases. There is also a single line at m_2^2. For instance, see the graph in L11. [**AC:**] insert graph for spectral density function

Problem 3 answer:

For reference, the action for ϕ^4-theory is

$$\begin{aligned} S &= \int d^D x \left(-\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{4!}\phi^4 \right) \\ &= \int d^D x \left(\frac{1}{2}\phi \left[\Box - m^2 \right] \phi - \frac{g}{4!}\phi^4 \right) \end{aligned} \tag{1.843}$$

where we integrated by parts on the first term using $A\partial_\mu B = -B\partial_\mu A$. We can then read off the propagator and interaction vertext to be[**AC:**] insert images later

$$\text{“propagator”} = \frac{i}{p^2 - m^2 + i\epsilon} \tag{1.844}$$

$$\text{“4-point vertex”} = -ig. \tag{1.845}$$

The $2 \to 2$ scattering amplitude $\mathcal{A}(p_3 p_4 | p_1 p_2)$ up to order g^2 can be written as

$$i\mathcal{A}(p_3 p_4 | p_1 p_2) = -ig + (-ig)^2 \left(\mathcal{F}(s) + \mathcal{F}(t) + \mathcal{F}(u) \right) + \mathcal{O}(g^3) \tag{1.846}$$

where the $\mathcal{F}$s are the integral function of the 1-loop corrections for the s, t, and u-channel with convention

$$s = -(p_1 + p_2)^2, \;\; t = -(p_1 - p_3)^2, \;\; u = -(p_1 - p_4)^2. \tag{1.847}$$

The masses of the incoming scalars are the same, but you could have them all be different.

We have three loop diagrams to compute, but luckily enough, scalars have so much symmetry (in that they essentially have an infinite $SO(D)$ symmetry about them) where we only need to compute the s-channel diagram, and the we can add the rest by switching s for t and u. Notice though that there is the constraint $s + t + u = 4m^2$ which arises due to conservation of momentum.

Therefore, the s-channel diagram is

$$\begin{aligned} \text{“s-channel”} &= \frac{(-ig)^2}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(k + p_1 + p_2)^2 - m^2 + i\epsilon} \\ &\equiv (-ig)^2 \cdot iV(p^2) \end{aligned} \tag{1.848}$$

where we defined $p \equiv p_1 + p_2$ for convenience but notice that $s = p^2$ in this convention, and the loop momentum is k^μ. The next two diagrams are identical, but replace s (i.e. p^2) with t and u. The entire amplitude can then be written as

$$i\mathcal{A} = -ig + (-ig)^2 \left(iV(s) + iV(t) + iV(u) \right) + \mathcal{O}(g^3). \tag{1.849}$$

We can compute $V(s)$ via dimensional-regularization (dim-reg) via $D = 4 - \delta$. This is in conjugation with the Feynman trick we have used previously. We therefore obtain

$$\begin{aligned}
V(s) &= \frac{i}{2}\int_0^1 dx \int \frac{d^D k}{(2\pi)^D}\frac{1}{(k^2 + 2xk\cdot p + xs - m^2)^2} \\
&= \frac{i}{2}\int_0^1 dx \int \frac{d^D k}{(2\pi)^D}\frac{1}{(\ell^2 + x(1-x)s - m^2)^2} && (\ell = k + xp) \\
&= -\frac{1}{2}\int_0^1 dx \int \frac{d^D \ell_E}{(2\pi)^D}\frac{1}{(\ell_E^2 - x(1-x)s + m^2)^2} && (\ell_E^0 = -i\ell^0) \\
&= -\frac{1}{2}\int_0^1 dx \frac{\Gamma(2-\frac{D}{2})}{(4\pi)^{D/2}}\frac{1}{(m^2 - x(1-x)s)^{2-D/2}}. && (1.850)
\end{aligned}$$

We then add two more copies of (1.850) but replace s with t and u and we are done with part (a).

For part (b) it is a little more tricky, and we must be careful with the algebra. The *counter-term* we want to include in our action is of the form

$$S \supset \int d^D x\, \frac{C}{\delta} g^2 \phi^4. \tag{1.851}$$

This produces its *own* corresponding Feynman diagram namely

$$\textbf{“counter-term”} = -ig^2\left(\frac{C}{\delta}\right). \tag{1.852}$$

The total amplitudes is then

$$i\mathcal{A} = -ig - ig^2\left(\frac{C}{\delta}\right) + i(-ig)^2\left(V(s) + V(t) + V(u)\right) + \mathcal{O}(g^3). \tag{1.853}$$

We can series expand in Mathematica the V-functions to find

$$\begin{aligned}
V(s) + V(t) + V(u) = \int_0^1 dx \Bigg(&\frac{3}{8\pi^2\delta} + \frac{1}{16\pi^2}\Bigg(-3\gamma_E + 6\log(2) + 3\log(\pi) - \log\Big(m^2 - s(1-x)x\Big) \\
&- \log\Big(m^2 - t(1-x)x\Big) - \log\Big(m^2 - u(1-x)x\Big)\Bigg)\Bigg) + \mathcal{O}(\delta) \\
= \frac{3}{8\pi^2\delta} &+ \frac{1}{16\pi^2}\int_0^1 dx \Bigg(-3\gamma_E + 3\log(4\pi) \\
&- \log\Big(m^2 - s(1-x)x\Big) - \log\Big(m^2 - t(1-x)x\Big) - \log\Big(m^2 - u(1-x)x\Big)\Bigg) + \mathcal{O}(\delta).
\end{aligned} \tag{1.854}$$

The constants in the equation, γ_E and $\log(4\pi)$ have no physical meaning, and so we can absorb them into our constant C as well. If we want the constants to cancel, then we can simply set them to the following when $s = t = u = 4m^2/3$

$$\begin{aligned} -ig^2\frac{C}{\delta} &= \frac{3}{8\pi^2\delta} + \frac{3}{16\pi^2}(\log(4\pi) - \gamma_E) \\ &\quad - \frac{3}{16\pi^2}\int_0^1 dx\left(\log\left(m^2 - \frac{4}{3}m^2(1-x)x\right)\right) + \mathcal{O}(\delta). \end{aligned} \tag{1.855}$$

Then, with the renormalized constant, we find the ugly looking, but finite with a lot of physical information, final result

$$\begin{aligned} i\mathcal{A} &= -ig \\ &\quad - \frac{ig^2}{32\pi^2}\int_0^1 dx\left(\log\left(\frac{m^2 - x(1-x)s}{m^2 - 4m^2(1-x)/3}\right) + \log\left(\frac{m^2 - x(1-x)t}{m^2}\right) + \log\left(\frac{m^2 - x(1-x)u}{m^2}\right)\right) + \mathcal{O}(g^3) \end{aligned} \tag{1.856}$$

The extra factors of $-\log(m^2)$ come from the fact that the original problem does not account for units of momenta, and so I fixed it. This problem is also in Peskin & Schroeder and, albeit, does a much more clear job than the lecture notes. I can show anyone how to do this on Mathematica.

1.8 Problem set 5:

1.8.1 Problem 1: Wigner group for massless fermions

In lecture, we have discussed massive spin-1/2 particle and the polarization of a spinor field $\hat{\psi}_\alpha(x)$. In this problem you will consider the case where the particle is massless.

Let $\left|\vec{k}, h\right\rangle$ label the 1-particle state of a massless particle (energy $\omega_{\vec{k}} = |\vec{k}|$) with helicity h, namely

$$\vec{J} \cdot \hat{\vec{k}} \left|\vec{k}, h\right\rangle = h \left|\vec{k}, h\right\rangle . \tag{1.857}$$

"spin-$\frac{1}{2}$" for a massless particle means that $h = \frac{1}{2}$ or $-\frac{1}{2}$. Note that h is invariant under Lorentz symmetry, and so it is possible that only the $h = +\frac{1}{2}$ state exists, or only the $h = -\frac{1}{2}$ state exists. The spinor polarization $u^h_\alpha(\vec{k})$ appears in the relation

$$\left\langle \vec{k}, h \right| \hat{\psi}_\alpha(x) \left|\Omega\right\rangle = \mathcal{Z}_\psi^{1/2} \frac{e^{-ik\cdot x}}{(2\pi)^{3/2}} u^h_\alpha(\vec{k}), \tag{1.858}$$

where $\mathcal{Z}_\psi$ is the field renormalization constant.

Now consider the reference momentum $k_R = (E, 0, 0, E)$, in which case the little group consists of Lorentz transformations of the form

$$W(a, b, \theta) = e^{iaA+ibB} e^{i\theta M_{12}}, \tag{1.859}$$

where

$$\begin{aligned} A &= M_{01} + M_{31} = -i \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \\ B &= M_{02} + M_{32} = -i \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \end{aligned} \tag{1.860}$$

and $(M_{\mu\nu})^\rho{}_\sigma = -i(\delta^\rho{}_\mu \eta_{\nu\sigma} - \delta^\rho{}_\nu \eta_{\mu\sigma})$. As seen in the lecture notes, it follows from Lorentz symmetry that $u^h_\alpha(k_R)$ obeys

$$u^h_\alpha(k_R) = (D(W))^* \, (R(W))_\alpha{}^\beta \, u^h_\beta(k_R), \tag{1.861}$$

where $D(W(a, b, \theta)) = e^{ih\theta}$.

(a) What is the matrix $R(W(a, b, \theta))$?

(b) By inspecting (5) with nonzero a or b, show that $\gamma^3 u^h = 2h\, u^h$. *(This means that if $\hat{\psi}_\alpha(x)$ obeys say $\gamma^3\hat{\psi} = \hat{\psi}$, it can only create a particle of helicity $+\frac{1}{2}$ and not $-\frac{1}{2}$.)*

1.8.2 Problem 2: 4-Fermi theory scattering

Consider in $D = 4$ spacetime dimensions a theory of Dirac fermion field $\psi_\alpha(x)$ with self-interaction described by the Lagrangian density

$$\mathcal{L} = -\bar{\psi}(\partial\!\!\!/ + m)\psi - \frac{1}{2} g \, (\bar{\psi}\gamma^\mu\psi)(\bar{\psi}\gamma_\mu\psi), \tag{1.862}$$

where g is a coupling constant.

(a) Calculate the tree level (i.e. order g) $2 \to 2$ scattering amplitude of a pair of fermions of the same type (i.e. both particles, as opposed to anti-particles). You can express the result in terms of the spinor polarizations. *(Be careful with the relative sign between different diagrams/Wick contractions!)*

(b) In the center-of-mass frame, calculate the total scattering cross section due to such an elastic process at tree level, for a pair of incoming particles of helicity $+\frac{1}{2}$[57], as a function of the total energy E.

Remark: in computing the cross section by summing over out-states, you can simplify your computation using the formulae

$$\begin{aligned} &\sum_\sigma u^\sigma(\vec{k})\,\bar{u}^\sigma(\vec{k}) = \frac{m - i\not{k}}{2\omega_{\vec{k}}}, \\ \text{or} \qquad &\sum_\sigma v^\sigma(\vec{k})\,\bar{v}^\sigma(\vec{k}) = \frac{-m - i\not{k}}{2\omega_{\vec{k}}}. \end{aligned} \tag{1.863}$$

In dealing with the in-state spinor polarizations, you may find it convenient to assume (without loss of generality) that the incoming particle momenta are in the x^3-direction, in which case $u^\sigma(k\hat{x}^3)$ can be obtained from $u^\sigma(0)$ by the Lorentz boost

$$u^\sigma(k\hat{x}^3) = \sqrt{\frac{m}{\omega_k}}\,\exp\Big(-i\nu_k s^{03}\Big)\,u^\sigma(0), \tag{1.864}$$

where $S^{03} = -\frac{i}{2}\,[\gamma^0, \gamma^3]$, and $\nu_k = \operatorname{arctanh}\Big(\frac{k}{\omega_k}\Big)$ is the rapidity of the particle.

[57] Helicity of a massive particle is defined in the same way as it was above. Unlike the massless case, massive particle states of different helicities can mix under Lorentz transformations.

Problem 1 answer:

For some background, recall that this problem focuses on the Little Group of *massless* spinors which does change the physics and Little Group structure since instead of polarizations of the particles, we will instead find a different quantity called *helicity* (which is just polarization for massless particles). The spinor polarization $u^h_\alpha(\vec{k})$ appears when we compute the 1-particle state of the spinor-field

$$\left\langle \vec{k}, h \right| \hat{\psi}_\alpha(x) \left| \Omega \right\rangle = \sqrt{Z_\psi} \frac{e^{-ik\cdot x}}{(2\pi)^{3/2}} u^h_\alpha(\vec{k}) \tag{1.865}$$

where Z_ψ is the field renormalization. Since these spinors are massless, we consider the reference momentum $k^\mu_R = (E, 0, 0, E)$ such that $k^2_R = E^2 - E^2 = 0$ as a null four-vector should behave. A lot of this problem will require reference to Weinberg's chapter 2.5 (starting on page 69) [1] (where in Weinberg they use $J^{\mu\nu}$ but here we use $M^{\mu\nu}$ which is primarily used to differentiate the generators of the Poincaré group when talking about massive or massless fields).

The Little Group element we are considering consists of Lorentz transformations of the form

$$W(a, b, \theta) = e^{iaA+ibB} e^{i\theta M_{12}} \tag{1.866}$$

which has a infinitesimal representation of

$$W(\theta, a, b)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu \tag{1.867}$$

where the infinitesimal rotations are

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & \theta & -a & a \\ -\theta & 0 & -b & -b \\ a & b & 0 & 0 \\ -a & -b & 0 & 0 \end{pmatrix} \tag{1.868}$$

and the Little Group element has a unitary representation of

$$U(W(a, b, \theta)) = 1 + iaA + ibB + i\theta M_{12} \tag{1.869}$$

where

$$A = M_{01} + M_{31} = -i \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad B = M_{02} + M_{32} = -i \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{1.870}$$

where the matrix M obeys

$$(M_{\mu\nu})^\rho{}_\sigma = -i(\delta^\rho_\mu \eta_{\nu\sigma} - \delta^\rho_\nu \eta_{\mu\sigma}). \tag{1.871}$$

At this point, we can continue following Weinberg's chapter and will find the same answer, but we will diverge from Weinberg and consider a different point of view that is sometimes

more useful. The matrices that were given prior are spacetime representations of the Little Group, but we will be using the *spinor* representation (for the spacetime one, just read Weinberg) also called the Weyl representation or Weyl spinors.

Recall that $S^{\mu\nu} = -\frac{i}{4}[\gamma^\mu, \gamma^\nu]$ that was defined previously in lecture L17. For massless spinors, we can decompose the Dirac spinor ψ from a four-dimensional "vector" into a two-component spinor comprised of two, two-component spinors becoming a left-handed and right-handed Weyl spinor. The Gamma matrices can also be decomposed in this manner too namely

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\beta}} \\ \overline{\sigma}^{\mu\alpha\dot{\beta}} & 0 \end{pmatrix} \tag{1.872}$$

where the μ is a spacetime index while $\alpha, \dot{\beta}$ are the spinor indices. The Pauli matrices are simply $\sigma^\mu = (I, \sigma^i)$ and similar for $\overline{\sigma}^\mu = (I, -\sigma^i)$. Based on the form of γ^μ, we can use them to find a new relation for $S^{\mu\nu}$ which creates a new 2×2 block namely

$$S^{\mu\nu} = \begin{pmatrix} (\sigma^{\mu\nu})^\beta_\alpha & 0 \\ 0 & (\overline{\sigma}^{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} \end{pmatrix} \tag{1.873}$$

where $\Sigma^{\mu\nu} \equiv (\sigma^{\mu\nu})^\beta_\alpha = \frac{i}{4}(\sigma^\mu\overline{\sigma}^\nu - \sigma^\nu\overline{\sigma}^\mu)^\beta_\alpha$ and similar for the right-handed spinor index version but with the overline-σs swapped. This is what allows us to write

$$A = M_{01} + M_{31} = K^1 + M_{31} = \frac{i}{2}\sigma^1 + \frac{1}{2}\sigma^2 \tag{1.874}$$

$$B = M_{02} + M_{32} = K^2 + M_{32} = \frac{i}{2}\sigma^2 + \frac{1}{2}\sigma^1 \tag{1.875}$$

since the Ms can be written in terms of S's, and the S's can be written in terms of $\sigma^{\mu\nu}$s, which can be reduced down to σ^μs ($M^{\mu\nu}$ obeys the same commutation relation as $S^{\mu\nu}$ and so there is a correspondence between them). Then, A and B have the form

$$A = \begin{pmatrix} 0 & 0 \\ i & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{1.876}$$

Notice that A and B are *nilpotent* i.e. $A^2 = I$ and thus their exponential expansion simplfies greatly to $e^{iaA} = 1 + iaA$ and $e^{ibB} = 1 + ibB$. The generator M_{12} is the rotation around the z-axis $\mathbf{J}^3 = \sigma^3/2$. The rotation in terms of Little Group elements is then

$$e^{i\theta\Sigma_{12}} = e^{i\theta\sigma^3/2} = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}. \tag{1.877}$$

The full form of $R(W)$ is then

$$R(W(a, b, \theta)) = e^{i\theta\sigma^3/2}(I + iaA)(I + ibB) \tag{1.878}$$

but the commutator between A and B are diagonal i.e. their commutator vanishes $[A, B] = 0$, and thus we can drop the a and b parts at the end. (Note: technically we should be writing our answer as $U(R(\theta))$ since it is a unitary representation of the rotations with

respect to the Little Group elements, but I am being sloppy). We can of course double check our answer using how spinors are invariant under Little Group transformations

$$u^h_\alpha(\vec{k}_R) = (D(W))^*(R(W))_\alpha{}^\beta u^h_\beta(\vec{k}_R) \tag{1.879}$$

with $D(W) = e^{ih\theta}$. Assume that u^h is an eigenvector of σ^3 with helicity h (and still reference momentum $\vec{k}_R$ but I am dropping that for now). Then, for $h = 1/2$ and $h = -1/2$ we can write the spinors as

$$u^+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } u^- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1.880}$$

which satisfies $\sigma^3 u^\pm = \pm u^\pm$. You can then check using D and R that (1.879) is satisfied.

Finally, for part (b), there are many ways to tackle this problem. Notice that if I act on the spinor u^h with A I get a back as the eigenvalue and similarly for B I get b back. The issue is that if we find one such set of non-zero eigenvalues of A and B, then we have an entire continuum. This can be seen by action on A and B via a unitary transformation of the $R(\theta)$ i.e. $U(R)AU^{-1}(R) = A\cos(\theta) - B\sin(\theta)$ and similar for B. However, from observation, massless particles do not have any continuous degree of freedom like θ. In order to avoid a continuous cascade of states, we must require that *physical* states are eigenvectors of A and B with $a = b = 0$. These states are then distinguished by the eigenvalue of the remaining generator

$$M_{12}u^h = hu^h. \tag{1.881}$$

This is the same as showing that for $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ acting on u^h gives

$$\gamma^5 u^h = \pm u^h = 2hu^h. \tag{1.882}$$

Problem 2 answer:

The Lagrangian we will be looking at the entire time is the 4-Fermi Lagrangian which, as you will see in QFT 2, is the effective field theory Lagrangian for weak interactions,

$$\mathcal{L} = -\overline{\psi}(\not{\partial} + m)\psi - \frac{1}{2}g(\overline{\psi}\psi)(\overline{\psi}\psi) \tag{1.883}$$

where there are implicit spinor indices on the 4-fermion interaction (typically in the literature, g here is G_F).

At tree level, there are only two diagrams as shown at the end of lecture 20 (L20). If you expand the partition function, $Z[J] = \int [D\overline{\psi}][D\psi]e^{-S_{\text{euclidean}}}$ then you find at order g the term

$$Z \supset \langle \psi_\alpha(x_1)\psi_\beta(x_2)\overline{\psi}^\gamma(x_3)\overline{\psi}^\delta(x_4)\rangle_{\text{connected}} \tag{1.884}$$

modulo any disconnected Feynman diagrams. A possible contraction is

$$\psi_\alpha(x_1)\psi_\beta(x_2)\overline{\psi}^\gamma(x_3)\overline{\psi}^\gamma(x_4)\left(-\frac{ig}{2}\right)\int d^4y\overline{\psi}^\xi(y)\psi_\xi(y)\overline{\psi}^\lambda(y)\psi_\lambda(y) \tag{1.885}$$

plus three other combinations, yet all three can be represented by two diagrams with an *effective vertex* [**AC:**] insert t and u diagrams later. Notice that there is **no** s-channel contribution since, if you can recall, the s-channel process corresponds to the incoming particles creating a virtual particle (and the opposite for the outgoing) while the t- and u-channel processes correspond to the exchange of momentum between particles. Our 4-fermion interaction does not allow the "destruction" of two fermions that combine into some other particle, and then come out (wait until you get to electroweak theory such as computing neutrino-lepton scattering).

The two Feynman diagrams are

$$\text{“}t\text{-channel”} = +ig\int d^4y\langle\psi_\alpha\overline{\psi}^\xi\rangle\langle\psi_\xi\overline{\psi}^\gamma\rangle\langle\psi_\beta\overline{\psi}^\lambda\rangle\langle\psi_\lambda\overline{\psi}^\delta\rangle \tag{1.886}$$

$$\text{“}u\text{-channel”} = -ig\int d^4y\langle\psi_\alpha\overline{\psi}^\xi\rangle\langle\psi_\xi\overline{\psi}^\delta\rangle\langle\psi_\beta\overline{\psi}^\lambda\rangle\langle\psi_\lambda\overline{\psi}^\gamma\rangle \tag{1.887}$$

We can be slightly lazy and set $Z_\psi = 1$ which allows our spinor-polarization factors to take the form

$$\langle\psi_\alpha\overline{\psi}^\xi\rangle = \frac{1}{(2\pi)^{3/2}}(\overline{u}^{\sigma_1})^\xi(\vec{k}_1) \tag{1.888}$$

$$\langle\psi_\xi\overline{\psi}^\gamma\rangle = \frac{1}{(2\pi)^{3/2}}u_\xi^{\sigma_3}(-\vec{k}_3). \tag{1.889}$$

These fermion legs come from the LSZ reduction of the 2-point fermion propagator and the momentum is taken on-shell $p_1^2 = m^2$ namely from

$$\langle\psi_\alpha(x_1)\overline{\psi}^\xi(y)\rangle = \int\frac{d^4p_1}{(2\pi)^4}\frac{i(\not{p}_1 + m)^\xi_\alpha}{p_1^2 - m^2 + i\epsilon}e^{-ip_1\cdot(x_1-y)}. \tag{1.890}$$

Lets do the t-channel diagram, which we will denote as $\mathcal{A}_t$, and then from which we can read off the u-channel diagram easily by swapping the $\gamma \leftrightarrow \delta$ terms. To make things

easier for our eyes (and me typing), we can use the notation of the outgoing fermion as $\overline{u} \to v$ which is more typical. If we plug-in 4-factors of the spinor-polarizations we get

$$\mathcal{A}_t = ig \int d^4y \left(\frac{1}{(2\pi)^{3/2}}\right)^4 u_\alpha^{\sigma_1}(\vec{k}_1) e^{-ik_1\cdot(x_1-y)} \overline{u}_\xi^{\sigma_3}(-\vec{k}_3) e^{-ik_3\cdot(y-x_3)} u_\beta^{\sigma_2}(\vec{k}_2) e^{-ik_3\cdot(x_2-y)} \overline{u}_\lambda^{\sigma_4}(-\vec{k}_4) e^{-ik_3\cdot(y-x_4)}. \tag{1.891}$$

Notice that the exponential term is really just $\exp(iy\cdot(k_1+k_2-k_3-k_4))$ which is not only a delta function after integrating over $\int d^4y$, but is the statement of conservation of momentum, namely we get $(2\pi)^4\delta^{(4)}(k_1+k_2-k_3-k_4)$ (with or without a factor of $(2\pi)^4$, I forget what our convention is sometimes). We are then just left with the spinor-polarizations. We should make the contractions more evident though that we did previously, and so I will group them by square brackets. The u-channel follows in a similar fashion

$$\mathcal{A}_t = \frac{ig}{(2\pi)^2}[\overline{u}^{\sigma_3}(-\vec{k}_3)u^{\sigma_1}(\vec{k}_1)][\overline{u}^{\sigma_4}(-\vec{k}_4)u^{\sigma_2}(\vec{k}_2)]\delta^{(4)}(k_1+k_2-k_3-k_4) \tag{1.892}$$

$$\mathcal{A}_u = \frac{ig}{(2\pi)^2}[\overline{u}^{\sigma_3}(-\vec{k}_3)u^{\sigma_2}(\vec{k}_2)][\overline{u}^{\sigma_4}(-\vec{k}_4)u^{\sigma_1}(\vec{k}_1)]\delta^{(4)}(k_1+k_2-k_3-k_4). \tag{1.893}$$

This is the end of part (a). Part (b) is a pain in the ass. Recall that the definition of the cross section for $2 \to 2$ scattering in the center-of-mass frame (COM)

$$\sigma_{2\to 2} = \frac{1}{2^7\pi^2E^2}\int d\hat{k}_1\, |\mathcal{A}|^2 \tag{1.894}$$

where for us $\mathcal{A} = \mathcal{A}_t + \mathcal{A}_u$. We can simplify the spin-sums using

$$\sum_\sigma u^\sigma(\vec{k})\overline{u}^\sigma(\vec{k}) = \frac{m - i\not{k}}{2\omega_k} \tag{1.895}$$

or

$$\sum_\sigma u(-\vec{k})\overline{u}(-\vec{k}) = \frac{-m - i\not{k}}{2\omega_k}. \tag{1.896}$$

The rest of this process I am going to do in Mathematica using FeynCalc.

Here is the code that will work which should give you an answer $\sigma \propto g^2s^2$:

```
Quit[]

NotebookDelete[Cells[EvaluationNotebook[], GeneratedCell -> True]]

$LoadAddOns = {"FeynHelpers"};

<< FeynCalc`

tchannel =
    SpinorUBar[k1, m] . SpinorV[k3, m]*
    SpinorUBar[k2, m] . SpinorV[k4, m]
uchannel =
```

```
    SpinorUBar[k1, m] . SpinorV[k4, m]*SpinorUBar[k2, m] . SpinorV[k3, m]

amp = I*g*tchannel - I*g* uchannel;
ampsquared = amp*ComplexConjugate[amp];

FCClearScalarProducts[];
SetMandelstam[s, t, u, k1, k2, k3, k4, m, m, m, m];

spinsumamp =
    1/2 ampsquared // FermionSpinSum[#] & // DiracSimplify //
    Simplify;(*1/4 due to spin-sum*)
COMamp =
    spinsumamp /. {t -> -((s - 4 m^2)/2) (1 - Cos[\[Theta]]),
        u -> -((-s - 4 m^2)/2) (1 + Cos[\[Theta]])} // Simplify

CrossSection =
    2*Pi*Integrate[COMamp*Sin[\[Theta]], {\[Theta], 0, \[Pi]}]
```

1.9 Problem set 6:

1.9.1 Problem 1: Scattering in QED at tree level

Consider QED described by the Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \overline{\psi}(\gamma^\mu D_\mu + m)\psi \tag{1.897}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $D_\mu\psi = \partial_\mu\psi + ieA_\mu\psi$. Here, e is the unit electric charge. We will fix the gauge at the level of the path integral by adding to the Lagrangian density

$$\Delta\mathcal{L} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{1.898}$$

(a) Calculate the $2 \to 2$ elastic scattering amplitude of a pair of electrons at tree level, namely at order e^2. Explain why your result is (hopefully) independent of ξ, even though the photon propagator depends explicitly on ξ.

(b) Show that to leading order to the non-relativistic limit (where all spatial momenta are much smaller than the electron mass m), the dependence on the electron spin drops out, and that the scattering amplitude is in agreement with that of Coulomb potential in the Born approximation.

Problem 1 answer:

The gauge-fixed QED Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \overline{\psi}(\gamma^\mu D_\mu + m)\psi - \frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{1.899}$$

Note that there is also the *renormalized* QED Lagrangian

$$\begin{aligned}\mathcal{L} = &-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \overline{\psi}(\gamma^\mu\partial_\mu + m)\psi - ieA_\mu\overline{\psi}\gamma^\mu\psi \\ &- \frac{1}{4}(Z_A - 1)F^{\mu\nu}F_{\mu\nu} - (Z_\psi - 1)\overline{\psi}(\gamma^\mu\partial_\mu + m)\psi \\ &- Z_\psi\delta_m\overline{\psi}\psi - ie(Z_\psi - 1)A_\mu\overline{\psi}\gamma^\mu\psi.\end{aligned} \tag{1.900}$$

where $m_b = m + \delta m$. You can further relate the renormalization parameters to the counterterms via $Z_i = 1 + \delta_i$. But, for practice, we will stick with (1.899). We can find the QED Feynman rules from the partition (generating) function(al) from

$$Z[\eta, \overline{\eta}, J_\mu] = \mathcal{N}\int[D\overline{\psi}][D\psi][DA_\mu]\exp\left(iS_{\text{QED}} + i\int d^4x\left(A_\mu J^\mu + \overline{\eta}\psi + \overline{\psi}\eta\right)\right). \tag{1.901}$$

The Feynman rules are [**AC:**] insert diagrams later when I finally (fucking) figure out Tikz

$$\text{“fermion propagator”} = -i\frac{(-i\not{k} + m)_\alpha{}^\beta}{k^2 + m^2 - i\epsilon} \tag{1.902}$$

$$\text{“photon propagator”} = \frac{-i}{k^2 - i\epsilon}\left(\eta^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2}\right) \tag{1.903}$$

$$\text{“3-vertex } \gamma\psi\overline{\psi} = -q(\gamma^\mu)_\beta{}^\alpha. \tag{1.904}$$

There are also the in-out states for fermions but we have two new in-out states for the photon

$$\text{“out photon”} = \frac{\sqrt{Z_A}}{(2\pi)^{3/2}\sqrt{2\left|\vec{k}\right|}}e_\mu^{h*}(k) \tag{1.905}$$

$$\text{“in photon”} = \frac{\sqrt{Z_A}}{(2\pi)^{3/2}\sqrt{2\left|\vec{k}\right|}}e_\mu^{h}(k) \tag{1.906}$$

For 4-fermion scattering in QED with electrons, there are two possibilities historically known as Møller scattering and Bhabha scattering. Møller scattering is two electrons coming in and two leaving with polarizations $e^-e^- \to e^-e^-$ (complex-conjugate for the positron version) while Bhabha scattering is two electrons come in and two also leaving but with differring polarizations $e^+e^- \to e^+e^-$. Bhabha scattering is really electron-positron scattering, and the question in part (a) is only asking for electron-scattering, so we will focus on Møller scattering. Bhabha scattering is directly related to muon scattering as well but in Møller scattering there is a t and u channel, but in electron-muon scattering there is only the s-channel (but they are all related by crossing-symmetries).

For electron-electron scattering, there is the t- and u-channel diagrams but no s-channel since this would imply that two electrons can come together and form a photon, but only an electron-positron can do this, instead, a photon can be exchanged between two electrons since from the view of the photon, one of the electrons appears to be a backwards moving electron aka a positron. For sake of brevity, we can take $Z_i = 1$ at tree level. Let m_e be the mass of the electron. The t- and u-channel diagrams are

$$\text{“}t\text{-channel”} = (-ie)^2 \left[u^{\sigma_3}(\vec{p}_3)\gamma_\nu u^{\sigma_1}(\vec{p}_1)\right] \frac{-i}{t - i\epsilon} \left(\eta^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{t}\right) \left[\overline{u}^{\sigma_4}(\vec{p}_4)\gamma_\mu u^{\sigma_2}(\vec{p}_2)\right] \tag{1.907}$$

$$\text{“}u\text{-channel”} = (-ie)^2 \left[u^{\sigma_3}(\vec{p}_3)\gamma_\nu u^{\sigma_2}(\vec{p}_2)\right] \frac{-i}{u - i\epsilon} \left(\eta^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{u}\right) \left[\overline{u}^{\sigma_4}(\vec{p}_4)\gamma_\mu u^{\sigma_1}(\vec{p}_1)\right] \tag{1.908}$$

where $q^\mu \equiv (p_1 - p - 4)^\mu$ implying the u-channel relation $u \equiv q^2 = (p_1 - p_2)^2$ and similar for t-channel, $t \equiv k^2 = (p_1 - p_3)^2$. From here, we like to be more modern, so let us put it into Mathematica. The final differential cross section yields the correct Coulomb potential, which is simply the Rutherford formula. Here is the code (if you want to know how to derive the Rutherford formula from the Born approximation check chapter 6 of Sakurai or just ask/email me).

```
Quit[]
NotebookDelete[Cells[EvaluationNotebook[], GeneratedCell -> True]]

$LoadAddOns = {"FeynHelpers"};

<< FeynCalc`

tchannel =
    Contract[(-I*e)^2*(SpinorUBar[p3, m] . GA[\[Mu]] .
    SpinorU[p1, m])*(-I*
    FAD[p1 -
      p3]*(MT[\[Mu], \[Nu]] - (1 - \[Xi]) ((FV[p1 - p3, \[Mu]] FV[
            p1 - p3, \[Nu]])/SP[p1 - p3])))*(SpinorUBar[p4, m] .
    GA[\[Nu]] . SpinorU[p2, m])] /. \[Xi] -> 1 /. D -> 4
uchannel =
    Contract[(-I*e)^2*(SpinorUBar[p3, m] . GA[\[Mu]] .
   SpinorU[p2, m])*(-I*
   FAD[p1 -
     p4]*(MT[\[Mu], \[Nu]] - (1 - \[Xi]) ((FV[p1 - p4, \[Mu]] FV[
            p1 - p4, \[Nu]])/SP[p1 - p4])))*(SpinorUBar[p4, m] .
   GA[\[Nu]] . SpinorU[p1, m])] /. \[Xi] -> 1 /. D -> 4

SetMandelstam[s, t, u, p1, p2, -p3, -p4, m, m, m, m];
```

```
amp = 1/I (tchannel - uchannel) // Simplify
ampsquared =
      amp*ComplexConjugate[amp] // FeynAmpDenominatorExplicit //
      FermionSpinSum[#] & // DiracSimplify // Simplify
 spinsumamp = 1/4 ampsquared // Simplify(*1/4 due to spin-sum*);
 COMamp =
      spinsumamp /. {t -> -((s - 4 m^2)/2) (1 - Cos[\[Theta]]),
       u -> -((-s - 4 m^2)/2) (1 + Cos[\[Theta]])} // Simplify
 Series[COMamp, {m, \[Infinity], 2}] // Simplify(*non-relativistic limit*)
 spinsumamp // ReplaceAll[#, {m -> 0}] & // Simplify(*to check massless lim
              ultrarelativistic*)
 ampusquared = 1/4 uchannel*ComplexConjugate[uchannel] // FeynAmpDenominat
// FermionSpinSum[#] & // DiracSimplify // Simplify
 amptsquared = 1/4*tchannel*ComplexConjugate[tchannel] // FeynAmpDenominat
// FermionSpinSum[#] & // DiracSimplify // Simplify
 interference = 1/4*(tchannel*ComplexConjugate[uchannel] +
ComplexConjugate[tchannel]*uchannel) // FeynAmpDenominatorExplicit //
FermionSpinSum[#] & // DiracSimplify // Simplify

 unpolarizedcrossection = 1/(64*\[Pi]^2*4*s)*(ampusquared + amptsquared -
interference) // Simplify
 nonrellimit = Series[unpolarizedcrossection, {m, Infinity, 1}] /. {t -> -(
m^2)/2) (1 - Cos[\[Theta]]), u -> -((-s - 4 m^2)/2) (1 + Cos[\[Theta]])} //
Expand
 ultrarellimit = Series[unpolarizedcrossection, {m, 0, 1}] /. {t -> -((s -
(1 - Cos[\[Theta]]), u -> -((-s - 4 m^2)/2) (1 + Cos[\[Theta]])}
```

You get to figure out why you can not simply add the interference cross section to the sum of the t- and u-channel, instead you must sum as

$$|\mathcal{M}|^2 = \frac{1}{4}\sum\left(|\mathcal{M}_t|^2 + |\mathcal{M}_u|^2 - |\mathcal{M}_t|^*|\mathcal{M}_u| - |\mathcal{M}_t||\mathcal{M}_u|^*\right). \tag{1.909}$$

2 Quantum Field Theory 2 (253b)

Welcome to the QFT 2 section of the notes! Since these notes have the entirety of QFT 1 attached to them, a brief review of the material from QFT 1 won't be provided, but just a quick discussion and a reference back to them will be made when referring back to important equations (unless it recieves a modification in which I will just re-write it completely). Now, in QFT 2, the outline for these notes are broken up into three primary sections:

1. The Analytic S-matrix 2.1:
   (a) bound states and resonances(found within amplitudes)
   (b) implications of causality and unitarity (on scattering amplitudes)
   (c) infrared divergences (*some* of Weinberg's theorems)
2. The Renormalization Group 2.2:
   (a) Bogoliubov-Parasiuk-Hepp-Zimmermann (BPHZ) renormalizability
   (b) the 1PI effective action
   (c) improved perturbation theory via Gell-Mann-Low equation
   (d) Wilson-Polchinski renormalization group
   (e) ϵ-expansion
3. Quantum Chromodynamics 2.3:
   (a) quantization of non-abelian gauge theory
   (b) Becchi-Rouet-Store-Tyutin (BRST) quantization formalism and Batalin-Vilkovisky (BV) quantization formalism
   (c) asymptotic freedom
   (d) operator product expansion (OPE) and deep inelastic scattering
   (e) $1/N$ expansion (large N-expansion)
   (f) chiral symmetry breaking and the effective theory of pions
   (g) anomalies
   (h) instantons

Let us now *very* briefly review some key results/thoughts from QFT 1, to which you can always look back and review above. Recall that a QM system with an Hilbert space $\mathcal{H}$ and Poincaré symmetry can be represented by a unitary operator $U(\Lambda, a)$ with an infinitesimal form (1.196) with the energy-momentum operator $\hat{P}^\mu$ and angular-momentum/boost operator $\hat{J}^{\mu\nu}$ built out of the stress-energy tensor $T^{\mu\nu}(x)$ (1.187). We assume that there is an (interacting) vacuum state that is Poincaré invariant. We also saw that there are local field operators $\hat{\Phi}_\alpha(x)$ where α is a Lorentz index (can contain spin, mass, momentum,

etc.) that transforms under Lorentz transformations with representations matrices of the Lorentz group or covering group (1.241).

We also saw that when assuming the Hilbert space is spanned by asymptotic states of particles, and in particular the 1-particle state can be decomposed via a spectral representation (1.256) leading to the *all* the results we found with the help of the Källen-Lehmann representation (1.257).

We also saw how there are various different types of Greens functions will all varying use (1.218), (1.217), and (1.214).

We also saw how scattering amplitudes, and thus observables like the cross-section, can be computed with the S-matrix technology we developed (1.311) by using the cluster property (1.321). We also have the method for computing the Lorentz-invariant amplitude that goes into the connected S-matrix diagrams (1.323) with (1.325) going into to (namely compute $\mathcal{A}$ and plug in and chug).

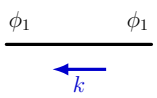
We now dive into the analytic S-matrix!

2.1 The Analytic S-Matrix

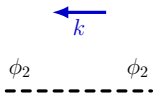
We will start our discussion of not just QFT 2 but of the S-matrix by considering a 2-scalar species theory

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi_1)^2 - \frac{1}{2}m_1^2\phi_1^2 - \frac{1}{2}(\partial_\mu \phi_2)^2 - \frac{1}{2}m_2^2\phi_2^2 - \frac{1}{2}g\phi_1^2\phi_2 \tag{2.1}$$

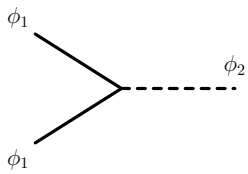
which is just $\lambda\phi^3$ theory but separated into two different scalar species. The theory is just some scalars, and so they have the following Feynman rules



$$= \frac{-i}{k^2 + m_1^2 - i\epsilon} \tag{2.2}$$



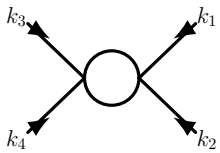
$$= \frac{-i}{k^2 + m_2^2 - i\epsilon} \tag{2.3}$$



$$= -ig \tag{2.4}$$

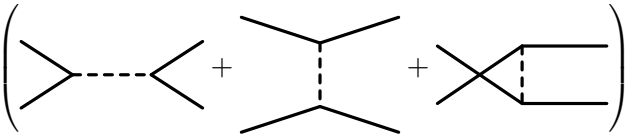
Let us consider the $2 \to 2$ scattering amplitude of the particle created by ϕ_1,

$$i\mathcal{A}(k_3, k_4; k_1, k_2) =$$



$$= i\mathcal{A}^{\text{tree}} + \mathcal{O}(g^4)$$

$$= i\Bigg($$



$$\Bigg) + \mathcal{O}(g^4) \tag{2.5}$$

where the tree level amplitude is specifically

$$i\mathcal{A}^{\text{tree}} = (-ig)^2 \frac{-i}{(k_1+k_2)^2 + m_2^2 - i\epsilon} + (k_2 \leftrightarrow -k_3) + (k_2 \leftrightarrow -k_4). \tag{2.6}$$

It will be convenient from here on out to define the following *Mandelstam variables*

$$s \equiv -(k_1+k_2)^2, \quad t = -(k_1-k_3)^2, \quad u = -(k_1-k_4)^2. \tag{2.7}$$

A quick discussion on the Mandelstam variables is in order. Our convention for the Mandelstam variables is solely based on what our convention is the momentum direction for incoming and outgoing particles, which of us, obey the following momentum conservation $(k_1+k_2)^\mu = (k_3+k_4)^\mu$. Then, out of these, you can show that with our convention and conservation of momentum, the Mandelstam variables obey

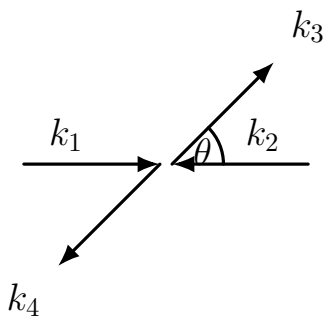
$$s + t + u = \sum_{i=4}^{4} m_i^2 \tag{2.8}$$

which, for 4-particle scattering of the ϕ_1 particles amounts to $s+t+u = 4m_1^2$. Remember that these particles on on-shell, and thus must obey $k^2 = -m_1^2$ (in mostly plus signature) (and similar for m_2^2, it's just that we are looking at the 4-point amplitude for ϕ_1). Also, something we won't show/explain here, but these three variables are the only unique variables one can defined for a scattering amplitude in $D = 4$ for a Poincaré invariant theory. In higher dimensions it gets more complicated. For higher point amplitudes, there are also similar variables that are typically defined as $s_{ij} = -(k_i+k_j)^2$.

Now, coming back to the tree amplitude, we can write it nicely in terms of the Mandelstam variables as

$$i\mathcal{A}^{\text{tree}} = ig^2 \left(\frac{1}{-s+m_2^2-i\epsilon} + \frac{1}{-t+m_2^2-i\epsilon} + \frac{1}{-u+m_2^2-i\epsilon} \right). \tag{2.9}$$

We will spend some time now interpreting this seemingly simple looking result. Note that we can also think of the Mandelstam variables in terms of the center of mass and total energy E and scattering angle θ based on the followig diagram:



to which we can find the following center of mass representation

$$s = E^2, \quad t = -\frac{E^2 - 4m_{1*}^2}{2}(1-\cos\theta), \quad u = -\frac{E^2 - 4m_{1*}^2}{2}(1+\cos\theta) \tag{2.10}$$

where it should be noted that the physical mass m_{1*} of the particle may differ from m_1 at higher orders in g as we saw in QFT 1, but we will ignore this for now.

Let us now consider the case of what happens when the mass of ϕ_2 obeys: $m_2 < 2m_1$. In general, if we want physically admissible (aka realizable or physical) kinematics, then we would require that $E \geq 2m_1$ which implies[58] that $s > m_2^2$ and that $t, u < 0$ to make $\mathcal{A}^{\text{tree}}$ always finite. If we analytically continue $\mathcal{A}^{\text{tree}}$ on the complex E-plane, or s-plane, we find that there is a pole at $E = m_2$. Namely, take the s-channel diagram, plug-in $s = E^2$, and then the pole is at $-E^2 + m_2^2 \implies E = m_2 < 2m_1$ (taking positive mass). The interpretation here is that the particle of mass m_2 is a *bound state* of a pair of particles of mass m_1. But can this be supported by quantum mechanical scattering such that we have similar results?

2.1.1 Scattering in Quantum Mechanics

Let us compare with a simple non-relativistic QM model where we will scatter in 1D with the corresponding Hamiltonian

$$H = \frac{p^2}{2m} + V(x) \tag{2.11}$$

for some potential $V(x)$ that has the following behavior (we will go over the other behaviors in it as well):

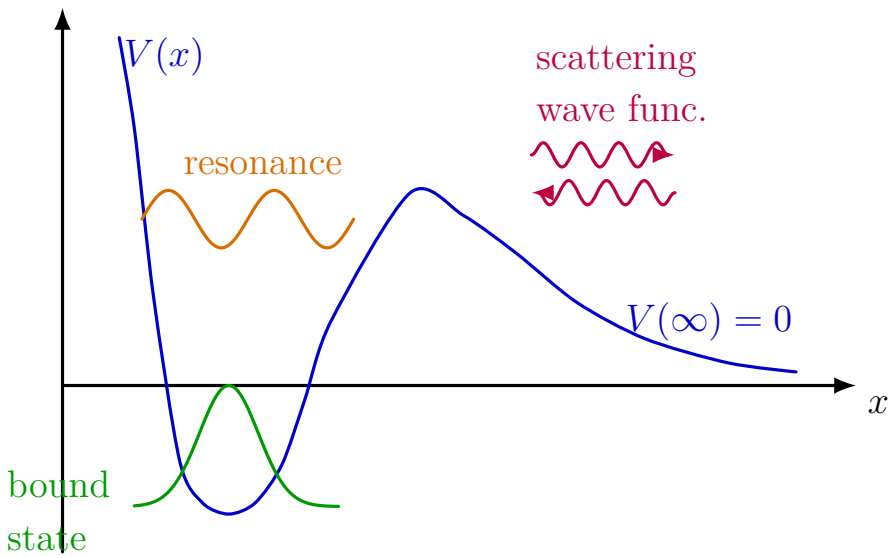


Figure 8. General behaviors for certain states that arise when scattering in QM based on the input of the wave function.

In the asymptotic region $x \to \infty$, we can describe a scattering wave function in the following form that we can call an *in-state*

$$\psi_k^{\text{in}} \sim e^{-ikx} + S(k)e^{ikx} \tag{2.12}$$

[58] Recall that $s = -(k_1 + k_2)^2 = -(k_1^2 + k_2^2 + 2k_1 \cdot k_2)$ and by the mass-shell $-s = 2m_1^2 + 2k_1 \cdot k_2 = E^2$, and thus we can impose the inequality.

where $S(k)$ is the scattering amplitudes that is analogous to partial wave amplitude in higher dimensions. The corresponding out state is then

$$\psi_k^{\text{out}} = (S(k))^{-1}\psi_k^{\text{in}}(x) \sim (S(k))^{-1}e^{-ikx} + e^{ikx} \tag{2.13}$$

where $k = \sqrt{2mE}$ and we assume that $V(\infty) = 0$. If the system omits a bound state, then the asymptotic wave function (again, $x \to \infty$) will be of the form

$$\psi(x) \sim e^{-\alpha x} \tag{2.14}$$

with $\alpha = \sqrt{-2mE} > 0$. A question arises though, "what is the relation between $S(k)$ and that of the bound states?"

If we do not impose the normalizability of the wave function in the asymptotic region $x \to \infty$, we can analytically continue $\psi_k^{\text{out}}(x)$ as a solution to the Schrödinger equation at complex k. We can analytically continue k as $k \to i\alpha$ and then the out-wave function becomes

$$\psi_{i\alpha}^{\text{out}}(x) \sim (S(i\alpha))^{-1}e^{\alpha x} + e^{-\alpha x} \tag{2.15}$$

where the first term must then vanish for a bound state to form. Therefore, for it to vanish, a bound state corresponds to a pole in the scattering matrix at $k = i\alpha$ for $\alpha > 0$ or in other words

$$\text{bound state} \leftrightarrow \text{pole of } S(k).$$

We can graph these in the complex plane for complex k and complex energy E:

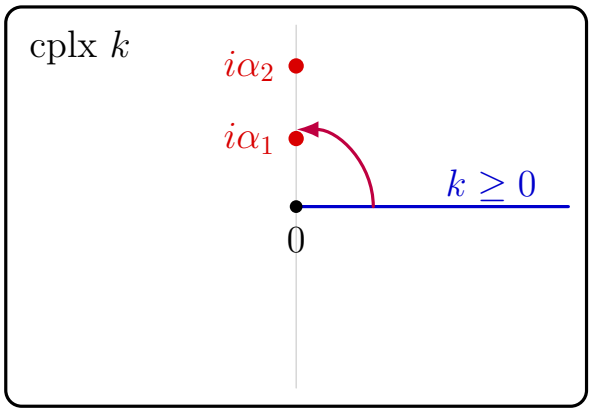


Figure 9. Graphing complex k. Notice that there is a physical branch cut that is described by the physical condition for $k \geq 0$ while there could be multiple bound state poles that are lined along the complex axis.

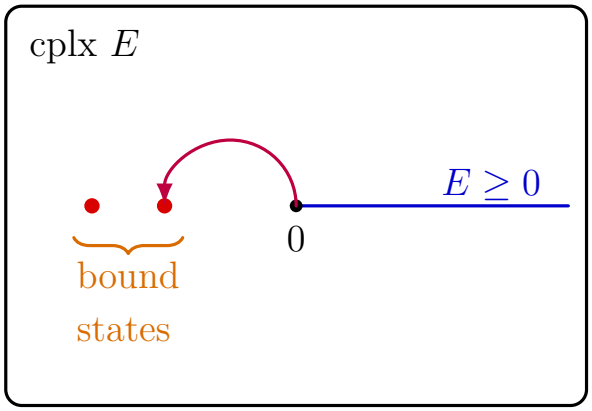


Figure 10. Here, recall that $E = k^2/2m$ or $k = \pm\sqrt{2mE}$, and so when we graph complex E, there are really two possible *sheets* for the information of the poles on the k-plane to be placed on the E-graph, thus the bound states get rotated.

The precise analog of this in QFT is the bound state pole in a *partial wave* amplitude (which we have see before (1.351). Recall the relation between 2-scalar-particle plane-wave basis and the partial wave basis in the center of mass (COM) frame (in the $D = 4$ case)

$$\langle \vec{k}_1, \vec{k}_2 | E, \vec{P} = 0, \ell, m\rangle = \delta^3(\vec{k}_1 + \vec{k}_2)\delta(\omega_{k_1} + \omega_{k_2} - E)\mathcal{N}(E)Y_{\ell m}(\hat{k}_1) \tag{2.16}$$

where for identical particles, $\mathcal{N}(E) = \sqrt{2E/(|\vec{k}_1|\omega_{k_1}\omega_{k_2})}$. In the COM frame with $\vec{P} = 0$, we can write the in-state in terms of the scattering matrix plus an inelastic part

$$|E, \ell, m\rangle^{\text{in}} = S_\ell(E)\,|E, \ell, m\rangle^{\text{out}} + \text{inelastic}. \tag{2.17}$$

Recall we can also write the Lorentz invariant amplitude in the partial wave basis (1.359)

$$i(2\pi)^4\mathcal{M} = (\mathcal{N}(E))^2\frac{1}{4\pi}\sum_{\ell=2\mathbb{Z}^+}^{\infty}(2\ell+1)P_\ell(\cos\theta)(S_\ell(E) - 1). \tag{2.18}$$

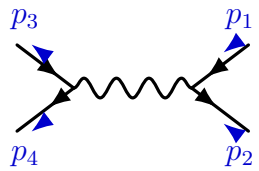
However, we can equivalently write this in terms of the amplitude $\mathcal{A}$

$$\begin{aligned} i\mathcal{A}(s,t) &= i(2\pi)^6E^2\mathcal{M} \\ &= \frac{8\pi E}{k}\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(\cos\theta)(S_\ell(E) - 1) \end{aligned} \tag{2.19}$$

where $E = 4(k^2 + m^2)$. A spin-ℓ bound state particle of mass M would correspond to a pole of $S_\ell(E)$ at $E = M < 2m$. We have seen that at tree level in the s-channel diagram that exchanged a massive ϕ_2

gives rise to a pole in the s-wave, which corresponds to $\ell = 0$ amplitude. An example of a spin-1 bound state, so $\ell = 1$, can be found in *scalar QED* which is taking regular QED and

exchanging the massive fermions for massive scalars. The particle-anti-particle scattering is



$$\propto \frac{(p_1 - p_2)^\mu (p_3 - p_4)_\mu}{s}$$

$$= \frac{4k^2 \cos\theta}{s}. \tag{2.20}$$

Now, referring back to (2.19) for $\ell = 1$, we can see that the Legendre polynomial turns into $P_1(\cos\theta) = \cos\theta$, and so this corresponds to a pole of $S_{\ell=1}$ at $s = 0$ implying that the photon can be viewed as a bound state! This seems very counter intuitive since the photon is massless, but nevertheless, there it is.

Now, returning back to our $\phi_1\phi_1 \to \phi_1\phi_1$ scattering, we still have the other possibility/case of when $m_2 > 2m_1$. In this case, from the s-channel diagram, there is a pole at $s = m_2^2$ (reading it off of the tree level amplitude) which is now in the physically admissible region implying that $\mathcal{A}^{\text{tree}}$ or $S^{\text{tree}}_{\ell=0}(E)$ diverges at $E = m_2$. But, unitarity demands that

$$|S_{\ell=0}(E)| \leq 1. \tag{2.21}$$

So seemingly there could a contradiction? The intuition here is that when $m_2 > 2m$ the "particle" created by ϕ_2 is unstable against decay into a pair of ϕ_1-particles, and so there is **no** stable particle of mass near m_2 but instead rather a resonance!

Let us revisit the simple non-relativistic QM model of the 1D scattering that we saw previously, but focus on the resonance this time with the following sketch:

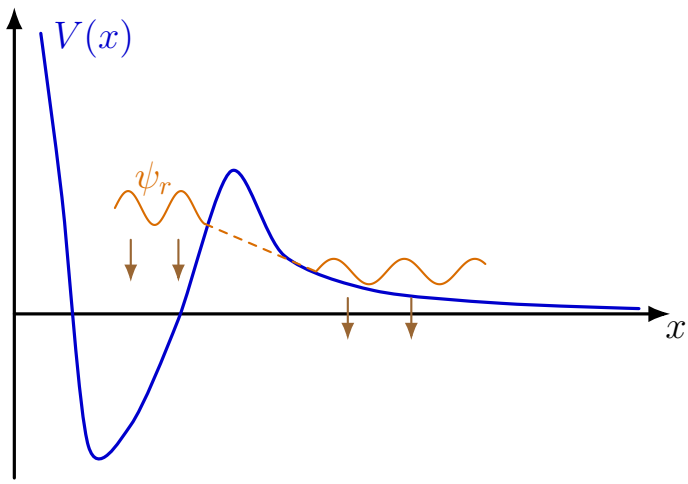


Figure 11. Here, we can see that the resonance wave function decays in time outside of the potential with only outgoing flux and no incoming flux. The arrow are just to show that the resonance is decaying downwards in the graph.

The wave function of a resonance "state" that decays uniformly in time is necessarily a non-renormalizable state due to conservation of probability. Consider one again the out-

state wave function and analytically continue it to complex $k = \sqrt{2mE}$. In this case, the out wave function is

$$\psi_k^{\text{out}}(x,t) \sim (S(k))^{-1} e^{-ikx-iEt} + e^{ikx-iEt} = e^{ikx-iEt} \tag{2.22}$$

where $S(k) \to 0$ since we only have the out state, and not the in one since we have a resonance (and this wave function is in the $x \to \infty$ limit). Now, suppose that $S(k)$ has a pole at k_* namely that

$$E = \frac{k_*^2}{2m} = \omega - i\gamma \tag{2.23}$$

for some $\gamma > 0$. Then, the out going wave function is

$$\psi_{k_*}^{\text{out}}(x,t) \sim e^{ik_*x - i\omega t - \gamma t} \tag{2.24}$$

in the asymptotic region. The pole then can be written as implies that $k_* = \sqrt{2m(\omega - i\gamma)}$. Then, for in the complex k-plane and complex E-plane, we get the following two graphs.

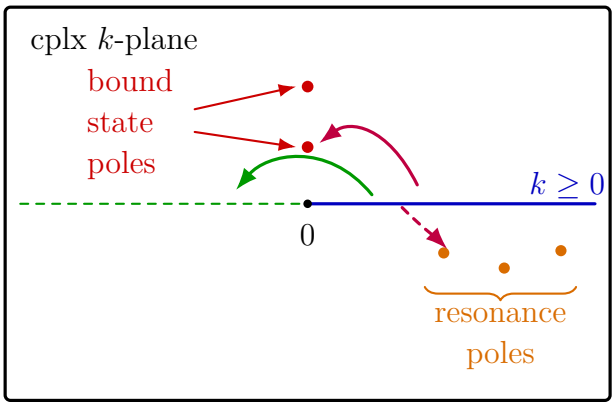


Figure 12. Here, we can see that the poles can take on different values. For $\gamma = 0$, we can have the normal bound state poles, while if $\gamma > 0$, then we will have a slight shift *off* of the real line which are where the resonances live.

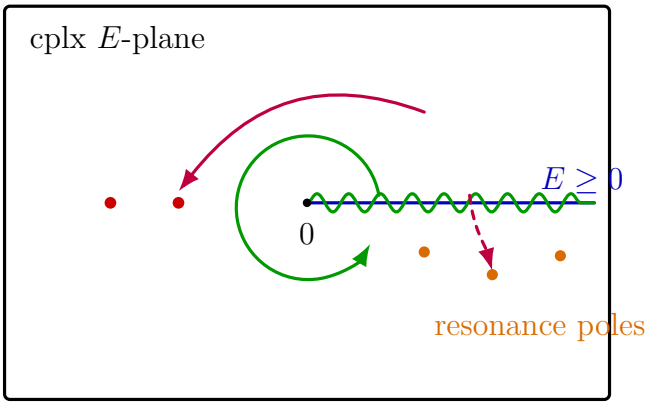
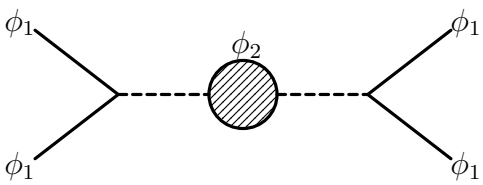


Figure 13. Here, when we split up the information/instead graph E from $E = k_*^2/2m$, we then see that the resonance poles are still in the same location, the bound state poles moved to the same spot we have seen before, and the green here is what happens when analytically continuing the momenta/energy. The resonance poles are on is called the *second sheet* or second Riemann sheet which is a common feature for exponential/logarithimic complex functions.

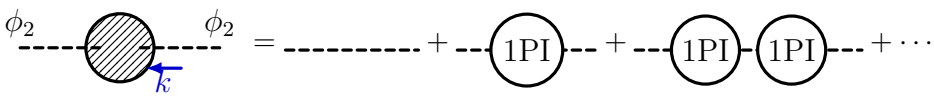
2.1.2 Poles, Branch Cuts, and Resonances in QFT

Let us now return back to computing $\phi_1\phi_1 \to \phi_1\phi_1$ scattering. We only considered the tree level calculation so far but we know that there are higher order loop corrections that can be ordered in terms the the 1PI diagrams that take into account the self energy of ϕ_2 which will contribute to the decay with of the resonance (as we will see). The general diagram in the s-channel for the 4-point scattering is given as



(2.25)

while we know that we can just look at the 1PI diagrams since the LSZ reduction will amputate the external legs for us just to find the on-shell matrix element given as

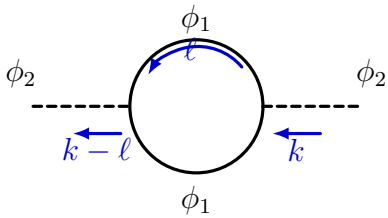


$$= \frac{-i}{k^2 + m_2^2 - i\epsilon - \Sigma(k)} \tag{2.26}$$

where $\Sigma(k)$ as we know contributes to the self-energy of the ϕ_2 field (contains the 1PI diagram calculations) which we can write as (to first loop order plus corrections)

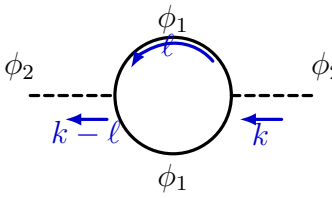
$$i\Sigma(k) =$$

$$=$$



$$+\mathcal{O}(g^4) \tag{2.27}$$

where the loop correction here can be computed in the way we saw in QFT 1 via dimensional regularization with $D = 4 - \epsilon$ and then using the Feynman trick to combine the denominators (B.1), but, in this case, let us just start in D-dimensions, and then apply a specific regulator later on (we will see why)



$$= \frac{(-ig)^2}{2}\int \frac{d^D\ell}{(2\pi)^D}\frac{-i}{\ell^2+m_1^2-i\epsilon}\frac{-i}{(k-\ell)^2+m_1^2-i\epsilon}$$

$$= \frac{g^2}{2}\int \frac{d^D\ell}{(2\pi)^D}\int_0^1 \frac{dx}{\left(\ell^2(1-x)+(k-\ell)^2x+m_1^2-i\epsilon\right)^2}. \tag{2.28}$$

Now, we are going to shift the loop momentum as $\ell^\mu \to \ell^\mu + xk^\mu$ since it will simplify the denominator by separating the loop momentum from the external momentum kinematics and leaves the measure unchanged. We will also perform a Wick rotation in order to have an analytic solution in complex ℓ^0 allowing us to perform the integration which in complex ℓ^0 has the following structure (when graphing (2.28):

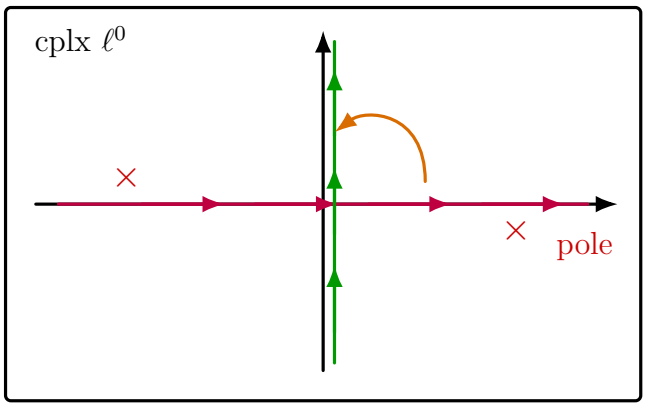
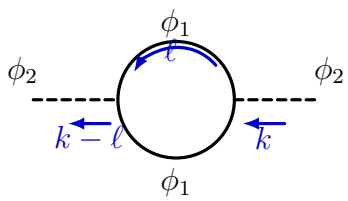


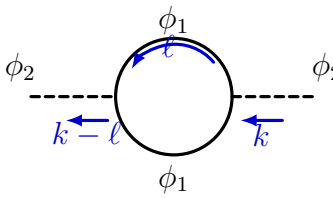
Figure 14. Here, the usual poles in ℓ^0 in the Lorentzian signature can be seen in their normal spots where the purple is our old integration (which would hit poles) while the green is our analytically continued (Wick rotated) function that won't hit any poles.

Then 1-loop integral then becomes



$$= \frac{ig^2}{2}\int \frac{d^D\ell_E}{(2\pi)^D}\frac{dx}{(\ell_E^2+k^2x(1-x)+m_1^2-i\epsilon} \tag{2.29}$$

where I have kept the $-i\epsilon$ just as a reminder of where it goes and we can track it along the way. We shall now specialize to $D=4$ case and regularize with a cutoff $|\ell_E|<\Lambda$. Of course, we know there is a more systematic way of doing this via dimensional regularization, but the simple momentum cutoff scheme suffices for our discussion at 1-loop order. We then have



$$= \frac{ig^2}{32\pi^2}\int_0^1 dx\left(\log\frac{\Lambda^2}{k^2x(1-x)+m_1^2-i\epsilon}+\mathcal{O}(\Lambda^{-2})\right). \tag{2.30}$$

The k-independent terms that come with the UV divergence in (2.30) can be absorbed into a re-definition of the mass parameter m_2. Equivalently, we can say that

$$(m_2^{\text{bare}})^2 = (m_2^R)^2+\delta m_2^2 \tag{2.31}$$

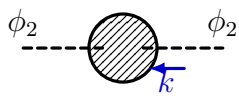
where m_2^R is finite, and we we recall that δm_2^2 is the counter term that serves to cancel the k-independent UV divergences in $\Sigma(k)$. We can write the self-energy corrections to ϕ_2 as a renormalized version

$$\Sigma(k) = \Sigma^R(k)+\delta m_2^2 \tag{2.32}$$

where the renormalized self-energy corrections function is

$$\Sigma^R(k) - \frac{g^2}{32\pi^2}\int_0^1 dx\ \log\frac{k^2x(1-x)+m_1^2-i\epsilon}{m_1^2} \tag{2.33}$$

which we can replace in the total self-energy graph with the renormalized mass



$$= \frac{-i}{k^2+(m_2^R)^2-i\epsilon-\Sigma^R(k)}. \tag{2.34}$$

With this result in hand, let us take a look at the analytic/singularity structure of the 4-point scattering (2.25) and see what we find. The specific contribution of (2.33) to the scattering amplitude $\mathcal{A}$ is

$$\mathcal{A}\supset \frac{g^2}{-s+(m_2^R)^2-i\epsilon+\frac{g^2}{32\pi^2}\int_0^1 dx\ \log\left(1-\frac{sx(1-x)}{m_1^2}-i\epsilon\right)} \tag{2.35}$$

where we recall that the momentum going into the loop k is really $k = k_1 + k_2$ (with appropriate 4-vectors of course), hence we can write it in terms of s. Notice that this logarithm has complex arguments when $1 - m_1^{-2} s x(1-x) = 0$ and given that $0 \leq x \leq 1$ (by the integration bounds) then $0 \leq x(1-x) \leq 1/4$, the argument of the logaithim is negative in the physical region of $s \in [4m_1^2, \infty)$. Here, with the help of the $-i\epsilon$ now (see why we kept it) will give us our convention for picking the physical branch choice. It should also be noted that s here then obeys

$$4m_1^2 \leq \frac{m_1^2}{x(1-x)} < s. \tag{2.36}$$

But more fundamentally, the value of $\Sigma(k)$ at real Lorentzian k^0 is determined by the Wick rotation/analytic continuation relating the Euclidean Greens function G_E to the time-ordered Greens function G. For real $k^0 > \sqrt{\vec{k}^2 + M^2}$, $\Sigma(k)$ is given by the analytic continuation from $k^0 \in i\mathbb{R}$ to positive real k^0-axis from above. Graphically this looks like the following in the complex k^0 plane:

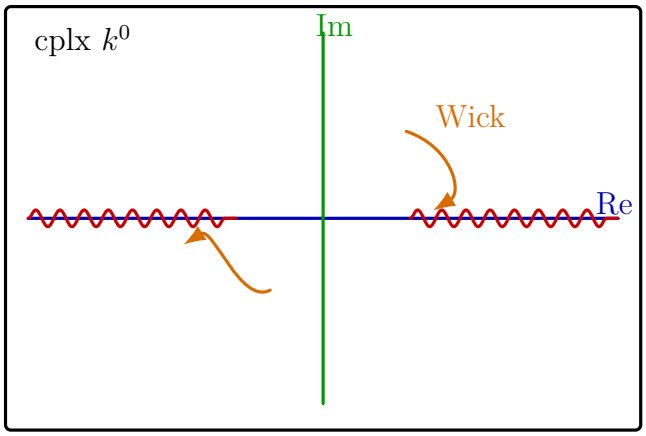


Figure 15. Since the Euclidean Greens function is along the imaginary time, we are Wick rotating back to the real line, which means we come down from above.

But now let us look at the graph for *complex* s of the renormalized self-energy (2.33). We know there is a branch cut at $s \geq 4m_1^2$ which is where the mass of $(m_2^R)^2$ will also be since it was shifted by the $\Sigma^R(k)$ contributions, and so the graph will look like this:

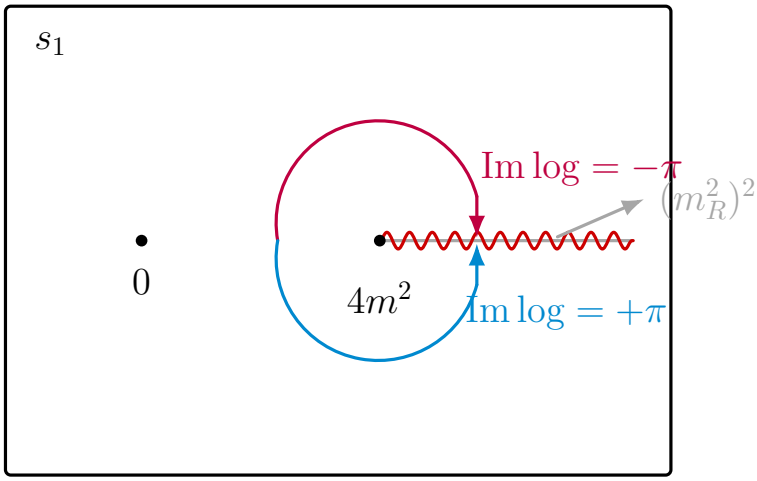


Figure 16. Here, there are two distinct ways for us to integrate over onto the branch cut which depends on the solution for the $\int dx$ integral.

It should also be noted that since we are dealing with a resonance which could be a *quasi-particle*, it's self-energy and specifically decay rate are directly related to the imaginary part of the self energy. We will take this for granted now and use it, but will see the derivation later on. Here, restricting ourselves to real $s > 4m_1^2$, the imaginary part of the renormalized self-energy contribution is (choosing the purple contour)

$$\mathrm{Im}\,\Sigma^R(k_1+k_2)\Big|_{\text{1-loop}} = -\frac{g^2}{32\pi^2}(-\pi)(x_+ - x_-) \tag{2.37}$$

where $x_\pm$ are solutions taken from $x(1-x) = m_1^2/s$ to which we get

$$\begin{aligned}\mathrm{Im}\,\Sigma^R(k_1+k_2)\Big|_{\text{1-loop}} &= \frac{g^2\pi}{32\pi^2}\left(\frac{1}{2}\left(1+\sqrt{\frac{s-4m_1^2}{s}}\right) - \frac{1}{2}\left(1-\sqrt{\frac{s-4m_1^2}{s}}\right)\right)\\ &= \frac{g^2\pi}{32\pi^2}\frac{\sqrt{s-4m_1^2}}{\sqrt{s}}.\end{aligned} \tag{2.38}$$

The imaginary part of Σ^R renders the amplitude $\mathcal{A}$ or $S_{\ell=0}(E)$ finite at all physical energies. This is one of the reasons we looked at only the imaginary part of Σ^R, but also as we will see later on, the imaginary part of the scattering amplitude is directly related to the real cross-section (while the self-energy is related to the decay rate of the resonance (quasi-particle) later on).

Let us now consider what happens to the amplitude (with the $\Sigma(k)$ correction) at weak coupling. In this regime, for s near $(m_2^R)^2$, the amplitude $\mathcal{A}$ for (2.25) is dominated by

$$\begin{aligned}\mathcal{A} &\approx \frac{g^2}{-s+(m_2^R)^2 - i\frac{g^2}{32\pi}\sqrt{\frac{s-4m_1^2}{s}}}\\ &\approx -16\pi i\sqrt{\frac{s}{s-4m_1^2}}(S_0(s)-1)\end{aligned} \tag{2.39}$$

where in the last line, we recalled this calculation is technically for $\ell = 0$ partial wave, and was able to use (2.19) where $E = \sqrt{s}$ and the exchanged momentum is $k^2 + 4m_1^2 = s$ to give the correct pre-factor. We can then read off the value for the S-matrix as

$$S_0(s) \approx \frac{-s + (m_2^R)^2 + i\frac{g^2}{32\pi}\sqrt{\frac{s-4m_1^2}{s}}}{-s + (m_2^R)^2 - i\frac{g^2}{32\pi}\sqrt{\frac{s-4m_1^2}{s}}}. \tag{2.40}$$

Graphically what this means is the following:

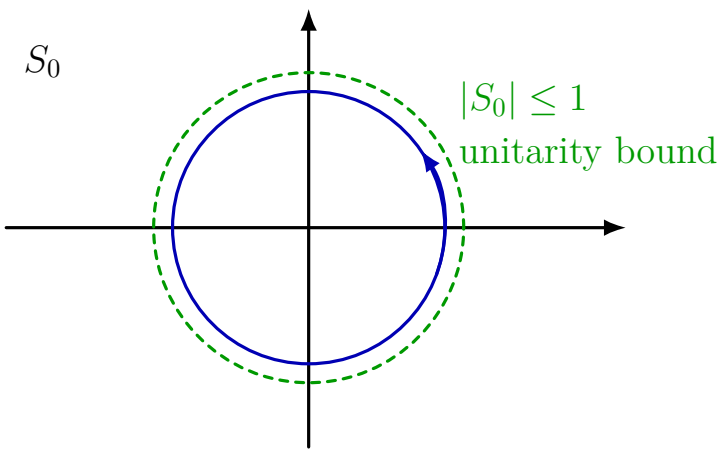
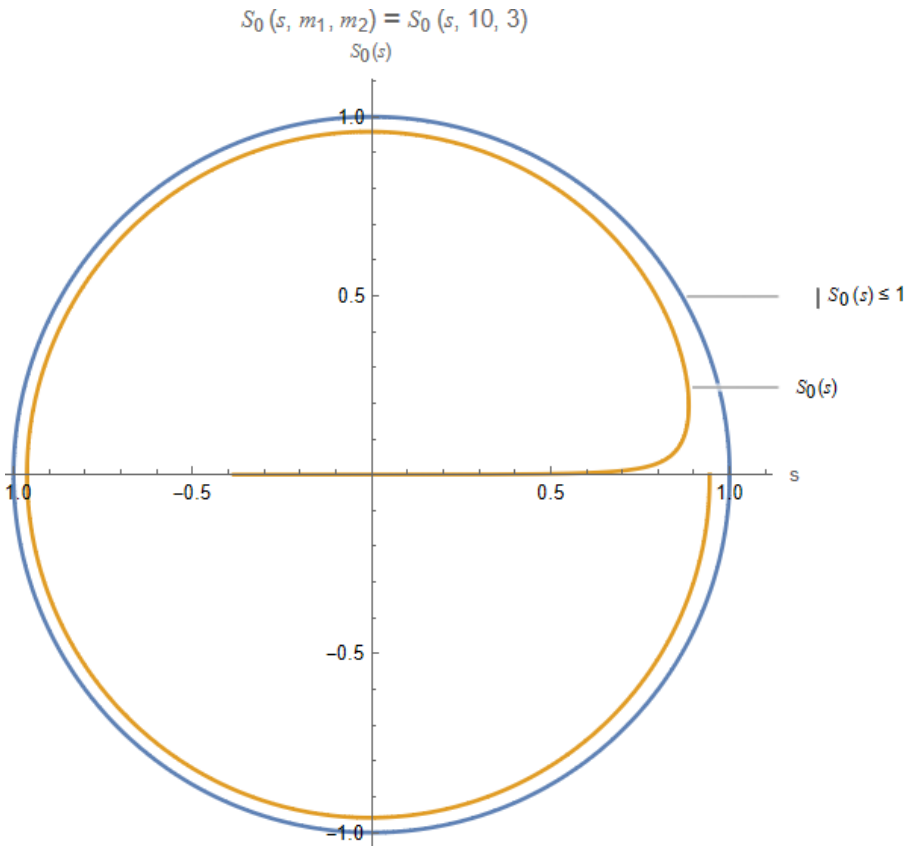


Figure 17. The blue line is the direction of the S-matrix in (2.40) while the green line is the unitarity bound.

Here, as s varies from below $(m_2^R)^2$ to above $(m_2^R)^2$, the phase of $S_0(s)$ changes by 2π. You can actually plot this in Mathematica using *PolarPlot*-command to find the following graph:



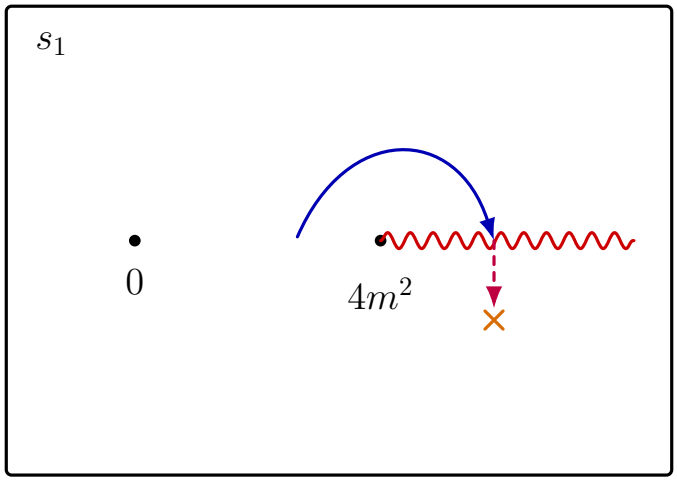
Here, the blue line is the unitarity condition, which is just plotting 1, and remember that this result was in the approximation regime of s near $(m_2^R)^2$ while also $s > 4m_1^2$, and thus if you pick too small of values for m_1 and m_2, the graph will actually diverge. We can also see the phase in action since we can see that the yellow line will continue to trace out the same shape every 2π. But now where is the singularity of $\mathcal{A}$ or S_0 associated with the ϕ_2-resonance on the complex s-plane? As we analytically continue $\sigma^R(k_1 + k_2)$ from real $s < 4m_1^2$ to the upper-half (UHP) complex s-plane namely $\operatorname{Im} s > 0$, we have

$$\operatorname{Im} \log \left(1 - \frac{sx(1-x)}{m_1^2} - i\epsilon\right) < 0 \tag{2.41}$$

and that $\operatorname{Im}(-s) < 0$. It follows that (2.25) has no pole in s on the UHP. Likewise, if we analytically continue in s to the lower-half plane (LHP) where $\operatorname{Im} s < 0$, we have the condition

$$\operatorname{Im} \log \left(1 - \frac{sx(1-x)}{m_1^2} - i\epsilon\right) > 0 \tag{2.42}$$

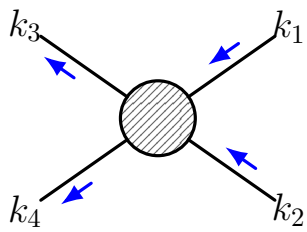
while $\operatorname{Im}(-s) > 0$. It once again follows that (2.25) has no pole in s on the LHP either! So where is the resonance pole that we saw in QM? The answer is that it is hiding on the *second-sheet* (in the same way that we got the second sheet by taking the energy $E = p^2/2m$ to get there, now we take $s = E^2$ and graph that)! The graph looks something like this, where we analytically continue through the branch cut condition to the second sheet finding the resonance pole in orange.



Let us now come back to the topic of *bound states* in QFT that we recently saw in our QM example. Here, we will investigate bound states as composite particles in QFT. Let us consider our old friend ϕ^4 theory in D-spacetime dimensions once again with corresponding Lagrangian

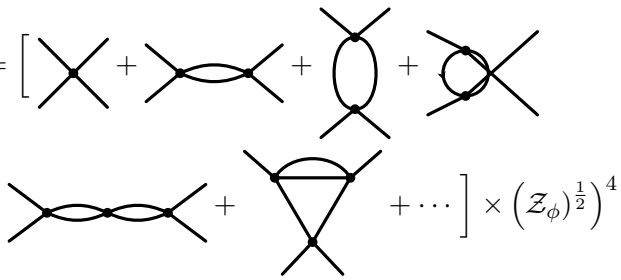
$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{1}{4!}g\phi^4. \tag{2.43}$$

Unlike the ϕ^3 theory we were looking at previously, ϕ_4 does not have a tree-level diagram, but only a contact diagram that is the usual Feynman rule $\propto ig$. Then, the order ($\mathcal{O}(g^2)$ correction actually starts at loop-level, and so to set convention again, recall that the 4-point amplitude has a diagram form of the following ([**AC:**] Yes, I know this looks weird, not sure why)

$i\mathcal{A} =$ 

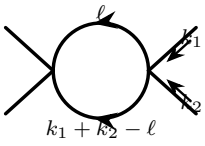
(2.44)

where the shaded region contains *all* possible diagrams and the amplitude is a function of $\mathcal{A} \equiv \mathcal{A}(s,t)$ with $s = -(k_1+k_2)^2$ and $t = -(k_1-k_3)^2$. If a pair of ϕ-particles form bound states, the latter should show up as poles of $\mathcal{A}$ at $s < 4m^2$. The most generic diagrams of (2.44) is a summation of the following form

$i\mathcal{A} =$ 

(2.45)

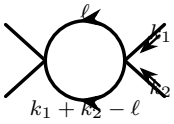
where the 1-loop t- and u-channel diagrams along with the renormalization factors do *not* affect the position of poles in s. Let us take the first of these diagrams and compute it (using the same techniques we have seen before: Feynman trick plus Wick rotation, no regularization for now)



$$= \frac{(-ig)^2}{2} \int \frac{d^D\ell}{(2\pi)^D} \frac{-i}{\ell^2 + m^2 - i\epsilon} \frac{-i}{(k_1 + k_2 - \ell)^2 + m^2 - i\epsilon}$$

$$= \frac{ig^2}{2} \int \frac{d^D\ell_E}{(2\pi)^D} \int_0^1 \frac{dx}{\ell_E^2 - sx(1-x) + m^2 - i\epsilon)^2}. \tag{2.46}$$

Let us specialize to the case of $D = 2$ where the momentum integral is exactly solvable

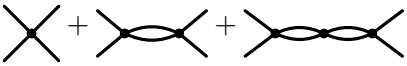


$$= \frac{ig^2}{8\pi} \int_0^1 \frac{dx}{-sx(1-x) + m^2 - i\epsilon} \tag{2.47}$$

and let us define the following function

$$f(s) \equiv \int_0^1 \frac{dx}{-sx(1-x) + m^2 - i\epsilon}. \tag{2.48}$$

Notice that growth of loop corrections can be seen as a multiple expression of bubble diagrams (in the s-channel) that can be re-summed in the following manner



$$+ \cdots = -ig \sum_{L=0}^{\infty} \left(-\frac{g}{8\pi} f(s) \right)^L = \frac{-ig}{1 + \frac{g}{8\pi} f(s)} \tag{2.49}$$

for L-loops. The question then becomes, is there a pole is this loop-summation found in $1 + gf(s)/8\pi \approx 0$? For s approaching $4m^2$ from below, let us shift s to (since there is still the mass-pole at the start of the branch cut) $s = 4m^2 - \delta$ where we take $\delta \to 0^+$. Then, the integral in $f(s)$ can be completed as[59]

$$f(s) \approx \frac{\pi}{m\sqrt{\delta}}. \tag{2.51}$$

Then, (2.49) will acquire a pole if $g < 0$ at $\delta \approx (g/8m)^2$. The interpretation for this this is that for weak and negative coupling g, there is a bound state of mass

$$m_{\text{bound}} \approx 2m - \frac{g^2}{256m^3}. \tag{2.52}$$

[59] When you fully do the integral you get

$$f(s) = \frac{4\tan^{-1}\left(\sqrt{\frac{4m^2}{\delta} - 1}\right)}{\sqrt{\delta(4m^2 - \delta)}} \approx \frac{\pi}{m\sqrt{\delta}} - \frac{1}{m^2} + \mathcal{O}(\delta) \tag{2.50}$$

but if the state is particularly heavy, we only need to consider the first term.

Now, you might worry that ϕ^4 theory with negative g is ill-defined (non-perturbatively) as the scalar potential $V(\phi) = m^2\phi^2/2 + g\phi^4/4!$ is not bounded from below, so that the perturbative vacuum is not the true vacuum state. Nonetheless, the same perturbative analysis applies to scalar theories with additional interactions such as ϕ^6 and ϕ^8 which can be re-summed into a general potential as

$$V(\phi) = \frac{m^2}{\beta}\cos(\beta\phi) \tag{2.53}$$

where $\beta = \sqrt{g}/m$ which is a class of scalar theories called the *Sine-Gordon model*.

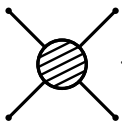
2.1.3 Bethe-Salpeter Equation (relativistic bound states)

More generally, we can consider probing a state $|B\rangle$ that we arbitrarily call a *bound state* of a composite particle using the following function

$$\langle\Omega|\, T\hat{\phi}(x_1)\hat{\phi}(x_2)\,|B\rangle \equiv \Phi_B(x_1, x_2) \tag{2.54}$$

where $\Phi_B(x_1, x_2)$ defines the *Bethe-Salpeter amplitude*. In a similar manner to the $2 \to 2$ scattering amplitude, we can consider the time ordered 4-point Greens function

$$\widetilde{G}(k_1, \cdots, k_4) = \int d^Dx_1 \cdots d^Dx_4\, e^{-i\sum_{j=1}^4 k_i\cdot k_j}\,\langle\Omega|\, T\hat{\phi}(x_1)\cdots\hat{\phi}(x_4)\,|\Omega\rangle$$

$=$  . (2.55)

The existence of a bound state $|B\rangle$ of some invariant mass M then implies that within $\widetilde{G}$ there exists a pole at $(k_1 + k_2)^2 = -M^2$ as we have seen above. We will now follow a similar derivation as the Källen-Lehmann spectral representation of a time-ordered 2-point function, just now it is fo a 4-point function.

The Fourier transformed 4-point function $\widetilde{G}$ will factorize when we take the momentum $k_1 + k_2 \to k_B$ and $k_B^0 = \sqrt{\vec{k}_B + M^2}$ to be on-shell into the following product of Bethe-Salpeter amplitudes

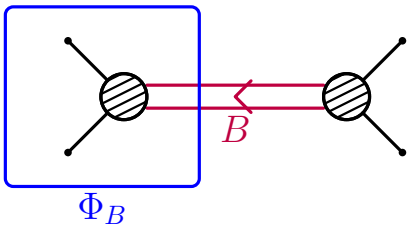
$$\widetilde{G} \to (2\pi)^D\delta^D(k_1 + \cdots k_4)\frac{-i}{(k_1 + k_2) + M^2 - i\epsilon}\widetilde{\Phi}_B(k_1, k_2)\widetilde{\Phi}_B^*(-k_3, -k_4) \tag{2.56}$$

$$\tag{2.57}$$

where we can define the Fourier transformed version of (2.54) as

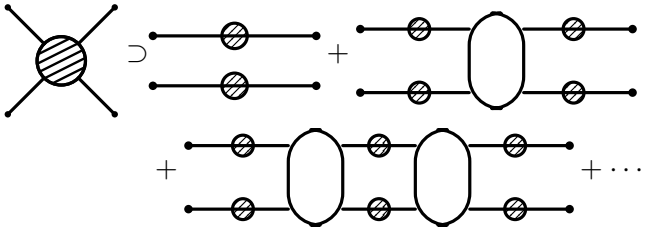
$$\begin{aligned}\Phi_B(x_1, x_2) &\equiv \langle\Omega|\, T\hat{\phi}(x_1)\hat{\phi}(x_2)\,\Big|B, \vec{k}_B\Big\rangle \\ &= \int \frac{d^Dk_1}{(2\pi)^D}\frac{d^Dk_2}{(2\pi)^D}e^{ik_1\cdot x_1 + ik_2\cdot x_2}(2\pi)^D\delta^D(k_1 + k_2 - k_B)\frac{\widetilde{\Phi}_B(k_1, k_2)}{(2\pi)^{\frac{D-1}{2}}\sqrt{2\omega_{k_B}}}\end{aligned} \tag{2.58}$$

In a Feynman diagram manner, to which we shall proceed from now on in this derivation, the Bethe-Salpeter amplitude/equation can be represented in the following manner



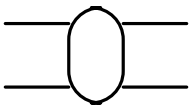
(2.59)

In the *2-particle approximation* that we shall take, the bound state pole comes from the *divergence* in summing over diagrams of the following type

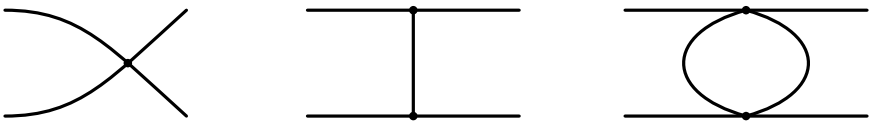


(2.60)

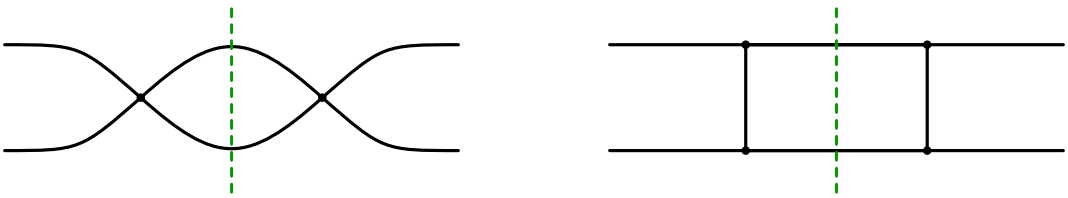
where diagrams like the following



denote diagrams that contain *no 2-particle cuts* which for example would contain diagrams like the following

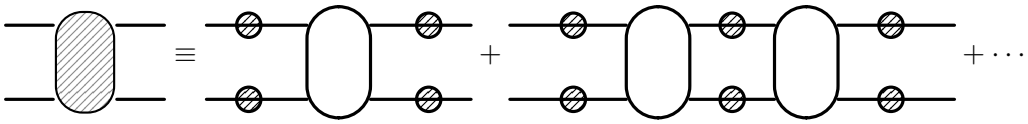


but we would discard the following types of diagrams



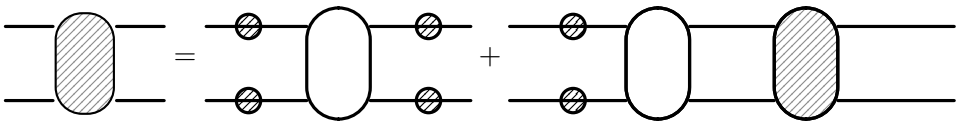
where a *cut-reducible diagram* would be one such that if you were to cut it vertically (i.e. insert some spacetime points in-between) then you would reduce the diagram to a lower point one. So the first cut diagram you see would reduce to two-contact terms or two 4-point functions, while if I tried, for instance, a *horizontal* cut, this would change the physics entirely and would become a completely disconnected diagram as seen above.

Notice that the "no 2-particle cuts" diagrams that are of the form



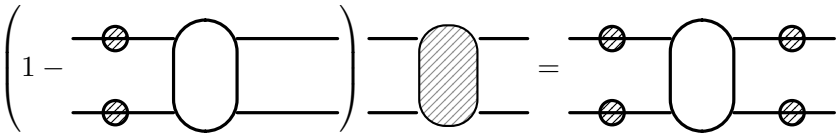
(2.61)

which obeys the following recursive relation



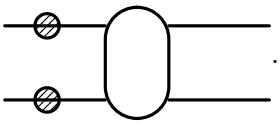
(2.62)

or, noticing that we can factorize out the shaded-box diagram to instead write

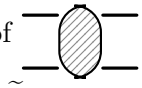


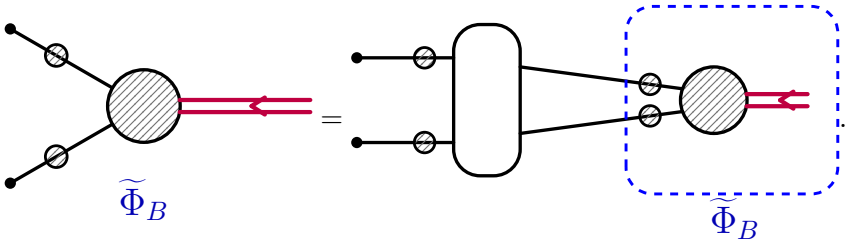
(2.63)

where we now define the following quantity

$\hat{V} \equiv$  .

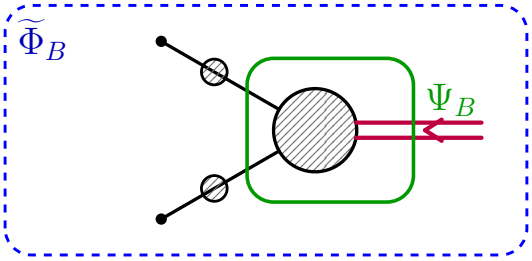
(2.64)

The residue of  at the bound state pole $P^2 = -M^2$ is proportional to $\widetilde{\Phi}_B$ which obeys $(1 - \hat{V})\widetilde{\Phi}_B = 0$ or diagramatically (where this has been written as $\widetilde{\Phi}_B = \hat{V}\widetilde{\Phi}_B$)



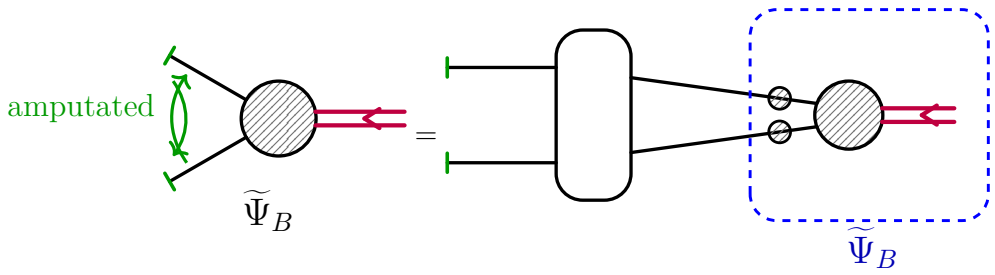
(2.65)

Or, in an equivalent manner, we can write this in terms of the *amputated* Bethe-Salpeter amplitude which we shall denote as Ψ_B that is defined as



(2.66)

which allows us to re-write (2.65) as



(2.67)

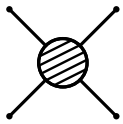
which (2.67) is commonly referred to as the *Bethe-Salpeter equation*.

Before moving onto our next topic, a few words on the Bethe-Salpeter equation should be made. This equation can be used to compute the relativistic bound state energy with corrections to many different systems. In the problem set, you will do this (if you'd like to of course) for the case of multiple scalars, but the equation was initially derived to compute the bound state energy for two nucleons exchanging a meson, but has wide spread applicability such as to computing the bound state of positronium. But, right after it's publication, three papers computing the corrections to the hyperfine splitting, fine structure constant, and lamb shift, showed off what it could do.

2.1.4 S-matrix bootstrap (& expectations)

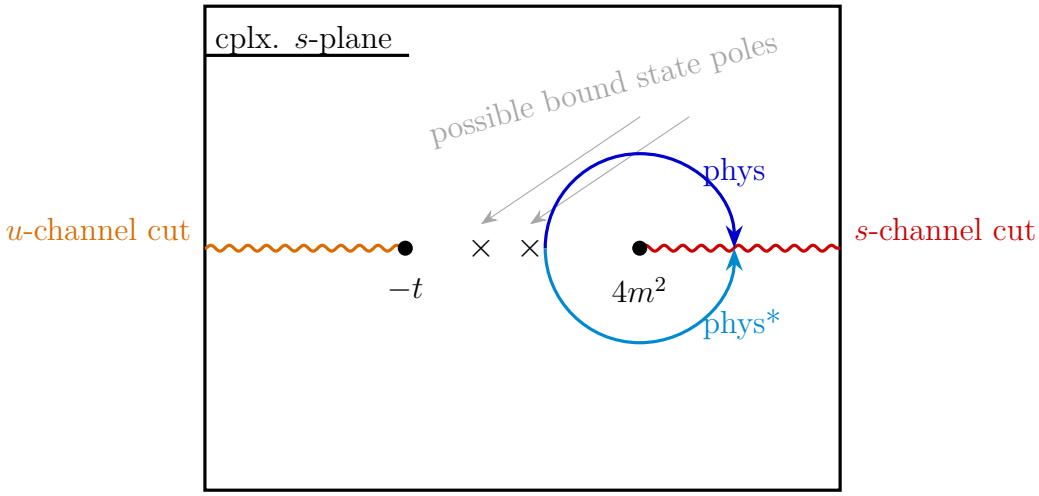
Moving onto our next topic, let us consider some general expectations of the amplitude $\mathcal{A}(s,t)$ when thinking about it's analytic properties that, as we have seen, are tied directly to obeying causality. We once again consider the $2 \to 2$ amplitude of identical massive scalar particles

$\mathcal{A}(s,t) =$



(2.68)

where s and t can be written in terms of the COM coordinates with $s = E^2 = 4(k^2 + m^2)$ and $t = -2k^2(1 - \cos\theta)$. The amplitude $\mathcal{A}$ can be analytically continued *away* from the physical region with $s, t \in \mathbb{R}$ and $s \geq 4m^2 - t$ with $t \leq 0$ which implies that (since the Mandelstam relation is $s + t + u = 4m^2$) $u = 4m^2 - s - t \leq 0$, which graphically can be given as follows if we graph the s-complex plane while holding t-fixed:



Notice that the usual analytic continuation into the UHP results in the physical amplitude or from above, while going from below onto the branch cut, in this case the s-channel cut, gives us the complex conjugate of the physical amplitude. The orange line here is the u-channel cut , and we included some possible bound state poles since we know they could exist in some general scalar theory. It should be noted that this diagram is just a general analysis of what we would expect from an amplitude for a given theory: simple poles that start at $(2m)^2$ since we have two-incoming particles that have a total energy of $s = E^2 = 4m^2$, and thus higher particle states/energies that are physical should extend from there (this is the branch cut), with possible bound state poles off of the s-channel, and then from the possible Mandelstam channels, they too can be found and taken into account. All of this is on the first sheet since to get to the next, we would instead need to graph the energy via $s = E^2$.

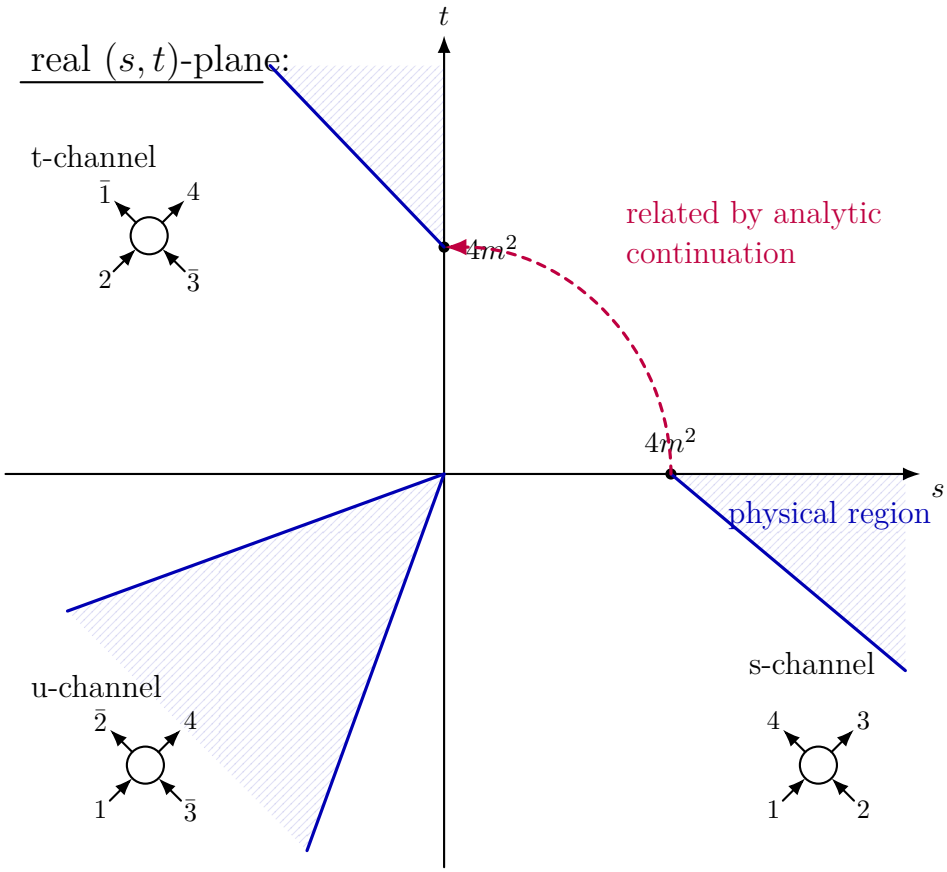
There are also some reality/conditions on the amplitude that we would expect some physical amplitude to obey.

The analytically continued amplitude $\mathcal{A}(s,t)$ is expected to obey the following property (due to reality condition on s and t being real)

$$\mathcal{A}(s^*, t^*) = (\mathcal{A}(s,t))^*. \tag{2.69}$$

In perturbation theory, this follows from the reality of the coupling constants in the Lagrangian being real.

We also expect the amplitude to obey a type of *crossing symmetry* regardless of particle type (which would just satisfy or simplfy the crossing relation). Superficially without a deep proof, this is a consequence of LSZ reduction and symmetry of the Greens functions. The crossing symmetry can be seen in *real* plane made by the variables s and t giving us the following plot (that was first done by Mandelstam)



Notice that there are three-distinct regions that are all related to each other via an analytic continuation from one channel to the next. The blue shaded regions are those of *physical* kinematics/amplitudes and that each type of diagram has been shown as well where a number with a bar across it denotes the anti-particle of it. It should be noted that this kind of diagram is called a *Mandelstam plot*.

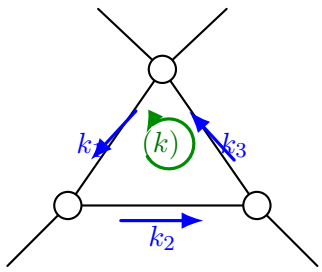
We would also expect, and do expect, that the amplitude obeys unitarity, namely as we have seen before, that the amplitude can be written in a partial wave basis

$$\mathcal{A}(s,t) = -16\pi i\sqrt{\frac{s}{s-4m^2}}\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(\cos\theta)(S_\ell(s)-1) \tag{2.70}$$

where the S-matrix obeys the unitarity condition $|S_\ell(s)| \leq 1$ for $s \geq 4m^2$ but must equal 1, $|S_\ell(s)| = 1$ for $4m^2 \leq s \leq M_{\text{inel}}^2$ where M_{inel} denotes the threshold of inelastic processes.

There is a caution however. **Not** all singularities of the amplitude (on the first sheet) have the interpretation of an intermediate physical state (bound state or multi-particle scattering states)! There are poles that are called *anomalous singularities* that produce *anomalous thresholds*. For example, we can consider this 4-point diagram at 1-loop triangle

diagram

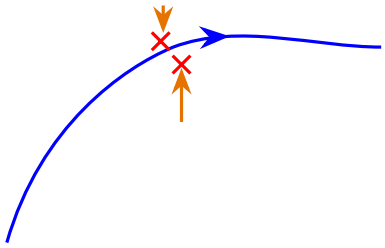


$$= \int d^D k \frac{(\text{polynomial in } k_i\text{'s})}{\prod_{i=1}^N (k_i^2 + m_i^2 - i\epsilon)}$$

$$= \int d^D k \int_0^1 \prod_{i=1}^N d\alpha_i \, \delta(\sum \alpha_i - 1) \frac{(\cdots)}{\left[\sum_{i=1}^N \alpha_i (k_i^2 + m_i^2) - i\epsilon\right]^N} \tag{2.71}$$

As a function of external momenta, (2.71) can be analytically continued so long as poles of the integrand stay away from the integration contour. Even if a pole hits the integration contour, we can still deform the α_i or k^μ contour to *evade* the pole generically.

A singularity can occur only if the contour *cannot* be deformed away from the pole, for example, if there are two poles that are on either sides of the contour and we try to pinch the contour such that it evades them, but can not:



This happens when all of the derivatives of the denominator with respect to to α_i and k^μ vanish, i.e. when

$$k_i^2 + m_i^2 = 0 \qquad i = 1, \ldots, N \tag{2.72}$$

$$\frac{\partial}{\partial k^\mu} \sum_{i=1}^N \alpha_i (k_i^2 + m_i^2) = 0 \tag{2.73}$$

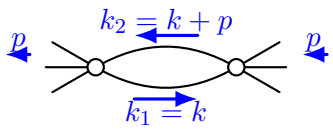
$$\tag{2.74}$$

where the set of equations in (2.73) are called *Landau equations*.

On the first Riemann sheet, in other words on the *physical sheet* (which should not be confused with the *physical region* since the latter refers to real kinematics only), singularities of the amplitude occur at solutions to Landau equation with $\alpha_i > 0$. For more information on this, I suggest reading [11] which is one of the hallmark books on the analytic S-matrix (with errors however) but [12] has a much more modern-friendly approach.

As an example of finding the singularity structure of such diagrams using Landau's

equations, we can consider the following diagram:



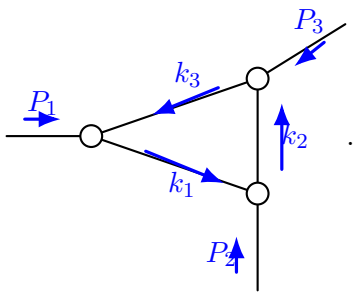
(2.75)

which has a singularity at $k^2 + m^2 = (k+p)^2 + m^2 = 0$ from the first Landau equation, and also

$$\alpha(k+p) + (1-\alpha)k = 0 \tag{2.76}$$

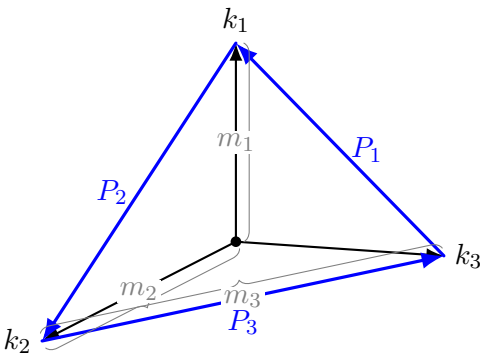
for some α such that $0 \leq \alpha \leq 1$. From these three equations, you can solve them for α, k, and p to find $\alpha = 1/2$, $k = -p/2$, and $p^2 = -4m^2$. We have seen this before at it is a hallmark of the 2-particle threshold. A pictorial way to see this singularity is the following. Think of $(k^0, \vec{k}^i)$ as a vector of some length m. From the second Landau equation, for a 2-momenta system we have $\alpha_1 k_1 + \alpha_2 k_2 = 0$ for $0 < \alpha_1, \alpha_2 < 1$, then then if the (magnitude) of the k's start at a point and go out their length m_1 and m_2, then this total length is denoted as p to which then $-p^2 = (m_1 + m_2)^2$.

Now consider the following triangle diagram:



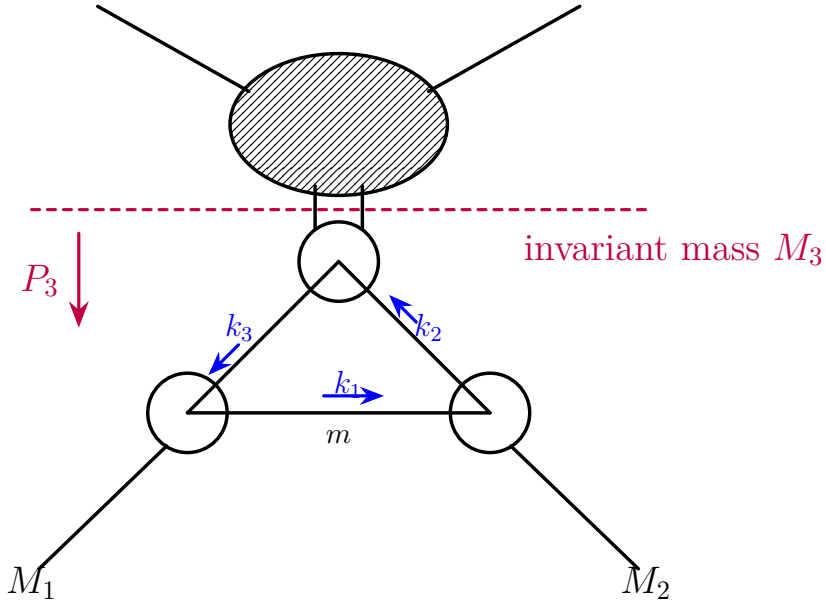
(2.77)

From Landau's equations in (2.73), a singularity will occur on the first sheet *if* we can arrange the internal momentum to obey the following structure:

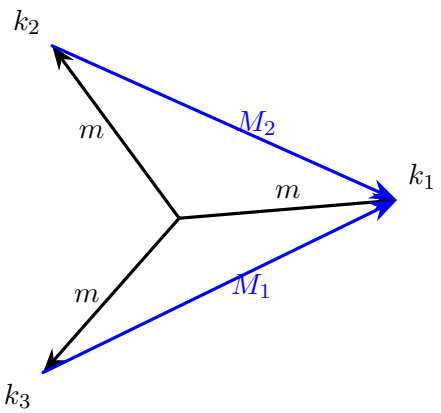


(2.78)

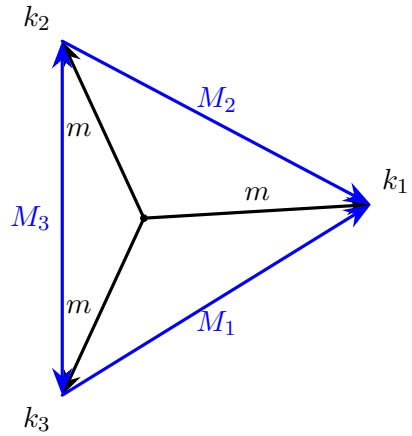
As an example of a diagram that cannot satisfy Landau's equations, we can also consider the following diagram where we extend the triangle diagram's external momentum P_3 to be on-shell that was originally cut and placed on-shell as $P_3^2 = -M_3^2$:



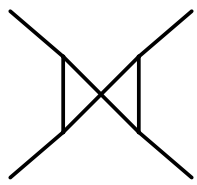
Here, if $M_1, M_2 < \sqrt{2}m$, then the only arrangment that the internal momentum vectors can take are



which visually, we can see that this cannot satisfy the Landau equations and thus implies that the amplitude is non-singular. On the other hand, if we instead had it such that $M_1, M_2 > \sqrt{2}m$, then we could have the following arrangment



Here, the amplitude acquires a Landau singularity at a specific value of $P_3^2 = -M_3^2$ (where M_3 is determined by M_1, M_2, and m) which does not admit the interpretation of a bound state. Another example that is worth noting are singularities that arise from diagrams such as



where the anomalous threshold singularity occurs in the Sine-Gordon model shown by Coleman and Thun in 1978 [13].

So, a lot of the analysis we performed before thinking about the analytic properties of a general amplitude $\mathcal{A}(s,t)$, were computed from perturbative scattering amplitudes, while everything we discussed in actually non-perturbative. And so what we really have right now that underlies this is the following working hypothesis: the kind of analytic structure of scattering amplitudes found in perturbation theory also applies to the non-perturbative amplitude *provided* that we interpret the on-shell internal propagators that give rise to Landau singularities as those of actual particles.

It should be noted however that only a small subset of such analyticity assumptions we have been making have ever been rigorously established based on robust axioms...

2.1.5 Lehmann Ellipse, Optical Theorem, and Froissart-Martin Bound

Ok, let us look now at some of the actual consequences on the amplitude (and it's corresponding behavior) that happens we impose analyticity, crossing symmetry, and (partial wave) unitarity since previously, we just stated the assumptions and then investigated general pole structure, but now lets see what the amplitude must obey in order to be considered *actually physical*.

For $2 \to 2$ scattering for some of the *lightest* particles in a given theory, let us write (recalling that the amplitude that is a function of $\mathcal{A}(s,t)$ can be written in the COM frame

and thus become a function of $\mathcal{A}(s,\theta)$)

$$x \equiv \cos\theta = 1 + \frac{2t}{s-4m^2} \tag{2.79}$$

If we fix a physical value of s (such that it is greater than $4m^2$), let us inspect the amplitude $\mathcal{A}(s,t)$ viewed as an analytic function of x only, $\mathcal{A}(x)$. For the rest of this discussion, we will be referring to the following diagram, that we will be filling in with details we we go:

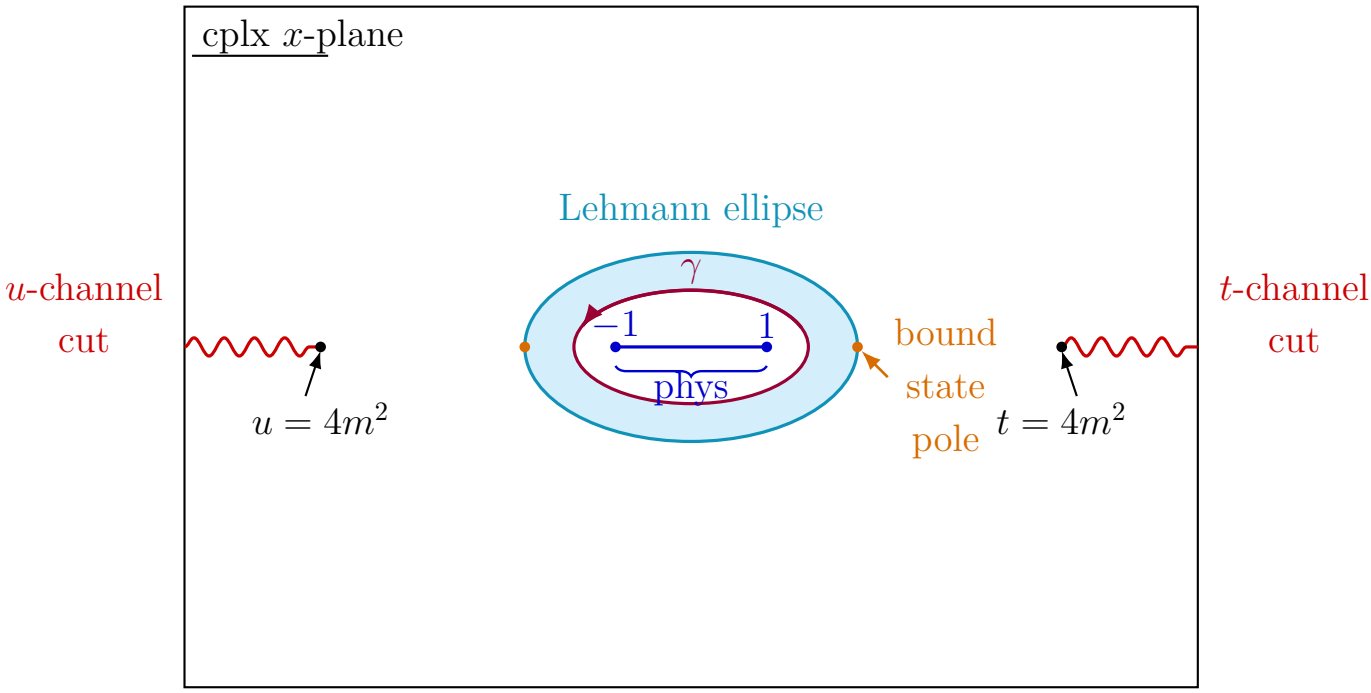


Figure 18. Graph of $\mathcal{A}(x)$ for complex x showing the Lehamann ellipse, the s,t-channel cuts, and the physical region, the mass-poles that start the branch cuts, and the bound state poles.

This physical value of x lies between -1 and 1 on the real axis given by the cosine function that is also related to the Mandelstam variables via the COM frame. We expect $\mathcal{A}(x)$ to be analytic on the complex x-plane away from he t-channel branch cut namely

$$x \in \left[\frac{s+4m^2}{s-4m^2}, \infty\right) \Leftrightarrow t \in [4m^2, \infty) \tag{2.80}$$

while for the u-channel cut has the following range

$$x \in \left(-\infty, -\frac{s+4m^2}{s-4m^2}\right], \tag{2.81}$$

and there are also possible poles that admit an interpretation as bound states in the t- and u-channels.

Thanks to complex analysis, we can always rewrite the amplitude as an integral over some fictitious pole at $z = x$ or in other words, reverse Cauchy's theorem

$$\mathcal{A}(x) = \oint_{\gamma} \frac{dz}{2\pi i} \frac{\mathcal{A}(z)}{z - x} \tag{2.82}$$

provided that γ encircles x and $\mathcal{A}(z)$ is analytic inside of the contour γ (the usefulness will be seen soon). Now let us consider the following identity that we kindly asked for from our fellow friendly mathematicians

$$\frac{1}{z - x} = \sum_{\ell=0}^{\infty} (2\ell + 1) P_\ell(x) Q_\ell(z) \tag{2.83}$$

where $P_\ell(x)$ is our usual Legendre polynomial but $Q_\ell(z)$ here is called the[60] *Legendre function of the second kind*. The identity in (2.83) holds when $|z + \sqrt{z^2 + 1}| > |x + \sqrt{x^2 - 1}|$. The RHS of (2.83) converges when $|z + \sqrt{z^2 + 1}| > |x + \sqrt{x^2 - 1}|$ holds true which can be seen when we go to large ℓ since

$$P_\ell(x) \sim \left(x + \sqrt{x^2 - 1}\right)^{\ell} \quad \text{and} \quad Q_\ell(z) \sim \left(z + \sqrt{z^2 - 1}\right)^{-\ell}. \tag{2.85}$$

Something that should be noted is that within the set of values where $\{z : |z + \sqrt{z^2 - 1}| = \text{constant}\}$ will form an ellipse on the complex plane with foci at ± 1.

Now, coming back to our integral in (2.82), we will take γ to be the largest ellipse with foci at ± 1 such that it encircles no singularity of $\mathcal{A}(z)$ for example, no touching of the first t-channel bound state pole. This largest ellipse is what is called the *Lehmann ellipse* written as

$$\begin{aligned} \mathcal{A}(x) &= \oint_{\gamma} \frac{dz}{2\pi i} \sum_{\ell=0}^{\infty} (2\ell + 1) P_\ell(x) Q_\ell(z) \mathcal{A}(z) \\ &= \sum_{\ell=0}^{\infty} (2\ell + 1) P_\ell(x) \oint_{\gamma} \frac{dz}{2\pi i} Q_\ell(z) \mathcal{A}(z) \end{aligned} \tag{2.86}$$

where in the first line we were able to exchange the summation over ℓ with the integral over z since the sum for x converges when inside the Lehmann ellipse, and thus we can exchange $\oint_\gamma$ and $\sum_\ell$. Now, in the second line, the partial wave expansion will converge for *unphysical* x provided that x lies inside the Lehmann ellipse, and so in 18, this unphysical x region would be the space between the physical blue up to the γ integral contour which, if the end of the γ contour has a point $x_0(s)$, then the range for unphysical values are $1 < x < x_0(s) \Leftrightarrow 0 < t < t_0$ where $t_0 = \min(4m^2, m_B^2)$ for a bound state mass m_B.

[60] The definition for the Legendre function of the second kind is defined as an integral over the regular Legendre function

$$Q_\ell(z) = \frac{1}{2} \int_{-1}^{1} dx \frac{P_\ell(x)}{z - x} = \int_{z + \sqrt{z^2 + 1}}^{\infty} \frac{d\zeta}{\zeta^{\ell + 1} \sqrt{1 - 2\zeta z + \zeta^2}}. \tag{2.84}$$

Also recall however that we can write the amplitude in terms of it's partial wave expansion (2.19) (which I shall recall for ease of access and comparison)

$$\mathcal{A}(x) = -16\pi i\sqrt{\frac{s}{s-4m^2}}\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)(S_\ell(s)-1) \tag{2.87}$$

which displays what the integration over $\mathcal{A}(z)$ should yield. We also have it that if x is real inside the Lehmann ellipse, then taking the imaginary part of the amplitude yields only the physical values the S-matrix can take

$$\operatorname{Im}\mathcal{A}(x) = 16\pi\sqrt{\frac{s}{s-4m^2}}\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)(1-\operatorname{Re}S_\ell(s)) \tag{2.88}$$

where the only values $1-\operatorname{Re}S_\ell(S)$ can take are $[0,2]$ namely the length of the physical part in 18. Also recall that we were able to put the elastic and inelastic cross sections in terms of the S-matrix (1.360)

$$\sigma_{2\to 2} = \frac{8\pi}{s-4m^2}\sum_{\ell=0}^{\infty}(2\ell+1)|S_\ell(s)-1|^2 \tag{2.89}$$

$$\sigma_{\text{inel}} = \frac{8\pi}{s-4m^2}\sum_{\ell=0}^{\infty}(2\ell+1)(1-|S_\ell(s)|)^2 \tag{2.90}$$

which when summed will give us the total cross section

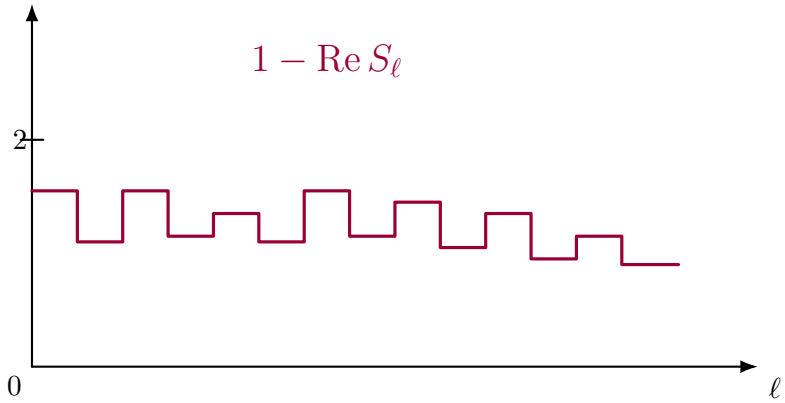
$$\begin{aligned}\sigma_{\text{tot}} &= \sigma_{2\to 2}+\sigma_{\text{inel}}\\ &= \frac{16\pi}{s-4m^2}\sum_{\ell=0}^{\infty}(2\ell+1)(1-\operatorname{Re}S_\ell(s))\\ &= \frac{1}{\sqrt{s(s-4m^2)}}\operatorname{Im}\mathcal{A}(x=1)\end{aligned} \tag{2.91}$$

where (2.91) is commonly referred to as the *optical theorem*.

Let us now consider the following comparison

$$\frac{1}{16\pi}\sqrt{\frac{s-4m^2}{s}}\operatorname{Im}\mathcal{A}(x) = \sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)(1-\operatorname{Re}S_\ell(s)). \tag{2.92}$$

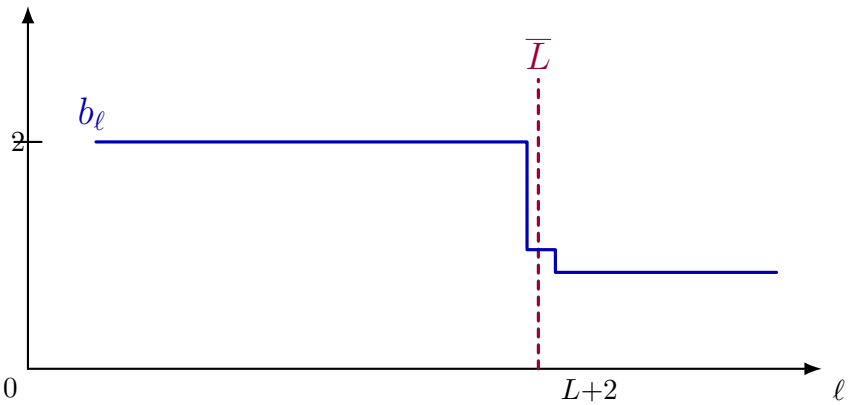
We can think of the purple term when graphed against the higher angular momentum states should be some sort of stair-case function that takes on different values for various ℓ's but never goes above 2:



We can also make a similar comparison with the total cross-section from (2.91) namely

$$\frac{s-4m^2}{16}\sigma_{\text{tot}} \equiv \sum_{\ell=2\mathbb{Z}^+}^{\infty} (2\ell+1) b_\ell \equiv \overline{L}^2. \tag{2.93}$$

This summation of b_ℓ must have the following behavior up to some point $\overline{L}$ when graphed against ℓ



We can relate the value at which b_ℓ falls off, namely $\overline{L}$ to that of the total cross-section via

$$\overline{L} \approx \sqrt{\frac{s-4m^2}{16\pi}\sigma_{\text{tot}}} \approx L \tag{2.94}$$

where L is some integer,and $\overline{L} \gg 1$ at high energies. Then, the optical theorem would imply

$$\sum_{\ell=0}^{\infty}(2\ell+1)(1-\mathrm{Re}\, S_\ell) = \sum_{\ell=0}^{\infty}(2\ell+1)b_\ell. \tag{2.95}$$

For values of x such that $x > 1$, $P_\ell(x)$ increases with ℓ, and thus for (2.95) we can place

the following inequality between them

$$\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)(1-\mathrm{Re}\,S_\ell) \geq \sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)b_\ell$$

$$\frac{1}{16\pi}\sqrt{\frac{s-4m^2}{s}}\,\mathrm{Im}\,\mathcal{A}(x) \geq \sum_{\ell=2\mathbb{Z}^+}^{L}(2\ell+1)P_\ell(x)\cdot 2. \tag{2.96}$$

For $x > 1$, we can rewrite the Legendre polynomial as

$$\begin{aligned} P_\ell(x) &= \frac{1}{\pi}\int_0^\pi dx\,(x+\cos\alpha\sqrt{x^2-1}) \\ &> C_\delta\left(1+(1-\delta)\sqrt{x^2-1}\right)^\ell \end{aligned} \tag{2.97}$$

where we shall define $y \equiv 1+(1-\delta)\sqrt{x^2-1}$ which is valid for all ℓ given an arbitrarily small fixed $\delta > 0$ and some constant $C_\delta > 0$. This result then implies the following for the imaginary part of the amplitude

$$\begin{aligned} \mathrm{Im}\,\mathcal{A}(x) &> \text{constant} \times \sum_{\ell=2\mathbb{Z}^+}^{L}(2\ell+1)y^\ell \\ &\gtrsim \text{constant} \times L\frac{y^L}{y-1}. \end{aligned} \tag{2.98}$$

For values of $s \gg m^2$, we have the following values $L \approx \sqrt{s\sigma_{\text{tot}}/16\pi}$, $y-1 \approx (1-\delta)\sqrt{4t/s}$, and $y^L \approx \exp\left((1-\delta)\sqrt{t\sigma_{\text{tot}}/4\pi}\right)$ which then gives the following inequality

$$\mathrm{Im}\,\mathcal{A}(x) > \text{constant} \times s\sqrt{\frac{\sigma_{\text{tot}}}{t}}e^{(1-\delta)\sqrt{\frac{t\sigma_{\text{tot}}}{4\pi}}} \tag{2.99}$$

which is valid for $1 < x < x_0(s) \Leftrightarrow 0 < t < t_0$.

For $x \lesssim x_0(s)$, *if* $\mathcal{A}(x)$ grows no faster than a polynomial in s at large s, then we can make the following statement

$$\mathrm{Im}\,\mathcal{A}(x) < C\cdot s^N \tag{2.100}$$

for some value of N. It then follows that we can bound the total cross section as

$$\sigma_{\text{total}} < \frac{(N-1)^2}{(1-\delta)^2}\frac{4\pi}{t_0}\left(\ln\frac{s}{M^2}\right)^2 \tag{2.101}$$

which is valid for $s \gg M^2$. (2.101) is known as the *Froissart-Martin bound*. Some things should be noted about this result. This bound in (2.101) is completely generic and is a general state about *all* amplitudes in (flat space) QFT! The original paper that came out in 1961 publishing this bound [14] gave the type of thinking and result that eventually lead to this paper [15] that, for deep inelastic scattering people, I surely hope you recognize...

2.1.6 Consequences of Causality on S-matrix (dispersion relations)

But ok, up to this point, when we restrict our amplitude to behave unitarity and only take on physical values within a certain domain, we were lead to the Lehmann ellipse and Froissart-Martin bound (the unitarity part came in when we wrote the amplitude in terms of the S-matrix and partial wave basis). We also used the analytic properties of the amplitude to derive these results as well in terms of identifying the pole structure. But we have not *really* taken into account causality... Taking into account causality will lead us to the concept of *polynomial boundedness in (complex) energy* for which we start off with a toy model as we have done previously.

Consider a toy model where we have some signal/function $f_{\text{out}}(t)$ that can be related to the in-signal via some matrix (so which we casually denote it as S)

$$f_{\text{out}}(t) = \int dt' \, S(t-t') f_{\text{in}}(t') \tag{2.102}$$

where we call $S(t-t')$ the "scattering amplitude." The scattering amplitude here can be written as a Fourier transform in frequency space as

$$\widetilde{S}(\omega) = \int dt \, e^{i\omega t} S(t). \tag{2.103}$$

If we enforce causality on $S(t)$, which for our case here is just ensuring that the times $t-t'$ is ordered in the sequence of events, then $S(t)$ must be supported at $t \geq 0$ which is equivalent to the statement that the Fourier transformed version $\widetilde{S}(\omega)$ is analytic for $\operatorname{Im}\omega > 0$. To see this, notice that we can write the complex exponential as

$$e^{i\omega t} = e^{iRt\cos\phi} e^{-Rt\sin\phi}. \tag{2.104}$$

Then,for $\omega = Re^{i\phi}$, if we have the imaginary piece to be greater than 0, the second real term will give us exponential suppression which is the implication that there is no exponential growth as $\omega \to \infty$ on the upper-half complex plane which is the same statement that $\widetilde{S}(\omega)$ is analytic in the UHP (in other words, real).

But, something very interesting here is that if we *did* have exponential growth as $\operatorname{Im}\omega \to \infty$, then this would imply a *time advance* and thus violation of causality! This way of thinking we will now translate over to QFT amplitudes by investigating the behavior of the $2 \to 2$ amplitude $\mathcal{A}(s,t)$ at large complex s [61].

For simplicity, we will assume we have a scattering theory of the lightest scalar particle (which implies there is no anomalous threshold), and *no* bound state and no self-coupling

[61] Something I wanted to mention here is that you might have picked up that the entire S-matrix program we have been looking into is just asking what happens when you make the kinematic variables like s and t be allowed to take on complex values. Well, you may be wondering what happens when you make not just the kinematics but the *angular momentum* ℓ complex as well. This is a subject that we will not have time to get into, but it is a subject called *Regge theory* and it is quite a beautiful one. Regge theory is what gave birth to Chew-Frautschi plots, which shows that when you graph the angular momentum to the mass squared, you found linear trajectories of hadrons, and it was this realization that lead to dual-models for hadrons and the strong force, and then eventually *string theory*.

pole either. So, all we will have are the simple poles for the total mass/energy of the scattering process and branch cuts. For our discussion, for now, please refer to the following figure:

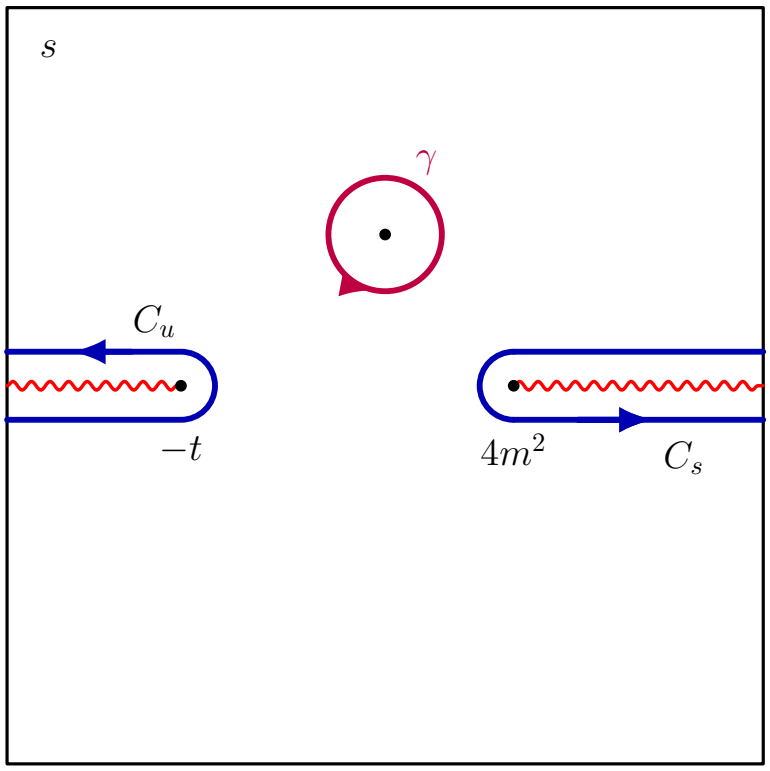


Figure 19. Here is a graph of $\mathcal{A}(s,t)$ for complex s showing the different contours over the u- and s-channels as well as a contour γ over some arbitrary point s'.

Let us fix t to be real for $t < 0$, and consider the analytic continuation of $\mathcal{A}(s,t)$ to the complex plane. We can consider re-writing the amplitude as a residue over some new arbitrary point s' (as we have done before) written as

$$\mathcal{A}(s,t) = \oint_{\gamma} \frac{ds'}{2\pi i} \frac{\mathcal{A}(s',t)}{s'-s}. \tag{2.105}$$

We want to deform the contour γ such that it instead becomes $-(C_u + C_s)$ contours or in other words, picks up integrating over the physical states. But, $\mathcal{A}(s',t)$ may grow as $s' \to \infty$ since we want the choice of s' to be arbitrary, but well under control. We will *assume* that the amplitude $\mathcal{A}(s,t)$ is polynomial bounded at large complex s for fixed $t < 0$ i.e.

$$\lim_{s\to\infty} \left| \frac{\mathcal{A}(s,t)}{s^N} \right| = 0 \tag{2.106}$$

for some large integer N. The value of N may depend on t[62]. Then, for a product of s_i

[62] A more rigorous treatment for polynomial bounded of the amplitude can be found in [16].

poles

$$\frac{\mathcal{A}(s,t)}{\prod_{i=1}^{N}(s-s_i)} \to 0 \tag{2.107}$$

as $s \to \infty$ for any $s_1, \cdots, s_N$. We can then write (2.105) as

$$\mathcal{A}(s,t) = \oint_{\gamma} \frac{ds'}{2\pi i} \frac{\prod_{i=1}^{N}(s-s_i)}{\prod_{j=1}^{N}(s'-s_j)} \frac{\mathcal{A}(s',t)}{s'-s}. \tag{2.108}$$

We now pick up C_n contours in the LHP that graphically looks like the following:

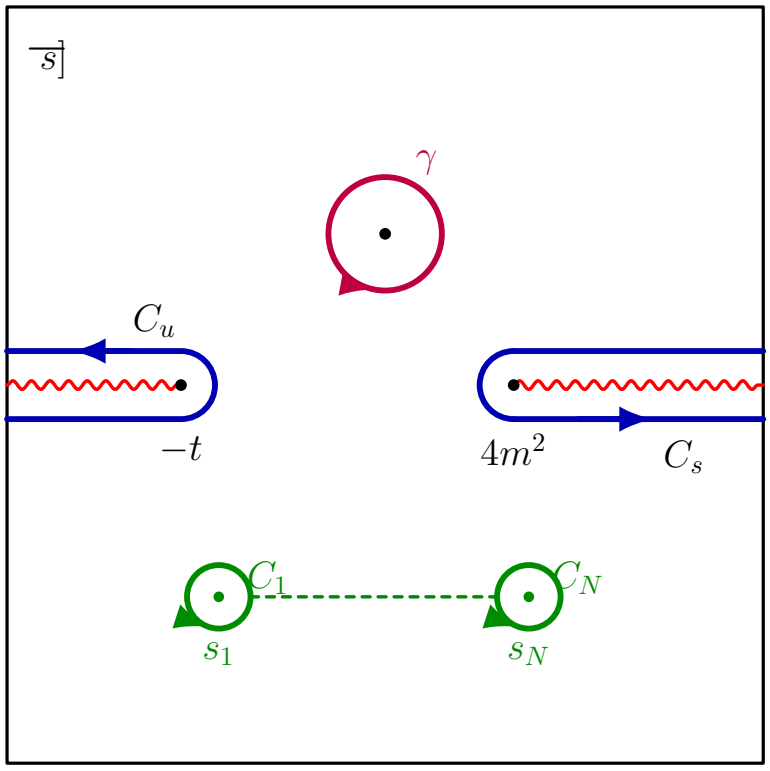


Figure 20. Here, the difference between 19 and now is that we have included C_i contours over s_i points in accordance with the limiting powers we have found.

Now, if we deform the γ contour to instead $-(C_u + C_s + \sum C_i)$ then we get

$$\begin{aligned}\mathcal{A}(s,t) &= \oint_{\gamma} \frac{ds'}{2\pi i} \frac{\prod_{i=1}^{N}(s-s_i)}{\prod_{j=1}^{N}(s'-s_j)} \frac{\mathcal{A}(s',t)}{s'-s}. \\ &= -\oint_{C_s+C_u} \frac{ds'}{2\pi i} \prod_{i=1}^{N} \frac{s-s_i}{s'-s_i} \frac{\mathcal{A}(s',t)}{s'-s} + \sum_{i=1}^{N} \prod_{j\neq 1} \frac{s-s_j}{s_i-s_j} \mathcal{A}(s_i,t)\end{aligned} \tag{2.109}$$

where this last term is the residue at $s' = s_i$ and we shall define the term as

$$P_{N-1}(s;t) \equiv \sum_{i=1}^{N} \prod_{j\neq 1} \frac{s-s_j}{s_i-s_j} \mathcal{A}(s_i,t) = \sum_{n=0}^{N-1} c_n(t) s^n \tag{2.110}$$

which is just some polynomial in s of degree $N-1$. We can write the first term in (2.109) as

$$-\oint_{C_s} \frac{ds'}{2\pi i}\mathcal{A}(s',t)(\cdots) = \int_{4m^2}^{\infty} \frac{ds'}{\pi} \operatorname{Im}\mathcal{A}(s',t)(\cdots). \tag{2.111}$$

We can rewrite the imaginary part of the amplitude in terms of something we shall define as the *discontinuity*[63] for real t and s as

$$\begin{aligned} 2i\ \operatorname{Im}\mathcal{A}(s,t) &= \mathcal{A}(s+i\epsilon,t) - \mathcal{A}(s-i\epsilon,t) \\ &\equiv \operatorname{disc}\mathcal{A}(s,t) \end{aligned} \tag{2.112}$$

where the discontinuity here is across the s-channel branch cut. We can then finally write the amplitude for $t<0$ in the following manner

$$\mathcal{A}(s,t) = P_{N-1}(s;t) + \int_{4m^2}^{\infty} \frac{ds'}{\pi}\frac{\operatorname{Im}\mathcal{A}(s',t)}{s'-s}\prod_{i=1}^{N}\frac{s-s_i}{s'-s_i} + \int_{4m^2}^{\infty} \frac{du'}{\pi}\frac{\operatorname{Im}\mathcal{A}(u',t)}{s'-s}\prod_{i=1}^{N}\frac{s-s_i}{s'-s_i} \tag{2.113}$$

where (2.113) is called an *N-time subtracted dispersion relation*. We use the wording *dispersion relation* here in the literal sense when comparing to Krönig-Kramers dispersion relations for light-media. In factor, to derive this expression for $N \geq 1$, we only need to assume the previously derived Froissart-Martin bound (2.101)

$$|\operatorname{Im}\mathcal{A}(s,t)| < C \cdot s^{N-\delta} \tag{2.114}$$

for large real s and some $\delta > 0$. Suppose that (2.113) is not true however, in other words, we need N' subtractions for $N' \geq N+1$. Then, the N'-time subtracted dispersion relation instead states that

$$\mathcal{A}(s,t) \sim C(t) \cdot s^{N'-1} \tag{2.115}$$

where $N'-1 \geq N$ when $s \to \infty$. If the $s^{N'-1}$ term in $P_{N'-1}$ happens to cancel against the order $s^{N'-1}$ term from $\int \operatorname{Im}\mathcal{A}(s',t)(\cdots)$, the leading large s behavior would be $s^{N'-2}$, contradicting the need for N' subtractions.

[63] For single variable functions, we can always write some complex function in terms of it's real and complex parts as $\operatorname{Re} f = (f+f^*)/2$ and it's imaginary component as $\operatorname{Im} f = (f-f^*)/2i$. For multivariable complex functions, these imaginary and complex definitions turns into discontinuities.

On the other hand, it follows from unitarity the following

$$\begin{aligned}
\operatorname{Im}\mathcal{A}(s,t=0) &= \sqrt{s(s-4m^2)}\sigma_{\text{tot}} \\
&\geq \sqrt{s(s-4m^2)}\sigma_{2\to 2} \\
&= 8\pi\sqrt{\frac{s}{s-4m^2}}\sum_{\ell}(2\ell+1)|S_\ell(s)-1|^2 \\
&= \frac{1}{64\pi}\sqrt{\frac{s-4m^2}{s}}\int_{-1}^{1} dx\,|\mathcal{A}(s,t=2k^2(x-1))|^2 \\
&= \frac{1}{32\pi\sqrt{s(s-4m^2)}}\int_{4m^2-s}^{0} dt\,|\mathcal{A}(s,t)|^2 \\
&\gtrsim (\text{positive constant})\times s^{2N-1}
\end{aligned} \tag{2.116}$$

where the last step is when $s\to\infty$, but this contradicts our assumption that $\operatorname{Im}\mathcal{A}s^{-(N-\delta)}\to 0$. The question then becomes, what value of N makes our assumption true? From the Froissart bound, for large s we have

$$\operatorname{Im}\mathcal{A}(s,t<0)\leq \operatorname{Im}\mathcal{A}(s,t=0)<C\cdot s(\log s)^2 \tag{2.117}$$

where the first inequality happens since recall that when the amplitude is written as in a partial wave basis as

$$\operatorname{Im}\mathcal{A}(s,t)=16\pi\sqrt{\frac{s}{s-4m^2}}\sum_{\ell}(2\ell+1)P_\ell(x)(1-\operatorname{Re}S_\ell) \tag{2.118}$$

the S-matrix term must obey $1-\operatorname{Re}S_\ell\geq 0$ and so then for $t<0$ we have $x=\cos\theta=1+2t(s-4m^2)^{-1}<1$ implying that $P_\ell(x)\leq P_\ell(1)=1$. From this inequality though we see that setting $N=2$ is good enough!

It then follows when setting $N=2$ that have the following bound on the amplitude

$$\lim_{s\to\infty}\left|\frac{\mathcal{A}(s,t)}{s^2}\right|=0 \tag{2.119}$$

for $t<0$. The $N=2$ can be extended to complex t in the discontinuity $|t|<|t_0|$. There is an idea here though: let us compare the Taylor series in t of both sides of the dispersion relation, and use the fact that

$$\partial_t^n\operatorname{Im}\mathcal{A}(s,t)\Big|_{t=0}\geq 0 \tag{2.120}$$

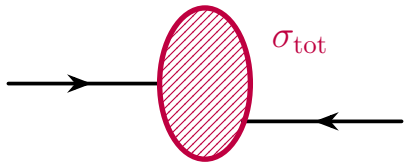
to show that the Taylor expansion in t commutes with dispersion integral, and that the same N-time subtraction extends to within the radius of convergence of the Taylor series in t. Thus, $N=2$ applies to the Froissart-Martin bound

$$\sigma_{\text{tot}}<\frac{1}{(1-\delta)^2}\frac{4\pi}{t_0}\left(\ln\frac{s}{M}\right)^2 \tag{2.121}$$

for $s \gg M^2$. Another point to make with $N = 2$ is that in D-dimensional spacetime, the analogous bound is

$$\sigma_{\text{tot}} < C \cdot (\ln s)^{D-2} \tag{2.122}$$

with, recall the following image for the total cross-section



and this bound is where the following phrase comes from

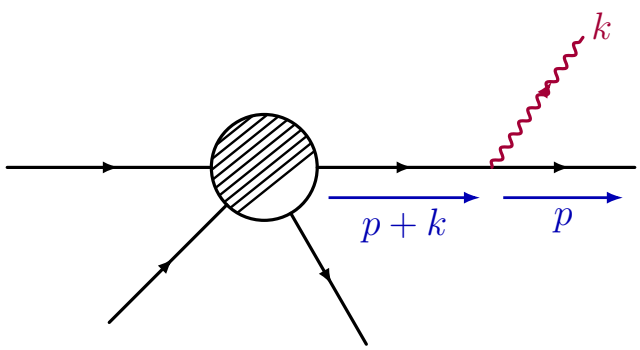
> "size [of the cross section area] cannot grow faster than log(energy)."

There are some caveats here though. We have assumed that all the particles present are massive. However, in a theory with massless particles, the total cross section may actually be infinite due to the (what is called collinear) divergence at zero scattering angle, as is the case in Coulomb/Rutherford scattering. A physically more meaningful observable would be the cross section excluding small scattering angles.

The idea that "size cannot grow faster than $\log(E)$" is actually violated in gravity. As an example, the Schwarzschild black hole horizon size grows as $\gamma_H \sim (G_N \cdot E)^{\frac{1}{D-3}}$, which we can see that as the horizon grows, and so the cross-section area for scattering of black holes, the energy grows faster than a logarithm. But, we won't be tackling gravity in this course. Instead, we will be looking at the point made previously, namely that when we include massless particle in our theory, they give rise to some weird behavior, and one such behavior we will investigate is that of *infrared divergences*.

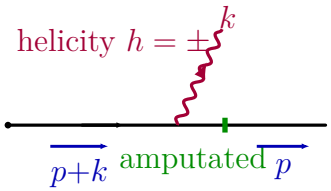
2.1.7 Soft Theorems and Infrared Divergences

The study of infrared divergences essentially starts with a study of QED, or technically and $D = 4$ Abelian gauge theory with charged particles, as we will look at what happens when the external legs of diagrams get an "extra" term attached namely the emission of an extra photon. The effect of *soft photon emission* is defined by the following diagram



(2.123)

where $k^0 = |\vec{k}|$ which is taken to be small in the lab frame. In scalar QED (so it is a scalar coupling to the photon and not fermions), we would have the following effect



$$= \frac{-i}{(p+k)^2 + m^2 - i\epsilon} iq(2p+k)^\mu e^{h*}_\mu(k)$$

$$= q\frac{p^\mu e^{h*}_\mu(k)}{p \cdot k - i\epsilon} \tag{2.124}$$

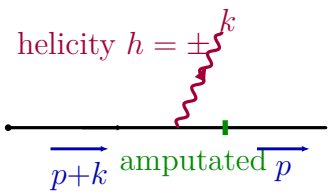
which diverges as the momentum of the photon goes to zero! Thus, we have a new divergence to worry about. Let us compare the amplitude $\mathcal{M}_{\beta|\alpha}$ for some initial state α and final state β to one with an extra soft photon emission denoted as $\mathcal{M}_{\beta;k,h|\alpha'}$ where $\alpha' \to \alpha$ when $k \to 0$ in which we have a rather simple relation when taking k^0 to zero

$$\mathcal{M}_{\beta;k,h|\alpha'} \to \mathcal{M}_{\beta\alpha} \sum_{n \in \alpha \cup \beta} \frac{q_n p^\mu_n e^{h*}_\mu}{\eta_n p_n \cdot k - i\epsilon} \tag{2.125}$$

where the summation here over $n \in \alpha \cup \beta$ denotes the sum over particle n in α and β, and η_n here is defined here such that it keep tracks of if $n \in \beta$ or $n \in \alpha$ namely

$$\eta_n = \begin{cases} + & n \in \beta \text{ out} \\ - & n \in \alpha \text{ in} \end{cases} \tag{2.126}$$

(2.125) is what is known as a *soft theorem* which for this case is particular for scalar QED. We can begin to do a similar calculation for spinor QED (or just the normal QED we are used to) with the same diagram

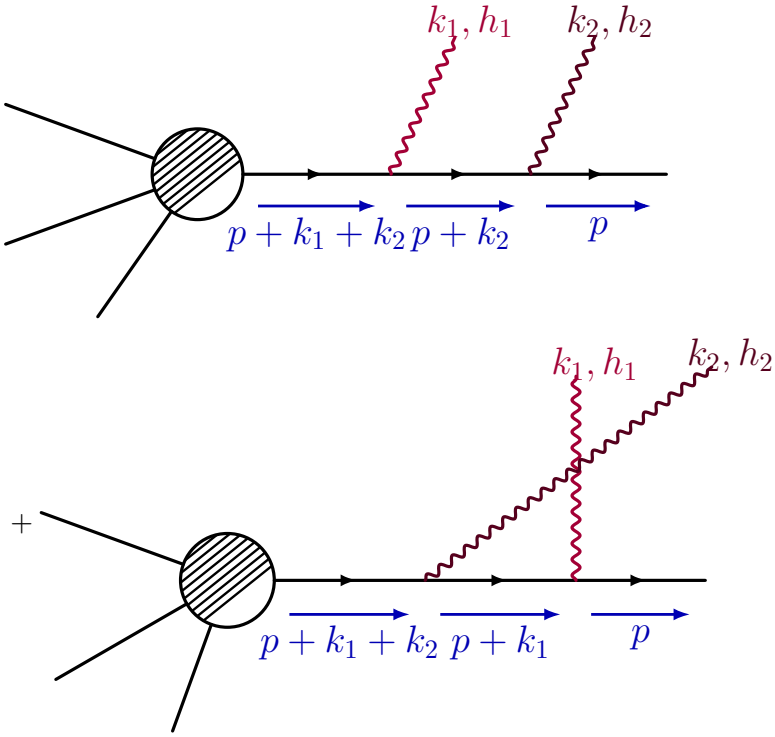


$$= \left[\overline{\mathcal{U}}^\sigma(-q\gamma^\mu)\frac{-i(-i(\not{p}+\not{k})+m)}{(p+k)^2+m^2-i\epsilon}\right]^\alpha e^{h*}_\mu(k)$$

$$\approx q\frac{p^\mu e^{h*}_\mu(k)}{p \cdot k - i\epsilon}\overline{\mathcal{U}}^{\sigma\alpha} \tag{2.127}$$

where the calculation went as follows: in the first line, since the particles are on-shell $(p+k)^2 + m^2 = p^2 + k^2 + m^2 + 2k \cdot p = -m^2 + 0 + m^2 + 2k \cdot p$ since the particles are on-shell as well as the photon! We also simplified the spinor algebra as $\overline{\mathcal{U}}^\sigma \gamma^\mu(-i\not{p} + m) = \overline{U}^\sigma(i\not{p} + m)\gamma^\mu - 2ip^\mu\overline{\mathcal{U}}^\sigma$ where the first term vanishes by the Dirac equation and we picked up a second term under the commutation relation. Notice that the spinor QED version here has the same divergence we found in (2.125) since $2p \cdot k$ term makes the propagator diverge as $k \to 0$. Therefore, the same soft photon relation we found in (2.124) holds. Now,

we will generalize to multiple soft photon emissions and see what happens[64]. Consider the following sum of diagrams that take into account the emissions of two soft photons:

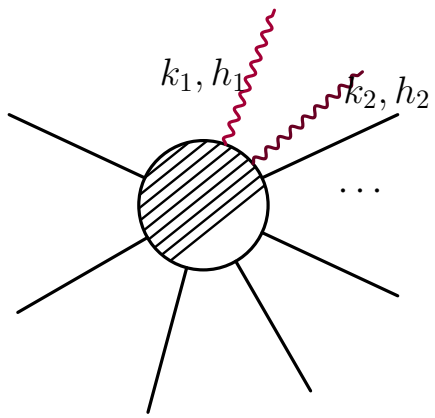


(2.128)

For the scalar QED case, in the $k_1, k_2 \to 0$ limit, the diagram with two soft photon emissions on the same external particle lines acquires the following factor

$$\frac{qp \cdot e^*_{h_2}(k_2)}{p \cdot k_2 - i\epsilon} \frac{qp \cdot e^*_{h_1}(k_1)}{p \cdot (k_1 + k_2) - i\epsilon} + (1 \leftrightarrow 2) = \frac{qp \cdot e^*_{h_1}(k_1) p \cdot e^*_{h_2}(k_2)}{(p \cdot k_1 - i\epsilon)(p \cdot k_2 - i\epsilon)} \tag{2.129}$$

and so we notice the following fact that in this case, the soft factors simply multiply! We can generalize this very nicely then for some n-number of photons emitted on external legs that we can think about as



(2.130)

[64] What follows is the proof given by Weinberg in 1965 who first derived these soft theorems and recognized the behavior of massless particles emitting on external legs. In fact, Weinberg's goal was to investigate the behavior of graviton emissions as legs and the photon was just a warm-up! We will not do the graviton example but it's implications are widespread and massive.

which in the soft-photon limit, this kind of diagram will enhance the amplitude by the following factor

$$\prod_{r=1}^{N}\sum_{n}\frac{q_n p_n\cdot e^*_{h_r}(k_r)}{\eta_n p_n\cdot k_r - i\epsilon} \tag{2.131}$$

where η_n is the same η we defined in our first soft photon example. But a question we can now ask is, "what is the effect of soft photon emissions on the transition probability?" Well, we can try to compute this directly in fact:

$$\frac{\Gamma_{\alpha+\beta+\text{soft photons}}}{\Gamma_{\alpha\to\beta}} = \sum_{N=0}^{\infty}\frac{1}{N!}\int\prod_{i=1}^{N}\frac{d^{D-1}\vec{k}_r}{(2\pi)^{D-1}2|\vec{k}_r|}\times\sum_{\{h_r=\pm\}}\left|\sum_n\frac{q_n p_n\cdot e_{h_r}(k_r)}{\eta_n p_n\cdot k_r}\right|. \tag{2.132}$$

To evaluate the polarization sum at the end of (2.132), we can use the following relation

$$\sum_{h_r=\pm} e_{h_R,\mu}(k_r)e^*_{h_r,\nu}(k_r) = \eta_{\mu\nu} + k_{r\mu}c_\nu + k_{r\nu}c_\mu \tag{2.133}$$

where again, c_μ is a constant such that $k_r\cdot c_r = -1$ and $c_r^2 = 0$. We then have the polarization sum becoming

$$\begin{aligned}\sum_{\{h_r=\pm\}}\left|\sum_n\frac{q_n p_n\cdot e_{h_r}(k_r)}{\eta_n p_n\cdot k_r}\right|^2 &= \sum_{n,m}\frac{\eta_n\eta_m q_n q_m}{p_n\cdot k_r p_m\cdot k_r}\left(p_n p_m + p_n\cdot k_r p_m\cdot c_r + p_m\cdot k_r p_n\cdot c_r\right)\\ &= \sum_{n,m}\frac{\eta_n\eta_m q_n q_m p_n\cdot p_m}{p_n\cdot k_r p_m\cdot k_r}\end{aligned} \tag{2.134}$$

where in the first line, the last two terms in the round brackets contributes zero since $p_n\cdot k_r p_m\cdot c_r + p_m\cdot k_r p_n\cdot c_r \propto \sum_n \eta_n q_n = 0$ by conservation of charge. Notice though that what we have must go inside of a loop integral, which will have a potential logarithmic divergence for $D=4$, which is the one we primary care about of course, so let us handle it. In $D=4$ we must evaluate

$$\int\frac{d^4\vec{k}}{(2\pi)^3 2|\vec{k}|^3}\sum_{n,m}\frac{\eta_n\eta_m q_n q_m p_n\cdot p_m}{(p_n^0-\vec{p}_n\cdot\hat{k})(p_m^0-\vec{p}_m\cdot\hat{k})}. \tag{2.135}$$

We can re-write the measure in terms of an angular integral over $\hat{k}$ and k^0 which is really ω here for the photon with $\omega\equiv|\vec{k}|$ and $\hat{k}\equiv\vec{k}/|\vec{k}|$ namely

$$\int\frac{d^4\vec{k}}{(2\pi)^3 2|\vec{k}|^3} = \frac{1}{2(2\pi)^3}\int\frac{d\omega}{\omega}\int d^2\hat{k}. \tag{2.136}$$

The angular integral over $\hat{k}$ gives

$$\begin{aligned}\int\frac{d^2\hat{k}}{(p_n^0-\vec{p}_n\cdot\hat{k})(p_m^0-\vec{p}_m\cdot\hat{k})} &= \int_0^1 dx\int\frac{d^2\hat{k}}{\left(p_n^0 x + p_m^0(1-x) - (\vec{p}_n x + \vec{p}_m(1-x))\cdot\hat{k}\right)^2}\\ &= \int_0^1 dx\int_0^{2\pi} d\theta\,\frac{2\pi\sin\theta}{(p_n^0 x + p_m^0(1-x) - |\vec{p}_n x + \vec{p}_m(1-x)|\cos\theta)^2}\\ &= \cdots\\ &= -\frac{2\pi}{p_n\cdot p_m}\frac{1}{\beta_{nm}}\log\frac{1+\beta_{nm}}{1-\beta_{nm}}\end{aligned} \tag{2.137}$$

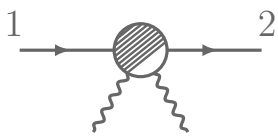
where β_{nm} here is defined as

$$\beta_{nm} \equiv \sqrt{1 - \frac{m_n^2 m_m^2}{(p_n \cdot p_m)^2}} \tag{2.138}$$

where the argument of the square root is greater than zero. Also, the result we found in (2.137) is monotonic in β_{nm}. The final result is then

$$\int \frac{d^4\vec{k}}{(2\pi)^3 2|\vec{k}|^3} \sum_{n,m} \frac{\eta_n \eta_m q_n q_m p_n \cdot p_m}{(p_n^0 - \vec{p}_n \cdot \hat{k})(p_m^0 - \vec{p}_m \cdot \hat{k})} = -\frac{1}{8\pi^2} \sum_{n,m} \frac{\eta_n \eta_m q_n q_m}{\beta_{nm}} \log \frac{1+\beta_{nm}}{1-\beta_{nm}} \int \frac{d\omega}{\omega}. \tag{2.139}$$

As an example of how this can be evaluated, we can consider the following diagram for $N = 2$ photons



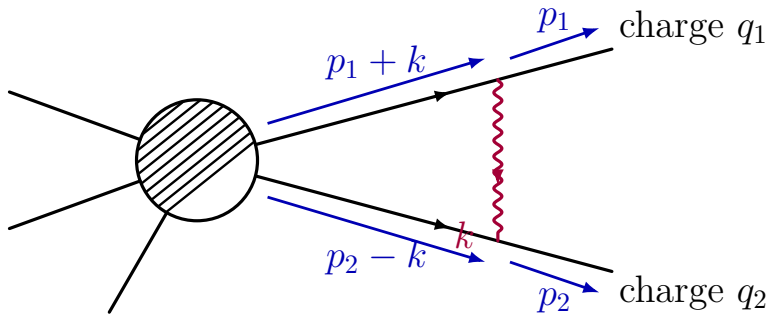
$$\Leftrightarrow \sum_{n,m}^{2} \eta_n \eta_m q_n q_m \frac{1}{\beta_{nm}} \log \frac{1+\beta_{nm}}{1-\beta_{nm}}$$

$$= 2q^2 \left(1 - \beta_{12} \log \frac{1+\beta_{12}}{1-\beta_{12}}\right) \leq 0. \tag{2.140}$$

The soft photon emission in (2.139) will enchance the transition probability by the following factor

$$\begin{aligned} \Gamma_{\alpha\to\beta} &\approx \sum_{N=0}^{\infty} \frac{A^N}{N!} \int_{\mu}^{E_T} \prod_{r=1}^{N} \frac{d\omega_r}{\omega_r} \\ &= e^{A \log(E_T/\mu)} \\ &= \left(\frac{E_T}{\mu}\right)^A. \end{aligned} \tag{2.141}$$

Note that for a sufficiently small energy E_T that below this energy, the soft photons are not observed in experimental apparatuses. But, what we had to introduce an *infrared cutoff* scale μ which is unphysical, and even worse yet, when $\mu \to 0$ it diverges! So what is going on now? Well, for now let us look at the effect of a soft photon when it itself creates a loop with respect to the external legs of the following form:



(2.142)

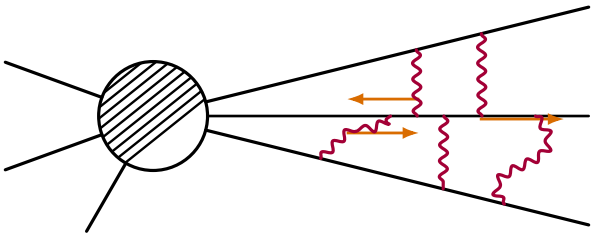
The inclusion of a virtual photon loop with a small $|\vec{k}|$ modifies this kind of diagram by multiplying by the following factor

$$\int_{\mu<|\vec{k}|<M} \frac{d^4k}{(2\pi)^4} \frac{q_1 p_1^\mu}{p_1 \cdot k - i\epsilon} \frac{q_2 p_2^\nu}{p_2 \cdot (-k) - i\epsilon} \frac{-i\eta_{\mu\nu}}{k^2 - i\epsilon} \tag{2.143}$$

where μ here is again the unphysical cutoff, M is taken to be a sufficiently small mass for soft approximation, and we are using the Feynman gauge (setting $\xi = 1$ in (1.608)) for the photon propagator. Similarly, a soft photon loop between two external lines of momenta p_n and p_m gives the factor $q_n q_m J_{mn}$ where

$$J_{mn} = -ip_n \cdot p_m \int\limits_{\mu<|\vec{k}|<M} \frac{d^4k}{(2\pi)^4} \frac{1}{(\eta_n p_n \cdot k - i\epsilon)(-\eta_m p_m \cdot k - i\epsilon)(k^2 - i\epsilon)}. \tag{2.144}$$

But more generally, if we have multiple soft photon loops such as this diagram:

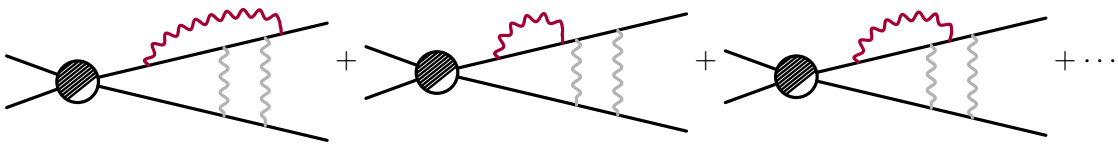


(2.145)

where the orange lines denote absorbing (truly) virtual soft photons in different orders will actually add up to a simple result. These N-number of virtual photon loops amounts to modifying the amplitude by the following factor

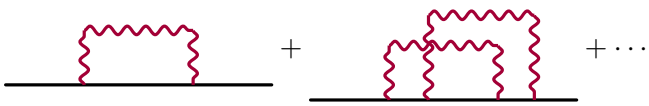
$$\sum_{N=0}^{\infty} \frac{1}{N!} \left(\frac{1}{2} \sum_{n \neq m} q_n q_m J_{nm} \right)^N = \exp\left(\frac{1}{2} \sum_{n \neq m} q_n q_m J_{nm} \right). \tag{2.146}$$

However, we are still missing two possible contributions to the question of soft photon emissions. The first are soft photon propagators that begin and end on the same charged particle line, for example:



(2.147)

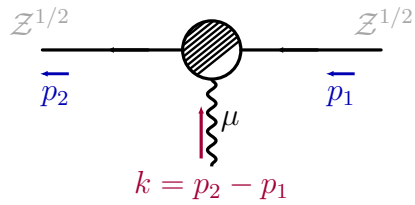
Secondly, we also need to include those virtual photons that arise from the renormalized fermion propagator since they include factors of the renormalization constant $\mathcal{Z}_n^{1/2}$ which are also IR-divergent! The types of diagrams we are talking about here are the following

$\Sigma =$  (2.148)

We can take into both of these kind of terms by computing the soft photon factors that arise from the electromagnetic (EM) form factor that we saw in QFT 1 (1.653) since we will see that the form factor must be IR divergent *free* and thus we can find a condition on what forces the soft photons to vanish when we include them. Without soft photons, the

EM form factor is

$$\langle \vec{p}_2 | \hat{j}^\mu(0) | \vec{p}_1 \rangle =$$



. (2.149)

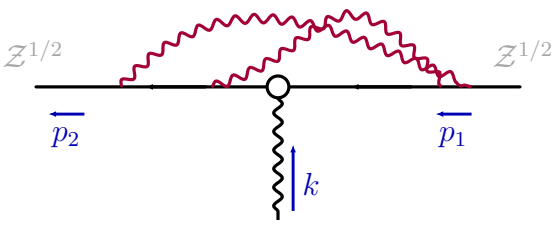
In scalar QED, the form factor becomes

$$\langle \vec{p}_2 | \hat{j}^\mu(0) | \vec{p}_1 \rangle = \frac{q}{(2\pi)^3 \sqrt{2\omega_{p_1}} \sqrt{2\omega_{p_2}}} (p_1 + p_2)^\mu F(k^2) \tag{2.150}$$

where we recall that at $k^2 = 0$ the form factor function becomes $F(0) = 1$. In spinor QED, we instead have

$$\langle \vec{p}_2, \sigma_2 | \hat{j}^\mu(0) | \vec{p}_1, \sigma_1 \rangle = \frac{iq}{(2\pi)^3} \overline{\mathcal{U}}^{\sigma_2}(p_2) \left(\gamma^\mu F(k^2) - \frac{i}{2m} (p_1 + p_2)^\mu G(k^2) \right) \mathcal{U}^{\sigma_1}(p_1) \tag{2.151}$$

where in this case for $k^2 = 0$, $F(0) + G(0) = 1$. The EM form factor in the $k^2 \to 0$ limit is free of IR divergence (something we won't prove here, but can be found in Weinberg's textbook in chapter 11.3.27 [1]). Recall previously that we have computed $G(0) = -q^2/8\pi^2 + \mathcal{O}(q^4)$ which was free of IR divergences, but on the other hand $F(k^2)$ does have IR divergences which you can check for by computing $dF/dk^2|_{k^2=0}$ which does have IR divergences. When we dress up the EM form factor with soft virtual photons, the form factor becomes



(2.152)

The soft virtual photons will contribute to the EM form factor (called a *soft factor*) as

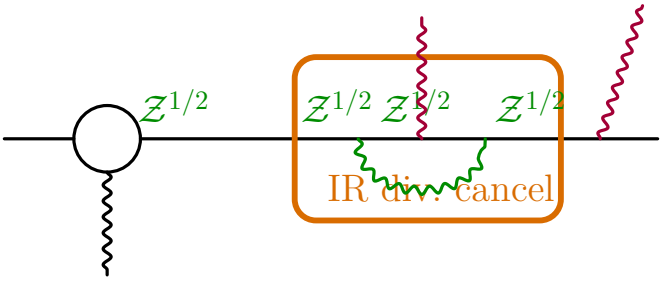
$$\sum_{N=0}^{\infty} \frac{1}{N!} (q^2 J_{12})^N = e^{q^2 J_{12}} \tag{2.153}$$

which must cancel again the IR divergence in the $(\mathcal{Z}^{1/2})^2$ in the $k \to 0$ limit!. To ensure this, we set the renormalization constant to the soft factor

$$\mathcal{Z}_{\text{IR}} = e^{-q^2 J_{12}|_{(p_2,\eta_2)=(p_1,-\eta_1)}} \tag{2.154}$$

where $\mathcal{Z}_{\text{IR}}$ denotes the IR divergent factor in $\mathcal{Z}$. To see how this will work in that there will be no further IR divergences from further look corrections to the soft vertices, consider

the following diagram

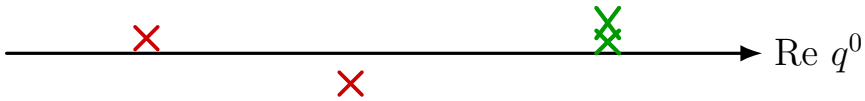


(2.155)

where we can see that in the orange box, we see that the IR divergences would then cancel when summing them up. Let us take a look at J_{12} and evaluate it explicitly namely

$$J_{12}\Big|_{(p_2,\eta_2,\eta_1)=(p_1,+,-)} = -ip_1^2 \int\limits_{\mu<|\vec{q}|<M} \frac{d^4q}{(2\pi)^4} \frac{1}{(q^2-i\epsilon)(-p_1\cdot q - i\epsilon)^2} \tag{2.156}$$

but in this integral that is noticeably different than the ones we have encountered so far, is that the q^0 integral can be evaluated from residues on either the UHP or LHP giving a real result since the pole structure resembles the following:



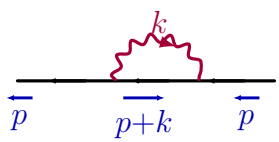
Let us compare this to our earlier formula for J_{nm} which formally, we can write, for example J_{11} as

$$J_{11} = -ip_1^2 \int\limits_{\mu<|\vec{q}|<M} \frac{d^4q}{(2\pi)^4} \frac{1}{(q^2-i\epsilon)(p_1\cdot q - i\epsilon)(-p_1\cdot q - i\epsilon)} \tag{2.157}$$

but in this case, the green poles for J_{11} there is one above and one below the $\mathrm{Re}\, q^0$ axis where the LHP one will contribute to the $\mathrm{Im}\, J_{11}$ divergence. But notice that the J_{12} and J_{11} case can be related to each other via

$$\mathrm{Re}\, J_{11} = -J_{12}\Big|_{(p_2,\eta_2)=(p_1,-\eta_1)}. \tag{2.158}$$

And so, we must have $\mathcal{Z}_{\mathrm{IR}} = e^{q^2\,\mathrm{Re}\,J_{11}}$. Let us check this explicitly with the 1-loop scalar QED case namely



$$\Leftrightarrow i\Sigma^{\text{1-loop}}(p) = \int \frac{d^4k}{(2\pi)^4} \frac{-i}{(p+k)^2+m^2-i\epsilon} \frac{-i}{k^2-i\epsilon} (iq)^2 (2p+k)^2. \tag{2.159}$$

The IR divergent contribution from (2.159) (from small k values) is

$$q^2 4p^2 \int\limits_{|\vec{k}|<M} \frac{d^4k}{(2\pi)^4} \frac{1}{p^2+m^2+2p\cdot k - i\epsilon} \frac{1}{k^2-i\epsilon}. \tag{2.160}$$

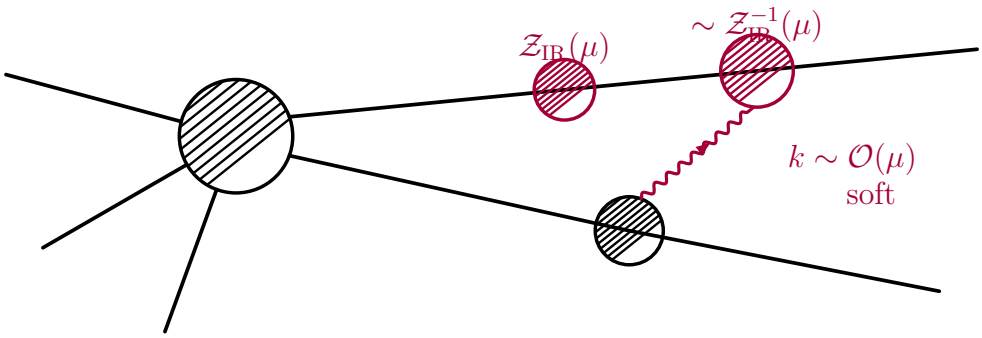
Now, recall that we have that the renormalization constant is related to the self-energy contribution Σ as

$$\mathcal{Z} = \left(1 - \frac{\partial \Sigma}{\partial p^2}\bigg|_{p^2=-m^2}\right)^{-1} \tag{2.161}$$

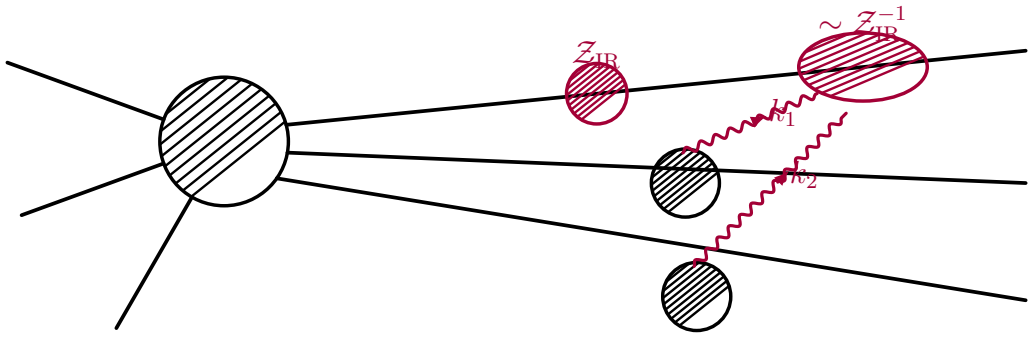
of which we can take the natural logarithm of to get, and then plug-in the self-energy diagram in (2.159) in the soft limit to find

$$\begin{aligned}
\ln \mathcal{Z} &= -\ln\left(1 - \frac{\partial \Sigma}{\partial p^2}\bigg|_{p^2=-m^2}\right) \\
&\sim iq^2 4p^2 \int\limits_{|\vec{k}|<M} \frac{d^4k}{(2\pi)^4} \frac{1}{(2p\cdot k - i\epsilon)^2} \frac{1}{k^2 - i\epsilon} \\
&= q^2 \operatorname{Re} J_{11}.
\end{aligned} \tag{2.162}$$

What all of this has finally taught us the following. Consider the following two classes of diagrams.



(2.163)



(2.164)

In the first one, this was a diagram where we learned that when photons are emitted from a general external leg, they contribute a factor of $\mathcal{Z}_{\mathrm{IR}}^{-1}(\mu)$ and in the second one, a similar effect occurs *even* with two photon momenta k_1, k_2. What this amounts to is that: IR divergences due to soft photon loops beginning and ending on the same external charged particle line cancel!

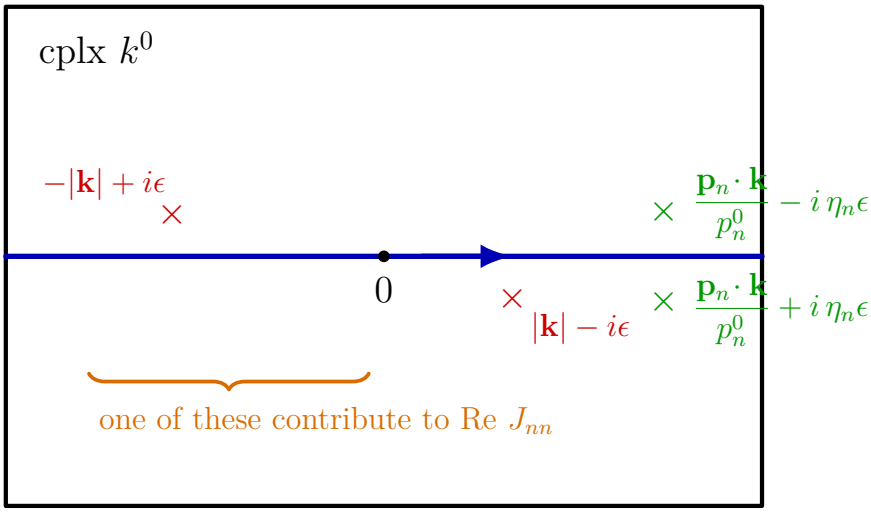
The final result is the following: soft photon loops modify the amplitude by the IR-divergent factor

$$e^{\frac{1}{2}\sum_{n\neq m} q_n q_m J_{nm}} \prod_n (\mathcal{Z}_n^{\mathrm{IR}})^{\frac{1}{2}} = (\text{phase}) \cdot e^{\frac{1}{2}\sum_{n,m} q_n q_m \operatorname{Re} J_{nm}} \tag{2.165}$$

where the $n-m$ current generated by the soft photons is

$$J_{nm} = -ip_n \cdot p_m \int\limits_{\mu<|\vec{k}|<M} \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2-i\epsilon)(\eta_n p_n \cdot k - i\epsilon)(-\eta_m p_m \cdot k - i\epsilon)}. \tag{2.166}$$

The pole structure of this integral is the following:



Here, the red poles are from the first denominator factor from the usual photon propagators, and the other two are the soft factors which are the green poles which may contribute *only* to $\operatorname{Im} J_{nm}$. The real part of J_{nm} then becomes

$$\begin{aligned}\operatorname{Re} J_{nm} &= -\eta_n\eta_m \frac{p_n \cdot p_m}{2} \int\limits_{\mu<|\vec{k}|<M} \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{|\vec{k}|^3 \cdot (p_n^0 - \vec{p}_n \cdot \hat{k})(p_m^0 - \vec{p}_m \cdot \hat{k})} \\ &= \frac{\eta_n\eta_m}{8\pi^2\beta_{nm}} \log\left(\frac{1+\beta nm}{1-\beta_{nm}}\right) \log\frac{M}{\mu}.\end{aligned} \tag{2.167}$$

Notice that the magnitude of the amplitude $|\mathcal{A}_{\alpha\to\beta}|$ contains the following IR divergent factor

$$e^{\frac{1}{2}\sum_{n,m} q_n q_m \operatorname{Re} J_{nm}} = e^{-\frac{1}{2}A\log\frac{M}{\mu}} = \left(\frac{\mu}{M}\right)^{\frac{A}{2}} \tag{2.168}$$

which does go to zero as $\mu \to 0$ (recall that $A > 0$ for generic kinematics). But hang on... apparently, all S-matrix elements between in- and out-states involving charged particles and a definite number of photons vanish?! Wait a second. However, restricting to states with finitely many photons is unnatural: experimentally, we cannot observe photons of energies below a certain threshold E_T since notice that the transition amplitude acquires the following factor

$$\Gamma_{\alpha\to\alpha+\text{soft}} \propto \left(\left(\frac{\mu}{M}\right)^{\frac{A}{2}}\right)^2 \times \left(\frac{E_T}{\mu}\right)^A \tag{2.169}$$

which becomes finite as $\mu \to 0$. The following remark should be made: in the above analysis, the IR regulator μ was introduced in a rather ad hoc manner. One might worry

about it's consistency or for example, why the same μ for virtual soft photons versus real soft photons emitted?

A more physical way to introduce the IR regulator would be to deform the QFT in a consistent manner, for example, by giving the photon a small mass via an *abelian Higgs model* which can be constructed in the following manner.

We shall introduce a complex scalar field $\rho(x)$ that is charged under $A_\mu(x)$ that acquires a non-zero vacuum expectation value $\langle\Omega|\,\rho(x)\,|\Omega\rangle = v$. The corresponding Lagrangian is

$$\mathcal{L} \supset |D_\mu \rho|^2 - V(\rho) \tag{2.170}$$

where in the kinetic term, we expand around $\rho = v$ which gives a term $-q^2|\nu|^2 A_\mu A^\mu$ which gives rise to a mass for the corresponding vector boson. The type of potential here that is associated with the Higgs-model is of the following:

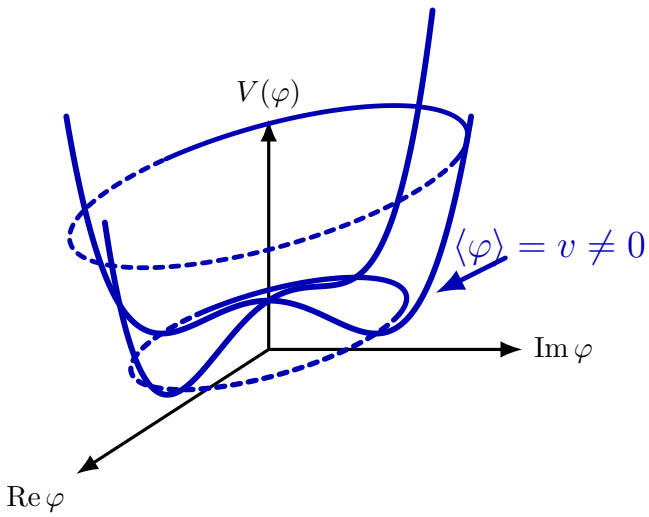


Figure 21. Here, we have graphed a typical "Mexican hat potential" that is common with Higgs-type models that is also used in symmetry breaking models.

To recap, we have just shown previously that the soft photons that come in the form of: self-energy diagrams of the scalars/fermions, the external legs that start and end on the same leg, and those that cross the external legs that give rise to loop contributions *all* vanish the physical limit of the experimental apparatus, and thus do not need to be worried about. But, as a *theorist*, well in fact we found that photons (and for that matter, any massless particle that is taken in the low energy limit) will contribute in fact an infinite number of themselves or soft-versions of themselves to any physical process, but yet in a highly controlled manner that is central to making any QFT consistent. Now, something that we will not get into here but highlights the importance of soft theorems is that when the massless particle is extended to include gravitons, then an attempt at making a consistent theory of quantum gravity can arise in limited cases. Part of this reason is that soft theorems are related to topics called *memory effects* and *asymptotic symmetry*.

It should also be noted that a result we found that was swept under the rug is that our photon soft theorem proved that the existence of massless spin-1 particles imply conservation of charge. If we extend to spin-2, we find two key results:

1. massless spin-2 particles imply gravity is universal, and
2. there are no interacting theories of massless particles of spin greater than 2

where the first state is simply the equivalence principle.

2.1.8 Quantum Effective Actions and *Naive* Renormalizability

We now shift gears entirely and will begin to look at some formalism with the path integral that will allow us to derive what are called *effective actions* or specifically the *quantum effective action*, which will set the stage to fully understand what renormalizability is. So, think of what we are about to do as a stepping stone to get to proving renormalizability for different theories.

Let us consider a QFT defined through the path integral

$$Z = \int [D\phi]\, e^{iS[\phi]} \tag{2.171}$$

where we shall call $S[\phi]$ the bare action, and as we know, the measure is defined through some regularization process such as using dimensional regularization. Recall also that to compute the correlation function for some operator $\mathcal{O}[\phi]$ as

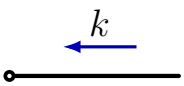
$$\langle \mathcal{O}[\phi] \rangle = \frac{1}{Z} \int [D\phi]\, e^{iS[\phi]} \mathcal{O}[\phi]. \tag{2.172}$$

Let us now define something called the *generating functional* $W[J]$ via the following insertion of a current-generating term

$$\begin{aligned} Z[J] &\equiv \int [D\phi]\, e^{iS[\phi] + i\int d^D x\, J(x)\phi(x)} \\ &\equiv e^{iW[J]} \end{aligned} \tag{2.173}$$

where the current term can not only be viewed as an extra term in the action but as a *new* 1-point vertex interaction of the form

$$i\widetilde{J}(k) = \tag{2.174}$$



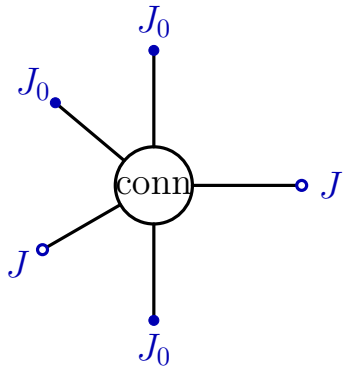
where the current term at the level of the action has the following Fourier transformation

$$\int d^D x\, J(x)\phi(x) = \int \frac{d^D k}{(2\pi)^D} \widetilde{J}(k)\widetilde{\phi}(-k). \tag{2.175}$$

We would like to be able to compute the contributions to $W[J]$ via perturbation theory which can be done as follows. The generating functional here, $W[J]$ is the sum of all

connected diagrams that we can write as

$$iW[J] = \sum$$



(2.176)

where the summation is over all connected Feynman diagrams. Now, we have that $Z[J]$ or $W[J]$ can capture all of the possible Greens functions via

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = \frac{1}{Z} \frac{\delta}{i\delta J(x_1)} \cdots \frac{\delta}{i\delta J(x_n)} Z[J] \Big|_{J=0} \tag{2.177}$$

which should feel very familiar from QFT via Wick's theorem and the QM examples we did but it won't become that. In relating this green function, recall that the ordering is determined by the choice of x^0-contour in $S[\phi]$ action in the path integral as seen in QFT 1. Now, note in particular that if we consider a *single* ϕ-correlation function (also sometimes referred to as *a single insertion* of ϕ) we would have

$$\langle \phi(x) \rangle = \frac{\delta}{i\delta J(x)} \ln Z[J] \Big|_{J=0} = \frac{\delta W}{\delta J(x)} \Big|_{J=0} . \tag{2.178}$$

Now let us consider the Legendre transform which we shall define here as[65]

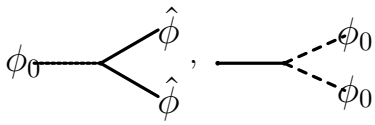
$$\Gamma[\phi_0] \equiv W[J] - \int d^D x \, \phi_0(x) J(x) \Big|_J \tag{2.179}$$

where J here is evaluated such that $\delta W/\delta J(x) = \phi_0(x)$. What we have written in (2.179) is referred to as the *1PI effective action* or the *quantum effective action*. Let us inspect the meaning of $\Gamma[\phi_0]$ in perturbation theory. First, notice that we can write the exponential of the quantum effective action as

$$\begin{aligned} e^{i\Gamma[\phi_0]} &= e^{W[J] - i\int \phi_0 J} \Big|_{\frac{\delta W}{\delta J} = \phi_0} \\ &= \int [D\phi] \, e^{iS[\phi] + i\int(\phi - \phi_0)J} \Big|_{\langle \phi \rangle_J = \phi_0} \\ &= \int [D\hat{\phi}] \, e^{iS[\phi_0 + \hat{\phi}] + i\int \hat{\phi} J} \Big|_{\langle \hat{\phi} \rangle_{J,\phi_0} = 0} \end{aligned} \tag{2.180}$$

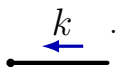
[65] You may be wondering why we are performing a Legendre transformation which is commonly used to switch between different conjugate coordinates or variables in a Lagrangian-Hamiltonian or thermodynamical system. But here in QFT, we will see that it's purpose is to be able to compute quantum effects with respect to *background fields* that will become very important as you will look at a particular on in the problem sets!

where in the second line, we inserted the 1-point function that we set equal to some vacuum value ϕ_0, and in the third line, we performed a change of variables by setting $\phi \equiv \phi_0 + \hat{\phi}$. But now, to stay consistent with our definition of the quantum effective action, we need to find a current J such that $\langle \hat{\phi} \rangle_{J,\phi_0} = 0$. For example, the action $S[\phi]$ contains the coupling (imagine $g\phi^3$ theory but set the coupling to $g = 1$) ϕ^3 vertex which has a corresponding Feynman diagram which, when we perform the same change of variables above, we instead get

$$S[\phi_0 + \hat{\phi}] \rightsquigarrow$$ 

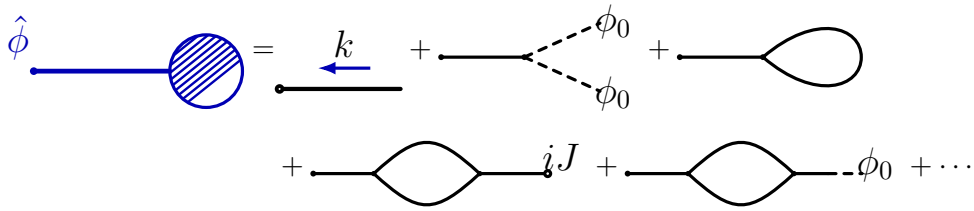
$$, \quad \text{etc.} \tag{2.181}$$

and then the (what we shall now official call) *tadpole diagram* with the source current becomes

$$\int \hat{\phi} J \rightsquigarrow \text{vertex}$$ 

$$. \tag{2.182}$$

Now with these rules, we can see what kind of J would produce $\langle \hat{\phi} \rangle_{J,\phi_0} = 0$ which is done explicitly by computing $\langle \hat{\phi} \rangle$. The tadpole diagram for the $\hat{\phi}$-field has the following expansion

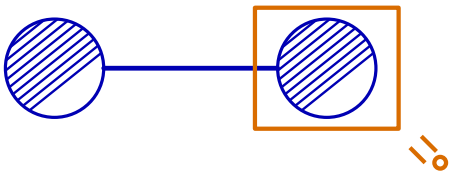


$$= 0 \tag{2.183}$$

which you can see all must equal zero *by choice* of J! So, from this viewpoint, J is not arbitrary but set by the cancellation of the total tadpole-like diagrams. The quantum effective action $i\Gamma[\phi_0]$ is computed by summing over all connected diagrams with *background field* ϕ_0 and source J, of which we have seen is chosen so that[66]

$$= 0. \tag{2.184}$$

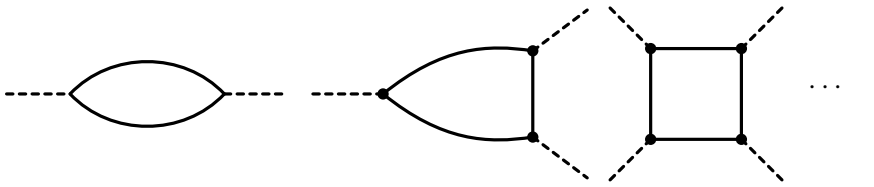
The effect of J is to remove all possible *1-particle cuts*, which is defined as the subset of diagrams of the form,



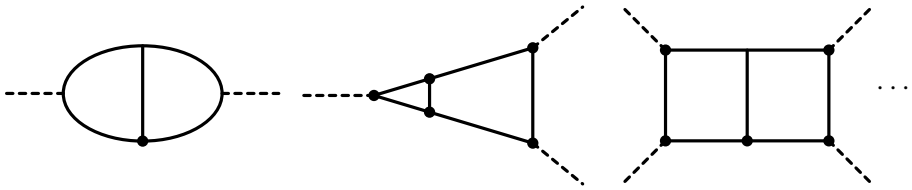
$$\tag{2.185}$$

[66] As a fun fact, the background "field" that plays this role in gravity, which is a non-renormalizable theory, is the cosmological constant, and the issue of said "constant" can be traced back to this formalism.

which sum up to zero. The only non-trivial contributions to the effective action $i\Gamma[\phi_0]$ come from the 1PI diagrams (that we already have seen many times before), for example at two, three, and four-point diagrams,



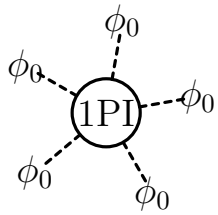
(2.186)



(2.187)

In other words, we can write the quantum effective action as

$$i\Gamma[\phi_0] = iS_{\text{kinetic}}[\phi_0] + \sum \frac{1}{n!}$$

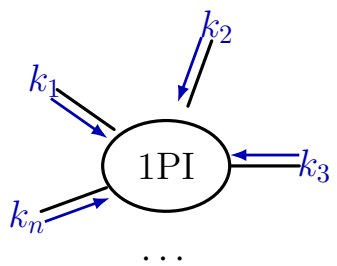


(2.188)

where the first term is only the regular kinetic action while the second term is the summation over all 1PI graphs with n-external background field lines. Note that there are different conventions for symmetry factors of graphs when ϕ_0 is viewed as an external line versus when it is part of the vertex itself. Or, more precisely, we can write the quantum effective action as

$$\Gamma[\phi] = \sum_n \int \prod_{i=1}^{n} \frac{d^D k_i}{(2\pi)^D} (2\pi)^D \delta^D(\sum k_i) \frac{1}{n!} \Gamma^{(n)}(k_1, \cdots, k_n) \widetilde{\phi}(k_1) \cdots \widetilde{\phi}(k_n) \tag{2.189}$$

where $\widetilde{\phi}(k_i)$ are the usual Fourier transform of scalar fields and here, $i\Gamma^{(n)}(k_1, \cdots, k_{n-1})$ is computed by the sum of n-point amputated 1PI diagrams of the form



(2.190)

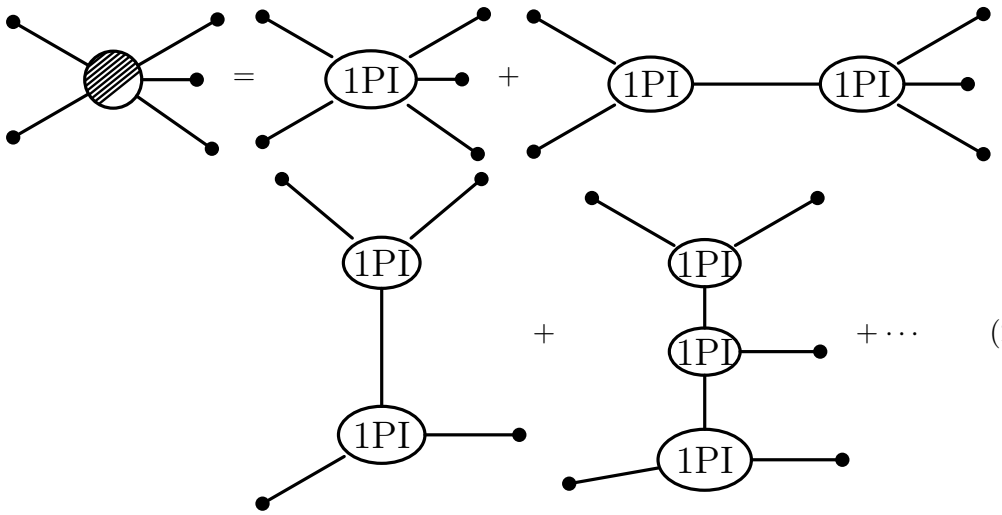
It should be noted that we will be working primarily with the effective action written as $\Gamma^{(n)}(k_i)$ except the only caveat here is that the momenta k_i's are off-shell, not on-shell...

Nonetheless, let us for example, consider the 2-point effective action which we can write as[67]

$$i\Gamma^{(2)}(k) = -i(k^2 + m^2) + \quad . \tag{2.191}$$

Note that the derivative expansion of $\Gamma^{(n)}$ in k_i is equivalent to the momentum expansion of *effective couplings* (which are the coefficients that are multiplied to the higher order operators we can include in the action). When there are loops of massless fields that contribute to $\Gamma^{(n)}$, the latter is typically not analytic in k_i's at $k_i = 0$.

the interpretation of $\Gamma[\phi]$ as the *quantum effective action* steps from that the *tree* diagrams formed out of Feynman vertices of $\Gamma[\phi]$ give the *exact* Greens functions namely



(2.192)

More formally, we can go from the effective action $\Gamma[\phi]$ to $W[J]$ via the inverse Legendre transform as

$$W[J] = \Gamma[\phi] + \int d^D x\, \phi(x) J(x) \Big|_{\frac{\delta\Gamma}{\delta\phi} = -J} \tag{2.193}$$

where this time, the last term ϕ is evaluated such that $\frac{\delta\Gamma}{\delta\phi} = -J$. We can check this directly. First, recall that the regular Legendre transform is written as

$$\Gamma[\phi] = W[J] - \int \phi J \Big|_{\frac{\delta W}{\delta J} = \phi} \tag{2.194}$$

and so then the derivative with respect to $\phi(x)$ of $\Gamma[\phi]$ is

$$\begin{aligned} \frac{\delta\Gamma}{\delta\phi(x)} &= \int dy\, \frac{\delta W}{\delta J(y)} \frac{\delta J(y)}{\delta\phi(x)} - J(x) - \int dy\, \phi(y) \frac{\delta J(y)}{\delta\phi(x)} \\ &= -J(x). \end{aligned} \tag{2.195}$$

[67] Take note that the quantum effective action for n-point vertices is only a function of $n-1$ momenta since we can always reduce the number of variables by one using conservation of momentum.

Note that we can write the generating functional equivalently as, when we restore $\hbar$,

$$\begin{aligned} iW[J] &= \lim_{\hbar\to 0} \hbar \log \int [D\phi]\, e^{\frac{i}{\hbar}\left(\Gamma[\phi]+\int \phi J\right)} \\ &= \text{sum over tree diagrams built from } \Gamma[\phi]. \end{aligned} \tag{2.196}$$

Also note that in the absence of a source J, we recover back a *true* vacuum expectation value

$$\phi \Big|_{\frac{\delta\Gamma}{\delta\phi}=0} = \frac{\delta W}{\delta J}\Big|_{J=0} = \langle\phi\rangle \tag{2.197}$$

or in other words, $\Gamma[\phi]$ is extremized at $\phi = \langle\phi\rangle$.

In principle, the quantum effective action $\Gamma[\phi]$ may be defined beyond perturbation theory but for this to be true, we would need $W[J]$ to be a *concave* functional for Legendre transform. As an example of such an analysis, we can consider the Euclidean version of the generating functional

$$Z[J] \equiv e^{-W[J]} = \int [D\phi]\, e^{-S[\phi]-\int \phi J}. \tag{2.198}$$

We shall assume there are only bosonic fields present, and thus the measure $[D\phi]$ has the nice property of being positive (semi-)definite. Consider the following re-writing of the generating functional by assuming that for some value t restricted to $0 \le t \le 1$, the generating functional becomes

$$\begin{aligned} e^{-W[tJ_1+(1-t)J_2]} &= \int [D\phi]\, e^{-S[\phi]-\int \phi(tJ_1+(1-t)J_2)} \\ &= \int [D\phi]\, e^{-t(S[\phi]+\int \phi J_1)-(1-t)(S[\phi]+\int \phi J_2)} \\ &\le \left(\int [D\phi]\, e^{-(S[\phi]+\int \phi J_1)}\right)^t \left(\int [D\phi]\, e^{-(S[\phi]+\int \phi J_2)}\right)^{(1-t)} \\ &= e^{-tW[J_1]-(1-t)W[J_2]} \end{aligned} \tag{2.199}$$

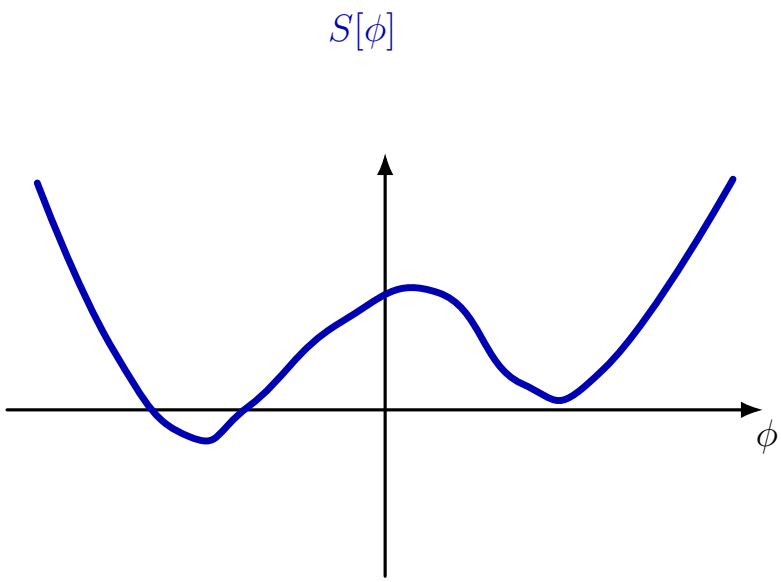
where in the second line, we used a result called *Hölder's inequality*. This seemingly simple result implies the following inequality

$$W[tJ_1 + (1-t)J_2] \ge tW[J_1] + (1-t)W[J_2] \tag{2.200}$$

which will allow us to determine whether or not W is concave, and thus give us a criterion for doing so. Then, if W obeys (2.200), W is in fact concave, and then our definition of

$$\Gamma[\phi] \coloneqq W[J] - \int \phi J \Big|_{\frac{\delta W}{\delta J}=\phi} \tag{2.201}$$

in in fact well-defined. Let us now look at an instructive toy model. Let us consider a "$D = 0$ QFT in the $\hbar \to 0$ limit." What we mean here is that we take $\phi \in \mathbb{R}$ and imagine an action that has the following behavior when graphed against the scalar field:



The partition function is related to the generating functional in the normal way (in the Euclidean signature)

$$Z[J] = e^{-W[J]} = e^{-S[\phi]-\phi J}\Big|_{\text{maximize w.r.t } \phi} \tag{2.202}$$

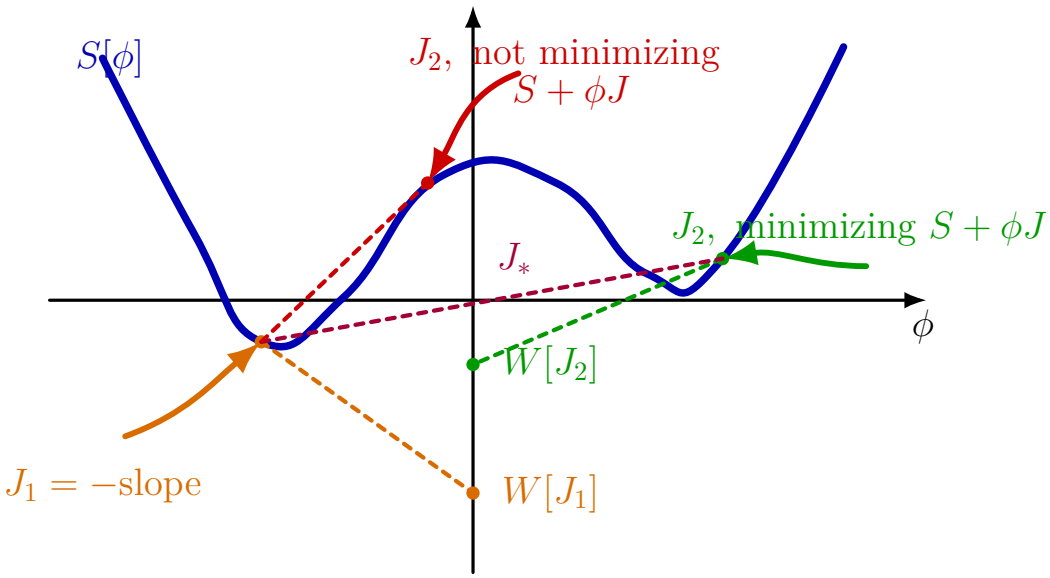
where the generating function is written as

$$W[J] = S[\phi] + \phi J\Big|_{\text{minimize w.r.t. } \phi} \tag{2.203}$$

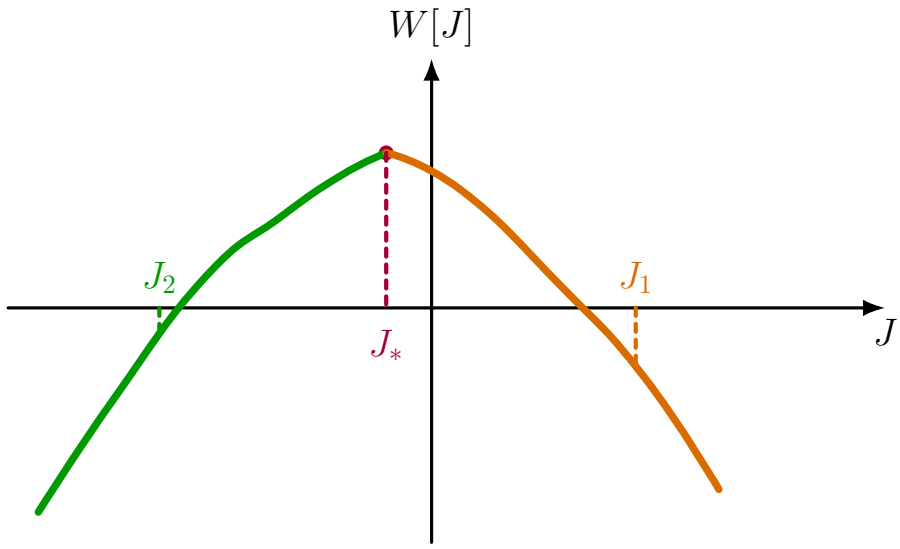
where in the partition function, the maximizing is of the full exponential, but if we look at just the generating functional, we must minimize the functional with respect to ϕ since the exponential is exponentially decaying in the Euclidean signature (minus sign). The minimizing condition can be found from

$$\frac{\partial S}{\partial \phi} + J = 0. \tag{2.204}$$

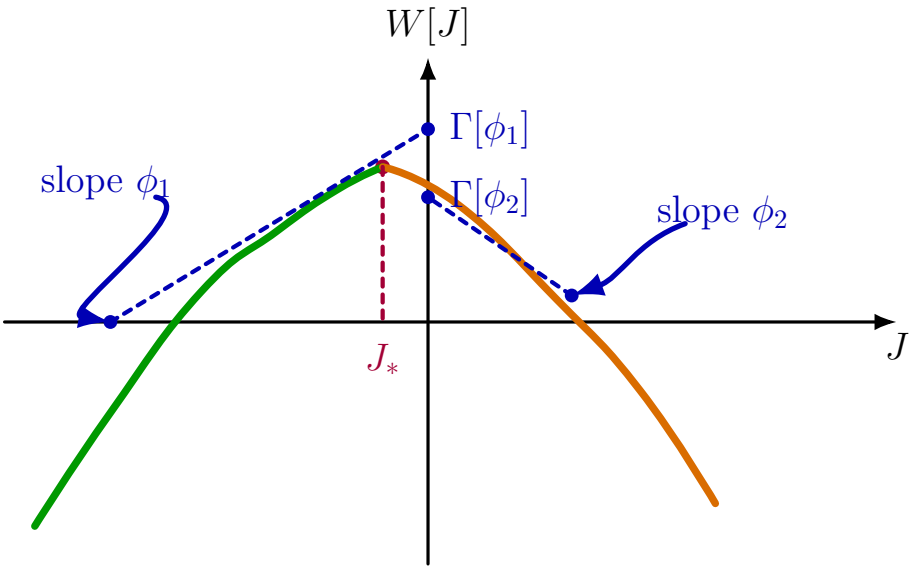
So, let us look at the graph 2.1.8 but more carefully, so consider the following:



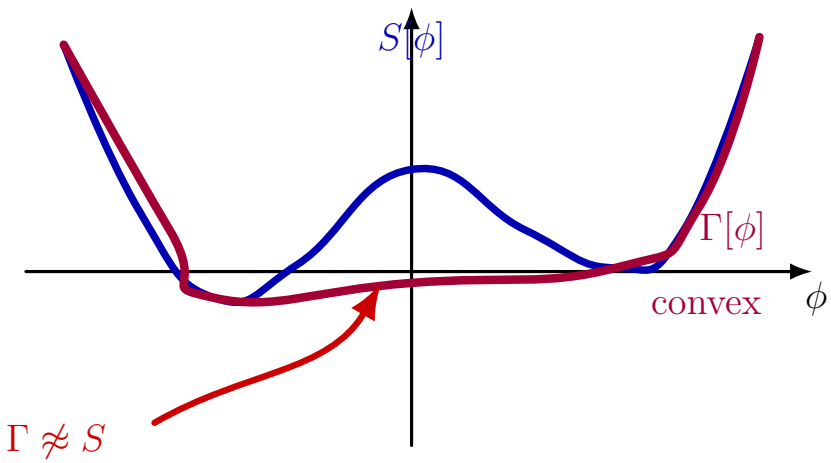
Here in figure 2.1.8, we have shown explictly where the minimizing and maximizing of the action occurs. We can then map these points onto the graph for $W[J]$ from which we will see that if the action has the following even (regardless of symmetry) behavior, the generating functional will have a concave behavior as in the following:



But now recall that we also have the relation between the generating functional and the *quantum* effective action, $W[J] = \Gamma[\phi] + \int \phi J$ from which we can also maximize via the condition $\partial\Gamma/\partial\phi + J = 0$. In doing so, we can plot those max/min tangent lines to find the following graph,



And now, we can Legendre transform *back* and then plot the *quantum* effective action against the regular action $S[\phi]$ to find the following:

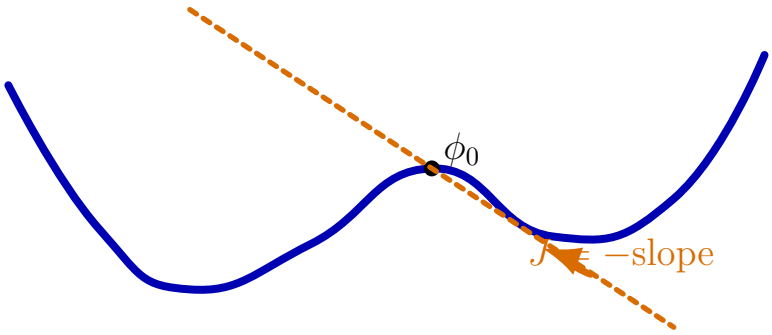


Notice that now, after we Legendre transform back this form of of $W[J]$, we now *do not* have a matching quantum action Γ to the regular action S i.e. $\Gamma \not\approx S$ which is apparently signaling to us that *perturbation theory breaks down*?! That is the interpretation we are lead to since the quantum action is not able to full replicate the dynamics of the original action after including quantum mechanics, and since we were able to define the quantum effective action in terms of perturbation theory, we have to conclude that our perturbation theory in the quantum action breaks down.

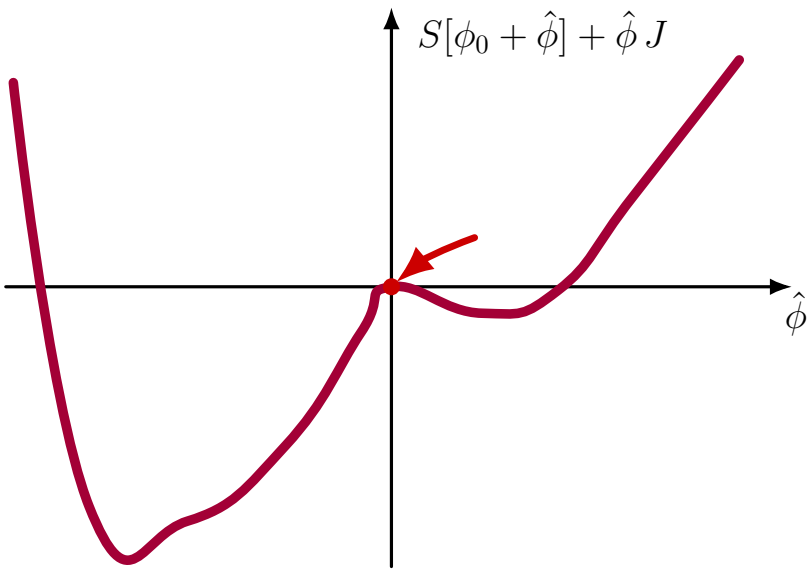
But, recall that the euclidean quantum effective action can be written as

$$e^{-\Gamma[\phi_0]} = \int [D\hat{\phi}]\, e^{-S[\phi_0+\hat{\phi}]-\int \hat{\phi} J}\Big|_{\langle\hat{\phi}\rangle_{J,\phi_0}=0} \tag{2.205}$$

where again, we choose J such that $\langle\hat{\phi}\rangle_{J,\phi_0} = 0$. Can (2.205) be evaluated using perturbation theory? In our toy model, we originally graphed the action as something like this:



The yellow dashed line that passes through ϕ_0 is just the condition that $J = -\text{slope}$, and we can subtract this line off to get $S + \hat{\phi}J$. If we however graph $S[\phi_0 + \hat{\phi}] + \hat{\phi}J$ we instead find



Where the red point on this graph is *not* positive definite with respect to the fluctuation $\hat{\phi}$ and thus we cannot use perturbation theory here!

This semi-instructive $D = 0$ QFT toy model was meant to serve as a systematic viewpoint to observe how to think about when perturbation theory can break down. This will serve as a viewpoint for attempting to define the condition for physical theories to be

renormalizable, and thus define the condition of *renormalizability*. This topic will be our next primary viewpoint, and thus we end our exposition of the S-matrix in QFT.

2.2 The Renormalization Group

Welcome to the renormalizability section! To get started, let us consider (you guessed it) a scalar theory defined via the Euclidean path integral

$$Z = \int [D\phi]\, e^{-\int d^D x\, \mathcal{L}_E} \tag{2.206}$$

with the following corresponding Euclidean *bare* Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2} m^2 \phi^2 + \sum_I g_I \mathcal{O}_I \tag{2.207}$$

Here, each $\mathcal{O}_I$ denotes some higher order scalar field operator that is a function of $\phi(x)$ and of it's corresponding derivatives (with proper contractions of the Lorentz indices) $\partial_\mu \phi(x)$, $\partial_\mu \partial_\nu \phi(x)$, and so on. The couplings here that are assigned to each operator $\mathcal{O}_I$ denoted g_I are *bare couplings* which include all counter terms.

Recall that the path integral measure is typically defined with a regularization scheme such as with a momentum cutoff $\widetilde{\phi}(k)$ with only including momentum states of $|k| \leq \Lambda$, or via dimensional regularization where we take generally $D = \text{integer} - \epsilon$.

But, we also know that physical observables, for example the mass spectrum of particles or scattering amplitudes, should be finite as we take the cutoff $\Lambda \to \infty$ or $\epsilon \to 0$ in either regularization scheme, while adjusting the bare couplings g_I accordingly as a function of Λ or ϵ.

But now let us pose the question, "is this possible for a Lagrangian with finitely many g_I's?" In essence, for a finite number of couplings or operators, can we absorb all divergences present?

2.2.1 BPHZ Renormalizability Theorem

Let us define a possible sufficient criterion: let us require that Euclidean Greens functions $\langle \phi_R(x_1) \cdots \phi_R(x_n) \rangle$ are finite where $\phi_R(x)$ is related to $\phi(x)$ by a rescaling of

$$\phi_R(x) = \mathcal{Z}_R^{-\frac{1}{2}} \phi(x) \tag{2.208}$$

where $\mathcal{Z}_R$ as we have seen is called the *renormalization constant* which may be finite in the limit that $\Lambda \to \infty$ or $\epsilon \to 0$. This is equivalent to demanding that the one-particle-irreducible (1PI) effective action $\Gamma[\phi]$ to be such that $\Gamma[\mathcal{Z}_R^{\frac{1}{2}} \phi_R]$ is a finite functional of $\phi_R(x)$.

We may equivalently work with ϕ_R as the field variable

$$S[\phi = \mathcal{Z}_R^{\frac{1}{2}} \phi_R] \equiv \widetilde{S}[\phi_R] \tag{2.209}$$

and then demand that

$$\Gamma[\phi = \mathcal{Z}_R^{\frac{1}{2}} \phi_R] = \widetilde{\Gamma}[\phi_R] \tag{2.210}$$

are to be finite. For example, we can consider our scalar theory (2.207) and re-write the kinetic term as (under integration by parts)

$$S[\phi] = \frac{1}{2}\int d^D x\, \phi(-\Box + m_0^2)\phi + \cdots \tag{2.211}$$

where m_0 here is the bare mass, to which the action in terms of the renormalized field would then be written as

$$\widetilde{S}[\phi_R] = \int d^D x\, \frac{1}{2}\phi_R(-\Box + m_R^2)\phi_R + (\mathcal{Z}_R - 1)\frac{1}{2}\phi_R(-\Box + m_R^2)\phi_R + \frac{1}{2}\delta m^2\, \phi_R^2 + \cdots \tag{2.212}$$

where the terms in blue are to be viewed all as *counter terms*. Generally, we write the renormalized action as

$$\widetilde{S}[\phi_R] = S_{\text{kin}}[\phi_R] + \int d^D x\, \sum_I g_I \mathcal{O}_I[\phi_R] \tag{2.213}$$

where the bare couplings g_I include all of the counter terms, and the quantum effective action can also be written generically as

$$\widetilde{\Gamma}[\phi_R] = S_{\text{kin}}[\phi_R] + \int d^D x\, \sum_I g_I^{\text{eff}} \mathcal{O}_i[\phi_R] \tag{2.214}$$

where now the couplings are g_I^{eff} which denote the *finite effective couplings* that are linearly related to the 1PI Greens functions of the well-defined field operator $\hat{\phi}_R$.

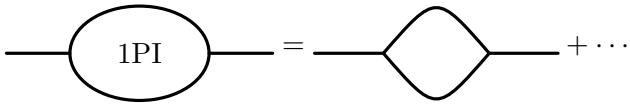
We would like g_I and therefore g_I^{eff} to be determined by a finite set of *physical couplings* that we will denote as λ_{phys} namely the bare couplings become functions of the physical couplings and the cutoff/regularization scheme dependent scale

$$g_I = g_I(\lambda_{\text{phys}}, \Lambda) \tag{2.215}$$

which may as well diverge in the limit that $\Lambda \to \infty$, but then with (2.215) we have the effective couplings to then be such that

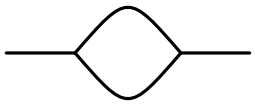
$$\lim_{\Lambda\to\infty} g_I^{\text{eff}}(\lambda_{\text{phys}}, \Lambda) = \text{finite} \tag{2.216}$$

for all I. Let us now inspect what kind of counter terms are need in the couplings $g_I(\lambda_{\text{phys}}, \Lambda)$ of which we will specialize to $D = 4$ and pick an action that has an interaction of the form $S[\phi] \supset \int g_3\phi^3$. The corrections to the propagator or the propagator's 1PI (since that is what matters) goes as



(2.217)

where the typical bubble diagram goes as

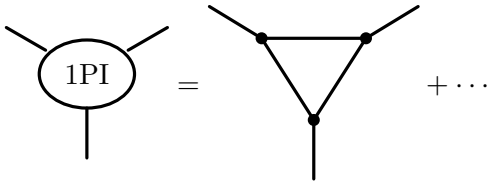


$$\sim g_3^2 \int d^4k\, \frac{1}{(k^2+m^2)^2} \sim g_3^2 \log\Lambda \tag{2.218}$$

if we consider a finite cutoff scale Λ. this divergent contribution to the mass term in the self-energy diagram are canceled by the mass counter term that we write as (just as previously)

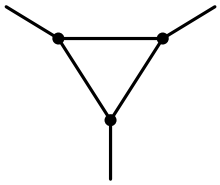
$$\rightsquigarrow \delta m^2 \sim g_3^2 \log \Lambda. \tag{2.219}$$

Moving up to 3-point diagrams (that we will soon enjoy very much [slight sarcasm]) the corresponding 1PI is



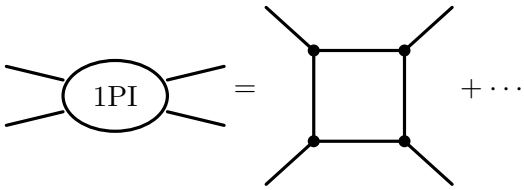
(2.220)

where the 1-loop triangle diagram typically behaves as



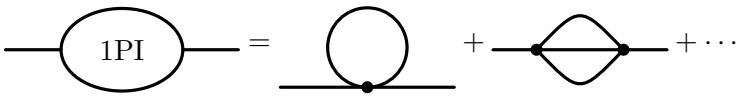
$$\sim \int d^4k \frac{1}{(k^2+m^2)^3} = \text{finite}. \tag{2.221}$$

Moving up now to the 1-loop of our favorite calculation, the 4-point function, the corresponding 1PI goes as



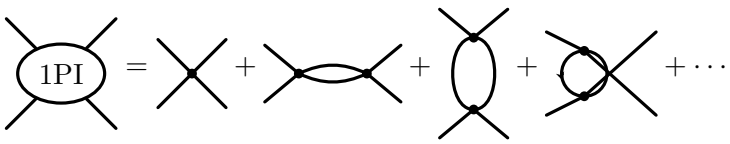
(2.222)

where the box diagram, similar to when going from the bubble to the triangle diagram, we get an extra factor of the propagator, contributing to the denominator, and thus it will be rendered finite when integrated. Therefore, we can conclude that we do *not* need any counter terms of the form $\delta g_4 \phi^4$, $\delta g_6 \phi^6$, etc. that needs to absorb divergences from these higher points diagrams for $g\phi^3$-theory. Let us move up to the other typical QFT toy model we use, $g_4\phi^4$-theory in $D = 4$. The 1PI's for the 2-, 3-, and 4-point diagrams will now change. the 2-point 1PI becomes



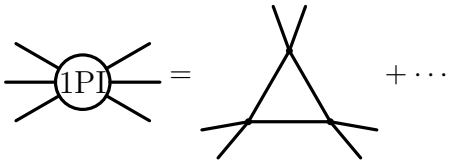
(2.223)

where the first diagram with scale as $\sim \Lambda^2$ while the second diagram (called the sunrise diagram) scales as $\sim \Lambda^2 \log \Lambda$, and thus we will need counter terms for these divergences. There is no 1PI 3-point function for ϕ^4-theory. Moving to the 4-point function we have



(2.224)

where the 1-loop corrections in the s-, t-, and u-channel all scale as $\sim \log \Lambda$. Now, let us consider a 6-point function that goes as



(2.225)

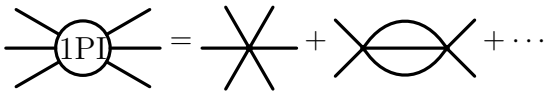
where now, we have found that at 6-points 1-loop, we find a finite loop result. A pattern you may be able to see is that when we consider higher order operators i.e. higher $n \geq 3$ on ϕ^n, we are trying to go to some n-point function at 1-loop that will eventually gives us a finite result. In our ϕ^4-theory, we can see that we not only need a mass counter term $\delta m^2\, \phi^2$ that goes as

$$\delta m^2 \sim g_\Lambda^2 + \mathcal{O}(g_4^2) \tag{2.226}$$

but also the following counter terms for the divergences that arise from the 4-point function and the *field* renormalization $\mathcal{Z}$ that will be used to absorb the logarithmic divergences in the 2-point function,

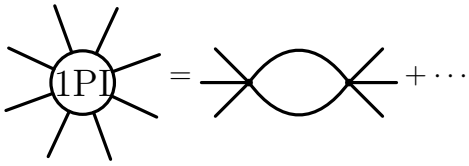
$$\begin{aligned} \delta\mathcal{Z}(\partial_\mu\phi)^2 &\Leftrightarrow \delta\mathcal{Z} \sim g_4^2 \log\Lambda \\ \delta g_4 \phi^4 &\Leftrightarrow \delta g_4 \sim g_4^2 \log\Lambda. \end{aligned} \tag{2.227}$$

We see that there are no counter terms need of the form $\delta g_6\, \phi^6$ or higher terms. But now let us go *even* higher in 1PI's such as $S[\phi] \supset \int g_6 \phi^6$. The 1PI for the 6-point and then 8-point functions are



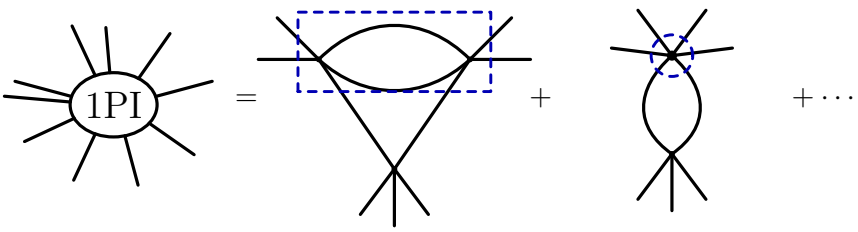
(2.228)

where the first term is just the contact term while the second term that is of loop order scales as $\sim \Lambda^2 \log\Lambda$. The 8-point function in a similar manner has a 1PI of the form



(2.229)

where the loop diagram goes as $\sim \log\Lambda$. We can see then that we need counter terms of the form $\delta g_8\, \phi^8$ of the form $\delta g_8 \sim g_6^2 \log\Lambda$. But let us keep playing this game! The 10-point 1PI function for the ϕ^6 theory at 2-loop order goes as[**AC:**] there are suppose to be 10 legs but tikz is hard



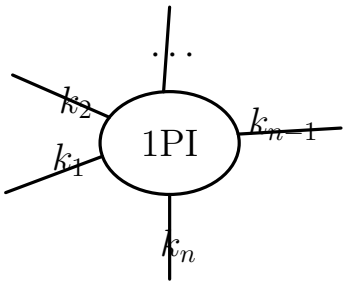
(2.230)

where these two diagrams scale as $\sim \log\Lambda$, and thus we need a counter term of the form $\delta g_{10}\,\phi^1 0$ with $\delta g_{10} \sim g_6^3 \log\Lambda$, and so forth and so forth, ad infinitum.

Superficially what is happening is that the necessary counter terms are dictated by dimensional analysis to which goes as the following (with $\hbar = c = 1$ of course). First, we know that the action is dimensionless, then with a D-dimensional measure that goes as $[d^D x] = M^{-D}$ and so if we consider the kinetic term of the Lagrangian for a scalar theory, with partial derivatives having $[\partial] = M$, then we have $M^0 = M^{-D} M^2 M^{2\Delta}$ where Δ is the unknown mass dimension of the scalar (and we have two of them). Solving for Δ gives $[\phi] = M^{\frac{D-2}{2}}$ or just $(D-2)/2$. In a similar manner, the general scalar operators $\mathcal{O}_I$ consists of n-ϕ's and ℓ number of derivatives thus will have a mass dimension of $d_I = \ell + n\frac{D-2}{2}$. The mass dimension of the couplings are $[g_I] = D - d_I$.

In correspondence to this, we can compute the effective couplings g_I^{eff} for each operator $\mathcal{O}_I \sim \phi\partial^{\ell_1}\phi\cdots\partial^{\ell_{n-1}}\phi$ via the following operation

$$g_I^{\text{eff}} \sim \partial_{k_1}^{\ell_1}\cdots\partial_{k_{n-1}}^{\ell_{n-1}}$$



$$\Big|_{k_i=0}. \tag{2.231}$$

If we assume a non-singular limit as $m \to 0$, the UV divergences in the 1PI diagram that contributes to g_I^{eff} takes the following form

$$g_I^{\text{eff}} \supset \prod_{J\in\text{1PI diagrams}} g_J \Lambda^{(D-d_I)-\sum_J (D-d_J)} \times (\log\Lambda)^{\leq \#\text{ loops}}. \tag{2.232}$$

We also need some potentially divergent counter term $\delta g_I\,\mathcal{O}_I$ in the action $\widetilde{S}[\phi]$ if the following is true

$$D - d_I - \sum_J (D - d_J) \geq 0. \tag{2.233}$$

If $d_I \leq D$ for all couplings $g_I\mathcal{O}_I$ appearing in the action $\widetilde{S}[\phi]$, then potentially the only counter terms for such couplings are needed to ensure a finite quantum effective action $\Gamma[\phi]$. This condition, if satisfied, is what we would called a *renormalizable theory.*

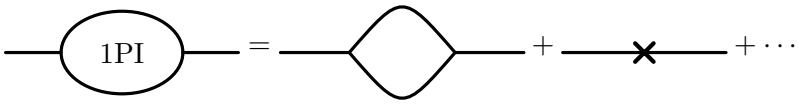
If $\widetilde{S}[\phi]$ contains *some* $g_I\mathcal{O}_I$ with $d_I > D$, then there will be *infinitely* many counter terms $\delta g_I\,\mathcal{O}_I$ needed with d_I being arbitrarily large. This then *cannot* produce a finite quantum effective action $\Gamma[\phi]$ with only finitely many terms in $\widetilde{S}[\phi]$. QFT's that obey this propery are called *non-renormalizable theories.* In other words, for these QFTs, we cannot pin down the theory entirely by measuring only a finite many number of λ_{phys}. But, this does not necessarily prove that such a QFT may not exist.

Let us understand in more detail now with an instructive model the cancellation of divergences in a *renormalizable theory* for example (since all the $D = 4$ ones are what people

talk about, let's do something completely different) $D = 6$ ϕ^3 theory. The corresponding *renormalized Lagrangian* in Euclidean signature is given as

$$\begin{aligned}\mathcal{L} =& \frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m_R^2\phi^2 + \delta\mathcal{Z}\left(\frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m_R^2\phi^2\right) + \frac{1}{2}\delta m^2\,\phi^2 \\ &+ \frac{1}{3!}g_R\phi^3 + \frac{1}{3!}\delta g\,\phi^3. \end{aligned} \tag{2.234}$$

The renormalized 1PI diagram with corresponding counter term can be computed as

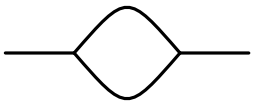


(2.235)

where the cutoff term goes as

$$= -\delta\mathcal{Z}(k^2 + m_R^2) - \delta m^2 \tag{2.236}$$

while the bubble diagram can be computed as



$$\begin{aligned} &= \frac{1}{2}g_R^2\int\frac{d^6\ell}{(2\pi)^6}\frac{1}{(\ell^2+m_R^2)((k-\ell)^2+m_R^2)} \\ &= \frac{g_R^2}{2}\int\frac{d^6\ell}{(2\pi)^6}\int_0^1 dx\,\frac{1}{(\ell^2+k^2x(1-x)+m_R^2)^2}. \end{aligned} \tag{2.237}$$

Using the cutoff regularization scheme, we would schematically have the following counter terms

$$\delta m^2 \sim g_R^2\Lambda^2 + g_R^2 m_R^2\log\Lambda \tag{2.238}$$

$$\delta\mathcal{Z} \sim g_R^2\log\Lambda \tag{2.239}$$

while in the dimensional regularization scheme with $D = 6-\epsilon$, we would normally find the following form for counter terms

$$\delta m^2 \sim g_R^2 m_R^2\frac{1}{\epsilon} \quad\text{and}\quad \delta\mathcal{Z} \sim g_R^2\frac{1}{\epsilon}. \tag{2.240}$$

It will be useful later on to inspect the explicit k-dependence of the 1-loop contribution to $\Sigma(k)$, and so let us finish this integral in D-dimensions, and then use $D = 6-\epsilon$ and expand around[68] $\epsilon \to 0$

$$\begin{aligned}\frac{g_R^2}{2}\int\frac{d^6\ell}{(2\pi)^6}\int_0^1 dx\,\frac{1}{(\ell^2+k^2x(1-x)+m_R^2)^2} &= \frac{g_R^2}{2}\frac{\Gamma\left(2-\frac{D}{2}\right)}{(4\pi)^{\frac{D}{2}}}\int_0^1 dx\,(k^2x(1-x)+m_R^2)^{\frac{D}{2}-2} \\ &= \frac{g_R^2}{2(4\pi)^2}\left(\frac{1}{\epsilon}\left(-\frac{8}{3}k^2 - 16m_R^2\right)\right. \\ &\quad\left. + \left(\frac{4}{3}k^2 + 8m_R^2\right)\log\left(k^2\right) + \#k^2 + \#m_R^2 + \cdots\right) \end{aligned} \tag{2.241}$$

[68] Note that Mathematica can handle these types of expansion of Gamma functions very nicely.

where the numbers # are just some finite constant and the $\cdots$ just denotes terms that vanish at large momentum k. To rid ourselves of the divergence, we match the ϵ^{-1} divergent pieces to the following counter terms and *choose*

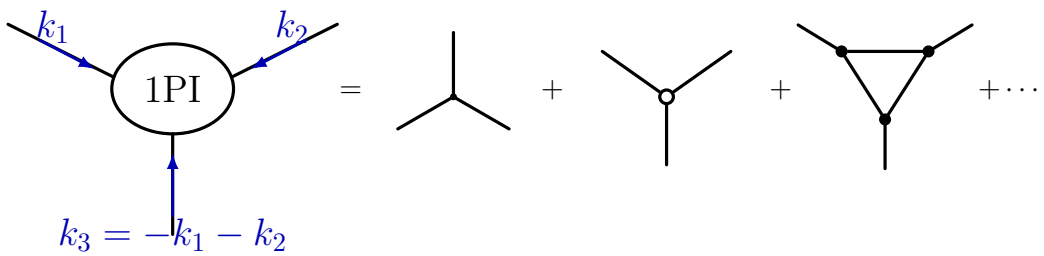
$$\delta\mathcal{Z}\Big|_{\mathcal{O}(g_R^2)} = \frac{g_R^2}{2(4\pi)^3}\left(-\frac{8}{3}\frac{1}{\epsilon} + \text{finite}\right) \tag{2.242}$$

$$\delta m_R^2\Big|_{\mathcal{O}(g_R^2)} = \frac{g_R^2 m_R^2}{2(4\pi)^3}\left(-\frac{40}{3}\frac{1}{\epsilon} + \text{finite}\right) \tag{2.243}$$

where for δm_R^2, this is evaluated on-shell where $k^2 = -m_R^2$, hence we get the factor of $-40/3$. Also, the "finite" part in the above differs depending on the choice of renormalization scheme. This choice will cancel the ϵ^{-1} terms which results in a finite $\Sigma(k)$ that behaves at large k as

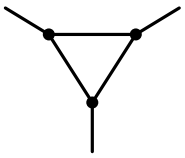
$$\Sigma(k)\Big|_{\mathcal{O}(g_R^2)} \sim (\alpha^2 k^2 + \beta)\log\left(k^2\right) \tag{2.244}$$

for when $k^2 \gg m_R^2$. Let us keep the same theory, but look at the 3-point function now which schematically goes as



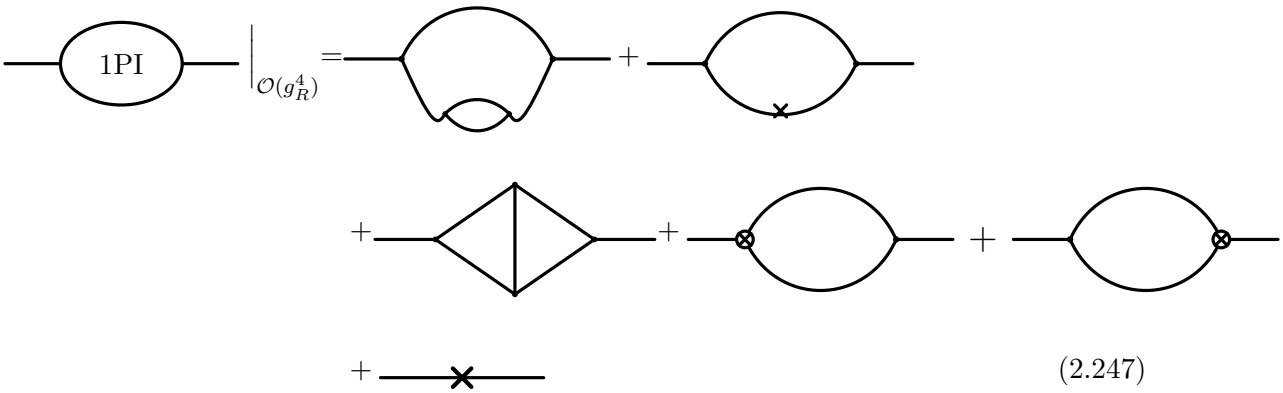
(2.245)

where this last diagram becomes



$$= -g_R^3 \int \frac{d^6\ell}{(2\pi)^6}\frac{1}{(\ell^2+m_R^2)((\ell+k_1)^2+m_R^2)((\ell+k_2)^2+m_R^2)}. \tag{2.246}$$

Here, we need a counter term of the form $\delta g \sim g_R^3 \log\Lambda$ in the cutoff reguarlization scheme or g_R^3/ϵ in dimensional regularization. But now what would happen at the next order in perturbation theory? Well, we can write the diagrams and counter terms as

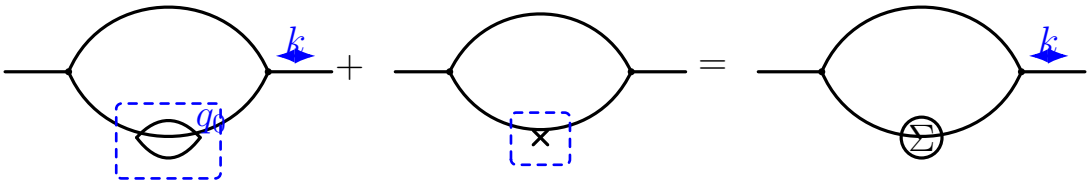


(2.247)

Here, the 2-loop counter term contribution is evaluated as

$$= -\delta\mathcal{Z}\Big|_{\mathcal{O}(g_R^4)}(k^2+m_R^2) - \delta m^2\Big|_{\mathcal{O}(g_R^4)}. \tag{2.248}$$

The 2-loop contribution and the corresponding counter term in (2.247) are both part of $G(q) = (q^2 + m_R^2 - \Sigma(q))^{-1}$ where q is the loop momentum of the initial bubble loop, and so we can instead write the sum of these two diagrams as a 1-loop 1PI itself of loop momentum q[69]



(2.249)

which can be written as

$$\begin{aligned}&\int \frac{d^6q}{(2\pi)^6}\frac{\Sigma(q)}{(q^2+m_R^2)((k-q)^2+m_R)^2}\\ &= \int \frac{d^6}{(2\pi)^6}\frac{\Sigma(q)}{(q^2+m_R^2)^2}\left(\frac{1}{(k-q)^2+m_R^2} - \frac{1}{q^2+m_R^2} - \frac{2k\cdot q - k^2}{(q^2+m_R^2)^2} - \frac{4(k\cdot q)^2}{(q^2+m_R^2)^3}\right)\end{aligned} \tag{2.250}$$

which by power counting we expect it to be a finite result! (Note that we used partial fractions here). Now let us verify this power counting explicitly using dimensional regularization. Using the Feynman trick, we can write (2.250) as

$$\int \frac{d^6q}{(2\pi)^6}\frac{\Sigma(q)}{(q^2+m_R^2)((k-q)^2+m_R)^2} = \int \frac{d^Dq}{(2\pi)^D}\int_0^1 dx \frac{2(1-x)\Sigma(q+kx)}{(q62+k^2x(1-x)+m_R^2)^3} \tag{2.251}$$

where notice now, we are forced to have the self-energy of the 1-loop result be dependent upon the Feynman parameterization in x. The divergent part of this q-integral that involves $\Sigma(q) \sim (\alpha q^2 + \beta)\log(q^2/M^2)$, can be evaluated/performed using the following identity

$$\begin{aligned}\int \frac{d^Dq}{(2\pi)^D}\frac{1}{(q^2+\Delta)^n} &= \frac{1}{(4\pi)^{\frac{D}{2}}}\frac{\Gamma\left(n-\frac{D}{2}\right)}{\Gamma(n)}\Delta^{\frac{D}{2}-n}\\ &\to \frac{\#}{\epsilon}\Delta^{3-n} + \text{finite}\end{aligned} \tag{2.252}$$

where in the last line we expanded around $D = 6 - \epsilon$ where we find a true result for $n = 2$ and $n = 3$. We can also use the following identity as well which takes a little more work

[69] [**AC:**] The first diagram should not have a line crossing through the bottom loop contribution, still need to fix.

to show and algebraically massage,

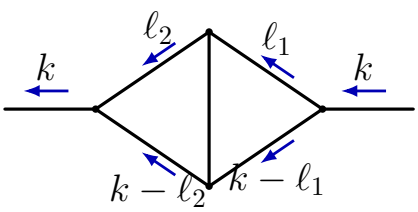
$$\begin{aligned}\int \frac{d^D q}{(2\pi)^D} \frac{\log(q^2+\Delta)}{(q^2+\Delta)^n} &= -\frac{1}{(4\pi)^{\frac{D}{2}}} \frac{\partial}{\partial n} \left(\frac{\Gamma\left(n-\frac{D}{2}\right)}{\Gamma(n)} \Delta^{\frac{D}{2}-n} \right) \\ &\sim \frac{\partial}{\partial \epsilon} \left(\frac{\#}{3} + \# \log \Delta \right) \Delta^{3-n} + \frac{\#}{3} \Delta^{3-n} + \text{finite} \\ &\sim \left(\frac{\#}{\epsilon^2} + \frac{\#}{\epsilon} \right) \Delta^{3-n} + \text{finite} \end{aligned} \tag{2.253}$$

where, despite we have divergences, the important note is that we have *no* divergences of the form[70] $\epsilon^{-1} \log \Delta$! These two integral identities leads to the following result for (2.250) in the expansion where $D = 6 - \epsilon$

$$\int \frac{d^D q}{(2\pi)^D} \frac{1}{(q^2+\Delta)^n} = A_\epsilon k^2 + B_\epsilon + \text{finite} \tag{2.254}$$

where $A_\epsilon = \alpha_1/\epsilon + \alpha_2/\epsilon_2$ and $B_\epsilon = \beta_1/\epsilon + \beta_2/\epsilon$ where α_1, α_2, β_1, and β_2 are some finite k-dependent terms. These divergences can be canceled by including in $\delta\mathcal{Z}$ and δm^2 counter terms of order g_R^4/ϵ^2. Note that *if* we had a divergence of the form $\sim \epsilon^{-1} \log(k^2)$, we would be in deep trouble as there is no counter term that can correspond to a local Lagrangian, that could cancel such a divergence. Luckily (?![71]?!) this did not occur for us.

Next, let us inspect the UV divergence of the following diagram for our ϕ^3 $D = 6$ theory



$$= \frac{g_R^2}{2} \int \frac{d^6\ell_1}{(2\pi)^6} \frac{d^6\ell_2}{(2\pi)^6} P(\ell_1)P(k-\ell_1)P(\ell_2)P(k-\ell_2)P(\ell_1-\ell_2) \tag{2.255}$$

where I have introduced a new way (that is very convenient) to write the propagators as some function $P(\ell)$ specifically

$$P(\ell) \equiv \frac{1}{k^2+m_R^2} = \int_0^\infty d\alpha \, e^{-\alpha(k^2+m_R^2)}. \tag{2.256}$$

This new way to write the propagator is often called *Schwinger parameterization* which we shall use now, and will allow us to evaluate the integrals quite nicely and isolate their UV

[70] Something that often is not discussed but should be is that if we have divergences of ϵ^{-1} times some number, it really does not matter and is fine for most purposes since ϵ is a constant, and if we want the potential from an amplitude for example, then the Fourier transform of a constant just vanishes. However, if we have a logarithm times a divergent ϵ, then this divergence is actually non-local, and thus must be renormalized immediately.

[71] This kind of behavior happens in gravity however and can lead to a quantum effective action that is non-local.

behavior more directly (albeit if it may seem odd to do so). The 2-loop contribution in (2.255) can then be written as

$$\int_0^\infty d\alpha_1 \cdots d\alpha_5 \, e^{-m_R^2 \sum_{i=1}^5 \alpha_i} \int \frac{d^6\ell_1}{(2\pi)^6} \frac{d^6\ell_2}{(2\pi)^6} e^{-\alpha_1 \ell_2^2 - \alpha_2 (k-\ell_2)^2 - \alpha_3 \ell_2^2 - \alpha_4 (k-\ell_2)^2 - \alpha_5 (\ell_1 - \ell_2)^2}$$
$$= \frac{1}{(4\pi)^6} \int_0^\infty d\alpha_1 \cdots d\alpha_5 \frac{e^{-m_R^2 \sum \alpha_i - k^2 Q}}{(\det A)^3} \tag{2.257}$$

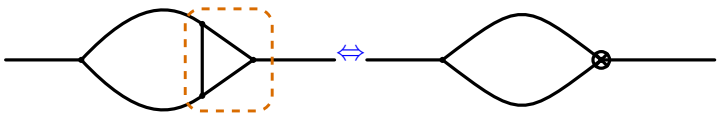
where we can write the argument of the exponential as a matrix space over the loop-momentum vector space $(\ell_1, \ell_2)^T$ which makes the matrix A take the following form

$$A = \begin{pmatrix} \alpha_1 + \alpha_2 + \alpha_5 & -\alpha_5 \\ -\alpha_5 & \alpha_3 + \alpha_4 + \alpha_5 \end{pmatrix} \tag{2.258}$$

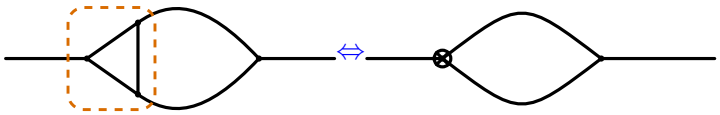
while the quantity Q is left over from completing the square

$$Q = \alpha_2 + \alpha_4 + (\alpha_2, \alpha_4) A^{-1} \begin{pmatrix} \alpha_2 \\ \alpha_4 \end{pmatrix} \tag{2.259}$$

where Q obeys $Q \geq 0$. The UV divergence in this calculation now comes from the term where $\det A \to 0$. The generic UV divergence of this form occurs in the following limits, either

$\alpha_1,\ \alpha_2,\ \alpha_5 \to 0$  (2.260)

where the two diagrams will be canceling against each other while in the other limit we have

$\alpha_3,\ \alpha_4,\ \alpha_5 \to 0$  (2.261)

where again, the counter term diagram and the boxed-loop divergence will cancel against each other. Consider re-writing (2.257) more generally as some function $\mathcal{F}$

$$\frac{1}{(4\pi)^6} \int_0^\infty d\alpha_1 \cdots d\alpha_5 \frac{e^{-m_R^2 \sum \alpha_i - k^2 Q}}{(\det A)^3} = \frac{1}{(4\pi)^6} \left[\prod_{i=1}^5 d\alpha_i \, e^{-m_R^2 \alpha_i} \right] \mathcal{F}(k^2; \alpha_1, \cdots, \alpha_5) \tag{2.262}$$

where we define the function $\mathcal{F}$ as

$$\mathcal{F} = \frac{e^{-k^2 Q}}{(\det A)^3}. \tag{2.263}$$

Notice that the function $\mathcal{F}$ is invariant under the following total rescaling of some scalar quantity ρ

$$\rho^6 \mathcal{F}(k^2/\rho^2; \rho\alpha_1, \cdots, \rho\alpha_5) = \mathcal{F}(k^2; \alpha_1, \cdots, \alpha_5). \tag{2.264}$$

Now, note that the effects of adding and amounts to replacing the function $\mathcal{F}$ with instead

$$\widetilde{\mathcal{F}} := \mathcal{F} - \lim_{\rho\to\infty} \rho^3 \mathcal{F}\left(\frac{k^2}{\rho^2}; \rho\alpha_1, \rho\alpha_2, \alpha_3, \alpha_4, \alpha_5\right) - \lim_{\rho\to\infty} \rho^3 \mathcal{F}\left(\frac{k^2}{\rho^2}; \alpha_1, \alpha_2, \rho\alpha_3, \rho\alpha_4, \alpha_5\right). \tag{2.265}$$

This rescaling of the Schwinger parameters by some constant ρ can be traced back to how the counter terms before the Schwinger parameterization affects the propagators in the sense that from (2.256), a shift in the propagators self-energy contribution when "un-integrating" in a sense to write it in terms of the Schwinger parameterization, the constants can be re-scaled and absorbed next to the variable α_i for each propagator. The re-defining of $\mathcal{F} \to \widetilde{\mathcal{F}}$ then amounts to subtracting off the divergences. The remaining divergences then in $\int \widetilde{\mathcal{F}}$ will come from the taking the limit of the Schwinger parameters simultaneously, $\alpha_1, \cdots, \alpha_5 \to 0$, and depends on k at most quadratically by power counting.

After canceling the divergence by adjusting the renormalization parameters $\delta\mathcal{Z}$ and δm^2, we are left with the following result

$$\int \left[\prod_i^5 d\alpha_i\, e^{-m_R^2\alpha_i}\right] \left(\widetilde{\mathcal{F}} - \widetilde{\mathcal{F}}|_{k^2=0} - \frac{\partial \widetilde{\mathcal{F}}}{\partial k^2}\bigg|_{k^2=0} \cdot k^2\right) \equiv \int \left[\prod_i^5 d\alpha_i\, e^{-m_R^2\alpha_i}\right] \mathcal{F}^R \tag{2.266}$$

where one can verify that the integral over $\mathcal{F}^R$ is finite. This analysis can be extended to show that *all* divergences in 1PI diagrams can be canceled by diagrams involving counter terms that correspond to nested subdiagrams, thereby proving perturbative renormalizability of naive-power-counting-renormalizable theories. The step we just took though is a a small portion of this which was proving something called the *Bogoliubov-Parasiuk-Hepp-Zimmermann theorem* or of course, the BPHZ theorem for short[72].

This endevour that we just went through though is *old* however, and still involves what would be called *naive-* or *old*-perturbation theory. This is becomes the condition of renormalizability relied on the total summing of diagrammatics in combination of power counting the number of loop momentum, to determine renormalizability. But, there are border cases of whether a theory is renormalizable or not, that actually this power counting method would fail to rule in or out such as in cases when there is a great deal of symmetries such as gauge symmetries that will actually restrict the form of the counter terms. Also, something we will see is that renormalizability is actually the condition that we perturb around a UV-finite fixed point, to which with this method of perturbation theory and renormalizability, can miss neighboring theories that would actually obey this.

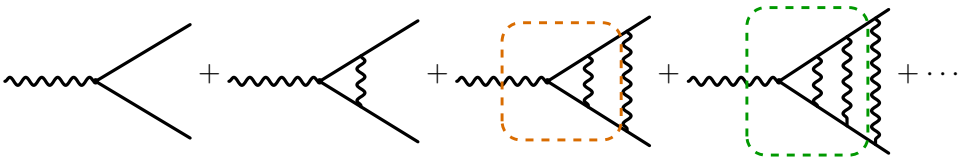
All of this talk essentially amounts to we will *finally* be looking into the start of the *renormalization group* (not exactly the *flow* part yet).

[72]The theorem is actually much easier to prove with new technology that we will look at extensively called *Wilson-Polchinski renormalization group flow* that we will see shorty.

2.2.2 The Renormalization Group

The *renormalization group* (RG) is a way of characterizing a QFT by *effective* descriptions at different energy/momentum scales, and a computational scheme for relating such effective descriptions at different scales.

A particular renormalization group we will look at is called the *1PI RG* which has a much more widely known name called the *Gell-Mann-Low* equation or the *Callan-Symanzik* equation[73]. The 1PI renormalization group equation is an improved version of perturbation theory in which we *recycle* computations of loop corrections in sub-diagrams, and perform a partial re-summation. For example, in the triangle diagram, we can have a loop expansion of the form



(2.267)

where we can see that the orange box is really just the first loop diagram, and then the green box is really just the nested loop diagram that came before it and so on and so on. This nesting and new perturbation style is accomplished by defining energy-scale-dependent effective/*renormalized* coupling $g(\mu)$ where μ is our new energy-scale, and studying how $g(\mu)$ varies with μ.

After the discovery of the Gell-Mann-Low equation (and others), a more advanced formulation of such a flow was invented/discovered called the *Wilsonian renormalization group* or namely the *Wilson-Polchinski* equation which allows us to define a quantity called the *Wilsonian effective action* denoted $S_\Lambda[\phi]$ by performing the functional integration over modes $\widetilde{\phi}(k)$ for $\Lambda < |k| < \Lambda_0$ where Λ is some finite energy scale while Λ_0 is the UV cutoff. This particular formulation allows us to study and see how the *action* $S_\Lambda[\phi]$ changes with Λ. This formulation also allows a direct path the to *Wilson-Polchinski flow* or in general a *renormalization group flow* which will be out topic *after* discussing the Gell-Mann-Low equation.

Let us begin (you guessed it) with a particular QFT example namely $D = 4$ ϕ^4-theory where the *bare* Euclidean Lagrangian is given as

$$\mathcal{L}_E = \frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2}m_0^2\phi^2 + \frac{1}{4!}g_0\phi^4 \tag{2.268}$$

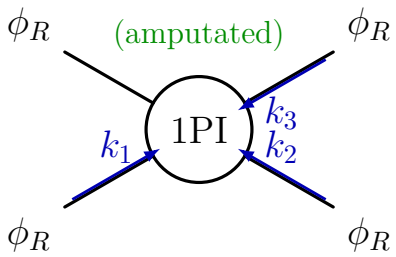
or equivalently we can write (2.268) in terms of the renormalziation field and mass (1.276) with $\phi_R = \mathcal{Z}_R^{-\frac{1}{2}}\phi$ as

$$\begin{aligned}\mathcal{L} =& \frac{1}{2}(\partial_\mu \phi_R)^2 + \frac{1}{2}m_R^2\phi_R^2 + (\mathcal{Z}_R - 1)\left(\frac{1}{2}(\partial_\mu \phi_R)^2 + \frac{1}{2}m_R^2\phi_R^2\right) \\ &+ \frac{1}{2}\delta m^2\,\phi_R^2 + \frac{1}{4!}g_R\phi_R^4 + \frac{1}{4!}\delta g\,\phi_R^4.\end{aligned} \tag{2.269}$$

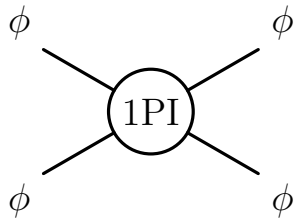
[73]There is also a version of the 1PI RG equation called t'Hooft-Weinberg RG, but that has different applicability since Weinberg creates a momentum dependent mass which can be useful in cured spacetimes.

Let us consider the 4-point effective 1PI 2.188 which we can write as

$$-\Gamma^{(4)}(k_1, k_2, k_3) =$$



$$= \left(\mathcal{Z}_R^{\frac{1}{2}}\right)^4 \times$$

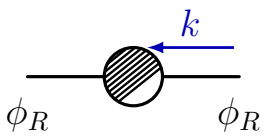


$$. \tag{2.270}$$

We would *define* the renormalized coupling as that taken from the effective 1PI 4-point function at zero momentum as

$$g_R = -\Gamma^{(4)}(k_1 = k_= k_3 = 0), \tag{2.271}$$

but this is just an arbitrary choice. For the mass scale μ, we could expect that the 4-point 1PI, $\Gamma^{(4)}\Big|_{k_i \sim \mathcal{O}(\mu)}$ for $i = 1, 2, 3, 4$ to characterize the effective coupling strength at scale μ. But this is not accurate as the normalization of the renormalized field ϕ_R is still arbitrary from when we look at the 1PI



$$= \frac{1}{k^2 + m_R^2 - \Sigma_R(k)} \tag{2.272}$$

Let us rescale the renormalized field ϕ_R by

$$\mathcal{Z}(\mu) \equiv \left(1 - \frac{\partial \Sigma_R}{\partial k^2}\right)^{-1}\Bigg|_{k^2=\mu^2} \tag{2.273}$$

(which is again in the Euclidean signature) such that the 1PI effective action $\Gamma[\phi_R]$ can be viewed as a *functional* of the renormalized field that is now *scale-dependent*

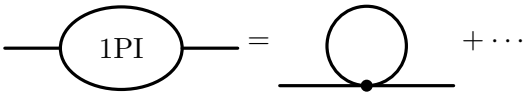
$$[\phi]_\mu \equiv (\mathcal{Z}(\mu))^{-\frac{1}{2}}\, \phi_R \tag{2.274}$$

which has a canonically normalized kinetic term at momentum $k \sim \mathcal{O}(\mu)$. With this, let us now *define* a scale-dependent *running coupling* as

$$g(\mu) = -\left(\mathcal{Z}(\mu)^{\frac{1}{2}}\right)^4 \Gamma^{(4)}(k_1, k_2, k_3)\Big|_{k_i^2=\mu^2; s_i=-\frac{4}{3}\mu^2} \tag{2.275}$$

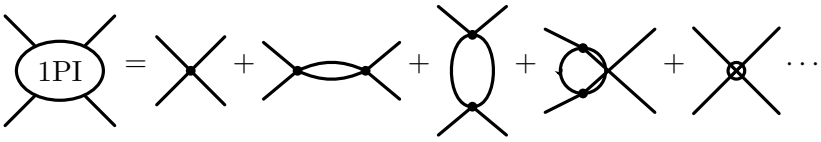
where the evaluation condition $s_i = -(4/3)\mu^2$ is shorthand for setting $s = t = u = -\frac{4}{3}\mu^2$. This particular choice is a *specific renormalization scheme*.

As a concrete example, let us consider the 1-loop example of the self-energy 1PI in ϕ^4 theory still (the action is (2.269))



(2.276)

where does not affect $\mathcal{Z}_R$ or $\mathcal{Z}(\mu)$ (but the 2-loop contributions will, which are in the problem sets). At 1-loop, $\mathcal{Z}_R = \mathcal{Z}(\mu) = 1$. Well ok, moving onto the 4-point 1PI we have



(2.277)

where the contact term contributes as $-g_R$, the s-, t-, and u-channel loop diagrams can all have loop momentum ℓ which we have computed many times, and the counter term contributes as δg. The running coupling, which again is defined from the effective action (2.275) where the effective action is just the renormalized 1PI (2.188), to find

$$-g(\mu) = -g_R - \delta g + 3 \cdot \frac{g_R^2}{32\pi^2} \int_0^1 dx \left[\log \frac{\Lambda^2}{m_R^2 - sx(1-x)} - 1 \right] \Big|_{s=-\frac{4}{3}\mu^2} + \mathcal{O}(g_R^3) \tag{2.278}$$

where the s-, t-, and u-channels all contribute equally, hence the factor of three in front. By our convention, the renormalized coupling is set to when the energy scale $\mu = 0$, $g_R = g(\mu = 0)$, and thus this fixes δg since

$$-g(0) = -g_R - \delta g + \frac{3g_R^2}{32\pi^2} \int_0^1 dx \left[\log \frac{\Lambda^2}{m_R^2} - 1 \right] \tag{2.279}$$

from which we can read off what δg must be in order to satisfy $g_R = g(\mu = 0)$. Our renormalized running coupling is then

$$g(\mu) = g_R - \frac{3g_R^2}{32\pi^2} \int_0^1 dx \, \log \frac{m_R^2}{m_R^2 + \frac{4}{3}\mu^2 x(1-x)} + \mathcal{O}(g_R^3). \tag{2.280}$$

This result now captures the effect that the 1-loop contributions will have on the coupling constant at some energy scale μ. We have thus obtained an unambiguous relation among the physical couplings! However, the 1-loop correction becomes large when $\mu \gg m_R$. One could of course compute more terms in perturbation theory, but they will also grow as $\mu \gg m_R$. Instead, let us compare the running coupling we just found $g(\mu)$ to the same running coupling but at a slightly different scale $g(\mu')$. When can perform this comparison by setting the renormalization g_R in $g(\mu')$ to be at $g(\mu)$ and then expanding to $g(\mu)$ powers instead we get

$$g(\mu') = g(\mu) - \frac{3}{32\pi^2} g(\mu)^2 \int_0^1 dx \, \log \frac{m_R^2 + \frac{4}{3}\mu^2 x(1-x)}{m_R^2 + \frac{4}{3}\mu'^2 x(1-x)} + \mathcal{O}((g(\mu))^3) \tag{2.281}$$

where the logarithm is for small $\mu' \sim \mu$, but notice that the perturbative expansion now is in terms of $g(\mu)$ *not* g_R which is much more reliable when μ' is close to μ!

2.2.3 RG: One of the most beautiful results ever

To get the most out of this relation, we can take $\mu' = \mu + d\mu$ which will allow us to write (2.281) as[74]

$$\frac{dg(\mu)}{d\log\mu} = \frac{3}{16\pi^2}(g(\mu))^2 \int_0^1 dx \, \frac{\frac{4}{3}\mu^2 x(1-x)}{m^2 + \frac{4}{3}\mu^2 x(1-x)} + \mathcal{O}((g(\mu))^3). \tag{2.282}$$

If we have $\mu \gg m$, then we can write the derivative as

$$\frac{dg(\mu)}{d\log\mu} = \frac{3}{16\pi^2}(g(\mu))^2 + \mathcal{O}((g(\mu))^3) \tag{2.283}$$

where (2.283) is called the *β-function* and is our first example of it. We can solve (2.283) as a differential equation at the two different μ-scale which amounts to solving the 1-loop renormalization group equation as

$$\frac{1}{g(\mu)} - \frac{1}{g(\mu')} = -\frac{3}{16\pi^2}\log\frac{\mu}{\mu'} \tag{2.284}$$

which implies that the running coupling changes as

$$g(\mu) = \frac{g(\mu')}{1 - \frac{3}{16\pi^2} g(\mu')\log\frac{\mu}{\mu'}}. \tag{2.285}$$

Equation (2.285) is such an achievement that it bares some time to be in awe of...

[74] Something I always need to keep on hand is that $d\ln x = \frac{dx}{x}$ for future use.

Anyway, (2.285) is a good approximation provided that $g(\mu'')$ is small for all $\mu < \mu'' < \mu'$ *even* if $g(\mu')\log(\mu/\mu')$ is large! (2.285) contains *all* order contributions in $g(\mu')$ or gR even though we only performed a 1-loop calculation! At the following scale

$$\mu_0 = \mu' e^{\frac{16\pi^2}{3}\frac{1}{g(\mu')}}, \tag{2.286}$$

the 1-loop renormalization group equation predicts that $g(\mu_0) = \infty$ with a corresponding graph.

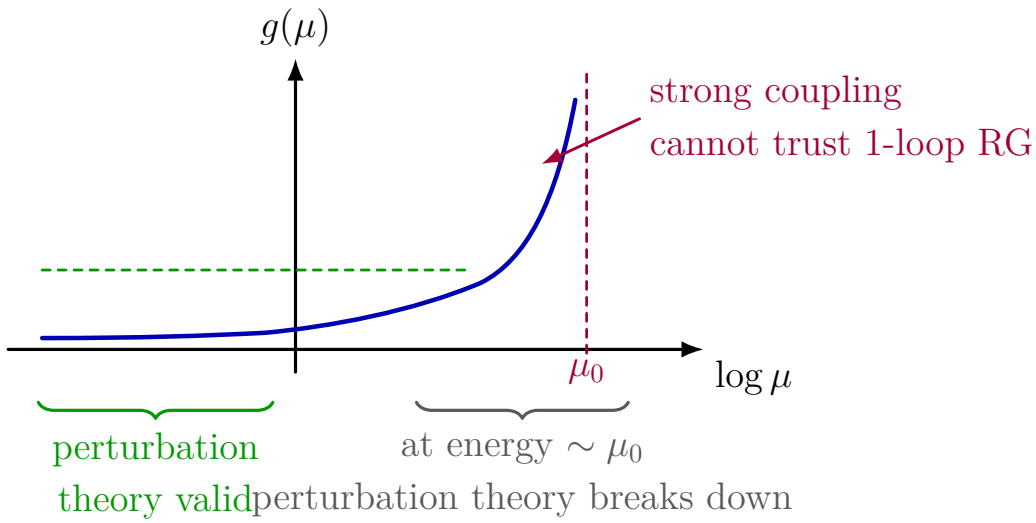


Figure 22. Graph of the running coupling for $g(\mu)\phi^4$ theory at 1-loop.

In this graph, the 1PI renormalized group equation shows that when the running coupling is at low (logarithimic) energy scales, then perturbation theory can be trusted and we can carry on with our calculations while any divergences tells us the energy scale at which we cannot trust perturbation theory anymore, and it breaks down at strong coupling.

We can equivalently write (2.285) given the energy scale μ_0 as

$$g(\mu) = \left(\frac{3}{16\pi^2}\log\frac{\mu_0}{\mu}\right)^{-1}. \tag{2.287}$$

So, instead of having a *coupling constant* that we are so used to namely $g\phi^4$, the $D = 4$ ϕ^4-theory is now characterized by amass scale μ_0. This process is what is known as *dimensional transmutation.*

The $D = 4$ ϕ^4-theory is weakly coupled at energy $E \ll \mu_0$ but strongly coupled at $E \sim \mu_0$. Perturbation theory is really an expansion in powers of $(\log\frac{\mu_0}{E})^{-1}$.

Experimentally here, the idea is to measure μ' and then you get μ_0 for free, or you can pick the energy scale μ' and then see at what point μ_0 your theory breaks down.

As we lose perturbative control for $E \gtrsim \mu_0$, it is unclear, at least based on our currently available definition of ϕ^4 theory, that the $D = 4$ version of ϕ^4 even *exists*as a QFT that obeys locality and/or microcausality at arbitrarily short distance scales, or even a well-defined QM system that admits exact Poincaré symmetry.

Let us now move onto formulating the 1PI renormalization group for some generic theory/Lagrangian. We can define a QFT via the (Euclidean) path integral

$$Z = \int [D\phi]\, e^{-\int d^D x Ł\mathcal{L}} \tag{2.288}$$

where the Euclidean Lagrangian can be written generically as the kinetic term plus higher order operators

$$\mathcal{L}_E = \mathcal{L}_{\text{kin}} + \sum_I g_I \mathcal{O}_I \tag{2.289}$$

where g_I are the bare couplings that contains counter terms including those due to field renormalization factors $\mathcal{Z}_R$. From the Euclidean Lagrangian, we know we can derive the 1PI (quantum) effective action perturbatively as (2.188)

$$\Gamma[\phi] = S_{\text{kin}} + \sum_n \frac{1}{n!} \int \Gamma^{(n)}(k_1, \cdots, k_{n-1}) \widetilde{\phi}(k_1) \cdots \widetilde{\phi}(k_n). \tag{2.290}$$

For operators $\mathcal{O}_I$ of the schematic form $\phi \partial^{\ell_1} x \phi \cdots \partial_x^{\ell_{n-1}} \phi$ (where there are n-ϕ's with the indices contracting in some way), we can write down an appropriate definition of the renormalized coupling $g_I(\mu)$ via through expanding $\Gamma^{(n)}(k_1, \cdots, k_{n-1})$ in k_i's around some momentum of order $k_i \sim \mathcal{O}(\mu)$ as

$$\Gamma^{(n)}(k_1 + \delta k_1, \cdots, k_{n-1} + \delta k_{n-1}) = \sum \Gamma^{(n;\ell_1,\cdots,\ell_{n-1})}(k_1, \cdots, k_{n-1}) \times \delta k_1^{\ell_1} \cdots \delta k_{n-1}^{\ell_{n-1}} \tag{2.291}$$

where on the momentum, the indices are contracted in some particular way. We then define the running coupling for I-operator as

$$\begin{aligned} g_I(\mu) &:= -\left[\mathcal{Z}(\mu)\right]^{\frac{n}{2}} \Gamma^{(n;\ell_1,\cdots,\ell_{n-1})} \Big|_{k_i \sim \mathcal{O}(\mu)} \\ &= g_I + \text{corrections in “naive/unimproved” perturbation theory”} \end{aligned} \tag{2.292}$$

where the scale $k_i \sim \mathcal{O}(\mu)$ is dependent on the choice of renormalization scheme (and the last line is more of a reminder) and we also have that the renormalization “constant” is written as

$$\mathcal{Z}(\mu) = \left(1 - \frac{\partial \Sigma}{\partial k^2}\right)^{-1} \Big|_{k^2 = \mu^2}. \tag{2.293}$$

Note that in the special case of the operator $I = (2,2)$ with form $g_{2,2}\phi\Box\phi$, where the quantum effective action becomes

$$\Gamma^{(2,2)} \Big|_{k^2=\mu^2} = \frac{\partial \Sigma}{\partial k^2} \Big|_{k^2=\mu^2} = 1 - (\mathcal{Z}(\mu))^{-1}, \tag{2.294}$$

the renormalized coupling is then given as

$$g_{2,2}(\mu) = -\mathcal{Z}(\mu)(1 - (\mathcal{Z}(\mu))^{-1}) = 1 - \mathcal{Z}(\mu). \tag{2.295}$$

Now, let us express the running coupling as $g_I(\mu')$ in terms of $g_I(\mu)$ for $\mu' = \mu + \delta\mu$. This is well-define in the limit of removing the UV regulator (either $\Lambda \to \infty$ or $\epsilon \to 0$) i.e.

free of UV divergences. There are also small logarithmic corrections for small $\delta\mu$. This then in turn, as with our example explored above, will give the β-function for the 1PI renormalization group equation of the form

$$\frac{dG_I(\mu)}{d\log\mu} = \beta_I(\{g_I(\mu)\};\mu). \tag{2.296}$$

Let us now define the dimensionless renormalized running couplings as

$$\lambda_I(\mu) \equiv \mu^{d_I - D} g_I(\mu). \tag{2.297}$$

For example, we can take $m_0^2 \equiv g_2^{\text{bare}}$ and then $g_2(\mu)$ can be written as

$$g_2(\mu) = \mu^{-2}\mathcal{Z}(\mu)\left(g_2^{\text{bare}} - \Sigma\Big|_{k^2=\mu^2}\right). \tag{2.298}$$

Then, we can rewrite the β-function in (2.296) in terms of the dimensionless running coupling as

$$\frac{d\lambda_I(\mu)}{d\log\mu} = \beta_I(\{\lambda_J(\mu)\}) \tag{2.299}$$

where there is no no further explicit μ-dependence in β_I. The job of the 1PI renormalization group is to improve perturbation theory in the following manner. In comparing physical observables at widely separated energy scales, the naive perturbation theory may break down due to "large logarithms" while the renormalization group-improved perturbation theory i.e. expansion in $g(\mu)$, the latter solved from renormalization group equation, may still be valid.

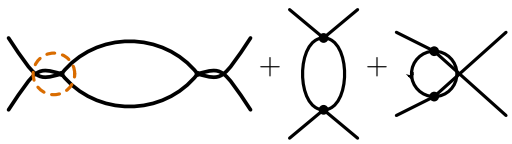
A few remarks are in order about (2.299). The first remark is that in formulating the renormalization group equation, we have assumed renormalizability in relating $g(\mu')$ to $g(\mu)$. Compare this to the "naive" perturbation theory in terms of bare coupling g_0 where we may have a generic form of

$$g(\mu) = g_0 - b_1 g_0^2 \log\frac{\Lambda}{\mu} - b_2 g_0^3 \log\frac{\Lambda}{\mu} - c_1 g_0^3\left(\log\frac{\Lambda}{\mu}\right)^2 + \mathcal{O}(g_0^4) \tag{2.300}$$

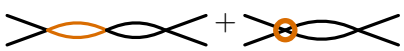
where the constants b_i and c_i are generic coefficients multiplied by the divergent terms. The corresponding β-equation is then computed as

$$\begin{aligned}\frac{dg(\mu)}{d\log\mu} &= b_1 g_0^2 + b_2 g_0^3 + 2c_1 g_0^3 \log\frac{\Lambda}{\mu} + \mathcal{O}(g_0^4)\\ &= b_1 g(\mu)^2 + b_2 g(\mu)^3 + 2(c_1 + b_1^2) g(\mu)^3 \log\frac{\Lambda}{\mu} + \mathcal{O}(g(\mu)^4)\end{aligned} \tag{2.301}$$

where the orange terms are ones that are conflicting with our renormalizability condition. In order to consistency with renormalizability requires us to set $c_1 + b_1^2 = 0 \implies c_1 = -b_1^2$. For example, in the $D = 4$ ϕ^4-theory we look at previously,



$$\sim b_1 g_0^2 \log\frac{\Lambda}{\mu} \tag{2.302}$$



$$+\text{ perms} \sim c_1 g_0^3\left(\log\frac{\Lambda}{\mu}\right)^2 + \text{subleading single log} \tag{2.303}$$

where we can see the first order loop term at the vertex in orange matches with the orange loop at 2-loop order and the same 1-loop in the second 2-loop order diagram ([**AC:**] That is supposed to be a loop, will fix later). From which, we can read off that $c_1 = -b_1^2$. Note that the 2-loop self-energy diagram (the "sunrise" diagram) only contributes a single logarithm to $\delta\mathcal{Z}$ in this example.

The second remark I would like to make is that in a different renormalization scheme coupling to $\widetilde{\lambda}(\mu)$ may be related to $\lambda(\mu)$ by a finite redefinition namely

$$\widetilde{\lambda}(\mu) = f(\lambda(\mu)) \tag{2.304}$$

where $f(\lambda)$ can be of the form $f(\lambda) = \lambda + \alpha\lambda^2 + \cdots$. Then the correspond β-function after this redefinition is

$$\frac{d\lambda(\mu)}{d\log\mu} = \beta(\lambda(\mu)) \Leftrightarrow \frac{d\widetilde{\lambda}(\mu)}{d\log\mu} = \widetilde{\beta}(\widetilde{\lambda}(\mu)) \tag{2.305}$$

of which we can relate the normal β-function to the redefinition one as

$$\begin{aligned}\beta(\lambda) &= \frac{\widetilde{\beta}(f(\lambda))}{f'(\lambda)} = (1 - 2\alpha\lambda + \cdots)\left(1 + \alpha\lambda^2\frac{\partial}{\partial\lambda} + \cdots\right)\widetilde{\beta}(\lambda) \\ &= \widetilde{\beta}(\lambda) + \alpha\lambda\left(\lambda\frac{\partial}{\partial\lambda} - 2\right)\widetilde{\beta}(\lambda) + \cdots .\end{aligned} \tag{2.306}$$

For a classically marginal coupling that is typically of them form

$$\beta(\lambda) = b_1\lambda^2 + b_2\lambda 62 + b_3\lambda^4 + \cdots \tag{2.307}$$

the redefinition of the β-function then takes the form

$$\widetilde{\beta}(\lambda) = \widetilde{b}_1\lambda^2 + \widetilde{b}_2\lambda^3 + \widetilde{b}_3\lambda^4 + \cdots \tag{2.308}$$

where $b_1 = \widetilde{b}_1$, $b_2 = \widetilde{b}_2$, but $b_3 \neq \widetilde{b}_3$ since $\widetilde{b}_3 = b_3 - \alpha b_2$.

2.2.4 Renormalized Operators

Let us now investigate the notion of *renormalized operators* in the context of the 1PI renormalization group. Consider an infinitesimal deformation of the action $S \to S + \delta S$ where the infinitesimally *deformtion* of action δS is of the form[75]

$$\delta S = \int d^D x \sum_I \delta g_I \cdot \mathcal{O}_I \tag{2.309}$$

in which corresponds to deforming the renormalized couplings $g_I(\mu)$ as

$$g_I(\mu) \rightsquigarrow g_I(\mu) + \delta g_I(\mu). \tag{2.310}$$

[75] I emphasize the wording of *deformed action* since when we get to the renormalization group flow, we will see that the inclusion of higher order operators in the action can be viewed as deforming the original kinetic action in the space of couplings and energies. With each deformation, we get a new action then based on the renormalization group flow formulation.

δg_I and $\delta g_I(\mu)$ are linearly related via the transformation

$$\delta g_I = \sum_J \delta g_J(\mu) N_{JI}(\mu). \tag{2.311}$$

We shall *define* the *renormalized operator*, denoted as $[\mathcal{O}]_\mu$ by rewriting the deformation of the action (2.309) as

$$\delta S = \int d^D x \sum_I \delta g_I(\mu)[\mathcal{O}_I]_\mu \Leftrightarrow [\mathcal{O}_I]_\mu = \sum_J N_{IJ}(\mu)\mathcal{O}_J. \tag{2.312}$$

In fact, the matrix $N_{IJ}(\mu)$ that transforms between couplings and running couplings are determined by the β-functions[76]

$$\begin{aligned} 0 \equiv \frac{d\delta g_I}{d\log\mu} &= \sum_J \frac{d}{d\log\mu}\left(\delta g_J(\mu) N_{JI}(\mu)\right) \\ &= \sum_J \delta\beta_J(g(\mu)) N_{JI} + \delta g_J(\mu)\frac{dN_{JI}(\mu)}{d\log\mu} \end{aligned} \tag{2.314}$$

which iiplies the following relation on the N-matrices

$$\frac{dN_{IJ}(\mu)}{d\log\mu} = -\sum_K \gamma_{IK}(g(\mu)) N_{KJ}(\mu) \tag{2.315}$$

where the γ_{IK} are defined as

$$\gamma_{IK} \equiv \frac{\partial\beta_K(g(\mu))}{\partial g_I(\mu)}. \tag{2.316}$$

If we can find an operator basis in which γ_{IK} is diagonal i.e. a basis such that $\gamma_{IK} = \delta_{IK}\gamma_I$, we can integrate (2.315) to obtain the following relation for the renormalized operator

$$[\mathcal{O}_I]_\mu = e^{-\int_{\mu'}^{\mu} \frac{d\widetilde{\mu}}{\widetilde{\mu}}\gamma_I(g(\widetilde{\mu}))}[\mathcal{O}_I]_{\mu'}. \tag{2.317}$$

This equation informs us how a given operator will change based on the energy scale being probed! We will see later on that in a *scale-invariant* QFT, $\gamma_I =$ constant and then (2.317) becomes

$$[\mathcal{O}_I]_\mu = \left(\frac{\mu}{\mu'}\right)^{-\gamma_I}[\mathcal{O}_I]_{\mu'}. \tag{2.318}$$

In this case, for these particular theories, the γ_I's are called the *anomalous dimension* of $\mathcal{O}_I$.

Let us explore a very special case. Consider the operator ϕ by itself. Then the deformation of the action is

$$\delta S = \int d^D x\, \delta g_1 \cdot \phi(x) \tag{2.319}$$

[76] Where

$$\delta\beta_J(g(\mu)) = \sum_K \frac{\partial\beta_J}{\partial g_K(\mu)}\delta g_K(\mu). \tag{2.313}$$

which has corresponding amputated 1PI diagram (called tadpole diagrams)

$$= \text{———} + \text{no further corrections} \tag{2.320}$$

where the tadpole term on the RHS contributes as ——— $= -\delta g_1$. The *renormalized running coupling* is then given as

$$\delta g_1(\mu) = -(\mathcal{Z}(\mu))^{\frac{1}{2}} \cdot \text{(1PI)——} = (\mathcal{Z}(\mu))^{\frac{1}{2}} \delta g_1. \tag{2.321}$$

We therefore have that the renormalized operator, $[\phi]_\mu$ can be written as

$$[\phi]_\mu = (\mathcal{Z}(\mu))^{-\frac{1}{2}} \phi. \tag{2.322}$$

The anomalous dimension of ϕ can be written as

$$\gamma_\phi = -\frac{d \log\Big((\mathcal{Z}(\mu))^{-\frac{1}{2}}\Big)}{d \log \mu} = \frac{1}{2} \frac{d \log \mathcal{Z}(\mu)}{d \log \mu}. \tag{2.323}$$

If we are working in a scaling invariant theory where γ_ϕ = constant, this implies that $\mathcal{Z}(\mu) \propto \mu^{2\gamma_\phi}$.

Now, we (or I guess I) should address all of the talk about a particular *renormalization scheme* and now present one here for the 1PI renormalization group called the *minimal subtraction scheme*. We will begin with the path integral formulation of QFT defined via dimensional regularization with $D = d - \epsilon$

$$S = S_{\text{kin}} - \int d^{d-\epsilon}x \, g_I^\epsilon \mathcal{O}_I(x) \tag{2.324}$$

where the kinetic term, for example, can go as the normal scalar one $\int d^{d-\epsilon}x \, (\partial_\mu \phi)^2/2$, the g_I^ϵ are bare couplings that depend upon the regularization procedure ϵ, and the operators here called *bare operators* are built out of the field ϕ and derivatives thereof. Dimensional analysis tells us that the mass-dimension of the scalar field is $[\phi] = \frac{d-\epsilon-2}{2}$, while the bare coupling has mass dimension $[g_I^\epsilon] = -\delta_I(\epsilon)$ (denotes the mass dimension) while the bare operators have mass dimension $[\mathcal{O}_I] = d - \epsilon + \delta_I(\epsilon)$ where $\delta_I(\epsilon) = \delta_I^{(0)} + \epsilon \delta_I^{(1)}$ by construction of the bare operators $\mathcal{O}_I$.

The minimal subtraction scheme (MS) is a renormalization scheme in which the dimensionless renormalized running couplings $\lambda_I(\mu)$ are defined such that the bare coupling g_I^ϵ, when expressed as a function of $\lambda_J(\mu)$ and ϵ, only involves negative powers of ϵ i.e. as

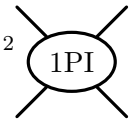
$$g_I^\epsilon = \mu^{-\delta(\epsilon)} \left(\lambda_I(\mu) + \sum_{n=1}^{\infty} \frac{1}{\epsilon^n} \mathcal{K}_I^{(n)}(\{\lambda_J(\mu)\}) \right) \tag{2.325}$$

where the last term we can call the *counter term* that contains no non-negative powers of ϵ. This definition removes the freedom/ambiguity in finite redefinition of $\lambda_I(\mu)$, at least in perturbation theory. To see how the minimal subtraction scheme works in practice, let us revisit $d = 4$ ϕ^4-theory with dimensional regularization $D = 4 - \epsilon$, with a coupling of the following form

$$\mathcal{L}_E \supset \frac{1}{4!} g^\epsilon \phi^4 \tag{2.326}$$

where the mass dimension of g^ϵ is $D - 4\frac{D-2}{2} = \epsilon$. In a generic renormalization scheme based on evaluating the 1PI diagram at momentum $\sim \mu$, we would define the renormalized dimensionless coupling $\widetilde{\lambda}(\mu)$ as

$$\mu^\epsilon \widetilde{\lambda}(\mu) \equiv -(\mathcal{Z}(\mu))^2$$



$$\Big|_{(k_i^2, s_i) = (\mathcal{O}(\mu^2), -\mu^2 a_{s_i})} \tag{2.327}$$

where the Mandelstam variables here are evaluated at $s = -\mu^2 a_s$, $t = -\mu^2 a_t$, and $u = -\mu^2 a_\mu$ for some constants a_s, a_t, and a_u that are of our choice. At 1-loop order for ϕ^4, we explicitly have

$$\begin{aligned}
\mu^\epsilon \widetilde{\lambda}(\mu) &= g^\epsilon - \frac{(g^\epsilon)^2}{2} \int \frac{d^{4-\epsilon}k}{(2\pi)^{4-\epsilon}} \int_0^1 dx \left(\frac{1}{(k^2 + \mu^2 a_s x(1-x))^2} + (a_s \to a_t) + (a_s \to a_u) \right) + \mathcal{O}(g^3) \\
&= g^\epsilon - \frac{(g^\epsilon)^2}{2} \mu^{-\epsilon} (4\pi)^{-2+\frac{\epsilon}{2}} \frac{\Gamma\left(\frac{\epsilon}{2}\right) \Gamma\left(1 - \frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)} \left(a_s^{-\frac{\epsilon}{2}} + a_t^{-\frac{\epsilon}{2}} + a_u^{-\frac{\epsilon}{2}} \right) + \mathcal{O}(g^3).
\end{aligned} \tag{2.328}$$

We can invert this relation to instead find an equation for g^ϵ as

$$\begin{aligned}
g^\epsilon &= \mu^\epsilon \left(\widetilde{\lambda}(\mu) + \frac{(\widetilde{\lambda})^2}{2} (4\pi)^{-2+\frac{\epsilon}{2}} \frac{\Gamma\left(\frac{\epsilon}{2}\right) \Gamma\left(1 - \frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)} \left(a_s^{-\frac{\epsilon}{2}} + a_t^{-\frac{\epsilon}{2}} + a_u^{-\frac{\epsilon}{2}} \right) + \mathcal{O}(\widetilde{\lambda}(\mu)^3) \right) \\
&= \mu^\epsilon \left(\widetilde{\lambda}(\mu) + (\widetilde{\lambda})^2 \left(\frac{3}{16\pi^2} \frac{1}{\epsilon} + \mathcal{O}(\epsilon^0) \right) + \mathcal{O}(\widetilde{\lambda}(\mu)^3) \right) \\
&= \mu^\epsilon \left(\lambda(\mu) + (\lambda(\mu))^2 \frac{3}{16\pi^2} \frac{1}{\epsilon} + \mathcal{O}(\lambda(\mu)^3) \right)
\end{aligned} \tag{2.329}$$

where in the second line, $\mathcal{O}(\epsilon^0)$ depends upon a_s, a_t, and a_u and that the third line is that under the minimal subtraction scheme, $\lambda(\mu)$ is defined such that it is equal in the last line. Working in the minimal subtraction scheme has the advantage of certain technical simplifications such as the nontrivial perturbative contributions to the β-functions are strictly of order ϵ^0. To see this, let us inspect the renormalization group equation in minimal subtraction scheme more closely. Namely, given that we have

$$g_I^\epsilon = \mu^{-\delta(\epsilon)} \left(\lambda_I(\mu) + \sum_{n=1}^{\infty} \frac{1}{\epsilon^n} \mathcal{K}_I^{(n)}(\{\lambda_J(\mu)\}) \right) \tag{2.330}$$

then the derivative is

$$0 \equiv \frac{d_I g^\epsilon}{d \log \mu} = -\delta_I \mu^{-\delta_I} \left(\lambda_I + \sum_{n \geq 1} \epsilon^{-n} \mathcal{K}_I^{(n)} \right) + \mu^{-\delta_I} \left(\beta_I^\epsilon + \sum_{n \geq 1} \epsilon^{-n} \frac{\partial \mathcal{K}_I^{(n)}}{\partial \lambda_J} \beta_J^\epsilon \right) \tag{2.331}$$

where the β-function is

$$\beta_I^\epsilon \equiv \frac{d\lambda_I(\mu)}{d \log \mu} \tag{2.332}$$

which is the β-function in $(d - \epsilon)$ dimensions. We do have an a priori expectation that β_I^ϵ is finite in but all-order expansion in ϵ i.e.

$$\beta_I^\epsilon = \sum_{m=0}^{\infty} \beta_I^{(m)}(\lambda) \epsilon^m = \lim_{\epsilon \to 0} \beta_I^{(0)}(\lambda) \equiv \beta_I(\lambda) \tag{2.333}$$

where in the limit that $\epsilon \to 0$, we recover the *true* β-function of the QFT. We can organize (2.331) in powers of ϵ as

$$-(\delta_I^{(0)} + \epsilon\delta_I^{(1)}) \left(\lambda_I + \sum_{n\geq 1} \epsilon^{-n} \mathcal{K}_I^{(n)} \right) + \sum_{m\geq 0} \epsilon^m \beta_I^{(m)} + \sum_{n\geq 1} \sum_{m\geq 0} \epsilon^{m-n} \frac{\partial \mathcal{K}_I^{(n)}}{\partial \lambda_J} \beta_J^{(m)} = 0. \quad (2.334)$$

For example, at order ϵ^1 we have the following relation as

$$-\delta_I^{(1)} \lambda_I + \beta_I^{(1)} + \sum_{n\geq 1} \frac{\partial \mathcal{K}_I^{(n)}}{\partial \lambda_J} \beta_J^{(n+1)} = 0 \quad (2.335)$$

while at order ϵ^2 we have the following relation

$$\beta_I^{(2)} + \sum_{n\geq 1} \frac{\partial \mathcal{K}_I^{(n)}}{\partial \lambda_J} \beta_J^{(n+2)} = 0 \quad (2.336)$$

and so one and so on. Note that with increasing n, the counter term $\mathcal{K}_I^{(n)}$ begins at higher order in λ. These relations can be satisfied in perturbation theory only if $\beta_I^{(m)} = 0$ for $m \geq 2$ and $\beta_I^{(1)} = \delta_I^{(1)} \lambda_I$. Thus,

$$\beta_I^{\epsilon} = \beta_I^{(0)}(\lambda) + \epsilon\delta_I^{(1)} \lambda_I \quad (2.337)$$

just as we had claimed prior.

The remaining equation at order $\epsilon^{\leq 0}$ determins $\mathcal{K}_I^{(n)}$ in terms of $\beta_I^{(0)}(\lambda) \equiv \beta_I(\lambda)$ through the recursive relations

$$\left(\beta_J \frac{\partial}{\partial \lambda_J} - \delta_I^{(0)} \right) \mathcal{K}_I^{(n)} = \delta_I^{(1)} \left(1 - \lambda_J \frac{\partial}{\partial \lambda_J} \right) \mathcal{K}_I^{(n+1)} \quad (2.338)$$

which is true for $n \geq 1$, and that we have

$$\beta_I - \delta_I^{(0)} \lambda_I = \delta_I^{(1)} \left(1 - \lambda_J \frac{\partial}{\partial \lambda_J} \right) \mathcal{K}_I^{(1)}. \quad (2.339)$$

Now, recall that the stress-energy tensor $T_{\mu\nu}(x)$ is the Noether current associated with translation symmetry. Namely, that given that the action $S[\phi]$ is invariant under $\phi(x) \mapsto \phi'(x)$ where $\phi'(x') \equiv \phi(x)$ for $x'^{\mu} = x^{\mu} + \delta a^{\mu}$ which has a transformation (or now what we can recognize as a deformation) $\delta\phi(x) = \phi'(x) - \phi(x) = -\delta a^{\mu} \partial_{\mu} \phi(x)$. Then, the Noether procedure promotes δa^{μ} to $\delta a^{\mu}(x)$ and then the transformation $\delta\phi(x)$ becomes

$$\delta\phi(x) = -\delta a^{\mu}(x) \partial_{\mu} \phi(x) + \partial_{\mu} \delta a_{\nu}(x) D^{\mu\nu} \phi(x) \quad (2.340)$$

where $D^{\mu\nu}$ is to be determined for a given theory, but now notice that this is no longer a symmetry variation namely (where the minus sign is by Euclidean convention)

$$S[\phi + \delta\phi] = S[\phi] - \int d^D x \, \partial_{\mu} \delta a_{\nu}(x) \cdot T^{\mu\nu}(x). \quad (2.341)$$

Note that the term in $\delta\phi(x)$ proportional to $\partial_\mu \delta a_\nu(x)$ only affects $T^{\mu\nu}$ by $D^{\mu\nu}\phi(x) \cdot \text{EOM}$. Now, consider the *scaling transformation* $\delta^\mu(x) = \delta a \cdot x^\mu$ which is in general not a symmetry, has the corresponding deformation of the action

$$\delta S[\phi] = -\int d^D x \, \delta a \cdot T^\mu{}_\mu(x) \tag{2.342}$$

while ϕ transforms by

$$\delta\phi(x) = -\delta a x^\mu \partial_\mu \phi(x) + \delta a \cdot D^\mu{}_\mu \phi(x). \tag{2.343}$$

We would like the massless free scalar action

$$S_{\text{kin}}[\phi] = \int d^D x \, \frac{1}{2}(\partial_\mu \phi)^2 \tag{2.344}$$

to be invariant under the above defined scaling transformation (2.343). This will be satisfied provided that we choose $D^{\mu\nu}$ to be such that the trace is (with $D = 4 - \epsilon$)

$$D^\mu{}_\mu = -\frac{D-2}{2} \tag{2.345}$$

which as a quick check,

$$\delta S_{\text{kin}}[\phi] = \delta a \int d^D x \, \partial^\mu \phi \left(-\partial_\mu \left(x^\nu \partial_\nu \phi - \frac{D-2}{2} \phi \right) \right) = 0. \tag{2.346}$$

Then, the scaling transformation (2.343) is equivalent to

$$\phi'(x') = (1 + \delta a)^{-\frac{D-2}{2}} \phi(x) \tag{2.347}$$

where we can take the transformation for the spacetime coordinate as $x' = (1 + \delta a)x$. More generally, an *operator* term $\mathcal{O}_I$ made out of $\phi(x)$ and it's derivatives of mass dimension $d_I(\epsilon) = d - \epsilon + \delta_I(\epsilon)$ transforms under this scaling symmetry/transformation as

$$\mathcal{O}'_I(x') = (1 + \delta a)^{-d_I(\epsilon)} \mathcal{O}_I(x). \tag{2.348}$$

Armed with this transformation, we can see what becomes of the energy-momentum tensor with the given action

$$S[\phi] = S_{\text{kin}}[\phi] + \int d^{d-\epsilon} x \sum_I g_I^\epsilon \mathcal{O}_I(x) \tag{2.349}$$

then under the scaling transformation rules (2.348) we get

$$\begin{aligned} S[\phi'] - S_{\text{kin}}[\phi'] &= \int d^{d-\epsilon} x \sum_I g_I^\epsilon \mathcal{O}'_I(x) \\ &= \int d^{d-\epsilon} x' \sum_I g_I^\epsilon \mathcal{O}'_I(x') \\ &= \int d^{d-\epsilon} x \, (1 + \delta a)^{d-\epsilon} \sum_I g_I^\epsilon (1 + \delta a)^{-(d-\epsilon+\delta_I)} \mathcal{O}_I(x) \\ \delta S[\phi] &= -\int d^{d-\epsilon} x \, \delta a \sum_I \delta_I(\epsilon) g_I^\epsilon \mathcal{O}_I(x). \end{aligned} \tag{2.350}$$

Therefore, up to a total derivative, we can read off the trace of the energy-momentum tensor as

$$T^{\mu}{}_{\mu}(x) = \sum_I \delta_I(\epsilon) g_I^\epsilon \mathcal{O}(x). \tag{2.351}$$

The piece in orange is not very illuminating, and there for it would be illuminating to re-express the right hand side in terms of renormalized quantities. Recall in the minimal subtraction scheme that

$$g_I^\epsilon = e^{-\delta_I(\epsilon)} \left(\lambda_I(\mu) + \sum_{n\geq 1} \epsilon^{-n} \mathcal{K}_I^{(n)}(\lambda(\mu)) \right) \tag{2.352}$$

which implies the following log-derivative as we have seen

$$0 = \frac{dg_I^\epsilon}{d\log\mu} = -\delta_I(\epsilon) g_I^\epsilon + \sum_J \frac{d\lambda_J(\mu)}{d\log\mu} \frac{\partial g_I^\epsilon}{\partial \lambda_J} \tag{2.353}$$

where $\partial g_I^\epsilon / \partial \lambda_J = \beta_J = \epsilon \delta_J^{(1)} \lambda_J$. Then, we can re-write the orange piece in (2.351) as

$$\delta_I(\epsilon) g_I^\epsilon = \sum_J \left(\beta_J + \epsilon \delta_J^{(1)} \lambda_J \right) \frac{\partial g_I^\epsilon}{\partial \lambda_J}. \tag{2.354}$$

Thus, the trace of the energy-momentum tensor takes the following form in terms or renormalized operators

$$T^{\mu}{}_{\mu} = \sum_J \left(\beta_J + \epsilon \delta_J^{(1)} \lambda_J \right) \frac{\partial}{\partial \lambda_J} \sum_I g_I^\epsilon \mathcal{O}_I \tag{2.355}$$

where the last term in blue is simply the renormalized operator $[\mathcal{O}_J]_\mu$. Now that the right hand side is expressed in terms of the renormalized coupling and operators, we can take the $\epsilon \to 0$ limit and write the trace as

$$T^{\mu}{}_{\mu} = \sum_J \beta_J \cdot [\mathcal{O}_J]_\mu \tag{2.356}$$

which holds as an operator identity. This result should really be striking. It is the statement that, even at the operator level i.e. in full quantum-fun, that, at least, the trace of the energy-momentum tensor can be written in terms of the summation of β-functions for corresponding operator $\mathcal{O}_J$. This result will not have much usage here except for a quick "discovery" in the next small paragraph/derivation, but it does have a great deal of usage in formulating theories of quantum gravity, string theory, and anomalies of which we will get an exposition of at the end of the lecture notes.

2.2.5 Discovering Conformal Field Theory

The real surprise of our renormalization group analysis is the existence of non-trivial fixed points that arise in the β-function, namely for points λ that satisfy $\beta_J(\lambda) = 0$ for *all* J at some non-zero dimensionless running coupling λ_I's. An equivalent statement of this using (2.356) is that the trace vanishes, $T^{\mu}{}_{\mu} = 0$ up to a total derivative. When this occurs, we have *enhanced symmetries* due to the Noether currents that are of the form

$$j_\mu(x) = T_{\mu\nu}(x) \varepsilon^\nu(x) \tag{2.357}$$

where the vector here $\varepsilon^\mu(x)$ obeys the following equation

$$\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu - \frac{2}{d}\eta_{\mu\nu}\partial_\rho \varepsilon^\rho = 0 \tag{2.358}$$

where $\varepsilon^\mu(x)$ is what is called a *conformal killing vector*. The constraint in (2.358) comes from the conservation of the Noether current

$$\begin{aligned} \partial_\mu j^\mu = \partial_\mu (T^{\mu\nu}\varepsilon_\nu) &= \frac{1}{2}T^{\mu\nu}(\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu) \\ &= \frac{1}{2}T^{\mu\nu}\left(\frac{2}{d}\eta_{\mu\nu}\partial_\rho \varepsilon^\rho\right) = 0 \end{aligned} \tag{2.359}$$

since the trace of the energy-momentum tensor vanishes, $T \equiv T^\mu{}_\mu 0$. Examples of conformal killing vectors are the following

$$\varepsilon^\mu(x) = \begin{cases} a^\mu & \to \text{translations} \\ b^{[\mu\nu]}x_\nu & \to \text{Lorentz} \\ cx^\mu & \to \text{scaling/dilatation} \\ c_\rho(2x^\mu x^\rho - x^2\eta^{\mu\rho}) & \to \text{special conformal symmetry} \end{cases} \tag{2.360}$$

Now, a QFT with an energy-momentum tensor that obeys $T^\mu{}_\mu = 0$, the QFT is what is called a *conformal field theory* (CFT). A remark is to be made here though. A priori, $\beta_I = 0 \implies T^\mu{}_\mu = \partial_\mu V^\mu$ for some V^μ. If V^μ is nontrivial, then the Noether current is $j_\mu = T_{\mu\nu}x^\nu - V_\mu$ is a conserved current that generates the scaling symmetry, but *no* special conformal symmetry. It turns out that while this occurs in certain free (massless) field theories, it does not occur in fully *interacting* QFTs.

Let us now look at a specific example (unlike the class of fix points we just saw that make up conformal field theories) of a nontrivial renormalization group fixed point. Let us consider ϕ^4-theory again in $D = 4 - \epsilon$ "dimensions" (and I put "dimensions" for a reason you shall soon see). Now, we shall view/interpret ϵ as a small but *finite* number *not* necessarily zero. We will work perturbatively in ϵ with the hope that the result may be extrapolated to $\epsilon = 1$ or even 2. This is what is known formally as the ϵ-*expansion*.

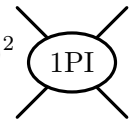
Recall that the Euclidean bare Lagrangian for ϕ^4-theory is given as

$$\mathcal{L}_E = \frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2}m_0^2\phi^2 + \frac{1}{4!}g_0\phi^4. \tag{2.361}$$

We will be interested in the possibility that there is no *mass gap* in the spectral density function of $\langle \phi(x)\phi(0)\rangle$[77]. In a given scheme, this may be accomplished by turning the bare mass m_0. But, caution should be given. this may or may not even be possible since there is the question about whether or not there are IR divergences or even states in the IR, and what if the effective coupling is strong in the IR? We may not even be able to trust perturbation theory then.

[77] By mass gap, we mean that when going from zero energy to the full mass-spectrum, there is no gap between the mass-pole and the curve from $\rho(k^2)$ function. This can be seen in the QFT 1 notes 1.2.3.

To address this question, we need to analyze the renormalization group equation. Recall that the renormalized coupling is given as

$$\lambda(\mu) = -\mu^{-\epsilon}(\mathcal{Z}(\mu))^2$$  $$\Big|_{k_i^2=\mu^2;s=t=u=-\frac{4}{3}\mu^2} \tag{2.362}$$

where the renormalization "constant" is

$$\mathcal{Z}(\mu) = \left(1 - \frac{\partial\Sigma}{\partial k^2}\right)\Big|_{k^2=\mu^2}. \tag{2.363}$$

We have already seen that at 1-loop for ϕ^4-theory $\mathcal{Z}(\mu) = 1$ and that the dimensionless running coupling constant goes as

$$\lambda(\mu) = \mu^{-\epsilon}\left(g_0 - \frac{3}{2}g_0^2(4\pi)^{-2+\frac{\epsilon}{2}}\frac{\Gamma\left(\frac{\epsilon}{2}\right)\Gamma\left(1-\frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\mu^2\right)^{-\frac{\epsilon}{2}} + \mathcal{O}(g_0^3)\right) \tag{2.364}$$

where it should be noted that here, ϵ is no longer infinitesimal! This running coupling leads to the following log-derivative (to get the β-function)

$$\frac{d\lambda(\mu)}{d\log\mu} = -\epsilon\mu^{-\epsilon}g_0 - 2\epsilon\mu^{-2\epsilon}g_0^2\left(-\frac{3}{2}\right)(4\pi)^{-2+\frac{\epsilon}{2}}\frac{\Gamma\left(\frac{\epsilon}{2}\right)\Gamma\left(1-\frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\right)^{-\frac{\epsilon}{2}} + \mathcal{O}(g_0^3). \tag{2.365}$$

To derive the renormalization group equation, we need to re-express the RHS of (2.365) om terms of $\lambda(\mu)$ as

$$\begin{aligned}\frac{d\lambda(\mu)}{d\log\mu} &= -\epsilon\lambda(\mu) - \epsilon(\lambda(\mu))^2\cdot\left(-\frac{3}{2}\right)(4\pi)^{-2+\frac{\epsilon}{2}}\frac{\Gamma\left(\frac{\epsilon}{2}\right)\Gamma\left(1-\frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\right)^{-\frac{\epsilon}{2}} + \mathcal{O}(\lambda(\mu)^3)\\ &= -\epsilon\lambda(\mu) - \epsilon(\lambda(\mu))^2\cdot\left(-\frac{3}{16\pi^2}\frac{1}{\epsilon} + \mathcal{O}(\epsilon^0)\right) + \mathcal{O}(\lambda(\mu)^3)\end{aligned} \tag{2.366}$$

from which we can read off the β-function as

$$\beta(\lambda) = -\epsilon\lambda) + \lambda^2\left(\frac{3}{16\pi^2} + \mathcal{O}(\epsilon)\right) + \mathcal{O}(\lambda^3). \tag{2.367}$$

The renormalization group fixed point can be found by solving $\beta(\lambda_*) = 0$ where λ_* denotes the fixed point value. The $\lambda = 0$ one is trivial, the one that is interesting is when

$$\lambda = \lambda_* = \frac{16\pi^2}{3}\epsilon + \mathcal{O}(\epsilon^2). \tag{2.368}$$

To visualize this fixed point, let us think about varying the scale μ as $\mu \to 0$ in which case, we can interpret the β-function as a renormalization group *flow*.

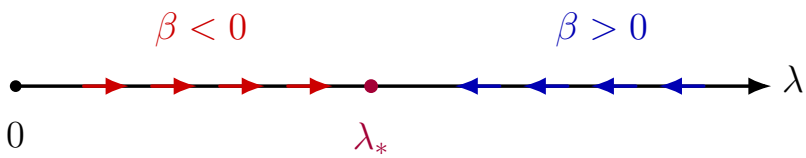


Figure 23. Renormalization group flow for ϕ^4-theory with fixed point $\lambda_* \approx 16\pi^2\epsilon/3$.

We can see that on the LHS of the fixed point λ_*, we will be approaching is, and similar for when on the RHS of λ_*, and thus when $\mu \to 0$, the β-function will flow towards λ_*.

Let us compare this to the bare coupling that has mass dimension ϵ with the "naive" dimensionless coupling where $\mu^{-\epsilon} g_0 \to \infty$ as $\mu \to 0$. However, the true coupling strength at scale μ is governed by $\lambda(\mu)$ and so in the same lmit,

$$\lambda(\mu \to 0) = \lambda_* \tag{2.369}$$

which is finite (at least for sufficiently small ϵ)!

Let us ask a question now though, does such a renormalization group fixed point exist for $\epsilon = 1, 2$, i.e. for ϕ^4 theory in $D = 3, 2$ dimensions? This cannot be rigorously established in the ϵ-expansion. Nonetheless, the answer is yes!

Now, all of this talk about renormalized operators and what not, we have yet to see how this impacts the observables we want to compute, such as correlation functions that can lead to S-matrix elements from which we can find cross sections and decay rates from. Well, we are almost there, but we need build up slowly and will in turn consider just a few different scalar operators that will come along with our ϕ^4-theory, and then from these operators, compute the correspond 2-point function or Greens function.

Specifically, let us inspect the properties of some field operators at the renormalization group fixed point. First, consider $\mathcal{O}_1 = \phi$ of which we have already seen that the γ_I factor is

$$\gamma_1 = \frac{1}{2} \frac{d \log \mathcal{Z}(\mu)}{d \log \mu} \tag{2.370}$$

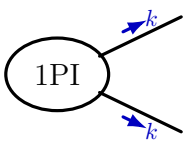
at 1-loop, and that for $\mathcal{Z}(\mu) = 1 \implies \gamma_1 = 0$ and thus $\mathcal{O}_1$ has no anomalous dimension at order ε^1.

Moving up in power, we can consider $\mathcal{O}_2 = \phi^2$ which has a corresponding deformation of the action

$$\delta S = \int d^{4-\epsilon} x \, \delta g_2 \cdot \phi^2(x) \tag{2.371}$$

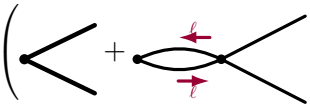
which has a corresponding running coupling of

$$\delta g_2(\mu) = -\mathcal{Z}(\mu) \cdot$$



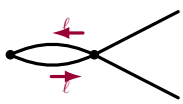
$$\Big|_{k^2=\mu^2}$$

$$= -\mathcal{Z}(\mu) \Bigg($$



$$\Bigg)\Bigg|_{k^2=\mu^2} . \tag{2.372}$$

But wait a second, that second diagram goes as



$$\propto \int \ell \frac{d^{4-\epsilon}\ell}{(\ell^2)^2} \tag{2.373}$$

which is IR-divergent [78]?! This means then that this is *not* a good definition of $\delta g_2(\mu)$. The reason being is that the spatial integration over $\int d^D x \, \langle \mathcal{O}(x) \cdots \rangle$ is IR divergent itself in perturbation theory when the mass dimension of the corresponding operator is $d_{\mathcal{O}} \leq D/2$.

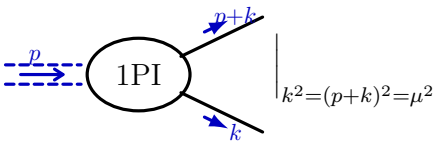
To define the appropriate renormalized operator in this case, we should consider the following deformation of the action instead

$$\delta S = \int d^D x \, \delta g_{\mathcal{O}}(x) \cdot \mathcal{O}(x) \tag{2.374}$$

where $\delta g_{\mathcal{O}}(x)$ is a spatially dependent "coupling" that can be thought of as a generalization of source J in the generating functional $W[J]$. Then, for $\mathcal{O}_2 = \phi^2$, we can consider

$$\begin{aligned} \delta S &= \int d^{4-\epsilon} x \, \delta g_2(x) \cdot \mathcal{O}_2(x) \\ &= \int \frac{d^{4-\epsilon} p}{(2\pi)^{4-\epsilon}} \delta \widetilde{g}_2(p) \cdot \widetilde{\mathcal{O}}_2(-p). \end{aligned} \tag{2.375}$$

Now, we shall define the renormalized coupling as

$$\delta \widetilde{g}_2(p) = -\mathcal{Z}(\mu) \cdot$$  (2.376)

where the blue dashed lines represent the injection of the momentum from the "current" as described earlier, and the renormalized operator $[\widetilde{\mathcal{O}}_2]_\mu$ is then defined as

$$\delta \widetilde{g}_2(p) \cdot \widetilde{\mathcal{O}}_2(-p) \equiv \delta \widetilde{g}_2(p;\mu) [\widetilde{\mathcal{O}}_2(-p)]_\mu \tag{2.377}$$

The anomalous dimension then of the $\mathcal{O}_2$ operator γ_2 is the *defined* via

$$\gamma_2(\mu) = \frac{d}{d \log \mu} \log \left(\frac{\delta \widetilde{g}_2(p;\mu)}{\delta \widetilde{g}_2(p)} \bigg|_{p^2=\mu^2} \right). \tag{2.378}$$

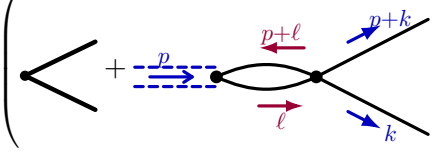
This definition is such that the renormalized operator is in the typical form we have seen prior,

$$[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|} \propto e^{-\int^{\mu} \frac{d\widetilde{\mu}}{\widetilde{\mu}} \gamma_2(\widetilde{\mu})} \cdot \widetilde{\mathcal{O}}_2(p). \tag{2.379}$$

[78] Remember that an IR divergence here after regulating the UV with dimensional regularization will appear when the lower bound of the integral is not is not well behaved, namely for $\int_\mu^\Lambda dk \, k^{-1} = \log(\Lambda/\mu)$, for dimensional regularization, the $\Lambda \to \infty$ is finite (or it would be if properly done) while the IR divergence here stems from $\mu \to 0$ which we have *not* regulated. [**AC:**] Also, nice.

The naive perturbation theory then at 1-loop to find $\delta\widetilde{g}_2(p;\mu)$ goes as

$$\delta\widetilde{g}_2(p;\mu) = -\Bigg($$



$$+ \text{ higher order}\Bigg)\Bigg|_{k^2=\mu^2}$$

$$= \delta\widetilde{g}_2(p)\left(1 - \frac{g_4}{2}\int \frac{d^{4-\epsilon}\ell}{(2\pi)^{4-\epsilon}}\frac{1}{\ell^2(p+\ell)^2} + \mathcal{O}(g_4^2)\right)$$

$$= \delta\widetilde{g}_2(p)\left(1 - \frac{g_4}{2}(4\pi)^{-2+\frac{\epsilon}{2}}\frac{\Gamma\left(\frac{\epsilon}{2}\right)\Gamma\left(1-\frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)}|p|^{-\epsilon} + \mathcal{O}(g_4^2)\right) \tag{2.380}$$

Then, to find γ_2, we take the log of the variation of the renormalized $\widetilde{g}_2(p;\mu)$ with respect to the bare running coupling $\widetilde{g}_2(p)$ in (2.380) which gives

$$\log\left(\frac{\delta\widetilde{g}_2(p;\mu)}{\delta\widetilde{g}_2(p)}\Bigg|_{p^2=\mu^2}\right) = -\frac{g_4\mu^{-\epsilon}}{2}(4\pi)^{-2+\frac{\epsilon}{2}}\frac{\Gamma\left(\frac{\epsilon}{2}\right)\Gamma\left(1-\frac{\epsilon}{2}\right)^2}{\Gamma(2-\epsilon)} + \mathcal{O}(g_4^2) \tag{2.381}$$

where the coupling g_4 on the RHS is the bare coupling. The γ_2 value is then

$$\begin{aligned}\gamma_2(\mu) &= \frac{d}{d\log\mu}log\left(\frac{\delta\widetilde{g}_2(p;\mu)}{\delta\widetilde{g}_2(p)}\Bigg|_{p^2=\mu^2}\right)\\ &= \epsilon g_4\mu^{-\epsilon}\left(\frac{1}{16\pi^2}\frac{1}{\epsilon} + \mathcal{O}(\epsilon^0)\right) + \mathcal{O}(g_4^2)\\ &= \lambda_4(\mu)\cdot\left(\frac{1}{16\pi^2} + \mathcal{O}(\epsilon)\right) + \mathcal{O}(g_4(\mu)^2)\end{aligned} \tag{2.382}$$

where in the second line we had the bare coupling, but then in the last line, we replaced it with it's dimensionless running coupling value. At the renormalization group fixed point that we found earlier

$$\lambda_4(\mu) = \lambda_{4*} = \frac{16\pi^2}{3}\epsilon + \mathcal{O}(\epsilon^2). \tag{2.383}$$

Then γ_2 becomes

$$\gamma_2(\mu) = \gamma_{2*} = \frac{\epsilon}{3} + \mathcal{O}(\epsilon^2) \tag{2.384}$$

and thus *finally* the properly renormalized operator for $\mathcal{O}_2 = \phi^2$ is

$$[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|} = \left(\frac{\mu}{\mu_0}\right)^{-\gamma_{2*}}\cdot\widetilde{\mathcal{O}}_2(p) \tag{2.385}$$

where the μ_0 is some UV regulator scale and $\widetilde{\mathcal{O}}_2(p)$ is the bare operator. The implications of the renormalized operators, and specifically this renormalized operator, on the Greens functions we now investigate.

At the renormalization group fixed poiint, correlators of renormalized operators cannon depend on any dimensionful parameter, and must be well-defined i.e. finite functions of the momenta involved. For example, one we saw previously, if we consider an operator $\mathcal{O}_2(x) = \phi^2(x)$ that has mass dimension $2-\epsilon$, the Fourier transform has

$$\widetilde{\mathcal{O}}_2(p) = \int d^{4-\epsilon}x\, e^{-ip\cdot x}\mathcal{O}(x) \tag{2.386}$$

has mass dimension -1 and the same holds for $[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|}$. Note that unlike $\widetilde{\mathcal{O}}_2(p)$, $[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|}$ has finite correlators.

Greens functions of $[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|}$ cannot depend on any dimensionalful parameter apart from p. By dimensional analysis, we have for example

$$\langle [\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|} [\widetilde{\mathcal{O}}_2(q)]_{\mu=|q|} \rangle \propto \delta^{4-\epsilon}(p+q) \cdot |p|^{-\epsilon}. \tag{2.387}$$

Now, using that $[\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|} = (\mu/\mu_0)^{-\gamma_{2*}} \widetilde{\mathcal{O}}_2(p)$, the 2-point correlator becomes

$$\mu_0^{2\gamma_{2*}} \langle \widetilde{\mathcal{O}}_2(p) \widetilde{\mathcal{O}}_2(q) \rangle \propto \delta^{4-\epsilon}(p+q) \cdot |p|^{-\epsilon+2\gamma_{2*}} \tag{2.388}$$

from which we can Fourier transform back to find the following result

$$\begin{aligned} \mu_0^{2\gamma_{2*}} \langle \mathcal{O}_2(x) \mathcal{O}_2(0) \rangle &\propto \int d^{4-\epsilon}p \, e^{ip\cdot x} |p|^{-\epsilon+2\gamma_{2*}} \\ &\propto \frac{1}{|x|^{2\Delta}} \end{aligned} \tag{2.389}$$

where $\Delta = 2 - \epsilon + \gamma_*$. The interpretation of this result is as follows. The operator $\mu_0^{\gamma_{2*}}$ is a well-defined local field operator in the QFT (CFT) with finite correlation functions. Here, Δ is the *mass dimension* of $\mu_0^{\gamma_{2*}} \mathcal{O}_2(x)$, and is a scaling dimension intrinsic to this operator in the CFT. The first remark I want to make on this result is that while $\mu_0^{\gamma_{2*}}$ is a local operator, the following

$$\begin{aligned} [\mathcal{O}_2(x)] &:= \int \frac{d^{4-\epsilon}p}{(2\pi)^{4-\epsilon}} e^{ip\cdot x} [\widetilde{\mathcal{O}}_2(p)]_{\mu=|p|} \\ &= \int \frac{d^{4-\epsilon}p}{(2\pi)^{4-\epsilon}} e^{ip\cdot x} \left(\frac{|p|}{\mu_0} \right)^{-\gamma_*} \widetilde{\mathcal{O}}_2(p) \end{aligned} \tag{2.390}$$

is **not** a local operator (due to the inverse p-dependent factor). The second remark I would like to make is that while we write $\mathcal{O}_2(x) = \phi^2(x)$ as a term in the bare Lagrangian density, the corresponding field operator cannot be viewed simply as $\lim_{x' \to x} \hat{\phi}(x')\hat{\phi}(x)$ since this limit is ill-defined since the corresponding Greens function is singular at $x' \to x$. This is why certain individuals ([**AC:**] such as myself admittedly...) will say that if you write down in a QFT system $\hat{\phi}^2(x)$ they will say this is nonsense, and instead if you want to write then, you instead need to write the normal ordered version of it $:\hat{\phi}(x)\hat{\phi}(x):$. The actual true operator relation takes the following form[79]

$$\mu_0^{\gamma_{2*}} \mathcal{O}_2(x) \propto \lim_{x' \to x} |x' - x|^{2\gamma_{1*} - \gamma_{2*}} \left[\hat{\phi}(x')\hat{\phi}(x) - \langle \hat{\phi}(x')\hat{\phi}(x) \rangle \right]. \tag{2.391}$$

More examples of operators in $(4-\epsilon)$-dimensional massless ϕ^4 theory are the following.

The equation of motion (which is just the massless Klein-Gordon equation, $\Box\phi = 0$) implies an operator relation of the form (implied by the mass dimension) $\mathcal{O}_3(x) = \phi^3(x) \propto \Box\phi(x)$ implies that $\mathcal{O}_3(x)$ is not independent from $\mathcal{O}_1(x)$ and thus the mass dimensions are

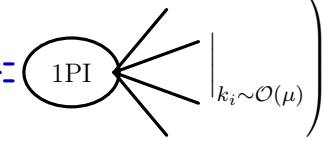
[79] We will not go through how to derive this rigorously, but in a CFT course you will, or a string theory course.

related as $\Delta_{\phi^3} = \Delta_\phi + 2$. These kind of operators where the the mass dimension of higher order operators are related to, say, their *parent*, are called *descendant operator*.

Let us now consider $\mathcal{O}_4(x) = \phi^4(x)$ and consider the deformation to the action as

$$\delta S = \int d^{4-\epsilon}x \, \delta g_4(x) \cdot \phi^4(x) \tag{2.392}$$

and we extra γ_4 from the following renormalized running coupling

$$\delta \widetilde{g}_4(p;\mu) = \delta \left(-(\mathcal{Z}(\mu))^2 \right.$$


$$\left. \right) \propto \delta \widetilde{g}_4(p) \cdot \mu^{\gamma_4} \tag{2.393}$$

for $p \sim \mathcal{O}(\mu)$ at renormalization group fixed point. In this case, we can set $p = 0$ and there is no IR divergence in agreement with earlier definition of anomalous dimension based on x-independent deformation of the running coupling. We can find the anomalous dimension in the usual way then via

$$\frac{d\delta g_4(\mu)}{d\log\mu} = \frac{\partial\beta(\lambda_4(\mu))}{\partial\lambda_4(\mu)} \cdot \delta g_4(\mu) = \gamma_4 \cdot \delta g_4(\mu). \tag{2.394}$$

We can find the corresponding renormalization group fixed point γ_{4*} via

$$\gamma_{4*} = \frac{\partial\beta(\lambda_4)}{\partial\lambda_4}\bigg|_{\lambda_{4*}} = \left(-\epsilon + \frac{3}{8\pi^2}\lambda_4 + \cdots\right)\bigg|_{\lambda_4} = \epsilon + \mathcal{O}(\epsilon^2) \tag{2.395}$$

where $\lambda_4 = 16\pi^2\epsilon/3 + \mathcal{O}(\epsilon^2)$. An attractive renormalization group fixed point corresponds to $\gamma_* > 0$, to which allows us to categorize the mass dimension of $\mathcal{O}_4(x)$ since

$$\Delta_{\phi^4} = 4 - \epsilon + \gamma_{4*} > d = 4 - \epsilon \tag{2.396}$$

and thus we classify this type of operator as an *irrelevant operator*. We can now ask the same question we asked at the very beginning of these escapades namely, "does the renormalized group fixed point of ϕ^4 theory in $(4-\epsilon)$-dimensions really exist for $\epsilon = 1, 2$? We can now confidently answer: yes! I table such values of the operator and the corresponding dimensional value below.

	Δ_ϕ	Δ_{ϕ^2}	Δ_{ϕ^4}
$d = 4 - \epsilon$	$1 - \frac{\epsilon}{2} + \mathcal{O}(\epsilon^2)$	$2 - \frac{2\epsilon}{3} + \mathcal{O}(\epsilon^2)$	$4 + \mathcal{O}(\epsilon^2)$
$d = 4 - 1 = 3$	$0.518\cdots$	$1.412\cdots$	$3.829\cdots$
$d = 4 - 2 = 2$	0.125	1	4

Table 1. Table displaying the different mass dimension values $\Delta_\mathcal{O}$ for various scalar operators $\mathcal{O}_I$ at different *integer* dimensions.

Now, let us make what *feels* like a complete change of topics, but as we will see, has some of it's most fundamental origins in QFT.

2.2.6 The Ising Model is just Scalar Field Theory (it really is and very pretty)!

The statistical Ising model is defined as a system of spin on D-dimensional lattice in *thermal equilibrium*. For example, in $D = 2$, we can model the Ising model on a square lattice denoted as Λ with points denoted as $(i, j) = x \in \Lambda$ where $i, j \in \mathbb{Z}$. Here is an image to think about the 2D lattice:

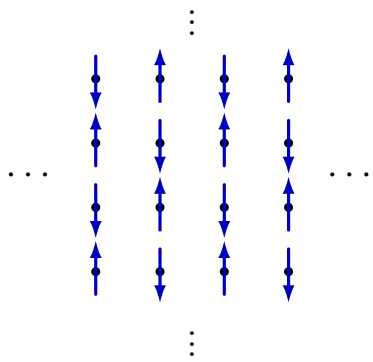


Figure 24. This is a 2D lattice for the Ising model where black points are inidividual points on the lattice, and the blue arrows are the direction of spin in $D = 2$ which is either up or down.

The corresponding Hilbert space for states living on the lattice can be written as $\mathcal{H} = \otimes_{x\in\Lambda} V_x$ where the vector space is written as $V_x = \text{span}\{|\uparrow\rangle, |\downarrow\rangle\} \simeq \mathbb{C}^2$. The corresponding Hamiltonian based on the Hilbert space is

$$H = -\sum_{\langle xx'\rangle} s_x s_{x'} \tag{2.397}$$

where s_x denotes the specific spin state at some lattice point x, and $\langle xx'\rangle$ denotes the neighboring sites x, x' for example, if $x = (i, j)$ then $x' = (i \pm 1, j)$ or $x' = (i, j \pm 1)$. this Hilbert space and Hamiltonian are unrelated to those of the QFT that will emerge later, they are just meant to serve as a way to get out footing when trying to derive the Ising model from QFT.

In thermal equilibrium, the probability of the system being in any energy eigenstate $|n\rangle$ is governed by the Boltzmann distribution

$$p_n = \frac{1}{Z} e^{-\beta E_n} \tag{2.398}$$

with the corresponding partition function $Z = \sum_n e^{-\beta E_n}$ where $\beta = 1/T$ and T is the temperature here. (Note that everything here are in units of where the Boltzmann constant is $k_B = 1$). Now, recall that Boltzmann's distribution maximizes the entropy $S = -\sum_n p_n \log p_n$ for given energy expectation value $\langle E\rangle = \sum_n p_n E_n$, and that the entropy is non-decreasing under probability evolution of the form (also known as the H-theorem)

$$\dot{p}_n = \sum_m \mathcal{V}_{nm}(p_m - p_n) \tag{2.399}$$

where $\mathcal{V}_{nm} = \mathcal{V}_{mn} \geq 0$.

The thermal expectation value of an operator $\mathcal{O}$ is given by

$$\langle \hat{\mathcal{O}} \rangle = \frac{1}{Z} \operatorname{Tr}_{\mathcal{H}} \left(e^{-\beta \hat{H}} \hat{\mathcal{O}} \right) \tag{2.400}$$

where the trace is over all possible states in the Hamiltonian, and then the partition function is $Z = \operatorname{Tr}_{\mathcal{H}} e^{-\beta \hat{H}}$. In the Ising model, take the operator to be of the general form $\hat{\mathcal{O}} = s_{x_1} \cdots s_{x_n}$, and so the observables are spin-correlation functions of the form $\langle s_{x_1} \cdots s_{x_n} \rangle$. In the low temperature limit where $T = \beta^{-1} \to 0$, the Ising model is dominated by ground states with a 2-fold degeneracy namely, $s_x = +1$ for all $x \in \Lambda$, or $s_x = -1$ for all $x \in \Lambda$. This low energy state of the system is called *spontaneous magnetization*.

At finite temperature T, thermal fluctuations have a tendency to wash out correlations between sites. In fact, spontaneous magnetization disappears for $T \geq T_C$ where T_C is called the *Curie temperature*.

For a general 2-point spin correlator, let us take one site to be at $x = (0, 0$ and another at some general point $y = (N, M)$ and define

$$\langle s_{0,0} s_{N,M} \rangle \equiv f_{N,M} \approx f(r) \tag{2.401}$$

where $r = \sqrt{N^2 + M^2}$ for $N, M \gg 1$ and let us take the temperature close to the Curie temperature. The function $f(r)$ here is essentially a measure of the correlation length between sites and for the three different temperature regimes, behaves in the following manner.

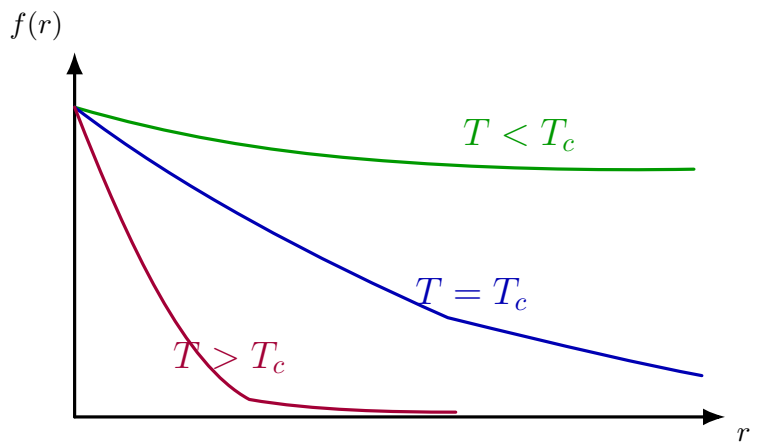


Figure 25. Graph of the correlation function $f(r)$ for temperature regimes $T < T_C$, $T = T_C$, and $T > T_C$.

In figure 24, the blue line can be computed and found to be $f(r) \sim \gamma^{-2\Delta}$ while the purple line can be modeled as $f(r) \sim e^{-r/\xi(T)}$ where $\xi(T) \sim (T - T_C)^{-\nu}$ where $\xi(T)$ is called the *correlation length*. The mass mass dimension and *correlation index* ν takes on different values for different dimensions. For $D = 2$, $\Delta = 1/8$ and $\nu = 1$ while for $D = 3$ $\Delta = 0.518 \cdots$ and $\nu = 0.62998 \cdots$ But ok, what does all of this even have to do with QFT?

Well, you might already be seeing similarities between this system and the scale-invariant QFT we were working with prior, and so I claim the following.

The first claim is that in the limit $T \to T_C$, spin correlators at large distance, namely $\langle s_{x_1} \cdots s_{x_n} \rangle$ for $|x_{ij}|$ of order $\xi(T) \sim (T - T_C)^{-\nu}$, become Euclidean Greens functions of a local operator $\sigma(x)$ in a D-dimensional, Poincaré invariant, QFT called *Ising field theory* (IFT). More precisely, this claim can be written as

$$\lim_{\substack{T \to T_C,\, L \to \infty \\ L/\xi(T) \to m \text{ (fixed)}}} (\mathcal{Z}_T(L))^{-\frac{n}{2}} \langle s_{[Lx_1]} \cdots s_{[Lx_n]} \rangle_{\text{Ising},T} = \langle \sigma(x_1) \cdots \sigma(x_n) \rangle_{\text{IFT},m} \tag{2.402}$$

where the operator $[L, x_n]$ (just the square brackets $[\cdot, \cdot]$) is equivalent to the nearest lattice point, and the mass here at the end of the IFT correlator denotes the mass gap of the IFT.

My second claim is that at $T = T_C$ where the correlation length goes as $\xi(T_X) = \infty$, long distance correlators of the Ising model are captured by IFT with mass gap $m = 0$ where this particular version is known as the *Ising CFT*. Furthermore, tht Ising CFT is equal to the renormalziation group fixed point of D-dimensional massless ϕ^4 theory, in essence the spin-operators here are really just $\sigma(x) \propto \phi(x)$ called the *spin-field*, and Δ becomes the scaling dimension of $\phi(x)$, and $D - \nu^{-1}$ is equal to the scaling dimension of $\phi^2(x)$.

But why should you even believe me? Well, the (what is called) *universality*, where details of the Ising lattice (for example a square or hexagon type of lattice) and Hamiltonian are unimportant at long distances, for temperature T close to T_C.

Now, to understand these claims, let us first rewrite the Ising model in the form of a lattice scalar field theory. In fact, we can consider a slightly generalized version of the Ising model where we will replace the sum-partition function we saw prior,

$$Z = \sum_{s_x = \pm 1, x \in \Lambda} e^{-\beta H[\{s_x\}]} \tag{2.403}$$

with the following new partition function

$$\begin{aligned} \widetilde{Z} &= \int \prod_{x \in \Lambda} \left(ds_x \, e^{-P(s_x)} \right) \cdot e^{-\beta H[\{s_x\}]} \\ &\equiv \int \prod_{x \in \Lambda} \prod ds_x \, e^{-S[\{s_x\}]}. \end{aligned} \tag{2.404}$$

The function we insert here, that differentiates it from our previously one, acts like a potential that has two minima between $s_x = \pm 1$ namely

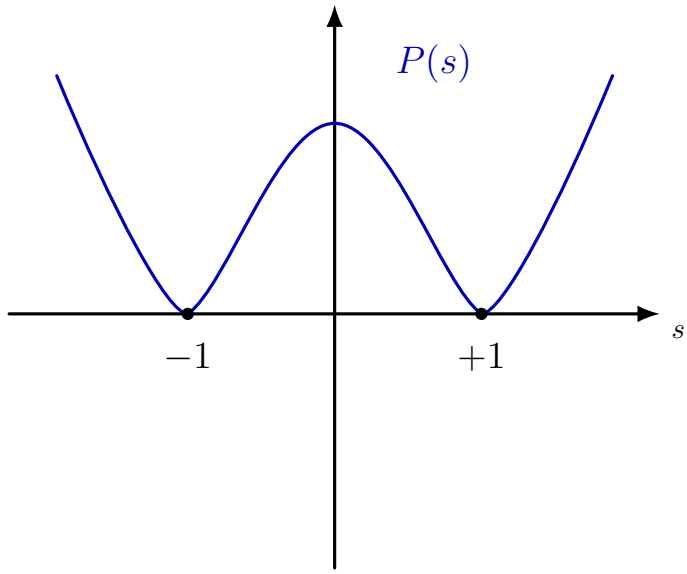


Figure 26. Graph of Ising model potential $P(s)$ with minima at the possible spin states.

Now, (2.404) looks like a euclidean path integral where the "Euclidean action" S is given as

$$\begin{aligned} S[\{s_x\}] &= \beta H[\{s_x\}] + \sum_{x\in\Lambda} P(s_x) \\ &= \frac{\beta}{2}\sum_{\langle xx'\rangle}(s_x - s_{x'})^2 + \sum_x (P(s_x) - \beta D s_x^2) \end{aligned} \tag{2.405}$$

where the first term is the kinetic term, and the last term in parenthesis is really the full potential (the D-here is the numbers of dimensions). We can rewrite this action (2.405) in a way that is more familiar from a field theory perspective by defining the Fourier transform of the spin-operators s_x as

$$s_x \equiv \int_B \frac{d^D\vec{k}}{(2\pi)^D}\widetilde{\phi}(\vec{k})e^{i\vec{k}\cdot\vec{x}} \tag{2.406}$$

where B here denotes a space called the *Brillouin zone* which is defined as $B : -\pi \leq k^\mu \leq \pi$ where μ is the usual spacetime index $\mu = 1, \cdots, D$, and thus $\widetilde{\phi}(\vec{k})$ is periodic under $k^\mu \to k^\mu + 2\pi$. The magnitude of the 4-vector, $|k^\mu| \leq \pi$ can be viewed as a UV cutoff.

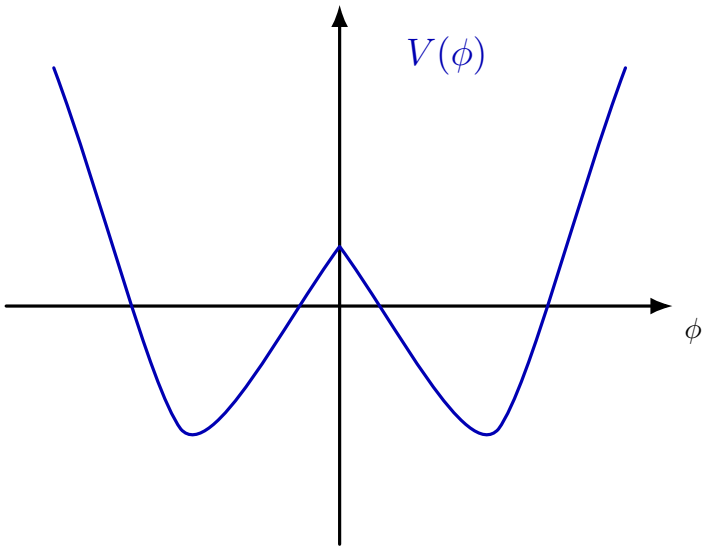
The kinetic term in (2.405) can then be rewritten as

$$\begin{aligned} \sum_{\langle xx'\rangle}(s_x - s_{x'})^2 &= \int_B \frac{d^D\vec{k}}{(2\pi)^D}\widetilde{\phi}(-\vec{k})\widetilde{\phi}(\vec{k})\sum_{\mu=1}^{D}\left|\, e^{ik_\mu} - 1\,\right|^2 \\ &= \int_B \frac{d^D\vec{k}}{(2\pi)^D}\widetilde{\phi}(-\vec{k})\widetilde{\phi}(\vec{k})\left(\vec{k}^2 - \frac{1}{12}\sum_{\mu=1}^{D}k_\mu^4 + \cdots\right). \end{aligned} \tag{2.407}$$

Then, with this in mind (and recall quickly that $k_\mu \to -i\partial_\mu$) the action (2.405) can be rewritten as

$$S[\phi] = \int d^D x \left(\frac{\beta}{2} \sum_{\mu=1}^{D} (\partial_\mu \phi)^2 + V(\phi) - \frac{\beta}{12} \sum_{\mu=1}^{D} (\partial_\mu \phi)^4 + \cdots \right) \tag{2.408}$$

where this last term in the action actually breaks the O(D)-symmetry that is present, but at long distances it is irrelevant. The potential $V(\phi)$ here is schematically of the following form:



Note that there is an inherent UV cutoff in our action (2.408) since $\widetilde{\phi}(k)$ is restricted to $|k_\mu| \leq \pi$.

The generalized statistical Ising model is therefore equivalent to a D-dimensional Euclidean scalar field theory with SO(D)-breaking higher derivative kinetic terms, and a non-Euclidean invariant UV cutoff. At distances $L \gg$ the lattice spacing, the Poincaré symmetry breaking effects are negligible.

On the other hand, at generic β (or T), at large distances there is also no nontrivial spin correlation function apart from spontaneous magnetization for $\beta > \beta_C = 1/T_C$. The only nontrivial long distances correlations arise when $\beta \to \beta_C$. In particular, at $\beta = \beta_C$, long distance spin correlators are the *same* (up to a rescaling of variables) as the renormalization group fixed point of D-dimension ϕ^4-theory for $D = 2, 3$ dimensions.

Note that the Ising path integral a priori differs from that of ϕ^4 theory in two important ways. The first is that the lattice action $S[\phi]$ (2.408), contains many more couplings such as ϕ^6 or $(\partial\phi)^4$ and so on. The second is that the lattice path integral comes with a finite UV cutoff $\Lambda \sim 1/a$ where a is the lattice spacing.

All of these connections can be summarized quite nicely with the following visual ([**AC:**] I can't make this one in Tikz yet, will take some time):

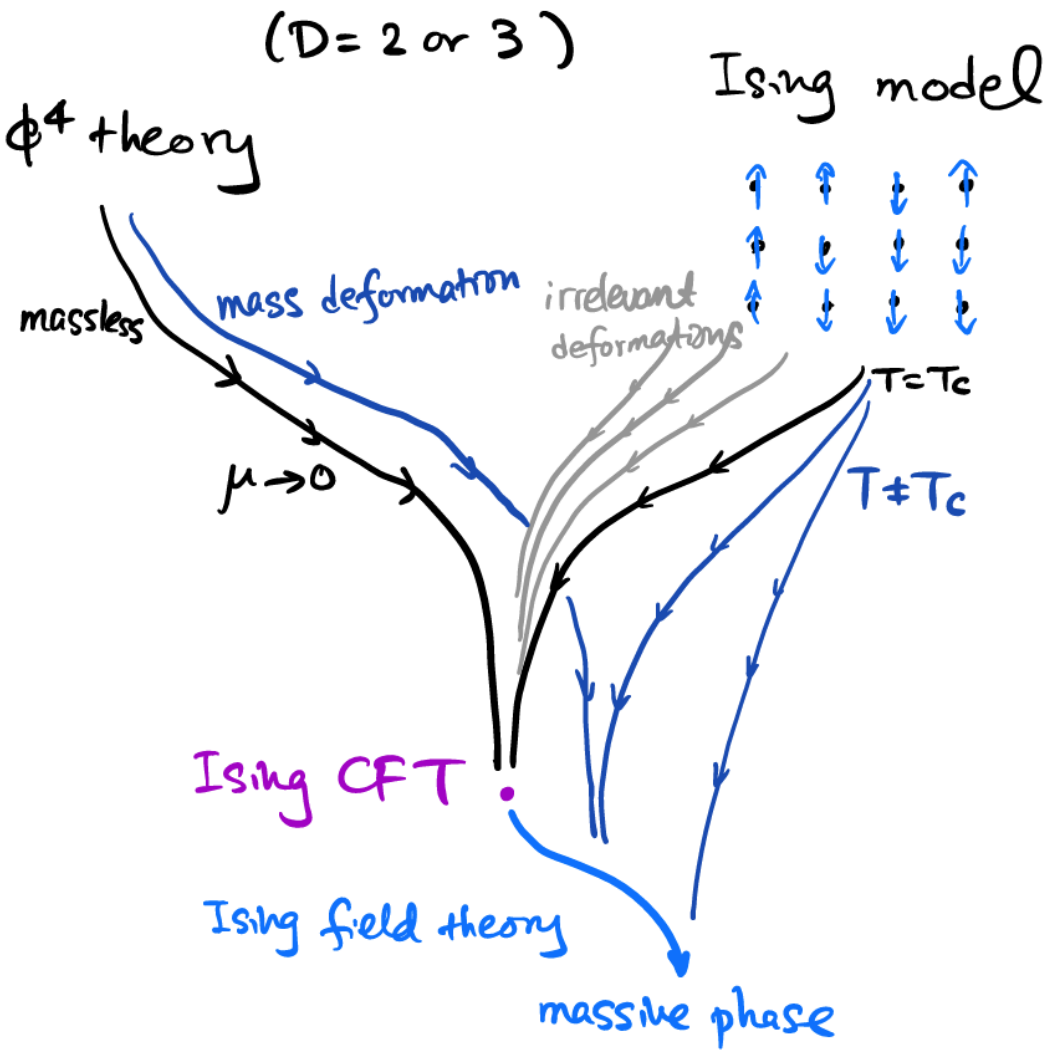


Figure 27. Renormalziation group flow graph relating ϕ^4-theory in $D = 2, 3$ dimensions to that of the Ising model with common points being Ising CFT.

The best way to prove this relation is with some intuition we will borrow for now, but look at very closely soon, is from the Wilsonian picture of the renormalization group (flow). This graph can be thought of as on the field-theory side, we are graphing vertically the value of μ, the energy-scale that we used to describe the physics of the running couplings and can be set to a given experimental value. In this case, we are taking it into the IR regime, which we can see on the statistical physics side of things, can be equated with a reaching the Curie temperature of a given system, namely the Ising model. The point at which these two regimes meet up is what we found to be Ising CFT, and thus beyond that, they are one of the same in Ising field theory.

The universality of Ising field theory can be understood from the fact that the Ising CFT has only two nontrivial local operators of scaling dimension $\Delta \leq D$, namely the

spin-field $\sigma(x) \leftrightarrow \phi(x)$, and the energy-operator $\varepsilon(x) \leftrightarrow \phi^2(x)$ (in relation to ϕ^4-theory).

Other local deformations of ϕ^4-theory amounts to inserting $e^{-\Delta S}$ into correlators where

$$\Delta S = \int d^D x \sum_I \Delta g_I \cdot \mathcal{O}_I \tag{2.409}$$

where the mass dimension of operator I is $\Delta_I > D$, and the dependence on the coupling Δg_I scales as $\sim \Delta g_I \cdot \mu^{-\Delta_I - D}$ disappears in the IR limit corresponding to $\mu \to 0$ (again, μ is the mass/energy scale of physical observable of interest).

On the other hand, the $\mathbb{Z}_s$-invariant energy-operator ε deformation has the form

$$\Delta S = \int d^D x \, \delta m^2 \cdot \varepsilon(x) \tag{2.410}$$

where here δm^2 is some function of T for $T \approx T_C$ (but I wrote it in a suggestive manner to relate back to field theory). We expect δm^2 to generically behave as $\delta m^2 \propto (T - T_C)$ for T close to T_C. The effective mass dimension of δm^2 is $D - \Delta_\varepsilon$, and thus implies the following correlation length

$$\xi(T) \propto (\delta m^2)^{-\frac{1}{D-\Delta_\varepsilon}} \propto (T - T_C)^{-\frac{1}{D-\Delta_\varepsilon}} \tag{2.411}$$

where $(D - \Delta_\varepsilon)^{-1}$ is one of the critical exponents.

2.2.7 Wilson-Polchinski and Renormalization Group Flow

Ok, we are now finally in a position to understand *Wilsonian effective field theory* and everything that comes along with it. As it currently stands, the Wilsonian(-Polchinski) perspective on the renormalization group and renormalization group flow is the most advanced and best understanding we have of QFT's and especially *effective field theories* which the most famous example of them all is the Standard Model (which is just a fancy EFT, it just so happens to be the EFT that governs our lives modulo gravity and well, a lot of other unanswered questions). And so I implore everyone to read Polchinski's paper from 1983 [17], as this is essentially the modern/today's perspective.

To discuss Wilsonian effective field theory, we will work in the Euclidean signature and we will illustrate this methodology with an example of a single scalar field, but do note that it is straightforward (in this regime) to generalize to multiple spinning fields.

The *Wilsonian effective action* denoted $S_\Lambda[\phi]$ is a functional of the following form

$$S_\Lambda[\phi] = S_{\text{kin},\Lambda}[\phi] + L_\Lambda[\phi] \tag{2.412}$$

where $S_{\text{kin},\Lambda}$ is the "kinetic term with a UV cutoff,"

$$S_{\text{kin},\Lambda} = \frac{1}{2} \int \frac{d^D k}{(2\pi)^D} \widetilde{\phi}(-k) \frac{k^2 + m^2}{\chi(k^2/\Lambda^2)} \widetilde{\phi}(k) \tag{2.413}$$

where $\chi(k^2/\Lambda^2)$ is a suitably chosen cutoff function that is of the general form:

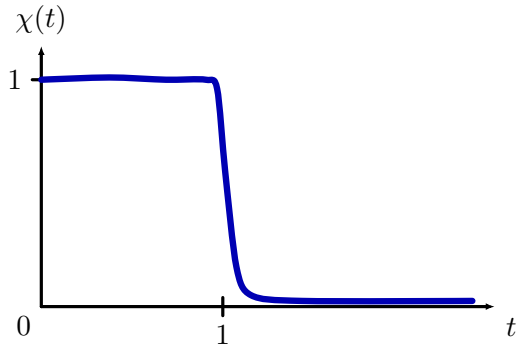
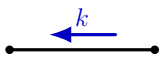


Figure 28. Cutoff function $\chi(t)$ utilized in the kinetic term of the Wilsonian action.

In perturbation theory, we will treat L_Λ as an interaction, and the corresponding Feynman rule for the propagator ϕ that can be read off from $S_{\text{kin},\Lambda}$ is



$$= \frac{\chi\left(\frac{k^2}{\Lambda^2}\right)}{k^2+m^2} \tag{2.414}$$

from which we can see that (2.414) agrees with the usual momentum space Feynman rule $(k^2+m^2)^{-1}$ for $|k|<\Lambda$ and falls off to 0 sufficiently fast for $|k|>\Lambda$.

Via the view of perturbation theory, the interaction piece L_Λ is a functional of the general form

$$L_\Lambda[\phi] = \sum \frac{1}{n!}\int \prod_{i=1}^{n-1} \frac{d^D k_i}{(2\pi)^D} L_\Lambda^{(n)}(k_1,\cdots,k_{n-1})\cdot\widetilde{\phi}(k_1)\cdots\widetilde{\phi}(k_{n-1})\widetilde{\phi}(-k_1-\cdots-k_{n-1}) \tag{2.415}$$

which you should notice has a very similar structure to the quantum effective action in (2.188). Greens functions are computed in the usual way from the path integral of which we will denote as the Wilsonian path integral as

$$Z_\Lambda[J] = \int [D\phi]\, e^{-S_\Lambda[\phi]-\int d^D x\, J(x)\phi(x)}. \tag{2.416}$$

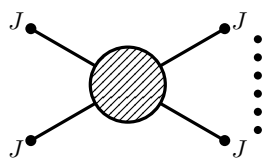
The idea now we want to investigate is that let us consider *moving* the cutoff Λ to a different cutoff such that it obeys $\Lambda'<\Lambda$ where Λ' is the new cutoff. But in doing so, we will be replacing $L_\Lambda \to L_{\Lambda'}$ but enforcing that the partitions functions remain the same $Z_{\Lambda'}[J]=Z_\Lambda[J]$ for $\widetilde{J}(k)$ supported at $|k|<\Lambda'$. The idea is that even if we move scales, the function we derive observables from (n-point Greens functions) should remain the same.

In other words, Wilsonian effective theories defined at different *floating cutoffs* Λ versus Λ' produce the ***same*** low energy/momentum observables.

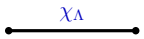
To see this, let us compare the propagators at the two different energy scales Λ and Λ' by literally $\pm\,\chi_{\Lambda'}$ on each side

$$\frac{\chi_\Lambda(k)}{k^2+m^2} = \frac{\chi_{\Lambda'}(k)}{k^2+m^2} + \frac{\chi_\Lambda(k)-\chi_{\Lambda'}(k)}{k^2+m^2} \tag{2.417}$$

where for shorthand $\chi_\Lambda(k) \equiv \chi(k^2/\Lambda^2)$. If we consider a general diagram such as



(2.418)

where the dots denote any n-number of external legs, within the shade/blob, we replace everything propagator usually 

with instead

(2.419)

where we will view this sum of propagagators as instead the sum of two *independent* propagating fields ϕ' and $\hat{\phi}$. We will then integrate out $\hat{\phi}$ to produce an effective theory of only ϕ' fields. Also, to get a sense of how the cutof function behaves for two different energy scales, we can refer to the graph below:

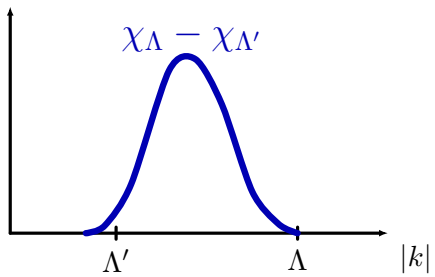


Figure 29. Graph of the difference between cutoff functions $\chi_\Lambda - \chi_{\Lambda'}$ with respect to the momentum $|k|$. Another way to think about this graph is that this difference is acting as a *window function* for which momenta we can probe (while still being finite).

More precisely, the integrating out of the $\hat{\phi}$ field goes as the following. We start with the separating out the partition function

$$Z[J] = \int [D\phi'] : e^{-S_{\text{kin},\Lambda'}[\phi']} \int [D\hat{\phi}]\, e^{-\hat{S}_{\text{kin}}[\hat{\phi}]} \times e^{-L_\Lambda[\phi'+\hat{\phi}] - \int J\cdot(\phi'+\hat{\phi})} \tag{2.420}$$

where the source term for $\hat{\phi}$ will drop out because the source $\widetilde{J}(k)$ is supported only at $|k| < \Lambda'$ by assumption. Note here that the kinetic action for $\hat{\phi}$ is given as

$$\hat{S}_{\text{kin}}[\hat{\phi}] = \frac{1}{2}\int \frac{d^D k}{(2\pi)^3} \widetilde{\hat{\phi}}(-k)\widetilde{\hat{\phi}}(k) \frac{k^2+m^2}{\chi_\Lambda(k) - \chi_{\Lambda'}(k)} \tag{2.421}$$

where we notice that (2.421) is quadratic in $\hat{\phi}$, and can thus be integrating out. We are then left with

$$Z[J] = \int [D\phi']\, e^{-S_{\text{kin},\Lambda'}[\phi'] - L_{\Lambda'}[\phi']} \tag{2.422}$$

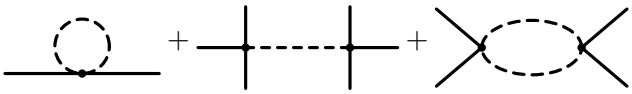
provided that the Λ' Lagrangian in the exponential can be written as

$$e^{-L_{\Lambda'}[\phi']} = \int [D\hat{\phi}] e^{-\hat{S}_{\text{kin}}[\hat{\phi}] - L_\Lambda[\phi' + \hat{\phi}]}. \tag{2.423}$$

For example, the L_Λ Lagrangian can give terms that look like

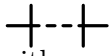
$L_\Lambda \supset$ $+ \cdots$ (2.424)

while the $L_{\Lambda'}$ Lagrangian will contain Feynman diagram such as

$L_{\Lambda'} \supset$  $\cdots$ (2.425)

If we take $\Lambda' = \Lambda - \delta\Lambda$ to first order in $\delta\Lambda$, the variation of the Lagrangian as $\delta L_\Lambda[\phi] = L_\Lambda[\phi] - \Lambda_{\Lambda'}[\phi]$, can be expressed in a form analogous to the renormalization group equation namely

$$\Lambda \frac{\partial}{\partial \Lambda} L_\Lambda[\phi] = \int d^D k \, \frac{(2\pi)^D}{k^2 + m^2} \Lambda \frac{\partial}{\partial \Lambda} \chi\left(\frac{k^2}{\Lambda^2}\right) \cdot \frac{1}{2} \left(\frac{\delta L_\Lambda}{\delta \widetilde{\phi}(k)} \frac{\delta L_\Lambda}{\delta \widetilde{\phi}(-k)} - \frac{\delta^2 L_\Lambda}{\delta \widetilde{\phi}(k) \delta \widetilde{\phi}(-k)} \right) \tag{2.426}$$

where in the parenthesis, the first terms contains the tree-level type of diagrams like  or any other number of external legs, while the second term contains loop diagrams with any number of external legs such as again with any number of external legs. Equation (2.426) is what is called the *Wilson-Polchinski renormalization group equation.*

As an example, to really show off the power of this formulation of the renormalization group, let us consider a massless scalar ϕ in $D = 4$ and suppose that we have the following interaction in the Wilsonian action

$$L_\Lambda[\phi] \supset \int d^4x \left(g_4 \phi^4 + g_6 \phi^6 \right) \tag{2.427}$$

where, based on the dimensions of the scalar field in $D = 4$, the dimensionless couplings are $\lambda_4 \equiv g_4$ and $\lambda_6 \equiv \Lambda^2 g_6$ since the scalar in $D = 4$ has mass dimension of M^1. We know that the corresponding β-equation will be of the following form

$$\Lambda \frac{d\lambda_4}{d\Lambda} = -a\lambda_6 + \cdots \tag{2.428}$$

$$\Lambda \frac{d\lambda_6}{d\Lambda} = 2\lambda_6 + b\lambda_4^2 \tag{2.429}$$

where the first term λ_6 represents a 6-point 1-loop term while the $b\lambda_4^2$ term represents the 6-point single exchange tree level diagram we saw prior. A *naive* truncation to (λ_4, λ_6) renormalization group flow to the IR (i.e. when $\Lambda \to 0$) looks like the following graph:

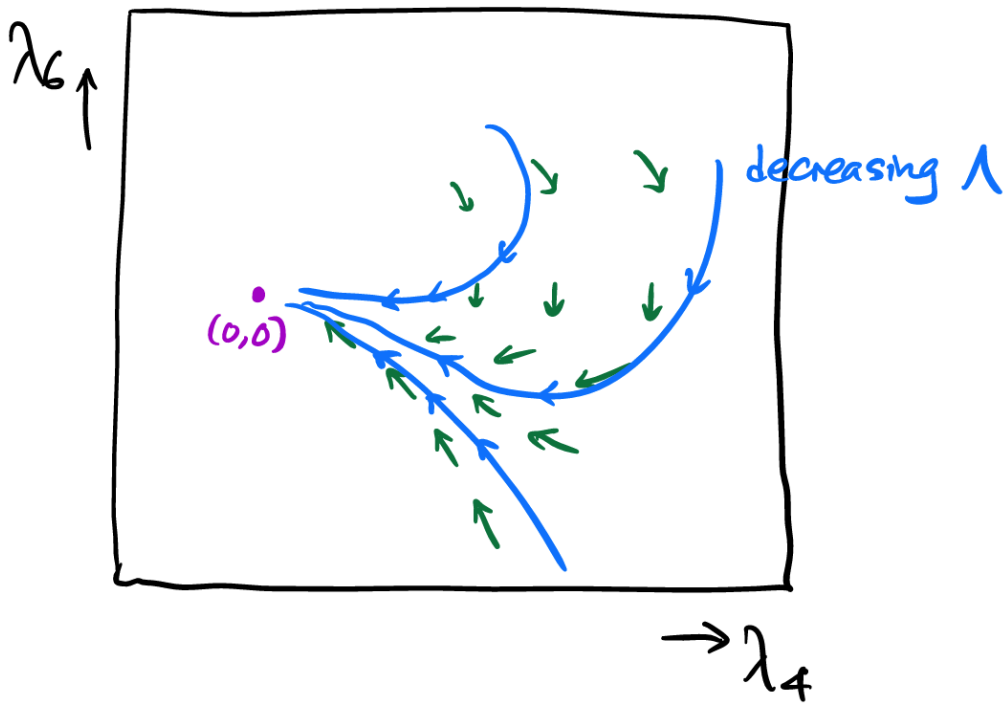


Figure 30. Graphical representation of the coupled differential equations in (2.429) showing the naive flow of the dimensionless couplings to the IR.

For small $\lambda_6 \sim \mathcal{O}(\lambda_4^2)$, the renormalization group equation is

$$\frac{d}{d\log\Lambda}\left(\lambda_6 + \frac{b}{2}\lambda_4^2\right) = 2\left(\lambda_6 + \frac{b}{2}\lambda_4^2\right) + \mathcal{O}(\lambda_4^3) \implies \lambda_6 + \frac{b}{2}\lambda_4^2 \propto \Lambda^2. \tag{2.430}$$

Now, if we have λ_6 quickly approach $-b\lambda_4^2/2$ as $\Lambda \to 0$, then

$$\frac{d}{d\log\Lambda}\lambda_4 \approx \frac{ab}{2}\lambda_4^2 + \mathcal{O}(\lambda_4^3) \tag{2.431}$$

as expected in the 1PI renormalization group equation. Now, in the Wilsonian effective theory framework, $S_\Lambda[\phi]$ at large Λ may or may not exist since the renormalization group is defined only for Λ. However, on the other hand, we can also rephrase *renormalizability* in the Wilsonian language in the following manner.

The regularized path integral of a renormalizable QFT defined with a UV cutoff Λ_0 may also be viewed as a special case of a Wilsonian path integral, where the Wilsonian effective action $S_{\Lambda_0}[\phi]$ coincides with the bare action. In particular, $L_{\Lambda_0}[\phi]$ contains the interaction terms whose coefficients are the bare couplings. The Wilson-Polchinski renormalization group equation then constructs $S_\Lambda[\phi]$ for $\Lambda < \Lambda_0$. Schematically what this looks like is that we have the bare Lagrangian which leads to under construction of the renormalization group

$$L_{\Lambda_0}[\phi] = \int g_I^{\text{bare}}\mathcal{O}_I[\phi] \rightsquigarrow L_\Lambda[\phi] = \int g_I(\Lambda)\mathcal{O}_I[\phi] + \text{many more terms} \tag{2.432}$$

where the operators in both cases are the same set of operators, and where $g_I(\Lambda)$ can be viewed as physical couplings, as physical observables such as Greens functions at momenta $|k| < \Lambda$ are computed as finite functions of $g_I(\Lambda)$ via the Wilsonian path integral with floating cutoff Λ.

We can denote $L_\Lambda[\phi]$ that results from the UV action as

$$L_\Lambda(g_I^{\text{bare}}, \Lambda_0)[\phi]. \tag{2.433}$$

Renormalizability is then equivalent to the statement that the following limit exists

$$\lim_{\Lambda_0 \to \infty} L_\Lambda(g_I^{\text{bare}}, \Lambda_0) \Big|_{\substack{g_I^{\text{bare}} \text{ chosen such that} \\ g_I(\Lambda) = g_I \text{ fixed}}}. \tag{2.434}$$

Let us illustrate this through the toy model of a truncated Wilsonian renormalization group equation with the same $D = 4$ ϕ^4-theory we saw prior with the Lagrangian

$$L_\Lambda[\phi] \supset \int g_4(\Lambda)\phi^4 + g_6(\Lambda)\phi^6 \tag{2.435}$$

where again we have the following dimensionless running couplings $\lambda_4(\Lambda) \coloneqq g_4(\Lambda)$ and $\lambda_6(\Lambda) \coloneqq \Lambda^2 \lambda_6(\Lambda)$. The toy-model we will be looking at is the following set of differential renormalization group equations

$$\frac{d\lambda_4}{d\log\Lambda} = \beta_4(\lambda_4, \lambda_6) \tag{2.436}$$

$$\frac{d\lambda_6}{d\log\Lambda} = 2\lambda_6 + \beta_6(\lambda_4, \lambda_6). \tag{2.437}$$

At the UV cutoff scale Λ_0, $\lambda_4(\Lambda_0) = \lambda_4^{\text{bare}}$ and $\lambda_6(\Lambda_0) = 0$ where the last one vanishes by assumption. We also have the dimensionless couplings at the physical scale Λ_R denoted as $\lambda_4(\Lambda_R)$ and $\lambda_6(\Lambda_R)$ with both being nontrivial and determined by λ_4^{bare}. The question of renormalizability then lies in determining, does the following limit exist:

$$\lim_{\Lambda \to \infty} \lambda_6(\Lambda_R) \Big|_{\lambda_4(\Lambda_R) = \lambda_4^R \text{ fixed}}. \tag{2.438}$$

This limiting procedure can be illustrated with the [**AC:**] beautiful picture given below:

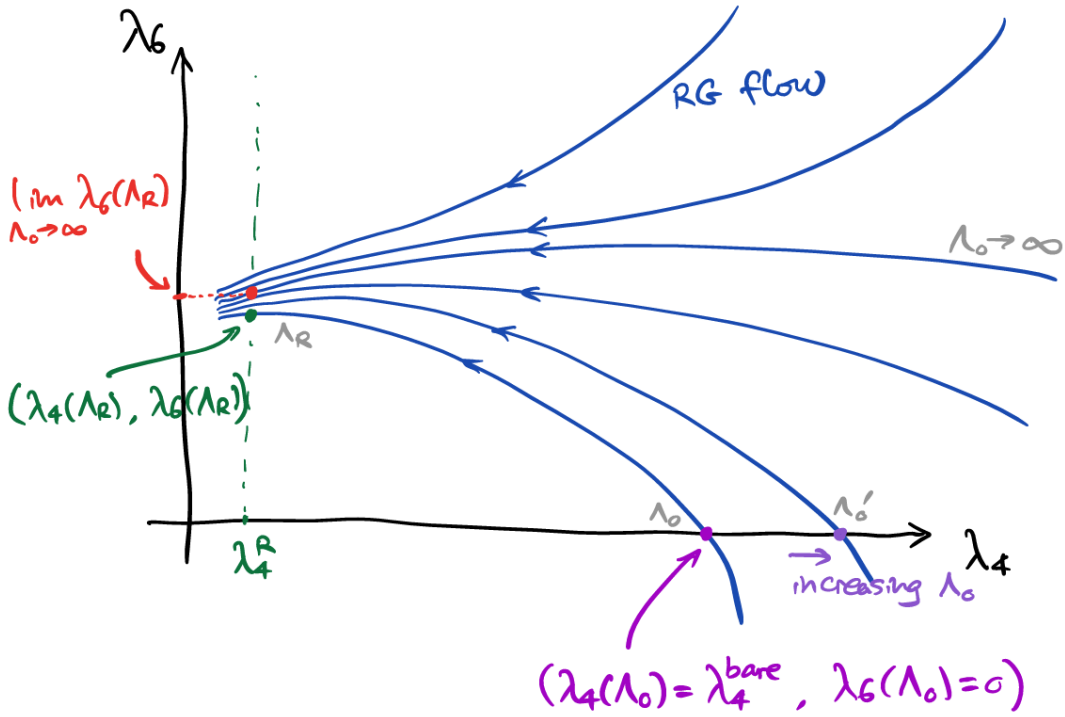


Figure 31. Graph of the renormalization group equations in (2.437) under the Wilson-Polchinski renormalization group flow.

This absolutely amazing graph displays how at the physically measurable energy scale Λ_R is met at which flow line/section and for what values of λ_4 and λ_6 would you find a theory of such! If we then follow the renormalization group flow lines, we can see which lines become *finite*, and thus renormalizable(!!) and which ones are not finite and diverge and are thus non-renormalizable! So, we learn that renormalization is not just a game of power counting theories and relation back the dimension of the coupling (all though this is still a good initial check, like how in gravity the coupling is Newton's constant which has mass M^{-2}, but that is not the whole point)!

Now, the $\lim_{\Lambda_0\to\infty} \lambda_6(\Lambda_R)$ exists provided that the derivatives $\partial\beta_6/\partial\lambda_6$ and $\partial\beta_4/\partial\lambda_4$ are sufficiently small along the renormalization group trajectory, and that $\beta_4(\lambda_4(\Lambda), \lambda_6(\Lambda))$ does not vary too fast with Λ, or in other words, have a parametrically small anomalous dimension. This explains why naively-power-counting renormalizable theories are *actually* (perturbatively) renormalizable.

Note that the same analysis applies to the situation where we allow *finite* nonzero $\lambda_6(\Lambda_0 \Leftrightarrow g_6^{\text{bare}} = \Lambda_0^{-2}\lambda_6(\Lambda_0)$, as is the case in typically lattice QFT action where Λ_0^{-1} is the lattice spacing. This is the logical foundation of lattice QFT.

This wraps up this section of QFT 2 notes, namely everything on the renormalization group. Before moving onto Yang-Mills theory and quantum chromodynamics, let us give a brief review of all that we went over and saw.

1. We first wrote down a condition or renormalizability that stemmed from ensuring that the effective running couplings are finite when we take the cutoff to infinity. This was fine for the time being but very primitive in the sense that we relied on power counting individual diagrams and mass dimensions of operators to which we had no footing on to show that this was all there was going on and could rely on it. Nonetheless, we saw the power of counter terms and computed a few and even saw what happens at the 2-loop order. This lead to the BPHZ theorem.

2. Next, we sought out something better and more intuitive namely the renormalization group which lead us to write down the Gell-Mann-Low equation for 1PI diagrams that allows us to derive quantum effective actions and also gave us the β-equation which allowed us to determine the behavior of a given theory at high and low energies for a certain operator at just 1-loop!

3. Armed with the β-function, we then investigated particular fixed points in the β-equation and discovered a flow of the sorts that allowed us to find out that fixed points in the renormalization group or β-equation pointed towards a new class of theories called conformal field theories. We will actually see that this does not only imply conformal theories, i.e. theories with scale invariance, but also asymptotic freedom! It should also be noted that fixed points in the UV are also theories that exhibit conformal symmetry. We then saw how this leads to a naturally a relation, almost a derivation, of Lattice theory in relation to the Ising model.

4. Finally, we saw how the renormalization group can instead be rephrased in terms of the action and Lagrangian themselves, and an interpretation of the renormalization group flow was established. What is so amazing about the Wilson-Polchinski flow is that the flow itself, of the running couplings and thus of the action/Lagrangian itself, has various different possibilities as it travels along the energy scale Λ from which only certain paths in the flow are realizable version *of said theory* that are finite when $\Lambda \to 0$. And then, we could even see at what energy scales we wanted to probe, Λ_R, would said theory exist, and if the values of the running couplings would be finite and thus observable (also what range)!

And now, time for some Yang-Mills theory.

2.3 Quantum Chromodynamics

The entire goal of formulating what is known as *Yang-Mills theory* is to formulate a theory of general *gauge fields*. Therefore, our primary motivation as *field theorists* will be wondering about how we can construct fields/particles that have an internal (local) gauge symmetry; not exactly to model some specific class of particles. The realization that these correspond to a particular physical theory will come later.

2.3.1 Group Theory Tools for Yang-Mills Theory

As in every case/class of particles, we start by formulating a classical theory of *multiple* gauge fields denoted as

$$A_{a\mu} \tag{2.439}$$

where a here denotes the internal[80] "gauge" index while μ here is the usual spacetime index vector taking on values $\mu = 0, 1, \cdots, D-1$. The simplest possible nonlinear *infinitesimal* gauge transformation is of the form[81]

$$\delta A_{a\mu}(x) = \partial_\mu \zeta_a(x) + f^{bc}{}_a A_{b\mu}(x)\zeta_c(x) \tag{2.440}$$

where $\zeta_a(x)$ are arbitrary functions of x^μ and $f^{bc}{}_a$ are called *structure constants* for the given Lie group at hand. Such a gauge transformation will closed[82] among themselves if we organize $A_{a\mu}$ into a Lie-algebra valued vector field, we can re-write the gauge field as

$$A_\mu(x) \equiv \sum_a A_{a\mu} t^a \tag{2.441}$$

where t^a are (for now) abstract generators of a Lie algebra where the t^a's obey

$$[t^a, t^b] = \sum_c i f^{ab}{}_c t^c. \tag{2.442}$$

Now, we can express the infinitesimal gauge transformation (2.440) as

$$\delta A_\mu(x) = \partial_\mu \zeta(x) - i[A_\mu(x), \zeta(x)] \tag{2.443}$$

where $\zeta(x) \equiv \sum_a \zeta_s(x) t^a$, or even better yet, the infinitesimal gauge transformation can be written in a *finite* form of gauge transformations

$$A_\mu(x) \mapsto A'_\mu(x) = g(x) A_\mu(x) g^{-1}(x) + i g(x) \partial_\mu g^{-1}(x) \tag{2.444}$$

[80] By internal gauge, what I really mean here is that the normal vector field here has been "dressed" up in connection with some group, specifically a Lie group, that characterized how the internal structure of the vector behaves. The best way I like to visualize this is that if we take a normal momentum vector, and make it the size of a car or just a box moving at you, then when it hits you, the momentum vector impacts on you, and we know how that goes. But, if we "dress-up" this box or car internally with some new gadgets that we will call a U(1) group, lets see what happens. The U(1) group is just a fancy circle, but we are placing it in a 3D space, and so if we partition said 3D space into little boxes, we fill each box with a circle or *element* of the U(1) group. Now, each element can only rotate clockwise or counter-clockwise. If the net amount of circles are rotating clockwise, then the overall object on the *outside* will spin as such, and vice versa for the other way, and if the net of the circles rotating all cancel out, then it is a polarization of 0 for the spin-1 car/box! But, if the car or box is massless, then well, that is a bit hard to visualize and I'll leave us there. But, if you are into differential geometry, a U(1) gauge field makes a honeycomb shape on a Minkowski spacetime. Also, I gave an example of U(1), but the analogy holds (somewhat).

[81] The structure constants here encodes the structure of the Lie algebra. The most common example in physics is that the structure constants for the Lie algebra of SO(3), often denoted as so(3) instead, is $i\hbar$ namely $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ for rotations.

[82] Here, the wording, "closed among themselves," is a reference to that the gauge transformations elements will never be outside of the group, i.e. always within the given elements of the group.

where $g(x)$ here is the exponentiation of the Lie algebra element

$$g(x) \equiv e^{i\sum_a \zeta_a(x)t^a}. \tag{2.445}$$

Note that due to the Baker-Campbell-Hausdorff (BCH) relation

$$\log\Big(e^X e^Y\Big) = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}[X,[X,Y]] - \frac{1}{12}[Y,[X,Y]] + \cdots \tag{2.446}$$

the product of (what we will poorly call) elements $g \cdot h$ is well-defined given that $g = e^{i\zeta_a t^a}$, $h = e^{i\eta_a t^a}$, and the Lie algebra commutator $[t^a, t^b] = if^{ab}{}_c t^c$ are well defined. In particular, given the inverse element $g^{-1}(x) = e^{-i\sum_a \zeta_a(x)t^a}$, and using the relation (2.446) written as

$$e^X Y e^Y = Y + [X,Y] + \frac{1}{2!}[X,[X,Y]] + \cdots + \frac{1}{n!}[X,[X,\cdots,[X,Y]]] + \cdots \tag{2.447}$$

where there are n-X's in the last commutator, we can express $g(x)A_\mu g^{-1}(x)$ entirely as a linear combination of the Lie algebra's generators t^a's. Thus, the transformed gauge field $A'_\mu(x)$ takes the form $A'_{a\mu}(x)t^a$ as needed.

Let us now define the corresponding *field strength* $F_{\mu\nu}$ as

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu] \\ &\equiv \sum_a F_{a\mu\nu} t^a. \end{aligned} \tag{2.448}$$

Unlike in Abelian gauge theory, $F_{\mu\nu}(x)$ is generally *not* gauge invariant. However, $F_{\mu\nu}$ transforms in a simple way under gauge transformation. If we rewrite the field strength as

$$F_{\mu\nu} = i\left[\partial_\mu - iA_\mu, \partial_\nu - iA_\nu\right] \tag{2.449}$$

and using that $\partial_\mu - iA'_\mu = g(x)(\partial_\mu - iA_\mu)g^{-1}(x)$ (where the multiplication here should be seen as differential operators), we can see that the field strength transforms as

$$F'_{\mu\nu}(x) = g(x) \cdot F_{\mu\nu}(x) \cdot g^{-1}(x). \tag{2.450}$$

To construct a gauge-invariant Lagrangian density, we need a notion of the "trace" on said Lie algebra, namely let us define the trace for *some* Lie algebra as

$$\mathrm{tr}\Big(t^a t^b\Big) \equiv d^{ab} \tag{2.451}$$

such that the trace is symmetric under $a \leftrightarrow b$, and have the typical trace-cyclicity property

$$\mathrm{tr}\Big([t^c, t^a]t^b\Big) = \mathrm{tr}\Big(t^a[t^b, t^c]\Big). \tag{2.452}$$

It then follows that under the gauge transformation

$$\mathrm{tr}\Big(gXYg^{-1}\Big) = \mathrm{tr}(XY) \tag{2.453}$$

for any X, Y that are elements of the Lie algebra. Note that the structure constant can equivalently be written as

$$f^{abc} \equiv \sum_e f^{ab}{}_e d^{ec} - -i\,\mathrm{tr}\Big([t^a, t^b]t^c\Big) \tag{2.454}$$

which also shows that f^{abc} is completely antisymmetric with respect to a, b, and c.

With a trace defined, we can write down the *Yang-Mills Lagrangian* that is already inherently gauge invariant,

$$\mathcal{L} = -\frac{1}{2g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}). \tag{2.455}$$

So far, we have not specified the normalization of the trace operation tr or d^{ab}. to construct a unitary theory, it will be necessary to take $d^{ab} \succ 0$, namely d^{ab} is a positive definite matrix. If so, we can choose a linear basis for t^a such that

$$\mathrm{tr}\Big(t^a t^b\Big) \equiv d^{ab} = \frac{1}{2}\delta^{ab} \tag{2.456}$$

where the factor of $1/2$ is the default convention.

Before moving on, a quick math note. It turns out that such a trace exists if and only if the Lie algebra takes the form of $\mathcal{Y} = \oplus_i \mathcal{Y}$ where each $\mathcal{Y}_i$ is either $\mathbb{R}$ (namely abelian) or "simple and compact" of the type $ABCDEFG$[83]. As an important example, let us take the Lie algebra A_1 namely su(2) where the Lie algebra generators are now

$$t^a = \frac{1}{2}\sigma^2 \tag{2.457}$$

with $a = 1, 2, 3$ and the trace here is the ordinary trace for 2×2 matrices.

More examples of simple, compact Lie algebra is where we take the Lie algebra to be $\mathcal{Y} = A_{N-1}$ or in other words, su(N) which is isomorphic to the space of $N \times N$ Hermitian traceless matrices. The elements are $\zeta(x) = \zeta_a(x)t^a \in \mathcal{Y}$ and the exponentiation of them are $g(x) = e^{i\zeta(x)} \in G = SU(N)$ which is the group of $N \times N$ unitary matrices with $\det = 1$.

Note that in $D = 4$, there is another gauge-invariant Poincaré-invariant, and power-counting renormalizable term one can include in the Yang-Mills Lagrangian namely a term called the *θ-term*

$$\Delta\mathcal{L} = \frac{\theta}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}). \tag{2.458}$$

This deformation to the Lagrangian/action $\Delta\mathcal{L}$ is firstly, odd under parity or time-reversal. Also, it can be written as a total derivative of an expression in A_μ, and thus does not affect the equations of motion *nor* quantum perturbation theory, namely there is no Feynman vertex corresponding to the θ-term.

However, A_μ need not vanish as $x \to \infty$ for a field configuration of *finite action*, and thus $\Delta S = \int d^4x\,\Delta\mathcal{L}$ can be non-vanishing for an admissible field configuration that contributes to the path integral.There are therefore *non-perturbative effects* that depend on θ (to which we will return to later)[84]. Let us now consider Yang-Mills theory but with

[83] For a quick rundown of what these letters mean, these capital letters refers to the classification of simple Lie algebra where the capital letters identifies their Dynkin diagram. Type A_n refers to special linear Lie algebra so(2n+1), type B_n refers to special orthonogal so(n), type C_n are symplectic Lie algebras sp(2n), and type D_n corresponds to the even-dimensional special orthogonal Lie algebra so(2n). The other letters EFG refer to the exceptional Lie algebras. [**AC:**] My favorite is E_8....

[84] As a side comment for now since we won't get into it, this θ-term here was first introduced by Ed Witten in 1988 introducing *topological quantum field theory*, and it is sometimes also referred to as a Chern-Simons term when written in terms of A_μ. It's primary phenomenological use is to explain the strong CP problem in QCD.

matter fields. Let us denote a general matter field $\psi(x)$ that transforms under a finite gauge transformation parameterized by

$$g(x) = e^{i\zeta_a(x)t^a} \in G \tag{2.459}$$

as the following transformation

$$\psi(x) \mapsto \psi'(x) = \rho(g(x)) \cdot \psi(x) \tag{2.460}$$

where $\psi(x)$ is a column vector while $\rho(g(x))$ is a matrix valued object and ρ plays the role of associating a Lie group element $g \in G$ to a matrix such that $\rho(g \cdot h) = \rho(g) \cdot \rho(h)$ where $g \cdot h$ denotes the groups multiplication operation and the second product on the RHS is matrix multiplication. We want to write the matter fields for specific purposes, and to that end, a *representation* denoted R of a group G is a map

$$\rho_R : G \mapsto GL(V) \tag{2.461}$$

where $GL(V)$ is the set of invertible linear maps $V \to V$, but ρ_R is such that $\rho_R(g \cdot h) = \rho_R(g) \cdot \rho_R(h)$ for all $g, h \in G$. Correspondingly, a represntation of the Lie algebra $\mathcal{Y}$ is defined as

$$\widetilde{\rho}_R : \mathcal{Y} \mapsto \mathrm{End}(V) \tag{2.462}$$

where $\mathrm{End}(V)$ denotes also a set of linear maps $V \to V$, and as such the two representations are related by

$$\rho_R(g = e^{i\zeta_a t^a}) = e^{i\zeta_a \widetilde{\rho}_R(t^a)}. \tag{2.463}$$

In particular with respect to the generators of the Lie algebra, the representation obeys

$$\widetilde{\rho}_R([t^a, t^b]) = \left[\widetilde{\rho}_R(t^a), \widetilde{\rho}_R(t^b)\right]. \tag{2.464}$$

Let us now then define a Lie algebra generator of a particular representation as $t^a_R \equiv \widetilde{\rho}_R(t^a)$ that obeys a similar commutator algebra

$$[t^a_R, t^b_R] = i f^{ab}{}_c t^c_R. \tag{2.465}$$

As a concrete example, let the Lie group be $G = SU(N)$ and thus the corresponding Lie algebra is $\mathcal{Y} = su(N)$. A particular representation that we will be paying attention to for SU(N) is the *fundamental representation*, also called the *defining representation*, which is denoted as (just notation) $R = \square$. The vector space is then $V = \mathbb{C}^N$ and the representation is $\rho_\square(g) = g$ as a unitary $N \times N$ matrix acting on V, which can be exchanged with $\widetilde{\rho}_\square(t^a) = t^a$.

There is also the *anti-fundamental representation* denoted[85] $R = \overline{\square}$ (also of SU(N) still). The vector space is $V = \mathbb{C}^N$ and the representation is $\rho_{\overline{\square}}(g) = g^*$ where g^* denotes the complex conjugate but *without* the transpose, and of course it can be exchanged for the Lie algebra representation $\widetilde{\rho}_{\overline{\square}}(t^a) = -(t^a)^*$.

[85] If you really zoom in or squint really hard, you can see there is a bar over the box...

The final representation we will be interested in is the *adjoint representation* denoted as $R = \text{adj}$ with vector space $V = \mathcal{Y} \simeq R^{N^2-1}$. The representation of the Lie group $\rho_{\text{adj}}(g)$ is a map $V \to V$ that acts on $v \in V$ a Hermitian traceless matrix by

$$(\rho_{\text{adj}}(g))(v) = gvg^{-1} \Leftrightarrow (\widetilde{\rho}_{\text{adj}}(t^a))(t^b) = [t^a, t^b] = if^{ab}{}_c t^c. \tag{2.466}$$

In other words, acting on the basis t^a of $\mathcal{Y}$, $\widetilde{\rho}_{\text{adj}}(t^a) \equiv t^a_{\text{adj}}$ is the $(N^2-1)\times(N^2-1)$ matrix

$$(t^a_{\text{adj}})_c{}^b = if^{ab}{}_c. \tag{2.467}$$

Coming back to the matter fields, a matter field $\psi(x)$ in a representation R transforms under a gauge transformation now as

$$\psi(x) \to \psi'(x) = \rho_R(g(x)) \cdot \psi(x) \tag{2.468}$$

with the corresponding gauge covariant derivative[86]

$$D_\mu \psi(x) \equiv \partial_\mu \psi(x) - iA_{a\mu}(x) t^a_R \cdot \psi(x) \tag{2.469}$$

which under a gauge transformation transforms as

$$D_\mu \psi(x) \to (D_\mu \psi)'(x) = \rho_R(g(x)) \cdot D_\mu \psi(x). \tag{2.470}$$

Now, for the *particular case* of QCD or *quantum chromodynamics*, we shall take the matter fields to be fermionic spinor fields $\psi_{I\alpha}(x)$ in the fundamental representation $R = \square$ of $G = SU(N)$ where the fermionic spinor fields have the following components

$$\psi_{Ii\alpha}(x) \tag{2.471}$$

where the i index here denotes the *gauge index* $i = 1, \cdots, N$, α is the usual spinor index we saw prior, and I is called the *flavor index* $I = 1, \cdots, N_f$ where N_f denotes the number of possible flavors. The anti-fermionic spinor field, or for short the conjugate spinor, is denoted $\overline{\psi}^\alpha_I(x)$ which is a representation that lives in the anti-fundamental group $R = \square$ of SU(N).

The most general Poincaré invariant, power-counting renormalizable Lagrangian that we can write in $D = 4$ dimensions is the following Lagrangian (which is the QCD Lagrangian)

$$\begin{aligned}\mathcal{L} = &-\frac{1}{2g^2}\,\text{tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{\theta}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\,\text{tr}(F_{\mu\nu}F_{\rho\sigma}) \\ &- \overline{\psi}_I \gamma^\mu D_\mu \psi_I - m_{IJ}\overline{\psi}_I \psi_J - \widetilde{m}_{IJ}\overline{\psi}_I \gamma_5 \psi_J\end{aligned} \tag{2.472}$$

where the second line can be equated to just the matter Lagrangian $\mathcal{L}_\psi$, and the m_{IJ}'s here are Hermitian mass matrices. We could alternatively define the left and right spinor fields via the chiral matrix as

$$\psi_{IL} = \frac{1+\gamma_5}{2}\psi_I \quad \text{and} \quad \psi_{IR} = \frac{1-\gamma_5}{2}\psi_I \tag{2.473}$$

[86] That again, communicates the gauge-field/spin bundle with the flat space Minkowski geometry, hence we can call it a covariant derivative, just in this case instead of abelian, it is an su(N) bundle.

and so the matter sector becomes

$$\mathcal{L}_\psi = -\overline{\psi}_{IL}\not{D}\psi_{IL} - \overline{\psi}_{IR}\not{D}\psi_{IR} - M_{IJ}\overline{\psi}_{IL}\psi_{JR} - M^*_{IJ}\overline{\psi}_{JR}\psi_{IL} \tag{2.474}$$

where M_{IJ} here is now a complex mass matrix. Both of these formulations have their respective uses. This is all we need from the classical side of things to write down Yang-Mills and classical QCD Lagrangian. It should be noted though that physically, there is **no** classical analog in nature of these actions unlike Maxwell's Lagrangian for example.

Ok, so how might we quantize such a gauge theory then? Well, to quantize Yang-Mills theory, one *could* (in principal) turn to canonical quantization. Canonical quantization proceeds in the same way as in Abelian gauge theory. One encounters first class constraints (vanishing Poisson bracket) that ought to be eliminated by choice of gauge, after which one can proceed with the canonical quantization prescription.

However, one often encounters ambiguities in IR regularization due to unfixed residual gauge redundancies, as well as complications due to Lorentz-non-covariance of gauge conditions. As you might be able to guess at this point, this is not the method/prescription we will use.

We will adopt a more systematic covariant approach to quantization by gauge fixing the path integral a la Faddeev-Popov (that we have seen prior in QFT 1 5 with quantizing the photon or abelian gauge field) and a method called *Becchi-Rouet-Stora-Tyutin* (BRST) quantization procedure. The introduction to these topics now will be very general, and then afterwards we will apply the methods.

2.3.2 Quantization of Yang-Mills Theory (Faddeev-Popov & BRST Quantization plus Cohomology)

We shall begin with a "formal" path integral of a gauge theory written as

$$Z \sim \int [D\phi]\, e^{iS[\phi]} \tag{2.475}$$

where ϕ here collectively denotes gauge fields and possibly matter fields. We will denote a general gauge transformation as

$$\phi \mapsto \phi^\zeta \tag{2.476}$$

where ϕ^ζ denotes the gauge-transformed field, under which the action and the measure are invariant under the gauge transformation ζ,

$$S[\phi^\zeta] = S[\phi] \quad \text{and} \quad D\phi^\zeta = D\phi \tag{2.477}$$

(you can think of $D \cdot \phi$ or just $D\phi$ as a covariant derivative being contracted over with the proper number of indices that come from the *general* matter field ϕ). Here, ζ represents an element of the group $\mathcal{G}$ of gauge transformations. Note that $\mathcal{G}$ is typically infinite-dimensional, not to be confused with the "gauge group" of Yang-Mills theory:

$$\mathcal{G} \simeq \{g(x) : G - \text{valued functions on } \mathbb{R}^{1,D-1} \text{ spacetime}\}. \tag{2.478}$$

We denote the elements of the gauge transformation group by ζ^α for α-coordinates on $\mathcal{G}$. For example, in Yang-Mills theory, $\zeta \equiv g(x) = e^{i\zeta_a(x)t^a}$, and we may take $\zeta^{(a,x)} \equiv \zeta_a(x)$ where $a = 1, \cdots, \dim G$ and $x \in \mathbb{R}^{1,D-1}$.

The idea of the Faddeev-Popov procedure is that the matter fields $\phi \sim \phi^\zeta$ represent *equivalent* field configurations. Therefore, we wish to avoid such over counting of equivalent fields within the measure by restricting the functional integral through a gauge-fixing condition[87]

$$F^A[\phi] = 0. \tag{2.479}$$

A graph to describe this general graph is given below:

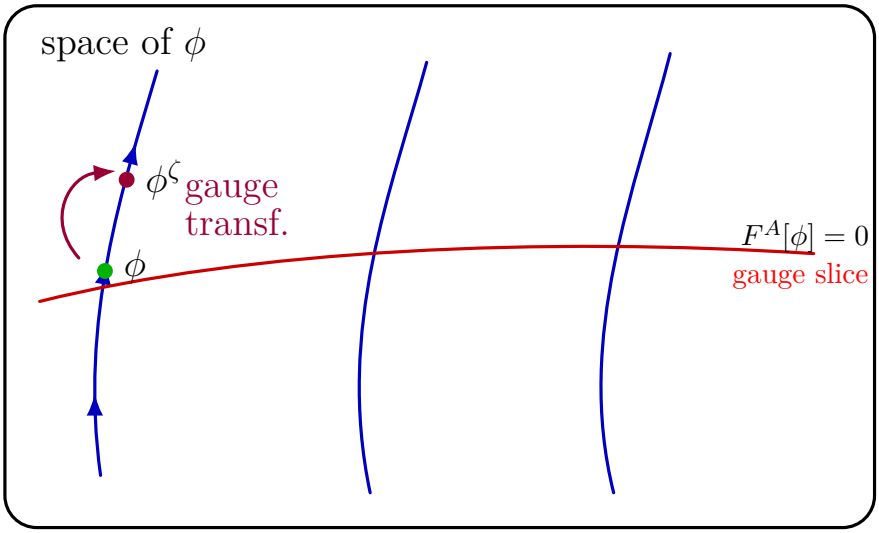


Figure 32. Here, each gauge orbit is an equivalent physical configuration for some matter field ϕ and ϕ^ζ where we account for only one configuration from each gauge orbit via the gauge slice $F^A = 0$.

But, we should restrict the functional integral in a way that respects gauge invariance of observables. This is achieved by a very cleverly constructed "1" which is at the heart of the Faddeev-Popov procedure,

$$1 = \int_{\mathcal{G}} D\zeta\, \delta(F^A[\phi^\zeta]) \cdot \det\left(\frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha}\right). \tag{2.480}$$

[87] We have already seen an example of F^A namely (1.579), $\partial_\mu A^\mu(x) - \varphi(x)$ for the Abelian photon.

When we insert this into the RHS of the path integral, we find

$$
\begin{aligned}
Z &\sim \int_{\mathcal{G}} D\zeta \int D\phi\, \delta(F^A[\phi^\zeta]) \det \frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha}\, e^{iS[\phi]} \\
&= \int_{\mathcal{G}} D\zeta \int D\phi^\zeta\, \delta(F^A[\phi^\zeta]) \det \frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha} e^{iS[\phi^\zeta]} \\
&= \int_{\mathcal{G}} D\zeta \int D\phi\, \delta(F^A[\phi]) \cdot \det \frac{\partial F^A[\phi^{\zeta'\circ\zeta^{-1}}]}{\partial \zeta'^\alpha}\bigg|_{\zeta'=\zeta} e^{iS[\phi]}
\end{aligned}
\tag{2.481}
$$

where in the third line we performed and inverse gauge transformation $\phi \to \phi^{\zeta^{-1}}$ of which the algebra goes $\phi^\zeta \to \phi^{\zeta^{-1}\circ\zeta} = \phi$, and we exchanged the integration over ϕ and the gauge transformation measure ζ. The factor involving ζ in the ϕ-integrand is

$$
\int_{\mathcal{G}} D\zeta\, \det \frac{\partial F^A[\phi^{\zeta'\circ\zeta^{-1}}]}{\partial \zeta'^\alpha}\bigg|_{\zeta'=\zeta} = \int_{\mathcal{G}} D\zeta \left(\det \frac{\partial(\zeta''\circ\zeta)^\alpha}{\partial \zeta''^\beta}\right)^{-1}\bigg|_{\zeta''=\mathrm{id}} \times \det\left(\frac{\partial F^A[\phi^{\zeta''}]}{\partial \zeta''^\alpha}\right)\bigg|_{\zeta''=\mathrm{id}}
\tag{2.482}
$$

where in going from the LHS to RHS, we set $\zeta' = \zeta''\circ\zeta$ since by the group algebra for the group of gauge transformations allows us to go back to the original gauge transformation (cycle property), and thus $\zeta' = \zeta \implies \zeta''\circ\zeta = \zeta \implies \zeta'' = \mathrm{id}$ where id is the identity element of the gauge transformation group. Also, the measure in purple is called the *Haar measure* on $\mathcal{G}$ denoted as $[D\zeta]_H$. The remaining term in blue is defined as the *Faddeev-Popov determinant* denoted as $\Delta_{\mathrm{FP}}[\phi]$. The Haar measure on the group of transformations $\mathcal{G}$ has the following defining property

$$
[D(\zeta\circ\zeta_1)]_H = [D\zeta]_H
\tag{2.483}
$$

which is valid for all fixed $\zeta_1 \in \mathcal{G}$. Formally, the integral over the Haar measure gives the volume of the gauge-space

$$
\int_{\mathcal{G}} [D\zeta]_H = \text{“volume of } \mathcal{G}\text{.”}
\tag{2.484}
$$

The appropriate definition for a QFT gauge theory is then

$$
\begin{aligned}
Z &= \frac{1}{\int_{\mathcal{G}}[D\zeta]_H} \int D\phi\, e^{iS[\phi]} \\
&= \int D\phi\, \delta(F^A[\phi]) \cdot \Delta_{\mathrm{FP}}[\phi] e^{iS[\phi]}.
\end{aligned}
\tag{2.485}
$$

Something that will be beneficial for us is that we can rewrite the Faddeev-Popov determinant by “un-doing” a *fermionic* Gaussian integral (since it is a whole power determinant not an integer)

$$
\begin{aligned}
\Delta_{\mathrm{FP}}[\phi] &= \det \frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha}\bigg|_{\zeta=\mathrm{id}} \\
&= \int Db_A\, Dc^\alpha\, e^{ib_A\cdot \det \frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha}\cdot c^\alpha}.
\end{aligned}
\tag{2.486}
$$

The fermionic fields introduced here, (b_A, c^α) are Grassmann-odd variables called *Faddeev-Popov ghosts*. Note that in the field theory context, $\alpha = (a, x)$ takes on continuous values, and A is an index that loosely speaking has the same "number of degrees of freedom," as α. Both a^α and b_A will be field variables in the sense that they depend on space-time coordinates for example, $c^\alpha \equiv c_a(x)$, and thus the integral measure $\int Db_A\, Dc^\alpha$, is a Grassmann-odd functional integral. The notion for summing over the indices $f_\alpha g^\alpha$ stands for

$$f_\alpha g^\alpha \equiv \sum\!\!\!\!\!\!\int d_\alpha\, f_\alpha g^\alpha. \tag{2.487}$$

For example, with $\alpha = (a, x)$ then the integral sum is equivalent to

$$f_\alpha g^\alpha \equiv \sum_a \int d^D x\, f_a(x) g^a(x). \tag{2.488}$$

In the definition of a QFT gauge theory (2.485), the gauge condition is implemented via the Dirac-delta, we know that we can undo the integration and write the delta function as

$$\delta(F^A[\phi]) = \int DB_A\, e^{iB_A F^A[\phi]} \tag{2.489}$$

where B_A are commuting field variables. Therefore, the total gauge-field path integral can equivalently be written as

$$Z = \int D\phi\, DB_A\, Db_A\, Dc^\alpha\, e^{i\left(S[\phi] + B_A F^A[\phi] + b_A \frac{\partial F^A[\phi^\zeta]}{\partial \zeta^\alpha}\Big|_{\zeta=\mathrm{id}} \cdot c^\alpha\right)}. \tag{2.490}$$

The total action can therefore be written as

$$S[\phi, B, b, c] = S[\phi] + S_{\text{G.F.}}[B, b, c] \tag{2.491}$$

where the last term is the gauge-fixing term with the sum of the $B_A F^A$ and ghost terms. But it should be noted that this total action is *not* gauge invariant as the gauge-fixing action $S_{\text{G.F.}}$ is not gauge invariant itself by construction. However, on the other hand, there is an emergent *fermionic global symmetry* known as the *BRST symmetry* which is a symmetry operator δ_B such that

$$\delta_B S[\Phi] \equiv S[\Phi + \delta_B \Phi] = S[\Phi] = 0 \tag{2.492}$$

where Φ denotes the matter and ghost fields $\Phi \equiv (\phi, B, b, c)$, and they have the corresponding transformation under the BRST operator δ_B:

$$\delta_B \phi = \epsilon c^\alpha \delta_\alpha \phi \qquad \text{with} \quad \delta_\alpha \phi \equiv \frac{\partial \phi^\zeta}{\partial \zeta^\alpha}\Big|_{\zeta=\mathrm{id}} \tag{2.493}$$

$$\delta_B B_A = 0 \tag{2.494}$$

$$\delta_B b_A == -\epsilon B_A \tag{2.495}$$

$$\delta_B c^\alpha = -\frac{1}{2}\epsilon \mathbb{F}^\alpha{}_{\beta\gamma} c^\beta c^\gamma \tag{2.496}$$

where the term $\epsilon c^\alpha \delta_\alpha \phi$ is the same as the infinitesimal gauge variation of ϕ except that the gauge parameter $\delta\zeta^\alpha$ is replaced by ϵc^α. Here, ϵ is a Grassmann-odd variational parameter, and hence ϵc^α is Grassmann-even as usual. Also, we have new structure constants with respect to the gauge-transformation group $\mathcal{G}$ written as $\mathbb{F}^\alpha{}_{\beta\gamma}$ that obeys the following commutator algebra

$$[\delta_\beta, \delta_\gamma] = \mathbb{F}^\alpha{}_{\beta\gamma}\delta_\alpha. \tag{2.497}$$

Now, given any functional $\mathcal{F}[\Phi]$, we will write the general transformation under the BRST symmetry operator δ_B as

$$\delta_B \mathcal{F}[\Phi] \equiv i\epsilon Q_B \cdot \mathcal{F}[\Phi] \tag{2.498}$$

where Q_B here takes the form of a first-order functional differential operator, and is sometimes called the *BRST charge*. However, a very key property of the BRST charge is that it is what is called *nilpotent*

$$Q_B^2 = 0. \tag{2.499}$$

As an example showing this property is true, consider that given $Q_B \cdot \phi = -ic^\alpha \delta_\alpha \phi$, then the square of Q_B^2 acting on the matter field ϕ is

$$\begin{aligned} Q_B^2 \cdot \phi &= Q_B \cdot (-ic^\alpha \delta_\alpha \phi) \\ &= -i(Q_B \cdot c^\alpha)\delta_\alpha \phi + ic^\alpha Q_B \cdot (\delta_\alpha \phi) \\ &= -i\left(\frac{i}{2}\mathbb{F}^\alpha{}_{\beta\gamma} c^\beta c^\gamma\right)\delta_\alpha \phi + ic^\alpha\left(-ic^\beta \delta_\beta \cdot (\delta_\alpha \phi)\right) \\ &= \frac{1}{2}\mathbb{F}^\alpha{}_{\beta\gamma} c^\beta c^\gamma \delta_\alpha \phi + c^\alpha c^\beta \delta_\beta \delta_\alpha \phi \\ &= 0 \end{aligned} \tag{2.500}$$

since $c^\alpha c^\beta = -c^\beta c^\alpha$. In a similar manner, you can also verify that $Q_B^2 \cdot c^\alpha = 0$ using the corresponding Jacobi identity for $\mathbb{F}^\alpha{}_{\beta\gamma}$ namely that $\mathbb{F}^\alpha{}_{\beta\gamma}\mathbb{F}^\gamma{}_{\rho\sigma} = 0$. To further show that the total action is invariant under δ_B, observe that the gauge fixing option can be re-written as

$$S_{\text{G.F.}} \equiv B_A F^A[\phi] + b_A \delta_\alpha F^A[\phi] c^\alpha = -iQ_B \cdot (b_A F^A[\phi]) \tag{2.501}$$

and therefore consequently, $Q_B \cdot S_{\text{G.F.}} = Q_B^2(\cdots) = 0$. Also, since we have that $Q_B \cdot S[\phi] = -ic^\alpha \delta_\alpha S[\phi] = 0$, the total gauge fixed action (2.491) is BRST-invariant or in other words, δ_B is a symmetry.

Continuing with our path integral quantization, as usual in the path integral, a quantum state $|\Psi\rangle$ is represented by a wave functional

$$\Psi\left[\phi, B, n, c\,\Big|_{\text{fixed time } x^0}\right]. \tag{2.502}$$

We will also promote the BRST charge to an operator $Q_B \to \hat{Q}_B$ defined as

$$\hat{Q}_B\,|\Psi\rangle := |Q_B \cdot \Psi\rangle \tag{2.503}$$

where the product here between $Q_B \cdot \Psi$ is the Q_B-variation of the wave functional Ψ at fixed time x^0. The insertion of a functional of a field variable, $\mathcal{F}[\phi]$, in the path integral

corresponds to acting on the state with an operator $\hat{\mathcal{F}}$. Then, the product operation between the BRST charge and functional of the field variable turns into commutator algebra under quantization

$$Q_B \cdot \mathcal{F}[\phi] \rightsquigarrow \left[\hat{Q}_B, \hat{\mathcal{F}}\right]. \tag{2.504}$$

Now, to investigate the available states to us under this formalism, we need to make a new definition. We shall define an *admissible (physical) state* $|\Psi\rangle$ is a state that obeys

$$\hat{Q}_B |\Psi\rangle = 0. \tag{2.505}$$

In essence, this is the requirement that a physical state must be invariant under the BRST charge/transformation. As a consequence, the transition amplitude $\langle\Psi_f| U(t_f, t_i) |\Psi_i\rangle$ between a pair of *admissible* states $|\Psi_i\rangle$ and $|\Psi_f\rangle$ is invariant with respect to deformations of the gauge fixing functional $F^A[\phi]$. The transition amplitude is therefore written as

$$\langle\Psi_f| U(t_f, t_i) |\Psi_i\rangle = \int [D\Phi(t)]_{t_i \leq t \leq t_f} \Psi_f^*[\Phi(t_f)]\, e^{iS[\Phi]} \Psi_i[\Phi(t_i)] \tag{2.506}$$

where the action used here is the gauge fixed action (2.491). Under a change of gauge condition $F^A[\phi] = 0$, in other words, we are making the following replacement

$$F^A[\phi] \to \widetilde{F}^A[\phi] = F^A[\phi] + \overline{\delta} F^A[\phi] \tag{2.507}$$

where $\overline{\delta} F^A[\phi]$ is some arbitrarily infinitesimal functional. The action then changed by

$$\overline{\delta} S[\Phi] = \overline{\delta} S_{\text{G.F.}} = -iQ_B \cdot (b_A \overline{\delta} F^A[\phi]) \tag{2.508}$$

and thus the transition amplitude changes as

$$\begin{aligned} \overline{\delta}\left(\langle\Psi_f| U(t_f, t_i) |\Psi_i\rangle\right) &= \int [D\Phi] \Psi_f^* e^{iS} \Psi_i\, Q_B(b_A \overline{\delta} F^A[\phi]) \\ &= \int [D\Phi]\, Q_B \cdot \left(\Psi_f^* e^{iS} \Psi_i\, b_A \overline{\delta} F^A[\phi]\right) \\ &= 0 \end{aligned} \tag{2.509}$$

where in the second to last line, $Q_B \cdot \Psi_f = Q_B \cdot \Psi_i = 0$ by assumption (that they are admissible states). The last line comes about by assuming the measure $[D\Phi]$ is also BRST-invariant). Note that a state of the form $\hat{Q}_B |\chi\rangle$ is admissible but in a *trivial* manner, namely that if the state goes as $|\Psi\rangle = \hat{Q}_B |\chi\rangle$ then of course $Q_B |\Psi\rangle = 0$ since $\hat{Q}_B^2 = 0$ or in other words, $\hat{Q}_B$ is nilpotent. In fact, the inner product between any admissible state $|\Psi\rangle$ and $\hat{Q}_B |\chi\rangle$ vanishes

$$\langle\Psi| \hat{Q}_B |\chi\rangle = \left(\langle\Psi|\hat{Q}_B\rangle\right) |\chi\rangle = 0 \tag{2.510}$$

where the state in blue vanishes under that $\langle\Psi|$ is an admissible state.

The statement then of BRST quantization is the following. *We declare*[88] that an admissible state $|\Psi\rangle$, i.e. a state such that $\hat{Q}_B |\Psi\rangle = 0$, and $|\Psi\rangle + \hat{Q}_B |\chi\rangle$ represent the

[88] I really wanted to say, "we the people under a more perfect quantization procedure..." but, anyway.

same physical state. That is, the Hilbert space of physical states, in the conventional quantum mechanical sense, is

$$\mathcal{H}_{\text{phys}} = \frac{\text{Ker}\,\hat{Q}_B}{\text{Im}\,\hat{Q}_B} \tag{2.511}$$

where $\text{Ker}\,\hat{Q}_B$ denotes the kernel of the BRST charge, and $\text{Im}\,\hat{Q}_B$ denotes the image of the BRST charge. The Hilbert space we wrote down *from the ground up(!)* in (2.511) is called the *BRST cohomology*. You will investigate/derive the BRST cohomology for Maxwell's theory (the photon) in the problem sets. To me, this result is just so amazing, it deserves the rest of the page.

Anyway, now let us finally apply all of this formalism to Yang-Mills theory, and quantize it properly to find a theory of the gluons (and later on for the quarks and quantum chromodynamics in total).

Applying the Faddeev-Popov procedure to the path integral of Yang-Mills theory, we will be using the following gauge-fixing functional (which is just a group-theoretic dressed up Landau gauge (1.578), also notice that $A = (a, x)$)

$$F^{(a,x)}[A_\mu] = \partial_\mu A^\mu_a(x), \tag{2.512}$$

which, under the infinitesimal gauge variation of the gauge field given as

$$\delta_\zeta A^\mu_a(x) = \partial^\mu \zeta_a(x) + f^{bc}{}_a A^\mu_b(x)\zeta_c(x) \tag{2.513}$$

the variation of the gauge condition varies by

$$\delta_\zeta F^{(a,x)}[A_\mu] = \partial_\mu \left(\partial^\mu \zeta_a + f^{bc}{}_a A^\mu_b \zeta_c\right). \tag{2.514}$$

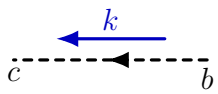
The Faddeev-Popov ghosts for Yang-Mills correspond to the following. The $c_a(x)$ ghost corresponds with the gauge parameters $\zeta_a(x)$ while the ghost $b^a(x)$ corresponds with the gauge conditions from $F^{(a,x)}[A]$. The ghost action for Yang-Mills is (take the ghost part of (2.485) and insert the gauge fixing function and compute the partial derivative)

$$\begin{aligned} S_{\text{gh}} &= \int d^4x\, b^a(x)\partial_\mu \left(\partial^\mu c_a(x) + f^{bc}{}_a A^\mu_b(x) c_c(x)\right) \\ &= -\int d^4x\, \partial_\mu b^a \left(\partial^\mu c_a + f^{bc}{}_a A^\mu_b c_c\right). \end{aligned} \tag{2.515}$$

The kinetic term for the ghosts b^a, c_a is identical to that of a complex scalar field practically *except* for the Grassmann-odd property of (b, c), and so the 2-point Greens function for the ghosts i.e. the propagator for the ghosts is derived via

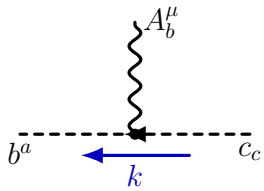
$$\begin{aligned} \langle\Omega| T\hat{b}^a(x)\hat{c}_{a'}(y) |\Omega\rangle &= -\langle\Omega| T\hat{c}_{a'}(y)\hat{b}^a(x) |\Omega\rangle \\ &= \delta^a_{a'} \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} \frac{-i}{k^2 - i\epsilon} \end{aligned} \tag{2.516}$$

and thus the Feynman rule for the propagator for the fermionic ghosts are



$$= \frac{-i\delta^a_{a'}}{k^2 - i\epsilon}. \tag{2.517}$$

Note that as in fermionic Feynman rules, a ghost loop such as, , will come with an overall factor of -1 due to the Grassmann-oddness. The interaction vertex between the gauge field and ghosts is



$$= -k_\mu f^{bc}{}_a. \tag{2.518}$$

Now, unlike in scalar QED, there is no 4-vertex of the form: . Now moving on to the other parts of the gauge-fixed Yang-Mills action.

The gauge fixing part of the action $S_{\text{G.F.}}$ contains the term

$$B_A F^A[\phi] \to \int d^4x\, B^a(x)\partial_\mu A^\mu_a(x). \tag{2.519}$$

We are free to add any terms to the B^A action as long as it does not affect the dynamics and is BRST-exact namely

$$\frac{g^2}{2}\xi \int d^4x\, B^a(x)B^a(x) \tag{2.520}$$

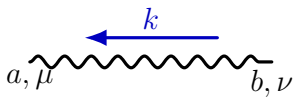
and thus with $B^a(x)$ quadratic in the exponential, we can perform the Gaussian integral and integrate out $B^a(x)$ in the path integral which results in the following deformation

$$\Delta S = -\frac{1}{2g^2\xi}\int d^4x\, \left(\partial_\mu A^\mu_a\right)^2. \tag{2.521}$$

This is exactly analogous to the Maxwell case with the ξ gauge, and thus we find something *extremely* similar

$$\langle\Omega|\, T\hat{A}_{a\mu}(x)\hat{A}_{b\nu}(y)\,|\Omega\rangle = \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} g^2 \frac{-i\delta_{ab}}{k^2 - i\epsilon}\left(\eta_{\mu\nu} + (\xi-1)\frac{k_\mu k_\nu}{k^2}\right) \tag{2.522}$$

which therefore has the corresponding Feynman rule



$$= g^2 \frac{-i\delta_{ab}}{k^2 - i\epsilon}\left(\eta_{\mu\nu} + (\xi-1)\frac{k_\mu k_\nu}{k^2}\right) \tag{2.523}$$

where the g^2 is due to our normalization convention of Yang-Mills kinetic term. There are also the 3-point and 4-point self-interaction vertices of the $A_{a\mu}$ gauge fields in Yang-Mills, but we will get to them later since for right now, as per usual, we want to investigate the 1PI renormalization group of QCD for *massless* quarks and see how quantized Yang-Mills behaves.

The strategy here for computing the 1PI renormalization group equation for QCD (of massless quarks) is to first compute the 1PI quantum effective action $\Gamma[A_\mu, \cdots]$ at 1-loop order using dimensional regularization scheme $D = 4 - \epsilon$. Then, we will use the minimal subtraction method as the renormalization scheme that defines the renormalized Yang-Mills coupling $g(\mu)$. Therefore, we will then be able to extract the renormalization group equation for $g(\mu)$, and therefore under Wilson-Polchinski, will find the renormalization group flow equation.

We have a naive expectation for this calculation though. Assuming quark fields are massless, all power-counting renormalizable couplings are controlled by a single coupling constant g due to gauge-invariance. This expectation is essentially correct, but it is not obvious why. In path integral quantization, gauge invariance is replaced by BRST invariance. Thereby, the possible counter terms to be included in the bare gauge-fixed action are restricted by BRST-invariance as opposed to gauge invariance.

A related complication is that the 1PI quantum effective action is not only gauge non-invariant, but is generally not BRST invariant either. Thiks is loosely speaking due to the fact that BRST variations are non-linear in the fields in general, and that $\langle Q_B \cdot \phi \rangle_{J_{\phi_0}} \neq Q_B \cdot \phi \Big|_{\phi=\phi_0}$ (for example, see the discussion in Weinberg's volume 2 textbook chapter 17.2 [2]). But hang on then, we broke gauge invariance in S, and now apparently Γ won't be a constructed gauge invariant action? Well, nonetheless, a simplification occurs for $\Gamma[A, \cdots]$ computed in the *background field gauge*.

In evaluating Γ, we begin by expanding the gauge field as[89]

$$A_\mu(x) = \widetilde{A}_\mu(x) + A'_\mu(x) \tag{2.524}$$

where the blue term denotes the *background field* (which we can think of as the *classical* dynamics of the gauge field) and the purple term are the fluctuations (which we can think of as the *quantum mechanical* dynamics/fluctuations). There is a similar background expansion for ψ, b, and c. To source the *classical* dynamics, we need to deform the action with a source term such as $\Delta S = \int d^4x\, j^\mu(x) A'_\mu(x)$ where we assume that $\langle A'_\mu(x) \rangle = 0$ with the interacting vacuum (makes sense, it is classical), which allows us to integrate out the classical field $A'_\mu(x)$ and the other classical fields as well to obtain the quantum effective action that is only in terms of the quantum-ness fields $\widetilde{A}_\mu(x)$. What this procedure looks like is the following.

We start with the definition of the quantum effective action

$$\begin{aligned} e^{i\Gamma[\widetilde{A},\widetilde{\psi},\cdots]} = &\int DA' D\psi' Db' Dc' \\ &\times \exp\left(i \left(S[\widetilde{A} + A', \widetilde{\psi} + \psi', \cdots] + S_{\text{G.F.}}[A', \psi', \cdots; \widetilde{A}] + \int j^\mu_A A'_\mu + j_\psi \cdot \psi' + \cdots \right) \right). \end{aligned} \tag{2.525}$$

The gauge condition for the classical background fields $A', \cdots$ may be chosen to depend upon the QM field $\widetilde{A}$. A particularly "nice" choice is

$$\begin{aligned} F^{(a,x)}[A'] &= \widetilde{D}_\mu A'^{\,\mu}_a(x) \\ &\equiv \partial_\mu A'^{\,\mu}_a + f^{bc}{}_a \widetilde{A}_{b\mu} A'^{\,\mu}_c. \end{aligned} \tag{2.526}$$

We observe that $F^{(a,x)}[A']$ is covariant with respect to the following transformation

$$\delta_{\text{NEW}} \widetilde{A}^\mu_a = \partial^\mu \zeta_a + f^{bc}{}_a \widetilde{A}^\mu_b \zeta_c \tag{2.527}$$

$$\delta_{\text{NEW}} A'^{\,\mu}_a = f^{bc}{}_a A'^{\,\mu}_b \zeta_c \tag{2.528}$$

where the last term in the first line we can attribute as "gauge-like" as if $\widetilde{A}$ were gauge fields as A' were matter fields transforming in the adjoint representation. Consequently, the gauge fixing action $S_{\text{G.F.}}$ is invariant with respect to the transformation (2.528) with

$$\delta_{\text{NEW}} \widetilde{b}^a = -f^{ab}{}_c \widetilde{b}^c \zeta_b \qquad \text{likewise with } \bar{b} \to b' \tag{2.529}$$

$$\delta_{\text{NEW}} \widetilde{c}_a = f^{bc}{}_a \widetilde{c}_b \zeta_c \qquad \text{likewise with } \bar{c} \to c' \tag{2.530}$$

[89] This procedure is extremely important and almost undeiable crucial for anyone getting into quantum-gravity dynamics and gravitational physics with a "sprinkle" of QFT on top.

and the full gauge-fixed action is invariant provided that the matter fields also transform according to

$$\delta_{\text{NEW}}\widetilde{\psi} = i\zeta_z t_R^a \cdot \widetilde{\psi} \tag{2.531}$$

and likewise with $\widetilde{\psi} \to \psi'$. The key point here is that δ_{NEW} does not mix $\widetilde{A}$ (and the other quantum fields) with the classical background A' (and others) at all.

After integrating out A' (and the rest of the classical background), we obtain the quantum effective action $\Gamma[\widetilde{A}, \cdots]$ that is invariant with respect to the new δ_{NEW} transformations of $\widetilde{A}, \cdots$, which take the form of gauge transformations *as if* b^a and c_a were also matter fields in the adjoint representation!

The conclusion we come to then is this. In the *background field gauge*, the 1PI effective action $\Gamma[\widetilde{A}, \widetilde{\psi}, \widetilde{b}, \widetilde{c}]$ is invariant under a gauge-like transformation δ_{NEW}. This constrains the possible terms in Γ similarly to how gauge-invariance constraints the Yang-Mills action. In particular, the operators of mass dimension dim ≤ 4 appearing in Γ, up to a field rescaling, are controlled by a single renormalized gauge coupling $g(\mu)$. To determine the corresponding renormalization group equation for $g(\mu)$, it suffices to consider $\Gamma[\widetilde{A}]$ with $\widetilde{\psi}$, $\widetilde{b}$, and $\widetilde{c}$ seto to zero.

Therefore, the total gauge fixed action of massless QCD (and we set $\theta = 0$) in the background field gauge with $\widetilde{A}$ turn on is then given as

$$\begin{aligned} S[A', \psi, b, c; \widetilde{A}] &= \int d^4x \left[-\frac{1}{4g^2} F_{a\mu\nu}F^{a\mu\nu} - \overline{\psi}_I \gamma^\mu D_\mu \psi_I - \frac{1}{2\xi g^2}\left(\widetilde{D}_\mu A_a'^{\,\mu}\right)^2 - \left(\widetilde{D}_\mu b^a\right)\left(D^\mu c_a\right)\right] \\ &= \int d^4x \left[-\frac{1}{4g^2}\left(\widetilde{F}_{a\mu\nu} + \widetilde{D}_\mu A'_{a\nu} - \widetilde{D}_\nu A'_{a\mu} + f^{bc}{}_a A'_{b\mu} A'_{c\nu}\right)^2 - \overline{\psi}_I \gamma^\mu \left(\widetilde{D}_\mu - iA'_{a\mu} t_R^a\right)\psi_I \right. \\ &\quad \left. - \frac{1}{2\xi g^2}\left(\widetilde{D}_\mu A_a'^{\,\mu}\right)^2 - \widetilde{D}_\mu b^a \left(\widetilde{D}^\mu c_a + f^{bc}{}_a A_b'^{\,\mu} c_c\right)\right] \end{aligned} \tag{2.532}$$

2.3.3 Deriving the QCD β-Function

Now, let us move onto computing the 1PI effective action and thus the 1-loop correction. The 1PI effective action of massless QCD in the background field gauge, restricting to the case of vanishing $\widetilde{\psi}$, $\widetilde{b}$, and $\widetilde{c}$, takes on the following form

$$e^{i\Gamma[\widetilde{A}]} = \int DA'D\psi DbDc \exp\left(iS[A', \psi.b, c; \widetilde{A}] + i\int j_{a\mu} A_a'^{\,\mu}\right)\Big|_{\substack{j \text{ chosen such that} \\ \langle A'\rangle = 0}} . \tag{2.533}$$

The 1-loop contribution to the effective action $\Gamma^{\text{1-loop}}[\widetilde{a}]$ comes from performing the Gaussian functional with the terms in the action that are quadratic in the fluctuation fields A', ψ, b, and c. *Just* the quadractic terms that arise from the action are the following

$$\begin{aligned} S^{(2)}(A', \psi, b, c; \widetilde{A}] &= \int d^4x \left[-\frac{1}{4g^2}\left(\widetilde{D}_\mu A'_{a\nu} - \widetilde{D}_\nu A'_{a\mu}\right)^2 - \frac{1}{2g^2} f^{bc}{}_a \widetilde{F}_a^{\mu\nu} A'_{b\mu} A'_{c\nu} \right. \\ &\quad \left. - \frac{1}{2\xi g^2}\left(\widetilde{D}_\mu A_a'^{\,\mu}\right)^2 - \overline{\psi}_I \gamma^\mu \widetilde{D}_\mu \psi_I - \left(\widetilde{D}_\mu b^a\right)\left(\widetilde{D}_\mu c_a\right)\right] \\ &\equiv \int d^4x \left[-\frac{1}{4g^2} A_a'^{\,\mu} (\mathbb{D}_A)_{a\mu, b\nu} A_b'^{\,\nu} + \overline{\psi}_I \mathbb{D}_\psi \psi_I + b^a \mathbb{D}_{\text{gh}} c_a \right] \end{aligned} \tag{2.534}$$

where we have separated out the action in (2.534) in terms of their corresponding differential operators namely

$$(\mathbb{D}_A)_{a\mu,b\nu} \equiv \left(-\widetilde{D}_\rho \widetilde{D}^\rho \eta_{\mu\nu} + \widetilde{D}_\nu \widetilde{D}_\mu - \frac{1}{\xi}\widetilde{D}_\mu \widetilde{D}_\nu\right)_{ab} \tag{2.535}$$

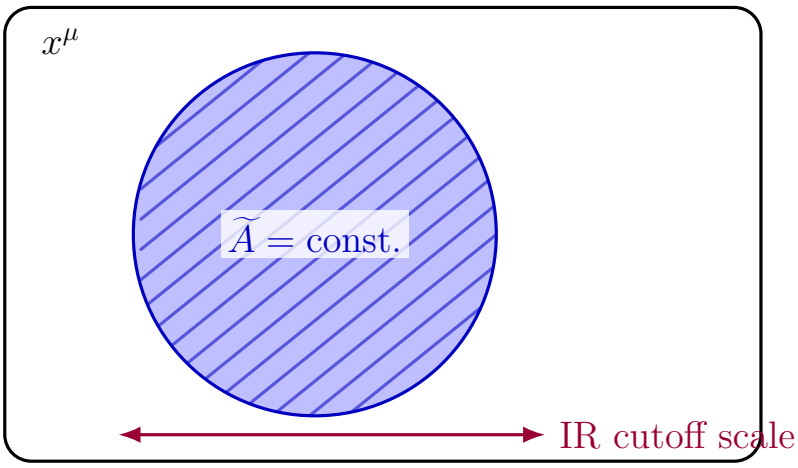
$$\mathbb{D}_\psi = -\gamma^\mu \widetilde{D}_\mu \tag{2.536}$$

$$\mathbb{D}_{\text{gh}} = \widetilde{D}_\mu \widetilde{D}^\mu. \tag{2.537}$$

Take note that $\mathbb{D}_\psi$ act on *each* fermionic flavor namely for all I in $I = 1, \cdots, N_f$. Written out this way, the action when plugging into the definition of the quantum effective action (2.534) and with the way we organized the differential operators with definition in (2.537), we can see that the action is split into a product of three Gaussian integrals. The gauge-field integration is bosonic, and thus will contribute $(\det \mathbb{D}_A)^{-1/2}$, the fermionic term (namely the massless quarks) will contribute as $(\det \mathbb{D}_\psi)^{N_f}$ for N_f flavors, while the ghost term will just contribute as a normal fermionic integral as $\det \mathbb{D}_{\text{gh}}$. The corresponding quantum effective action is then

$$i\Gamma^{\text{1-loop}}[\widetilde{A}] = \log \frac{(\det \mathbb{D}_\psi)^{N_f} (\det \mathbb{D}_{\text{gh}})}{(\det \mathbb{D}_A)^{\frac{1}{2}}}. \tag{2.538}$$

Note that in the Euclidean signature, we would instead get $-\Gamma^{\text{1-loop}}$ instead of an overall factor of i. The evaluation of these functional determinants for general $\widetilde{A}$ is complicated. For our purpose of extracting the renormalized coupling $g(\mu)$ in the minimal subtraction scheme, it suffices to consider a *nearly* constant quantum fluctuating field $\widetilde{A}_{a\mu}$. More precisely, we will evaluate $\Gamma^{\text{1-loop}}[\widetilde{A}]$ on a (Euclidean) background gauge field configuration of the following form:



The idea here is that in spacetime, the quantum fluctuations turn off when the spacetime points go to infinity $\widetilde{A} \to 0$ as $x^\mu \to \infty$ but is a constant value within some region L. This region also defines the IR-cutoff scale $L \sim \mu^{-1}$ regulating it from below. The Euclidean effective action then becomes[90]

$$-\Gamma^{\text{1-loop}}[\widetilde{A}] = N_f \operatorname{Tr}\log \mathbb{D}_\psi + \operatorname{Tr}\log \mathbb{D}_{\text{gh}} - \frac{1}{2}\operatorname{Tr}\log \mathbb{D}_A \tag{2.539}$$

[90] There is an *extremely* handy identity where for some matrix A, $\log\det(A) = \operatorname{tr}(\log(A))$.

where each trace is defined over the appropriate configuration space of field variables. We will now show how to evaluate one of these kinetic operators, and leave the rest of the problem sets[91]. For a general kinetic operator $\mathbb{D}_\alpha{}^\beta$ acting on a set of field variables $\varphi_\alpha(x)$, the trace can be evaluated as

$$\begin{aligned}\operatorname{Tr} f(\mathbb{D}) &= \int d^4x \, \langle x| \operatorname{tr} f(\mathbb{D}) |x\rangle \\ &= \int d^4x \int \frac{d^4k}{(2\pi)^D} \langle x| \operatorname{tr} f(\mathbb{D}) |k\rangle \langle k|x\rangle \end{aligned} \tag{2.540}$$

where $\langle k|x\rangle = e^{ik\cdot x}$. For a constant $\widetilde{A}$, the relevant differential operator $\mathbb{D}$ has no explicit x-dependence and thus

$$\mathbb{D}_\alpha{}^\beta |k\rangle = \mathbb{M}_\alpha{}^\beta(k) |k\rangle \implies \operatorname{tr} f(\mathbb{D}) = \int d^4x \int \frac{d^4k}{(2\pi)^D} \operatorname{tr} f(\mathbb{M}(k)). \tag{2.541}$$

And thus, the 1-loop effective action becomes

$$-\Gamma^{\text{1-loop}}[\widetilde{A}] = \int d^4x \int \frac{d^4k}{(2\pi)^D} \left(N_f \operatorname{tr}\log \mathbb{M}_\psi(k) + \operatorname{tr}\log \mathbb{M}_{\text{gh}}(k) - \frac{1}{2} \operatorname{tr}\log \mathbb{M}_A(k) \right). \tag{2.542}$$

For example, the fermionic (quark) eigenvalue of the differential operator is

$$\mathbb{M}_\psi(k) = i(\not{k} - \gamma^\mu \widetilde{A}_{a\mu} t^a_R). \tag{2.543}$$

This is a matrix of size $4 \cdot \dim R$ acting on $\psi_{i\alpha}$ where again $\alpha = 1, \cdots, 4$ is the spinor index while $i = 1, \cdots, \dim R$ is the gauge index. Then, using (2.541) for the quark differential operator becomes

$$\begin{aligned}\int \frac{d^4k}{(2\pi)^4} \operatorname{tr}\log \mathbb{M}_\psi(k) =& (\widetilde{A}\text{-independent, regularization scheme dependent constant}) \\ &- \int \frac{d^4k}{(2\pi)^4} \sum_{n=1}^{\infty} \frac{1}{n} \operatorname{tr}\left(\frac{\not{k}}{k^2} \not{\widetilde{A}} \right)^n . \end{aligned} \tag{2.544}$$

Note that here, the trace is over spinor and gauge indices, and the UV divergence occurs for $n \leq 4$ in dimensional regularization. The IR divergence occurs for when $n \geq 4$ with a cuttoff of $L \sim \mu^{-1}$. Let us compare this to the bare Euclidean Yang-Mills Lagrangian

$$\begin{aligned}\mathcal{L}_E &= \frac{1}{2g^2} \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) \\ &\supset \text{quartic term} - \frac{1}{2g^2} \operatorname{tr}\left([A_\mu, A_\nu][A^\mu, A^\nu]\right). \end{aligned} \tag{2.545}$$

We can therefore extract $g(\mu)$ from the $\widetilde{A}^4$ order term in $\Gamma[\widetilde{A}]$. working in dimensional regularization with $D = 4 - \epsilon$, and expanding out to $n = 4$, the $\mathbb{M}_\psi$ eigenvalue can be written as

$$\int \frac{d^Dk}{(2\pi)^D} \operatorname{tr}\log \mathbb{M}_\psi(k) \Big|_{\widetilde{A}^4} = -\frac{1}{4} \operatorname{tr}\left(\widetilde{A}_{\mu_1} \widetilde{A}_{\mu_2} \widetilde{A}_{\mu_3} \widetilde{A}_{\mu_4} \right) \int \frac{d^Dk}{(2\pi)^D} \frac{\operatorname{Tr}(\not{k}\gamma^{\mu_1}\not{k}\gamma^{\mu_2}\not{k}\gamma^{\mu_3}\not{k}\gamma^{\mu_4})}{(k^2)^4}. \tag{2.546}$$

[91] Hehehe, but don't worry, the details can be found in Weinberg's textbook chapter 17.4 and 17.5 [2].

Now, the spinor-trace term can be rewritten as

$$\frac{\mathrm{Tr}(\not{k}\gamma^{\mu_1}\not{k}\gamma^{\mu_2}\not{k}\gamma^{\mu_3}\not{k}\gamma^{\mu_4})}{(k^2)^4} = \frac{k^{\nu_1}k^{\nu_2}k^{\nu_3}k^{\nu_4}}{(k^2)^4}\,\mathrm{Tr}(\gamma_{\nu_1}\gamma^{\mu_1}\cdots\gamma_{\nu_4}\gamma^{\mu_4}). \tag{2.547}$$

We can further reduce this combination of momentum 4-vectors by exploiting their SO(D) symmetry

$$\frac{k^{\nu_1}k^{\nu_2}k^{\nu_3}k^{\nu_4}}{(k^2)^4} = \frac{1}{(k^2)^2}\frac{1}{D(D+2)}\left(\delta^{\nu_1\nu_2}\delta^{\nu_3\nu_4} + (2\leftrightarrow 3) + (2\leftrightarrow 4)\right). \tag{2.548}$$

We can use this combination of the Euclidean metrics to contract with the trace over the gamma matrices. Since this combination of gamma matrices and indices may be new, let us go through this explicitly:

$$\begin{aligned}
&\Big(\delta^{\nu_1\nu_2}\delta^{\nu_3\nu_4} + (2\leftrightarrow 3) + (2\leftrightarrow 4)\Big)\,\mathrm{Tr}(\gamma_{\nu_1}\gamma^{\mu_1}\cdots\gamma_{\nu_4}\gamma^{\mu_4})\\
&\quad=2\,\mathrm{Tr}\left(\gamma_\alpha\gamma^{\mu_1}\gamma^\alpha\gamma^{\mu_2}\gamma_\beta\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\right) + \mathrm{Tr}\left(\gamma_\alpha\gamma^{\mu_1}\gamma_\beta\gamma^{\mu_2}\gamma^\alpha\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\right)\\
&\quad=2(2-D)^2\,\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_4}) + 2\delta^{\mu_2\alpha}\,\mathrm{Tr}\Big(\gamma_\alpha\gamma^{\mu_1}\gamma_\beta\gamma^{\mu_2}\gamma^\alpha\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\Big)\\
&\qquad-2\delta^\alpha_\beta\,\mathrm{Tr}\Big(\gamma_\alpha\gamma^{\mu_1}\gamma^{\mu_2}\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\Big) + 2\delta^{\mu_1\alpha}\,\mathrm{Tr}\Big(\gamma_\alpha\gamma_\beta\gamma^{\mu_2}\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\Big)\\
&\qquad-D\,\mathrm{Tr}\Big(\gamma^{\mu_1}\gamma_\beta\gamma^{\mu_2}\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\Big)\\
&\quad=(2(2-D)^2 - 2(2-D))\,\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_4}) + 2(2-D)\,\mathrm{Tr}(\gamma^{\mu_2}\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_4})\\
&\qquad+(2-D)\,\mathrm{Tr}\Big(\gamma^{\mu_1}\gamma_\beta\gamma^{\mu_2}\gamma^{\mu_3}\gamma^\beta\gamma^{\mu_4}\Big)\\
&\quad=(2(2-D)^2 - 2(2-D) - (2-D)^2)\,\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_4}) + 2(2-D)\,\mathrm{Tr}(\gamma^{\mu_2}\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_4})\\
&\qquad+2(2-D)\,\mathrm{Tr}(\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_2}\gamma^{\mu_4})\\
&\quad=8\,\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_4}) - 4\,\mathrm{Tr}(\gamma^{\mu_2}\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_4})\\
&\qquad-4\,\mathrm{Tr}(\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_2}\gamma^{\mu_4}) + \mathcal{O}(\epsilon).
\end{aligned} \tag{2.549}$$

Ok, so what the hell just happened. In the first line, this was just contracting all possible indices from the Euclidean metrics. Next in the first term, we can contract over like indices by moving the gamma matrices next to each other in the trace as long as we pick up the appropriate minus sign. For instance, moving a gamma matrix one gives (-1) while moving twice gives $(-1)^2$ and so on. So in the first line, we contract the α and β indices in each term. Then, we simplified in the next line (next equal sign) using the anti-commutator for the gamma matrices (1.388) in D-dimensions. We essentially kept doing this with all contracted indices and grouped liked terms until the end when there were none left. The final line is substituting $D = 4-\epsilon$ and then expanding out to finite order in ϵ. The most elegant part of this is that now, that summation of traces we just worked out simplifies to

$$\begin{aligned}
&8\,\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_4}) - 4\,\mathrm{Tr}(\gamma^{\mu_2}\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_4}) - 4\,\mathrm{Tr}(\gamma^{\mu_1}\gamma^{\mu_3}\gamma^{\mu_2}\gamma^{\mu_4})\\
&\qquad= 4\,\mathrm{Tr}\left([\gamma^{\mu_1},\gamma^{\mu_2}][\gamma^{\mu_3},\gamma^{\mu_4}]\right)\\
&\qquad= -64\left(\delta^{\mu_1\mu_3}\delta^{\mu_2\mu_4} - \delta^{\mu_1\mu_4}\delta^{\mu_2\mu_3}\right).
\end{aligned} \tag{2.550}$$

We can use this final result to contract over with the traced over gauge fields $\widetilde{A}_{\mu_i}$ which are being traced over for some representation R,

$$-64\,\mathrm{tr}_R(\widetilde{A}_{\mu_1}\cdots\widetilde{A}_{\mu_4})\cdot(\delta^{\mu_1\mu_3}\delta^{\mu_2\mu_4}-\delta^{\mu_1\mu_4}\delta^{\mu_2\mu_3}) = -32\,\mathrm{tr}_R\left([\widetilde{A}_{\mu_1},\widetilde{A}_{\mu_2}][\widetilde{A}^{\mu_1},\widetilde{A}^{\mu_2}]\right) \tag{2.551}$$

up to order $\mathcal{O}(\epsilon)$ (just dropping it for now). Thus, we finally have

$$\int\frac{d^Dk}{(2\pi)^D}\,\mathrm{tr}\log\mathbb{M}_\psi(k)\,\Big|_{\widetilde{A}^4} = \left(\frac{1}{3}\,\mathrm{tr}_R\left([\widetilde{A}_\mu,\widetilde{A}_\nu][\widetilde{A}^\mu,\widetilde{A}^\nu]\right)+\mathcal{O}(\epsilon)\right)\int\frac{d^{4-\epsilon}k}{(2\pi)^{4-\epsilon}}\frac{1}{(k^2)^2}. \tag{2.552}$$

The momentum integral we have left over is IR divergent. The IR divergence is of course expected since we assumed constant $\widetilde{A}$ without μ-dependence. The renormalized Yang-Mills coupling should be extracted from $\Gamma[\widetilde{A}]$ evaluated on a background field configuration $\widetilde{A}$ of momentum scale μ, for example as the one given in figure 2.3.3. To extract the μ-dependence, it suffices to modify the earlier calculation by introducing an IR cutoff $|k|<\mu$ namely

$$\begin{aligned}\int d^Dx\int\frac{d^Dk}{(2\pi)^D}\,\mathrm{tr}\log\mathbb{M}_\psi(k)\,\Big|_{\widetilde{A}^4} &= \int d^Dx\frac{1}{3}\,\mathrm{tr}_R\left([\widetilde{A}_\mu,\widetilde{A}_\nu][\widetilde{A}^\mu,\widetilde{A}^\nu]\right)\cdot\int\limits_{|k|>\mu}\frac{d^{4-\epsilon}k}{(2\pi)^{4-\epsilon}}\frac{1}{(k^2)^2}+\text{finite}\\ &= \int d^Dx\,\frac{1}{3}\,\mathrm{tr}_R\left([\widetilde{A}_\mu,\widetilde{A}_\nu][\widetilde{A}^\mu,\widetilde{A}^\nu]\right)\frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}+\text{finite}.\end{aligned} \tag{2.553}$$

Completing this term via the formal gauge invariance, δ_{NEW} symmetry, of $\Gamma[\widetilde{A}]$ in the background field gauge, we obtain a contribution to the quantum effective action of

$$\Gamma[\widetilde{A}]\supset\int d^Dx\left(-\frac{1}{3}\,\mathrm{tr}_R(\widetilde{F}_{\mu\nu}\widetilde{F}^{\mu\nu})\cdot\frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}+\text{finite}\right) \tag{2.554}$$

from which *each* of the N_f flavors of the quark fields will contribute. In a similar manner to the derivation we just did (again, which is found in the problem sets), the contributions from the Faddeev-Popov ghost determinant and the functional determinant for A' are

$$\mathrm{Tr}\log\mathbb{D}_{\mathrm{gh}} = \int d^Dx\,\frac{1}{12}\,\mathrm{tr}_{\mathrm{adj}}(\widetilde{F}_{\mu\nu}\widetilde{F}^{\mu\nu})\cdot\frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}+\text{finite} \tag{2.555}$$

$$\mathrm{Tr}\log\mathbb{D}_A = \int d^Dx\,\left(-\frac{5}{3}\right)\mathrm{tr}_{\mathrm{adj}}(\widetilde{F}_{\mu\nu}\widetilde{F}^{\mu\nu})\cdot\frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}+\text{finite}. \tag{2.556}$$

If we compare to the expect form of the Euclidean 1PI quantum effective action evaluated on such $\widetilde{A}$ fluctuations,

$$\Gamma[\widetilde{A}] = \frac{1}{4g^2(\mu)}\int d^4x\,\widetilde{F}_{a\mu\nu}\widetilde{F}^{a\mu\nu}+(\text{higher order terms in } g(\mu) \text{ or } \widetilde{A}) \tag{2.557}$$

we can read off the relationship between the coupling here, and the contributions we can sum above with respect to the quantum effective action

$$-\frac{1}{4g^2(\mu)} = -\frac{1}{4g_0^2}+\frac{b}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}+\text{finite} \tag{2.558}$$

where the coefficient b here is

$$b = -\frac{N_f}{3}C(R) + \left(-\frac{1}{2}\left(-\frac{5}{3}\right) + \frac{1}{12}\right)C(\text{adj}). \tag{2.559}$$

Here, the $C(R)$ and $C(\text{adj})$ factors are called *Casimirs* and specifically *quadratic Casimir* function. The Casimir for representation R and adj are written as

$$\text{tr}_R(t^a t^b) = C(R)\delta^{ab} \tag{2.560}$$

$$\text{tr}_{\text{adj}}(t^a t^b) = if^{ac}{}_d \cdot if^{bd}{}_c = C(\text{adj})\delta^{ab}. \tag{2.561}$$

For $G = SU(N)$ we can take the representation $R = \square$ or the anti-fundamental representation $\overline{\square}$, then $C(R) = 1/2$ and $C(\text{adj}) = N$. Thus,(2.559) becomes

$$\begin{aligned} b &= \frac{11}{12}C(\text{adj}) - \frac{N_f}{3}C(R) \\ &= \frac{11}{12}N - \frac{1}{6}N_f. \end{aligned} \tag{2.562}$$

In our usual notation $\lambda^2(\mu) \equiv \mu^\epsilon g^2(\mu)$. We can therefore invert (2.558) to write it in terms of the bare coupling

$$\frac{1}{g_0^2}\mu^{-\epsilon}\left(\frac{1}{\lambda^2(\mu)} + \frac{b}{2\pi^2}\frac{1}{\epsilon} + \text{finite} + \text{higher order in } \lambda(\mu)\right) \tag{2.563}$$

where in the minimal subtraction scheme the "finite" part drops out. The renormalization group equation is then

$$\begin{aligned} 0 &= \frac{d}{d\log\mu}\frac{1}{g_0^2} \\ &= \mu^{-\epsilon}\left(\frac{d}{d\log\mu}\frac{1}{\lambda^2(\mu)} - \epsilon\left(\frac{1}{\lambda^2(\mu)} + \frac{b}{2\pi^2}\frac{1}{\epsilon}\right) + \text{higher order in } \lambda\right) \\ \frac{d}{d\log\mu}\frac{1}{\lambda^2(\mu)} &= \frac{b}{2\pi^2} + \mathcal{O}(\lambda^2) \end{aligned} \tag{2.564}$$

from which we can write in an equivalent manner

$$\frac{d}{d\log\mu}\lambda(\mu) = \beta(\lambda(\mu)) \tag{2.565}$$

or namely the QCD beta function for (massless quarks) is[92]

$$\begin{aligned} \beta(\lambda) &= -\frac{b}{4\pi^2}\lambda^3 + \mathcal{O}(\lambda^5) \\ &= -\frac{\lambda^3}{16\pi^2}\left(\frac{11}{3}N - \frac{2}{3}N_f\right). \end{aligned} \tag{2.566}$$

[92] If your name was Hugh Politzer and David Gross with Frank Wilczek and wrote down (2.566) in 1973, then you would be winning the Nobel Prize in physics as *they* did in 1973! They actually wrote down the form we had prior to picking a specific representation for SU(N) and kept it in terms of Casimirs and including complex scalars in their calculation. Fun fact, 't Hooft had derived this result in 1972 but never published it! 't Hooft thought the result was useless actually since it never addressed the confinement problem! I **highly** recommend reading 't Hooft's account of physics during this time [18] since for me, it just brought tears to my eyes to be reading his own thoughts on the subject; thoughts coming from such a giant in theoretical physics.

(2.566) is in fact the famous β-function for massless QCD. We can solve the 1-loop renormalization group equation as

$$\frac{d}{d\log\mu}\frac{1}{\lambda^2(\mu)} = \frac{b}{2\pi^2} \implies \frac{1}{\lambda^2(\mu)} = \frac{1}{\lambda^2(\mu_0)} + \frac{b}{2\pi^2}\log\frac{\mu}{\mu_0} \tag{2.567}$$

or in other words,

$$\lambda^2(\mu) = \left(\frac{b}{2\pi^2}\log\frac{\mu}{\Lambda_{\text{QCD}}}\right)^{-1} \tag{2.568}$$

where the scale Λ_{QCD} is given as

$$\Lambda_{\text{QCD}} \equiv \mu_0 e^{-\frac{2\pi^2}{b\lambda^2(\mu_0)}} \tag{2.569}$$

where recall that μ_0 is the energy scale at which we are probing, and Λ_{QCD} is a characteristic mass scale. For corresponding graph for $\lambda(\mu)$ is the following with a striking feature.

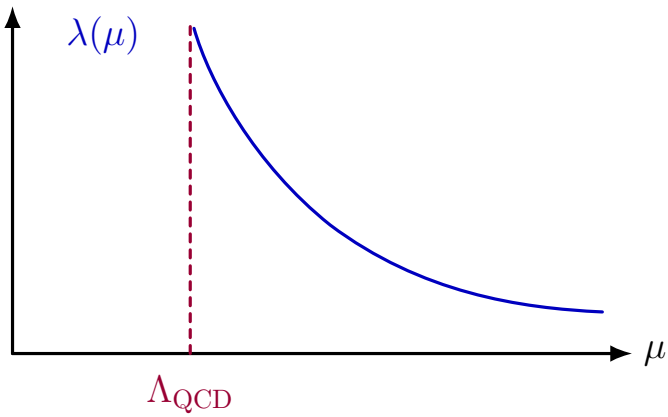


Figure 33. Graph of (2.558), namely the running coupling for QCD.

We can immediately see a striking feature as compared to our previous graph of the β-function 22. For $b > 0$, the 1-loop renormalization group equation indicates that QCD is strongly coupled in the IR while weakly coupling in the UV. This is the defining feature, and literal definition, of a theory having *asymptotic freedom*. So then we can ask, "what happens in the IR?"

As $\lambda(\mu)$ becomes large, higher loop corrections to the renormalization group equation will become important. Schematically, the QCD beta function at the next loop order takes on the form

$$\beta(\lambda) = b_1\lambda^3 + b_2\lambda^5 + \cdots \tag{2.570}$$

where $b_1 = -b/4\pi^2$. An important fact to recall is that b_1 and b_2 are both independent of the renormalization scheme (or just scheme independent). We can then look at the following two different behaviors that the 2-loop correction would have.

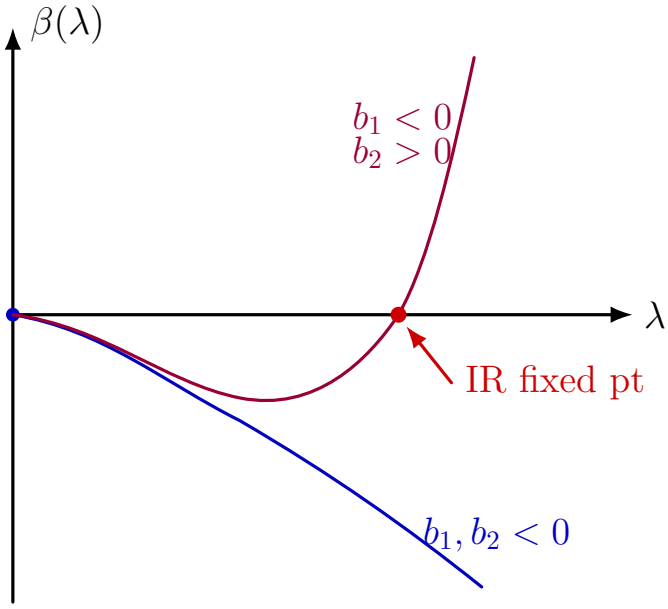


Figure 34. 2-loop correction to QCD β-function.

The 2-loop renormalization group equation predicts an IR fixed point when $b_1 < 0$ and $b_2 > 0$. Furthermore, if $|b_2| \ll |b_1|$,the fixed point value of λ is small, but then can we trust the 2-loop approximation near the fixed point?

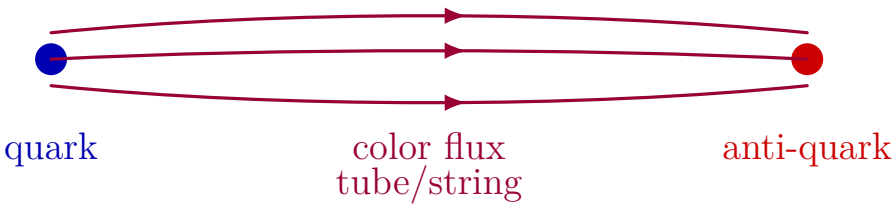
2.3.4 QCD Infrared Fixed Points and Flux Tubes/Strings

Such an IR fixed point of massless QCD occurs when

$$\frac{34N^2}{13N^2-3} < N_f < \frac{11}{2}N \tag{2.571}$$

and is believed to give rise to a CFT at which if the fixed point is achieved, the fixed point is called the *Bankz-Zaks fixed point*. This is not relevant for the real world QCD however where the number N_f of light quarks is ouside of the inequality window (2.571). We do have the following expectation though for real world dynamics. For N_f not too large, the strong coupling in the IR leads to a phenomenon called *confinment*, namely that there are no particles created by A_μ (gluons) or ψ (quarks). The particles are created by gauge-invariant local operators like $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$, and are *color neutral*, for example like mesons, baryons, or glueballs.

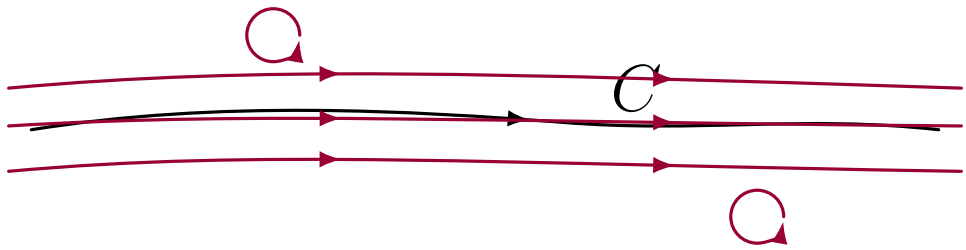
The gluons and quarks exists as *jets* or *particles with color flux strings attached*, which diagramatically appears as the following:



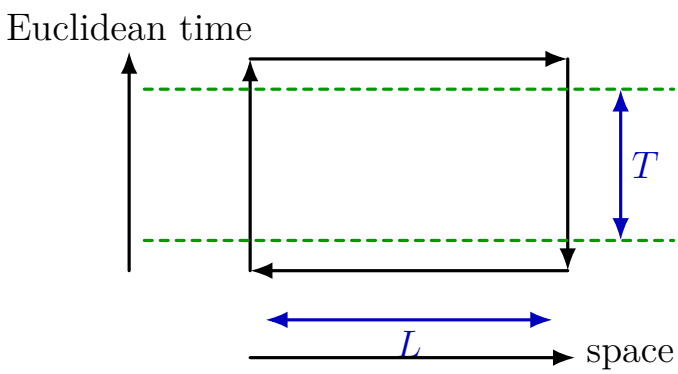
These flux tubes can create such a state with non-local gauge invariant operators of the form $\overline{\psi}(x_2)W_R(C)\psi(x_1)$ where $W_R(C)$ is a Wilson operator defined as

$$W_R(C) = P\, e^{i\int_C A_{a\mu}(x)t_R^a dx^\mu} \tag{2.572}$$

where P is the path-ordering exponential and C is a path in spacetime going from x_1 to x_2 and is commonly referred to as a *Wilson line*. The state $W_R(C)\,|\Omega\rangle$ is generally a state corresponding to a color flux string[93] extended roughly along the Wilson path C with all sorts of excitations, or which a cartoon-ish diagram is the following:



In *pure* Yang-Mills theory, in other words QCD without quarks, the color string created by $W_R(C)$ for $R = \square$ or the anti-fundamental representation $R = \overline{\square}$ cannot break or split. The color string carries a $\mathbb{Z}_N$ valued *string charge*. The latter is a charge of a *1-form symmetry* known as the *center symmetry*. The flux string has a tension of $\gamma \sim \Lambda_{\text{QCD}}^2$. The expectation value for such an operator can be computed as follows. For a closed path in Euclidean spacetime of the form:



The green dashed lines represent the state of a flux tube of length L that begins and ends on color sources or "classical quarks." The expectation value then of the Wilson operator for a flux tube is then, when we approximate for large time T,

$$\langle W_R(C)\rangle \sim e^{-V(L)\cdot T} \tag{2.573}$$

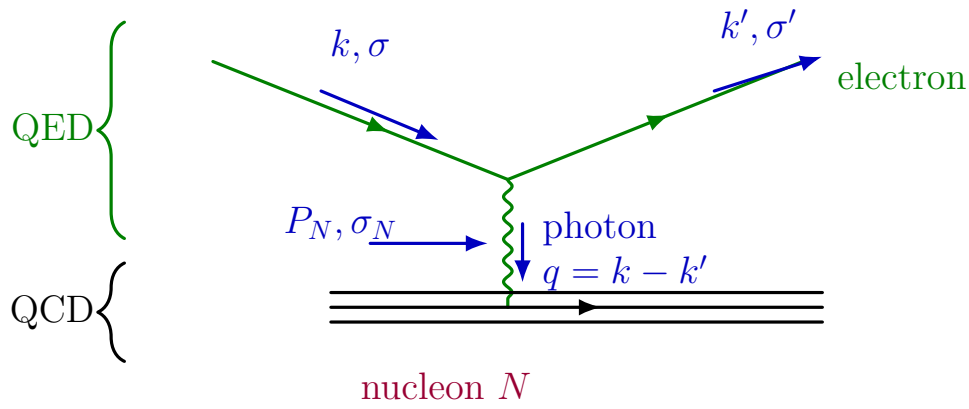
where $V(L)$ is the volume of spacetime and T is the total time. Confinement is reached when $V(L) \to \infty$ as $L \to \infty$, and we would expect that for $L, T \gg \Lambda_{\text{QCD}}^{-1}$, that the volume reaches a multiple of the string tension $V(L) \to \gamma L$.

[93] This discovery of flux tubes or flux strings was one of the first reasons to start formulating a theory of strings (along with evidence from the S-matrix for a dual resonance model).

2.3.5 Deep Inelastic Scattering (Theory)

Ok, so all of that was about the IR limit and interesting behavior of QCD at low energies, but what is perturbative QCD then good for (we still have the high energy behavior to explore)? For large values of μ, the coupling $\lambda(\mu)$ is small. We can therefore reliably calculate Greens functions of gauge-invariant local operators, such as $\operatorname{tr} F_{\mu\nu}^2$, at short distances i.e. large momentum.

Experimentally, the signature of asymptotic freedom can be found in deep inelastic scattering (DIS) experiments. Diagramatically, they are experiments of the following form:



Deep inelastic scattering is an experiment setup such that there is an incoming electron that scatters off a generic hadronic state that is not necessarily a 1-particle state in which communication between the two is mediated by the photon. And so we are combining the two primary theories we have into one experiments: QED for the electron plus photon and QCD for the photon-hadron interaction.

To leading order in he electric charge, with respect to the photon, the coupling of QCD to electromagnetism id described by the following term

$$\Delta\mathcal{L} = A_\mu^{\mathrm{EM}} J^\mu \tag{2.574}$$

in the Lagrangian density where the current here is of the form

$$J^\mu = i \sum_{I=1}^{N_f} e_I \overline{\psi}_I \gamma^\mu \psi_I \tag{2.575}$$

where the e_I here is the electric charge of the I-th quark type. To write down the scattering amplitude of (2.3.5), we setup and incoming state of the electron plus a nucleon N that then transitions into an electron plus some generic hadronic state H. This, to leading order in the electric charge, takes on the form

$$\mathcal{A}(k', \sigma'; H|k, \sigma; N) = \overline{\mathcal{U}}_{\sigma'}(k') e \gamma^\mu \mathcal{U}_\sigma(k) \frac{-i}{q^2 - i\epsilon} \langle H| J^\mu(0) |N\rangle \tag{2.576}$$

where the momentum here is $q = k - k'$ and the term in purple is a form factor in QCD. An observable of interest we can take away from this amplitude is the differential cross

section, in the nucleon N's rest frame, that includes all possible hadronic states H within a given range of energy transfer $\nu \equiv -q \cdot p_N/m_N$. Specifically,

$$\begin{aligned}\frac{d^2\sigma}{d\Omega d\nu}\bigg|_{\substack{\text{avg. over}\\ e^-\&N\text{ spins}}} &= \frac{|\vec{k}'|^2}{\nu_e}\frac{d|\vec{k}'|^2}{d\nu}\sum_H (2\pi)^4\delta^4(p_H - p_N - q)\cdot\frac{1}{4}\sum_{\sigma,\sigma_N}\sum_{\sigma'}|\mathcal{A}(k',\sigma';H|k,\sigma;N)|^2 \\ &= (\text{known factor})_{\mu\nu}\frac{1}{2}\sum_{\sigma_N}\sum_H (2\pi)^4\delta^4(p_H - p_N - q)\,\langle N|\,J^\nu(0)\,|H\rangle\,\langle H|\,J^\mu(0)\,|N\rangle\end{aligned} \tag{2.577}$$

where the known factor is just the tree-level QED calculation and we have assumed that $\mathbb{1} = \sum_I |H\rangle\langle H|$. We can also rewrite the double summation in terms of a 2-point function as

$$\frac{1}{2}\sum_{\sigma_N}\sum_H (2\pi)^4\delta^4(p_H - p_N - q)\,\langle N|\,J^\nu(0)\,|H\rangle\,\langle H|\,J^\mu(0)\,|N\rangle = \int d^4x e^{-iq\cdot x}\frac{1}{2}\sum_{\sigma_N}\langle N|\,J^\nu(x)J^\mu(0)\,|N\rangle \tag{2.578}$$

where notice that the two current are *not* time-ordered. From this, we then define the 2-current (a tensorial current) of the summation over possible hadrons N

$$W^{\mu\nu}(q,P_N) \equiv \frac{p_N^0}{m_N}\int d^4x\, e^{-iq\cdot x}\frac{1}{2}\sum_N \langle N|\,J^\nu(x)J^\mu(0)\,|N\rangle \tag{2.579}$$

where $W^{\mu\nu}$ is a Lorentz covariant quantity that obeys current conservation $q_\mu W^{\mu\nu} = q_\nu W^{\mu\nu} = 0$. This implies (as we have seen previously with the Ward identity (1.634) and form factors (1.653)) that the only possible form $W^{\mu\nu}$ can take is one constructed out of Lorentz covariant expression that cancel out when dotted in with q_μ

$$\begin{aligned}W^{\mu\nu}(q,p) = &\left(\eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}\right)W_1(\nu,q^2) \\ &+ \left(p^\mu - \frac{p\cdot q}{q^2}q^\mu\right)\left(q^\nu - \frac{p\cdot q}{q^2}q^\nu\right)W_2(\nu,q^2)\end{aligned} \tag{2.580}$$

where again $\nu \equiv -q\cdot p/m$. This 2-current or (what we will now officially call the) *hadronic tensor* has the following properties. First, it is symmetric and Hermitian $W^{\mu\nu} = W^{\nu\mu} = (W^{\mu\nu})^*$. Secondly, the form factor scalars/coefficients are strictly positive $W_1, W_2 \geq 0$. Also, assuming that N is the lightest hadron with the given quantum numbers $-p_H^2 \geq m_N$, then this implies the Lorentz invariant quantity ν obeys the following inequality

$$\nu \equiv -\frac{q\cdot p_N}{m_N^2} \geq \frac{q^2}{2m_N}. \tag{2.581}$$

All of this implies that the hadronic tensor is supported at $\nu \geq q^2/2m$. Take note that if we are in the proton's rest frame then ν reduces to the energy lost by the electron in the exchange.

But, we still have the summation over the possible spins of the hardonic states N in (2.578), and in order to have a well-defined 2-point function it must be time ordered, and

so if we include the spin σ_N and properly time order we get

$$\begin{aligned}
T^{\mu\nu}(q,p_N) \equiv & \frac{p_N^0}{m_N}\int d^4x\, e^{-iq\cdot x}\frac{1}{2}\sum_{\sigma_N}\langle N|\, TJ^\nu(x)J^\mu(0)\,|N\rangle \\
&= \frac{p_N^0}{m_N}\frac{1}{2}\sum_{\sigma_N}\int_{-\infty}^{\infty}\frac{d\omega}{\omega+i\epsilon}\left(e^{-iq\cdot x+i\omega n\cdot x}\langle N|\, J^\nu(x)J^\mu(0)\,|N\rangle + e^{iq\cdot x-i\omega n\cdot x}\langle N|\, J^\mu(0)J^\nu(x)\,|N\rangle\right) \\
&= \int_{-\infty}^{\infty}\frac{d\omega}{\omega+i\epsilon}\left(W^{\mu\nu}(q-\omega n,p_N)+W^{\nu\mu}(-q-\omega n,p_N)\right)
\end{aligned} \tag{2.582}$$

which if we take the real result of implies the following

$$\operatorname{Re} T^{\mu\nu}(q,p_N) = \frac{1}{2}\left(W_N^{\mu\nu}(q,p_N)+W_N^{\nu\mu}(-q,p_n)\right) \tag{2.583}$$

where the last term will vanish for $\nu > 0$ and $q^2 > 0$. We won't go into how this formula is used and what not, but in essence (2.582) is all you now need to compute the cross section for DIS and leads into the parton model and *parton distribution functions*. Beyond this, it is primarily phenomenological work, but if you look into the regime where the exchanged momenta is larger than the mass of the parton, then you will find a result referred to as the *Callan-Gross relation* which tests that quarks have spin 1/2.

2.3.6 Operator Product Expansion (for QCD)

For now, we will dive into another topic with respect to deep inelastic scattering, and that is of the *operator product expansion* (OPE). Essentially,the OPE is the position space version of the low energy expansion used in deriving effective field theories. We will briefly show off the tool here and show it's usage for QCD theory (and then move onto a completely different topic).

The behavior in the regime of high momentum transfer $|q| \gg \Lambda_{\text{QCD}}$, where deep inelastic physics is located, is governed by the property of the product operator $J^\mu(x)J^\nu(0)$at $|x| \ll \Lambda_{\text{QCD}}^{-1}$. The operator product expansion can be defined as (well, defined in very loose terms)

$$\mathcal{O}_i(x)\mathcal{O}_j(y) \approx \sum_\ell F_{ij}^\ell(x)\mathcal{O}_\ell(0) \tag{2.584}$$

where the approximation here is valid at least for small x, or in momentum space the OPE can be written as

$$\int d^Dx\, e^{-iq\cdot x}T\left[\mathcal{O}_i(x)\mathcal{O}_j(0)\right] \approx \sum_\ell U_{ij}^\ell(q)\mathcal{O}_i(0). \tag{2.585}$$

In QCD, the OPE we are concerned with is that of the product of currents that take on the form

$$\begin{aligned}
&\int d^4x\, e^{-iq\cdot x}T\left[J^\mu(x)J^\nu(0)\right] \\
&\qquad \sim \left(\eta^{\mu\nu}f_1(q^2)q_{\mu_1}\cdots q_{\mu_s}\mathcal{O}^{\mu_1\cdots\mu_s}(0) + f_s(q^2)q_{\mu_1}\cdots q_{\mu_2}\mathcal{O}^{\mu\nu\mu_1\cdots\mu_s}\right)\Big|_{\substack{\text{projection}\\ \text{to } q^\perp}}
\end{aligned} \tag{2.586}$$

where the operators $\mathcal{O}^{\mu_1\cdots\mu_s}$ are general spin-s operators that, without loss of generality, are symmetric and traceless. The combination of f's and q's here are called *Wilson coefficients* (which are the same ones you would find at the level of the action (after Fourier transforming) for effective field theories. In the $q^2 \to \infty$ limit, the RHS of (2.586) is dominated by the operator $\mathcal{O}^{\mu_1\cdots\mu_s}$ of lowest scaling dimension at a given operator spin s, for example

$$\mathcal{O}^{\mu_1\cdots\mu_s} = \begin{cases} \overline{\psi}_I \gamma^{(\mu_1} \overleftrightarrow{D}^{\mu_2} \cdots \overleftrightarrow{D}^{\mu_s)} \psi_J \Big|_{\text{traceless}} \\ F_{a\nu}{}^{(\mu_1} \overleftrightarrow{D}^{\mu_2} \cdots \overleftrightarrow{D}^{\mu_{s-1}} F_a{}^{\mu_s)\nu} \Big|_{\text{traceless}} \end{cases} \tag{2.587}$$

where this class of operators are called *twist-2 operators*. The twist of an operator is defined as the dimension of the operator minus the spin. If we denote the twist as t, and each operator here has mass dimension $\Delta_{\mathcal{O}} = 2 + s - 2r$ for r number of box derivatives and s-spin, then $\tau_s(i) \equiv 2 + 2r = d_s(i) - s$. But since the operators with extra factors of $\Box$ are suppressed, the OPE will be dominated by operators with the lowest twist.

The anomalous dimension of $\mathcal{O}^{\mu_1\cdots\mu_s}$ controls the scaling behavior of the hadronic tensor $W^{\mu\nu}(q, p_N)$ at large q^2 and ν as

$$W^{\mu\nu}(q, p_N) \sim \omega^S \left(\log \frac{|q|}{\Lambda_{\text{QCD}}} \right)^{-\frac{2\pi^2}{b}\tau_1} \tag{2.588}$$

where ω here is called the *Bjorken scaling* and $\omega \equiv 2m_n\nu/q^2$. The correspond anomalous dimension is then

$$\gamma_{\mathcal{O}}(\mu) = 2 + s + \tau_1 g^2(\mu) + \cdots \tag{2.589}$$

where this last term corresponds to the 1-loop anomalous dimension. For more details, I recommend reading Weinberg's chapter 20 [2].

This wraps up our talk on pure QCD dynamics and effects, and now for the rest of the lectures in QFT 2 are not exactly "miscellaneous topics," but rather more of extremely important physical and theoretical aspects that we can now only explore in full detail after all of the material from QFT 1 and QFT 2. The remaining topics are anomalies (and details therein) and spontaneous symmetry breaking of which will give us just a plethora of strong-force physics it almost deserves it's entire section since it gives us the effective theory of the pions.

2.3.7 Anomalies and 't Hooft Anomalies

We start off our discussion of what will be eventually called (and understood as) anomalies with a warm-up example of a $D = 2$ massless Dirac fermion theory

$$S = -\int d^2x \, \overline{\psi}\gamma^\mu \partial_\mu \psi \tag{2.590}$$

where the fermions here are written as $\psi = (\psi_1, \psi_2)^T$ (which is a column vector, I am just lazy and typing it as a row vector transposed) and the 2×2 gamma matrices only take on the values $\mu = 0, 1$ that still obey $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ with the complex conjugate being

$(\gamma^\mu)^\dagger = \gamma_\mu$, and the conjugate of the spinor field has the usual definition $\overline{\psi} \equiv i\psi^\dagger\gamma^0$. There is a clear U(1) *global* symmetry that the action (2.590) obeys under which ψ transforms as

$$\psi \mapsto e^{i\alpha}\psi \tag{2.591}$$

where $\alpha =$ any constant, with a corresponding Noether current

$$j^\mu = i\overline{\psi}\gamma^\mu\psi \tag{2.592}$$

to which you can check that it does obey the typical *classical* conservation law $\partial_\mu j^\mu = 0$. At the *classical* level, the conservation $\partial_\mu j^\mu = 0$ follows from the equations of motion. But, we have seen that at the QFT level back in QFT 1, as an operator, $\partial_\mu \hat{j}^\mu = 0$ will instead hold as an operator equation namely

$$\langle \partial_\mu \hat{j}^\mu(x)\hat{\mathcal{O}}(y)\cdots\rangle = 0 \tag{2.593}$$

modulo any possible contact terms supported at $y = x$ and any other coincident spacetime points. However, in the action (2.590) there is also a *chiral* symmetry which we shall denote as[94] $U(1)_A$:

$$\psi \mapsto e^{i\alpha\gamma}\psi \tag{2.594}$$

where $\gamma \equiv \gamma^0\gamma^1$ is the $D = 2$ chirality matrix similar to the one we saw prior in (1.421). The chiral matrix here still obeys $\gamma^2 = \mathbb{1}$ and $\gamma^\dagger = \gamma$. Note that while the kinetic term $\overline{\psi}\gamma^\mu\partial_\mu\psi$ is invariant under $U(1)_A$, a mass term that is proprotional to $\propto \overline{\psi}\psi$ would have broken the U(1) chiral symmetry, hence our restriction to only the massless case. The corresponding conserved Noether current is then

$$j_A^\mu = i\overline{\psi}\gamma^\mu\gamma. \tag{2.595}$$

Let us now see if *this* particular current is conserved as an operator equation (i.e. quantum mechnically). We shall do so by starting with the 2-point function of the chiral current with the non-chiral U(1) current

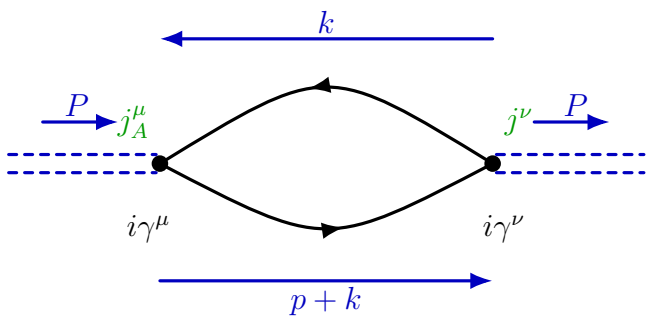
$$\langle\Omega|\, T\hat{j}_A^\mu(x)\hat{j}^\nu(y)\,|\Omega\rangle \tag{2.596}$$

with corresponding momentum space version

$$\int d^2x\, e^{-ip\cdot x}\,\langle\Omega|\, T\hat{j}_A^\mu(x)\hat{j}^\nu(0)\,|\Omega\rangle \tag{2.597}$$

[94] You might be wondering why we are calling $U(1)_A$ a chiral U(1) symmetry and yet denoting it with an A. Well, this is because the other common name for this U(1) symmetry is the *axial* symmetry.

which we know with it being in momentum space, we can compute the correlation function using Feynman diagram(s)



$$= -\int \frac{d^2k}{(2\pi)^2} \frac{\mathrm{Tr}\left[i\gamma^\nu(\not{p}+\not{k})i\gamma^\mu\gamma\not{k}\right]}{(k^2-i\epsilon)((p+k)^2-i\epsilon)} \tag{2.598}$$

where the minus sign comes from the Fermion loop. Since our particular interest is whether or not the 2-point function is conserved, we instead want to compute

$$\partial_\mu \langle\Omega| T\hat{j}_A^\mu(x)\hat{j}^\nu(0)|\Omega\rangle \tag{2.599}$$

which naively we would assume is zero due to the consevation of $\hat{j}_A^\mu$. But, what about contact terms that are supported at $y = x$? The Fourier transform of this *possibly* conserved 2-point function (2.599) is going to be similar to the one we had prior, just with an extra factor of ip_μ that we can move into the trace $\mathrm{Tr}[\cdots]$ and evaluate using the Feynman trick, and then dimensional regularization,

$$\begin{aligned}
\int d^2x\, e^{-ip\cdot x}\partial_\mu \langle\Omega| T\hat{j}_A^\mu(x)\hat{j}^\nu(0)|\Omega\rangle &= i\int \frac{d^2k}{(2\pi)^2}\frac{\mathrm{Tr}\left[\gamma^\nu(\not{p}+\not{k})\not{p}\gamma\not{k}\right]}{(k^2-i\epsilon)((p+k)^2-i\epsilon)} \\
&= i\int_0^1 dx \int \frac{d^2k}{(2\pi)^2}\frac{\mathrm{Tr}\left[\gamma^\nu(\not{k}+\not{p}(1-x))\not{p}\gamma(\not{k}-\not{p}x)\right]}{(k^2+p^2x(1-x)-i\epsilon)^2} \\
&= i\int_0^1 dx \int \frac{d^2k}{(2\pi)^2}\frac{\mathrm{Tr}\left(\gamma^\nu\not{k}\not{p}\gamma\not{k}\right) - x(1-x)p^2\mathrm{Tr}\left(\gamma^\nu\gamma\not{p}\right)}{(k^2+p^2x(1-x)-i\epsilon)^2}.
\end{aligned} \tag{2.600}$$

Here, we need to evaluate the two traces that are present. The blue term is the nicest and can be written in terms of the totally antisymmetric tensor

$$\mathrm{Tr}\left(\gamma^\nu\gamma\not{p}\right) = -2\epsilon^{\mu\nu}p_\mu \tag{2.601}$$

where $\epsilon^{01} = -\epsilon^{01} = 1$, but the purple term is the tricky one since there is a potential logarithmic divergence. To handle this we can use dimensional regularization (yes, even in a free theory!) with $D = 2-\epsilon$ but we will *maintain* that $\gamma = \gamma^0\gamma^1$[95]. Then, the purple

[95] This is one of the caveats with dimensional regularization that shows we need something better to handle divergences of Feynman diagrams since dimensional regularization cannot handle dimensionally-uniquely defined quantities such as the chiral matrix.

term becomes

$$\begin{aligned}
\mathrm{Tr}\Big(\gamma^\nu \not{k}\not{p}\gamma\not{k}\Big) &= k_\mu k_\rho \,\mathrm{Tr}\Big(\gamma^\mu\gamma^\nu\gamma^\rho\not{p}\gamma\Big) \\
&= \left(\frac{1}{D}k^2\eta_{\mu\rho}\right)\mathrm{Tr}\Big(\gamma^\mu\gamma^\nu\gamma^\rho\not{p}\gamma\Big) \\
&= \frac{2-D}{D}k^2\,\mathrm{Tr}\Big(\gamma^\nu\not{p}\gamma\Big) \\
&= \frac{2-D}{D}k^2\,(2\epsilon^{\mu\nu}p_\mu)
\end{aligned} \tag{2.602}$$

where we can see that this term would have vanished in $D=2$ and we would have missed this contribution! Thus, the integral in (2.600) becomes

$$\begin{aligned}
i\int_0^1 dx \int \frac{d^{2-\epsilon k}}{(2\pi)^{2-\epsilon}} \frac{\frac{\epsilon}{2-\epsilon}k^2 + x(1-x)p^2}{(k^2+p^2x(1-x)-i\epsilon)^2}\,(2\epsilon^{\mu\nu}p_\mu) &= -\int_0^1 dx\left(\frac{1}{2\pi}\frac{1}{\epsilon}\frac{\epsilon}{2}+\frac{1}{4\pi}+\mathcal{O}(\epsilon)\right)(2\epsilon^{\mu\nu}p_\mu) \\
&= -\frac{1}{\pi}\epsilon^{\mu\nu}p_\mu
\end{aligned} \tag{2.603}$$

where in the first line we Wick rotated which gave us that positive sign in the numerator. Now wait a second... (2.603) is a finite answer up to all orders in ϵ?! Well ok ok, let's go back to the correlator and see what this means.

In position space, the result (2.603) amounts to[96]

$$\partial_\mu \langle\Omega| T\hat{j}^\mu_A(x)\hat{j}^\nu(y)|\Omega\rangle = \frac{i}{\pi}\epsilon^{\mu\nu}\partial_\mu\delta^2(x-y). \tag{2.605}$$

In contrast, the analogous result for the computation but with two currents of the same U(1) group (namely without the insertion of γ) gives (so they are both non-chiral currents)

$$\begin{aligned}
\int d^2x\, e^{-ip\cdot x}\partial_\mu\langle\Omega| T\hat{j}^\mu(x)\hat{j}^\nu(y)|\Omega\rangle &= -\int_0^1 dx\left(\frac{1}{2\pi}\frac{1}{\epsilon}\frac{\epsilon}{2}-\frac{1}{4\pi}+\mathcal{O}(\epsilon)\right)\mathrm{Tr}\Big(\gamma^\nu\not{p}\Big) \\
&= 0
\end{aligned} \tag{2.606}$$

where it should be noted, dimensional regularization is need to arrive at this answer! (Also, $\mathrm{Tr}\Big(\gamma^\nu\not{p}\Big) = 2p^\nu$). What is the significance then of this answer? Consider a massless Dirac fermion coupled to a background U(1) gauge field A_μ with the corresponding action

$$S = -\int d^2x\,\overline{\psi}\gamma^\mu D_\mu\psi \tag{2.607}$$

where $D_\mu = \partial_\mu - iA_\mu$. This new action can be viewed as a free massless Dirac fermion that was deformed by the inclusion of

$$\Delta S = \int d^2x\, j^\mu(x)A_\mu(x). \tag{2.608}$$

[96] If you are still working in Euclidean space, the analogous result holds in Euclidean space as

$$\langle\partial_\mu j^\mu_A j^\nu(y)\rangle = -\frac{1}{\pi}\epsilon^{\mu\nu}\partial_\mu\delta^2(x-y). \tag{2.604}$$

In this new theory with background field A_μ turned on, the time-ordered Greens function involving $\partial_\mu \hat{j}^\mu_A(x)$ are related to the original one, with $A_\mu = 0$ theory, by

$$\begin{aligned}
\partial_\mu \langle T \hat{j}^\mu_A(x) \cdots \rangle_{\text{NEW}} &= \partial_\mu \langle T^\mu_A(x) e^{i \int d^2y\, \hat{j}^\nu(x) A_\nu(x)} \cdots \rangle_{\text{OLD}} \\
&= i \partial_\mu \hat{j}^\mu_A(x) \int d^2y\, \hat{j}^\nu(y) A_\nu(y) \\
&= -\frac{1}{\pi} \int d^2y\, \epsilon^{\mu\nu} \partial_\mu \delta^2(x-y) A_\nu(y) \\
&= \frac{1}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu}(x)
\end{aligned} \tag{2.609}$$

where the expectation value only contained contact terms, hence we were able to evaluate it directly. The time-ordered correlators of $\partial_\mu \hat{j}^\mu(x)$ would vanish if it were not for this integrated contact term! Consequently, we can identify that the correct operator equation for the chiral current is

$$\partial_\mu \hat{j}^\mu_A(x) = \frac{1}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu} \tag{2.610}$$

as an operator equation in the new theory. Note that under the transformation $\psi \mapsto e^{i\alpha\gamma}\psi$, the action S with nontrivial background field A_μ is still invariant. *However*, $\hat{j}^\mu_A(x)$ is no longer a conserved current! And thus we arrive at (2.610) which is known as the *chiral anomaly*.

But wait, why is there an anomaly in the first place? Our quantization procedure, or equivalently regularization scheme, does not respect the $U(1)_A$ symmetry of the action $\int [D\psi]\, e^{iS[\psi;A]}$ where it is the *measure* that is not invariant under $\mathrm{U}(1)_A$. On the other hand, the quantum violation of $U(1)_A$ is "mild" in the following sense.

Suppose a U(1)-symetry corresponds to a conserved current $\hat{j}^\mu$ with corresponds conservation equation $\partial_\mu \hat{j}^\mu(x) = 0$ and corresponding Noether charge $\hat{Q} = \int dx^1\, j^0(x^0, x^1)$. The symmetry transformation of a local operator $\hat{\mathcal{O}}(x)$ is given by

$$\hat{\mathcal{O}}(x) \mapsto \hat{\mathcal{O}}'(x) = U(\alpha) \hat{\mathcal{O}}(x) (U(\alpha))^{-1} \tag{2.611}$$

with $U(\alpha) = e^{-i\alpha\hat{Q}}$. Another way to formulate this transformation of the operator is to think about it in the following graph since the symmetry transformations are after all encoded in the corresponding Noether charge.

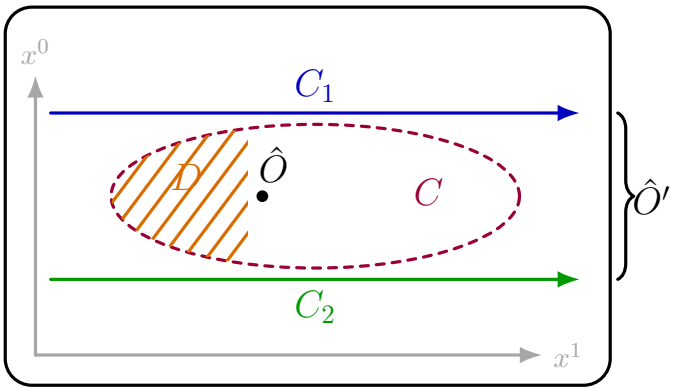


Figure 35. Integral transform of an operator via the U(1) symmetry.

The transformation of an operator via the U(1) symmetry can instead be written as a time ordering operator of contour around an operator with some "spread" or *distribution* (which is another way we should think of general QFT operators)

$$\hat{\mathcal{O}}'(x) = T\, e^{-i\alpha \int_{C_1 - C_2} j^0 dx^1} \hat{\mathcal{O}}. \tag{2.612}$$

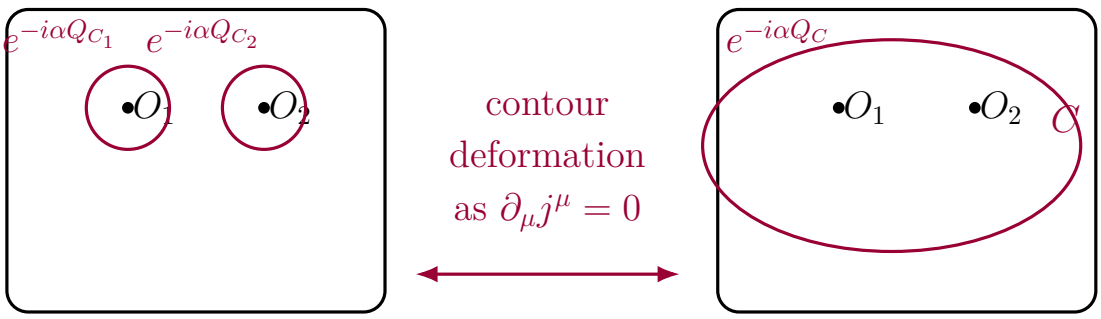
We can replace the difference in contours by deforming them to get $C_1 - C_2 = C$, and then perform Stokes theorem where the boundary of the region D is $\partial D = C$,

$$\hat{Q}_C = \int_C j^\mu n_\mu ds = \int_D \partial_\mu j^\mu d^2x. \tag{2.613}$$

A similar calculation can be done in the Euclidean signature with $\hat{\mathcal{O}}' = e^{-i\alpha \hat{Q}_C} \hat{\mathcal{O}}$. But now suppose that we have an insertion of two operators that transform under U(1) in a Greens function. We know that Greens functions are invariant under the symmetry transformation in the sense that

$$\langle \mathcal{O}_1 \mathcal{O}_2 \cdots \rangle = \langle \mathcal{O}_1 \mathcal{O}_2 \cdots \rangle \tag{2.614}$$

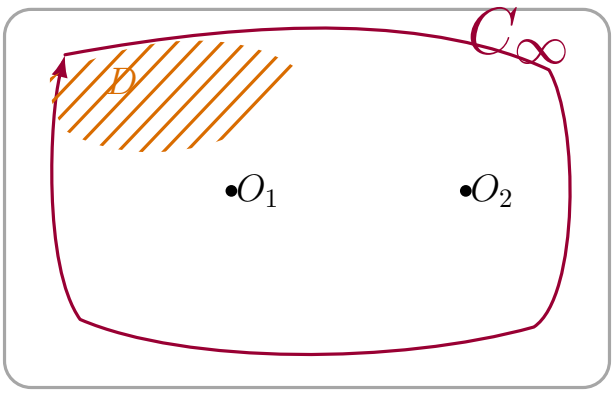
which in terms of graphs, has the following equivalent descriptions ([**AC:**] come back later to clean up):



In the second graph here, the new contour C can move off to infinity and we can therefore drop an overall $e^{i\alpha Q_C}$. Now, for the anomalous chiral "symmetry" $U(1)_A$ which acts as

$$\psi \mapsto \psi' = e^{i\alpha\gamma}\psi, \quad \mathcal{O} \mapsto \mathcal{O}', \quad \text{and} \quad \hat{\mathcal{O}}(x) = Te^{-i\alpha \hat{Q}_C} \hat{\mathcal{O}}(x) \tag{2.615}$$

where the charge here $\hat{Q}_C = \oint_C d^\mu_A n_\mu ds$ where C is an infinitesimal contour enclosing x. We can perform the same contour deformation argument for $U(1)_A$ as we did for $U(1)$ which now leads to (but remember, we can not make the same final argument of vanishing contour since that was dependent upon $\partial_\mu j^\mu = 0$) the same first graph as above with the 2+ operator insertions, but now we are left with a general spacetime region D given as



From which in terms of the Greens functions can be written as

$$\langle T\hat{\mathcal{O}}_1'\hat{\mathcal{O}}_2'\cdots\rangle = \langle Te^{-i\alpha(\hat{Q}_{C_1}+\hat{Q}_{C_2}+\cdots)}\hat{\mathcal{O}}_1\hat{\mathcal{O}}_2\cdots\rangle \tag{2.616}$$

where the exponential with the charges can be rewritten as $e^{-i\alpha(\hat{Q}_{C_1}+\hat{Q}_{C_2}+\cdots)} = e^{-i\alpha(\hat{Q}_\infty-\Delta Q)}$ where the total change in charge is

$$\begin{aligned}\Delta Q &= \hat{Q}_\infty - (\hat{Q}_{C_1}+\hat{Q}_{C_2}+\cdots)\\ &= \int_{D=\text{entire spacetime}} d^2x\,\partial_\mu\hat{j}_A^\mu\\ &= \frac{1}{2\pi}\int d^2x\,\epsilon^{\mu\nu}F_{\mu\nu}(x)\end{aligned} \tag{2.617}$$

which implies the following change in the Greens function correlator

$$\langle T\mathcal{O}_1\mathcal{O}_2\cdots\rangle = e^{i\alpha\Delta Q}\langle T\mathcal{O}_1\mathcal{O}_2\cdots\rangle. \tag{2.618}$$

The effect of this chiral "symmetry" transformation amounts to a deformation or shifting of the action by

$$\Delta S = \alpha\Delta Q = \frac{\alpha}{2\pi}\int d^2x\,\epsilon^{\mu\nu}F_{\mu\nu}(x) \tag{2.619}$$

of which what we have found is the $D=2$ analog of the Yang-Mills θ-term.

Now let us turn to the $D=4$ massless Dirac fermion case where the Lagrangian is just the massless Dirac Lagrangian

$$\mathcal{L} = -\overline{\psi}\gamma^\mu\partial_\mu\psi \tag{2.620}$$

with the following two U(1) symmetries:

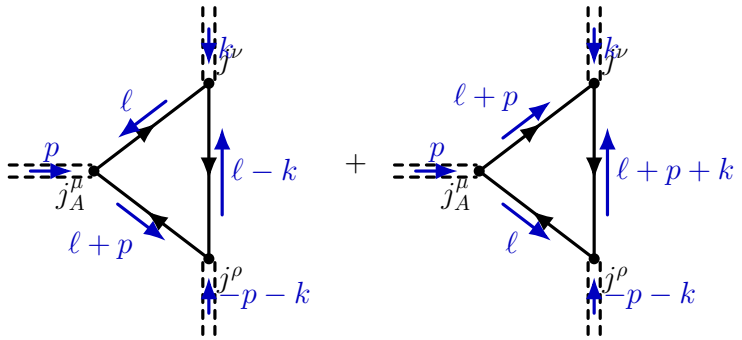
$$U(1): \quad j^\mu = i\overline{\psi}\gamma^\mu\psi \tag{2.621}$$

$$U(1)_A: \quad j_A^\mu = i\overline{\psi}\gamma^\mu\gamma_5\psi \tag{2.622}$$

where the chiral matrix here is $\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$. To find the anomalous effect, we will consider the 3-point function[97]

$$\langle\Omega|\, T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(z)\,|\Omega\rangle \tag{2.624}$$

which after Fourier transforming can be written in terms of the following two Feynman diagrams:



(2.625)

We are particularly interested in the contact term (which in this case is the first 1-loop contribution) which can be written as

$$\begin{aligned}
&\int d^4x\, e^{-ip\cdot x}\int d^4y\, e^{-ik\cdot y}\frac{\partial}{\partial x^\mu}\langle\Omega|\, T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(0)\,|\Omega\rangle \\
&\quad = -ip_\mu\int\frac{d^4\ell}{(2\pi)^4}\frac{\mathrm{Tr}\left[i\gamma^\mu\gamma_5\not{\ell}\, i\gamma^\nu(\not{\ell}-\not{k})i\gamma^\rho(\not{\ell}+\not{p})\right]}{(\ell^2-i\epsilon)((\ell-k)^2-i\epsilon)((\ell+p)^2-i\epsilon)} + (\nu\leftrightarrow\rho, k\leftrightarrow -p-k) \\
&\quad == i\int\frac{d^4\ell}{(2\pi)^4}\frac{\mathrm{Tr}\left[\not{p}\gamma_5(\not{\ell}\gamma^\nu(\not{\ell}-\not{k})\gamma^\rho(\not{\ell}+\not{p})\right]}{\ell^2(\ell-k)^2(\ell+p)^2} + (\nu\leftrightarrow\rho, k\leftrightarrow -p-k).
\end{aligned} \tag{2.626}$$

Now, just off of power counting alone, we could suspect there is a potential logarithmic divergence present in the integral, and so let us evaluate this with dimensional regularization $D = 4-\epsilon$. We can split the D-dimensional loop momentum ℓ^μ according to $\ell^\mu + \ell^\mu_\parallel + \ell^\mu_\perp$ where the first term is the $D = 4$ part while the last term (the perpendicular one) is the extra $-\epsilon$-dimension part. But, in doing this we will still maintain that the chiral matrix remains the same. As a consequence of doing this, $\not{\ell}_\parallel$ anti-commutes with γ_5 whereas $\not{\ell}_\perp$ commutes with γ_5. For instance, taking the $\not{p}\gamma_5$ term inside the trace can be re-written as

$$\not{p}\gamma_5 = (\not{\ell}+\not{p})\gamma_5 + \gamma_5\not{\ell} - 2\gamma_5\not{\ell}_\perp. \tag{2.627}$$

The purple and green terms once we put them back into the integral take the form

$$\int\frac{d^4\ell}{(2\pi)^4}\frac{(\ell+p)^2\,\mathrm{Tr}(\gamma_5\not{\ell}\gamma^\nu(\not{\ell}-\not{k})\gamma^\rho)}{\ell^2(\ell-k)^2(\ell+p)^2} \tag{2.628}$$

$$\int\frac{d^4\ell}{(2\pi)^4}\frac{\ell^2\,\mathrm{Tr}\Big(\gamma_5\gamma^\nu(\not{\ell}-\not{k})\gamma^\rho(\not{\ell}+\not{p})\Big)}{\ell^2(\ell-k)^2(\ell+p)^2} \tag{2.629}$$

[97] Or of course you can consider the Euclidean analog

$$\langle j^\mu_A(x)j^\nu(y)j^\rho(z)\rangle. \tag{2.623}$$

fro which after including the contributions from $(\nu \leftrightarrow \rho)$ and $(k \leftrightarrow -p-k)$, these terms will cancel. This just leave us with the perpendicular term above, and we get the following

$$\begin{aligned}\int d^4x\, e^{-ip\cdot x} \int d^4y\, e^{-ik\cdot y} \frac{\partial}{\partial x^\mu} \langle\Omega| T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(0)|\Omega\rangle \\ =2i\int \frac{d^D\ell}{(2\pi)^D} \frac{\mathrm{Tr}\Big[\gamma_5 \not{\ell}_\perp \not{\ell}\gamma^\nu(\not{\ell}-\not{k})\gamma^\rho(\not{\ell}+\not{p})\Big]}{\ell^2(\ell-k)^2(\ell+p)^2}.\end{aligned} \tag{2.630}$$

We now just have to handle the algebra in the numerator which goes as follows (with expanding $\ell = \ell_\perp + \ell_\parallel$)

$$\begin{aligned}-\ell_\perp^4\,\mathrm{Tr}(\gamma_5\gamma^\nu\gamma^\rho) + \ell_\perp^2\,\mathrm{Tr}\Big(\gamma_5\gamma^\nu(\not{\ell}_\parallel - \not{k})\gamma^\rho(\not{\ell}_\parallel + \not{p}) + \gamma_5\not{\ell}_\parallel\gamma^\nu\gamma^\rho(\not{\ell}_\parallel - \not{p}) + \gamma_5\not{\ell}_\parallel\gamma^\nu(\not{\ell}_\parallel - \not{k})\gamma^\rho\Big) \\ = \ell_\perp^2\cdot(-4i)\epsilon^{\nu\rho\alpha\beta}\Big(-(\ell_\parallel - k)_\alpha(\ell_\parallel - p)_\beta + \ell_{\parallel\alpha}(\ell_\parallel + p)_\beta - \ell_{\parallel\alpha}(\ell_\parallel - k)_\beta\Big) \\ = \ell_\perp^2(-4i)\epsilon^{\nu\rho\alpha\beta}k_\alpha p_\beta.\end{aligned} \tag{2.631}$$

This then turns the integral into

$$\begin{aligned}\int d^4x\, e^{-ip\cdot x} \int d^4y\, e^{-ik\cdot y} \frac{\partial}{\partial x^\mu} \langle\Omega| T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(0)|\Omega\rangle \\ =8\epsilon^{\nu\rho\alpha\beta}k_\alpha p_\beta \int \frac{d^D\ell}{(2\pi)^D}\frac{\ell_\perp^2}{\ell^2(\ell-k)^2(\ell+p)^2} + (\nu\leftrightarrow\rho, k\leftrightarrow -k-p) \\ = \frac{1}{2\pi^2}\epsilon^{\nu\rho\alpha\beta}p_\alpha p_\beta\end{aligned} \tag{2.632}$$

where the integral was performed as (after using the Feynman trick

$$\int \frac{d^D\ell}{(2\pi)^D}\frac{\ell_\perp^2}{(\ell^2+\mu^2)^3} = \frac{1}{(2\pi)^4}(2\pi^2)\frac{1}{\epsilon}\frac{-\epsilon}{4} + \mathcal{O}(\epsilon) = -\frac{1}{32\pi^2} \tag{2.633}$$

where $\ell_\perp^2 = (D-4)\ell^2/D$. Or in other words,

$$\int d^4x\, e^{-ip\cdot x} \int d^4y\, e^{-ik\cdot y} \frac{\partial}{\partial x^\mu} \langle\Omega| T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(0)|\Omega\rangle = \frac{1}{2\pi^2}\epsilon^{\nu\rho\alpha\beta}p_\alpha k_\beta. \tag{2.634}$$

We can Fourier transform back to position space to instead find

$$\partial_\mu \langle\Omega| T\hat{j}^\mu_A(x)\hat{j}^\nu(y)\hat{j}^\rho(0)|\Omega\rangle = -\frac{1}{2\pi^2}\epsilon^{\nu\rho\alpha\beta}\partial_\alpha\delta^4(x)\partial_\beta\delta^4(y). \tag{2.635}$$

Now consider a new theory (similar to what we did in the $D=2$ case) where we now couple the massless Dirac fermion to a *background* U(1) gauge field A_μ via the deformation to the action as

$$\Delta S = \int d^4x\, A_\mu(x)j^\mu(x) \tag{2.636}$$

which has the following affect on the derivative of the Greens function as

$$\partial_\mu\langle T\hat{j}^\mu_A(x)\cdots\rangle_{\mathrm{NEW}} = \partial_\mu\langle T\hat{j}^\mu_A(x)e^{i\int d^4y\, A_\nu(y)\hat{j}^\nu(y)}\cdots\rangle_{\mathrm{old}} \tag{2.637}$$

where again, this product only contains contact terms and when we expand to second order and perform proper Wick contractions we find

$$\partial_\mu \hat{j}^\mu_A(x)\left(-\frac{1}{2}\int d^4y d^4z\, \hat{j}^\nu(y)\hat{j}^\rho(z)A_\nu(y)A_\rho(z)\right) = \frac{1}{4\pi^2}\epsilon^{\nu\rho\alpha\beta}\partial_\alpha A_\nu(x)\partial_\beta A_\rho(x)$$
$$= -\frac{1}{16\pi^2}\epsilon^{\nu\rho\alpha\beta}F_{\mu\nu}F_{\rho\sigma}. \tag{2.638}$$

Once again, the chiral "symmetry" is anomalous in the sense that the current actually obeys the following operator equation

$$\partial_\mu \hat{j}^\mu_A(x) = -\frac{1}{16\pi^2}\epsilon^{\nu\rho\alpha\beta}F_{\mu\nu}F_{\rho\sigma}. \tag{2.639}$$

A remark needs to be made about this result. Even though we simply extracted this anomaly from a specific contact term that arose at order $\mathcal{O}(A^2)$, the result is in fact *exact*, and can be reproduced by inspecting the non-invariant of the suitably regularized path integral measure $[D\psi]$ with respect to the chiral "symmetry" transformation. This method is called the *Fujikawa method* by looking at the measure and can be found in chapter 22.2 of Weinberg's textbook [2]. Another method for recovering anomalies that is not seen in most textbooks however is called the *Nielson and Ninomiya* derivation, which instead looks at particle production of fermions and chiral fermions to deduce the anomaly that I recommend reading [19].

So far, we have seen that the chiral symmetry of a classical action of massless fermions may be rednered a non-symmetry in the quantum theory when the fermion fields are coupled to gauge fields either as background or dynamical fields. This notion of anomaly does not indicate any inconsistency of the QFT however. Rather, it iplies that the associated Noether current $\hat{j}^\mu_A$ while still a well-defined local operator, is not conserved, but rather obeys a modified conservation relation, namely

$$\partial_\mu \hat{j}^\mu_A = -\frac{1}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{2.640}$$

There is a *different notion* of anomaly however called a *'t Hooft anomaly* which refers to a **true** global symmetry of a QFT that *would become* anomalous if coupled to a dynamical gauge field via a deformation

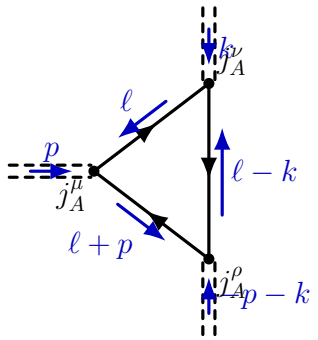
$$\Delta S = \int d^D x\, j^\mu(x)A_\mu(x), \tag{2.641}$$

leading to an inconsistent gauge theory that does not exist as a QFT. As an example of such a theory, we can consider again in $D = 4$ a theory of free massless Dirac fermions $\mathcal{L} = -\overline{\psi}\not{D}\psi$ with symmetry currents

$$j_\mu = i\overline{\psi}\gamma_\mu\psi \quad \text{and} \quad j^A_\mu = i\overline{\psi}\gamma_\mu\gamma_5\psi. \tag{2.642}$$

Previously, we analyzed the time-ordered Greens function that contained a single insertion of the chiral current. Here however we will now consider the time-ordered Greens function

of *three* chiral currents namely $\langle \partial_\mu j_A^\mu(x) j_A^\nu(y) j_A^\rho(z)\rangle$ which diagrammatically goes as (the red γ^5 are the new contributions from the two chiral currents)



$$\sim \int \frac{d^D\ell}{(2\pi)^D} \frac{\mathrm{Tr}\Big(\gamma_5 \not{\ell}_\perp \not{\ell} \gamma^\nu \gamma_5 (\not{\ell} - \not{k}) \gamma^\rho \gamma_5 (\not{\ell} + \not{p})\Big)}{\ell^2 (\ell - k)^2 (\ell + p)^2}. \tag{2.643}$$

We can evaluate the trace in the following way

$$\begin{aligned}
&\mathrm{Tr}\left(\gamma_5 \not{\ell}_\perp \not{\ell} \gamma^\nu \gamma_5 (\not{\ell} - \not{k}) \gamma^\rho \gamma_5 (\not{\ell} - \not{p})\right) \\
&= \ell_\perp^2 (-4i) \epsilon^{\nu\rho\alpha\beta} \left(-(\ell_\parallel - k)_\alpha (\ell_\parallel + p)_\beta + (-1) \ell_{\parallel\alpha} (\ell_\parallel + p)_\beta - \ell_{\parallel\alpha} (\ell_\parallel - k)_\beta\right) \\
&= \ell_\perp^2 (-4i) \epsilon^{\nu\rho\alpha\beta} (k - 2\ell)_\alpha p_\beta.
\end{aligned} \tag{2.644}$$

We can write the last term in red in a semi-fancy manner as

$$\ell_\alpha \equiv \frac{1}{3} \left(\ell_\alpha + (\ell - k)_\alpha + (\ell + p)_\alpha\right) + \frac{(k - p)_\alpha}{3}. \tag{2.645}$$

This first term in blue when evaluated with the rest of the integral that is of the form $\int d^D\ell \, \ell_\perp^2 (\ell^2 (\ell - k)^2 (\ell + p)^2)^{-1} (\cdots)$, then this is invariant under permutation on $(k, p, -k-p)$ which implies that there is no finite contribution which must be linear in k and p. Thus, we can make the following replacement

$$\begin{aligned}
\epsilon^{\nu\rho\alpha\beta} (k - 2\ell)_\alpha p_\beta &\to \epsilon^{\nu\rho\alpha\beta} \left(2 - 2\frac{k - p}{3}\right)_\alpha p_\beta \\
&= \frac{1}{3} \epsilon^{\nu\rho\alpha\beta} k_\alpha p_\beta
\end{aligned} \tag{2.646}$$

which gives us the final result

$$\partial_\mu \langle \Omega | T j_A^\mu(x) j_A^\nu(y) j_A^\rho(z) | \Omega \rangle = \frac{1}{3} \left(-\frac{1}{2\pi^2}\right) \epsilon^{\nu\rho\alpha\beta} \partial_\alpha \delta^4(x - z) \partial_\beta \delta^4(y - z). \tag{2.647}$$

We therefore come to the following interpretation. If we couple the fermion fields to a background U(1) gauge field via the *axial* current

$$\Delta S = \int d^4x \, j_A^\mu(x) A_\mu(x). \tag{2.648}$$

To make sense of this result, consider the following effective action $\Gamma[A]$

$$\begin{aligned}\Gamma[A+d\zeta]-\Gamma[A] &= \int d^4x\, \partial_\mu\zeta(x)\langle j_A^\mu(x)\rangle_{\text{new}} \\ &= -\int d^4x\, \zeta(x)\langle \partial_\mu j_A^\mu(x) e^{i\int d^4y\, j_A^\mu(y)A_\nu(y)}\rangle_{\text{old}} \\ &= \frac{1}{2}\int d^4x\, \zeta(x)\int d^4y d^4z\, A_\nu(y)A_\rho(z)\langle \partial_\mu j_A^\mu(x) j_A^\nu(y) j_A^\rho(z)\rangle \\ &= \int d^4x\, \zeta(x)\frac{1}{3}\frac{1}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}(x)F_{\rho\sigma}(x). \end{aligned} \tag{2.649}$$

But now notice the key different here compared to our prior result, $\Gamma[A]$ is *not* gauge invariant, and we cannot promote A_μ to a dynamical gauge field! Nonetheless, the 't Hooft anomaly is a useful characterization of the symmetry generated by $\hat{j}_A$ intrinsic to the original ungauged QFT.

2.3.8 Spontaneous Symmetry Breaking

It is possible for the vacuum state of a QFT to not be invariant with respect to a global symmetry. This is slightly surprising when we consider the following toy model. Consider a QM system with a single degree of freedom with Hamiltonian

$$H = p^2 + V(x) \tag{2.650}$$

where the potential behaves as $V(x) = V(-x)$ with two degenerate minima $x = \pm a$. This is graphed as the following[98]:

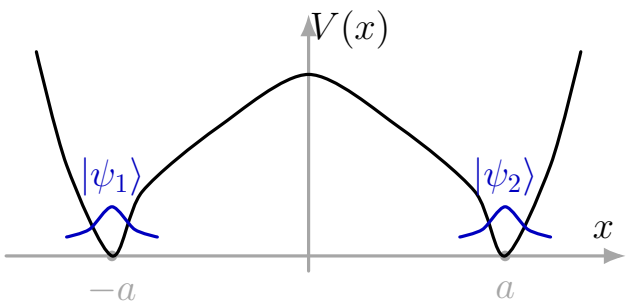


Figure 36. Double well potential $V(x)$ displaying the two eigenstates.

The Hamiltonian's corresponding eigenstates are $|\psi_\pm\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle \pm |\psi_2\rangle)$. Typically, the state $|\psi_+\rangle$ is the true ground state (since it is $\mathbb{Z}_2$-invariant) while $|\psi_-\rangle$ has exponentially small excitation energy $\Delta E \sim e^{-S_0/\hbar}$ where S_0 is the action of a Euclidean *tunneling* solution to $-\partial_\tau^2 x + V''(x) = 0$ with $x(\pm\infty) = \pm a$.

In a general QFT with this type of potential however, tunneling between different minima is suppressed by $e^{-S_0/\hbar}$ and so the action is $S_0 \propto V$, the spatial volume (which does diverge $V = \infty$). Suppose there is a U(1) global symmetry represented by the unitary

[98] It is a double well potential but I am not that good with tikz...

operator $U(\alpha) = e^{-i\alpha\hat{Q}}$ where we assume that $[U(\alpha), \hat{P}_\mu] = 0$. If a vacuum state $|\Omega\rangle$ is not invariant i.e. the vacuum does not obey

$$U(\alpha)\,|\Omega\rangle \neq |\Omega\rangle \Leftrightarrow \hat{Q}\,|\Omega\rangle \neq 0, \tag{2.651}$$

then we obtain a family of vacua by the symmetry transformation

$$U(\alpha)\,|\Omega\rangle \equiv |\Omega, \alpha\rangle\,. \tag{2.652}$$

We then claim the following: there exists a basis of vacua $|\Omega_i\rangle$ such that for any *local* operator $\hat{A}(x)$ that

$$\langle\Omega_i|\,\hat{A}(x)\,|\Omega_j\rangle = 0 \tag{2.653}$$

unless $i = j$. (2.653) is called *superselection sectors*. This is actually a natural consequence of causality. Consider that for any two local operators $\hat{A}(x)$ and $\hat{B}(y)$ we have the following expectation value

$$\begin{aligned}\langle\Omega_i|\,\hat{A}(x)\hat{B}(y)\,|\Omega_j\rangle = &\sum_k \langle\Omega_i|\,\hat{A}(0)\,|\Omega_k\rangle\,\langle\Omega_k|\,\hat{B}(0)\,|\Omega_j\rangle \\ &+ \int\limits_{\substack{\text{excited}\\ \text{states}}} d\alpha\,\langle\Omega_i|\,\hat{A}(0)\,|\alpha\rangle\,\langle\alpha|\,\hat{B}(0)\,|\Omega_j\rangle\,e^{iP_\alpha\cdot(x-y)}\end{aligned} \tag{2.654}$$

where this last term vanishes in the limit that $|x-y| \to \infty$. Then, for spacelike separation, namely for $(x-y)^2 > 0$, the commutator is $[\hat{A}(x), \hat{B}(y)] = 0$ when we take the $(x-y)^2 \to \infty$ limit. This then leads to the following result $A_{ik}B_{kj} = B_{ik}A_{kj}$ where $A_{ij} \equiv \langle\Omega_i|\,\hat{A}(0)\,|\Omega_j\rangle$. This implies that we can diagonalize the *vacuum matrices* (A_{ij}) and (B_{ij}) simultaneously.

More generally, we can find a basis of vacua $|\Omega_i\rangle$ such that A_{ij} is *diagonal* for any local operator $\hat{A}(x)$. The cluster property of Greens functions holds in each vacuum $|\Omega_i\rangle$ (1.304). Suppose that the vacuum state $|\Omega\rangle$ is not invariant with respect to the symmetry generated by the corresponding Noether charge $\hat{Q} = \int d^{D-1}\vec{x}\,\hat{j}^0(t,\vec{x})$ i.e. $\hat{Q}\,|\Omega\rangle \neq 0$. We therefore make the next claim.

We claim that $\hat{j}^\mu(x)\,|\Omega\rangle$ contains the 1-particle state of a massless scalar particle known as the *Nambu-Goldstone boson*.

One way we see this is to consider a charged scalar field operator $\hat{\phi}(x)$, say of charge -1 namely

$$[\hat{Q}, \hat{\phi}(x)] = -\hat{\phi}(x). \tag{2.655}$$

Note that then for the conjugate scalar $\hat{\phi}^\dagger(x)$ (since it is a charged scalar it does have an anti-particle) has the opposite charge

$$[\hat{Q}, \hat{\phi}^\dagger(x)] = +\hat{\phi}^\dagger(x). \tag{2.656}$$

And now we suppose that the expectation value of the charged scalar with respect to the vacuum does not vanish, $\langle\phi\rangle \equiv \langle\Omega|\,\hat{\phi}(x)\,|\Omega\rangle \neq 0$.

Note that if the symmetry is preserved by the vacuum $|\Omega\rangle$ namely $\hat{Q}\,|\Omega\rangle = 0$ then

$$\langle\phi\rangle = -\,\langle\Omega|\,[\hat{Q}, \hat{\phi}(0)]\,|\Omega\rangle = 0. \tag{2.657}$$

Then, by the same derivation of Källen-Lehmann spectral representation (1.257), we have

$$\langle\Omega|\,[\hat{j}^\mu(x),\hat{\phi}(y)]\,|\Omega\rangle = \frac{\partial}{\partial x_\mu}\int_0^\infty d\mu^2\,\rho_\phi(\mu^2)\left[\Delta_+(x-y;\mu^2)-\Delta_+(y-x;\mu^2)\right] \tag{2.658}$$

where

$$\Delta_+(x;\mu^2) \equiv \int \frac{d^Dk}{(2\pi)^{D-1}} e^{ik\cdot x}\theta(k^0)\delta(k^2+\mu^2) \tag{2.659}$$

and as before, the spectral function $\rho_\phi(\mu^2)$ is defined via (there are two now thanks to the anti-particle/opposite charge)

$$\int d\alpha\,\langle\Omega|\,\hat{j}^\mu(0)\,|\alpha\rangle\,\langle\alpha|\,\hat{\phi}(0)\,\langle\Omega|\,\delta^D(k-P_\alpha) = \frac{i}{(2\pi)^{D-1}}k^\mu\rho_\phi(-k^2)\theta(k^0) \tag{2.660}$$

$$\int d\alpha\,\langle\Omega|\,\hat{\phi}(0)\,|\alpha\rangle\,\langle\alpha|\,\hat{j}^\mu(0)\,|\Omega\rangle\,\delta^D(k-P_\alpha) = \frac{i}{(2\pi)^{D-1}}k^\mu\tilde{\rho}_\phi(-k^2)\theta(k^0) \tag{2.661}$$

where $\tilde{\rho}_\phi = -\rho_\phi$ by causality (since $[\hat{j}(x),\hat{\phi}(y)]=0$ for spacelike separation $(x-y)^2>0$). Now, since the current here is conserved $\partial_\mu\hat{j}^\mu(x)=0$ implies that $\mu^2\rho_\phi(\mu^2)=0$. But, this does **not** imply that $\rho_\phi=0$! Instead, the true implication can be found by

$$\begin{aligned}\int d^{D-1}\vec{x}\,\langle\Omega|\,[\hat{j}^0(x),\hat{\phi}(y)]\,|\Omega\rangle &= \langle\Omega|\,[Q,\hat{\phi}]\,|\Omega\rangle = -\langle\phi\rangle \\ &= \int d^{D-1}\vec{x}\int d\mu^2\,\rho_\phi(\mu^2)\left(-\frac{\partial}{\partial x^0}\right)\left(\Delta_+(x-y;\mu^2)-\Delta_+(y-x;\mu^2)\right)\Big|_{x^0=y^0} \\ &= i\int d\mu^2\,\rho_\phi(\mu^2)\end{aligned} \tag{2.662}$$

(where the paranthesis with the derivative is equal to $i\delta^{D-1}(\vec{x}-\vec{y})$) which implies that the spectral function (which holds the information on the mass spectrum) is equal to

$$\rho_\phi(\mu^2) = i\langle\phi\rangle\delta(\mu^2). \tag{2.663}$$

This contribution to the spectral function $\rho_\phi(\mu^2)$ can only come from some state $|\alpha\rangle$ with

$$\langle\Omega|\,\hat{j}^\mu(0)\,|\alpha\rangle\langle\alpha|\,\hat{\phi}(0)\,|\Omega\rangle \neq 0 \tag{2.664}$$

where the blue term implies that $|\alpha\rangle$ cannot be the vacuum while the cyan term cannot equal zero and implies that $|\alpha\rangle$ can only be a scalar (spinless) particle. Therefore, we instead rewrite the $|\alpha\rangle$ state as $|\alpha\rangle \to \left|B,\vec{k}\right\rangle$ with $k^0=|\vec{k}|$ and this state is called the *Nambu-Goldstone boson*. The corresponding 1-particle state can then be ground via the expectation value of the Nambu-Golstone state with the vacuum with the current and scalar field namely

$$\langle\Omega|\,\hat{j}^\mu(0)\left|B,\vec{k}\right\rangle = i\frac{Fk^\mu}{(2\pi)^{\frac{D-1}{2}}\sqrt{2k^0}} \tag{2.665}$$

$$\left\langle B,\vec{k}\right|\hat{\phi}(0)\,|\Omega\rangle = \frac{\mathcal{Z}_\phi^{\frac{1}{2}}}{(2\pi)^{\frac{D-1}{2}}\sqrt{2k^0}} \tag{2.666}$$

from which we can read off using (2.664) what the relation between the constant F and the renormalization constant to the vacuum is namely

$$iF\mathcal{Z}_{\phi}^{\frac{1}{2}} = -\langle\phi\rangle. \tag{2.667}$$

This seemingly simple result has massive implications. In essence, (2.667) gives us a relation between the renormalization constant of the Nambu-Goldstone boson to that of the ground state for a symmetry breaking system.

A simple model that displays this behavior of spontaneous symmetry breaking is that of a complex scalar

$$\mathcal{L} = -\partial_\mu\phi^*\partial^\mu\phi - V(|\phi|) \tag{2.668}$$

with a potential of the following form (very similar to that of the 1D double-well):

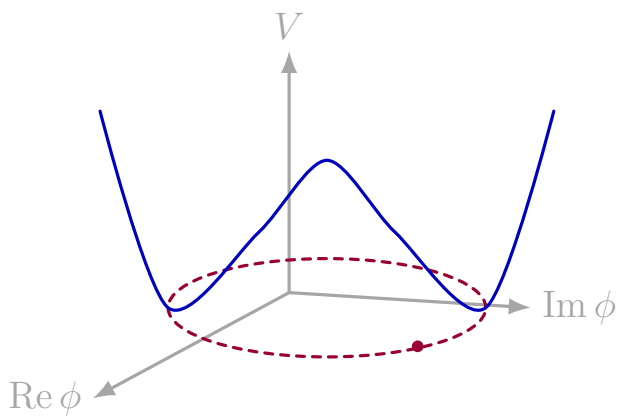


Figure 37. Sketch of a 3D double-well potential that is commonly referred to as a *Mexican-hat potential* due to the shape it traces out.

Note that the purple point on the circle denotes/traces out a class of minima or called a family of U(1) symmetry breaking vacua with $\langle\phi\rangle \neq 0$. The corresponding U(1) current is

$$j_\mu = -i\phi^*\overleftrightarrow{\partial}_\mu\phi. \tag{2.669}$$

We can reprameterize the field variable in terms of the spacetime position and the angle of which it takes with respect to the minima or family of U(1) minima as

$$\phi(x) \equiv r(x)e^{i\theta(x)}. \tag{2.670}$$

What this implies for the Lagrangian is

$$\mathcal{L} = -\partial_\mu r\partial^\mu r - V(r) - r^2\partial_\mu\theta\partial^\mu\theta \tag{2.671}$$

with corresponding change in the current $j_\mu = 2r^2\partial_\mu\theta$. There is of course a corresponding U(1) Noether charge $Q = \int d\vec{x}j^0 = -\int d\vec{x}2r^2\dot{\theta}$ where $2r^2\dot{\theta} = \pi_\theta$, the conjugate momentum. The symmetry generator or charge operator generates the following symmetry transformation on the $\hat{\theta}(x)$ field $[\hat{Q}, \hat{\theta}(x)] = i$. The Noether charge $\hat{Q}$ generates shift symmetry on

$\theta(x)$. Also, the potential $V(r)$ is minimized at $r = v$ (which is greater than zero) and so it would be instructive to rewrite r in terms of the shifted minima plus the field dynamics

$$r(x) \equiv v + \rho(x). \tag{2.672}$$

The Lagrangian then becomes

$$\begin{aligned}\mathcal{L} = &- \partial_\mu \rho \partial^\mu \rho - \frac{1}{2} V''(v) \rho^2 + \text{constant} \\ &- v^2 \partial_\mu \theta \partial^\mu \theta + (\text{interaction terms involving } \rho \text{ and } \partial_\mu \theta)\end{aligned} \tag{2.673}$$

where the θ-kinetic term is massless and is the field creating/being the Nambu-Goldstone boson. An **extremely key** feature of spontaneous symmetry breaking is that if we deform the theory in such a way that the symmetry is *explicitly* broken slightly, for example we add to the scalar potential as $\Delta V(\phi) = -\varepsilon \operatorname{Re}(\phi)$ for $\varepsilon > 0$, then there is a unique vacuum *no matter how small* ε is. Namely, take the potential in figure 37 and send $V \to V + \Delta V$, and then the purple point is a unique vacuum $\langle \phi \rangle$.

In the limit $\varepsilon \to 0$, this vacuum state approaches a specific vacuum in the family of degenerate vacua of the un-deformed theory. In this example, it is the one in which $\langle \phi \rangle$ is real and positive.

Bringing this all the way back to QCD with massless quarks 2.3, there is a non-anomalous global symmetry

$$U(1)_{\text{non-chiral}} \times SU(N_f)_L \times SU(N_f)_R \tag{2.674}$$

where $SU(N_f)_L \in U_L$ and $SU(N_f)_R \in U_R$. The corresponding transformations for the quarks/left-right fermions are

$$\psi_{IL} \to (U_L)_I{}^J \quad \text{and} \quad \psi_{IR} \to (U_R)_I{}^J \psi_{JR}. \tag{2.675}$$

It is believed[99] that the vacuum of QCD is only invariant under

$$U(1)_{\text{non-chiral}} \times SU(N_f)_{\text{non-chiral}}. \tag{2.676}$$

An operator that transforms non-trivially under the chiral $SU(N_f)$ whose vacuum expectation value could indicate the spontaneous breaking of chiral symmetry, is the *quark mass operator*

$$\Phi_J{}^I = \overline{\psi}_L{}^I \psi_{JR} \quad \text{with} \quad \langle \Phi_J{}^I \rangle \equiv A_J{}^I \neq 0. \tag{2.677}$$

[99] The words, "it is believed..." are used literally since it is known that QCD in the non-chiral sector has an invariant global symmetry, but as we will see, the chiral aspect of the global group gives rise to pseudoscalar mesons, and for $N_f = 3$, all eight of the pseudoscalar mesons of the quark model become Goldstone boson. The actual masses of the mesons are found via *chiral perturbation theory*. It is unknown for certain cases whether or not some phases of quark matter the full chiral symmetry is restored. Also, if the symmetry breaking is misaligned (which is a topic covered in the axion-section of these notes 4) then the global symmetry can become enlarged and new quark-states can be produced such as candidates for *dark matter* namely a dark sector SU(3).

Under $SU(N_f)_L \times SU(N_F)_R$ (where their product is in (U_L, U_R) which is represented by unitary operator $\hat{U}$) the transformation of the quark mass operator is

$$\hat{\Phi}_J{}^I \to \hat{U}\hat{\Phi}_J{}^I\hat{U}^{-1} = (U_R)_J{}^M\hat{\Phi}_M{}^K(U_L^\dagger)_K{}^I. \tag{2.678}$$

Given a vacuum state $|\Omega\rangle$, the unitary operator will generate a new vacuum state $\hat{U}\,|\Omega\rangle \equiv |\Omega\rangle_U$ which has the following expectation value under the quark mass operator

$$_U\langle\Omega|\,\hat{\Phi}_J{}^I\,|\Omega\rangle_U = \langle\Omega|\,\hat{U}^{-1}\hat{\Phi}_J{}^I\hat{U}\,|\Omega\rangle = \left(U_R^{-1}\cdot A\cdot U_L\right)_J{}^I. \tag{2.679}$$

We can choose (U_L, U_R) such that $U_R^{-1}AU_L$ is diagonal, and thus without loss of generality, we assume that A it itself diagonal. Assuming that the vacuum is invariant under a *non-chiral* $SU(N_f)$, $A_J{}^I$ must be proportional to the identity matrix namely $A_J{}^I = A\delta_J{}^I$. In this vacuum, the unbroken non-chiral $SU(N_f)$ corresponds to (U_L, U_R) with $U_L = U_R$. A few remarks are in order.

The first remark is that the non-chiral $SU(N_f)$ cannon be spontaneously broken in the absence of θ-angle. this can be argued by deforming to the theory with nonzero quark masses, and consider the Euclidean path integral computation of the vacuum expectation value (vev)

$$\langle\Phi_J{}^I\rangle = \frac{1}{Z}\int[DA]_{\text{G.F.}}e^{-S_{\text{YM}}[A]} \times \int D\psi\, e^{-\int\overline{\psi}(\not{D}_A+m)\psi}\Phi_J{}^I \tag{2.680}$$

where G.F. denotes the gauge-fixed measure. The first integral over $[DA]$ is defined with a positive UV regulator which is real and positive when $\theta = 0$. The second term can be integrated exactly to give

$$\int D\psi\, e^{-\int\overline{\psi}(\not{D}_A+m)\psi}\Phi_J{}^I = \det(\not{D}_A + m)\langle\overline{\psi}_L{}^I\psi_{JR}\rangle_A. \tag{2.681}$$

The determinant here is positive(!) with eigenvalues of Euclidean $\not{D}_A$ that comes in pairs $\pm i\lambda$ with $\lambda \in \mathbb{R}$. The last expectation value is the trace of the fermion propagator in the background gauge field. This term can be shown to uniformly approach $\propto \delta_J{}^I$ in the limit where all quark masses are equal for any Euclidean gauge field configuration. This result is called the *Vafa-Witten theorem* (from 1984 [20, 21]).

The second remark I would like to make is that given $\langle\Phi_J{}^I\rangle = A\delta_J{}^I$ may be complex a priori. One may attempt to rotate it's phase with axial $U(1)_A$ transformation on ψ_L and ψ_R, but recall that $U(1)_A$ is anomalous, and that a $U(1)_A$ "non-symmetry" rotation amounts to shifting the θ-angle.

To summarize then, QCD with N_f massless Dirac fermions (the quarks) has a global symmetry structure of

$$U(1) \times SU(N_f)_L \times SU(N_f)_R \tag{2.682}$$

where only the non-chiral diagonal portion of $SU(N_f)_L$ is preserved by the vacuum. The broken chiral $SU(N_f)$ generators are in correspondence with $N_f^2 - 1$ species of Nambu-Goldstone boson or in other words, *pions*.

In particular, since $\Phi_J{}^I \equiv \overline{\psi}_L{}^I\psi_{JR}$ transforms non trivially under the chiral $SU(N_f)$, the 1-particle state $\hat{\Phi}_J{}^I\,|\Omega\rangle$ contains, namely has nonzero overlap with, 1-pion states.

2.3.9 Effective Field Theory of Pions

After having found the pions, we now seek an *effective field theory* of pions[100], in which pions are created by fundamental scalar fields that appear as field variables in an effective Lagrangian. We shall assume a $SU(N_f)_L \times SU(N_f)_R$ global symmetry is spontaneously broken to diagonal $SU(N_f)$.

Consider an $SU(N_f)$-valued scalar field $U(x)$ that transforms under $U(U_L, U_R) \in SU(N_f)_L \times SU(N_f)_R$ by the following transformation rule

$$U(x) \mapsto U_R U(x) U_L^{-1}. \tag{2.683}$$

By spontaneous symmetry breaking, the vev of this scalar field is

$$\langle \Omega | \, U(x) \, | \Omega \rangle = U_0 \in SU(N_f). \tag{2.684}$$

The symmetries preserved by the vacuum $|\Omega\rangle$ correspond to (U_L, U_R) that obey $U_R U_0 U_L^{-1} = U_0$ or in other words $U_L = U_0^{-1} U_R U_0$. This is exactly what we mean by the diagonal part of $SU(N_f)$.

We will *assume* that there is a Wilsonian effective Lagrangian of the scalar fields $U(x)$ that comes with a floating cutoff Λ along with a suitable regularization scheme. As in any Wilsonian effective Lagrangian, there can be infinitely many couplings that involve arbitrarily many derivative and powers of fields that are Λ-dependent.

We expect that such a description is valid so long as $\Lambda \lesssim \Lambda_{\text{QCD}}$, such that the pions are the only possible excitations at energies below Λ.

In the low energy limit, which is equivalent to saying the fields are slowly varying fields, the Lagrangian is dominated by terms with the fewest number of derivatives. Also, no potential term is possible by our symmetry assumption.

At the two-derivative order, the only possible Lagrangian is

$$\mathcal{L} = -\frac{F^2}{16} \operatorname{Tr}\Big(\partial_\mu U \partial^\mu U^\dagger\Big) \tag{2.685}$$

where possible higher order terms are

$$(\operatorname{Tr} \partial_\mu U \partial^\mu U^\dagger), \quad \operatorname{Tr}\Big(\partial_\mu U \partial_\nu U^\dagger\Big) \operatorname{Tr}\Big(\partial^\mu U \partial^\nu U^\dagger\Big), \quad \operatorname{Tr}\Big(\partial_\mu U \partial^\mu U^\dagger \partial_\nu U \partial^\nu U^\dagger\Big) \tag{2.686}$$

and so on and so on.

A quick remark is in order before we proceed. The $SU(N_f) \times SU(N_f)$-invariant terms can be built out of traces of products of derivatives of U and $U^\dagger$ alternatively. There is an exceptional term however, known as the *Wess-Zumino-Witten* (WZW) term [22, 23, 23][101]

[100] This is characteristically different than the *quantum* effective action because we are constructing the action, not just starting with it and computing the corresponding 1PI's. But in effect, they are related via a Fourier transform (actually the sum of quantum effective actions i.e. their 1PI's matched the effective action).

[101] Witten's paper is particularly clear on the formulation of this term and also Weinberg chapter 19.8 and 22.7 go into particular depth as to how the homotopy groups of the global symmetries relate to the ssymmetry left over after spontaneous symmetry breaking.

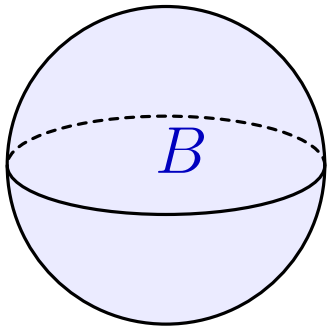
that is not of the above form,but exists for $N_f \geq 3$. The WZW term takes the form

$$\Delta S_{\text{WZW}} = n \cdot \int_B \omega \tag{2.687}$$

where ω here is topological in nature and can be written as[102]

$$\omega = -\frac{i}{240\pi^2} \operatorname{Tr}(U{-}1dU)^5) \tag{2.689}$$

where now we have extended to a five dimensional auxiliary space we shall denote as B. A way to think about this is with the following sketch:



Here, we have a fully-shaded or filled in sphere that is called a ball and in this case, a 5-ball that we denote B. This is our axillary space. To get back to normal 4D spacetime, we instead want the boundary of this ball ∂B = spacetime, which is where our usual spacetime does live (in this mathematical picture). The WZW term (2.689) turns out to be well-defined up to shifts by integer multiple of 2π for $N_f \geq 3$ in which case, $e^{i\Delta S_{\text{WZW}}}$ is well-defined for integer n. Consideration of gauging $U(1) \supset SU(N_f)_{\text{diag}}$ which leads to an anomaly in chiral $SU(N_f)$, can be used to fix the overall constant $n = N_c$ where N_c is the number of colors. *This* is not our focus though; back to the pions (2.685).

Let us specialize to the case of $N_f = 2$ where the scalars (pions) are then $U(x) \in SU(2)$, and as can be seen in the problem sets, we can parameterize $U(x)$ in terms of unconstrained field variables $\vec{\zeta}(x) = (\zeta_1(x), \zeta_2(x), \zeta_3(x)$ which allows us to rewrite the pion field as

$$U(x) = \frac{(1 + i\vec{\zeta} \cdot \vec{\sigma})^2}{1 + \zeta^2}. \tag{2.690}$$

The $\vec{\zeta}$ vector here can be viewed as a stereographic coordinates on $SU(2) \simeq S^3$ via the following image/projection:

[102]This term is actually deceiving since it is written in terms of a 5-five form. The full terms is actually written as

$$n \int_B \omega = n \int_B d^5x\, \omega = n \int_B d^5x\, - \frac{i}{240\pi^2} \epsilon^{\mu\nu\rho\sigma\tau} \operatorname{Tr}\Big(U^\dagger \frac{\partial U}{\partial x^\mu} U^\dagger \frac{\partial U}{\partial x^\nu} U^\dagger \frac{\partial U}{\partial x^\rho} U^\dagger \frac{\partial U}{\partial x^\sigma} U^\dagger \frac{\partial U}{\partial x^\tau} \Big) \tag{2.688}$$

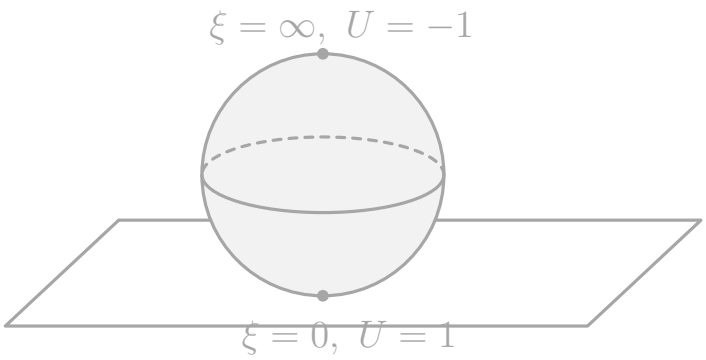


Figure 38. Stereographic projection of the scalar field $U(x)$ under group structure $SU(2)$ to vectors $\vec{\zeta}$.

It then becomes a matter of (tedious) algebra to verify that the chiral Lagrangian can be expressed in terms of $\vec{\zeta}$ as

$$\begin{aligned}\mathcal{L} &= -\frac{F^2}{16}\,\mathrm{Tr}\Big(\partial_\mu U \partial^\mu U^{-1}\Big) \\ &= -\frac{F^2}{2}\frac{\partial_\mu \vec{\zeta}\cdot\partial^\mu\vec{\zeta}}{(1+\zeta^2)^2}\end{aligned} \tag{2.691}$$

where (2.691) is commonly referred to as the *non-linear sigma model*. Under infinitesimal $SU(2)_L \times SU(2)_R$ symmetry transformation

$$\begin{aligned}U &\mapsto U_R U U_L^{-1} \\ &= U + i\vec{\theta}_R\cdot\frac{\vec{\sigma}}{2}U - iU\vec{\theta}_L\cdot\frac{\vec{\sigma}}{2}\end{aligned} \tag{2.692}$$

where $U_{L/R} = \exp\Big(i\vec{\theta}_{L/R}\cdot\frac{\vec{\sigma}}{2}\Big)$. We now defined $\vec{\theta}_{V/A} \equiv \frac{1}{2}(\vec{\theta}_R \pm \vec{\theta}_L)$ which then implies the infinitesimal transformation of $\vec{\zeta}$ is

$$\delta\vec{\zeta} = -\vec{\theta}_V \times \vec{\zeta} + \frac{1}{2}\vec{\theta}_A(1-\zeta^2) + \vec{\zeta}(\vec{\theta}_A\cdot\vec{\zeta}) \tag{2.693}$$

where the blue term holds the non-chiral symmetry that acts linearly in behavior while the purple term is the chiral symmetry that behaves non-linearly. It will be convenient to work with a re-scaled derivative

$$\vec{D}_\mu \equiv \frac{\partial_\mu\vec{\zeta}}{1+\zeta^2} \tag{2.694}$$

which has the following transformation rule

$$\delta\vec{D}_\mu = -\vec{\theta}_V \times \vec{D}_\mu - (\vec{\zeta}\times\vec{\theta}_A)\times\vec{D}_\mu. \tag{2.695}$$

The higher derivative corrections to the chiral Lagrangian can be organized as

$$\mathcal{L} = -\frac{F^2}{2}\vec{D}_\mu\vec{D}^\mu - \frac{c_4}{4}(\vec{D}_\mu\cdot\vec{D}^\mu)^2 - \frac{c_4'}{4}(\vec{D}_\mu\cdot\vec{D}_\nu)(\vec{D}^\mu\cdot\vec{D}^\nu) + \cdots \tag{2.696}$$

We can put the kinetic term into a canonical normalized by redefining the *pion field* as $\vec{\pi} \equiv F\vec{\zeta}$, and the effective Lagrangian in terms of $\vec{\pi}$ then expands as

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \vec{\pi})^2 + \frac{1}{F^2}\vec{\pi}^2(\partial_\mu \vec{\pi})^2 + \cdots \tag{2.697}$$

$$- \frac{c_4}{4F^4}((\partial_\mu \vec{\pi})^2)^2 - \frac{c_4'}{4F^4}(\partial_\mu \vec{\pi} \cdot \partial_\nu \vec{\pi})^2 + \cdots \tag{2.698}$$

Note that F has mass dimension M^1, and the dimensionless effective coupling would then be $\sim$ energy$/F$.

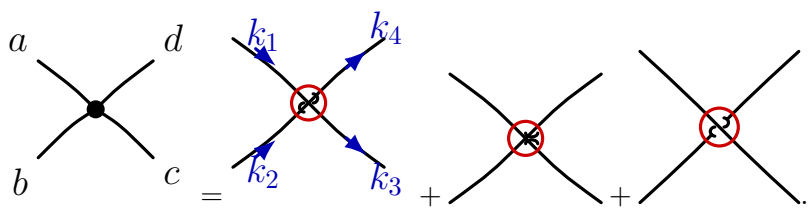
Let us now inspect the $2 \to 2$ scattering amplitude of pions in this framework, in other words, *chiral perturbation theory* (recall that the vectors here are over the SU(2) group, and so the external legs have group indices) via the 4-point function:

(2.699)

where our Mandelstam variables are the usual $s = -(k_1 + k_2)^2$, $t = -(k_1 - k_4)^2$, and $u = -(k_1 - k_3)^2$. The amplitude corresponding to (2.699) can be written as

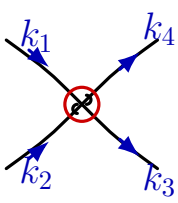
$$\mathcal{A}_{(cd|ab)} = \delta_{ab}\delta_{cd}\mathcal{A}(s,t) + \delta_{ac}\delta_{bd}\mathcal{A}(u,t) + \delta_{ad}\delta_{bc}\mathcal{A}(t,s). \tag{2.700}$$

At order E^2/F^2, the only contributions to the amplitude are from the *effective* tree diagrams[103]



(2.701)

The first diagram will contribute to the $\mathcal{A}(s,t)$ amplitude as

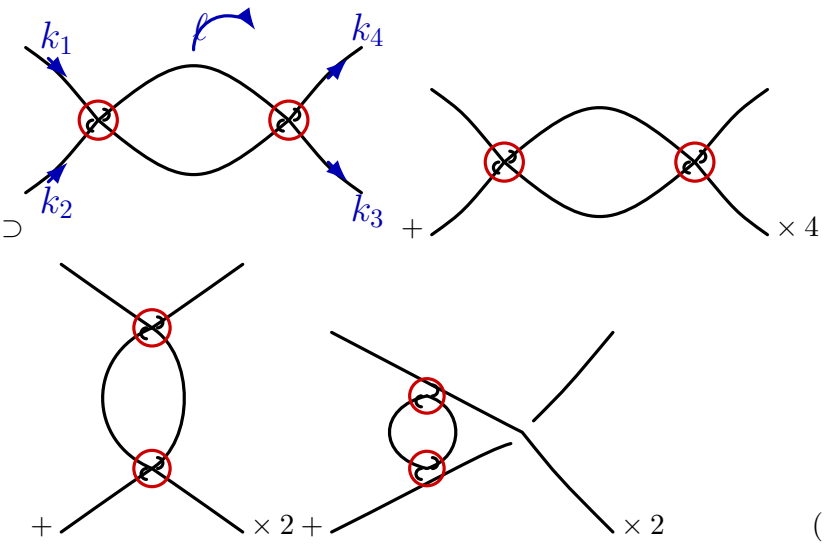


$$= \frac{4}{F^2}(-k_1 \cdot k_2 - k_3 \cdot k_4) = \frac{4s}{F^2}. \tag{2.702}$$

At next order, E^4/F^4, there are two types of contributions. The first is from 1-loop diagrams built out of the 2-derivative vertices. The second type of contribution comes tree diagrams built out of the 4-derivative vertices. But, it suffices to consider diagrams that

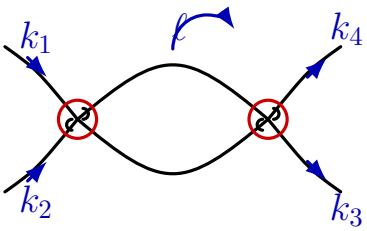
[103] [**AC:**] The tree diagrams are supposed to show a sort of "crossing legs" since none of them actually hit, they are just effective vertices, and so the first one is the s-channel, the second is t-channel, and the third is u-channel.

contribute to $\delta_{ab}\delta_{cd}\mathcal{A}(s,t)$ for now, namely

$\delta_{ab}\delta_{cd}\mathcal{A}(s,t) \supset$ 

(2.703)

As an example, the first diagram above evaluates to the following



$$= \left(\frac{4}{F}\right)^2 \cdot \frac{3}{2} \int^{\Lambda} \frac{d^4\ell_E}{(2\pi)^4} \frac{(-k_1\cdot k_2 - \ell\cdot(k_1+k_-\ell))(-k_3\cdot k_4 - \ell\cdot(k_1+k_2-\ell)}{\ell^2(k_1+k_2-\ell)^2} \tag{2.704}$$

where the red number 3 denotes the sum/loop index while the blue number 2 denotes the symmetry factor. Also, Λ here is our Wilsonian cutoff. The loop momentum integral contains a power divergence of the order $\sim \Lambda^4$ that comes from the $\Lambda^2 s$ term, but also a logarithmic divergence that is of the order $\sim s^2 \log(\Lambda^2/-s)$. In the Wilsonian framework, these are not divergences as we take Λ to be finite since $\Lambda \lesssim \Lambda_{\text{QCD}}$.

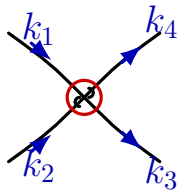
However, we should adopt a Wilsonian cutoff scheme that respects the underlying global symmetries. A priori, a simple cutoff on the Fourier modes of $\vec{\pi}(x)$ may not respect the nonlinearly realized chiral SU(2).

For instance, the appearance of Λ^4 contribution to $\mathcal{A}(s,t)$ would generate an effective potential that violates the chiral SU(2) symmetry. This problem can be fixed by either working with a symmetry-preserving scheme, for example dimensional regularization, or by including suitable local counter terms to cancel the power "divergences."

Modulo the power "divergences," which should be canceled in the "correct" scheme or equivalently with the appropriate counter terms, the 1-loop contribution to $\mathcal{A}(s,t)$ at order E^4/F^4 is

$$\mathcal{A}(s,t) \supset \frac{1}{F^4}\left(\frac{s^2}{2\pi^2}\log\frac{\Lambda^2}{-s^2} + \frac{u^2-s^2+3t^2}{12\pi^2}\log\frac{\Lambda^2}{-t} + \frac{t^2-s^2+3u^2}{12\pi^2}\log\frac{\Lambda^2}{-u}\right). \tag{2.705}$$

In addition, we also have at this order the contribution from tree diagrams involving the 4-derivative vertices with coefficients c_4 and c_4'



$$= \frac{c_4}{2}s^2 - \frac{c_4'}{2}(t^2+u^2). \tag{2.706}$$

We can now absorb the remaining Λ-dependence into c_4 and c_4' via

$$c_4 = c_4(\mu) + \frac{2}{3\pi^2}\log\frac{\Lambda^2}{\mu^2} \tag{2.707}$$

$$c_4' = c_4'(\mu) + \frac{4}{3\pi^2}\log\frac{\Lambda^2}{\mu^2} \tag{2.708}$$

which gives us the final result

$$\begin{aligned}\mathcal{A}(s,t) =& \frac{4s}{F^2} + \frac{1}{F^4}\left(\frac{s^2}{2\pi^2}\log\frac{\mu^2}{-s} + \frac{u^2-s^2+3t^2}{12\pi^2}\log\frac{\mu^2}{-t} + \frac{t^2-s^2+3u^2}{12\pi^2}\log\frac{\mu^2}{-u}\right.\\ &\left. - \frac{c_4(\mu)}{2}s^2 - \frac{c_4'(\mu)}{2}(t^2+u^2)\right) + \mathcal{O}(F^{-6}).\end{aligned} \tag{2.709}$$

Here μ is an arbitrary finite renormalization scale. While we do not know the value of c_4 and c_4' based on our symmetry consideration, we see that remarkably chiral perturbation theory still makes unambiguous predictions on the structure of the low-energy/momentum expansion of $\mathcal{A}(s,t)$! Of course though, a few remarks are in order.

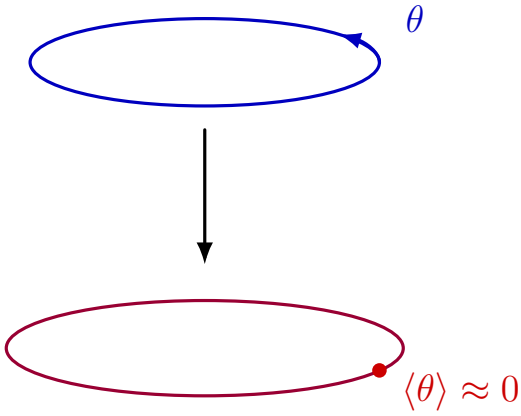
The first remark is that in real world QCD, the quark fields are massive, although up and down quarks are light, and their mass terms may be viewed as a small deformation of QCD, with gauge group SU(3), with $N_f = 2$ flavors of massless quarks by the term

$$\Delta\mathcal{L} = -m_u\overline{u}u - m_d\overline{d}d \tag{2.710}$$

where we have denoted $\psi_1 = u$ and $\psi_2 = d$. In terms of the pion $\Phi_J{}^I = \overline{\psi}_{:}{}^I\psi_{JR}$, we can write the quark mass term as

$$\Delta\mathcal{L} = -\frac{m_u+m_d}{2}\operatorname{tr}\Big(\Phi+\Phi^\dagger\Big) - \frac{m_u-m_d}{2}\operatorname{tr}\Big(\sigma_3(\Phi+\Phi^\dagger)\Big). \tag{2.711}$$

But what is this terms effect on the pion mass? Consider the following example of spontaneously broken U(1):



The effective Lagrangian is $\mathcal{L} = -\frac{R^2}{2}(\partial_\mu\theta)^2$ but with the U(1) breaking the Lagrangian will pick up a deformation of $\Delta\mathcal{L} = u\cos\theta$ and therefore when we expand θ the Lagrangian becomes

$$\mathcal{L} + \Delta\mathcal{L} = -\frac{R^2}{2}(\partial_\mu\theta)^2 - \frac{u}{2}\theta^2 + \cdots \tag{2.712}$$

The *pseudo*-Nambu-Goldstone boson created by $\theta(x)$ now acquires a mass m of

$$m^2 \approx \frac{u}{R^2} \tag{2.713}$$

to leading order at small u. The same analysis is applicable to deformation of the chiral Lagrangian of pions, except that we should be careful in comparing the normalization of kinetic terms. Recall that $U(x) = (1+i\vec{\zeta}\cdot\vec{\sigma})^2(1+\zeta^2)^{-1}$, transforms under $SU(2)_L \times SU(2)_R$ the same way as $\Phi_J{}^I$ namely $\Phi \mapsto U_R\Phi U_L^{-1}$. In particular, the $U(1) \supset SU(2)$ corresponding to $\vec{\zeta} = (0,0,\zeta_3)$ is parameterized by $U = 1 + 2i\zeta_3\sigma_3 + \cdots$, which should be identified with

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \tag{2.714}$$

or in other words, $\theta = 2\zeta_3 + \cdots$

Comparing the effective Lagrangian

$$\mathcal{L} = -\frac{F^2}{2}(\partial_\mu\zeta_3)^2 + \cdots \tag{2.715}$$

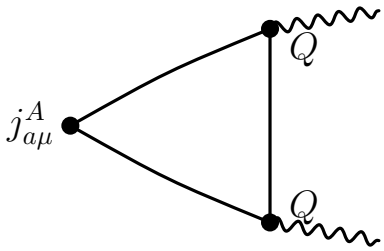
we should identify $R = F/2$. This leads to the pion mass term $-\frac{1}{2}m_{ab}^2\pi_a\pi_b$ with the mass matrix

$$\begin{aligned} m_{ab}^2 &= \delta_{ab}\frac{\langle\Delta\mathcal{L}\rangle}{(F/2)^2} \\ &= \delta_{ab}\frac{4}{F^2}(m_u + m_d)\langle -\frac{\overline{\psi}\psi}{2}\rangle. \end{aligned} \tag{2.716}$$

For reference, the real world pion masses are $m_{\pi^0} \approx 135\,\text{MeV}$ and $m_{\pi^\pm} \approx 140\,\text{MeV}$.

The other remark that should be made is that still in the approximation of vanishing u and d quark masses, the coupling of QCD to electromagnetic U(1) gauge field leads to a

new anomaly in the chiral SU(2) global symmetry. Namely, we can consider the following anomaly triangle diagram



(2.717)

where the corresponding electric charge matrix is

$$Q = e \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix} \tag{2.718}$$

where u has charge $2e/3$ while the d quark has charge $-e/3$. Also, note that the current has an isospin index $a = 1, 2, 3$. The corresponding current anomaly is

$$\partial_\mu j_a^{A\mu} = -\frac{N_c}{16\pi^2} \operatorname{tr}\left(\frac{\sigma_a}{2} Q^2\right) \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} \tag{2.719}$$

where N_c is the number of colors, and the trace here is non-vanishing only for $a = 3$ which gives $\operatorname{tr}(\sigma_3 Q^2/2) = e^2/6$ (also, the field strengths here are the electromagnetic field strength).

This anomaly must also be present in the pion effective theory based on the chiral Lagrangian. Recall that $j^A_{a\mu}$ is such that

$$\vec{\theta}_A \cdot \vec{j}^\mu_A = -\frac{\partial \mathcal{L}}{\partial \partial_\mu \vec{\zeta}} \cdot \delta_A \vec{\zeta} \tag{2.720}$$

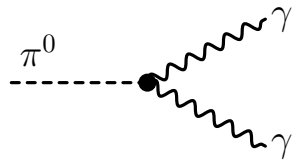
which implies that the current can be written as

$$\begin{aligned} \vec{j}^\mu_A &= -\frac{1}{2}(1 - \zeta^2) \frac{\partial \mathcal{L}}{\partial \partial_\mu \vec{\zeta}} - \vec{\zeta}\left(\vec{\zeta} \cdot \frac{\partial \mathcal{L}}{\partial \partial_\mu \vec{\zeta}}\right) \\ &= \frac{F}{2} \partial_\mu \vec{\pi} + \cdots \end{aligned} \tag{2.721}$$

The anomaly due to the coupling to electromagnetism can be reproduced by the deformation of the action with the term

$$\Delta S = \int d^4x \, \frac{N_c e^2}{48\pi^2 F} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} \pi_3(x) \tag{2.722}$$

where $\pi_3(x)$ is usually denoted as the π_0-field, and with this term in the effective action, it leads to the decay of π^0 into a pair of photon



(2.723)

which leads to the following decay rate

$$\Gamma(\pi^0 \to 2\gamma) \sim \frac{N_c^2 \alpha^2}{F^2} m_{\pi^0}^3 \tag{2.724}$$

where F is also known as the *pion decay constant.*

2.4 Problem set 1:

2.4.1 Problem 1: QM S-matrix example

In class we discussed the poles in the analytic continuation of the scattering amplitude that are associated with bound states and resonances respectively. In this problem you will investigate the analytic structure of the scattering amplitude in the simplest non-trivial model of quantum scattering: a non-relativistic particle scattering off of a δ-function potential in front of a hard wall, described by the Hamiltonian $H = \frac{p^2}{2m} + V(x)$ with,

$$V(x) = \begin{cases} \infty & x < 0 \\ \lambda\delta(x-a) & x > 0 \end{cases} \tag{2.725}$$

Here $a > 0$ and the coefficient λ may take either sign.

(b) Calculate the scattering amplitude $S(E)$ (as a function of energy E) defined as usual through the asymptotic form of the (in-state) wave function

$$\psi^{in}(x) \sim e^{-ikx} + S(E)e^{ikx}, \;\; x \geqq 0. \tag{2.726}$$

(b) Now consider the analytic continuation of $S(E)$ to the complex E-plane, and find the locations of the poles that correspond to bound states and resonances respectively, in the case of negative λ as well as the case of positive λ. Describe what happens to the positions of these poles if we dial λ continuously from negative to positive values.

2.4.2 Problem 2: Solving the Bethe-Salpeter equation

Consider a 4-dimensional relativistic quantum field theory with two massive scalar field ϕ_1, ϕ_2 of identical mass m, weakly coupled through a massless scalar field φ, described by the Lagrangian density,

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu\phi_1)^2 - \frac{1}{2}m\phi_1^2 - \frac{1}{2}(\partial_\mu\phi_2)^2 - \frac{1}{2}m\phi_2^2 - \frac{1}{2}(\partial_\mu\varphi)^2 - \frac{1}{2}g\varphi\phi_1^2 - \frac{1}{2}g\varphi\phi_2^2. \tag{2.727}$$

In this problem, you will study the spectrum of bound states of a ϕ_1-particle with a ϕ_2-particle, in the ladder approximation, through the Bethe-Salpeter amplitude

$$\Phi_B(x_1, x_2) = \langle\Omega| \, \mathbf{T}\hat{\phi}(x_1)\hat{\phi}(x_2) \, |B\rangle \, . \tag{2.728}$$

For $|B\rangle$ a momentum eigenstate of four-momentum P^μ, with $P^2 = -M^2$, normalized in the usual convention for a 1-particle states $\langle B, \mathbf{P}|B, \mathbf{P'}\rangle = \delta^3(\mathbf{P} - \mathbf{P'})$, we can write Φ_B as

$$\Phi_B(x_1, x_2) = \int \frac{d^4k}{(2\pi)^4} e^{i\left(\frac{P}{2}+k\right)\cdot x_1 + i\left(\frac{P}{2}-k\right)\cdot x_2} \frac{\tilde{\Phi}_B(k, P)}{(2\pi)^{\frac{3}{2}}\sqrt{2P_B^0}}. \tag{2.729}$$

Note: as we will be primarily interested in the mass spectrum, namely admissible values of M, the normalization convention of $\tilde{\phi}_B(k,P)$ will be inessential.

(a) Show that in the ladder approximation, the Bethe-Salpter equation can be written as,

$$\left(\left(\frac{P}{2}+k\right)^2+m^2\right)\left(\left(\frac{P}{2}-k\right)^2+m^2\right)\tilde{\Phi}_B(k,P)=g^2\int\frac{d^4q}{(2\pi)^4}\frac{-i}{(q-k)^2-i\epsilon}\tilde{\Phi}_B(q,P). \tag{2.730}$$

By analytic continuation, we may equivalently work with the Wick rotated version of the equation,

$$\left(\left(k_E+\frac{iM}{2}\hat{e}_4\right)^2+m^2\right)\left(\left(k_E-\frac{iM}{2}\hat{e}_4\right)^2+m^2\right)\chi(k_E)=g^2\int\frac{d^4q_E}{(2\pi)^4}\frac{1}{(q_E-k_E)^2}\chi(q_E), \tag{2.731}$$

where we have set $P=(M,\mathbf{0})$, $\hat{e}_4$ stands for the unit vector in the x_4-direction in the Euclidean space $\mathbb{R}^4$, and the inner produce of k_E, q_E etc. are understood to be Euclidean inner products. The Wick rotated Bethe-Salpeter amplitude is now denoted $\chi(k_E)$.

(b) Show that the following (clever) change of variables

$$k_E^\mu=\sqrt{m-\frac{M^2}{4}}\left(\hat{e}_4^\mu+2\frac{\tilde{k}^\mu-\hat{e}_4^\mu}{(\tilde{k}-\hat{e}_4)^2}\right), \tag{2.732}$$

$$\chi(k_E)\equiv((\tilde{k}-\hat{e}_4)^2)^3\tilde{\chi}(\tilde{k}), \tag{2.733}$$

the Wick rotated Beth-Salpeter equations can be put in the manifestly $SO(4)$ invariant form,

$$\left(m^2(\tilde{k}^2+1)^2-M^2\tilde{k}^2\right)\tilde{\chi}(\tilde{k})=g^2\int\frac{d^4\tilde{q}}{(2\pi)^4}\frac{\tilde{\chi}(\tilde{q})}{(\tilde{q}-\tilde{k})^2}. \tag{2.734}$$

For the purpose of solving this equation, from now on the tilde on the variables k and q will be omitted.

(c) By the $SO(4)$ rotational symmetry of the Wick rotated Bethe-Salpeter equation, without loss of generality we can consider a solution with definite rotational quantum mechanics, which is proportional to the $SO(4)$ spherical harmonics $\mathcal{Y}_{n\ell m}(\hat{k})$,

$$\tilde{\chi}(k)=R_n(k^2)\mathcal{Y}_{n\ell m}(\hat{k}). \tag{2.735}$$

Show that the equation for $R_n(k^2)$ reduces to,

$$\left(m^2(x+1)^2-M^2x\right)R_n(x)=\frac{g^2}{16\pi^2(n+1)}\times \tag{2.736}$$
$$\int_0^\infty dy\left(x^{-\frac{n}{2}-1}y^{\frac{n}{2}+1}\theta(x-y)+x^{\frac{n}{2}}y^{-\frac{n}{2}}\theta(y-x)\right)R_n(y).$$

(d) If we assume that the binding energy $2m-M$ is small, the factor $(m^2(x+1)^2-M^2x)$ multiplying $R_n(x)$ on the LHS of the above equation is small at $x=1$, and we expect $R_n(x)$ to be sharply peaked at $x=1$. Thus, assuming the RHS of the above equation is dominated by the integral near $y=1$, find the approximate expression for $R_n(x)$ and solve for the possible values of M under this approximation.

Problem 1 answer:

Answer (a): First, we have two regions of interest, that where the particle is somewhere in the region $x < a$ which on its left is an infinitly high wall (region 2) and on the right is the delta-potential. While in region 1, this is when the particle is $x > a$ and can travel off to $+\infty$.

First, lets circumvent the delta function by integrating over the total Schrödinger equation by some small amount $\pm\epsilon$ as,

$$0 = \int_{a-\epsilon}^{a+\epsilon} dx \left(E + \frac{\hbar^2}{2m}\frac{d}{dx^2} + \lambda\delta(x-a) \right) \psi(x) \tag{2.737}$$

$$= 2\epsilon E\psi(a) + \frac{\hbar^2}{2m}\left(\psi'(a+\epsilon) - \psi'(a-\epsilon)\right) + \lambda\psi(a), \tag{2.738}$$

where $\psi'(x) \equiv d\psi(x)/dx$. Now, in the limit $\epsilon \to 0$ we have,

$$-\frac{\hbar^2}{2m}\left(\psi_1'(a) - \psi_2'(a)\right) + \lambda\psi(a) = 0, \tag{2.739}$$

with ψ_1 corersponding to region 1 and ψ_2 corresponds to region 2.

In order to maintain contiuity across the delta function, the first boundary condition can be written as,

$$\psi_1'(a) = \psi_2'(a) + \frac{2M\lambda}{\hbar^2}\psi(a). \tag{2.740}$$

We have already been given the wave function corresponding to region 1 ($\psi_1 \equiv \psi_{in}$),

$$\psi_1 = e^{-ikx} + S(E)e^{ikx}. \tag{2.741}$$

For region 2, we can pick $\psi_2 = A\sin(kx)$ since when $x = 0 \implies \psi_2(0) = 0$. Note, k is the **same** in both regions.

Now, since

$$\psi_1(a) = \psi_2(a) \implies A\sin(ka) = e^{-ika} + S(E)e^{ika}, \tag{2.742}$$

the delta-function boundary condition gives (equation 6),

$$Ak\cos(ka) + \frac{2M\lambda}{\hbar^2}A\sin(ka) = -ik\left(e^{-ika} - S(E)e^{ika}\right). \tag{2.743}$$

Now, if we take equation nine and divide by equation eight,

$$k\cot(ka) + \frac{2M\lambda}{\hbar^2} = -ik\left(\frac{e^{-ika} - S(E)e^{ika}}{e^{-ika} + S(E)e^{ika}}\right) \tag{2.744}$$

$$= -ik\left(\frac{e^{-2ika} - S(E)}{e^{-2ika} + S(E)}\right) \tag{2.745}$$

$$\left(k\cot(ka) + \frac{2M\lambda}{\hbar^2}\right)e^{-2ika} + ike^{-2ika} = \left(ik - k\cot(ka) - \frac{2M\lambda}{\hbar^2}\right)S(E) \tag{2.746}$$

$$S(E) = \frac{\left(k\cot(ka) + \frac{2M\lambda}{\hbar^2} + ik\right)}{\left(ik - k\cot(ka) - \frac{2M\lambda}{\hbar^2}\right)}e^{-2ika} \tag{2.747}$$

$$= \left(\frac{\cot(ka) + \beta + i}{i - \cot(ka) - \beta}\right)e^{-2ika}, \tag{2.748}$$

where in the last line we defined $\beta = 2M\lambda/(\hbar^2 k)$. Then, if we recall that the energy of the incoming particle is $E = p^2/2m = \hbar^2 k^2/2m$, then $k = \sqrt{2mE}$ when $\hbar = 1$. As a physical check, as $k \to 0$, then $S(k) \to 1$ which is what we expect from the unitarity condition $|S(k)| \leq 1$.

Answer (b): We can see that in equation 14 we do have a phase shift and have poles at $\cot(ka) = i \pm \beta$ where it is minus if $\lambda > 0$ and plus if $\lambda < 0$. Since this is a phase shift, then the corresponding s-wave phase shift is (equation 6.148 for $l = 0$ of Sakurai's text 3rd edition, [**AC:**] sakurai)

$$\delta_0(k) = \tan^{-1}((\beta + \cot(ka))) - ka \tag{2.749}$$

$$= \cot^{-1}(\cot(ka) + \beta) - ka. \tag{2.750}$$

The bound state poles correspond to poles at negative energies. This would be the pole of $\cot(ka) = i(1 - 2\lambda\sqrt{M/2E})$ (by letting $E \to -E$). The poles that correspond to resonances are when we have poles *near* positive energies in the complex plane. These would be slightly below the physical line of $k \geq 0$ (the real axis), and would only accessible on the second sheet if going from top down on the contour. These would be the $\lambda \in \mathbb{R}^+$ poles as described above.

For a more clear and graphical interpretation, when plotting $k(E)$ we only have the positive real axis representing the branch cut (physical points) but with both the $\pm E$ values lines along it. We also must integrate clockwise starting from the branch cut and integrating over a contour that begins downwards. However, the the k plane, the branch cut then splits into two distinct halfs, the $-E$ values will be on the left-hand side of the k plane while the $+E$ values will be on the right hand side of the k-plane on the real axis. The lower-half-plane in this graph then represent where the resonances live.

Problem 2 answer:

Answer (a): First, a little background (mainly for myself for future reference), the original Bethe-Salpeter equation [**AC:**] BS was derived for the case of two charged electrons in a bound state (equation 19 of [**AC:**] BS). So, we can write the analog of that equation for our scalar model. For generality, I will assume, until the end of this part of the problem, that the masses are *different*, i.e $m_1 \neq m_2$, and that the mediator scalar does have mass, μ. Our interaction Lagrangian is then,

$$\mathcal{L}_{int} = \varphi g(\phi_1^\dagger \phi_1 + \phi_2^\dagger \phi_2). \tag{2.751}$$

The analog for our model to equation 19 of [**AC:**] BS is,

$$\left(\Box_{x_1} + m_1^2\right)\left(\Box_{x_2} + m_2^2\right)\Phi_P(x_1, x_2) + \int d^4z_1 d^4z_2 V(x_1, x_2; z_1, z_2)\Phi_P(z_1, z_2) = 0, \tag{2.752}$$

where our $\Phi_P(x_1, x_2)$ is equivalent to $\Phi_B(x_1, x_2)$ above with $B \to P$, the total momentum of the bound state (in most of the literature however, I have noticed they use P and usually $\chi_P(x_1, x_2)$). V is our kernel that shall be taken in the Born approximation, or in this case, the ladder approximation, where we don't take into account crossed-ladder diagram. (Note, there is also the conjugate of equation 23 above that we never mentioned, but does exist. We just won't need it for calculations).

Now, by translational invariance, $\Phi_P(x_1 + a, x_2 + a) = e^{-iP\cdot a}\Phi_P(x_1, x_2)$. We can then introduce a new coordinate $x \equiv x_1 - x_2$ but this is arbitrary. Since x is arbitrarily defined, we can define two new arbitrary parameters η_1, η_2 such that,

$$\eta_1 + \eta_2 = 1, \;\; X = \eta_1 x_1 + \eta_2 x_2, \;\; x_1 = X + \eta_2 x, \;\; x_2 = X - \eta_1 x. \tag{2.753}$$

This allows us to write the Bethe-Salpeter amplitude as $\Phi_P(x_1, x_2) = e^{-iP\cdot X}\Phi_b(x)$ (and its respective conjugate). Note these are form-factors, not necessarily wave functions.

You can then solve the B-S (Bethe-Salpeter) amplitude in coordinate space as done in [**AC:**] textbook (equation 10.26), but we want our amplitude in momentum space. So, in a similar manner as above (used to illustrate the direction we are heading) we can define the following total momenta as,

$$P = p_1 + p_2, \;\; p = \eta_2 p_2 - \eta_1 p_1 \implies p_1 = \eta_1 P + p, \;\; p_2 = \eta_2 P - p. \tag{2.754}$$

Since there is no notion of relative momenta in NRQM (non-relativistic quantum mechanics [**AC:**] sakurai), we can choose η_1 and η_2 to be,

$$\eta_{1,2} = \frac{m_{1,2}}{m_1 + m_2}. \tag{2.755}$$

We then have the following equations used to solving the Bethe-Salpeter amplitude,

$$\Phi_P(p) = \int e^{ip\cdot x}\Phi_P(x) d^4x \tag{2.756}$$

$$0 = \left((\eta_1 P + p)^2 - m_1^2\right)\left((\eta_2 P - p)^2 - m_2^2\right)\Phi_P(p) + \int \frac{d^4p'}{(2\pi)^4} V(p', p; P)\Phi_P(p') \tag{2.757}$$

(and the conjugate equation). Now, in the ladder approximation, the kernel for a massive propagator can be evaluated as,

$$V = -\frac{ig^2}{(p-p')^2 - \mu^2 + i\epsilon}. \tag{2.758}$$

Then, the B-S equation becomes,

$$\left(\left(\frac{m_1}{m_1+m_2}P_0 + p_0\right)^2 - m_1^2\right)\left(\left(\frac{m_2}{m_1+m_2}P_0 - p_0\right)^2 - m_2^2\right)\Phi_P(p) - \frac{ig^2}{(2\pi)^4}\int \frac{d^4p'\Phi_P(p')}{(p-p')^2 - \mu^2 + i\epsilon} = 0, \tag{2.759}$$

where now if we take the special case of $m_1 = m_2 = m$ and $\mu^2 = 0$ and writing it in a more compact manner (i.e not expanding to see the temporal terms, but this will be useful for the Wick roation), we have,

$$\left(\left(\frac{P}{2} + p\right)^2 + m^2\right)\left(\left(\frac{P}{2} - p\right)^2 + m^2\right)\Phi_P(p) - \frac{-ig^2}{(2\pi)^4}\int \frac{d^4p'\Phi_P(p')}{(p-p')^2 + i\epsilon} = 0. \tag{2.760}$$

(I should note that p' here is the exact same as q above and similar for p and k above, just different notation). If we were concerned about normalization procedures, then at the end of section 10.2.1 of [**AC:**] textbook the normalization condition is given.

Now, in order to Wick rotate, instead of performing the operation right away, we should take a look to see if when we Wick rotate we encounter any singularities (because if we did, we couldn't complete the Wick rotation). In his original paper on the procedure, Wick gave the full details of our scalar model in [**AC:**] wick bs. Here, we will do a similar analysis based on [**AC:**] textbook.

Set $X = 0$ (as defined in equation 24). Then, we can break up Φ_P into modes and separate the temporal component via a step-function defined as,

$$\Phi_P(x) = \theta(x^0)f(x,P) + \theta(-x^0)g(x,P), \tag{2.761}$$

where f, g are functions such that,

$$f(x,P) = \langle 0|\,\phi_1(\eta_2 x)\phi_2(-\eta_1 x)\,|P\rangle = \int \frac{d^4p}{(2\pi)^4}e^{-ip\cdot x}f(p;P), \tag{2.762}$$

$$g(x,P) = \langle 0|\,\phi_2(-\eta_2 x)\phi_1(\eta_2 x)\,|P\rangle = \int \frac{d^4p}{(2\pi)^4}e^{-ip\cdot x}g(p;P). \tag{2.763}$$

We can write the step-function in the momentum basis via a Fourier transformation as,

$$\theta(x^0) = -\frac{1}{2\pi i}\int \frac{d\omega\, e^{-i\omega x^0}}{\omega + i\epsilon}, \tag{2.764}$$

allowing us to write the decomposed B-S amplitude as,

$$\Phi_P(p) = -\frac{1}{2\pi i}\int dq_0 \frac{f(q_0,\mathbf{p},P)}{p^0 - q^0 + i\epsilon} + \frac{1}{2\pi i}\int dq_0 \frac{g(q_0,\mathbf{p},P)}{p^0 - q^0 - i\epsilon}. \tag{2.765}$$

(Note that there is also the conjugate of the B-S amplitude that we are not worrying about, but the definitions follow in a similar manner).

We can perform the analytic Wick rotation given that f and g have support in q^0 (our complex momentum variable). This can be seen to be the case by inserting a unity of n-states into the definition of f,

$$f(q,P) = \int d^4x e^{iq\cdot x} \sum_n \langle 0| \phi_1(0) |n\rangle \langle n| \phi_2(0) |P\rangle \tag{2.766}$$

$$= \sum_n (2\pi)^4 \delta^4(q - p_n + \eta_1 P) \langle 0| \phi_1(0) |n\rangle \langle n| \phi_2(0) |P\rangle , \tag{2.767}$$

where we performed integration over x to pick up q from the first exponentil, p_n from $\phi_1(0)|n\rangle$ and $\phi_2(0)|P\rangle$ within the delta function. Similar reasoning for $g(p,P)$.

To determine if any poles exist over the integration contour, notice that if particle 1 is to be stable, then in $f(q,P)$ we must require $p_n^2 \geq m_1^2$ and $p_n^0 > 0$, which means that $f(q,P)$ will vanish unless $(\eta_1 P + q)^2 \geq m_1^2$ and $\eta_1 P_0 + q_0 > 0$ (based on the delta function in equation 38). There is a similar situation for g. Integrating over f and g in Φ_P (and $\bar{\Phi}_P$ respectively), the integral extends over ω_+ to $+\infty$ (for f or f^*) with g being over $-\infty$ to ω_- with,

$$\omega_\pm = \pm\sqrt{m_1^2 + (\eta\mathbf{P} \pm \mathbf{p})^2} \mp \eta_1 P_0. \tag{2.768}$$

Thus, the Wick rotation can be performed if $\omega_- < \omega_+$. In the COM-frame (center of mass) we have $\mathbf{P} = 0$ and $\eta_{1,2} = m_{1,2}/(m_1+m_2)$ implies $\omega_+ = \sqrt{m_1^2 + \mathbf{p}^2} - m_1 P_0/(m_1+m_2) > 0$ and $\omega_- = m_2 P_0/(m_1+m_2) - \sqrt{m_2^2 + \mathbf{p}^2} < 0$. However, there is a consequence, we have a gap between the two cuts in the Φ_P and $\bar{\Phi}_P$ integral representation.

Now, if we stay in the COM-frame, with the previous $\eta_{1,2}$ as defined, as well as stay in the ladder approximation with the kernel defined in equation 29, then the general (unequal mass) B-S amplitude, equation 30, can be Wick rotated by an analytic continuation of p^0 that picks up a phase $p^0 \to p^0 e^{i\theta}$ with $0 \leq \theta \leq \frac{\pi}{2}$ (similar for the conjugate equation), then when we plug in the f-g mode representation for Φ_P as defined in equation 32-34 for the two cases $p^0 > 0$ and $p^0 < 0$, we can Wick rotate with a Euclidean signature. Again, emphasizing that since $\omega_- < 0 < \omega_+$, we encounter no singularities via the analytic continuation, then we find the following [**AC:**] wick bs,

$$\left(\left(p_0 - \frac{im_1}{m_1+m_2}P_0\right)^2 + \mathbf{p}^2 + m_1^2\right)\left(\left(p_0 + \frac{im_1}{m_1+m_2}P_0\right)^2 + \mathbf{p}^2 + m_1^2\right)\Phi_P(p) \tag{2.769}$$
$$- \frac{g^2}{(2\pi)^4}\int \frac{d^4p'\,\Phi_P(p')}{(p-p')^2 + \mu^2} = 0.$$

Finally, in the case of $m_1 = m_2 = m$ and $\mu^2 = 0$, we have the final result (recalling $P = (iP_0, \mathbf{0})$),

$$\left(\left(p - \frac{iP}{2}\right) + m^2\right)\left(\left(p + \frac{iP}{2}\right) + m^2\right)\Phi_P(p) - \frac{g^2}{(2\pi)^4}\int \frac{d^4p'\,\Phi_P(p')}{(p-p')^2} = 0. \tag{2.770}$$

Answer (b): There are actually two ways to put the Wick rotated B-S equation for, what will now be called, the Wick-Cutkosky model (W-C) into a manifestly SO(4) invariant

form. One way is the one suggested, the other is to use a stereographic projection onto a higher dimensional sphere. This is analgous to solving the hydrogen atom in QM. I will show both.

To start, we will project onto a $D = 5$ sphere. To lessen the clutter, let us define a unitless quantity [**AC:**] BS,

$$\lambda = \frac{g^2}{16m^2\pi}. \tag{2.771}$$

Then, the coefficient infront of the second term on the LHS of equation 41 becomes λ/π^3.

Let $P = (P_0, \mathbf{0})$. To perfrom the stereographic projection $\exists p \in \mathbb{R}^4$ to a vector z in $D = 5$ space, choose a sphere with a radius of $\frac{1}{2}\sqrt{1 - P^2/4}$ with a 4D projection along p and polar angle ζ such that $\tan(\theta/2) = |p|/\sqrt{1 - P^2/4}$. Graphically, this looks like the following.

Figure 39. Graph of a stereographic projection of a point p onto a 5-sphere.

The other polar angles describing S^5 are (β, θ, φ) leading to $p \cdot P = |p||P|\cos(\beta)$ and the B-S/W-C model, equation 41, becomes,

$$\left(\left(p - \frac{iP}{2}\right) + m^2\right)\left(\left(p + \frac{iP}{2}\right) + m^2\right) = \frac{1}{\cos^4(\zeta/2)}\left(1 - \frac{P^2}{4}\right)\left(1 - \frac{P^2}{4} + \frac{P^2}{4}\sin^2(\zeta)\cos^2(\beta)\right). \tag{2.772}$$

If $d\Omega_4$ is the solid angle differential, on the 5-sphere we can normalize this to $\int d\Omega_4 = 8\pi^2/3$ implying that our momentum integrand in equation 41 becomes,

$$\frac{d^4p'}{(p - p')^2} = \frac{1}{8}\left(1 - \frac{P^2}{4}\right)\frac{\cos^2(\zeta/2)}{\cos^6(\zeta/2)}\frac{d\Omega_4'}{1 - \cos(\alpha)}, \tag{2.773}$$

where α is now the angle made between z and z' (the vector we are projecting to as stated previously) where their product is $z \cdot z' = ((1 - P^2)/4)\cos(\alpha)$. If we define a new function $K(z)$ as,

$$K(z) = \frac{1}{\cos^6(\zeta/2)}\Phi_P, \tag{2.774}$$

then we find the following analog of equation 41,

$$\left(1 - \frac{P^2}{4} + \frac{P^2}{4}\sin^2(\zeta)\cos^2(\beta)\right)K(z) = \frac{\lambda}{8\pi^3}\int \frac{d\Omega_4'}{1 - \cos(\alpha)}K(z'). \tag{2.775}$$

In the general case of $0 < P^2 < 4$, equation 49 has an SO(4) invariance. This is the case since $\sin(\zeta)\cos(\beta) \propto \text{proj}(z)$ along P. This allows us to first project to a unit sphere on $\mathbb{R}^5$ then be projected back to Euclidean $\mathbb{R}^4$ but along P. Define $\cos(\gamma) = \sin(\zeta)\cos(\beta)$ and $\tilde{\Phi}(q) = \cos^6(\gamma/2)K(z)$. Plugging in these definitions into equation 49 gives,

$$\left(m^2q^4 + 2m^2q2\left(1 - \frac{P^2}{2}\right) + m^2\right)\tilde{\Phi}(q) = \frac{\lambda}{\pi^3}\int \frac{d^4q'}{(q - q')^2}\tilde{\Phi}(q'). \tag{2.776}$$

This is the exact same as equation 44, the desired solution, just with an expanding $(\tilde{k}+1)^2$ term and recall that $P = (P_0) \implies P^2 = +M^2$ under the Euclidean metric (where our q is the k in the original prompt).

The other way, and much quicker, is to simply expand equation 42 and 43 in terms of, what we will use p and q which are[**AC:**] nakanishi 3,

$$p = \sqrt{m^2 - \frac{P^2}{4}}\left(n + 2\frac{q-n}{q^2 - 2q\cdot n + 1}\right) \tag{2.777}$$

$$\Phi(p) = \left(1 - \frac{P^2}{4}\right)^4 (q^2 - 2q\cdot n + 1)\tilde{\Phi}(q). \tag{2.778}$$

As you can see, in the variables I am using, these two are equivalent. The n in my above transformations is the unit vector along the temporal axis. The choice of n is made since for $n \geq 1$ it will serve as an index for the representation of SO(4) of dimension n^2. (Note: in [**AC:**] textbook they set $m^2 = 1$ which has been a great reference but I believe I restored all the m^2 properly).

Answer (b): To put equation 50 in terms of spherical harmonics, we can utilize the identity given, translated into the variables we are using,

$$\frac{1}{(q-q')^2} = \frac{2\pi^2}{q_>^2}\sum_{n=1}^{\infty}\left(\frac{q_<}{q_>}\right)^{n-1}\frac{1}{n}\sum_{\ell,m}\mathcal{Y}_{n\ell m}(\hat{q})\mathcal{Y}^*_{n\ell m}(\hat{q}). \tag{2.779}$$

Consider defining the radial amplitude as,

$$\tilde{\Phi}(\hat{q}) = \frac{R_n(q^2)}{|q|^{n+1}(q^4m^2 + 2q^2(1-P^2/2) + m^2)}\mathcal{Y}_{n\ell m}(\hat{q}) \tag{2.780}$$

(which we may recognize was read off of equation 50). If we plug both equation 55 and 56 into equation 50, and defining $x = q^2$, we have,

$$R_n(x) = \frac{\lambda}{n\pi}\int_0^\infty dx'\left(\theta(x-x') + \left(\frac{x}{x'}\right)^n\theta(x'-x)\right)\frac{R_n(x')}{m^2x^{2\prime} + 2x'(m^2 - P^2/2) + m^2}, \tag{2.781}$$

which is equivalent to equation 54 (given $y \to x'$). Actually, an equivalent view, and maybe easier to handle, is the differential equation of 54 given in equation 10.66 of [**AC:**] textbook. (For preference, there is another version of this same problem where Wick [**AC:**] wick bs and Cutkosky [**AC:**] cutkosky bs used instead $z = (x-1)/(x+1)$ and the algebra initially seems worrisome, but simplifies much better in the end since they defined $R_n(x) = (1-z)^{-n}g_n(z)$).

Answer (d):

The case we are referring to is physical when the total momentum is when $P^2 = 4$. In this threshold limit the kernel of equation 57 becomes,

$$\frac{1}{\pi\left(m^2x^2 + 2x(m^2 - P^2/4) + m^2\right)} \equiv \frac{1}{\pi\left(m^2(x-1)^2 + x(m^2 - P^2)\right)} \overset{P^2\to 4}{\simeq} \frac{\delta(x-1)}{\sqrt{4m^2 - P^2}}. \tag{2.782}$$

From the integral form of $R_n(x)$ equation 57, we have the following,

$$R_n(x) \simeq \frac{\lambda}{n\sqrt{4-P^2}} R_n(1) \left(\theta(x-1) + x^n \theta(1-x)\right). \tag{2.783}$$

If we demand consistency that both sides agree when $x = 1$, we can set $\theta(0) = \frac{1}{2}$ in which we then require $n\sqrt{4m^2 - P^2} = \lambda$, which is the hydrogen spectrum. We can put this in terms fo the binding energy where $P^2 = (2 - B/m)^2$ then in this approximation,

$$B = \frac{m}{2}\frac{\lambda^2}{2n^2} + \cdots, \tag{2.784}$$

where $m/2$ is the reduced mass and λ is still as defined in equation 45. Then an approximation to the mass M (that of P) is

$$M = 2 - \frac{\lambda^2}{4n^2}. \tag{2.785}$$

As a final note, this approximation yields a function without any nodes ($K = 0$) corresponding to no limit for the eigenvalues of $K \geq 1$. This is outlined in detail by Nakanishi in [**AC:**] nakanishi 1, [**AC:**] nakanishi 2, and [**AC:**] nakanishi 3 (as a general overview).

2.5 Problem set 2: Non-linear sigma model

2.5.1 Problem 1: Non-linear sigma model

In Lagrangian mechanics, the configuration space need not admit a linear (i.e. vector space) structure, and different parameterizations of the configuration space via generalized coordinates lead to equivalent Euler-Lagrange equations. A basic example of this is the three-dimensional rigid body, whose rotational configuration space is SO(3), and one may formulate its Euler-Lagrange equation in terms of Euler angles, or any other parameterization of SO(3). In this problem you will investigate the Lagrangian formulation of a quantum field theory in which the fields parameterize a nonlinear configuration space.

Consider a theory of three scalar fields $\phi = (\phi_1, \phi_2, \phi_3)$ in four-dimensional spacetime with Lagrangian density,

$$\mathcal{L} = -\frac{M^2}{4}\,\mathrm{tr}\Big(\partial_\mu U \partial^\mu U^\dagger\Big), \;\; U(x) \equiv \exp(i\sigma\cdot\phi(x)). \tag{2.786}$$

Here $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ are Pauli matrices, M is a constant parameter, and $U(x)$ is a 2×2 unitary matrix.

(a) Expand the Lagrangian density (1) to quartic order in ϕ and derive the Feynman rules of up to 4-point vertices. Compute the tree-level $2 \to 2$ scattering amplitude of a pair of particles created by ϕ_a, ϕ_b into a pair of particles created by ϕ_c, ϕ_d, for arbitrary a, b, c, d. You should express your answer in terms of the independent Mandelstam variables, say s and t.

(b) We could consider an alternative parameterization of the SU(2) matrix $U(x)$,

$$U(x) \equiv \frac{1}{1+\zeta(x)\cdot\zeta}\,(1+i\sigma\cdot\zeta)^2\,, \tag{2.787}$$

in terms of the scalar fields $\zeta = (\zeta_1, \zeta_2, \zeta_3)$. Expand (1) to quartic order in ζ and calculate the same amplitude as in (a) but now for particles created by ζ fields. How does your result compare to that of (a)?

(c) Expand the above tree-level $2 \to 2$ amplitude in terms of the partial wave amplitudes $S_\ell(s)$ for all non-negative integer ℓ. How does your answer compare with the unitarity bound on $S_\ell(s)$?

2.5.2 Problem 2: *Properly* renormalized QED

Recall that the QED Lagrangian density can be expressed in terms of the renormalized vector potential $A_\mu(x)$ and electron field $\psi_\alpha(x)$ as

$$\mathcal{L} = -\frac{1}{4}Z_A F_{\mu\nu}F^{\mu\nu} - Z_\psi\bar{\psi}\left(\gamma^\mu D_\mu + m_0\right)\psi, \tag{2.788}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $D_\mu\psi = \partial_\mu\psi + ieA_\mu\psi$. Here e is the unit (physical) electric charge, m_0 is the bare (not physical) mass, Z_A and Z_ψ are the field renormalization constants, as usual defined such that the 1-photon component of $F_{\mu\nu}(x)\,|\Omega\rangle$ and the 1-electron component of $\psi_\alpha(x)\,|\Omega\rangle$ are normalized as in free field theories. In this problem

we will carry out perturbation theory by splitting the above Lagrangian as,

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \text{int} \tag{2.789}$$

$$\mathcal{L}_{\text{free}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}\left(\gamma^\mu\partial_\mu + m_R\right)\psi \tag{2.790}$$

$$\mathcal{L}_{\text{int}} = -ieA_\mu\bar{\psi}\gamma^\mu\psi - \frac{1}{4}(Z_A - 1)F_{\mu\nu}F^{\mu\nu} - (Z_\psi - 1)\bar{\psi}(\gamma^\mu D_\mu + m_R)\psi - Z_\psi\delta m\bar{\psi}\psi, \tag{2.791}$$

and view $\mathcal{L}_{\text{free}}$ as the free theory Lagrangian density, treating all terms in $\mathcal{L}_{\text{int}}$ as interactions that are of order e or higher. Here $m_R = m_0 - \delta m$ is a finite quantity that we refer to as the renormalized mass, which a priori need not agree with the physical mass m of the electron. For the rest of this problem, we will *choose* m_R to agree with the physical mass m.

The time-ordered 2-point function of the electron field takes the form

$$\langle\Omega|\,\mathbf{T}\hat{\psi}_\alpha(x)\hat{\bar{\psi}}^\beta(y)\,|\Omega\rangle = -i\int\frac{d^4k}{(2\pi)^4}e^{ik\cdot(x-y)}\left((i\not{k} + m - i\epsilon - \Sigma(k))^{-1}\right)^\beta_\alpha, \tag{2.792}$$

where the self-energy $\Sigma(k)$ is a matrix with spinor indices. Note that the self-energy of the electron field is a priori gauge-dependent, since $\hat{\psi}_\alpha(x)$ is not a gauge-invariant operator. You can adopt Feynman gauge for this problem.

Calculate $\Sigma(k)$ at order e^2, and determine Z_ψ as well as δm at this order. You can regularize the UV divergence using dimensional regularization (setting spacetime dimension $D = 4 - \epsilon$), and regularize the IR divergence by cutting off the photon loop momentum at small mass scale μ. You should find in your final result that $\Sigma(k)$ is free of UV divergence (due to the way we defined Z_ψ and δm), but it does contain an IR divergence.

2.5.3 Problem 3: Spontaneous Particle Production and 4-Photons can scatter

Consider scalar QED defined by the action

$$S[A,\varphi] = \int d^4x\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - |D_\mu\varphi|^2 - m^2|\varphi|^2\right), \tag{2.793}$$

where φ is a complex scalar field, $D_\mu\varphi \equiv \partial_\mu\varphi - iqA_\mu\varphi$. We will define an effective action $S_{\text{eff}}[A]$ (this is similar to, but not the same as, the 1PI effective action introduced in class) by performing the functional integral over φ,

$$e^{iS_{\text{eff}}[A]} = \int[D\varphi D\varphi^*]e^{iS[A,\varphi]}. \tag{2.794}$$

Evidently, the RHS of the integral above can be evaluated as a Gaussian integral, resulting in the effective action (up to an inconsequential constant shift)

$$S_{\text{eff}}[A] = \int d^4x\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\log\Big(\det\Big(-D^\mu D_\mu + m^2\Big)\Big)\right), \tag{2.795}$$

where the $\det(\cdots)$ stands for the determinant of a linear functional.

(a) Using $\log(\det) = \mathrm{tr}(\log)$, and the formula

$$\log\left(\frac{H}{\mu}\right) = -\int_0^\infty \frac{ds}{s}\left(e^{-sH} - e^{-2\mu}\right), \tag{2.796}$$

calculate the RHS of the effective action in the case of a constant magnetic field $F_{12} \equiv B_3$. From the result, deduce the correction to the energy density of a constant magnetic field due to coupling to charged scalar field.

(b) Extend your result of (a) to the case of a constant electric field $F_{01} \equiv E_1$ by a suitable analytic continuation.

(c) If you are sufficiently careful with the analytic continuation in (b), you should find that in the presence of a constant electric field, S_{eff} is complex. Calculate the imaginary part of S_{eff} in this case, and from the result deduce the probability of the electri field decaying via pair production of charged particles per unit spatial volume per unit time.

Problem 1 answer:

Answer (a): First, consider the exponential expansion of $U(x)$ in equation one,

$$U(x) = \exp(i\sigma \cdot \phi(x)) = 1 + i\sigma \cdot \phi + i^2(\sigma \cdot \phi)^2 + \cdots , \tag{2.797}$$

and similar for $U^\dagger$. To handle the spin matrices, we note that partial derivatives view them as constants and can be pulled out of the operation, and we can use two properties ([**AC:**] sakurai chapter 3),

$$(\sigma \cdot \mathbf{a})(\sigma \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\sigma \cdot (\mathbf{a} \times \mathbf{b}) \tag{2.798}$$

$$\sigma_i^2 = 1, \tag{2.799}$$

where equation three implies that when $\mathbf{a} = \mathbf{b} \implies (\sigma \cdot \mathbf{a})^2 = |\mathbf{a}|^2$. Then, up to fourth order in ϕ (terms that go as $\sim \phi^2(\partial\phi)^2$ or ϕ^4 or some combination of the field and its derivatives) we have,

$$(\partial_\mu U)(\partial^\mu U^\dagger) = -i^2\sigma^2(\partial\phi)^2 - 2i\sigma \cdot \phi(\partial\phi)^2 + 4\phi^2(\partial\phi)^2 + 2i\sigma \cdot \phi(\partial\phi)^2 + \cdots \tag{2.800}$$

$$= (\partial_\mu\phi)^2 + 4\phi^2\left(\partial_\mu\phi\right)^2 + \cdots \tag{2.801}$$

which gives the following Lagrangian,

$$\mathcal{L} = \frac{M^2}{4}\left(|\partial_\mu \boldsymbol{\phi}|^2 + 4\left(\boldsymbol{\phi} \cdot \partial_\mu \boldsymbol{\phi}\right)^2 + \cdots\right) \tag{2.802}$$

(where we reinstated the component notation of ϕ denoted as bold-face symbols). To find the Feynman diagram rules, notice in the first term of equation 7 we can perform integration by parts (similar to deriving the rules from the KG-equation for a scalar field) to get $\sim \boldsymbol{\phi}\partial_\mu\partial^\mu\boldsymbol{\phi}$ which when we Fourier transform gives, $+\boldsymbol{\phi}k^2\boldsymbol{\phi}$. Then, when we plug into the partition function we can read off the Feynman rules multiplied by δ^{ij} due to $\boldsymbol{\phi}$ having multiple components. We then can perform a similar operation on the fourth order component of the Lagrangian in equation seven using the orthogonal condition for $SO(3)$ namely that if for an ith component non-linear sigma model, $n^i(x) = (\phi_1, \phi_2, ..., \phi_{N-1})$ are constrained via $\sum_{n=i}^{N} |n^i(x)|^2 = 1$. The Feynman rules are then presented in equation 8 and 9 below (the graphs are taken from [**AC:**] pands).

$$= \frac{iM^2}{p^2}\delta^{ij} \tag{2.803}$$

$$= -\frac{4i}{M^2}((p_1 + p_2) \cdot (p_3 + p_4)\delta^{ij}\delta^{kl} + (p_1 + p_3) \cdot (p_2 + p_4)\delta^{ik}\delta^{jl} + (p_1 + p_4) \cdot (p_2 + p_3)\delta^{il}\delta^{jk}). \tag{2.804}$$

(Note that the a, b, c, d particles that was asked for in the question are the exact same as the i, j, k, l indices we are using. We explain what these really mean later.) In terms of Mandelstam variables, if $s = -(p_1 + p_2)^2$, $t = -(p_1 - p_3)^2$, and $u = -(p_1 - p_4)^2$, then equation 9 becomes (with the use of Mathematica),

$$\text{vertex} = -\frac{4i}{M^2}\left(s\delta^{ij}\delta^{kl} + t\delta^{ik}\delta^{jl} + u\delta^{il}\delta^{jk}\right). \tag{2.805}$$

(We could also physically/heuristically derive the above vertex by writing down the diagram, and then recall that the amplitude has a crossing symmetry where the vertex then holds the sum of the crossing symmetry. For reference, the lecture notes on Feburary third on page 2). Then, to tree order, (i.e. no loops since this would involve a second order vertex) our amplitude is simply that of equation 10,

$$i\mathcal{A} = -\frac{4i}{M^2}\left(s\delta^{ij}\delta^{kl} + t\delta^{ik}\delta^{jl} + u\delta^{il}\delta^{jk}\right). \tag{2.806}$$

We can simplify this further to find an amplitude of the form $\mathcal{A} = \mathcal{A}(s,t)$ by using the relation between Mandelstam variables for massless interactions, $s+t+u=0 \implies -(s+t)=u$ giving,

$$i\mathcal{A} = -\frac{4i}{M^2}\left(s\left(\delta^{ij}\delta^{kl} - \delta^{il}\delta^{jk}\right) + t\left(\delta^{ik}\delta^{jl} - \delta^{il}\delta^{jk}\right)\right). \tag{2.807}$$

In light of this, we can then interpret the delta functions telling us the channel of which the particles are coming from as given in the diagram of equation 9, or specifically *what* particles are incoming and outgoing. For example, they can be pions $\pi_\pm$ or π_0 as described in [**AC:**] schwartz.

Answer (b): To expand equation 12 to fourth order, note that $(1+i\sigma\cdot\zeta)^2 = 1-\zeta^2+2i\sigma\cdot\zeta$ with the conjugate being $1-\zeta^2-2i\sigma\cdot\zeta$ and that $(1+\zeta^2)^{-1} \approx 1-\zeta^2+\mathcal{O}(\zeta^4)$. Then,

$$U(x) \approx \left(1-\zeta^2\right)\left(1-\zeta^2+2i\sigma\cdot\zeta\right) + \mathcal{O}(\zeta^4). \tag{2.808}$$

Then, once we compute the derivative of $U(x)$ and multiply their conjugate products, we get the following terms,

$$\partial_\mu U(x)\partial^\mu U^\dagger \approx 4\zeta^2\left(\partial\zeta\right)^2 + \left(\partial_\mu\zeta\right)^2 + \mathcal{O}\left(\zeta^{4>}\right), \tag{2.809}$$

where $\mathcal{O}(\zeta^{4>})$ denotes terms that are greater than fourth order. This is exactly what we got to fourth order for the SO(3) parameterization in part (1a). Then, we will get to the same amplitude calculations as above (i.e. the same physics).

Answer (c): The equation we want in order to find the expansion of non-negative ℓ of $S_\ell(E)$ is given on page two of the lecture notes from January twenty-seventh namely (or section 5.3 of [**AC:**] textbook)

$$i\mathcal{A}(s,t) = i(2\pi)^6 E^2\mathcal{M} \tag{2.810}$$

$$= \frac{8\pi E}{p}\sum_{\ell=0}^{\infty}(2\ell+1)P_\ell(x)\left(S_\ell(E)-1\right), \tag{2.811}$$

where $x=\cos(\theta)$ and p is in the COM frame and with $p=\sqrt{E/4-m^2}$. If we then want to solve for $S_\ell(E)$ and find the first few terms of ℓ, recall the orthogonal relation for Legendre polynomials with index ℓ and ℓ',

$$\int_{-1}^{1} dx P_\ell(x)P_{\ell'}(x) = \frac{2}{2\ell+1}\delta_{\ell,\ell'}. \tag{2.812}$$

Then, if we multiply both sides of equation 16 by $P_{\ell'}$ and integrate over x we have,

$$i\int_{-1}^{1} dx\mathcal{A}(s,t)P_{\ell'}(x) = \frac{8\pi E}{p}\int_{-1}^{1} dx\sum_{\ell=0}^{\infty}(2\ell+1)P_{\ell'}(x)P_{\ell}(x)\left(S_{\ell}(E)-1\right) \tag{2.813}$$

$$= \frac{8\pi E}{p}\sum_{\ell=0}^{\infty}\int_{-1}^{1} dx(2\ell+1)P_{\ell'}(x)P_{\ell}(x)\left(S_{\ell}(E)-1\right) \tag{2.814}$$

$$= \frac{8\pi E}{p}\sum_{\ell=0}^{\infty}\left(\frac{2}{2\ell+1}\delta_{\ell',\ell}\right)\left(S_{\ell}(E)-1\right) \tag{2.815}$$

$$= \frac{16\pi E}{p}\sum_{\ell'=0}^{\infty}\left(S_{\ell'}(E)-1\right). \tag{2.816}$$

In equation 20 we assumed that we can exchange the integral with the sum with no affect on the final result; equation 21 is the substitution of the orthogonal relation for Legendre polynomials and finally equation 22 is simplifying and using the delta function.

Now, recall the first few expansions of the Legendre polynomial, $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}\left(3x^2-1\right)$, and $P_3(x) = \frac{1}{2}\left(5x^3-3x\right)$. A good resource, and gives the first few terms of P_ℓ is on this Wolfram page here: Legendre Polynomials. We can see that we only need to integrate over these few polynomials since only $\ell' = 1$ gives a non-zero result of 2, while ℓ' for $\ell' = \{0,2,3,4\}$ gives integrals of zero by symmetry. So, we will only give ℓ' for 0 and 1. These are,

$$S_{\ell'=0}(E) = 1 \tag{2.817}$$

$$S_{\ell'=1}(E) = \frac{-ip}{2\pi M^2 E}\left[s\left(\delta^{ij}\delta^{kl}-\delta^{il}\delta^{jk}\right)+t\left(\delta^{ik}\delta^{jl}-\delta^{il}\delta^{jk}\right)\right]+1. \tag{2.818}$$

We can see that for spinless particles, the $\ell' = 0$ gives the correct bound on the unitarity bound on the S-matrix as long as $0 \leq s \leq M^2_{\text{inelastic}}$. However notice that in the second term in equation 24 we have a non-unitary term signaling a break down of the theory which could change when we include loop order terms that then have poles that correspond to either bound or resonance states. For example, the next order term in equation 9, since this is essentially a 4-point "interaction" vertex would be of the order,

$$\sim (\text{vertex})\left(\frac{\delta_{ij}}{p^2}\right)\left(\frac{\delta_{jl}}{k+p}\right), \tag{2.819}$$

with an integral over the loop integral (the above is completely general but if we assume something along the lines that the incoming momentum is of p, and one of the propagators has momentum k, then the other propagator picks up $k+p$ and the vertex terms and denominator simplifies greatly).

Problem 2 answer:

Answer: (Note: much of what I will derive comes from a mixture of Weinberg for formalism [**AC:**] weinberg text, Ryder for integral identities [**AC:**] ryder, and Peskin & Schroeder for computational help [**AC:**] pands).

I would like to start from the ground up with this problem starting with the bare-Lagrangian and then constructing the renormalized QED Lagrangian that will be the base of our our calculations.

The steps for this calculation are as follows: absorb the field strength renormalization, split the Lagrangian into a renormalized version and counter term version, specify renormalization conditions that relates physical quantities, and finally compute the amplitudes we are looking for with the Feynman diagrams from the renormalized QED Lagrangian.

The bare QED Lagrangian is,

$$\mathcal{L}_0 = -\frac{1}{4}\left(F_{\mu\nu}\right)^2 + \bar{\psi}\left(i\partial\!\!\!/ - m_0\right)\psi - e_0\bar{\psi}\gamma^\mu\psi A_\mu, \tag{2.820}$$

where m_0 and e_0 is the bare mass and coupling constant respectively (or the bare electron charge in this particular case), and the interaction term comes from expanding the gauge-derivative. We would then expect the following general renormalized corrections for the electron and photon propagator to take the following forms,

$$= \frac{iZ_2}{p\!\!\!/ - m} + \cdots \text{ and } = \frac{-iZ_3 g_{\mu\nu}}{q^2}, \tag{2.821}$$

where we are working in the (as suggested) Feynman gauge (i.e. $\xi = 1$), and Z_2 is the field renormalization of the spinor propagator (Z_ψ) and Z_3 for gauge field propagator (Z_A). As stated previously, for our first step we can eliminate the renormalization constants Z_2 and Z_3 from the tree level propagators by defining

$$\psi = Z_2^{1/2}\Psi_R \text{ and } A^\mu = Z_3^{1/2}A_R^\mu. \tag{2.822}$$

Then our bare Lagrangian becomes,

$$\mathcal{L} = -\frac{1}{4}Z_3\left(F_R^{\mu\nu}\right)^2 + Z_2\bar{\psi}_R\left(i\partial\!\!\!/ - m_0\right)\psi - e_0 Z_2 Z_3^{1/2}\bar{\psi}_R A_{\mu,R}. \tag{2.823}$$

Let $e_0 Z_2 Z_3^{1/2} = eZ_1$, and let m be the observable mass. Then we can define the following,

$$\delta_3 = Z_3 - 1,\ \delta_2 = Z_2 - 1,\ \delta_m = Z_2 m_0 - m,\ \delta_1 = Z_1 - 1 = \left(\frac{e_0}{e}\right)Z_2 Z_3^{1/2} - 1. \tag{2.824}$$

The, we can write our Lagrangian as,

$$\begin{aligned}\mathcal{L} = &-\frac{1}{4}\left(F_R^{\mu\nu}\right)^2 + \bar{\psi}_R\left(i\partial\!\!\!/ - m\right)\psi_R - e\bar{\psi}\gamma^\mu\psi_R A_{\mu,R} \\ &-\frac{1}{4}\delta_3\left(F_R^{\mu\nu}\right)^2 + \bar{\psi}_R\left(i\delta_2\partial\!\!\!/ - \delta_m\right)\psi_R - e\delta_1\bar{\psi}_R\gamma^\mu\psi_R A_{R,\mu}.\end{aligned} \tag{2.825}$$

This is equivalent to equations 27-29 with the given definitions for the δ's in equation 35. From here, we can now read off the *renormalized* Feynman rules for QED. From the

first line of equation 36 we find the normal QED Feynman diagram rules. The photon propagator (denoted $P(\text{particle}, \text{momentum})$ and from ν to μ), fermion propagator, and vertex from QED are all the same,

$$P(\text{photon}, q) = \frac{-ig_{\mu\nu}}{q^2 + i\epsilon}, \quad P(\text{fermion}, p) = \frac{i}{\not{p} - m + i\epsilon}, \quad \text{vertex} = -ie\gamma^\mu, \tag{2.826}$$

(where we have implemented the $+i\epsilon$ procedure to ensure analyticity). (Note, we are not taking into account any polarization of the photon since we will not be dealing with terms resembling $A_\mu |\mathbf{p}\rangle$). The counter term Feynman diagrams then come from the second line of equation 23. From the δ_3 term we can find the counter term for the photon propagator in the following manner. Notice we can write the square of the field strength as

$$-\frac{1}{4}(F_{\mu\nu})^2 = -\frac{1}{2}A_\mu \left(-\partial^2 g^{\mu\nu} + \partial^\mu \partial^\nu\right) A_\nu \tag{2.827}$$

$$\overset{\text{Fourier}}{\Longrightarrow} -\frac{1}{2}\tilde{A}_\mu(q)\left(k^2 g^{\mu\nu} - k^\mu k^\nu\right)\tilde{A}_\nu(q), \tag{2.828}$$

(where tildes denote the Fourier transform of the field in question). Then, with the factor of i coming from the action, we find the following counter term for the photon propagator,

$$= -i\left(g^{\mu\nu}q^2 - q^\mu q^\nu\right)\delta_3. \tag{2.829}$$

We find something similar for the fermion propagator and vertex counter terms given as,

$$P_{\text{c.c}}(\text{fermion}, p) = i\left(\not{p}\delta_2 - \delta_m\right) \text{ and } \text{vertex}_{\text{c.c}} = -ie\gamma^\mu\delta_1. \tag{2.830}$$

Our goal then turns into finding $\delta_3, \delta_2, \delta_m$, and δ_1. To figure out how to find these counter term coefficients, we need to recall from QFT 1 (253a) the amplitudes for the two point amplitude of the photon propagator, the 2 point amplitude of the fermion propagator, and the vertex amplitude written as,

$$i\Pi^{\mu\nu}(q) = i\left(g^{\mu\nu}q^2 - q^\mu q^\nu\right)\Pi(q^2) \text{ and } -i\Sigma(\not{p}) \text{ and } -ie\Gamma^\mu(p', p) \tag{2.831}$$

where Γ^μ is the (amputated) vertex function. Our four conditions then are the following ([**AC:**] pands, section 10.3),

$$\Sigma(\not{p} = m) = 0, \ \frac{d}{d\not{p}}\Sigma(\not{p})\Big|_{\not{p}=m} = 0, \ \Pi(q^2 = 0) = 0, \ -ie\Gamma^\mu(p' - p = 0) = -ie\gamma^\mu. \tag{2.832}$$

The first condition of equation 43 fixes the electron mass. The second and third fix the radius of the residues of the electron and photon propagators at 1. The fourth condition fixes the electron's charge (coupling constant) as e. We can start with the first condition which leads us to computing the electron's self energy using the diagram below as reference.

Figure 40. Feynman diagram for the electron self energy. Taken from [**AC:**] pands.

From figure 1, re can read take off the external propagators and write the second order in e^2 matrix as (giving a small mass to the photon of μ^2 in order to handle the IR-divergences as suggested in the problem),

$$-i\Sigma_2(p) = (-ie)^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu \frac{i(\not{k} + m_0)}{k^2 - m_0^2 + i\epsilon} \gamma_\mu \frac{-i}{(p-k)^2 - \mu^2 + i\epsilon}, \tag{2.833}$$

where we already contracted over the gamma matrix, $g^{\mu\nu}\gamma_\nu$. We can write the denominator using Feynman parameters as,

$$\frac{1}{(k^2 - m_0^2 + i\epsilon)\left((p-k)^2 - \mu^2 + i\epsilon\right)} = \int_0^1 dx \frac{1}{\left[x\left(k^2 - m_0^2 + i\epsilon\right) + (1-x)\left((p-k)^2 - \mu^2 + i\epsilon\right)\right]^2} \tag{2.834}$$

$$= \int_0^1 dx \frac{1}{\left[k^2 - 2xk\cdot p + xp^2 - x\mu^2 - (1-x)m_0^2 + i\epsilon\right]^2}. \tag{2.835}$$

We can then complete the square by defining $\ell \equiv k - xp$. We can then regularize the integral by letting $d = 4 - \epsilon$ via dimensional regularization, we can then perform the integral over $d^d\ell$ and with the following identity,

$$\int \frac{d^d\ell_E}{(2\pi)^d} \frac{1}{\left(\ell_E^2 + \Delta\right)^2} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma\left(2 - \frac{d}{2}\right)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{2-\frac{d}{2}}, \tag{2.836}$$

where Δ is the re-defined mass term, $\Delta = -x(1-x)p^2 + x\mu^2 + (1-x)m_0^2$. We also need to handle the Dirac algebra in equation 44 since we have terms like $\gamma^\mu k_n u \gamma^\nu \gamma_\mu = \gamma^\mu k_\nu \gamma^\nu \gamma^\alpha g_{\alpha\mu}$. So consider the following identities,

$$\gamma^\mu\gamma^\nu\gamma_\mu = -(2-\epsilon)\gamma^\nu, \quad \gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 4g^{\nu\rho} - \epsilon\gamma^\nu\gamma^\rho, \quad \gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu + \epsilon\gamma^\nu\gamma^\rho\gamma^\sigma. \tag{2.837}$$

This separates the integral in equation 31 into a part that we need to deal with and one that disappears which has the form of,

$$\sim \frac{\gamma^\mu\left(\gamma\cdot k\right)\gamma_\mu}{\left((k-p)^2 + p^2(x(1-x) - m^2x)\right)} \to 0. \tag{2.838}$$

This integral goes to zero by symmetry since the numerator is odd in k but even in k in the denominator.

Then when we plug equations 47 and 48 into 44 we find,

$$-i\Sigma_2(p) = \frac{-ie^2}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma\left(2 - \frac{d}{2}\right)\left((4-\epsilon)m - (2-\epsilon)x\not{p}\right)}{\left((1-x)m^2 + x\mu^2 - x(1-x)p^2\right)^{2-d/2}}. \tag{2.839}$$

If we then utilize the first condition of equation 43, we can pull out a factor of m in the numerator and simplify the denominator as (notice that p^2 can be put on the same footing

of $\not{p}$ since $\gamma^2 = 4$),

$$m\delta_2 - \delta_m = \Sigma_2(m) = \frac{e^2 m}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma\left(d - \frac{d}{2}\right) \cdot (4 - 2x - \epsilon(1-x))}{((1-x)^2 m^2 + x\mu^2)^{2-d/2}}. \tag{2.840}$$

To find δ_m we need δ_2 which corresponds to determining condition number two of equation 43 (using equation 50 to take the derivative with respect to $\not{p}$ or $\gamma^\mu p_\mu$),

$$\delta_2 = \frac{d}{d\not{p}} \Sigma(\not{p}) \Big|_{\not{p}=m} \tag{2.841}$$

$$= \frac{-e^2}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma\left(2 - \frac{d}{2}\right)}{((1-x)^2 m^2 + x\mu^2)^{2-d/2}} \times \tag{2.842}$$

$$\times \left((2-\epsilon)x - \frac{\epsilon}{2} \frac{2x(1-x)m^2}{(1-x)^2 m^2 + x\mu^2} (4 - 2x - \epsilon(1-x)) \right).$$

Then, δ_m is $m\delta_2 - \Sigma_2(m)$ by taking equation 53 and subtracting off equation 50. Notice that the second term (the long fraction) in the parenthesis is finite as $\epsilon \to 0$.

To determine Z_3 (which is Z_A that we are asked to find in the problem) this comes from the third condition of the photon propagator. To do this, lets save us some time by recalling the result for $\Pi_2(q^2)$ and then set $q^2 = 0$,

$$\Pi_2(q^2) = \frac{-e^2}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma\left(2 - \frac{d}{2}\right)(8x(1-x))}{(m^2 - x(1-x)q^2)^{2-d/2}} \tag{2.843}$$

$$\delta_3 \overset{q^2=0}{=} \frac{-e^2}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma\left(2 - \frac{d}{2}\right)(8x(1-x))}{(m^2)^{2-d/2}}. \tag{2.844}$$

(To derive the above, what we would actually need to derive is the following)

$$i\Pi_2^{\mu\nu}(a) = -4ie^2 \int_0^1 dx \int \frac{d^4\ell_E}{(2\pi)^4} \tag{2.845}$$

$$\times \frac{-\frac{1}{2}g^{\mu\nu}\ell_E^2 + g^{\mu\nu}\ell_E^2 - 2x(1-x)q^\mu q^\nu + g^{\mu\nu}(m^2 + x(1-x)q^2)}{(\ell_E^2 + \Delta)^2}, \tag{2.846}$$

(where $\Delta = m^2 - x(1-x)q^2$.)

We could for completeness compute the value of the electron vertex function, the last condition of equation 30, but this is to find the form factors F_1 that allows us to determine $\delta_1 = Z_1 - 1$. As one can check, by using either [**AC:**] pands or in [**AC:**] weinberg text, we would find that the vertex condition explicitly gives $\delta_1 = \delta_2 \implies Z_1 = Z_2$ at order α. (This implies the Ward identity for diagrams without counterterm vertices i.e. $q_\mu \Gamma^\mu = 0$).

To wrap up, we determined the self-energy up to order e^2, $\sigma_2(p)$ in equation 38. We notice that in equation 50 when $\mu^2 \to 0$ we have a divergence by power counting in powers

of x that solely depends on the number of d dimensions, specifically an IR-divergence. We also determined the field strength renormaliztion of ψ, Z_2, which can be found either with $\delta_2 = Z_2 - 1$ or $\delta_m = Z_2 m_0 - m$, where we found δ_2 in equation 53. We found δ_m once we found δ_2 and $\Sigma_2(m)$ given in equation 51. Finally, we found Z_3 (Z_A) in equation 55.

Problem 3 answer:

Answer (a): We will start off this problem with a little history and background. This problem refers to what the quantum mechanical effects are of a strong electromagnetic field (in this case a constant magnetic and electric field). The effect was predicted by Schwinger in which we now call it "Schwinger pair production" [**AC:**] schwinger. Schwinger built his work from Euler and Heisenberg who computed various Lagrangians for QED given a strong electromagnetic field (a review of which can be found here [**AC:**] HE overview) now dubbed Heisenberg-Euler Lagrangians. Weisskopf computed a similar quantity but for (what we now call) scalar-QED [**AC:**] weisskopf. However, Weisskopf's version of the computation was purely in the Hamiltonian version of the theory since it was in 1936. However, in this version of the problem, we will be using the path integral formulation. (Note that Schwinger used proper time to derive most results, which we shall relate to here and there).

Lets first understand how we go to equation 58. Notice that the Lagrangian density in equation 56 can be written, after an integration by parts, as

$$\mathcal{L} = -\frac{1}{4}\left(F_{\mu\nu}\right)^2 - \varphi^*\left(D^2 + m^2\right)\varphi. \tag{2.847}$$

Now, the full expression we want to compute is,

$$\int [DA_\mu]\, e^{iS_{\text{eff}}[A_\mu]} = \int [DA_\mu]\,[D\varphi^*]\,[D\varphi]\exp\left(i\int d^4x\left(-\frac{1}{4}\left(F_{\mu\nu}\right)^2 - \varphi^*\left(D^2+m^2\right)\varphi\right)\right). \tag{2.848}$$

To perform this integral over $[D\varphi^*]\,[D\varphi]$ we can use the following Gaussian identity [**AC:**] cayley,

$$\int [D\varphi^*]\,[D\varphi]\exp\left(i\int d^4x\left(\varphi^* B\varphi + J\cdot B\right)\right) = \frac{\mathcal{N}}{\det(B)}e^{iJB^{-1}J}, \tag{2.849}$$

where B is our Green's operator. For us, $B = -D^2 - m^2$ to get (multiply out the negative),

$$\int [DA_\mu]\, e^{iS_{\text{eff}}} = \mathcal{N}\int [DA_\mu]\exp\left(i\int d^4x\left(-\frac{1}{4}\left(F_{\mu\nu}\right)^2\right)\right)\left[\det\left(-D^2-m^2\right)\right]^{-1}, \tag{2.850}$$

which if we drop the integration over the gauge field and divide out the field strength term we have,

$$\exp\left(iS_{\text{eff}}[A_\mu] + i\int d^4x\left(\frac{1}{4}\left(F_{\mu\nu}\right)^2\right)\right) = \mathcal{N}\left[\det\left(-D^2-m^2\right)\right]^{-1}. \tag{2.851}$$

We still need the action and the effective correction by itself. We can use the properties of logs to bring out the (infinite) constant over to the LHS in the form of $\log(\mathcal{N})$, and then apply the given identity relating determinants and traces of logs,

$$iS_{\text{eff}}[A_\mu] + \frac{i}{4}\int d^4x\left(F_{\mu\nu}\right)^2 - \log(\mathcal{N}) = -\log\left(\det\left(-D^2-m^2\right)\right) \tag{2.852}$$

$$= -\operatorname{tr}\left(\log\left(-D^2-m^2\right)\right). \tag{2.853}$$

Then the effective action is,

$$iS_{\text{eff}}[A_\mu] = -\frac{i}{4}\int d^4x\,(F_{\mu\nu})^2 - \text{tr}\Big(\log\Big(-D^2 - m^2\Big)\Big) + \log(\mathcal{N}). \tag{2.854}$$

The trace can be computed in position space since the trace has the nice property of being independent of the basis of our choice giving,

$$iS_{\text{eff}}[A_\mu] = \int d^4x\left(-\frac{i}{4}\,(F_{\mu\nu})^2 - \langle x|\log\Big(-D^2 - m^2\Big)|x\rangle\right) + \log(\mathcal{N}). \tag{2.855}$$

It will now be easier, and give us more physical insight, if we take a page from Schwinger by computing the derivative with respect to m^2,

$$\frac{d}{dm^2}\langle x|\log\Big(-D^2 - m^2\Big)|x\rangle = -\langle x|\left(-D^2 - m^2\right)^{-1}|x\rangle \tag{2.856}$$

$$= i\int_0^\infty ds e^{-ism^2}\langle x|e^{-i\hat{H}s}|x\rangle\,, \tag{2.857}$$

where the Hamiltonian is $\hat{H} = -\hat{\pi}^2 = -(\hat{\mathbf{p}} - eA(\hat{x}))^2$. If we then integrate with respect to m^2 in equation 70 (i.e. $\int dm^2$) then we pick up a negative $1/s$ multiple. This puts the effective Lagrangian density then as (and restoring $i\epsilon$ term),

$$\mathcal{L}_{\text{eff}} = -\frac{1}{4}\,(F_{\mu\nu})^2 - i\int_0^\infty \frac{ds}{s}e^{-s\epsilon}e^{-ism^2}\langle x|e^{-i\hat{H}s}|x\rangle + \text{constant}, \tag{2.858}$$

where the constant term is both the integration constant from dm^2 and the $\log(\mathcal{N})$. Now we need to determine what the expectation value in the position basis of $e^{-i\hat{H}s}$ is. There seems to be two general ways. One is Schwinger's method, which we have been utilizing here and there but not fully, that utilizes a proper time formalism in conjugation with Heisenberg's equations (your goal then is to solve equation 2.35 in [**AC:**] schwinger). The other way is more quantum mechanical (and a little more illuminating) by inserting a complete basis of state $|\psi_n\rangle$ and have $\hat{H}$ operator on them. This gives an eigenvalue equation that resembles the calculation of Landau levels (for a quick review, and a resource we will use, check Weinberg's quantum mechanics text [**AC:**] weinberg qm). This is the method we follow.

Consider the following,

$$\int d^4x\,\langle x|e^{-i\hat{H}s}|x\rangle = \int d^4x\sum_n \langle x|\psi_n\rangle\langle\psi_n|e^{-i\hat{H}s}|x\rangle \tag{2.859}$$

$$= \int d^4x\sum_n |\psi(x)|^2 e^{-iE_n s}, \tag{2.860}$$

where $e^{-i\hat{H}s}|\psi_n\rangle = e^{-iE_n s}|\psi_n\rangle$.

Now, given that $F_{12} = B_3 = \partial_1 A_2 - \partial_2 A_1 \implies A_y = B\hat{x}$ (a magnetic field in the z-direction and with $\mu = (0,1,2,3) \leftrightarrow (t,x,y,z)$). The Hamiltonian in component form is

then,

$$\hat{H} = (\hat{p} - eA(\hat{x}))^2 \tag{2.861}$$

$$= \left(p_x\hat{x} + p_z\hat{z} + p_t\hat{t} + (p_y eA_y)\,\hat{y}\right)^2 \tag{2.862}$$

$$= \left(-p_t^2 + p_x^2 + p_z^2 + (p_y - eB\hat{x})^2\right) \cdot \mathbf{1}_{4\times 4}. \tag{2.863}$$

We can break this eigenvalue equation into two distinct parts. At least for p_t, p_y, and p_z we get the following eigenstate,

$$\psi_n^{p_{t,y,z}} = \chi_n\left(x - \frac{p_y}{eB}\right) e^{ip_t\cdot t - ip_y\cdot y - ip_z\cdot z}. \tag{2.864}$$

This reduces our problem to solving the eigenstate for $p_x^2 + (eB\hat{x})^2$. Notice that our Schrödinger equation is $(p_x^2 + (p_y - eB_z x)^2)\Psi = E\Psi$ or,

$$\left(p_x^2 + \frac{1}{2}\omega^2 m^2 (x - x_0)^2\right)\Psi = E\Psi, \tag{2.865}$$

with $x_0 = -p_y/(eB)$, and we can recognize that the LHS is exactly a discrete simple harmonic oscillator and is exactly our χ_n [**AC:**] schwartz. The energy eigenvalues are then [**AC:**] weinberg qm,

$$E_n^{p_{t,y,z}} = -p_t^2 + p_z^2 + eB(2n+1). \tag{2.866}$$

These are just the Landau levels that describes the motion of a particle (usually an electron, in our case a scalar) in a uniform magnetic field. Then, when we plug everything into equation 72,

$$\int d^4x\,\langle x|\,e^{-i\hat{H}s}\,|x\rangle = 2\int d^4x \frac{dp_t dp_y dp_z}{(2\pi)^3}\sum_{n=0}^{\infty}\left|\chi_n\left(x - \frac{p_y}{eB}\right)\right| \exp\Big(is\left((p_t^2 - p_z^2) - e(2n+1)B\right)\Big) \tag{2.867}$$

(note the factor of 2 infront is due to $\hat{H}$ being block-diagonal). Now, in order to perform the integration, suppose we are within a box of length L [**AC:**] schwartz. Since the SHO eigenstates are normalized, the integral over dx gives 1. The integrals over dt, dy, and dz each give a multiply of L for a total of L^3. With the p_y integral as well, the integrand depends upon $(x - p_y/eB)$ which if equate to zero to determine the bounds gives the following inequality, $x - p_y/eB \to L - p_y/eB = 0 \implies p_y < eBL$. So when we integrate over p_y, we also pick up a factor of eBL. Then we have,

$$\int d^4x\,\langle x|\,e^{-i\hat{H}s}\,|x\rangle = \frac{2eBL^4}{(2\pi)^3}\int_{-\infty}^{\infty} dp_z dp_t e^{i(p_t^2 - p_z^2)s}\sum_{n=0}^{\infty} e^{-ies(2n+1)B}. \tag{2.868}$$

Notice the sum on the RHS is just a geometric series with a constant multiple of e^{-iesB} where the sum turns into $\sum_n e^{-an} = \sum_n (1/e^a)^n = 1/(e^a - 1)$. From the integration we get a factor of $(\sqrt{\pi/is})^2$ (written deliberately to show two Gaussian integrations performed)

to get

$$\int d^4x \, \langle x| \, e^{-i\hat{H}s} \, |x\rangle = -\frac{eBL^4}{8\pi^3}\frac{1}{s^2}\frac{e^{-iesB}}{e^{iesB}-1} \tag{2.869}$$

$$= -i\frac{eBL^4}{8\pi^2}\frac{1}{s}\frac{1}{\sinh(esB)}. \tag{2.870}$$

Before we plug this into equation 71, notice that if we had constant parallel magnetic and electric fields then $(F_{\mu\nu})^2 = 2(B^2 - E^2)$. If only a magnetic field is present then we just have $2B^2$. Our final effective action then for a constant magnetic field becomes,

$$\mathcal{L}_{\text{eff}} = -\frac{1}{2}B^2 + \frac{eB}{8\pi^2}\int_0^\infty \frac{ds}{s}e^{s\epsilon}e^{-ism^2}\frac{1}{s}\frac{1}{\sinh(esB)}. \tag{2.871}$$

Answer (b): Recall from classical electromagnetism (or even optics) that on the imaginary plane, if either the electric or magnetic field is propagating along the real axis, the former can be thought of along the imaginary axis. In this sense, consider the analytic continuation of the magnetic field to the electric field via $B \to iE$. With this, and the fact that $\sin(ix) = i\sinh(x)$ equation 84 becomes,

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}E^2 + \frac{eE}{8\pi^2}\int_0^\infty \frac{ds}{s^2}e^{s\epsilon}e^{-ims^2}\frac{1}{\sin(esE)}. \tag{2.872}$$

Answer (c): Before we continue onto the probability for pair production calculation, notice we have made a crucial error in equation 85: we do not have any counter terms at all and the Lagrangian is unrenormalized. Schwartz in his text [**AC:**] schwartz determines what these should be (in general) in equation 33.73 of which if we trace the calculation back (since it is spinor QED based) the real part of the cosine in the fraction comes from the spinors and should be set to zero to satisfy scalar QED (Schwinger also finds the renormalized version but in his own manner in equation 3.53 of [**AC:**] schwartz). Following Schwartz, the renormalized calculation for the effective action with a constant electric field in the background is,

$$\mathcal{L}_{\text{eff}} = \frac{1}{2}E^2 - \frac{1}{8\pi^2}\int_0^\infty \frac{ds}{s^3}e^{i\epsilon s}e^{-sm^2}\left(\frac{eEs}{\sin(eEs)} - 1 - \frac{(eEs)^2}{6}\right). \tag{2.873}$$

(Note: you can also compute the exact same quantity by computing $\delta^2\mathcal{L}_{\text{eff}}/\delta(m^2)^2$ and then integrating twice).

There are two ways to compute the probability of pair production: one from optical theorem[**AC:**] pands and another from (of course) Schwinger. In the optical theorem approach, notice that we can apply this without any worry to our case since scalar QED is unitary. In this sense, we can using either equation 7.50 from Peskin and Schroeder [**AC:**] pands or a more convient form is equation 7.67 given as,

$$\text{Im}\mathcal{M}(p \to p) = -\frac{1}{2}\sum_f \int d\Pi_f |\mathcal{M}(p \to p)|^2. \tag{2.874}$$

However, as I said previously, Schwinger has another (much faster) version[**AC:**] schwartz,[**AC:**] schwinger. Notice if no pair production was to occur with an external background field, $\langle A| S |S\rangle = 1$ for some S-matrix. Since our S-matrix is constant $S = e^{iS_{\text{eff}}}$. Then, $|\langle A| e^{iS_{\text{eff}}} |A\rangle|^2 = |e^{iS_{\text{eff}}}|^2$. Then,

$$|e^{iS_{\text{eff}}}|^2 = e^{iS_{\text{eff}}} e^{-iS^*_{\text{eff}}} = e^{iS_{\text{eff}} - iS^*_{\text{eff}}} = e^{-2\text{Im}[iS_{\text{eff}}]} = e^{2VT\text{Im}\mathcal{L}_{\text{EH}}}. \tag{2.875}$$

Thus, $2\text{Im}\mathcal{L}_{\text{EH}}$ is the probability per unit time and volume that any number of pairs are created. Now, notice in equation 86 that we have poles when integrating over s namely for when E is real and when s is $s_n = n\pi/(eE)$ (i.e, when does $\sin(eEs) = 0$, for integer values of π, hence the factor of n). We also do not have a pole at $s = 0$. The imaginary part of equation 86 only omits the sine term. Then, when we perform the contour integral via residue theorem, we get a factor of 2π (not the i since we are taking the imaginary part). When we perform the integration we also omit the $e^{i\epsilon s}$ term since there is nothing to regulate anymore. Notice a factor of s then cancels. We then have the following sum,

$$2\text{Im}\mathcal{L}_{\text{eff}} = \frac{1}{4\pi^2} \sum_{n=1}^{\infty} \left(\frac{eE}{n}\right)^2 \exp\left(-\frac{n\pi m^2}{eE}\right). \tag{2.876}$$

Surprisingly, this is the exact same result as we would get for the spinor case which is due to the behavior of the cotangent function that is original derived in [**AC:**] schwartz. The only time this result of scalar QED diverges away from the spinor case is when the electric and magnetic fields are parallel and we pick up a factor of $(-1)^{n+1}$ term in our sum.

2.6 Problem set 3:

2.6.1 Problem 1: 2-loop β-function for ϕ^3 theory

In class we considered ϕ^4 theory in $4-\epsilon$ dimensions, where ϵ is viewed as a small but finite expansion parameter, and found an interacting infrared RG fixed point. In this problem you will investigate an analogous RG fixed point of the perturbative ϕ^3 theorem in $6-\epsilon$ dimensions, whose bare Euclidean action takes the form

$$S_E = \int d^{6-\epsilon}x \left(\frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2}m_0^2\phi^2 + \frac{1}{3!}g_0\phi^3 \right). \tag{2.877}$$

The bare mass m_0 will be chosen such that there is no mass gap in the spectrum of states created by $\hat{\phi}(x)$ acting on the vacuum.

As introduced in class, the dimensionless renormalized running coupling $\lambda(\mu)$ is defined as,

$$\lambda(\mu) = -\mu^{-\delta(\epsilon)} \left(Z(\mu)\right)^{3/2} \Gamma^{(3)}(k_1, k_2)|_{k_1^2=k_2^2=(k_1+k_2)^2=\mu^2}. \tag{2.878}$$

Here μ is a mass scale variable $-\delta(\epsilon)$ is the mass dimension of g_0, $\Gamma^{(3)}(k_1, k_2)$ is computed by the amputated 1PI diagram with momenta $k_1, k_2, -k_1-k_2$ assigned to the three external lines, and $Z(\mu)$ is related to the self-energy $\Sigma(k)$ via,

$$Z(\mu) = \left(1 - \frac{\partial \Sigma}{\partial k^2}\right)^{-1} \Bigg|_{k^2=\mu^2}. \tag{2.879}$$

(a) Determine the beta function $\beta(\lambda)$ appearing in the RG equation $\frac{d\lambda(\mu)}{d\log(\mu)} = \beta(\lambda(\mu))$ at 1-loop order.

(b) You should find an interacting RG fixed point at a complex value of λ. Determine the latter to the leading nontrivial order in the ϵ-expansion.

2.6.2 Problem 2: Gross-Neveu Fermion Model

Consider a theory of massless fermionic spinor fields in two-dimensional spacetime, define by the bare (Lorentzian) action

$$S[\psi, \bar{\psi}] = \int d^2x \left(-\sum_{i=1}^{N} \bar{\psi}_i \gamma^\mu \partial_\mu \psi_i + \frac{1}{2}g_0^2 \left(\sum_{i=1}^{N} \bar{\psi}_i \psi_i \right)^2 \right). \tag{2.880}$$

Here $\psi_i (i = 1, ..., N)$ are a set of N fermonic fields that transform as 2-component complex spinors with respect to the $1+1$ dimensional Lorentz symmetry. The 2D version of gamma matrices $\gamma^\mu (\mu = 0, 1)$ are 2×2 constant matrices that obey $-(\gamma^0)^2 = (\gamma^1)^2 = 1, \{\gamma^0, \gamma^1\} = 0$. We have also defined $\bar{\psi}_i = i\psi_i^\dagger \gamma^0$ as usual.

The chirality matrix is defined as $\gamma \equiv \gamma^0\gamma^1$ (this is the 2D analog of γ_5 we ave seen in 4D). Observe that the action above is invariant under the "chiral symmetry" $\psi_i \mapsto \gamma\psi_i$, while a mass term for the fermion fields would have violated the chiral symmetry.

(a) The renormalized coupling $g(\mu) = g_0 +$ loop corrections is defined through the four-fermion 1PI diagram with external momenta set at mass scale μ as usual, and obeys

in RG equation of the form $\frac{dg(\mu)}{d\log(\mu)} = \beta(g(\mu))$,. Compute the beta function $\beta(g)$ at 1-loop order. Is this theory weakly or strong coupled at low energies?

(b) Next, we consider an equivalent formulation of the same theory by writing the path integral as

$$Z = \int [D\bar{\psi}_i D\psi_i D\sigma] e^{i\tilde{S}(\psi,\bar{\psi},\lambda)}, \tag{2.881}$$

where $\sigma(x)$ is a scalar field, and the new action $\tilde{S}$ is

$$\tilde{S}[\psi,\bar{\psi},\sigma] = \int d^2x \left[-\sum_{i=1}^{N} \bar{\psi}_i \partial^\mu \gamma_\mu \psi_i - \sum_{i=1}^{N} \sigma \bar{\psi}_i \psi_i - \frac{1}{2g_0}\sigma^2 \right]. \tag{2.882}$$

Integrating out $\sigma(x)$ gives back the original fermion theory. On the other hand, integrating out ψ_i, $\bar{\psi}_i$ produces an effective action $S_{\text{eff}}[\sigma]$,

$$e^{iS_{\text{eff}}[\sigma]} = \int [D\bar{\psi}_i D\psi_i] e^{i\tilde{S}[\psi,\bar{\psi},\sigma]}. \tag{2.883}$$

Evaluate $S_{\text{eff}}[\sigma]$ for a constant field configuration $\sigma(x) = \sigma_0$. The answer should be of the form $S_{\text{eff}} = -\int d^2x V_{\text{eff}}(\sigma_0)$, where the "effective potential" $V_{\text{eff}}(\sigma_0)$ is hopefully finite when expressed as a function of the renormalized coupling $g(\mu)$ defined in part (a).

(c) You may minimize V_{eff} with respect to σ_0 to find an approximate vacuum expectation value of the field $\sigma(x)$. Note that in the formulation of the partition function defined above, the chiral symmetry acts by $\psi_i \mapsto \gamma\psi_i, \sigma \mapsto -\sigma$. Is $V_{\text{eff}}(\sigma)$ invariant under this symmetry? What about the vacuum expectation value of σ?

Problem 1 answer:

Answer (a): First, recall that there are a few different ways to determine the β-function to determine how the renormalization group flow behaves. One way is to use the definition of the dimensionless renormalized running coupling $\lambda(\mu)$ as defined above and plug-in to the following formula,

$$\frac{d\lambda(\mu)}{d\log(\mu)} = \beta(\lambda(\mu)), \tag{2.884}$$

where μ is some mass scale of the theory. Or, there is the "brute-force" way utilizing the Callan-Symanzik equation, the Gell-Mann-Low equation (which is technically the one above), the 't Hooft-Weinberg equation, or the Georgi-Politzer equation, which has many different names, each referring to the different scheme used in renormalization (such as minimal-substraction scheme or the on-shell physical renormalization scheme). This is can be stated as (for the case of QED with coupling e)[**AC:**] schwartz,

$$\left(\mu\frac{\partial}{\partial\mu} + \frac{n}{2}\gamma_3 + \frac{m}{2}\gamma_2 + \beta\frac{\partial}{\partial e_R} + \gamma_m m_R\frac{\partial}{\partial m_R}\right) G_{n,m} = 0, \tag{2.885}$$

or in the case of ϕ^4 theory (written in different notation with a mass scale M [**AC:**] pands),

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial\lambda}n\gamma(\lambda)\right) G^{(n)}(\{x_i\}; M, \lambda) = 0. \tag{2.886}$$

For our particular case, we will be using the Gell-Mann-Low equation since it is particularly efficient in its computations since we only need to compute the 1-loop counter terms for the ϕ^3 interaction, and then we immediately get the β function out. The 1-loop counter terms will be computed utilizing the minimal-subtraction scheme (denoted MS) for our renormalization scheme. (Note: to ensure our calculation is correct, the β-function has been computed already by Macfarlane and Woo in 1974 [**AC:**] phi 3 using both the Gell-Man-Low equation *and* the 't Hooft-Weinberg formula, from which we can compare our calculation to theirs. They actually go to two loop for those interested).

So, our plan is the following: compute the 3-vertex interaction to one loop order including the correction term for the 3-vertex. With this, and that $Z(\mu) = 1$ at one loop (given in lecture), we can compute the β-function using equation 4 (Gell-Mann-Low equation).

Consider writing the Lagrangian density in equation one as,

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu\phi\right)^2 - \frac{1}{2}m^2\phi^2 - \frac{1}{3!}g\mu^{3-d/2}\phi^3, \tag{2.887}$$

with m and g the renormalized (finite) constants, but does not have to be the physical values, and $d = 6 - \epsilon$ i.e. $\mu^{\epsilon/2}$ to make g dimensionless. Then, consider the following counter terms to the Lagrangian,

$$\mathcal{L}_{\text{counter term}} = \frac{1}{2}(\delta Z)(\partial_\mu\phi)^2 = \frac{1}{2}\delta m^2\phi^2 - \frac{1}{3!}\delta g\mu^\epsilon\phi^3, \tag{2.888}$$

then the total renormalized Lagrangian is,

$$\mathcal{L}_{\text{renormalized}} = \frac{1}{2}\left(1 + \delta Z\right)\left(\partial_\mu\phi\right)^2 - \frac{1}{2}(m^2 + \delta m^2)\phi^2 - \frac{1}{3!}(g + \delta g)\mu^\epsilon\phi^3. \tag{2.889}$$

In order to get back to the bare Lagrangian consider the following re-definitions of the scalar field, mass, and coupling constant as,

$$\phi_0 = (1+\delta Z)^{1/2}\phi, \;\; m_0^2 = (1+\delta Z)^{-1}(m^2+\delta m^2), \;\; g_0 = \mu^\epsilon (1+\delta Z)^{-3/2}(g+\delta g), \quad (2.890)$$

where each factor of ϕ has normally picked up a factor of $Z^{1/2}$. These new definitions then give us the the typical, renormalized Lagrangian as given in equation one but with ϕ_0, m_0, and g_0.

We can then read off the Feynman rules. The propagator with momentum p^μ and vertex are respectively,

$$\frac{i}{p^2 - m^2 + i\epsilon} \text{ and } -ig\mu^{\epsilon/2}, \quad (2.891)$$

while the counter terms for the propagator and vertex are

$$ip^2\delta Z - \delta m^2 \text{ and } -i\delta g\mu^{\epsilon/2}. \quad (2.892)$$

(Notice that we have chosen both ϵ as our dimensional regularization and analytic contour coefficient, but they are of course different).

Now, to find the renormalized running coupling λ as defined in equation two, we need to compute the 3-vertex up to 1-loop order (as asked for). The 3-vertex will then contain a regular vertex term, $-ig\mu^{\epsilon/2}$ plus the counter term plus the 1-loop contribution. We then have the total 3-vertex being,

$$i\Gamma^{(3)}(k_1,k_2) = -i(g+\delta g)\mu^{\epsilon/2} + \quad (2.893)$$

$$(-ig)^3\mu^{3\epsilon/2}\int \frac{d^d\ell}{(2\pi)^d}\frac{i^3}{(\ell^2-m^2+i\epsilon)((\ell+k^2)^2-m^2+i\epsilon)((\ell-k_1)^2-m^2+i\epsilon)}. \quad (2.894)$$

Now, in order to compute the 1-loop integral, recall the following formula for the epsilon expansion of the gamma function and the Feynman trick for three terms in the denominator [**AC:**] schwartz,

$$\Gamma(\epsilon - n) = \frac{(-1)^n}{n!}\left(\frac{1}{\epsilon} + \psi(n+1) + \mathcal{O}(\epsilon)\right) \quad (2.895)$$

$$\frac{1}{ABC} = 2\int_0^1 dx \int_0^{1-x} dx \frac{1}{(A+(B-A)x+(C-A)z)^3}. \quad (2.896)$$

We shall be referring off of the following diagram,

Figure 41. Labeled 1-loop diagram for ϕ^3 theory in $d=6$

Then, the 1-loop integral term in equation 14 becomes,

$$
\begin{aligned}
i\Gamma^{(3)}_{1-\text{loop}} &= (-ig)^3\mu^{3\epsilon/2}i^3\int\frac{d^d\ell}{(2\pi)^d}2\int_0^1\int_0^{1-x}dz\times \\
&\times\frac{1}{(\ell^2-m^2+((\ell+k_2)^2-\ell^2)x+((\ell-k_1)^2-\ell^2)z+i\epsilon)^3} \qquad (2.897)\\
&= -(-ig)^3\mu^{3\epsilon/2}\frac{\pi^{d/2}}{(2\pi)^d}\Gamma\left(\frac{\epsilon}{2}\right)\int_0^1 dx\int_0^{1-x}dz\times \\
&\times\frac{1}{(m^2-k_2^2x-k_1^2z+k_2^2x^2+k_1^2z^2)^{\epsilon/2}-2xzk_1\cdot k_2} \qquad (2.898)\\
&= \frac{-ig^3\mu^{\epsilon/2}}{2^d\pi^{d/2}}\left(\frac{2}{\epsilon}+\cdots\right)\int_0^1 dx\int_0^{1-x}dz\left(1-\frac{\epsilon}{2}\ln\left(\frac{\Delta}{\mu^2}\right)\right), \qquad (2.899)
\end{aligned}
$$

where $\Delta = m^2 - k_2^2x - k_1^3z + k_2^2x^2 + k_1^2z^2 - 2xzk_1\cdot k_2$ and μ^2 is some mass scale (which will soon become the renormalization scale). Then the total 3-point vertex is

$$
\Gamma^{(3)}(k_1,k_2) = -\mu^\epsilon\left(g+\delta g+\frac{g^3}{2(4\pi)^3}\left(\frac{1}{\epsilon}+\mathcal{O}(\epsilon^0)\right)\right). \qquad (2.900)
$$

We can then choose the counter term to be

$$
\delta g = -\frac{g^3}{2(4\pi)^3}\left(\frac{1}{\epsilon}+c_g\right), \qquad (2.901)
$$

which would then give the 3-point vertex to one loop order to be,

$$
\Gamma^{(3)} = -\mu^\epsilon g\left(1+\frac{g^2}{2(4\pi)^3}\left(-c_g+\mathcal{O}(\epsilon^0)\right)\right). \qquad (2.902)
$$

Or, to relate it back directly to our definition of the beta-function, we can, given the two on-shell external legs, write the 3-vertex as,

$$
\begin{aligned}
\Gamma^{(3)}(k_1,k_2) &= -\mu^\epsilon\left(g+\delta g+\frac{g^3}{2(4\pi)^3}\left(\frac{1}{\epsilon}-\log\left(-\frac{(k_1+k_2)^2}{4\pi\mu^2}\right)-\gamma+3\right)\right) \qquad (2.903)\\
&= -\mu^\epsilon\left(1+\frac{g^2}{2(4\pi)^3}\left(-c_g-\log\left(-\frac{(k_1+k_2)^2}{4\pi\mu^2}\right)-\gamma+3\right)\right). \qquad (2.904)
\end{aligned}
$$

Under $\bar{\text{M}}$S (where Schwartz has a great section on it in chapter 18 and chapter 21 [**AC:**] schwartz) given $c_g = -\gamma + \log(4\pi)$,

$$
\gamma^{(3)}(k_1,k_2) = -\mu^\epsilon\left(1+\frac{g^2}{2(4\pi)^3}\left(-\log\left(-\frac{(k_1+k_2)^2}{\mu^2}\right)+3\right)\right). \qquad (2.905)
$$

Now, given that $Z(\mu) = 1$ at 1-loop order, all we have to do is evaluate the 3-vertex function at $(k_1+k_2)^2 = \mu^2$ which gives $\log(-1)$. However, note that $-1 = e^{i\pi(2n+1)} \implies \log(-1) = i\pi(2n+1)\ \forall n\in\mathbb{Z}$ as given by the Euler equation. Then we have,

$$
\lambda = g\left(1+\frac{i\pi g^2(2n+1)}{2(4\pi)^3}\right), \qquad (2.906)
$$

or if we let $n = 1$ then,

$$\lambda = \left(g + \frac{3ig^3}{128\pi^2} \right). \tag{2.907}$$

Then, our β function simply becomes,

$$\beta = -\frac{3ig^3}{128\pi^3}, \tag{2.908}$$

(if we refer back to the vertex function before completely solving for λ).

Answer (b): Since we have already done most of the work for this, we can just read off the fixed point as the following (i.e. as $\beta(\lambda_*) = 0 + \mathcal{O}(\epsilon^0)$ what makes this true)

$$\lambda_* = +\frac{3g^3 i\pi}{2(4\pi)^3} - \mathcal{O}(\epsilon^0). \tag{2.909}$$

Problem 2 answer:

Answer (a): First, as it has been evident in all my homeworks, I would like to start off by giving a little background to the theory, to understand what we are dealing with physically. This model was first formulated by David Gross and André Neveu in 1974 in order to give a toy model of the interactions of QCD [**AC:**] gross, as well as a theory with (hidden) spontaneous symmetry breaking (as we shall see later in part (b)). Otherwise, our task for part (a) is to compute the beta function.

First, notice the following facts about the $d = 2$ gamma matrices. In $d = 2$, $\gamma^0 = \sigma^2$ and $\gamma^1 = i\sigma^1$ where σ^i are the Pauli-matrices. Then, $\gamma^5 \equiv \gamma = \sigma^3$ and $\{\gamma^5, \gamma^\mu\} = 0$ since $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$.

Now, lets check a few claims, as to whether the action in equation 30 really is invariant under the chirality matrix. Notice that $\bar{\psi}_i \partial\!\!\!/ \psi_i \to (\gamma^5)^2 \bar{\psi}_i \partial\!\!\!/ \psi_i = \bar{\psi}_i \partial\!\!\!/ \psi_i$. And in a similar manner, $(\bar{\psi}_i \psi_i)^2 \to (\gamma^5)^2 (\bar{\psi}_i \psi_i)^2 = (\bar{\psi}_i \psi_i)^2$. Then, the Gross-Neveu action (GN for short) is invariant under γ^5.

Notice that this theory is renormalizable since the coupling constant is dimensionless ($[g] = 0$).

Now, in order to find $g(\mu)$ we need to compute the 4-vertex 1-loop correction. Notice that since we are working with an N-number species of fermions, and that they are spinors, we have two sets of indices to worry about, i, j or Latin indices to be used to denote the channels of the incoming or outgoing fermions and their count (number of species), while α, β or Greek indices to represent the $d = 2$ spinor components.

Now, to compute the 4-vertex, we need the Feynman diagram rules. The vertex can be written as (we will be working in the Lorentzian signature so we pick up a factor of i in the action of equation 30),

$$ig_0^2 \left(\delta_{ij} \delta_{k\ell} \epsilon_{\alpha\beta} \epsilon_{\gamma\delta} + \delta_{i\ell} \delta_{jk} \epsilon_{\delta\alpha} \epsilon_{\beta\gamma} \right), \tag{2.910}$$

while the propagator with momentum k^μ can be written as

$$\left(\frac{i}{k\!\!\!/} \right)_{\beta\alpha} \cdot \delta_{ij}. \tag{2.911}$$

(As a note, notice that the two-point function plus its counter term yields 0 since there is a term of $\int d^2k \, \mathrm{tr}(1/k\!\!\!/) = 0$).

Now, to compute the 4-vertex, notice that there are 3-channels we need to worry about, the t, u, and s-channel. If we compute one channel, then we can figure out the other two.

Then, consider the t-channel diagram,[104]

$$= i\Gamma^{(4)}(q, p_2, p_3) \tag{2.912}$$

$$= (ig_0^2)^2 \int \frac{d^dq}{(2\pi)^d} \operatorname{tr}\left(\frac{i}{\not{q} + \not{p}_3} \cdot \frac{i}{\not{q} + \not{p}_1}\right) \tag{2.913}$$

$$= g_0^4 \int_0^1 dx \int \frac{d^dq}{(2\pi)^d} \frac{(\not{q} + x\not{p}_3)(\not{q} + (1-x)\not{p}_1)}{(q^2 + x(1-x)t)^2} \tag{2.914}$$

$$= g_0^4 \int_0^1 dx \int \frac{d^dq}{(2\pi)^d} \frac{q^2 + x(1-x)\not{p}_3\not{p}_1}{(q^2 + x(1-x)t)^2} \tag{2.915}$$

$$= g_0^4 \int_0^1 dx \int \frac{d^dq}{(2\pi)^d} \left(\frac{1}{q^2 + x(1-x)t} - \frac{x(1-x)t}{(q^2 + x(1-x)t)^2}\right), \tag{2.916}$$

where $t = (p_1 - p_3)^2 = -2p_1 \cdot p_3$. Two notes: first is that we can not use on-shell renormalization or zero-momentum subtraction since $s = 4m^2 = t = u = 0$. Second, this calculation is closely related to equation 10.32 of Peskin and Schroeder when computing the loop integral for a 2-point function. With this in mine, in the last equation we used the massless Dirac equations namely $\not{p}_i u(p_i) = 0$ and $\bar{u}(p_j)\not{p}_j = 0$ for fermions and anti-fermions respectively. Also notice that the second term of equation 37 is finite, while the first term is divergent. So, we only will be taking a look at the first term.

There is also the s and u-channel diagrams as stated previously that are

$$i\Gamma_s = (ig_0^2)^2 N \int \frac{d^dq}{(2\pi)^d} \operatorname{tr}\left(\frac{i}{-q + \not{p}_1} \cdot \frac{i}{\not{q} + \not{p}_2}\right) \tag{2.917}$$

$$i\Gamma_u = (ig_0^2)^2 N \int \frac{d^dq}{(2\pi)^d} \operatorname{tr}\left(\frac{i}{\not{q} + \not{p}_1} \cdot \frac{i}{\not{q} + \not{p}_4}\right). \tag{2.918}$$

Now, under dimensional regularization let $d = 2(1-\epsilon)$ with $g_0^2 = g^2\mu^{2\epsilon}$. Then, under $\overline{\text{MS}}$ ([**AC:**] schwartz chapter 18) equation 37 (only the divergent part) becomes

$$i\Gamma^{(4)}(q, p_1, p_3) = -2(N-1)g^4\mu^{4\epsilon} \int_0^1 dx \int \frac{d^dq}{(2\pi)^d} \left(\frac{1}{q^2 + x(1-x)t}\right) \tag{2.919}$$

$$= 2(N-1)g^4\mu^{4\epsilon} \left(\frac{i}{(4\pi)^{1-\epsilon}}\right) \int_0^1 dx \Gamma(\epsilon)(x(1-x)(-t))^{-\epsilon} \tag{2.920}$$

$$= 2(N-1)g^4\mu^{4\epsilon} \left(\frac{i}{(4\pi)^{1-\epsilon}}\right) \Gamma(\epsilon) \left(\frac{\Gamma(1-\epsilon)\Gamma(1-\epsilon)}{\Gamma(2-2-\epsilon)}\right) (-t)^{-\epsilon}, \tag{2.921}$$

where we used the Euler-integral of first kind (which is also called the beta-function),

$$B(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1}dt = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(2-2\epsilon)}. \tag{2.922}$$

[104] I am still trying to figure out how to use Tikz and other Feynman LaTeXpackages.

If we expand the gamma function via epsilon expansion using equation 15 we have,

$$i\Gamma^{(4)} = i(N-1)g^4\mu^{4\epsilon}\left(\frac{1}{4\pi}\right)(-t)^{-\epsilon}\left(\frac{1}{\epsilon} - \gamma + \ln(4\pi)\right) \tag{2.923}$$

We can then let the counter term be

$$i\Gamma^{(4)}_{\text{counter term}}l = i(N-1)g^4\mu^{4\epsilon}\left(\frac{1}{2\pi}\right)\left(\frac{1}{\epsilon} - \gamma + \ln(4\pi)\right), \tag{2.924}$$

in order to cancel the epsilon expansion. Then, with the tree-level vertex, 1-loop correction, and the counter term, as $\epsilon \to 0$ we have the following 4-vertex,

$$i\Gamma^{(4)}(t) = i\Gamma^{(4)}_0 + i\Gamma^{(4)}_{\text{1-loop}} + i\Gamma^{(4)}_{\text{counter term}} \tag{2.925}$$

$$= ig^2\left(1 - \frac{(N-1)g^2}{2\pi}\ln\left(\frac{-t}{\mu^2}\right)\right) \implies \tag{2.926}$$

$$\Gamma^{(4)}(t) = g^2\left(1 - \frac{(N-1)g^2}{2\pi}\ln\left(\frac{-t}{\mu^2}\right)\right) + \mathcal{O}(g^6). \tag{2.927}$$

Then, via the Callan-Symanzik equation (knowing that at 1-loop order $\gamma = 0$),

$$\mu\frac{\partial\Gamma^{(4)}}{\partial\mu} = \frac{(N-1)g^4}{\pi} + \mathcal{O}(g^6) = -\beta(g)(2g + \mathcal{O}(g^3)) \implies \beta(g) = \frac{-(N-1)g^3}{2\pi}. \tag{2.928}$$

A few features we can comment on is that notice if only one species of fermions is present then $\beta = 0$ which implies that the coupling constant is independent of of the scale, i.e we have a scale invariant theory. Otherwise when $\beta(g) < 0$ then the coupling decreases as the energy scale increases.

Answer (b): Consider the Lorentzian partition function,

$$Z = \int[D\bar{\psi}][D\psi][D\sigma]\exp\left(i\int d^2x\left(\bar{\psi}_i i\partial\!\!\!/\psi_i - \frac{1}{2g^2}\sigma^2 - \sigma\bar{\psi}_i\psi_i\right)\right). \tag{2.929}$$

Notice that

$$\int[D\sigma]\exp\left(-i\int d^2x\left(\frac{1}{2g^2}\sigma^2 - \sigma\bar{\psi}_i\psi_i\right)\right) = \mathcal{N}\exp\left(i\int d^2x\frac{1}{2}g_0^2\left(\bar{\psi}_i\psi_i\right)^2\right), \tag{2.930}$$

where we are integrating over a Gaussian with respect to σ; the identity can be found in [**AC:**] cayley. Then we have the following,

$$Z = \int[D\bar{\psi}][D\psi]\exp\left(i\int d^2x\left(i\bar{\psi}_i\partial\!\!\!/\psi_i + \frac{1}{2}g_0^2(\bar{\psi}_i\psi_i)^2\right)\right), \tag{2.931}$$

showing that this is indeed the correct effective action. Then to find the effective potential for σ where we will be evaluating the background field as a constant value as σ_0. To do so, consider integrating over the fermions (Grassmann numbers from the view of the integral)

using the identity in [AC:] cayley,

$$\int [D\bar{\psi}][D\psi] \exp\left(i \int d^2x \left(i\bar{\psi}_i \partial\!\!\!/ \psi_i - \sigma \bar{\psi}_i \psi_i\right)\right) = \left(\det(i\partial\!\!\!/ - \sigma)\right)^N \tag{2.932}$$

$$= \left(\det\left(\partial^2 + \sigma^2\right)\right)^N \tag{2.933}$$

$$= \exp\left(\mathcal{N} \int \frac{d^2k}{(2\pi)^2} \log\left(-k^2 + \sigma^2\right)\right) \tag{2.934}$$

$$= \mathcal{N} \int \frac{d^dk_E}{(2\pi)^d} \log\left(k_E^2 + \sigma^2\right) \tag{2.935}$$

$$= \mathcal{N} \int \frac{d^dk_E}{(2\pi)^d} \left(\frac{\partial}{\partial\alpha} \frac{1}{k_E^2 + \sigma^2}\right)\Big|_{\alpha=0} \tag{2.936}$$

$$= -i\mathcal{N} \frac{\Gamma(-d/2)(\sigma^2)^{d/2}}{(4\pi)^{d/2}}. \tag{2.937}$$

In equation 56, this was a Gaussian integral where we can re-write the first term in the exponential as $\bar{\psi}_i(i\partial\!\!\!/ - \sigma)\psi_i$. Then in equation 58, in order to diagonalize the operator by Fourier transforming into momentum space, performed a Wick rotation to equation 59, and then re-wrote the logarithm in a manner that resembles a propagator, which we know how to integrate over, leading us to equation 61. However, we really need to regularize this integral in equation 59. To do so, we can use dimensional regularization by letting $d = 2 - \epsilon$ and then send $\epsilon \to 0$ as,

$$\mathcal{N} \int \frac{d^dk_E}{(2\pi)^d} \log\left(k_E^2 + \sigma^2\right) = \frac{i\mathcal{N}\sigma^2}{4\pi} \left(\frac{2}{\epsilon} - \gamma + \log(4\pi) - \log\left(\sigma^2\right) + 1\right), \tag{2.938}$$

where we used the epsilon expansion in the gamma function. Then we have the following effective potential via $\overline{\text{MS}}$ scheme,

$$V_{\text{eff}}(\sigma) = \frac{1}{2g_0^2}\sigma^2 + \frac{\mathcal{N}}{4\pi}\sigma_2 \left(\log\left(\frac{\sigma^2}{\mu^2} - 1\right)\right). \tag{2.939}$$

underline**Answer (c):** To minimalize the potential we simply compute the derivative of the effective potential with respect to σ_0 to give,

$$0 = \frac{\partial V_{\text{eff}}}{\partial \sigma_0} = \frac{1}{g^2}\sigma_0 + \frac{\mathcal{N}}{2\pi} \log\left(\frac{\sigma_0^2}{\mu^2}\right), \tag{2.940}$$

which implies the following expectation value for the ground state (solving the above equation),

$$\sigma_0 \implies \{\sigma_0\} = \pm\mu \exp\left(-\pi/g^2\mathcal{N}\right). \tag{2.941}$$

This is entirely dependent on the subtraction point (renormalization scales or as dictated in (a)).

Notice that since $V_{\text{eff}}(\sigma_0)$ is of power σ^2, so V_{eff} is invariant under $\mathbb{Z}_2$ symmetry, $\sigma \to -\sigma$, while the vacuum depends upon $\mathcal{N}$, and thus correlated to a $1/\mathcal{N}$ loop expansion. (In other words, the higher order of loops corresponds the higher order of $1/N$ expansion. So, if the loop expansion is even in $(\bar{\psi}_i\psi_i)^{\mathcal{N}}$ then $V_{\text{eff}}(\sigma_0)$ is invariant under chiral symmetry).

2.7 Problem set 4:ϕ^4-theory in $D = 6$

Consider the massless ϕ^4 theory in $D = 4 - \epsilon$ dimensions, defined by the bare Euclidean Lagrangian density,

$$\mathcal{L}_E = \frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2} m_0^2 \phi^2 + \frac{1}{4!} g_0 \phi^4, \tag{2.942}$$

where the bare mass m_0 is chosen such that the mass gap vanishes. As discussed in class, an appropriate definition of the dimensionless renormalized coupling $\lambda(\mu)$ is

$$\lambda(\mu) = -\mu^{-\epsilon} (Z(\mu))^2 \Gamma^{(4)}(k_1, k_2, k_3)_{k_i^2 = \mu^2, s=t=u=-\frac{4}{3}\mu^2}, \tag{2.943}$$

where $\Gamma^{(4)}(k_1, k_2, k_3)$ is computed by the 1PI diagram with four external momenta $k_1, k_2, k_3, -k_1 - k_2 - k_3$, and $Z(\mu)$ is related to the self-energy $\Sigma(k)$ by

$$Z(\mu) = \left(1 - \frac{\partial \Sigma}{\partial k^2}\right)^{-1} \Bigg|_{k^2 = \mu^2}. \tag{2.944}$$

The renormalied operator $[\phi]_\mu$ at mass scale μ is defined as

$$[\phi]_\mu = (Z(\mu))^{-1/2} \phi. \tag{2.945}$$

In the RG-improved perturbation theory, we compute Green functions at momenta scale μ, or equivalently distance scales μ^{-1}, through the perturbative expansion of Green functions of $[\phi]_\mu$ with respect to $\lambda(\mu)$, truncated to a certain order in $\lambda(\mu)$, and then substitute $\lambda(\mu)$ with its value determined through the RG equation truncated to the relevant loop order.

Recall that in class, we have studied in the RG equation for $\lambda(\mu)$ at 1-loop order, and found the RG fixed point for λ up to order ϵ in the ϵ-expansion. We have also seen that at 1-loop order, $\Sigma(k)$ is independent of k and thus does not contribute to $Z(\mu)$.

2.7.1 Problem 1:

(a) Calculate $\Sigma(k)$ (in the absence of mass gap) at 2-loop order in the naive perturbation theory, and then determine $Z(\mu)$ in the RG-improved perturbation theory by expressing

$$\gamma_\phi(\mu) \equiv \frac{1}{2} \frac{d \log(Z(\mu))}{d \log(\mu)} \tag{2.946}$$

as a function of $\lambda(\mu)$, keeping up to order λ^2 terms, and using the solution for $\lambda(\mu)$ from the 1-loop RGE of the latter.

(b) Calculate the momentum space 2-point function of the renormalized operator $[\tilde{\phi}(p)]_\mu$

$$\langle [\tilde{\phi}(p)]_\mu [\tilde{\phi}(q)]_\mu \rangle \tag{2.947}$$

in the RG-improved perturbation theory, with the choice $\mu = |p|$ and the expansion of the 2-point correlation function in $\lambda(\mu)$ truncated to order λ^2.

(c) How does the 2-point function $\langle \phi(x)\phi(0) \rangle$ behave at large $|x|$? Determine the scaling dimension of the operator $\phi(x)$ at the infrared RG fixed point to order ϵ^2.

Problem 1 answer:

Answer (a): (We will be working in the Euclidean signature, so any time there is a -1, replace it with an i for the Lorentzian signature). It looks like that our goal is to compute the self-energy up to two-loop order via perturbation theory, and then determine what $Z(\mu)$ is from that. Then, given $Z(\mu)$ we are suppose to derive the dimensionaless parameter γ_ϕ corresponding to the coefficient in the Callan-Symanzik equation for the field-renormalization as a function of the renormalized coupling $\lambda(\mu)$ given in equation 2, where we are to compute the 4-point function but only up to 1-loop order (thankfully). We will start with the loop correction calculations for the self=energy.

The Feynman diagrams for the self-energy and 4-point function up to two-loop order (for both, although we won't use the 4-point 2-loop diagrams) are written in figure one.

Figure 42. 1PI diagrams for the self-energy and 4-point function in ϕ^4 theory.

The first diagram is just the typical propagator in ϕ^4 theory given as $(p^2 - m^2 + i\epsilon)^{-1}$ (I have included the $i\epsilon$ term for analytic continuity, but for the rest of the calculations I will drop it). The second diagram is the first loop while the last two diagrams are the only two-loop contributions we have (the last one is called the *sunset* diagram). (Note: we are neglecting the symmetry factors here in our loop calculations. For example, there should be a factor of $\frac{1}{2}$ in-front of the 1-loop, a factor of $\frac{1}{4}$ in front of the first 2-loop, and a factor of $\frac{1}{6}$ in front of the sunset diagram). For the 1-loop we have the following (which we cite, not derive)[**AC:**] ryder,

$$\text{“1-loop”} = -g_0 \int \frac{d^D p}{(2\pi)^D} \frac{1}{p^2 + m^2} = \frac{g_0 (m^2)^{(D-2)/2}}{(4\pi)^{D/2}} \Gamma\left(1 - \frac{D}{2}\right). \tag{2.948}$$

But, we know the gamma function is divergent for $D = 4$ (and as I have been reading, mainly for even dimension systems). So, we can define a dimensionless coupling $g = g_0 m^{D-4} \implies g = g_0 \mu^{D-4} = g_0 \mu^{-\epsilon}$ with under dimensional regularization $D = 4 - \epsilon$. Then the 1-loop becomes,

$$\text{“1-loop”} = -\frac{m^2 g}{(4\pi)^2} \left(\frac{4\pi\mu^2}{m^2}\right)^{\epsilon/2} \Gamma\left(\frac{\epsilon}{2} - 1\right). \tag{2.949}$$

We can expand in terms of epsilon to contain the IR-divergence as,

$$\left(\frac{4\pi\mu^2}{m^2}\right)^{\epsilon/2} = 1 + \frac{\epsilon}{2} \log\left(\frac{4\pi\mu^2}{m^2}\right) + \mathcal{O}(\epsilon). \tag{2.950}$$

To perform the ϵ-expansion on the gamma function, for general n it can be given as [**AC:**] pands

$$\Gamma(\epsilon - n) = \frac{(-1)^n}{n!} \left(\frac{1}{\epsilon} + \psi(n+1) + \frac{\epsilon}{2}\left[\frac{\pi^2}{3} + \psi^2(n+1) - \psi'(n+1)\right] + \mathcal{O}(\epsilon^3)\right), \tag{2.951}$$

where $\psi(z)$ is the Euler digamma function given by $\psi(z) \equiv \Gamma'(z)/\Gamma(z)$. If we plug this into equation 7 we have (keeping up to order ϵ)

$$\text{“1-loop”} = -\frac{m^2 g}{(4\pi)^2}\frac{(-1)^n}{n!}\left(1+\frac{\epsilon}{2}\log\left(\frac{4\pi\mu^2}{m^2}\right)+\mathcal{O}(\epsilon)\right)\left(\frac{2}{\epsilon}+\psi(n+1)+\mathcal{O}(\epsilon)\right) \tag{2.952}$$

$$= \frac{m^2 g}{(4\pi)^2}\left(\frac{2}{\epsilon}+\log\left(\frac{4\pi\mu^2}{m^2}\right)+\psi(2)+\mathcal{O}(\epsilon)\right), \tag{2.953}$$

where we let $n = 1$ and sent $\epsilon \to \epsilon/2$ in equation 9. Notice that the reside in the 1-loop equation is proportional to m^2, but independent of μ.

Now consider the 2-loop diagram,

$$\text{“2-loop”} = g^2 \int \frac{d^D p_1}{(2\pi)^D}\frac{d^D p_2}{(2\pi)^D}\frac{1}{p^2+m^2}\frac{1}{(p_2+m^2)^2}. \tag{2.954}$$

Notice that the p_1 integral is just the 1-loop integral, so we know how to handle that. But, in order to compute the p_2 integral we need to be more careful. In general, it can be computed using Schwinger's proper time reparameterization method [**AC:**] schwartz. In general consider,

$$\frac{1}{(p^2+2p\cdot q+m^2)^a} = \frac{1}{\Gamma(a)}\int_0^\infty d\tau\, \tau^{a-1}e^{-\tau(p^2+2p\cdot q+m^2)} \implies \tag{2.955}$$

$$\frac{1}{\Gamma(a)}\int_0^\infty d\tau\, \tau^{a-1}\int \frac{d^D p}{(2\pi)^D}e^{-\tau(p^2+2p\cdot q+m^2)} = \frac{1}{\Gamma(a)}\int_0^\infty d\tau\, \tau^{a-1}e^{-\tau(m^2-q^2)}\int\frac{d^D p}{(2\pi)^D}e^{-\tau p^2}. \tag{2.956}$$

The momentum integral in equation 14 is equal to $(1/4\pi\tau)^{D/2}$. The τ integral gives,

$$\int_0^\infty \tau^{a-D/2-1}e^{-\tau(m^2+q^2)} = \frac{\Gamma(a-D/2)}{(m^2-q^2)^{a-D/2}}. \tag{2.957}$$

Then our general integral becomes

$$\int \frac{d^D p}{(2\pi)^D}(p^2+2p\cdot q+m^2)^{-a} = \frac{\Gamma(a-D/2)}{(4\pi)^{D/2}\Gamma(a)}\frac{1}{(m^2-q^2)^{a-D/2}}. \tag{2.958}$$

Then, our p_2 integral becomes,

$$g_0\int\frac{d^D p_2}{(2\pi)^D}\frac{1}{(p_2^2+m^2)^2} = \frac{g\mu^\epsilon}{(4\pi)^{D/2}}\frac{\Gamma(2-D/2)}{\Gamma(2)}\frac{1}{(m^2)^{2-D/2}} \tag{2.959}$$

$$= \frac{g}{(4\pi)^2}\left(\frac{4\pi\mu^2}{m^2}\right)^{\epsilon/2}\Gamma\left(\frac{\epsilon}{2}\right) \tag{2.960}$$

$$= \frac{g}{(4\pi)^2}\left(\frac{2}{\epsilon}+\psi(1)+\log\left(\frac{4\pi\mu^2}{m^2}\right)+\mathcal{O}(\epsilon)\right), \tag{2.961}$$

then we have the following "2-loop" correction as the product of the loops,

$$\text{“2-loop} = -\frac{m^2 g^2}{(4\pi)^4}\left(\frac{4}{\epsilon^2} + \frac{2}{\epsilon}(\psi(1)+\psi(2)) + \frac{4}{\epsilon}\log\left(\frac{4\pi\mu^2}{m^2}\right) + \mathcal{O}(\epsilon)\right). \tag{2.962}$$

Finally, we have to compute the "sunset" diagram. This can be written as

$$\text{“sunset”} = \lambda^2 \int \frac{d^D p_1}{(2\pi)^D}\frac{d^D p_2}{(2\pi)^D}\frac{1}{p_1^2+m^2}\frac{1}{p_2^2+m^2}\frac{1}{(q+p_1+p_2)^2+m^2}, \tag{2.963}$$

where q is the incoming momentum. Now, this is a typical integral encountered that we can cite from either [AC:] pands or [AC:] ryder since there is an appendix for common integrals. However, we can make this integral a little easier by considering the following line of reasoning. Since $\partial^\mu/\partial^\nu = \delta^\mu_\nu$ and that we can write the following integral,

$$\int \frac{d^D p}{(2\pi)^D} p_\mu p_\nu f(p^2) = \frac{1}{D}\int \frac{d^D p}{(2\pi)^D}\delta_{\mu\nu} p^2 f(p^2), \tag{2.964}$$

then we can consider inserting the following "1″:

$$1 = \frac{1}{2D}\left(\frac{\partial p_1^\mu}{\partial p_1^\mu} + \frac{\partial p_2^\mu}{\partial p_2^\mu}\right). \tag{2.965}$$

We can plug this into the sunset diagram that allows us to separate the integral as

$$\text{“sunset”} = \frac{-g_0^2}{D-3}\int \frac{d^D p_1}{(2\pi)^D}\frac{d^D p_2}{(2\pi)^D}\frac{3m^2 + q\cdot(q+p_1+p_2)}{(p_1^2+m^2)(p_2^2+m^2)((q+p_1+p_2)^2+m^2)^2} \tag{2.966}$$

$$= -\frac{g_0^2}{D-3}\left(3m^2 A(q) + B(q)\right), \tag{2.967}$$

where we have

$$A(q) = \int \frac{d^D p_1}{(2\pi)^D}\frac{d^D p_2}{(2\pi)^D}\frac{1}{(p_1^2+m^2)(p_2^2+m^2)((q+p_1+p_2)^2+m^2)^2} \tag{2.968}$$

$$B(q) = \int \frac{d^D p_1}{(2\pi)^D}\frac{d^D p_2}{(2\pi)^D}\frac{q^\mu(q+p_1+p_2)_\mu}{(p_1^2+m^2)(p_2^2+m^2)((q+p_1+p_2)^2+m^2)^2}. \tag{2.969}$$

The $A(q)$ integral can be done by either looking it up [AC:] ryder, or can be computed directly. The steps would be writing the integral in via the Feynman technique, integrate out the momentum, expand in ϵ via dimensional regularization with $D = 4-\epsilon$, use Euler's integral formula given as

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} = \int_0^1 dy\, y^{\alpha-1}(1-y)^{\beta-1}, \tag{2.970}$$

perform the ϵ-expansion to finally get,

$$A(q) = \frac{1}{(4\pi)^4}\left(\frac{2}{\epsilon^2} + \frac{1}{\epsilon}\left[1 + 2\psi(1) + 2\log\left(\frac{4\pi}{m^2}\right)\right] + \mathcal{O}(\epsilon)\right). \tag{2.971}$$

For the $B(q)$, the integral can be re-written since the denominator has the same factor in the numerator so,

$$B(q) = -\frac{q^\mu}{2}\frac{\partial}{\partial q^\mu}A(q). \tag{2.972}$$

Then, we follow the same steps for computing $A(q)$ but evaluate the partial derivative when appropriate. Then we get,

$$B(q) = \frac{q^2}{(4\pi)^4}\frac{1}{2\epsilon} + \mathcal{O}(\epsilon). \tag{2.973}$$

Then all together we get the full expression for the sunset diagram

$$\text{“sunset”} = -\frac{g^2m^2}{(4\pi)^4}\left(\frac{6}{\epsilon^2} + \frac{6}{\epsilon}\left[\frac{3}{2} + \psi(1) + \log\left(\frac{4\pi\mu^2}{m^2}\right)\right] + \frac{q^2}{2m^2\epsilon} + \mathcal{O}(\epsilon)\right). \tag{2.974}$$

Finally, the total 2-loop self-energy computation becomes,

$$\text{“1PI”} = (p^2+m^2)^{-1} + \text{“1-loop”} + \text{“2-loop”} + \text{“sunset”} \tag{2.975}$$

$$= \frac{1}{p^2+m^2-\Sigma(p^2)} \implies \tag{2.976}$$

$$\Sigma(p^2) = \frac{m^2g}{(4\pi)^2}\left(\log\left(\frac{4\pi\mu^2}{m^2}\right) + \psi(2)\right) \tag{2.977}$$

$$- \frac{m^2g^2}{(4\pi)^2}\left(\frac{2(\psi(1)+\psi(2))}{(4\pi)^2} + \frac{4}{(4\pi)^2}\log\left(\frac{m^2}{4\pi\mu^2}\right) - 2 + \frac{6}{(4\pi)^2}\left(\frac{3}{2} + \psi(1) + \log\left(\frac{4\pi\mu^2}{m^2}\right)\right)\right)\frac{1}{\epsilon} \tag{2.978}$$

$$- \frac{5m^2g^2}{128\pi^4}\frac{1}{\epsilon^2} + \mathcal{O}(\epsilon^3). \tag{2.979}$$

Before we move onto finding the renormalization factor $Z(\mu)$ (which is the field strength renormalization, better denoted Z_ϕ) we also need, as the problem asks for, the 4-point function to 1-loop order. This is just the first two diagrams of the 4-point function in

figure 1. The 1-loop integral we have computed many times. It is therefore the following,

$$\text{“1-loop”} = -g_0^2 \int \frac{d^D p}{(2\pi)^D} \frac{1}{(p^2+m^2)} \frac{1}{(p+k)^2+m^2} \tag{2.980}$$

$$= g_0^2 \int_0^1 dx \int \frac{d^D p}{(2\pi)^D} \frac{1}{((p^2+m^2)x + ((p_k)^2+m^2)(1-x))^2} \tag{2.981}$$

$$= g_0^2 \int_0^1 dx \int \frac{d^D p}{(2\pi)^D} \frac{1}{(p^2+2p\cdot kx + k^2x + m^2)^2} \tag{2.982}$$

$$= \frac{g_0^2}{(4\pi)^{D/2}} \frac{\Gamma(2-D/2)}{\Gamma(2)} \int_0^1 dx \frac{1}{(k^2x(1-x)+m^2)^{2-D/2}} \tag{2.983}$$

$$= \frac{g^2\mu^\epsilon}{(4\pi)^2} \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \left(\frac{4\pi\mu^2}{k^2x(1-x)+m^2}\right)^{\epsilon/2} \tag{2.984}$$

$$= \frac{g^2\mu^\epsilon}{(4\pi)^2} \left(\frac{2}{\epsilon} + \psi(1)_{\mathcal{O}}(\epsilon)\right) \left(1 + \frac{\epsilon}{2} \int_0^1 dx \log\left(\frac{4\pi\mu^2}{k^2x(1-x)+m^2}\right) + \mathcal{O}(\epsilon)\right) \tag{2.985}$$

$$= \frac{g^2\mu^\epsilon}{(4\pi)^2} \left(\frac{2}{\epsilon} + \psi(1) + \int_0^1 dx \log\left(\frac{4\pi\mu^2}{k^2x(1-x)+m^2}\right) + \mathcal{O}(\epsilon)\right). \tag{2.986}$$

Now, we are in a position to find $Z(\mu)$. Under the minimal substraction scheme, denoted MS [**AC:**] schwartz, we can write the field strength renormalization as

$$Z_\phi(g,\epsilon) = 1 + c_\phi = 1 - \frac{1}{12}\frac{g^2}{(4\pi)^2}\frac{1}{\epsilon}, \tag{2.987}$$

while the self-energy up to two-loop order gives the mass-squared renormalization as

$$Z_g(g,\epsilon) = 1 + c_g = 1 + \frac{3g}{(4\pi)^2}\frac{1}{\epsilon} + \frac{g^2}{(4\pi)^4}\left(\frac{9}{\epsilon^2} - \frac{3}{\epsilon}\right), \tag{2.988}$$

where c_g and c_ϕ are the counter terms with respect to the self-energy and 4-point functions. (These counter terms can be read off by finding the divergent terms in equation 37 and 44 up to ϵ^2 order). (Note: in a lot of literature the renormalization constants $Z(\mu)$ are computed for N-field components, so set $N = 1$ to get our result).

We can then compute $\gamma_\phi(g)$ (which can obtained via chain rule in equation 5) becomes,

$$\gamma_\phi(g) = \frac{1}{12}\frac{g^2}{(4\pi)^2}. \tag{2.989}$$

This is related to the Callan-Symanzik equation as

$$\left(\mu\frac{\partial}{\partial\mu} + \beta(g)\frac{\partial}{\partial g} + n\gamma_\phi(g) + \gamma_{\mathcal{O}}(g)\right) G^{(n,\mathcal{O})}(p,k) = 0, \tag{2.990}$$

where the last term in the Callan-Symanzik equation relates to our renormalized operator we will investigate in part (b). We can also easily find $\gamma_{\mathcal{O}}$ with our counter terms above,

$$\gamma_{\mathcal{O}} = Z_{\mathcal{O}}\mu\frac{\partial}{\partial\mu}\left(\frac{n}{2}c_\phi - c_{\mathcal{O}}\right). \tag{2.991}$$

Answer (b): The renormalized operator is defined in equation 4. Notice that for an n-point Green's function, we can write for the general renormalized Green's function as [**AC:**] pands

$$G^{(n,\mathcal{O})}(p;k) = \langle\tilde{\phi}(p_1)\cdots\phi(\tilde{p}_i)\tilde{\mathcal{O}}(k)\rangle = Z_\phi^{-n/2}Z_{\mathcal{O}}^{-1}\langle\tilde{\phi}_0(p_1)\cdots\tilde{\phi}_0(p_i)\tilde{\mathcal{O}}_0(k)\rangle. \tag{2.992}$$

However, for our situation, it is much more simplified since we only have two operators to worry about. So equation 50 becomes,

$$\langle[\tilde{\phi}(p)]_\mu[\tilde{\phi}(q)]_\mu\rangle = Z(\mu)\langle\phi(p)\phi(q)\rangle. \tag{2.993}$$

(Note: there should be a factor of $(2\pi)^4\delta(\Sigma_i p_i)$ in order for it to be compatible with S-matrix calculations, but we are not worried about that here). Notice that the two point function is exactly that of the self-energy computation we found in equation 34 and 35. So, our job is done. Where it is worth trying to simplify is when we Fourier transform back to positions space and make a few assumptions we shall in part (c).

Answer (c): Notice that if we Fourier transform equation 52, and place one of our momentum states at $y = 0$ to get the desired two-point function we have (where $G(p)$ is the 2-loop self-energy Green's function)

$$\langle\phi(x)\phi(0)\rangle = \int\frac{d^4p}{(2\pi)^4}e^{ip\cdot x}Z(\mu)G(p). \tag{2.994}$$

If we are going to large *correlation distances* i.e. large $|x|$ then we will be in the IR-range. In this range, we can disregard the m^2. This allows us to re-write our Green's function as

$$G(p) \approx \frac{1}{p^2 - \Sigma(k^2)} \tag{2.995}$$

$$= \frac{1}{p^2 - g\Pi_1(p^2) - g^2\Pi_2(p^2)}, \tag{2.996}$$

where Π_1 and Π_2 denote our 1-loop and 2-loop self-energy calculation respectively. We can take the very strong assumption that the Green's function behaves as a power law $G(p) \approx p^{-(2-\epsilon+\gamma_\phi)}$ where γ_ϕ is now our anomalous dimension. This allows us to solve the integral above given that

$$\int d\Omega e^{ip\cdot x} = 4\pi\frac{\sin(p|x|)}{p|x|}. \tag{2.997}$$

We are then left with only the radial integral to compute, where if we let $u = p|x|$ to perform u-substitution where our bounds for the integral are $0 < \epsilon - \gamma < 2$, we have the following

$$\langle\phi(x)\phi(0)\rangle \sim |x|^{-(4-\epsilon+\gamma)}. \tag{2.998}$$

For our particular case, the scaling dimension is then computed as,

$$\Delta_\phi = 2 - \frac{\epsilon}{2} + \frac{\gamma}{2} = 2 - \frac{\epsilon}{2} + \frac{1}{24}\frac{g^2}{(4\pi)^2}. \tag{2.999}$$

Our answer is *almost* in agreement with what was given at the end of lecture on March 10th but we are multiplied by an extra factor of 2 initially. This seems to be the case so I should change my definition, since in $D = 3$ and we let $\epsilon = g = 1$, using our equation 58 (with a 1 not a 2) we have $\Delta_\phi = 0.5003$.

2.8 Problem set 5:

2.8.1 Problem 1: $D = 0$ Yang-Mills Theory

Consider a 0-dimensional analog of Yang-Mills theory where the spacetime is replaced by a point and the gauge field variable[105] is replaced by an $N \times N$ unitary matrix U, with the Euclidean action

$$S(U) = -\lambda \operatorname{tr}\Big(U + U^{-1}\Big). \tag{2.1000}$$

The $U(N)$ "gauge" transformation on U takes the form

$$U \mapsto gUg^{-1}, \tag{2.1001}$$

where g is an arbitrary $N \times N$ unitary matrix. The analog of the path integral for the partition function is

$$Z = \frac{1}{\operatorname{vol}(U(N))} \int\limits_{U(N)} dU\, e^{-S(U)}. \tag{2.1002}$$

where dU is the Haar measure on $U(N)$, with the defining property $dU = d(gU) = d(Ug)$ for any unitary g and $\operatorname{vol}(U(N)) \equiv \int_{U(N)} dU$. Note that the overall normalization convention of the Haar measure drops out of equation 2, and that Z is unambiguously defined as a function of λ.

(a) Any unitary matrix U can be brought to a diagonal one by a gauge transformation of the form in equation 2, and thus we can partially fix the gauge redundancy by imposing the condition

$$U_{ab} = \delta_{ab} e^{i\theta_a}, \;\; a, b = 1, \cdots, N, \tag{2.1003}$$

where $\theta_1, \cdots, \Theta_N$ are unconstrained real variables. Following the Faddeev-Popov procedure, find the Faddeev-Popov determinant Δ associated with the gauge condition in equation 4 as a function of $\boldsymbol{\theta} = (\theta_1, \cdots, \theta)$, such that equation 3 is equal to

$$Z_N = C_N \prod_{a=1}^{N} \int\limits_0^{2\pi} d\theta_a\, \Delta(\boldsymbol{\theta}) e^{2\lambda \sum_{a=1}^{N} \cos(\theta_a)}, \tag{2.1004}$$

for some λ-independent constant C_N.

(b) In the case $N = 2$, we can parameterize the $U(2)$ matrix U as

$$U \equiv e^{i\alpha} \frac{(1 + i\boldsymbol{\sigma} \cdot \mathbf{x})^2}{1 + \mathbf{x}^2}, \tag{2.1005}$$

and write the Haar measure as

$$dU = f(\alpha, \mathbf{x}) d\alpha d^3\mathbf{x}. \tag{2.1006}$$

Determine the function $f(\alpha, \mathbf{x})$ (up to an overall normalization), and verify that equation 3 is equivalent to equation 5 by directly comparing the integration over $(\alpha, \mathbf{x})$ in the former to the integration over (θ_1, θ_2) in the latter (numerically if you are so inclined).

[105] More precisely, the unitary matrix U here is the analog of a closed Wilson loop in Yang-Mills theory.

2.8.2 Problem 2: BRST Cohomology for Maxwell's Theory

In 253a we introduced a gauged fixed form of the free Maxwell theory, whose action is

$$S_{\text{naive}} = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right]. \tag{2.1007}$$

Following the recipe of the BRST formalism introduced in class, we can now understand this through the BRST-invariant form

$$S = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + B \partial_\mu A^\mu + \frac{\xi}{2} B^2 + b \partial^\mu \partial_\mu c \right], \tag{2.1008}$$

where the BRST variations of the field variables $A_\mu(x)$, $B(x)$, $b(x)$, $c(x)$ take the form

$$\delta_B A_\mu(x) = \epsilon \partial_\mu c(x),\ \delta_B B(x) = 0,\ \delta_B b(x) = -\epsilon B(x),\ \delta_B c(x) = 0, \tag{2.1009}$$

or equivalently, we may perform the Gaussian functional integration over $B(x)$, leaving the action

$$\tilde{S} = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + b \partial^\mu \partial_\mu c \right], \tag{2.1010}$$

and the BRST variations of the field variables now take the form

$$\delta_B A_\mu(x) = \epsilon \partial_\mu c(x),\ \delta_B b(x) = \frac{\epsilon}{\xi} \partial_\mu A^\mu(x),\ \delta_B c(x) = 0. \tag{2.1011}$$

In this problem, you will analyze the Hilbert space of physical states in the BRST formalism, in the Feynman gauge $\xi = 1$.

(a) You should observe that in the Feynman gauge, the part of the action involving A_μ takes the form identical to that of 4 massless free scalar. Describe the full space of states in terms of a Fock basis constructed using creation operators for A_μ and b, c fields.

(b) Find an explicit expression of the BRST operator $\hat{Q}_B$ in terms of the creation and annihilation operators for A_μ, b, c.

(c) Show that the BRST cohomology classes describing the physical 1-particle state are precisely in correspondence with the 1-photon states of either positive or negative helicity, nothing more and nothing less.

Problem 1 answer:

Answer (a): Notice that our gauge group $\mathcal{G}$ is $\mathcal{G} = U(N)$. Then elements in $\mathcal{G}$ are of the form $g = \exp(i\epsilon)$ where $\epsilon = \alpha \cdot T$ where α is our gauge parameter (coordinate) of the gauge-space and T are our generators for the group $U(N)$. Notice that since g is of dimension $N \times N$ then so is ϵ. Under the gauge transformation defined in equation two we have,

$$U' = (1 + i\epsilon)U(1 - i\epsilon) \implies \tag{2.1012}$$

$$U' - U = \delta U = i[\epsilon, U]. \tag{2.1013}$$

Then, under the constraint over the diagonal elements of our unitary matrices in equation 4, δU becomes,

$$\delta U_{ab} = i\delta_{ab}\left(e^{i\theta_a} - e^{i\theta_b}\right)\alpha_a. \tag{2.1014}$$

Notice that we still have a summation over gauge index a referencing back to our gauge-fixing in equation 4. To find the Faddeev-Popov determinant, we first need the Jacobian which is computed as

$$J = \frac{\partial\theta_a}{\partial\alpha_b}\bigg|_{\alpha=0} = -i\delta_{ad}\left(e^{i\theta_a} - e^{i\theta_b}\right)\delta_{bd}, \tag{2.1015}$$

which gives the following determinant,

$$\Delta(\boldsymbol{\theta}) = \det\left(J^{-1}\right) = \prod_{a\neq b}^{N}(e^{i\theta_a} - e^{i\theta_b}). \tag{2.1016}$$

We can re-write this product in the following manner to make it a more tractable computation for part (b),

$$\Delta(\boldsymbol{\theta}) = \prod_{a<b}^{N}\left(e^{i\theta_a} - e^{-\theta_b}\right)\left(e^{-i\theta_a} - e^{-i\theta_b}\right) \tag{2.1017}$$

$$= \prod_{a<b}^{N} 2i\sin(\theta_a - \theta_b) \tag{2.1018}$$

$$= 2^{N(N-1)/2} i^{N(N-1)} \det(\sin(\theta_a - \theta_b))_{a,b=1,\cdots N}. \tag{2.1019}$$

Now, to modify equation three to put it in terms of equation 5 first notice that

$$\mathrm{tr}(U + U^{-}1) = \sum_{a=1}^{N}\left(e^{i\theta_a} + e^{-i\theta_a}\right) \tag{2.1020}$$

$$= 2i\sum_{a=1}^{N}\cos(\theta_a). \tag{2.1021}$$

We then want to insert a factor of "one" that resembles

$$1 = \int_0^{2\pi} d\theta_a \ \det\left(\left|\ \frac{\delta F[\theta_a]}{\delta\alpha_a}\ \right|\right)\delta(\delta_{ab}e^{i\theta_a}). \tag{2.1022}$$

With these substitutions, equation three is independent of U, so we can integrate out the gauge-volume which cancels with the denominator leaving us with

$$Z_N = C_N \prod_{a=1}^{N} \int_0^{2\pi} d\theta_a \, 2^{N(N-1)/2} i^{N(N-1)} \det(\sin(\theta_a - \theta_b)) \exp\left(2\lambda \sum_{a=1}^{N} \cos(\theta_a)\right). \tag{2.1023}$$

Part (b): Notice we can evaluate the differential dU directly as

$$dU = \frac{\partial U}{\partial \alpha} d\alpha + \frac{\partial U}{\partial \mathbf{x}} d\mathbf{x} \tag{2.1024}$$

$$= e^{i\alpha} \left(\frac{i(1 + i\boldsymbol{\sigma} \cdot \mathbf{x}^2)}{1 + \mathbf{x}^2}) + \frac{2(1 + i\boldsymbol{\sigma} \cdot \mathbf{x})(i\boldsymbol{\sigma})(1 - i\boldsymbol{\sigma} \cdot \mathbf{x}) d\mathbf{x}}{(1 + \mathbf{x}^2)^2} \right) \tag{2.1025}$$

We can then use this to compute the Jacobian (i.e. $J = J(\alpha, \mathbf{x}) \equiv f(\alpha, \mathbf{x})$) since we have the partial derivative above. This gives us $f(\alpha, \mathbf{x}) \sim 1/(1 + \mathbf{x}^2)^3$. To get the full partition function in equation three, notice we can write the action as

$$S(U) = -\lambda \operatorname{tr}\left(U + U^{-1}\right) \tag{2.1026}$$

$$= -\lambda \operatorname{tr}\left(e^{i\alpha} \frac{(1 + i\boldsymbol{\sigma} \cdot \mathbf{x})^2}{1 + \mathbf{x}^2} + e^{-i\alpha} \frac{(1 - i\boldsymbol{\sigma} \cdot \mathbf{x})^2}{1 + \mathbf{x}^2} \right) \tag{2.1027}$$

$$\equiv S(\alpha, \mathbf{x}). \tag{2.1028}$$

Then the partition function becomes

$$Z \sim \int_0^{2\pi} d\alpha \int d^3\mathbf{x} \, \frac{1}{(1 + \mathbf{x}^2)^3} e^{-S(\alpha, \mathbf{x})}. \tag{2.1029}$$

Now, both this partition function and the one we found in equation 17 look quite difficult for an analytic solution. We can do this in *Mathematica* where we will integrate over a finite bounds on Z and the usual continuous bounds on Z_N while picking a value for $\lambda = 0.1$. The code is below along with their values.

Figure 43. *Mathematica* code for computing Z and Z_N.

Now obviously we (I) did something wrong. I am not sure what though. If for the first integral I increase the bounds any large, I get back an error that the behavior is undefined for large bounds. But when I increase them, the whole number doesn't change. And for the second integral, technically, I don't even need the $\sin(\theta_a - \theta_b)$ factors since their multiple gives 1. The factor of two comes from our definition above for $N = 2$. Although if that factor is gone, then we are at least closer with $Z_N = 20.3$, but even then they are still far off.

Problem 2 answer:

Answer (a): First we should verify the claim that just the action S_A containing the gauge field is identical (well, congruent in some sense) to a scalar field with four massless scalars. Lets write down a field theory with four massless scalars first. If $i = \{1, 2, 3, 4\}$ then

$$S[\phi] = \int d^4x \, \frac{1}{2} \left(\partial_\mu \phi_i\right)^2 . \tag{2.1030}$$

Then we can relate this to the gauge-field part of the action in the following manner,

$$S[A_\mu] = \int d^4x \, \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \left(\partial_\mu A^\mu\right)^2\right) \tag{2.1031}$$

$$= \int d^4x \, \left(-\frac{1}{2} \left(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu\right) - \frac{1}{2} \left(\partial_\mu A^\mu\right)^2\right) \tag{2.1032}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Then in light of this, we can mirror a Fourier expansion of the gauge field modes as the expansion of the scalar modes. We can mirror off of equation 4.131 of Peskin and Schroeder's [**AC:**] pands by first defining $a_\mu(\mathbf{p}) = a^r_{\mathbf{p}} \epsilon^r_\mu(\mathbf{p})$ where $\epsilon^\mu_r(\mathbf{p})$ is our polarization vector and the basis of polarization is denoted by $r = \{0, 1, 2, 3\}$. Then we can write the Fourier expansion of A_μ as,

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(a_\mu(\mathbf{p}) e^{-ip\cdot x} + a^\dagger_\mu(\mathbf{p}) e^{ip\cdot x}\right) . \tag{2.1033}$$

These creation and annihilation operators obey the following commutation relations,

$$\left[a_\mu(\mathbf{p}), a^\dagger_\nu(\mathbf{k})\right] = (2\pi)^3 \delta^{(3)} \left(p - k\right) \eta_{\mu\nu}, \tag{2.1034}$$

while the rest of the commutator relationships are zero.

We can do something similar for the ghost fields by mirroring the Fourier expansion of spinors since they are Grassmann variables after all (equation 3.92 and 3.96 from [**AC:**] pands). The Fourier modes for the ghosts with appropriate creation and annihilation operators are

$$b(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(b(\mathbf{p}) e^{-ip\cdot x} b^\dagger(\mathbf{p}) e^{ip\cdot x}\right) \tag{2.1035}$$

$$c(x) = \int \frac{d^3p}{(2\pi)^3} \left(c(p) e^{-ip\cdot x} + c^\dagger(p) e^{ip\cdot x}\right) , \tag{2.1036}$$

with the following commutation relations

$$\left[b(p), b^\dagger(k)\right] = \left[c(p), c^\dagger(k)\right] = (2\pi)^3 \delta^{(3)}(p - k), \tag{2.1037}$$

and the rest are zero.

To then build the Fock space with respect to the vacuum state $|\Omega\rangle$, we first want to establish that for the annihilation operators on the vacuum obey,

$$a_\mu(p) \, |\Omega\rangle = b(p) \, |\Omega\rangle = c(p) \, |\Omega\rangle = 0. \tag{2.1038}$$

Then to construct the Fock space, this is the entire Hilbert space of states such that the states are constructed out of successive creation operators applied [AC:] textbook. Since we have three species of particles, the entire Fock space is then the following tensor product,

$$\mathcal{F} = \mathcal{F}_{A_\mu} \otimes \mathcal{F}_b \otimes \mathcal{F}_c. \tag{2.1039}$$

To construct $|n\rangle$-states, we can use the following formula (and then generalize)

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |\Omega\rangle \,, \tag{2.1040}$$

(I can only find a reference for this equation in Nakaraha's text [AC:] nakahara in the beginning of chapter 1). Then, for the gauge fields we can create Fock space states in the following manner,

$$\left| n_{a_{\mu,1}(k_1),\cdots,a_{\mu,N}(k_N)} \right\rangle = \prod_{i=1}^{N} \frac{\left(a^\dagger_{\mu,i}(k_i)\right)^n}{\sqrt{n_{\mu,i}!}} |\Omega\rangle \tag{2.1041}$$

We find something similar for the ghost states, but will tensor them together to keep them separate from what we expect are the only physical fields,

$$|n_b(k_1), n_c(k_2), \cdots, n_b(k_{N_b}), n_c(k_{N_c})\rangle = \prod_{i=1}^{N_b} \frac{\left(b^\dagger(k_i)\right)^n}{\sqrt{n_b!}} |\Omega\rangle \otimes \prod_{i=1}^{N_c} \frac{\left(c^\dagger(k_i)\right)^n}{\sqrt{n_c!}} |\Omega\rangle \tag{2.1042}$$

where N_b and N_c are the number of b and c-ghost fields.

Then the full Fock space are the tensor product of equation 42 and 43.

Answer (b): To find an explicit form of the BRST operator, there is really only one way that comes to mind which is to find each individual Q_B operators for each field from the BRST symmetries defined above, then Fourier transform them back to find $Q_B(k)$.

Recall that the definition of the BRST charge is for some matter field(s) Φ is $Q_B\Phi = \epsilon\delta\Phi$ where ϵ is some Grassmann number. Then we would expect $Q_B A_\mu = \partial_\mu c(x)$, $Q_B b(x) = \partial_\mu A^\mu(x)$, and $Q_B c(x) = 0$. All three of these obey $Q_B^2 = 0$ (nilpotent) with the equations of motion for $c(x)$ and $A_\mu(x)$ for both (the third is obvious). Thus, at least intuitively, we wouldn't expect any $b(x)$ ghost fields to be in our expression for Q_B.

Now, let's Fourier transform the BRST symmetries. These become

$$\delta_B A_\mu(p) = \epsilon \int d^3x\, e^{ik\cdot x} \partial_\mu c(x) = \epsilon k_\mu c(x) \tag{2.1043}$$

$$\delta_B b(x) = \epsilon \int d^3x\, e^{ip\cdot x} \partial_\mu A^\mu = \epsilon k^\mu a_\mu(x), \tag{2.1044}$$

where the last equation follows from equation 34. Then, if the Fourier transform of the BRST charge is $Q_B = \int d^3k\, Q_B(k)$ we have

$$Q_B = \int d^3k \, \left(k^\mu a_\mu(k) c^\dagger(k) + k^\mu a^\dagger_\mu(k) c(k) \right). \tag{2.1045}$$

This feels correct, but lets check by computing

$$[Q_B(k), A_\mu(p)] = k^\nu \left[a_\nu(k)c^\dagger(k), A_\mu(p)\right] + k^\rho \left[a^\dagger_\rho(k)c(k), A_\mu(p)\right] \tag{2.1046}$$
$$= k^\nu (2\pi)^3 (2\omega_{\mathbf{k}})\eta_{\mu\nu}\delta^{(3)}(k-p)c(k) \tag{2.1047}$$
$$= k_\mu c(k), \tag{2.1048}$$

which is what we expect for the Fourier transformed version of the BRST charge Q_B. Then equation 46 is our BRST operator.

Answer (c): (I should clarify, the polarization state of choice is of course characterized by the helicity. I am a bit sloppy with the language here). First, lets completely forget about the part of the question asking on the BRST cohomology classes, and instead focus on the particle states. First, recall that we defined $a_\mu = \epsilon_\mu a_{\mathbf{p}}$ (and similar for the creation operator). Then for the one particle state of A_μ, $|k, \mu\rangle$ with k momentum and polarization index μ (more like a compact index) we have,

$$Q_B |k, \mu\rangle = k_\mu \epsilon^\mu_r c^\dagger(k) |\Omega\rangle = 0, \tag{2.1049}$$

where we used the commutation relation of equation 35 with the definition of Q_B in equation 46 to get the factor of $c^\dagger(k)$. Then, to get $Q_B |k, \mu\rangle = 0$ we used the fact that $k_\mu \epsilon^\mu_r = 0$. Thus, by definition, this is an admissible (physical) state.

Then similar for the ghosts with one particle states $|k_b\rangle = b^\dagger(k)|\Omega\rangle$ and $|k, c\rangle = c^\dagger(k)|\Omega\rangle$ we have,

$$Q_B |k, b\rangle = Q_B b^\dagger(k) |\Omega\rangle \neq 0 \tag{2.1050}$$
$$Q_B |k, c\rangle = Q_B c^\dagger(k) |\Omega\rangle \neq 0, \tag{2.1051}$$

which both do not equal zero since nothing within our definition of Q_B doesn't give us a way to utilize either the commutation relations or anything along those lines. Thus, we only have the one state above, and thus our only admissible state in the entire system with two polarizations.

To relate this back to cohomology, recall from some basic algebraic topology (not even that, but basics of groups, and I recommend Hatcher [**AC:**] hatcher) that for some group G, there may exist a group homomorphism ϕ between another group H (typically a subgroup) $\phi : G \to H$. Then, the kernel of ϕ is defined as $\text{kernel}(\phi) = \{g | g \in G \wedge \phi(g) = 1_G\}$ where 1_G denotes the unit of the group. The image of the group homomorphism is defined as $\text{image}(\phi) = \{\phi(g) | g \in G\} \subset H$ but does not have to be $\subseteq H$.

We can then view Q_B as our "group homomorphism" between physical states and all of our Fock space. Then, the kernel of Q_B, $\text{kernel}(Q_B)$ is only those states that are admissible, i.e. equation 50. The image of Q_B, $\text{image}(Q_B)$ is the entire Fock space. Then, when we modulo the kernel of Q_B with the image of Q_B, we only get out the physical state as described in equation 50. (For example, this is the exact same thing as $\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2$).

You can go more in depth about the physical states, as described in [**AC:**] pands where due to the properties of Q_B being nilpotent, it breaks the Hilbert space into three sectors, but luckily we are not asked about that here.

2.9 Problem set 6: 1-loop β-function for QCD

2.9.1 Problem 1:

In this problem you will finish the computation of the one-loop β-function of QCD with $SU(N)$ gauge fields and N_f flavors of massless Dirac fermions in a presentation R, as was analyzed in class through the 1PI effective action computed in the background field gauge via dimensional regularization (in $D = 4 - \epsilon$ spacetime dimensions), with the renormalized Yang-Mills coupling $g(\mu)$ defined in the minimal subtraction scheme.

In the background field gauge, the total (bare) action including Faddeev-Popov ghosts is

$$\begin{aligned} S[A', \psi, b, c; \bar{A}] = \int \Big[& -\frac{1}{4g^2}\left(\tilde{F}_{a\mu\nu} + \tilde{D}_\mu A'_{a\nu} - \tilde{D}_\nu A'_{a\mu} + f_a^{bc} A'_{b\mu} A'_{c\nu}\right)^2 \\ & - \bar{\psi}_i \gamma^\mu \left(\tilde{D}_\mu - i A'_{a\mu} t_R^a\right) \psi_i \\ & - \frac{1}{2\xi g^2}\left(\tilde{D}_\mu A'^{\mu}_a\right)^2 - \left(\tilde{D}_\mu b^a\right)\left(\tilde{D}^\mu c_a + f_a^{bc} A'^{\mu}_b c_c\right) \Big], \end{aligned} \tag{2.1052}$$

where the original gauge field is written as $A_{a\mu}(x) = \tilde{A}_{a\mu}(x) + A'_{a\mu}(x)$ with $\tilde{A}_{a\mu}(x)$ being a background gauge field, while $A'_{a\mu}(x), \psi_i(x), b^a(x)$ and $c_a(x)$ are fluctuation fields to be integrated out in obtaining the 1PI effective action[106]. $\tilde{D}_\mu$ is the gauge covariant derivative with respect to the background gauge field $\tilde{A}$, in the adjoint representation when acting on A', b, c, and in the representation R when acting on ψ.

The one-loop contribution to the *Euclidean* 1PI effective action $\Gamma^{\text{1-loop}}[\tilde{A}]$ is computed through the functional determinant

$$e^{-\Gamma^{\text{1-loop}}[\tilde{A}]} = \frac{\left(\det(\mathbb{D}_\psi)\right)^N \left(\det(\mathbb{D}_{\text{gh}})\right)}{\left(\det(\mathbb{D}_A)^{1/2}\right)} \tag{2.1053}$$

for a Euclidean background gauge field $\tilde{A}_{a\mu}(x)$, where

$$\mathbb{D}_\psi = -\gamma^\mu \tilde{D}_\mu, \ \text{(in the representation } R) \tag{2.1054}$$

$$\mathbb{D}_{\text{gh}} = \tilde{D}_\mu \tilde{D}^\mu \ \text{(in the adjoint representation)} \tag{2.1055}$$

$$[\mathbb{D}_A]_{a\mu,b\nu} = \left(-\tilde{D}_\rho \tilde{D}^\rho \eta_{\mu\nu} + \tilde{D}_\nu \tilde{D}_\mu - \frac{1}{\xi}\tilde{D}_\mu \tilde{D}_\nu\right)\delta_{ab} + f_c^{ab}\tilde{F}_{c\mu\nu}. \tag{2.1056}$$

For a constant background field configuration $\tilde{A}$, the functional determinant in equation two is formally evaluated as

$$\Gamma^{\text{1-loop}}[\tilde{A}] = -\int d^D x \int \frac{d^D k}{(2\pi)^D}\left(N_f \operatorname{tr}(\log(\mathbb{M}_\psi(k))) + \operatorname{tr}(\log(\mathbb{M}_{\text{gh}}(k))) - \frac{1}{2}\operatorname{tr}(\log(\mathbb{M}_A)(k))\right) \tag{2.1057}$$

where $\mathbb{M}(k)$ is the momentum space version of the kinetic operator $\mathbb{D}$, e.g

$$\mathbb{M}_\psi = i\gamma^\mu \otimes \left(k_\mu - \tilde{A}_{a\mu} t_R^a\right), \tag{2.1058}$$

[106] As usual, in deriving the 1PI effective action, linear terms in the fluctuation fields are added to the bare action so that the tadpoles i.e. 1-point functions of fluctuation fields vanish.

where t_R^a is the generator of $SU(N)$ in the representation R. As explained in class, the IR divergence in the k-integral on the RHS of equation four is cut off at the renormalization scale μ, corresponding to a background field $\tilde{A}$ that is constant in a domain of size $L \sim 1/\mu$ but falls off outside of the domain.

In class, we have computed the constribution from the functional determinant of ψ to the singular part of $\tilde{A}^4$ term in the Euclidean one-loop effective Lagrangian,

$$-\int\limits_{|k|>\mu} \frac{d^D k}{(2\pi)^D} \operatorname{tr}(\log(\mathbb{M}_\psi(k)))\Big|_{\tilde{A}^4} \sim \frac{1}{3} \operatorname{tr}\Big(\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}\Big)_R \frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}, \tag{2.1059}$$

where on the RHS we have omitted terms that are finite in the $\epsilon \to 0$ limit as they do not affect the on-loop beta function (and do not appear in the one-loop renormalized coupling in the minimal subtraction scheme either).

Your task is to compute the analogous contribution from the functional determinant of A' and the Faddeev-Popov ghosts b, c, in the $\xi = 1$ gauge. Show that their respective contributions to the singular part of the Euclidean effective Lagrangian is

$$-\int\limits_{|k|>\mu} \frac{d^D k}{(2\pi)^D} \operatorname{tr}\Big(\log(\mathbb{M})_{\text{gh}}(k)\Big)\Big|_{\tilde{A}^4} \sim -\frac{1}{12} \operatorname{tr}\Big(\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}\Big)_{\text{adj}} \frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}, \tag{2.1060}$$

$$+\frac{1}{2}\int\limits_{|k|>\mu} \frac{d^D k}{(2\pi)^D} \operatorname{tr}(\log(\mathbb{M}_A(k)))\Big|_{\tilde{A}^4} \sim \frac{5}{6} \operatorname{tr}\Big(\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}\Big)_{\text{adj}} \frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon} \tag{2.1061}$$

which result in the one-loop β-function claimed in the lecture. Here $\operatorname{tr}_{\text{adj}}$ stands for the trace in the adjoint representation.

Problem 1 answer:

The lecture notes appear to be based on Weinberg's derivation in chapter 17.4 and 17.5 [**AC:**] weinberg text where instead of using dimensional regularization, we take a matrix multiplication approach in what I can only describe as using Schwinger's methodology that we have previously seen in our computation of Euler-Heisenberg Lagrangian to compute the probability for pair production (i.e. putting our operators in terms of logarithms and traces in order to evaluate). There is another version however that is put forward by Peskin and Schroeder in chapter 16.6 [**AC:**] pands where in the derivation, everything is the same up until the point of evaluating the determinants where instead we take a more Feynman approach by writing out the Feynman diagrams and then computing those. Obviously, both scenarios lead to a correct computation of the beta function, our final goal. In this homework, I will take the stance of Weinberg, while the use of Peskin and Schroeder will be used where Weinberg is not entirely clear. A major difference though between what we did in class and to what I will do is that our notation will be very different. Instead of the tildes to denote the background field, we will simply separate the field A as $A \to A' + A$ where A' will denote the quantum gauge field fluctuations and A will for now on be the *classical* background gauge field. Our sign convention will be Lorentzian.

Consider the action in equation one. Let $A_{a\mu} =$ constant while we set all the other classical analogues of our fields are 0, $\psi = b = c = 0$. Then, our entire Lagrangian can be written as

$$\mathcal{L} = \mathcal{L} + \mathcal{L}_f + \mathcal{L}_{\text{gh}} \tag{2.1062}$$

where

$$\mathcal{L} = -\frac{1}{4}\left(F_{a\mu\nu} + \bar{D}_\mu A'_{a\nu} - \bar{D}_\nu A'_{a\mu} + f_{abc}A'_{b\mu}A'_{c\nu}\right)^2 - \bar{\psi}'\left(\bar{\not{D}} - it_a\not{A}'_a + m\right)\psi' \tag{2.1063}$$

$$\mathcal{L}_f = -\frac{1}{2\xi}\left(f_a f_a\right) = -\frac{1}{2\xi}\left(\bar{D}_\mu A^{\mu'}_a\right)^2 \tag{2.1064}$$

$$\mathcal{L}_{\text{gh}} = -\left(\bar{D}_\mu b'_a\right)\left(\bar{D}^\mu c'_a - f_{abc}c'_b A^{\mu'}_c\right). \tag{2.1065}$$

To compute the one-loop effective action, we will only care about terms that are quadratic in the quantum field fluctuation terms (anything higher corresponds to higher loop corrections as given in [**AC:**] pands). We can write the quadratic terms as

$$\begin{aligned}\mathcal{L}_{\text{quad}} = &-\frac{1}{4}\left(\bar{D}_\mu A'_{a\nu} - \bar{D}_\nu A'_{a\mu}\right)^2 - \frac{1}{2}F^{\mu\nu}_a f_{abc}A'_{b\mu}A'_{c\nu} \\ &- \overline{\psi}'\left(\overline{\not{D}} + m\right)\psi' - \frac{1}{2\xi}\left(\overline{D}_\mu A'^{\mu}_a\right)^2 - \left(\overline{D}_\mu b'_a\right)\left(\overline{D}^\mu c'_a\right).\end{aligned} \tag{2.1066}$$

We can plug this Lagrangian into our action, and then put it in terms of operators that will become our determinants to solve,

$$S_{\text{quad}} \equiv \int d^4x\, \mathcal{L}_{\text{quad}} \tag{2.1067}$$

$$= -\frac{1}{2}\int d^4x d^4y\, A'^{\mu}_a(x)\mathbb{D}^A_{a\mu,b\nu} - \int d^4x d^4y\, A'^{\mu}_a(x)\mathbb{D}^{\psi}_{k,l}\psi'_l(y) - \int d^4x d^4y\, b'_a(x)\mathbb{D}^{\text{gh}}_{a,b}c'_b(y). \tag{2.1068}$$

We can write out the operators $\mathbb{D}$ in an explicit form where $\partial_{x,\lambda} \equiv \partial/\partial x^\lambda$ as,

$$\begin{aligned}
\mathbb{D}^A_{a\mu,b\nu} &\equiv \eta_{\mu\nu}\left(-\delta_{ca}\partial_{x,\lambda} + f_{cda}A^\lambda_d(x)\right)\left(\delta_{db}\partial_{\lambda,y} + f_{deb}A_{e\lambda}(y)\right)\delta^{(4)}(x-y) \\
&\quad - \left(-\delta_{da}\partial_{\mu,x} + f_{dca}A_{d\nu}(x)\right)\left(-\delta_{cb}\partial_{\lambda,y} + f_{ceb}A_{e\mu}(y)\right)\delta^{(4)}(x-y) + F_{c\mu\nu}(x)f_{cab}\delta^{(4)}(x-y) \\
&\quad \frac{1}{\xi}\left(-\delta_{ca}\partial_{\mu,x} + f_{cda}A_{d\mu}(x)\right)\left(-\delta_{cb}\partial_{\lambda,y} + f_{ceb}A^\lambda_e(y)\right)\delta^{(4)}(x-y) \qquad (2.1069) \\
\mathbb{D}^{\text{gh}}_{a,b} &\equiv \left(-\delta_{ca}\partial_{\lambda,x} + f_{cda}A_{d\lambda}(x)\right)\left(-\delta_{cb}\partial_{\lambda,y} + f_{ceb}A^\lambda_e(y)\right)\delta^{(4)}(x-y) \qquad (2.1070) \\
\mathbb{D}^\psi_{k,l} &\equiv \left(-\gamma^\mu\partial_{\mu,y} - it_a\not{A}_a(y) + m\right)_{kl}\delta^{(4)}(x-y). \qquad (2.1071)
\end{aligned}$$

The minus signs in front of the partial derivatives with respect to the spacetime coordinates x^μ and y^μ drop out when we perform integration by parts when putting the action in a form that allows us to get the operators above in the form of a determinant.

The one-loop effective action then becomes

$$\begin{aligned}
e^{i\Gamma_{\text{1-loop}}[A]} &= \mathcal{A}\int_{\text{1PI}}[DA'][D\psi][D\overline{\psi}'][Db][Dc]e^{iS_{\text{quad}}[A',\psi',\overline{\psi}',b'^a,c_a;A]} \qquad (2.1072) \\
&= \mathcal{A}\left(\det\mathbb{D}^A\right)^{-1/2}\left(\det\mathbb{D}^\psi\right)\left(\det\mathbb{D}^{\text{gh}}\right) \qquad (2.1073)
\end{aligned}$$

where $\mathcal{A}$ is some normalization constant we do not need to care about, and notice in the determinant carrying the fermionic fields (non-ghosts) that we are only considering one species. If we were to work in the $\text{SU}(N_f)$ group, then we would have a power of N_f. This will actually come into play when computing the Casimir coefficients at the end, so we can not worry about this for now.

From here, there are two methods to find the forms $\mathbb{D}$: Feynman graph method and Fourier transforming equations 18-20 and computing the matrix algebra. Here, we will follow the method in class.

Notice that for a general operator $\mathbb{D}$ that is dependent upon two spacetime coordinates x and y with corresponding momentum q and p, the Fourier transform is

$$\begin{aligned}
\mathbb{D}_{q\cdots,p,\cdots} &= \int\frac{d^4x}{(2\pi)^4}e^{-iq\cdot x}\int\frac{d^4y}{(2\pi)^4}e^{ip\cdot y}\mathbb{D}_{x\cdots,y\cdots} \qquad (2.1074) \\
&= \delta^{(4)}(p-q)\mathbb{M}_{\cdots,\cdots}(q). \qquad (2.1075)
\end{aligned}$$

We already have computed $\mathbb{M}_\psi(k)$ in class, so we will just state that result when we require it. We then move on to computing $\mathbb{M}^A$ and $\mathbb{M}^{\text{gh}}$. Fourier transforming each term in equation 18 and 19 term by term we find the following

$$\begin{aligned}
\mathbb{M}^A_{a\mu,b\nu}(q) &= \eta_{\mu\nu}\left(-iq_\lambda\delta_{ca} + A_{\delta\lambda}F_{cda}\right)\left(iq^\lambda\delta_{cb} + A^\lambda_e f_{deb}\right) \\
&\quad - \left(-iq_\nu\delta_{ca} + A_{d\nu}f_{cda}\right)\left(iq_\mu\delta_{cb} + A_{e\mu}f_{ceb}\right) + F_{c\mu\nu}f_{cab} \\
&\quad + \left(-q_\mu\delta_{ca} + A_{d\mu}f_{cda}\right)\left(iq_\nu\delta_{cb} + A_{e\nu}f_{ceb}\right) + \epsilon - \text{terms} \qquad (2.1076) \\
\mathbb{M}^\psi_{kl}(q) &= \left(i\not{q} - it_a\not{A}_a + m\right)_{kl} + \epsilon - \text{terms} \qquad (2.1077)
\end{aligned}$$

where we have

$$F_{a\mu\nu} = f_{abc}A_{b\mu}A_{c\nu} \tag{2.1078}$$

$$D_\lambda F_{c\mu\nu} = f_{dea}f_{abc}A_{e\lambda}A_{b\mu}A_{c\nu}. \tag{2.1079}$$

Now, notice we can solve the effective action $\Gamma_{\text{1-loop}}$ in terms of logarithms as

$$i\Gamma_{\text{1-loop}}[A] = -\frac{1}{2}\ln\det\mathbb{D}_A + \ln\det\mathbb{D}_\psi + \ln\det\mathbb{D}_{\text{gh}} \tag{2.1080}$$

$$= -\frac{1}{2}\operatorname{Tr}\ln\mathbb{D}_A + \operatorname{Tr}\ln\mathbb{D}_\psi + \operatorname{Tr}\ln\mathbb{D}_{\text{gh}} \tag{2.1081}$$

$$= \delta^{(4)}(p-p)\int d^4q\,\left(-\frac{1}{2}\operatorname{tr}\ln\mathbb{M}^A(q) + \operatorname{tr}\ln\mathbb{M}^\psi(q) + \operatorname{tr}\ln\mathbb{M}_{\text{gh}}(q)\right). \tag{2.1082}$$

The lower case trace denotes the typical trance over finite matrices, and not a specific representation. For a the general Fourier transformed operator $\mathbb{M}$ we can decompose it into terms that contain either no contributions of A, first order, or second order in A as $\mathbb{M} = \mathbb{M}_0 + \mathbb{M}_1 + \mathbb{M}_2$. Then, up to fourth order in A we can write the trace of the logarithm of $\mathbb{M}$ as

$$\operatorname{tr}\ln\mathbb{M}\Big|_{A^4} = \operatorname{tr}\left(-\frac{1}{2}\left(\mathbb{M}_0^{-1}\mathbb{M}_2\right)^2 + \left(\mathbb{M}_0^{-1}\mathbb{M}_1\right)^2\mathbb{M}_0^{-1}\mathbb{M}_2 - \frac{1}{4}\left(\mathbb{M}_0^{-1}\mathbb{M}_1\right)^4\right). \tag{2.1083}$$

This was found by insert factors of ϵ and ϵ^2 into our expansion of $\mathbb{M}$, differentiate four times with respect to x, then divide by the symmetry factor of 4! and set $\epsilon \to 0$.

To compute $\mathbb{M}^A$, we need all the components given above. These are,

$$\left(\mathbb{M}_0^A(q)\right)^{-1}_{a\mu,b\nu} = \delta_{ab}\eta_{\mu\nu}\left(q^2 - i\epsilon\right)^{-1} \tag{2.1084}$$

$$\left(\mathbb{M}_1^A(q)\right)^{-1}_{a\mu,b\nu} = 2i\eta_{\mu\nu}q^\lambda f_{abc}A_{c\lambda} \tag{2.1085}$$

$$\left(\mathbb{M}_2^A(q)\right)^{-1}_{a\mu,b\nu} = \left(\eta_{\mu\nu}\mathcal{A}_\lambda\mathcal{A}^\lambda - \mathcal{A}_\nu\mathcal{A}_\mu - \mathcal{A}_\mu\mathcal{A}_\nu\right)_{ab} + F_{c\mu\nu}f_{cab} \tag{2.1086}$$

where we have defined

$$(\mathcal{A}_\lambda)_{ab} = -if_{abc}A_{c\lambda} \tag{2.1087}$$

$$[\mathcal{A}_\lambda, \mathcal{A}_\rho]_{ab} = f_{acb}f_{dec}A_{c\lambda}A_{e\rho} \tag{2.1088}$$

$$= f_{adb}F_{d\rho\lambda}. \tag{2.1089}$$

Notice that for these functions above will be integrals of some general function $f(q^2)$ that have either two or four insertions of momentum q^μ. Then can generally be written as

$$\int d^4q\,q^\mu q^\nu f(q^2) = \frac{1}{4}\eta^{\mu\nu}\int d^4q\,q^2 f(q^2) \tag{2.1090}$$

$$\int d^4q\,q^\mu q^\nu q^\rho q^\sigma f(q^2) = \frac{1}{24}\left(\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\right)\int d^4q\,(q^2)^2 f(q^2). \tag{2.1091}$$

Notice if we were in D-dimensions, then we would instead have a factor of $1/(D)$ and $1/6D$ due to the traces over the metric. Then, the integral over the traces of each of the $\mathbb{M}_i^A$ components (expanded version of equation 31) gives the following three factors

$$\int d^4q \ \mathrm{tr}\Big(\Big(\mathbb{M}_0^A(q)^{-1}\mathbb{M}_2^A(q)\Big)^2\Big) = \int d^4q \ \frac{4}{(q^2-i\epsilon)^2}\Big(\mathrm{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta\Big) + f_{cab}c_{dab}F_{c\mu\nu}F_d^{\mu\nu}\Big) \tag{2.1092}$$

$$\int d^4q \ \mathrm{tr}\Big(\Big(\mathbb{M}_0^A(q)^{-1}\mathbb{M}_1^A(q)\Big)^2 \mathbb{A}_0^A(q)^{-1}\mathbb{M}_2^A(q)\Big) = \int d^4q \ \frac{4}{(q^2-i\epsilon)^2}\mathrm{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta\Big) \tag{2.1093}$$

$$\int d^4q \ \mathrm{tr}\Big(\Big(\mathbb{M}_0^A(q)^{-1}\mathbb{M}_1^A(q)\Big)^4\Big) = \frac{8}{3}\int \frac{d^4q}{(q^2-i\epsilon)^2}\mathrm{tr}\Big(2\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta + \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\Big). \tag{2.1094}$$

Notice we can only simplify a few of these products due to the commutation relation given in equation 37, but they can be added and simplified to give

$$\int d^4q \ \mathrm{tr}\ln\mathbb{M}^A(q)\Big|_{A^4} = \frac{2}{3}\int \frac{d^4q}{(q^2-i\epsilon)^2}\Big(\mathrm{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta - \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\Big) - f_{cab}f_{dab}F_{c\mu\nu}F_d^{\mu\nu}\Big) \tag{2.1095}$$

$$= -\frac{5}{3}\int \frac{d^4q}{(q^2-i\epsilon)^2}f_{cab}f_{dab}F_{c\mu\nu}F_d^{\mu\nu}. \tag{2.1096}$$

The ghost momentum matrix has the following components,

$$\Big(\mathbb{M}_0^\psi(q)\Big)_{kl}^{-1} = \Big(q^2-i\epsilon\Big)_{kl}^{-1} \tag{2.1097}$$

$$\Big(\mathbb{M}_1^\psi(q)\Big)_{kl} = -i\,(t_a\not{A}_a)_{kl} \tag{2.1098}$$

$$\Big(\mathbb{M}_2^\psi(q)\Big)_{kl} = 0. \tag{2.1099}$$

These in term give the following giant (enormous) integral performed with gamma technology found in the appendix of [**AC:**] pands,

$$\int d^4q \ \mathrm{tr}\ln\mathbb{M}^\psi(q)\Big|_{A^4} = -\frac{1}{4}\int d^4q \ \mathrm{tr}\Big(\Big((i\not{q}+m)^{-1}t_a\not{A}_a\Big)^4\Big) \tag{2.1100}$$

$$\overset{\text{UV-limit}}{=} -\frac{1}{4}\mathrm{tr}(t_at_bt_ct_d)A_{a\mu}A_{b\nu}A_{c\rho}A_{d\sigma}\int d^4q\frac{\mathrm{tr}\Big(\not{q}\gamma^\mu\not{q}\gamma^\nu\not{q}^\rho\not{q}\gamma^\sigma\Big)}{(q^2-i\epsilon)^4}$$

$$\times\ \mathrm{tr}\Big(2\gamma_\lambda\gamma^\mu\gamma^\lambda\gamma^\nu\gamma_\eta\gamma^\rho\gamma^\eta\gamma^\sigma + \gamma_\lambda\gamma^\mu\gamma_\eta\gamma^\nu\gamma^\rho\gamma^\eta\gamma^\sigma\Big) \tag{2.1101}$$

$$= \frac{1}{2}\int \frac{d^4q}{(q^2-i\epsilon)^2}\mathrm{tr}([t_a,t_b][t_c,t_d])A_{a\mu}A_{b\nu}A_c^\mu A_d^\nu \tag{2.1102}$$

$$= \frac{1}{2}\int \frac{d^4q}{(q^2-i\epsilon)^2}F_{c\mu\nu}F_d^{\mu\nu}\,\mathrm{tr}(t_ct_d) \tag{2.1103}$$

where the large trace of gamma functions is computed to be

$$-64\eta^{\mu\rho}\eta^{\nu\sigma} + 32\eta^{\mu\nu}\rho^{\rho\sigma} + 32\eta^{\mu\sigma}\eta^{\nu\rho}. \tag{2.1104}$$

Finally, to find the ghost matrix we first have the following zeroth, first, and second order terms containing A as

$$\left(\mathbb{M}_0^{\text{gh}}(q)\right)_{ab}^{-1} = \delta_{ab}\left(q^2 - i\epsilon\right)^{-1} \tag{2.1105}$$

$$\left(\mathbb{M}_1^{\text{gh}}(q)\right)_{ab} = -2\left(\mathcal{A}^\lambda\right)_{ab} q_\lambda \tag{2.1106}$$

$$\left(\mathbb{M}_2^{\text{gh}}(q)\right)_{ab} = \left(\mathcal{A}^\lambda \mathbb{A}_\lambda\right)_{ab}. \tag{2.1107}$$

These contributions in the $\mathbb{M}^{\text{gh}}(q)$ integral then gives

$$\int d^4q\ \text{tr}\Big(\left(\mathbb{M}_0^{\text{gh}}(q)\mathbb{M}_2^{\text{gh}}(q)\right)^2\Big) = \int \frac{d^4q}{(q^2-i\epsilon)^2}\,\text{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta\Big) \tag{2.1108}$$

$$\int d^4q\ \text{tr}\Big(\left(\mathbb{M}_0^{\text{gh}}(q)^{-1}\mathbb{M}_1^{\text{gh}}(q)\right)^2 \mathbb{M}_0^{\text{gh}}(q)^{-1}\mathbb{M}_2^{\text{gh}}(q)\Big) = \int \frac{d^4q}{(q^2-i\epsilon)^2}\,\text{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta\Big) \tag{2.1109}$$

$$\int d^4q\ \text{tr}\Big(\left(\mathbb{M}_0^{\text{gh}}(q)^{-1}\mathbb{M}_1^{\text{gh}}(q)\right)^4\Big) = \frac{2}{3}\int \frac{d^4q}{(q^2-i\epsilon)^2}\,\text{tr}\Big(2\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta + \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\Big). \tag{2.1110}$$

When plugging everything together we get the following contribution form the ghost-matrix,

$$\int d^4q\ \text{tr}\ln\mathbb{M}^{\text{gh}}(q)\Big|_{A^4} = \frac{1}{6}\int \frac{d^4q}{(q^2-i\epsilon)^2}\,\text{tr}\Big(\mathcal{A}^\lambda\mathcal{A}_\lambda\mathcal{A}^\eta\mathcal{A}_\eta - \mathcal{A}^\lambda\mathcal{A}^\eta\mathcal{A}_\lambda\mathcal{A}_\eta\Big) \tag{2.1111}$$

$$= \frac{1}{12}\frac{d^4q}{(q^2-i\epsilon)^2} f_{cab}f_{dab}F_{c\mu\nu}F_d^{\mu\nu}. \tag{2.1112}$$

The total one-loop contribution now becomes

$$\gamma^{\text{1-loop}}[A]\Big|_{A^4} = -\frac{i}{(2\pi)^4}\int \frac{d^4q}{(q^2-i\epsilon)^2}\int d^4x\ F_{c\mu\nu}F_d^{\mu\nu}\left[\left(\frac{5}{6}+\frac{1}{12}\right)f_{cab}f_{dab} - \frac{1}{3}\,\text{tr}(t_ct_d)\right] \tag{2.1113}$$

where the first term containing the structure constants gives a factor of $g^2C_1\delta_{cd}$ while the second term gives $g^2C_2\delta_{cd}$ where C_1 is the Casimir coefficient in the adjoint representation giving $C_1 = N$ and the second Casimir coefficient is in the fundamental representation giving $C_2 = N_f/2$ for N_f number of fermionic species stated by our group SU(N). When plugging in everything together we get the following one-loop effective action

$$\Gamma_{\text{1-loop}}[A]\Big|_{A^4} = -\frac{ig^2}{(2\pi)^4}\int \frac{d^4q}{(q^2-i\epsilon)^2}\int d^4x\ F_{c\mu\nu}F_d^{\mu\nu}\left(\frac{11}{12}C_1 - \frac{1}{3}C_2\right). \tag{2.1114}$$

If we renormalized our coupling as $g_R = (c_g+1)^{-1/2}g$, then our counter term is

$$c_g = \frac{4ig^2}{(2\pi)^4}\int \frac{d^4q}{(q^2-i\epsilon)^2}\left(\frac{11}{12}N - \frac{1}{6}N_f\right). \tag{2.1115}$$

We can solve the q-momentum integral by Wick rotating, imposing a UV-bound of Λ and an IR-cutoff of μ, and then integrating gives

$$\int \frac{d^4q}{(q^2 - i\epsilon)^2} = i \int_0^\infty dq \, \frac{2\pi^2 q^3}{q^4} \tag{2.1116}$$

$$= 2\pi^2 i \int_\mu^\Lambda \frac{dq}{q} \tag{2.1117}$$

$$= 2\pi^2 i \ln\left(\frac{\Lambda}{\mu}\right). \tag{2.1118}$$

This allows us to write g_R fully, which when expanded and simplified gives

$$g_R = g\left(1 + \frac{g^2}{4\pi^2} \ln\left(\frac{\Lambda}{\mu}\right)\left(\frac{11}{12}N - \frac{1}{6}N_f\right) + \mathcal{O}(g^4)\right). \tag{2.1119}$$

This then gives us the following beta function

$$\beta(g) = \frac{d}{d\log\mu} g_R = \mu \frac{d}{d\mu} g_R \tag{2.1120}$$

$$= -\frac{g^3}{4\pi^2}\left(\frac{11}{12}N - \frac{1}{6}N_f\right). \tag{2.1121}$$

2.10 Final Project: Introduction to C-Theorems

Abstract: We review and present an in-depth analysis of Zomoldchikov's original c-theorem. We start with a review of the renormalization group flow, and review the basic facts from conformal field theory. Next we derive, in-detail, Zomoldchikov's original derivation. This leads us to attempt to understand the physical ramifications of the c-theorem, in which re-deriving the c-theorem via a spectral representation will prove useful. We then finish by commenting on efforts made to extend the c-theorem to dimensions greater than two.

2.10.1 Review of Essential Material

In this section, we give a brief overview of the necessary material towards understanding Zomoldchikov's c-theorem. The theorem is built from the beta-function of the renormalization group flow, and utilizes facts from the fixed points of the beta-function that correspond to conformal field theories. Thus, we will give a brief overview of the renormalization group flow from the perspective of the Wilsonian effective field that will give us the idea the group flow *loses information* when flowing from ultraviolet (UV) energies to infrared (IR) energies. Then we will only review the necessary details from conformal field theory that will help us throughout the derivation[107].

Renormalization group flow

Nothing new about the renormalization group flow will be introduced in this section, but this, more or less, will be a review of the Wilsonian effective action given from the perspective of losing information along the flow. This will aid to our understanding about how Zomoldchikov's c-theorem fits in with the beta-function, and the renormalization group flow.

Consider the partition function for some scalar theory in d-dimensions with a Euclidean signature,

$$Z = \int [D\phi]\, e^{-S_E[\phi] + \int d^d x\, J(x)\phi(x)} \tag{2.1122}$$

where $J(x)$ is our current (source) of a given theory of scalars. We can impose a UV-momentum cutoff on the modes of the Fourier transformed scalar fields that are non-renormalizable by re-writting the path integral measure in the partition function as

$$Z = \int [D\phi]_{|k|<\Lambda} e^{-S_E[\phi] + \int d^D x\, J(x)\phi(x)} \tag{2.1123}$$

where the measure can be explicitly written as

$$[D\phi]_{|k|<\Lambda} \equiv \prod_{|k|<\Lambda} d\phi(k). \tag{2.1124}$$

We can then define the Wilsonian effective action as

$$e^{-S_E^{\text{eff}}[\phi;\Lambda]} = \int [D\phi]_{|k|<\Lambda} e^{-S_E[\phi]}. \tag{2.1125}$$

[107] For an information perspective on the renormalization group flow Peskin and Schroeder chapter 12.2 has a section on the Wilsonian effective action [**AC:**] pands. While for conformal field theory (CFT) the "yellow-pages" is the best resource for CFTs [**AC:**] cft.

In other words, we integrate out the higher energy modes that are above our UV-cutoff. Our path integral $Z[J]$ then only includes those modes below the UV-cutoff Λ. In this manner, both regularization and renormalization are achieved in this picture [**AC:**] pands. The cutoff removes UV-divergences from the path integral, but physical quantities can not depend on our UV cutoff Λ. So, the couplings are rescaled in the effective action in such a manner to cancel out any explicit dependence; i.e. integrating out UV modes impose a Λ-dependence on the couplings that is quantified via the beta-function,

$$\beta(g) \equiv \frac{\partial g}{\partial \ln \Lambda} = \Lambda \frac{\partial g}{\partial \Lambda}, \tag{2.1126}$$

where g is our *running coupling*. Notice the difference between typical renormalization schemes; for example, in dimensional regularization or using momentum-cutoff methods are purely computational tools. However, under the Wilsonian approach, the cutoff corresponds to a physical scale. For example, in high energy systems, the cutoff corresponds to the energy scales at which the effective action breaks down.

In the above prescription, our coupling within the beta-function turns into a *running coupling* as previously stated. To understand the renaming of the coupling, consider lowering the UV-cutoff $\Lambda \to b\Lambda$ where $b \in \mathbb{R}$ such that $b < 1$. The Fourier modes of some scalar field with momentum k, $\phi(k)$ now lie between $b\Lambda < |k| < \Lambda$, and will be integrated out in the effective action, $S_{\text{eff}}[\phi; b\Lambda]$; while the measure runs over $|k| < b\Lambda$. (Note that the source in the path integral for $|k| > b\Lambda$ is $J(k) = 0$). We then have an associated process: if we continue to reduce the UV-cutoff such that we now have $b_1\Lambda$ with $b_1 < b < 1$, from our current standpoint is equivalent to integrating out all modes above $b_2\Lambda$. Note however that the UV-modes are gone, but their effect is still found in the renormalization parameters. It is the beta-function of how these renormalizable parameters depend on the coupling as we lower the cutoff scale or equivalently integrating out more and more high-energy modes in the process.

Notice that along this flow, we are integrating out degrees of freedom (dof) going from UV→IR. This is evidence that the Wilsonian effective action is a form of *coarse graining* (i.e. the details are "smoothed out"). Of course then with any course graining process we have a loss of information that is irreversible, or for our particular scenario, an irreversible flow. Thus, there is no "backward" flow. A key point through is that this loss of information is undetectable to any low-energy observer since the renormalization prescription we have developed so that the correlation functions below the cutoff are preserved. Thus, we denote the actions arising from this process as *effective field theories*.

This is everything we need to know to understand the c-theorem. We will not need to use the Callan-Simanchik equation extensively from renormalization group flow (only when looking at Zomoldchikov's original derivation briefly), nor will we need to compute any three or four point functions as it is common among beta-function computations [108].

[108] We will be computing many two point functions however but in the context of a quantum field theory with a conformal symmetry (conformal field theory). For a great introduction an understanding of how to compute the beta-function from a given field theory we recommend the following resources: [**AC:**] collins,[**AC:**] pands,[**AC:**] schwartz,[**AC:**] textbook, and [**AC:**] weinberg text.

The main point of the this section was a qualitative understanding to help enhance our understanding of c-theorems.

A few details on conformal field theories

To understand the c-theorem, we will need to have an understanding of the basics of conformal symmetry, how it works on conjugation with quantum field theories, how to express field theories in $D = 2$ in terms of complex coordinates, and we will need an understanding of what the central charge is. Everything we build-up here is self-contained but can be found in the "yellow pages," [**AC:**] cft.

Recall that after the analysis of the renormalization group, we find the existence of nontrivial fixed point, i.e. [**AC:**] lecture notes

$$\beta_J(g_*) = 0 \tag{2.1127}$$

for all J at some non-zero g_*'s. An equivalent expression is stating that the stress-energy tensor is traceless, $T^\mu_\mu = 0$[109]. When this occurs, we have enhanced symmetries due to the Noether currents of the form $j_\mu(x) = T_{\mu\nu} \cdot \epsilon^\nu(x)$. To determine the constraint on $\epsilon^\nu(x)$, we can consider what the infinitesimal coordinate transformation is on a general metric $g_{\mu\nu}$ given the transformation is given as $x^\mu \to x^{\mu'} = x^\mu + \epsilon^\mu$. This gives

$$\delta g_{\mu\nu} = -\left(\partial^\lambda \epsilon_\mu g_{\lambda\nu} + \partial^\lambda \epsilon_\nu g_{\lambda\mu} - \epsilon^\lambda \partial_\lambda g_{\mu\nu}\right). \tag{2.1128}$$

If we want a conformal transformation around a flat metric, then set $g_{\mu\nu} \to \delta_{\mu\nu}$, where $\delta_{\mu\nu}$ is our (Euclidean) flat metric. This implies $\delta g_{\mu\nu} = 0$ giving

$$\begin{aligned} \partial^\lambda \epsilon_\mu \delta_{\lambda\nu} + \partial^\lambda \epsilon_\nu \delta_{\lambda\mu} &= \epsilon^\lambda \partial_\lambda \delta_{\mu\nu} \\ \partial_\nu \epsilon_\mu + \partial_\mu \epsilon_\nu &= \frac{2}{d}(\epsilon^\lambda \partial_\lambda)\delta_{\mu\nu}. \end{aligned} \tag{2.1129}$$

If we operate on both sides with ∂_μ then we get

$$\Box \epsilon_\nu + \left(1 - \frac{2}{d}\right) \partial_\nu(\partial_\lambda \epsilon^\lambda) = 0. \tag{2.1130}$$

Now consider operating $\Box \equiv \partial_\alpha \partial^\alpha$ on equation 2.1129 to get

$$\partial_\mu \Box \epsilon_\nu + \partial_\nu \Box \epsilon_\mu = \frac{2}{d}\delta_{\mu\nu} \Box(\partial_\lambda \epsilon^\lambda). \tag{2.1131}$$

Then if we add 2.1130 with 2.1131 we find the following constraint equation on $\epsilon^\nu(x)$

$$(\delta_{\mu\nu}\Box + (d-2)\partial_\mu \partial_\nu)\left(\partial_\lambda \epsilon^\lambda\right) = 0. \tag{2.1132}$$

Equation 2.1132 is called the conformal Killing vector field equations and ϵ^ν is the conformal Killing vector. Common examples of conformal Killing vectors that obey equation 2.1132 and what symmetries they represent are the following:

$$\epsilon^\mu(x) = \begin{cases} a^\mu & \text{translation} \\ \frac{1}{2}\left(b^{\mu\nu} - b^{\nu\mu}\right) x_\nu & \text{Lorentz} \\ c x^\mu & \text{scaling/dilatation} \\ c_\rho \left(2x^\mu x^\rho - x^2 \delta^{\mu\rho}\right) & \text{special conformal symmetry} \end{cases} \tag{2.1133}$$

[109] Up to a total derivative.

Moving towards the field theory part of conformal field theories, suppose we determined to have a fixed point in the beta function for some field theory that corresponds to a $D = 2$ conformal field theory. Then, we can express the field theory in terms of complex coordinates $z^\mu = (z^0, z^1)$ where $z^{0/1} = x \pm iy$. Suppose we change coordinate systems $z^\mu \to w^\mu(x)$. Then, the metric transforms like a typical rank-2 tensor,

$$g_{\mu\nu} \to \frac{\partial w^\mu}{\partial z^\alpha} \frac{\partial w^\nu}{\partial^\beta} g^{\alpha\beta}. \tag{2.1134}$$

Recall that for a general metric, the conformal transformation is defined as $g'_{\mu\nu}(x') = \Lambda(x) g_{\mu\nu}(x)$ with $\Lambda(x)$ some function that essentially scales the metric. Then $g'_{\mu\nu}(w) \propto g_{\mu\nu}(z)$. This can be given explicitly by the Cauchy-Rieman equation for holomorphic functions,

$$\frac{\partial w^1}{\partial z^0} = \frac{\partial w^0}{\partial z^1} \text{ and } \frac{\partial w^0}{\partial z^0} \text{ or } \frac{\partial w^1}{\partial z^0} = -\frac{\partial w^0}{\partial z^1} \text{ and } \frac{\partial w^0}{\partial z^0} = \frac{\partial w^1}{\partial z^1}. \tag{2.1135}$$

We can then use complex coordinates z and $\overline{z}$ with transition rules given as

$$z = z^0 + iz^1 \;\; z^0 = \frac{1}{2}(z + \overline{z}) \tag{2.1136}$$

$$\overline{z} = z^0 - iz^1 \;\; z^1 = \frac{1}{2i}(z - \overline{z}) \tag{2.1137}$$

$$\partial_z = \frac{1}{2}(\partial_0 - i\partial_1) \;\; \partial_0 = \partial_z + \partial_{\overline{z}} \tag{2.1138}$$

$$\partial_{\overline{z}} = \frac{1}{2}(\partial_0 + i\partial_1) \;\; \partial_1 = i(\partial_z - \partial_{\overline{z}}). \tag{2.1139}$$

The corresponding metric then becomes $g_{\mu\nu} = \frac{1}{2}\text{antidiag}(1,1)$ and $g^{\mu\nu} = 2\text{antidiag}(1,1)$. For example the antisymmetric tensor in holomorphic form is $\epsilon_{\mu\nu} = \frac{1}{2}\text{antidiag}(i,-i)$ and $\epsilon^{\mu\nu} = 2\text{antidiag}(-i,i)$.

This all sets up the basis of conformal field theory calculations, especially the structure of the metric since factors of two will be important.

This allows us to write all of our favorite field theories in terms of holomorphic coordiantes. For example, and one that will become important for us, is that of a free massless scalar. The action is [110]

$$S = \frac{1}{2}\int d^2x \, \partial_\mu\phi\partial^\mu\phi. \tag{2.1140}$$

The two point correlation function in $D = 2$ can be solved exactly. The top of the partition function becomes

$$\frac{1}{2}\int d^2x d^2y \, \phi(\mathbf{x}) A(\mathbf{x},\mathbf{y}) \phi(\mathbf{y}) \tag{2.1141}$$

where $A(\mathbf{x},\mathbf{y}) = \delta(\mathbf{x}-\mathbf{y})(-\partial^2)$. This can be solved as a Green's function denoted $K(\mathbf{x},\mathbf{y})$ where $A^{-1} = K$ and written as

$$(-\partial_x^2)K = \delta(x-y). \tag{2.1142}$$

[110] We can multiply by a renormalization factor g, but for simplicity we set $g = 1$.

We can integrate the above over x within a disk D of radius r centered around y. If we let our integration variable be r' then we get

$$1 = 2\pi \int_0^r dr'\, r' \cdot \left(-\frac{1}{r'} \frac{\partial}{\partial r'} \left(r' \frac{dK}{dr'} \right) \right) \tag{2.1143}$$

$$= 2\pi \left(-r \frac{dK}{dr} \right) \implies \tag{2.1144}$$

$$K(r) = -\frac{1}{2\pi} \ln(r). \tag{2.1145}$$

Then we have

$$\langle \phi(x)\phi(y) \rangle = -\frac{1}{4\pi} \ln(x-y)^2. \tag{2.1146}$$

We can write this in terms of holomorphic coordinates to get

$$\langle \phi(z,\overline{z})\phi(w,\overline{w}) \rangle = -\frac{1}{4\pi} \left(\ln(z-w) + \ln(\overline{z}-\overline{w}) \right). \tag{2.1147}$$

We can seperate the holomorphic from the antiholomorphic components by computing their corresponding ∂_z and ∂_w as

$$\langle \partial_z \phi(z,\overline{z}) \partial_w \phi(w,\overline{w}) \rangle = -\frac{1}{4\pi} \frac{1}{(z-w)^2} \tag{2.1148}$$

and similar for the antiholomorphic component.

Now recall that for some operators $\mathcal{O}_1(x)$ and $\mathcal{O}_2(x)$ the operator product expansion (OPE) is given as

$$\lim_{x \to y} \mathcal{O}_1(x)\mathcal{O}_2(y) = \sum_n C_n(x-y)\mathcal{O}_n(x), \tag{2.1149}$$

where C_n are Wilson coefficients. Then, we can write the OPE for equation 27 as

$$\partial\phi(z)\partial\phi(w) \sim -\frac{1}{4\pi} \frac{1}{(z-w)^2}. \tag{2.1150}$$

The energy-momentum tensor for the massless scalar is

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} \partial_\alpha \phi \partial^\alpha \phi \tag{2.1151}$$

or in terms of holomorphic coordinates

$$T(z) = -2\pi \lim_{w \to z} \left(\partial\phi(z)\partial\phi(w) - \langle \partial\phi(z)\partial\phi(w) \rangle \right) \tag{2.1152}$$

$$= -2\pi :\partial\phi\partial\phi: . \tag{2.1153}$$

Then the OPE with the operator $\partial\phi$ will be

$$T(z)\partial\phi(w) = -2\pi :\partial\phi(z)\partial\phi(z): \partial\phi(w) \tag{2.1154}$$

$$\sim -2\pi :\partial\phi(z)\partial\phi(z): \partial\phi(w) \tag{2.1155}$$

$$\sim \frac{\partial\phi(z)}{(z-w)^2} \tag{2.1156}$$

$$\sim \frac{\partial\phi(w)}{(z-w)^2} + \frac{\partial_w^2 \phi(w)}{(z-w)} \tag{2.1157}$$

where in equation 36 we expanded $\partial\phi(z)$ around $z = w$. This then allows us to see how to compute $T(z)T(w)$ which is given in general as

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{(z-w)} \tag{2.1158}$$

where c is called the *central charge*. The above case is completely general for any theory. For example, if we continued our calculation for the free scalar field, then we would find $c = 1$. The central charge is determined by the short distance behavior of the theory. Thus the central charge is somehow an extensive measure of the number of degrees of freedom of particular system.

We also need to know an important identity, that is the commutator between two operators that are an integral over space at fixed time of some field $a(z)$ or $b(w)$ given as

$$[A(z), B(w)] = \oint_0 dw \oint_w dz\, a(Z)b(w). \tag{2.1159}$$

We are now finally in a position to compute the Virasoro algebra. The conformal generators that form the Virasora algebra come from the mode expansion of the energy-momentum tensor,

$$T(z) = \sum_{n\in\mathbb{Z}} z^{-(n+2)} L_n \tag{2.1160}$$

where L_n are the generators defined as

$$L_n = \frac{1}{2\pi i}\oint dz\, z^{n+1}T(z), \tag{2.1161}$$

and similar for the antiholomorphic mode expansion. Then, the commutator of the generators becomes

$$[L_n, L_m] = \oint \frac{dzdw}{(2\pi i)^2}\left(z^{n+1}T(z)w^{m+1}T(w) - w^{m+1}T(w)z^{n+1}T(z)\right) \tag{2.1162}$$

$$= \oint \frac{dzdw}{(2\pi i)^2}\left((z^{n+1} - w^{n+1})T(w)w^{m+1}T(z)\right) \tag{2.1163}$$

$$= \oint \frac{dzdw}{(2\pi i)^2}\left((z-w)\sum_{k=0}^{n} z^{n-k}w^k T(w)w^{m+1}T(z)\right) \tag{2.1164}$$

$$= \oint \frac{dzdw}{(2\pi i)^2}\left((z-w)\sum_{k=0}^{n} z^{n-k}w^k w^{m+1}\left(\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{(z-w)}\right)\right) \tag{2.1165}$$

$$= \frac{1}{2\pi i}\oint_0 dw\, w^{m+1}\left(\frac{1}{12}c(n+1)n(n-1)w^{n-2} + 2(n+1)w^n T(w) + w^{n+1}\partial T(w)\right) \tag{2.1166}$$

$$= \frac{c}{12}n(n^2-1)\delta_{n+m,0} + 2(n+1)L_{m+n} - \frac{1}{2\pi i}\oint_0 dw(n+m+2)w^{n+m+1}T(w) \tag{2.1167}$$

$$= \frac{c}{12}n(n^2-1)\delta_{n+m,0} + (n-m)L_{m+n}, \tag{2.1168}$$

and similar for when both generators are antiholomorphic, but when one is holomorphic and the other not, then the commutator is zero.

In this section, we have developed all the machinery we will need to fully understand the derivation and physical meaning behind the c-theorem. In particular, the OPE of the stress energy tensor is extremely useful. The derivation of the Virasora algebra is also helpful since the c in the c-theorem is identified with the central charge of the field theory as well shall see in two different ways: via Zomoldchikov's original derivation and another more physically relevant derivation via spectral representation.

2.10.2 $D = 2$ C-Theorem

In this section, we re-derive in detail and with explanations Zomoldchikov's original derivation of the c-theorem, and then give another, more physically relevant, derivation of the same theorem via spectral representation of the two-point correlation functions we shall see soon.

Zomoldchikov's original derivation Our present derivation is identical to the one found in Zomoldhikov's original paper [**AC:**] Zamolodchikov:1986gt.

Suppose that we have an action $S(g, \Lambda)$ in the Euclidean signature with a local density expressed as $S = \int d^d x\, \mathcal{L}(g, \Lambda, x)$ where $\mathcal{L}$ is our Lagrangian for a field theory, Λ is our UV-cutoff, and g are our coupling constants which may be an (infinite) set of parameters $g = \{g_1, g_2, \cdots\}$. A basic assumption is that there exists a single-parameter group of motions in the space of coupling constants g denoted as Q such that under the group homomorphism transforms as $R_t Q \to Q$, such that a field theory describable by an action $S(R_t g, e^t a)$ is equivalent to the original theory with the action $S(\Lambda, g)$ in the sense that all the correlation function of the two theories agree at scales $x \gg e^t a$ $(t > 0)$. x here is our scaling parameter and t is some constant used for rescaling.

The components of the vector field that generate the renormalization group as stated previously are the beta-functions, [111]

$$\beta_i(g) = \frac{dg_i}{dt}. \tag{2.1169}$$

As we have shown in section 1.1, some of the information on the UV-behavior of the field theory is lost under renormalization transformations with $t > 0$. Then, the motion of the space Q under the influence of the renormalization group is an irreversible process.

We shall restrict ourselves to field theories in $D = 2$ and prove the following statements about the renormalization group:

1. There exists a function $c(g) \geq 0$ such that we have the following,

$$\frac{dc}{dt} \equiv \beta^i(g) \frac{\partial c(g)}{\partial g^i} \leq 0. \tag{2.1170}$$

   The equality is reached only at fixed points of the renormalization group (i.e. if g_* is a fixed point then $\beta^i(g_*) = 0$) [112].

[111] Note that the explicit form of the beta-function, a vector field, can be written explicitly as $\beta^i(g) = \Lambda(dg^i/d\Lambda)$ where Λ is our UV-cutoff scale.

[112] An example is the Wilson-Fisher fixed point in ϕ^4 theory in $d - \epsilon$ dimensions [**AC:**] schwartz.

2. The fixed points are stationary for $c(g)$; i.e. $\beta^i(g) = 0$ implies $\partial c/\partial g^i = 0$. At fixed points, the 2D field theory has an infinite conformal symmetry. The generators of the symmetry are L_n as defined in equation 40, and form the algebra in equation 47.

3. The value of $c(g)$ at the fixed point g_* is the same as the corresponding central charge given in equation 47.

To prove these statements, we will utilize the following properties of field theories: Euclidean invariance, the existence of the energy-momentum tensor $T_{\mu\nu}$ and unitarity of the field theory. Euclidean invariance, or spatial symmetries, in the local field theory lead to the existence of the symmetric energy-momentum tensor $T_{\mu\nu}(x)$ that satisfies the following conservation equation, $\partial^\mu T_{\mu\nu} = 0$. We can put this in terms of holomorphic coordinates as

$$\begin{aligned} 0 &= \partial^\mu \langle T_{\mu\nu} X\rangle \\ &= g^{\mu\alpha}\partial_\alpha \langle T_{\mu\nu} X\rangle \\ &= g^{\mu z}\partial_z \langle T_{\mu\nu} X\rangle + g^{\mu\overline{z}}\partial_{\overline{z}} \langle T_{\mu\nu} X\rangle \\ &= g^{\overline{z}z}\partial_z \langle T_{\overline{z}\nu} X\rangle + g^{z\overline{z}}\partial_{\overline{z}} \langle T_{z\nu} X\rangle \\ &= \partial_z \langle T_{\overline{z}\nu} X\rangle + \partial_{\overline{z}} \langle T_{z\nu} X\rangle \end{aligned} \tag{2.1171}$$

where X is some operator.

We can construct three correlation functions that are invariant under rotations within the space Q as

$$F(t) = z^4 \langle T(x)T(0)\rangle \tag{2.1172}$$

$$H(t) = z^3\overline{z} \langle T(x)\Theta(0)\rangle \tag{2.1173}$$

$$G(t) = z^2\overline{z}^2 \langle \Theta(x)\Theta(0)\rangle \tag{2.1174}$$

where we have defined $\Theta = \beta^i(g)\Phi_i$, $T(x) = T_{zz}$, the parameter t is $t = \log(z\overline{z})$ (scaled by our UV-cutoff Λ suppressed for now), and Φ_i are defined as the scalar local fields $\Phi_i(g, x) = \partial_i \mathcal{L}(g, \Lambda, x) \equiv (\partial/\partial g_i)\mathcal{L}$ [113]. An equivalent definition of the functions in equations 51-53 are

$$C(g) = 2z^4 \langle T(x)T(0)\rangle \tag{2.1175}$$

$$H_i(g) = z^2 x^2 \langle T(x)\Phi_i(0)\rangle \tag{2.1176}$$

$$G_{ij}(g) = x^4 \langle \Phi_i(x)\Phi_j(0)\rangle \tag{2.1177}$$

where $x^2 = z\overline{z}$.

To prove the three assertions above, we can define the following "c-function,"

$$c(g) = C(g) + 4\beta^k H_k - 6\beta^i\beta^j G_{ij}. \tag{2.1178}$$

[113] Notice this is the exact same definition when we consider a deformation of the action via $S \to S + \delta S$ where $\delta S = \int d^d x \sum_I \delta g_I \mathcal{O}_I$ for an I number of operators $\mathcal{O}$[**AC:**] lecture notes.

We need to put this in useful forms to prove the assertions, so consider letting $X = T_{zz}(0)$ in equation 50. Then we have

$$\partial_z \langle T_{\overline{z}z} T(0)\rangle + \partial_{\overline{z}} \langle T_{zz} T(0)\rangle = \partial_z \left(-\frac{H(t)}{z^3 \overline{z}}\right) + \partial_{\overline{z}} \left(\frac{F(t)}{z^4}\right) \tag{2.1179}$$

$$= \frac{3H(t)}{z^4 \overline{z}} - \frac{\dot{H}(t)}{z^3 \overline{z}} \frac{\partial t}{\partial z} + \frac{\dot{F}(t)}{z^4} \frac{\partial t}{\partial \overline{z}} \tag{2.1180}$$

$$= \frac{1}{z^4 \overline{z}} \left(\dot{F} + 3H - \dot{H}\right) \tag{2.1181}$$

$$= 0 \tag{2.1182}$$

from which we get the following differential equation $\dot{F} = -3H + \dot{H}$. (Note, we were able to commute T and T_{zz} if we pick a coordinate system appropriately such that they do so).

Recall equation 49 that we can expand the derivative as $d/dt = \beta^k \partial_k$ and we can write $H(t) = \beta^i H_i$. If we plug these into the differential equation we find the first of two renormalization group differential equations,

$$\beta^k \partial_k F = \frac{1}{2} \beta^k \partial_k C \tag{2.1183}$$

$$= -3\beta^i H_i + \beta^k \partial_k (\beta^i H_i). \tag{2.1184}$$

Now we pick $X = T_{z\overline{z}}(0) = \Theta(0)$. This will give a differential equation for H:

$$0 = \partial_z \langle T_{\overline{z}z} T_{z\overline{z}}(0)\rangle + \partial_{\overline{z}} \langle T_{zz} T_{z\overline{z}}(0)\rangle \tag{2.1185}$$

$$= \partial_z \left(\frac{G(t)}{z^2 \overline{z}^2}\right) - \partial_{\overline{z}} \left(\frac{H(t)}{z^3 \overline{z}}\right) \tag{2.1186}$$

$$= \left(-2\frac{G(t)}{z^3 \overline{z}^2} + \frac{\dot{G}(t)}{z^2 \overline{z}^2} \frac{dt}{dz}\right) - \frac{H(t)}{z^3 \overline{z}^2} + \frac{\dot{H}(t)}{z^3 \overline{z}} \frac{dt}{dz} \tag{2.1187}$$

$$= z^{-3} \overline{z}^{-2} \left(-2G + \dot{G} + H - \dot{H}\right) \tag{2.1188}$$

$$\dot{H} - H = \dot{G} - 2G. \tag{2.1189}$$

Using the same relations previously but now including $G = \beta^i \beta^j G_{ij}$ we get our second renormalization group differential equation

$$\beta^k \partial_k \left(\beta^i H_i\right) - \beta^i H_i = \beta^k \beta^k \partial_k (\beta^i \beta^j G_{ij}) - 2\beta^i \beta^j G_{ij}. \tag{2.1190}$$

Now we are in a place to start proving Zomoldchikov's assertions. Assertion one, or equation 49, can be proven in the following manner.

Starting from our definition of $c(g)$ in equation 57, first notice that we can expand equation 69 as

$$\beta^i \beta^k \partial_k G_{ij} + \beta^j (\partial_i \beta^k) G_{jk} + \beta^j (\partial_j \beta^k) G_{ik} = \beta^k \partial_k H_i + \partial_i \beta^k H_k - H_i + 2\beta^k G_{ik}. \tag{2.1191}$$

Then, if we plug this in and equation 63 into $c(g)$ while acting on $c(g)$ with $\beta^k \partial_k$ we get

$$\beta^k \partial_k c(g) = \beta^k \partial_k C(g) + \beta^k \partial_k (\beta^i H_i) - 6\beta^k \partial_k (\beta^i \beta^j G_{ij}) \tag{2.1192}$$

$$= -6\beta^i H_i + 6\beta^k \partial_k (\beta^i H_i) - 6\beta^k \partial_k (\beta^i \beta^j G_{ij}). \tag{2.1193}$$

The last term can be expanded as

$$\beta^k\partial_k(\beta^i\beta^j G_{ij}) = \beta^k(\partial_k\beta^i)\beta^j G_{ij} + \beta^k\beta^k(\partial_k\beta^j)G_{ij} + \beta^k\beta^i\beta^j\partial_k G_{ij} \tag{2.1194}$$
$$= \beta^j(\partial_j\beta^k)\beta^i G_{ki} + \beta^i\beta^j(\partial_i\beta^k)G_{jk} + \beta^k\beta^i\beta^j\partial_k G_{ij}. \tag{2.1195}$$

Notice both equation 73 and 74 all have dummy indices, so we can permute the indices in the first equation as $j \to i, i \to k$ and $k \to j$, and in the second equation as $j \to k, k \to i$ and $i \to j$. These gives the following equation

$$\beta^k(\beta^i H_i) - \beta^i H_i + 2\beta^i\beta^k G_{ik} = \beta^i\left(\beta^j(\partial_j\beta^k)G_{ki} + \beta^j(\partial_i\beta^k)G_{jk} + \beta^k\beta^j\partial_k G_{ij}\right) \tag{2.1196}$$
$$= \beta^i\left(\beta^k\partial_k H_i + \partial_i\beta^k H_k - H_i + 2\beta^k G_{ik}\right) \tag{2.1197}$$
$$= \beta^i\beta^k\partial_k H_i + \beta^i\partial_i\beta^k H_k - \beta^i H_i + 2\beta^i\beta^k G_{ik} \tag{2.1198}$$
$$= \beta^i\beta^k\partial_k H_i + \beta^k\partial_k\beta^i H_i - \beta^i H_i + 2\beta^i\beta^k G_{ik}. \tag{2.1199}$$

If we plug this back into 72, and treat G_{ij} as the metric we have

$$\beta^i\partial_i c(g) = -6\beta^i H_i + 6\beta^k\partial_k(\partial^i H_i) - 6\beta^k\partial_k(\beta^i\beta^j G_{ij}) \tag{2.1200}$$
$$= 6\beta^i H_i + 6\beta^k\partial_k(\beta^i H_i) - 6\left(\beta^k(\beta^i H_i) - \beta^i H_i + 2\beta^i\beta^k G_{ik}\right) \tag{2.1201}$$
$$= -12\beta^i\beta^k G_{ik}. \tag{2.1202}$$

Then, from the perspective of field theory, G_{ij} must be positive definite due to unitarity [114], or from the perspective of the metric, it must be positive definite. We can then instead have an inequality of sign $\leq$. Thus we have prove assertion one.

To prove assertion two, this follows directly from our end equation of proving assertion one. We can rewrite equation 81 as

$$\beta^i\left(\partial_i c(g) + 12\beta^k G_{ik}\right) = 0 \implies \tag{2.1203}$$
$$\partial^k c(g) = -12\beta^k. \tag{2.1204}$$

Notice if $\beta^k(g) = 0$ this implies that $\partial_i c(g) = 0$. Thus proving assertion two.

Finally to prove assertion three, let us evaluate $c(g)$ at a fixed point g_*. Then, $\beta^i(g_*) = 0$. This implies the $c(g)$ function can be written as

$$c(g_*) = C(g_*) \tag{2.1205}$$
$$= 2z^4\langle T(z)T(0)\rangle\Big|_{g=g_*} \qquad \text{Equation 54} \tag{2.1206}$$
$$= 2z^4\left(\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}\right)\Big|_{w=0} \qquad \text{Equation 37} \tag{2.1207}$$
$$= c + z^2 T(0) - \partial T(0) z^3 \tag{2.1208}$$
$$= 2z^4\left(\frac{c}{2}\frac{1}{z^4}\right) \qquad \text{Limit } z \to \infty \tag{2.1209}$$
$$= c \tag{2.1210}$$

[114] Otherwise if we did not impose unitarity, the correlation functions would not make sense

Thus, the c-function is directly related to the central charge, and we have proven assertion three.

Alternative (quicker) derivation of the c-theorem

The derivation provided above was in the original spirit of [**AC:**] Zamolodchikov:1986gt. However, there is a much more straightforward and simplified derivation *only* utilizing the scaling parameter $t = \ln(z\overline{z})$. Notice that we can constantly exchange $d/dt \Leftrightarrow \beta_k \partial^k$ via the chain rule and how we defined the beta-function in equation 48. However, a lot of computational hours can be saved by keeping everything in terms of t. In this manner, the c-function in equation 57 becomes

$$c(t) = C(g) + 4H(t) - 6G(t) \tag{2.1211}$$

where we can think of $H(t)$ as a vector and $G(t)$ is a (metric) tensor. Then, the prove of assertions proceed in the following manner. Recall the definitions of F, H, and G in equation 51,52 and 53. We will rewrite them to show the dependence of the mass scale μ in relation to $z\overline{z}$:

$$\langle T(z,\overline{z})T(0,0)\rangle = \frac{F(z\overline{z}\mu)}{z^4} \tag{2.1212}$$

$$\langle \Theta(z,\overline{z})T(0,0)\rangle = \frac{G(z\overline{z}\mu)}{z^3\overline{z}} \tag{2.1213}$$

$$\langle \Theta(z,\overline{z})\Theta(0,0)\rangle = \frac{H(z\overline{z}\mu)}{z^2\overline{z}^2}. \tag{2.1214}$$

Then, given $\dot{F} = z\overline{z}dF/d(z\overline{z})$ we have the following relation from equation 57 by computing $\dot{c}$,

$$\dot{c} = -\frac{3}{4}H \leq 0. \tag{2.1215}$$

Equation 94 reduces to the central charge when we place ourselves at a fixed point since $\Theta = 0$ at the fixed point. By dimensional analysis, $\langle \Theta(z,\overline{z})\Theta(0,0)\rangle \propto 1/(z^2\overline{z}^2) \implies \langle \Theta(z,\overline{z})\Theta(0,0)\rangle = (c)/(z^2\overline{z}^2)$ where c is some constant that we can call the central charge. Unitarity of the theory states that H is positive definite, giving the inequality sign. This gives all three relations.

Examples of c-functions

A quick application of the c-theorem is by calculating how the central charge changes between two fixed points in the renormalization group flow. Starting from equation 81,

and recalling that $\beta^i\partial_i = d/dt$ we have

$$\beta^i\partial_i c = 12\beta^i\beta^k G_{ik} \tag{2.1216}$$

$$\frac{dc}{dt} = -12G(t) \tag{2.1217}$$

$$-\int_{c(0)}^{c(\infty)} dc = 12\int d(\ln(z\overline{z}))\,\langle\Theta(x)\Theta(0)\rangle(z\overline{z})^2 \tag{2.1218}$$

$$= 12\int dr\,\langle\Theta(x)\Theta(0)\rangle r^4 \tag{2.1219}$$

$$= 24\int dr\,r^3\langle\Theta(x)\Theta(0)\rangle \tag{2.1220}$$

$$c(0) - c(\infty) = \Delta c = 24\int dr\,r^3\langle\Theta(x)\Theta(0)\rangle \tag{2.1221}$$

In equation 100 we integrated from an UV-fixed point and $r = z\overline{z}$.

We can then apply this to the Ising model. Let the action of the Ising model be

$$S = \int d^2x\left(\psi\partial_{\overline{z}}\psi + \overline{\psi}\partial_z\overline{\psi}\right). \tag{2.1222}$$

We now perturb the action with a mass term $-im\overline{\psi}\psi$. We can then calculate the following two point function,

$$\langle\Theta(x)\Theta(0)\rangle = \left(\frac{m^2}{2\pi}\right)^2\left(K_1^2(mr) - K_0^2(mr)\right) \tag{2.1223}$$

where K_0 and K_1 are Bessel functions. When we plug this into equation 100 we find that first $\Delta c = 1/2$ implying that $c_1 = 1/2 \implies c_2 = 0$ which is the central charge of a purely massive field theory in two dimensions. WE thus see that this mass perturbation generates an renormalization group flow to a theory which is a purely massive theory and therefore have checked explicitly that $c = 0$ for a massive theory. A clear application then of the c-theorem is the ability to calculate the values of c for other theories.

In this section, we stated, and then explicitly proved all three statements of Zomoldchikov's c-theorem. We also gave a quick example of where the c-theorem and therefore the c-function can be applied as a measure of how the central charge changes between two fixed points. However, recall that the central charge can be viewed as the number of degrees of freedom of a system that is a renormalizable quantum field theory in two dimensions (for our particular case), and in third assertion, we showed that the c-function and central charge are directly related. So, when we showed that the c-function is monotonically decreasing along the renormalization group flow, this can be viewed as coarse-graining as stated in section 1.1 (i.e. via the Wilsonian approach to the renormalization group). To make this a more rigorous fact, we can prove the c-theorem again but in Källen-Lehman or spectral representation of the two-point functions. This is what we do next.

Physical meaning of the c-theorem via Källen-Lehmann representation

Recall that the idea of a spectral representation consists in constraining the form of a two-point correlator by enforcing Poincaré invariance of the propagating intermediate states [115]. These belong to the physical Hilbert space of the theory and correspond to the representations of the Poincaré group. Let the propagation of states have an intermediate mass μ given by the propagator of a free scalar particle of the same mass. The construction of the spectral representation of a two-point correlator is first done by finding a basis of the Hilbert space of the theory (for physical states), then use it to produce an identity operator, and finally insert the identity into the correlator.

In d-dimensional spacetime with a Euclidean signature, the Hilbert space is constructed out of the operator $\Box = -\partial_\mu \partial^\mu$ with corresponding eigenfunctions,

$$\Box \left|\mathbf{p}, \mu\right\rangle = -\mu^2 \left|\mathbf{p}, \mu\right\rangle \text{ and } \phi_{\mu,\mathbf{p}}(x) = \langle x|\mathbf{p}, \mu\rangle \tag{2.1224}$$

where for brevity let $\phi_{\mathbf{p}} \equiv \phi_{\mu,\mathbf{p}}$, and we have the quantum numbers $P_\mu = (\mathbf{p}, p_d = -ip_0)$ where $p_0 = \sqrt{\mathbf{p}^2 + \mu^2}$ where we will take the positive energy values. These eigenfunctions form a complete $(d-1)$ dimensional basis of the Hilbert space with proper definitions of delta functions such that the normalization of the scalar field is given as

$$\phi_{\mathbf{p}}(x, x^d) = \frac{e^{i\mathbf{x}\cdot\mathbf{p} + x^d p_0}}{\sqrt{2p_0(2\pi)^{d-1}}}. \tag{2.1225}$$

This space gives a unitarity representation of the Poincaré group. The momentum operator is $P_\mu = -i\partial_\mu$ and the eigenvalue of the Casimir is $P^2 = -\mu^2$. The projection operator on representations of squared mass is built then as

$$\mathcal{P} = \int d^{d-1}\mathbf{p} \ \left|\mathbf{p}, \mu\right\rangle \left\langle \mathbf{p}, \mu\right| . \tag{2.1226}$$

The sum over all representations of the Poincaré group gives the identity

$$I = \int d\mu^2 \, \mathcal{P}. \tag{2.1227}$$

To see how this works, consider the two-point propagator of a free scalar particle of mass μ which can be solved by inserting equation 99,

$$\begin{aligned}
G(x, \mu) &\equiv \langle \phi(x)\phi(0)\rangle && (2.1228)\\
&= \langle \phi(x)\mathcal{P}\phi(0)\rangle && (2.1229)\\
&= \theta(x^0)\langle \phi(0, x)e^{x^d H}\mathcal{P}\phi(0)\rangle + \theta(-x^0)\langle \phi(0)e^{x^d H}\mathcal{P}\phi(0, x)\rangle && (2.1230)\\
&= \int \frac{d^d p}{(2\pi)^d} \frac{e^{ipx}}{p^2 + \mu^2}. && (2.1231)
\end{aligned}$$

[115] The majority of this derivation can be found in [AC:] Cappelli:1990yc section 2 and appendix A. Notes specifically on spectral representations can be found in [AC:] pands and [AC:] weinberg text.

To find the spectral density, consider rewriting the projection operator on Poincaré representations as

$$I = \int d\mu^2 \, \mathcal{P} \tag{2.1232}$$

$$= \int d\mu^2 \, \delta_+(P^2 + \mu^2) \tag{2.1233}$$

$$= \int d\mu^2 \int\limits_{\text{Minkowski}} d^d p \, \delta(p^2 + \mu^2) \delta^{(d)}(P - p) \theta(p_0) \tag{2.1234}$$

Let us insert this into a correlator of an arbitrary scalar field $\langle A(x) A(0) \rangle$ for $x^d < 0$,

$$\langle A(x) A(0) \rangle = \langle 0 | \, A(0) A(x) \, | 0 \rangle \tag{2.1235}$$

$$= \langle 0 | \, A(0) e^{iP \cdot x} A(0) \, | 0 \rangle \tag{2.1236}$$

$$\int d\mu^2 \; \langle 0 | \, A(0) \mathcal{P} e^{iP \cdot x} A(0) \, | 0 \rangle \tag{2.1237}$$

$$= \int d\mu^2 \int\limits_{\text{Minkowski}} d^d p \, e^{ipx} \delta(p^2 + \mu^2) \theta(p_0) \, \langle 0 | \, A(0) \delta^{(d)}(P - p) A(0) \, | 0 \rangle \, . \tag{2.1238}$$

The integral over p factorizes as

$$\int\limits_M d^d p \, e^{ipx} \delta(p^2 + \mu^2) \theta(p_0) = \int d^{d-1}\mathbf{p} \, e^{i\mathbf{p} \cdot \mathbf{x}} \int\limits_0^\infty dp_0 \, e^{p_0 x^d} \delta(-p_0^2 + \mu^2 + \mathbf{p}^2) \tag{2.1239}$$

$$= \int d^{d-1}\mathbf{p} \, e^{i\mathbf{p} \cdot \mathbf{x}} \int\limits_{-\infty}^\infty \frac{dp_d}{2\pi} e^{ip_d x^d} \frac{\theta(-x^d)}{p_d^2 + \mathbf{p}^2 + \mu^2} . \tag{2.1240}$$

We therefore have a spectral representation given the spectral measure is

$$\rho_A(\mu^2) = (2\pi)^{d-1} \, \langle 0 | \, A(0) \delta^{(d)}(P - p) A(0) \, | 0 \rangle \, \Big|_{p^2 = -\mu^2} . \tag{2.1241}$$

We can now move onto finding a spectral density representation of the c-theorem by considering the two-point correlator of the stress energy tensor. For the a massless scalar field theory, the momentum representation is given as

$$T_{\mu\nu}(p) = - \Big(g_{\mu\nu} p^2 - p_\mu p_\nu \Big) \, . \tag{2.1242}$$

Then,

$$\langle T_{\mu\nu}(x) T_{\rho\sigma}(0) \rangle = \frac{\pi}{3} \int\limits_0^\infty d\mu \, c(\mu) \int \frac{d^2 p}{(2\pi)^2} e^{ipx} \frac{(g_{\mu\nu} p^2 - p_\mu p_\nu)(g_{\rho\sigma} p^2 - p_\rho p_\sigma)}{p^2 + \mu^2} . \tag{2.1243}$$

Since we are in two dimensions, we only have one Lorentz invariant made out of p_μ with four indices that is compatible with the energy-momentum tensor. Thus, we are left with one unknown scalar function of the intermediate mass scale μ, namely the spectral density that we now denote as $c(\mu)$ (for reasons that will become apparent soon). The density also depends on a mass scale denoted Λ and dimensionless coupling g_i.

The proof of the c-theorem goes on by establishing the properties of $c(\mu)$:

1. Reflection positivity of the Euclidean field theorym i.e. unitarity of the Hilbert space implies $c(\mu) \geq 0$.

2. The dimension of $d\mu c(\mu)$ vanishes since the dimension of the stress energy tensor is 2. Then this spectral density is a dimensionless measure of degrees of freedom.

3. The form of $c(\mu)$ in a scale-invariant field theory is completely fixed by its dimensionality. There are no scale in the theory, thus we must have either $c(\mu) = c_0\delta(\mu)$ or $c(\mu) \propto 1/\mu$. Only the first behavior is allowed since the second gives an IR-divergence in the representation at $\mu = 0$.

4. Then, scale invariance is equivalent to conformal invariance in two dimensions. Consider substituting a delta function in the spectral representation of the two-point function between the trace components of the energy-momentum tensor where $\Theta(x) \equiv T^{\mu}_{\mu}$,

$$\langle \Theta(x)\Theta(0)\rangle = \frac{\pi}{3}\int_0^\infty d\mu\, c(\mu) \int \frac{d^2p}{(2\pi)^2} e^{ipx} \frac{(p^2)^2}{p^2+\mu^2} = -\frac{\pi}{3} c_0 \partial^2 \delta^{(2)}(x), \tag{2.1244}$$

which vanishes for $|x| \neq 0$. Thus the traces annihilates the vacuum, so the trace is zero, thus the theory is conformally invariant.

5. If the theory is not scale invariant the general form of $c(\mu)$ is $c(\mu) = c_0\delta(\mu) + c_1(\mu, \Lambda)$ where c_1 is supported away from $\mu = 0$ and depends on the mass scale Λ of the given theory.

6. Consider the two point function of the energy-momentum tensor that we have worked with previously (equation 51) but now written explicitly. In the limit as $z \to 0$ we have

$$\langle T_{zz}(z)T_{zz}(0)\rangle \to \frac{1}{2z^4}\int_0^\infty d\mu\, c(\mu) \tag{2.1245}$$

while in the limit $z \to \infty$ we have

$$\langle T_{zz}(z)T_{zz}(0)\rangle \to \frac{1}{2z^4}\lim_{\epsilon\to 0}\int_0^\epsilon d\mu\, c(\mu). \tag{2.1246}$$

According the the renormalization group these asymptotic limits of the massive theory are described by the UV and IR scale-invariant theories that lie at the end points of the renormalization group trajectory. Since these correlators are conformally invariant, we can identify the coefficients of $1/2z^4$ as the central charges of the UV and IR behavior,

$$c_{\text{UV}} = \int_0^\infty d\mu\, c(\mu) \text{ and } c_{\text{IR}} = \lim_{\epsilon\to 0}\int_0^\infty d\mu\, c(\mu). \tag{2.1247}$$

Then, we can write the spectral density as

$$c(\mu) = c_{\text{IR}}\delta(\mu) + c_1(\mu,\Lambda) \text{ and } c_{\text{UV}} = \int_0^\infty d\mu\, c(\mu) = c_{\text{IR}} + \int_0^\infty d\mu\, c_1(\mu). \tag{2.1248}$$

By positivity of the spectral measure $c(\mu)$, the end of the proof follows $c_{\rm UV} \geq c_{\rm IR}$. In other words, the central charge decreases for conformal theories connected by a renormalization group trajectory, and it is constant if the theory is conformal along the flow.

7. Conversely if $c_{\rm UV} = c_{\rm IR}$, then the theory is conformally invariant. If we use the wave equation satisfied by the propagtor, we can write the two-point function (equation 116) as

$$\langle \Theta(x)\Theta(0)\rangle = \frac{\pi}{3}\int_0^\infty d\mu\, \mu^4 c_1(\mu) \int \frac{d^2p}{(2\pi)^4}\frac{e^{ipx}}{p^2+\mu^2} \tag{2.1249}$$

for $|x| \neq 0$. If $c_{\rm UV} = c_{\rm IR}$, then equation 120 implies $c_1 = 0$ by positivity (unitarity). Then, this correlator vanishes for $|x| \neq 0$.

As we can see above, this approach, expressing the central charge via spectral representation, makes explicit the unitarity constraints on the field theory and implements a nice physical picture of the renormalization group flow. In other words, the shape of the density function $c(\mu)$ gives the "number" of degrees of freedom of the theory as a function of the mass scale, and it shows the decoupling of massive states in the IR limit. This gives a physical picture of the loss of degrees of freedom (and information) along the renormalization group flow. Examples of where $c(\mu)$ can be exactly computed is for the Majorana fermon [**AC:**] Cappelli:1990yc and for the bosonic field theory. For Majorana fermions we can find the $c_{\rm UV} = 1/2$ and for bosonic field theory we have $c_{\rm UV} = 1$ (as we found above).

2.10.3 Discussion of $D = 4$ or Even Dimensional C-Theorem (A-Theorem)

Obviously, we would like to extend c-theorems to dimensions higher than two, or at least for four dimensions. The first attempt was made by John Cardy in 1988 [**AC:**] Cardy:1988cwa. Cardy showed that for even dimensions, under a suitable construction, there exists a c-function which at one-loop order is descreasing along renormalization group trajectories and is stationary at renormalization group fixed points.

Then, Cardy's derivation was extended to non-abelian gauge theories by Osborn [**AC:**] Osborn:1989td.

Then, Cappelli, Friedan, and Latorre in 1990 computed the 2D and attempted to generalize to D-dimensions using the spectral representation that we gave a hint of above [**AC:**] Cappelli:1990yc. The paper also showed the relationship between the c-theorem and curved spacetime of the conformal field theory.

Then, closer to the present day, Komargodski and Adam Schwimmer have shown the relationship between spontaneously broken symmetries, namely the conformal symmetry. They find a relationship between the S-matrix and Euler anomalies in the UV and IR establishing the four dimensional c-theorem also called the a-theorem [**AC:**] Komargodski:2011vj.

3 Quantum Field Theory 3 (253c, conformal field theory)

3.1 Homework 1

3.1.1 Problem 1: Energy-Momentum Tensors and Conformal Invariance

The goal of the problem is to become familiar with the energy-momentum tensor $T_{\mu\nu}$ namely for a massless scalar and fermion fields, as well as for Maxwell theory,

$$\mathcal{L} = -\frac{1}{2}\partial^\mu\phi\partial_\mu\phi \tag{3.1}$$

$$\mathcal{L} = -\overline{\psi}\gamma^\mu\partial_\mu\psi \tag{3.2}$$

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{3.3}$$

where the field strength is $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. We will also check for each case whether or not $T_{\mu\nu}$ is traceless modulo the equation of motion.

Lets first consider the massless scalar field, and perturb the field via $\tilde{\delta}\phi(x) = -\varepsilon^\mu(x)\partial_\mu\phi(x)$ under which the action changes as

$$\begin{aligned}\tilde{\delta}S &= \int d^Dx\,[\partial^\mu\phi\partial_\mu(\varepsilon^\nu(x)\partial_\nu\phi)] \\ &= \int d^Dx\left[\partial_\mu\varepsilon^\nu\partial^\mu\phi\partial_\nu\phi + \varepsilon^\nu\partial_\nu\left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi\right)\right] \\ &= \int d^Dx\,\partial_\mu\varepsilon^\nu(x)T^\mu{}_\nu(x)\end{aligned} \tag{3.4}$$

where the energy-momentum tensor is then

$$T^\mu{}_\nu(x) = \partial^\mu\phi\partial^\nu\phi - \frac{1}{2}\delta^\mu{}_\nu\partial^\alpha\phi\partial_\alpha\phi. \tag{3.5}$$

We can further check to see if (3.5) is traceless with the knowledge that on-shell the scalar field obeys $\Box\phi = 0$,

$$\begin{aligned}T &= \eta^{\mu\nu}T_{\mu\nu} \\ &= (1 - D/2)\,(\partial\phi)^2.\end{aligned} \tag{3.6}$$

Notice then that for $D = 2$, the energy-momentum tensor of a massless scalar is traceless, but not for $D = 4$.

For the massless fermion, we can perform the same variation under translations via $\tilde{\delta}\psi(x) = -\varepsilon^\nu\partial_\nu\psi(x)$. The variation of the action gives

$$\begin{aligned}\tilde{\delta}S &= \int d^Dx\ -\gamma^\mu\partial_\mu\overline{\psi}(\delta\psi) + \text{c.c} \\ &= \int d^Dx\ -\gamma^\mu\partial_\mu\overline{\psi}\,(-\varepsilon^\nu\partial_\nu\psi(x)) + \text{c.c} \\ &= \int d^Dx\,\partial_\mu\varepsilon^\nu(x)\,\gamma^\mu\overline{\psi}\partial_\nu\psi(x) + \text{c.c.}\end{aligned} \tag{3.7}$$

The energy momentum tensor is then

$$T^{\mu\nu} = \overline{\psi}\gamma^\mu\partial^\nu\psi + \text{c.c.} \tag{3.8}$$

We can then see that if we take the trace, with the knowledge that the on-shell condition (equations of motion) for the fermion is $\gamma^\mu\partial_\mu\psi = 0$, then $T = 0$. Therefore in D-dimensions, the massless fermion is always (classically) conformally invariant.

Finally, for the (abelian) gauge field, we will use the usual formula for finding the symmetry energy-momentum tensor

$$T_{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial^\mu A_\alpha)}\partial_\nu A_\alpha - \eta_{\mu\nu}\mathcal{L}. \tag{3.9}$$

The variation of the kinetic term F^2 with respect to $\partial_\mu A_\nu$ omits $4F^{\mu\nu}$ (hence the 1/4 on the kinetic term). The energy-momentum tensor is then

$$T_{\mu\nu} = -F_{\mu\alpha}F_\nu{}^\alpha + \frac{1}{4}\eta_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \tag{3.10}$$

which we cannot reduce further since the on-shell condition here is $\partial_\mu F^{\mu\nu} = 0$. The trace is similar to that of the scalar namely

$$T = \left(\frac{D}{4} - 1\right)F_{\mu\nu}F^{\mu\nu}. \tag{3.11}$$

Therefore, in *only* $D = 4$, the free abelian gauge theory is conformally invariant. This also fits nicely with the known fact that classically, Maxwell's equations are scale invariant.

3.1.2 Problem 2: Weyl Invariance

In this problem, we are considering a free massless scalar field in D-dimensions along some *background* metric $g_{\mu\nu}$ given by the following covariant action[116]

$$S[\phi; g] = -\frac{1}{2}\int d^Dx\sqrt{-\det g}\left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + aR\phi^2\right). \tag{3.12}$$

We want to show (3.12) is invariant under the following Weyl transformation

$$g_{\mu\nu}(x) \mapsto e^{2\omega(x)}g_{\mu\nu}(x) \text{ and } \phi(x) \mapsto e^{\lambda\omega(x)}\phi(x). \tag{3.13}$$

Now, something new that many people may have not seen (if you have only taken a QFT course and not a GR course), is that the Ricci scalar under a Weyl transformation will be tricky since R itself can be expressed in terms of the Ricci tensor, which in itself is dependent upon the metric. The calculation is tedious, but if you attempt it on your own, for a general conformal transformation Ω, the Ricci tensor transforms as[117]

$$\tilde{R}_{\mu\nu} = R_{\mu\nu} - (D-2)\nabla_\mu\nabla_\nu(\ln\Omega) - g_{\mu\nu}\nabla_\rho\nabla^\rho\ln\Omega + (D-2)\nabla_\mu\ln\Omega\nabla_\nu\ln\Omega - (D-2)g_{\mu\nu}\nabla_\rho\ln\Omega\nabla^\rho\ln\Omega \tag{3.14}$$

[116] The notation here on the action $S[\phi; g]$ says that the dynamical fields ϕ are along some background (not necessarily fixed) $g_{\mu\nu}$.

[117] Note that these transformations can also be found in Polchinski's volume one of string theory.

of which we can trace over with $\Omega = e^{2\omega(x)}$ to give the transformed Ricci scalar

$$\tilde{R} = e^{-2\omega(x)} \left[R - 2(D-1)\nabla^2\omega - (D-2)(D-1)(\nabla\omega)^2 \right] \tag{3.15}$$

where $\nabla^2 = g^{\mu\nu}\nabla_\mu\nabla_\nu$. To transform the kinetic term notice that the derivative behaves as

$$\partial_\mu\tilde{\phi} = e^{\lambda\omega} \left(\partial_\mu\phi + \lambda\phi\partial_\mu\omega \right). \tag{3.16}$$

Notice that the determinant of the metric transforms as $\sqrt{-g} \to e^{D\omega}\sqrt{-g}$. The scalar kinetic term then becomes

$$\sqrt{-\tilde{g}}\tilde{g}^{\mu\nu}\partial_\mu\tilde{\phi}\partial_\nu\tilde{\phi} = \sqrt{-g}e^{\omega(D-2+2\lambda)}g^{\mu\nu} \left(\partial_\mu\phi\partial_\nu\phi + \lambda\phi(\partial_\mu\phi\partial_\nu\omega + \partial_\mu\omega\partial_\nu\phi) + +\lambda^2\phi^2\partial_\mu\omega\partial_\nu\omega \right). \tag{3.17}$$

Notice that we have an exponential infront that we can set to be *exactly* one if $\lambda = 1 - D/2$. The typical CFT notation is namely that a scalar field has a scalar dimension of $\Delta = \frac{D-2}{2}$ and thus transforming as $\phi \mapsto e^{\Delta\omega}\phi$ and therefore $\lambda = -\Delta$ (sign convention). We also have two terms that are new that we will need to get rid of. The cross-term after tracing over with $g^{\mu\nu}$ is simply $2\lambda\phi\partial^\mu\phi\partial_\mu\omega$ while second term gives $\lambda^2\phi^2(\nabla\omega)^2$

The interaction term trasnforms in a similar manner

$$\sqrt{-\tilde{g}}\tilde{R}\tilde{\phi}^2 = \sqrt{-g}e^{\omega(D-2+2\lambda)} \left(R - 2(D-1)\nabla^2\omega - (D-2)(D-1)(\nabla\omega)^2 \right) \phi^2. \tag{3.18}$$

Notice we have the same conformal factor as prior. We stll have two extra terms however that we need to deal with. However, we can group these two terms together with those leftover in the kinetic term of the scalar to find two equations we want to drop out. There are two variables, λ and a. We know that $\lambda = 1 - D/2$, and therefore

$$a = \frac{D-2}{4(D-1)}. \tag{3.19}$$

This value of a ensures conformal invariance. Notice in $D = 4$, this gives the famous $\xi = 1/6$ value.

To derive the energy-momentum tensor, we use the general formula

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S}{\delta g^{\mu\nu}}. \tag{3.20}$$

The variation of the weight i $\delta(\sqrt{-g}) = -\frac{1}{2}g_{\mu\nu}\delta g^{\mu\nu}$ with the rest being straightforward since the variation of the Ricci tensor is zero (take a look at your favorite GR book). This gives the on-shell energy momentum tensor

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - a\left(g_{\mu\nu}\Box - \partial_\mu\partial_\nu\right)\phi^2. \tag{3.21}$$

We are being sloppy about covariant and partial derivatives since I already put the covariant derivatives into partial derivatives which occurs in going to Minkowski space. To check conservation of the current, $\partial_\mu T^{\mu\nu} = 0$, notice that in Minkowski space, $\Box\phi = 0$. By observation, we can see that re-arranging the terms in the energy-momentum tensor with use of the on-shell condition gives us a (classically) conserved current.

To check to see if $T_{\mu\nu}$ is traceless, notice that when we trace over with the Minkowski metric $\eta^{\mu\nu}$, the first term is simply $(\partial\phi)^2$, while the second term is $-\frac{D}{2}(\partial\phi)^2$ while the a term is $-a(D\Box\phi^2 - \Box\phi^2)$. The trace is then

$$T = \left(1 - \frac{D}{2}\right)(\partial\phi)^2 - a(D-1)\Box\phi^2. \tag{3.22}$$

To determine $\Box\phi^2$ notice that

$$\Box\phi^2 = 2\phi\Box\phi + 2(\partial\phi)^2 = 2(\partial\phi)^2 \tag{3.23}$$

(it is just chain rule with a mixture of $\Box\phi = 0$). The trace then becomes

$$T = \left[\left(1 - \frac{D}{2}\right) - 2a(D-1)\right](\partial\phi)^2. \tag{3.24}$$

Solving for a we find that $a = (D-1)/4(D-2)$ once again.

3.1.3 Problem 3: Conformal Killing Vectors and Poincaré Algebra

The first part of this problem is to construct a conserved *traceless* energy-momentum tensor *operator* $\hat{T}_{\mu\nu}(x)$, which is a straightforward process, but extremely tedious. The Lagrangian we will be using is typical massless scalar one in (3.1). The energy-momentum tensor is the same, except that we now normal order the operators

$$\hat{T}_{\mu\nu}(x) = :\partial_\mu\hat{\phi}(x)\partial_\nu\hat{\phi}(x): - \frac{1}{4}\eta_{\mu\nu} :\partial_\alpha\hat{\phi}(x)\partial^\alpha\hat{\phi}(x): . \tag{3.25}$$

The scalar field operator along with its corresponding conjugate momenta have the following equal time commutator relations

$$\left[\hat{\phi}(\mathbf{x},t), \hat{\phi}(\mathbf{y},t)\right] = [\hat{\pi}(\mathbf{x},t), \hat{\pi}(\mathbf{y},t)] = 0 \tag{3.26}$$

$$\left[\hat{\phi}(\mathbf{x},t), \hat{\pi}(\mathbf{y},t)\right] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{3.27}$$

which arises from the commutation relations of the creation and annihilation operators

$$\left[\hat{a}(\mathbf{p}), \hat{a}^\dagger(\mathbf{k})\right] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{k}) \tag{3.28}$$

while the rest are zero.

The proper way to derive the follow commutation relations,

$$\left[\hat{D}, \hat{P}_\mu\right] = -\hat{P}_\mu \tag{3.29}$$

$$\left[\hat{D}, \hat{K}_\mu\right] = -i\hat{K}_\mu \tag{3.30}$$

$$\left[\hat{P}_\mu, \hat{K}_\nu\right] = 2i\left(\eta_{\mu\nu}\hat{D} - \hat{J}_{\mu\nu}\right), \tag{3.31}$$

is to use the equal time commutation relations stated above, but a trick is to have the commutator relations above act on some test function f (as I am sure you learned in

quantum mechanics) and go from there. After doing one the rest are trivial, and so the first one (3.29) goes as

$$\begin{aligned}\left[\hat{D}, \hat{P}_a\right] f(x) &= x^a \partial_a(\partial_b f) - \partial_b(x^a \partial_a f) \\ &= x^a(\partial_a \partial_b f) - \delta_b{}^a \partial_a f - x^a(\partial_a \partial_b f) \\ &= i(i\partial_b f) \\ &= i\hat{P}_b f \end{aligned} \tag{3.32}$$

(I felt like using Latin indices instead). The rest follow in a similar fashion.

3.1.4 Problem 4: State-Operator Correspondence and Constructing CFTs

Starting with the action (3.12) where $a = (D-2)/4(D-1)$ which in $D = 4$ becomes $a = 3/8$, but notice that $R = (D-1)(D-2)/L^2 \implies 6/L^2$ and thus the action is

$$\begin{aligned} S[\phi; g] &= -\frac{1}{2}\int d^D x \sqrt{-\det g}\left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \frac{D-2}{4(D-1)}\frac{(D-1)(D-2)}{L^2}\phi^2\right) \\ &= -\frac{1}{2}\int d^d x \sqrt{-\det g}\left((\partial\phi)^2 + m^2\phi^2\right) \end{aligned} \tag{3.33}$$

where $g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ and

$$m^2 = \left(\frac{D-2}{2L}\right)^2. \tag{3.34}$$

In $D = 4$, $m = 1$ (really nice). The equation of motion is $(\Box + m^2)\phi(x) = 0$ and thus on S^3, $\Box$ becomes simply $-\nabla^2$, the Laplacian on S^{D-1}. The Hamiltonian (for the relativistic particle on S^{D-1}) is the

$$\hat{H} = \sqrt{-\nabla^2 + m^2} = \sqrt{-\nabla^2 + 1} \tag{3.35}$$

in $D = 4$. We can have the Hamiltonian act on $\psi \in L^2(S^3)$ (where $L^2(S^{D-1})$ are all square-integrable functions on a Hilbert space in S^{D-1}). In $D = 4$, $L^2(S^3)$ is spanned by wave functions labeled by (ℓ, m) which is the same as in $D = 3$[118]. Therefore, the functions are labeled by (ℓ, m) with $\ell \in \mathbb{Z}^+$ and $m = -\ell, \ell+1, \ldots, \ell$ with $-\nabla^2 = \ell(\ell+2)$ (in $L = 1$ units for now). The Hamiltonian is then

$$\hat{H} = \sqrt{\ell(\ell+2)+1} = \ell + 1. \tag{3.36}$$

On S^3, the degeneracy for each ℓ has dimension $(\ell+1)^2$. We can then perform the trace of the 1-particle state partition function to give (restoring units)

$$\mathrm{Tr}_{\text{1-particle}}\, e^{-\beta\hat{H}} = \sum_{\ell=0}^{\infty}(\ell+1)^2 e^{-\beta(\ell+1)/L}. \tag{3.37}$$

We now want to find all linearly independent local operators (at one point) of scaling dimension $\Delta \leq 4$ for a free massless quantum scalar field theory in $D = 4$. In lecture, we saw that the *letter* partition function for D-dimensions is

$$f(q) = \mathrm{Tr}_{\text{letters}}\, q^\Delta = q^{\frac{D-2}{2}}\frac{1-q^2}{(1-q)^D} \tag{3.38}$$

[118] For a reference, and references therein, please see [24].

where the first term accounts for all "scalar" operators (i.e. any and all traced-over index operators, so $\partial_\mu\phi\partial^\mu\phi$ but not $\partial_\mu\phi\partial_\nu\phi$ which both have the same conformal weight), the numerator in the second term modulo's out operators of the form $\Box\phi$ due to the equations of motion $\Box\phi = 0$, while the denominator accounts for D-different $\partial_\mu\phi$ operators.

If we only consider scalar operators, then the only ones we have are the following.

1. $\Delta = 0$: nothing since $\Delta_\phi = (D-2)/2 = 1$.
2. $\Delta = 1$: there is only ϕ (the primary operator).
3. $\Delta = 2$ there is ϕ^2, but technically these must be written as $:\phi^2:$ otherwise they make absolutely no sense. We can take a derivative of the primary to get $\Delta_{\partial_\mu\phi} = 2$, but it would not be considered a scalar operator.
4. $\Delta = 3$: there is again only $:\phi^3:$ for scalar operators since there is no combination of derivatives that is either a total derivative or non-scalar.
5. $\Delta = 4$: we can imagine $\Box\phi^2$ but this is simply $\Box\phi^2 = \partial_\mu\phi\partial^\mu\phi$, but also any product of the previous scalar operators but primarily $:\phi^4:$.

To see the 1-to-1 correspondence directly, notice that for the ground state, $\ell = 0$ that the letter partition function can be written as

$$f(q) = q\frac{1-q^2}{(1-q)^4} \implies f(e^{-\beta}) = e^{-\beta}\frac{1-e^{-2\beta}}{(1-e^{-\beta})^4} \tag{3.39}$$

which is the closed form of

$$\sum_{\ell=0}^{\infty}(\ell+1)^2 e^{-\beta(\ell+1)} \tag{3.40}$$

in units of $L = 1$. The sum can be performed in the following manner

$$\sum_{\ell=0}^{\infty}(\ell+1)^2 e^{-\beta(\ell+1)} = x\sum_{\ell=0}^{\infty}(\ell+1)^2 x^\ell \tag{3.41}$$

where I defined $x \equiv e^{-\beta}$. Notice the $(\ell+1)^2$ term can be expanded to give three power series summations that are well known,

$$\sum_{\ell=0}^{\infty} x^\ell = \frac{1}{1-x} \tag{3.42}$$

$$\sum_{\ell=0}^{\infty} \ell x^\ell = \frac{x}{(1-x)^2} \tag{3.43}$$

$$\sum_{\ell=0}^{\infty} \ell^2 x^\ell = \frac{x(1+x)}{(1-x)^3}. \tag{3.44}$$

Therefore, after expanding out the product we get the following closed form

$$\sum_{\ell=0}^{\infty}(\ell+1)^2 e^{-\beta(\ell+1)} = \frac{e^{-\beta}(1+e^{-\beta})}{(1-e^{-\beta})^3}. \tag{3.45}$$

You may get worried that this does not match exactly what we have for $f(e^{-\beta})$, but fear not since the numerator and the denominator can be factorized to give (in $D = 4$) $f(q) = q\frac{1+q}{(1-q)^3}$ which does match our closed form summation (do the q-manipulation prior to substituting $e^{-\beta}$ which makes it easier to see). This confirms directly the 1-to-1 correspondence between states and operators.

3.2 Homework 2

3.2.1 Problem 1: Unitarity Bound

The point of this problem is to bound the conformal weight (that we will denote as Δ_{ϕ_s}) for a general, non-scalar primary rank-s symmetric traceless tensor representation of SO(d) rotation (i.e. spin-s) $\phi_{\mu_1\ldots\mu_s}$ that is symmetric in all indices with $\phi_{\mu_1\ldots\mu_s}\delta^{\mu_1\mu_2} = 0$.

To derive the unitarity bound for a spin-s primary operator which we collectively denote as ϕ_a where $a \equiv \{\mu_1, \ldots \mu_s\}$, let us first define a working norm on $\mathcal{H}_{S^{D-1}}$ as $\langle\phi_a|\phi_b\rangle = \delta_{ab}$. Then, consider the same setup of using the requirement of positive norm states on $\mathcal{H}_{S^{D-1}}$ but now with $|\phi_a\rangle$

$$\begin{aligned}\left(\hat{P}_\mu |\phi_a\rangle\right)^\dagger \hat{P}_\nu |\phi_b\rangle &= \langle\phi_a| \hat{K}_\mu \hat{P}_\nu |\phi_b\rangle \\ &= \langle\phi_a| \left[\hat{K}_\mu, \hat{P}_\nu\right] |\phi_b\rangle \\ &= -2i \langle\phi_a| \eta_{\mu\nu}\hat{D} - \hat{J}_{\mu\nu} |\phi_b\rangle \,. \end{aligned} \tag{3.46}$$

So far, everything is exactly as in lecture. However, now we must consider what the inner product is of $\langle\phi_a| \hat{J}_{\mu\nu} |\phi_b\rangle$. If we only looked at the trace as we did in lecture, then our result would only hold for $s = 0$. What we are really doing now is finding out what is the maximum eigenvalue of the matrix $\Delta \geq \text{MaxEigen}(\langle\phi_a| \hat{J}_{\mu\nu} |\phi_b\rangle)$. The operator $\hat{J}_{\mu\nu}$ encodes the information of both boosts and rotations, of which we could factorize out into two different matrices, schematically, $J \sim B \cdot R$. The boosts give us a Lorentz transformed spin-state which we don't care about at the moment, but the rotations we know from quantum mechanics as the angular momentum quantum numbers (those $J^2 = j(j+1)$). However, when operating on a spin-state that has an SO(d) symmetry, we need to count the way we normally do for accounting for all the spin-states: $s-1$, $s = 0$, and s. This gives us three *Casimir* numbers where the usual Casimir for representation s is $s(s+d-2)$ which we sum to get the correct bound. This gives us

$$\Delta_{\phi_a} \geq s + d - 2. \tag{3.47}$$

The energy-momentum tensor is a special case of this, since it is traceless, and obeys the conversation equation. $T^{\mu\nu}$ is a spin-2 quantity, giving us the immediate consequence that $\Delta_T \geq d$.

3.2.2 Problem 2: Two-Point Energy-Momentum Correlator

The goal is to completely constrain the two-point function of the energy-momentum tensor that was defined in the PSET as

$$\langle T_{\mu\nu}(x) T_{\rho\sigma}(y)\rangle \equiv G_{\mu\nu,\rho\sigma}(x,y) \tag{3.48}$$

up to a normalization constant that we will call c_T (as was in the lecture notes). Just as in lecture, we can consider how $T_{\mu\nu}(x)$ transforms under conformal transformations namely

$$\tilde{T}_{\mu\nu}(\tilde{x}) = \left(\det \frac{\partial \tilde{x}}{\partial x}\right)^{-\frac{\Delta-2}{D}} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial x^\beta}{\partial \tilde{x}^\nu} T_{\alpha\beta}(x). \tag{3.49}$$

Under scale invariance, $\tilde{x}^\mu(x) = \lambda^{-1} x^\mu$, each energy-momentum tensor transforms as $\tilde{T}_{\mu\nu}(\tilde{x}) = \lambda^d T_{\mu\nu}(\lambda^{-1}x)$. Therefore, we can write (3.48) as

$$\langle \tilde{T}_{\mu\nu}(\tilde{x}) \tilde{T}_{\rho\sigma}(0)\rangle = \lambda^{2d} \langle T_{\mu\nu}(x) T_{\rho\sigma}(0)\rangle \tag{3.50}$$

which implies the following form

$$\langle T_{\mu\nu}(x) T_{\rho\sigma}(0)\rangle = \frac{c}{|x|^{2d}} \mathcal{I}_{\mu\nu,\rho\sigma} \tag{3.51}$$

up to a constant c. We need to determine the tensor structure next in $\mathcal{I}_{\mu\nu,\rho\sigma}$.

To determine the tensor structure, we can first use the properties of $T_{\mu\nu}$. First, since $T_{\mu\nu}$ is symmetric, then $\mathcal{I}_{\mu\nu,\rho\sigma}$ is symmetric in $(\mu\nu)$ and $(\rho\sigma)$. The energy-momentum tensor is also traceless, and thus we would expect that

$$\delta^{\mu\nu} \mathcal{I}_{\mu\nu,\rho\sigma} = 0. \tag{3.52}$$

There is also the conservation of the energy-momentum tensor which is a statement about transversality implying that

$$x^\mu \mathcal{I}_{\mu\nu,\rho\sigma} = 0. \tag{3.53}$$

Now, if we want to check invariance under special conformal transformations, as we did in lecture, a quicker way is to understand that there are only two tensor structures available to us that form spin-2: the inversion tensor and the Euclidean metric $\delta_{\mu\nu}$ the former being

$$I_{\mu\nu} \equiv \delta_{\mu\nu} - \frac{2x_\mu x_\nu}{x^2}. \tag{3.54}$$

Let us write down then the most general ansatz possible for such a tensor

$$\mathcal{I}_{\mu\nu,\rho\sigma}(x) = A\delta_{\mu\nu}\delta_{\rho\sigma} + B\delta_{\mu\rho}\delta_{\nu\sigma} + C\delta_{\mu\sigma}\delta_{\nu\rho} + DI_{\mu\rho}(x)I_{\nu\sigma}(x) + EI_{\mu\sigma}(x)I_{\nu\rho}(x). \tag{3.55}$$

Since $T_{\mu\nu}$ is symmetric, then $\mu \leftrightarrow \nu \implies B = C$, and similarly for $(\rho\sigma)$. This also implies that $D = E$ since $I_{\mu\nu}$ is also symmetric.

Under the condition that $T = 0$, notice that $\delta^{\mu\nu} I_{\mu\nu} = d - 2$. Finally, under transversality, notice that $x^\mu I_{\mu\nu}(x) = -x_\nu$. These conditions gives us constraints on the remaining coefficients namely that $A = -\frac{1}{d}$, $B = 0$, and $D = 1/2$ namely

$$\mathcal{I}_{\mu\nu,\rho\sigma}(x) = \frac{1}{2}\left(I_{\mu\rho}(x)I_{\nu\sigma}(x) + I_{\mu\sigma}(x)I_{\nu\rho}(x)\right) - \frac{1}{d}\delta_{\mu\nu}\delta_{\rho\sigma}. \tag{3.56}$$

We then completely constrain the two-point function.

Now, we compute the same 2-point function but explicitly for the free massless scalar field, the free massless Dirac fermion field, and the free Maxwell theory in $D = 4$.

For the free massless scalar with action $S = -\frac{1}{2}\int d^D x\ \partial_\mu\phi\partial^\mu\phi$, we first need the Green's function which, from the equations of motion $\Box\phi = 0 \implies \Box G(x,y) = \delta^{(4)}(x-y)$ which has a solution of $G(r) = (4\pi^2)^{-1}r^{-2}$ which we can set $r^\mu \equiv x^\mu - y^\mu$ via translation invariance, allows all derivatives we will see to be in terms of r only. Also, recall that the energy momentum tensor is

$$T_{\mu\nu}(x) = \partial_\mu\phi\partial_\nu\phi(x) - \frac{1}{2}\eta_{\mu\nu}\partial^\rho\phi\partial_\rho\phi(x). \tag{3.57}$$

The two-point function then of the energy-momentum tensor is all possible wick contractions modulo contact terms. We then have

$$\begin{aligned}\langle T_{\mu\nu}(x)T_{\alpha\beta}(y)\rangle =&\langle\partial_\mu\phi\partial_\nu\phi(x) - \frac{1}{2}\eta_{\mu\nu}\partial^\rho\phi\partial_\rho\phi(x))(\partial_\alpha\phi\partial_\beta\phi(x) - \frac{1}{2}\eta_{\alpha\beta}\partial^\sigma\phi\partial_\sigma\phi(x))\rangle\\ =&(\partial_\mu\partial_\alpha G)(\partial_\nu\partial_\beta G) + (\partial_\mu\partial_\beta G)(\partial_\nu\partial_\alpha G) - \eta_{\alpha\beta}(\partial_\mu\partial^\sigma G)(\partial_\nu\partial_\sigma G)\\ &- \eta_{\mu\nu}(\partial_\alpha\partial^\rho G)(\partial_\beta\partial_\rho G) + \frac{1}{2}\eta_{\mu\nu}\eta_{\alpha\beta}(\partial_\rho\partial_\sigma G)(\partial^\rho\partial^\sigma G)\end{aligned} \tag{3.58}$$

where we used the fact that $\langle\partial_\mu\phi\partial_\nu\phi\rangle = \partial_\mu\partial_\nu\langle\phi\phi\rangle = \partial_\mu\partial_\nu G$. We then just need derivatives of the Green's functions. The first derivative is[119]

$$\partial_\mu G(r) = \frac{1}{4\pi^2}\frac{\partial}{\partial r^\mu}(r_\nu r^\nu)^{-2} = \frac{1}{4\pi^2}(-1)\frac{1}{r^4}r_\mu = -\frac{r_\mu}{4\pi^2 r^4}. \tag{3.59}$$

The second derivative is then

$$\begin{aligned}\partial_\alpha\partial_\mu G(r) &= -\frac{1}{4\pi^2}\partial_\alpha\left(\frac{r_\mu}{r^4}\right) = -\frac{1}{4\pi^2}\left(\frac{\eta_{\mu\alpha}}{r^4} + r_\mu(-1)\frac{2r^2 r_\alpha}{r^8}\right)\\ &= -\frac{1}{4\pi^2}\frac{1}{r^4}\left(\eta_{\mu\alpha} - 2\frac{r_\mu r_\alpha}{r^2}\right)\\ &= -\frac{1}{4\pi^2}\frac{1}{r^4}I_{\mu\alpha}(r)\end{aligned} \tag{3.60}$$

where we defined the inversion tensor as [**AC:**] Fix the second order derivative, there should be a factor of 4 on the second term, not 2.

$$I_{\mu\alpha}(r) \equiv \eta_{\mu\nu} - 2\frac{r_\mu r_\alpha}{r^2}. \tag{3.61}$$

We now plug-in the derivatives of the Green's functions into the 2-point correlator to find

$$\begin{aligned}\langle T_{\mu\nu}(x)T_{\alpha\beta}(y)\rangle &= \left(\frac{1}{4\pi^2 r^4}\right)^2 I_{\mu\alpha}I_{\nu\beta} - \left(\frac{1}{4\pi^2 r^4}\right)^2 I_{\mu\beta}I_{\nu\alpha} - \eta_{\alpha\beta}\left(\frac{1}{4\pi^2 r^4}\right)^2 I_\mu{}^\sigma I_{\nu\sigma}\\ &- \eta_{\mu\nu}\left(\frac{1}{4\pi^2 r^4}\right)^2 I_\alpha{}^\rho I_{\beta\rho} + \frac{1}{2}\left(\frac{1}{4\pi^2 r^4}\right)^2 \eta_{\mu\nu}\eta_{\alpha\beta}I_{\rho\sigma}I^{\rho\sigma}\\ &= \frac{1}{16\pi^4 r^8}\left(I_{\mu\alpha}I_{\nu\beta} - I_{\mu\beta}I_{\nu\alpha} - \eta_{\alpha\beta}\eta_{\mu\nu} - \eta_{\mu\nu}\eta_{\alpha\beta} + \frac{1}{2}\eta_{\mu\nu}\eta_{\alpha\beta}\cdot 4\right)\\ &= \frac{1}{16\pi^4 r^8}\left(I_{\mu\alpha}I_{\nu\beta} - I_{\mu\beta}I_{\nu\alpha}\right)\end{aligned} \tag{3.62}$$

[119] Note that $\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \frac{\partial r^\alpha}{\partial x^\mu}\frac{\partial}{\partial r^\alpha} = \delta^\alpha_\mu\frac{\partial}{\partial r^\alpha} = \frac{\partial}{\partial r^\mu}$, thus we can interchange the x-derivatives, but the y-derivatives will pick up a -1. But, since partial derivatives commute, and every term is an even products of derivatives, then we can neglect the minus since each term will pick up an overall $(-1)^2$.

3.2.3 Problem 3: Constructing Conformally Invariant Local Operators

We are working with a theory of free-massless scalar fields with an elementary operator $\phi(x)$. Part (a) of the question is simply asking us, "what are the possible local operators built out of the primary operator $\phi(x)$ that are conformal primaries?" The condition of being a conformal primary (and thus *not* a descendant) is to be an operator that is traceless and has no total derivatives. For example, suppose we wrote down the operator $:\partial_\mu\phi\partial_\nu\phi:$. The trace is then $:(\partial\phi)^2:$ which does not vanish. However, if we subtract off this term then the operator

$$\mathcal{O}^{(2,2)}_{\mu\nu} =:\partial_\mu\phi\partial_\nu\phi: -\frac{1}{d}\delta_{\mu\nu} :(\partial\phi)^2: \tag{3.63}$$

is in fact traceless, has no total derivatives (i.e. modulo $\Box\phi = 0$), and is thus a conformal primary. Let us categorize them as given in the example (3.63).

The general scaling (conformal) dimension for the primary operator $\phi(x)$ is

$$\Delta_\phi = \frac{d-2}{2}. \tag{3.64}$$

We know from lecture that each addition derivative on the primary operator adds an extra weight. However, we are also considering (when there is no index on the operators asked of us) $:\phi^2:$, and thus really we have weight $2\Delta_\phi = d-2$. If we have N-derivatives, the the general conformal weight for any spin-ℓ SO(d) representation of the desired operator is $(d-2)+N$.

Something to clarify is that conformally primary operators are annihlated by K_μ, which is equivalent to containing no total derivatives and is traceless. Enforcing this will find only symmetric-traceless towers of "even-spin" operators omitting new primary operators.

Therefore, the final classification for finding primary operators out of the square of the primary operator is the following:

1. Label by the number of total derivatives N.

2. Project to spin-ℓ irreducible representations by symmetrizing all indices and enforcing tracelessness (i.e. subtract off the trace) which schematically is

$$\mathcal{O}_{\mu_1\ldots\mu_\ell}(x) =:\partial_{(\mu_1}\ldots\partial_{\mu_k}\phi\partial_{\mu_{k+1}}\ldots\partial_{\mu_\ell}\phi: -\text{traces} \tag{3.65}$$

   where the symmetrize operation is defined as $A_{\alpha\beta} \equiv \frac{1}{2}(A_{\alpha\beta} + A_{\beta\alpha})$.

3. Remove all total derivatives (i.e. anything of the form $\phi^n\Box\phi$).

4. Only those even in ℓ should exists where ℓ is subject to $\ell \leq N$ and $\ell \equiv N \mod 2$.

5. Each primary operator produced this way will have scaling dimension $\Delta = (d-2)+N$.

As a few examples, consider labeling each primary operator as $\mathcal{O}^{(N,\ell)}_{\mu_1\ldots\mu_N}$. At $N=\ell$, we only have $:\phi^2:$ (no derivatives) and that is it (obviously spin-0 since no Lorentz indices).

At $N=1$, there are no combination of derivatives that will allow for a traceless operator.

At $N = 2$, we can use the condition $\ell \equiv \mod 2$ and start at $\ell = 0$. We could attempt such a feat by writing (i.e. products of the primary operator $\phi(x)$ with two derivatives)

$$:\partial_\mu \phi \partial^\mu \phi: \tag{3.66}$$

which does have two derivatives, no free Lorentz indices, but is it traceless, no. If we tried to remedy this, by say instead considering $\partial_\mu(\phi\partial^\mu\phi)$, we run into the same issue since subtracting off the trace is the equation of motion which vanishes. Thus, there is nothing at $(N, \ell) = (2, 0)$.

There is no odd ℓ (since it could never be traceless).

At $\ell = 2$, consider first the product $:\partial_\mu\phi\partial_\nu\phi:$. The trace is $:(\partial\phi)^2:$ where $(\partial\phi)^2 \equiv \partial_\alpha\phi\partial^\alpha\phi$. If we subtract off the trace then we have

$$\mathcal{O}^{(2,2)}_{\mu\nu} =:\partial_\mu\phi\partial_\nu\phi: -\frac{1}{d}\delta_{\mu\nu} :(\partial\phi)^2: \tag{3.67}$$

which is a valid conformal primary operator. (3.67) has scaling dimension $\Delta = d - 2 + 2 = d$[120].

This logic continues onwards.

Now, for part (b), in computing the operator product expansion (OPE) of $\phi(x)\phi(y)$ (take $y \to 0$) as a sum over the previously derived primaries and their conformal descendants.

Recall that the general structure of the OPE is

$$\phi_1(x)\phi_2(0) = \sum_i \left[C^i_{12}(x)\Phi_i(0) + \text{descendants of } \Phi_i(0)\right] \tag{3.68}$$

where $\Phi_i(0)$ are primary operators built from the primary, and a priori, the sum is over a complete basis of primaries (that we built previously), organized according to increasing scaling dimension Δ_i.

The two-point function is

$$\langle\phi(x)\phi(0)\rangle = \frac{1}{|x|^{d-2}} \tag{3.69}$$

where the scaling dimension shifted to $2\Delta_\phi = d - 2$. We can then write out schematically what the OPE would look like

$$\phi(x)\phi(0) = \frac{1}{|x|^{d-2}} + \sum_{N,\ell}^{\infty} c_{N,\ell}|x|^N \hat{x}^\ell \mathcal{O}^{(N,\ell)}_N(0) + \text{descendants}. \tag{3.70}$$

The first few terms are

$$\phi(x)\phi(0) = \frac{1}{|x|^{d-2}} + c_{0,0} :\phi^2: (0) + c_{2,2}|x|^2\hat{x}^{\mu\nu} :\partial_\mu\phi\partial_\nu\phi - \frac{1}{d}\delta_{\mu\nu}(\partial\phi)^2: (0) + \cdots + \text{descendants} \tag{3.71}$$

where $\hat{x}^{\mu\nu} = x^\mu x^\nu/x^2 - \delta^{\mu\nu}/d$ (structure constant that is traceless).

The coefficients of the OPE can be found by computing the correlation function of of $\langle\phi(x)\phi(0)\Phi_i(y)\rangle$ where $\Phi_i(y)$ are the conformal primaries used in the OPE expansion, and then read off the coefficients via matching (take $y \to 0$ limit and ensure the coefficient is then finite). In performing this process, $c_{0,0} = 1$, while $c_{2,2} = (d-2)/2(d-1)$.

[120] Note that we essentially constructed the energy-momentum tensor of the free-massless scalar without the correction term $aR\phi^2$

3.3 Homework 3

This homework set is on the 3D $O(N)$ quantum field theory written as

$$S[\phi_i, \sigma] = \int d^3x \left[\frac{1}{2} \sum_{i=1}^{N} (\partial^\mu \phi_i)(\partial_\mu \phi_i) + \frac{i}{2} \sigma \left(\sum_{i=1}^{N} \phi_i^2 - \frac{N}{g} \right) \right]. \tag{3.72}$$

For part (a), we derive the effective action which we can write in terms of the partition function

$$\int [D\sigma] e^{-S_{\text{eff}}[\sigma]} = \int [D\sigma][D\phi_i] e^{-S[\phi_i, \sigma]} \tag{3.73}$$

where we will integrate out N-ϕ_i. When we plug-in (3.72) into (3.73), first notice that the term $S \supset -\int d^3x\, \sigma \frac{iN}{2g}$ is just a constant when integrating over $[D\phi_i]$. We then have in the exponential

$$\exp\left(\frac{1}{2} \int d^3x \sum_{i=1}^{N} \left[(\partial_\mu \phi_i)^2 + \frac{i\sigma}{2} \phi_i^2 \right] \right). \tag{3.74}$$

We can integrate by parts on the kinetic term in the usual manner to get the $\int d^3x$ integral into the form

$$\int [D\phi] \exp\left(-\frac{1}{2} \int d^3x\, \phi A \phi \right) = (\det A)^{-1/2} \tag{3.75}$$

where $A \equiv -\Box + i\sigma$. We also can make use of the fact that

$$\exp\left(\sum_{i=1}^{N} \cdots \right) = \exp(\cdots)^N \tag{3.76}$$

which handles the N-summation and means we simply take the N-th power of (3.75). Note that the effective action is stuck in the exponential, and so we must take the logarithm of each side. This gives us the term $\ln \det(A) = \operatorname{Tr} \ln(A)$. This gives us the following effective action

$$S_{\text{eff}}[\sigma] = \frac{iN}{2g} \int d^3x\, \sigma(x) - \frac{N}{2} \operatorname{Tr} \left[\ln(-\Box + i\sigma) \right] \tag{3.77}$$

where $\Box \equiv \delta^{\mu\nu} \partial_\mu \partial_\nu$. Notice that the effective action is proportional to N, allowing us to perform the *large N-expansion* which is simply making a saddle-point approximation (it is the large N that allows us to perform the approximation).

In the saddle-point approximation, assume that at large N, we have a solution $\sigma(x) \to \overline{\sigma}$ namely

$$Z = \int [D\sigma] e^{-S_{\text{eff}}[\sigma]} \to e^{-S_{\text{eff}}[\sigma]} \tag{3.78}$$

where we take $N \to \infty$. $\overline{\sigma}$ is chosen such that $\delta S / \delta \sigma = 0$. Diagrammatically, the trace-log part is a 1-loop contribution to the "classical" contribution in (3.77). The variation of the first term in (3.77) evaluated as $\sigma(x) = \overline{\sigma}$ is simply $iN/2g$ while the second term is

$$-\frac{N}{2} \frac{\partial}{\partial \sigma} \operatorname{Tr}[\ln(-\Box + i\sigma)] \Big|_{\sigma(x) = \overline{\sigma}} = -\frac{N}{2} \operatorname{Tr} \left[\frac{i}{-\Box + i\overline{\sigma}} \right]. \tag{3.79}$$

We then get the following relation between the coupling g and the trace over A

$$\frac{1}{g} = \mathrm{Tr}\left[\frac{1}{-\Box + i\overline{\sigma}}\right]. \tag{3.80}$$

We can Fourier transform the trace, and to make it more recognizable, let $M^2 \equiv i\overline{\sigma}$ which then becomes

$$\frac{1}{g} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{p^2 + M^2} \tag{3.81}$$

where $-\Box = p^2$. We can perform the integral under dimensional regularization (dim-reg) with $d = 3 - \epsilon$ which can be found in the back of any QFT textbook giving

$$\int \frac{d^dp}{(2\pi)^d} \frac{1}{p^2 + M^2} = \frac{M^2}{4\pi} \tag{3.82}$$

where we have a finite integral after taking $\epsilon \to 0$. We therefore have the following relationship (keeping the positive root)

$$M = \frac{4\pi}{g} \tag{3.83}$$

or, in terms of $\overline{\sigma}$, $\overline{\sigma} = -i(4\pi/g)^2$.

For part (b), when we generalize to include a source term $\mathcal{L} \supset J_i\phi_i$, our effective action then becomes

$$S_{\text{eff}}[\sigma, J_i] = \frac{iN}{2g}\int d^3x\, \sigma(x) - \frac{N}{2}\,\mathrm{Tr}\ln(-\Box + i\sigma) - \frac{1}{2}\sum_{i=1}^{N}\int d^3x d^3y\, J_i(x)(-\Box + i\sigma)^{-1}_{xy} J_i(y) \tag{3.84}$$

where the new source term arises from the inclusion of the currents, and then re-computing the effective action Gaussian integral namely use the following identity (pulled from Wikipedia)

$$\int d^dx\, e^{-\frac{1}{2}\phi_i A\phi_i + B\phi_i} = \sqrt{\frac{(2\pi)^n}{\det A}} e^{\frac{1}{2}BA^{-1}B}. \tag{3.85}$$

The source term in (3.84) contains the propagator for $\langle\phi_i\phi_j\rangle$ with a "mass-term" $M^2 \equiv i\overline{\sigma}$ namely the Green's function for the $O(N)$ scalars is

$$\langle\phi_i(x)\phi_j(y)\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{\delta_{ij}\, e^{ip\cdot(x-y)}}{p^2 + M^2} \tag{3.86}$$

which is the two-point function in the large-N limit since the action is proportional to N.

From the loop-contribution in the effective action, we can solve for when the $O(N)$ theory is gapless, i.e. when $M = 0$. From part (a), we have the relationship $M = 4\pi/g$. Therefore, when $g \to \infty$ then $M = 0$, implying that $\overline{\sigma} = 0$. Therefore, $O(N)$ is gapless only in the limit that $g \to \infty$ at leading order in $1/N$.

For part (c), we are now working with the *critical* $O(N)$ model which si defined when the theory is gapless, which we found in part (b) to be when $g \to \infty$ or when $\langle\sigma\rangle \equiv \overline{\sigma} \to 0$. Therefore, we are in the simple regime when we expand σ as $\sigma(x) \equiv \langle\sigma\rangle + \sigma'(x)$. Recall

that the 1-loop interaction term is of the form $\mathrm{Tr}\ln(A+B)$ where $A \equiv -\Box$ and $B \equiv i\sigma'(x)$, which we can expand to higher orders as

$$\mathrm{Tr}\ln(A+B) = \mathrm{Tr}\ln(A) + \mathrm{Tr}\Big(A^{-1}B\Big) - \frac{1}{2}\mathrm{Tr}\Big(A^{-1}BA^{-1}B\Big) + \cdots \tag{3.87}$$

where A^{-1} is the massless propagator. The second term can be written as a trace over $\int d^3x\,|x\rangle\,\langle x|$ namely

$$\mathrm{Tr}\Big(A^{-1}B\Big) \propto \int d^3x\,\sigma'(x)\,\langle x|\,A^{-1}\,|x\rangle \tag{3.88}$$

which diverges, but in the gapless limit, the divergence is cancelled by $\frac{iN}{2g}\int d^3x\,\sigma'(x)$ (from the gap equation) leaving no term linear in $\sigma'(x)$ (which was the whole point in order to cancel any tadpoles).

The second order term has two insertions of the propagator and two insertions of $\sigma'(x)$ namely, at the level of the effective action at second order,

$$S^{(2)}_{\text{eff}} = \frac{N}{4}\,\mathrm{Tr}(i\sigma' G_0 i\sigma' G_0) \tag{3.89}$$

where G_0 is A^{-1}. We can Fourier transform The effective action to get

$$\begin{aligned} S^{(2)}_{\text{eff}} &= -\frac{N}{4}\int \frac{d^3p}{(2\pi)^3}\sigma'(p)\sigma'(-p)\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(k+p)^2} \\ &= \frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\sigma'(p)\Gamma^{(2)}(p)\sigma'(-p) \end{aligned} \tag{3.90}$$

where we identified the second order effective potential. The integral has a well-known solution in $d=3$ (for example see [3] Appendix B)

$$\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(k+p)^2} = \frac{1}{8\pi}\frac{1}{|p|} \tag{3.91}$$

where $|p|$ is some number. The effective potential is then $\Gamma = -N/(8\pi|p|)$. Notice that the Fourier transformed quadratic action is the propagator of σ and thus we have (in momentum space)

$$\langle\sigma'(p)\sigma'(-p)\rangle = [\Gamma^{(2)}(p)]^{-1} = -\frac{8\pi|p|}{N} \tag{3.92}$$

which shows the $1/N$ expansion, and is valid when $N\to\infty$ and $\overline{\sigma}=0$. In position space, the Fourier transform of the two-point correlation function is then

$$\langle\sigma'(x)\sigma(0)\rangle \sim \frac{1}{N}\frac{1}{|x|^4}. \tag{3.93}$$

To determine the scaling dimensions, recall that in d-dimensions, a scalar operator has conformal weight $\Delta_\phi = (d-2)/2$ and thus in $d=3$, $\Delta_\phi = 1/2$.

In a general 3D CFT, the scaling dimension for an operator $\mathcal{O}(x)$ with scaling dimension $\Delta_{\mathcal{O}}$ is of the form

$$\langle\mathcal{O}(x)\mathcal{O}(0)\rangle \sim \frac{1}{|x|^{2\Delta_{\mathcal{O}}}}. \tag{3.94}$$

If we compare with our two-point correlation function, then $\Delta_{\sigma'} = 2$.

Now, recall that in $d = 3$ for a scale invariant theory, the three-point correlation function is constrained purely by the conformal weight Δ_i up to some constant that we have determine previously before. A general three-point correlation function for some operator $\mathcal{O}(x_i)$ behaves as

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\rangle = \frac{C_{\mathcal{O}}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta_3}|x_{23}|^{\Delta_2+\Delta_3-\Delta_1}|x_{31}|^{\Delta_3+\Delta_1-\Delta_2}} \tag{3.95}$$

where $|x_{ij}| \equiv |x_i - x_j|$. Given that $\Delta_{\phi_i} = 1/2$ and $\Delta_{\sigma'} = 2$, then the three-point correlation function is

$$\langle \phi_i(x_1)\phi_j(x_2)\sigma'(x_3)\rangle = \frac{\delta_{ij}\cdot C}{|x_{23}|^2|x_{31}|^2}. \tag{3.96}$$

We can also get here by performing Wick contractions after inserting $\sigma'(x) \sim \phi_k^2(x_3)$ and then sum over k-index. Also, use the fact that $\langle \phi_i(x_1)\phi_j(x_2)\rangle \propto \delta_{ij}|x_1 - x_2|^{-1}$ in $d = 3$.

For the three-point function of $\langle \sigma^3\rangle$ (schematically written), is found in the exact same manner using (3.95). At leading N, the three-point correlation function is

$$\langle \sigma'(x_1)\sigma'(x_2)\sigma'(x_3)\rangle = \frac{C_{\sigma^3}}{|x_{12}|^2|x_{23}|^2|x_{31}|^2}. \tag{3.97}$$

The coefficient C_{σ^3} can be found by computing the 1-loop triangle-diagram of ϕ-loops with three insertions of $\sigma'\phi^2$.

Finally, extending our calculation to higher order in $1/N$ expansion $\langle \phi_i(x)\phi_j(0)\rangle$ and $\langle \sigma'(x)\sigma'(0)\rangle$ as well as derive the scaling dimensions of the operators ϕ_i and σ' for higher order in the critical $O(N)$ model (i.e. previously we found the conformal weight Δ at order $1/N^0$, and now we go to the 1-loop result at order $1/N$).

Recall that the propagator for ϕ_i is given by

$$\langle \phi_i(x)\phi_j(0)\rangle_{(0)} = \delta_{ij}\frac{C}{|x|} \tag{3.98}$$

implying that $\Delta_\phi = 1/2$ since the 2-point function scales as $|x|^{2\Delta}$. We now need to compute the $\Gamma_\phi^{(2)}$ for the 1PI two-point function of ϕ. This is just the 1-loop integral in $d = 3$ dimensions scaled by $1/N$ which we know the result being $(8/3\pi^2)N^{-1}$. The conformal weight is then shifted and is captured by the *anomalous dimension*

$$\eta = 2(\Delta_\phi - 1/2) \tag{3.99}$$

but from our 1-loop result,

$$\eta = \frac{8}{3\pi^2}\frac{1}{N} + \mathcal{O}(1/N^2). \tag{3.100}$$

The conformal dimension then in $d = 3$ is

$$\Delta_\phi = \frac{d-2}{2} + \frac{\eta}{2} = \frac{1}{2} + \frac{4}{3\pi^2}\frac{1}{N} + \cdots \tag{3.101}$$

We can perform a similar analysis for the 1-loop 1PI result for $\langle\sigma'\sigma\rangle$ namely the 1-loop result is $-32/3\pi^2$ implying the following conformal dimensions

$$\Delta_{\sigma'} = 2 - \frac{32}{3\pi^2}\frac{1}{N} + \cdots \tag{3.102}$$

For more information, please see Coleman, Jackiw, and David Politzer's work [25] and a review on the large-N expansion in [26].

3.4 Homework 4

3.4.1 Problem 1: Four-Point Correlation Function

The goal is to compute the 4-point correlation function as a sum over conformal blocks given schematically in the form

$$\langle\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle = |x_{12}|^{-2\Delta_\phi}|x_{34}|^{-2\Delta_\phi}\sum_{\Delta,s}\lambda^2_{\Delta,s}g_{\Delta,s}(u,v) \tag{3.103}$$

where for brevity, we will be denoting $\phi(x_1) \equiv \phi_1$ from now on. The entirety of this first problem can be found in [27], and therefore I will outline the general procedure for computing the 4-point correlation function via conformal blocks, while the expansion of the coefficients can be easily implemented into Mathematica.

The u and v variables in (3.103) are the conformally invariant cross ratios

$$u = \frac{|x_{12}||x_{34}|}{|x_{13}||x_{24}|} \text{ and } v = \frac{|x_{14}||x_{23}|}{|x_{13}||x_{24}|}. \tag{3.104}$$

We know that there exists a partial wave decomposition of the correlation function g

$$g(u,v) = \sum_{\mathcal{O}} f^2_{\mathcal{O}} G_{\mathcal{O}}(u,v) \tag{3.105}$$

where G are the conformal blocks of the primary operators created out of products of ϕ. The function g must satisfy the crossing relation

$$v^\Delta g(u,v) = u^\Delta g(v,u). \tag{3.106}$$

The coordinates we will be working with are the complex coordiantes z and $\overline{z}$ related to u and v as $u = z\overline{z}$ and $v = (1-z)(1-\overline{z})$. The mapping of x_{1-4} is that z is the distance between x_1 and x_2, we take x_3 to be equal to one, and $x_4 \to \infty$. Then, using Weyl transformations, we can map ϕ along the circumference of a cylinder with the "in" and "out" states being mapped to $\pm\infty$ namely on $\mathbb{R}\times S^{d-1}$. The flat space four point function with points assigned in this configuration then maps to the cylinder matrix element

$$\langle\phi|\,\phi(\tau_3,\mathbf{n}_3\phi)(\tau_2,\mathbf{n}_2)\,|\phi\rangle\,. \tag{3.107}$$

We can take $\mathbf{n}_2\cdot\mathbf{n}_3 = \cos(\theta)$ and $\theta = \arg z$.

To now determine the form of g, we can insert a complete basis of energy eigenstates into the correlator

$$g(u,v) = \sum_E |z|^E \langle\phi|\,\phi(0,\mathbf{n}_3)\,|E\rangle\,\langle E|\,\phi(0,\mathbf{n}_2\,|\phi\rangle\,. \tag{3.108}$$

The Euclidean time exponent we are used to is hidden hear, $\exp(-E(\tau_3 - \tau_2)) = |z|^E$. To determine the structure of the functions that enter the energy sum, notice that only symmetric traceless tensor multiplets of spin $j \geq 0$ can couple to the considered correlator. The right matrix element is $\langle E, \{\mu_1, \ldots, \mu_j\}| \, \phi(0, \mathbf{n}_2) \, |\phi\rangle$, must be a rank-$j$ symmetric traceless tensor constructed out of the vector $\mathbf{n}_2$ that is fixed upto a constant.

The left matrix element is fixed up to a constant, and is thus proportional to the Gegenbauer polynomials. Therefore,

$$g(u, v) = 1 + \sum p_{E,j} |z|^E C_j^\nu(\cos(\theta) \tag{3.109}$$

where $p_{E,j} \geq 0$.

If the conformal primary operators have dimension $\Delta + n$, and spin levels for level n, $j = \ell + n, \ell - 2, \ldots, \max(\ell - n, \ell + n \mod 2)$, the conformal block must have the following expansion (skipping details)

$$G_{\Delta,\ell} = \sum_{n=0}^{\infty} |z|^{\Delta+n} \sum_j A_{n,j} \frac{C_j^\nu(\cos(\theta)}{C_j^\nu(1)} \tag{3.110}$$

where $A_{n,j}$ are positive coefficients yet to be determined. In [27] a recursion is found for the coefficients.

3.4.2 Problem 2: $T(z)T(y)$-OPE

We are now working with 2-dimensional conformal field theories with a massless scalar field being described by the action

$$S = \frac{1}{2} \int d^2x \, \partial^\mu \phi \partial_\mu \phi \equiv \int d^2z \, \partial\phi\overline{\partial}\phi \tag{3.111}$$

where the notation here is $z = x^1 + ix^2$, $\overline{z} = x^1 - ix^2$, $\partial \equiv \partial_z$, $\overline{\partial} \equiv \partial_{\overline{z}}$, $d^2z \equiv 2dx^1dx^2$, and the metric is purely off-diagonal $\eta_{ab} = \text{antidiag}(1/2, 1/2)$, and $\eta^{ab} = \text{antidiag}(2, 2)$.

For part (a) of the problem, we are given the energy-momentum tensors to be

$$T = \gamma \, :\partial\phi\partial\phi: \tag{3.112}$$

$$\widetilde{T}(\overline{z}) = \widetilde{\gamma} \, :\overline{\partial}\phi\overline{\partial}\phi: \, . \tag{3.113}$$

We are asked to find γ and $\tilde{\gamma}$. First, recall that in $d = 2$, the propagator for a massless scalar is

$$G(z, w) \equiv \langle \phi(z)\phi(w) \rangle = -\ln|z - w|^2. \tag{3.114}$$

We also notice that when performing the OPE of $T(z)T(0)$, we will have Wick contractions over second order derivatives of the propagator, and thus,

$$\langle \partial_z\phi\partial_w\phi \rangle \equiv \partial_z\partial_w G(z, w) = -\partial_z \left(\frac{-1}{|z - w|} \right) = \frac{1}{(z - w)^2}. \tag{3.115}$$

The TT-OPE is then (take $w \to 0$ at the end)

$$T(z)T(w) = \gamma^2 \left(2\langle \partial_z\phi\partial_w\phi \rangle + \cdots \right) = \frac{2\gamma^2}{(z - w)^4} + \cdots \tag{3.116}$$

where the $\cdots$ denotes less-singular terms. We also know from lecture that the general form for any $2d$ CFT has a TT-OPE of the form

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \sum_{n\geq 0} z^n \Psi_n(0). \tag{3.117}$$

We can therefore compare the two and read-off and match the coefficients to find

$$2\gamma^2 = \frac{c}{2} \implies \gamma = \pm\frac{\sqrt{c}}{2} \tag{3.118}$$

but for the massless scalar, $c = 1$, and so $\gamma = \pm 1/2$. To determine the sign, we can compute another OPE, the energy-momentum tensor with that of a vertex operator which allows us to find the conformal weight, and to obey unitarity, $h \geq 0$, we can select the sign on γ then. Consider $V_k = e^{ik\phi}$ which has an OPE with $T(z)$ as

$$\begin{aligned} T(Z)V_k(w) &= \gamma\left(2\langle\partial\phi(z)ik\phi(w)\rangle :\partial\phi V_k: (w) - k^2\langle\partial\phi(z)\partial\phi(w)\rangle V_k(w)\right) \\ &= \frac{-\gamma k^2}{(z-w)^2}V_k(w) + \frac{\partial V_k(w)}{z-w} + \cdots \end{aligned} \tag{3.119}$$

from which we can see that $h = -\gamma k^2$, and to ensure $h \geq 0$, we take $\gamma = -1/2$. In a similar fashion with the exact same arguments, $\widetilde{\gamma} = -1/2$. In doing this calculation, we also verified that the TT for the massless scalar, has the universal form we quoted prior. But to be specific, the OPE for the massless scalar of TT is

$$T(z)T(W) = \gamma^2\left(:\partial\phi(z)\partial\phi(z)\partial\phi(w)\partial\phi(w): + 4\langle\partial\phi(z)\partial\phi(w)\rangle :\partial\phi(z)\partial\phi(w): + 2\langle\phi(z)\partial\phi(w)\rangle^2\right). \tag{3.120}$$

The last term is simple since it is the square of the propagator,

$$2\gamma^2\langle\partial\phi(z)\partial\phi(w)\rangle^2 = \frac{2\gamma^2}{(z-w)^4}. \tag{3.121}$$

To find the singular behavior for the normal ordered terms, we can expand $\partial\phi(z)$ around w,

$$\partial\phi(z) = \partial\phi(w) + (z-w)\partial^2\phi(w) + \cdots \tag{3.122}$$

which implies that the normal ordered product is

$$:\partial\phi(z)\partial\phi(w): = :\partial\phi(w)\partial\phi(w): + (z-w) :\partial^2\phi(w)\partial\phi(w): + \cdots \tag{3.123}$$

We can relate the normal ordered products back to $T(w)$ as

$$:\partial\phi(w)\partial\phi(w): = \frac{T(w)}{\gamma} \quad\text{and}\quad :\partial^2\phi(w)\partial\phi(w): = \frac{\partial T(w)}{2\gamma}. \tag{3.124}$$

Thus, after collecting terms we find

$$T(z)T(w) = \frac{2\gamma^2}{(z-w)^4} + \frac{4\gamma}{(z-w)^2}T(w) + \frac{2\gamma}{z-w}\partial T(w) + \text{regular} \tag{3.125}$$

which matches the universal TT-OPE.

Moving onto part (b), we are considering local operators built out of derivatives of ϕ as

$$\mathcal{O}_{n_1,\cdots,n_k;\widetilde{n}_1,\cdots\widetilde{n}_\ell} =:\partial^{n_1}\phi\cdots\partial^{n_k}\phi\overline{\partial}^{\widetilde{n}_1}\phi\cdots\overline{\partial}^{\widetilde{n}_\ell}\phi: \tag{3.126}$$

where $n_1 \geq n_2 \geq \cdots \geq n_k 1$ and similar for $\widetilde{n}_\ell$. Here, the scaling dimension is $\Delta = h+\overline{h} \leq 4$ for this problem. Since we are working with scales, we must have $h = \overline{h}$, and every n-th power derivative will contribute to the weight thus for example, $\partial^n\phi$ has weight $(h,\widetilde{h}) = (n,0)$, and similar for $\widetilde{n}$. Note that ϕ by itself is not a primary operator since any contractions with the OPE of T would generate logarithmic singular terms. We also must be careful to *not* generate "spinning" operators out of scalar fields since the spin is $s = h - \widetilde{h} \implies h = \widetilde{h}$ (and we only want $s = 0$ operators; they also wouldn't be Virasoro primaries).

Therefore, at $\Delta = 2$, we have only one possibility

$$\mathcal{O}_2 =:\partial\phi\overline{\partial}\phi: \tag{3.127}$$

which has weight $(1,1)$. To generate $\Delta = 4$, we have two possible operators to pick from, $\partial^2\phi$ and $(\partial\phi)^2$ [121]. We therefore have the following combinations all with weight $(2,2)$

$$\mathcal{O}_4^{(1)} =:\partial^2\phi\overline{\partial}\phi: \tag{3.128}$$

$$\mathcal{O}_4^{(2)} =:\partial^2\phi(\overline{\partial}\phi)^2: \tag{3.129}$$

$$\mathcal{O}_4^{(3)} =:(\partial\phi)^2\overline{\partial}^2\phi: \tag{3.130}$$

$$\mathcal{O}_4^{(4)} =:(\partial\phi)^2(\overline{\partial}\phi)^2: \tag{3.131}$$

where the last one has four-fields in it that is similar to $T\overline{T}$.

Now we will find all linear combinations of operators of the same form in (3.126) for $\Delta \leq 4$ that are primaries with respect to the Virasoro algebra. The definition for this that we will use is that any operator that is a Virasoro primary must have an OPE with $T(z)$ such that there are no z^{-4} nor z^{-3} poles present in the OPE.

There are a few things to notice however to not make this a tedious calculation. If an operator contains more than two factors of $\partial^m\phi$, then when contracting with the two-derivative terms inside of T, this will generate a z^{-4} poles in the OPE. If the operator $\mathcal{O} \supset \partial^m\phi$ such that $m \geq 2$, then a contraction of the form

$$\langle\partial\phi(z)\partial^m\phi(w)\rangle \propto \frac{(-1)^m m!}{(z-w)^{m+1}} \tag{3.132}$$

will generate a pole of z^{-3} or worse (thus also not a primary). Thus the only way to avoid these fourth and third order poles is for the operator $\mathcal{O}$ to contain only one copy of $\partial\phi$. Thus, based on our list prior, the only operator that would satisfiy being a Virasoro primary is

$$\mathcal{O}_2 =:\partial\phi\overline{\partial}\phi: \tag{3.133}$$

[121] Notice that these do not vanish by the equations of motion since in complex coordinates, the equation of motion for a free scalar is $\partial\overline{\partial}\phi = 0$.

which is the usual kinetic term.

Now lets repeat this process but including products of ϕ in $\mathcal{O}$ namely

$$\mathcal{O}_{n,m} =:\phi\partial^{n_1}\phi\cdots\partial^{n_k}\overline{\partial}^{\widetilde{n}_1}\phi\cdots\overline{\partial}^{\widetilde{n}_\ell}\phi: \tag{3.134}$$

with arbitrary $m \geq 0$. Here, we could consider terms such as

$$\langle\phi(z)\partial\phi(w)\rangle = \partial_w G(z,w) = \frac{1}{(z-w)}. \tag{3.135}$$

To facilitate this analysis, consider defining a holomorphic block P such that P contains only derivatives of ϕ and similar for $\widetilde{P}$. Then the operator in question is $\mathcal{O}_m =:\phi^m P(\partial^n\phi)\widetilde{P}(\overline{\partial}\phi):$. Let $h = \deg(P)$ and $\widetilde{h} = \deg(\widetilde{P})$. The OPE we must compute is $T(z)\mathcal{O}_m(w,\overline{w})$ to determine if there are poles greater than third order. We already know that one derivative of the propagator $G(z,w)$ generates a single pole. Thus, if we take ϕ^m with $m \geq 0$ and contract with one of the derivatives term $T \supset \partial\phi$ then the net contribution would be

$$T(z)\mathcal{O}(w) \supset \frac{m}{z-w} :\phi^{m-1}\partial\phi P\widetilde{P}: (w). \tag{3.136}$$

Initially this is great since there is no third or fourth order pole, but the $\phi^{m-1}\partial\phi$ is **not** proportional to the original operator unless $m = 0$. Thus, there is no integer $m \geq 1$ such that this new operator $\mathcal{O}_m$ can become a Virasoro primary operator *except* for $m = 0$, which was the case we were already dealing with prior.

3.5 Homework 5

3.5.1 Problem 1: Partition Function and Local Operators of Massless Scalar

For this problem, we will once again be considering the free massless scalar field in $D = 2$ with the same action we used in complex coordinates in the past homework namely $S = \int d^2z\, \partial\phi\overline{\partial}\phi$, but now we are considering it's partition function

$$Z(\tau) = \mathrm{Tr}_{\mathcal{H}} \exp\left[2\pi i\tau\left(L_0 - \frac{c}{24}\right) - 2\pi i\overline{\tau}\left(\widetilde{L}_0 - \frac{\widetilde{c}}{24}\right)\right] \tag{3.137}$$

where L_0 is the $n = 0$ Virasoro generator, c is the conformal charge, the the trace is performed over the Hilbert space $\mathcal{H}$ of local operators at the origin. Modular invariance would require that

$$Z(\tau) = Z(\tau+1) = Z(-1/\tau). \tag{3.138}$$

For the first part, we will calculate Z from local operators built out of derivatives of ϕ that we have already seen of the form (re-stated for reference)

$$\mathcal{O}_{n_1,\cdots,n_k;\widetilde{n}_1,\cdots\widetilde{n}_\ell} =:\partial^{n_1}\phi\cdots\partial^{n_k}\phi\overline{\partial}^{\widetilde{n}_1}\phi\cdots\overline{\partial}^{\widetilde{n}_\ell}\phi: . \tag{3.139}$$

We must also show that the resulting partition function of $\tau, \overline{\tau}$ is *not* modular invariant.

Firstly, given the general family of operators we are working with, notice that each derivative-operator $\partial^n\phi$ has a corresponding state via the state-operator correspondence

(mapped to how it is generated as well). Namely, suppose that we are looking at a descendent at level m, which is obtained via

$$\alpha_{-1}^{k_1}\alpha_{-2}^{k_2}\cdots|0\rangle \Leftrightarrow m = \sum_{n=1}^{\infty} nk_n. \tag{3.140}$$

Each sequence $\{k_n\}$ that solves the above equation is a partition of the integer m. We can denote the number of partitions as $p(m)$.

Now, let us think about what the generating function would be for such partitions. We can consider a single oscillator of frequency n

$$\mathrm{Tr}\Big(q^{na^\dagger a}\Big) = \sum_{k_n}^{\infty} q^{nk_n} = \frac{1}{1-q^n} \tag{3.141}$$

where $q \equiv e^{2\pi i\tau}$. But, different n's for an oscillator will act independently, and so the level $m = \sum nk_n$ will turn into a product over oscillator and such partitions,

$$\mathrm{Tr}_{\mathrm{all}}(q^{L_0}) = \prod_{n=1}^{\infty}\frac{1}{1-q^n} = \sum_{m=0}^{\infty} p(m)q^m \tag{3.142}$$

which has been known since Euler

$$\sum_{m\geq 0} p(m)q^m = \prod_{n\geq 1}(1-q^n)^{-1}. \tag{3.143}$$

How this is related to our CFT partition function is that the normal ordering L_0 (the n-products of $\alpha_{-n}^{k_n}$), it is shifted by $L_0 - \frac{1}{24}$, shifting our above answer by a product of $q^{1/24}$

$$\mathrm{Tr}\, q^{L_0-\frac{1}{24}} = q^{-1/24}\sum_{m\geq 0} p(m)q^m = q^{-1/24}\prod_{n\geq 1}(1-q^n)^{-1} = [\eta(\tau)]^{-1} \tag{3.144}$$

where we have used the Dedekind eta function. The anti-holomorphic part gives a similar answer but with $\overline{\tau}$ which will gives us a partition function for *all* derivative-type operators on the torus

$$Z_{\mathcal{O}}(\tau,\overline{\tau}) = [\eta(\tau)]^{-2}. \tag{3.145}$$

However this Z is *not* modular invariant. For the "translation" invariance the eta-function becomes $\eta(\tau+1) = e^{i\pi/12}\eta(\tau)$, and thus under a T-transformation, Z is invariant. But, for the inversion or S-transformation, the eta-function becomes $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$. Therefore, $Z \to |\tau|^{-1}Z$, which is not invariant under the full condition for having modular invariance.

Now for part (b), we will consider the theory of a *compact* free scalar with ϕ periodic $\phi + 2\pi R$ which is equivalent to ϕ namely $\phi \cong \phi + 2\pi R$. This shift symmetry is a gauge redundancy, and so our set of local operators become of the form

$$\mathcal{O}_{m|n_1,\cdots,n_k;\widetilde{n}_1,\cdots,\widetilde{n}_\ell} =: e^{im\phi/R}\partial^{n_1}\phi\cdots\partial^{n_k}\phi\overline{\partial}^{\widetilde{n}_1}\phi\cdots\overline{\partial}^{\widetilde{n}_\ell}\phi: \tag{3.146}$$

which is well-defined for any integer m. We now must calculation the contribution from these operators to the partition function Z, and again check modular invariance.

The first thing we must do is analyze how the two-point function with the insertion of these "vertex-like" operators $V_m \sim e^{im\phi/R}$, first recall that the two-point correlation function is $\langle\phi(z)\phi(0)\rangle = -(2\pi)^{-1}\ln|z|^2$. Then, the two-point function for the vertex operators is

$$\langle e^{ik\phi(z)}e^{-ik\phi(0)}\rangle = |z|^{k^2/2\pi} \tag{3.147}$$

where $k = m/R$, and we can read off that the weight is

$$h_m = \frac{k^2}{4\pi} = \frac{m^2}{4\pi R^2} \tag{3.148}$$

since the scaling behavior of the two-point function for general fields Φ is $\langle\Phi_a(z)\Phi_b(0)\rangle \sim |z|^{-2(h_a+h_b)}$. Thus, insertions of the vertex operator will shift L_0 by an amount h_m. The counting of the descendents at fixed m remains the same as in part (a). Thus, the only change is shifting $L_0 \to h_m + r$ where r are the oscillator levels. But r factorizes away from the total trace, the partition function simply becomes

$$\begin{aligned} Z_m(\tau,\overline{\tau}) &= q^{h_m}\overline{q}^{h_m}[\eta(\tau)]^{-2} \\ &= [\eta(\tau)]^{-2}\sum_{m\in\mathbb{Z}} q^{\frac{m^2}{4\pi R^2}}\overline{q}^{\frac{m^2}{4\pi R^2}} \end{aligned} \tag{3.149}$$

where in the last line, the target space is identified via $\phi \sim \phi + 2\pi R$, and thus the momentum label m must be an integer.

But, is this new partition function invariant under modular transformations? Well, under T (i.e. $\tau \to \tau + 1$), $q \to e^{2\pi i}q$ for every term in the sum of $e^{im^2/2\pi R^2}$, which, unless the integer of m matches with the factor of π's to cancel out by Euler's formula, $Z(\tau) \neq Z(\tau+1)$. Under S transformation (i.e. $\tau \to -1/\tau$), The eta-function picks up a factor of $|\tau|$, but the Poisson resummation of the Gaussian produces another Gaussian, but with a factor of R^2, and will not cancel the extra τ. Thus, this partition function is *not* invariant under modular transformations.

Moving onto part (C), we now once again work with different local operators, but we now extend the family of them even further out to include the following

$$\mathcal{O}_{m,m'|n_1,\cdots,n_k;\widetilde{n}_1,\cdots,\widetilde{n}_\ell} =: e^{im\phi/R + im'2\pi R(\phi_+ - \phi_-)}\partial^{n_1}\phi\cdots\partial^{n_k}\phi\overline{\partial}^{\widetilde{n}_1}\phi\cdots\overline{\partial}^{\widetilde{n}_\ell}\phi: \tag{3.150}$$

where ϕ_+ and ϕ_- are the holomorphic and anti-holomorphic components of ϕ with their respective OPEs

$$\phi(z,\overline{z})\phi_+(0) =: \phi(z,\overline{z})\phi_+(0): -\frac{1}{4\pi}\log(z) \tag{3.151}$$

$$\phi(z,\overline{z})\phi_-(0) =: \phi(z,\overline{z})\phi_-(0): -\frac{1}{4\pi}\log(\overline{z}). \tag{3.152}$$

With these, we need to show that the OPE between ϕ and $\mathcal{O}$ is single-valued for integer m, m' up to shifts of ϕ by multiples of $2\pi R$.

First, notice that near the orign, the expansion of $\phi_+(z)$ is

$$\phi_+(z) = \phi_+(0) + z\partial\phi_+(0) + \cdots \tag{3.153}$$

and similar for $\phi_-(z)$. Notice that when we multiply $\phi(z,\overline{z})$ by the vertex operator in $\mathcal{O}_{m,m'}$ and move fields inside the normal ordering brackets via Wick's theorem, we generate two types of terms. The first is a single contraction between ϕ_+ or ϕ_- and the exponential which generates a logarithmic singularity and thus a logarithmic branch cut as $z \to 0$. The second piece is completely regular and no contraction (i.e. they are analytic terms). Thus, we do not need to be concerened about the derivative terms in $\mathcal{O}_{m,m'}$, and only need to focus on the vertex term. Thus, we can write $\mathcal{O}_{m,m'}$ as

$$\mathcal{O}_{m,m'}(0) =:e^{i\left(\frac{m}{R}+2\pi Rm'\right)\phi_+(0)+i\left(\frac{m}{R}-2\pi Rm'\right)\phi_-(0)}: \tag{3.154}$$

where $m, m' \in \mathbb{Z}$. To clean things up, let $a \equiv m/R + 2\pi Rm'$ and $b \equiv m/R - 2\pi Rm'$. The OPE with ϕ is then

$$\begin{aligned}\phi(z,\overline{z})\mathcal{O}_{m,m'}(0) &= (\phi_+(z) + \phi_-(\overline{z})) :e^{ia\phi_+(0)+ib\phi_-(0)}: \\ &= \left(-\frac{a}{4\pi}\ln z - \frac{b}{4\pi}\ln\overline{z}\right)\mathcal{O}_{m,m'}(0) + :\phi(z,\overline{z})\mathcal{O}_{m,m'}(0): .\end{aligned} \tag{3.155}$$

To reduce this OPE further, consider what happens under the gauge $\phi \sim \phi + 2\pi R$. If we take z once around the origin, then the logarithmic term becomes $\ln z \to \ln(z) + 2\pi i$ and $\ln(\overline{z}) \to \ln(\overline{z}) - 2\pi i$. Looking only at the singular parts of the OPE, the wrapping (or *monodromy*) picks up

$$\delta = -\frac{a}{4\pi}(2\pi i) + \frac{b}{4\pi}(2\pi i) = -i(a-b)/2 = -i(2\pi Rm'). \tag{3.156}$$

The OPE then becomes

$$\phi(e^{2\pi i}z, e^{-2\pi i}\overline{z})\mathcal{O}_{m,m'}(0) = \left(\phi(z,\overline{z}) - 2\pi Rm'\right)\mathcal{O}_{m,m'}(0) + \text{regular} \tag{3.157}$$

Due to our gauge redundancy and the configuration space it creates, this shift in the OPE is gauge-trivial whenever $m' \in \mathbb{Z}$, and similar for m. Thus, for every pair of m, m', the operator $\mathcal{O}_{m,m'}$ has an OPE with ϕ that is single-valued in the compact theory. This ensures that when we widened the family of operators, then we are allowed to include them (since the OPE implies a well-defined theory by showing $\phi \sim \phi + 2\pi R$ are equivalent states/operators).

Finally for part (d), we will once again compute the partition function but for this compact free scalar that has an enlarged operator family (for the torus we are on). We have become pretty good at this, so first notice that the only change from the operator $\mathcal{O}$ compared to the rest is form of the vertex operators namely,

$$\mathcal{O}_{m,m'}(0) =:\exp\left(i\frac{m}{R}\phi(0) + 2\pi iRm'\phi_+(0) - 2\pi iRm'\phi_-(0)\right): \tag{3.158}$$

where we can split the left and right movers into their respective momenta $k_L = m/R + 2\pi Rm'$ and $k_R = m/R - 2\pi Rm'$, and with corresponding weights

$$h_{m,m'} = \frac{k_L^2}{4\pi} \text{ and } \overline{h}_{m,m'} = \frac{k_R^2}{4\pi}. \tag{3.159}$$

Thus, the trace over states (or operators) will include a shift in the weights as in part (b), and we are thus left with (where m' is now the *winding* number)

$$Z(\tau,\overline{\tau}) = [\eta(\tau)]^{-2} \sum_{m,m'\in\mathbb{Z}} q^{\frac{1}{4\pi}k_R^2} \overline{q}^{\frac{1}{4\pi}k_L^2}. \tag{3.160}$$

Under modular transformation, the T transformation sends the difference in the conformal weights h as $h_{m,m'} - \overline{h}_{m,m'} = mm' \in \mathbb{Z}$, which does not change the sum. While for the S transformation it is slightly more involved with the Poisson-resummation formula namely let $\tau_2 = \operatorname{Im}\tau > 0$. Then

$$\sum_{m\in\mathbb{Z}} e^{-\pi\frac{\tau_2}{\pi}k_R^2} e^{2\pi i\tau_1 k_R^2/4\pi} \to \frac{R}{\sqrt{\tau_2}} \sum_{n\in\mathbb{Z}} e^{-\pi\frac{R^2}{\tau_2}(n-\tau_1 2\pi R m')^2} e^{2\pi i n m'}. \tag{3.161}$$

It is this prefactor that allows us to cancel the factor from the Dedekind-eta function implying the partition function is invariant under S transformations. Thus, *this* partition function for the family of operators we were considering on a $\mathbb{Z}_2$ torus is invariant under modular transformations.

3.5.2 Problem 2: Partition Function and Local Operators of Massless Fermions

Now we will consider a free Majorana fermion in $2D$ with components $(\psi_+, \psi_-) \equiv (\psi, \widetilde{\psi})$ that obey the following OPEs

$$\psi(z)\psi(0) = :\psi(z)\psi(0): + \frac{1}{z} \tag{3.162}$$

$$\widetilde{\psi}(z)\widetilde{\psi}(0) = :\widetilde{\psi}(z)\widetilde{\psi}(0): + \frac{1}{\overline{z}} \tag{3.163}$$

$$\psi(z)\widetilde{\psi}(0) = :\psi(z)\widetilde{\psi(0)}: . \tag{3.164}$$

Since the fermiond fields are anti-commuting, the OPEs obey

$$\psi(z_1)\psi(z_2) = -\psi(z_2)\psi(z_1) \tag{3.165}$$

which implies the following relation

$$:\psi(z)\psi(0):= \sum_{n=1}^{\infty} \frac{z^n}{n!} :\partial^n\psi(0)\psi(0): . \tag{3.166}$$

For part (a), we will in essence be doing the same thing as in the first problem by computing the partition function Z via tracing over a family of operators of the form

$$\mathcal{O}_{n_1,\cdots,n_k;\widetilde{n}_1,\cdots,\widetilde{n}_\ell} = :\partial^{n_1}\psi\cdots\partial^{n_k}\psi\overline{\partial}^{\widetilde{n}_1}\widetilde{\psi}\cdots\overline{\partial}^{\widetilde{n}_\ell}\widetilde{\psi}: \tag{3.167}$$

and then verify that Z is invariant under $\tau \to -1/\tau$ but not $\tau \to \tau + 1$.

From the state-operator correspondence, n-derivatives of ψ maps onto an occupied mode that contributes $r = n + \frac{1}{2}$ to L_0 for $n \in \mathbb{N}$. Since a Majorana mode can be used *at most once*, the level-h descendants are in a 1-to-1 correspondence with partitions of

the integer h into distinct positive integers telling us how to compute the trace. The holomorphic trace is then

$$\sum_{h=0}^{\infty} q^{h-\frac{1}{48}} = q^{-1/48} \prod_{n=0}^{\infty} (1 + q^{n+\frac{1}{2}}) \equiv \chi(\tau) \tag{3.168}$$

where $q = \exp(2\pi i\tau)$ and $c = 1/2$ hence the shift in L_0 is changed to $L_0 - 1/48$ with similar counting for the anti-holomorphic sector $\widetilde{\psi}$. The partition function is then a product of $\chi(\tau)\chi^*(\overline{\tau})$, which we can re-write in terms of Jacobi-θ functions and eta-Dedekind functions (since their modular properties are well known, and thus makes are job easier)

$$\begin{aligned} Z_1(\tau, \overline{\tau}) = |\chi(\tau)|^2 &= \left| q^{-1/48} \prod_{n=0}^{\infty} (1 + q^{n+1/2}) \right|^2 \\ &= |\eta(\tau)|^{-2} |\vartheta_3(\tau)|^2 \end{aligned} \tag{3.169}$$

where we know the η-function but the Jacobi function is $\vartheta_3 = \sum_{m\in\mathbb{Z}} q^{m^2/2}$. The modularity properties of the Dedekin and Jacobi function are[122]

$$\begin{aligned} \eta(-1/\tau) &= \sqrt{-i\tau}\eta(\tau), \;\; \eta(\tau+1) = e^{i\pi/12}\eta(\tau) \\ \vartheta_3(-1/\tau) &= \sqrt{-i\tau}\vartheta_3(\tau), \;\; \vartheta_3(1+\tau) = \vartheta_4(\tau). \end{aligned} \tag{3.170}$$

Thus, under S-transformations, clearly factors cancel. However under T-transformations, Z_1 is not invariant since $\vartheta_3 \neq \vartheta_4$ and thus Z_1 is not invariant under modular transformations.

For part (b), we will now define $Z_2(\tau) \equiv Z_1(\tau+1)$ and $Z_3(\tau) \equiv Z_1(1-1/\tau)$, from which we will verify that $Z_3(\tau) = Z_3(\tau+1)$. For this, we will need more modular transformation rules, namely we will need the ones from part (a) with the $\eta(\tau)$ function but the ϑ functions transform as

$$\vartheta_4(\tau+1) = \vartheta_4(\tau), \;\; \vartheta_4(-1/\tau) = \sqrt{-i\tau}\vartheta_2(\tau), \;\; \vartheta_2(\tau+1) = e^{i\pi/4}\vartheta_2(\tau). \tag{3.171}$$

Therefore, notice that under $\tau \to \tau + 1$ Z_2 becomes

$$Z_2(\tau) = Z_1(\tau+1) = \frac{|\vartheta_3(\tau+1)|^2}{|\eta(\tau+1)|^2} = \frac{|\vartheta_4(\tau)|^2}{|\eta(\tau)|^2}. \tag{3.172}$$

For Z_3, notice that under S from part (a), $Z_1(-1/\tau) = Z_1(\tau)$ while under T $\vartheta_3(1-1/\tau) = \vartheta_4(-1/\tau)$. Thus, with the identities listed prior,

$$Z_3(\tau) = \frac{|\vartheta_2(\tau)|^2}{|\eta(\tau)|^2}. \tag{3.173}$$

Z_3 under T-transformation is invariant as well since each of the phases obtained in the number and denominator simply turn into 1 due to the $|\cdots|$ operation. Thus, $Z_3(\tau+1) = Z_3(\tau)$.

[122] For reference, $\vartheta_4(z,q) \equiv \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} e^{2niz}$.

Finally, we are onto part (c) where we define the full partition function as

$$Z(\tau) \equiv \frac{1}{2}(Z_1(\tau) + Z_2(\tau) + Z_3(\tau)) \tag{3.174}$$

which we must show is modular invariant, and that if we interpret Z as the partition function of a CFT, the spectrum of Virasoro primaries of such a putative CFT consists of precisely those of conformal weights $(0,0)$, $(1/16, 1/16)$, and $(1/2, 1/2)$.

From part (a) and (b), under T transformation $Z_1 \leftrightarrow Z2$ and $Z_3 \leftrightarrow Z_3$. Thus the sum of $Z(\tau)$ is unchanged. Under S transformations $Z_1 \to Z_1$, $Z_2 \to Z_3$, and $Z_3 \to Z_2$, and thus the sum is also invariant.

Thus, $Z(\tau)$ is invariant under modular transformation. To show the CFT content of Z, consider defining the following "characters"

$$\chi_0 = \frac{\vartheta_3 + \vartheta_4}{2\eta}, \;\; \chi_{1/2} = \frac{\vartheta_3 - \vartheta_4}{2\eta}, \;\; \chi_{1/16} = \frac{\vartheta_2}{2\eta}. \tag{3.175}$$

Under the identity $|a+b|^2 + |a-b|^2 = 2(|a|^2 + |b|^2)$, notice the following

$$|\chi_0|^2 + \left|\chi_{1/2}\right|^2 = \frac{|\vartheta_3|^2 + |\vartheta_4|^2}{2|\eta|^2} \tag{3.176}$$

$$\left|\chi_{1/16}\right|^2 = \frac{|\vartheta_2|^2}{4|\eta|^2}. \tag{3.177}$$

Thus,

$$\frac{|\vartheta_3|^2 + |\vartheta_4|^2 + |\vartheta_2|^2}{|\eta|^2} = 2\left(|\chi_0|^2 + \left|\chi_{1/2}\right|^2 + \left|\chi_{1/16}\right|^2\right). \tag{3.178}$$

This is the *Ising partition function* built from three irreducible Virasoro characters. The spectrum then contains three primaries, namely the $h = 0$ χ_0 primary, the energy ε with weight $h = 1/2$, and the spin σ with weight $h = 1/16$.

4 Quantum Field Theory 3 (253c', axions)

4.1 Introduction

4.2 Quantum Mechanics of a Particle on a Circle

4.3 U(1) Gauge Theory

4.4 Chern-Simons Terms

4.5 Non-Abelian Gauge Theory

4.6 Chiral Anomaly

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4.13 Constraints on Axion Parameter Space

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4.15 Problem Sets

4.15.1 (PSET 1) Problem 1: Particle on a Circle/Poisson Resummation

Greg Moore's TASI lectures, "Introduction to Chern-Simons Theories," §2.1 discuss the finite temperature Euclidean propagator for the quantum mechanics of a particle on a circle. Two different formulas, (2.19) and (2.23), are given, which are superficially different but asserted to be related via the Poisson summation formula. Here you will work out the details and then do the exercise Moore included.

(a) *Poisson summation formula.* This formula asserts that the sum of a (sufficiently well-behaved) function at integer values is the same as the sum of its Fourier transform at integer values, i.e.,

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{k=-\infty}^{\infty} \hat{f}(k), \text{ where } \hat{f}(k) \equiv \int_{-\infty}^{\infty} f(x) e^{-2\pi i k x} dx. \tag{4.1}$$

To see why this is true, notice that $F(x) \equiv \sum_{n=-\infty}^{\infty} f(x+n)$ is a periodic function, so it can be expanded in a Fourier series. The usual formula for the coefficients in a Fourier series leads almost directly to the desired identity; fill in the details.

Answer: Consider a function [28] $F(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ such that $f : \mathbb{R} \to \mathbb{C}$. F is a periodic function in the sense that $f(x+1) = f(x)$, while the entire sum of *all* possible values of n creates $F(x)$ [123].Since F is periodic, it is a function that

[123] f is really a function on a circle, $S^1 = \mathbb{R}/\mathbb{Z} \to \mathbb{C}$.

behaves as $F : \mathbb{R}/\mathbb{Z} \to \mathbb{C}$. Since $R/\mathbb{C} \cong U(1)$, we can break-up $F(x)$ into irreducible representation (of $U(1)$ over some parameter t),

$$F(x) = \sum_m \int_0^1 dt\, e^{-2\pi i m t} F(t) e^{2\pi i m x} \tag{4.2}$$

where

$$\hat{F}(\rho_m) = \int_0^1 dt\, e^{-2\pi i m t} F(t). \tag{4.3}$$

Notice that we can rewrite (4.2) in the following manner

$$\begin{aligned} F(x) &= \sum_m e^{2\pi i m x} \int_0^1 \sum_{m'} f(m'+t) e^{-2\pi i m t} dt \qquad && \left(F(x) = \sum_n f(x+n)\right) \\ &= \sum_m e^{2\pi i m x} \int_{-\infty}^{\infty} e^{-2\pi i m t} f(t) dt \qquad && \left(f(t) = \sum_{m'} f(m'+t)\right). \end{aligned} \tag{4.4}$$

Then, setting $x \to 0$ of (4.4) we find (after re-labeling the summation index)

$$\sum_m f(m) = \sum_{w \in \mathbb{Z}} \int_{-\infty}^{\infty} e^{-2\pi i t w} f(t) dt \tag{4.5}$$

which is the Poisson resummation formula.

(b) Use the Poisson summation formula and the general formula for a Gaussian integral to show that

$$\sum_{n=-\infty}^{\infty} e^{-An^2 - Bn - C} = \sqrt{\frac{\pi}{A}} \sum_{k=-\infty}^{\infty} e^{-\tilde{A}k^2 - \tilde{B}k^2 - \tilde{C}} \tag{4.6}$$

where the coefficents are related as

$$\tilde{A} = \frac{\pi^2}{A} \qquad \tilde{B} = -\frac{i\pi B}{A} \qquad \tilde{C} = C - \frac{B^2}{4A}. \tag{4.7}$$

This resummation trick is often useful; for instance, it appears in many contexts where n plays the role of a momentum quantum number and k plays the role of a winding quantum number.

<u>Answer:</u> Consider the following sum

$$\begin{aligned} \lim_{x \to 0} \sum_{n \in \mathbb{Z}} e^{-A(x+n)^2 - B(x+n) - C} &= \sum_{m \in \mathbb{Z}} \left(\int_{-\infty}^{\infty} dy\, e^{-Ay^2 - By - C} e^{-2\pi i y m} \right) e^{2\pi i m x} \\ &= \sqrt{\frac{\pi}{A}} \sum_{m \in \mathbb{Z}} \exp\left(\frac{B^2}{4A} - C + \frac{imB\pi}{A} - \frac{m^2\pi^2}{A} \right) \\ &= \sqrt{\frac{\pi}{A}} \sum_{m \in \mathbb{Z}} e^{-\tilde{A}m^2 - \tilde{B}m - \tilde{C}} \end{aligned} \tag{4.8}$$

where in the first line we used the Poisson resummation formula (4.5), the second line is the general Gaussian formula integration i.e. the integration over the reals of $\int_{\mathbb{R}} e^{-Ax^2-Bx-C} = \sqrt{\pi/A}\exp(B^2/4A - C)$ shifted in coefficients, and in the final line, equation (4.8), we re-label the coefficients as

$$\tilde{A} = \frac{\pi^2}{A} \qquad \tilde{B} = -\frac{i\pi B}{A} \qquad \tilde{C} = C - \frac{B^2}{4A}. \tag{4.9}$$

We then take the $x \to 0$ limit of the first line of (4.8) to get the desired result.

(c) Show that the relationship between (2.19) and (2.23) in Moore's lecture notes is precisely of this form.

Answer: Moore's (2.19) is

$$\langle\phi_1| e^{-\beta H_{\mathcal{B}}} |\phi_1\rangle = \frac{1}{2\pi}\sum_{m\in\mathbb{Z}} e^{-\frac{\beta}{2I}(m-\mathcal{B})^2 + im(\phi_1-\phi_2)}. \tag{4.10}$$

From (4.10), we can read off what A, B, and C should be

$$A = +\frac{\beta}{2I} \qquad B = -\left[i(\phi_1-\phi_2) + \frac{\mathcal{B}\beta}{I}\right] \qquad C = +\frac{\mathcal{B}^2\beta}{2I}. \tag{4.11}$$

Then, using the transformation rules given in (4.7) when performing the Poisson transformation of (4.10) (and, well, *Mathematica* to do the algebra) we find

$$\langle\phi_1| e^{-\beta H_{\mathcal{B}}} |\phi_1\rangle = \sqrt{\frac{I}{2\pi\beta}}\sum_{w\in\mathbb{Z}} e^{-\frac{2\pi^2 I}{\beta}\left(w+\frac{\phi_2-\phi_1}{2\pi}\right)^2 - 2\pi i\mathcal{B}\left(w+\frac{\phi_2-\phi_1}{2\pi}\right)} \tag{4.12}$$

which is exactly Moore's (2.23).

(d) Solve the exercise in Moore's lectures on page 15-16 i.e., for odd integer $2\mathcal{B}$ find the asymptotic expression for the matrix element in the $\beta \to \infty$ limit. (He has dropped an overall factor of $1/2\pi$, and you will find that his phase factor $\exp(i(\phi_1-\phi_2)/2)$ is a typo–it should be replaced by something $\mathcal{B}$-dependent.)

Answer: If we series expand (4.12) around $\beta \to \infty$ then directly I find

$$\langle\phi_1| e^{-\beta H_{\mathcal{B}}} |\phi_1\rangle = \sqrt{\frac{I}{2\pi\beta}} e^{-2\pi i\mathcal{B}\left(\frac{\phi_2-\phi_1}{2\pi}+w\right)} + \mathcal{O}(\beta^{-1})^{3/2}. \tag{4.13}$$

I suppose we can forget about the $\beta^{-1/2}$ term since it already suppresses the answer, and then replace the equation with an approximate symbol. But, to get the ground state energy, $e^{-\beta E}$, this is just the $\sim e^w$ term, and to get the cosine we have to manipulate the phase shift. If we let $\Delta\phi \equiv \phi_1 - \phi_2$ then we can replace the exponential containing the phase with $2\cos(\Delta\phi) = e^{i\Delta\phi} + e^{-i\Delta\phi}$ which gives us a $\sim (e^{i\Delta\phi} + 1)e^{-\beta E_{\text{ground}}}$, of which we can just drop the factor of $1/\sqrt{2\pi\beta}$ and we get the following expansion for $\beta \to \infty$

$$\langle\phi_1| e^{-\beta H_{\mathcal{B}}} |\phi_1\rangle \sim \sqrt{\frac{I}{2\pi}} e^{i(\phi_1-\phi_2)\mathcal{B}} \cos((\phi_1-\phi_2)/2) e^{-\beta E_{\text{ground}}} + \cdots \tag{4.14}$$

(e) *Optional: I don't actually know the answer–please tell me if you do!* Moore's discussion of the partition function in §2.1.2 invokes Riemann theta functions ϑ, which have nice modular properties under $SL(2,\mathbb{Z})$ transformation of τ. Modular transformations usually arise when we discuss a torus but here we are talking about a particle on a circle. Why are modular properties showing up? Is there a hidden torus in the problem?

Answer: Honestly, I don't know much about theta functions (only what I needed to get through string theory). But, based on these notes that I used for this first part of the problem [28], the Riemann ϑ-function are complex functions on $\mathcal{H}_g \times \mathbb{C}^g$ where $\mathcal{H}$ is the Siegel upper half-space. But notice in equation (D.2.2) of [28] that there is a relationship when shifting the values of $\vartheta[\delta](\zeta|\Omega)$ which oddly resembles the form of the exponentials we were working with. Otherwise, I really have no idea.

4.15.2 Problem 2: Chern-Simons terms in 2+1 dimensions

Consider a 3d U(1) gauge theory with both a Maxwell kinetic term and a Chern-Simons term

$$S = \int \left(-\frac{1}{2e^2} F \wedge \star F + \frac{k}{4\pi} A \wedge dA \right). \tag{4.15}$$

For the purposes of this problem, you can ignore the subtleties associated with defining the Chern-Simons terms for fields that are not single-valued, and find equations of motion in the usual way from the action given above.

(a) Derive the equation of motion for the gauge field, and show that it implies that the field strength obeys a massive wave equation,

$$\left(\Box + m_A^2 \right) F^{\mu\nu} = 0, \tag{4.16}$$

for some constant m_A. What is m_A in term of the parameters e and k?

Answer: If we stay with the differential forms, and not the component version of (4.15), then we are going to vary S_{Maxwell} and $S_{\text{Chern-Simons}}$. The Maxwell term when computing the variation of the action becomes

$$\begin{aligned}
\delta S_{\text{Maxwell}} - \frac{1}{2e^2} \int \delta \left(F \wedge \star F \right) \\
&= -\frac{1}{2e^2} \int \left(\delta F \wedge \star F + F \wedge \delta(\star F) \right) \\
&= -\frac{1}{2e^2} \int \left(\delta F \wedge \star F + F \wedge \star(\delta F) \right) \qquad (\delta(\star F) = \star(\delta F)) \\
&= -\frac{1}{e^2} \int \delta F \wedge \star F
\end{aligned} \tag{4.17}$$

where in the last line, recall that the inner product of forms is defined as $\int_M \omega \wedge \star\eta$ over some manifold M. One of the properties of the inner product over integration is that it is symmetry

$$\int_M \omega \wedge = \int_M \eta \wedge \star\omega \tag{4.18}$$

which is how (4.17) came to be. We can complete the variation of (4.17) by recalling that $F = dA$ and us integration by parts, $A \wedge dB = -dA \wedge B$ (assuming proper boundary conditions) which turns (4.17) into

$$\delta S_{\text{Maxwell}} = \frac{1}{e^2} \int \delta A \wedge d \star F. \tag{4.19}$$

The variation of the Chern-Simons term follows in a similar manner

$$\begin{aligned} \delta S_{\text{Chern-Simons}} &= \frac{k}{4\pi} \int \delta \left(A \wedge dA \right) \\ &= \frac{k}{4\pi} \int \left(\delta A \wedge dA + A \wedge \delta(dA) \right) \\ &= \frac{k}{2\pi} \int \delta A \wedge F. \end{aligned} \tag{4.20}$$

Then, under $\delta S = 0$, the sum of (4.19) and (4.20) gives the equation of motion

$$d \star F + \frac{ke^2}{2\pi} F = 0. \tag{4.21}$$

Notice that in D-dimensions, $[A] = (D-2)/2$ implying that the field strength has units of $D/2$. The units then of $F \wedge \star F$ in $D = 3$ is 3. This implies that the units of $[e] = 0$. Applying the same analysis to k gives $[k] = 1$. So, we can identify $m_A = ke^2/2\pi$ as an *induced mass*, but not a true physical mass. To find the form of (4.21) in a wave-equation format, consider acting on the LHS of (4.21) with the *codifferential* of the form define as [124] $\star d \star$. Then, using the Laplace-de Rham operator $\Delta = d\delta + \delta d$, we can write the codifferential of $d \star F$ as

$$\delta(d \star F) = \Delta \star F - d\delta \star F. \tag{4.22}$$

However, $\star F$ is a 1-form (since the Hodge star takes p-forms and makes them $(d-p)$-forms), thus $\star dF = 0$ (since F is a 2-form, and d sends p-forms to $(p+1)$-forms, making the Hodge star of it vanish). Then $\delta d \star F = \Delta \star F$. We also have $\delta F = \star d \star F$ and so from (4.21) $\delta F = -m_A \star F$. Putting the pieces together we then get

$$(\Delta - m_A^2) F = 0 \tag{4.23}$$

where we did have $\star F$ but this is just the dual of F, so we can replace them allowing us to generalize to (4.23). Then, in index notation, for some p-form ω_{μ_p}, $(\Delta\omega)_{\mu_1 \cdots \mu_p} = -\Box \omega_{\mu_1 \cdots \mu_p}$ where $\Box$ is the d'Alembertian operator (in Minkowski space). With the definition of a 2-form $F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$ we can write the Laplace-de Rham on F as $(\Delta F)_{\mu\nu} = -\Box F_{\mu\nu}$. We can then write (4.23) as

$$(\Box + m_A^2) F_{\mu\nu} = 0. \tag{4.24}$$

[124] We use the recommended textbook, "Differential Forms in Algebraic Topology," by Raoul Bott and Loring W. Tu

(b) How many independent, propagating polarization states does a massless gauge field (with $k = 0$) have in 2+1 dimensions? What about the gauge field with a Chern-Simons mass, $k \neq 0$? Explain your answers, and comment on how this differs from the familiar $3 + 1$ dimensional case.

Answer: To find the physical polarization modes, for $m_A = 0$ (or equivalently $k = 0$) we have a massless wave equation in $d = 3$. The gauge field A_μ has 3 degrees of freedom (dof). Via the equation of motion, we get the constraint equation $\partial_\mu F^{\mu\nu} = 0$ eliminating 1 dof. We also have gauge invariance under $A_\mu \to A_\mu + \partial_\mu \alpha(x)$ eliminating another degree of freedom giving us a total of 1 propagating degree of freedom the photon.

Now lets turn on the Chern-Simons term via $k \neq 0$. The Chern-Simons term is gauge-invariant, and therefore does not add any new constraint equation. Naively, if we treated this as an actual mass term, then from $SO(2)$ group the states would form, we would expect 2 degrees of freedom, but this is **not** the case. Due to the Chern-Simons term being gauge invariant, the spectrum is still $SO(1)$, and we still only have 1 propagating degree of freedom, it just so happens that the photon now picks up an induced mass, and as $e \to \infty$, it becomes infinitely heavy, and no modes of propagation can occur.

(c) Suppose that we compactify the theory on a circle $x^3 \cong x^3 + 2\pi L$, and consider an ansatz where we turn on a constant $A_3 \neq 0$ so that

$$\varphi \equiv \int_0^{2\pi L} dx^3 \, A_3 \neq 0. \tag{4.25}$$

Show that in the diensionally-reduced 1+1 dimensional theory, the massless gauge field has a 2d theta term

$$\frac{\theta}{2\pi} \int F. \tag{4.26}$$

How is θ related to φ? Explain how the periodicity of the 2d coefficient θ is compatible with the 3d origin of the term.

Answer: To dimensionally reduce the action (4.15), we can consider the following ansatz where $A_3 \neq 0$

$$\frac{\varphi}{2\pi L} = A_3 \tag{4.27}$$

(or written as $A = \varphi dx^3/(2\pi L)$). Writing the Maxwell part of the action in terms of indices, we can write out the components and then integrate out dx^3. Namely,

$$S_{\text{Maxwell}} = -\frac{1}{4e^2} \int d^3x \left((F_{01})^2 + 2(F_{03})^2 + 2(F_{13})^2 \right) \tag{4.28}$$

where $F_{01} = \partial_0 A_1 - \partial_1 A_0$, $F_{03} = \partial_0 A_3$, and $F_{13} = \partial_1 A_3$ and any terms of the form $\partial_3 A_\mu$ vanish since we are assuming independence of x^3.

In a similar manner the Chern-Simons term becomes

$$
\begin{aligned}
S_{\text{Chern-Simons}} &= \frac{k}{4\pi}\int d^3x\, \epsilon^{\mu\nu\alpha} A_\mu \partial_\nu A_\alpha \\
&= \frac{k}{4\pi}\int d^3x \left(\epsilon^{013} A_0 \partial_1 A_3 + \epsilon^{130} A_1 \partial_3 A_0 + \epsilon^{301} A_3 \partial_0 A_1\right) \\
&= \frac{k}{4\pi}\int d^3x \left(A_0 \partial_1 A_3 + A_3 \partial_0 A_1\right) && (\epsilon^{013} = 1 \text{ and } \partial_3 A_\mu = 0) \\
&= \frac{k}{4\pi}\int d^3x \left(A_0 \partial_1 A_3 - A_1 \partial_0 A_3 + A_3(\partial_0 A_1 - \partial_1 A_0)\right) && \left(\int -\text{by parts}\right) \\
&= \frac{k}{2\pi}\int d^3x\, A_3 F_{01} \\
&= \frac{k}{2\pi}\int d^3x \left(\frac{\varphi}{2\pi L}\right) F_{01} && (A_3 = \varphi/2\pi L) \\
&= \frac{k}{2\pi}\int dx^2\, F_{01} && (\varphi/2\pi L = A_3) \\
&= \frac{\theta}{2\pi}\int F && (4.29)
\end{aligned}
$$

where on the last line we define $\varphi \equiv \theta$, but that was because we "normalized" the ansatz initially, and instead if we just used (4.25), then we would instead have to define $\theta \equiv 2\pi L\varphi$ which is slightly more interesting (also in the last line we recognized that $F = F_{01} dx^0 \wedge dx^1$).

4.15.3 Problem 3: Chern-Simons terms, winding, and integration by parts

Chern-Simons terms must be careully defined when fields are not globally defined. For example, we have argued that given compact scalar fields $\theta \cong \theta + 2\pi$ and $\sigma \cong \sigma + 2\pi$, a 1d Chern-Simons term should be interpreted as a sum over integrals on non-overlapping coordinate intervals with corrections for discontinuities at the interval boundaries:

$$
\oint \frac{\theta}{2\pi} d\sigma = \sum_i \left[\int_{x_{i-1,i}}^{x_{i,i+1}} \frac{\theta_i}{2\pi} d\sigma_i - n_{i,i+1}\sigma(x_{i,i+1})\right] \tag{4.30}
$$

(a) Given this definition, what can you say about the relationship between $\oint \frac{\theta}{2\pi} d\sigma$ and $\oint \frac{\sigma}{2\pi} d\theta$? Does integration by parts work as we ordinarily expect?

Answer: Following a similar discussion as in lecture (9/17/2024), consider the following integral taking into account their discontinuities

$$
\oint \frac{\theta}{2\pi} d\sigma = \sum_i \left(\int_{x_{i-1}}^{x_i} \frac{\theta_i}{2\pi} d\sigma_i - n_{i,i+1}\sigma(x_{i,i+1})\right) \tag{4.31}
$$

$$
\oint \frac{\sigma}{2\pi} d\theta = \sum_i \left(\int_{x_{i-1}}^{x_i} \frac{\sigma_i}{2\pi} d\theta_i - n_{i,i+1}\theta(x_{i,i+1})\right). \tag{4.32}
$$

To determine whether or not (4.31) and (4.32) are equivalent, consider the different between then

$$\oint \frac{\theta}{2\pi} d\sigma - \oint \frac{\sigma}{2\pi} d\theta = \sum_i \left(\int_{x_{i-1}}^{x_i} \left(\frac{\theta_i}{2\pi} d\sigma_i - \frac{\sigma_i}{2\pi} d\theta_i \right) - n_{i,i+1} \left(\sigma(x_{i,i+1}) - \theta(x_{i,i+1}) \right) \right). \tag{4.33}$$

We can apply integration by parts (without worry) on the continuous integral terms in (4.33) which omits a total derivative that only contributes on the boundary

$$\oint \frac{\theta}{2\pi} d\sigma - \oint \frac{\sigma}{2\pi} d\theta = - \left(\frac{\theta\sigma}{2\pi} \right)_{\text{at boundary}} - n_{i,i+1} \left(\sigma(x_{i,i+1}) - \theta(x_{i,i+1}) \right). \tag{4.34}$$

θ and σ are periodic up to a factor of $2\pi n_{\theta,\sigma}$ for $n \in \mathbb{Z}$. We can then substitute these values at the boundary term giving

$$\oint \frac{\theta}{2\pi} d\sigma - \oint \frac{\sigma}{2\pi} d\theta = n_\theta \sigma(0) + n\sigma\theta(0) + 2\pi n_\theta n_\sigma. \tag{4.35}$$

Equation (4.35) shows then that the 2 integrals are in fact no the same but by taking their difference, are related by the initial values and their respective winding numbers of θ and σ. It is from our use of integration by parts to arrive to (4.35) that integration by parts does not work as we would normally expect. In the presence of discontinuities, the fields introduce boundary terms that we had to take into account.

(b) Along similar lines, investigate the relationship between the 3d Chern-Simons terms $\int_M A \wedge dB$ and $\int_M B \wedge dA$ for U(1) gauge fields A and B, where M is a closed 3-manifold.

Answer: Suppose that we have some gauge fields A and B on some manifold M but we don't know the exact topology of M. Consider then that $d(A \wedge B) = dA \wedge B - A \wedge dB$. Then, the difference between the integrals of $B \wedge dA$ and $A \wedge dB$ are given as

$$\int_M A \wedge dB - \int_M B \wedge dA = - \int_M d(A \wedge B). \tag{4.36}$$

If $A \wedge B$ is globally defined and smooth, then the RHS of (4.36) is 0. Then the integral would be equal. If $A \wedge B$ is not globally well-defined due to the topology of M (or discontinuities that arise such as in the first part of this problem), then the RHS of (4.36) does not vanish. The integral then gathers the global topological information such as the winding numbers of A and B. The implication of not globally well-defined, also leads us to not being able to use integration by parts.

4.15.4 (PSET 2) Problem 1: Baryon and Lepton Number Symmetries of the Standard Model

The Standard Model has a classical *baryon number* symmetry $U(1)_B$ under which only the quark fields are charged, convientionally normalized as $B[q] = +1/3$, $B[\overline{u}] = -1/3$. It also has a classical *lepton number* symmetry $U(1)_L$ under which only the lepton fields are

charged, conventionally normalized as $L[\ell] = +1$, $L[\overline{e}] = L[\overline{\nu}] = -1$, where $\overline{\nu}$ is a singlet Weyl fermion we can add to have Dirac neutrino masses (i.e. the hermitian conjugate of a "right-handed neutrino" field).

In all of the formulas related to $U(1)$ gauge theory and anomalies that we have quoted in class, we have assumed that $U(1)$ gauge fields are normalized so that the minimal charge is 1, so that under a $U(1)$ transformation $e^{i\alpha}$, a field of charge q picks up a phase $e^{iq\alpha}$. This does not make sense for the conventional normalization of baryon number, where we have charges of 1/3; instead we either have to rescale the charge so the minimal charge is $\hat{B} = 1$ and we pick up a phase $e^{i\hat{B}\alpha}$, or say that we have defined the charge so that a field of baryon number B picks up a phase $e^{3iB\alpha}$. These are equivalent with $\hat{B} = 3B$.

A similar issue arises for hypercharge, where the conventional normalization has the smallest hypercharge in the Standard Model being 1/6 instead of 1.

Keeping track of these normalization issues is a large part of correctly working out this problem.

Fine print: for the purposes of this problem, the Standard Model gauge group is $SU(3)_C \times SU(2)_L \times U(1)_Y$, not a quotient thereof, and we assume that the quark doublet q has the smallest possible hypercharge. It is worthwhile to consider the same problem if the Standard Model gauge group has $\mathbb{Z}_2$, $\mathbb{Z}_3$, or $\mathbb{Z}_6$ quotient, or if the minimal unit of hypercharge is even smaller, but doing so is not a required part of the problem.

Hint: do not forget that there are three generations of each particle in the Standard Model, that QCD colors count as extra copies of a field from the viewpoint of $SU(2)$ and $U(1)$, and that an $SU(2)$ doublet counts as two copies of a field from the viewpoint of $SU(3)$ and $U(1)$.

(a) Show that if we apply a baryon number transformation $e^{i\alpha} \in U(1)_B$, the chiral anomaly leads to a shift in our action (as in Fujikawa's calculation) by terms proportional to $\mathrm{tr}(W \wedge W)$ (the $SU(2)_L$ instanton number) and $F_Y \wedge F_Y$ (the $U(1)_Y$ instanton number) but not any term proportional to $\mathrm{tr}(G \wedge G)$ (the $SU(3)_C$ instanton number). Work out the correct coefficients.

<u>Answer:</u> The general form for a chiral anomaly with respect to $SU(N)$ gauge theories for fermions with representation R of $SU(N)$ will have an action changed as

$$\delta S = \frac{1}{8\pi^2} \int (2I(R)\alpha(x)\,\mathrm{tr}\,(F \wedge F) \tag{4.37}$$

where F in this case is the 2-form of $F^a_{\mu\nu}$. The form of (4.37) arises from either computing the triangle diagrams for the insertion of 3-currents or from Fujikawa's method of variation under the path integral's differential. As an example, if we want to determine how $SU(2)$ transforms under a transformation from $U(1)_B$, then we would compute either $\partial_\mu \langle J^\mu_B(x) J^\nu_{\mathrm{SU}(2)}(y) J^\alpha_{\mathrm{SU}(2)}(z)\rangle$ corresponding to two triangle diagrams, or we would shift the differential of $[\mathcal{D}\psi\mathcal{D}\psi^\dagger]$ by $e^{i\alpha} \in U(1)_B$ to find δS. The Lagrangian, in either scenario, is just comprised of the kinetic terms of the three gauge groups we are considering

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}\,\mathrm{tr}\,(W_{\mu\nu}W^{\mu\nu}) - \frac{1}{2}\,\mathrm{tr}\,(G_{\mu\nu}G^{\mu\nu})\,. \tag{4.38}$$

The particle content of (4.38) comprises of charged fermions, the quark doublet, the up and down quarks, the lepton doublet and the electron. We *could* include the right handed neutrino but this has never been observed, and thus it is in the table, but *not* included in our calculations.

Table 2. Quantum numbers for Standard Model $SU(3)_C \times SU(2)_L \times U(1)_Y$.

Field	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$	B	L	Multiplicity
Q_L	3	**2**	$\frac{1}{6}$	$\frac{1}{3}$	0	$N_g \cdot N_c = 9$
u_R	**3**	1	$\frac{2}{3}$	$\frac{1}{3}$	0	9
d_R	**3**	1	$-\frac{1}{3}$	$\frac{1}{3}$	0	9
L_L	**1**	**2**	$-\frac{1}{2}$	0	1	3
e_R	**1**	**1**	-1	0	1	3
ν_R	**1**	**1**	0	0	1	3

To start, lets see how the path integral is transformed under $e^{i\alpha B_f} \in U(1)_B$. The partition function with the path integral can be written as

$$Z = \int [D\psi][D\psi^\dagger] e^{iS_f}. \tag{4.39}$$

Let $D\phi_n(x) = \lambda_n \phi_n(x)$ for a complete basis of states $\{\phi_f\}$. Then $\psi_f(x) = \sum_n a_{f,n}\psi_n(x)$ which implies that the differential changes as $D\phi_n(x) = \prod_{f,n} da_{f,n}$. Then $a_{f,n} \to e^{i\alpha B_f} a_{f,n}$ implying that the measure changes as

$$[D\psi] = \prod_{f,n} d\left(e^{i\alpha B_f} a_{f,n}\right) = [D\psi] \prod_f e^{i\alpha B_f \sum_n} = J[D\psi] \tag{4.40}$$

where the Jacobian J can be found as $J = e^{i\delta S} \implies \delta S = -i \ln J$ or

$$J = \exp\left(i\alpha \sum_f B_f \sum_n \int d^4x \phi_n^\dagger(x)\phi_n(x)\right). \tag{4.41}$$

The Jacobian has a divergent sum however which we can regularize in the following manner

$$\sum_n \phi_n^\dagger(x)\phi_n(x) = \lim_{M\to\infty} \sum_n \phi_n^\dagger(x) e^{-\lambda^2/M^2} \phi_n(x). \tag{4.42}$$

The variation of the action δS can then be written as

$$\delta S = -i \ln J = -i\alpha \sum_f B_f \lim_{M\to\infty} \int d^4x \, \mathrm{tr}\left(\phi_n^\dagger e^{-D^\dagger D/M^2} \phi_n(x)\right) \tag{4.43}$$

where, since we are primarily working with Weyl-fermions, (taken from equation 10.106 in Schwartz) $D^\dagger D = D^2 - \frac{1}{2}\sigma^{\mu\nu}F_{\mu\nu}$. The trace of δS then becomes in the

limit of $M \to \infty$ at second order,

$$\delta S = \alpha \sum_f B_f \left(\frac{1}{32\pi^2} \int d^4x \, \mathrm{tr}_R \left(F \wedge F \right) \right). \tag{4.44}$$

(4.44) can then be extended to $SU(2)$ and $SU(3)$. We can start by computing the coefficients that would go infront of the $SU(3)$ anomaly, since this is the simplest. By the quantum numbers in the table for $SU(3)$, with respect to the baryon number, the contribution vanishes.

For the $U(1)_Y$ group however the anomaly for a single generation is

$$\sum_L BY^2 - \sum_R BY^2 = \frac{1}{3} \left(6 \cdot \frac{1}{6^2} - 3 \left(\frac{2}{3} \right)^2 - 3 \cdot \left(-\frac{1}{3} \right)^2 \right) = -\frac{1}{2} \tag{4.45}$$

for the $U(1)_B$ showing directly the symmetry is violated for $U(1)_B$. The $SU(2)$ has a coefficient of $9/2$ (I think, I just know that they both should vanish under B_L).

(b) Repeat the above exercise for $U(1)_L$.

Answer: A similar counting process can occur for the $U(1)_L$ group where for $SU(3)$, we can see based on the tablet that the sum of the terms vanishes. While for $U(1)_Y$, we get $-1/3$ and for $SU(2)_L$ we get $-9/2$.

(c) Your results so far imply that baryon and lepton number symmetries have ABJ anomalies in the Standard Model. What about the combinations $B + L$ and $B - L$ (in the conventional normaization)?

Answer: Notice that for $B + L$, none of the anomalies for $F \wedge F$ or $\mathrm{tr}(W \wedge W)$ vanish, but for $B - L$ both the $U(1)_Y$ and the $SU(2)_L$ vanish implying that there is one global symmetry in the standard model namely the $U(1)_B - U(1)_L$ symmetry is conserved (as an IR effect),

(d) For certain discrete choices of α, the anomalous term is nonzero in the action but trivial in $\exp(iS)$. Explain how this allows for certain discrete subgroups of baryon or lepton number to be gauged. Which subgroups of $B, L, B + L$, and $B - L$ are gauge-able?

Answer: Since the anomalies are integer-valued topological terms, the shift giving δS under $e^{i\alpha}$ is just some proportional factor of α times an integer. For the baryon and lepton number, $\mathbb{Z}_3$ discrete group are gauge-able since the anomalies cancel modulo 2π with $\alpha = 2\pi/3$. For $B - L$ the discrete group is $\mathbb{Z}_6$ with $\alpha = \pi/3$. However, there is no discrete group for $B + L$ since the anomalies will never cancel.

(e) Comment on the implications of these possible discrete gauge symmetries for the stability of the proton or the existence of Dirac neutrino masses.

Answer: For grand unified theories such as $SU(5)$ and $SO(10)$, baryon number is not conserved such as the original example from Georgi and Glashow, "Unity of All Elemntary-Particle Forces," which is entirely anomaly free. However, these unification schemes often predict proton decay (which we have never observed).

4.15.5 Problem 2: Descent Formalism for Axion Couplings

Consider an axion θ, i.e. a scalar field with gauge invariance $\theta \cong \theta + 2\pi n$ (with $n \in \mathbb{Z}$), which couples to an $SU(N)$ gauge field A with field strength $F = dA - iA \wedge A$. The axion coupling that we are interested in is the Chern-Simons coupling

$$S_{\text{CS}} = \frac{k}{8\pi^2} \int \theta \, \text{tr}(F \wedge F), \quad k \in \mathbb{Z}, \tag{4.46}$$

where we take the trace in the fundamental representation with $\text{tr}\left(T^a T^b\right) = \delta^{ab}/2$.

To define this coupling for general field configurations (i.e. θ can have winding or equivalently is define by a $\mathbb{Z}$=bundle over spacetime, and the $SU(N)$ gauge fields are in general bundle as well), we can use the descent formalism starting from the 5d quantity

$$\Omega_5 = \frac{1}{4\pi} d\theta \wedge \text{tr}(F \wedge F). \tag{4.47}$$

Our goal is to work out the descent relation. I will prodive the first few steps.

$$\Omega_5 = d\omega_4, \quad \text{where} \quad \omega_4 \equiv \frac{1}{4\pi} \theta \, \text{tr}(F \wedge F) \tag{4.48}$$

and then we have a gauge variation

$$\Omega_4 \equiv \delta\omega_4 = \frac{1}{2} n \, \text{tr}(F \wedge F) \tag{4.49}$$

under the gauge transformation $\theta \mapsto \theta + 2\pi n$. Then we look for ω_3 by solving

$$\Omega_4 = d\omega_3 \quad \text{where} \quad \omega_3 = \frac{1}{2} n \, \text{tr}\left(A \wedge dA - i\frac{2}{3} A \wedge A \wedge A\right), \tag{4.50}$$

and we recognize ω_3 as a multiple of the 3d Chern-Simons form as mentioned in class.

Your next step is to find the gauge variation of ω_3. Recall that A transforms as

$$A \mapsto g^{-1} A g + i g^{-1} dg \tag{4.51}$$

with g an element of $SU(N)$. When we studied descent relations for $U(1)$ gauge fields, we has $A \mapsto A + d\alpha$ where the group element was $e^{i\alpha}$. The analogue here is to write

$$g = \exp(i\alpha^a T^a), \tag{4.52}$$

where T^a are generators of $SU(N)$ in the fundamental representation and α^a are real-valued parameters. Like α in the $U(1)$ case, these have some periodicity properties that allow them to shift by integer multiples of some basic quantity. The trickiest part of this calculation is to understand how to get this periodicity right in general. It is often convenient to use a basis of generators such that $\text{tr}\left(T^a T^b\right) = \delta^{ab}/2$, but then you are not guaranteed that all α^a have the same period. (You can play with the Gell-Mann matrices for $SU(3)$ to see this issue explicitly.)

Your assignment: keep deriving descent relations down the chain, $\Omega_3 = \delta\omega_3$, $\Omega_3 = d\omega_2$, $\Omega_2 = \delta\omega_2$,...until you hit Ω_0 and stop. (You might want to first do the $U(1)$ case $d\theta \wedge F \wedge F$ as a simpler warm-up exercise, to get the idea, but it is not the assignment).

Answer: To reiterate what we are doing (for my own sake), we can consider the 5d quantity

$$\Omega_5 = \frac{1}{4\pi} d\theta \wedge \mathrm{tr}(F \wedge F) \tag{4.53}$$

which we can construct out of a 4-form via the differential operator d which takes a p-form and raises it to a $p+1$-form namely $d\omega_4$ from which we can read off ω_4

$$\omega_4 = \frac{1}{4\pi} \theta \, \mathrm{tr}(F \wedge F). \tag{4.54}$$

The gauge variation labeled δ shifts $\theta \to \theta + 2\pi n$ namely

$$\delta\omega_4 = \Omega_4 = \frac{n}{2} \mathrm{tr}(F \wedge F) \tag{4.55}$$

and so on and so forth. So, we are left with ω_3 and need to find $\delta\omega_3 = \Omega_3$. However, an nicer way is to write the infinitesimal transformation instead by expanding out $g \approx 1 + \epsilon$ where ϵ will have some gauge-structure which gives $A \to A + D\epsilon$ where $D\epsilon = d\epsilon - i[A, \epsilon]$ (taken from Nakahara, but hopefully I got conventions correct). Then

$$\delta\omega_3 = n \, \mathrm{tr}(\epsilon F \wedge F) - d(n \, \mathrm{tr}(\epsilon F \wedge A)). \tag{4.56}$$

We can then read off $\Omega_3 = \delta\omega_3 = d\omega_2$ where $\omega_2 = n \, \mathrm{tr}(\epsilon F)$. To compute $\delta\omega_2$ which gives

$$\delta\omega_2 = n \, \mathrm{tr}(D\epsilon \wedge F) = -n \, \mathrm{tr}(\epsilon DF) = 0 \tag{4.57}$$

where we have the Bianchi $FD = 0$. Thus $\delta\omega_2 = d\omega_1$ where $\omega_1 = -\frac{1}{2} n \, \mathrm{tr}(\epsilon d\epsilon)$. For $\delta\omega_1$

$$\delta\omega_1 = -\frac{1}{2} n \, \mathrm{tr}(D\epsilon \wedge d\epsilon) = \frac{1}{2} n \, \mathrm{tr}(\epsilon[\epsilon, A]) = d\omega_0 \tag{4.58}$$

where $\omega_0 = \frac{1}{6} n \, \mathrm{tr}(\epsilon^3)$ with the final variation is $\delta\omega_0 0 \Omega_0$ which ends the descent chain.

4.15.6 Problem 3: Properties of the BPST Instanton

(a) Decompose a point $a^\mu \in \mathbb{R}^4$ into a scalar and three-vector $(x^0, \vec{x})$, and denote the Euclidean distance of the point from the origin by $|x| \sqrt{|x^0|^2 + \vec{x} \cdot \vec{x}}$. Show that the function

$$g : x^\mu \mapsto \frac{x^0 + i\vec{x} \cdot \boldsymbol{\sigma}}{|x|}, \tag{4.59}$$

where σ are the Pauli-matrices, defines a bijective map from rays in $\mathbb{R}^4$ into elements of $SU(2)$.

Answer: To show that $g(x^\mu)$ defines a bijective map from rays in $\mathbb{R}^4$ into elements of $SU(2)$, we first need to show that $g(x^\mu)$ is an element of $SU(2)$ which, if an element $g \in SU(2)$ then it is unitary i.e. $gg^\dagger = g^\dagger g = 1$ and that $\det g = 1$ (or the identity matrix).

First recall that the Pauli matrices σ_i satisfy $\sigma_i \sigma_j = \delta_{ij} I i \epsilon_{ijk} \sigma_k$ where I is the identity matrix. The Hermitian adjoint of $g(x^\mu)$ is

$$g(x^\mu)^\dagger = \left(\frac{x^0 I + i\vec{x} \cdot \boldsymbol{\sigma}}{|x|} \right)^\dagger = \frac{x^0 I - i\vec{x} \cdot \boldsymbol{\sigma}}{|x|}. \tag{4.60}$$

This allows us to verify unitarity which gives

$$g(x^\mu)^\dagger g(x^\mu) = \frac{(x^0)^2 + \vec{x}^2}{|x|^2} I = I \tag{4.61}$$

since $|x|^2 + (x^0)^2 + \vec{x}^2$. We then need to verify the determinant of $g(x^\mu)$. In computing the determinant of $g(x^\mu)$, since $|x|^{-2}$ is just a constant, we can pull it out, and must instead evaluate

$$\det\Big(x^0 I + i\vec{x}\cdot\boldsymbol{\sigma}\Big) = (x^0)^2 + \vec{x}^2 = |x|^2. \tag{4.62}$$

Thus $\det g(x^\mu) = 1$ and therefore $g(x^\mu) \in SU(2)$.

We also need to show that g is injective (one-to-one). Suppose that $g(x) = g(y)$ (where now $g(x^\mu) \equiv g(x)$ for shorthand). Then

$$\frac{x^0 + i\vec{x}\cdot\boldsymbol{\sigma}}{|x|} = \frac{y^0 + i\vec{y}\cdot\boldsymbol{\sigma}}{|y|}. \tag{4.63}$$

We can cross multiply and equate the scalar and vector parts separately with $|y|x^0 = |x|y^0$ and $|y|\vec{x} = |x|\vec{y}$ which implies that $x^\mu/|x| = y^\mu/|y|$ implying that x^μ and y^μ are along the same ray, thus g is injective.

Finally to show that g surjective (onto), consider an element $U \in SU(2)$ which can be expressed as $U = a^0 I + ia^i\sigma_i$ for $a^0, a^i \in \mathbb{R}$ and $(a^0)^2 + a^i a^i = 1$. Set $x^\mu = (a^0, a^i)$ so $|x| = 1$. Then

$$g(x^\mu) = \frac{a^0 I + ia^i\sigma_i}{1} = U. \tag{4.64}$$

Therefore, since g is both injective and surjective, it defines a dbijective map froms rays in $\mathbb{R}^4$ into elements of $SU(2)$.

(b) Compute he field strength $F = dA - iA\wedge A$ of the field configuration

$$A = \frac{|x|^2}{|x|^2 + \rho^2}\Big(-ig^{-1}dg\Big) \tag{4.65}$$

and check that this is a self-dual field configuration.

This field configuration manifestly approaches a pure gauge configuration as $|x| \to \infty$; that is why F decreases so rapidly at large distances.

(Optional: investigate the relationship between this formula and the expression for the BPST instanton presented in class and in my TASI lecture notes, involving the 't Hooft symbols. The formulation presented here follows Sidney Coleman's lecture "The uses of instantons," which can be found in the book *Aspects of Symmetry*.

<u>Answer:</u> Let $f(x) = |x|^2/(|x|^2 + \rho^2)$ such that (4.65) becomes

$$A = f(x)(-ig^{-1}dg) \tag{4.66}$$

with g defined previously. Given $F = dA - iA\wedge A$, then

$$\begin{aligned} F = &= dA - iA\wedge A = df\wedge A_0 + ifA_0\wedge A_0 - if^2 A_0\wedge A_0 \\ &= df\wedge A_0 + if(1-f)A_0\wedge A_0 \end{aligned} \tag{4.67}$$

where $A_0 = - - ig^{-1}dg$. We can write df explicitly as

$$df = \frac{2\rho^2 x_\mu dx^\mu}{(|x|^2 + \rho^2)^2} \tag{4.68}$$

allowing us to compute $dg \wedge A_0$. We can then check that the field is self-dual under the Hodge dual of F namely $F_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$. The gauge-field strength can be put into the form of 't Hooft symbols

$$F^a_{\mu\nu} = -\frac{4\rho^2}{(x^2+\rho^2)^2}\eta_{a\mu\nu} \tag{4.69}$$

where

$$\eta_{a\mu\nu} = \epsilon_{a\mu\nu} + \delta_{a\mu}\delta_{0\nu} - \delta_{a\nu}\delta_{0\mu}. \tag{4.70}$$

The 't Hooft symbols already satisfy the self-dual relation

$$\eta_{a\mu\nu} = \frac{1}{2}\epsilon_{a\mu\nu}\eta_{a\rho\sigma} \tag{4.71}$$

implying that $F = \star F$.

(c) Based on your expression for F, explain why ρ can be interpreted as the size of the solution. (We have omitted the other 4 moduli that appeared in the solution presented in class, but it is clear that we can obtain these solutions by translations of the solution above).

Answer: Given that $F \propto \rho^2/(x^2+\rho^2)^2$, then F is strong when $x^2 \sim \rho^2$ and decreases rapidly when $x^2 \gg \rho^2$. Also notice that when $x^2 \ll \rho^2$ then $F \propto 1/\rho^2$. Then the field strength is strongest towards ρ while F decreases when $x \ll \rho$ as $F \propto 1/x^4$.

Therefore when ρ is small, the field is concentrated at a small point or region around the origin implying the instanton is localized in a small location. while for large ρ the instanton is more diffuse. Therefore, the region where the gauge field is non-trivial is when ρ is large and this distance ρ can be interpreted as the size of the instanton. In our case, the instanton is centered at the origin, $x^\mu + 0 = 0$ effectively removing the other four position moduli.

(d) Compute $\int \mathrm{tr}(F \wedge F)$ for this configuration.

Answer: We can write the field strength as $F^a_{\mu\nu} = C(x)\eta_{a\mu\nu}$ where $C(x) = -4\rho^2/(x^2+\rho^2)^2$. Then, in the fundamental representation of $SU(2)$, $\mathrm{tr}\left(T^aT^b\right) = \delta^{ab}/2$ the trace of the field strength is simply

$$\begin{aligned}\mathrm{tr}(F \wedge F) &= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}d^4x \\ &= \frac{C(x)^2}{4}\epsilon^{\mu\nu\rho\sigma}\left(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho} + \epsilon_{\mu\nu\rho\sigma}\right)d^4x\end{aligned} \tag{4.72}$$

where

$$\eta_{a\mu\rho}\eta_{b\rho\sigma} = \delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho} + \epsilon_{\mu\nu\rho\sigma}. \tag{4.73}$$

The contractions with the delta functions and Levi-Civita symbols all vanish but $\epsilon_{\mu\nu\rho\sigma}\epsilon^{\mu\nu\rho\sigma} = 24$ thus giving an overall factor of $6C(x)^2 d^4x$. If we substitute $C(x)$ back into the integral, and compute over $d^4x = \Omega_4 r^3 dr$ and set $|x| = r$ then the integral becomes (with $\Omega_4 = 2\pi^2$)

$$I = \int \mathrm{tr}(F \wedge F) = 192\pi^2 \int_0^\infty \frac{\rho^4 r^4 dr}{(r^2+\rho^2)^4}. \tag{4.74}$$

At this point, we can through the integral into *Mathematica* which omits that $I = 16\pi^2$ thus showing that

$$\int \mathrm{tr}(F \wedge F) = 16\pi^2. \tag{4.75}$$

If we of course had assigned $k = -I/8\pi^2$ then we get $k = -2$ indicating an anti-instanton.

(e) Now, a more conceptual question: we expect that $\int \mathrm{tr}(F \wedge F)$ is a topological invariant of a gauge bundle (independent of any specific choice of connection for a given bundle). We also expect that A can be single-valued only for a trivial bundle, because a non-trivial bundle necessarily has non-trivial transition function. For a trivial bundle, we must have $\int \mathrm{tr}(F \wedge F) = 0$. But, the formula above looks single-valued, and has a non-zero $\int \mathrm{tr}(F \wedge F)$. Why is this not a contradiction?

You could give a one line answer that simply identifies the logical gap in the preceding paragraph, but there is a slightly longer but more enlightening answer that makes the analogy between the gauge bundle constructed here and the familiar Dirac monopole construction more clear.

Answer: The gauge potential appears single-valued and is globally defined over $\mathbb{R}^4$, corresponding to a non-trivial gauge bundle due to the behavior at infinity. Under the mapping of $g(x)$ which has a non-zero winding number, this reflect a non-trivial element of the homotopy group $\pi_3(SU(2)) = \mathbb{Z}$. This implies a non-trivial topology of the gauge bundle over S^4 even though we have a globally defined A. Thus there is no contradiction.

4.15.7 Problem 4: Chiral Lagrangian and the QCD Axion

(a) Derive the vacuum energy of 2-flavor QCD as function of the parameter $\overline{\theta}$. First, carry out a chiral rotation of a quark field, e.g. $u_L \mapsto u_L e^{i\theta}$, to remove the $\theta\,\mathrm{tr}(F \wedge F)$ term from the action. This places θ in the phase of the determinant of the quark mass matrix M. Then, use the chiral Lagrangian for pions focusing on the term

$$\mathcal{L} \supset \mu^3 \,\mathrm{tr}\left(MU^\dagger M^\dagger U\right) \tag{4.76}$$

that gives mass to the pions. Integrate out the pion fields taking into account the phase $\overline{\theta}$ contained in $\det(M)$. You should be able to derive the standard formula

$$V_{\mathrm{QCD}}(\overline{\theta}) = -m_\pi^2 f_\pi^2 \sqrt{1 - \frac{4m_u m_d}{(m_u+m_d)^2}\sin^2(\overline{\theta}/2)}. \tag{4.77}$$

Answer: The QCD action with the topological term contains

$$\theta \int d^4x \frac{g^2}{32\pi^2} F^a_{\mu\nu} \tilde{F}^{\mu\nu a}. \tag{4.78}$$

We can remove it via a chiral rotation of the up quark $u_L \to u_L e^{i\theta}$, but due to the chiral anomaly, a chift in θ induces a quark mass M with phase $M \to M e^{i\theta}$. The physical parameter $\overline{\theta}$ is then $\overline{\theta} = \theta + \arg\det M$. Then (following Schwartz) we can write the mass term of the chiral Lagrangian as

$$\mathcal{L} \supset = \frac{f_\pi^2}{4} \operatorname{tr}\Big(\chi U^\dagger + U\chi^\dagger\Big) \tag{4.79}$$

wehere $\chi = 2B_0 M$. Then the mass matrix is

$$M = \operatorname{diag}(m_u e^{i\alpha_u}, m_d e^{i\alpha_d}) \tag{4.80}$$

with $\alpha_u + \alpha_d = \overline{\theta}$. Let $\alpha_u = \phi_u$ and $\alpha_d = \overline{\theta} - \phi_u$.

The potential energy can be written as

$$V(u) = -\frac{f_\pi^2}{2} B_0 \operatorname{Re}\operatorname{tr}\Big(MU^\dagger\Big). \tag{4.81}$$

To integrate out the pion fields while treating U a constant, substitute $U = e^{i\phi\tau_3}$. We then want to minimize the potential to find the vacuum energy which gives the relation

$$\frac{dV}{d\phi_u} = 0 \implies m_u \sin(\phi_u) - m_d \sin\Big(\overline{\theta} - \phi_u\Big) = 0 \implies \tan(\phi_u) = \frac{m_d \sin\overline{\theta}}{m_u + m_d \cos\Big(\overline{\theta}\Big)}. \tag{4.82}$$

We can then simplify the potential energy at the minimum by using trigonometric identities and the *Gell-Mann-Oakes-Renner relation* $m_\pi^2 = B_0(m_u + m_d)$ giving

$$V_{\text{QCD}}(\overline{\theta}) = -m_\pi^2 f_\pi^2 \sqrt{1 - \frac{4m_u m_d}{(m_u + m_d)^2} \sin^2(\overline{\theta}/2)}. \tag{4.83}$$

(b) Promoting $\overline{\theta}$ to a dynamical axion field a via $a = f\overline{\theta}$ gives a quantitative estimateof the axion mass as a function of its decay constant f. That is, give a formula as well as a numerical value normalized to a specific choice, say $f = 10^{12}$GeV.

Answer: The mass of the axion can be found by computing

$$m_a^2 = \frac{\partial^2 V}{\partial a^2} \tag{4.84}$$

evaluating the axion at $a = 0$. The mass after computing the second derivative of the QCD potential is

$$m_a^2 = \frac{1}{2}\left(\frac{f_\pi m_\pi}{f}\right)^2 \frac{m_u m_d}{(m_u + m_d)^2}. \tag{4.85}$$

If we estimate $f \approx 10^{12}$GeV with known values for: $m_\pi \approx 135$MeV, pion decay $f_\pi \approx 92$MeV, the up quark mass $m_u \approx 2.2$MeV, the down quark mass $m_d \approx 4.7$MeV, and the axion decay constant f as quoted above, then $m_a \approx 4 \times 10^{-6}$eV.

(c) How large is the axion's quartic self-coupling λ? Give an analytic formula as well as a numerical estimate normalized to the same choice of f.

Answer: Let $\epsilon = 2m_u m_d/(m_u + m_d)^2$ for convience. The potential can be approximated for small a/f giving and expansion as

$$V(a) \approx= -m_\pi^2 f_\pi^2 \left(1 - \frac{\epsilon}{4}\left(\frac{a}{f}\right)^2 + \frac{\epsilon^2}{64}\left(\frac{a}{f}\right)^4 + \cdots\right) \tag{4.86}$$

which will allow us to read off terms proportional to a typical quartic coupling of the form $\lambda a^4/4!$. We can read this off to be $\lambda = m_a^2/24f^2$. This can also be found by computing the fourth order derivative of the potential and dividing by 4!. Based on our previous numerical values, we can estimate $\lambda \approx 6.67 \times 10^{-55}$.

(d) Derive the effective coupling to photons that a QCD axion inherits through its mixing with the pion. Estimate the lifetime for an axion to decay to two photons through this mixing.

Answer: To find the effective coupling of the QCD axions with the coupling to photos via mixing with pions, we can consider an induced mass term in the chiral Lagrangian given by the mixing angle

$$\theta = \frac{m_d - m_u}{m_u + m_d}\frac{f_\pi}{f_a}. \tag{4.87}$$

We can then write down the axion-photon coupling as

$$g_{a\gamma\gamma} = \frac{\alpha}{4\pi f_a}\frac{m_d - m_u}{m_u + m_d}. \tag{4.88}$$

If $f_a = 10^{12}$GeV then $g_{a\gamma\gamma} \approx= 3.33 \times 10^{-20}\text{MeV}^{-1}$. The decay rate can then be written as

$$\Gamma(a \to \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi} \approx 3.54 \times 10^{-76}\text{MeV}. \tag{4.89}$$

This corresponds to an axion lifetime of $\tau = \hbar/\Gamma \approx 1.86 \times 10^{54}$s.

(e) (Optional, somewhat open-ended.) We have been discussing the effective theory of pions, but one could also ask about the role of the η'. This was investigated in the paper by Witten [29] and by di Vecchia and Venenziano [30] who argued that one should extend the chiral Lagrangian to include a term of the form

$$V(U) = \frac{V_0}{N}\left(-i\log(\det(U)) - \theta\right)^2 \tag{4.90}$$

with N the number of colors (3 for real-world QCD) and V_0 a parameter of order the confinement scale. Explore the effects of incorporating the η' into the theory along with axions and pions.

Answer: N/A

4.15.8 (PSET 3) Problem 1: Axion Electrodynamics

Consider the theory of an axion field interacting with the photon field as a perturbative QFT. In particular, for this problem we can take the axion field $a(x)$ and the photon field $A(x)$ to be *canonically normalized*, rather than the normalizations we have been using in class that are based on 2π flux quantization. This is more convenient because we are not dealing with any topological questions in this problem. We will also work in Minkowski spacetime.

Our effective action then takes the form

$$S = \int d^4x \left(\frac{1}{2}(\partial a)^2 - \frac{1}{4}F^2 + \frac{1}{4}g_{a\gamma\gamma} a F_{\mu\nu}\tilde{F}^{\mu\nu} - V(a) - A_\mu J^\mu \right) \tag{4.91}$$

where $F^2 \equiv F_{\mu\nu}F^{\mu\nu}$, $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$, and J^μ is the electromagnetic current.

1. Derive the equations of motion for a and A from this action, and write them in non-relativistic language in terms of $\vec{E}$, $\vec{B}$, ρ, and $\vec{J}$. Here, the electric field vector has component $E_i = F_{0i}$ and the magnetic field has $B_i = -\frac{1}{2}\epsilon_{ijk}F^{jk}$. The axion-coupled Maxwell's equations, in this form, and with the quadractic approximation $V(a) = \frac{1}{2}m_a^2 a^2$, guide most experimental searches for axions.

   Answer: To compute the equations of motion we can compute the variation of the action in (4.91) with respect to δa and δA_ν. With respect to the axion field this yields

$$\delta S = \int d^4x \left(-\Box a + \frac{1}{4}gF_{\mu\nu}\tilde{F}^{\mu\nu} - \frac{dV(a)}{da} \right) \delta a \tag{4.92}$$

   which implies the following equation of motion

$$\Box a + \frac{dV(a)}{da} = \frac{g}{4}F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{4.93}$$

   In a similar manner we can compute the variation with respect to the photon but this requires a little more algebra. Recall that $a \equiv a(x)$ and if we then had the following situation

$$(\partial_\mu A_\nu \partial_\alpha A_\beta - \partial_\mu A_\nu \partial_\beta A_\alpha)\, a(x)\epsilon^{\mu\nu\alpha\beta} = -\left(A_\nu \partial_\alpha A_\beta - A_\nu \partial_\beta A_\alpha\right)(\partial_\mu a(x))\epsilon^{\mu\nu\alpha\beta}\delta A_\nu \tag{4.94}$$

   where we used integration by parts to pull out a δA_ν variation. This is the term that arises when varying the $F\tilde{F}a$ term. What remains from (**??**) is

$$-2\tilde{F}^{\mu\nu}(\partial_\mu a(x))\delta A_\nu \tag{4.95}$$

   where we used the definition of the dual field strength tensor. The total variation of δA_ν then follows to give

$$\delta S = \int d^4x \left(-\partial_\mu F^{\mu\nu} + g\partial_\mu a(x)\tilde{F}^{\mu\nu} + J^\nu \right) \delta A_\nu \tag{4.96}$$

   which implies the following equation of motion for the photon field

$$\partial_\mu F^{\mu\nu} - J^\nu = g\partial_\mu a(x)\tilde{F}^{\mu\nu}. \tag{4.97}$$

To find the Maxwell-axion equations, lets first decompose the axion equation of motion (4.93) since that is the easiest. The $F\tilde{F}$-density just gives (chapter 33 of [3]) $-4\vec{E}\cdot\vec{B}$. Then, leaving the box-operator in place, we have (with $V(a) = m_a^2 a^2/2$)

$$\left(\Box + m_a^2\right) a(x) = -g\vec{E}\cdot\vec{B}. \tag{4.98}$$

The *true* set of Maxwell-axion equations arises from (4.97), which requires a little more algebra than the previous equation of motion. The components of the field strength are

$$F^{0i} = E^i,\ \ F^{ij} = -\epsilon^{ijk}B_k,\ \ \tilde{F}^{0i} = -B^i,\ \ \tilde{F}^{ij} = -i\epsilon^{ijk}E_k \tag{4.99}$$

where the diagonal components of F and $\tilde{F}$ all vanish due to its antisymmetric nature. Notice that if we set $\nu = 0$ in (4.97) then we can set $\mu = 0$ or $\mu = i$, but for $\mu = 0$ the equation of motion omits a vacuum, and something we do not care about (at the moment). But for $\mu = j$, then we get [31]

$$\partial_j F^{j0} - \rho = g\partial_j a(x)\tilde{F}^{j0} \tag{4.100}$$

which if we plug in the components gives

$$\boldsymbol{\nabla}\cdot\vec{E} = \rho - g\vec{B}\cdot\boldsymbol{\nabla}a(x) \tag{4.101}$$

which is just Gauss' law. In a similar manner, for $\nu = j$, we can recover Ampere's Law

$$\boldsymbol{\nabla}\times\vec{B} - \dot{\vec{E}} = \vec{j} - \frac{g}{2}\left(\vec{E}\dot{a}(x) + \vec{E}\times\boldsymbol{\nabla}a\right) \tag{4.102}$$

where we have been using $J^\mu = (\rho, \vec{j})$. There is also the Faraday and Gauss' law for magnetism but we only want to focus on the axion-photon dynamics.

2. A different possible definition of the electric field in this theory is given by taking $\boldsymbol{\mathcal{E}}$ to be the canonically conjugate momentum to the field $\vec{A}$. This reduce to $\vec{E}$ when $a = 0$. Derive an expression for $\boldsymbol{\mathcal{E}}$. Does an electron feel an acceleration determined by $\boldsymbol{\mathcal{E}}$ or by $\vec{E}$?

   (In the quantum theory, it is $\boldsymbol{\mathcal{E}}$ that has quantized fluxes, which leads to the Witten effect: magnetic monopoles acquire an axion-dependent electric charge).

   Answer: To find the canonical conjugate momentum with respect to $\vec{A}$, we can be a little-tricky when computing the derivative of the Lagrangian, namely

$$\mathcal{E} = \frac{\partial\mathcal{L}}{\partial\dot{\vec{A}}} = \frac{\partial\mathcal{L}}{\partial\vec{E}}\frac{\partial\vec{E}}{\partial\dot{\vec{A}}}. \tag{4.103}$$

   The validation for this is because $\vec{E}$ and $\vec{B}$ ahve a dependence on $\vec{A}$, but only $\vec{E}$ depends linearly on the time-derivative of $\vec{A}$ namely $\vec{E} = \dot{\vec{A}} - \boldsymbol{\nabla}A_0$ and $\vec{B} = \nabla\times\vec{A}$. The Lagrangian in (4.91) in component form is

$$\mathcal{L} = \frac{1}{2}(\partial a) + \frac{1}{2}(\vec{E}^2 - \vec{B}^2) - ga\vec{E}\cdot\vec{B} - V(a) - A_\mu J^\mu. \tag{4.104}$$

The conjugate momentum is straightforward then since $\partial\vec{E}/\partial\dot{\vec{A}} = 1$ and

$$\mathcal{E} = \frac{\partial\mathcal{L}}{\partial\vec{E}} = \vec{E} - ga\vec{B}. \tag{4.105}$$

Recall that the force (or acceleration) that an electron feels is governed by Lorentz's force law

$$m\frac{d\vec{v}}{dt} = e(\vec{E} + \vec{v}\times\vec{B}). \tag{4.106}$$

So on the surface, it appears that $\mathcal{E}$ does not enter, but we could imagine a photon traveling along an axion-like background field such that $\vec{E} \sim \cos(\rho_a t)$ for some background value of the axion ρ_a.

3. Treat $a(t)$ as a slowly evolving classical field, and work out the equations of motion for modes of the electromagnetic field with definite momentum $\vec{k}$ and helicity ± 1. You should find that, in the time-varying axion background, certain modes become unstable (i.e. they grow exponentially). A number of interesting cosmological models rely on this instability.

   Answer: If $a(x^\mu) \to a(t)$ then all we have left from our Maxwell-axion electrodynamic equations are the time derivatives of the axion field, and so (4.101) and (4.102) changes as

$$\boldsymbol{\nabla}\cdot\vec{E} = \rho \tag{4.107}$$

$$\boldsymbol{\nabla}\times\vec{B} - \dot{\vec{E}} = \vec{j} - \frac{g}{2}\left(\vec{E}\cdot a\right). \tag{4.108}$$

   To make our lives easier (and to solve the differential equations above, which also include Gauss' law for magnetism and Faraday's law not shown) we can turn off the current $\vec{j}$ and ρ.

   We can writ ethe modes of the electromagnetic field as

$$\vec{E}(\vec{r},t) = \int \frac{d^3k}{(2\pi)^3}\vec{E}_k(t)e^{i\vec{k}\cdot\vec{r}} \tag{4.109}$$

$$\vec{B}(\vec{r},t) = \int \frac{d^3k}{(2\pi)^3}\vec{B}_k(t)e^{i\vec{k}\cdot\vec{r}}. \tag{4.110}$$

   From Gauss' law for both electric fields and magnetism, when we plug in our mode expansion (**??**) then we are left with transverse field propagation $\vec{k}\cdot\vec{E}_k = 0$ and $\vec{k}\cdot\vec{B}_k = 0$. We need to define helicity states which we choose to be circularlly polarized with vectors $\epsilon^\pm(\vec{k})$. This gives us a helicity basis to expand our mode functions $\vec{E}_k(t)$ and $\vec{B}_k(t)$ along into $\pm$ modes. The equations of motion in terms of the mode functions and performing Fourier tranformations with $\vec{k}\leftrightarrow i\boldsymbol{\nabla}$ giving

$$\dot{B}_k^\pm = \mp ikE_k^\pm \tag{4.111}$$

$$\dot{E}_k^\pm = \pm ikB_k^\pm - \frac{g}{2}\dot{a}B_k^\pm \tag{4.112}$$

where E_k and B_k are the components of the electric and magnetic fields in the helicity basis.

Notice that if we take the time derivative of (4.112), this allows us to substitute the time derivative of the magnetic field given in (4.111) giving us

$$\ddot{E}_k^{\pm} = -k^2 E_k^{\pm} + \frac{gi}{2} k\dot{a} E_k^{\pm} - \frac{g}{2}\ddot{a} B_k^{\pm}. \tag{4.113}$$

But, we have a *slowly*-time evolving axion field, and so we can discard $\ddot{a}(t)$. In doing so, let us also define the following coupling dependent quantity

$$\omega_a(t) \equiv \frac{g}{2}\dot{a} \tag{4.114}$$

which allows us to write down the following second order equation in the electric field

$$\ddot{E}_k^{\pm} + k^2 E_k^{\pm} \mp 2ik\omega_a(t) E_k^{\pm} = 0. \tag{4.115}$$

This is a wonderful differential equation that we can solve with typical methods. Let us guess that $E_k^{\pm}(t) = E_k^{\pm}(0)e^{-i\omega t}$. Then we get the dispersion relation $\omega^2 = k^2 \mp 2ik\omega_a(t)$. The dispersion relation can be approximated to be $\omega \approx k \mp i\omega_a(t)$.

From ω, the solution $E_k^{\pm}(t)$ picks up an axion-dependent part which for the plus mode gives a solution that has exponential growth, while the minus mode gives exponential decay of the form $\sim e^{\pm\omega_a t}$. The axion field then drives the electric field to have a preferred mode that *should* be observable.

4. What is the lifetime for the decay $a \to \gamma\gamma$ in terms of the axion mass m_a and the coupling $g_{a\gamma\gamma}$? For the KSVZ axion ($E/N = 0$) with $f/k_G = 10^{12}$Gev, what is the axion's lifetime (in years)?

   Answer: The decay rate for a scalar field into two massless particles is generic and can be written as

$$\Gamma(a \to \gamma\gamma) = \frac{g^2 m_a^3}{64\pi}. \tag{4.116}$$

   For the KSVZ model (from the lecture notes)

$$g = -\frac{\alpha}{2\pi f_a}\left(\frac{2}{3}\frac{4+z}{1+z}\right) \tag{4.117}$$

   where $z = m_u/m_d \approx 0.56$. Then, if the mass of the axion is determined by $m_a(\sqrt{z}/(1+z))f_\pi m_\pi/f_a$ then $m_a \sim 6 \times 10^{-6}$eV. The decay rate is then $\Gamma \sim 5 \times 10^{-66}$eV which we can invert to get the lifespan namely $\tau \sim 4 \times 10^{42}$years.

4.15.9 Axion couplings in the SU(5) GUT

In the SU(5) GUT model, the gauge group SU(5) is higgs-ed to the Standard Model gauge group in the form $[\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ through the vev of an adjoint scalar Φ with

$$\langle\Phi\rangle = v_{\mathrm{GUT}} \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}. \tag{4.118}$$

1. How do the generators for $\mathrm{SU}(3)_C$, $\mathrm{SU}(2)_L$, and $\mathrm{U}(1)_Y$ embed in SU(5)? (That is, write them as a traceless hermitian 5×5 matrices). Be careful about how the standard normalizations of $\mathrm{U}(1)_Y$ relates to the generator you construct within SU(5).

   Answer: I do not know the GUT groups and decompositions well, but these two papers helped a lot, [32, 33] especially the supplemental material found in [32]. The original SU(5) model by Georgi-Glashow is also a nice read [34].

   The content of $SU(3)_C$ that has generators of t^a can be found in the upper-left 3×3 block of the SU(5) matrix (4.118). Namely,

$$T^a_{\mathrm{SU}(3)} = \begin{pmatrix} t^a & 0_{3\times2} \\ 0_{2\times3} & 0_{2\times2} \end{pmatrix} \tag{4.119}$$

   where the generators satisfy $\mathrm{tr}\left(t^a t^b\right) = \delta^{ab}/2$. In a similar manner, SU(2) is embedded in SU(5) in the lower right 2×2 block with similar 0-matrices as in the SU(3) case. Finally, the hyperchage U(1) group is embedded as a diagonal matrix in SU(5) namely

$$Y = \sqrt{\frac{3}{5}}\mathrm{diag}(1/3, 1/3, 1/3, -1/2, -1/2). \tag{4.120}$$

   (A lot of this group algebra was with the help of [35]).

2. Now, suppose that in the UV theory we have an axion coupled as $\frac{k}{8\pi^2}\theta\,\mathrm{tr}_{\mathrm{SU}(5)}\,(F\wedge F)$. Below the scale of GUT breaking, this matches on to couplings

$$\frac{k_G}{8\pi^2}\theta\,\mathrm{tr}(G\wedge G) + \frac{k_W}{8\pi^2}\theta\,\mathrm{tr}(W\wedge W) + \frac{k_B}{8\pi^2}\theta B\wedge B. \tag{4.121}$$

   (Here, B is the hypercharge field strength in the standard normalization where the Standard Model quark doublet field has hypercharge 1/6). Express k_G, k_W, and k_B in terms of k.

   Answer: To find the relationship of k onto k_W, k_B, and k_G, we can compute the trace of their respective generators with respect to SU(5) and using their respective algebra conditions namely $\mathrm{tr}\left(t^A t^B\right) = \delta^{AB}/2$ for SU(3) for example. Over SU(5), the trace yields the same algebra conditions except over the hypercharge U(1) group gives just 1/2.

   We can decompose the SU(5) field strength in terms of the SU(3), SU(2), and U(1) gauge fields, each with a factor of 1/2 for normalization purposes. In doing so, we can read off the values of k with relation to each field strength. I find that $k_G = 2$, $K_B = 6$ and$k_W = -9$.

3. Show that below the electroweak scale, we obtain an axion coupling $\frac{k_F}{8\pi^2}\theta F\wedge F$ (with F the photon field strength in the conventional normalization where the electric charge is -1), where $k_F = \frac{1}{2}k_W + k_B$.

   Answer: This is given by the way the axion couples to the W and B-fields namely

$$\frac{\theta}{8\pi^2}\left(\frac{k_W}{2}F_{\mu\nu}\wedge F^{\mu\nu} + k_B F_{\mu\nu}\wedge F^{\mu\nu}\right) \tag{4.122}$$

which occurs when we go below the electroweak scale. This omits the $k_F = \frac{1}{2}k_W + k_B$ relation.

4. Finally, check that your results imply that $E/N = 8/3$ in the conventional notation where $E/N = 2k_G/k_F$.

   Answer: Given that $N \in 1/2$ and $E \in 1/3$ which are related to the full standard model prior to electroweak symmetry breaking (modulo $\mathbb{Z}_6$), we can find that $k_F = k_w/2 + k_B = 3/2$. Then $E/N = 2k_G/k_F = 8/3$.

4.15.10 Axion potential from extra dimensions

Consider the theory of an axion θ arising from a 5d gauge field A on a circle $x^5 \cong x^5 + 2\pi R$ through the ansatz $A = \frac{\theta}{2\pi R}dx^5$.

1. Check that the 5d scalar mass M and charge q has Kaluza-Klein modes of mass

$$m_n^2 = M^2 + \frac{1}{R^2}\left(n + \frac{q\theta}{2\pi}\right)^2. \tag{4.123}$$

   Answer: The physical system we are considering is a 5d charged scalar field. I will use capital Latin letters to denote 5d indexes while the typical Greek letters for 4d components. In essence, we have 5d scalar QED with action (taking inspiration from Polchinski's chapter 8 [4])

$$S = \int d^5x \left(-\frac{1}{4}F_{MN}F^{MN} + (D_M\phi)^*(D^M\phi) - M^2\phi^*\phi\right) \tag{4.124}$$

   where the gauge-derivative is

$$D_M\phi = \partial_M\phi + iqA_M\phi. \tag{4.125}$$

   For the purpose of find the Kaluza-Klein (KK) mass modes m_n^2, we will simply drop the kinetic (F^2) term in the calculations that will follow.

   Since we have a gauge field on a circle with $x^5 \cong x^5 + 2\pi R$ with ansatz $A_5 = \theta/2\pi R$ (in component form, not differential-form "form") we can expand the 5d scalar field as well as

$$\phi(x^5, x^\mu) = \sum_{n=-\infty}^{\infty} e^{inx^5/R}\phi_n(x^\mu). \tag{4.126}$$

   Then the value of $D_M\phi$ with (4.126) is

$$\begin{aligned} D_M\phi &= (\partial_\mu + \partial_5 + iqA_5)\phi \\ &= \sum_{n=-\infty}^{\infty} e^{inx^5/R}\left(\partial_\mu + \frac{in}{R} + \frac{iq\theta}{2\pi R}\right)\phi_n(x^\mu) \end{aligned} \tag{4.127}$$

   where we take A_μ components to be zero with a similar result for $(D_M\phi)^*$. If we let $a = in/R + iq\theta/(2\pi R)$ with similar form for a^*, then the 5d kinetic term of the scalar

field in the action becomes

$$
\begin{aligned}
S &= \int d^5x\, (D_M\phi)^*(D^M\phi) \\
&= \int d^4x \int_0^{2\pi R} dx^5 \sum_{n,m} e^{ix^5(m-n))/R} \left(\partial_\mu\phi_n^* + a^*\phi_n^*\right)\left(\partial^\mu\phi_m + a\phi^m\right). \qquad (4.128)
\end{aligned}
$$

Notice that

$$
\int_0^{2\pi R} dx^5\, e^{ix^5(m-n)/R} = 2\pi R\delta_{mn}. \qquad (4.129)
$$

We can then use the δ_{nm} in (4.129) to simplify down to one single sum over n. If we also recall that we will *not* have any cross terms (that look like a current) since $A_\mu = 0$. This gives us then the 4d action (including the mass term M^2)

$$
S = \int d^4x \sum_n \left(\partial_\mu\phi^*\partial^\mu\phi - \left(M^2 + \left[\frac{n}{R} + \frac{q\theta}{2\pi R}\right]^2\right)\right). \qquad (4.130)
$$

We can then read off the KK mass modes which matches (4.123).

2. We can obtain an effective potential $V(\theta)$ by integrating out the tower of massive KK modes. (Integrating out a heavy field whose mass depends on the value of a light scalar field to obtain an effective potential for the scalar is a standard calculation, often referred to a the *Coleman-Weinberg potential*.) Express this effective potential as a sum over KK modes and then perfom Poisson summation (see problem set 1) to rewrite it as a different sum

$$
V(\theta) = \sum_n V_n(\theta) = \sum_w \tilde{V}_w(\theta).. \qquad (4.131)
$$

Show that in the Poisson resummed expression, each term $\tilde{V}(\theta)$ is manifestly periodic in θ.

Which terms in this sum have the biggest effect on the axion potential if $2\pi MR \ll 1$?

(Ths sum is divergent, but the divergence is associated with the $w = 0$ piece that renormalizes the cosmological constant, and so does not affect the θ-dependence.)

Answer: Starting from core basics to derive the effective potential from the effective action recall that the partition function for a charged scalar field in Euclidean signature is (using chapters 33 and 34 from [3])

$$
Z = \int [D\phi][D\phi^*]e^{-S[\phi,\phi^*]}. \qquad (4.132)
$$

At 1-loop level, the effective action Γ is

$$
\Gamma = S_{\text{classical}} + \frac{i}{2}\ln\det\left(-D^2 + M^2\right) \qquad (4.133)
$$

where $D^2 \equiv D_M D^M$. The effective potential that contains the θ-dependent part lies within the natural log-determinant part since $\Gamma = -\int d^4x\, V_{\text{eff}}(\theta)$. We of course can transform the determinant into a trace. We also already know the form of $-D_M D^M + M^2 = -\Box^2 + m_n^2$ as we found above. Then

$$\operatorname{tr}\ln\left(-\Box^2 + m_n^2\right) = \frac{1}{2}\sum_{n=-\infty}^{\infty}\int \frac{d^4k}{(2\pi)^4}\ln\left(k^2 + m_n^2\right) \tag{4.134}$$

after Fourier transforming. Notice though that in (4.134) we have a highly divergent term in the k^2, so if we only want the θ-part, lets just subtract off the same propagator but evaluated at $\theta \to 0$ i.e. insert a $k^2 + m_n^2|_{\theta=0}$.

Recall that we can convert to Schwinger proper time s that allows us to integrate over the momentum differential d^4k

$$\ln A = -\int_0^\infty \frac{ds}{s} e^{-sA}. \tag{4.135}$$

Inserting (4.135) into (4.134) we have

$$V_{\text{eff}}(\theta) = -\frac{1}{2}\int_0^\infty \frac{ds}{s}\int \frac{d^4k}{(2\pi)^4}\sum_{n=-\infty}^{\infty}\left(e^{-s(k^2+m_n^2)} - e^{-s(k^2+m_n^2|_{\theta=0})}\right). \tag{4.136}$$

Notice we can now integrate over the momentum (by looking in the appendix of you favorite QFT book) namely

$$\int \frac{d^4k}{(2\pi)^4} e^{-sk^2} = \frac{1}{(4\pi s)^2}. \tag{4.137}$$

With (4.137) we can reduce (4.136) to

$$V_{\text{eff}}(\theta) = -\frac{1}{2}\int_0^\infty \frac{ds}{s}\frac{e^{-sM^2}}{(4\pi s)^2}\sum_n\left(e^{-\frac{s}{R^2}(n+q\theta/2\pi)^2} - e^{-sn^2/R^2}\right) \tag{4.138}$$

where we substituted the value of m_n^2.

Recall that the Poisson resummation formula is given as (for $c = 0$)

$$\sum_{n=-\infty}^{\infty} e^{-a(n+b)^2} = \sqrt{\frac{\pi}{a}}\sum_{n=-\infty}^{\infty} e^{-\pi^2k^2/a+2\pi ikb}. \tag{4.139}$$

For us, let $\alpha \equiv q\theta/2\pi$, then with respect to the (4.139), $a = s/R^2$ and $b = \alpha$. The θ-independent part of (4.138) is similarly but with $b = 0$. The difference in (4.138) then omits a cosine namely

$$\sum_n\left(e^{-\frac{s}{R^2}(n+\alpha)^2} - e^{-sn^2/R^2}\right) = 2R\sqrt{\pi s}\sum_{k=1}^{\infty} e^{-\pi^2R^2k^2/s}\cos(2\pi k\alpha). \tag{4.140}$$

With (4.140), (4.138) then becomes (after some algebra)

$$V_{\text{eff}}(\theta) = -\frac{R}{(4\pi)^2}\int_0^\infty \frac{ds}{s^{5/2}} e^{-sM^2} \sum_{k=1}^\infty e^{-\pi^2 R^2 k^2/s} \cos(2\pi k\alpha). \tag{4.141}$$

The proper-time integral is sadly a Bessel function with index 5/2 giving us

$$V_{\text{eff}}(\theta) = -\frac{2R}{(4\pi)^2}\sum_{k=1}^\infty \left(\frac{\pi R k}{M}\right)^{5/2} K_{5/2}(2\pi R k m)\cos(2\pi k\alpha). \tag{4.142}$$

In the small mass limit, $2\pi MR \ll 1$, this corresponds to small x in the Bessel function $K_{5/2}(x) \sim 8/(15x^{5/2})$. This simplifies our result drastically since we just get some constant multiplied by a cosine giving (restoring the value of α)

$$V_{\text{eff}}(\theta) \propto \sum_{k=1}^\infty \cos(q\theta k). \tag{4.143}$$

The sum is divergent, as it was hinted to prior, and notice that at $q = 0$ we get a constant that renormalizes (apparently) the cosmological constant.

3. Consider the Euclidean worldline effective action for the charged field in 5d with worldline γ along the curve $x^\mu(\lambda)$

$$S = m\int_\gamma d\lambda \sqrt{\frac{dx_\mu}{d\lambda}\frac{dx^\mu}{d\lambda}} + i\int_\gamma qA, \tag{4.144}$$

and show that a worldline winding w times around the circle gives a saddle point for this action. What is the action evaluated on this saddle? How does this relate to the computation in part (b)?

<u>Answer:</u> Notice that if we paramterize the worldline curve from $\lambda \in [0, s]$for some point s then $x^\mu(\lambda) =$ constant while $x^5(\lambda) = x^5(0) + 2\pi R w\lambda/s$. This ensures periodicity of the x^5 coordinate since $x^5(s) - x^5(0) = 2\pi R w$ giving us a particle with winding w around a circle. Taking x^μ constant along the worldline, we are only left with the $dx^5/d\lambda$ derivative from (4.144) namely

$$\begin{aligned}\int_\gamma d\lambda\sqrt{\frac{dx_M}{d\lambda}\frac{dx^M}{d\lambda}} &= \int_0^s d\lambda\sqrt{\left(\frac{dx^\mu}{d\lambda}\right)^2 + \left(\frac{dx^5}{d\lambda}\right)} \\ &= \int_0^s d\lambda\left|\frac{dx^5}{d\lambda}\right| = \left|x^5(s) - x^5(0)\right| \\ &= 2\pi R w.\end{aligned} \tag{4.145}$$

The $\int_\gamma qA$ term follows in a similar manner with the ansatz previously given

$$\begin{aligned} a\int_\gamma A_M dx^M &= q\int_{x^5(0)}^{x^5(s)} A_5 dx^5 \\ &= \frac{q\theta}{2\pi R}(x^5(s)-x^5(0)) \\ &= q\theta w. \end{aligned} \tag{4.146}$$

The action then evaluated on the saddle point is

$$S = 2\pi m R w + q\theta w. \tag{4.147}$$

Notice that the Eucldiean action is then simply

$$e^{-S} = e^{-(2\pi m R w + q\theta w)}. \tag{4.148}$$

This matches the result we previously found for the effective potential by expanding out the cosine in (4.143) with an extra mass term (that is hidden in the proportionality in (4.143)).

4.15.11 (PSET 4) Problem 1: Conservation of Entropy in an Expanding Universe

A key ingredient in our estimate of the axion dark matter abundance was that both the number density of axion particles n_a and the entropy density s of the Standard Model plasma scales as $1/a(t)^3$ in an expanding universe with scale factor $a(t)$, and hence the ratio n_a/s is conserved. The goal of this problem is to explain by (in thermal equilibrium– in a setting where chemical potentials are all zero) we have $s \propto 1/a(t)^3$.

First, we recall that for a particle of mass m_i in thermal equilibrium, which for momentum p has energy $E_i(p)^2 = p^2 + m_i^2$, the energy density and pressure are given by

$$\rho_i = \frac{g}{2\pi^2}\int_0^\infty \frac{E_i(p)}{\exp(E_i(p)/T)\pm 1} p^2 dp \tag{4.149}$$

$$p_i = \frac{g}{2\pi^2}\int_0^\infty \frac{p^2}{3E_i(p)}\frac{1}{\exp(E_i(p)/T)\pm 1} p^2 dp. \tag{4.150}$$

Here, g is a count of degrees of freedom (e.g. spin states), the sing ± 1 is chosen to be $+$ for bosons and $-$ for fermions, and the total energy density and pressure are given by summing over the species i that are in thermal equilibrium.

1. Show directly from (4.149), (4.150) that

$$\frac{dp}{dT} = \frac{1}{T}(p+\rho). \tag{4.151}$$

Answer: We can first compute the derivative of pressure with respect to temperature first (dropping the i-index for now) that gives

$$\begin{aligned}\frac{dp}{dT} &= \frac{g}{2\pi^2}\int_0^\infty \frac{p^2 dp}{3E(p)}\frac{d}{dT}\left(\frac{1}{e^{E/T}\pm 1}\right)\\ &= \frac{g}{2\pi^2}\int_0^\infty \frac{p^2 dp}{3}\frac{1}{T^2}\frac{e^{E/T}}{(e^{E/T}\pm 1)^2}.\end{aligned} \tag{4.152}$$

Notice however, that if we define the boson-fermion distribution to be

$$f(E) = \frac{1}{e^{E/T}\pm 1} \tag{4.153}$$

then we can write the exponentials in the integrand as

$$f(E)(1\mp f(E)) = \frac{e^{E/T}}{(e^{E/T}\pm 1)^2}. \tag{4.154}$$

Another trick we can use is by noticing that

$$\frac{\partial f(E)}{\partial E} = -\frac{1}{T}f(E)(1\mp f(E)) \implies -T\frac{\partial f(E)}{\partial E} = f(E)(1\mp f(E)). \tag{4.155}$$

Then we can write dp/dT as

$$\frac{dp}{dT} = \frac{g}{2\pi^2}\int_0^\infty \left(-\frac{1}{T}\frac{\partial f(E)}{\partial E}\right)\frac{p^4}{3}dp. \tag{4.156}$$

We can integrate by parts where the boundary term vanishes since $f(E)\to 0$ as $p\to\infty$ where the integrand changes as

$$-\frac{1}{T}\frac{\partial f(E)}{\partial E} \to +\frac{1}{T}f(E)\frac{\partial}{\partial E}\frac{p^4}{3}. \tag{4.157}$$

We can chain-rule the derivative as

$$\frac{\partial}{\partial E}\left(\frac{p^4}{3}\right) = \frac{dp}{dE}\frac{d}{dp}\left(\frac{p^4}{3}\right) = \frac{E}{p}\frac{4p^3}{3}. \tag{4.158}$$

The final form of the derivative of pressure we will use is then

$$\frac{dp}{dT} = \frac{g}{2\pi^2}\int_0^\infty \frac{4p^2 E}{3T}f(E)dp. \tag{4.159}$$

Lets now head over to the $(p+\rho)/T$ term to see what we find:

$$\frac{1}{T}(p+\rho) = \frac{g}{2\pi^2 T}\int_0^\infty \left(\frac{p^4}{3E(p)} + p^2 E(p)\right)f(E)dp. \tag{4.160}$$

It then comes down to showing that

$$\frac{dp}{dT} - \frac{1}{T}(p+\rho) = \frac{g}{2\pi^2 T}\int_0^\infty f(E)\left(\frac{4p^2E}{3} - \frac{p^4}{3E} - p^2E\right)dp = 0. \tag{4.161}$$

The p's and E's in the brackets can be integrated by parts twice, once with respect to E and then another with respect to p which will finally omit a total derivative (assuming everything vanishes on the boundary as it will thanks to the way $f(E)$ behaves at $p \to \infty$. Thus, proving (4.151).

2. Next, consider an alternative derivation. Use the thermodynamic relation

$$dU = TdS - PdV \tag{4.162}$$

with $U = \rho V$, to compute $\partial S(T,V)/\partial V$ and $\partial S(T,V)/\partial T$. Then use the relation

$$\frac{\partial}{\partial T}\frac{\partial S(T,V)}{\partial V} = \frac{\partial}{\partial V}\frac{\partial S(T,V)}{\partial T} \tag{4.163}$$

to deduce (4.151).

Answer: We don't even need the Maxwell-type relations in (4.163). Instead, consider defining first $s \equiv S/V$. Then with the first law we can derive

$$\begin{aligned} Td(sV) &= d(\rho V) + PdV \\ TsdV + TVds &= \rho dV + Vd\rho + PdV. \end{aligned} \tag{4.164}$$

Notice that ρ and s only depend upon T, not volume, allowing us to write

$$(Ts - \rho - P)dV + V\left(T\frac{ds}{dT} - \frac{d\rho}{dT}\right)dT = 0. \tag{4.165}$$

The two brackets must both vanish, and therefore with respect to the dV differential gives us the equation

$$s = \frac{\rho + P}{T}. \tag{4.166}$$

3. Show that the Friedmann equation $\dot{\rho} = -3H(\rho + p)$ can be rewritten in the form

$$\dot{p}(t) = \frac{1}{a(t)^3}\frac{d}{dt}\left(a(t)^3(\rho(t) + p(t))\right), \tag{4.167}$$

then use (4.151) to rewrite (4.167) as

$$\frac{d}{dt}\left(\frac{a(t)^3}{T}(\rho + p)\right) = 0. \tag{4.168}$$

Answer: Actually, it is much easier to first use $sT = \rho + P$ and plug into the Friedmann equation (really the continuity equation on a FLRW background)

$$\frac{d\rho}{dt} = -3H(\rho + P) = -3HTs. \tag{4.169}$$

We can use the vanishing of the second brackets from the prior question to deduce

$$\frac{ds}{dT} = \frac{1}{T}\frac{d\rho}{dT} \implies \frac{ds}{dt} = -3Hs \tag{4.170}$$

since we can drop the dT and just divide out by dt. Then consider the following

$$\begin{aligned}\frac{ds}{dt} + 3Hs &= 0 \\ &= \frac{ds}{dt}a^3 + 3Hsa^3 \\ &= \frac{ds}{dt}a^3 + 3\dot{a}a^2 s \implies \\ \frac{d(sa^3)}{dt} &= 0.\end{aligned} \tag{4.171}$$

This means that the total entropy is conserved in equilibrium and that entropy evolves as $s \propto a^{-3}$.

4. Use (4.151) and (4.162) and integrate to argue that

$$S(T, V) = \frac{V}{T}(\rho + p) + S_0 \tag{4.172}$$

with S_0 as an additive constant. Conclude that (4.168) can be interpreted as the conservation of S or equivalently, the statement that $s = S/V$ scales as $1/a(t)^3$.

Answer: We already did this prior in the last question.

Some comments: This derivation follows Steven Weinberg's book Gravitation and Cosmology. Most of what I am asking you to do is to fill in algebra steps that he omits, although he does give a few more details. I ask you to try to work this out on your own before consulting the book if you are stuck. (If you're stuck on part (a), you won't find his book helpful, but I can give you a hint.)

The conservation of s also holds when some species exit thermal equilibrium. For example, at some point in the early universe neutrinos decouple from the plasma of photons and electrons. They maintain a thermal equilibrium distribution, initially at the same temperature. Subsequently, electrons become nonrelativistic and annihilate into photons (the inverse process being Boltzmann suppressed and inefficient). Entropy conservation means that the photons inherit entropy from the electrons, and thus become hotter. The total entropy of photons and neutrinos, which are now out of equilibrium with each other and at different temperatures, matches the original entropy that was carried by all three species before the neutrinos decoupled.

The conservation of s does not hold during reheating, when relativistic Standard Model particles are produced from the decay of a heavy scalar ϕ. In this case, the particles initially produced have momentum $m\phi/2$ (in the case of 2-body decays) and only subsequently interact with each other and thermalize.

Answer:

4.15.12 Problem 2: Failure of Misalignment for Spin-1 Dark Matter

For spin-0 dark matter, a field with intial value ϕ_i and a mass m stores a potential energy density $\frac{1}{2}m_i^2\phi_i$ until the Hubble scale drops to $H \sim m$, at which case the field begins to oscillate and this store energy density is converted to non-relativistic matter. This is the misalignement mechanism, as we have discussed.

Consider a massive spin-1 particle described by the Proca action

$$S = \int d^4x\sqrt{-g}\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu\right). \tag{4.173}$$

The massive spin-1 field has three propagating degrees of freedom. Its time component A_t does not have a kinetic term and can be viewed as an auxillary field, not associated with a propagating particle. The spatial component A_i correspond to physical particles.

Explain why this action does not allow the field A_i to store energy in the same way as a spin-0 field, and hence the misalignement mechanism fails to efficiently produce spin-1 dark matter. (Hint: we are considering an expanding universe; be careful about all the ways that the metric enters the calculation).

Answer: We can consider a flat FLRW universe (where $k = 0$) implying our metric in the mostly plus signature is $g_{\mu\nu} = \text{diag}(-1, a(t)^2\delta_{ij})$. When performing the variation of the action in (4.173) with respect to the "Proca field" $A_\mu(x)$, the kinetic term varies as

$$\delta(\sqrt{-g}F_{\mu\nu}F^{\mu\nu}) = -2\nabla_\mu\left(\sqrt{-g}F^{\mu\nu}\right)\delta A_\nu(x) \tag{4.174}$$

where we performed integration by parts. Notice the inclusion of the metric weight, $\sqrt{-g}$ inside the covariant derivative. The equation of motion is then

$$\nabla_\mu\left(\sqrt{-g}F^{\mu\nu}\right) - m^2\sqrt{-g}A^\nu. \tag{4.175}$$

The metric weight in terms of the FLRW metric is $\sqrt{-g} = a(t)^3$. There are three degrees of freedom, two transverse modes and one longitudinal mode. Consider that during inflation, spacetime blows up a small patch of space, meaning we can assume that the Proca field is uniformaly distributed and picks a particular polarization. This implies that in the cosmic frame $\nabla_i A^\mu = 0$, and the time component in the equation of motion (4.175) implies that $A_0 = 0$ as long as $m \neq 0$. Then, the spatial components of $A_i \equiv \mathbf{A}(x,t)$ then obeys the following equation of motion

$$\ddot{\mathbf{A}} + 3H\dot{\mathbf{A}} + m^2\mathbf{A} = 0. \tag{4.176}$$

Consider a time when $H \gg m$, then the equation of motion simplifies to $\ddot{\mathbf{A}} + H\mathbf{A} \approx 0$. We can solve this by having $\dot{\mathbf{A}} = B/a(t)$ which gives the solution $\mathbf{A}(t) \propto 1/a(t)$. If we then have the energy density of the Proca field to be $\rho = \langle m^2A^2\rangle$ then $\rho \propto 1/a(t)^2$. This is only true however for the longitudinal mode. The two transverse modes are instead $\propto 1/a(t)^4$ and behave like radiation. In order for the misalignment mechanism to occur, we require $\rho \propto 1/a^3$ which occurs for the massive scalar field. But, we instead have $1/a(t)^2$, and therefore at the point $H \sim m$, will be diluted too much and not produce enough dark matter (i.e when we are able to observe dark matter abundances when $H \ll m$, then we won't see the correct amount of dark matter).

4.15.13 Problem 3: Misalignment Production of Axions with Early Matter Domination

Suppose that a QCD axion is discovered experimentally. We could initially learn the mass of the axion and (under the assumption that axions make up 100% of the dark matter) the strength $g_{a\gamma\gamma}$ of its coupling to photons. For a QCD axion, we can deduce the value of f/k_G from the mass, and we expect that $g_{a\gamma\gamma}$ is an $\mathcal{O}(1)$ factor multiplying $\alpha/(2\pi f/k_G)$. This provides a nice cross-check on the value of f/k_G inferred from the mass. For the purposes of this problem, let us assume that $k_G = 1$ to avoid cluttering our notation.

In the simplest misalignment scenario, where the axion begins oscillating in a radiation-dominated universe, knowing the axion mass and assuming that the axion constitutes all of the dark matter tells us the initial misalignment angle θ_i.

Consider a more complex scenario where the early universe is matter dominated, with a scalar field Φ of mass M_Φ oscillating until it decays with with Γ_Φ. (This might be the inflaton, or it might be some other modulus field–possibly the scalar superpartner of the axion itself, for example). Instead of the decay width, it is useful to think in terms of a "reheating temperature" which we can define to be

$$T_{\rm RH} = \left(\frac{90}{\pi^2 g_*(T_{\rm RH})}\right)^{1/4} \sqrt{\Gamma_\Phi M_{\rm pl}} \tag{4.177}$$

which is approximately the temperature that the Standard Model plasma has at the time that Φ has decayed away. If the axion begins oscillating when Φ still dominates the energy density of the universe, our misalignment calculation changes. Effectively, the axion dark matter abundance is diluted by the entropy dumped into the Standard Model by Φ decays. Furthermore, the initial temperature of the Standard Model plasma before Φ decays is zero, so the axion mass at the time that oscillations begin is not suppressed by the usual $(T/T_{\rm QCD})^{-n/2}$ factor.

1. Make use of simplifying assumptions, e.g. instantaneous transition from a phase dominated by Φ oscillations to a Standard Model plasma at the time when $3H = \Gamma_\Phi$, instantaneous oscillations of the axion at the time $3H = m_a(T)$, and a quadratic potential (or a cosine potential including anharmonicity corrections you might find in the literature) to compute the axion relic abundance as a function of the reheating temperature $T_{\rm RH}$ and of the initial misalignment angle θ_i. Plot the curve in the $(\theta_i, T_{\rm RH})$ plane for which an axion has the correct relic abundance, assuming specific choices of f such as $f = 10^{12}$GeV and $f = 10^{14}$GeV. The interesting range of temperatures to plot is between BBN, and the temperature at which the axion would start oscillating in a radiation dominated phase i.e. roughly MeV to GeV.

   Answer: (Note that much of the material of these answers comes from [36], [37], and [38] for axion misalignment mechanism but [39] for perturbations in inflation).

   First, lets understand the scenario previously described prior to the question. We now consider a matter dominated universe where the scale factor behaves as $a(t) \propto t^{2/3}$ implying that $H(t) = \dot{a}/a = 2/3t$. If the axion field starts to oscillate when $H \approx m_a$,

then the time this is done is at $t_{\rm osc} = 2/3m_a$. Similarly, if the decay is when $H \approx \Gamma_\Phi$ then the time of "reheating" is $t_{\rm RH} = 2/3\Gamma_\Phi$.

This allows us to find a ratio of the scale factor during reheating to that of the scale factor during oscillations namely

$$\frac{a_{\rm RH}}{a_{\rm osc}} = \left(\frac{m_a}{\Gamma_\Phi}\right)^{2/3}. \tag{4.178}$$

The number density when the axion starts oscillating is [38]

$$n_a(t_{\rm osc}) \approx \frac{1}{2} m_a(T_{\rm osc}) f^2 \theta_i^2 \tag{4.179}$$

or

$$n_a(t_{\rm RH}) = n_a(t_{\rm osc}) \left(\frac{a_{\rm osc}}{a_{\rm RH}}\right)^3 = \frac{1}{2} f^2 \theta_i^2 \left(\frac{\Gamma_\Phi}{m_a}\right)^2. \tag{4.180}$$

We can find the entropy density during reheating namely

$$s(T_{\rm RH}) = \frac{2\pi^2}{45} g_*(T_{\rm RH}) T_{\rm RH}^3 \tag{4.181}$$

which we can find the entropy-density ratio n_a/s. This can be put in terms of the reheating temperature by solving for the decay width Γ_Φ from the definition of the reheating temperature above giving

$$\frac{n_a}{s} = \frac{45}{4} \frac{g_* f^2 \theta_i^2 T_{\rm RH}^5}{m_a M_{\rm pl}^4}. \tag{4.182}$$

The present day value of the axion energy density given as $\rho_{a,0} = m_a(n_a/s)s_0$ can then be used to find the axion density namely

$$\Omega_a h^2 = \frac{45}{4} \frac{g_* f^2 \theta_i^2 T_{\rm RH}^5 s_0}{M_{\rm pl}^4} \frac{1}{\rho_c/h^2} \tag{4.183}$$

where ρ_c is the critical density. We know that $\Omega_a h^2 \approx 0.12$, and so we can find the misalignment angle to be

$$\theta_i \approx \sqrt{\frac{0.12 M_{\rm pl}^4}{C f^2 T_{\rm RH}^5}} \tag{4.184}$$

where $C \approx 3.097 \cdot 10^9 \text{GeV}^{-1}$.

Now, for our current scenario where $3H = \Gamma_\phi$, we can follow a similar procedure but with differing values of f and the potential.

We have the a similar oscillating time $t_{\rm osc} = 2/3H_{\rm osc} = 2/m_a$ since we are taking $3H = m_a$. The time of reheating is then $t_{\rm RH} = 2/\Gamma_\Phi$. The ratio of $a_{\rm RH}/a_{\rm osc}$ remains the same. The axion number density also stays the same. With the new values for the oscillating time and decay rate, we can immediately write down the number density to entropy ratio as

$$\frac{n_a}{s} = \frac{f^2 \theta_i^2 \pi^2 g_* T_{\rm RH}^5}{180 m_a^2 M_{\rm pl}^4}. \tag{4.185}$$

The present day abundance is then

$$\Omega_a h^2 = \frac{f^2\theta_i^2\pi^2 g_* T_{\rm RH}^5 s_0}{180 m_a M_{\rm pl}^4(\rho_c/h^2)}. \tag{4.186}$$

We can group the constants into a factor of C again namely

$$C = \frac{\pi^2 g_* s_0}{180(\rho_c/h^2)} \approx 1.51 \cdot 10^8 {\rm GeV}^{-1}. \tag{4.187}$$

Then, the misalignment angle is

$$\theta_i = \sqrt{\frac{0.12 m_a M_{\rm pl}^4}{C f^2 T_{\rm RH}^5}}. \tag{4.188}$$

We can plot θ_i for the two different values for f in the following manner using *Mathematica*.

```
pl=1.22*^19; (*Planck mass in GeV*)
gStar=10;       (*Effective degrees of freedom at reheating*)
s0=2891;        (*Entropy density today in cm^-3*)
rhoC=1.05*^-5;(*Critical density in GeV/cm^3*)

CC=(\[Pi]^2*gStar*s0)/(180*rhoC); (*Units:GeV^-1*)

ma[f_]:=(6*^-15)*(10^12/f); (*Axion mass in GeV*)

thetaI[T_,f_]:=Sqrt[(0.12*ma[f]*Mpl^2)/(CC*f^2*T^5)] (*Corrected Mpl^2*)

Tmin=1*^-3; (*1 MeV*)
Tmax=1;     (*1 GeV*)
Tvalues=Table[10^(Log10[Tmin]+i*(Log10[Tmax/Tmin]/1000)),{i,0,1000}];

f1=1*^12; (*f=1e12 GeV*)
thetaIValues1=thetaI[#,f1]&/@Tvalues;

f2=1*^14; (*f=1e14 GeV*)
thetaIValues2=thetaI[#,f2]&/@Tvalues;

ListLogLogPlot[{Transpose[{Tvalues,thetaIValues1Real}],
    Transpose[{Tvalues,thetaIValues2Real}]},
    PlotRange->{{Tmin,Tmax},{1*^-3,Pi}},
    PlotLabels->None,AxesLabel->
    {"Reheating Temperature Subscript[T, R H] (GeV)","Initial
    Misalignment Angle Subscript[\[Theta], i]"},
    PlotLegends->{"f==10^12 GeV","f==10^14 GeV"},
```

```
Frame->True,
FrameLabel->{"Reheating Temperature Subscript[T, R H] (GeV)",
"Initial Misalignment Angle Subscript[\[Theta], i]"},
GridLines->Automatic, Joined->True, PlotStyle->{Blue, Red},
BaseStyle->{FontSize->12}]
```

This generates the following graph.

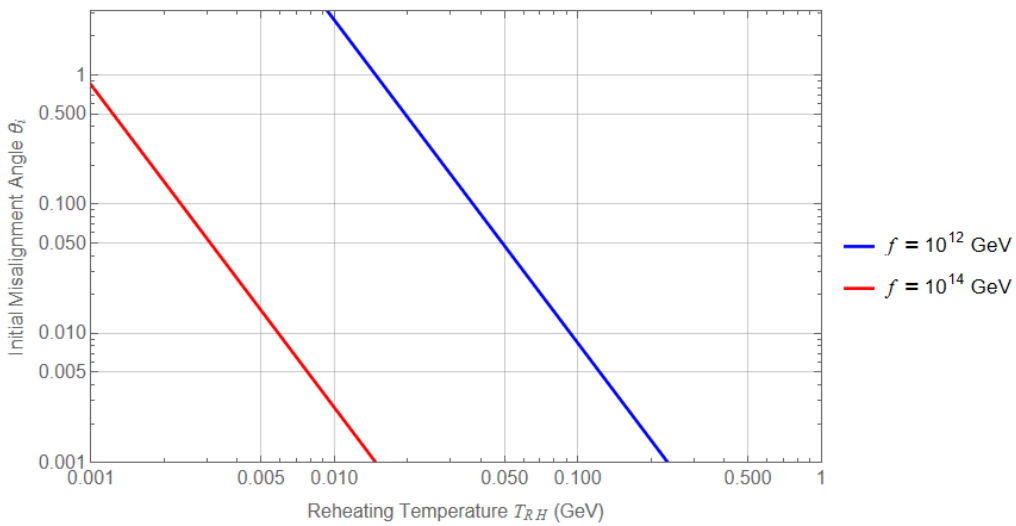

Figure 44. Reheating temperature at two different values of f.

2. Repeat the calculation with fewer simplifying assumptions. For example, use the correct shape of the axion potential and the most precise lattice estimates of the temperature dependence, and numerically solve the coupled equations for the evolution of Φ, the axion field, and the energy density of the Standard Model plasma instead of patching together instantaneous transitions. (This is somewhat open-ended. I would like you to at least think through some of what is required to do this properly, and consider whether you can add improvements to your calculation in part (a), but really doing the full calculation properly is fairly elaborate and should be considered optional. If an experiment really does find an axion, something along these lines would be one of the first things I would aim to publish to follow up.)

   Answer: Now we perform a similar analysis but with less assumptions and our potential changes to that of QCD namely (replace $\chi(T)$, the topological susceptibility, with $f^2 m_a^2(T)$ in equation (43) of [38])

$$V(a,T) = f^2 m_a^2(T)\,(1-\cos(a/f))\,. \tag{4.189}$$

   For $T \gtrsim T_{\rm QCD}$ when the mass is proportional to the temperature as [36]

$$m_a(T) \propto T^{-n} \tag{4.190}$$

with $n \approx 4$ while for $T \lesssim T_{\rm QCD}$ the mass of the axion approaches its value at zero temperature defined as $m_{a,0}$.

Moving onto the equations of motion, we have the equation of motion for the scalar Φ namely given as

$$\ddot{\Phi} + 3H\dot{\Phi} + \frac{dV_\Phi}{d\Phi} = 0 \tag{4.191}$$

where the potential is given as $V = \frac{1}{2}M_\Phi^2\Phi^2$ with an energy density of the time-kinetic term plus the potential

$$\rho_\Phi = \frac{1}{2}\dot{\Phi}^2 + V_\Phi(\Phi). \tag{4.192}$$

For the axion field, denoted as $a(t)$, we have a similar equation of motion

$$\ddot{a} + 3H\dot{a} + \frac{dV(a,T)}{da} = 0 \tag{4.193}$$

with $V(a,T)$ defined previously.

We also need the evolution of the energy density given by the continuity equation (we have already derived above) in relation to the decay rate as

$$\dot{\rho}_r + 3H\rho_r = \Gamma_\Phi\rho_\Phi \tag{4.194}$$

with the pressure being negligible (another assumption in our calculations). The first Friedmann equation gives us the relation between the Hubble parameter and the sum of energy densities as

$$H^2 = \frac{8\pi}{3M_{\rm pl}^2}(\rho_\Phi + \rho_r + \rho_a). \tag{4.195}$$

The energy density of the axion is found similar to that of the scalar field, $\rho_a = \dot{a}^2/2 + V(a,T)$. The evolution of the radiation density can be given in terms of the temperature and effective number of (relativistic) degrees of freedom as [39]

$$\rho_r = \frac{\pi^2}{30}g_*(T)T^4. \tag{4.196}$$

To solve the equations of motion numerically, consider the following code.

```
Mpl = 1.22*^19; (*Planck mass in GeV*)
gStar = 10.75;  (*effective degrees of freedom at low temperatures \
taken from Baumann*)

f = 1*^12;
ma0 = 6*^-15;

MPhi = 1*^13;
GammaPhi = 1*^2;

(*initial conditions*)
```

```
Phi0 = Mpl;
PhiDot0 = 0;
ThetaI = 2;
a0 = f*ThetaI;
aDot0 = 0;

(*time settings*)
tStart = 1*^-25;
tEnd = 1*^4;

(*define temperature-dependent axion mass m_a(T)*)
maT[T_] := If[T > 0.15, ma0*(0.15/T)^4, ma0];

(*define the axion potential derivative dV/da*)
dVda[a_, T_] := f*maT[T]^2*Sin[a/f];

(*define hubble parameter H*)
Hubble[Phi_, PhiDot_, a_, aDot_, rhoR_] :=
  Module[{rhoPhi, rhoA, rhoTotal},
   rhoPhi = (1/2) PhiDot^2 + (1/2) MPhi^2 Phi^2;
   rhoA = (1/2) aDot^2 + f^2*maT[T[rhoR]]^2*(1 - Cos[a/f]);
   rhoTotal = rhoPhi + rhoA + rhoR;
   Sqrt[rhoTotal/(3*Mpl^2)]];

T[rhoR_] := (rhoR*30/(Pi^2*gStar))^(1/4);

eqns = {Phi''[t] + (3*Hubble[Phi[t], Phi'[t], a[t], a'[t], rhoR[t]])*
      Phi'[t] + MPhi^2*Phi[t] == 0,
   a''[t] + (3*Hubble[Phi[t], Phi'[t], a[t], a'[t], rhoR[t]])*a'[t] +
     dVda[a[t], T[rhoR[t]]] == 0,
   rhoR'[t] +
     4*Hubble[Phi[t], Phi'[t], a[t], a'[t], rhoR[t]]*rhoR[t] ==
    GammaPhi*((1/2) Phi'[t]^2 + (1/2) MPhi^2 Phi[t]^2)};

initConds = {Phi[tStart] == Phi0, Phi'[tStart] == PhiDot0,
   a[tStart] == a0, a'[tStart] == aDot0, rhoR[tStart] == 1*^-20 };

vars = {Phi, a, rhoR};

sol = NDSolve[Join[eqns, initConds], vars, {t, tStart, tEnd},
   MaxSteps -> Infinity, Method -> Automatic, AccuracyGoal -> 10,
   PrecisionGoal -> 10];
```

```
PhiSol[t_] := Phi[t] /. sol[[1]];
PhiDotSol[t_] := Phi'[t] /. sol[[1]];
aSol[t_] := a[t] /. sol[[1]];
aDotSol[t_] := a'[t] /. sol[[1]];
rhoRSol[t_] := rhoR[t] /. sol[[1]];
TSol[t_] := T[rhoRSol[t]];

rhoPhiSol[t_] := (1/2) PhiDotSol[t]^2 + (1/2) MPhi^2 PhiSol[t]^2;
rhoASol[t_] := (1/2) aDotSol[t]^2 +
   f^2*maT[TSol[t]]^2*(1 - Cos[aSol[t]/f]);
HubbleSol[t_] :=
  Hubble[PhiSol[t], PhiDotSol[t], aSol[t], aDotSol[t], rhoRSol[t]];

Plot[Log10[Evaluate[rhoPhiSol[t]]], {t, tStart, tEnd},
 PlotRange -> All,
 AxesLabel -> {"t (s)", "Log10[\[Rho]_\[CapitalPhi]]"},
 PlotLabel -> "Energy Density of \[CapitalPhi] over Time"]
Plot[Log10[Evaluate[rhoRSol[t]]], {t, tStart, tEnd}, PlotRange -> All,
  AxesLabel -> {"t (s)", "Log10[\[Rho]_r]"},
 PlotLabel -> "Radiation Energy Density over Time"]
Plot[Log10[Evaluate[rhoASol[t]]], {t, tStart, tEnd}, PlotRange -> All,
  AxesLabel -> {"t (s)", "Log10[\[Rho]_a]"},
 PlotLabel -> "Axion Energy Density over Time"]

na[t_] := (rhoASol[t])/maT[TSol[t]];

comovingNa[t_] :=
 na[t]*Exp[
   3*NIntegrate[HubbleSol[\[Tau]], {\[Tau], tStart, t},
     MaxRecursion -> 10]]

naFinal = na[tEnd];
rhoA0 = ma0*naFinal;
rhoC = 1.05*^-5;
OmegaAh2 = rhoA0/rhoC;

Print["Axion Relic Abundance \[CapitalOmega]_a h^2 = ", OmegaAh2]
```

This code produces the following three graphs.

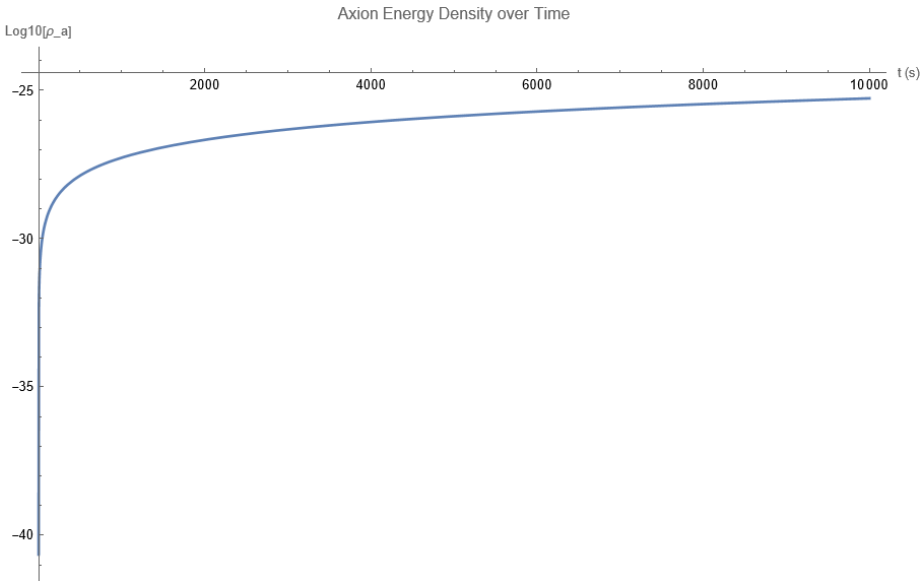

Figure 45. Energy density of the axion over time.

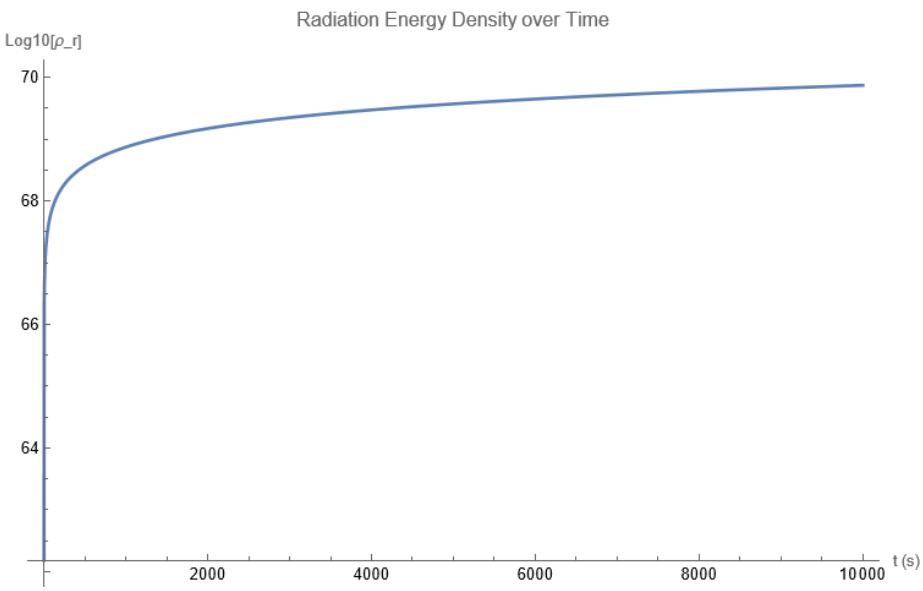

Figure 46. Energy density of radiation over time.

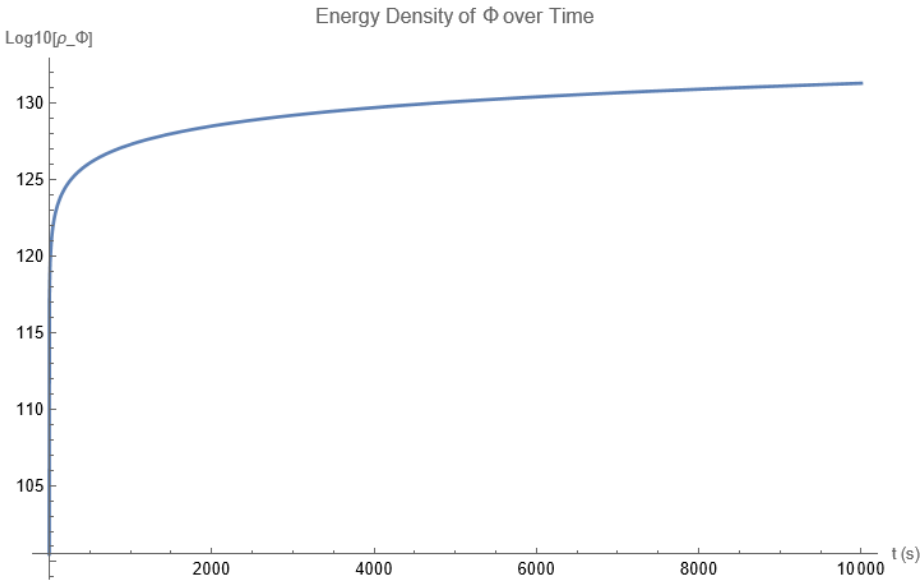

Figure 47. Energy density of the scalar field over time.

4.15.14 (PSET 5) Problem 1: Axion-fermion Interactions at Low Energies

Suppose that an axion has derivative interactions with a Standard Model fermion, like the electron. For example, we could begin with a coupling

$$\frac{c_{e_R}}{f}\partial_\mu a e_R^\dagger \sigma^\mu e_R \tag{4.197}$$

to the right-handed electron e_R. You might additionally include a similar coupling to the left-handed lepton doublet ℓ.

(a) Explain why it is reasonable to view the axion interaction, below the scale of electroweak symmetry breaking, as having the axial-vector form

$$\frac{c_e}{f}\partial_\mu a \overline{e}\gamma^\mu\gamma_5 e \tag{4.198}$$

where e is the Dirac fermion field for the electron.

Answer: Recall that after electroweak symmetry breaking (EWSB) that the Higgs acquires a vacuum expectation value that will generate mass terms coupling the left-handed electron e_L and the right-handed electron e_R into something of the form $\sim m_e \overline{e}_L e_R$. We can therefore talk about a single Dirac electron $e = e_R + e_L$. We can always pick whichever L and R-version we want by using the projection operators $P_{L,R} = (1 \mp \gamma_5)/2$. The axial current can then be written as

$$\overline{e}\gamma^\mu\gamma_5 e = \overline{e}_L\gamma^\mu e_L - \overline{e}_R\gamma^\mu e_R. \tag{4.199}$$

We can relate the right-handed electron e_R back to the interaction in (4.197) as

$$e_R^\dagger\sigma^\mu e_R = -\overline{e}_r\gamma^\mu e_R = -\overline{e}\gamma^\mu P_R e. \tag{4.200}$$

We can therefore write the original interaction as

$$\frac{c_{e_R}}{f}\partial_\mu a e_R^\dagger \sigma^\mu e_R = -\frac{c_{e_R}}{f}\partial_\mu a \bar{e}\gamma^\mu\gamma_5 e + \text{(left-handed terms)}. \tag{4.201}$$

(b) Axion dark matter experiments search for non-relativistic axions. Derive that in the non-relativistic limit, the axion interacts with the electron's spin.

Answer: Consider writing the Dirac electron in terms of its two, 2-component spinors, $e = (\phi, \chi)^T$ where we know that χ in the non-relativistic limit can be expressed in terms of ϕ as [3]

$$\chi \approx \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{2m_e}\phi. \tag{4.202}$$

In order to evaluate the fermion-current $\bar{e}\gamma^\mu\gamma_5 e$ consider first the $\mu = 0$ component

$$\bar{e}\gamma^0\gamma_5 e = \phi^\dagger\chi - \chi^\dagger\phi \tag{4.203}$$

where $\gamma^5 = \text{antidiag}(I, I)$ where $I \equiv I_{2\times 2}$. The $\mu = i$ component is

$$\bar{e}\gamma^i\gamma^5 e = \phi^\dagger\sigma^i\phi + \chi^\dagger\sigma^i\chi \approx \phi^\dagger\sigma^i\phi \tag{4.204}$$

in the non-relativistic limit, also, $\phi^\dagger\sigma^i\phi$ is the electron's spin-density.

In considering the derivative coupling of the axion, consider that in wave-dark matter, the axion fields behaves classically and so we can approximate the axion as $a(t) = a_0\cos(m_a t)$. The time derivative is then $\dot{a} \approx -m_a a(t)$. However, the spatial derivative of a will couple more strongly to the spin-components of e, and therefore we should consider that as well. We can also include a more general wave equation for the axion of the form $a = a_0\cos(m_a t - \mathbf{k}\cdot\mathbf{x})$ with derivative $\partial_i a = k_i a_0 \sin(m_a t - \mathbf{k}\cdot\mathbf{k}) \ll \dot{a}$.

The interaction in the non-relativistic limit can then be written as

$$H_{\text{int}} \approx -\frac{c_e}{f}\partial_i a \phi^\dagger\sigma^i\phi \tag{4.205}$$

since the terms that are of the form $\dot{a}\bar{e}\gamma^0\gamma^5 e$ are supressed due to $\bar{e}\gamma^0\gamma^5 e$ being small in the non-relativistic limit. In a more recognizable form, the interaction becomes

$$H_{\text{int}} \approx -\frac{c_e}{f}(\boldsymbol{\nabla} a)\cdot(\phi^\dagger\boldsymbol{\sigma}\phi) \tag{4.206}$$

Recall that the spin-operator is $\mathbf{S} = \boldsymbol{\sigma}/2$ and therefore

$$H_{\text{int}} \approx -\frac{2c_e}{f}(\boldsymbol{\nabla} a)\cdot(\phi^\dagger\mathbf{S}\phi). \tag{4.207}$$

4.15.15 Problem 2: Axion-photon Oscillations

The axion-photon coupling has the form $a\mathbf{E}\cdot\mathbf{B}$, so in an external $\mathbf{B}$-field, this behaves as a mixing between the axion and photons that have polarization perpendicular to $\mathbf{B}$. Consider the case of a uniform $\mathbf{B}$-field, and waves propagating with momentum orthogonal to $\mathbf{B}$. These waves are, in general, a superposition of an axion and a photon.

(a) Solve for propagating states of definite frequency ω. This should reduce to diagonalizing a 2×2 matrix where the off-diagonal entries depend on the axion-photon mixing. The diagonal entries should depend on the axion mass m_a, and (if the oscillations are happening in a medium instead of in a vacuum) on the plasma frequency ω_p of the photon.

Answer: Recall from problem set three the Maxwell-axion equations of motion which gives a modified Ampére's law, the typical Faraday's law, and the axion equation of motion

$$\boldsymbol{\nabla} \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = g\mathbf{B}_0 \frac{\partial a}{\partial t} \tag{4.208}$$

$$\boldsymbol{\nabla} \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0 \tag{4.209}$$

$$\left(\partial_t^2 - \nabla^2 + m_a^2\right) a = -g\mathbf{E} \cdot \mathbf{B}_0 \tag{4.210}$$

where $g \equiv g_{a\gamma\gamma}$.

Consider placing the wave propagation along the x axis with momentum $\mathbf{k} = k\hat{x}$, with a background magnetic field $\mathbf{B}_0 = B_0\hat{z}$, with an electric field polarized in the y-direction, $\mathbf{E} = E_y\hat{y}$ (therefore everything is perpendicular to each other making the solution nice).

We can combine Ampére's law with Faraday's law to find a second order differential equation in E, with a coupling between B_0 and a, as well as the axion equation of motion which both become

$$\partial_x^2 E_y - \partial_t^2 E_y + \omega_p^2 E_y = -gB_0 \partial_{x,t}^2 a \tag{4.211}$$

$$(\partial_t^2 - \partial_x^2 + m_a^2)a = -gB_0 E_y \tag{4.212}$$

where I included the plasma-frequency by hand assuming that $J^\mu = (\omega_p, \mathbf{0})$. We want plane-wave solutions, and so we would expect them to be of the form

$$E_y(x,y) = E_{0,y} e^{i(\mathbf{x}\cdot\mathbf{k} - \omega t)}, \text{ and } a(x,t) = a_0 e^{i(\mathbf{x}\cdot\mathbf{k} - \omega t)}. \tag{4.213}$$

When plugging in the plane-wave solutions I find that

$$(\omega^2 - k^2 - \omega_p^2)E_y + gB_0 k\omega a = 0 \tag{4.214}$$

$$(\omega^2 - k^2 - m_a^2)a + gB_0 E_y = 0. \tag{4.215}$$

These two equations can be expressed as a matrix

$$\begin{pmatrix} \Delta_\gamma & gB_0 k\omega \\ gB_0 & \Delta_a \end{pmatrix} \begin{pmatrix} E_y \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{4.216}$$

where $\Delta_\gamma = \omega^2 - k^2 - \omega_p^2$ and $\Delta_a = \omega^2 - k^2 - m_a^2$. We can see that the off-diagonal components depends upon the coupling between the axion-photon, as hoped for. We

can then diagonalize the matrix where we will find solutions to the frequency ω from the determinant of the matrix namely

$$\Delta_\gamma \Delta_a - (gB_0)^2 k\omega = 0. \tag{4.217}$$

Mathematica spits out something that is rather gross, so I will leave this as is for solving for ω.

(b) Use this result to derive the oscillation probability $\mathcal{P}_{\gamma\to a}(L)$ for a photon to turn into an axion after propagating a distance L. In the relativistic limit with small coupling, your expression should reduce to the form

$$\mathcal{P}_{\gamma\to a}(L) \approx \left(\frac{g_{a\gamma\gamma}B_e}{q}\right)^2 \sin^2\left(\frac{qL}{2}\right) \tag{4.218}$$

where $q \approx (m_a^2 - \omega_p^2)/2\omega$ is the difference between the momentum that a pure axion mode of frequency ω and a pure photon mode of frequency ω would have (in the absence of mixing).

<u>Answer:</u> Consider the photon field $A(x)$ and the axion field $a(x)$ both propagating along the x-direction with some general evolution described by the Schrödinger-like equation

$$i\frac{d}{dx}\begin{pmatrix} A(x) \\ a(x) \end{pmatrix} = \mathbf{H}\begin{pmatrix} A(x) \\ a(x) \end{pmatrix}. \tag{4.219}$$

We found the form of the effective Hamiltonaian prior $\mathbf{H}$ namely

$$\mathbf{H} = \begin{pmatrix} \Delta_{11} & \Delta_{a\gamma\gamma} \\ \Delta_{a\gamma\gamma} & \Delta_{22} \end{pmatrix} \tag{4.220}$$

where we have the following (effective) mass terms (simplified from the previous part) $\Delta_{11} = -\omega_p^2/2\omega$, $\Delta_{22} = -m_a^2/2\omega$ and $\Delta_{a\gamma\gamma} = gB_0/2$.

The difference between the momentum of a pure axion mode and a photon mode is therefore given by the difference in diagonal components $q = \Delta_{22} - \Delta_{11} = (m_a^2 - \omega_p^2)/2\omega$.

For small mixing, i.e. $\Delta_{a\gamma\gamma} \ll q$, the off-diagonal components are perturbations and the solutions for the photon and axion field will be in terms of sines and cosines

$$A(x) = A_0\cos(\Delta L/2) - ia_0\sin(\Delta L/2) \tag{4.221}$$

$$a(x) = a_0\cos(\Delta L/2) - iA_0\sin(\Delta L/2) \tag{4.222}$$

where $\Delta \equiv \sqrt{q^2 + 4\Delta_{a\gamma\gamma}^2} \approx q$, and we can always place $x \to L$. Therefore, the probability of finding a photon as a pure state, which implies $a_0 = 0$, a distance L is

$$\mathcal{P}_{\gamma\to a}(L) = |a(L)|^2 = \left(\frac{2\Delta_{a\gamma\gamma}}{\Delta}\right)^2 \sin^2(\Delta L/2). \tag{4.223}$$

In the limit that $\Delta \approx q$ and where $\Delta_{a\gamma\gamma} = gB_0/2$ the probability is

$$\mathcal{P}_{\gamma\to a}(L) \approx \left(\frac{gB_0}{q}\right)^2 \sin^2\left(\frac{qL}{2}\right). \tag{4.224}$$

5 Supersymmetry

These lecture notes are constructed from my own study, and thus understanding of what the subject/topic/study of *supersymmetry* entails. Therefore, I am proceeding in making these as if I would be actually teaching a course on supersymmetry, and thus I would hope that these notes, and the problem sets I make at the end, could be considered a 253c"-type of course when compared with the subjects above.

Prelude: my primary purpose in learning supersymmetry is to first understand how it was originally formulated and for what purpose. For example, we can ask questions such as:

1. How does supersymmetry (SUSY) evade very famous no-go theorems?

2. What questions in physics can SUSY answer and give understanding to in the year 2026 after recent LHC runs, i.e., no signs of supersymmetry?

3. Why should I still care about supersymmetry in 2026? Is it purely string-y motivated or are there good reasons and models to do so?

I also want to understand how supersymmetry particles play a role in scattering amplitudes at the energies at which SUSY is probed? Namely, since my primary study is cosmology, at what energy levels does supersymmetry play a primary role in cosmology? What can SUSY-models answer in cosmology that we are currently missing? How does SUSY fit into typical amplitude calculations? What are supergraphs? How does SUSY fit into spinor-helicity variables and what is the current state of research with them?

But, there are also two primary formal aspects of supersymmetry and constructing supersymmetric models. Namely, how does one construct *classical* supersymmetric models and Lagrangians, and from just the classical aspects, what physics can we extract? When I want to discuss the quantum mechanical effects of supersymmetry, what changes, how, and why? What are some quantum aspects of SUSY that makes it (apparently) amazing? Are their quantum mechanical aspects of SUSY that are pertinent and probable in cosmology today? Also, what in the world is supergravity and why should a cosmologists care?

Thus, after this long line of questioning, we will be answering the following questions (which will correspond to the number of sections in these notes):

1. How are classical supersymmetric Lagrangians constructed for different number of supercurrents $\mathcal{N}$ such that they avoid known no-go theorems, and then how are they constraints by today's experimental observations and thus fit in with known parameters of the standard model? What questions does supersymmetry answer?

2. How does supersymmetry affects observables such as the amplitude (namely the cross-section and decay rate from the amplitude) at what energies? How are amplitudes that contain matter sectors with supersymmetry computed? Namely, what are supergraphs and spinor-helicity variables?

3. What are the quantum-aspects of supersymmetry and what can it answer? What is the Seiberg-Witten theorem? What are supersymmetric index theorem and how does it tie in with Morse theory?

4. What are the different types of supergravity formulations and what implications does it have? Are they just the low-energy effective field theories for string theory? How does supergravity tie into the specific holographic correspondance of AdS/CFT?

5. What are the implications for supersymmetry in cosmology? Where does it primarily come into play? Just during inflation, or what about the percentage of possible SUSY particles left over *after* inflation that could affect baryogenesis and the quark phases of matter?

5.1 Motivation for Supersymmetry & Construction

5.2 Observables of Supersymmetry

5.3 Quantum Supersymmetry

5.4 Supergravity & Holography

5.5 Cosmology & Supersymmetry

5.6 Problem Sets:

6 String Theory 1

6.1 Introduction: Why String Theory?

6.2 Bosonic String: a First Pass

6.3 Conformal Field Theory in $D = 2$

6.4 Quantization of the Bosonic String

6.5 S-matrix for the Bosonic String

6.6 Superstring theory Algebra, States, and Effective Actions (SUGRA)

6.7 Anomaly Cancellations and UV-Completness

6.8 Amplitudes with Superstrings

6.9 Superstring theory in $D = 4$ (Calabi-Yau Compactifications)

6.10 Problem sets

6.10.1 Problem set 1: Bosonic string theory, differential geometry, and mass spectrum

Polchinski 1.3 For world-sheets with boundary, show that

$$\chi = \frac{1}{4\pi} \int_M d\tau d\sigma \sqrt{-\gamma} R + \frac{1}{2\pi} \int_{\partial M} ds\, k \tag{6.1}$$

is Weyl-invariant. Here ds is the proper time along the boundary in the metric γ_{ab}, and k is the geodesic curvature of the boundary

$$k = \pm t^a n_b \nabla_a t^b \tag{6.2}$$

where t^a is a unit vector tangent to the boundary and n^a is an outward pointing unit vector orthogonal to t^a. The upper sign is for a Lorentz world-sheet and the lower sign for a Euclidean world-sheet.

Answer: The first part of χ has already been done for us by Polchinski since $\sqrt{-\gamma'}R' = \sqrt{-\gamma}(R - 2\nabla^2\omega)$. We would only need to show that the addition to the Euler characteristic $\delta\chi$ is invariant under Weyl rescaling.

But lets check this anyways (this can be skipped if seen as unnecessary to do). First notice that $\gamma_{ab} \to \gamma'_{ab} = e^{2\omega(\sigma)}\gamma_{ab}$ under Weyl-rescaling. Lets also recall a few facts about the Ricci tensor, namely how it is constructed. Here are the necessary ingrediants:

$$R = \gamma^{ab} R_{ab} = \gamma^{ab} R^c_{acb} \tag{6.3}$$

$$R^a_{bcd} = \partial_c \Gamma^a_{db} - \partial_d \Gamma^a_{cb} + \Gamma^a_{cd}\Gamma^d_{ba} - \Gamma^c_{bd}\Gamma^d_{ca} \tag{6.4}$$

$$\Gamma^a_{bc} = \frac{1}{2} g^{ad} \left(\partial_b g_{cd} + \partial_c g_{bf} - \partial_d g_{bc}\right). \tag{6.5}$$

Then under Weyl-rescaling, the connections γ^a_{bc} changes as

$$\Gamma'^c_{bc} = \Gamma^a_{bc} + \gamma^{ad}\left(\gamma_{cd}\partial_b\omega + \gamma_{bf}\partial_c\omega - \gamma_{bc}\partial_d\omega\right) \tag{6.6}$$

$$= \Gamma^a_{bc} + W^a_{bc} \tag{6.7}$$

where in the last line we defined W^a_{bc} for convenience denoting the "Weyl" part. Notice that within the Ricci scalar after Weyl transforming we will have terms $R' \supset \{\partial_n\omega, \partial_n\omega\partial_m\omega, \partial_n\partial_m\omega\}$, anything without ω will form $\sqrt{-\gamma}R$. We can then write down a general expression

$$\sqrt{-\gamma'}R' = \sqrt{-\gamma}(R + a\nabla_d\omega\nabla^d\omega + b\nabla^2\omega) \tag{6.8}$$

for some coefficient $a, b \in \mathbb{R}$. Putting our focus on finding b, we can find this coefficient by first writing down $\nabla^2\omega$,

$$\nabla^2\omega = \nabla_a(\partial^a\omega) = \left(\partial_a\partial^a + \Gamma^a_{ab}\partial^b\right)\omega \tag{6.9}$$

$$= \partial^2\omega + \mathcal{O}(\partial\omega^3) \tag{6.10}$$

where anything of third order or higher in ω won't be present in R. We then need the transformed Riemann tensor. Out of the Riemann tensor, only terms such as $\partial_c W^a_{db}$ and $\partial_d W^a_{cb} \in R'$ since they are up to second order derivatives (anything higher we can discard). Then,

$$R' = e^{-2\omega}\gamma^{ab}\left(\partial_c W^c_{bc} - \partial_b W^c_{ca}\right). \tag{6.11}$$

Then, $\gamma^{ab}\partial_c W^c_{ba} = (2-D)\partial^2\omega$ and $\gamma^{ab}\partial_b W^c_{ca} = D\partial^2\omega$ in D-dimensions (where we will set $D = 2$ later). Then,

$$\sqrt{-\gamma'}R' = -2(D-1)\sqrt{-\gamma}2\partial^2\omega \tag{6.12}$$

implying that $b = -2(D-1)$ or in $D = 2$, $b = -2$.

To fix a, notice that since ω is a scalar function of σ then $a\nabla^b\omega\nabla_b\omega = a\partial^b\omega\partial_b\omega$. So we need to find terms such as $\partial_a\omega\partial_b\omega \in R'$. These will come from

$$R' = \gamma'^{ab}\left(\Gamma'^c_{cd}\Gamma'^e_{ba} - \Gamma'^c_{be}\Gamma'^e_{ca}\right) + \cdots \tag{6.13}$$

$$= e^{-2\omega}\gamma^{ab}\left(W^c_{ce}W^e_{ba} - W^c_{be}W^e_{ca}\right) + \cdots \tag{6.14}$$

Now notice that the Weyl-transformed parts will give

$$W^c_{ce}W^e_{ba} = 2D\partial_a\omega\partial_b\omega - D\gamma_{ab}\partial^c\omega\partial_c\omega \tag{6.15}$$

$$W^c_{be}W^e_{ca} = (D+2)\partial_a\omega\partial_b\omega - 2\gamma_{ab}\partial^c\omega\partial_c\omega. \tag{6.16}$$

When we plug these two relations into equation 14 we find that a must be

$$a = -(D-1)(D-2) \tag{6.17}$$

or in $D = 2 \implies a = 0$. Then for general dimension D we find

$$\sqrt{-\gamma'}R' = \sqrt{-\gamma}\left(R - 2(D-1)\nabla^2\omega - (D-2)(D-1)\partial_b\omega\partial^b\omega\right) \tag{6.18}$$

or in $D = 2$, we find the one Polchinski stated, i.e. $\sqrt{-\gamma}(R - 2\nabla^2\omega)$.

Now, under the integral, the Ricci part of the Euler-characteristic becomes

$$\frac{1}{4\pi}\int\limits_M d\tau d\sigma\sqrt{-\gamma}\left(R - 2\nabla^2\omega\right) = \frac{1}{4\pi}\left(\int\limits_M d\tau d\sigma\sqrt{-\gamma}R - 2\int\limits_M d\tau d\sigma\partial^a\left(\nabla_a\sqrt{-\gamma}\omega\right)\right) \tag{6.19}$$

where we used equation 1.2.32, $\sqrt{-\gamma}\nabla_a v^a = \partial_a(\sqrt{-\gamma}v^a)$. The last term will vanish in the closed string case since it is an integral of a total divergence, and we have no boundary. But, in general or in the open string case, we can instead apply Stoke's theorem where the differential will change as $d\tau d\sigma \to dsn^a(-\gamma)^{-1/2}$ or in other words,

$$-\frac{1}{2\pi}\int\limits_M d\tau d\sigma\partial^a\left(\nabla_a\sqrt{-\gamma}\omega\right) = -\frac{1}{2\pi}\int\limits_{\partial M} ds\, n^a\partial_a\omega. \tag{6.20}$$

Now, given that $k = pmt^a n_B\nabla_a t^b$, this measures how far some curve on ∂M is away from being a geodesic. t^a and n^a are unit vectors, so we will assume they have been properly normalized, but this is dependent upon the metric since $t^a t_a = t^a t^b\gamma_{ab} = \mp 1$ for the tangent vector (and similar for the normali vectors n_a) where the minus is for timelike boundaries and the plus is for spacelike. Under Weyl rescaling the condition on the tangent vector implies that the tangent vector must change as $t^{a'} = t^a e^{-\omega}$ since

$$t^{a'}t^{b'}\gamma'_{ab} = t^a t^b e^{2\omega}\gamma_{ab} \tag{6.21}$$

and similar for n^a (i.e. $n^{a'} = e^{-\omega}n^a$). Then k transforms as

$$k' = \pm t'^a n'_b\nabla'_a t'^b \tag{6.22}$$

$$= \pm t^a n_b\left(\partial_a\left(e^{-\omega}t^b\right) + \Gamma'^b_{ac}t'^c\right). \tag{6.23}$$

We know how Γ^a_{bc} transforms already given in equation 7, then the product with the tangent and normal vector becomes

$$n_b\Gamma'^b_{ac}t'^c = n_b\Gamma^b_{ac}(e^{-\omega}t^c) + n_a(\partial_c\omega)(e^{-\omega}t^c) + (\partial_c\omega)(e^{-\omega}n_a t^c) + e^{-\omega}n_b t^b(\partial_a\omega) - n_b\gamma_{ac}(\partial^b\omega)(e^{-\omega}t^c) \tag{6.24}$$

where the fourth term will disappear since $t^a n_a = 0$. We can then gather terms to find

$$k' = e^{-\omega}k \pm e^{-\omega}t^a n_b\left(-\gamma_{ac}(\partial^b\omega)t^c\right) \tag{6.25}$$

$$= e^{-\omega}k \pm e^{-\omega}n_b\left(-(\partial^b\omega)\gamma_a c t^c t^a\right) \tag{6.26}$$

$$= e^{-\omega}(k + n^a\partial_a\omega). \tag{6.27}$$

Before we start the integral, lets not forget the differential $ds = \sqrt{\gamma_{ab} dx^a dx^b}$ which transforms under Weyl-rescaling as $ds' = e^{\omega} ds$. Then the k-integral transforms as

$$\frac{1}{2\pi} \int_{\partial M} ds' \, k' = \frac{1}{2\pi} \int_{M} ds e^{\omega} \left(e^{-\omega} (k + n^a \partial_a \omega) \right) \tag{6.28}$$

$$= \frac{1}{2\pi} \int_{\partial M} ds \, k + \frac{1}{2\pi} \int_{\partial M} ds \, n^a \partial_a \omega. \tag{6.29}$$

This exactly cancels the integral in equation 20, showing that the Euler-Characteristic is Weyl invariant.

Polchinski 1.8 In the following exercises one has the usual open or closed string boundary conditions (Neumann or periodic) on X^{μ} for $\mu = 0, ..., 24$ but a different boundary condition on X^{25}. Each of these has an important physical interpretation, and will be developed in detail in chapter 8. Find the mode expansion, the mass spectrum, and (for the closed string) the constraint from σ-translation invariance in terms of the occupation numbers. In some cases you need the result of exercise 1.5.

Closed strings with

$$X^{25}(\tau, \sigma + \ell) = X^{25}(\tau, \sigma) + 2\pi R \tag{6.30}$$

where R is a constant. This is a winding string in toroidal (periodic) compactification. In this case p^{25} must be a multiple of $1/R$.

Answer: Recall that the general solution with index μ is given in equation 1.4.4 of Polchinski, but for our particular case, we can split the μ index into $\mu = \{i, 25\}$ where $i = 0, \cdot 24$ where this changes equation 1.4.4 into

$$X^i(\tau, \sigma) = x^i + \frac{p^i}{p^+} \tau + i \sqrt{\frac{\alpha'}{2}} \sum_{n=-\infty, n \neq 0}^{\infty} \left(\frac{\alpha_n^i}{n} \exp\left(\frac{-2\pi i n(\sigma + c\tau)}{\ell} \right) + \frac{\tilde{\alpha}_n^i}{n} \exp\left(\frac{2\pi i n(\sigma - c\tau)}{\ell} \right) \right) \tag{6.31}$$

where the center-of-mass coordinates are defined as

$$x^i = \frac{1}{\ell} \int_0^{\ell} d\sigma \, X^i(\tau, \sigma) \tag{6.32}$$

$$p^i(\tau) = \int_0^{\ell} d\sigma \Pi^i(\tau, \sigma) \tag{6.33}$$

$$= \frac{p^+}{\ell} \int_0^{\ell} d\sigma \, \partial_\tau X^i(\tau, \sigma). \tag{6.34}$$

Recall that the bosonic strings equation of motion is given in equation 1.2.25, but after using reparameterization invariance to enforce the conformal gauge, which then using Weyl invariance allows us to set $\omega(\sigma) = 0$, we found equation 1.3.21. Notice that this also comes from the fact that the trace of the energy-momentum tensor is

0. But in general, the solution to the string wave equation can be written in terms of left and right-moving modes written as a superposition as

$$X^\mu(\sigma,\tau) = X^\mu_L(\tau+\sigma) + X^\mu_r(\tau-\sigma). \tag{6.35}$$

Since we are with a closed string, notice two things: first, we have the following boundary condition on $i = 0, \cdot, 24$ indices, $X^i(\tau, \sigma+\ell) = X^\mu(\tau,\sigma)$ imposed on the mode expansion of equation 31. Two, since we are working with the closed string, we can save ourselves some confusion by letting $\ell = 2\pi$, since this is the length of the string (up so some modulo). But for $\mu = 25$, we have the boundary condition in equation 30. Consider re-writing our coordinates $\sigma^\pm = \tau \pm \sigma$ [**AC:**] Kiritsis:2019npv then the boundary condition becomes (where again, $2\pi = \ell$ here)

$$X^{25}_L(\sigma^+ + 2\pi) - X^{25}_L(\sigma^+) = X^{25}_R(\sigma^-) - X^{25}_R(\sigma^- - 2\pi) + 2\pi R. \tag{6.36}$$

Then when we apply ∂_{σ^+} onto the above, we find the constraint that

$$\partial_{\sigma^+} X^{25}_L(\sigma^+ + 2\pi) = \partial_{\sigma^+} X^{25}_L(\sigma^+). \tag{6.37}$$

This means that $\partial_\sigma X^{25}_L$ is a periodic function with respect to σ^-. This is also the case for $\partial_{\sigma^+} X^{25}_R$. Since $\partial_{\sigma^+} X^{25}_L$ and $\partial_{\sigma^+} X^{25}_R$ are periodic functions of σ^+ and σ^-, then we can write them as Fourier expansion as (equation 2.1.58 and 2.1.59 of [**AC:**] Green:1987sp)

$$\partial_{\sigma^+} X^{25}_L = \sum_{n=-\infty}^{\infty} \tilde{\alpha}^{25}_n e^{-in\sigma^+} \text{ and } \partial_{\sigma^-} X^{25}_R = \sum_{n=-\infty}^{\infty} \alpha^{25}_n e^{-in\sigma^-}. \tag{6.38}$$

Integrating the equations in 38 means we must have a constant being the center-of-mass position, and up to a linear term in σ^+ taken to be the momenta, with a factor of i/n integrating the exponential. But, with our term linear in σ^+, we will not want to label the momenta the same since it appears that due to the boundary condition in equation 30. This looks like

$$X^{25}_L(\sigma^+) = \frac{1}{2}x^{25}_0 + \frac{\alpha'}{2}p^{25}_L\sigma^+ + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{\tilde{\alpha}^{25}_n}{n}e^{-in\sigma^+} \tag{6.39}$$

$$X^{25}_R(\sigma^-) = \frac{1}{2}x^{25}_0 + \frac{\alpha'}{2}p^{25}_R\sigma^- + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{\alpha^{25}_n}{n}e^{-in\sigma^-}. \tag{6.40}$$

These are not the final modes expansion yet since we have not implemented the boundary conditions yet in equation 36. By shear observation, notice we will have term cancel on both sides leaving an equation of the form

$$p^{25}_L = p^{25}_R + \frac{2R}{\alpha'} \tag{6.41}$$

or restoring the factors of ℓ we have

$$p^{25}_L = p^{25}_R + \frac{4\pi R}{\alpha'\ell}. \tag{6.42}$$

Just for fun, taking a look at chapter 8 of [**AC:**] Polchinski:1998rq in comparison with equation 8.2.7, we have a winding mode number of $w = 4\pi/e \ll$ or 2.

Coming back to the problem, if we add now equation 39 and 40 with our momentum in equation 42 we find

$$X^{25}(\tau,\sigma) = \alpha_0^{25} + \frac{1}{2}\alpha' \left(p_R^{25}(\tau-\sigma) + (p_R^{25}(4\pi R/\alpha'\ell)(\tau+\sigma))\right) + i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \frac{e^{-in\tau}}{n} \left(\tilde{\alpha}_n e^{-in\sigma} + \alpha_n e^{in\sigma}\right) \tag{6.43}$$

where it is clear it would make sense if we set $p_R^{25} \to p^{25}$, but we shall leave it. This is the mode expansion.

To find the mass-spectrum, notice if we simplify the mode expansion, we just pick-up a factor of σR. This is a constant factor in the eyes of the commutator, so these do not change as in equation 1.3.25a (Polchinski) rewritten here for clarity

$$\left[\alpha_m^i, \alpha_n^j\right] = \left[\tilde{\alpha}_m^i, \tilde{\alpha}_n^j\right] = m\delta^{ij}\delta_{m,-n} \tag{6.44}$$

while the other combinations of the Fourier modes are 0. Now, lets use the amss formula (in the lightcone gauge)

$$M^2 = p_\mu p^\mu = 2p^+p^- - p^ip^i \tag{6.45}$$

where the Hamiltonian is $H = p^-$. Notice the momenta defined in equation 33 (or 34) is invariant under our change of boundary condition (i.e. not affect by the σR term since its a temporal derivative). Then, we only need to find out how the Hamiltonian changes written as (equation 1.3.19 of Polchinski)

$$H = \frac{\ell}{4\pi\alpha' p^+} \int_0^\ell d\sigma \left(2\pi\alpha'\Pi^i\Pi^i + \frac{1}{2\pi\alpha'}\partial_\sigma X^i \partial_\sigma X^i\right). \tag{6.46}$$

The last term $\sim (\partial_\sigma X^i)^2$ will be the only one to change. This will give us a factor of $R^2/(2p^+\alpha'^2)$ or $\pi^2R^2/\alpha'^2\ell^2$, and will thus modify the Hamiltonian of the open string in equation 1.3.30 by this term but we must not forget the $\tilde{\alpha}$ Fourier modes as well to find

$$H = \frac{p^ip^i}{2p^+} + \frac{1}{p^+\alpha'}\left(\sum_{n=1}^{\infty}\left(\alpha_{-n}^i\alpha_n^i + \tilde{\alpha}_{-n}^i\tilde{\alpha}_n^i\right) - 2\right) + \frac{\pi^2R^2}{\alpha'^2\ell^2} \tag{6.47}$$

where we have already used the fact that the critical dimension is $D = 26$ and the normal ordering constant is $A = -1$. Notice that N and $\tilde{N}$ pop-up, and so the mass formula in the total 26 dimensions is (where we have now set $\ell = 2\pi$)

$$M_{d=26}^2 = \frac{2}{\alpha'}\left(N + \tilde{N} - 2\frac{R^2}{2\alpha'}\right). \tag{6.48}$$

But, for the observers living in our $D = 25$ dimensional space who can not see the 26-dimension we have compactified, the mass formula then becomes

$$M_{d=25}^2 = \frac{2}{\alpha'}\left(N + \tilde{N} - 2\frac{R^2}{2\alpha'} + \frac{\alpha' n^2}{2R^2}\right) \tag{6.49}$$

where we have plugged in the value of p^{25}. Notice here that the winding momentum compactified around the circle of some radius R with a momentum of n/R that it contributes to the mass on the larger dimensions (which is quite interesting).

Finally, the level-matching constraint that arises from the σ-translation invariance cane be found by computing the total conserved momentum (equation 1.4.10 of Polchinski)

$$P = -\int_0^\ell d\sigma\, \Pi^i \partial_\sigma X^i \tag{6.50}$$

$$= -\frac{2\pi}{\ell}(N - \tilde{N} + p^{25}R). \tag{6.51}$$

The level-matching condition then becomes

$$N - \tilde{N} = p^{25}R = n \tag{6.52}$$

where we used $p^{25} = n/R$.

Polchinski 2.5 Derive the classical Noether theorem in the form usually found in textbooks. That is, assume that

$$S[\phi] = \int d^d\sigma\, \mathcal{L}\left(\phi(\sigma), \partial_a\phi(\sigma)\right) \tag{6.53}$$

and ignore the variation of the measure. Invariance of the action implies that the variation of the Lagrangian density is a total derivative

$$\delta\mathcal{L} = \epsilon\partial_a\mathcal{K}^a \tag{6.54}$$

under a symmetry transformation (2.3.1). Then the classical result, with conventional string normalization, is

$$j^a = 2\pi i\left(\frac{\partial\mathcal{L}}{\partial\phi_{\alpha,a}}\frac{1}{\epsilon}\delta\phi_\alpha - \mathcal{K}^a\right). \tag{6.55}$$

Derive the Lagrangian equations of motion and show that they imply that this current is conserved. Show that it agrees with equation (2.3.4).

<u>Answer:</u> Since $\mathcal{L} = \mathcal{L}(\phi_\alpha, \partial_a\phi_\alpha)$ then the total variation is

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_\alpha}\delta\phi_\alpha + \frac{\partial\mathcal{L}}{\partial(\partial_a\phi_\alpha)}\partial_a(\delta\phi_\alpha). \tag{6.56}$$

Then since we require the action to be invariant under $\delta\phi_\alpha$, i.e. $\delta S = 0$ then

$$\delta S = \int d^d\sigma\left(\frac{\partial\mathcal{L}}{\partial\phi_\alpha}\delta\phi_\alpha + \frac{\partial\mathcal{L}}{\partial(\partial_a\phi_\alpha)}\partial_a(\delta\phi_\alpha)\right) \tag{6.57}$$

$$= \int d^d\sigma\left(\frac{\partial\mathcal{L}}{\partial\phi_\alpha}\delta\phi_\alpha - \partial_a\left(\frac{\partial\mathcal{L}}{\partial(\partial_a\phi_\alpha)}\right)\delta\phi_\alpha + \text{Boundary term}\right) \tag{6.58}$$

where given a proper boundary condition, we recover the Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial \phi_\alpha} - \partial_a \left(\frac{\partial \mathcal{L}}{\partial(\partial_a \phi_\alpha)} \right) = 0. \tag{6.59}$$

But, if the boundary term does not vanish, i.e. $\delta\phi_\alpha$ assuming it is a symmetry, then $\delta\mathcal{L}$ is a total derivative i.e. $\delta\mathcal{L} = \epsilon\partial_a\mathcal{K}^a$ where $\mathcal{K}^a$ is assumed to be of the form $\mathcal{K}^a(\phi_\alpha(\sigma), \partial_a\phi_\alpha, \cdots)$. Then to show the current stated in equation 55 is conserved consider the following

$$\begin{aligned}
\partial_a j^a &= 2\pi i \partial_a \left(\frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \frac{1}{\epsilon} \delta\phi_\alpha - \mathcal{K}^a \right) && (6.60)\\
&= 2\pi i \left(\partial_a \left(\frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \frac{1}{\epsilon} \delta\phi_\alpha \right) - \frac{1}{\epsilon} \left(\frac{\partial \mathcal{L}}{\partial\phi_\alpha} \delta\phi_\alpha - \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \partial_a(\delta\phi_\alpha) \right) \right) && (6.61)\\
&= 0 && (6.62)
\end{aligned}$$

where in equation 61 we used the relation $\epsilon^{-1}\delta\mathcal{L} = \partial_a\mathcal{K}^a$. This show that j^a is indeed a conserved current. But does this current also obey equation 2.3.1 (in Polchinski).

Recall that we deriving the Ward identity, we let $\epsilon \to \rho(\sigma)\delta\phi_\alpha(\sigma)$ (equation 2.3.3, or similarly $\delta\phi_\alpha \to \rho(\sigma)\delta\phi_\alpha(\sigma)$). Then in a similar manner in the total variation in equation 56,

$$\begin{aligned}
\delta\mathcal{L} &= \frac{\partial \mathcal{L}}{\partial\phi_\alpha} \delta(\phi_\alpha\rho(\sigma)) + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \partial_a(\delta\phi_\alpha\rho(\sigma)) && (6.63)\\
&= \frac{\partial \mathcal{L}}{\partial\phi_\alpha} \delta(\phi_\alpha\rho(\sigma)) + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \left(\rho(\sigma)\partial_a\delta\phi_\alpha + \delta\phi_\alpha\partial_a\rho(\sigma) \right) && (6.64)\\
&= \left(\frac{\partial \mathcal{L}}{\partial\phi_\alpha} \delta\phi_\alpha + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \partial_a\delta\phi\alpha \right) \rho(\sigma) + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} (\partial_a\rho(\sigma))]delta\phi_\alpha. && (6.65)
\end{aligned}$$

If we want $\delta\mathcal{L} = \epsilon\partial_a\mathcal{K}^a$, then

$$\epsilon\partial_a\mathcal{K}^a = \frac{\partial \mathcal{L}}{\partial\phi_\alpha} \delta\phi_\alpha + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \partial_a\delta\phi\alpha. \tag{6.66}$$

This implies that the variation of the action then becomes

$$\begin{aligned}
\delta S &= \int d^d\sigma \left(\epsilon\partial_a\mathcal{K}^a\rho + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \delta\phi_\alpha\partial_a\rho \right) && (6.67)\\
&= \int d^d\sigma \left(-\epsilon\mathcal{K}\partial_a\rho(\sigma) + \frac{\partial \mathcal{L}}{\partial(\partial_a\phi_\alpha)} \delta\phi_\alpha\partial_a\rho \right) + \text{Boundary term}(\rho) && (6.68)\\
&= \frac{\epsilon}{2\pi i} \int d^d\sigma\, j^a\partial_a\rho(\sigma) && (6.69)
\end{aligned}$$

where in equation 68 we identified the quantity in parenthesis as the current j^a. Then, given equation 2.3.2, $[d\phi'] \exp(-S[\phi']) = [d\phi] \exp(-S[\phi])$, equation 69 is in agreement with equation 2.3.4 in flat space under no variation of the differential measure (equation 69 is exactly equation 2.3.4 since we can expand $\delta S = S' - S$ and divide out the S to get the factor of 1 in equation 2.3.4).

Polchinski 2.6 Consider the flat Euclidean metric δ_{ab}, in d-dimensions. An infinitesimal coordinate transformation $\delta\sigma^a = v^a(\sigma)$ changes the metric by $\delta g_{ab} = -\partial_a v_b - \partial_b v_a$. Determine the most general $v^a(\sigma)$ that leaves the metric invariant up to a local rescaling. There are $\frac{1}{2}(d+1)(d+2)$ independent solutions in $d > 2$, and an infinite number in $d = 2$. [Although the result is given in many places in the literature, the author has not found a simple derivation].

Answer: There are a lot of sources on this problem, namely using *thee* textbook for conformal field theory [**AC:**] DiFrancesco:1997nk, and so I am not sure how far this problem wants me to go, so I will stop at identifying the explicit for of $v^a(\sigma)$ (which are given in [**AC:**] Becker:2006dvp,DiFrancesco:1997nk,Kiritsis:2019npv).

Recall that for a general metric γ_{ab}, the metric transforms as 2-index tensor

$$\gamma'_{ab}(\sigma') = \frac{\partial\sigma^c}{\partial\sigma^{a'}}\frac{\partial\sigma^d}{\partial\sigma^{b'}}\gamma_{cd}. \tag{6.70}$$

Now under infinitesimal coordinate transformations, $\sigma^a \to \sigma^{a'} = \sigma^a + v^a$ the metric varies as

$$\delta\gamma_{ab} = \gamma'_{ab} - \gamma_{ab} = -\left(\partial^c v_a \gamma_{cb} + \partial^c v_b \gamma_{ca}\right) - v^c \partial_c \gamma_{ab}. \tag{6.71}$$

If we let $\gamma_{ab} \to \delta_{ab}$, i.e. we have a flat (Euclidean signature) metric, then the $\partial_c\gamma_{ab}$ term drops out. For this to be a conformal transformation around the flat metric, we can simplify equation 71 by bringing in the metric δ_{ab} to lower the indices, and then set it equal to the original metric transformation, which after plugging in the infinitesimal transformation gives the following relation

$$\partial_a v_b + \partial_b v_a = \frac{2}{d}\partial_c v^c \delta_{ab}. \tag{6.72}$$

To find more constraints on v_a, we can act with ∂^a on both sides above which allows us to combine two terms with a common factor of $\partial_c(\partial^a v_a)$ to give

$$\Box v_a + \left(1 - \frac{2}{d}\right)\partial_a\left(\partial_b v^b\right) = 0. \tag{6.73}$$

If we now do the same operation but with a second order operator $\Box \equiv \partial_a\partial^a$ on 72 we find

$$\partial_a \Box v_b + \partial_b \Box v_a - \frac{2}{d}\delta_{ab}\Box\left(\partial^c v_c\right) = 0. \tag{6.74}$$

We can then write these two equations as in a superposition to get a condition of $v_a(\sigma)$ to get

$$\left(\delta_{ab}\Box + (d-2)\partial_a\partial_b\right)\partial^c v_c = 0. \tag{6.75}$$

Already at $d = 2$, this is a special case since the constraint equation implies an infinite conformal symmetry. In other words, at $d = 2$ there exists infinitely many dimensions of local conformal deformations. If we are at $d > 2$, then the parameter $\partial_a v^a$ can at most be linear in σ^a but v^a can be second order in σ^a. This is because the differential operator acting on it is non-degenerate. We can write the most general vector v^a as

$$v_a = a_a + b_{ab}\sigma^b + c_{abc}\sigma^b\sigma^c \tag{6.76}$$

for some constant matrices $a_a, b_{ab}, c_{a,b,c} \in \mathbb{R}$. Notice c_{abc} is symmetric in the last two indices. Then, the following possibilities are allowed for v^a ([**AC:**] DiFrancesco:1997nk,Kiritsis:2019npv)

$$v^a = a^a \qquad \text{(translation)} \tag{6.77}$$

$$v^a = \omega^a_b v^b \qquad \text{(rotations)} \tag{6.78}$$

$$v^a = \lambda v^a \qquad \text{(scale transformations)} \tag{6.79}$$

$$v^a = b^a \sigma^2 - 2\sigma^a (b_b \sigma^b) \qquad \text{(special conformal transformations)} \tag{6.80}$$

Problem 5 (a) Calculate the Noether conserved currents for the $SO(D-1,1)$ Lorentz transformation global symmetry of the classical Polyakov action.

Answer: First, recall that the Polyakov action in a Minkowski background geometry is given as (note that in this case, both the metric for the worldsheet and background geometry have the same symbol $\eta^{\mu\nu}$ and η^{ab} which are only distinguishable by Greek or latin indices)

$$S_P = -\frac{1}{4\pi\alpha'} \int_M d^2\sigma \eta^{ab} \partial_a X^\mu \partial_b X^\mu \tag{6.81}$$

where $T = 1/2\pi\alpha'$. To find the Noether current associated with $SO(D-1,1)$ Lorentz transformation, we can transform the X^μ fields as $X^{\mu'} = \Lambda^\mu_\nu X^\nu + X^\mu$ which we can plug into the Polyakov action to get

$$\delta S = -\frac{1}{2\pi\alpha'} \int d^2\sigma \left(X^\nu \partial^a X^\alpha \eta_{\mu\alpha} (\partial_\alpha \Lambda^\mu_\nu) + \Lambda^\mu_\alpha \partial_a X^\alpha \partial^a X^\nu \right). \tag{6.82}$$

The last term can be integrated by parts to get the same $\partial_a \Lambda^\mu_\nu$ factor which then gives the current [**AC:**] Green:1987sp

$$J^{\mu\nu}_a = \frac{1}{2\pi\alpha'} \left(X^\mu \partial_a X^\nu - X^\nu \partial_a X^\mu \right). \tag{6.83}$$

This current is conserved due to the equation of motion $\partial^a \partial_a X^\mu = 0$.

(b) Show that in the lightcone gauge, the Lorentz generators are

$$M^{ij} = x^{[i} p^{j]} - i \sum_{n=1}^{\infty} \frac{1}{n} \alpha^{[i}_{-n} \alpha^{j]}_n \tag{6.84}$$

$$M^{i+} = x^i p^+ - \tau p^i \tag{6.85}$$

$$M^{-+} = x^- p^+ - \tau H \tag{6.86}$$

$$M^{i-} = x^i H - x^- p^i - \frac{i}{\sqrt{2\alpha'} p^+} \left(\sum_{n=1}^{\infty} \frac{1}{n} \left(\alpha^i_{-n} L_n - L_{-n} \alpha^i_n \right) \right) + B \frac{p^i}{p_+}. \tag{6.87}$$

The Virasoro generators $L_n = \frac{1}{2} \sum_{n' \in \mathbb{Z}} :\alpha_{n'} \alpha_{n-n'}:$, where the :'s indicate normal ordering, placing oscillator mode annihilation operators after creation operators, and $\alpha^i_0 = \sqrt{2\alpha'} p^i$. B is an unfixed normal ordering constant, along with A in the expression $H = (2p^+\alpha')^{-1}(L_0 + A)$ from class.

Answer: The Lorentz generators are the conserved charges found by integrating over the conserved current, in this case given by equation 83. Since the charge is conserved, we can let $a = 0$ allowing us to perform the following integral

$$M^{\mu\nu} = \oint d\sigma \, J_0^{\mu\nu}. \tag{6.88}$$

To do this where the Fourier modes are present, recall that the mode expansion for the string is

$$X^\mu(\sigma,\tau) = x^\mu + \frac{p^\mu}{p^+}\tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n\neq 0} \left(\frac{\alpha_n^\mu}{n} \exp\left(-\frac{2\pi i n(\sigma+\tau)}{\ell} \right) + \frac{\tilde{\alpha}_n^\mu}{n} \exp\left(\frac{2\pi i n(\sigma-\tau)}{\ell} \right) \right). \tag{6.89}$$

Since we set $a = 0$, the current reduces to $J_0^{\mu\nu} = X^\mu \dot{X}^\nu - X^\nu \dot{X}^\mu$. The τ-derivatives then only leave the momentum term and the Fourier modes without the factor of i/n in equation 89. When we plug X^μ and $\dot{X}^\nu$ into the current, we must utilize the commutators give in equation 44 (since we assign an n so one of the X's and an m on the other), and the cross-terms of α and $\tilde{\alpha}$ cancel after an integration over σ.

Notice however that our answer will not be normal ordered since this makes the final part of the problem much easier. We then find ([**AC:**] Green:1987sp equation 2.3.18)

$$M^{\mu\nu} = x^\mu p^\nu - x^\nu p^\mu - i \sum_{n=1}^{\infty} \frac{1}{n} \left(\alpha_{-n}^\mu \alpha_n^\nu - \alpha_{-n}^\nu \alpha_n^\mu + \tilde{\alpha}_{-n}^\mu \tilde{\alpha}_n^\nu - \tilde{\alpha}_{-n}^\nu \tilde{\alpha}_n^\mu \right) \tag{6.90}$$

where we can rewrite this in terms of spin $S^{\mu\nu}$ and $\tilde{S}^{\mu\nu}$ ([**AC:**] Polchinski:1998rq equation 1.3.4) and $\ell^{\mu\nu} = x^\mu p^\nu - x^\mu p^\nu$. Notice that our sum starts at $n = 1$ since in the lightcone gauge $\alpha_n^+ = 0$. In the lightcone gauge then, we can write the components of the generator $M^{\mu\nu}$ given that $x^+ = \tau$, $H = p^-$ we have

$$M^{+-} = x^+ H - x^- p^+ \tag{6.91}$$

$$M^{\pm i} = x^\pm p^i - x^i p^\pm \tag{6.92}$$

$$M^{ij} = x^i p^j - x^j p^i - i \sum_{n=1}^{\infty} \frac{1}{n} \left(\alpha_{-n}^i \alpha_n^j - \alpha_{-n}^j \alpha_n^i + \tilde{\alpha}_{-n}^i \tilde{\alpha}_n^j - \tilde{\alpha}_{-n}^j \tilde{\alpha}_n^i \right). \tag{6.93}$$

If we did normal order then the factor of B would arise, but this is not necessary to show the final point of the problem.

(c) Show that $M^{\mu\nu}$ satisfies the $SO(D-1,1)$ Lie algebra commutator relations, but only when $D = 26$ and all normal ordering constants are set to the correct value. In particular, $A = -1$.

It may save some effort to first calculate some smaller commutators, like those of the form $[M,\alpha]$, $[M,x]$, $[L,\alpha]$, and $[L,L]$.

Answer: Recall that in the lightcone gauge $[x^-,p^+] = -i$, $[x^i,p^j] = i\delta^{ij}$ and the same commutators for the Fourier modes in equation 44 where $i,j = 2,\cdots,D-$

1. Recall that in the lightcone gauge $p^+ = \ell/2\pi\alpha'$ (equation 1.3.17). In units of $c = 1$, recall that we have the relation that the Virasora constraints relate the Fourier modes X^i to the transverse modes where this is given in general equation 1.3.21 in Polchinski, but written in terms of lightcone coordinates we get

$$-\partial_- X^- = \partial_- X^i \partial_- X^i \tag{6.94}$$

$$\partial_+ X^- = \partial_+ X^i \partial_+ X^i. \tag{6.95}$$

The second relation gives the following equation that allows us to write the momentum in terms of Fourier modes

$$\frac{(p^i)^2 - 2p^- p^+}{2(p^+)^2} + \frac{2\pi}{\ell}\sqrt{\alpha'}2\sum_{n\neq 0}\left(\frac{p^i\alpha^i_n}{p^+} - \alpha^-_n\right)\exp\left(-\frac{2\pi i n}{\ell}\sigma^+\right) = \frac{2\pi^2\alpha'}{\ell^2}\sum_{n,m\neq 0}\alpha^i_n\alpha^i_m\exp\left(-\frac{2\pi i(n+m)}{\ell}\sigma^+\right) \tag{6.96}$$

which gives the following two relations

$$p^- = \frac{(p^i)^2}{2p^+} + \frac{1}{p^+\alpha'}\sum_{n\neq 0}\alpha^i_{-n}\alpha^i_n \tag{6.97}$$

$$\alpha^-_n = \frac{p^i\alpha^i_n}{p^+} + \frac{1}{\sqrt{2\alpha'}p^+}\sum_{n,k\neq 0}\alpha^i_{n-k}\alpha^i_k \tag{6.98}$$

and similar for $\tilde{\alpha}^i_n$. If we let [**AC:**] Green:1987sp

$$\alpha^i_0 = \sqrt{\frac{\alpha'}{2}}p^i \text{ and } \alpha^-_0 = \sqrt{\frac{\alpha'}{2}}p^- \tag{6.99}$$

then we can write equation 98 as

$$\alpha^-_n = \frac{1}{\sqrt{2\alpha'}}\frac{1}{p^+}\sum_{k=-\infty}^{\infty}\alpha^i_{n-m}\alpha^i_m. \tag{6.100}$$

Let us now return to the issue or normal ordering. Let us first normal order our p^- equation, equation 97 which gives the following

$$p^- = \frac{(p^i)^2}{2p^+} + \frac{1}{p^+\alpha'}\left(\sum_{n=1}\left(\alpha^i_{-n}\alpha^i_n + \tilde{\alpha}^i_{-n}\tilde{\alpha}^i_n\right) + 2A\right) \tag{6.101}$$

where $A = \frac{D-2}{2}\sum_{n=1}^{\infty} n$ as given in equation 1.3.31 in Polchinski [**AC:**] Polchinski:1998rq. The normal ordering version of α^-_n is also

$$\alpha^-_n = \frac{1}{2p^+}\left(\sum_{i=1}^{D-2}\sum_{m=-\infty}^{\infty} :\alpha^i_{n-m}\alpha^i_m: - A\delta_n\right). \tag{6.102}$$

With these, we can compute $[M^{i-}, M^{j-}]$ which we expect to vanish if Lorentz invariance is to hold. If this is not the case, then we are left with an anomaly. In the lightcone gauge notice that $S^{\mu+} = S^{+\mu} = 0$, but S^{i-} is cubic in oscillators in the lightcone gauge on α_n^-. Then if the anomaly is to arise, it must be quadratic in the Fourier (oscillator) modes of the form [**AC:**] Green:1987sp

$$\left[M^{i-}, M^{j-}\right] = -\frac{1}{(p^+)^2} \sum_{m=1}^{\infty} \Delta_m \left(\alpha^i_{-m}\alpha^j_m - \alpha^j_{-m}\alpha^i_m\right). \tag{6.103}$$

To find Δ_m, notice that $[x^-, (p^+)^{-1}] = i(p^+)^{-2}$ and if we let $S^j = p^+ S^{j-}$ then $[x^i, S^j] = -iS^{ij}$ giving

$$\left[J^{i-}, J^{j-}\right] = -(p^+)^{-2}\left(2ip^+\alpha_0^- S^{ij} - \left[S^i, S^j\right] - iS^i p^j + iE^j p^i\right) \tag{6.104}$$

where the term in the parenthesis can be defined as C_{ij}. To then find Δ_m, we can compute the matrix elements of $\langle 0|\, \alpha^k_m C^{ij} \alpha^l_{-m}\, |0\rangle = m^2(\delta^{ik}\delta^{jl} - \delta^{jk}\delta^{il})\Delta_m$. With the commutators of the modes in equation 44, and using the relation

$$\left[p^+\alpha^-_m, p^+\alpha^-_n\right] = (m-n)p^+\alpha^-_{m+n} + \left(\frac{D-2}{12}(m^3 - m) + 2Am\right)\delta_{m+n} \tag{6.105}$$

which gives the following relation

$$(p^+)^2 \langle 0|\, \alpha^-_m \alpha^-_{-m}\, |0\rangle = \frac{D-2}{12}(m^3 - m) + 2Am \tag{6.106}$$

and that taken as an identity

$$p^+ \langle 0|\, \alpha^-_m \sum_{n=1}^{m} \frac{1}{n}\alpha^j_{-n}\alpha^l_{n-m}\, |0\rangle = p^j p^l + \delta^{jl}\frac{m(m-1)}{2} \tag{6.107}$$

we get out

$$\Delta_m = m\left(1 - \frac{D-2}{24}\right) + \frac{1}{m}\left(A + \frac{D-2}{24}\right). \tag{6.108}$$

This shows that if Δ_m is to vanish, then $D = 26$ and $A = -1$ in order for Lorentz invariance to occur.

6.10.2 Problem set 2: CFT, OPEs, and Operator-State Theorem

Polchinski 2.1 Verify that

$$\partial\overline{\partial} \ln |z|^2 = \partial\frac{1}{z} = \overline{\partial}\frac{1}{z} = 2\pi\delta^{(2)}(z, \overline{z}) \tag{6.109}$$

(a) by use of the divergence theorem (2.1.9);

<u>Answer:</u> Recall that the divergence theorem is given by (equation 2.1.9 in Polchinski [4])

$$\int_M d^2z \left(\partial_z v^z + \partial_{\overline{z}} v^{\overline{z}}\right) = i \oint_{\partial M} \left(v^z d\overline{z} - v^{\overline{z}} dz\right) \tag{6.110}$$

where M is some complex region (and similarly its boundary ∂M). Now in order to show equation one, let $f(z,\overline{z})$ be some analytic test function. Recall that the case when $z \neq 0$, $\partial\overline{\partial}\ln|z|^2 = 0$ (as stated by Polchinski), so let $z = \overline{z} = 0$. Then if we integrate over the LHS of equation one we find

$$\int_M d^2z\, f(z,\overline{z})\partial\overline{\partial}\ln|z|^2 = f(0,0)\int_M d^2z\,\partial\overline{\partial}\ln|z|^2 \qquad (z=\overline{z}=0) \tag{6.111}$$

$$= if(0,0)\int_M d^2z\,\partial\overline{\partial}\ln|z|^2 \quad (d^2z = 2d\sigma_1\sigma_2 \equiv idz\wedge d\overline{z}) \tag{6.112}$$

$$= i^2 f(0,0)\oint_{\partial M} d\overline{z}\,\overline{\partial}\ln|z|^2 \qquad \text{(Divergence Theorem)} \tag{6.113}$$

$$= -2\pi f(0,0)\oint_{\partial M}\frac{dz}{2\pi i}\frac{1}{z} \tag{6.114}$$

$$= 2\pi f(0,0). \qquad \text{(Residue Theorem)} \tag{6.115}$$

Then, this implies equation one.

Notice we could have also shown this by building-up a clever "$\mathbb{1}$",

$$\mathbb{1} = \oint_{\partial M}\frac{dz}{2\pi i}\frac{1}{z} = -\frac{1}{2\pi i}\int_M dz\wedge\overline{z}\,\overline{\partial}\frac{1}{z}. \tag{6.116}$$

Also note that $\overline{\partial}\ln|z|^2 = 2z/(2\overline{z}) = 2/\overline{z}$ and similar for z.

(b) by regulating the singularity and then taking a limit.

Answer: We can find inspiration for a regulator denoted ϵ as Polchinski did in equation 1.3.33 by performing the limit of ϵ as it goes to zero. Then by performing the same integral in equation 3 we find

$$\int d^2z f(z,\overline{z})\partial\overline{\partial}\ln|z|^2 = \lim_{\epsilon\to 0} f(0,0)\int d^2z\,\partial\overline{\partial}\ln\Big(|z|^2+\epsilon\Big) \tag{6.117}$$

$$= \lim_{\epsilon\to 0} f(0,0)\int d^2z\,\partial\left(\frac{z}{|z|^2+\epsilon}\right) \tag{6.118}$$

$$= \lim_{\epsilon\to 0} f(0,0)\int d^2z\frac{\epsilon}{(|z|^2+\epsilon)^2} \tag{6.119}$$

$$= 2\lim_{\epsilon\to 0} f(0,0)\int_0^R dr\, r\int_0^{2\pi} d\theta\,\frac{\epsilon}{(r^2+\epsilon)^2} \tag{6.120}$$

$$= 2\pi f(0,0)\lim_{\epsilon\to 0}\left(1-\frac{\epsilon}{R^2+\epsilon}\right) \tag{6.121}$$

$$= 2\pi f(0,0) \tag{6.122}$$

where in equation 10 I used $\overline{\partial}\ln|z|^2 = (2\sigma_1 + 2i\sigma_2)/(\sigma_1^2 + \sigma_2^2)$ which simplifies to $2/\overline{z}$, and equation 13 was a basic u-substitution problem.

Polchinski 2.3 The expectation value of a product of exponential operators on the plane is

$$\left\langle \prod_{i=1}^{n} :e^{ik_i\cdot X(z_i,\overline{z}_i)}: \right\rangle = iC^X(2\pi)^D\delta^{(D)}\left(\sum_{i=1}^{n} k_i\right)\prod_{\substack{i,j=1\\ i<j}} |z_{ij}|^{\alpha' k_i\cdot k_j} \tag{6.123}$$

with C^X a constant. This can be obtained as a limit of the expectation value (6.2.17) on the sphere, which we will obtain by several methods in chapter 6.

(a) Show that the leading behavior as one vertex operator approaches another is in agreement with the OPE (2.2.14).

<u>Answer:</u> Recall that equation 2.2.14 is

$$:e^{ik_1\cdot X(z,\overline{z})}::e^{ik_2\cdot X(0,0)}:= |z|^{\alpha' k_1\cdot k_2}\ :e^{i(k_1+k_2)\cdot X(0,0)}\left(1+\mathcal{O}(z,\overline{z})\right): . \tag{6.124}$$

To do this question, we could be a little cheap by multiplying to both sides of equation 2.2.14 by a factor of

$$\left\langle :\prod_{i=3}^{n} e^{ik_i\cdot X(z_i,\overline{z}_i)}: \right\rangle \tag{6.125}$$

but really this question is getting at showing the generalized version equation 15, reproduces equation 2.2.14. To do this, lets expand the RHS of equation 15

$$\left\langle \prod_{i=1}^{n} :e^{ik_i\cdot X(z_i,\overline{z}_i)}: \right\rangle = iC^X(2\pi)^D\delta^{(D)}\left(k_1+k_2+\sum_{i=3}^{n} k_i\right)|z_{12}|^{\alpha' k_1\cdot k_2}\prod_{j=3}^{n}|z_{1j}|^{\alpha' k_1\cdot k_j}|z_{2j}|^{\alpha' k_2\cdot k_j} \tag{6.126}$$

$$\times \prod_{\substack{i,j=3\\ i<j}}^{n} |z_{ij}|^{\alpha' k_i\cdot k_j}$$

$$= iC^X(2\pi)^D\delta^{(D)}\left(\sum_{i=3}^{n} k_i\right)|z_{12}|^{\alpha' k_1\cdot k_2}\prod_{j=3}^{n}|z_{2j}|^{\alpha'(k_1+k_2)\cdot k_j}\prod_{\substack{i,j=3\\ i<j}}^{n}|z_{ij}|^{\alpha' k_i\cdot k_j} \tag{6.127}$$

$$= |z_{12}|^{\alpha' k_1\cdot k_2}\left\langle :e^{i(k_1+k_2)\cdot X(z_2,\overline{z}_2)}::\prod_{j=3}^{n} e^{ik_i\cdot X(z_i,\overline{z}_i)}: \right\rangle . \tag{6.128}$$

Notice that equation 20 is the same as equation 16 since if we expand the LHS of equation 18 we get a product of normal ordered product $e^{ik_1X}\cdot e^{ik_2X(0,0)}$ times the same $j=3$ product on the RHS of equation 20.

(b) As implied by equation (2.2.13), the expectation value is $|z_{12}|^{\alpha' k_1\cdot k_2}$ times a function that is smooth as $z_1 \to z_2$. For $n=3$, work out the explicit expansion

of this smooth function in powers of z_{12} and $\overline{z}_{12}$. From the ratio of the large-order terms find the radius of convergence.

Answer: At $n = 3$ equation 15 becomes

$$\left\langle \prod_{i=1}^{3} :e^{ik_i\cdot X(z_i,\overline{z}_i)}: \right\rangle = iC^X(2\pi)^D\delta^{(D)}\left(k_1+k_2+k_3\right)\prod_{\substack{i,j=1\\ i<j}}^{3}|z_{ij}|^{\alpha' k_i\cdot k_j} \tag{6.129}$$

where the product is given as

$$\prod_{\substack{i,j=1\\ i<j}}^{3}|z_{ij}|^{\alpha' k_i\cdot k_j} = |z_{12}|^{\alpha' k_1\cdot k_2}|z_{13}|^{\alpha' k_1\cdot k_3}|z_{23}|^{\alpha' k_2\cdot k_3} \tag{6.130}$$

and the LHS becomes (for completeness)

$$\left\langle :e^{ik_1\cdot X(z_1,\overline{z}_1)}::e^{ik_2\cdot X(z_2,\overline{z}_2)}::e^{ik_3\cdot X(z_3,\overline{z}_3)}: \right\rangle . \tag{6.131}$$

To get the ratios we want, we need to handle $|z_{13}|$. We can do this if we travel to $|z_{12}|$ then to $|z_{23}|$ which is just doing a $\pm z_2$ in the sum of $|z_{13}|$ i.e $|z_{13}| = |z_{12}| + |z_{23}|$. Then, along with the fact that $|z| = \sqrt{z\overline{z}}$ then equation 22 becomes

$$|z_{12}|^{\alpha' k_1\cdot k_2}|z_{13}|^{\alpha' k_1\cdot k_3}|z_{23}|^{\alpha' k_2\cdot k_3} = |z_{12}|^{\alpha' k_1\cdot k_2}\left|\frac{z_{12}}{z_{23}}+1\right|^{\alpha' k_1\cdot k_3}|z_{23}|^{\alpha' k_1\cdot(k_2+k_3)} \tag{6.132}$$

$$= |z_{12}|^{\alpha' k_1\cdot(k_2+k_3)}\left(1+\frac{z_{12}}{z_{23}}\right)^{\alpha' k_2\cdot k_2}\left(1+\frac{\overline{z}_{12}}{\overline{z}_{12}}\right)^{\alpha' k_1\cdot k_2}|z_{23}|^{\alpha' k_1\cdot(k_2+k_3)} \tag{6.133}$$

To handle the expressions of the form $(1+z)^n$, this is just a binomial expansion of the general form (taken from Wikipedia admittedly)

$$(x+y)^r = \sum_{k=0}^{\infty}\binom{r}{k}x^{r-k}y^k = \sum_{k=0}^{\infty}C_k(r)x^{r-k}y^k \tag{6.134}$$

where $C_k(r)$ is the choose function of the form $r!/(k!(r-k)!)$ (which we will use later). This allows us to put the expressions in paranthesis in equation 25 as

$$\left(1+\frac{z_{12}}{z_{23}}\right)^{\alpha' k_2\cdot k_2} = \sum_{k=0}^{\infty}C_k(\alpha' k_1\cdot k_3)\left(\frac{z_{12}}{z_{13}}\right)^k . \tag{6.135}$$

Then the RHS of equation 25 turns into

$$|z_{12}|^{\alpha' k_1\cdot(k_2+k_3)}|z_{23}|^{\alpha' k_1\cdot(k_2+k_3)}\sum_{k=0}^{\infty}C_k(\alpha' k_1\cdot k_3)\left(\frac{z_{12}}{z_{13}}\right)^k\sum_{n=0}^{\infty}C_n(\alpha' k_1\cdot k_3)\left(\frac{\overline{z}_{12}}{\overline{z}_{13}}\right)^n . \tag{6.136}$$

This is explicit expansion in terms of z_{12} and $\overline{z}_{12}$ powers as asked for.

To find the radius of convergence, apparently it does not matter if we perform the ratio a_{k+1}/a_k or a_k/a_{k+1} as long as you take the absolute value and perform the limit $k \to \infty$ (it's just a matter of reading off the correct radius).[125] With a_k the series expansion given in equation 27 then we find

$$\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty}\left|\frac{C_{k+1}(\alpha' k_1\cdot k_3)}{C_k(\alpha' k_1\cdot k_3)}\frac{(z_{23})^k}{(z_{23})^{k+1}}\right| \tag{6.137}$$

$$= \lim_{k\to\infty}\left|\frac{1}{z_{23}}\frac{1}{k+1}\frac{(\alpha' k_1\cdot k_3)\cdots(\alpha' k_1\cdot k_3)-(k+1)}{\alpha' k_1\cdot k_3\cdot\alpha' k_1\cdot k_3 - k}\right| \tag{6.138}$$

$$= \lim_{k\to\infty}\left|\frac{1}{z_{23}}\frac{\alpha' k_1\cdot k_3-(k+1)}{k+1}\right| \tag{6.139}$$

$$= \lim_{k\to\infty}\left|\frac{1}{z_{23}}\left(\frac{\alpha' k_1\cdot k_3}{k+1}-1\right)\right| \tag{6.140}$$

$$= \frac{1}{|z_{23}|} \tag{6.141}$$

and similar for the $\overline{z}_{23}$ term. The radius of convergence is then $R = |z_{23}|$ and $R = |\overline{z}_{23}|$ for the second series. Then the entire radius of convergence for the series expansion given in equation 28 is $|z_{23}|$. (Apologies for the abuse of notation where $|z|$ can denote the absolute value and/or the magnitude of the complex value, and in the series sum above a k without an index is just a sum index while a k with k_i (or similar) is the momentum).

(c) Give a general proof that the free-field OPE (2.2.4) is convergent inside the dashed circle in figure 2.1.

Answer: Polchinski actually outlines the argument in section 2.9 for a general set of operators. (There is also a nice outline of this in [**?**]).

Recall equation 2.2.4

$$X^\mu(z_1,\overline{z}_1)X^\nu(z_2\overline{z}_2) = -\frac{\alpha'}{2}\eta^{\mu\nu}\ln|z_{12}|^2 + :X^\nu X^\mu(z_2,\overline{z}_2): \tag{6.142}$$

$$\sum_{k=1}^{\infty}\frac{1}{k!}\left((z_{12})^k :X^\nu\partial^k X^\mu(z_2,\overline{z}_2): + (\overline{z}_{12})^k :X^\nu\overline{\partial}^k X^\mu(z_2,\overline{z}_2):\right). \tag{6.143}$$

This is an operator equation, and thus we are able to insert a product of operators $\mathcal{O}(w_1)\cdots\mathcal{O}_n(w_n)$ as a string of time-ordered correlation functions. We are then left with a series expansion in powers of z_{12} and $\overline{z}_{12}$ with coefficients the expectation values involving $z_2, \overline{z}_2$ and w. Due to the conformal symmetry present, the correlation function will only depend on the distance between points (such as with the OPEs)

$$\langle :X^\nu\partial^k X^\mu(z_2,\overline{z}_2)\mathcal{O}_1(w_1)\cdots := f_k\left(|w_i - z_2|, |w_i - w_j|\right)$$

[125] The first ratio came from my real analysis text by Abbott and the second comes from Wikipedia. I am going with Abbott's definition but the answers are the same.

for some functional f. We can express the functional in terms of their conformal weights due to the rescaling present since $z \to \lambda z$ (and similar for $\overline{z}$) then the operators transform as $\mathcal{O} \to \lambda^{-h}\mathcal{O}$. Then the functional is equal to

$$\prod_{i,j\neq i} |w_i - z_2|^{-(k+h_i)} |w_i - w_j|^{-(h_i - h_j)} g_k(w_i, z_2) \tag{6.144}$$

where g is some conformally invariant function and k is the scaling dimension.

Now, referring back to figure 2.1, there are four operator insertions with a contour drawn around z_1, z_2 and thus f can only be a function of $|w_1 - z_2|$ and $|w_2 - z_2|$ (where w_i can just be some other measuring point). The function f is then given as [**?**]

$$f_k(|w_i - z_2|, |w_i - w_j|) = c_k |w_1 - z_2|^{-(k+h_1)} |w_2 - z_2|^{-(k+h_2)} |w_2 - w_1|^{-(h_2+h_1)}. \tag{6.145}$$

Now our reasoning from part (b) can finally come into play. The coefficients that are constant in k given in 34 we don't care about. The large order terms, based off of part (b), are given as

$$\left(\frac{z_{12}}{|w_i - z_2|}\right)^k. \tag{6.146}$$

The limit of the normal of this series ratio will be less iff $z_{12} < \min(|w_1 - z_2|, |w_2 - z_2|)$. The OPE converges then within the dashed lines of figure 2.1.

There is also a similar argument, that is much shorter and more physical, given in [**?**].

Polchinski 2.7 (a) By computing the relevant OPEs, confirm the weights stated in equation (2.4.17) and determine which operators are tensors.

<u>Answer:</u> Lets start with $T(z)X(0,0)$

$$\begin{aligned}
T(z)X^\mu(0,0) &= -\frac{1}{\alpha'} :\partial X^\nu \partial X_\nu: (z) X^\mu(0,0) && (6.147)\\
&= -\frac{2}{\alpha'} \partial \overbracket{X^\nu(z) X^\mu}(0) \partial X_\nu(z) && (6.148)\\
&= \frac{2}{\alpha'} \partial \left(-\frac{\alpha'}{2} \eta^{\mu\nu} \ln |z|^2\right) \partial X_\nu(z) && (6.149)\\
&\sim \frac{1}{z} \partial X_\nu(z) && (6.150)
\end{aligned}$$

which if we compare with equation 2.4.16 X^μ has weight $h = 0$, which since the OPE in equation 42 has the structure of a primary field is a tensor. Note we have a factor of 2 in equation 40 due to the Wick contraction (and hence the insertion into equation 41) which will appear consistently throughout all of the following OPEs.

Next is $T(z)\partial X^\mu(0,0)$ via a trick. Let X^μ be at w and consider the following:

$$T(z)\partial X^\mu(w) = -\frac{1}{\alpha'} :\partial X^\nu \partial X_\nu: (z)\partial X^\mu(w) \tag{6.151}$$

$$= \partial_w \left(\frac{\partial X^\mu(w)}{z-w} \right) \tag{6.152}$$

$$= \frac{\partial X^\mu(w)}{(z-w)^2} + \frac{\partial^2 X^\mu}{z-w} \tag{6.153}$$

$$\sim \frac{\partial X^\mu(0)}{z^2} + \frac{\partial^2 X^\mu(0)}{z}. \tag{6.154}$$

We can then read off from equation 46 that ∂X^μ has weight $h = 1$, and thus the antiholomorphic part, $\overline{\partial} X^\mu$ has weight $\tilde{h} = 1$. Both of these are then tensors as well due to the structure of their OPEs.

We can perform the same trick as we did previously for $\partial^2 X^\mu$:

$$T(z)\partial^2 X(0) = \partial_w \left(\frac{\partial X^\mu(w)}{(z-w)^2} + \frac{\partial^2 X^\mu}{z-w} \right) \Big|_{w=0} \tag{6.155}$$

$$\sim \frac{2\partial X^\mu(0)}{z^3} + \frac{2\partial^2 X^\mu}{z^2} + \frac{\partial^3 X^\mu}{z} \tag{6.156}$$

which shows that $\partial^2 X^\mu$ is a non-primary field which implies it is not a tensor. But, it does have weight of $h = 2$.

Finally, the OPE for $:e^{ik\cdot X}::$

$$T(z) :e^{ik\cdot X(w)}: = -\frac{1}{\alpha'} :\partial X^\nu \partial X_\nu: (z) :\sum_{n=0}^{\infty}: \frac{i^n}{n!}(k\cdot X)^n \tag{6.157}$$

$$= -\frac{1}{\alpha'} \Bigg(\sum_{n=0}^{\infty} 2\frac{ni^n}{n!} k_\nu \partial \overbracket{X^\mu(z) X^\nu}(w) :(k\cdot X)^{n-1} \partial X_\mu(z): +$$

$$\left(\sum_{n=0}^{\infty} 2\frac{n(n-1)i^n}{n!} k_\nu k_\sigma \overbracket{\partial X^\mu(z) \partial X_\mu(z) X^\nu(w) X^\sigma}(w) :(k\cdot X)^{n-2}: \right) \Bigg) \tag{6.158}$$

$$= -\frac{1}{\alpha'} \Bigg(2ik_\nu \sum_{n=1}^{\infty} \left(-\frac{\alpha'}{2} \frac{\eta^{\mu\nu}}{z-w} \right) \frac{i^{n-1}}{(n-1)!} :(k\cdot X)^{n-1} \partial X_\mu:$$

$$+ k_\nu k_\sigma i^2 \sum_{n=2}^{\infty} \frac{i^{n-2}}{(n-2)!} :(k\cdot X)^{n-2}: \left(-\frac{\alpha'}{2} \frac{\delta^\nu_\mu}{z-w} \right) \left(-\frac{\alpha'}{2} \frac{\eta^{\mu\nu}}{z-w} \right) \Bigg) \tag{6.159}$$

$$= \left(\frac{ik_\mu \partial X^\mu}{z-w} + \frac{k^2\alpha'}{4} \frac{1}{(z-w)^2} \right) :e^{ik\cdot X(w)}: \tag{6.160}$$

$$\sim \frac{\alpha' k^2}{4} \frac{:e^{ik\cdot X(w)}:}{(z-w)^2} + \frac{\partial :e^{ik\cdot X(w)}:}{z-w} \tag{6.161}$$

which implies that the operator $:e^{ik\cdot X(w)}:$ has weight $h = \alpha' k^2/4$ and by symmetry arguments $h = \tilde{h}$. Note in the last line we Taylor expanded ∂X^μ. Also,

since a pole on the order of z^{-3} is not present, then $:e^{ik\cdot X(w)}:$ is a primary field and thus a tensor.

(b) Do this for the same operators in the linear dilaton theory.

Answer: In the linear dilaton theory, the component of the energy-momentum tensor we need is given in equation 2.5.1a as

$$T(z) = -\frac{1}{\alpha'} :\partial X^\mu \partial X_\mu: + V_\mu \partial^2 X^\mu \tag{6.162}$$

where V_μ is some fixed D-vector. The OPEs we need to compute are with respect to X^μ, ∂X^μ, $\partial^2 X^\mu$ and $:e^{ik\cdot X(w)}:$ since the weights $(h, \tilde{h})$ from ∂X^μ will be the same as well as $:e^{ik\cdot X(w)}:$ (just change all the holomorphic coordinate to anti-holomorphic and then we are done).

I will do the $T(z)X^\mu(0)$ calculation explicitly, but notice that we are just adding a factor of $V_\mu \partial^2 X^\mu$ to the OPE, so we really only need to calculate $V_\mu \partial^2 X^\mu$ part of $T(z)$ with whatever operator we choose.

$T(z)X^\mu(0)$ becomes the following:

$$\begin{aligned}
T(z)X^\mu(0) &= \left(-\frac{1}{\alpha'}\partial X^\nu \partial X_\nu(z) + V_\nu \partial^2 X^\nu\right) X^\mu(0) && (6.163)\\
&= -\frac{2}{\alpha'}\partial \overbracket{X^\mu(0) X^\mu}(0) \partial X_\nu(z) + V_\nu \partial^2 \overbracket{X^\nu(z) X^\mu}(0) && (6.164)\\
&= -\frac{2}{\alpha'}\partial \left(-\frac{\alpha'}{2}\eta^{\mu\nu} \ln |z|^2\right) \partial X_\nu(z) + V_\nu \partial^2 \left(-\frac{\alpha'}{2}\eta^{\mu\nu} \ln |z|^2\right) && (6.165)\\
&\sim \frac{\alpha'}{2}\frac{V^\mu}{z^2} + \frac{\partial X^\mu(0)}{z} && (6.166)
\end{aligned}$$

Notice that now X^μ is not a tensor anymore, and has ill-defined weights.

Now to compute $T(z)\partial X^\mu$, we can perform the same trick as in part (a), but also only take into account the extra factor of V^μ, since everything else will be the same. This gives a factor of z^3 showing the tensor is no longer primary which can be seen in its full form

$$T(z)\partial X^\mu(0) = \sim \frac{\alpha' V^\mu}{z^3} + \frac{\partial X^\mu(0)}{z^2} + \frac{\partial^2 X^\mu(0)}{z}. \tag{6.167}$$

$\partial X^m u(0)$ still has $h = 1$, but is no longer a tensor/primary field. This is also true of $\overline{\partial} X^\mu(0)$ since $\tilde{h} = 1$.

Taking another derivative with respect to w in equation 59 (just let $w = 0$ at the end) to find the $\partial^2 X^\mu(0)$ weight, we get a z^{-4} term still showing it is not a tensor field, and since the V^μ on equation 59 is not attached to the z^{-1} factor, it will keep the same weight of $(h, \tilde{h}) = (2, 0)$ just as in part (a).

Finally, the OPE with $:e^{ik\cdot X(w)}:$ seems a little tricky, but since $V_\mu \partial^2 X^\mu$ term in $T(z)$ is a second order derivative, it will only contribute to the z^{-2} pole in equation 51 giving us an extra factor being added on of the form $i\alpha' V_\mu k^\mu / 2z^2$ giving us the following weights

$$(h, \tilde{h}) = \frac{\alpha'}{4}\left(k^2 + 2iV_\mu k^\mu, k^2 + 2iV_\mu k^\mu\right) \tag{6.168}$$

showing it is still a primary field, but with complex weights. (No idea how to interpret that).

Polchinski 2.9 Derive the central charges for the linear dilaton, bc , and $\beta\gamma$ CFTs by working out the TT OPEs.

Answer: Recall that the normal TT is given in equation (2.4.25) as

$$T(z)T(0) \sim \frac{c}{2z^4} + \frac{2}{z^2}T(0) + \frac{1}{z}\partial T(0) \tag{6.169}$$

where the central charge is identified with the number of dimensions D. Notice that only terms on the order of z^{-4} contribute to the central charge by arguments in Polchinski. Then, the only terms we need to care about are those of fourth order in derivatives between the product of $T(z)T(0)$ (this will be true for all of the following OPEs). The term that does this in the linear dilaton CFT is the following

$$T(z)T(0) \supset V_\mu V_\nu \partial^2 \overline{X^\mu(z)\partial^2 X^\nu}(0) \tag{6.170}$$

$$= V_\mu V_\nu \partial_z^2 \partial_w^2 \left(-\frac{\alpha}{2}\eta^{\mu\nu}\ln|z|^2\right)\Big|_{w=0} \tag{6.171}$$

$$= V^2\left(\frac{6\alpha'}{2z^4}\right). \tag{6.172}$$

When we combine this with the regular term we get for the OPE of $T(z)T(0)$ we find

$$T(z)T(0) \supset \frac{1}{z^4}\left(D + 6\alpha' V^2\right) \tag{6.173}$$

implying that $c = D + 6\alpha' V^2 = \tilde{c}$.

For the bc conformal field theory the antiholomorphic energy-momentum tensor is $\tilde{T}(\overline{z}) = 0 \implies \tilde{c} = 0$. Then we only need to worry about $T(z)$ For the bc theory this is given as

$$T(z) = (1-\lambda) :\partial(bc): -\lambda :b\partial c: . \tag{6.174}$$

Then, just as above, the central charge will only come from terms of fourth order derivatives, or with double contractions,

$$\begin{aligned}
T(z)T(0) &= (1-\lambda)^2 :\partial bc(z)::\partial bc(0): +\lambda^2 :b\partial c(z)::b\partial c(0): - \\
&\quad \lambda(1-\lambda)\left(:\partial bc(z)::b\partial c(0): + :b\partial c(z)::\partial bc(0):\right) && (6.175)\\
&\sim (1-\lambda)^2 \partial \overline{b(z)\partial c}(0)\overline{c(z)b}(0) + \lambda^2 \overline{b(z)\partial c}(0)\partial\overline{c(z)b}(0) \\
&\quad - \lambda(1-\lambda)\left(\partial \overline{b(z)\partial c}(0)\overline{c(z)b}(0) + \overline{b(z)c}(0)\partial\overline{c(z)\partial b}(0)\right) + \cdots && (6.176)\\
&= (1-\lambda)^2\left(-\frac{1}{z^4}\right)\left(\frac{1}{z^4}\right) + \lambda^2\left(\frac{1}{z^4}\right)\left(-\frac{1}{z^4}\right) \\
&\quad - \lambda(1-\lambda)\left(\left(-\frac{2}{z^3}\right)\left(\frac{1}{z}\right) + \left(\frac{1}{z^2}\right)\left(-\frac{2}{z^3}\right)\right) + \cdots && (6.177)\\
&= \frac{1}{z^4}\left(-(1-\lambda)^2 - \lambda^2 + 4\lambda(1-\lambda)\right) && (6.178)\\
&= \frac{1}{z^4}\left(-3(2\lambda-1)^2 + 1\right) && (6.179)
\end{aligned}$$

which is the exact same central charge c as given in equation 2.5.12.

Finally for the $\beta\gamma$ CFT we have the exact same $T(z)$ as given in the bc CFT equation 66, just swap the bc for $\beta\gamma$. But, the main difference is that now $\beta\gamma$ are commuting fields. This implies that $\beta(z_1)\gamma(z_2) = -1/z_{12}$ but $\gamma(z_2)\beta(z_1) = +1/z_{12}$ (just let $z_2 = 0$ to get the correct forms for the $T(z)T(0)$ OPE). This means we need to just track the minus sign through our previous derivation to find we pick up an overal minus sign in the central charge giving $c = 3(2\lambda - 1)^2 - 1$ (and still $\tilde{c} = 0$).

Polchinski 2.11 Evaluate the central charge in the Virasoro algebra for X^μ by calculating

$$L_m \left(L_{-m} \left|0;0\right\rangle\right) - L_{-m} \left(L_m \left|0;0\right\rangle\right). \tag{6.180}$$

Answer: The entirety of this problem rests on summation indices and the following commutator from chapter one when we quantized the bosonic string

$$[\alpha^\mu_n, \alpha^\nu_m] = n\eta^{\mu\nu}\delta_{m,n} \tag{6.181}$$

and how it can be manipulated by raising and lowering indices.

The question is asking to compute $[L_m, L_{-m}]\left|0;0\right\rangle$. Recall that the Virasoro generators can be written in terms of the bosonic string's creation and annhilation operators as

$$L_m = \frac{1}{2}\sum_{n=-\infty}^{\infty} \alpha^\mu_{m-n}\alpha_{n,\mu}. \tag{6.182}$$

Then when we plug these modes into equation 72 we first find

$$[L_m, L_{-m}]\left|0;0\right\rangle = \frac{1}{4}\sum_{n,k=-\infty}^{\infty}\left(\alpha^\mu_{m-n}\alpha_{n,\mu}\alpha^\nu_{-m-k}\alpha_{k,\nu} - \alpha^\mu_{-m-k}\alpha_{k,\mu}\alpha^\nu_{m-n}\alpha_{n,\nu}\right)\left|0;0\right\rangle. \tag{6.183}$$

Notice that terms proportional to $\alpha^\mu_{m-n}\alpha_{n,\mu}$ will annhilate on the vacuum since $\alpha_{n,\mu}$ will, but the term will only total disappear if α^μ_{m-n} will as well which will only happen when $n > 0$ or $m > n$, but if $n < 0$ then $\alpha_{n,\mu}$ will not. So the second term in equation 75 disappears. We then just need to keep inputting the commutator in equation 73 until we are left with terms without oscillators. This is done as the following

$$\begin{aligned}
\alpha^\mu_{m-n}\alpha_{n,\mu}\alpha^\nu_{-m-k}\alpha_{k,\nu} &= \alpha^\mu_{m-n}\left(\alpha^\nu_{-m-k}\alpha_{n,\mu} + n\eta^\nu_\mu\delta_{n,-m-k}\right)\alpha_{k,\nu} && (6.184)\\
&= \alpha^\mu_{m-n}\alpha^\nu_{-m-k}\alpha_{n,\mu}\alpha_{k\nu} + n\eta^\nu_\mu\delta_{n,-m-k}\alpha^\mu_{m-n}\alpha_{k,\nu} && (6.185)\\
&= \left(\alpha^\nu_{-m-k}\alpha^\mu_{-m-n} + (m-n)\delta_{n,-k}\eta^{\mu\nu}\right)\left(n\eta_{\mu\nu}\delta_{k,n} + \alpha_{k,\nu}\alpha_{n,\mu}\right)\\
&\quad + n\eta^\nu_\mu\delta_{n,-m-k}\left(\alpha_{k,\nu}\alpha^\mu_{m-n} + (m-n)\delta_{m-n,-k}\delta^\mu_\nu\right) && (6.186)\\
&= n(m-n)\delta_{n,-k}\delta_{k,n}\eta^{\mu\nu}\eta_{\mu\nu} + n(m-n)\eta^\nu_\mu\eta^\mu_\nu\delta_{m-n,-k}\delta_{n,-m-k} && (6.187)\\
&= 2D(-k)(m+k) && (6.188)
\end{aligned}$$

where in equation 79 we dropped the terms that would annihilate on the vacuum. What we are left with are two sums of the form $\sum k$ and $\sum k^2$ that have well known

forms being $m(m+1)/2$ and $m(m+1)(m+1)/6$ respectively. When added together we find the following commutator

$$[L_m, L_{-m}]\,|0;0\rangle = \frac{D}{12}m(m^2-1) \tag{6.189}$$

Polchinski 3.6 On dimensional grounds, a current that is conserved on a flat world-sheet has the possible anomaly

$$\nabla_a j^a = aR. \tag{6.190}$$

By comparing the Weyl transformations of the two sides, show that the sum of the coefficients of z^{-3} in Tj and of $\overline{z}^{-3}$ in $\tilde{T}\tilde{J}$ is $4a$.

<u>Answer:</u> If we treat the current here as an arbitrary holomorphic function, then we would expect the general expansion of the energy-momentum tensor together with the current to be (equation (2.4.11) of Polchinski)

$$T(z)j(w) \sim \sum_{n=0}^{\infty} \frac{A^n}{(z-w)^{n+1}} j^n(w). \tag{6.191}$$

where we are only keeping the singular terms. A_n are some coefficients that are to be determined, specifically we would want A_3 (for example, $A_3 = 1 - 2\lambda$ in the bc CFT).

Now, the Weyl transformation we know the Ricci scalar transforms as $\delta R = -2\nabla^2\omega$, then under the Weyl transformation the current should be invariant but it may be equal to the central charge c giving the following relation from equation 82 $aR = -c\nabla^2\omega$. Then if $a = c/3R$ and given that $c = -12a$ from equation (3.4.15), then the A_3 sum in equation 83 gives $A_3 = 4a$. (Note: there is a floating $v(z)$ around attached to the current, but if we are using the definition of the current, then it is just absorbed into j).

Something to note though is that if we are only working with the free bosonic string, then we would never find a term on the order of z^{-3} since we would have 4 products of ∂X.

These kind of anomalies could only occur when $\partial X^\mu(z)$ is not a primary field such as in the case of linear dilaton theory.

(Note: the following derivation is really a re-wording of Polchinski's introduction to the Weyl anomaly in section 3.4 worded differently in terms of contact terms and speaking directly about how the number of operators must be odd in order to have the Weyl-anomaly occur; or at least, this is how it appears to me.)

Another way of thinking about this problem is that in order to get a term on the order of z^{-3}, we would need to require an odd number of field variables. So, for example, consider the following OPE that keeps tracks of contact terms using $\overline{\partial}(1/z) = 2\pi\delta^{(2)}(z)$ (what we used in problem 2.1 of Polchinski),

$$\overline{\partial}T(z)T(0) = 2\pi\left(-\frac{c}{12}\partial^3\delta^{(2)}(z) - 2\partial\delta^{(2)}(z)T(0) + \delta^{(2)}(z)\partial T(0)\right), \tag{6.192}$$

(this is equation 3.4.13 of Polchinski). This is the typical OPE of $T(z)T(0)$ but taking a $\overline{\partial}$ on the first term to give an odd number of powered poles. Now recall the divergence of the energy-momentum tensor due to the Ward identity with associated Poincaré symmetry [**?**]

$$\left(\overline{\partial}T_{zz} + \partial_z T_{\overline{z}z}\right) T_{zz}(0) = 2\pi \left(-2\partial\delta^{(2)}(z)T(0) + \delta^{(2)}\partial T(0)\right). \tag{6.193}$$

From this, we can integrate once with respect to z we find

$$T_{\overline{z}z}T_{zz}(0) = 2\pi \left(\frac{c}{12}\right) \partial^2\delta^{(2)}(z). \tag{6.194}$$

We can for generality sake move $T(0) \to T(w)$ and also act on the above with $\overline{partial}$ (to get that z^{-3} term keeping in mind about the OPE)

$$T_{z\overline{z}}(z,w)\overline{\partial}T(w) = -2\pi\frac{c}{12}\partial^2\overline{\partial}\delta^{(2)}(z-w). \tag{6.195}$$

We can set $\overline{\partial}T_{zz}$ with $-\partial T_{z\overline{z}}$ since the product $\left(\overline{\partial}T_{zz} + \partial_z T_{\overline{z}z}\right)(w,\overline{w})T_{zz}(z)$ contains only contact term proportional to $T_{z\overline{z}}$ and $\partial T_{z\overline{z}}$. If we integrate with respect to w once we then have

$$T_{z\overline{z}}(z)T_{z\overline{z}}(w) = -2\pi\frac{c}{12}\partial\overline{\partial}\delta^{(2)}(z-w). \tag{6.196}$$

This would be the z^{-3}-order pole that we are asked to find, in order to compare to the Weyl-transformed version of $\nabla_a j^a = aR$. Since $J^\mu = \nabla_\nu T^{\mu\nu}$, can find the transformed version of $T_{z\overline{z}}$ from equation 88 by integrating over the Weyl-transformed metric where $\delta\gamma_{ab} = 2\delta\omega\gamma_{ab}$ changing $T_{z\overline{z}}$ as

$$\delta T'_{z\overline{z}}(z) = -\frac{1}{\pi}\int d^2w\ \delta\omega(w)T_{z\overline{z}}(z)T_{z\overline{z}}(w) \tag{6.197}$$

$$= \frac{c}{6}\partial\overline{\partial}\delta\omega(z). \tag{6.198}$$

Then since $R = -2\nabla^2\omega$ under a Weyl transformation, the covariant form of the previous equation is $\delta T' = -\frac{c}{12}\delta R'$. This is just the infinitesimal version of equation (3.4.15) of Polchinski. From either case, we can still see we get the factor of $c/12$ as shown.

6.10.3 Problem set 3: Amplitudes in CFT: torus, Möbius strip, and n-puncture spheres

Polchinski 4.3 By writing out terms, demonstrate the Jacobi identity

$$\{[Q_B, L_m], b_n\} - \{[L_m, b_n], Q_B\} - [\{b_n, Q_B\}, L_m] = 0. \tag{6.199}$$

This is the sum of cyclic permutations, with commutators or anticommutators as appropriate, and minus signs from anticommutation. Use the Jacobi identity and the (anti)commutators (2.6.24) and (4.3.6) of b_n with L_n and Q_B to show that $\{[Q_B, L_m], b_n\}$ vanishes when the total central charge is zero. This implies that

$[Q_B, L_m]$ contains no c modes. However, this commutator has $N^g = 1$, and so it must vanish: the charge Q_B is conformally invariant. Now use the $Q_B Q_B b_n$ Jacobi identity in the same way to show that Q_B is nilpotent when the total central charge is zero.

Answer: First, equation (2.6.24) of the text is given as

$$[L_m, \mathcal{O}_n] = ((h-1)m - n)\mathcal{O}_{m+n} \tag{6.200}$$

and equation (4.3.6) is the following

$$\{Q_B, b_m\} = L_m^m + L_m^g = L_m. \tag{6.201}$$

Since we are working in the convention of $\lambda = 2$ then the weight of the ghost mode b_n turns into $h_b = \lambda = 2$ making equation (2.6.24) (or equation 2) into $(m-n)b_{m+n}$ when substituting $\mathcal{O}_n \to b_n$. Then the considering the first term of the Jacobi identity

$$\begin{aligned}
\{[Q_B, L_m], b_n\} &= Q_B L_m b_n - L_m Q_B b_n + b_n Q_B L_m - b_n L_m Q_B && (6.202)\\
&= Q_B L_m b_n + L_m b_n Q_B - Q_B b_n L_m - b_n L_m Q_B + [L_m, L_n] && (6.203)\\
&= \{[L_m, b_n], Q_B\} - [L_m, L_n] && (6.204)\\
&= \{(m-n)b_{m+n}, Q_B\} - \{L_m, L_n\} && (6.205)\\
&= (m-n)L_{m+n}^M - \left(\frac{c}{12}(m^3 - m)\delta_{m,-n} + (m-n)L_{m+n}^m\right) && (6.206)\\
&= \frac{c}{12}(m^3 - m)\delta_{m,-n}. && (6.207)
\end{aligned}$$

We now see that in order for this to vanish, we must require the *total* central charge to vanish i.e. $c = c_{\text{gh}} + c_{\text{m}} = 0$ meaning if $c_{\text{gh}} = -26$ then $c_{\text{m}} = -c_{\text{gh}}$.

To show the nilpotency of the BRST charge Q_B consider the following

$$\begin{aligned}
[\{Q_B, Q_B\}, b_n] &= -[\{b_n, Q_B\}, Q_B] - [\{Q_B, b_n\}, Q_B] && (6.208)\\
&= -[L_n, Q_B] - [L_n, Q_B] && (6.209)\\
&= -2[L_n, Q_B]. && (6.210)
\end{aligned}$$

Knowing that Q_B is conformally invariant, then $[L_n, Q_B]$ must vanish. This implies Q_B^2 has no c-ghost modes, and has ghost number of 2 which implies it must vanish. This leads us to saying Q_B is nilpotent iff $c = 0$.

Polchinski 4.5 (a) Carry out the BRST quantization for the first two levels of the closed string explicitly.

Answer: Following the discussion by Polchinski [4] from pages 134-137 (especially 136-137) the closed string must satisfiy the b-ghost modes from the left and right movers written as

$$b_0 |\psi\rangle = \tilde{b_0} |\psi\rangle = 0 \tag{6.211}$$

or equivalently

$$L_0 |\psi\rangle = \tilde{L}_0 |\psi\rangle = 0 \tag{6.212}$$

for a physical state $|\psi\rangle$ where L_0 is given by

$$L_0 = \frac{\alpha'}{4}(p^\mu p_\mu + m^2) \text{ and } \tilde{L}_0 = \frac{\alpha'}{4}(p_\mu p^\mu + m^2) \tag{6.213}$$

with the following mass condition relation to mode-level (equations (4.3.32a,b))

$$\frac{\alpha'}{4}\tilde{m}^2 = \sum_{n=1}^{\infty} n \left(\tilde{N}_{bn} + \tilde{N}_{cn} + \sum_{\mu=0}^{25} \right) - 1 \tag{6.214}$$

where $\tilde{N}_{bn}$ and $\tilde{N}_{cn}$ are the particle count for the bc ghosts with a similar mass equation for m^2 (just removes the tildes). The closed string inner product of the ground state to satisfy hermiticity of the ghost zero modes must take the following form (4.3.18b)

$$\langle 0; k| \, \tilde{c}_0 c \, |0, k'\rangle = i(2\pi)^{26}\delta^{(26)}(k - k'), \tag{6.215}$$

but just as Polchinski stated at the top of page 135, we need to reduce to inner product that ignore the X^0 matter field zero modes and the ghost zero modes.

Mimicking the open string spectrum calculation, at $N = \tilde{N} = 0$ the state is given by $|0, 0; \mathbf{k}\rangle$ where the physical states are denoted by $\left|N, \tilde{N}; \mathbf{k}\right\rangle$ giving

$$\frac{\alpha'}{4}\tilde{m}^2 = -1 \implies -\mathbf{k}^2 = -\frac{4}{\alpha'} \tag{6.216}$$

where we used the mass-shell condition $\mathbf{k}^2 + m^2 = 0$. The BRST charge Q_B is given in equation (4.3.7) showing that the ground state is closed BRST state since $Q_B \, |0, 0; \mathbf{k}\rangle = 0$ since $L_0, \tilde{L}_0 \in Q_B$, and thus be equation 14 becomes 0.

Now recall that for a state to be an exact state under BRST quantization it satisfies $|\psi\rangle = Q_B \, |\chi\rangle$ which are modulo out of the physical spectrum. Then, if $|0, 0; \mathbf{k}\rangle$ were BRST exact, then for some $|\chi\rangle$ such that $|0, 0; \mathbf{k}\rangle = Q_B\chi$, but this is the lowest state of the closed bosonic string implying no $|\chi\rangle$ exists or worded differently, no exact states at $N = \tilde{N} = 0$, i.e. each invariant state corresponds to a cohomology class (the closed string tachyon).

Now for the $\tilde{N} = N = 1$ sates, wew want to construct a state out of all the possibilities between $\tilde{\alpha}^\mu_{-n}, \alpha^\mu_{-n}, c, \tilde{c}, b, \tilde{b}$. This can be written as (for some coefficients $A, \tilde{A}, B, C \in \mathbb{C}$)

$$\begin{aligned} |1, 1; \mathbf{k}\rangle = \Big(& \alpha^\mu_{-1} e_{\mu\nu} \alpha^\nu_{-1} + \beta_\mu \alpha^\mu_{-1} \tilde{b}_{-1} + \tilde{\beta}_\mu \tilde{\alpha}^\mu_{-1} b_{-1} + \gamma_\mu \alpha^\mu_{-1} \tilde{c}_{-1} + \tilde{\gamma}_\mu \tilde{\alpha}^\mu_{-1} c_{-1} \\ & A b_{-1} \tilde{c}_{-1} + \tilde{A} c_{-1} \tilde{b}_{-1} + B b_{-1} \tilde{b}_{-1} + C c_{-1} \tilde{c}_{-1} \Big) \, |0, 0; \mathbf{k}\rangle \end{aligned} \tag{6.217}$$

where $e_{\mu\nu}$ is the tensor fields, γ^μ, β^μ are the vector fields, and the A, B, C are our scalar fields. There are a total of $(26 \times 26) + (4 \times 26) + 4$ total states (where I grouped

them in terms of tensor, vector, and scalar fields). Then via the hermiticity condition in equation (4.3.17) the inner product becomes (for the products that don't vanish)

$$\langle 1,1;\mathbf{k}|1,1;\mathbf{k}\rangle = \left(e^*_{\mu\nu}e^{\mu\nu} + \beta^*_\mu\gamma^\mu + \gamma^*_\mu\beta^\mu + \tilde{\beta}^*_\mu\tilde{\gamma}^\mu + \tilde{\gamma}^*_\mu\tilde{\beta}^\mu + A^*\tilde{A} + \tilde{A}^*A + B^*C + C^*B \right) \langle 0,0;\mathbf{k}|0,0;\mathbf{k}\rangle \,. \tag{6.218}$$

The tensor modes can be broken into e_{00} and $e_{ij} = 1, ..., 25$. Those of positive norm states are e_{ij} and e_{00}, and a linear combination of the vector states $\beta_\mu, \gamma_\mu, \tilde{\gamma}_\mu$ and $\tilde{\beta}_\mu$, and half of the ghost (scalar) states. This gives a total of $(25 \times 25) + 1 + (2 \times 26) + 2$ (which goes as $e_{ij} + e_0 0 +$ vectors $+$ scalar combination) positive norm states. But, we have a count of $(2 \times 25) + (2 \times 26) + 2$ negative norm states leftover that we need to discard by the appropriate conditions.

For the modes to be BRST closed under both (anti)holomorphic BRST operators gives the following constraints (similar to equation (4.3.27))

$$0 = Q_B\,|1,1;\mathbf{k}\rangle \tag{6.219}$$

$$= \left(\sqrt{\frac{\alpha'}{2}}(c_{-1}k_\mu\alpha^\mu_1 + c_1k_\mu\alpha^\mu_{-1}) + \alpha^\mu_{-1}\alpha_{\mu,1}c_0 - b_{-1}c_0c_1 - c_{-1}c_0b_1 - c_0 \right) |1,1;\mathbf{k}\rangle \tag{6.220}$$

$$= \sqrt{\frac{\alpha'}{2}} \Big(k^\mu e_{\mu\nu}\tilde{\alpha}^\nu_{-1}c_{-1} + k^\mu\beta_\mu c_{-1}\tilde{b}_{-1} + k^\mu\gamma_\mu c_{-1}\tilde{c}_{-1} + (\tilde{\beta}_\mu\alpha^\mu_{-1})(k_\nu\alpha^\nu_{-1})$$
$$Ak_\mu\alpha^\mu_{-1}\tilde{c}_{-1} + B\tilde{b}_{-1}k_\mu\alpha^\mu_{-1} \Big) |0,0;\mathbf{k}\rangle \tag{6.221}$$

$$0 = \tilde{Q}_B\,|1,1;\mathbf{k}\rangle \tag{6.222}$$

$$= \sqrt{\frac{\alpha'}{2}} \Big(\alpha^\mu_{-1}e_{\mu\nu}k^\nu\tilde{c}_{-1} + k_\mu\tilde{\beta}^\mu\tilde{c}_{-1}b_{-1} - k^\mu\tilde{\gamma}_\mu c_{-1}\tilde{c}_{-1} + (\beta_\mu\alpha^\mu_{-1})(k_\nu\alpha^\nu_{-1})$$
$$\tilde{A}k_\mu\tilde{\alpha}^\mu_{-1}c_{-1} + Bb_{-1}k_\mu\tilde{\alpha}^\mu_{-1} \Big) |0,0;\mathbf{k}\rangle \,. \tag{6.223}$$

From equation 24 and 26 we can read off what we must require for the state $|1,1;\mathbf{k}\rangle$ to be a physical state. We must require $e_{\mu\nu}k^\nu = 0$ which is just the polarization-momentum relation we are used to from QFT; $A = \tilde{A} = 0$ meaning the ghost coefficients are 0; $k^\mu\tilde{\gamma}_\mu = k^\mu\tilde{\gamma}_\mu = 0$ and $\beta_\mu = \tilde{\beta}_\mu = 0$ and $B = 0$, which is just saying the ghost modes have 0 norm.

This lowers the spectrum mode count 24×24 which are just the transverse positive modes confirming what Polchinski said on page 137. All these states are exact states since they were derived from $Q_B\,|1,1;\mathbf{k}\rangle = Q_B(\cdots)\,|0,0;\mathbf{k}\rangle = 0$. These 24×24 positive modes come across as the graviton, dilaton, and the 2-form field.

Polchinski 5.3 (a) Show from the OPE with the BRST current that the vertex operator $\tilde{c}c\mathcal{V}_m$ is BRST-invariant for $\mathcal{V}_m$ a $(1,1)$ matter current.

<u>Answer:</u> For simplicity sake, lets only focus on the holomorphic sector of the calculation (since the antiholomorphic sector or right moving modes is just dou-

ble the writing). Notice that the vertex operator in question is one that arises when changing equation (5.3.9) in Polchnski to that of the closed bosonic string. To modify equation (5.3.9) to be used for the closed bosonic string, we would just need so insert a factor of

$$\prod_{(a,i)\in f} \tilde{c}^a(\hat{\sigma}_i) \tag{6.224}$$

which we can see gives the vertex operator in question (note: this is what was exactly described at the bottom of page 160).

Now, to check for BRST invariance of the operator, we need to first compute the OPE between the BRST current j_B with $\tilde{c}c\mathcal{V}_m$. Then, we need to perform the contour integration over the OPE giving $\delta_B(\tilde{c}c\mathcal{V}_m)$ which will hopefully be zero. Recall that the (quantum)[126] BRST current j_B is given as (equation (4.3.3))

$$j_B = cT^m + :bc\partial c: + \frac{3}{2}\partial^2 c \tag{6.225}$$

and similar for $\tilde{j}_B$. The following OPE's for the bc (conformal) ghost system will also be helpful

$$b(z)c(w) \sim \frac{1}{z-w} \text{ and } c(z)b(w) \sim \frac{1}{z-w} \tag{6.226}$$

and for a general matter-field (operator) $\mathcal{O}(z)$ with weight h that is primary the OPE with T^m is given as (equation (2.4.15))

$$T(z)\mathcal{O}(w,w) \sim \frac{h}{(z-w)^2}\mathcal{O}(w) + \frac{\mathcal{O}(w)}{z-w}. \tag{6.227}$$

Something to keep in mind (since it completely missed mine the first time around) is that the ghost fields and matter fields will never interact. Another way to phrase this is that the insertion of the ghost field operators are in a different sector from the matter field operators in field theory space. Thus they will never have an OPE and can be commuted through each other. We can then calculate the OPE between j_B and $c\tilde{c}\mathcal{V}_m$

$$j_B(\tilde{c}c\mathcal{V}_m) = \left(cT^m + :bc\partial c: + \frac{3}{2}\partial^2 c\right)\tilde{c}c\mathcal{V}_m \tag{6.228}$$

$$= c\tilde{c}c(T^m\mathcal{V}_m) + (-1)^3 c\partial c\tilde{c}(bc)\mathcal{V}_m + \frac{3}{2}\partial^2 c\tilde{c}c\mathcal{V}_m \tag{6.229}$$

$$\sim c\tilde{c}c\left(\frac{\mathcal{V}_m}{(z-w)^2} + \frac{\partial\mathcal{V}_m}{z-w}\right) - \frac{c\partial c\tilde{c}\mathcal{V}_m}{z-w} \tag{6.230}$$

where in equation 32 I dropped any regular (non-pole containing) term and the factor of $(-1)^3$ comes from the Grassmannaility of the b and c ghosts anticommuting. Notice that the second and third term are really just the bi-products of

[126] I say quantum since we the factor of $\frac{3}{2}\partial^2 c$ is inserted by hand to make it be a primary field of weight $(1,0)$ but this last term actually disappears when we are not on a circle.

$c\mathcal{V}/(z-w)^2$ when taking the partial derivative ∂_w. Then, this reverse-product rule gives minus that of the first term, equating the singular terms to zero. Then it is trivial to compute

$$\delta_B(\tilde{c}c\mathcal{V}) = \oint_{C_w} \frac{dz}{2\pi i}(\text{regular terms from } j_B(\tilde{c}c\mathcal{V}_m)) \tag{6.231}$$

$$= 0. \tag{6.232}$$

Thus the closed string vertex operator $\tilde{c}c\mathcal{V}_m$ is invariant under BRST transformations.

(b) Show that the vertex operator satisfies equation (5.4.5).

Answer: We can use the exact same reasoning as utilized in part (a) but with less work since Polchinski has already given us what the OPE between the BRST current and any matter-field operator $\mathcal{O}_m$ is (equation (4.3.4c))

$$j_B(z)\mathcal{O}_m(w) \sim \frac{h}{(z-w)^2}c\mathcal{O}_m(w) + \frac{1}{(z-w)}\left(h(\partial c)\mathcal{O}_m(w) + c\partial\mathcal{O}_m(w)\right). \tag{6.233}$$

Then for our particular case of $\mathcal{V}_m$ with weights $(h, \tilde{h}) = (1, 1)$ (again, only caring about the left-movers since to include the right-movers is just double the writing and putting tildes everywhere)

$$j_B(z)\mathcal{V}_m(w) \sim \frac{c\mathcal{V}_m(w)}{(z-w)^2} + \frac{1}{z-w}\left((\partial c)\mathcal{V}_m(w) + c\partial_w\mathcal{V}_m(w)\right). \tag{6.234}$$

Now lets perform the w-contour over equation 36,

$$\delta_B\mathcal{V}_m = i\epsilon \oint_{C_w} \frac{dz}{2\pi} j_B(z)\mathcal{V}_m(w) \tag{6.235}$$

$$= i\epsilon \oint_{C_w} \frac{dz}{2\pi}\left(\frac{c\mathcal{V}_m(w)}{(z-w)^2} + \frac{1}{z-w}\left((\partial c)\mathcal{V}_m(w) + c\partial_w\mathcal{V}_m(w)\right)\right) \tag{6.236}$$

$$= i\epsilon \oint \frac{dz}{2\pi}\frac{1}{(z-w)^2}\partial(c\mathcal{V}_m) \tag{6.237}$$

$$= i\epsilon\partial(c\mathcal{V}_m). \tag{6.238}$$

Before ending the problem that I should note the only reason I (well, we) are able to find the BRST invariance of an operator this way is thanks to the operator-state correspondence. Otherwise I would have to compute $\{Q_B, \tilde{c}c\mathcal{V}_m\}$ and I prefer integrals over (anti)commutators.

Polchinski 6.2 For the linear dilaton CFT, $\Phi = V_\mu X^\mu$, the X^μ zero-mode path integral diverges. This reflects the fact that the coupling is diverging in some direction. Consider therefore imaginary $\Phi = ib_\mu X^\mu$; this is unphysical but has technical applications (one could instead take complex momenta).

(a) Generalize the calculation that gave (6.2.17) to this case. Show that in the end the only changes are in the Weyl dependence (reflecting a change in the dimension of the exponential) and in the spacetime momentum conservation.

Answer: Firstly, lets understand what we are actually trying to do since the wording of the question is a little convoluted. What we are asked to do is to see on does the Weyl dependence changes in $\omega(\sigma)$ when the vertex operator changes in the presence of the background field being the linear dilaton. The vertex operator thus changes to

$$V(k) =:e^{ik\cdot X(\sigma)+i\alpha'\Phi}: \tag{6.239}$$

where $\Phi = V_\mu X^\mu$ for the typical linear dilaton, but we will instead be considering $\Phi = ib_\mu X^\mu$.

Recall that equation (6.2.17) from Polchinski is given as

$$A^n_{S^2}(k,\sigma) = iC^X_{S^2}(2\pi)^d\delta^{(d)}(\sum_i k_i)\exp\left(-\frac{\alpha'}{2}\sum_i k_i^2\omega(\sigma_i)\right)\prod_{i,j=1;i<j}^{n}|z_{ij}|^{\alpha' k_i\cdot k_j} \tag{6.240}$$

But, let us go back a bit further in how we arrived at equation (6.2.17) from the product of tachyon vertex operators (6.2.11)

$$A^n_{S^2}(k,\sigma) =\left\langle \left[e^{ik_1\cdot X(\sigma_1)}\right]_r\left[e^{ik_2\cdot X(\sigma_2)}\right]_r\cdots\left[e^{ik_n\cdot X(\sigma_n)}\right]_r\right\rangle . \tag{6.241}$$

Notice if I change the strings momentum from $k_i \to k_i + ib_\mu$ then we can trace this through the derivation to notice that the two factors that change are the delta function (momentum conservation)

$$\delta^{(d)}\sum_i k_i \to \delta^{(d)}\sum_i (k+\alpha' ib)_i \tag{6.242}$$

and the term inside the exponential changes as (the α' is to keep units consistent)

$$\exp\left(-\frac{\alpha'}{2}\sum_i k_i^2\omega(\sigma_i)\right) \to \exp\left(-\frac{\alpha'}{2}\sum_i (k+ib)_i^2\omega(\sigma_i)\right) \tag{6.243}$$

$$= \exp\left(-\frac{\alpha'}{2}\sum_i (k^2+b^2)_i\omega(\sigma_i)\right) \tag{6.244}$$

thus showing that the momentum conservation and the Weyl-dependency does change as the problem states. (Note that this is if I take the correct complex product of what we mean of $(a+ib)^2$ but if not then really the answer is $k^2 + 2ib - b^2$ in the exponential).

(b) Generalize the calculation of the same quantity using holomorphicity. Show that the result is the same as in the regular X^μ CFT but with an extra *background charge* operator $\exp(-2ib\cdot X)$ taken to infinity.

Answer: To generalize the calculation using holomorphicity, all we have to do is to replace the momentum k_i^μ with the new complex momentum *due to the background field carrying charge* b $k_i^\mu + ib^\mu$. Then we just need to follow the steps from equation (6.2.24) until equation (6.2.31). Since we just need to carry around a constant *complex* factor, we just need to identify where we need to be careful.

Notice that equation (6.2.26) seems to not be conserved anymore unless we place ourselves in a weird complex-momentum space giving

$$A^n_{S^2}(k,\sigma)\sum_{i=1}^{n}(k+i\alpha' b)_i^\mu = 0. \tag{6.245}$$

This still allows us to argue at the bottom of page 173 that momentum is still conserved since $b \neq b(w)$ and so the pole structure does not change and then by holomorphicity at the patch u, $u = 0$ and will vanish.

Finally, skipping steps until equation (6.2.31) we have

$$A^n_{S^2}(k,\sigma) \propto \delta^{(d)}(\sum_i (k+i\alpha' b)_i) \prod_{\substack{i,j=1\\ i<j}}^{n} |z_{ij}|^{\alpha' k_i\cdot k_j} \exp(-\alpha' b\cdot X(\sigma_i)\omega(\sigma_i)) \tag{6.246}$$

where we need to send the curvature i.e $\omega(\sigma_i) = 0$ or in other word we take the background charge operator to infinity, which in this context perform the same job.

Polchinski 6.13 (a) Find the $PSL(2,\mathbb{C})$ transformation that takes three given points $z_{1,2,3}$ into chosen positions $\hat{z}_{1,2,3}$.

Answer: We want some group element $p \in \mathrm{PSL}(2,\mathbb{C})$ where $\mathrm{PSL}(2,\mathbb{C}) = \mathrm{SL}(2,\mathbb{C})/\mathbb{Z}_2$. Let $p = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where if $p \in \mathrm{PSL}(2,\mathbb{C})$ then this implies the elements of p obey $ad - bc = 1$ for $\exists a,b,c,d \in \mathbb{C}$. We know from equation (6.1.6) of Polchinski that p creates the following map for complex coordinates z

$$p : z \to \frac{az+b}{cz+d} \tag{6.247}$$

which is the group action. We then want $p : z_{1,2,3} \to \hat{z}_{1,2,3}$ such that (for our benefits of fixing string vertices in amplitudes) $z_{1,2,3} \to \{0,1,\infty\}$ and $\hat{z}_{1,2,3} \to \{0,1,\infty\}$. To do so lets use the fact that since $\mathrm{PSL}(2,\mathbb{C})$ is a group it must have closure, thus there exists a $p_1, p_2 \in \mathrm{PSL}(2,\mathbb{C})$ such that $p = p_1 \circ p_2$ where we can now separate the $z_{1,2,3}$ and $\hat{z}_{1,2,3}$ maps as $p_1 : z_{1,2,3} \to \{0,1,\infty\}$ and $p_2 : \hat{z}_{1,2,3} \to \{0,1,\infty\}$.

For p_1 we would expect $z_1 \to 0$ which implies (as seen from the map given in equation 49) $az + b = 0$. Similarly, if $z_3 \to \infty$ then $cz + d \to= 0$ when $z = z_3$. Then we can consider the general map for p_1

$$p_1 : z \mapsto \frac{\alpha(z-z_1)}{\beta(z-z_3)}. \tag{6.248}$$

Now, if $z_2 \to 1$ then $z = z_2$ giving $p(z_2) = \alpha(z_2 - z_1)/\beta(z_2 - z_3) = 1$. This implies that $\alpha = z_2 - z_3$ and $\beta = z_2 - z_1$. Then p_1 becomes

$$p_1 = \frac{(z_2 - z_2)z + (-z_1)(z_2 - z_3)}{(z_2 - z_1)z + (-z_3)(z_2 - z_1)} \tag{6.249}$$

where we can identify $z_2 - z_3 \equiv a$, $(-z_1)(z_2 - z_3) \equiv b$ and so on.
Similarly for p_2, this will actually be the inverse of the map just found for p_1 so we can write it as

$$p_2^{-1} : z \to \frac{(\hat{z}_2 - \hat{z}_2)z + (-\hat{z}_1)(\hat{z}_2 - \hat{z}_3)}{(\hat{z}_2 - \hat{z}_1)z + (-\hat{z}_3)(\hat{z}_2 - \hat{z}_1)}. \tag{6.250}$$

Due to the constraints of $\mathrm{PSL}(2, \mathbb{C})$, i.e. since $ad - bc = 1 \Leftrightarrow \det p = 1$, the inversion of the matrix elements for a general p is simply

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \to \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \tag{6.251}$$

Then we can write p_2 (with the previous identification of a, b, c, d in finding p_1)

$$p_2 = \frac{(-\hat{z}_3)(\hat{z}_2 - \hat{z}_1)z - (-\hat{z}_1)(\hat{z}_2 - \hat{z}_3)}{-(\hat{z}_2 - \hat{z}_3)z + (\hat{z}_2 - \hat{z}_3)}. \tag{6.252}$$

Then $p = p_1 \circ p_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where we have the following coefficients

$$a = \hat{z}_1\hat{z}_2(z_1 - z_2) + \hat{z}_2\hat{z}_3(z_2 - z_3) + \hat{z}_2\hat{z}_3(z_3 - z_1) \tag{6.253}$$
$$b = \hat{z}_1\hat{z}3z_2(z_1 - z_3) + \hat{z}_1\hat{z}_2z_3(z_2 - z_1) + \hat{z}_2\hat{z}_3z_1(z_3 - z_2) \tag{6.254}$$
$$c = \hat{z}_3(z_2 - z_1) + \hat{z}_2(z_1 - z_3) + \hat{z}_1(z_3 - z_2) \tag{6.255}$$
$$d = \hat{z}_1z_1(z_2 - z_3) + \hat{z}_3z_3(z_1 - z_3) + \hat{z}_2z_2(z_3 - z_1). \tag{6.256}$$

(b) Verify the Möbius invariance results (6.7.3)-(6.7.7). Show that to derive (6.7.5) it is sufficient that L_1 and $\tilde{L}_1$ annihilate the operators.
Answer: Equation (6.7.3) is given as

$$\langle \mathcal{A}_i(0,0)\rangle_{S^2} = \langle \mathcal{A}_i'(0,0)\rangle_{S^2} = \gamma^{-h_i}\overline{\gamma}^{-\tilde{h}_i}\langle \mathcal{A}_i(0,0)\rangle_{S^2} \tag{6.257}$$

where α has weights $(h_i, \tilde{h}_i)$. The Möbius transformation is given as a scalaing $z' = \gamma z$ for this case. Recall that for an operator $\mathcal{A}$ of weight given previously, there is an arbitrary mapping $z \to w(z)$ and $\overline{z} \to \overline{w}(\overline{z})$ that transforms the operator $\mathcal{A}$ as (equation (2.4.15))

$$\mathcal{A}_i' = \left(\frac{\partial w}{\partial z}\right)^{h_i} \left(\frac{\partial \overline{w}}{\partial \overline{z}}\right)^{\tilde{h}_i} \mathcal{A}_i. \tag{6.258}$$

Then, with the given transformation above, we simply pick up a factor of $1/\gamma$ when transforming raised to a power of h_i which when simplified gives back equation 59 since $\partial z/\partial z' = 1/\gamma$.

For equation (6.7.4), this has a Möbius transformation of $a = 1$, $\beta = -z_2$, $c = 0$ and ∓ 1 which implies that $z_1 \to z_1 - z_2$ and $z_2 \to 0$. Then

$$\langle \mathcal{A}_i(z_1, \overline{z}_1)\mathcal{A}_j(z_2, \overline{z}_2)\rangle = \langle \mathcal{A}_i(z_1 - z_2, \overline{z}_1 - \overline{z}_2)\mathcal{A}_j(0,0)\rangle_{S^2}. \tag{6.259}$$

Now by the same arguments given prior for the Möbius invariance of equation (6.7.3), rescale z as $z \to z' = z(z_1 - z_2)^{-1}$, which can also be done with a Möbius transformation, but by the same transformation rules prior (i.e. equation 60). We then get

$$\langle \mathcal{A}_i(z_1, \overline{z}_2)\mathcal{A}_j(z_2, \overline{z}_2)\rangle_{S^2} = z_{12}^{-h_i+h_j}\overline{z}_{12}^{-\tilde{h}_i-\tilde{h}_j}\langle \mathcal{A}_i(1,1)\mathcal{A}_j(0,0)\rangle_{S^2} \tag{6.260}$$

or Polchinski's (6.7.4).

Via the analysis of the 2-point function at the top of page 199, $z \to z' = \epsilon(z - z_1)(z - z_2) + \mathcal{O}(\epsilon^2)$. This can be reproduced from a Möbius transformation as

$$\begin{aligned} z' = \frac{az+b}{cz+d} &= \frac{1(z) + (-z_1)}{(-\epsilon)z + (1 + \epsilon z_2)} && (6.261)\\ &= \frac{z - z_1}{z + \epsilon(z_2 - z)} && (6.262)\\ &= (z - z_1)(1 - \epsilon(z_2 - z)) + \mathcal{O}(\epsilon^2) && (6.263)\\ &= z - z_1\epsilon(z - z_1)(z - z_2) + \mathcal{O}(\epsilon^2). && (6.264) \end{aligned}$$

Now recall that $\exists \mathcal{O}_p(z, \overline{z})$ that the infinitesimal conformal transformation is given as $\delta z = \epsilon(z)$ and $\delta\overline{z} = \overline{\epsilon}(\overline{z})$. So, if $\mathcal{O}_p$ has weights $(h_p, \tilde{h}_p)$ then it implies that $\mathcal{O}_p$ transforms as

$$\delta\mathcal{O}_p = -h_p(\partial_z\epsilon(z))\mathcal{O}_p(z,\overline{z}) - \epsilon(z)\partial_z\mathcal{O}_p(z,\overline{z}) - \tilde{h}_p(\partial_{\overline{z}})\overline{\epsilon}\mathcal{O}_p - \overline{\epsilon}(\overline{z})\partial_{\overline{z}}\mathcal{O}_p. \tag{6.265}$$

From previously we then have $\epsilon(z) = \epsilon(z^2 - (z_1 + z_2)z + z_1z_2)$ and similarly for $\overline{\epsilon}(\overline{z})$. Then the transformation of $\mathcal{O}_p$ becomes

$$\begin{aligned} \delta\mathcal{O}_p = &-h_p\epsilon(2z - (z_1 + z_2))\mathcal{O}_p - \epsilon(z^2 - (z_1 + z_2)z + z_1z_2)\partial\mathcal{O}_p\\ &- \tilde{h}_p\overline{\epsilon}(2\overline{z} - (\overline{z}_1 + \overline{z}_2))\mathcal{O}_p - \overline{\epsilon}(\overline{z}^2 - (\overline{z}_1 + \overline{z}_2)\overline{z} + \overline{z}_1\overline{z}_2)\overline{\partial}\mathcal{O}_p \end{aligned} \tag{6.266}$$

which is evaluated at $z \to z_1$ and $\overline{z} \to \overline{z}_1$. But, $z = z_1$ and $\overline{z} = \overline{z}_1$ which implies $\partial\overline{O}_p$ and $\overline{\partial}\mathcal{O}_p$ vanish. Then the transformation becomes

$$\delta\mathcal{O}_p = -h_p\epsilon(z_1 - z_2)\mathcal{O}_p - \tilde{h}_p\overline{\epsilon}(\overline{z}_1 - \overline{z}_2)\mathcal{O}_p. \tag{6.267}$$

Similar for $\delta\mathcal{O}_q$ (just replace all the p's with q's). Then the change in the correlation is

$$\begin{aligned} \langle(\delta\mathcal{O}_p)\mathcal{O}_q + \mathcal{O}_p(\delta\mathcal{O}_q)\rangle &= \langle(-h_p\epsilon(z_1 - z_2)\mathcal{O}_p - \tilde{h}_p\overline{\epsilon}(\overline{z}_1 - \overline{z}_2)\mathcal{O}_p)\mathcal{O}_q\\ &\quad + \mathcal{O}_q(-h_q\epsilon(z_2 - z_1)\mathcal{O}_q - \tilde{h}_q\overline{\epsilon}(\overline{z}_2 - \overline{z}_1)\mathcal{O}_q)\rangle && (6.268)\\ &= \langle\epsilon\mathcal{O}_p\mathcal{O}_q((-h_p + h_q)(z_1 - z_2) + (-\tilde{h}_p + h_q)(\overline{z}_1 - \overline{z}_2))\rangle. && (6.269) \end{aligned}$$

From this, we can tell that this must vanish to leading order in ϵ. This can only occur if $h_p = h_q$ (and similarly $\tilde{h}_p = \tilde{h}_q$) **or** $\langle \mathcal{O}_p \mathcal{O}_q \rangle = 0$. This is exactly equation (6.7.5).

Now with respect to equation (6.7.6), notice that under translation invariance, the 3-point correlator can only contain the difference between measuring points/insertions in z_{ij} which must also be consistent under rescalings $z \to \gamma z$ given as

$$\mathcal{O}_{p_1}\mathcal{O}_{p_2}\mathcal{O}_{p_3} \to \gamma^{-h_1-\tilde{h}_1}\gamma^{-h_2-\tilde{h}_2}\gamma^{-h_3-\tilde{h}_3}\mathcal{O}_{p_1}\mathcal{O}_{p_2}\mathcal{O}_{p_3}. \tag{6.270}$$

Then the only product of functions that obeys the above transformation are the products of z_{ij} with the physical condition $i < j$ giving all combination i.e. equation (6.7.6).

Finally with respect to equation (6.7.7), notice that we have 4 positions (i.e. z_i with $i = \{1, 2, 3, 4\}$) and can construct 4Choose2 = 6 Möbius invariant corss-ratios given as 1 particular fusion (below equation (6.7.7) of Polchinski)

$$z_c = \frac{z_{12}z_{34}}{z_{13}z_{24}}, \tag{6.271}$$

which is invariant under Möbius transformations (i.e. functions of difference, homogeneous of degree 0, same index in numerator and denominator). By argument previously used for equation (6.7.6), only allowed form via scaling and up to a constant no longer holds since a Möbius invaraitn function can multiply the scaling factors, but this is the only vagueness allowed which was implies in equation (6.7.7). Then, equation (6.7.7) is the most general form allowed for 4-point tensor.

6.10.4 Problem set 4: Quantization of the Bosonic String and the S-matrix

Polchinski 5.2 Repeat the steps in the previous problem for paths forming a closed loop. In order to fixed the residual gauge symmetry, introduce a fake vertex operator by differentiating with respect to m^2. The answer is given in equation (7.3.9). If you do both exercises 5.1 and 5.2, discuss the relation between the results for the moduli measure.

<u>Answer:</u> In order to put the point particle (or matter fields) around a circle, we will impose the boundary conditions $X_f^\mu = X(0) = X_0$ integrate to $X_i^\mu = X(1) = X_0$. We can then write the partition function in the following manner

$$Z_{S^2}(m^2) = \int_{X_0}^{X_0} \frac{[dXde]}{2\text{Vol}(\text{diff} \times \text{trans})} e^{-S_m[X,e]} \tag{6.272}$$

where the matter action is

$$S_m[X, e] = \frac{1}{2}\int_0^1 d\tau \left(e^{-1}\dot{X}^\mu \dot{X}_\mu + em^2\right). \tag{6.273}$$

Let me explain the gauge-volume we are dividing out. Recall that the worldline parameter τ is periodic $\tau \in [0,1]$ that describes the circle. Periodicity can also be described as $X^\mu(\tau) = X^\mu(\tau+1)$ and similar for $e(\tau)$. Now S_m is diffeomorphism invariant but not necessarily the einbein $e(\tau)$. As discussed in chapter 5.1 if $l = \int_0^1 d\tau e(\tau)$ then under a diffeomorphism transformation $\tau \to \tau'(\tau)$ implies that $\tau'(1) = l$ (equation (5.1.4)). So we can not fix both the gauge and coordinate region. Thus, l acts as the modulus for the particle on the circle, keeping track of circles not diffeomorphism equivalent. After gauge fixing, there is a residual global symmetry (page 146 of Polchinski) in the form of translation or in other words, free to pick the origin of the circle. This can be described as $\tau \to \tau + v \mod 1$ (equation (5.1.6)), and gauge transformation not fixed by a choice of metric. Because of this we divide by Vol(translation) $= l$. But there is also a reflection symmetry $\tau \to -\tau$ with Vol(Reflections) $= 2$ (hence the factor of 2).

Following Polchinski, lets define the inner product on the space of functions with unit interval as

$$(f,g) = \int_0^1 d\tau\, e(\tau) f(\tau) g(\tau). \tag{6.274}$$

Then the action can be rewritten as

$$S_m[X,e] = \frac{1}{2}\int_0^1 \left(e \cdot e^{-1}\dot{X}^\mu e^{-1}\dot{X}_\mu\right) + \frac{1}{2}lm^2 \tag{6.275}$$

$$= \frac{1}{2}\left(e^{-1}\dot{X}^\mu, e^{-1}\dot{X}_\mu\right) + \frac{1}{2}lm^2. \tag{6.276}$$

Let $\hat{e}$ be the fiducial gauge, then the Faddeev-Popov determinant is defined as (equation (5.3.3))

$$1 = \Delta_{FP}(e)\int dl[d\zeta]\delta(e - \hat{e}^\zeta) \tag{6.277}$$

with the gauge parameter ζ and the integral over dl is exactly like the string integral over all of moduli space $\int_F d^\mu t$. Then the partition function is

$$Z_{S^1} = \int_{X_0}^{X_0} \frac{[dXde]}{2\text{Vol}(\text{diff} \times \text{trans})}\Delta_{FP}(e)\int dl[d\zeta]\delta(e - \hat{e}^\zeta)e^{-S_m[X,e]} \tag{6.278}$$

$$= \int_0^\infty dl \int_{X_0}^{X_0} \frac{[dX^\zeta de^\zeta]}{2\text{Vol}(\text{diff} \times \text{trans})}\Delta_{FP}(\hat{e})e^{-S_m[X^\zeta,e^\zeta]}. \tag{6.279}$$

Since the Faddeev-Popov determinant is gauge-invariant over the measure as shown in chapter 3.3, we can drop the gauge invariant paramter. The integration over $[de]$ then gives Vol(diff) and Vol(trans) $= l$ giving

$$Z_{S^1} = \int_0^\infty \frac{dl}{2l}\int_{X_0}^{X_0}[dX]\Delta_{FP}(e)e^{-S_m[X,e]}. \tag{6.280}$$

Following chapter 3.3, just as in equation (3.3.16) consider the expansion of the einbein $\tau \to \tau^\zeta = \tau + e^{-1}\zeta$ and the modulus transforms as $l \to l + \delta l$. If l is fixed then

$$\delta e = e^\zeta(\tau) - e(\tau) = e^\zeta(\tau^\zeta - e^{-1}\zeta) - e(\tau) \tag{6.281}$$
$$= e^\zeta(\tau^\zeta) - e^{-1}\zeta\partial_\tau e(\tau) - e(\tau) = e^\zeta(\tau^\zeta) - \zeta\partial\ln(e(\tau)) - e(\tau). \tag{6.282}$$

Given that $e^\zeta(\tau^\zeta)d\tau^\zeta = e(\tau)d\tau$ we have

$$e^\zeta(\tau^\zeta) = e(\tau\frac{d\tau}{d\tau^\zeta}) = e(\tau)\frac{d}{d\tau}(\tau - e^{-1}\zeta) = e(\tau)(1 + e^{-2}\partial(e\zeta) - e^{-1}\partial\zeta) \tag{6.283}$$
$$= e(\tau) + \zeta\partial\ln e - \partial\zeta. \tag{6.284}$$

Thus

$$\delta e = e + \zeta\partial\ln e - \partial\zeta - \zeta\partial\ln e - e = -\partial\zeta. \tag{6.285}$$

Then adding a small change in the modulus would change this to

$$\delta e = -\partial\zeta + \partial_l e\delta l. \tag{6.286}$$

Note that $\tau^\zeta \in [0,1]$ which implies $\zeta(\tau = 0) = \zeta(\tau = 1) = 0$. then the Faddeev-Popov determinant changes as

$$\Delta_{FP}^{-1}(e) = \int dl[d\zeta]\delta(e - \hat{e}^\zeta) = \int dl[d\zeta]\delta(\partial\zeta - \partial_l\delta l). \tag{6.287}$$

We can rewrite the Δ_{FP}^{-1} in terms of an exponential in order to insert into the partition integral as

$$\Delta_{FP}^{-1}(e) = \int dl[d\zeta]\delta(e - \hat{e}^\zeta) \tag{6.288}$$
$$= \int dl[d\zeta d\beta]\exp\left(2\pi i\int d\tau\beta(\partial\zeta - \partial_l e\delta l)\right) \tag{6.289}$$
$$= \int dl[d\zeta d\beta]\exp\left(2\pi i\int d\tau\beta(e^{-1}\partial\zeta - e^{-1}\partial_l e\delta l)\right) \tag{6.290}$$
$$= \int dl[d\zeta d\beta]\exp\left(2\pi i(\beta, e^{-1}\partial\zeta - e^{-1}\partial_l e\delta l)\right). \tag{6.291}$$

We can invert the Faddeev-Popov determinant via the ghost-Grassmann method by introducing 3 Grassmann variables ξ, c, and b and evaluate the determinant at $\hat{e}$ by fidicual gauge choice

$$\Delta_{FP}(\hat{e}) = \int d\xi[dcdb]\exp\left(\frac{1}{4\pi}\left(b, \hat{e}^{-1}\partial c - \hat{e}^{-1}\partial_l\hat{e}\xi\right)\right). \tag{6.292}$$

Notice this is exactly (up to a minus sign) equation (5.3.8). Performing the ξ integration up to a constant gives

$$\Delta_{FP}(\hat{e}) = \int[dcdb]\frac{1}{4\pi}\left(b, \hat{e}^{-1}\partial_l\hat{e}\right)\exp\left(\frac{1}{4\pi}\left(b, \hat{e}^{-1}\partial c\right)\right). \tag{6.293}$$

If $S_g[b,c,\hat{e}] = -\frac{1}{4\pi}(b,\hat{e}^{-1}\partial c)$ then the partition function becomes

$$Z_{S^1}(m^2) = \int_0^\infty \frac{dl}{2l} \int_{X_0}^{X_0} [dX] \int_{c(1)=0}^{c(1)=0} [dcdb] \frac{1}{4\pi} \left(b, \hat{e}^{-1}\partial_l \hat{e}\right) e^{-S_m[X,\hat{e}]-S_g[b,c\hat{e}]}. \tag{6.294}$$

Note that $\zeta(0) = \zeta(1) = 0$ or in other words $c(0) = c(1) = 0$ for the c-ghost integration. To evaluate $Z_{S^1}(m^2)$, select the einbein fiducial gauge to be $\hat{e} = l$. Then the inner product defined in equation 3 becomes

$$(f,g) = l \int_0^1 d\tau\ f \cdot g. \tag{6.295}$$

Consider now how a ghost field f would obey the Laplacian (equivalent to the analysis from equation (5.3.12) in Polchinski, pages 157-158),

$$\Delta_l f = -\frac{1}{l^2}\partial^2 f = \nu^2 f \tag{6.296}$$

on a circle the l^{-2} is for normalization and ν are the eigenvalues. There are 2 possible set of solutions, $\sin(\nu\tau/l)$ and $\cos(\nu\tau/l)$. But since they must be periodic with respect to τ, $\tau \to \tau + 1$, requires $\nu/l = 2\pi k$ for $k \in \mathbb{N}$[127]. Then the eigenfunction have the following eigenvalues

$$\Delta_l \sin(2\pi\tau j) = -\frac{1}{l^2}\partial^2 \sin(2\pi\tau j) = \frac{4\pi^2 j^2}{l^2}\sin^2(2\pi j\tau) \tag{6.297}$$

$$\Delta_l \cos(2\pi j\tau) = -\frac{1}{l^2}\partial^2 \cos(2\pi j\tau) = \frac{4\pi^2 j^2}{l^2}\cos(2\pi j\tau) \tag{6.298}$$

Both of which point to having eigenvalues

$$\nu_j^2 = \frac{4\pi^2 j^2}{l^2}. \tag{6.299}$$

We can expand the b and c ghosts as

$$b = \frac{1}{\sqrt{l}}b_0 + \sqrt{\frac{2}{l}}\sum_{j=1}^\infty b_j \cos(2\pi j\tau) \text{ and } c = \sqrt{\frac{2}{l}}\sum_{j=1}^\infty c_j \sin(2\pi j\tau). \tag{6.300}$$

The c-ghost has **no** c_0-mode since $c(0) = c(1) = 0$. the ghost action now becomes

$$S_g[b,c,\hat{e}] = -\frac{1}{4\pi}\left(b, \hat{e}^{-1}\partial c\right) \tag{6.301}$$

$$= -\frac{1}{4\pi}\int_0^1 d\tau \left(\frac{1}{\sqrt{l}}b_0 + \sqrt{\frac{2}{l}}\sum_{j=1}^\infty b_j \cos(2\pi j\tau)\right)\partial\left(\sqrt{\frac{2}{l}}\sum_{k=1}^\infty c_k \sin(2\pi k\tau)\right) \tag{6.302}$$

$$= -\frac{1}{\sqrt{2}l}\int_0^1 d\tau \left(b_0 + \sqrt{2}\sum_{j=1}^\infty b_j \cos(2\pi j\tau)\right)\sum_{k=1}^\infty k c_k \cos(2\pi k\tau). \tag{6.303}$$

[127] Where I define $\mathbb{N}$ as the set of all positive integers not including 0.

The first and second integrals are given as

$$\int_0^1 d\tau\, b_0 \sum_{k=1}^{\infty} k c_k \cos(2\pi k\tau) = 0 \text{ and } \int_0^1 d\tau\, \cos(2\pi j\tau)\cos(2\pi k\tau) = \frac{1}{2}\delta_{jk}. \tag{6.304}$$

This gives us the following result

$$-\frac{1}{\sqrt{2}l}\left(\sum_{j,k}^{\infty} b_j \sqrt{2}k\frac{1}{2}c_k\delta_{jk}\right) = \frac{1}{2l}\sum_{j=1}^{\infty} j b_j c_j. \tag{6.305}$$

The b-ghost insertion is

$$\frac{1}{4\pi}\left(b, \hat{e}^{-1}\partial_l \hat{e}\right) = \frac{l}{4\pi}\int_0^1 d\tau\, b l^{-1}\partial_l l = \frac{1}{4\pi}\int_0^1 d\tau\, b \tag{6.306}$$

$$= \frac{1}{4\pi}\int_0^1 d\tau \left(\frac{b_0}{\sqrt{l}} + \sqrt{\frac{2}{l}}\sum_{j=1}^{\infty} b_j \cos(2\pi j\tau)\right) = \frac{1}{4\pi\sqrt{l}}b_0. \tag{6.307}$$

The ghost contribution to the partition function then becomes

$$Z_g = \int \prod_{j=1}^{\infty}[dc_j db_0 db_j]\frac{1}{4\pi\sqrt{l}}b_0 \exp\left(\frac{1}{2l}\sum_{j=1}^{\infty} j b_j c_j\right) = \frac{1}{4\pi\sqrt{l}}\prod_{j=1}^{\infty}\frac{j}{2l}. \tag{6.308}$$

Given the eigenvalue of v_j^2,

$$Z_g = \frac{1}{4\pi\sqrt{l}}\prod_{j=1}^{\infty}\prod_{j=1}^{\infty}\frac{v_j}{4\pi^2} = \frac{1}{4\pi\sqrt{l}}\sqrt{\det{}'\left(\frac{\Delta_l}{16\pi^2}\right)} \tag{6.309}$$

where the $\det'$ excludes the zero-mode.

To work out the matter part of the path-integral, we cna use the same eigenfunctions from the ghost system but in order to do so we must split the matter fields X^μ into classical and quantum fields (exactly like the background field method). For the classical part, the equations of motion are $\ddot{X}^\mu = 0$ with general solution

$$X_{cl}^\mu = x_0^\mu + (x_1 - x_0)^\mu \tau \tag{6.310}$$

with $x_0 = X^\mu(0)$ and x_1 is determined by boundary conditions. We now expand X^μ as $X^\mu = X_{cl}^\mu + \chi^\mu$. The quantum field must also be periodic under $\tau \to \tau + 1$ that satisfies

$$\chi^\mu(0) = X^\mu(0) - X_{cl}^\mu(0) = X^\mu(0) - x_0 = 0 \tag{6.311}$$

and similar for $\chi^\mu(1)$. The quantum fields will then have the same expansion as the ghost fields

$$X^\mu = C_{cl}^\mu + \sqrt{\frac{2}{l}}\sum_{j=1}^{\infty} x_j^\mu \sin(2\pi j\tau). \tag{6.312}$$

The matter action in the fiducial gauge $\hat{e} = l$ becomes

$$S_m[X,e] = \frac{1}{2}\int_0^1 d\tau \frac{1}{l}\partial\left(X^\mu_{cl} + \sqrt{\frac{2}{l}}\sum_{j=1}^\infty x^\mu_j \sin(2\pi j\tau)\right)^2 + lm^2 \tag{6.313}$$

$$= \frac{1}{2l}\int_0^1 d\tau \left[\left(x^\mu_1 - x^\mu_0 + 2\pi\sqrt{\frac{2}{l}}\sum_{j=1}^\infty j x^\mu_j \cos(2\pi j\tau)\right)^2 + l^2m^2\right] \tag{6.314}$$

and using the previous trigonometric integrals we find

$$S_m[X,e] = \frac{1}{2l}\left[(x^\mu_1 - x^\mu_0)^2 + l^2m^2 + 4\pi^2\frac{2}{l}\sum_{j=1}^\infty j^2 x_j^2 \frac{1}{2}\right] \tag{6.315}$$

$$= \frac{(x_1-x_0)^2}{2l} + \frac{lm^2}{2} + \frac{\pi^2}{l^2}\sum_{j=1}^\infty j^2 x_j^2. \tag{6.316}$$

Then the matter part of the partition function is

$$Z_X = \exp\left(-\frac{(x_1-x_0)^2}{2l} + \frac{lm^2}{2}\right)\int \prod_{\mu=1}^D \prod_{j=1}^\infty dx^\mu_j \exp\left(-\frac{\pi^2}{l^2}\sum_{j=1}^\infty j^2 x_j^2\right). \tag{6.317}$$

The dx^μ_j integral is just a Gaussian giving

$$\exp\left(-\frac{(x_1-x_0)^2}{2l} + \frac{lm^2}{2}\right)\left(\prod_{j=1}^\infty \sqrt{\frac{\pi}{\pi^2 j^2/l^2}}\right)^d = \exp\left(-\frac{(x_1-x_0)^2}{2l} - \frac{lm^2}{2}\right)\left(\prod_{j=1}^\infty \frac{\pi j^2}{l^2}\right)^{-d/2} \tag{6.318}$$

$$= \det{}'(\Delta_l)^{-d/2}\left(\prod_{j=1}^\infty \frac{\nu_j^2}{4\pi}\right)^{-d/2}. \tag{6.319}$$

Then the partition function of a point-particle on a circle is

$$Z_{S^1}(m^2) = \int_0^\infty \frac{dl}{2l} Z_X S_g \tag{6.320}$$

$$= \int_0^\infty \frac{dl}{2l}\exp\left(-\frac{(x_1-x_0)^2}{2l} - \frac{1}{2}lm^2\right)\det{}'\left(\frac{\Delta_l}{4\pi}\right)^{-d/2}\frac{1}{4\pi\sqrt{l}}\det{}'\left(\frac{\Delta_l}{16\pi^2}\right)^{1/2} \tag{6.321}$$

$$= C\int_0^\infty \frac{dl}{l^{3/2}}\exp\left(-\frac{(x_1-x_0)^2}{2l} - \frac{1}{2}lm^2\right)\det{}'(\Delta_l)^{1-d/2}. \tag{6.322}$$

Notice that the determinant at the end depends upon the modulus and thus in order to determine it we will have to regularize it since we can assume it will give a divergence. (The C is also some constant we won't care about).

To regularize $\det'$, we can use Pauli-Villars regularization given in the appendix section A.1 (page 340) by adding a large frequency Ω (called the "regulator oscillator") (equation (A.1.60))

$$\det{}' \Delta_l \to \frac{\det' \Delta_l}{\det'(\Delta_l + \omega^2)} = \prod_{j=1}^{\infty} \frac{\pi^2 j^2}{\pi^2 j^2 + \Omega^2 l^2} = \frac{\Omega l}{\sinh(\omega l)}. \tag{6.323}$$

Notice as $\Omega \to \infty$ that $\sinh(\Omega l) \to \exp(\Omega l/2)$. This then gives

$$\frac{\det' \Delta_l}{\det'(\Delta_l + \Omega^2)} \sim 2\Omega l e^{-\Omega l}. \tag{6.324}$$

The partition function then becomes

$$Z_{S^1}(m^2) = C \int_0^\infty \frac{dl}{l^{3/2}} \exp\left(-\frac{(x_1 - x_0)^2}{2l} - \frac{1}{2} l m^2\right) \left(2\Omega l e^{-\Omega l}\right)^{(1-d)/2} \tag{6.325}$$

$$= \frac{C}{(2\Omega)^{(d-1)/2}} \int_0^\infty dl\, l^{-d/2-1} \exp\left(-\frac{(x_1 - x_0)^2}{2l} - \frac{1}{2} l m^2\right). \tag{6.326}$$

To get rid of $\Omega^{(1-d)/2}$, since it is a multiplicative factor, if we performed a field re-definition as $X^\mu \to \sqrt{Z} X^\mu$ for some Z that is also infinite. The term $\propto e^{-l/2}$ divergence can be removed via a counter-term proportional to $l^2 A$ for some A giving

$$\exp\left(-\frac{(x_1 - x_0)^2}{2} - \frac{l}{2}\left(m^2 - (d-1)\Omega + 2A^2\right)\right). \tag{6.327}$$

If $m \to m_{\text{phys}}$ then $A = (d-1)\Omega$. $Z_{S^1}(m^2)$ now becomes

$$Z_{S^1}(m^2) \propto \int_0^\infty dl\, l^{-d/2-1} \exp\left(-\frac{(x_1 - x_0)^2}{2l} - \frac{1}{2} l m^2\right). \tag{6.328}$$

Let $x \equiv x_1 - x_0$ and Fourier transform x^2 to give

$$\int_0^\infty dl\, l^{-d/2-1} \exp\left(-\frac{l m^2}{2}\right) \int d^D x \exp\left(ik \cdot x - \frac{x^2}{2l}\right). \tag{6.329}$$

Now we perform the Gaussian integration after completing the square

$$\hat{Z}_{S^1}(k) \propto \int_0^\infty dl\, l^{-d/2-1} \exp\left(-\frac{l m^2}{2}\right) \int d^D x \exp\left(\frac{-(x - ilk)^2 + l^2 + k^2}{2l}\right) \tag{6.330}$$

$$= \int_0^\infty dl\, l^{-d/2-1} \exp\left(-\frac{l m^2}{2}\right) \exp\left(-\frac{l k^2}{2}\right) (2\pi l)^{d/2} \tag{6.331}$$

$$= (2\pi)^{d/2} \int_0^\infty \frac{dl}{l} \exp\left(-\frac{l(k^2 + m^2)}{2}\right). \tag{6.332}$$

To get all the possible particle states we dived by $2/V_d$ to get (and over all the possible momentum states via integration)

$$Z_{S^1}(m^2) = V_d \int \frac{d^d k}{(2\pi)^d} \int_0^\infty \frac{dl}{2l} \exp\Big(-l(k^2+m^2)/2\Big). \tag{6.333}$$

Polchinski 6.10 Carry out the integral in equation (6.6.5) to obtain the Virasoro-Shapiro amplitude (6.6.10). The relation

$$|z|^{-2a} = \frac{1}{\Gamma(a)} \int_0^\infty dt\, t^{a-1} \exp(-tz\overline{z}) \tag{6.334}$$

is useful. (Reference: Green, Schwarz, & Witten (1987), section 7.2; note that their d^2z is $\frac{1}{2}$ times our.)

Answer: Recall that equation (6.6.5) is

$$J(s,t,u) = \int_{\mathbb{C}} d^2 z_4 |z_4|^{-\frac{\alpha' u}{2}-4} |1-z_4|^{-\frac{\alpha' t}{2}-4} \tag{6.335}$$

where we have the following Mandelstam relation $s+t+u=-16/\alpha'$. We will use this to obtain the Virasoro-Shapiro amplitude (equation (6.6.10)) later. But for now, let $2a \equiv -\frac{\alpha' u}{2}+4$ and $2b \equiv -\frac{\alpha' t}{2}+4$ to make things easier for us. Then equation (6.6.5) becomes $J(s,t,u) \to J(a,b)$ which is

$$J(a,b) = \int_{\mathbb{C}} d^2 z\, |z|^{-2a} |1-z_4|^{-2b} \tag{6.336}$$

where for simplicity we are letting $z_4 \to z$. From the integral definition of the Gamma function (found either on Wolfram or Wikipedia)

$$\Gamma(a) = \int_0^\infty dx\, x^{a-1} e^{-x} \tag{6.337}$$

$$= \int_0^\infty dt\, (t|z|^2)^{a-1} e^{-t|z|^2} \qquad (x \to t|z|^2) \tag{6.338}$$

$$= \int_0^\infty dt\, |z|^2 t^{a-1} |z|^{2a-2} e^{-t|z|^2} \tag{6.339}$$

$$= |z|^{2a} \int_0^\infty dt\, t^{a-1} e^{-t|z|^2} \tag{6.340}$$

$$\implies |z|^{-2a} = \frac{1}{\Gamma(a)} \int_0^\infty dt\, t^{a-1} e^{-t|z|^2}. \tag{6.341}$$

Now, we can switch over from z's to real coordinates via $z = x + iy$ which will split up the $J(a,b)$ integral into x and y over all the reals, but we also need two "dummy" variables t and u when using the $|z|^{-2a}$ relation (since we will have two factors of the Gamma function namely $\Gamma(a)$ (which will be assigned a t) and $\Gamma(b)$ (which will be assigned a u)). The integral now becomes

$$J(a,b) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^\infty dtdu\, t^{a-1}u^{b-1} \int_{-\infty}^{\infty} 2dxdy\, e^{-t(x^2+y^2)}e^{-u((1-x)^2+y^2)} \tag{6.342}$$

$$= \frac{2}{\Gamma(a)\Gamma(b)} \int_0^\infty dtdu\, t^{a-1}u^{b-1} \int_{-\infty}^{\infty} dxdy\, e^{-(t+u)x^2+2ux}e^{-(t+u)y^2}e^{-u}. \tag{6.343}$$

In the way written, it is clear that we have a Gaussian in the y-integral, and another Gaussian in the x-integral after completing a square. The y-integral gives

$$\int_{-\infty}^{\infty} dy\, e^{-(t+u)y^2} = \sqrt{\frac{\pi}{y+u}} \tag{6.344}$$

while the x-integral gives

$$\int_{-\infty}^{\infty} dx\, e^{-(t+u)((x-u/(t+u))^2-u^2/(t+u)^2)} = \sqrt{\frac{\pi}{t+u}}e^{u^2/(t+u)}. \tag{6.345}$$

We can now substitute these in and simplify $J(a,b)$ to get

$$J(a,b) = \frac{2}{\Gamma(a)\Gamma(b)} \int_0^\infty dtdu\, u^{b-1}\left(\frac{\pi}{t+u}\right)e^{u^2/(t+u)}e^{-u} \tag{6.346}$$

$$= \frac{2\pi}{\Gamma(a)\Gamma(b)} \int_0^\infty dtdu\, \frac{t^{a-1}u^{b-1}}{t+u}e^{=tu/(t+u)}. \tag{6.347}$$

If we now define $x \equiv ut/(t+u)$ and $\lambda = u/(t+u)$ with their corresponding inverse $t = x/\lambda$ and $u = x/(1-\lambda)$, when transforming coordinates we will pick up the following Jacobian

$$J = \left|\frac{\partial(t,u)}{\partial(x,\lambda)}\right| \tag{6.348}$$

$$= \det\begin{bmatrix} 1/\lambda & -x\lambda^2 \\ 1/(1-\lambda) & x/(1-\lambda)^2 \end{bmatrix} \tag{6.349}$$

$$= \frac{x}{\lambda^2(1-\lambda)^2} \tag{6.350}$$

which gives the following $J(a,b)$ integral

$$J(a,b) = \frac{2}{\Gamma(a)\Gamma(b)} \int_0^1 d\lambda \int_0^\infty dx \frac{x}{\lambda^2(1-\lambda)^2} \frac{\left(\frac{x}{\lambda}\right)^{a-1}\left(\frac{x}{1-\lambda}\right)^{b-1}}{\frac{x}{\lambda(1-\lambda)}} e^{-x} \tag{6.351}$$

$$= \frac{2}{\Gamma(a)\Gamma(b)} \int_0^1 d\lambda \int_0^\infty dx x^{1+a-1+b-1-1} \lambda^{-a+1-2+1}(1-\lambda)^{-b+1-2+1} e^{-x} \tag{6.352}$$

$$= \frac{2}{\Gamma(a)\Gamma(b)} \int_0^1 d\lambda\, \lambda^{-a}(1-\lambda)^{-b} \int_0^\infty dx\, x^{a+b-2} e^{-x}. \tag{6.353}$$

This last term is just another Gamma function with a different power (specifically $a+b-1$) giving us the following integral

$$J(a,b) = \frac{2}{\Gamma(a)\Gamma(b)} \Gamma(a+b-1) \int_0^1 d\lambda\, \lambda^{-a}(1-\lambda)^{-b} \tag{6.354}$$

$$= 2\pi \frac{\Gamma(a+b-1)}{\Gamma(a)\Gamma(b)} B(1-a, 1-b) \tag{6.355}$$

where we have used the definition of the Euler-beta function defined as

$$B(x,y) = \int_0^1 dt\, t^{x-1}(1-t)^{y-1} = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}. \tag{6.356}$$

So we finally get the following integral

$$J(a,b) = 2\pi \frac{\Gamma(a+b+1)\Gamma(1-a)\Gamma(1-b)}{\Gamma(a)\Gamma(b)\Gamma(b-a+2)} \tag{6.357}$$

$$= 2\pi B(1-a, 1-b, a+b-1) \tag{6.358}$$

where the 3-dimensional Euler-beta function is now defined as

$$B(x,y,z) = \frac{\Gamma(x)\Gamma(y)\Gamma(z)}{\Gamma(x+y)\Gamma(x+z)\Gamma(y+z)}. \tag{6.359}$$

This allows us to write the following 4-point tachyon amplitude in critical dimension

$c = 26$ where $C_{S^2} = 8\pi/\alpha' g_c^2$

$$S_{S^2} = i g_c^4 C_{S^2} (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right) \int_{\mathbb{C}} d^2 z |z|^{-\alpha' u/2 - 4} |1 - z|^{-\alpha' t/2 - 4} \tag{6.360}$$

$$= \frac{8\pi i g_c^2}{\alpha'} (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right) J \left(\frac{\alpha' 4}{4} + 2, \frac{\alpha' t}{4} + 2 \right) \tag{6.361}$$

$$= \frac{8\pi i g_c^2}{\alpha'} (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right) 2\pi B(1 - \alpha' u/4, -1 - \alpha' t/4, \alpha' u/4 + \alpha' t/4 - 3) \tag{6.362}$$

$$= \frac{8\pi i g_c^2}{\alpha'} (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right) 2\pi B(-\alpha_c(u), \alpha_c(t), 1 + \alpha_c(u) + \alpha_c(t)) \tag{6.363}$$

$$= \frac{8\pi i g_c^2}{\alpha'} (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right) C(-\alpha_c(u), -\alpha_c(t)) \tag{6.364}$$

where $\alpha_c(x) = \alpha' x 4 + 1$ and $C(x, y) = 2\pi B(x, y, 1 - x - y)$.

Polchinski 6.11 (a) Verify the amplitude (6.6.14)

Answer: For clarity later on, here is equation (6.6.14)

$$\begin{aligned} S_{S^2}(k_1, e_1; k_2, k_3) &= g_c^2 g_c' e^{-2\lambda} e_{1\mu\nu} \langle : \tilde{c} c \partial X^\mu \overline{\partial} X^\nu e^{i k_1 \cdot X}(z_1, \overline{z}_1) :: \tilde{c} c e^{i k_2 \cdot X}(z_2, \overline{z}_2) :: \tilde{c} c e^{i k_c \cdot X}(z_3, \overline{z}_3) : \rangle \\ &= -\frac{\pi i \alpha'}{2} g_c' e_{1\mu\nu} k_{23}^\mu k_{23}^\nu (2\pi)^{26} \delta^{(26)} \left(\sum_i k_i \right). \end{aligned} \tag{6.365}$$

In order to recreate this, first, since we are on the sphere, we need to understand its moduli space in order to determine the number of ghost insertions we would need. As a quick recap from chapter 5.2, 6.1, and 6.2 from Polchinski, recall that the moduli space is defined as $\mathcal{M}_r = \mathcal{G}_r/(\text{diff} \times \text{Weyl})$ where $\mathcal{G}_r$ is the space of all metrics. The formalism for extracting what $\mathcal{M}_r$ is for the sphere was developed and it can be shown as in the text, that the number of conformal killing vectors (CKV) is related to the number of moduli via the Riemann-Roch theorem. In particular for the sphere it has 0-moduli but 6 CKVs.

Based on the discussion of chapter 5.3 that leads into chapter 6.3 on the (bc) CFT, by the Riemann-Roch theorem, we need 6 ghosts (c-ghosts) to be inserted into the path integral for it to not vanish. For the sphere, it was then found in equation (6.3.5) how to find the expectation value. In other words, since the Euler-number of S^2 is $3\chi = \kappa - \mu = 6 - 0 = 6 \implies \chi = 2$. Then $e^{-\lambda\chi}$ in the path integral in equation (5.3.1) becomes $e^{-2\lambda}$. In total then the 2-point correlator is then equation (6.6.14) (the first line).

The matter portion of $S_{S^2}(k_1, e_1; k_2, k_3)$ denotes $S_{S^2}^X$ is given in general by equa-

tion (6.2.19) where for our 3-point case, let $d = 26$, $n = 3$ (and take $\omega \to 0$)

$$S^X_{S^2} = \langle :\partial X^\mu \overline{\partial} X^\nu e^{ik_1\cdot X}(z_1,\overline{z}_1)::e^{ik_2\cdot X}(z_2,\overline{z}_2)::e^{ik_3\cdot X}(z_3,\overline{z}_3):\rangle \tag{6.366}$$

$$= iC^X_{S^2}(2\pi)^{26}\delta^{(26)}(\sum_i k_i)|z_{12}|^{\alpha' k_1\cdot k_2}|z_{13}|^{\alpha' k_1\cdot k_3}|z_{23}|^{\alpha' k_2\cdot k_3}$$

$$\times\left[-\frac{i\alpha'}{2}\left(\frac{k_2^\mu}{z_1-z_2}+\frac{k_3^\mu}{z_1-z_3}\right)\right]\left[-\frac{i\alpha'}{2}\left(\frac{k_2^\nu}{\overline{z}_1-\overline{z}_2}+\frac{k_3^\nu}{\overline{z}_1-\overline{z}_3}\right)\right] \tag{6.367}$$

(Note: I am disregarding the antiholomorphic portion for now since it will contribute essentially all the same terms just with a tilde). The action ca then be written as

$$S_{S^2}(k_1,e_1;k_2,k_3) = -\frac{i\alpha'^2}{4}g_c^2 g_c' e^{-2\lambda}C^X_{S^2}S^g_{S^2}(2\pi)^{26}\delta^{(26)}\left(\sum_{i=1}^3 k_i\right)\epsilon_{1\mu\nu}$$

$$\times|z_{12}|^{\alpha' k_1\cdot k_2+2}|z_{13}|^{\alpha' k_1\cdot k_3+2}|z_{23}|^{\alpha' k_2\cdot k_3+2}\left(\frac{k_2^\mu}{z_{12}}+\frac{k_3^\mu}{z_{13}}\right)\left(\frac{k_2^\nu}{\overline{z}_{12}}+\frac{k_3^\nu}{\overline{z}_{13}}\right) \tag{6.368}$$

where the ghost action part came from equation (6.3.4) (or equation (6.3.5)) written as

$$\langle :c(z_1)\tilde{c}(z_1)::c(z_2)\tilde{c}(z_2)::c(z_3)\tilde{c}(z_3):\rangle = C^g_{S^2}|z_{12}|^2|z_{13}|^2|z_{23}|^2. \tag{6.369}$$

To continue, first let $C_{S^2} = e^{-2\lambda}C^X_{S^2}C^g_{S^2}$ for convience. Now, we can simplify using the momentum relations being on-shell. Since the gauge bosons are massless, $k_1^2 = 0$ while the tachyons give $k_2^2 = k_3^2 = +4/\alpha'$ which implies $k_1\cdot k_2 = k_1\cdot k_3 = 0$ and $k_2\cdot k_3 = -4/\alpha'$ since $k_1^2 = (k_2+k_3)^2 = k_2^2+k_3^2+2k_2\cdot k_3 = \frac{8}{\alpha'}+2k_1\cdot k_2$ which implies that $k_2\cdot k_3 = -4/\alpha'$. Then the amplitude now becomes

$$S_{S^2}(k_1,e_1;k_2,k_3) = -\frac{i\alpha'^2}{4}g_c^2 g_c' C_{S^2}(2\pi)^{26}\delta^{(26)}\left(\sum_{i=1}^3 k_i\right)e_{1\mu\nu}$$

$$\times|z_{12}|^2|z_{13}|^2|z_{23}|^{-2}\left(\frac{k_2^\mu}{z_{12}}+\frac{k_3^\mu}{z_{13}}\right)\left(\frac{k_2^\nu}{\overline{z}_{12}}+\frac{k_3^\nu}{\overline{z}_{13}}\right). \tag{6.370}$$

We can use the momentum conservation $k_3 = -k_1 - k_2$ along with the polarization vector relation to momentum of k_1 i.e. $e_{1\mu\nu}k_1^\mu = 0$ to reduce the momentum term as

$$e_{1\mu\nu}\left(\frac{k_2^\mu}{z_{12}}+\frac{k_3^\mu}{z_{13}}\right)z_{12}z_{13}z_{23}^{-1} = \frac{1}{2}e_{1\mu\nu}\left(\frac{k_2^\mu-k_1^\mu-k_3^\mu}{z_{12}}+\frac{k_3^\mu-k_1^\mu-k_2^\mu}{z_{13}}\right)z_{12}z_{13}z_{23}^{-1} \tag{6.371}$$

$$= \frac{1}{2}e_{1\mu\nu}k_{23}^\mu(z_{13}-z_{12})z_{23}^{-1} \tag{6.372}$$

$$= \frac{1}{2}e_{1\mu\nu}k_{23}^\mu z_{23}z_{23}^{-1} \tag{6.373}$$

$$= \frac{1}{2}e_{1\mu\nu}k_{23}^\mu \tag{6.374}$$

(where $k_{ij} = k_i - k_j$). We get a similar term for $\overline{z}_{ij}$,

$$e_{1\mu\nu}\left(\frac{k_2^\nu}{\overline{z}_{12}} + \frac{k_3^\nu}{\overline{z}_{13}}\right)\overline{z}_{12}\overline{z}_{13}\overline{z}_{23}^{-1} = \frac{1}{2}e_{1\mu\nu}k_{23}^\nu. \tag{6.375}$$

Then we find

$$\begin{aligned} S_{S^2}(k_1, e_1; k_2, k_3) &= -\frac{i\alpha'^2}{16}g_c^2 g_c' C_{S^2}(2\pi)^{26}\delta^{(26)}\left(\sum_{i=1}^{3} k_i\right)e_{1\mu\nu}k_{23}^\mu k_{23}^\nu \\ &= -\frac{i\alpha'\pi}{2}g_c'(2\pi)^{26}\delta^{(26)}\left(\sum_{i=1}^{3} k_i\right)e_{1\mu\nu}k_{23}^\mu k_{23}^\nu \end{aligned} \tag{6.376}$$

where we used $C_{S^2} = 8\pi/(\alpha' g_c^2)$ to get exactly equation (6.6.14).

(b) Verify the relation (6.6.15) between vertex operator normalizations.

Answer: Prior to equation (6.6.15) (which is $g_c' = (2/\alpha')g_c$) in chapter 6.6 we know from factorization that we can write the 4-point tachyon closed string amplitude into a combination of 3-string amplitudes of all possible intermediate states. This was used to get the relation of equation (6.6.8) $C_{S^2} = 8\pi/\alpha' g_c^2$ with $C_{S^2} = e^{-2\lambda}C_{S^2}^X C_{S^2}^g$ and g_c is the closed string tachyon vertex operator. From (6.6.6) we know the mass states exist at the poles $\alpha' s, \alpha' t, \alpha' u = -4, 0, 4, 8, \cdots$. The net pole is for the massless intermediate state. So if we look at the behavior of the factorized 4-point tachyon amplitude as $s \to 0$ we can link the 4-point amplitude to two amplitudes of the massless closed string with 2 tachyonic closed string, and thus their vertex couplings.

As $s \to 0$ we have

$$\lim_{s\to 0} S_{S^2}(k_1; k_2; k_3; k_4) = 0 \lim_{s\to 0}\sum_e \int \frac{d^{26}k}{(2\pi)^{26}}\frac{S_{S^2}(k, e; k_1; k_2)S_{S^2}(-k, e; k_3, k_4)}{-k^2 + i\epsilon}. \tag{6.377}$$

Using equation (6.6.10) to focus on the 4-point amplitude given as

$$S_{S^2}(k_1; k_2; k_3; k_4) = \frac{8\pi i g_c^2}{\alpha'}(2\pi)^{26}\delta^{(26)}\left(\sum_i k_i\right)C(-\alpha_c(t), -\alpha_c(u)) \tag{6.378}$$

where $C(a, b)$ given in equation (6.6.11) or explicitly as

$$C(-\alpha_c(t), -\alpha_c(u)) = 2\pi\frac{\Gamma(-\alpha_c(s))\Gamma(-\alpha_c(t))\Gamma(-\alpha_c(u))}{\Gamma(1+\alpha_c(s))\Gamma(1+\alpha_c(t))\Gamma(1+\alpha_c(u))}. \tag{6.379}$$

Notice in the $s \to 0$ channel part,

$$\lim_{s\to 0}\frac{\Gamma(-\alpha_c(s))}{\Gamma(1+\alpha_c(s))} = \lim_{s\to 0}\frac{\Gamma(-\alpha' s/4 - 1)}{\Gamma(\alpha' s/4 + 2)} \tag{6.380}$$

$$= -\lim_{s\to 0}\Gamma(-\alpha' s/4) \tag{6.381}$$

$$= \frac{4}{\alpha' s} \tag{6.382}$$

with $\Gamma(x) = \Gamma(x-1)\Gamma(x-1)$ and $\lim_{\epsilon\to 0}\Gamma(\epsilon) = \epsilon^{-1} + \text{finite}$. Now from $\alpha'(s+t+u) = -16$ this implies $\alpha' t = -\alpha' u - 16$ and likewise $\alpha_c(t) = \alpha' t/4 + 1 = -\alpha' u/4 - 4 + 1 = -\alpha_c(u) - 2$. Then,

$$\frac{\Gamma(-\alpha_c(t))\Gamma(-\alpha_c(u))}{\Gamma(1+\alpha_c(t))\Gamma(1+\alpha_c(u))} = \frac{(\alpha_c(u)+1)\Gamma(\alpha_c(u)+1)(-\alpha_c(u)-1)\Gamma(-\alpha_c(u)-1)}{\Gamma(-\alpha_c(u)-1)\Gamma(1+\alpha_c(u)+1)} \tag{6.383}$$

$$= (\alpha_c(u)+1)(-\alpha_c(u)-1) = -(\alpha_c(u)+1)^2 \tag{6.384}$$

$$= -(\alpha' u/4 + 2)^2. \tag{6.385}$$

Now if $s \to 0$ then $t \to 0$ since $s \leq t$, and then since $\alpha'(s+t+u) = -16$ then $\alpha' u = -16$ so

$$\frac{\Gamma(-\alpha_c(t))\Gamma(-\alpha_c(u))}{\Gamma(1+\alpha_c(t))\Gamma(1+\alpha_c(u))} = -(-4+2)^2 = -4. \tag{6.386}$$

Then all together

$$\lim_{s\to 0} C(-\alpha_c(t), -\alpha_c(u)) = 2\pi 4 \frac{4}{\alpha' s} = -\frac{32\pi}{\alpha' s} \tag{6.387}$$

and the s-pole of the 40tachyon amplitude is

$$LHS = \lim_{s\to 0} S_{S^2}(k_1; k_{2;3}; k_4) = -\frac{256\pi^2 i g_c^2}{\alpha'^2}(2\pi)^{26}\delta^{(26)}\left(\sum_i k_i\right)\frac{1}{2}. \tag{6.388}$$

The right hand side becomes

$$RHS = i\lim_{s\to 0}\sum_e \int \frac{d^{26}k}{(2\pi)^{26}} \frac{S_{S^2}(k,e;k_1,k_2)S_{S^2}(-k,e;k_3,k_4)}{-k^2+i\epsilon} \tag{6.389}$$

$$= i\lim_{s\to 0}\sum_e \int \frac{d^{26}k}{(2\pi)^{26}}\left[-\frac{i\pi\alpha'}{2} g_c' e_{\mu\nu} k_{12}^\mu k_{12}^\nu (2\pi)^{26}\delta^{(26)}(k_1+k_2+k)\right]$$
$$\times\left[-\frac{\pi i\alpha'}{2} g_c' e_{\rho\sigma} k_{34}^\rho k_{34}^\sigma (2\pi)^{26}\delta^{(26)}(k_3+k_4-k)\right]\cdot\frac{1}{-k^2+i\epsilon} \tag{6.390}$$

$$= -\lim_{s\to 0}\sum_e \frac{\pi^2\alpha'^2 g_c^2}{4}(2\pi)^{26}\delta^{(26)}(k_1+k_2+k_3+k_4)e_{\mu\nu}e_{\rho\sigma}\frac{k_{12}^\mu k_{12}^\nu k_{34}^\rho k_{34}^\sigma}{-(k_3+k_4)^2+i\epsilon}. \tag{6.391}$$

Since $s = -(k_1+k_2)^2 = -(k_3+k_4)^2$ we can write the right hand side as

$$RHS = -\lim_{s\to 0}\sum_e \frac{\pi^2\alpha' g_c^2}{4} e_{\mu\nu}e_{\rho\sigma}k_{12}^\mu k_{12}^\nu k_{23}^\rho k_{23}^\sigma \tag{6.392}$$

or in other words

$$g_c^2 = \frac{\alpha'^4 g_c^4}{1064}\sum_e e_{\mu\nu}e_{\rho\sigma}k_{12}^\mu k_{12}^\nu k_{23}^\rho k_{23}^\sigma. \tag{6.393}$$

To figure out the RHS relation notice that $k_1 - k_2 = k_1 + k_2 - 2k_2 = k - 2k_2$ and similarly $k_3 - k_4 = -k - 2k_4$ and since $e_{\mu\nu}k^\mu = e_{\mu\nu}k^\nu = 0$ implies that

$$\sum_e e_{\mu\nu}e_{\rho\sigma}k_{12}^\mu k_{12}^\nu k_{23}^\sigma k_{23}^\rho = 16\sum_e e_{\mu\nu}e_{\rho\sigma}k_2^\mu k_2^\nu k_4^\rho k_4^\sigma. \tag{6.394}$$

Now, similar to QED, the completeness relation here is

$$\sum_e e_{\mu\nu}e_{\rho\sigma} = g_{\mu\rho}g_{\nu\sigma} \implies \sum_e e_{\mu\nu}e_{\rho\sigma}k_{12}^{\mu}k_{12}^{\nu}k_{23}^{\sigma}k_{23}^{\rho} = 16(k_2\cdot k_4)^2. \tag{6.395}$$

But $t = -(k_1+k_3)^2 = -(k_2+k_4)^2 = k_2^2 + k_4^2 + 2k_2\cdot k_4 = \frac{8}{\alpha'} + 2k_2\cdot k_4$. Then in the limit $s\to 0$ we have $t\to 0$ which implies $k_2\cdot k_4 = -4/\alpha'$. This then gives

$$\sum_e e_{\mu\nu}e_{\rho\sigma}k_{12}^{\mu}k_{12}^{\nu}k_{23}^{\sigma}k_{23}^{\rho} = \frac{256}{\alpha'^2}. \tag{6.396}$$

Plugging this in we get

$$g_c^2 = \frac{\alpha'^4 g_c'^2}{1064}\frac{256}{\alpha'^2} \implies g_c^2 = \frac{\alpha'^2}{4}g_c'^2 \tag{6.397}$$

or in other words

$$g_c' = \frac{2}{\alpha'}g_c. \tag{6.398}$$

(c) Complete the field theory calculation and verify the relation (6.6.18) between g_c and the gravitational coupling.

<u>Answer:</u> $\mathbf{S}$ in equation (3.7.20) is given as

$$\mathbf{S} = \frac{1}{2\kappa_D^2}\int d^Dx\sqrt{-G}e^{-2\Phi}\left(-\frac{2(D-26)}{3\alpha'} + \mathbf{R} - \frac{1}{12}H_{\mu\nu\lambda}H^{\mu\nu\lambda} + 4\partial_\mu\Phi\partial^\mu\Phi + \mathcal{O}(\alpha')\right) \tag{6.399}$$

where anything boldface represents the spacetime version (or in Einstein frame). The full action we are looking at includes the tachyon action[128] given as

$$\mathbf{S}_T = -\frac{1}{2}\int d^{26}x\sqrt{-G}e^{-2\tilde{\Phi}}\left(G_{\mu\nu}\partial^\mu T\partial^\nu T - \frac{4}{\alpha'}T^2\right). \tag{6.400}$$

Let us perturb the metric $\tilde{G}_{\mu\nu} = \eta_{\mu\nu} + 2\kappa e_{\mu\nu}f$ with $e^2 = 1$. To do the perturbation, recall from graviton dynamics that when we perturb the metric as $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ the weight of the integral in the Einstein-Hilbert action becomes

$$\sqrt{-g} = 1 + \frac{\kappa}{2}h + \cdots \tag{6.401}$$

We can then just map $\kappa\to 2\kappa f$ and the graviton field as $h_{\mu\nu}\to e_{\mu\nu}$ to get the weight in the tachyon action to be

$$\sqrt{-G} = 1 + \kappa e f + \cdots \tag{6.402}$$

Then the tachyon Lagrangian becomes

$$\mathcal{L} = -\frac{1}{2}(\partial T)^2 + \frac{2}{\alpha'}T^2 + \mathcal{L}_{int} + \mathcal{O}(f^2) \tag{6.403}$$

[128] I do have one question which is how are the string-tachyon action constructed? Are they constructed by hand based on what we know from its mass spectrum (hence the negative kinetic term)?

where the interaction Lagrangian is

$$\mathcal{L}_{int} = \kappa e_{\mu\nu} f \partial^\mu T \partial^\nu T + \kappa e f \left(\frac{2}{\alpha'} T^2 - \frac{1}{2} \partial_\mu T \partial^\mu T \right). \tag{6.404}$$

Note that we are now using the $\eta_{\mu\nu}$ to raise and lower indices and to perform tensor products. Notice that the trace of $e_{\mu\nu}$ will vanish on shell since in the amplitude calculation the term is in it of the form

$$-i\kappa e \left(\frac{4}{\alpha'} + k_2 \cdot k_3 \right) (2\pi)^{(26)} \delta^{(26)}(k_1 + k_2 + k_3) = 0 \tag{6.405}$$

where we Fourier transformed the terms proportional to $\kappa e f$ in the kinetic Lagrangian to get the vertex rule. (The term vanishes because $k_2 \cdot k_3 = -4/\alpha'$). But the first term gives

$$2i\kappa e_{\mu\nu} k_2^\mu k_4^\nu (2\pi)^{26} \delta^{(26)}(k_1 + k_2 + k_3) = -\frac{i\kappa}{2} e_{\mu\nu} k_{23}^\mu k_{23}^\nu \delta^{(26)}(k_1 + k_2 + k_3) \tag{6.406}$$

since $e_{1\mu\nu} k_!^\mu = 0$ (similar to part (b) of the calculation). If we now compare with equation (6.6.14) we find $\kappa - \pi\alpha' g_c'$.

Polchinski 7.1 Fill in the steps leading to the expectation value (7.2.4) of exponential operators on the torus. Show that it has the correct transformation under $\tau \to -1/\tau$.

Answer: The calculation (as Polchinski said in the text as well) mimics that performed in section 6.2 (pages 170-171) on calculating the tree-level amplitude on a sphere. So, lets take some steps from that section, and then substitute the proper renormalized Green's function when necessary.

We need to know the behavior of the vertex operator(s), so consider in full form the renormalized version of the vertex operator (where a renormalized operator with respect to the symmetry of the world sheet (diff $\times$ Weyl) equation (3.6.5))

$$\left[e^{ik_i \cdot X(\sigma_i)} \right]_r = \exp\left(\frac{1}{2} \int d^2\sigma d^2\sigma' \Delta(\sigma, \sigma') i^2 k_i \cdot k_i \delta^{(2)}(\sigma - \sigma_i) \delta^{(2)}(\sigma - \sigma_i) \right) e^{ik_i \cdot X(\sigma_i)} \tag{6.407}$$

$$= \exp\left(-\frac{1}{2} \int d^2\sigma d^2\sigma' \, \frac{\alpha'}{2} \ln\left(d^2(\sigma, \sigma') \right) k_i^2 \delta^{(2)}(\sigma - \sigma_i) \delta^{(2)}(\sigma - \sigma_i) \right) e^{ik_i \cdot X(\sigma_i)} \tag{6.408}$$

$$= \exp\left(\frac{-\alpha'}{4} \ln\left(d^2(\sigma_i, \sigma_i) \right) k_i^2 \right) e^{ik_i \cdot X(\sigma_i)} \tag{6.409}$$

where $\Delta(\sigma, \sigma') = \frac{\alpha'}{2} \ln(d^2(\sigma, \sigma'))$ and $d^2(\sigma, \sigma')$ is just the geodesic distance. Then, to get equation (6.2.13) the amplitude becomes

$$A_{S^2}^n(k, \sigma) = \left\langle \left[e^{ik_1 \cdot X(\sigma_1)} \right]_r \cdots \left[e^{ik_n \cdot X(\sigma_n)} \right]_r \right\rangle \tag{6.410}$$

$$= \exp\left(-\frac{\alpha'}{4} \sum_{i=1}^n k_i^2 \ln\left(d^2(\sigma_i, \sigma_i) \right) \right) \left\langle \exp\left(i \sum_{j=1}^n k_j \cdot K(\sigma_j) \right) \right\rangle. \tag{6.411}$$

Notice that we can put the last exponential into a generating functional form with the insertion of a delta function,

$$\left\langle \exp\left(i\sum_{j=1}^{n} k_j \cdot K(\sigma_j)\right) \right\rangle = \left\langle \exp\left(i\int d^2\sigma \sum j = 1^n k^\mu \delta^{(2)}(\sigma = \sigma_j) X_\mu(\sigma)\right) \right\rangle \tag{6.412}$$

$$= \left\langle \exp\left(i\int d^2\sigma J^\mu(\sigma) X_\mu(\sigma)\right) \right\rangle \tag{6.413}$$

where $J^\mu(\sigma) = \sum_{j=1}^{n} k_j^\mu \delta^{(2)}(\sigma - \sigma_j)$ (this is just equation (6.2.12) from Polchinski but he just tells you what is it, but this way it feels must more physical and natural). Then the partition function (equation 6.2.6) becomes (without any of the prefactors for simplicity for now)

$$Z[J] \propto \exp\left(-\frac{\alpha'}{4}\sum_{i=1}^{n} j_i^2 \ln\left(d^2(\sigma_i, \sigma_i)\right) - \frac{1}{2}\int d^2\sigma J^\mu(\sigma) J_\mu(\sigma') G'(\sigma, \sigma')\right). \tag{6.414}$$

The last exponential simplifies to the following

$$\exp\left(-\sum_{i\neq j=1}^{n} k_i \cdot k_j G'(\sigma_i, \sigma_j) - \frac{1}{2}\sum_{i=1}^{n} k_i^2 G'(\sigma_i, \sigma_i)\right) \tag{6.415}$$

to give equation (6.2.13)

$$A_{S^2}^n(k, \sigma) = iC_{S^2}^X (2\pi)^d \delta^{(d)}(\sum_i k_i) \exp\left(-\sum_{i\neq j=1}^{n} k_i \cdot k_j G'(\sigma_i, \sigma_j) - \frac{1}{2}\sum_{i=1}^{n} k_i^2 G_r'(\sigma_i, \sigma_i)\right) \tag{6.416}$$

where the renormalized Greeen's function is originally given as

$$G_r'(\sigma, \sigma') = G'(\sigma, \sigma) + \frac{\alpha'}{2}\ln\left(d^2(\sigma, \sigma')\right) \tag{6.417}$$

(this is equation (6.2.15)) and the geodesic distance is given in equation (3.6.9) repeated here for clarity

$$d^2(\sigma_1, \sigma_2) = (\sigma_1 - \sigma_2)^2 e^{2\omega(\sigma)} = |z_{12}|^2 e^{2\omega(\sigma)}. \tag{6.418}$$

To understand the renoamlizable Green's function G_r, consider its behavior at 2 coincident (measuring) points

$$\lim_{w'\to w} G'(w, \overline{w}; w', \overline{w}') = \lim_{w'\to w} G'(w, \overline{w}; w'\overline{w}') + \frac{\alpha'}{2}\ln|w - w'|^2 + \alpha'\omega(w) \tag{6.419}$$

where we are keeping the Weyl-factor in place for now and not sending it to 0 until the end and G' is given in equation (7.2.3) (repeated here for clarity and ease-of-access)

$$G'(w, \overline{w}; w', \overline{w}') = -\frac{\alpha'}{2}\ln\left|\vartheta\left(\frac{w - w'}{2\pi}\,\middle|\,\tau\right)\right|^2 + \alpha'\frac{\mathrm{Im}\{w - w'\}^2}{4\pi\tau_2} + k(\tau, \overline{\tau}). \tag{6.420}$$

To understand this limit, consider the perturbation of ϑ_1 around $\nu = 0$

$$\vartheta_1(\nu,\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} \sum_{m=1}^{\infty} \frac{(2\pi i n \nu)^m}{m!} \tag{6.421}$$

$$= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} \sum_{m=1}^{\infty} \frac{(2\pi i n)^m}{m!}(1+\nu+\nu^2) \tag{6.422}$$

$$= \partial_\nu \vartheta_1(0,\tau) + \mathcal{O}(\nu^2) \tag{6.423}$$

where $\partial_\nu \vartheta_1(\nu,\tau) = \vartheta_1'$. Then the limit becomes

$$\lim_{w'\to w} G'(w,\overline{w};w',\overline{w}') = -\frac{\alpha'}{2}\left(\ln|w-w'| + \ln\left|\frac{\partial_\nu \varphi_1(0,\tau)}{2\pi}\right|^2 + \mathcal{O}(|w-w'|^2)\right). \tag{6.424}$$

This implies that the renormalized Green's function becomes

$$\lim_{w\to w'} G'_r = -\frac{\alpha'}{2}\left(\ln\left|\frac{\partial_\nu \varphi_1(0,\tau)}{2\pi}\right|^2 + \alpha'\omega(w)\right). \tag{6.425}$$

Then the amplitude becomes (using the trick $\ln\Big(ae^b\Big) = \ln(a) + \ln\Big(e^b\Big) = \ln(a) + b$)

$$A^n_{T^2}(k,\sigma) \propto \exp\Bigg(-\sum_{i<j=1}^{n} k_i\cdot k_j \left(-\frac{\alpha'}{2}\ln\left|\vartheta_1\left(\frac{w_{ij}}{2\pi},\tau\right)\right|^2 + \alpha'\frac{(\mathrm{Im}\{w_{ij}\})^2}{4\pi\tau_2}\right)$$
$$-\frac{1}{2}\sum_{i=1}^{n} k_i^2\left(-\frac{\alpha'}{2}\left(\ln\left|\frac{\partial_\nu\vartheta_1(0,\tau)}{2\pi}\right|^2 + \alpha'\frac{(\mathrm{Im}\{w_{ij}\})^2}{4\pi\tau_2}\right)\right)\Bigg) \tag{6.426}$$

$$= \exp\Bigg(\sum_{i<j=1}^{n} \alpha' k_i\cdot k_j \ln\left(\left|\vartheta_1\left(\frac{w_{ij}}{2\pi},\tau\right)\right| \exp\left(-\alpha'\frac{\mathrm{Im}\{w_{ij}\}^2}{4\pi\tau_2}\right)\right) \tag{6.427}$$
$$-\frac{1}{2}\sum_{i=1}^{n}\alpha' k_i^2\omega(w_i) + \sum_{i=1}^{n}\frac{\alpha'}{2}k_i^2\ln\left|\frac{\partial_\nu\vartheta_1(0,\tau)}{2\pi}\right|\Bigg)$$

$$= \exp\Bigg(\sum_{i<j=1}^{n} \ln\left(\left|\vartheta_1\left(\frac{w_{ij}}{2\pi},\tau\right)\right| \exp\left(-\alpha'\frac{\mathrm{Im}\{w_{ij}\}^2}{4\pi\tau_2}\right)\right)^{\alpha' k_i\cdot k_j}$$
$$-\frac{1}{2}\sum_{i=1}^{n}\alpha' k_i^2\omega(w_i) - \sum_{i=1}^{n}\ln\left|\frac{2\pi}{\partial_\nu\vartheta_1(0,\tau)}\right|^{\frac{\alpha'}{2}k_i^2}\Bigg) \tag{6.428}$$

$$= \exp\left(-\frac{1}{2}\sum_{i=1}^{n}\alpha' k_i^2\omega(w_i)\right)\prod_{i<j=1}^{n}\left(\left|\vartheta_1\left(\frac{w_{ij}}{2\pi},\tau\right)\right| \exp\left(-\alpha'\frac{\mathrm{Im}\{w_{ij}\}^2}{4\pi\tau_2}\right)\right)^{\alpha' k_i\cdot k_j}$$
$$\times\prod_{i=1}^{n}\left|\frac{2\pi}{\partial_\nu\vartheta_1(0,\tau)}\right|^{-\frac{\alpha'}{2}k_i^2} \tag{6.429}$$

Via momentum conservation we have the following relation

$$2\sum_{i<j}^{n} k_i\cdot k_j + \sum_{i=1}^{n} k_i^2 = \sum_{i=j=1}^{n} k_i\cdot k_j \tag{6.430}$$

which must equal 0 to give the relationship $k_i^2 = -2k_i \cdot k_j$. This simplifies the final product to be

$$\prod_{i=1}^{n} \left| \frac{2\pi}{\partial_\nu \vartheta_1(0,\tau)} \right|^{-\frac{\alpha'}{2}k_i^2} = \prod_{i<j=1}^{n} \left| \frac{2\pi}{\partial_\nu \vartheta_1(0,\tau)} \right|^{\alpha' k_i \cdot k_j} \tag{6.431}$$

which simplifies the amplitude (with the correct coefficients) becomes

$$\begin{aligned} A_{T^2}^n(k,\sigma) = iC_{T^2}^X (2\pi)^d \delta^{(d)}(\sum_i k_i) \exp\left(-\frac{1}{2}\sum_i \alpha' k_i^2 \omega(w_i)\right) \\ \times \prod_{i<j=1}^{n} \left| \frac{2\pi}{\partial_\nu \vartheta_1(0,\tau)} \vartheta_1\left(\frac{w_{ij}}{2\pi},\tau\right) \exp\left(-\alpha' \frac{(\mathrm{Im} w_{ij})^2}{4\pi\tau_2}\right) \right|^{\alpha' k_i \cdot k_j} \end{aligned} \tag{6.432}$$

This is exactly equation (7.2.4) if we take the flat space limit i.e $\omega(w) \to 0$.

Now, in order to show the modular invariance asked for, $\tau \to -1/\tau$ recall how we defined just ϑ first

$$\vartheta(\nu,\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n \nu} \tag{6.433}$$

which can also be written as the *theta function with characteristics* (pages 214-215 in Polchinski)

$$\vartheta \begin{bmatrix} a \\ b \end{bmatrix} (\nu,\tau) = \sum_{n=-\infty}^{\infty} \exp\Big(\pi i (n+a)^2 \tau + 2\pi i (n+a)(\nu+b)\Big). \tag{6.434}$$

For ϑ_1 this is negative of the above at $a = b = 1/2$. Then consider the derivative of ϑ_1

$$\partial_\nu \vartheta_1(\nu|\tau) = 2\pi i \left(n + \frac{1}{2}\right) \vartheta_1(\nu|\tau). \tag{6.435}$$

Then under the modular transformation, we get a τ factor in the exponentials that appear as $|\exp(\pi i (n+1/2)^2/\tau)|^2 = 1$ leaving the amplitude invariant under modular transformations.

Polchinski 7.4 Evaluate $\langle \partial_w X^\mu(w) \partial_w X_\mu(0) \rangle$ on the torus by representing it as a trace. Show that the result agrees with equation (7.2.16).

<u>Answer:</u> For clarity at the end of the problem lets write down equation (7.2.16) to know what we are trying to find

$$Z(\tau)^{-1} \langle \partial_w X^\mu(w) \partial_w X_\mu(0) \rangle = \frac{\alpha' d}{2} \frac{\vartheta_1 \partial_w^2 \vartheta_1 - \partial_w \vartheta_w \partial_w \vartheta_1}{\vartheta_1^2} + \frac{\alpha' d}{8\pi\tau_2}. \tag{6.436}$$

Consider writing the second-order derivative of the 2-point function as

$$\langle \partial_w X^\mu(w) \partial_w X_\mu(0) \rangle = \lim_{z\to 0} \langle \partial_w X^\mu(w) \partial_z X^\mu(z) \rangle \tag{6.437}$$

$$= \lim_{z\to 0} \partial_w \partial_z \langle X^\mu(w) X_\mu(z) \rangle. \tag{6.438}$$

Then we only need to compute the 2-point string amplitude on a torus (which we could denote the amplitude as $\langle \cdots \rangle_{T^2}$ but will drop for now).

As discussed on page 209, we can form the torus with modular τ by taking a field theory on a circle and evolve by Euclidean time for $2\pi\tau_2$ which translates in $\sigma^1 \to 2\pi\tau_1$ and identify the ends. The time evolution is governed by the Hamiltonian H written as

$$H = L_0 + \tilde{L}_0 - \frac{1}{24}(c + \tilde{c}), \tag{6.439}$$

and the spatial translation is governed by the momentum operator written as

$$P = L_0 - \tilde{L}_0 \tag{6.440}$$

with the "identifiying ends" is just a gluing of circles together via a trace as the *Hilbert space states*.

This gets us the partition function equation (7.2.5) from Polchinski, but with any field theory, we know to get $\langle X^\mu(w) X_\mu(z) \rangle$ we just insert these two operators into the trace,

$$\begin{aligned}
\langle X^\mu(\sigma) X_\mu(\sigma') \rangle &= \mathrm{tr}\big(X^\mu(\sigma) X_\mu(\sigma') Z(\tau)\big) && (6.441)\\
&= \mathrm{tr}\big(X^\mu(\sigma) X_\mu(\sigma') \exp(2\pi i \tau_1 P - 2\pi\tau_2 H)\big) && (6.442)\\
&= \mathrm{tr}\Big(X^\mu(\sigma) X_\mu(\sigma') q^{L_0} \overline{q}^{\tilde{L}_0} (q\overline{q})^{-d/24}\Big) && (6.443)
\end{aligned}$$

where we used equation (7.2.5) for the partition function $Z(\tau)$, and note that the lowercase trace here still denotes the typical trace over Hilbert space states and not over anything else.

Now recall the solution of the closed string $X^\mu(\sigma)$ given in equation (1.4.4)

$$X^\mu(\tau, \sigma) = x^\mu + \alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \left(\frac{\alpha_n^\mu}{n} e^{-in(\tau - \sigma)} + \frac{\tilde{\alpha}_n^\mu}{n} e^{-in(\tau + \sigma)} \right). \tag{6.444}$$

In order to compute the trace, for now, we only need to consider the exicted states constructed out of the vacuum built out of raising operators denoted as $|E\rangle$ which will only include the right-moving (non-tilde oscillators) since they it will be easy to incorporate them later. The state $|E\rangle$ can be written as

$$|E\rangle = \prod_{\mu=0}^{d} \prod_{n=1}^{\infty} \frac{(\alpha_{-n}^\mu)^{E_{\mu,n}}}{\sqrt{(n^{E_{\mu,n}})(E_{\mu,n})!}} |0\rangle. \tag{6.445}$$

Note that this spectrum is simply equation (1.4.7) with $E_{\mu,n} \in \mathbb{N}$. Now if $|E\rangle$ has λ_n-excitations (eigenvalues) at level n, then given $(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$ the raising and lower operators behave

$$\begin{aligned}
\alpha_{-n}^i |E\rangle &= \sqrt{n(\lambda_n + 1)}\, |\lambda_1, \cdots, \lambda_{n+1}\rangle && (6.446)\\
\alpha_n |E\rangle &= \sqrt{n\lambda_n}\, |\lambda_1, \cdots, \lambda_{n-1}, \cdots \lambda_{n+1}\rangle && (6.447)
\end{aligned}$$

which are just the rules for a simple harmonic oscillator.

Lets now go term by term in the product of $X^\mu(\sigma)X_\mu(\sigma')$ (which I should have noted we are working in real coordinates for now). The first term we have is $x_\mu x^\mu \subset X^\mu(\sigma)X_\mu(\sigma')$ which when we take the derivative $\partial_w\partial_w$ will vanish so no need to care about it.

For the $p^\mu p_\mu \subset X^\mu(\sigma)X_\mu(\sigma')$ we can rewrite it in terms of oscillators as

$$p^2 = -M^2 = \frac{4}{\alpha'}\sum_{i>0}\alpha_i\alpha_{-i} \tag{6.448}$$

which we shall return to later since the real issue is the following product

$$\propto \sum_{E,m,n}\langle E|\,\frac{\alpha_n^\mu}{n}\frac{\alpha_{m,\mu}}{m}e^{-in(\tau-\sigma)}e^{-im(\tau'-\sigma')}q^{L_0}\overline{q}^{-\tilde{L}_0}(q\overline{q})^{-d/24}\,|E\rangle\,. \tag{6.449}$$

This is a sum over E, n and m all of which we have to perform. There is also a similar term for $\tilde{\alpha}$. Notice that unless $m \to -n$ (or vice versa) the string expectation value will disappear since α_n^i acting on the vacuum will disappear. Then, dropping the $\overline{q}^{-\tilde{L}_0}(q\overline{q})^{-d/24}$ term for now we want to compute

$$-\sum_{E,n}\langle E|\,\frac{\alpha_n^\mu\alpha_{-n,\mu}}{n^2}e^{-in(\tau-\sigma)e^{in(\tau'-\sigma')}q^{L_0}}\,|E\rangle\,. \tag{6.450}$$

Consider first the computation of $\sum_E\langle E|\,q^{L_0}\,|E\rangle$. Recall equation (2.7.10) for $m = 0$,

$$L_0 = \sum_n \alpha_{-n}\alpha_n \tag{6.451}$$

where we dropped the creation annhilation normal ordering. Now consider in the basis $\{|\lambda_n = 0\rangle, |\lambda_n = 1\rangle, \cdots\}$ then with $q^{\sum_n \alpha_{-n}\alpha_n}$ in matrix form becomes (with picking only one state)

$$q^{\alpha_{-n}\alpha_n} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & q^n & 0 & \cdots \\ 0 & 0 & q^{2n} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{6.452}$$

which is just within the subspace of Hilbert string states. Then the trace of the matrix is simply

$$\mathrm{tr}(q^{\alpha_{-n}\alpha_n}) = \sum_{i=0}^{\infty} q^{in} = (1-q^n)^{-1} \tag{6.453}$$

since this is just a geometric series with the ratio of q. In the full space of states, the entire sum over n, $\sum_n \alpha_{-n}\alpha_n$ we just have a product instead $\prod_{n=0}^{\infty}(1-q^n)^{-1}$ which Polchinski tells us is related to the eta-Dedekind function $\prod_{n=0}^{\infty}(1-q^n)^{-1} = q^{1/24}|\eta(\tau)|^{-1}$.

Lets now split up the sum over all states into $n > 0$ and $n < 0$. For the case of $n > 0$ we have the following computation

$$\sum_{\{m_i\}} \langle \lambda_1 = m_1, \cdots | \frac{\alpha_n \alpha_{-n}}{n^2} e^{-in(\tau-\sigma)} e^{in(\tau'-\sigma')} q^{L_0} | \lambda_1 = m_1, \cdots \rangle \tag{6.454}$$

$$= e^{-in[(\tau-\sigma)-(\tau'-\sigma')]} | \lambda_1, \cdots \rangle \frac{\alpha_n \alpha_{-n}}{n^2} q^{L_0} | \lambda_1, \cdots \rangle \tag{6.455}$$

$$= e^{-in[(\tau-\sigma)-(\tau'-\sigma')]} \frac{1}{n} \sum_{\{m_i\}} (q^{1m_1} q^{2m_2} \cdots) \langle \lambda_1, \cdots | \alpha_n \alpha_{-n} | \lambda_1, \cdots \rangle \tag{6.456}$$

$$= e^{-in[(\tau-\sigma)-(\tau'-\sigma')]} \frac{1}{2} \sum_{\{m_i\}} (q^{nm_n}(m_n+1)) \left(\prod_{m_j \neq m_n}^{\infty} q^{jm_j} \right). \tag{6.457}$$

Notice that for the q^{jm_j} sums over m_j, this is just $(1-q^j)^{-1}$ so we have then the total product

$$\prod_{m_j \neq m_n}^{\infty} q^{jm_j} = \prod_{j \neq n}^{\infty} (1-q^j)^{-1}. \tag{6.458}$$

We then find

$$\sum_{\{m_j\}} (q^{nm_n}(m_n+1)) \left(\prod_{j \neq n}^{\infty} (1-q^j)^{-1} \right). \tag{6.459}$$

The first sum is simply the derivative of the geometric series since

$$\frac{d}{dq} \left(\sum_m q^{mn} \right) = \sum_m q^{mn} m = \frac{q^n}{(1-q^n)^2} \tag{6.460}$$

and then the second sum is $\sum_{m_n=0} q^{nm_n} = (1-q^n)^{-1}$. All together we now have

$$e^{-in[(\tau-\sigma)-(\tau'-\sigma')]} \frac{1}{n} \left(\prod_{j \neq n} (1-q^j)^{-1} \right) \left(\frac{q^n}{(1-q^n)^2 + (1-q^n)^{-1}} \right) \tag{6.461}$$

$$= \sum_{n=1}^{\infty} e^{-in[(\tau-\sigma)-(\tau'-\sigma')]} \frac{1}{n} \left(\prod_j (1-q^j)^{-1} \right) \left(\frac{q^n}{1-q^n} + 1 \right) \tag{6.462}$$

where in the last line we are now taking the sum over $n > 0$ and we pulled out a common factor of $(1-q^n)$ and absorbed it into the product which allows us to take the product over all of j. To make the result easier that will allow us to put the amplitude in terms of ϑ_1 functions, consider the following relations

$$\sum_{n=1}^{\infty} \frac{1}{n} e^{inx} = -\log\left(1 - e^{ix}\right) \tag{6.463}$$

$$\sum_{m=1}^{\infty} \frac{1}{m} \frac{x^m}{1-y^m} = -\sum_{n=9}^{\infty} \log(1-xy^n). \tag{6.464}$$

Now let $\Delta\sigma = (\tau' - \tau) - (\sigma' - \sigma)$ for convience. Then

$$\sum_{n=1}^{\infty} e^{-in\Delta\sigma} \frac{1}{n} \left(\prod_j (1-q^j)^{-1} \right) \left(\frac{q^n}{1-q^n} + 1 \right) \tag{6.465}$$

$$= - \left(\prod_j (1-q^j)^{-1} \right) \left(\log\left(1 - e^{-i\delta\sigma}\right) + \sum_{m=0}^{\infty} \log\left(1 - q^{m+1} e^{-i\Delta\sigma}\right) \right). \tag{6.466}$$

Note however that this is *only* for $n > 0$, but what about $n < 0$? The setup is is the exact same, but given that α_{-n} hits the state *after* q^{L_0} is a lowering operator. Then the product $\alpha_n \alpha_{-n}$ brings out a factor of $-nm_{-n}$ rather then $n(m_n + 1)$. Thus

$$\sum_{\{m_j\}} \langle \lambda_1 = m_1, \cdots | \frac{\alpha_n \alpha_{-n}}{n^2} e^{-in(\tau-\sigma)} e^{-in(\tau'-\sigma')} q^{L_0} | \lambda_1 = m_1, \cdots \rangle \tag{6.467}$$

$$= e^{-in\Delta\sigma} \frac{1}{n^2} \sum_{\{m_j\}} (q^{1m_1} q^{2m_2} \cdots) \langle \lambda_1, \cdots | \alpha_n \alpha_{-n} | \lambda_1, \cdots \rangle \tag{6.468}$$

$$= -e^{-in\Delta\sigma} \frac{1}{n} \sum_{\{m_j\}} (q^{nm_{-n}m_{-n}}) \left(\prod_{m_j \neq m_{-n}} q^{jm_j} \right) \tag{6.469}$$

$$= -e^{-in\Delta\sigma} \frac{1}{n} \left(\frac{q^{-n}}{(1-q^{-n})^2} \right) \left(\prod_{j \neq -n}^{\infty} (1-q^j)^{-1} \right) \tag{6.470}$$

$$= -e^{-in\Delta\sigma} \frac{1}{n} \left(\frac{q^{-n}}{1-q^{-n}} \right) \left(\prod_j (1-q^j)^{-1} \right) \tag{6.471}$$

and then we can use the same logarithm rules but with $n \to -n$. We still have to deal with the p^2 term which contributes $\sim \alpha_i \alpha_{-i}$ and this is the exact same calculation but without the n^{-2} nor exponential $\exp(-in\Delta\sigma)$ factor giving

$$\prod_j (1-q^j)^{-1} n \left(\frac{q^n}{1-q^n} + 1 \right). \tag{6.472}$$

Notice that the sum $\sum_n (q^n n/(1-q^n) + n)$ is divergent implying we need to regularize this sum. This can be done via zeta-regularization since

$$\sum_{n=1}^{\infty} n = \zeta(-1) = -\frac{1}{12} \tag{6.473}$$

(which of course is just *so* obvious...). Now for the q^n term, recall that $q = \exp(2\pi i\tau)$ (given below equation (7.2.5)) so,

$$\frac{nq^n}{1-q^n} = \frac{ne^{2\pi in\tau}}{1-e^{2\pi in\tau}} = -\frac{1}{2\pi i} \frac{1}{1-e^{2\pi in\tau}} \partial_\tau (1 - e^{2\pi in\tau}) \tag{6.474}$$

$$= -\frac{1}{2\pi i} \partial_\tau \log(1-q^n). \tag{6.475}$$

This will keep the sum finite. Notice though that we (well I) have been neglecting the L-movers ($\tilde{\alpha}$ terms) but this can easily be found by sending all $\tau + \sigma \to \tau - \sigma$ in all the previous expression (or if performed in the complex plane send $z \to \overline{z}$).

Since $\prod_j (1 - q^j)^{-1} = q^{1/24} |\eta(\tau)|^{-1}$, these are independent of the worldsheet coordinates (τ, σ), so we wil treat them as a normalization up to an arbitrary factor. But notice if we divide by the partition function $Z(\tau)$, then this cancels the $|\eta(\tau)|^{-2}$ we needed to cancel, which is exactly what Polchinskin does ($Z(\tau)$ as given in equation (7.2.9)). Then the partition function $Z(\tau) \propto |\eta(\tau)|^{-2}$ hold the information about the disconnected Feynman diagrams.

To get Polchinski's ϑ_1 result, consider what we got for the oscillator trace

$$\sum_{m=1}^{\infty} \log\Big(1 - q^m e^{-\Delta\sigma}\Big) + \log\Big(1 - q^m e^{-i\Delta\sigma}\Big). \tag{6.476}$$

Since we will take ∂_w's, lets add a term $\log(1 - q^m)$ to get the following sum

$$\sum_{m=1}^{\infty} \Big(\log\Big(1 - q^m e^{i\Delta\sigma}\Big) + \log\Big(1 - q^m e^{-i\Delta\sigma}\Big) + \log(1 - q^m)\Big) \tag{6.477}$$

$$= \log\left(\prod_{m=1}^{\infty} (1 - q^m e^{i\Delta\sigma})(1 - q^m e^{-i\Delta\sigma})(1 - q^m)\right). \tag{6.478}$$

Notice that this is exactly equation (7.2.34) from Polchinski or specifically equation (7.2.38c) for ϑ_1,

$$\vartheta_1(\nu, \tau) = 2 \exp\left(\frac{\pi i \tau}{4}\right) \sin(\pi\nu) \prod_{m=1}^{\infty} (1 - q^m)(1 - zq^m)(1 - z^{-1}q^m) \tag{6.479}$$

where $z = \exp(2\pi i \nu)$. Then we can notice the inside of the log becomes

$$\log(\vartheta_1(\nu, \tau)) - \log(\sin(\pi\nu)) + f(\tau) \tag{6.480}$$

where we have added $f(\tau)$ since $\partial_w f(\tau) = 0$ and choose $z = \exp(i\Delta\sigma)$ implying that $\nu = \Delta\sigma/2\pi$.

When we include the L-modes we get a $\log\Big(\overline{\vartheta}_1\Big)$ term giving an overall factor of $\log(|\vartheta_1|^2)$. Thus the propagator is

$$Z^{-1}(\tau)\langle X^\mu(w) X_\mu(z)\rangle = -\frac{\alpha' d}{2} \log\left(\left| \vartheta_1\left(\frac{z - w}{2\pi}, \tau\right) \right|^2\right) + \alpha' d \frac{\mathrm{Im}((z - w)^2)}{4\pi\tau_2} + f(\tau) \tag{6.481}$$

which is exactly equation (7.2.3) but written explicitly. Now we simply take two derivatives of ∂_w where the $f(\tau)$ term disappears and the only terms tricky is the $\log(x)$ term since the second order derivative of just the $\log(x)$ term becomes

$$-\frac{\alpha'}{2}\left(2\partial_w^2 \vartheta_1 \vartheta_1^{-1} - 2\partial_w \vartheta_1 \partial_w \vartheta_1 \vartheta_1^{-2}\right) \tag{6.482}$$

which if we pull out a common factor of $1/\vartheta_1$ we recover equation (7.2.16).

Polchinski 7.12 Repeat the previous exercise for the torus, showing in this case that the vacuum amplitude is modular-invariant only for $D = 26$ and $D' = 24$ (Shapiro, 1972).

Answer: This problem seems that we are just to finish Polchinski's caluclation of the torus contribution to the vacuum 1-loop amplitude for the closed bosonic string where there are two ways to do this. The long version is the beginning of chapter 7.3 that ends at equation (7.3.8) where we start with the general action in the partition function, include proper number of ghosts based on moduli and CKVs, and so on. Polchinski actually did this in [**?**] with the answer given in equation 27 of his paper (where his $f(q)$ is exactly our $\eta(q)$)[129]. But this problem is asking us to find Z_{T^2} from the point particle action[130].

The way I see this derivation is that we can view the closed string running in a loop (which generates a torus) is just a field theory on a circle that has length $2\pi t$, and the field theory is evolved by the Hamiltonian over a periodic Euclidean time of $2\pi i t$. As stated in problem 7.4, the time translations are given by the Hamiltonian and the position translations are given by the momentum operator. The partition function by the Coleman-Weinberg formula then, which just states that the vacuum expectation value is the trace over the exponential of the Hamiltonian, can be written as

$$Z_{T^2}(\tau) = \mathrm{Tr}\left[e^{2\pi i \mathrm{Re}(\tau) P} e^{-2\pi \mathrm{Im}(\tau) H}\right]. \tag{6.483}$$

To relate the point-particles to the closed string, we can relate the total energy (or on-shell condition) to that of the string,

$$p^2 + m^2 = \frac{2}{\alpha'}(L_0 + \tilde{L}_0). \tag{6.484}$$

We also have the momentum operator given as $P = L_0 + \tilde{L}_0$ and the Hamiltonian is $H = L_0 + \tilde{L}_0 - (c + \tilde{c})/24$. We also want to impose the physical constraint that the left and right-moving ground state oscillators should obey $L_0 = \tilde{L}_0$ which we can insert into the partition function via a delta function (equation (7.3.11))

$$\delta(L_0 - \tilde{L}_0) = \int_{-\infty}^{\infty} d\theta \, e^{2\pi i \theta (L_0 - \tilde{L}_0)}. \tag{6.485}$$

The partition function is then

$$Z_{T^2} = iV_D \int \frac{d^D p}{(2\pi)^D} \int_0^{\infty} \frac{dt}{2t} \int_{-\infty}^{\infty} \mathrm{Tr}' \left[\exp\Big(-4\pi t (L_0 + \tilde{L}_0)/\alpha'\Big) \exp\Big(2\pi i \theta (L_0 - \tilde{L}_0)\Big)\right]. \tag{6.486}$$

Let $\tau = \tau_1 + i\tau_2$ and $2t = \alpha' \tau_2$ and $\theta = \tau_1$. Then τ is in the upper-half-plane $\mathcal{H}_2$. This transforms the partition function as

$$Z_{T^2} = \frac{iV_D}{4} \int \frac{d^D p}{(2\pi)^D} \int_{H_2} \frac{d^2\tau}{\tau_2} \mathrm{Tr}' \left[\exp\Big(2\pi i \tau L_0 - 2\pi i \bar{\tau} \tilde{L}_0\Big)\right] \tag{6.487}$$

[129] The paper that Polchinski cites is [**?**] which does give the correct answer in equation (7) but in a completely different manner then what we are used to.

[130] A helpful textbook for this calculation that was used is [**?**].

where we only take the trace over the physical states. This is because if we did the calculation over the entire Hilbert space, we would over count and pick up a factor of $|\eta(\tau)|^{-4}$ which would need to be canceled by a factor of $|\eta(\tau)|^4$, but this skips that step. We can write the partition function in terms of $q's$ as

$$\mathrm{Tr}'\left[q^{L_0-c/24}\overline{q}^{\tilde{L}_0-\tilde{c}/24}\right]. \tag{6.488}$$

To evaluate the $q's$, these are over N_{in} states where N_{in} denotes the number of oscillators presents in the physical states at level n, and i is the spatial direction. They can be evaluated as

$$\sum_{i\in\mathcal{H}_{\text{phys}}} q^{\alpha' m_i^2} = \left(\prod_{i=1}^{D'}\prod_{n=1}^{\infty}\sum_{N_{in}=0}^{\infty}\right) q^{-1+\sum_{n=1}^{\infty} nN_{in}} \tag{6.489}$$

$$= q^{-1}\prod_{i=1}^{D'}\prod_{n=1}^{\infty}\frac{1}{1-q^n} \tag{6.490}$$

$$= q^{D'/24-1}\eta(\tau)^{-D'} \tag{6.491}$$

and similar for $\overline{q}$. Now lets insert a factor of

$$\exp\Big(-\alpha' p^2 s/2\Big). \tag{6.492}$$

Notice that this insertion into the partition function Z_{T^2} that we will pick up a pole structure when performing the D-dimensional momentum integral with a shifted pole structure of $\sim (\alpha'(s+\pi\tau)/2)^{-(D-24)/2}$ if $D'=24$. Notice if $D=26$ there is a pole structure that is simple but only at $D=26$ (which is what we wanted since it implies a propagator). We also require even power of $\eta(\tau)$ in order to recover modular invariance, thus if $D'=24$ and $D=26$, then we still have modular invariance. The partition function then becomes

$$Z_{T^2} = \frac{iV_{26}}{(4(2\pi)^{26}(\alpha')^{13})}\int_{\mathcal{H}_2}\frac{d^2\tau}{\tau_2^2}\frac{1}{\tau_{12}|\eta(\tau)|^{48}}. \tag{6.493}$$

6.10.5 Problem set 5: R & NS algebras & superstring amplitudes

Polchinski 10.4 Obtain the R and NS algebras (10.2.11) from the OPE.

Answer: There are three algebras here, the purely bosonic, purely fermionic, and then the mixed operator algebra. The bosonic algebra, $[L_m, L_n]$ was already performed in chapter 2 namely equation (2.6.18) by Polchinski himself which for clarity starts out as

$$[L_m, L_n] = \oint\frac{dz_2}{2\pi i}\mathrm{Res}_{z_1\to z_2} z_1^{m+1}T_B(z_1)z_2^{n+1}T_B(z_2) \tag{6.494}$$

which is given in equation (10.2.11a) or from chapter 2 equation (2.6.19) in volume 1 (the steps to get that algebra are the exact same we take here). The fermionic

algebra can be computed as (again, all thanks to the operator-state correspondence)

$$\{G_r, G_s\} = \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_1^{r+1/2} z_2^{s+1/2} T_F(z_2) T_F(z_2) \tag{6.495}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_1^{r+1/2} z_2^{s+1/2} \left(\frac{2c}{3z_{12}^3} + \frac{2}{z_{12}} T_B(z_2) \right) \tag{6.496}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_1^{r+1/2} z_2^{s+1/2} \left(+ \frac{c}{3} \partial_1^2 + 2T_B(z_2) \right) \frac{1}{z_{12}} \tag{6.497}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_2^{s+1/2} \left(\frac{c(r+1/2)(r-1/2)}{3} z_1^{r-3/2} + 2T_B(z_2) z_1^{r+1/2} \right) \frac{1}{z_{12}} \tag{6.498}$$

$$= \oint \frac{dz_2}{2\pi i} \left(\frac{c(4r^2-1)}{12} z_2^{s+1/2+r-3/2} + 2T_B(z_2) z_2^{s+1/2+r+1/2} \right) \tag{6.499}$$

$$= \oint \frac{dz_2}{2\pi i} \left(\frac{c(4r^2-1)}{12} z_2^{s+r-1} + 2T_B(z_2) z_2^{s+r+1} \right) \tag{6.500}$$

$$= \frac{c}{12}(4r^2-1)\delta_{r,-s} + 2L_{r+s} \tag{6.501}$$

where in equation 3 I used the OPE of equation (10.1.15c) and in the last line I used equation (2.6.29). The mixing of the fermionic and bosonic generators commutator can be found in a similar fashion as

$$[L_m, G_r] = \oint \frac{dz_2}{2\pi i} \mathrm{Res} z_1 \to z_2 T_B(z_1) T_F(z_2) \tag{6.502}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_1^{m+1} z_2^{r+1/2} \left(\frac{3}{2z_{12}^2} T_F(z_2) + \frac{1}{z_{12}} \partial_2 T_F(z_2) \right) \tag{6.503}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} z_1^{m+1} z_2^{r+1/2} \left(-\frac{3}{2} \partial_1 \frac{1}{z_{12}} T_F(z_2) + \frac{1}{z_{12}} \partial_2 T_F(z_2) \right) \tag{6.504}$$

$$= \oint \frac{dz_2}{2\pi i} \mathrm{Res}_{z_1 \to z_2} \left(+ \frac{3z_2^{r+1/2}}{2} \partial_1 z_1^{m+1} - \partial_2 z_2^{r+3/2+m} \right) \frac{T_F(z_2)}{z_{12}} \tag{6.505}$$

$$= \oint \frac{dz_2}{2\pi i} \left(\frac{3(m+1)}{2} z_2^{r+1/2+m} - z_2^{m+1+r-1/2}(r++m+3/2) \right) T_F(z_2) \tag{6.506}$$

$$= \oint \frac{dz_2}{2\pi i} \left(\frac{3(m+1)}{2} - (m+r+3/2) \right) z^{r+m+1/2} T_F(z_2) \tag{6.507}$$

$$= \frac{m-2r}{2} G_{m+r} \tag{6.508}$$

where integration by parts is scattered throughout and I used the OPE for $T_B T_F$ given in equation (10.1.15b).

Polchinski 10.13 (a) Verify the operator products (10.5.22)

Answer: Recall that the BRST current for the superstring is given in equation (10.5.21)

$$j_B = cT_B^m + \gamma T_F^m + bc\partial c + \frac{3}{4}(\partial c)\beta\gamma + \frac{1}{4}c(\partial\beta)\gamma - \frac{3}{4}c\beta\partial\gamma - b\gamma^2. \tag{6.509}$$

The OPEs of the ghost fields we need are given in equation (2.5.22) (the $\beta\gamma$ CFT) and equation (2.5.9) (the bc CFT). Then the OPE of the BRST current with $b(0)$ and $\beta(0)$ are (recall that the $\beta\gamma$ and bc CFT's are in different sectors and therefore do not omit singular OPEs)

$$j_B(z)b(0) \sim \frac{1}{z}T_B(z) - \frac{1}{z^2}bc(z) - \frac{1}{z}b\partial c(z) - \frac{3}{4z^2}\beta\gamma + \frac{1}{4z}(\partial\beta)\gamma(z) - \frac{3}{4z}\beta\partial\gamma(z) \tag{6.510}$$

$$\sim \cdots \frac{1}{z}\left(T_B^m(0) - (\partial b)c(0) - 2b\partial c(0) - \frac{1}{2}(\partial\beta)\gamma(0) - \frac{3}{2}\beta(\partial\gamma)(0)\right) \tag{6.511}$$

$$= \cdots \frac{1}{z}T_B(0) \tag{6.512}$$

where the entire bosonic energy-momentum tensor $T_B(z)$ is given in equation (101.21a). In a similar fashion

$$j_B(z)\beta(0) \sim \frac{1}{z}T_F^m(z) + \frac{3}{4z}(\partial c)\beta(z) + \frac{1}{4z}c(\partial\beta)(z) + \frac{3}{4z^2}c\beta(z) - \frac{2}{z}b\gamma(z) \tag{6.513}$$

$$\sim \frac{1}{z}\left(T_F^m(0) + \frac{3}{2}(\partial c)\beta(0) + c(\partial\beta)(0) - 2b\gamma(0)\right) \tag{6.514}$$

$$= \cdots + \frac{1}{z}T_F(0) \tag{6.515}$$

where the fermionic energy-momentum tensor was given in equation (10.1.21b).

(b) Using the Jacobi identity as in exercise 4.3, verify the nilpotence of Q_B.

Answer: Now to verify the nilpotency of the superstring BRST charge Q_B, this is *exactly* the same setup and steps as was performed in problem 4.3 (from homework 3). We will even use the same anticommutation relation

$$\{Q_B, b_m\} = L_m \tag{6.516}$$

but also the other *commutator* relation (equation (10.5.23))

$$[Q_B, \beta_r] = G_r. \tag{6.517}$$

Really, we just need the last relation and the way we did problem 4.3 but every time we see a commutator switch the brackets (and vice versa) to get the following relation

$$\{[Q_B, Q_B], \beta_r\} = -\{[\beta_r, Q_B], Q_B\} - \{[Q_B, \beta_r], Q_B\} \tag{6.518}$$

$$= -2\{G_r, Q_B\} \tag{6.519}$$

where I used equation 24 to simplify the commutators. Recalling that Q_B is conformally invariant, then the anticommuatator must vanish. This implies Q_B^2 has no c-ghost modes. Then Q_B is nilpotent iff the total central charge vanishes, $c = 0$.

Polchinski 12.8 Calculate the tree-level type II amplitude with one NS-NS boson and two R-R bosons.

Answer: The 3-point amplitude with type II superstrings (only closed strings) will be similar to the 3-point amplitude in the bosonic case in the sense that we will need three sets of ghost-antighost pairs within the amplitude, but also need to worry about fixing the ghost-anomaly on the sphere which is -2. The anomaly comes from the fact that we are now dealing with "supercoordinates" (since we are now in superspace[131]) with coordinates $(\theta, \overline{\theta})$. We will want to fix these two coordinates, and thus will need to have an overall charge of $(-2,-2)$ from the picture of the vertex operators ([5], page 110). The R-R vertex operator contributes $(-\frac{1}{2},-\frac{1}{2})$ each while the NS-NS vertex operator contributes $(-1,-1)$. The 3-point amplitude is then

$$\mathcal{A}^{(3)}_{S^2} = \langle c\tilde{c}\mathcal{V}_1^{(-\frac{1}{2},-\frac{1}{2})}(z_1)c\tilde{c}\mathcal{V}_2^{(-\frac{1}{2},-\frac{1}{2})}(z_2)c\tilde{c}\mathcal{V}_3^{(-1,-1)}(z_3)\rangle_{S^2} \tag{6.520}$$

where the vertex operators are given as

$$\mathcal{V}_i^{(-\frac{1}{2},-\frac{1}{2})}(z) = g_c e^{-\phi/2-\tilde{\phi}/2}\Theta_\alpha\tilde{\Theta}_\beta e^{ik_i\cdot X}F^{\alpha\beta} \tag{6.521}$$

$$\mathcal{V}_i^{(-1,-1)}(z) = g_c e^{-\phi-\tilde{\phi}}\psi^\mu\psi^\nu e^{ik_i\cdot X}\zeta_{\mu\nu} \tag{6.522}$$

where $F_{\alpha\beta}$ and $\zeta_{\mu\nu}$ are related to the momentum-polarization vectors as $2ig_{\text{YM}}k_{[\mu}e_{\nu]} \simeq F_{\mu\nu}$ (to convert to an effective action later on if we wanted to, page 116). Then, we will need the following correlation functions already computed for us: the 6-ghost amplitude (equation (6.3.4) from volume 1), the general bosonic vertex correlation function on a sphere (volume 1), the 3-point super-ghost amplitude on a sphere (equation (12.4.6a)), and the 3-point amplitude of two bosons and 1 fermion (equation (12.4.6c)) (of course all generalized to the closed string case). They are the following

$$\begin{aligned}\langle c(z_1)\tilde{c}(z_1)c(z_2)\tilde{c}(z_2)c(z_3)\tilde{c}(z_3)\rangle &= C^{gh}_{S^2}z_{12}z_{13}z_{23}\overline{z}_{12}\overline{z}_{13}\overline{z}_{23} \\ &= C^{gh}_{S^2}|z_{12}z_{13}z_{23}|^2\end{aligned} \tag{6.523}$$

$$\left\langle \prod_{i=1}^{3} :e^{ik_i\cdot X}: \right\rangle_{S^2} = C^X_{S^2}\prod_{i<j}|z_{ij}|^{\alpha' k_i\cdot k_j}\delta^{(10)}\left(\sum_i k_i\right) \tag{6.524}$$

$$\left\langle e^{-\phi/2-\tilde{\phi}/2}(z_1)e^{-\phi/2-\tilde{\phi}/2}(z_2)e^{-\phi-\tilde{\phi}}(z_3)\right\rangle_{S^2} = C^{sgh}_{S^2}|z_{12}|^{-1/2}|z_{13}z_{23}|^{-1} \tag{6.525}$$

$$\langle\Theta_\beta(z_1)\Theta_\gamma(z_2)\psi^\mu(z_3)\rangle_{S^2} = z_{12}^{-3/4}z_{13}^{-1/2}z_{23}^{-1/2}\frac{1}{\sqrt{2}}\left(C\Gamma^\mu\right)_{\alpha\gamma}. \tag{6.526}$$

If we then let all the constants be wrapped up (no pun intended since we are not doing compactification) into the constant C_{S^2} such that

$$C_{S^2} = C^{sgh}_{S^2}C^X_{S^2}C^{gh}_{S^2} \tag{6.527}$$

[131] Not to get confused with Wheeler's superspace.

then we find the 3-point amplitude to be

$$\mathcal{A}_{S^2}^{(3)} = C_{S^2}\delta^{(10)}\left(\sum_i k_i\right) F_1^{\alpha\beta}F_2^{\gamma\delta}\zeta_{\mu\nu}\left(C\Gamma^\mu\right)_{\alpha\gamma}\left(C\Gamma^\mu\right)_{\beta\delta}|z_{12}|^{-\frac{1}{2}-\frac{3}{2}-2}|z_{23}|^{-1-1+2}|z_{13}|^{-1-1+2}\prod_{i<j}|z_{ij}|^{\alpha' k_i\cdot k_j}. \tag{6.528}$$

Wonderfully, the z_{ij} dependence drops out (as we would have hoped) and we are left with

$$\mathcal{A}_{S^2}^{(3)} = C_{S^2}\delta^{(10)}\left(\sum_i k_i\right) F_1^{\alpha\beta}F_2^{\gamma\delta}\zeta_{\mu\nu}\left(C\Gamma^\mu\right)_{\alpha\gamma}\left(C\Gamma^\nu\right)_{\beta\delta}. \tag{6.529}$$

A Common Gaussian Integrals

Here are a list of common integrals we will *practically* encounter at what feels like every step of the way for the QM path integral.

$$\int d^n x \, \exp\left(-\frac{1}{2} x \cdot A \cdot x + J \cdot x\right) = \sqrt{\frac{(2\pi)^n}{\det A}} \exp\left(\frac{1}{2} J \cdot A^{-1} \cdot J\right) \tag{A.1}$$

$$\int d^n x \, \exp\left(-\frac{i}{2} x^T \cdot A \cdot x + i J^T \cdot x\right) = \sqrt{\frac{(2\pi i)^n}{\det A}} \exp\left(-\frac{i}{2} J^T \cdot A^{-1} \cdot J\right) \tag{A.2}$$

However, these are only valid for QM, not field-valued integrals, but there are of course analogous Gaussian integrals for when there are differential operators in the argument.

B Tips, Tricks, and Regularization of D-Dimensional Integrals

B.1 Feynman Trick

The "Feynman trick" is typically necessary for computing any loop diagram. For an integrand with two denominators, simply use

$$\frac{1}{AB} = \int_0^1 \frac{dx}{(xA + (1-x)B)^2} \tag{B.1}$$

If there are more than 2-denominators then the general Feynman trick is

$$\frac{1}{A_1 \cdots A_n} = (n-1)! \int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-2}} \frac{dx_{n-1}}{(A_1 x_{n-1} + A_2(x_{n-2} - x_{n-1}) + \cdots + A_n(1 - x_1))^n}. \tag{B.2}$$

As an *extremely* common example is that of 2-propagators, which corresponds to a loop diagram with some incoming leg-momenta p

$$\begin{aligned} \int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2} \frac{1}{(k-p)^2} &= \int \frac{d^4k}{(2\pi)^4} \int_0^1 \frac{dx}{(k^2 + x((k-p)^2 - k^2))^2} \\ &= \int_0^1 dx \int \frac{d^4k}{(2\pi)^4} \frac{1}{((k - xp)^2 - \Delta)^2} \end{aligned} \tag{B.3}$$

where $\Delta = -p^2 x(1-x)$, and we can see this way that if we shift $k \to k + xp$ which leaves the integral only dependent upon k^2 and the measure is invariant.

B.2 Schwinger Parameters

Schwinger parameters, which were used in (2.256) are based on the following mathematical identity which is valid for when $\text{Im}\, A > 0$

$$\frac{i}{A}\int_0^\infty ds\, e^{isA} \tag{B.4}$$

$$\left(\frac{i}{A}\right)^2 = \int_0^\infty ds\, s e^{isA} \tag{B.5}$$

and so on by computing derivatives with respect to A on each side. The Schwinger parameter has an added benefit that it can be interpreted as the proper times of the particles in question as they travel along their paths in the Feynman graphs. Another useful relation from Schwinger parameters is that for two denominators

$$\frac{1}{AB} = -\int_0^\infty ds \int_0^\infty dt\, e^{isA+itB} \tag{B.6}$$

which can be generalized for any powers of A and B as (after a bit of elbow grease)

$$\frac{1}{A^n B^m} = \frac{\Gamma(n+m)}{\Gamma(n)\Gamma(m)}\int_0^\infty ds \frac{s^{m-1}}{(A+Bs)^{n+m}}. \tag{B.7}$$

B.3 Dimensional Regularization

This is integration technique is single handedly the most important tool for modern physics developed by 't Hooft and Veltmann in 1972. Here, I will present some details on performing this class of D-dimensional *scalar* integrals, and then show how to reduce any even power of $k^\mu \cdots k^\nu$ to scalar integrals.

The general class of integrals we care about are of the form

$$\mathcal{I} = \int \frac{d^D k}{(2\pi)^D} \frac{f(k)}{(k^2 - \Delta + i\epsilon)} \tag{B.8}$$

where $f(k)$ is some function of the loop-momentum, and $+i\epsilon$ is only present for doing Lorentizan signature integrals, but if you Wick rotate, as seen in lectures (1.32), you can drop the Feynman prescription.

Let us first take a look at the D-dimensional measure. We can always parametrize the momentum into spherical coordinates and so $d^D k = d\Omega_D\, k^{D-1} dk$ where $d\Omega_D$ is the differential solid angle for an D-dimensional sphere. Explicitly, it can be written as

$$d\Omega_D = \sin^{D-2}(\phi_{D-1}) \sin^{D-3}(\phi_{D-2}) \cdots \sin(\phi_2) d\phi_1 \cdots d\phi_{D-1} \tag{B.9}$$

where ϕ_i's are the angles for the ith axis with $0 \leq \phi_1 < 2\pi$ and $0 \leq \phi_i < \pi$ for the rest. For example, in $D = 3$, we get the usual volume element for a 2D-surface $d\Omega_3 = d\cos\theta\, d\phi$.

The actual integration over $\int d\Omega_d$ gives back the $D-1$ dimensional areas of the D-balls with unit radius, namely

$$V_D = \Omega_D \int_0^R dr\, r^{D-1} = \Omega_D \frac{1}{D} R^D. \tag{B.10}$$

For non-integer dimensions, you have to not square the Gaussian function but instead put it to the power of D to derive a similar result

$$(\sqrt{\pi})^D = \left(\int_{-\infty}^{\infty} dx\, e^{-x^2} \right)^D = \int d\Omega_D \int_0^{\infty} dr\, r^{D-1} e^{-r^2} = \frac{1}{2}\Gamma(d/2) \int d\Omega_D \tag{B.11}$$

and thus

$$\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \tag{B.12}$$

Also, the Gamma function arises so many times, which is quite funny since it is just a special function that is the analytic continuation of the factorial operation $n!$ defined as $\Gamma(x) = (x-1)!$. But since it arises, it also is the object that gives us the typical ϵ^{-1} poles in dimensional-regularization since if you series expand around $x = 0$ you find

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon) \tag{B.13}$$

where γ_E is the Euler-Mascheroni constant. The D-dimensional Euclidean version of the class of integrals we are considering can be evaluated to be found to be

$$\int dk_E \frac{k_E^a}{(k_E^2 + \Delta)^b} = \Delta^{\frac{a+1}{2} - b} \frac{\Gamma\left(\frac{a+1}{2}\right)\Gamma\left(b - \frac{a+1}{2}\right)}{2\Gamma(b)} \tag{B.14}$$

which, the most general answer to any of our loop diagrams for even powers of k (since odd powers vanish) in loop momentum is

$$\frac{d^D k}{(2\pi)^D} \frac{k^{2a}}{(k^2 - \Delta)^b} = i(-1)^{a-b} \frac{1}{(4\pi)^{\frac{D}{2}}} \frac{1}{\Delta^{b-a-\frac{D}{2}}} \frac{\Gamma\left(a + \frac{D}{2}\right)\Gamma\left(b - a - \frac{D}{2}\right)}{\Gamma(b)\Gamma\left(\frac{D}{2}\right)}. \tag{B.15}$$

A very important note though is that if you are dealing with coupling constants in loop integrals, for example the electron charge coupling e has mass dimension

$$[e] = \frac{4-D}{2}, \tag{B.16}$$

then a very hand technique is to write e in a dimenionless form

$$e \to \mu^{\frac{4-D}{2}} e. \tag{B.17}$$

For example, if we have our loop above for $b = 2$ and $a = 0$, then the integral becomes

$$\int \frac{d^4 k}{(2\pi)^4} \frac{e^2}{(k^2 - \Delta + i\epsilon)^2} \to \mu^{4-D} \frac{ie^2}{(4\pi)^{\frac{D}{2}}} \Gamma\left(\frac{4-D}{2}\right) \left(\frac{1}{\Delta}\right)^{2-\frac{D}{2}} \tag{B.18}$$

which has the following expansion around $\epsilon \to 0$

$$\mu^{4-D} \frac{ie^2}{(4\pi)^{\frac{D}{2}}} \Gamma\left(\frac{4-D}{2}\right) \left(\frac{1}{\Delta}\right)^{2-\frac{D}{2}} = \frac{ie^2}{16\pi^2} \left(\frac{2}{\epsilon} + \ln \frac{4\pi e^{-\gamma_E} \mu^2}{\Delta} + \mathcal{O}(\epsilon) \right). \tag{B.19}$$

C Some Handy Dirac-Gamma Matrix Technology

C.1 Dirac Algebra

The Dirac matrices are such that they satisfiy $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, and the chirality matrix in even dimensions is typically defined as the product of all gamma matrices times a factor of i, like in 4-dimensions

$$\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 \tag{C.1}$$

which eventually leads to $\{\gamma^5, \gamma^\mu\} = 0$.

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